Group Project

Use Operations Research to Solve Problem
A case study of ICM 2024

Course: Operations Research (1001)

Tutor: Dr. Peimin CHEN

Ziheng LYU	2330036105
Zeming CHEN	2330036010
Jiaming LIU	2330036089
Xin WU	2330036159
Ang MA	2330036106

Contents

1 Problem Background **2**

1.1 Problem Statement . . . 3

1.2 Assumption . . . 3

2 Data Preparation **3**

3 Question 1 **5**

3.1 Basic Network Model . . . 5

3.2 General Description . . . 5

3.3 Modeling . . . 5

3.3.1 Objective Function . . . 5

3.3.2 Optimization Objective . . . 7

3.3.3 Dynamic Weight Adjustment . . . 7

3.3.4 Fitness . . . 8

3.3.5 Constraints . . . 8

3.3.6 Optimal Value . . . 8

4 Question 2 **8**

4.1 Modeling . . . 8

4.1.1 Lake Level Change . . . 8

4.1.2 Discretization . . . 9

4.1.3 Downstream relationship and controlling . . . 9

4.1.4 Optimal target of Lake level . . . 9

4.1.5 Objective function . . . 10

4.1.6 Optimization of inflows and outflows . . . 10

4.1.7 Gradient descent method . . . 10

4.1.8 Iteration process of the gradient descent . . . 11

5 Sensitive Analysis **11**

5.1 RMSE Model . . . 11

5.2 Multi-Linear Regression Model . . . 13

6 Advantages and promotion **14**

A Code **14**

Abstract

The Great Lakes are not only contained 20 percent of the world's freshwater, but they also have a tremendous impact on the economy, climate, and ecosystems of the United States and Canada. However, it's difficult to control the flow of water due to its complicated dynamics. Meanwhile, stakeholders are not satisfied with the current situation and the ever-changing climate has raised uncertainties. Hence, in this study, we analyzed the issue and found the optimal solution.

For task 1, we utilize **multi-objective optimization** with **genetic algorithm**. Initially, we create weighted sum of the multiple stakeholder objective function. Then, we dynamically adjust weight for each contraints in order to encounter each extreme conditions. Ultimately, under genetic algorithm, we found the optimal water levels for Great Lakes with 32 situations.

For task 2, we design an **optimal water level control system** for remaining the future optimal water level. Meanwhile, we utilize partial differential equation of water level changes with inflow and outflow (which related to a dam and locks in 2 rivers) and **discretization** to apply numerical analysis. With objective function, we minimize difference between current lake level and ideal lake level. Ultimately, **gradient descent** is used to obtain optimal water net flow.

For task 3, we conducted a sensitivity analysis on the model parameters and performed multiple linear regression on the changes in the parameters. We also achieved function evaluation and stability analysis through derivation.

For the further advanced development, we have already achieved advantages and promotion for our models' performance which may greatly benefit related projects.

Key Words: multi-objective optimization; genetic algorithm; optimal water level control system; discretization; gradient descent; sensitive analysis; linear regression

1 Problem Background

The Great Lakes (including Lake Superior, Lake Michigan-Huron, Lake St. Clair, Lake Erie, Lake Ontario) locate between the U.S. and Canada, mutually form the largest fresh water lake unity in the world, covering large amounts of cities which have different climates. It is widely applied in multiple fields, attracting many stakeholders. Therefore, it's vital to control the lake level to maximize values for all the stakeholders.

Lake level depends on inflow and outflow, which are affected by precipitation, evaporation, climate change, ice jams. In addition, two important dams, the Soo Locks in Sault Ste. Marie and the Moses-Saunders Dam in Cornwall, control the flow of water in the Great Lakes basin, adjusting the flow to prevent water levels from becoming overflow or lower-flow. Low water levels may obstruct shipping, while high water levels may cause flooding, affecting surrounding residents, the community and natural environment. Water level management may face continuous changes of precipitation, evaporation, meet different demands conflicts of stakeholders.

1.1 Problem Statement

The International Joint Commission (IJC) requests our team to assist in developing a model to manage the water levels of the Great Lakes and optimize the flow control strategies. It requires:

1. Determine the optimal water levels for the Great Lakes, considering demands of different stakeholders (the cost and values of each stakeholder may be different).
2. Design an algorithm to maintain the optimal lake level for the Great Lakes by defining and apply appropriate inflow and outflow data.
3. The sensitive analysis of algorithm to changes in environmental conditions (e.g., precipitation, winter snowfall, ice jams).

1.2 Assumption

To simplify the problem in order to replicate real live scenarios, we established several assumption. The following are the assumptions we made:

Assumption 1 The Lake Michigan and Lake Huron is viewed as one water body.

Assumption 2 Consider the historical average water level data as the ideal water level.

Assumption 3 The inflow from one lake is equal to the outflow from the next lake.

Assumption 4 The dam's outflow regulation operates without any delay.

Assumption 5 The optimal water level is consistent throughout the year

Assumption 6 Do not consider other runoff outside the Great Lakes

Assumption 7 The downstream river flow from the lake equal to the outflow controlled by the dam.

2 Data Preparation

Based on the dataset titled `Problem_D_Great_Lakes.xlsx` given by ICM (International Network Control Modelers), several issues should be considered. Firstly, the required data is dispersing across 11 different worksheets, making unified analysis difficult. What's more, each worksheet has a different structure, which complicates systematic data processing. Additionally, due to some worksheets contain placeholder values and missing values, accuracy of the analysis may be impacted.

In order to address these potential issues, the following approach was taken: To begin with, importing the `'openpyxl'` library to extract specific rows (rows 7 to 30) from each

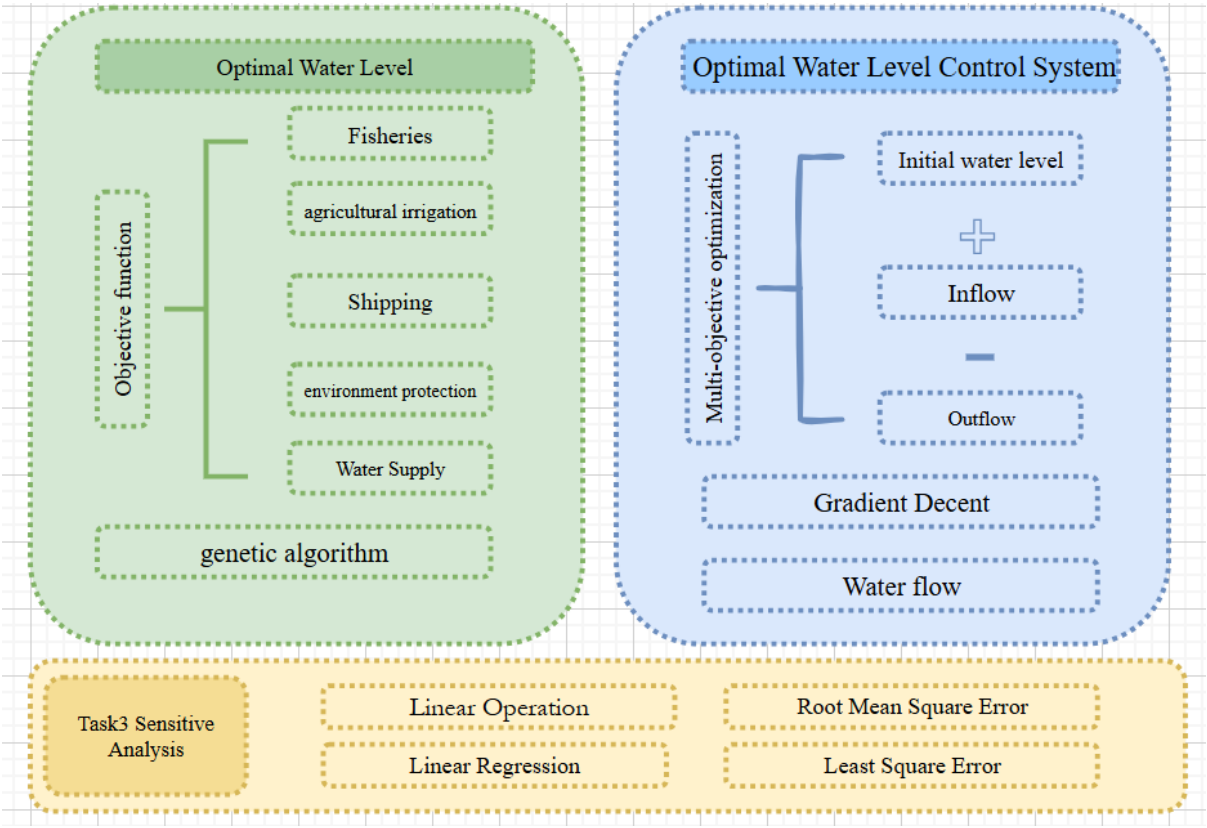


Figure 1: Work Diagram

worksheet and set the second row as the column header. Next, saving the extracted data as individual CSV files, employing the `'pandas'` library to merge all the CSV files into a single dataset, and adding a new column named "Source" to track the originating worksheet of each row. Finally, in order to standard the dataset, using `'NaN'` to replace all placeholder values, and using the means from each numeric column computing based on groups to fill all the missing values.

Through python, we can calculate the mean, maximum and minimum values of each lake. Additionally, the values of w_{ecology} and w_0 are based on the sources [1] [2] [3]. The results are shown in the table below:

	Lake Superior	Lake Michigan-Huron	Lake Erie	Lake Ontario
w_{ideal}	183.55	176.33	174.28	74.83
w_{min}	182.79	175.57	173.73	74.28
w_{max}	183.88	177.45	175.14	75.91
w_{ecology}	183.1	176.1	174.0	74.6
w_0	183.0	176.0	174.0	75.0

Table 1: Result

[1] https://www.glfc.org/pubs/lake_committees/common_docs/LS_2020_FishCommunitySurveyReport.pdf
[2] https://www.glfc.org/lake-superior-committee.php
[3] https://news.umich.edu/ambitious-new-pollution-targets-needed-to-protect-lake-erie-from-massive-dead-zone/

3 Question 1

3.1 Basic Network Model

The Great Lakes contains 5 lakes, which are Lake Superior, Lake Michigan, Lake Huron, Lake Erie and Lake Ontario. Following figure (Figure 2) showcases the specific location of each lake. Initially, we create network model for great lakes. By virtue of shared waters

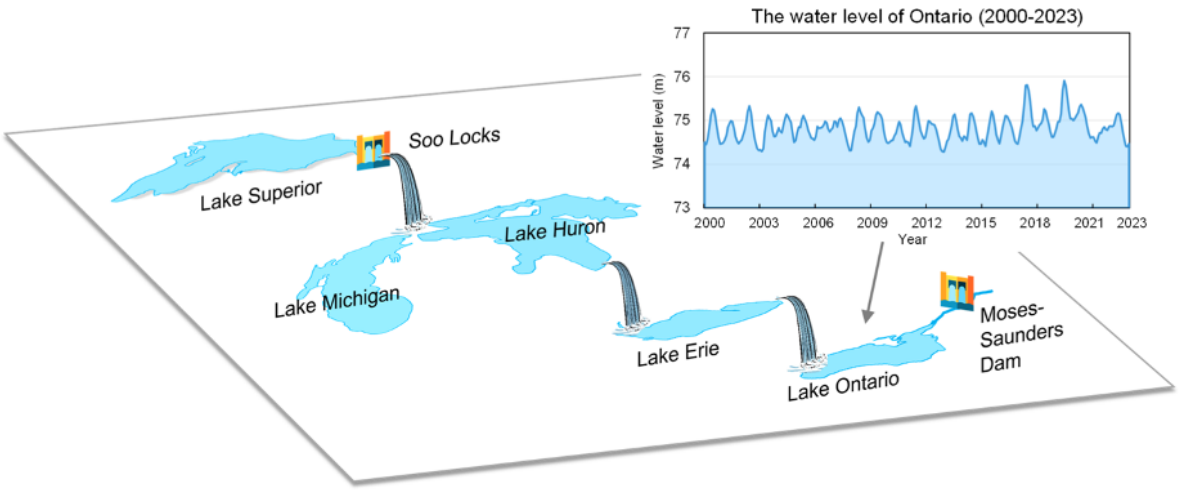


Figure 2: Lake Locations

between Michigan and Huron Lake, an aggregated node is illustrated. Hence, we have 4 lake nodes and 5 branches as follows (Figure 3):

3.2 General Description

In this section, we are going to determine the optimal water level of each lake. The above network model demonstrates that we can focus on only 4 lakes.

Meanwhile, previous assumption has already claimed that the historical average water level data as the ideal water level.

3.3 Modeling

This is model is a multi-objective optimization problem which is used a genetic algorithm to solve.

3.3.1 Objective Function

With 4 lakes $(L_1, L_2, \ldots, L_4)$, we denote each lake's optimal water level a w_i. The objective function for lakes is a weighted sum of multiple stakeholder objectives.

Each lake L_i's objective function is:

$$f_i(w_i) = \alpha_1 f_1(w_i) + \alpha_2 f_2(w_i) + \alpha_3 f_3(w_i) + \alpha_4 f_4(w_i) + \alpha_5 f_5(w_i) \tag{1}$$

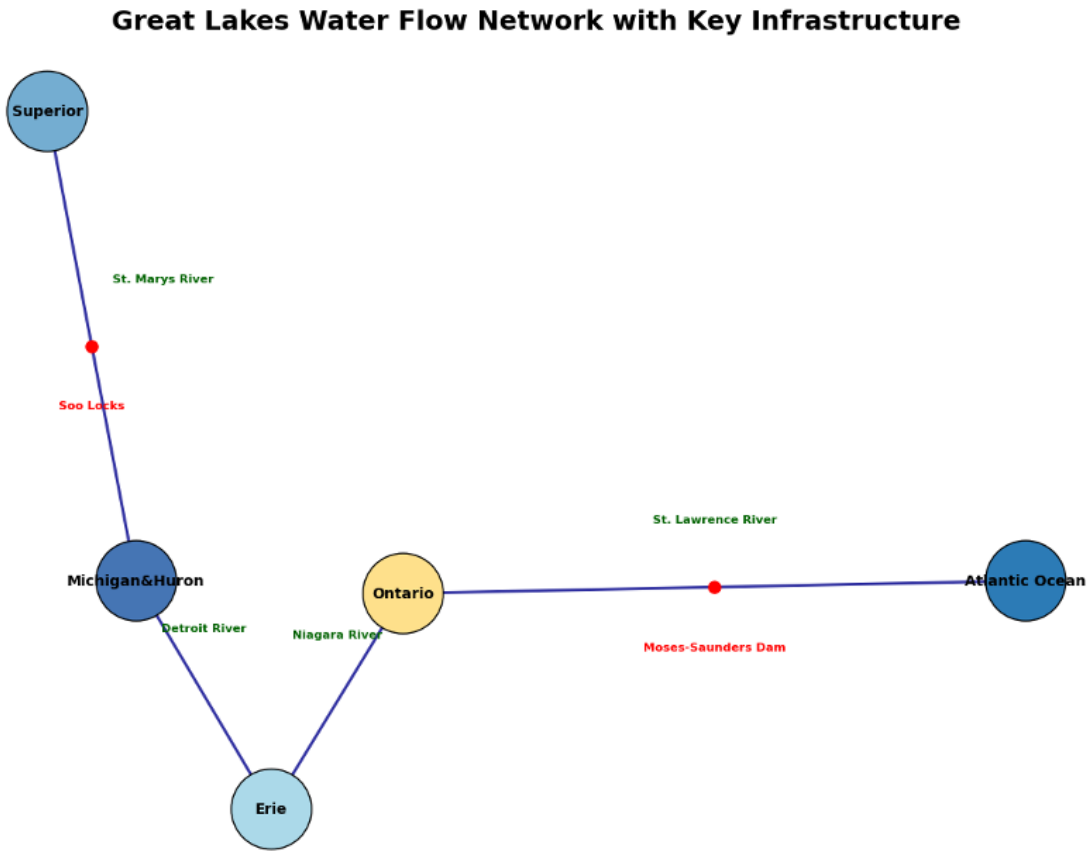


Figure 3: Great Lakes Water Flow Network with Key Infrastructure

where $f_1(w_i)$ is the fisheries objective function; $f_2(w_i)$ is the shipping objective function; $f_3(w_i)$ is the water supply objective function; $f_4(w_i)$ is the agricultural irrigation objective function; $f_5(w_i)$ is the environmental protection objective function.

For detailed explanation for $f_n(w_i)$ is demonstrated as follows

$f_1(w_i)$ Fisheries prefer water levels close to the ideal water level in fishing field w_0, so the objective function is a deviation as follows

$$f_1(w_i) = (w_i - w_0)^2 \tag{2}$$

$f_2(w_i)$ Shipping requires water levels to stay within the minimum $w_{\min}$ and maximum $w_{\max}$. The objective function is the cost of deviation from the boundaries

$$f_2(w_i) = \max(0, w_{\min} - w_i) + \max(0, w_i - w_{\max}) \tag{3}$$

$f_3(w_i)$ Water supply asks the water level as close as to the ideal water level w_{ideal}

$$f_4(w_i) = \max(0, w_{\min} - w_i) + \max(0, w_i - w_{\max}) \tag{4}$$

$f_5(w_i)$ Water level should stay close to ecological ideal water level w_{ecology}

$$f_5(w_i) = (w_i - w_{\text{ecology}})^2 \tag{5}$$

Hence, we have the ultimate objective function

$$f_i(w_i) = \alpha_1(w_i - w_0)^2 + \alpha_2(\max(0, w_{\min} - w_i) + \max(0, w_i - w_{\max}) + \alpha_3(w_i - w_{\text{ideal}})^2 \\ +\alpha_4(\max(0, w_{\min} - w_i)) + \alpha_4(\max(0, 2_{\min} - w_i))) + \max(0, w_i - w_{\max})) + \alpha_5(w_i - w_{\text{ecology}})^2 \tag{6}$$

3.3.2 Optimization Objective

For all 4 lakes, we intend to figure out the optimal water level combination $\mathbf{w} = [w_1, w_2, \ldots, w_4]$ by minimizing the total weighted objective function

$$F(\mathbf{w}) = \sum_{i=1}^{N} f_i(w_i) \tag{7}$$

3.3.3 Dynamic Weight Adjustment

Nevertheless, conditions illustrated above may altered frequently in reality. Therefore, we utilize dynamic weight adjustment to adjust weight α_i depending on the changes. The initial weights are $\alpha_1^0, \alpha_2^0, \alpha_3^0, \alpha_4^0, \alpha_5^0$. The adjustment rules are

Drought Water supply and agricultural irrigation raises

$$\alpha_3 = \alpha_3^0 + \Delta\alpha, \alpha_4 = \alpha_4^0 + \Delta\alpha, \alpha_5 = \alpha_5^0 - \Delta\alpha \tag{8}$$

Flooding Environment protection weight raises but water supply and agricultural irrigation falls

$$\alpha_3 = \alpha_3^0 + \Delta\alpha, \alpha_4 = \alpha_4^0 + \Delta\alpha, \alpha_5 = \alpha_5^0 - \Delta\alpha \tag{9}$$

Environment Policy Positive Change Environment protection increases

$$\alpha_5 = \alpha_5^0 + \Delta\alpha, \alpha_3 = \alpha_3^0 + \Delta\alpha, \alpha_2 = \alpha_2^0 - \Delta\alpha \tag{10}$$

Social Events Water supply and environment protection increases

$$\alpha_5 = \alpha_5^0 + \Delta\alpha, \alpha_3 = \alpha_3^0 + \Delta\alpha, \alpha_2 = \alpha_1^0 - \Delta\alpha \tag{11}$$

Economic Crisis Shipping and Fisheries falls

$$\alpha_2 = \alpha_2^0 - \Delta\alpha, \alpha_1 = \alpha_1^0 - \Delta\alpha, \alpha_4 = \alpha_4^0 + \Delta\alpha, \alpha_3 = \alpha_3^0 + \Delta\alpha \tag{12}$$

With the adjustment, we should do the normalization:

$$\alpha_i = \frac{\alpha_i}{\sum_{j=1}^{5} \alpha_j} \quad \text{for} \quad i = 1, 2, 3, 4, 5 \tag{13}$$

3.3.4 Fitness

Genetic Algorithm requires fitness as the negative of the objective function, which judges each combination performance

$$\text{Fitness}(\mathbf{w}) = -F(\mathbf{w}) = -\sum_{i=1}^{N} f_i\left(w_i\right) \tag{14}$$

3.3.5 Constraints

The water level w_i should within the following range

$$w_{min} \leq w_i \leq w_{max} \text{ for all } i = 1, 2, \ldots, N \tag{15}$$

If w_i is not in the range, it should be using clamping

$$w_i = min\left(max\left(w_i, w_{min}\right), w_{max}\right) \tag{16}$$

3.3.6 Optimal Value

With each iteration of each generation, a new set of solution will be generated. If the current solution is no longer dominated by any other solution and cannot be improved further (that is, the fitness is not significantly improved), it can be considered that the optimal solution has been reached or is close to the optimal solution. Above algorithm can be solved in python codes given.

Ultimately, the optimal for each lake can be drawn as follows and corresponding CSV file has be given in compressed file.

4 Question 2

In this question of scenario where we need to control dams to manage water flow, we develop a complete algorithms, which is used to adjust to ideal lake level based on the current lake level. According to the question 1, we have already got stable and optimal lake level H_{target} , and inflow and outflow through 2 dams. Therefore, we define the basic equation of water level change, discretize the water level change, set relationship between inflow and outflow. Then we define objective function, adjust current water level approximately equal to ideal lake level, use gradient descent to minimize the difference.

4.1 Modeling

4.1.1 Lake Level Change

Each lake level is affected by inflow and outflow, the basic equation of water level change is defined as:

$$\frac{dH_i(t)}{dt} = \text{Inflow}_i(t) - \text{Outflow}_i(t) \tag{17}$$

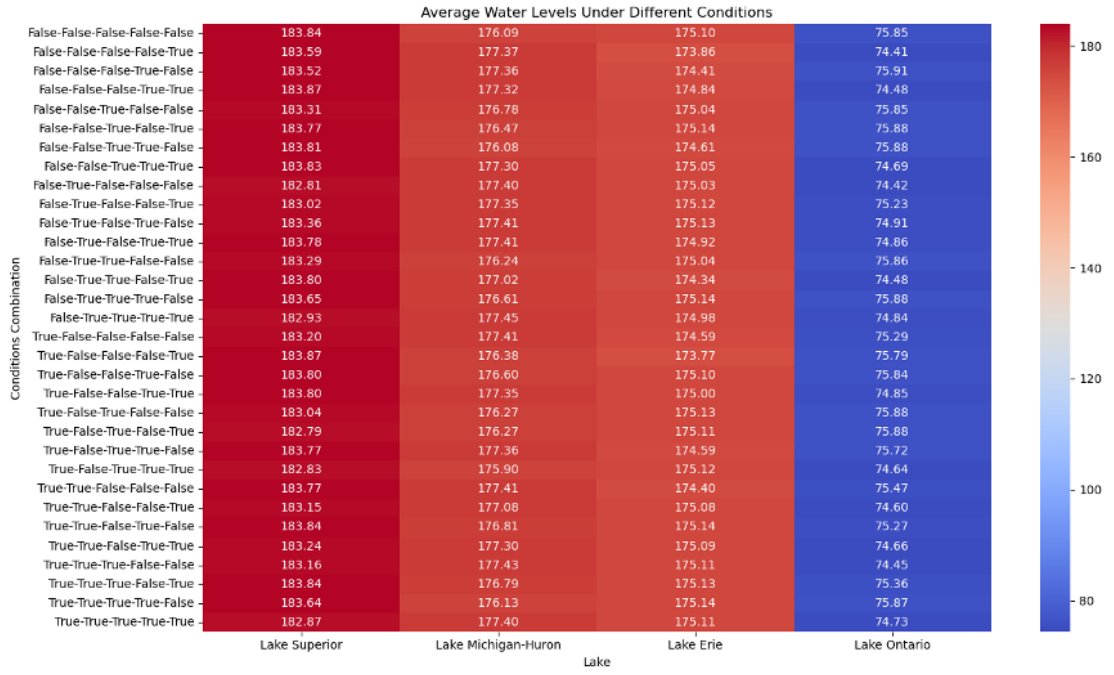


Figure 4: Average Water Levels Under Different Conditions

4.1.2 Discretization

In order to facilitate the practical application of numerical calculation, we use discretization method to transfer water level equation

$$H_i(t+1) = H_i(t) + (\text{Inflow}_i(t) - \text{Outflow}_i(t))\Delta t \tag{18}$$

where $H_i\,(t)$ is the water level of lake i at time $t\,(i \in \{A, B, C, D\})$. $\text{Inflow}_i(t)$ is the inflow rate to lake i. $\text{Outflow}_i(t)$ is the outflow rate from lake i. Δt is the time step, which in months.

4.1.3 Downstream relationship and controlling

There are no backflow flows in the process from Lake Superior (H_A) to Lake Michigan & Huron(H_B), and Lake Michigan & Huron (H_B) to Lake Erie (H_c), and Lake Erie (H_C) to Lake Ontario (H_D). The relationships of inflow and outflow are shown below

$$\begin{aligned} \text{Inflow}_B(t) &= \text{Outflow}_A(t) \\ \text{Inflow}_C(t) &= \text{Outflow}_B(t) \\ \text{Inflow}_D(t) &= \text{Outflow}_C(t) \end{aligned} \tag{19}$$

4.1.4 Optimal target of Lake level

Each target lake level is known:

- The target lake level of H_A is a The target lake level of H_B is b The target lake level of H_C is c The target lake level of H_D is d

4.1.5 Objective function

To adjust current lake level approximately close to target lake level, we define an objective function to balance and minimize the lake level difference. This function refers to the sum of the squares of the error between the lake's current level and the target lake level.

$$\mathcal{J} = \sum_{i=1}^{4} \sum_{t=1}^{T} \left(H_i(t) - H_i^{\text{target}} \right)^2 \tag{20}$$

where i is Lake Superior, Lake Marchigan-Hurion, Lake Erie, Lake Ontario; $H_i(t)$ is the water level of lake i at time t; H_i^{target} is the target water level for lake i; and T is the time steps

4.1.6 Optimization of inflows and outflows

Inflows and outflows are the optimized control variables which are required to be adjusted to minimize objective function.

4.1.7 Gradient descent method

Gradient Descent is an iterative optimization algorithm that updates parameters in the direction opposite to the gradient of the objective function to minimize it. It is advantaged at simplicity, wide applicability, and controllable convergence speed which is suitable to optimize inflows and outflows. We continuously update the flow rate by calculating the gradient of the inflow by calculating the objective function $\mathcal{J}$

Firstly, we need to calculate the partial derivatives of the objective function $\mathcal{J}$ with respect to each flow variable. According to the definition and purpose of the objective function, we can calculate the gradients of inflow and outflow separately.

$$\begin{aligned} \frac{\partial \mathcal{J}}{\partial \text{ Inflow}_i(t)} &= -2 \left(H_i(t) - H_{\text{target},i} \right) \cdot \frac{\partial \ H_i(t)}{\partial \text{ Inflow}_i(t)} \\ \frac{\partial \ \mathcal{J}}{\partial \text{ Outflow}_i(t)} &= -2 \left(H_i(t) - H_{\text{target},i} \right) \cdot \frac{\partial \ H_i(t)}{\partial \text{ Outflow}_i(t)} \end{aligned} \tag{21}$$

By Chain rule, we can get the partial derivatives by the equation of lake level change.

By the Chain Rule, the partial derivatives of the water level with respect to inflow and outflow, calculated by the equation of lake level change are

$$\begin{aligned} \frac{\partial \ H_i(t)}{\partial \text{ Inflow}_i(t)} &= \Delta t \\ \frac{\partial \ H_i(t)}{\partial \text{ Outflow}_i(t)} &= -\Delta t \end{aligned} \tag{22}$$

With updating rules

$$\begin{aligned} \text{Inflow}_i(t) &= \text{Inflow}_i(t) - \alpha \frac{\partial \mathcal{J}}{\partial\ \text{Inflow}_i(t)} \\ \text{Outflow}_i(t) &= \text{Outflow}_i(t) - \alpha \frac{\partial\ \mathcal{J}}{\partial\ \text{Outflow}_i(t)} \end{aligned} \tag{23}$$

where α is the learning rate, controlling the step size for each update.

4.1.8 Iteration process of the gradient descent

Initialization Set initial lake level $H_{\text{initial}} = x$, initial inflow $\text{Inflow}_i\,(t)$, and initial outflow $\text{Outflow}_i\,(t)$.

Calculate objective function Calculate the current objective function J, which refers to the sum of the squares of the error between the lake's current level and the target lake level.

Calculate the gradients Calculate the partial derivatives of the objective function about inflow and outflow.

Updating results Updating inflows and outflows by the gradients

Iteration Repeatedly process step 2 (calculate objective function) to 4 (Updating results), until the objective function completely convergences.

This algorithm includes three core mechanism: activate equation of lake level change, objective function to define the difference between current lake level and target lake level, iteration process by utilizing Gradient descent. It can systematically control water level timely and efficiently.

5 Sensitive Analysis

After tackling with the Question 4 by establishing the Network Model with the five lakes, our group is focusing on the Sensitivity Analysis of the Model.

5.1 RMSE Model

We previously defined the model of water level rise per unit time $v_{i,t}$ as

$$v_i = \frac{dh_i}{dt} = I_i - O_i + A_i \tag{24}$$

where h_i is the current water level of lake i; I_i, O_i, and A_i is the inflow, outflow, adjustment changing lake i water level.

And each lake is demanded for consumers according to the different uses and dependencies. Here we define the demand of water per unit water level

$$D_i = a_i \cdot (h_i - b_i)^2 + c_i \tag{25}$$

where a_i, b_i, c_i is the adjustment parameter to describe the demand (priority) of each lake.

And we maybe should also adjust the water level manually in order to make the maximum revenue of each water level, so there is also a corresponding adjustment function

$$A_i = k \cdot \left(h_i^{\text{target}} - h_i\right) \tag{26}$$

where h_i^{target} is the target water level of lake i, and k is the adjustment rate (coefficient).

Therefore, the question became to find the maximum profit (Total Revenue minus Total Expense), i.e., this is our first Linear Operation

$$z_1 : \ \max \sum_i \int_t (\text{Rev} \cdot D_i - \text{Exp} \cdot A_i) \ dt \tag{27}$$

And this question requires us to establish the Sensitivity Analysis, therefore, we choose the Root Mean Square Error (RMSE) of the water level control performance as the sensitivity index, which is a commonly used indicator to measure the prediction accuracy of the model

$$\text{RMSE}(\hat{\theta}) = \sqrt{\text{MSE}(\hat{\theta})} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{\theta}_i - \theta_i)^2} \tag{28}$$

where θ_i is the original value, $\hat{\theta}_i$ is the new value with slight change $\theta + \delta$.

Here we consider the three possible influence the environment variables: Inflow, Outflow, Adjustment. We establish the relationship between system performance (RMSE) and environmental condition variables. Assuming the model is a linear relationship:

$$\text{RMSE}\left(\alpha I_i + \beta O_i + \gamma A_i + \epsilon\right) \tag{29}$$

where α, β, γ are the variables need to determine, and ϵ is the random error.

By finite difference method, we estimate the sensitivity of coefficients a, b, c to RMSE. Assume we take $\delta = 0.01$ as an example:

Inflow α

$$\text{Sensitivity}_\alpha = \frac{\text{RMSE}(\alpha + \delta, \beta, \gamma) - \text{RMSE}(\alpha - \delta, \beta, \gamma)}{2\delta} \tag{30}$$

Outflow β

$$\text{Sensitivity}_\beta = \frac{\text{RMSE}(\alpha, \beta + \delta, \gamma) - \text{RMSE}(\alpha, \beta - \delta, \gamma)}{2\delta} \tag{31}$$

Adjustment γ

$$\text{Sensitivity}_\gamma = \frac{\text{RMSE}(\alpha, \beta, \gamma + \delta) - \text{RMSE}(\alpha, \beta, \gamma - \delta)}{2\delta} \tag{32}$$

And our goal is to minimize the sensitivity so that we can get a stable model

$$z_2 : \ \min \left\{ \text{RMSE} \left[\alpha \int_t I_i \ dt + \beta \int_t O_i \ dt + \gamma \int_t \left(h_i^{\text{target}} - h_i \right) \ dt + \int_t \epsilon \ dt \right] \right\} \tag{33}$$

with given historical data to train

$$\mathcal{J}^{\text{hist}} \left(I, O, h \right) = \frac{\partial h}{\partial I_i} \int_t I_i \ dt + \frac{\partial h}{\partial O_i} \int_t O_i \ dt + \frac{\partial h}{\partial A_i} \int_t A_i \ dt \tag{34}$$

After inference, we found that it is essential to make the estimation of the important functions: I_i, O_i, and A_i. Here we would like the use the Linear Regression with the method called Least Square Error (LSE) to fit the model.

5.2 Multi-Linear Regression Model

We consider a basic regression model where there is linear predictor variables and the regression function is linear. The model can be stated as follows:

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \varepsilon_i \tag{35}$$

where Y_i is the value of the response variable in the i-th trial, β_0, β_1, $\ldots$, β_n are parameters, X_i is a known constant, namely, the value of the predictor variable in the i-th trial, and ε_i is a random error term with mean $E\left\{\varepsilon_i\right\} = 0$ and variance $\sigma^2\left\{\varepsilon_i\right\} = \sigma^2$.

To find "precise" estimators of the regression parameters β_0, β_1, $\cdots$, β_n, we employ the method of least squares considers the deviation of Y_i from its expected value

$$Q = \sum_{i=1}^{m} \left(Y_1 - \beta_0 - \sum_{j=1}^{n} \beta_j X_{i,j} \right)^2 \tag{36}$$

And the values of $\beta_j, \ j = 1, 2, \ldots, n$ that minimize Q can be derived by differentiating with respect to β_j

$$\begin{aligned} \frac{\partial Q}{\partial \beta_0} &= -2 \sum_{i=1}^{m} \left(Y_i - \beta_0 - \sum_{j=1}^{n} \beta_j X_{i,j} \right) \\ \frac{\partial Q}{\partial \beta_k} &= -2 \sum_{i=1}^{m} X_{i,k} \left(Y_i - \beta_0 - \sum_{j=1}^{n} \beta_j X_{i,j} \right) \end{aligned} \tag{37}$$

where $k = 1, 2, \ldots, n$.

We then set these partial derivatives equal to zero, using b_j to denote the particular values of β_j that minimize Q

$$\begin{aligned} -2 \sum_{i=1}^{m} \left(Y_i - b_0 - \sum_{j=1}^{n} b_j X_{i,j} \right) &= 0 \\ -2 \sum_{i=1}^{m} X_{i,k} \left(Y_i - b_0 - \sum_{j=1}^{n} b_j X_{i,j} \right) &= 0 \end{aligned} \tag{38}$$

where $k = 1, 2, \ldots, n$.

Here we want to simply the calculation, we redefine the variables

$$\mathbf{X} = \begin{bmatrix} X_{11} & X_{12} & \cdots & X_{1n} \\ X_{21} & X_{22} & \cdots & X_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ X_{m1} & x_{m2} & \cdots & X_{mn} \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_m \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad \mathbf{e} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_m \end{bmatrix}$$

Therefore, in the least-square method, we can calculate the parameter vector by minimize the sum of mean squared error (MSE)

$$\beta = \arg\min_{\beta} Q\left(D, \beta\right) = \arg\min_{\beta} \sum_{i=1}^{m} \left(\beta \cdot \mathbf{X}_i - Y_i\right)^2 \tag{39}$$

Noe we do the partial derivative and let them be zero

$$\frac{\partial L\left(D, \beta\right)}{\partial \beta} = \frac{\partial \left(Y^{\mathsf{T}} Y - Y^{\mathsf{T}} X \beta - \beta^{\mathsf{T}} X^{\mathsf{T}} Y + \beta^{\mathsf{T}} X^{\mathsf{T}} \beta\right)}{\partial \beta} = -2X^{\mathsf{T}} Y + 2X^{\mathsf{T}} X \beta = 0 \tag{40}$$

i.e., we can get the Estimated Regression Function is

$$\mathbf{b} = \left(\mathbf{X}^{\mathsf{T}} \mathbf{X}\right)^{-1} \mathbf{X}^{\mathsf{T}} \mathbf{Y} \tag{41}$$

6 Advantages and promotion

Our model effectively utilize genetic algorithms to solve multi-objective optimization functions, completely considering the various demands of stakeholders, finally obtaining the optimal lake level. Optimal water level control system combines dynamic equation of water level changes and gradient descent to minimize the difference between the current lake level and ideal lake level, iterating process to allow water level to gradually approach the target level and ensure the stability and optimization of lake water levels.

To enhance model's adaptability to environmental conditions, sensitive analysis with RMSE is applied to maintain stable performance and creditability in complex natural environment. Least squares regression is also used to optimize the prediction of water flow, accurately manage inflow and outflow.

For further application and promotion, this model can be utilized in multi-lake system, reservoirs, wetlands, rivers and other water systems. Additionally, our models can be used in industrial field such as managing resources and energy, factory manipulation, chemical machine controlling.

Through deep reinforcement learning and more detailed inspection and optimization, the universality and efficiency of the model will also be improved for sustainable resources management.

A Code

[1]:
```
import numpy as np
import pandas as pd
import random
import matplotlib.pyplot as plt
import seaborn as sns
from itertools import product

# Parameters for each lake (data typically derived from observations or␣
 ↪historical records)
lake_info = {
    'Lake Superior': {'w_ideal': 183.35, 'w_min': 182.79, 'w_max': 183.88,␣
 ↪'w_ecology': 183.1, 'w_0': 183.0},
    'Lake Michigan-Huron': {'w_ideal': 176.33, 'w_min': 175.57, 'w_max': 177.45,␣
 ↪'w_ecology': 176.1, 'w_0': 176.0},
    'Lake Erie': {'w_ideal': 174.28, 'w_min': 173.73, 'w_max': 175.14,␣
 ↪'w_ecology': 174.0, 'w_0': 174.0},
    'Lake Ontario': {'w_ideal': 74.83, 'w_min': 74.28, 'w_max': 75.91,␣
 ↪'w_ecology': 74.6, 'w_0': 75.0}
}

# Initial weights (default weights)
initial_weights = {
    'f1': 0.1,  # Fisheries
    'f2': 0.1,  # Shipping
    'f3': 0.3,  # Water Supply
    'f4': 0.2,  # Agricultural Irrigation
    'f5': 0.3   # Environmental Protection
}

def adjust_weights(drought=False, flood=False, policy_change=False,␣
 ↪social_event=False, economic_crisis=False):
    """
    Dynamically adjusts weights based on different conditions
    """
    adjusted_weights = initial_weights.copy()

    if drought:
        adjusted_weights['f3'] += 0.1
        adjusted_weights['f4'] += 0.1
        adjusted_weights['f5'] -= 0.05

    if flood:
        adjusted_weights['f5'] += 0.15
        adjusted_weights['f3'] -= 0.1
        adjusted_weights['f4'] -= 0.1

    if policy_change:
```

```
        adjusted_weights['f5'] += 0.2
        adjusted_weights['f3'] += 0.15
        adjusted_weights['f2'] -= 0.05

    if social_event:
        adjusted_weights['f5'] += 0.1
        adjusted_weights['f3'] += 0.1
        adjusted_weights['f1'] -= 0.1

    if economic_crisis:
        adjusted_weights['f2'] -= 0.15
        adjusted_weights['f1'] -= 0.1
        adjusted_weights['f4'] += 0.1
        adjusted_weights['f3'] += 0.05

    # normalization
    total_weight = sum(adjusted_weights.values())
    for key in adjusted_weights:
        adjusted_weights[key] /= total_weight

    return adjusted_weights

def objective(w, lake, weights):
    w_ideal = lake_info[lake]['w_ideal']
    w_min = lake_info[lake]['w_min']
    w_max = lake_info[lake]['w_max']
    w_ecology = lake_info[lake]['w_ecology']
    w_0 = lake_info[lake]['w_0']

    # Stakeholder objective functions
    f1 = (w - w_0)**2  # Fisheries
    f2 = max(0, w_min - w) + max(0, w - w_max)  # Shipping
    f3 = (w - w_ideal)**2  # Water supply
    f4 = max(0, w_min - w) + max(0, w - w_max)  # Agricultural irrigation
    f5 = (w - w_ecology)**2  # Environmental protection

    # Weighted total objective function
    total_cost = weights['f1'] * f1 + weights['f2'] * f2 + weights['f3'] * f3 +␣
↪weights['f4'] * f4 + weights['f5'] * f5
    return total_cost


def fitness(individual, weights):
    total_fitness = 0
    for lake, water_level in zip(lake_info.keys(), individual):
        total_fitness += objective(water_level, lake, weights)
    return total_fitness
```

```
# Generate initial population
def generate_population(pop_size):
    population = []
    for _ in range(pop_size):
        individual = [random.uniform(lake_info[lake]['w_min'],␣
 ↪lake_info[lake]['w_max']) for lake in lake_info]
        population.append(individual)
    return population

# Selection operation (roulette wheel selection)
def selection(population, weights):
    fitness_values = [fitness(individual, weights) for individual in population]
    total_fitness = sum(fitness_values)
    prob = [f/total_fitness for f in fitness_values]
    selected = np.random.choice(range(len(population)), size=len(population),␣
 ↪p=prob)
    return [population[i] for i in selected]

# Single-point crossover
def crossover(parent1, parent2):
    crossover_point = random.randint(1, len(parent1) - 1)
    child1 = parent1[:crossover_point] + parent2[crossover_point:]
    child2 = parent2[:crossover_point] + parent1[crossover_point:]
    return child1, child2

# Mutation operation
def mutate(individual, mutation_rate):
    if random.random() < mutation_rate:
        mutation_point = random.randint(0, len(individual) - 1)
        new_value = random.uniform(lake_info[list(lake_info.
 ↪keys())[mutation_point]]['w_min'],
                                   lake_info[list(lake_info.
 ↪keys())[mutation_point]]['w_max'])
        individual[mutation_point] = new_value
    return individual

# Main Genetic Algorithm
def genetic_algorithm(pop_size, generations, mutation_rate, weights):
    population = generate_population(pop_size)

    for generation in range(generations):
        # Selection
        selected_population = selection(population, weights)

        # Crossover
        new_population = []
```

```
        for i in range(0, len(selected_population), 2):
            parent1, parent2 = selected_population[i], selected_population[i+1]
            child1, child2 = crossover(parent1, parent2)
            new_population.append(child1)
            new_population.append(child2)

        # Mutation
        for individual in new_population:
            individual = mutate(individual, mutation_rate)
        population = new_population

        # Print the best solution of the current generation
        best_individual = min(population, key=lambda ind: fitness(ind, weights))
        print(f"Generation {generation + 1} Best Solution: {best_individual}␣
↪Fitness: {fitness(best_individual, weights)}")

    # Return the best solution
    best_individual = min(population, key=lambda ind: fitness(ind, weights))
    return best_individual

def run_all_weight_combinations(pop_size, generations, mutation_rate):
    # Generate all combinations of conditions (32 in total)
    conditions = list(product([True, False], repeat=5))
    condition_labels = ['drought', 'flood', 'policy_change', 'social_event',␣
↪'economic_crisis']

    results = []
    for i, condition in enumerate(conditions):
        print(f"Simulation {i+1}/32 with conditions: {dict(zip(condition_labels,␣
↪condition))}")

        # Adjust weights based on current condition
        weights = adjust_weights(
            drought=condition[0],
            flood=condition[1],
            policy_change=condition[2],
            social_event=condition[3],
            economic_crisis=condition[4]
        )


        optimal_solution = genetic_algorithm(pop_size, generations,␣
↪mutation_rate, weights)

        result_entry = dict(zip(condition_labels, condition))
        result_entry.update({
            'Lake Superior': optimal_solution[0],
```

```
            'Lake Michigan-Huron': optimal_solution[1],
            'Lake Erie': optimal_solution[2],
            'Lake Ontario': optimal_solution[3]
        })
        results.append(result_entry)
        print(f"Optimal Solution for Simulation {i+1}: {optimal_solution}")
        print("-" * 50)


    df = pd.DataFrame(results)
    return df

# Set simulation parameters
pop_size = 30
generations = 500
mutation_rate = 0.1


all_results = run_all_weight_combinations(pop_size, generations, mutation_rate)

print("\nFinal DataFrame of All 32 Simulations:")
print(all_results)

all_results.to_csv('simulation_results.csv', index=False)
def plot_water_levels(df):
    lakes = ['Lake Superior', 'Lake Michigan-Huron', 'Lake Erie', 'Lake Ontario']


    melted_df = df.melt(id_vars=['drought', 'flood', 'policy_change',␣
↪'social_event', 'economic_crisis'],
                        value_vars=lakes,
                        var_name='Lake',
                        value_name='Water Level')


    melted_df['Lake'] = pd.Categorical(melted_df['Lake'], categories=lakes,␣
↪ordered=True)

    # Boxplot
    plt.figure(figsize=(12, 6))
    sns.boxplot(x='Lake', y='Water Level', data=melted_df)
    plt.title('Water Levels Across Lakes (32 Conditions)')
    plt.ylabel('Water Level (m)')
    plt.xlabel('Lake')
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()
```

```
    # Heatmap: Group by conditions and calculate mean water levels
    condition_cols = ['drought', 'flood', 'policy_change', 'social_event',␣
↪'economic_crisis']
    avg_levels = df.groupby(condition_cols)[lakes].mean().reset_index()
    avg_levels_melted = avg_levels.melt(id_vars=condition_cols, var_name='Lake',␣
↪value_name='Avg Water Level')


    pivot_df = avg_levels_melted.pivot_table(index=['drought', 'flood',␣
↪'policy_change', 'social_event', 'economic_crisis'],
                                             columns='Lake', values='Avg Water␣
↪Level')
    pivot_df = pivot_df[lakes]


    plt.figure(figsize=(14, 8))
    sns.heatmap(pivot_df, annot=True, fmt='.2f', cmap='coolwarm')
    plt.title('Average Water Levels Under Different Conditions')
    plt.xlabel('Lake')
    plt.ylabel('Conditions Combination')
    plt.tight_layout()
    plt.show()


plot_water_levels(all_results)
```

```
Simulation 1/32 with conditions: {'drought': True, 'flood': True,
'policy_change': True, 'social_event': True, 'economic_crisis': True}
Generation 1 Best Solution: [182.79996059552604, 176.26184007562597,
174.88120053756893, 74.5074730402007] Fitness: 0.7012354475460643
Generation 2 Best Solution: [183.09860735905568, 176.2848891294817,
174.18725052865952, 74.5074730402007] Fitness: 0.08880677383813834
Generation 3 Best Solution: [183.5755610512445, 176.90478462194642,
173.88540916411216, 75.2766240175159] Fitness: 0.8994669251772045
Generation 4 Best Solution: [183.3906508259968, 176.02311838877208,
174.56256846584606, 75.16772128317885] Fitness: 0.453205442357621
Generation 5 Best Solution: [182.81324702767753, 175.77808783254125,
174.68576094394137, 75.01065607928942] Fitness: 0.7113400871926696
Generation 6 Best Solution: [183.59806290196133, 175.59849127060323,
174.85079071351726, 75.01065607928942] Fitness: 1.0165752002530986
Generation 7 Best Solution: [183.47419016144892, 177.11173287792977,
174.71950908525255, 75.04107935951976] Fitness: 1.1668848795684705
Generation 8 Best Solution: [183.55785935203866, 176.85449306056083,
174.85079071351726, 75.04107935951976] Fitness: 1.0175762802544857
Generation 9 Best Solution: [183.3723521698123, 176.33001477913862,
175.0405996613251, 75.5101950082136] Fitness: 1.3520038261432732
```

```
Generation 10 Best Solution: [183.59806290196133, 177.4025617228677,
173.74633022820876, 74.72133859032468] Fitness: 1.4822131459979229
Generation 11 Best Solution: [183.3723521698123, 177.06814053837047,
174.8689515730518, 74.72133859032468] Fitness: 1.1271954639657402
Generation 12 Best Solution: [182.9654933776168, 177.06814053837047,
174.0658733175603, 75.5101950082136] Fitness: 1.3084140857345044
Generation 13 Best Solution: [182.9654933776168, 176.79318650709598,
175.0405996613251, 75.01065607928942] Fitness: 1.1698939986762953
Generation 14 Best Solution: [183.59806290196133, 177.34986571069052,
174.85079071351726, 75.01065607928942] Fitness: 1.7697017092907197
Generation 15 Best Solution: [182.9654933776168, 177.34986571069052,
174.35237349060998, 75.5101950082136] Fitness: 1.8184001297479842
Generation 16 Best Solution: [183.59806290196133, 177.28157805282152,
175.0405996613251, 75.01065607928942] Fitness: 1.9039842812739498
Generation 17 Best Solution: [183.47087145969368, 177.28157805282152,
175.0405996613251, 75.01065607928942] Fitness: 1.8383243126252822
Generation 18 Best Solution: [183.72346404104786, 177.34986571069052,
173.97107964179898, 75.5101950082136] Fitness: 1.9486185211772011
Generation 19 Best Solution: [183.59806290196133, 177.34986571069052,
174.85079071351726, 74.59317372482074] Fitness: 1.6836419668381588
Generation 20 Best Solution: [183.41477378511556, 177.34986571069052,
175.01251258482046, 75.51040549516539] Fitness: 2.399291792550132
Generation 21 Best Solution: [183.59806290196133, 177.34986571069052,
175.0405996613251, 75.5101950082136] Fitness: 2.5274858441637256
Generation 22 Best Solution: [183.41477378511556, 177.34986571069052,
175.01251258482046, 74.50017218596962] Fitness: 1.838833678831891
Generation 23 Best Solution: [183.53756420726916, 175.6491830392437,
175.01251258482046, 75.8141859817019] Fitness: 2.15586416667737
Generation 24 Best Solution: [183.41477378511556, 176.5332377753402,
175.01251258482046, 75.8141859817019] Fitness: 1.9045125231284148
Generation 25 Best Solution: [183.41477378511556, 176.5332377753402,
175.0405996613251, 75.8141859817019] Fitness: 1.9471995568216867
Generation 26 Best Solution: [182.89404717151302, 175.94202546591498,
175.01251258482046, 75.8141859817019] Fitness: 1.9604735075475732
Generation 27 Best Solution: [183.72346404104786, 177.28157805282152,
175.0405996613251, 74.56162138447083] Fitness: 1.9154505286220602
Generation 28 Best Solution: [183.20983724262527, 177.34986571069052,
175.0405996613251, 74.56162138447083] Fitness: 1.8378938999448287
Generation 29 Best Solution: [183.41477378511556, 176.5332377753402,
175.01251258482046, 75.57157312971776] Fitness: 1.4839412494589086
Generation 30 Best Solution: [183.72346404104786, 176.5332377753402,
175.01251258482046, 75.57157312971776] Fitness: 1.661633369260461
Generation 31 Best Solution: [183.72346404104786, 176.5332377753402,
175.01251258482046, 75.15778080311345] Fitness: 1.1771343617841936
Generation 32 Best Solution: [183.41477378511556, 176.29088518861576,
175.0405996613251, 74.51259738473158] Fitness: 0.7939197680868357
Generation 33 Best Solution: [183.41477378511556, 176.29088518861576,
175.0405996613251, 74.56162138447083] Fitness: 0.7817710089583497
```

Generation 34 Best Solution: [183.85152199441052, 177.34986571069052,
174.42843149201457, 74.32660966588617] Fitness: 1.6367848148349466
Generation 35 Best Solution: [183.41477378511556, 177.28157805282152,
175.01251258482046, 74.32660966588617] Fitness: 1.7907753214350328
Generation 36 Best Solution: [183.41477378511556, 177.34986571069052,
175.0405996613251, 74.7025043298604] Fitness: 1.8536583939684936
Generation 37 Best Solution: [183.41477378511556, 177.34986571069052,
175.0405996613251, 74.60292162599077] Fitness: 1.858601171017355
Generation 38 Best Solution: [183.41477378511556, 177.34986571069052,
175.10539606797673, 74.60292162599077] Fitness: 1.9622381579482417
Generation 39 Best Solution: [183.85152199441052, 177.34986571069052,
175.10539606797673, 74.60292162599077] Fitness: 2.261583800640931
Generation 40 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.62880024043871] Fitness: 2.1550274516452457
Generation 41 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.62880024043871] Fitness: 2.1550274516452457
Generation 42 Best Solution: [183.85152199441052, 177.34986571069052,
173.93334910088333, 74.60292162599077] Fitness: 1.499339537247333
Generation 43 Best Solution: [183.50404742536577, 177.34986571069052,
173.93334910088333, 74.60292162599077] Fitness: 1.2345928791963665
Generation 44 Best Solution: [183.62071792049377, 177.34986571069052,
175.0405996613251, 74.60292162599077] Fitness: 1.9590121831543812
Generation 45 Best Solution: [183.41477378511556, 177.34986571069052,
175.0405996613251, 74.60292162599077] Fitness: 1.858601171017355
Generation 46 Best Solution: [183.41477378511556, 177.34986571069052,
175.10539606797673, 74.37290512595472] Fitness: 2.0386377930268997
Generation 47 Best Solution: [183.43740704266332, 177.34986571069052,
175.10539606797673, 74.37290512595472] Fitness: 2.0461167427098403
Generation 48 Best Solution: [183.41477378511556, 177.34986571069052,
175.10539606797673, 74.37290512595472] Fitness: 2.0386377930268997
Generation 49 Best Solution: [183.43740704266332, 177.34986571069052,
174.337992104172, 74.42110794081839] Fitness: 1.2573468562890353
Generation 50 Best Solution: [183.41477378511556, 177.34986571069052,
174.337992104172, 75.48781842480368] Fitness: 1.7487163130618577
Generation 51 Best Solution: [183.41477378511556, 177.34986571069052,
175.0405996613251, 75.50448712937299] Fitness: 2.43360065594639
Generation 52 Best Solution: [183.85152199441052, 176.68282254053253,
175.10539606797673, 75.8141859817019] Fitness: 2.4488303878940756
Generation 53 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.96189787035283] Fitness: 2.219942881785773
Generation 54 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.87748993629772] Fitness: 2.1855005035654305
Generation 55 Best Solution: [183.41477378511556, 177.34986571069052,
175.0405996613251, 75.20297198868701] Fitness: 2.0862227058765197
Generation 56 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 75.20297198868701] Fitness: 2.3855683485692087
Generation 57 Best Solution: [183.85152199441052, 176.19735627822885,
175.0405996613251, 75.8141859817019] Fitness: 2.1648029516997833

```
Generation 58 Best Solution: [183.41477378511556, 177.15148294465433,
175.10539606797673, 75.8141859817019] Fitness: 2.7069116333123375
Generation 59 Best Solution: [183.20811956943177, 177.15148294465433,
175.10539606797673, 75.78338362613529] Fitness: 2.620250510055251
Generation 60 Best Solution: [183.85152199441052, 177.34986571069052,
174.20774421604688, 75.8141859817019] Fitness: 2.560725739132111
Generation 61 Best Solution: [183.85152199441052, 177.34986571069052,
174.20774421604688, 75.8141859817019] Fitness: 2.560725739132111
Generation 62 Best Solution: [183.41477378511556, 177.15148294465433,
174.92492582638874, 75.78338362613529] Fitness: 2.377168128786665
Generation 63 Best Solution: [183.34408324265382, 177.34986571069052,
174.20774421604688, 75.78338362613529] Fitness: 2.184687380488976
Generation 64 Best Solution: [183.46560816555012, 177.34986571069052,
175.0405996613251, 75.36655517728065] Fitness: 2.2733747001702636
Generation 65 Best Solution: [182.91607053652191, 176.80538561366936,
175.0405996613251, 75.8141859817019] Fitness: 2.2133245327357702
Generation 66 Best Solution: [183.46560816555012, 177.34986571069052,
174.92492582638874, 74.41449228268584] Fitness: 1.7653792851168235
Generation 67 Best Solution: [182.91607053652191, 177.34986571069052,
175.0405996613251, 74.41449228268584] Fitness: 1.9725295264934781
Generation 68 Best Solution: [182.91607053652191, 177.15148294465433,
174.92492582638874, 75.8141859817019] Fitness: 2.494215087225013
Generation 69 Best Solution: [183.85152199441052, 176.78019413694372,
175.10539606797673, 75.8141859817019] Fitness: 2.5336563505212712
Generation 70 Best Solution: [183.71620891668195, 176.49820681705867,
175.0284899888706, 75.8141859817019] Fitness: 2.0827117602577765
Generation 71 Best Solution: [183.71620891668195, 176.49820681705867,
175.10539606797673, 75.3132691485305] Fitness: 1.4474804493237616
Generation 72 Best Solution: [182.82858502353517, 177.15148294465433,
175.0405996613251, 75.3132691485305] Fitness: 1.9580365774473387
Generation 73 Best Solution: [183.85152199441052, 177.34986571069052,
174.2080051650687, 74.45459602768122] Fitness: 1.5055798818431936
Generation 74 Best Solution: [183.85152199441052, 177.34986571069052,
174.2080051650687, 74.62705056904012] Fitness: 1.463941461583789
Generation 75 Best Solution: [182.91607053652191, 177.34986571069052,
173.94777691336455, 75.78338362613529] Fitness: 2.28818787424951
Generation 76 Best Solution: [183.85152199441052, 176.0767422746464,
175.0405996613251, 75.8141859817019] Fitness: 2.1826436354955474
Generation 77 Best Solution: [183.85152199441052, 177.34986571069052,
173.79203670122433, 75.31612762934314] Fitness: 1.9062370757488498
Generation 78 Best Solution: [183.47682688286073, 177.34986571069052,
173.79203670122433, 75.8141859817019] Fitness: 2.3838295103870406
Generation 79 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.83340308818622] Fitness: 2.172366665296769
Generation 80 Best Solution: [183.85152199441052, 176.5556963498178,
175.0405996613251, 75.49069083768943] Fitness: 1.7205547897334488
Generation 81 Best Solution: [183.85152199441052, 176.5556963498178,
174.54246439734055, 75.8141859817019] Fitness: 1.7025357018276832
```

```
Generation 82 Best Solution: [183.1651165819798, 177.34986571069052,
175.0405996613251, 74.78268723002832] Fitness: 1.8378045370857687
Generation 83 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.78268723002832] Fitness: 2.161379133245066
Generation 84 Best Solution: [183.85152199441052, 176.5556963498178,
175.0405996613251, 75.49069083768943] Fitness: 1.7205547897334488
Generation 85 Best Solution: [183.1651165819798, 176.06904073455038,
175.0405996613251, 75.49069083768943] Fitness: 1.3227010775563368
Generation 86 Best Solution: [183.1651165819798, 176.06904073455038,
175.0405996613251, 75.78338362613529] Fitness: 1.8020669199692438
Generation 87 Best Solution: [183.85152199441052, 176.55942706503035,
174.6683992696751, 75.78338362613529] Fitness: 1.746166226409489
Generation 88 Best Solution: [183.85152199441052, 176.55942706503035,
174.6683992696751, 75.49069083768943] Fitness: 1.266800383996582
Generation 89 Best Solution: [183.85152199441052, 176.55942706503035,
174.6683992696751, 75.49069083768943] Fitness: 1.266800383996582
Generation 90 Best Solution: [183.1651165819798, 176.55942706503035,
175.0405996613251, 75.78338362613529] Fitness: 1.8784835974950949
Generation 91 Best Solution: [183.1651165819798, 176.07353962052625,
175.0405996613251, 75.8141859817019] Fitness: 1.859882654687313
Generation 92 Best Solution: [182.79126388809985, 177.34986571069052,
175.0405996613251, 74.88635500300218] Fitness: 2.0284879914175127
Generation 93 Best Solution: [182.818535673002, 177.34986571069052,
175.0405996613251, 74.50354955724647] Fitness: 1.9997395007692613
Generation 94 Best Solution: [182.79126388809985, 177.34986571069052,
175.0405996613251, 74.50354955724647] Fitness: 2.019769468248714
Generation 95 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.97671715492551] Fitness: 2.2272502501592473
Generation 96 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.65354308505773] Fitness: 2.153309803205587
Generation 97 Best Solution: [182.95485728522004, 175.88213023082173,
175.0405996613251, 75.8141859817019] Fitness: 2.0029198989512342
Generation 98 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.65022850776792] Fitness: 2.1534790220562403
Generation 99 Best Solution: [183.85152199441052, 177.34986571069052,
174.90269381754928, 74.65022850776792] Fitness: 1.95686940149824
Generation 100 Best Solution: [183.85152199441052, 175.7601260985381,
174.24043230186896, 75.78338362613529] Fitness: 1.6025800592204718
Generation 101 Best Solution: [183.85152199441052, 176.2826390533584,
174.24043230186896, 75.8141859817019] Fitness: 1.4806737445417246
Generation 102 Best Solution: [183.85152199441052, 176.2826390533584,
174.24043230186896, 75.8141859817019] Fitness: 1.4806737445417246
Generation 103 Best Solution: [183.85152199441052, 176.96932876018255,
175.0405996613251, 75.52039157319831] Fitness: 2.1448262426112255
Generation 104 Best Solution: [183.85152199441052, 175.93289939391894,
173.9911105740633, 75.8141859817019] Fitness: 1.5603161319415535
Generation 105 Best Solution: [183.85152199441052, 175.93289939391894,
173.9911105740633, 75.8141859817019] Fitness: 1.5603161319415535
```

Generation 106 Best Solution: [183.85152199441052, 176.63985747982343,
174.04864933238113, 75.8141859817019] Fitness: 1.6248738817394341
Generation 107 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.31892702125116] Fitness: 2.2654147585030042
Generation 108 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.31892702125116] Fitness: 2.2654147585030042
Generation 109 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.31892702125116] Fitness: 2.2654147585030042
Generation 110 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.31892702125116] Fitness: 2.2654147585030042
Generation 111 Best Solution: [183.85152199441052, 176.71939528995614,
175.0405996613251, 74.31892702125116] Fitness: 1.3885595592810696
Generation 112 Best Solution: [183.85152199441052, 176.71939528995614,
175.0405996613251, 74.31892702125116] Fitness: 1.3885595592810696
Generation 113 Best Solution: [183.85152199441052, 175.62108883733546,
175.0405996613251, 74.31892702125116] Fitness: 1.4885203891250307
Generation 114 Best Solution: [183.85152199441052, 177.34986571069052,
174.64660722366742, 74.31892702125116] Fitness: 1.7901900120113907
Generation 115 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.76654731048687] Fitness: 2.158807335650861
Generation 116 Best Solution: [183.85152199441052, 176.2028335670804,
175.0405996613251, 74.31892702125116] Fitness: 1.1779962240879105
Generation 117 Best Solution: [183.4471522472166, 177.34986571069052,
175.1237745459736, 74.43381378089877] Fitness: 2.0512532589633716
Generation 118 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.64883862324376] Fitness: 2.15355558428484
Generation 119 Best Solution: [183.85152199441052, 177.3463535453137,
173.77621341906124, 75.8141859817019] Fitness: 2.6634548222284034
Generation 120 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.67458286084617] Fitness: 2.152674869887634
Generation 121 Best Solution: [183.85152199441052, 176.51262181587722,
175.0405996613251, 75.58383976096485] Fitness: 1.8342280761754979
Generation 122 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.63934525713383] Fitness: 2.154167088230443
Generation 123 Best Solution: [183.85152199441052, 176.18326308629622,
174.96893907690412, 75.8141859817019] Fitness: 2.0593667178816233
Generation 124 Best Solution: [183.85152199441052, 177.34986571069052,
175.0405996613251, 74.67458286084617] Fitness: 2.152674869887634
Generation 125 Best Solution: [183.85152199441052, 176.84703586220388,
175.0405996613251, 75.39277421769306] Fitness: 1.8317206551224938
Generation 126 Best Solution: [183.33229158236264, 176.44625732126195,
175.0405996613251, 75.39277421769306] Fitness: 1.221714175900513
Generation 127 Best Solution: [183.68267043627813, 177.34986571069052,
174.63434769846575, 75.17703581777435] Fitness: 1.722720804712342
Generation 128 Best Solution: [183.85152199441052, 176.23843438013438,
175.0405996613251, 75.53682518740025] Fitness: 1.6918738067380468
Generation 129 Best Solution: [183.33229158236264, 177.34986571069052,
174.4910727751572, 75.53682518740025] Fitness: 1.8707209611355617

```
Generation 130 Best Solution: [183.85152199441052, 176.9118086578241,
174.0625452602074, 75.8141859817019] Fitness: 1.8804397675096904
Generation 131 Best Solution: [182.89885672786218, 177.34986571069052,
174.04169979375794, 75.8141859817019] Fitness: 2.333403765414267
Generation 132 Best Solution: [183.22709910850557, 177.34986571069052,
174.04169979375794, 75.8141859817019] Fitness: 2.237544564801441
Generation 133 Best Solution: [183.22709910850557, 177.34986571069052,
174.04169979375794, 75.0984573375622] Fitness: 1.2870712485880167
Generation 134 Best Solution: [183.85152199441052, 176.75195333706583,
175.0405996613251, 75.8141859817019] Fitness: 2.4037437810891307
Generation 135 Best Solution: [183.85152199441052, 176.75195333706583,
175.0405996613251, 75.8141859817019] Fitness: 2.4037437810891307
Generation 136 Best Solution: [183.85152199441052, 177.34986571069052,
174.47299809859416, 75.8141859817019] Fitness: 2.6518387004402717
Generation 137 Best Solution: [183.85152199441052, 176.19860687553924,
175.0405996613251, 75.8141859817019] Fitness: 2.1647486005481906
Generation 138 Best Solution: [183.85152199441052, 176.19860687553924,
175.0405996613251, 75.8141859817019] Fitness: 2.1647486005481906
Generation 139 Best Solution: [182.99768862948343, 176.19860687553924,
175.0405996613251, 75.8141859817019] Fitness: 1.8847810626351278
Generation 140 Best Solution: [182.95215425391754, 177.34986571069052,
175.0405996613251, 74.34131002051538] Fitness: 1.9921293729943303
Generation 141 Best Solution: [182.95215425391754, 176.17128841818806,
175.0405996613251, 75.8141859817019] Fitness: 1.9067500829717348
Generation 142 Best Solution: [183.85152199441052, 176.19860687553924,
175.0405996613251, 74.34131002051538] Fitness: 1.1646708699637391
Generation 143 Best Solution: [182.95215425391754, 176.19860687553924,
175.0405996613251, 75.8141859817019] Fitness: 1.9049524182767894
Generation 144 Best Solution: [182.95215425391754, 176.19860687553924,
175.0405996613251, 75.8141859817019] Fitness: 1.9049524182767894
Generation 145 Best Solution: [182.98539090532668, 176.19860687553924,
175.0405996613251, 75.8141859817019] Fitness: 1.8898785093379793
Generation 146 Best Solution: [182.98539090532668, 176.19860687553924,
175.0405996613251, 75.8141859817019] Fitness: 1.8898785093379793
Generation 147 Best Solution: [183.85152199441052, 176.19860687553924,
175.0405996613251, 75.35931226149718] Fitness: 1.4589802176327593
Generation 148 Best Solution: [183.85152199441052, 176.2557467687094,
175.0405996613251, 75.35931226149718] Fitness: 1.4593567031280466
Generation 149 Best Solution: [182.95215425391754, 176.62558501760293,
175.0405996613251, 75.8141859817019] Fitness: 2.0431195780905167
Generation 150 Best Solution: [182.95215425391754, 177.18750624219396,
175.0405996613251, 75.35931226149718] Fitness: 1.9954851786176868
Generation 151 Best Solution: [183.85152199441052, 177.34986571069052,
174.4570807765242, 74.50223935185988] Fitness: 1.5711929697643539
Generation 152 Best Solution: [182.90617949795717, 176.270406461701,
175.0405996613251, 75.8141859817019] Fitness: 1.9303002231335
Generation 153 Best Solution: [183.3764594043871, 175.9219583405761,
175.0405996613251, 75.8141859817019] Fitness: 1.932071463469838
```

Generation 154 Best Solution: [182.95215425391754, 177.34986571069052, 174.6186380644716, 75.24119299137945] Fitness: 1.6623365514938926
Generation 155 Best Solution: [182.90617949795717, 176.4525495679721, 175.0405996613251, 75.8141859817019] Fitness: 1.9734352541114957
Generation 156 Best Solution: [182.95215425391754, 177.04889980861535, 175.0405996613251, 75.46306410881304] Fitness: 1.9126071934706874
Generation 157 Best Solution: [182.95215425391754, 176.888925889157, 175.0405996613251, 75.8141859817019] Fitness: 2.2841541776935133
Generation 158 Best Solution: [182.95215425391754, 177.34986571069052, 174.08363163779774, 75.8141859817019] Fitness: 2.299719368388904
Generation 159 Best Solution: [183.85152199441052, 177.15518066271562, 174.08363163779774, 75.8141859817019] Fitness: 2.2160277632684746
Generation 160 Best Solution: [183.85152199441052, 177.33964307796543, 175.0405996613251, 75.53148496067604] Fitness: 2.7520004709630475
Generation 161 Best Solution: [183.85152199441052, 176.41141791349224, 175.0405996613251, 75.8141859817019] Fitness: 2.1945466676154846
Generation 162 Best Solution: [183.85152199441052, 176.41141791349224, 175.0405996613251, 75.8141859817019] Fitness: 2.1945466676154846
Generation 163 Best Solution: [183.85152199441052, 175.7216412146891, 174.2684416873468, 75.8141859817019] Fitness: 1.69889240163188
Generation 164 Best Solution: [183.85152199441052, 175.7216412146891, 174.2684416873468, 75.8141859817019] Fitness: 1.69889240163188
Generation 165 Best Solution: [183.1489901767057, 176.41141791349224, 175.0405996613251, 75.8141859817019] Fitness: 1.8730808510887726
Generation 166 Best Solution: [183.1489901767057, 176.41141791349224, 175.0405996613251, 75.8141859817019] Fitness: 1.8730808510887726
Generation 167 Best Solution: [183.38801977810195, 177.33964307796543, 175.0405996613251, 75.8141859817019] Fitness: 2.9252972209107173
Generation 168 Best Solution: [183.38801977810195, 177.33964307796543, 175.0405996613251, 75.8141859817019] Fitness: 2.9252972209107173
Generation 169 Best Solution: [183.38801977810195, 177.34986571069052, 174.38458079498167, 75.8141859817019] Fitness: 2.3010124493979456
Generation 170 Best Solution: [183.85152199441052, 176.1780668446561, 175.0405996613251, 75.8141859817019] Fitness: 2.165980878218583
Generation 171 Best Solution: [183.85152199441052, 177.34986571069052, 174.38458079498167, 75.8141859817019] Fitness: 2.6080661561030207
Generation 172 Best Solution: [183.38801977810195, 177.34986571069052, 175.0405996613251, 75.8141859817019] Fitness: 2.944949579145108
Generation 173 Best Solution: [183.85152199441052, 177.34986571069052, 175.0405996613251, 75.3602869816587] Fitness: 2.547368028414772
Generation 174 Best Solution: [183.73765518009432, 177.34986571069052, 174.26739807040119, 75.3602869816587] Fitness: 1.7565113050996852
Generation 175 Best Solution: [183.73765518009432, 177.34986571069052, 175.0405996613251, 74.380535525525] Fitness: 2.1208003741416777
Generation 176 Best Solution: [183.51675500591725, 177.34986571069052, 175.0405996613251, 74.59627733094516] Fitness: 1.900170683164132
Generation 177 Best Solution: [183.38801977810195, 177.34986571069052, 175.0405996613251, 74.59627733094516] Fitness: 1.8518278703397513

Generation 178 Best Solution: [183.51675500591725, 177.34986571069052,
175.0405996613251, 74.380535525525] Fitness: 1.9716466478872607
Generation 179 Best Solution: [183.38801977810195, 177.34986571069052,
175.0405996613251, 74.66717637183584] Fitness: 1.8457581246163148
Generation 180 Best Solution: [183.85152199441052, 176.09553321689094,
175.0405996613251, 75.8141859817019] Fitness: 2.178224148017602
Generation 181 Best Solution: [183.85152199441052, 176.09553321689094,
175.0405996613251, 75.35090781993895] Fitness: 1.462753052850759
Generation 182 Best Solution: [183.38801977810195, 177.34986571069052,
174.55210575548872, 75.8141859817019] Fitness: 2.3953079251977156
Generation 183 Best Solution: [183.38801977810195, 177.34986571069052,
174.55210575548872, 75.8141859817019] Fitness: 2.3953079251977156
Generation 184 Best Solution: [182.85068604598297, 177.34986571069052,
174.55210575548872, 75.8141859817019] Fitness: 2.5003002540446575
Generation 185 Best Solution: [182.85068604598297, 177.34986571069052,
173.93369507078245, 75.8141859817019] Fitness: 2.3912121712839665
Generation 186 Best Solution: [183.51675500591725, 177.34986571069052,
174.55210575548872, 75.8141859817019] Fitness: 2.4436507380220966
Generation 187 Best Solution: [183.51675500591725, 177.34986571069052,
174.55210575548872, 75.8141859817019] Fitness: 2.4436507380220966
Generation 188 Best Solution: [183.51675500591725, 177.34986571069052,
174.55210575548872, 75.24107590318322] Fitness: 1.6125124765696595
Generation 189 Best Solution: [183.29431545387376, 177.34986571069052,
175.0405996613251, 74.49659297737563] Fitness: 1.8576151743197116
Generation 190 Best Solution: [183.51675500591725, 177.34986571069052,
174.2622494911865, 74.49659297737563] Fitness: 1.2408787859606791
Generation 191 Best Solution: [182.92647844859485, 177.34986571069052,
174.2622494911865, 74.49659297737563] Fitness: 1.2527348509300484
Generation 192 Best Solution: [183.85152199441052, 176.03530025416936,
175.0405996613251, 75.8141859817019] Fitness: 2.194530064386629
Generation 193 Best Solution: [183.29431545387376, 177.10918240806166,
175.0405996613251, 75.29145342019896] Fitness: 1.7318217459871286
Generation 194 Best Solution: [183.85152199441052, 175.86953022413095,
175.0405996613251, 75.29145342019896] Fitness: 1.4908672818811748
Generation 195 Best Solution: [183.29431545387376, 177.10918240806166,
175.0405996613251, 74.30133146708943] Fitness: 1.537091728204838
Generation 196 Best Solution: [183.51675500591725, 177.10918240806166,
175.0405996613251, 74.30133146708943] Fitness: 1.6027566071745452
Generation 197 Best Solution: [183.29431545387376, 176.2631620798866,
175.0405996613251, 75.8141859817019] Fitness: 1.8412084831294697
Generation 198 Best Solution: [183.85152199441052, 175.86953022413095,
175.0405996613251, 75.06816814834055] Fitness: 1.3001902909263001
Generation 199 Best Solution: [183.29431545387376, 175.86953022413095,
175.0405996613251, 75.06816814834055] Fitness: 0.9758145180758984
Generation 200 Best Solution: [183.51675500591725, 177.05431779695692,
174.7368270135046, 75.06816814834055] Fitness: 1.1361776308505607
Generation 201 Best Solution: [183.85152199441052, 175.86953022413095,
175.0405996613251, 74.78536490463004] Fitness: 1.1813645159725303

```
Generation 202 Best Solution: [183.29431545387376, 175.86953022413095,
175.0405996613251, 75.8141859817019] Fitness: 1.9471430354402712
Generation 203 Best Solution: [183.47776996269445, 175.8963935722523,
175.0405996613251, 74.28322710802045] Fitness: 1.0161980328762015
Generation 204 Best Solution: [183.29431545387376, 175.86953022413095,
175.0405996613251, 75.8141859817019] Fitness: 1.9471430354402712
Generation 205 Best Solution: [183.29431545387376, 175.86953022413095,
175.0405996613251, 75.8141859817019] Fitness: 1.9471430354402712
Generation 206 Best Solution: [183.29431545387376, 175.86953022413095,
175.0405996613251, 75.8141859817019] Fitness: 1.9471430354402712
Generation 207 Best Solution: [183.29431545387376, 175.86953022413095,
175.0405996613251, 75.8141859817019] Fitness: 1.9471430354402712
Generation 208 Best Solution: [183.017778419339, 176.48516870832054,
175.07228375563787, 75.10869352828678] Fitness: 1.0419585366684285
Generation 209 Best Solution: [183.29431545387376, 177.34986571069052,
174.62845494583232, 75.35486662089059] Fitness: 1.7260046532012527
Generation 210 Best Solution: [183.29431545387376, 175.86953022413095,
175.07228375563787, 75.8141859817019] Fitness: 1.996920107766579
Generation 211 Best Solution: [183.29431545387376, 175.86953022413095,
174.871939840552, 75.8141859817019] Fitness: 1.7111340470474534
Generation 212 Best Solution: [183.29431545387376, 175.64087051169545,
175.0405996613251, 74.4560495483538] Fitness: 1.0749053642891404
Generation 213 Best Solution: [183.85152199441052, 175.86953022413095,
175.0405996613251, 74.4560495483538] Fitness: 1.2157785955603206
Generation 214 Best Solution: [183.017778419339, 175.86953022413095,
175.0405996613251, 75.68380405352345] Fitness: 1.745221131265775
Generation 215 Best Solution: [183.29431545387376, 175.64087051169545,
173.93813846852407, 75.8141859817019] Fitness: 1.4703612866574214
Generation 216 Best Solution: [183.29431545387376, 176.43547028789828,
174.9251629610254, 75.8141859817019] Fitness: 1.7116229549814912
Generation 217 Best Solution: [183.55938684467895, 175.64087051169545,
174.871939840552, 75.49470973126316] Fitness: 1.4498061022230644
Generation 218 Best Solution: [183.29431545387376, 177.34986571069052,
174.1343088790456, 75.8141859817019] Fitness: 2.2324440606102054
Generation 219 Best Solution: [183.29431545387376, 175.86953022413095,
175.07228375563787, 75.8141859817019] Fitness: 1.996920107766579
Generation 220 Best Solution: [183.29431545387376, 175.86953022413095,
175.07228375563787, 75.8141859817019] Fitness: 1.996920107766579
Generation 221 Best Solution: [183.29431545387376, 175.86953022413095,
175.07228375563787, 75.8141859817019] Fitness: 1.996920107766579
Generation 222 Best Solution: [183.29431545387376, 175.86953022413095,
175.07228375563787, 75.8141859817019] Fitness: 1.996920107766579
Generation 223 Best Solution: [183.29431545387376, 175.86953022413095,
175.07228375563787, 75.13426918952722] Fitness: 1.0731335747138897
Generation 224 Best Solution: [183.85152199441052, 177.34986571069052,
175.07228375563787, 75.09312702278646] Fitness: 2.347522894516796
Generation 225 Best Solution: [183.29431545387376, 176.8969652895753,
175.07228375563787, 75.8141859817019] Fitness: 2.278580141242943
```

```
Generation 226 Best Solution: [183.85152199441052, 176.5713449769075,
175.07228375563787, 75.8141859817019] Fitness: 2.317811992601489
Generation 227 Best Solution: [183.5034919869192, 176.7137360844986,
175.07228375563787, 75.8141859817019] Fitness: 2.1551359054959303
Generation 228 Best Solution: [183.29431545387376, 177.34986571069052,
174.231412214533, 75.8141859817019] Fitness: 2.2395787498457294
Generation 229 Best Solution: [183.85152199441052, 176.21045006192608,
174.52544954749283, 75.8141859817019] Fitness: 1.5965024731058857
Generation 230 Best Solution: [183.85152199441052, 176.21045006192608,
174.52544954749283, 75.8141859817019] Fitness: 1.5965024731058857
Generation 231 Best Solution: [183.29431545387376, 176.5713449769075,
175.07228375563787, 75.8141859817019] Fitness: 1.9934362197510873
Generation 232 Best Solution: [183.29431545387376, 176.5713449769075,
175.07228375563787, 75.8141859817019] Fitness: 1.9934362197510873
Generation 233 Best Solution: [183.85152199441052, 175.61823577444406,
174.88956593300844, 75.67440056381213] Fitness: 2.0097956756513824
Generation 234 Best Solution: [183.29431545387376, 177.34986571069052,
175.07228375563787, 74.73522178822195] Fitness: 1.8804915728548826
Generation 235 Best Solution: [183.29431545387376, 176.0255645301486,
175.07228375563787, 75.8141859817019] Fitness: 1.9231508444224463
Generation 236 Best Solution: [183.29431545387376, 176.0255645301486,
175.07228375563787, 75.8141859817019] Fitness: 1.9231508444224463
Generation 237 Best Solution: [183.85806119132886, 176.39816411942707,
175.07228375563787, 75.8141859817019] Fitness: 2.247167457379397
Generation 238 Best Solution: [183.85806119132886, 175.90810389980606,
175.07228375563787, 75.8141859817019] Fitness: 2.306142136232917
Generation 239 Best Solution: [183.85806119132886, 175.90810389980606,
175.07228375563787, 75.02196266436007] Fitness: 1.3060290223342987
Generation 240 Best Solution: [183.85806119132886, 175.90810389980606,
173.7619697151137, 75.8141859817019] Fitness: 1.6836454750314567
Generation 241 Best Solution: [183.2558565116506, 175.90810389980606,
175.07228375563787, 75.77549886778385] Fitness: 1.8982203781861986
Generation 242 Best Solution: [183.2558565116506, 176.393495788393,
175.07228375563787, 75.8141859817019] Fitness: 1.9116919039664932
Generation 243 Best Solution: [183.85152199441052, 176.19854225524034,
175.07228375563787, 75.47457080778436] Fitness: 1.6540401110845164
Generation 244 Best Solution: [182.86742971847653, 177.34986571069052,
175.07228375563787, 75.30885218559749] Fitness: 2.3267717103846457
Generation 245 Best Solution: [182.86742971847653, 176.35078945205996,
175.07228375563787, 75.8141859817019] Fitness: 2.0151215864459324
Generation 246 Best Solution: [183.29431545387376, 176.16082445426073,
175.07228375563787, 75.8141859817019] Fitness: 1.8929750105222714
Generation 247 Best Solution: [182.91416292825141, 176.16082445426073,
175.07228375563787, 75.8141859817019] Fitness: 1.9771043777690231
Generation 248 Best Solution: [183.29431545387376, 177.34986571069052,
175.07228375563787, 74.99991362919617] Fitness: 1.964845554095846
Generation 249 Best Solution: [183.51446669229665, 176.3637701203195,
175.07228375563787, 75.8141859817019] Fitness: 1.9710883913843906
```

```
Generation 250 Best Solution: [182.86742971847653, 177.34986571069052,
174.15625984480153, 75.8141859817019] Fitness: 2.344214314925847
Generation 251 Best Solution: [182.86742971847653, 177.34986571069052,
174.15625984480153, 75.8141859817019] Fitness: 2.344214314925847
Generation 252 Best Solution: [183.29431545387376, 175.87558826291001,
175.07228375563787, 75.8141859817019] Fitness: 1.9932772454143926
Generation 253 Best Solution: [183.82027852099293, 177.2078143133302,
174.15625984480153, 75.09507049168265] Fitness: 1.3148281928998664
Generation 254 Best Solution: [183.85152199441052, 177.2078143133302,
174.15625984480153, 75.09507049168265] Fitness: 1.3471018220721338
Generation 255 Best Solution: [182.99179997972695, 177.34986571069052,
174.15625984480153, 75.39696692943237] Fitness: 1.6186492056606026
Generation 256 Best Solution: [183.82027852099293, 176.72411247064414,
174.307128093553, 75.81922898864794] Fitness: 1.685447914624521
Generation 257 Best Solution: [183.51446669229665, 177.34986571069052,
174.15625984480153, 75.39696692943237] Fitness: 1.6363900096185944
Generation 258 Best Solution: [183.29431545387376, 177.34986571069052,
174.15625984480153, 75.39696692943237] Fitness: 1.5718324554788292
Generation 259 Best Solution: [183.51446669229665, 176.72411247064414,
174.75255408787802, 75.77549886778385] Fitness: 1.6719418039983278
Generation 260 Best Solution: [183.53733409701283, 176.72411247064414,
174.75255408787802, 75.81922898864794] Fitness: 1.7670500762722736
Generation 261 Best Solution: [183.82873946549688, 176.262215039755,
175.07228375563787, 75.8141859817019] Fitness: 2.191598499371159
Generation 262 Best Solution: [183.82873946549688, 176.262215039755,
175.07228375563787, 75.53706858765226] Fitness: 1.7194091686551327
Generation 263 Best Solution: [183.53733409701283, 176.262215039755,
175.07228375563787, 75.53706858765226] Fitness: 1.4947588847406401
Generation 264 Best Solution: [183.29431545387376, 177.34986571069052,
173.964316011909, 75.10529307800816] Fitness: 1.3133227568433812
Generation 265 Best Solution: [182.99179997972695, 175.7612881430739,
175.07228375563787, 75.53706858765226] Fitness: 1.6472410567414189
Generation 266 Best Solution: [182.99179997972695, 177.34986571069052,
175.07228375563787, 74.47577204163309] Fitness: 1.9611864192420099
Generation 267 Best Solution: [182.99179997972695, 175.7612881430739,
174.75255408787802, 75.53706858765226] Fitness: 1.2238720649269865
Generation 268 Best Solution: [183.82027852099293, 177.34986571069052,
174.75255408787802, 75.03131152787391] Fitness: 1.8515533343675854
Generation 269 Best Solution: [183.82027852099293, 177.39780664524415,
174.75255408787802, 75.03131152787391] Fitness: 1.9461067885019083
Generation 270 Best Solution: [183.82027852099293, 177.39780664524415,
174.75255408787802, 75.03131152787391] Fitness: 1.9461067885019083
Generation 271 Best Solution: [183.82027852099293, 177.10267970058268,
175.07228375563787, 74.80473442384368] Fitness: 1.7581243092521917
Generation 272 Best Solution: [183.53733409701283, 177.39780664524415,
175.07228375563787, 74.80473442384368] Fitness: 2.0615951702211146
Generation 273 Best Solution: [182.99179997972695, 177.39780664524415,
175.07228375563787, 74.80473442384368] Fitness: 2.0323853672953622
```

```
Generation 274 Best Solution: [183.82027852099293, 177.34986571069052,
175.07228375563787, 74.80473442384368] Fitness: 2.1831173066574006
Generation 275 Best Solution: [183.82027852099293, 177.39780664524415,
174.65981551905904, 74.40556801419453] Fitness: 1.816647874393477
Generation 276 Best Solution: [183.82027852099293, 175.66388338342512,
174.65981551905904, 75.77549886778385] Fitness: 1.8627922495639184
Generation 277 Best Solution: [183.82027852099293, 176.78938497101962,
175.07228375563787, 75.53706858765226] Fitness: 1.9841796205395836
Generation 278 Best Solution: [182.99179997972695, 177.39780664524415,
174.65981551905904, 75.53706858765226] Fitness: 2.1332041784933313
Generation 279 Best Solution: [183.53733409701283, 177.39780664524415,
174.2450396960953, 75.53706858765226] Fitness: 1.9402641219058046
Generation 280 Best Solution: [183.29668374896818, 177.39780664524415,
175.07228375563787, 74.49541150628838] Fitness: 2.002574132581982
Generation 281 Best Solution: [183.1240072617476, 177.39780664524415,
175.07228375563787, 74.87542820039263] Fitness: 2.0118239571118886
Generation 282 Best Solution: [182.99179997972695, 177.39780664524415,
175.07228375563787, 74.87542820039263] Fitness: 2.051583532234155
Generation 283 Best Solution: [183.53733409701283, 177.39780664524415,
174.65981551905904, 75.53706858765226] Fitness: 2.1624139814190837
Generation 284 Best Solution: [183.53733409701283, 176.19860717690588,
175.07228375563787, 75.83579535206499] Fitness: 2.0085304605270773
Generation 285 Best Solution: [183.82027852099293, 176.54182389059096,
175.07228375563787, 75.8141859817019] Fitness: 2.2686733328569995
Generation 286 Best Solution: [183.82027852099293, 176.54182389059096,
175.07228375563787, 75.8141859817019] Fitness: 2.2686733328569995
Generation 287 Best Solution: [182.83659421112634, 176.54182389059096,
175.07228375563787, 75.8141859817019] Fitness: 2.1082995709479766
Generation 288 Best Solution: [182.99179997972695, 176.54182389059096,
175.07228375563787, 75.8141859817019] Fitness: 2.0233879393606387
Generation 289 Best Solution: [183.82027852099293, 176.31984071269187,
175.07228375563787, 75.8141859817019] Fitness: 2.189711137892753
Generation 290 Best Solution: [183.5574481450996, 177.39780664524415,
174.67842363445294, 75.8141859817019] Fitness: 2.662311126282084
Generation 291 Best Solution: [183.82027852099293, 177.39780664524415,
175.07228375563787, 74.60736110439115] Fitness: 2.269421312559008
Generation 292 Best Solution: [183.23250114660584, 175.94105664449384,
174.67842363445294, 75.8141859817019] Fitness: 1.4582335691391406
Generation 293 Best Solution: [182.8641110928957, 175.94105664449384,
174.67842363445294, 75.8141859817019] Fitness: 1.5750830293954459
Generation 294 Best Solution: [182.8641110928957, 175.94105664449384,
175.07228375563787, 75.83579535206499] Fitness: 2.113939613184133
Generation 295 Best Solution: [183.76331768889233, 176.16669588947133,
175.07228375563787, 75.8141859817019] Fitness: 2.129945043261692
Generation 296 Best Solution: [182.8641110928957, 177.1782438160652,
175.07228375563787, 75.23068845362857] Fitness: 1.9438702998143351
Generation 297 Best Solution: [183.1240072617476, 177.34986571069052,
173.83702100006926, 75.8141859817019] Fitness: 2.3181559947051253
```

```
Generation 298 Best Solution: [183.82027852099293, 177.1782438160652,
173.83702100006926, 75.8141859817019] Fitness: 2.2970113914355954
Generation 299 Best Solution: [183.1240072617476, 177.1782438160652,
174.83695339241396, 75.8141859817019] Fitness: 2.349635928644076
Generation 300 Best Solution: [183.38635893452224, 177.39780664524415,
173.84963239051092, 75.8141859817019] Fitness: 2.4161223507813605
Generation 301 Best Solution: [183.82027852099293, 175.90045118293415,
173.83702100006926, 75.83579535206499] Fitness: 1.6471358570469832
Generation 302 Best Solution: [183.82027852099293, 177.39780664524415,
173.83702100006926, 74.42961382593114] Fitness: 1.6528378193966216
Generation 303 Best Solution: [182.9399076790415, 177.39780664524415,
173.83702100006926, 74.42961382593114] Fitness: 1.431346929065035
Generation 304 Best Solution: [182.9399076790415, 176.36021509588934,
173.84963239051092, 75.8141859817019] Fitness: 1.303577353501688
Generation 305 Best Solution: [183.82027852099293, 176.73586641950652,
173.83702100006926, 75.8141859817019] Fitness: 1.7405852209136181
Generation 306 Best Solution: [183.82027852099293, 176.73586641950652,
173.83702100006926, 75.8141859817019] Fitness: 1.7405852209136181
Generation 307 Best Solution: [183.02596908701923, 177.39780664524415,
175.07228375563787, 74.48008947229634] Fitness: 2.0410847348771015
Generation 308 Best Solution: [183.51517592550817, 176.12772809776556,
175.07228375563787, 75.47209839731781] Fitness: 1.3985165111527644
Generation 309 Best Solution: [182.9399076790415, 176.1736832053876,
175.07228375563787, 75.83579535206499] Fitness: 2.004704091388974
Generation 310 Best Solution: [183.43489681288344, 176.1736832053876,
175.07228375563787, 75.83579535206499] Fitness: 1.9657291385533457
Generation 311 Best Solution: [183.43489681288344, 176.1736832053876,
175.07228375563787, 75.8141859817019] Fitness: 1.9233751183835754
Generation 312 Best Solution: [183.00340736038436, 176.1736832053876,
175.07228375563787, 75.8141859817019] Fitness: 1.9338649327391606
Generation 313 Best Solution: [183.00340736038436, 177.39780664524415,
175.07228375563787, 74.75976600427987] Fitness: 2.0199408751335555
Generation 314 Best Solution: [183.82027852099293, 177.39780664524415,
175.07228375563787, 74.75976600427987] Fitness: 2.2699169039451856
Generation 315 Best Solution: [183.43489681288344, 177.39780664524415,
175.07228375563787, 74.75976600427987] Fitness: 2.0094510607779705
Generation 316 Best Solution: [183.82027852099293, 177.39780664524415,
175.07228375563787, 74.75976600427987] Fitness: 2.2699169039451856
Generation 317 Best Solution: [183.43489681288344, 177.39780664524415,
175.07228375563787, 74.75976600427987] Fitness: 2.0094510607779705
Generation 318 Best Solution: [183.43489681288344, 177.39780664524415,
175.07228375563787, 74.75976600427987] Fitness: 2.0094510607779705
Generation 319 Best Solution: [183.82027852099293, 176.27443211701703,
173.83162580630196, 75.86481363386584] Fitness: 1.6209761100076352
Generation 320 Best Solution: [183.43489681288344, 177.39780664524415,
175.07228375563787, 75.50311005360594] Fitness: 2.582596544844238
Generation 321 Best Solution: [183.72880867714315, 175.90986483468546,
175.07228375563787, 75.8141859817019] Fitness: 2.181088270790406
```

Generation 322 Best Solution: [183.72880867714315, 175.90986483468546,
175.07228375563787, 75.8141859817019] Fitness: 2.181088270790406
Generation 323 Best Solution: [183.72880867714315, 175.90986483468546,
175.07228375563787, 75.86481363386584] Fitness: 2.2815769111315998
Generation 324 Best Solution: [182.844678351082, 177.39780664524415,
175.07228375563787, 74.34461333912967] Fitness: 2.196248459071036
Generation 325 Best Solution: [183.82027852099293, 176.0496022236527,
174.4317039624301, 75.86481363386584] Fitness: 1.6360343709808518
Generation 326 Best Solution: [183.82027852099293, 176.0496022236527,
174.4317039624301, 75.8141859817019] Fitness: 1.535545730639658
Generation 327 Best Solution: [183.82027852099293, 176.0496022236527,
174.4317039624301, 75.86481363386584] Fitness: 1.6360343709808518
Generation 328 Best Solution: [183.82027852099293, 176.0496022236527,
174.04030727373265, 75.8141859817019] Fitness: 1.4711292489023942
Generation 329 Best Solution: [183.43489681288344, 177.32750945873096,
175.02481478633868, 74.37065215401066] Fitness: 1.875901416050862
Generation 330 Best Solution: [183.43489681288344, 177.39780664524415,
175.02481478633868, 74.37065215401066] Fitness: 2.0132008256159817
Generation 331 Best Solution: [183.43489681288344, 177.39369873591966,
175.02481478633868, 74.37065215401066] Fitness: 2.0049444957705203
Generation 332 Best Solution: [183.82027852099293, 176.7188552215195,
175.02481478633868, 75.8141859817019] Fitness: 2.3182589714626802
Generation 333 Best Solution: [183.43489681288344, 176.39075867497723,
174.4317039624301, 75.8141859817019] Fitness: 1.2732359559470519
Generation 334 Best Solution: [183.82027852099293, 177.41421593453754,
174.4317039624301, 74.32430301490585] Fitness: 1.7353694323984854
Generation 335 Best Solution: [182.99485600935546, 176.06685532306705,
175.07228375563787, 75.8141859817019] Fitness: 1.956172687173718
Generation 336 Best Solution: [183.43489681288344, 176.24244695651947,
175.02481478633868, 75.8141859817019] Fitness: 1.8476417473695075
Generation 337 Best Solution: [183.43489681288344, 177.39780664524415,
175.02481478633868, 75.12391375694267] Fitness: 2.097931507137361
Generation 338 Best Solution: [183.00340736038436, 176.56357193348848,
175.08088386650022, 75.8141859817019] Fitness: 2.044784891920285
Generation 339 Best Solution: [183.43489681288344, 176.56357193348848,
175.08088386650022, 75.8141859817019] Fitness: 2.0342950775646997
Generation 340 Best Solution: [183.00340736038436, 176.32653438623024,
175.02481478633868, 75.8141859817019] Fitness: 1.8669473846345728
Generation 341 Best Solution: [183.43489681288344, 176.56357193348848,
175.08088386650022, 74.62136841612919] Fitness: 0.9380401176418461
Generation 342 Best Solution: [183.3572970738827, 177.38173437403785,
175.07228375563787, 74.62136841612919] Fitness: 1.9535490229603607
Generation 343 Best Solution: [183.43489681288344, 177.41421593453754,
175.07228375563787, 74.62136841612919] Fitness: 2.0406083595857414
Generation 344 Best Solution: [183.3572970738827, 177.38173437403785,
175.07228375563787, 74.62136841612919] Fitness: 1.9535490229603607
Generation 345 Best Solution: [183.82027852099293, 176.55053165030796,
175.07228375563787, 74.62136841612919] Fitness: 1.177237661707363

Generation 346 Best Solution: [183.82027852099293, 177.39780664524415,
174.15301987512507, 74.62136841612919] Fitness: 1.5229622370345701
Generation 347 Best Solution: [183.43489681288344, 177.41421593453754,
174.15301987512507, 74.62136841612919] Fitness: 1.2957653734046184
Generation 348 Best Solution: [183.43489681288344, 177.41421593453754,
174.15301987512507, 75.17051945370976] Fitness: 1.4974863748748373
Generation 349 Best Solution: [183.43489681288344, 176.51389662815546,
175.07228375563787, 75.73759502057727] Fitness: 1.8499579036041691
Generation 350 Best Solution: [183.43489681288344, 177.39780664524415,
174.07435754091094, 75.17051945370976] Fitness: 1.4677654655835628
Generation 351 Best Solution: [183.43489681288344, 177.39780664524415,
175.07228375563787, 74.79957040110855] Fitness: 2.016138304257733
Generation 352 Best Solution: [183.43489681288344, 176.87986202788187,
175.07228375563787, 75.8141859817019] Fitness: 2.2907163682926384
Generation 353 Best Solution: [182.8543766342629, 177.29576821558842,
175.07228375563787, 75.8141859817019] Fitness: 2.995345219773111
Generation 354 Best Solution: [182.8543766342629, 177.41421593453754,
175.07228375563787, 74.54514547936607] Fitness: 2.1417362159109574
Generation 355 Best Solution: [183.43489681288344, 177.39780664524415,
175.07228375563787, 74.31617336450772] Fitness: 2.118724644252908
Generation 356 Best Solution: [183.43489681288344, 176.79202260620272,
175.07228375563787, 74.94815129626448] Fitness: 1.1599663293170266
Generation 357 Best Solution: [183.43489681288344, 175.9576826836282,
174.28905472294466, 75.8141859817019] Fitness: 1.2558060685746502
Generation 358 Best Solution: [182.84844798077148, 177.29576821558842,
175.07228375563787, 74.94815129626448] Fitness: 1.960724602970526
Generation 359 Best Solution: [183.43489681288344, 177.39780664524415,
175.07228375563787, 74.94815129626448] Fitness: 2.0650921047290565
Generation 360 Best Solution: [183.43489681288344, 177.39780664524415,
175.07228375563787, 74.94815129626448] Fitness: 2.0650921047290565
Generation 361 Best Solution: [183.24244193194193, 176.83362337188066,
175.07228375563787, 75.8141859817019] Fitness: 2.205755626576718
Generation 362 Best Solution: [182.8543766342629, 176.83362337188066,
175.07228375563787, 75.8141859817019] Fitness: 2.328777237892493
Generation 363 Best Solution: [183.20182945465152, 177.41421593453754,
175.07228375563787, 74.34472048123764] Fitness: 2.1008325118208298
Generation 364 Best Solution: [182.84844798077148, 176.64746224014732,
175.02481478633868, 75.8141859817019] Fitness: 2.09366651651142
Generation 365 Best Solution: [182.84844798077148, 176.64746224014732,
175.07228375563787, 74.3009378529089] Fitness: 1.1924752813115922
Generation 366 Best Solution: [182.84844798077148, 175.82222408910033,
174.56449448016477, 75.8141859817019] Fitness: 1.560782219752055
Generation 367 Best Solution: [183.20182945465152, 175.59442922308546,
175.02481478633868, 75.8141859817019] Fitness: 2.1523729251229593
Generation 368 Best Solution: [182.84844798077148, 177.41421593453754,
174.56449448016477, 75.54878490632935] Fitness: 2.1835880368973797
Generation 369 Best Solution: [182.8480294683961, 175.7279362749053,
174.56449448016477, 75.54878490632935] Fitness: 1.178621644926392

```
Generation 370 Best Solution: [182.84844798077148, 177.41421593453754,
174.56449448016477, 75.54878490632935] Fitness: 2.1835880368973797
Generation 371 Best Solution: [182.84844798077148, 176.07786444761913,
175.07228375563787, 75.8141859817019] Fitness: 2.0315507982414704
Generation 372 Best Solution: [183.20182945465152, 176.50006370712214,
175.07228375563787, 75.8141859817019] Fitness: 1.952928737802928
Generation 373 Best Solution: [182.84844798077148, 177.41421593453754,
174.47646000490266, 74.7769703348346] Fitness: 1.489472481599635
Generation 374 Best Solution: [182.84844798077148, 175.5940484876417,
175.07228375563787, 75.8141859817019] Fitness: 2.3528407013610986
Generation 375 Best Solution: [182.84844798077148, 177.41421593453754,
175.07228375563787, 75.02925977180611] Fitness: 2.2332032047722365
Generation 376 Best Solution: [182.84844798077148, 177.39780664524415,
175.07228375563787, 75.02925977180611] Fitness: 2.1999342252349736
Generation 377 Best Solution: [182.84844798077148, 176.30245824075192,
174.10952807466649, 75.8141859817019] Fitness: 1.2745865909303418
Generation 378 Best Solution: [182.84844798077148, 176.30245824075192,
174.10952807466649, 75.8141859817019] Fitness: 1.2745865909303418
Generation 379 Best Solution: [182.84844798077148, 177.41421593453754,
174.6917746661877, 74.82760774405527] Fitness: 1.6638718726715298
Generation 380 Best Solution: [183.56939437971428, 177.41421593453754,
175.07228375563787, 74.90898484757881] Fitness: 2.143762816981434
Generation 381 Best Solution: [182.84844798077148, 177.41421593453754,
174.10952807466649, 75.8141859817019] Fitness: 2.4848214071105295
Generation 382 Best Solution: [182.84844798077148, 177.41421593453754,
175.07228375563787, 75.14926608306935] Fitness: 2.3170559883262225
Generation 383 Best Solution: [183.68518558078034, 175.57469905066603,
174.20316304971885, 75.8141859817019] Fitness: 1.680495533673073
Generation 384 Best Solution: [182.84844798077148, 176.27734288332974,
175.07228375563787, 75.8141859817019] Fitness: 2.0159129423385123
Generation 385 Best Solution: [183.68518558078034, 177.41421593453754,
174.10952807466649, 75.5411077435252] Fitness: 2.066597993884946
Generation 386 Best Solution: [183.56939437971428, 175.57469905066603,
174.6917746661877, 75.8141859817019] Fitness: 1.8598374920460885
Generation 387 Best Solution: [182.84844798077148, 177.41421593453754,
174.10952807466649, 74.99806533834419] Fitness: 1.4712569389774677
Generation 388 Best Solution: [182.84844798077148, 176.59721326244204,
175.07228375563787, 75.8141859817019] Fitness: 2.133229373680954
Generation 389 Best Solution: [183.68518558078034, 176.28252023959183,
175.07228375563787, 75.8141859817019] Fitness: 2.0644441800138558
Generation 390 Best Solution: [183.80493194429252, 177.41421593453754,
174.42437494093772, 75.8141859817019] Fitness: 2.706417526739351
Generation 391 Best Solution: [182.84844798077148, 175.97704639942276,
175.07228375563787, 75.72998629024889] Fitness: 1.9080108765533368
Generation 392 Best Solution: [182.84844798077148, 175.97704639942276,
175.07228375563787, 75.72998629024889] Fitness: 1.9080108765533368
Generation 393 Best Solution: [183.08087162224635, 176.46563136449922,
175.07228375563787, 75.8141859817019] Fitness: 1.9566836364644797
```

Generation 394 Best Solution: [182.84844798077148, 177.41421593453754,
174.18009019237107, 75.8141859817019] Fitness: 2.485403137359802
Generation 395 Best Solution: [183.80493194429252, 177.41421593453754,
174.6167462625155, 75.72998629024889] Fitness: 2.6745248919240985
Generation 396 Best Solution: [182.84844798077148, 177.41421593453754,
174.4419753046915, 75.8141859817019] Fitness: 2.5621875897476714
Generation 397 Best Solution: [183.37554649441287, 177.41421593453754,
175.07228375563787, 75.72998629024889] Fitness: 2.9619811319276743
Generation 398 Best Solution: [182.84844798077148, 176.63685878191757,
175.07228375563787, 75.8141859817019] Fitness: 2.1599868838836347
Generation 399 Best Solution: [182.84844798077148, 177.41421593453754,
175.07228375563787, 75.68765138221012] Fitness: 2.9970754457721522
Generation 400 Best Solution: [182.84844798077148, 176.13981187772146,
175.07228375563787, 75.72998629024889] Fitness: 1.8619982306097271
Generation 401 Best Solution: [183.80493194429252, 177.41421593453754,
175.07228375563787, 75.0764998363026] Fitness: 2.416340778376324
Generation 402 Best Solution: [182.84844798077148, 176.71494337529208,
175.00420295695673, 75.68765138221012] Fitness: 1.8837950000209824
Generation 403 Best Solution: [182.84844798077148, 176.79568303314724,
175.07228375563787, 75.8141859817019] Fitness: 2.2941990686695872
Generation 404 Best Solution: [183.68518558078034, 177.41421593453754,
175.08624149044363, 74.30729985498189] Fitness: 2.3202745354015435
Generation 405 Best Solution: [183.68518558078034, 177.41421593453754,
175.08624149044363, 74.63027962085256] Fitness: 2.202408922932812
Generation 406 Best Solution: [183.68518558078034, 176.79568303314724,
175.00420295695673, 75.68765138221012] Fitness: 2.0054556838169697
Generation 407 Best Solution: [183.1888631361287, 176.79568303314724,
175.00420295695673, 75.72998629024889] Fitness: 1.9066921954144225
Generation 408 Best Solution: [183.1888631361287, 176.79568303314724,
175.00420295695673, 75.68765138221012] Fitness: 1.832147522316827
Generation 409 Best Solution: [182.84844798077148, 177.41421593453754,
175.08624149044363, 74.63402689630048] Fitness: 2.1540619517929085
Generation 410 Best Solution: [182.84844798077148, 177.41421593453754,
174.76662804111828, 75.379218744423] Fitness: 2.1382897493418227
Generation 411 Best Solution: [183.83362186522504, 177.41421593453754,
174.1710730179212, 74.63402689630048] Fitness: 1.569320947029889
Generation 412 Best Solution: [183.80493194429252, 175.9180062156877,
175.08624149044363, 75.72998629024889] Fitness: 2.1114756676384174
Generation 413 Best Solution: [183.80493194429252, 176.71432601319015,
175.00420295695673, 75.57977813795276] Fitness: 1.8602926423359014
Generation 414 Best Solution: [183.80493194429252, 175.6804170647232,
175.08624149044363, 75.57977813795276] Fitness: 2.0336159027411567
Generation 415 Best Solution: [183.80493194429252, 175.6804170647232,
175.00420295695673, 75.57977813795276] Fitness: 1.9063077249489708
Generation 416 Best Solution: [183.80493194429252, 175.6804170647232,
174.3752434045298, 75.72998629024889] Fitness: 1.564189890858048
Generation 417 Best Solution: [183.80493194429252, 176.23753587038559,
175.08624149044363, 75.57977813795276] Fitness: 1.7811390204587292

Generation 418 Best Solution: [182.84844798077148, 177.04211288601059,
175.00420295695673, 75.72998629024889] Fitness: 2.3258135495708734
Generation 419 Best Solution: [182.80405563105785, 176.0405545171155,
175.07228375563787, 75.72998629024889] Fitness: 1.9156338361794876
Generation 420 Best Solution: [183.58904089396114, 177.41421593453754,
174.05257432734575, 75.8141859817019] Fitness: 2.4720549877476583
Generation 421 Best Solution: [183.80493194429252, 177.41421593453754,
175.07228375563787, 74.87568569923894] Fitness: 2.314983766795342
Generation 422 Best Solution: [183.58904089396114, 177.41421593453754,
175.07228375563787, 74.87568569923894] Fitness: 2.1433862496273233
Generation 423 Best Solution: [183.45629898699195, 175.8451782015964,
175.08624149044363, 75.8141859817019] Fitness: 2.074096808744494
Generation 424 Best Solution: [183.58904089396114, 177.41421593453754,
174.32476280010744, 75.57977813795276] Fitness: 2.0867677820267647
Generation 425 Best Solution: [183.80493194429252, 176.9445825327404,
175.08624149044363, 75.57977813795276] Fitness: 2.2268521881044103
Generation 426 Best Solution: [183.80493194429252, 176.69459296684218,
175.08624149044363, 74.97958418329098] Fitness: 1.3560257878633826
Generation 427 Best Solution: [183.80493194429252, 176.04976739867172,
175.07228375563787, 75.57977813795276] Fitness: 1.7843134185205993
Generation 428 Best Solution: [183.80493194429252, 176.06836196697205,
175.07228375563787, 75.8141859817019] Fitness: 2.187073511196992
Generation 429 Best Solution: [183.80493194429252, 175.82763311790742,
175.07228375563787, 75.8661032286094] Fitness: 2.4038042309985017
Generation 430 Best Solution: [183.80493194429252, 175.82763311790742,
175.07228375563787, 75.8661032286094] Fitness: 2.4038042309985017
Generation 431 Best Solution: [183.80493194429252, 175.78294280615222,
175.00420295695673, 75.57977813795276] Fitness: 1.8198955035569497
Generation 432 Best Solution: [183.58904089396114, 176.0151422696572,
175.07228375563787, 75.25705185275453] Fitness: 1.2164435723525846
Generation 433 Best Solution: [183.482594397672, 175.78294280615222,
175.07228375563787, 75.57977813795276] Fitness: 1.6979346661497399
Generation 434 Best Solution: [183.51934603967268, 175.78294280615222,
175.07228375563787, 75.8141859817019] Fitness: 2.122792929633616
Generation 435 Best Solution: [183.482594397672, 176.15037577042384,
175.08624149044363, 75.57977813795276] Fitness: 1.5587337633773612
Generation 436 Best Solution: [183.51934603967268, 175.78294280615222,
175.07228375563787, 75.8141859817019] Fitness: 2.122792929633616
Generation 437 Best Solution: [183.51934603967268, 175.58263473723866,
175.07228375563787, 75.8141859817019] Fitness: 2.3084079650108933
Generation 438 Best Solution: [183.80493194429252, 177.41421593453754,
175.07228375563787, 74.37237727924568] Fitness: 2.3647122297990233
Generation 439 Best Solution: [183.80493194429252, 177.2415835425583,
175.07228375563787, 74.99105921855617] Fitness: 2.0378801824240798
Generation 440 Best Solution: [183.80493194429252, 177.41421593453754,
174.3571547617403, 75.15243262365901] Fitness: 1.7681098160378068
Generation 441 Best Solution: [183.58904089396114, 177.41421593453754,
175.07228375563787, 74.99105921855617] Fitness: 2.1931694206253978

Generation 442 Best Solution: [183.58904089396114, 176.69900519883933,
175.07228375563787, 75.8141859817019] Fitness: 2.188831079093603
Generation 443 Best Solution: [183.58904089396114, 177.41421593453754,
175.07228375563787, 74.99105921855617] Fitness: 2.1931694206253978
Generation 444 Best Solution: [183.80493194429252, 177.41421593453754,
175.07228375563787, 74.61385800254999] Fitness: 2.2866591638773457
Generation 445 Best Solution: [183.58904089396114, 177.26048704162403,
175.07228375563787, 74.99105921855617] Fitness: 1.899586373285617
Generation 446 Best Solution: [183.80493194429252, 177.0146782646128,
175.07228375563787, 75.08794420648245] Fitness: 1.7453638965975933
Generation 447 Best Solution: [183.0800823292914, 177.41421593453754,
175.07228375563787, 74.80740019058807] Fitness: 2.036349280132592
Generation 448 Best Solution: [183.0800823292914, 177.41421593453754,
175.07228375563787, 74.80740019058807] Fitness: 2.036349280132592
Generation 449 Best Solution: [183.58904089396114, 177.41421593453754,
175.07228375563787, 74.80740019058807] Fitness: 2.1246709930879617
Generation 450 Best Solution: [183.80493194429252, 177.41421593453754,
174.49294127121317, 75.40561526398793] Fitness: 2.0937318180488784
Generation 451 Best Solution: [183.0800823292914, 177.41421593453754,
175.07228375563787, 74.80740019058807] Fitness: 2.036349280132592
Generation 452 Best Solution: [183.58904089396114, 175.9828006941282,
175.07228375563787, 74.64829455447426] Fitness: 0.9460647298078955
Generation 453 Best Solution: [183.58904089396114, 177.41421593453754,
175.07228375563787, 74.80740019058807] Fitness: 2.1246709930879617
Generation 454 Best Solution: [183.0800823292914, 177.41421593453754,
175.07228375563787, 74.80740019058807] Fitness: 2.036349280132592
Generation 455 Best Solution: [183.3481835022498, 177.41421593453754,
175.07228375563787, 74.81002724300455] Fitness: 2.0280968883443524
Generation 456 Best Solution: [183.80493194429252, 176.74927230675857,
174.7451849004861, 75.46396400333536] Fitness: 1.3977139964743217
Generation 457 Best Solution: [183.3481835022498, 177.41421593453754,
175.07228375563787, 74.80740019058807] Fitness: 2.0275247296128374
Generation 458 Best Solution: [183.80493194429252, 176.40540078455112,
174.09724211166755, 75.87730991944403] Fitness: 1.577448188945814
Generation 459 Best Solution: [183.58904089396114, 176.54110306035508,
174.80298892293987, 75.87730991944403] Fitness: 1.83918362967339
Generation 460 Best Solution: [183.58904089396114, 176.54110306035508,
174.80298892293987, 75.90464211961579] Fitness: 1.8958460616197357
Generation 461 Best Solution: [183.58904089396114, 176.54110306035508,
174.80298892293987, 75.8661032286094] Fitness: 1.8163212136917015
Generation 462 Best Solution: [183.06476434747114, 176.0924074955158,
175.07228375563787, 75.8661032286094] Fitness: 2.028592898539278
Generation 463 Best Solution: [183.06476434747114, 176.0924074955158,
175.07228375563787, 75.86697031040744] Fitness: 2.0303541201950654
Generation 464 Best Solution: [183.80493194429252, 176.58298932464575,
175.07228375563787, 75.87730991944403] Fitness: 2.4033298797791645
Generation 465 Best Solution: [183.58904089396114, 177.41421593453754,
175.07228375563787, 74.480124070199] Fitness: 2.1459371352193046

Generation 466 Best Solution: [183.58904089396114, 176.72468150089503,
175.07228375563787, 75.86697031040744] Fitness: 2.31520049945429
Generation 467 Best Solution: [183.80493194429252, 177.41421593453754,
175.07228375563787, 75.08725890637994] Fitness: 2.4237223495799296
Generation 468 Best Solution: [183.58904089396114, 177.41421593453754,
175.07228375563787, 75.08725890637994] Fitness: 2.2521248324119107
Generation 469 Best Solution: [183.80493194429252, 176.40489568680275,
174.93021644119005, 75.87730991944403] Fitness: 2.110959718931673
Generation 470 Best Solution: [183.80493194429252, 176.40489568680275,
174.93021644119005, 75.87730991944403] Fitness: 2.110959718931673
Generation 471 Best Solution: [183.80493194429252, 176.40489568680275,
174.93021644119005, 75.87730991944403] Fitness: 2.110959718931673
Generation 472 Best Solution: [183.58904089396114, 177.41421593453754,
175.07228375563787, 74.85012043416293] Fitness: 2.1354433525845264
Generation 473 Best Solution: [183.58904089396114, 177.41421593453754,
175.07228375563787, 74.85012043416293] Fitness: 2.1354433525845264
Generation 474 Best Solution: [183.58904089396114, 176.57729889894838,
175.07228375563787, 75.66861920146282] Fitness: 1.8378338262535245
Generation 475 Best Solution: [183.58904089396114, 176.57729889894838,
175.07228375563787, 75.66861920146282] Fitness: 1.8378338262535245
Generation 476 Best Solution: [183.58904089396114, 176.22257286974923,
175.07228375563787, 75.66861920146282] Fitness: 1.730441503393823
Generation 477 Best Solution: [183.58904089396114, 177.41421593453754,
174.93021644119005, 74.89129540464279] Fitness: 1.9390346887867564
Generation 478 Best Solution: [183.36206461329374, 177.41421593453754,
173.8771316367935, 75.32100016988056] Fitness: 1.6814673288552167
Generation 479 Best Solution: [183.12138950808213, 177.41421593453754,
173.81330605852054, 75.23892108098157] Fitness: 1.6260736030305092
Generation 480 Best Solution: [183.2551486365285, 176.34110104027087,
175.07228375563787, 75.86697031040744] Fitness: 2.003601030693345
Generation 481 Best Solution: [183.2551486365285, 176.34110104027087,
175.07228375563787, 75.86697031040744] Fitness: 2.003601030693345
Generation 482 Best Solution: [183.51348233668952, 177.41421593453754,
173.94595105831618, 75.66861920146282] Fitness: 2.1921486842641613
Generation 483 Best Solution: [183.58904089396114, 177.41421593453754,
175.07228375563787, 74.60434256757144] Fitness: 2.116245362170699
Generation 484 Best Solution: [183.80493194429252, 176.68472522445012,
175.07228375563787, 75.18456992890628] Fitness: 1.4663689783739766
Generation 485 Best Solution: [183.58904089396114, 177.41421593453754,
173.96096405991543, 75.18456992890628] Fitness: 1.610388397901628
Generation 486 Best Solution: [183.58904089396114, 177.41421593453754,
175.07228375563787, 74.39339515869551] Fitness: 2.18234945874819
Generation 487 Best Solution: [183.58904089396114, 176.68472522445012,
174.38673754022108, 75.86697031040744] Fitness: 1.5894224178533902
Generation 488 Best Solution: [183.58904089396114, 176.48774969250655,
174.67027002480597, 75.86697031040744] Fitness: 1.6557142057935144
Generation 489 Best Solution: [183.58904089396114, 176.7036335335839,
175.07228375563787, 75.87730991944403] Fitness: 2.318591662845854

```
Generation 490 Best Solution: [183.58904089396114, 177.24108213550275,
173.76126970880108, 75.86697031040744] Fitness: 2.365554446578729
Generation 491 Best Solution: [183.58904089396114, 177.24108213550275,
174.000735144344, 75.86697031040744] Fitness: 2.259232789335246
Generation 492 Best Solution: [183.5261239372916, 177.41421593453754,
173.76126970880108, 75.86697031040744] Fitness: 2.658343487674787
Generation 493 Best Solution: [183.44144946611536, 177.41421593453754,
175.0579094902911, 74.8854037487103] Fitness: 2.0525759933210086
Generation 494 Best Solution: [183.58904089396114, 177.41421593453754,
175.0579094902911, 74.55714934540859] Fitness: 2.1016142578843833
Generation 495 Best Solution: [183.58904089396114, 177.41421593453754,
174.57037657492714, 74.64813788293105] Fitness: 1.5258592121058407
Generation 496 Best Solution: [183.58904089396114, 176.8813640852495,
173.76126970880108, 75.87730991944403] Fitness: 1.8699629376046785
Generation 497 Best Solution: [183.58904089396114, 177.07148114555116,
174.81563061471135, 75.86697031040744] Fitness: 2.3626248875354925
Generation 498 Best Solution: [183.58904089396114, 175.97995950116385,
175.07228375563787, 75.86697031040744] Fitness: 2.1505269312263415
Generation 499 Best Solution: [183.58904089396114, 177.07148114555116,
174.81563061471135, 75.33530714947793] Fitness: 1.524597304310849
Generation 500 Best Solution: [183.58904089396114, 175.97995950116385,
175.07228375563787, 75.33530714947793] Fitness: 1.3124993480016975
Optimal Solution for Simulation 1: [183.58904089396114, 175.97995950116385,
175.07228375563787, 75.33530714947793]
--------------------------------------------------
Simulation 2/32 with conditions: {'drought': True, 'flood': True,
'policy_change': True, 'social_event': True, 'economic_crisis': False}
Generation 1 Best Solution: [182.95832333386818, 176.44539722448576,
174.56635606969974, 74.31885780421185] Fitness: 0.4386109712613309
Generation 2 Best Solution: [182.95832333386818, 176.44539722448576,
174.20060466907688, 75.43328541923519] Fitness: 0.604750710477872
Generation 3 Best Solution: [183.84650264128578, 175.62492844170842,
174.4151727321347, 74.97614545805253] Fitness: 0.7990454020612564
Generation 4 Best Solution: [183.75065050292076, 177.0566568580368,
174.09996523935487, 74.31885780421185] Fitness: 1.0263514092108488
Generation 5 Best Solution: [183.75065050292076, 176.54802587263316,
174.8909513152566, 74.97614545805253] Fitness: 0.9486792660151411
Generation 6 Best Solution: [183.75065050292076, 176.25479867604372,
174.4151727321347, 75.8401023816416] Fitness: 1.4485911463343635
Generation 7 Best Solution: [183.84650264128578, 176.04598850315904,
174.8909513152566, 75.65860080350912] Fitness: 1.6633929014612718
Generation 8 Best Solution: [183.84650264128578, 177.0056060191574,
175.10330377761736, 75.43212809816991] Fitness: 2.1732090296787776
Generation 9 Best Solution: [183.84650264128578, 176.90076246445264,
173.94247794225396, 75.65860080350912] Fitness: 1.5871557456255898
Generation 10 Best Solution: [183.84650264128578, 176.71623784786053,
173.94247794225396, 75.65860080350912] Fitness: 1.4003860315070495
Generation 11 Best Solution: [183.84650264128578, 176.71623784786053,
```

173.94247794225396, 75.65860080350912] Fitness: 1.4003860315070495
Generation 12 Best Solution: [183.84650264128578, 176.37842107915614,
175.11121393370405, 75.8499077917731] Fitness: 2.3275257727788796
Generation 13 Best Solution: [183.84245347716467, 176.37842107915614,
175.11121393370405, 75.8499077917731] Fitness: 2.3232439297904657
Generation 14 Best Solution: [183.7863640960935, 176.37842107915614,
174.8909513152566, 75.8499077917731] Fitness: 1.9444678626011938
Generation 15 Best Solution: [183.7863640960935, 176.37842107915614,
173.87116278856627, 75.8499077917731] Fitness: 1.5062017574920896
Generation 16 Best Solution: [183.84650264128578, 176.26833691763937,
174.0924967463077, 75.8499077917731] Fitness: 1.492418448951208
Generation 17 Best Solution: [183.84650264128578, 176.26833691763937,
174.0924967463077, 75.8499077917731] Fitness: 1.492418448951208
Generation 18 Best Solution: [183.84650264128578, 176.64071841175016,
175.10330377761736, 74.73481811388868] Fitness: 1.3506937659270055
Generation 19 Best Solution: [183.48288737296932, 176.64071841175016,
175.11121393370405, 75.8499077917731] Fitness: 2.1867766305559027
Generation 20 Best Solution: [183.84245347716467, 176.64071841175016,
175.10330377761736, 75.8499077917731] Fitness: 2.445078100687609
Generation 21 Best Solution: [183.84710732056274, 176.8126105930416,
175.11121393370405, 75.8499077917731] Fitness: 2.6135138793931043
Generation 22 Best Solution: [183.84650264128578, 176.10552738076896,
175.11121393370405, 75.8499077917731] Fitness: 2.308980722089931
Generation 23 Best Solution: [183.84650264128578, 176.10552738076896,
175.11121393370405, 75.86465619831199] Fitness: 2.3373980008485526
Generation 24 Best Solution: [183.84650264128578, 176.10552738076896,
174.8909513152566, 75.79017025854935] Fitness: 1.875311707097986
Generation 25 Best Solution: [183.84650264128578, 176.10552738076896,
175.11121393370405, 75.45934401537448] Fitness: 1.6883580197298456
Generation 26 Best Solution: [183.1829090962297, 176.0659776393276,
175.11121393370405, 75.65950601907115] Fitness: 1.6452732027285455
Generation 27 Best Solution: [183.84650264128578, 176.10552738076896,
174.83943984190796, 75.79017025854935] Fitness: 1.811609564646421
Generation 28 Best Solution: [183.1829090962297, 177.4400017461042,
174.45957837880763, 75.79017025854935] Fitness: 2.4126310007935077
Generation 29 Best Solution: [183.75065050292076, 176.42048083332486,
175.11121393370405, 75.8499077917731] Fitness: 2.247395878538109
Generation 30 Best Solution: [183.1829090962297, 177.4400017461042,
174.45957837880763, 74.94640551436002] Fitness: 1.4745222676831025
Generation 31 Best Solution: [183.06524401817398, 177.4400017461042,
174.45957837880763, 74.94640551436002] Fitness: 1.4913725820225368
Generation 32 Best Solution: [183.1829090962297, 176.606620162974,
175.11121393370405, 74.85768747158843] Fitness: 1.0221218063087052
Generation 33 Best Solution: [183.1829090962297, 175.76900679437966,
175.11121393370405, 74.85768747158843] Fitness: 1.040809688340809
Generation 34 Best Solution: [183.1829090962297, 176.58147646597754,
175.11121393370405, 74.85768747158843] Fitness: 1.0056590413341293
Generation 35 Best Solution: [183.17467739429392, 175.76900679437966,

175.11121393370405, 75.8499077917731] Fitness: 2.1204388812765482
Generation 36 Best Solution: [183.1829090962297, 176.87278014893053,
174.4399028876965, 75.8499077917731] Fitness: 1.6103091695501615
Generation 37 Best Solution: [183.1829090962297, 175.9049188133154,
175.10330377761736, 74.94727889041893] Fitness: 0.9753846448832988
Generation 38 Best Solution: [183.1829090962297, 176.87278014893053,
174.3590969886101, 75.8499077917731] Fitness: 1.5730980613501782
Generation 39 Best Solution: [183.17467739429392, 176.87278014893053,
174.3590969886101, 75.8499077917731] Fitness: 1.5735262025224912
Generation 40 Best Solution: [183.1829090962297, 176.70473174467588,
175.10330377761736, 75.8499077917731] Fitness: 2.162665017286244
Generation 41 Best Solution: [182.86578904856867, 176.70473174467588,
175.10330377761736, 75.8499077917731] Fitness: 2.2607877356347155
Generation 42 Best Solution: [183.75065050292076, 175.9148253096762,
175.10330377761736, 75.8499077917731] Fitness: 2.2626944294555056
Generation 43 Best Solution: [183.29313764661268, 177.0896041025244,
174.60211456215222, 75.8499077917731] Fitness: 2.004874455323476
Generation 44 Best Solution: [183.8605573945108, 177.4400017461042,
174.4362447262021, 74.57884930690219] Fitness: 1.7763402186733057
Generation 45 Best Solution: [183.8605573945108, 177.4400017461042,
174.4362447262021, 74.57884930690219] Fitness: 1.7763402186733057
Generation 46 Best Solution: [183.7890762361714, 175.57609734764694,
175.10330377761736, 75.23348386072573] Fitness: 1.6923325138051621
Generation 47 Best Solution: [183.75065050292076, 177.4400017461042,
173.81511691307236, 74.45362866037779] Fitness: 1.7032719742604114
Generation 48 Best Solution: [183.8605573945108, 175.83857077152186,
175.10330377761736, 75.8499077917731] Fitness: 2.4130373055661805
Generation 49 Best Solution: [183.8605573945108, 175.83857077152186,
175.10330377761736, 75.8499077917731] Fitness: 2.4130373055661805
Generation 50 Best Solution: [183.8605573945108, 175.83857077152186,
175.10330377761736, 75.8499077917731] Fitness: 2.4130373055661805
Generation 51 Best Solution: [183.8605573945108, 175.83857077152186,
175.10330377761736, 75.8499077917731] Fitness: 2.4130373055661805
Generation 52 Best Solution: [183.75065050292076, 175.83857077152186,
175.10330377761736, 75.8499077917731] Fitness: 2.303935674529334
Generation 53 Best Solution: [183.1829090962297, 177.4400017461042,
175.10330377761736, 74.96274089613881] Fitness: 2.187631348725675
Generation 54 Best Solution: [183.24874387934878, 177.4400017461042,
175.10330377761736, 74.96274089613881] Fitness: 2.1882706579922604
Generation 55 Best Solution: [183.17467739429392, 177.4400017461042,
174.77627627237732, 75.8499077917731] Fitness: 2.785585450872654
Generation 56 Best Solution: [183.75065050292076, 176.52971224049068,
175.10330377761736, 75.8499077917731] Fitness: 2.2842859535339604
Generation 57 Best Solution: [183.1829090962297, 176.52971224049068,
175.10330377761736, 75.8499077917731] Fitness: 2.041311745832803
Generation 58 Best Solution: [183.1829090962297, 176.6281250401824,
174.0162649129285, 75.8499077917731] Fitness: 1.3122920756075622
Generation 59 Best Solution: [183.1829090962297, 176.6281250401824,

175.10330377761736, 75.43106195400003] Fitness: 1.4475730614986575
Generation 60 Best Solution: [183.1829090962297, 176.6281250401824,
173.8913976675309, 75.8499077917731] Fitness: 1.3475397493501697
Generation 61 Best Solution: [183.1829090962297, 175.73482296779022,
175.10330377761736, 75.8499077917731] Fitness: 2.1326344491833
Generation 62 Best Solution: [183.75065050292076, 176.82385787528213,
174.97555045625325, 74.82538780380736] Fitness: 1.2355971557074554
Generation 63 Best Solution: [183.75065050292076, 176.82385787528213,
175.10330377761736, 75.43106195400003] Fitness: 1.8617454689632438
Generation 64 Best Solution: [183.8605573945108, 176.82385787528213,
175.10330377761736, 75.4820375478373] Fitness: 2.035021103513272
Generation 65 Best Solution: [183.1829090962297, 176.83787841160972,
174.84984794748343, 75.8499077917731] Fitness: 1.928689189449248
Generation 66 Best Solution: [183.8605573945108, 177.31420141325594,
175.11121393370405, 75.23004980181477] Fitness: 2.482197996279417
Generation 67 Best Solution: [183.8605573945108, 176.57626354131642,
175.10330377761736, 75.8499077917731] Fitness: 2.420681224464647
Generation 68 Best Solution: [183.8605573945108, 177.31420141325594,
174.0939420234071, 75.23004980181477] Fitness: 1.6694350762802173
Generation 69 Best Solution: [183.8605573945108, 177.4400017461042,
174.0939420234071, 75.23004980181477] Fitness: 1.9159830931477912
Generation 70 Best Solution: [183.8605573945108, 176.57626354131642,
175.10330377761736, 74.74808550448084] Fitness: 1.3229051075064548
Generation 71 Best Solution: [183.3453667917654, 176.48157707669654,
175.11121393370405, 75.83573318135936] Fitness: 2.0175495218737534
Generation 72 Best Solution: [183.3453667917654, 176.48157707669654,
175.11121393370405, 75.8499077917731] Fitness: 2.0445195646028007
Generation 73 Best Solution: [183.8605573945108, 175.89546885967752,
175.10330377761736, 75.8499077917731] Fitness: 2.381346949579706
Generation 74 Best Solution: [183.8605573945108, 175.89546885967752,
175.10330377761736, 75.8499077917731] Fitness: 2.381346949579706
Generation 75 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 75.11684367048332] Fitness: 2.380122520979235
Generation 76 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.90187013215454] Fitness: 2.269713221917629
Generation 77 Best Solution: [183.8605573945108, 176.39631780993477,
175.10330377761736, 75.8499077917731] Fitness: 2.335179807572086
Generation 78 Best Solution: [183.8605573945108, 176.39631780993477,
175.10330377761736, 75.75224637680728] Fitness: 2.1561538870049706
Generation 79 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.91529605238598] Fitness: 2.2743537366745796
Generation 80 Best Solution: [183.3037310134649, 177.4400017461042,
175.10330377761736, 74.91529605238598] Fitness: 2.175535569652538
Generation 81 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.91529605238598] Fitness: 2.2743537366745796
Generation 82 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.91529605238598] Fitness: 2.2743537366745796
Generation 83 Best Solution: [183.8605573945108, 177.31420141325594,

175.10330377761736, 74.91529605238598] Fitness: 2.2743537366745796
Generation 84 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 75.13644976181892] Fitness: 2.3940247662713876
Generation 85 Best Solution: [183.8605573945108, 176.84765152427624,
175.10330377761736, 75.48084310036133] Fitness: 2.058631796360511
Generation 86 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.74944874264771] Fitness: 2.238096087941098
Generation 87 Best Solution: [183.8605573945108, 177.31420141325594,
174.65882222762556, 74.84269133938949] Fitness: 1.6914124327593543
Generation 88 Best Solution: [183.8605573945108, 177.31420141325594,
174.00738741748773, 75.0839020237481] Fitness: 1.5688811189262273
Generation 89 Best Solution: [183.8605573945108, 177.31420141325594,
174.00738741748773, 75.0839020237481] Fitness: 1.5688811189262273
Generation 90 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.84269133938949] Fitness: 2.2528393029037206
Generation 91 Best Solution: [183.8605573945108, 176.363024449709,
175.10330377761736, 74.84269133938949] Fitness: 1.2423518084153018
Generation 92 Best Solution: [183.8605573945108, 176.363024449709,
175.10330377761736, 74.84269133938949] Fitness: 1.2423518084153018
Generation 93 Best Solution: [183.8605573945108, 177.31420141325594,
174.65882222762556, 75.23280032598929] Fitness: 1.9102283556507615
Generation 94 Best Solution: [183.8605573945108, 177.31420141325594,
174.65882222762556, 75.23280032598929] Fitness: 1.9102283556507615
Generation 95 Best Solution: [183.8605573945108, 177.31420141325594,
174.65882222762556, 75.61847213014286] Fitness: 2.3758860674381865
Generation 96 Best Solution: [183.8605573945108, 177.31420141325594,
174.65882222762556, 74.88869496049615] Fitness: 1.704024567000373
Generation 97 Best Solution: [183.8605573945108, 177.31420141325594,
174.65882222762556, 74.88869496049615] Fitness: 1.704024567000373
Generation 98 Best Solution: [183.8605573945108, 177.31420141325594,
174.47848561676025, 75.61847213014286] Fitness: 2.2419999830633053
Generation 99 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 75.02986396519172] Fitness: 2.326172805314817
Generation 100 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.8067666231014] Fitness: 2.2454430635627207
Generation 101 Best Solution: [183.8605573945108, 177.31420141325594,
174.38836349101732, 75.8499077917731] Fitness: 2.5938543477593043
Generation 102 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.74415180251872] Fitness: 2.237693518084866
Generation 103 Best Solution: [183.8605573945108, 176.15068276901263,
174.65882222762556, 75.8499077917731] Fitness: 1.744153796488462
Generation 104 Best Solution: [183.8605573945108, 176.15068276901263,
174.65882222762556, 75.8499077917731] Fitness: 1.744153796488462
Generation 105 Best Solution: [183.8605573945108, 176.15068276901263,
174.65882222762556, 75.8499077917731] Fitness: 1.744153796488462
Generation 106 Best Solution: [183.18235970137727, 177.31420141325594,
175.10330377761736, 75.13581061939695] Fitness: 2.041510681175077
Generation 107 Best Solution: [183.8605573945108, 177.31420141325594,

174.62423811456082, 74.74415180251872] Fitness: 1.6463899992039122
Generation 108 Best Solution: [183.8605573945108, 176.0573306853075,
174.13439679124517, 75.8499077917731] Fitness: 1.5203042852310245
Generation 109 Best Solution: [183.8605573945108, 176.60003297050852,
175.10330377761736, 75.42960081538274] Fitness: 1.778542253889506
Generation 110 Best Solution: [183.8605573945108, 175.81162917893593,
175.10330377761736, 75.42960081538274] Fitness: 1.77245699243941
Generation 111 Best Solution: [183.8605573945108, 175.81162917893593,
175.10330377761736, 75.42960081538274] Fitness: 1.77245699243941
Generation 112 Best Solution: [183.8605573945108, 177.31420141325594,
174.04604364358605, 75.42960081538274] Fitness: 1.8827540005397896
Generation 113 Best Solution: [183.8605573945108, 177.31420141325594,
174.04604364358605, 75.42960081538274] Fitness: 1.8827540005397896
Generation 114 Best Solution: [183.8605573945108, 177.31420141325594,
174.04604364358605, 75.42960081538274] Fitness: 1.8827540005397896
Generation 115 Best Solution: [183.2270045556692, 177.31420141325594,
174.27324857446038, 75.8499077917731] Fitness: 2.2015769553342324
Generation 116 Best Solution: [183.28821040691057, 176.4959847581252,
175.10330377761736, 75.8499077917731] Fitness: 2.0282790059600906
Generation 117 Best Solution: [183.16392105660123, 175.9307022837929,
175.10330377761736, 75.42960081538274] Fitness: 1.3560419385400566
Generation 118 Best Solution: [183.16392105660123, 177.31420141325594,
174.52709803689976, 75.42960081538274] Fitness: 1.6628197224968324
Generation 119 Best Solution: [183.8605573945108, 177.31420141325594,
174.52709803689976, 75.42960081538274] Fitness: 2.0137377679367905
Generation 120 Best Solution: [183.8605573945108, 176.4959847581252,
175.10330377761736, 75.42823843052646] Fitness: 1.7167484596148515
Generation 121 Best Solution: [183.8605573945108, 176.4959847581252,
175.10330377761736, 75.42823843052646] Fitness: 1.7167484596148515
Generation 122 Best Solution: [183.8605573945108, 176.4959847581252,
175.10330377761736, 75.5473442318252] Fitness: 1.8728944442720916
Generation 123 Best Solution: [183.80783926778574, 177.31420141325594,
175.10330377761736, 74.51881677231746] Fitness: 2.20903156054669
Generation 124 Best Solution: [183.8605573945108, 176.57718610583788,
175.10330377761736, 74.51881677231746] Fitness: 1.3493703433129502
Generation 125 Best Solution: [183.8605573945108, 176.57718610583788,
175.10330377761736, 74.51881677231746] Fitness: 1.3493703433129502
Generation 126 Best Solution: [183.1097102409178, 175.73934696966862,
175.10330377761736, 74.51881677231746] Fitness: 1.0650168283689674
Generation 127 Best Solution: [183.8714747322502, 175.9421377356669,
175.10330377761736, 75.12651797714216] Fitness: 1.4224564623224796
Generation 128 Best Solution: [183.8605573945108, 177.26210199149776,
175.10330377761736, 74.51881677231746] Fitness: 2.1694932464635928
Generation 129 Best Solution: [183.63296340861004, 177.31420141325594,
175.10330377761736, 74.93855437713019] Fitness: 2.07949798847952
Generation 130 Best Solution: [183.70247556433924, 177.31420141325594,
175.10330377761736, 74.93855437713019] Fitness: 2.1325264355197113
Generation 131 Best Solution: [182.8626167297145, 175.65350112838416,

175.10330377761736, 74.93855437713019] Fitness: 1.2486471547384912
Generation 132 Best Solution: [183.8605573945108, 175.65350112838416,
175.10330377761736, 74.93855437713019] Fitness: 1.5007719773038608
Generation 133 Best Solution: [183.29393235610186, 177.31420141325594,
175.10330377761736, 75.42823843052646] Fitness: 2.329826633893945
Generation 134 Best Solution: [183.8605573945108, 177.31420141325594,
174.30852333114615, 75.8499077917731] Fitness: 2.5638818978414837
Generation 135 Best Solution: [183.8605573945108, 177.38757482247445,
175.10330377761736, 74.36003598739113] Fitness: 2.473744434460971
Generation 136 Best Solution: [183.28327438897043, 177.31420141325594,
175.10330377761736, 75.3466600299912] Fitness: 2.2351251893373085
Generation 137 Best Solution: [183.8605573945108, 176.821547657613,
174.37044879696825, 75.8499077917731] Fitness: 1.874587148617605
Generation 138 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.83998857504501] Fitness: 2.252208027437715
Generation 139 Best Solution: [183.8605573945108, 176.57994433224243,
174.37044879696825, 74.83998857504501] Fitness: 0.5898777574364465
Generation 140 Best Solution: [183.8605573945108, 177.31420141325594,
174.37044879696825, 74.83998857504501] Fitness: 1.5026484880230777
Generation 141 Best Solution: [183.8605573945108, 176.77768692480385,
174.37044879696825, 75.8499077917731] Fitness: 1.830842113754158
Generation 142 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.36087974843487] Fitness: 2.3326710316563197
Generation 143 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.36087974843487] Fitness: 2.3326710316563197
Generation 144 Best Solution: [183.8605573945108, 177.29930085081307,
175.10330377761736, 74.36087974843487] Fitness: 2.305215475528534
Generation 145 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.36087974843487] Fitness: 2.3326710316563197
Generation 146 Best Solution: [183.8605573945108, 177.29930085081307,
175.10330377761736, 74.36087974843487] Fitness: 2.305215475528534
Generation 147 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.36087974843487] Fitness: 2.3326710316563197
Generation 148 Best Solution: [183.8605573945108, 176.47855783799466,
175.10330377761736, 74.36087974843487] Fitness: 1.3644670519791082
Generation 149 Best Solution: [183.84585408664543, 177.31420141325594,
175.10330377761736, 74.36087974843487] Fitness: 2.316908944948825
Generation 150 Best Solution: [183.8605573945108, 177.29930085081307,
175.10330377761736, 74.36087974843487] Fitness: 2.305215475528534
Generation 151 Best Solution: [183.8605573945108, 177.31420141325594,
174.7233936423896, 75.49687649048896] Fitness: 2.2634311961201186
Generation 152 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.84886415285703] Fitness: 2.2543267220010152
Generation 153 Best Solution: [182.97109450701313, 177.31420141325594,
174.10027539496932, 75.8499077917731] Fitness: 2.2305749907375727
Generation 154 Best Solution: [183.8605573945108, 175.94949961248113,
174.10027539496932, 75.8499077917731] Fitness: 1.5561833475215952
Generation 155 Best Solution: [183.8605573945108, 176.35050045085194,

175.10330377761736, 75.8499077917731] Fitness: 2.3220295282977954
Generation 156 Best Solution: [183.8605573945108, 177.42756403050822,
175.10330377761736, 74.93902546793511] Fitness: 2.5042874224795253
Generation 157 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.69967130983925] Fitness: 2.2361580843798663
Generation 158 Best Solution: [182.985502013962, 177.31420141325594,
175.10330377761736, 74.95769444031632] Fitness: 1.9802926537148648
Generation 159 Best Solution: [183.8605573945108, 175.99664566394137,
175.10330377761736, 75.8499077917731] Fitness: 2.3383227381109606
Generation 160 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.95769444031632] Fitness: 2.2909806352295123
Generation 161 Best Solution: [183.8605573945108, 175.9848729904525,
175.10330377761736, 75.49687649048896] Fitness: 1.7704280591909258
Generation 162 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.95769444031632] Fitness: 2.2909806352295123
Generation 163 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.95769444031632] Fitness: 2.2909806352295123
Generation 164 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.97474823153948] Fitness: 2.298513332783842
Generation 165 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.70583198128274] Fitness: 2.23617401634403
Generation 166 Best Solution: [183.8605573945108, 176.09135578659652,
173.73651438231747, 75.8499077917731] Fitness: 1.6376107604780872
Generation 167 Best Solution: [183.8605573945108, 177.31420141325594,
173.73651438231747, 74.93309682953989] Fitness: 1.6050715466130003
Generation 168 Best Solution: [183.8605573945108, 175.62214824492924,
173.73651438231747, 74.93309682953989] Fitness: 0.8521788708970108
Generation 169 Best Solution: [183.17087613990066, 177.31420141325594,
173.99478628618482, 74.70583198128274] Fitness: 1.098001032904341
Generation 170 Best Solution: [183.8605573945108, 177.31420141325594,
173.99478628618482, 74.70583198128274] Fitness: 1.4494129055073608
Generation 171 Best Solution: [183.8605573945108, 177.31420141325594,
174.77286183217993, 74.3899845408602] Fitness: 1.868082923051164
Generation 172 Best Solution: [183.86115322191253, 177.31420141325594,
174.77286183217993, 74.3899845408602] Fitness: 1.8687292520964813
Generation 173 Best Solution: [183.19750390869822, 177.31420141325594,
175.10330377761736, 74.63699507038443] Fitness: 1.8870340550785814
Generation 174 Best Solution: [183.17087613990066, 177.31420141325594,
175.10330377761736, 74.3899845408602] Fitness: 1.965456816038503
Generation 175 Best Solution: [183.8605573945108, 177.31420141325594,
174.7731652177791, 74.86490940507503] Fitness: 1.8100329192459048
Generation 176 Best Solution: [183.8605573945108, 177.38817751224195,
175.10330377761736, 74.3899845408602] Fitness: 2.4586549125234947
Generation 177 Best Solution: [183.17087613990066, 177.31420141325594,
174.77286183217993, 74.78360963324549] Fitness: 1.4416179551515287
Generation 178 Best Solution: [183.8605573945108, 176.81636794448613,
173.90425081005452, 75.8499077917731] Fitness: 1.8582597851116867
Generation 179 Best Solution: [182.84357951654485, 177.31420141325594,

175.10330377761736, 74.3899845408602] Fitness: 2.076067894593861
Generation 180 Best Solution: [183.20130069332777, 177.31420141325594,
175.10330377761736, 74.3899845408602] Fitness: 1.9642443172947572
Generation 181 Best Solution: [182.84357951654485, 177.31420141325594,
175.10330377761736, 74.87379068695894] Fitness: 2.0201782972749984
Generation 182 Best Solution: [182.84357951654485, 177.31420141325594,
174.62873694249006, 75.8499077917731] Fitness: 2.5074334897564032
Generation 183 Best Solution: [183.8605573945108, 175.9688009282734,
175.10330377761736, 75.8499077917731] Fitness: 2.348461781560126
Generation 184 Best Solution: [182.84357951654485, 175.9688009282734,
175.10330377761736, 75.8499077917731] Fitness: 2.107660987512464
Generation 185 Best Solution: [183.8605573945108, 175.9688009282734,
175.10330377761736, 75.8499077917731] Fitness: 2.348461781560126
Generation 186 Best Solution: [183.27997058451774, 175.9688009282734,
175.10330377761736, 75.8499077917731] Fitness: 1.9998542475143417
Generation 187 Best Solution: [182.84357951654485, 176.29684886658666,
175.10330377761736, 75.8499077917731] Fitness: 2.070277135129067
Generation 188 Best Solution: [183.8605573945108, 175.9688009282734,
175.10330377761736, 75.77909698829033] Fitness: 2.217072053631065
Generation 189 Best Solution: [182.84357951654485, 176.22905296186283,
175.10330377761736, 74.8955407982566] Fitness: 0.995165332455719
Generation 190 Best Solution: [183.29508022188477, 175.9688009282734,
175.10330377761736, 75.77909698829033] Fitness: 1.8704168208004543
Generation 191 Best Solution: [183.8605573945108, 175.9688009282734,
175.10330377761736, 74.66509989806158] Fitness: 1.2499398036265508
Generation 192 Best Solution: [182.84357951654485, 175.9688009282734,
175.10330377761736, 75.8499077917731] Fitness: 2.107660987512464
Generation 193 Best Solution: [183.7726934222108, 176.18651706591584,
175.10330377761736, 75.8499077917731] Fitness: 2.214799468110585
Generation 194 Best Solution: [182.84357951654485, 176.33119331643204,
175.10330377761736, 75.8499077917731] Fitness: 2.076735098993806
Generation 195 Best Solution: [182.84357951654485, 176.33119331643204,
175.10330377761736, 75.8499077917731] Fitness: 2.076735098993806
Generation 196 Best Solution: [183.1144274839279, 176.5616281227746,
175.10330377761736, 75.8499077917731] Fitness: 2.066635352746036
Generation 197 Best Solution: [183.1144274839279, 176.5616281227746,
175.10330377761736, 75.8499077917731] Fitness: 2.066635352746036
Generation 198 Best Solution: [183.8605573945108, 176.2705579725833,
175.10330377761736, 75.8499077917731] Fitness: 2.3074627814113575
Generation 199 Best Solution: [183.8605573945108, 176.22480393954785,
175.10330377761736, 75.8499077917731] Fitness: 2.303918296079118
Generation 200 Best Solution: [183.8605573945108, 177.38817751224195,
173.99815180148724, 75.8499077917731] Fitness: 2.6900783841269407
Generation 201 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.6387611054329] Fitness: 2.2394049832641603
Generation 202 Best Solution: [182.87275741699733, 177.38817751224195,
173.99815180148724, 75.8499077917731] Fitness: 2.4321680809916173
Generation 203 Best Solution: [183.8605573945108, 177.38817751224195,

175.10330377761736, 74.9187185955776] Fitness: 2.417370977035543
Generation 204 Best Solution: [182.8887602290919, 177.38817751224195,
175.10330377761736, 74.6893802244307] Fitness: 2.111367172479579
Generation 205 Best Solution: [183.8605573945108, 177.38817751224195,
174.68949779937842, 74.67793062962404] Fitness: 1.8451350660827637
Generation 206 Best Solution: [182.8887602290919, 176.96008877335572,
173.9245002465236, 75.8499077917731] Fitness: 1.7490876863752285
Generation 207 Best Solution: [183.8605573945108, 177.38817751224195,
174.40269595739707, 74.5445240019391] Fitness: 1.662993980575394
Generation 208 Best Solution: [182.8887602290919, 177.38817751224195,
175.10330377761736, 74.5445240019391] Fitness: 2.1317068902078047
Generation 209 Best Solution: [182.8887602290919, 177.38817751224195,
174.95960455620002, 74.5445240019391] Fitness: 1.9141811958180892
Generation 210 Best Solution: [183.8605573945108, 177.31420141325594,
175.10330377761736, 74.5445240019391] Fitness: 2.2566122772755093
Generation 211 Best Solution: [182.8887602290919, 176.8248688535026,
174.00508127941, 75.8499077917731] Fitness: 1.5720219435636769
Generation 212 Best Solution: [183.8605573945108, 176.25262841245097,
175.10330377761736, 75.8499077917731] Fitness: 2.305658075783325
Generation 213 Best Solution: [182.8887602290919, 176.8248688535026,
175.10330377761736, 75.8499077917731] Fitness: 2.3608980958520043
Generation 214 Best Solution: [182.8887602290919, 176.8248688535026,
175.10330377761736, 75.8499077917731] Fitness: 2.3608980958520043
Generation 215 Best Solution: [182.8887602290919, 176.8248688535026,
175.10330377761736, 75.8499077917731] Fitness: 2.3608980958520043
Generation 216 Best Solution: [182.8887602290919, 176.8248688535026,
175.10330377761736, 75.8499077917731] Fitness: 2.3608980958520043
Generation 217 Best Solution: [182.8887602290919, 176.8248688535026,
174.50787058399845, 75.8499077917731] Fitness: 1.6837047985969218
Generation 218 Best Solution: [182.8887602290919, 176.8248688535026,
174.50787058399845, 75.8499077917731] Fitness: 1.6837047985969218
Generation 219 Best Solution: [183.8605573945108, 176.8248688535026,
174.50787058399845, 75.8499077917731] Fitness: 1.950396409546598
Generation 220 Best Solution: [182.8887602290919, 176.16146035966915,
175.10330377761736, 75.8499077917731] Fitness: 2.0380784293389804
Generation 221 Best Solution: [183.8605573945108, 176.16146035966915,
175.10330377761736, 75.8499077917731] Fitness: 2.304770040288657
Generation 222 Best Solution: [183.52161158229418, 176.38891317619448,
175.10330377761736, 75.8499077917731] Fitness: 2.0610482504308116
Generation 223 Best Solution: [183.52161158229418, 176.38891317619448,
173.81805761801894, 75.8499077917731] Fitness: 1.338138632415199
Generation 224 Best Solution: [183.52161158229418, 176.38891317619448,
173.81805761801894, 75.8499077917731] Fitness: 1.338138632415199
Generation 225 Best Solution: [183.8605573945108, 176.26552015970492,
173.81805761801894, 75.8499077917731] Fitness: 1.5839919588820783
Generation 226 Best Solution: [183.8605573945108, 177.38817751224195,
174.62887207156766, 74.40729159589208] Fitness: 1.8625115100746472
Generation 227 Best Solution: [183.8605573945108, 177.38817751224195,

174.62887207156766, 74.77488065198047] Fitness: 1.7950504171338164
Generation 228 Best Solution: [183.8605573945108, 177.38817751224195,
175.10330377761736, 74.76303034400085] Fitness: 2.3811281870498853
Generation 229 Best Solution: [183.8605573945108, 177.39028999657626,
175.10330377761736, 75.32421343388556] Fitness: 2.705579981010091
Generation 230 Best Solution: [183.18585962960634, 176.49880511103228,
175.10330377761736, 75.8499077917731] Fitness: 2.025059537812788
Generation 231 Best Solution: [183.18585962960634, 176.49880511103228,
175.10330377761736, 75.8499077917731] Fitness: 2.025059537812788
Generation 232 Best Solution: [183.8605573945108, 177.38817751224195,
174.64736477787093, 75.32421343388556] Fitness: 2.1298515594026295
Generation 233 Best Solution: [183.79230022428428, 177.38817751224195,
174.64736477787093, 75.32421343388556] Fitness: 2.0597254157766525
Generation 234 Best Solution: [183.8605573945108, 176.90723661235623,
175.10330377761736, 75.49450041396118] Fitness: 2.1436903860257885
Generation 235 Best Solution: [183.8605573945108, 176.59967356796562,
175.12601489102485, 74.4570817736223] Fitness: 1.423353668359104
Generation 236 Best Solution: [183.8605573945108, 176.59967356796562,
175.12601489102485, 75.0880265657857] Fitness: 1.4983879046489121
Generation 237 Best Solution: [183.8605573945108, 177.38817751224195,
174.52500861334718, 75.02856305894221] Fitness: 1.8012867935339814
Generation 238 Best Solution: [183.8605573945108, 177.38817751224195,
174.52500861334718, 75.1583091518963] Fitness: 1.886105297190113
Generation 239 Best Solution: [183.8605573945108, 177.01255669901067,
174.25989358459242, 75.8499077917731] Fitness: 2.067105186275975
Generation 240 Best Solution: [183.8605573945108, 177.01255669901067,
174.25989358459242, 75.8499077917731] Fitness: 2.067105186275975
Generation 241 Best Solution: [183.8605573945108, 177.38817751224195,
174.25989358459242, 74.43340074332336] Fitness: 1.6527741633883923
Generation 242 Best Solution: [183.8605573945108, 177.38817751224195,
174.25989358459242, 74.43340074332336] Fitness: 1.6527741633883923
Generation 243 Best Solution: [183.8605573945108, 176.13044795304978,
175.10330377761736, 75.8499077917731] Fitness: 2.307625551233674
Generation 244 Best Solution: [182.88573912340206, 176.74601631212798,
174.9278847858211, 75.8499077917731] Fitness: 2.024836671220836
Generation 245 Best Solution: [182.88573912340206, 176.74601631212798,
173.8939790945295, 75.8499077917731] Fitness: 1.529023853698134
Generation 246 Best Solution: [183.8605573945108, 175.57290004992896,
174.08950627608948, 75.8499077917731] Fitness: 1.8328645739359866
Generation 247 Best Solution: [183.8605573945108, 176.88404894148135,
174.4888289174051, 75.8499077917731] Fitness: 2.0029238137983167
Generation 248 Best Solution: [183.8605573945108, 177.38817751224195,
174.05199537145506, 74.69174986391788] Fitness: 1.5817390185991507
Generation 249 Best Solution: [183.8605573945108, 176.74601631212798,
174.08950627608948, 75.8499077917731] Fitness: 1.7512512295205322
Generation 250 Best Solution: [183.8605573945108, 176.74601631212798,
174.08950627608948, 75.8499077917731] Fitness: 1.7512512295205322
Generation 251 Best Solution: [183.8605573945108, 177.38817751224195,

174.08950627608948, 74.39081979635897] Fitness: 1.6586654633033269
Generation 252 Best Solution: [183.8605573945108, 176.88404894148135,
174.21771102646588, 75.59246300348451] Fitness: 1.4613137966156433
Generation 253 Best Solution: [183.7132116442171, 177.38817751224195,
174.21771102646588, 75.59246300348451] Fitness: 2.1051721074563696
Generation 254 Best Solution: [183.8605573945108, 176.88404894148135,
174.28231605588542, 75.8499077917731] Fitness: 1.9126183254630709
Generation 255 Best Solution: [183.7132116442171, 177.38817751224195,
175.10330377761736, 74.60661261309953] Fitness: 2.2437288684202
Generation 256 Best Solution: [183.8605573945108, 177.38817751224195,
175.10330377761736, 74.60661261309953] Fitness: 2.3853980056510475
Generation 257 Best Solution: [183.8605573945108, 176.88404894148135,
175.10330377761736, 75.11760798945035] Fitness: 1.7369112575106627
Generation 258 Best Solution: [183.8605573945108, 176.08959386327913,
174.77386002192603, 74.60661261309953] Fitness: 0.7739774759999093
Generation 259 Best Solution: [183.8605573945108, 175.74983190431345,
175.10330377761736, 75.63691461293097] Fitness: 2.1032585109239705
Generation 260 Best Solution: [183.8605573945108, 175.74983190431345,
175.10330377761736, 74.78694007589165] Fitness: 1.3797497807250978
Generation 261 Best Solution: [183.00393395200334, 177.38817751224195,
173.7778186605508, 75.04449688169589] Fitness: 1.457755607772423
Generation 262 Best Solution: [183.8605573945108, 177.1239216148787,
174.0458116294254, 75.8499077917731] Fitness: 2.2174546107479083
Generation 263 Best Solution: [183.45450954469982, 176.13668240965606,
175.10330377761736, 75.8499077917731] Fitness: 2.0040561022443084
Generation 264 Best Solution: [183.00393395200334, 177.11216719491796,
175.10330377761736, 75.25876673424736] Fitness: 1.8371628369939443
Generation 265 Best Solution: [183.45450954469982, 177.38817751224195,
175.10330377761736, 75.25876673424736] Fitness: 2.3341429083804055
Generation 266 Best Solution: [183.45450954469982, 177.38817751224195,
175.12601489102485, 74.88730359390668] Fitness: 2.141466387786312
Generation 267 Best Solution: [183.8605573945108, 177.38817751224195,
173.89649890217564, 75.8499077917731] Fitness: 2.7198753306298444
Generation 268 Best Solution: [183.8605573945108, 176.59002643733226,
175.12601489102485, 75.8499077917731] Fitness: 2.4669709153855246
Generation 269 Best Solution: [183.0957725116559, 177.38817751224195,
174.569966042527, 75.8499077917731] Fitness: 2.501566584414153
Generation 270 Best Solution: [183.8605573945108, 176.14542546305134,
175.10330377761736, 75.8499077917731] Fitness: 2.306046340253834
Generation 271 Best Solution: [183.8605573945108, 176.14542546305134,
175.10330377761736, 75.8499077917731] Fitness: 2.306046340253834
Generation 272 Best Solution: [183.8605573945108, 177.38817751224195,
174.5816382927015, 75.07053972341103] Fitness: 1.866254098021004
Generation 273 Best Solution: [183.8605573945108, 175.94115652521958,
175.10330377761736, 74.72007823386782] Fitness: 1.2604951791186267
Generation 274 Best Solution: [183.8605573945108, 175.94115652521958,
175.10330377761736, 75.8499077917731] Fitness: 2.3598061817571976
Generation 275 Best Solution: [183.8605573945108, 177.38817751224195,

175.12601489102485, 74.72007823386782] Fitness: 2.415767926246617
Generation 276 Best Solution: [183.64843332161985, 177.38817751224195,
175.12601489102485, 74.72007823386782] Fitness: 2.2232669712082185
Generation 277 Best Solution: [183.8605573945108, 177.38817751224195,
175.12601489102485, 74.72007823386782] Fitness: 2.415767926246617
Generation 278 Best Solution: [182.83467536086928, 177.38817751224195,
175.10330377761736, 75.24564600125186] Fitness: 2.3896632854047004
Generation 279 Best Solution: [183.64843332161985, 177.38817751224195,
175.10330377761736, 75.24564600125186] Fitness: 2.43245927937805
Generation 280 Best Solution: [183.8605573945108, 175.8012998516829,
174.5816382927015, 75.8499077917731] Fitness: 1.8113559927267262
Generation 281 Best Solution: [182.83467536086928, 176.13839945286625,
174.84681854147672, 75.8499077917731] Fitness: 1.7072939649233756
Generation 282 Best Solution: [183.64843332161985, 176.14294183307177,
175.12601489102485, 75.24564600125186] Fitness: 1.2987198545506522
Generation 283 Best Solution: [183.64843332161985, 176.14294183307177,
174.84681854147672, 75.8499077917731] Fitness: 1.7496317133080077
Generation 284 Best Solution: [183.25384339965996, 176.04107830472503,
174.5816382927015, 75.8499077917731] Fitness: 1.348369450072339
Generation 285 Best Solution: [183.49389020653211, 176.04107830472503,
174.5816382927015, 75.8592854681346] Fitness: 1.4319559219851727
Generation 286 Best Solution: [183.49389020653211, 176.04107830472503,
174.5816382927015, 75.8499077917731] Fitness: 1.4139289547982614
Generation 287 Best Solution: [183.64843332161985, 176.14294183307177,
175.10330377761736, 75.8499077917731] Fitness: 2.1137813977504347
Generation 288 Best Solution: [183.64843332161985, 176.14294183307177,
174.8469410533071, 75.8499077917731] Fitness: 1.7497794788689742
Generation 289 Best Solution: [183.64843332161985, 177.38817751224195,
174.2972341353201, 75.8499077917731] Fitness: 2.5097864526763183
Generation 290 Best Solution: [183.64843332161985, 177.38817751224195,
174.98600802962466, 75.8499077917731] Fitness: 3.104911299116454
Generation 291 Best Solution: [183.64843332161985, 177.38817751224195,
174.98600802962466, 75.8499077917731] Fitness: 3.104911299116454
Generation 292 Best Solution: [183.64843332161985, 177.38817751224195,
174.98600802962466, 74.3834006936677] Fitness: 2.0894669726373185
Generation 293 Best Solution: [182.83467536086928, 175.9599436315329,
175.10330377761736, 75.00333986570959] Fitness: 1.0931266778461337
Generation 294 Best Solution: [182.83467536086928, 175.9599436315329,
175.10330377761736, 75.8499077917731] Fitness: 2.116660921537168
Generation 295 Best Solution: [183.64843332161985, 177.38817751224195,
173.8641288777983, 75.00333986570959] Fitness: 1.5169438474038117
Generation 296 Best Solution: [183.64843332161985, 177.38817751224195,
173.8641288777983, 75.00333986570959] Fitness: 1.5169438474038117
Generation 297 Best Solution: [183.67512139616434, 177.38817751224195,
174.98600802962466, 75.00333986570959] Fitness: 2.101472168478132
Generation 298 Best Solution: [183.64843332161985, 177.38817751224195,
175.12601489102485, 75.00333986570959] Fitness: 2.299043730155755
Generation 299 Best Solution: [183.8695782375283, 176.13485144428716,

174.88510388804247, 75.8499077917731] Fitness: 2.000216323133825
Generation 300 Best Solution: [183.64843332161985, 176.5553586094964,
173.7578194863258, 75.8499077917731] Fitness: 1.5262262407001126
Generation 301 Best Solution: [183.64843332161985, 176.5553586094964,
174.98600802962466, 75.50102312565976] Fitness: 1.4688280887599823
Generation 302 Best Solution: [183.64843332161985, 175.75745079550487,
175.12601489102485, 75.8499077917731] Fitness: 2.312576092115104
Generation 303 Best Solution: [183.8695782375283, 176.62124244905894,
174.98600802962466, 75.8499077917731] Fitness: 2.280194397747831
Generation 304 Best Solution: [183.64843332161985, 175.75745079550487,
175.10330377761736, 74.70468209190683] Fitness: 1.1754496282171205
Generation 305 Best Solution: [183.82420706747445, 176.19710016052082,
175.12601489102485, 75.51621910133156] Fitness: 1.7566232096798458
Generation 306 Best Solution: [183.8695782375283, 176.62124244905894,
175.12601489102485, 74.70468209190683] Fitness: 1.3982631842146616
Generation 307 Best Solution: [183.09076191700157, 176.62124244905894,
174.98600802962466, 75.8499077917731] Fitness: 1.929506288594672
Generation 308 Best Solution: [183.64843332161985, 176.4156203538885,
175.12601489102485, 75.8499077917731] Fitness: 2.1867950351747725
Generation 309 Best Solution: [183.64843332161985, 176.4156203538885,
175.12601489102485, 75.8499077917731] Fitness: 2.1867950351747725
Generation 310 Best Solution: [183.64843332161985, 176.4156203538885,
174.39058510624344, 75.8499077917731] Fitness: 1.4083426096577965
Generation 311 Best Solution: [183.64843332161985, 175.861581355044,
175.12601489102485, 74.6766704838382] Fitness: 1.1454924036105758
Generation 312 Best Solution: [183.66626890128447, 177.38817751224195,
175.12601489102485, 74.6766704838382] Fitness: 2.2367693397264574
Generation 313 Best Solution: [183.04712349929505, 176.7685981431756,
175.12601489102485, 75.8499077917731] Fitness: 2.278686319918711
Generation 314 Best Solution: [183.2253200345469, 177.38817751224195,
174.22617690099582, 75.8499077917731] Fitness: 2.3333925258713064
Generation 315 Best Solution: [183.2253200345469, 177.38817751224195,
175.12601489102485, 75.8499077917731] Fitness: 3.1625870788067068
Generation 316 Best Solution: [182.98232518055357, 176.31569611394067,
175.12601489102485, 75.8499077917731] Fitness: 2.04241613338243
Generation 317 Best Solution: [183.85855078802808, 176.2449481091953,
175.12601489102485, 75.84218381369159] Fitness: 2.3256672253978503
Generation 318 Best Solution: [183.85855078802808, 176.2449481091953,
174.72485772045192, 75.84218381369159] Fitness: 1.7892958144963658
Generation 319 Best Solution: [183.85855078802808, 176.2449481091953,
174.72485772045192, 75.84218381369159] Fitness: 1.7892958144963658
Generation 320 Best Solution: [183.8695782375283, 176.00219455150813,
174.32923111307025, 75.8499077917731] Fitness: 1.5811765919863374
Generation 321 Best Solution: [183.482869136992, 176.09713208908494,
175.12601489102485, 75.8499077917731] Fitness: 2.059368227122027
Generation 322 Best Solution: [183.85855078802808, 177.38817751224195,
175.12601489102485, 74.63327971801186] Fitness: 2.417142907988732
Generation 323 Best Solution: [183.85855078802808, 175.99442953258333,

175.12601489102485, 75.8499077917731] Fitness: 2.374438606157505
Generation 324 Best Solution: [183.8695782375283, 177.38817751224195,
175.12601489102485, 74.63327971801186] Fitness: 2.4291640136296273
Generation 325 Best Solution: [183.87839952480476, 177.38817751224195,
175.12601489102485, 75.12757965477043] Fitness: 2.5865814458255585
Generation 326 Best Solution: [183.8695782375283, 176.55372013006667,
175.12601489102485, 75.42553736589548] Fitness: 1.7920025823472836
Generation 327 Best Solution: [183.8695782375283, 177.38817751224195,
174.5688899427313, 75.62395816088467] Fitness: 2.462390078766754
Generation 328 Best Solution: [183.8695782375283, 176.55481209806314,
175.12601489102485, 75.8499077917731] Fitness: 2.4550326198824854
Generation 329 Best Solution: [183.4842400442978, 177.38817751224195,
174.18747360775762, 75.8499077917731] Fitness: 2.390475653332487
Generation 330 Best Solution: [183.8695782375283, 175.73972759127355,
175.12601489102485, 75.8499077917731] Fitness: 2.528295343315137
Generation 331 Best Solution: [183.8695782375283, 177.38817751224195,
174.2204556269847, 74.46478252333263] Fitness: 1.6417481885358085
Generation 332 Best Solution: [183.4280660596762, 177.38817751224195,
174.2204556269847, 74.46478252333263] Fitness: 1.318839317670627
Generation 333 Best Solution: [183.4280660596762, 175.73972759127355,
175.12601489102485, 75.62395816088467] Fitness: 1.8153473329491308
Generation 334 Best Solution: [183.13238445190063, 176.21572561148406,
175.12601489102485, 75.84218381369159] Fitness: 1.9787531795526272
Generation 335 Best Solution: [183.16706420797834, 177.38817751224195,
175.12601489102485, 74.4730117320548] Fitness: 2.1077113067446924
Generation 336 Best Solution: [183.0796704616732, 175.73972759127355,
175.05091807658133, 75.8499077917731] Fitness: 2.0590998942268985
Generation 337 Best Solution: [183.16706420797834, 177.38817751224195,
175.12601489102485, 74.36121284421085] Fitness: 2.1606457904728273
Generation 338 Best Solution: [183.13238445190063, 176.5980688526339,
175.12601489102485, 75.84218381369159] Fitness: 2.1098314235542617
Generation 339 Best Solution: [183.16706420797834, 177.10810815119282,
175.12601489102485, 75.53388984794833] Fitness: 2.153435704561595
Generation 340 Best Solution: [183.16706420797834, 176.16674678355682,
175.12601489102485, 75.84218381369159] Fitness: 1.9760826292726752
Generation 341 Best Solution: [183.8695782375283, 176.5980688526339,
175.12601489102485, 75.22048747741503] Fitness: 1.6071936728942804
Generation 342 Best Solution: [183.8695782375283, 177.38817751224195,
174.35798312393376, 75.01476417712799] Fitness: 1.7151596287092266
Generation 343 Best Solution: [183.16706420797834, 177.38817751224195,
175.12601489102485, 75.22048747741503] Fitness: 2.2890359722215634
Generation 344 Best Solution: [183.16706420797834, 177.38817751224195,
173.80464563253602, 75.53388984794833] Fitness: 1.8886627039260113
Generation 345 Best Solution: [183.16706420797834, 177.10810815119282,
175.12601489102485, 74.88619236908963] Fitness: 1.6041438660159066
Generation 346 Best Solution: [183.16706420797834, 177.38817751224195,
175.13789337290103, 74.88619236908963] Fitness: 2.1128089953627383
Generation 347 Best Solution: [183.16706420797834, 177.38817751224195,

175.05091807658133, 74.59614071046371] Fitness: 1.9527035506152
Generation 348 Best Solution: [183.56097357102368, 177.38817751224195,
175.05091807658133, 74.59614071046371] Fitness: 2.053819355049026
Generation 349 Best Solution: [183.8695782375283, 176.18747849737966,
175.12601489102485, 75.8499077917731] Fitness: 2.3509882563304414
Generation 350 Best Solution: [183.60586161742802, 177.38817751224195,
175.05091807658133, 74.59614071046371] Fitness: 2.081755998878704
Generation 351 Best Solution: [183.8695782375283, 176.18747849737966,
175.12601489102485, 75.8499077917731] Fitness: 2.3509882563304414
Generation 352 Best Solution: [183.8695782375283, 176.75237293196867,
175.05091807658133, 75.8499077917731] Fitness: 2.483176960709893
Generation 353 Best Solution: [183.8695782375283, 177.05656811944047,
175.12601489102485, 75.35712519043378] Fitness: 2.2194670935690257
Generation 354 Best Solution: [183.8695782375283, 175.65576568316004,
175.05091807658133, 75.8499077917731] Fitness: 2.4779327882530118
Generation 355 Best Solution: [183.60313972955095, 177.38817751224195,
174.32945788139193, 75.35712519043378] Fitness: 1.7475345921678063
Generation 356 Best Solution: [183.60313972955095, 177.38817751224195,
174.32945788139193, 75.35712519043378] Fitness: 1.7475345921678063
Generation 357 Best Solution: [183.60313972955095, 177.38817751224195,
175.12601489102485, 74.73960857219919] Fitness: 2.19281188799921
Generation 358 Best Solution: [183.80685444217949, 176.15797076273472,
175.05091807658133, 75.8499077917731] Fitness: 2.165901077166616
Generation 359 Best Solution: [183.8695782375283, 176.05251774824657,
175.05091807658133, 75.8499077917731] Fitness: 2.248439303252841
Generation 360 Best Solution: [183.8695782375283, 176.05251774824657,
175.05091807658133, 75.8499077917731] Fitness: 2.248439303252841
Generation 361 Best Solution: [183.60313972955095, 177.38817751224195,
174.43518465851554, 75.8499077917731] Fitness: 2.5342711329573318
Generation 362 Best Solution: [183.8695782375283, 177.38817751224195,
175.05091807658133, 74.55735887281203] Fitness: 2.3217475378962322
Generation 363 Best Solution: [183.60313972955095, 177.38817751224195,
175.05091807658133, 74.28399358253596] Fitness: 2.2158194715903496
Generation 364 Best Solution: [183.60313972955095, 176.72972014582572,
175.05091807658133, 75.8499077917731] Fitness: 2.229058076659206
Generation 365 Best Solution: [183.60313972955095, 176.72972014582572,
175.05091807658133, 75.30769474943337] Fitness: 1.4359799851675334
Generation 366 Best Solution: [183.60313972955095, 177.38817751224195,
174.44133980745195, 74.28399358253596] Fitness: 1.5829462278151936
Generation 367 Best Solution: [183.80685444217949, 175.90264324021314,
174.67076231100464, 75.8499077917731] Fitness: 1.7712602660640149
Generation 368 Best Solution: [183.8674930346434, 175.90264324021314,
175.05091807658133, 75.68204906250715] Fitness: 2.0041215187132697
Generation 369 Best Solution: [183.07280755086998, 177.38817751224195,
175.05091807658133, 74.77727855578652] Fitness: 1.9624775324439163
Generation 370 Best Solution: [183.8674930346434, 177.38817751224195,
175.05091807658133, 74.77727855578652] Fitness: 2.3070403011991565
Generation 371 Best Solution: [183.60313972955095, 177.38817751224195,

175.05091807658133, 74.54856106895859] Fitness: 2.0901840120039195
Generation 372 Best Solution: [183.40118964747268, 177.38817751224195,
175.12601489102485, 74.41168271686972] Fitness: 2.163094948967617
Generation 373 Best Solution: [183.60313972955095, 177.38817751224195,
174.15121045033698, 74.54856106895859] Fitness: 1.3736204750975256
Generation 374 Best Solution: [183.60313972955095, 177.38817751224195,
175.12601489102485, 74.41168271686972] Fitness: 2.2614327536765066
Generation 375 Best Solution: [183.60313972955095, 177.38817751224195,
175.12601489102485, 74.41168271686972] Fitness: 2.2614327536765066
Generation 376 Best Solution: [183.60313972955095, 176.07257147518013,
175.05091807658133, 75.8499077917731] Fitness: 2.0100678707039523
Generation 377 Best Solution: [183.60313972955095, 176.07257147518013,
175.05091807658133, 75.8499077917731] Fitness: 2.0100678707039523
Generation 378 Best Solution: [183.60313972955095, 176.07257147518013,
175.05091807658133, 75.8499077917731] Fitness: 2.0100678707039523
Generation 379 Best Solution: [183.76889806713035, 177.0478433986274,
175.12601489102485, 74.56385045863982] Fitness: 1.762051313378707
Generation 380 Best Solution: [183.86535893427936, 177.38817751224195,
175.05091807658133, 74.90916371773508] Fitness: 2.335922970681521
Generation 381 Best Solution: [182.92349241496916, 177.38817751224195,
174.41174730124686, 74.90916371773508] Fitness: 1.3985812864994487
Generation 382 Best Solution: [183.8674930346434, 177.38817751224195,
174.5732759681643, 75.33073747185155] Fitness: 2.08407115896412
Generation 383 Best Solution: [183.60313972955095, 177.38817751224195,
174.5732759681643, 75.8499077917731] Fitness: 2.6219660912656413
Generation 384 Best Solution: [183.86535893427936, 177.38817751224195,
174.41174730124686, 75.8985474566733] Fitness: 2.8467485869722893
Generation 385 Best Solution: [183.8674930346434, 177.38817751224195,
175.05091807658133, 74.69114185996912] Fitness: 2.3023013174744205
Generation 386 Best Solution: [183.8674930346434, 177.38817751224195,
175.05091807658133, 74.69114185996912] Fitness: 2.3023013174744205
Generation 387 Best Solution: [182.8448888772804, 177.38817751224195,
175.05091807658133, 74.71905635105358] Fitness: 2.053323587825299
Generation 388 Best Solution: [183.25657459338362, 177.38817751224195,
175.05091807658133, 74.69114185996912] Fitness: 1.9438614025946463
Generation 389 Best Solution: [183.8674930346434, 177.38817751224195,
175.05091807658133, 74.69114185996912] Fitness: 2.3023013174744205
Generation 390 Best Solution: [183.25657459338362, 177.38817751224195,
173.818827996246, 75.7805551519258] Fitness: 2.2746012899791115
Generation 391 Best Solution: [183.86535893427936, 177.38817751224195,
175.05091807658133, 75.32447662191177] Fitness: 2.6236103369544828
Generation 392 Best Solution: [183.00581081648897, 177.38817751224195,
175.05091807658133, 74.70634561577542] Fitness: 1.976735853507042
Generation 393 Best Solution: [183.86535893427936, 177.38817751224195,
174.74078749835297, 75.01631096249902] Fitness: 1.9832558867640133
Generation 394 Best Solution: [183.59638515539476, 177.38817751224195,
174.74078749835297, 75.01631096249902] Fitness: 1.7497543497387242
Generation 395 Best Solution: [183.8674930346434, 176.7888479043832,

175.05091807658133, 74.45061329553639] Fitness: 1.4682237968335299
Generation 396 Best Solution: [183.8674930346434, 177.38817751224195,
174.74078749835297, 75.01631096249902] Fitness: 1.9855906849846783
Generation 397 Best Solution: [183.8674930346434, 177.38817751224195,
175.05091807658133, 74.95238633704574] Fitness: 2.3547958255890675
Generation 398 Best Solution: [183.8674930346434, 175.8066551037516,
175.05091807658133, 75.74643449624236] Fitness: 2.168271356259179
Generation 399 Best Solution: [183.8674930346434, 175.8066551037516,
175.05091807658133, 75.74643449624236] Fitness: 2.168271356259179
Generation 400 Best Solution: [183.86535893427936, 175.9041217112859,
175.05091807658133, 75.8499077917731] Fitness: 2.298940119050904
Generation 401 Best Solution: [183.8674930346434, 175.9041217112859,
175.05091807658133, 75.8499077917731] Fitness: 2.3012749172715683
Generation 402 Best Solution: [183.8674930346434, 175.9041217112859,
175.05091807658133, 75.74643449624236] Fitness: 2.1120962183172107
Generation 403 Best Solution: [183.86535893427936, 176.58299515634667,
173.8058567094396, 75.8499077917731] Fitness: 1.7135717575779252
Generation 404 Best Solution: [183.2025351626596, 176.07795491392056,
175.05091807658133, 75.8499077917731] Fitness: 1.880185312946047
Generation 405 Best Solution: [183.2025351626596, 177.38817751224195,
175.05091807658133, 74.73885205056068] Fitness: 1.9431972707431568
Generation 406 Best Solution: [183.8674930346434, 177.0595105860604,
175.05091807658133, 74.98553534732407] Fitness: 1.8094070434180025
Generation 407 Best Solution: [183.56241084536185, 177.38817751224195,
175.05091807658133, 74.73885205056068] Fitness: 2.046645337519941
Generation 408 Best Solution: [183.8674930346434, 176.11283301314882,
175.05091807658133, 75.23817946288565] Fitness: 1.3749170567887188
Generation 409 Best Solution: [183.2025351626596, 176.96018413040747,
174.16584643446566, 75.88652533023598] Fitness: 1.703095309065489
Generation 410 Best Solution: [183.2025351626596, 176.96018413040747,
174.16584643446566, 74.87567327730967] Fitness: 0.5576326307289284
Generation 411 Best Solution: [183.2025351626596, 176.96018413040747,
174.16584643446566, 74.87567327730967] Fitness: 0.5576326307289284
Generation 412 Best Solution: [183.2025351626596, 176.11283301314882,
175.05091807658133, 75.8499077917731] Fitness: 1.8740347905480719
Generation 413 Best Solution: [183.8674930346434, 176.11283301314882,
175.05091807658133, 75.35801318824608] Fitness: 1.494130907366228
Generation 414 Best Solution: [183.8674930346434, 177.38817751224195,
174.39331221790727, 74.39070496575218] Fitness: 1.7261394681066722
Generation 415 Best Solution: [183.8674930346434, 176.3164562954929,
175.05091807658133, 75.8499077917731] Fitness: 2.2387986706261147
Generation 416 Best Solution: [183.8674930346434, 176.3164562954929,
175.05091807658133, 75.8499077917731] Fitness: 2.2387986706261147
Generation 417 Best Solution: [183.2025351626596, 176.3164562954929,
175.05091807658133, 75.8499077917731] Fitness: 1.878597531619398
Generation 418 Best Solution: [183.8674930346434, 176.3164562954929,
175.05091807658133, 74.83912544090921] Fitness: 1.1550435342675718
Generation 419 Best Solution: [183.64745661044265, 177.38817751224195,

175.01474794822263, 74.86655927793416] Fitness: 2.069393512805801
Generation 420 Best Solution: [183.8674930346434, 176.12544859446632,
175.05091807658133, 75.73771829751605] Fitness: 2.0282110472885413
Generation 421 Best Solution: [182.81466995726274, 177.38817751224195,
175.05091807658133, 74.94565671630507] Fitness: 2.12200687751762
Generation 422 Best Solution: [183.8674930346434, 177.002910933102,
173.9645835209503, 75.8499077917731] Fitness: 2.067094307638512
Generation 423 Best Solution: [183.8674930346434, 177.38817751224195,
173.9645835209503, 75.8499077917731] Fitness: 2.705573636232541
Generation 424 Best Solution: [183.86215773739528, 176.45627453919414,
175.01474794822263, 75.8499077917731] Fitness: 2.221289124874903
Generation 425 Best Solution: [183.8674930346434, 175.78344743546506,
175.05442251729832, 75.88652533023598] Fitness: 2.4498105758354765
Generation 426 Best Solution: [183.8674930346434, 176.59879433588446,
175.05091807658133, 75.88652533023598] Fitness: 2.43068516220566
Generation 427 Best Solution: [183.86215773739528, 176.82895011839156,
175.05091807658133, 74.79268674841357] Fitness: 1.4576645502801644
Generation 428 Best Solution: [182.81466995726274, 176.02559835517215,
175.05091807658133, 75.88652533023598] Fitness: 2.094638023726414
Generation 429 Best Solution: [183.60994943593937, 176.59879433588446,
175.05091807658133, 75.88652533023598] Fitness: 2.203737212792032
Generation 430 Best Solution: [183.8674930346434, 176.82895011839156,
174.55162613424514, 75.88652533023598] Fitness: 2.0630827272904027
Generation 431 Best Solution: [183.8674930346434, 177.38817751224195,
174.48956845158034, 74.67828793768089] Fitness: 1.697291991738134
Generation 432 Best Solution: [183.22882214954402, 177.38817751224195,
174.48956845158034, 74.67828793768089] Fitness: 1.3373396439955325
Generation 433 Best Solution: [183.8674930346434, 177.38817751224195,
174.48956845158034, 74.67828793768089] Fitness: 1.697291991738134
Generation 434 Best Solution: [183.8674930346434, 176.35688499254783,
175.05091807658133, 75.88652533023598] Fitness: 2.3191489630200044
Generation 435 Best Solution: [183.86215773739528, 177.38817751224195,
174.32474615981147, 74.28463514309976] Fitness: 1.7576327862930752
Generation 436 Best Solution: [183.8674930346434, 176.21041410991742,
175.05091807658133, 75.88652533023598] Fitness: 2.2990215321174645
Generation 437 Best Solution: [183.6418118898889, 177.38817751224195,
175.05091807658133, 74.82705095842282] Fitness: 2.1105527468076897
Generation 438 Best Solution: [183.8674930346434, 176.21041410991742,
175.05091807658133, 75.88652533023598] Fitness: 2.2990215321174645
Generation 439 Best Solution: [183.8674930346434, 176.35688499254783,
175.05091807658133, 75.88652533023598] Fitness: 2.3191489630200044
Generation 440 Best Solution: [183.8674930346434, 176.35688499254783,
175.05091807658133, 75.88652533023598] Fitness: 2.3191489630200044
Generation 441 Best Solution: [183.6418118898889, 177.00642183938396,
174.34242311377233, 75.88652533023598] Fitness: 1.9572379258178454
Generation 442 Best Solution: [183.8674930346434, 177.00642183938396,
174.34242311377233, 75.88652533023598] Fitness: 2.1621009111137948
Generation 443 Best Solution: [183.8674930346434, 177.00642183938396,

174.34242311377233, 75.88652533023598] Fitness: 2.1621009111137948
Generation 444 Best Solution: [183.6418118898889, 175.69760174176233,
175.05091807658133, 75.88652533023598] Fitness: 2.3054304683868905
Generation 445 Best Solution: [183.6418118898889, 177.00642183938396,
173.88739295987222, 75.88652533023598] Fitness: 1.9635264146193392
Generation 446 Best Solution: [183.8674930346434, 177.00642183938396,
173.88739295987222, 75.88652533023598] Fitness: 2.1683893999152883
Generation 447 Best Solution: [183.8674930346434, 177.38817751224195,
175.05091807658133, 75.16606896213017] Fitness: 2.4823029722836734
Generation 448 Best Solution: [183.47664976928795, 177.00642183938396,
174.61390189107857, 75.88652533023598] Fitness: 2.0217130620744515
Generation 449 Best Solution: [183.47664976928795, 177.00642183938396,
175.05091807658133, 75.88652533023598] Fitness: 2.538273465164176
Generation 450 Best Solution: [183.8674930346434, 176.899851526525,
175.05091807658133, 75.88652533023598] Fitness: 2.705712411862578
Generation 451 Best Solution: [183.6418118898889, 177.00642183938396,
175.05091807658133, 74.91356170559627] Fitness: 1.5011574354081985
Generation 452 Best Solution: [183.0266895872179, 177.38817751224195,
175.05091807658133, 74.50806240263299] Fitness: 2.001056802612406
Generation 453 Best Solution: [183.8674930346434, 177.38817751224195,
175.05091807658133, 74.50806240263299] Fitness: 2.3333021219209376
Generation 454 Best Solution: [183.8674930346434, 177.38817751224195,
175.05091807658133, 74.50806240263299] Fitness: 2.3333021219209376
Generation 455 Best Solution: [182.82718230709028, 177.3829990529618,
175.05091807658133, 74.50806240263299] Fitness: 2.0849568572654635
Generation 456 Best Solution: [183.8674930346434, 177.3829990529618,
174.90115836821653, 74.50806240263299] Fitness: 2.110211938650658
Generation 457 Best Solution: [183.6418118898889, 177.3829990529618,
175.05091807658133, 75.11254518805595] Fitness: 2.2281359026528054
Generation 458 Best Solution: [183.6418118898889, 177.3829990529618,
175.05091807658133, 74.50806240263299] Fitness: 2.1182169591360682
Generation 459 Best Solution: [183.6418118898889, 177.3829990529618,
173.8277982560888, 74.50806240263299] Fitness: 1.473718093044417
Generation 460 Best Solution: [183.8674930346434, 176.74154982349532,
175.05091807658133, 75.88652533023598] Fitness: 2.5422657253923386
Generation 461 Best Solution: [183.8674930346434, 175.94384000927792,
174.19533251911798, 75.88652533023598] Fitness: 1.6412644785353416
Generation 462 Best Solution: [182.86668997883345, 177.3829990529618,
175.05091807658133, 74.98737207355065] Fitness: 2.0982107812589943
Generation 463 Best Solution: [183.6418118898889, 176.0721781517472,
175.05091807658133, 75.88652533023598] Fitness: 2.1079599947423455
Generation 464 Best Solution: [182.86668997883345, 177.3829990529618,
174.19533251911798, 74.98737207355065] Fitness: 1.3853293401744549
Generation 465 Best Solution: [183.6418118898889, 177.38817751224195,
174.19533251911798, 74.98737207355065] Fitness: 1.4527179656767633
Generation 466 Best Solution: [183.6418118898889, 176.0721781517472,
175.05091807658133, 75.88652533023598] Fitness: 2.1079599947423455
Generation 467 Best Solution: [183.47204190504294, 177.38817751224195,

175.05091807658133, 74.98737207355065] Fitness: 2.0674364498855504
Generation 468 Best Solution: [183.4351138243239, 177.3829990529618,
175.05091807658133, 74.98737207355065] Fitness: 2.0422228339826303
Generation 469 Best Solution: [183.8674930346434, 177.38817751224195,
173.79873604839682, 74.98737207355065] Fitness: 1.7409757472857508
Generation 470 Best Solution: [183.8674930346434, 177.38817751224195,
175.05091807658133, 74.98737207355065] Fitness: 2.370462392057252
Generation 471 Best Solution: [182.8055040165688, 177.3829990529618,
175.05091807658133, 74.45216932723073] Fitness: 2.119774989824191
Generation 472 Best Solution: [183.8674930346434, 177.38817751224195,
175.05091807658133, 74.45216932723073] Fitness: 2.353897242306946
Generation 473 Best Solution: [183.6418118898889, 177.29498456596266,
175.05091807658133, 74.45216932723073] Fitness: 1.9719084754860305
Generation 474 Best Solution: [183.8674930346434, 177.38817751224195,
175.05091807658133, 74.5607494144007] Fitness: 2.31865565148634
Generation 475 Best Solution: [183.6103219864717, 177.38817751224195,
173.83878257859863, 75.34772048120882] Fitness: 1.7740351580331986
Generation 476 Best Solution: [183.6103219864717, 176.9024222656835,
173.83878257859863, 75.88652533023598] Fitness: 1.8322055390321672
Generation 477 Best Solution: [183.8674930346434, 177.38817751224195,
174.22828080897654, 74.45216932723073] Fitness: 1.6458815313064543
Generation 478 Best Solution: [183.8674930346434, 177.21714655831872,
174.68345169628995, 74.5607494144007] Fitness: 1.5490330917959778
Generation 479 Best Solution: [183.8674930346434, 176.65907118230112,
175.05091807658133, 74.5607494144007] Fitness: 1.319264322331118
Generation 480 Best Solution: [183.8674930346434, 176.65907118230112,
175.05091807658133, 75.5791798226164] Fitness: 1.945199490603069
Generation 481 Best Solution: [183.68810190314616, 176.49856653366945,
175.05091807658133, 75.88652533023598] Fitness: 2.202876777711147
Generation 482 Best Solution: [183.8674930346434, 177.38817751224195,
175.05091807658133, 74.76907020200251] Fitness: 2.3060556492569053
Generation 483 Best Solution: [183.68810190314616, 176.7091936904684,
175.05091807658133, 75.88652533023598] Fitness: 2.3442357228854114
Generation 484 Best Solution: [183.68810190314616, 176.7091936904684,
175.05091807658133, 75.88652533023598] Fitness: 2.3442357228854114
Generation 485 Best Solution: [183.68810190314616, 177.38817751224195,
175.05091807658133, 74.5959065043656] Fitness: 2.141692895165619
Generation 486 Best Solution: [183.68810190314616, 177.38817751224195,
175.05091807658133, 74.32077315012492] Fitness: 2.2530577850680875
Generation 487 Best Solution: [182.8316172795227, 177.38817751224195,
175.05091807658133, 74.88919646433203] Fitness: 2.0907328982841555
Generation 488 Best Solution: [183.68810190314616, 177.38817751224195,
174.290062520852, 74.32077315012492] Fitness: 1.5590430111091935
Generation 489 Best Solution: [183.8674930346434, 177.38817751224195,
173.74093832231995, 74.7154323533333] Fitness: 1.706938446064036
Generation 490 Best Solution: [183.76794322214343, 176.2290633644818,
175.04382956483076, 75.88652533023598] Fitness: 2.1878490169629745
Generation 491 Best Solution: [183.68810190314616, 176.41577221060948,

```
175.04382956483076, 75.88652533023598] Fitness: 2.1566370645811244
Generation 492 Best Solution: [183.76794322214343, 176.2290633644818,
175.04382956483076, 75.88652533023598] Fitness: 2.1878490169629745
Generation 493 Best Solution: [183.68810190314616, 176.9242023932103,
174.54721525047225, 75.88652533023598] Fitness: 1.9974048679265786
Generation 494 Best Solution: [183.68810190314616, 176.2290633644818,
175.04382956483076, 75.88652533023598] Fitness: 2.1189163425097983
Generation 495 Best Solution: [183.68810190314616, 177.22251066796684,
175.04382956483076, 74.54425418247364] Fitness: 1.8371954796452963
Generation 496 Best Solution: [183.76794322214343, 176.2290633644818,
175.04382956483076, 75.88652533023598] Fitness: 2.1878490169629745
Generation 497 Best Solution: [183.68810190314616, 176.1014991505022,
175.05091807658133, 75.88652533023598] Fitness: 2.137471281081148
Generation 498 Best Solution: [183.76794322214343, 177.38817751224195,
175.05091807658133, 74.70692275831404] Fitness: 2.2014139277353038
Generation 499 Best Solution: [183.76794322214343, 177.38817751224195,
175.05091807658133, 74.70692275831404] Fitness: 2.2014139277353038
Generation 500 Best Solution: [183.76794322214343, 177.38817751224195,
174.31951559428364, 75.11114303791804] Fitness: 1.6563302005002547
Optimal Solution for Simulation 2: [183.76794322214343, 177.38817751224195,
174.31951559428364, 75.11114303791804]
--------------------------------------------------
Simulation 3/32 with conditions: {'drought': True, 'flood': True,
'policy_change': True, 'social_event': False, 'economic_crisis': True}
Generation 1 Best Solution: [183.41433624875856, 176.30713993834118,
174.4487876270089, 74.86852649743116] Fitness: 0.20496636297738738
Generation 2 Best Solution: [183.23304375101637, 176.30713993834118,
174.4487876270089, 74.37362775517488] Fitness: 0.24110808493157726
Generation 3 Best Solution: [183.43604148926843, 176.30713993834118,
174.4487876270089, 75.59125602247406] Fitness: 0.85527743171332
Generation 4 Best Solution: [183.43604148926843, 176.30713993834118,
174.4487876270089, 75.67759469693016] Fitness: 0.991143675184166
Generation 5 Best Solution: [183.78234614264673, 176.44203237837024,
173.86196790475185, 75.67759469693016] Fitness: 1.2338679844143847
Generation 6 Best Solution: [182.91667441725235, 176.07145027667048,
174.76191945033617, 75.53060547906558] Fitness: 1.0595687726760366
Generation 7 Best Solution: [183.78234614264673, 176.44501340880873,
173.86196790475185, 75.2628584960259] Fitness: 0.6976716493710415
Generation 8 Best Solution: [183.74737209747073, 176.07145027667048,
173.86196790475185, 75.67759469693016] Fitness: 1.1685087836736565
Generation 9 Best Solution: [183.78234614264673, 176.44203237837024,
174.76191945033617, 75.00907231171988] Fitness: 0.7924308706952679
Generation 10 Best Solution: [183.78234614264673, 176.44203237837024,
174.76191945033617, 75.2628584960259] Fitness: 0.9777189707731735
Generation 11 Best Solution: [183.78234614264673, 176.44203237837024,
174.76191945033617, 74.75078295454819] Fitness: 0.7157704319510885
Generation 12 Best Solution: [182.98681873853556, 176.26026044567618,
174.76191945033617, 75.53060547906558] Fitness: 1.0161184945616082
```

```
Generation 13 Best Solution: [183.6317832875707, 176.26026044567618,
174.76191945033617, 75.67759469693016] Fitness: 1.3442994998139892
Generation 14 Best Solution: [183.74737209747073, 176.80024203933195,
174.76191945033617, 75.44318107365547] Fitness: 1.3955214179647255
Generation 15 Best Solution: [183.7741376149611, 176.3091146393026,
174.76191945033617, 75.2628584960259] Fitness: 0.9314051761180493
Generation 16 Best Solution: [183.6317832875707, 177.3656320490554,
174.76191945033617, 75.11261498113927] Fitness: 1.8221342791657902
Generation 17 Best Solution: [183.78234614264673, 177.32427393327285,
174.76191945033617, 74.87894615224943] Fitness: 1.752875413829845
Generation 18 Best Solution: [183.6317832875707, 175.69827782057015,
173.87230448197533, 75.67759469693016] Fitness: 1.2727445844201726
Generation 19 Best Solution: [183.78234614264673, 176.0645067369821,
173.87230448197533, 75.53564297952416] Fitness: 0.9814867163121941
Generation 20 Best Solution: [183.6317832875707, 175.69827782057015,
173.80378352787625, 75.67759469693016] Fitness: 1.30628312412614
Generation 21 Best Solution: [183.41433624875856, 177.32427393327285,
174.42273711980422, 75.25584817282626] Fitness: 1.4777835715544367
Generation 22 Best Solution: [183.6317832875707, 175.69827782057015,
174.76191945033617, 75.71565851547948] Fitness: 1.6224535813110605
Generation 23 Best Solution: [183.78234614264673, 175.69827782057015,
175.06060410559402, 75.25584817282626] Fitness: 1.5365689884364093
Generation 24 Best Solution: [183.78234614264673, 177.3656320490554,
174.76191945033617, 74.48177717187525] Fitness: 1.8489479908374507
Generation 25 Best Solution: [183.78234614264673, 177.3656320490554,
174.76191945033617, 74.48177717187525] Fitness: 1.8489479908374507
Generation 26 Best Solution: [183.6317832875707, 177.3656320490554,
174.17588588717825, 75.12189065648721] Fitness: 1.4898014797789871
Generation 27 Best Solution: [183.78234614264673, 177.23075449394082,
174.42273711980422, 75.67759469693016] Fitness: 2.091544340679913
Generation 28 Best Solution: [183.3685215265787, 177.32427393327285,
175.13129639484694, 74.9183527128049] Fitness: 2.024612308876074
Generation 29 Best Solution: [183.3685215265787, 177.32427393327285,
175.13129639484694, 74.9183527128049] Fitness: 2.024612308876074
Generation 30 Best Solution: [183.78234614264673, 177.32427393327285,
173.84876809709613, 75.56582785922524] Fitness: 2.0796432272949286
Generation 31 Best Solution: [183.78234614264673, 177.32427393327285,
174.0745853502466, 75.67759469693016] Fitness: 2.1898366395863125
Generation 32 Best Solution: [183.78234614264673, 177.32427393327285,
174.48152246356625, 74.49858976048432] Fitness: 1.5284065006282928
Generation 33 Best Solution: [183.6317832875707, 177.32427393327285,
174.48152246356625, 74.49858976048432] Fitness: 1.402681609073752
Generation 34 Best Solution: [183.78234614264673, 177.32427393327285,
173.74221426705498, 75.23164273639925] Fitness: 1.7468757564875002
Generation 35 Best Solution: [183.78234614264673, 177.32427393327285,
173.74221426705498, 75.23164273639925] Fitness: 1.7468757564875002
Generation 36 Best Solution: [183.78234614264673, 177.32427393327285,
173.74221426705498, 75.23164273639925] Fitness: 1.7468757564875002
```

Generation 37 Best Solution: [183.78234614264673, 177.32427393327285,
173.76486677637507, 75.67759469693016] Fitness: 2.298619966434738
Generation 38 Best Solution: [183.6317832875707, 177.32427393327285,
175.13129639484694, 75.2283412139579] Fitness: 2.345732677498486
Generation 39 Best Solution: [183.86873000765394, 177.32427393327285,
174.16347961473318, 75.67759469693016] Fitness: 2.2780497423255985
Generation 40 Best Solution: [183.6317832875707, 175.92632639885193,
175.13129639484694, 75.67759469693016] Fitness: 1.919336129917951
Generation 41 Best Solution: [183.3226585680946, 177.32427393327285,
175.13129639484694, 75.67759469693016] Fitness: 2.7768496034644716
Generation 42 Best Solution: [183.78234614264673, 175.64977097820864,
175.13129639484694, 75.67759469693016] Fitness: 2.2399885324787796
Generation 43 Best Solution: [183.78234614264673, 176.43845759484915,
175.13129639484694, 75.67759469693016] Fitness: 2.0258609838367247
Generation 44 Best Solution: [183.78234614264673, 176.43845759484915,
175.13129639484694, 75.67759469693016] Fitness: 2.0258609838367247
Generation 45 Best Solution: [183.86873000765394, 176.25698025001878,
175.13129639484694, 75.67759469693016] Fitness: 2.071342927601021
Generation 46 Best Solution: [183.78234614264673, 176.4416419989293,
175.13129639484694, 75.39510150793915] Fitness: 1.629474051856401
Generation 47 Best Solution: [183.78234614264673, 176.4416419989293,
175.13129639484694, 75.83618752345511] Fitness: 2.309566913527326
Generation 48 Best Solution: [183.78234614264673, 176.4416419989293,
175.13129639484694, 75.65631786119161] Fitness: 1.99247663361931
Generation 49 Best Solution: [183.86873000765394, 176.8379553025263,
175.13129639484694, 75.39510150793915] Fitness: 2.010844165114563
Generation 50 Best Solution: [183.78234614264673, 176.28843788467134,
175.13129639484694, 75.39510150793915] Fitness: 1.5878628867187343
Generation 51 Best Solution: [182.90337554915925, 176.28843788467134,
175.13129639484694, 75.39510150793915] Fitness: 1.3956429300139412
Generation 52 Best Solution: [182.90337554915925, 176.28843788467134,
175.13129639484694, 75.4155756278572] Fitness: 1.4199243676517552
Generation 53 Best Solution: [182.90337554915925, 176.28843788467134,
175.13129639484694, 75.67759469693016] Fitness: 1.7932989928981364
Generation 54 Best Solution: [183.06902848436488, 176.28843788467134,
175.13129639484694, 75.67759469693016] Fitness: 1.729541017761318
Generation 55 Best Solution: [183.86873000765394, 176.28843788467134,
175.13129639484694, 75.67759469693016] Fitness: 2.074971685494641
Generation 56 Best Solution: [182.90337554915925, 176.63677626828414,
175.13129639484694, 75.67759469693016] Fitness: 1.9454252161012544
Generation 57 Best Solution: [182.86795419496406, 175.8553184729259,
175.13129639484694, 75.67759469693016] Fitness: 1.9102001674799396
Generation 58 Best Solution: [183.45098777212053, 177.32427393327285,
175.13129639484694, 75.67759469693016] Fitness: 2.814461015040574
Generation 59 Best Solution: [183.86873000765394, 177.32427393327285,
174.01256547981404, 75.67759469693016] Fitness: 2.2880739694886536
Generation 60 Best Solution: [183.86873000765394, 177.01084036304178,
175.13129639484694, 75.67759469693016] Fitness: 2.6191108344162823

Generation 61 Best Solution: [183.86873000765394, 177.32427393327285,
175.13129639484694, 75.14751065829346] Fitness: 2.494788606151159
Generation 62 Best Solution: [183.32350353603402, 177.32427393327285,
175.13129639484694, 75.67759469693016] Fitness: 2.777006103430854
Generation 63 Best Solution: [183.86873000765394, 176.3034215931288,
175.13129639484694, 75.67759469693016] Fitness: 2.0772889209127947
Generation 64 Best Solution: [183.49781625873405, 175.94632160314347,
175.13129639484694, 75.67759469693016] Fitness: 1.8306465799104452
Generation 65 Best Solution: [183.86873000765394, 175.61457995388918,
175.13129639484694, 75.67759469693016] Fitness: 2.3635282142218785
Generation 66 Best Solution: [183.86873000765394, 175.61457995388918,
174.7210332909656, 75.67759469693016] Fitness: 1.808864283562249
Generation 67 Best Solution: [183.45098777212053, 177.3214647329207,
174.9837473372329, 74.84747869935339] Fitness: 1.7929901209420127
Generation 68 Best Solution: [183.45098777212053, 175.61457995388918,
175.13129639484694, 75.67759469693016] Fitness: 2.048071913871807
Generation 69 Best Solution: [183.45098777212053, 176.56046861311455,
174.0519378873672, 75.67759469693016] Fitness: 1.0125774503988552
Generation 70 Best Solution: [183.86873000765394, 176.03047686212054,
175.13129639484694, 75.67759469693016] Fitness: 2.094654861294644
Generation 71 Best Solution: [183.25749980177642, 177.3214647329207,
175.13129639484694, 75.11493135656254] Fitness: 2.1044506818004995
Generation 72 Best Solution: [183.86873000765394, 176.41384022155017,
175.13129639484694, 75.04571524856698] Fitness: 1.4034118361880334
Generation 73 Best Solution: [183.86873000765394, 177.32427393327285,
174.7517969822767, 75.11493135656254] Fitness: 1.9483132065897237
Generation 74 Best Solution: [183.45098777212053, 177.32427393327285,
174.7517969822767, 75.11493135656254] Fitness: 1.632856906239652
Generation 75 Best Solution: [183.45098777212053, 176.41384022155017,
175.13129639484694, 75.67759469693016] Fitness: 1.7906253817826046
Generation 76 Best Solution: [183.45098777212053, 176.41384022155017,
175.13129639484694, 75.67759469693016] Fitness: 1.7906253817826046
Generation 77 Best Solution: [183.45098777212053, 176.41384022155017,
175.13129639484694, 75.11493135656254] Fitness: 1.1319722560255416
Generation 78 Best Solution: [183.86873000765394, 176.41384022155017,
175.13129639484694, 75.11493135656254] Fitness: 1.4474285563756135
Generation 79 Best Solution: [183.40969113513734, 176.7880199704205,
175.13129639484694, 74.41587716470293] Fitness: 1.2958323061283665
Generation 80 Best Solution: [183.40969113513734, 176.22675891298007,
175.13129639484694, 75.67759469693016] Fitness: 1.7388330837940236
Generation 81 Best Solution: [183.45098777212053, 177.3214647329207,
175.13129639484694, 74.74226995975124] Fitness: 2.009189192250389
Generation 82 Best Solution: [183.45098777212053, 177.3214647329207,
175.13129639484694, 74.74226995975124] Fitness: 2.009189192250389
Generation 83 Best Solution: [183.45098777212053, 177.32427393327285,
175.13129639484694, 74.74226995975124] Fitness: 2.0145057390454415
Generation 84 Best Solution: [183.45098777212053, 177.3214647329207,
175.13129639484694, 74.74226995975124] Fitness: 2.009189192250389

Generation 85 Best Solution: [183.86873000765394, 175.9697795937007,
175.13129639484694, 75.50250531193625] Fitness: 1.8532727570609353
Generation 86 Best Solution: [183.86873000765394, 175.9697795937007,
174.24902024154161, 75.67759469693016] Fitness: 1.2752175331438338
Generation 87 Best Solution: [183.26661920464048, 175.66404382044084,
175.13129639484694, 75.67759469693016] Fitness: 1.9554639418104651
Generation 88 Best Solution: [183.45098777212053, 177.32427393327285,
175.13129639484694, 74.63310445636802] Fitness: 2.0176201588303053
Generation 89 Best Solution: [183.45098777212053, 176.40579670779798,
175.13129639484694, 75.67759469693016] Fitness: 1.7878311850011936
Generation 90 Best Solution: [183.86873000765394, 176.6452360348399,
174.01773341798227, 75.67759469693016] Fitness: 1.3905225633825953
Generation 91 Best Solution: [183.45098777212053, 176.6452360348399,
174.01773341798227, 75.67759469693016] Fitness: 1.0750662630325236
Generation 92 Best Solution: [183.07493219895065, 177.3214647329207,
174.25311460237083, 75.67759469693016] Fitness: 1.9381778431207906
Generation 93 Best Solution: [183.64867414511878, 177.0383277851755,
174.07676859547453, 75.67759469693016] Fitness: 1.603454456897082
Generation 94 Best Solution: [183.64867414511878, 177.0383277851755,
175.13129639484694, 75.31415933641826] Fitness: 1.9675689384707538
Generation 95 Best Solution: [183.64867414511878, 177.0383277851755,
174.01773341798227, 75.67759469693016] Fitness: 1.6114490817018938
Generation 96 Best Solution: [183.55827099712948, 175.6154899242946,
175.13129639484694, 75.31415933641826] Fitness: 1.6132916018149408
Generation 97 Best Solution: [183.55827099712948, 175.6154899242946,
175.13129639484694, 75.2746648437111] Fitness: 1.5738667835254265
Generation 98 Best Solution: [183.55827099712948, 175.6154899242946,
174.1637596761685, 75.2746648437111] Fitness: 0.7220164370866586
Generation 99 Best Solution: [183.45098777212053, 177.3214647329207,
174.1637596761685, 75.2746648437111] Fitness: 1.4311652131735766
Generation 100 Best Solution: [183.64867414511878, 177.3214647329207,
174.4502363743443, 75.07225531206244] Fitness: 1.4701482159900185
Generation 101 Best Solution: [183.45098777212053, 176.21852964908024,
175.13129639484694, 75.67759469693016] Fitness: 1.7537256771836436
Generation 102 Best Solution: [183.45098777212053, 176.21852964908024,
175.13129639484694, 75.67759469693016] Fitness: 1.7537256771836436
Generation 103 Best Solution: [183.45098777212053, 176.10080553361138,
175.13129639484694, 75.67759469693016] Fitness: 1.7626664886055547
Generation 104 Best Solution: [182.99665982731352, 176.21852964908024,
175.13129639484694, 75.17855605925959] Fitness: 1.1348524361293166
Generation 105 Best Solution: [182.99665982731352, 176.21852964908024,
175.13129639484694, 75.4457284350822] Fitness: 1.4095699538073667
Generation 106 Best Solution: [182.99665982731352, 176.02495918445135,
175.13129639484694, 75.4457284350822] Fitness: 1.4366939828692749
Generation 107 Best Solution: [182.99665982731352, 176.21852964908024,
175.13129639484694, 75.67759469693016] Fitness: 1.7458929378188435
Generation 108 Best Solution: [183.45098777212053, 176.02495918445135,
175.13129639484694, 75.67759469693016] Fitness: 1.7808497062455517

```
Generation 109 Best Solution: [182.99665982731352, 176.02495918445135,
175.13129639484694, 75.67759469693016] Fitness: 1.7730169668807516
Generation 110 Best Solution: [183.45098777212053, 176.02495918445135,
175.13129639484694, 75.67759469693016] Fitness: 1.7808497062455517
Generation 111 Best Solution: [183.55827099712948, 175.8304680832276,
175.13129639484694, 75.67759469693016] Fitness: 1.9247973907849467
Generation 112 Best Solution: [183.84159219907116, 177.32427393327285,
173.74321816128085, 74.84969782635352] Fitness: 1.5889523753964347
Generation 113 Best Solution: [183.64867414511878, 176.58484480049503,
175.08611115868405, 75.75484773585445] Fitness: 2.045624961580741
Generation 114 Best Solution: [183.55827099712948, 177.22784981415668,
173.74321816128085, 75.75484773585445] Fitness: 2.0965347818805014
Generation 115 Best Solution: [183.55827099712948, 175.99005645198375,
175.0608689360601, 75.67759469693016] Fitness: 1.7298520133625679
Generation 116 Best Solution: [183.55827099712948, 177.3214647329207,
175.13129639484694, 74.958813920723] Fitness: 2.1155225149442844
Generation 117 Best Solution: [183.55827099712948, 176.0939461728008,
175.13129639484694, 75.67759469693016] Fitness: 1.8167421242690949
Generation 118 Best Solution: [183.55827099712948, 176.0939461728008,
175.13129639484694, 75.67759469693016] Fitness: 1.8167421242690949
Generation 119 Best Solution: [183.55827099712948, 177.3214647329207,
174.74727711190812, 74.42033244192665] Fitness: 1.6014557035014043
Generation 120 Best Solution: [182.99665982731352, 177.3214647329207,
174.13905298419078, 75.67759469693016] Fitness: 1.9484523245301586
Generation 121 Best Solution: [183.55827099712948, 177.32427393327285,
174.3869429932888, 75.03992682477538] Fitness: 1.3653872662543969
Generation 122 Best Solution: [183.64867414511878, 176.03297945810783,
174.13905298419078, 75.67759469693016] Fitness: 1.0380798613015656
Generation 123 Best Solution: [182.99665982731352, 176.53130243561847,
175.13129639484694, 75.67759469693016] Fitness: 1.836071417573828
Generation 124 Best Solution: [182.99665982731352, 176.53130243561847,
175.13129639484694, 75.67759469693016] Fitness: 1.836071417573828
Generation 125 Best Solution: [183.35483086905427, 176.53130243561847,
175.13129639484694, 75.67759469693016] Fitness: 1.8131043238431421
Generation 126 Best Solution: [183.35483086905427, 177.3214647329207,
175.13129639484694, 74.84345775919832] Fitness: 1.9935130815395519
Generation 127 Best Solution: [182.99665982731352, 177.3214647329207,
175.13129639484694, 74.6982811439556] Fitness: 2.0001854633652494
Generation 128 Best Solution: [183.55827099712948, 177.3214647329207,
175.13129639484694, 74.6982811439556] Fitness: 2.0608497976861067
Generation 129 Best Solution: [183.64867414511878, 175.8029584229543,
175.13129639484694, 74.6982811439556] Fitness: 1.201367598666975
Generation 130 Best Solution: [182.99665982731352, 175.8029584229543,
175.13129639484694, 74.6982811439556] Fitness: 1.0810622452802539
Generation 131 Best Solution: [182.99665982731352, 176.20507364750947,
174.18920054602302, 75.75484773585445] Fitness: 1.0282582728141558
Generation 132 Best Solution: [183.64867414511878, 176.05101549883148,
175.13129639484694, 74.6982811439556] Fitness: 1.08485162458642
```

Generation 133 Best Solution: [183.55827099712948, 176.05101549883148,
175.13129639484694, 74.6982811439556] Fitness: 1.025210605520556
Generation 134 Best Solution: [183.55827099712948, 177.3214647329207,
175.13129639484694, 74.55508900322876] Fitness: 2.0797173270540408
Generation 135 Best Solution: [182.99665982731352, 177.3214647329207,
175.13129639484694, 74.55508900322876] Fitness: 2.0190529927331835
Generation 136 Best Solution: [182.99665982731352, 176.38444837002146,
175.13129639484694, 75.75484773585445] Fitness: 1.9053756384720497
Generation 137 Best Solution: [182.99665982731352, 176.20156908334565,
175.13129639484694, 75.75484773585445] Fitness: 1.8779973152163794
Generation 138 Best Solution: [183.86996829899306, 175.75102252253845,
175.13129639484694, 75.39693407979445] Fitness: 1.848918484688743
Generation 139 Best Solution: [182.99665982731352, 176.20156908334565,
175.13129639484694, 75.75484773585445] Fitness: 1.8779973152163794
Generation 140 Best Solution: [182.99665982731352, 177.3214647329207,
174.71973378356583, 75.39693407979445] Fitness: 1.8498318321790739
Generation 141 Best Solution: [182.99665982731352, 176.20156908334565,
175.13129639484694, 75.75484773585445] Fitness: 1.8779973152163794
Generation 142 Best Solution: [182.82160159396568, 176.20846004109782,
175.13129639484694, 75.57798619115601] Fitness: 1.6803211778648286
Generation 143 Best Solution: [182.82160159396568, 176.20846004109782,
175.13129639484694, 75.75484773585445] Fitness: 1.9682133147319845
Generation 144 Best Solution: [182.82160159396568, 177.3214647329207,
174.0491313162629, 75.57798619115601] Fitness: 1.8880823286588424
Generation 145 Best Solution: [182.82160159396568, 175.9909313740986,
175.13129639484694, 75.75484773585445] Fitness: 2.0068111735896803
Generation 146 Best Solution: [182.82160159396568, 175.9909313740986,
175.13129639484694, 75.57798619115601] Fitness: 1.7189190367225247
Generation 147 Best Solution: [182.94771890779919, 177.3214647329207,
174.0491313162629, 75.38833868031267] Fitness: 1.5679783415163513
Generation 148 Best Solution: [182.82160159396568, 175.9909313740986,
175.13129639484694, 75.38833868031267] Fitness: 1.4690281911515783
Generation 149 Best Solution: [182.82160159396568, 177.3214647329207,
174.0491313162629, 75.38833868031267] Fitness: 1.6381914830878963
Generation 150 Best Solution: [182.98444276213044, 177.32427393327285,
174.34283792347347, 75.38833868031267] Fitness: 1.5920624853005378
Generation 151 Best Solution: [182.98444276213044, 177.3214647329207,
174.34283792347347, 75.38833868031267] Fitness: 1.5867459385054854
Generation 152 Best Solution: [183.86996829899306, 177.3214647329207,
174.34283792347347, 75.38833868031267] Fitness: 1.9067967789959472
Generation 153 Best Solution: [183.86996829899306, 176.61649023782368,
175.13129639484694, 74.93624662310107] Fitness: 1.45824830432315
Generation 154 Best Solution: [183.86996829899306, 176.15265289920973,
175.13129639484694, 74.93624662310107] Fitness: 1.3169358167170269
Generation 155 Best Solution: [183.14617195021677, 177.3214647329207,
174.34283792347347, 75.38833868031267] Fitness: 1.5461489493650822
Generation 156 Best Solution: [183.15296178831383, 175.60132356219935,
175.13129639484694, 75.75484773585445] Fitness: 2.1491645593206936

Generation 157 Best Solution: [183.86996829899306, 177.3214647329207,
175.13129639484694, 74.93624662310107] Fitness: 2.3702415250383404
Generation 158 Best Solution: [183.86996829899306, 176.41120815108843,
175.13129639484694, 75.75484773585445] Fitness: 2.2387917334874885
Generation 159 Best Solution: [183.53355481224077, 176.60531776396306,
175.13129639484694, 75.75484773585445] Fitness: 2.0606638073844668
Generation 160 Best Solution: [182.86885447160068, 175.60132356219935,
175.13129639484694, 75.77713557043897] Fitness: 2.286671179268525
Generation 161 Best Solution: [183.86996829899306, 176.85449897781166,
175.13129639484694, 74.41001908157733] Fitness: 1.7000802842306058
Generation 162 Best Solution: [183.4793828353779, 176.78398954418844,
175.13129639484694, 74.41001908157733] Fitness: 1.3219895098760264
Generation 163 Best Solution: [183.86996829899306, 176.0719645972266,
175.13129639484694, 75.10855635382268] Fitness: 1.4222178504233125
Generation 164 Best Solution: [183.86996829899306, 176.78398954418844,
175.13129639484694, 74.41001908157733] Fitness: 1.626732174751871
Generation 165 Best Solution: [182.86885447160068, 177.3214647329207,
174.56228051947278, 75.0012992978097] Fitness: 1.442559579581189
Generation 166 Best Solution: [182.86885447160068, 177.3214647329207,
174.56228051947278, 74.41001908157733] Fitness: 1.4414451449425902
Generation 167 Best Solution: [183.86996829899306, 177.3214647329207,
173.9841195573407, 75.37356492980167] Fitness: 1.86790615961675
Generation 168 Best Solution: [182.86885447160068, 177.3214647329207,
174.56228051947278, 75.0012992978097] Fitness: 1.442559579581189
Generation 169 Best Solution: [183.86996829899306, 176.01672216730938,
175.13129639484694, 75.37356492980167] Fitness: 1.677808904686306
Generation 170 Best Solution: [182.86885447160068, 177.3214647329207,
174.56228051947278, 74.28180099145007] Fitness: 1.5192634575383221
Generation 171 Best Solution: [183.86996829899306, 176.42403895331907,
175.13129639484694, 75.75484773585445] Fitness: 2.243418439985654
Generation 172 Best Solution: [183.86996829899306, 176.42403895331907,
175.13129639484694, 75.75484773585445] Fitness: 2.243418439985654
Generation 173 Best Solution: [183.86996829899306, 176.71220033571478,
175.13129639484694, 75.37356492980167] Fitness: 1.8660238895028947
Generation 174 Best Solution: [183.59936690430308, 176.71220033571478,
175.13129639484694, 75.32612868366887] Fitness: 1.5756197375015808
Generation 175 Best Solution: [183.59936690430308, 177.3214647329207,
173.92215744749458, 75.75484773585445] Fitness: 2.202258531665828
Generation 176 Best Solution: [183.86996829899306, 176.3331742129449,
175.13129639484694, 74.85926928610058] Fitness: 1.3034875151801721
Generation 177 Best Solution: [183.59936690430308, 177.3214647329207,
175.13129639484694, 74.73988364441584] Fitness: 2.0872705838583197
Generation 178 Best Solution: [183.59936690430308, 177.3214647329207,
175.13129639484694, 74.62057990591009] Fitness: 2.0921794825227455
Generation 179 Best Solution: [183.07170543110826, 177.3214647329207,
173.92215744749458, 75.75484773585445] Fitness: 2.0934062036395034
Generation 180 Best Solution: [183.07170543110826, 176.59129935428933,
175.13129639484694, 75.75484773585445] Fitness: 1.9817656396184873

Generation 181 Best Solution: [183.07170543110826, 177.3214647329207,
175.13129639484694, 74.62057990591009] Fitness: 1.9833271544964215
Generation 182 Best Solution: [183.86996829899306, 176.19760510424354,
175.13129639484694, 75.75484773585445] Fitness: 2.2026937089889422
Generation 183 Best Solution: [183.86996829899306, 175.63864914247355,
175.13129639484694, 75.4577347690878] Fitness: 2.0202204027346116
Generation 184 Best Solution: [183.86996829899306, 175.63864914247355,
175.13129639484694, 74.50733943180366] Fitness: 1.5731093866202552
Generation 185 Best Solution: [183.86996829899306, 176.13054343935255,
175.13129639484694, 75.75484773585445] Fitness: 2.207286741888425
Generation 186 Best Solution: [183.86996829899306, 175.93206734928214,
175.13129639484694, 75.75484773585445] Fitness: 2.265475066161353
Generation 187 Best Solution: [183.12257834445847, 176.6402557769369,
175.13129639484694, 75.75484773585445] Fitness: 2.0058066298120454
Generation 188 Best Solution: [183.12257834445847, 176.6402557769369,
175.13129639484694, 75.75484773585445] Fitness: 2.0058066298120454
Generation 189 Best Solution: [183.86996829899306, 177.3214647329207,
174.14115566592443, 75.75484773585445] Fitness: 2.4054234694262013
Generation 190 Best Solution: [183.86996829899306, 177.3214647329207,
175.13129639484694, 74.69346070345152] Fitness: 2.324919359050996
Generation 191 Best Solution: [183.59936690430308, 177.3214647329207,
175.13129639484694, 74.69346070345152] Fitness: 2.08631788574128
Generation 192 Best Solution: [183.8343250469265, 177.3214647329207,
174.12183615197515, 74.69346070345152] Fitness: 1.433453012779365
Generation 193 Best Solution: [183.86996829899306, 176.10580372364433,
175.13129639484694, 74.89069958868629] Fitness: 1.3068031432529819
Generation 194 Best Solution: [183.86996829899306, 177.3214647329207,
175.13129639484694, 74.79631899907793] Fitness: 2.3319420228557375
Generation 195 Best Solution: [183.51231073467403, 176.5140308602384,
175.13129639484694, 75.75484773585445] Fitness: 1.9946818630368761
Generation 196 Best Solution: [183.86996829899306, 175.7153160198025,
174.93183123581392, 75.75484773585445] Fitness: 2.099926750504947
Generation 197 Best Solution: [183.86996829899306, 176.5140308602384,
175.13129639484694, 75.75484773585445] Fitness: 2.2836985956520715
Generation 198 Best Solution: [183.86996829899306, 176.5140308602384,
175.13129639484694, 75.75484773585445] Fitness: 2.2836985956520715
Generation 199 Best Solution: [183.86996829899306, 176.5140308602384,
175.13129639484694, 75.75484773585445] Fitness: 2.2836985956520715
Generation 200 Best Solution: [183.86996829899306, 176.5140308602384,
175.13129639484694, 75.29922678458017] Fitness: 1.6495155974873503
Generation 201 Best Solution: [183.86996829899306, 176.5140308602384,
175.13129639484694, 75.75484773585445] Fitness: 2.2836985956520715
Generation 202 Best Solution: [183.86996829899306, 176.5140308602384,
175.13129639484694, 75.75484773585445] Fitness: 2.2836985956520715
Generation 203 Best Solution: [183.86996829899306, 176.09612957249806,
175.13129639484694, 75.75484773585445] Fitness: 2.2125986463116982
Generation 204 Best Solution: [183.86996829899306, 177.3214647329207,
175.13129639484694, 74.79285990696599] Fitness: 2.3314149189582865

Generation 205 Best Solution: [182.8368434281378, 177.3214647329207,
174.73045556546975, 75.75484773585445] Fitness: 2.4687480216574325
Generation 206 Best Solution: [183.12335027953173, 175.6514329761295,
175.13129639484694, 75.75484773585445] Fitness: 2.1039181199305497
Generation 207 Best Solution: [183.101678713868, 177.3214647329207,
174.5936702401781, 75.75484773585445] Fitness: 2.2354284625296112
Generation 208 Best Solution: [183.101678713868, 177.32474956922903,
174.24702839004271, 75.75484773585445] Fitness: 2.069720538803272
Generation 209 Best Solution: [183.101678713868, 177.32474956922903,
175.13129639484694, 75.27252021619742] Fitness: 2.250105776832389
Generation 210 Best Solution: [183.86996829899306, 177.3214647329207,
174.02362072586158, 75.75484773585445] Fitness: 2.4143512398508347
Generation 211 Best Solution: [183.86996829899306, 177.38764300955037,
174.02362072586158, 74.84212325222639] Fitness: 1.625739170926672
Generation 212 Best Solution: [183.0217719793302, 175.71868624182108,
175.13129639484694, 75.75484773585445] Fitness: 2.0690448417545184
Generation 213 Best Solution: [183.0217719793302, 177.38764300955037,
174.09193333372212, 75.75484773585445] Fitness: 2.201760686010276
Generation 214 Best Solution: [183.0217719793302, 176.37739760736096,
175.13129639484694, 75.75484773585445] Fitness: 1.8945837351069632
Generation 215 Best Solution: [183.86996829899306, 176.51403955316388,
174.78465290238316, 75.75484773585445] Fitness: 1.79639059977048
Generation 216 Best Solution: [183.0217719793302, 176.37739760736096,
175.13129639484694, 75.75484773585445] Fitness: 1.8945837351069632
Generation 217 Best Solution: [183.86996829899306, 176.5918022330902,
174.43466677899622, 75.75484773585445] Fitness: 1.5565259738024608
Generation 218 Best Solution: [183.86996829899306, 176.5918022330902,
174.43466677899622, 75.75484773585445] Fitness: 1.5565259738024608
Generation 219 Best Solution: [183.86996829899306, 176.35043877873468,
174.43466677899622, 75.75484773585445] Fitness: 1.447640400314423
Generation 220 Best Solution: [183.86996829899306, 176.35043877873468,
174.43466677899622, 75.75484773585445] Fitness: 1.447640400314423
Generation 221 Best Solution: [183.86996829899306, 176.94779510938537,
174.63228768108846, 75.75484773585445] Fitness: 2.0329116362460393
Generation 222 Best Solution: [183.86996829899306, 176.94779510938537,
174.63228768108846, 75.75484773585445] Fitness: 2.0329116362460393
Generation 223 Best Solution: [183.86996829899306, 176.94779510938537,
174.63228768108846, 75.75484773585445] Fitness: 2.0329116362460393
Generation 224 Best Solution: [183.86996829899306, 176.2362717429051,
175.13129639484694, 75.75484773585445] Fitness: 2.203504653451898
Generation 225 Best Solution: [183.86996829899306, 176.0292798880238,
175.13129639484694, 75.75484773585445] Fitness: 2.2286451208200386
Generation 226 Best Solution: [183.108571366272, 176.0292798880238,
175.13129639484694, 75.75484773585445] Fitness: 1.8734864659987394
Generation 227 Best Solution: [183.108571366272, 176.0292798880238,
175.13129639484694, 75.75484773585445] Fitness: 1.8734864659987394
Generation 228 Best Solution: [183.108571366272, 177.38764300955037,
174.18423373743792, 75.75484773585445] Fitness: 2.1816411746359337

```
Generation 229 Best Solution: [183.86996829899306, 176.30115490255906,
175.13129639484694, 75.75484773585445] Fitness: 2.210550431051633
Generation 230 Best Solution: [183.86996829899306, 176.30115490255906,
175.13129639484694, 75.75484773585445] Fitness: 2.210550431051633
Generation 231 Best Solution: [183.86996829899306, 176.30115490255906,
175.13129639484694, 75.75484773585445] Fitness: 2.210550431051633
Generation 232 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.38075402918224] Fitness: 2.542320960783829
Generation 233 Best Solution: [183.741009856718, 177.38764300955037,
175.13129639484694, 74.38075402918224] Fitness: 2.4131565846451064
Generation 234 Best Solution: [183.05674524352455, 175.8258575917421,
175.13129639484694, 75.75484773585445] Fitness: 1.980324091673893
Generation 235 Best Solution: [183.05674524352455, 176.69282795120833,
175.13129639484694, 75.75484773585445] Fitness: 2.060721652461612
Generation 236 Best Solution: [183.05674524352455, 177.38764300955037,
175.13129639484694, 74.38075402918224] Fitness: 2.198649834338409
Generation 237 Best Solution: [183.741009856718, 177.38764300955037,
175.13129639484694, 74.38075402918224] Fitness: 2.4131565846451064
Generation 238 Best Solution: [183.05674524352455, 176.67768544804795,
175.13129639484694, 75.75484773585445] Fitness: 2.048403054457485
Generation 239 Best Solution: [183.39694837285518, 176.67768544804795,
175.13129639484694, 75.75484773585445] Fitness: 2.0560086822030943
Generation 240 Best Solution: [183.741009856718, 175.99080702837247,
175.13129639484694, 75.44571368371048] Fitness: 1.6435405243713355
Generation 241 Best Solution: [183.05674524352455, 176.3429822523728,
175.13129639484694, 75.75484773585445] Fitness: 1.8751981489923
Generation 242 Best Solution: [183.05674524352455, 177.38764300955037,
174.08559325995134, 75.80506591583251] Fitness: 2.2832466471628323
Generation 243 Best Solution: [183.05674524352455, 177.38764300955037,
174.08559325995134, 75.80506591583251] Fitness: 2.2832466471628323
Generation 244 Best Solution: [183.39694837285518, 177.38764300955037,
174.08559325995134, 74.86155970525373] Fitness: 1.2868976657818167
Generation 245 Best Solution: [183.39694837285518, 177.38764300955037,
174.36182183385222, 75.22790114904824] Fitness: 1.5428793540806318
Generation 246 Best Solution: [183.39694837285518, 176.60532129741864,
175.13129639484694, 75.80506591583251] Fitness: 2.093891181437355
Generation 247 Best Solution: [183.39694837285518, 177.38764300955037,
174.3352288640704, 75.75484773585445] Fitness: 2.234581498425149
Generation 248 Best Solution: [183.39694837285518, 177.38764300955037,
173.78192764671442, 75.75484773585445] Fitness: 2.2989039729000855
Generation 249 Best Solution: [183.39694837285518, 177.38764300955037,
175.13129639484694, 74.48200674812999] Fitness: 2.1594484217653727
Generation 250 Best Solution: [183.39694837285518, 177.38764300955037,
174.36182183385222, 75.75484773585445] Fitness: 2.244538637980513
Generation 251 Best Solution: [183.741009856718, 177.38764300955037,
173.78192764671442, 75.75484773585445] Fitness: 2.5058050954611737
Generation 252 Best Solution: [183.39694837285518, 175.85505130823555,
175.13129639484694, 75.75484773585445] Fitness: 1.9699418940020847
```

Generation 253 Best Solution: [183.86996829899306, 175.85505130823555,
175.13129639484694, 75.75484773585445] Fitness: 2.306007392701895
Generation 254 Best Solution: [183.86996829899306, 176.4963335957697,
175.13129639484694, 75.75484773585445] Fitness: 2.2746947597477596
Generation 255 Best Solution: [183.741009856718, 177.38764300955037,
175.13129639484694, 75.17423578316635] Fitness: 2.511114222666183
Generation 256 Best Solution: [183.39694837285518, 175.90838231359675,
174.8133778465513, 75.75484773585445] Fitness: 1.4861476425736901
Generation 257 Best Solution: [183.741009856718, 176.42694111723506,
174.57716185185916, 75.75484773585445] Fitness: 1.4336241004364392
Generation 258 Best Solution: [183.741009856718, 177.38764300955037,
175.13129639484694, 74.39524770661853] Fitness: 2.405392454419965
Generation 259 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.46470275162527] Fitness: 2.502284036557562
Generation 260 Best Solution: [183.741009856718, 175.85006293475843,
175.13129639484694, 75.75484773585445] Fitness: 2.1798144537805846
Generation 261 Best Solution: [183.46750422647818, 177.38764300955037,
175.13129639484694, 74.46470275162527] Fitness: 2.1923186601393194
Generation 262 Best Solution: [183.46750422647818, 177.38764300955037,
175.13129639484694, 74.46470275162527] Fitness: 2.1923186601393194
Generation 263 Best Solution: [183.46750422647818, 177.38764300955037,
174.26354242702098, 75.80506591583251] Fitness: 2.331195072867404
Generation 264 Best Solution: [183.46750422647818, 177.38764300955037,
175.13129639484694, 74.31530940110727] Fitness: 2.2718403535756866
Generation 265 Best Solution: [183.07204788254774, 177.38764300955037,
175.13129639484694, 74.31530940110727] Fitness: 2.2342697742006234
Generation 266 Best Solution: [183.741009856718, 176.303548052094,
175.13129639484694, 75.80506591583251] Fitness: 2.173175582382836
Generation 267 Best Solution: [183.741009856718, 176.303548052094,
175.13129639484694, 75.80506591583251] Fitness: 2.173175582382836
Generation 268 Best Solution: [183.741009856718, 176.303548052094,
175.13129639484694, 75.80506591583251] Fitness: 2.173175582382836
Generation 269 Best Solution: [183.1829784434237, 177.38764300955037,
175.13129639484694, 75.80506591583251] Fitness: 3.114720935401515
Generation 270 Best Solution: [183.46750422647818, 177.38764300955037,
175.13129639484694, 74.52639943370127] Fitness: 2.170497606953482
Generation 271 Best Solution: [183.86996829899306, 175.64762449902346,
175.13129639484694, 75.80506591583251] Fitness: 2.5564902508415503
Generation 272 Best Solution: [183.31712200935107, 176.01254517882202,
175.13129639484694, 75.80506591583251] Fitness: 1.969801717877005
Generation 273 Best Solution: [183.3073840098791, 176.05953341565098,
175.13129639484694, 75.80506591583251] Fitness: 1.954777179906908
Generation 274 Best Solution: [183.3073840098791, 177.38764300955037,
175.13129639484694, 74.47489852977438] Fitness: 2.1411711286281037
Generation 275 Best Solution: [183.34144694602503, 177.38764300955037,
175.13129639484694, 74.47489852977438] Fitness: 2.147556987223433
Generation 276 Best Solution: [183.3073840098791, 177.38764300955037,
174.28003559849023, 75.78170225381425] Fitness: 2.245080538566862

Generation 277 Best Solution: [183.86996829899306, 176.11310598986873,
174.28003559849023, 75.78170225381425] Fitness: 1.424839594842441
Generation 278 Best Solution: [183.3073840098791, 177.38764300955037,
174.28003559849023, 75.31117303799505] Fitness: 1.574698446466857
Generation 279 Best Solution: [183.10294435272442, 176.52918838259973,
175.13129639484694, 75.80506591583251] Fitness: 2.029093720950363
Generation 280 Best Solution: [183.86996829899306, 176.52918838259973,
174.58668746045865, 75.80506591583251] Fitness: 1.708839094888289
Generation 281 Best Solution: [183.86996829899306, 176.52918838259973,
174.58668746045865, 75.80506591583251] Fitness: 1.708839094888289
Generation 282 Best Solution: [183.86996829899306, 176.52918838259973,
174.58668746045865, 75.80506591583251] Fitness: 1.708839094888289
Generation 283 Best Solution: [183.86996829899306, 176.52918838259973,
174.25508501777935, 75.80506591583251] Fitness: 1.5440710125728154
Generation 284 Best Solution: [183.46750422647818, 176.52918838259973,
174.25508501777935, 75.80506591583251] Fitness: 1.2341056361545735
Generation 285 Best Solution: [183.46750422647818, 176.52918838259973,
174.2493999381436, 75.80506591583251] Fitness: 1.232903314180422
Generation 286 Best Solution: [183.46750422647818, 177.38764300955037,
174.15407217793768, 74.83288155422423] Fitness: 1.3052112367761017
Generation 287 Best Solution: [183.05014771018844, 176.52918838259973,
174.2162531822281, 75.78170225381425] Fitness: 1.152014051161872
Generation 288 Best Solution: [183.86996829899306, 176.07866183018885,
174.58668746045865, 75.80506591583251] Fitness: 1.6330691116267935
Generation 289 Best Solution: [183.86996829899306, 176.52918838259973,
175.13129639484694, 74.99876767207199] Fitness: 1.431658603435359
Generation 290 Best Solution: [183.86996829899306, 175.86510538377632,
174.76265014075284, 75.80506591583251] Fitness: 1.880159095304088
Generation 291 Best Solution: [183.86996829899306, 175.86510538377632,
174.07202976307238, 75.0418624784886] Fitness: 0.6126077861576776
Generation 292 Best Solution: [183.86996829899306, 175.86510538377632,
174.07202976307238, 75.37104245833322] Fitness: 0.8922070396637054
Generation 293 Best Solution: [183.46750422647818, 176.83395752394404,
175.13129639484694, 75.0418624784886] Fitness: 1.3907554132456053
Generation 294 Best Solution: [183.46750422647818, 176.52918838259973,
175.13129639484694, 75.80506591583251] Fitness: 2.0732597113769495
Generation 295 Best Solution: [183.46750422647818, 175.58715926562573,
175.13129639484694, 75.80506591583251] Fitness: 2.306605821443483
Generation 296 Best Solution: [183.27943453298627, 176.6851254074784,
175.13129639484694, 75.80506591583251] Fitness: 2.128635884072675
Generation 297 Best Solution: [183.46750422647818, 176.724001816282,
175.13129639484694, 75.80506591583251] Fitness: 2.212402841464799
Generation 298 Best Solution: [183.46750422647818, 176.724001816282,
175.13129639484694, 75.80506591583251] Fitness: 2.212402841464799
Generation 299 Best Solution: [183.86996829899306, 177.38764300955037,
174.9195594353731, 74.55908538962017] Fitness: 2.149682261357618
Generation 300 Best Solution: [183.46750422647818, 176.724001816282,
175.13129639484694, 75.80506591583251] Fitness: 2.212402841464799

Generation 301 Best Solution: [183.46750422647818, 177.38764300955037,
175.13129639484694, 74.76352294192692] Fitness: 2.146587278647896
Generation 302 Best Solution: [183.86345812202788, 176.78669575165944,
174.6592084783309, 75.80506591583251] Fitness: 1.9600591190488457
Generation 303 Best Solution: [183.86345812202788, 175.86716648728913,
174.6592084783309, 75.80506591583251] Fitness: 1.7696115490709683
Generation 304 Best Solution: [183.86996829899306, 176.28884925650343,
174.9195594353731, 75.80506591583251] Fitness: 1.9782294596255556
Generation 305 Best Solution: [183.86996829899306, 177.38764300955037,
174.30713703553, 74.30113664805465] Fitness: 1.7657086258502972
Generation 306 Best Solution: [183.86996829899306, 176.28884925650343,
175.13129639484694, 75.76299159700979] Fitness: 2.223197977166396
Generation 307 Best Solution: [183.86996829899306, 176.28884925650343,
175.13129639484694, 75.62973843815838] Fitness: 1.9995373636832148
Generation 308 Best Solution: [183.86996829899306, 176.28884925650343,
173.79366526075003, 75.80506591583251] Fitness: 1.5412550636839435
Generation 309 Best Solution: [183.86996829899306, 177.38764300955037,
173.90777668180107, 75.19307344828574] Fitness: 1.843341785940344
Generation 310 Best Solution: [183.86996829899306, 176.40142949917077,
173.90777668180107, 75.19307344828574] Fitness: 0.6917632110221548
Generation 311 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.63258290925528] Fitness: 2.4579913399103615
Generation 312 Best Solution: [183.86996829899306, 177.38764300955037,
174.23848358564223, 74.63258290925528] Fitness: 1.6154796191158354
Generation 313 Best Solution: [183.86996829899306, 176.80133737811073,
175.13129639484694, 75.57100731729254] Fitness: 2.205853001255829
Generation 314 Best Solution: [183.41275832088292, 177.38764300955037,
175.0163405604778, 74.63258290925528] Fitness: 1.9429002896237133
Generation 315 Best Solution: [183.86996829899306, 175.8020795518077,
174.66673571503338, 75.80506591583251] Fitness: 1.8243764866181702
Generation 316 Best Solution: [183.86996829899306, 175.7979914091158,
174.66673571503338, 75.80506591583251] Fitness: 1.827175046531686
Generation 317 Best Solution: [183.41275832088292, 176.98020593379718,
175.13129639484694, 75.57100731729254] Fitness: 2.0826246750110533
Generation 318 Best Solution: [183.33164493264073, 175.7979914091158,
175.13129639484694, 75.80506591583251] Fitness: 2.0811884197335337
Generation 319 Best Solution: [183.33164493264073, 175.7979914091158,
175.13129639484694, 75.57100731729254] Fitness: 1.6916283899427884
Generation 320 Best Solution: [183.86996829899306, 175.7373764627121,
175.13129639484694, 75.46738013056081] Fitness: 1.9462926437616643
Generation 321 Best Solution: [183.86996829899306, 176.35381626741952,
173.7369264322399, 75.57100731729254] Fitness: 1.1992919770973893
Generation 322 Best Solution: [183.86996829899306, 176.35381626741952,
173.7369264322399, 75.57100731729254] Fitness: 1.1992919770973893
Generation 323 Best Solution: [183.86996829899306, 176.35381626741952,
173.7369264322399, 75.57100731729254] Fitness: 1.1992919770973893
Generation 324 Best Solution: [183.86996829899306, 176.55075294736176,
175.13129639484694, 75.57100731729254] Fitness: 2.005906024989945

Generation 325 Best Solution: [183.86996829899306, 177.38764300955037,
174.4807410037754, 75.57100731729254] Fitness: 2.34160613531545
Generation 326 Best Solution: [182.98593394277057, 177.38764300955037,
175.13129639484694, 74.42214426798725] Fitness: 2.2004632367398216
Generation 327 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.42214426798725] Fitness: 2.521090574117533
Generation 328 Best Solution: [183.51908627793162, 177.38764300955037,
175.13129639484694, 74.42214426798725] Fitness: 2.2355373778493046
Generation 329 Best Solution: [183.86996829899306, 177.07591898192717,
174.55347494452255, 75.80506591583251] Fitness: 2.2372496710783016
Generation 330 Best Solution: [183.86996829899306, 177.07591898192717,
174.55347494452255, 75.80506591583251] Fitness: 2.2372496710783016
Generation 331 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.47600028636245] Fitness: 2.4978065049662628
Generation 332 Best Solution: [182.98593394277057, 176.10641692412074,
174.69526561532692, 75.80506591583251] Fitness: 1.4015726972917044
Generation 333 Best Solution: [183.86996829899306, 177.38764300955037,
174.55347494452255, 75.80506591583251] Fitness: 2.7791502868326488
Generation 334 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.50307637350014] Fitness: 2.4879546483425643
Generation 335 Best Solution: [183.86996829899306, 177.38764300955037,
173.9559906699472, 74.50307637350014] Fitness: 1.6598018944377484
Generation 336 Best Solution: [183.86996829899306, 176.45582924524598,
175.13129639484694, 75.80506591583251] Fitness: 2.3474755216463388
Generation 337 Best Solution: [183.86996829899306, 176.45582924524598,
175.13129639484694, 75.80506591583251] Fitness: 2.3474755216463388
Generation 338 Best Solution: [183.86996829899306, 176.45582924524598,
175.13129639484694, 75.80506591583251] Fitness: 2.3474755216463388
Generation 339 Best Solution: [183.86996829899306, 176.61442701032612,
174.28946517093547, 75.80506591583251] Fitness: 1.6054850839815087
Generation 340 Best Solution: [183.86996829899306, 175.5910816480437,
175.13129639484694, 75.01413174332393] Fitness: 1.6687693302598916
Generation 341 Best Solution: [183.86996829899306, 176.72048976353258,
175.13129639484694, 75.17401086821873] Fitness: 1.6809664623959089
Generation 342 Best Solution: [183.86996829899306, 176.61442701032612,
175.13129639484694, 75.80506591583251] Fitness: 2.4362026608235636
Generation 343 Best Solution: [183.86996829899306, 176.72048976353258,
175.13129639484694, 75.80506591583251] Fitness: 2.519291279460275
Generation 344 Best Solution: [183.5276587541478, 176.72048976353258,
175.13129639484694, 75.80506591583251] Fitness: 2.2382315083916593
Generation 345 Best Solution: [183.86996829899306, 176.2312120235179,
175.13129639484694, 75.80506591583251] Fitness: 2.294648074473364
Generation 346 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.69194051130373] Fitness: 2.4537438817076196
Generation 347 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.69194051130373] Fitness: 2.4537438817076196
Generation 348 Best Solution: [183.5942766152516, 177.38764300955037,
175.13129639484694, 75.80506591583251] Fitness: 3.236522321852089

Generation 349 Best Solution: [183.86996829899306, 177.38764300955037,
173.8253618557276, 75.80506591583251] Fitness: 2.7025749243752006
Generation 350 Best Solution: [183.74523884461055, 176.5966556958728,
175.13129639484694, 75.80506591583251] Fitness: 2.2987679331257267
Generation 351 Best Solution: [183.86996829899306, 176.31147853919015,
175.13129639484694, 75.67517989225864] Fitness: 2.0760945784115106
Generation 352 Best Solution: [183.74523884461055, 176.5966556958728,
175.13129639484694, 75.80506591583251] Fitness: 2.2987679331257267
Generation 353 Best Solution: [183.74523884461055, 176.5966556958728,
175.13129639484694, 75.80506591583251] Fitness: 2.2987679331257267
Generation 354 Best Solution: [183.74523884461055, 176.5966556958728,
175.13129639484694, 75.80506591583251] Fitness: 2.2987679331257267
Generation 355 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.45581691863137] Fitness: 2.5059574970785565
Generation 356 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.57623715163433] Fitness: 2.467539689093762
Generation 357 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.98442055385239] Fitness: 2.5198887319228
Generation 358 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.98442055385239] Fitness: 2.5198887319228
Generation 359 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.98442055385239] Fitness: 2.5198887319228
Generation 360 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.71690739583994] Fitness: 2.453738748090916
Generation 361 Best Solution: [183.86996829899306, 175.88728540146792,
175.13129639484694, 75.80506591583251] Fitness: 2.379215089023978
Generation 362 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.42926929982077] Fitness: 2.517728408201823
Generation 363 Best Solution: [183.86996829899306, 175.75380056158798,
175.13129639484694, 74.42926929982077] Fitness: 1.5052639830137458
Generation 364 Best Solution: [183.86996829899306, 176.48095464093922,
175.13129639484694, 75.80506591583251] Fitness: 2.3586942480612922
Generation 365 Best Solution: [183.86996829899306, 176.48095464093922,
175.13129639484694, 75.80506591583251] Fitness: 2.3586942480612922
Generation 366 Best Solution: [183.86996829899306, 176.48095464093922,
175.13129639484694, 75.80506591583251] Fitness: 2.3586942480612922
Generation 367 Best Solution: [183.86996829899306, 176.48095464093922,
175.13129639484694, 75.80506591583251] Fitness: 2.3586942480612922
Generation 368 Best Solution: [183.86996829899306, 176.48095464093922,
175.13129639484694, 75.80506591583251] Fitness: 2.3586942480612922
Generation 369 Best Solution: [183.86996829899306, 176.48095464093922,
175.13129639484694, 75.80506591583251] Fitness: 2.3586942480612922
Generation 370 Best Solution: [183.86996829899306, 176.48095464093922,
175.13129639484694, 75.80506591583251] Fitness: 2.3586942480612922
Generation 371 Best Solution: [183.86996829899306, 176.48095464093922,
175.13129639484694, 75.80506591583251] Fitness: 2.3586942480612922
Generation 372 Best Solution: [183.86996829899306, 176.2135254329776,
175.13129639484694, 75.80506591583251] Fitness: 2.2941146033372615

Generation 373 Best Solution: [183.86996829899306, 176.31722043531138,
175.13129639484694, 75.80506591583251] Fitness: 2.3047888436493817
Generation 374 Best Solution: [183.86996829899306, 176.48095464093922,
175.13129639484694, 75.65247312590459] Fitness: 2.0942048194909537
Generation 375 Best Solution: [183.86996829899306, 176.48095464093922,
175.13129639484694, 75.80506591583251] Fitness: 2.3586942480612922
Generation 376 Best Solution: [183.86996829899306, 176.57165618940877,
175.13129639484694, 75.80506591583251] Fitness: 2.4080827701089262
Generation 377 Best Solution: [183.86996829899306, 175.92716154342875,
175.13129639484694, 74.84176813931472] Fitness: 1.3502688887228833
Generation 378 Best Solution: [183.86996829899306, 176.57165618940877,
174.1504662631315, 75.80506591583251] Fitness: 1.5555611213105434
Generation 379 Best Solution: [183.86996829899306, 175.92716154342875,
175.13129639484694, 75.80506591583251] Fitness: 2.359150998373518
Generation 380 Best Solution: [183.86996829899306, 176.02575551275316,
174.91801747974975, 75.80506591583251] Fitness: 1.997198323316987
Generation 381 Best Solution: [183.63505755712288, 176.360195539252,
175.13129639484694, 75.80506591583251] Fitness: 2.1003204294816835
Generation 382 Best Solution: [183.63505755712288, 176.360195539252,
175.13129639484694, 75.80506591583251] Fitness: 2.1003204294816835
Generation 383 Best Solution: [182.93703879050523, 176.2380766205152,
175.13129639484694, 75.80506591583251] Fitness: 1.995234709405545
Generation 384 Best Solution: [182.9999414461786, 176.8186297720873,
175.13129639484694, 75.80506591583251] Fitness: 2.2872716026895836
Generation 385 Best Solution: [183.62889897971374, 177.38764300955037,
173.88917146120062, 74.29087727974996] Fitness: 1.5748122296669347
Generation 386 Best Solution: [183.86996829899306, 177.38764300955037,
173.88917146120062, 74.29087727974996] Fitness: 1.7933979419464772
Generation 387 Best Solution: [183.62889897971374, 177.38764300955037,
174.91801747974975, 75.80506591583251] Fitness: 2.9359429029680384
Generation 388 Best Solution: [183.0284474729006, 177.38764300955037,
174.92035362585074, 75.80506591583251] Fitness: 2.822179154018691
Generation 389 Best Solution: [183.86996829899306, 176.107446407694,
174.91801747974975, 75.80506591583251] Fitness: 1.9781280308359928
Generation 390 Best Solution: [183.0284474729006, 176.107446407694,
174.91801747974975, 75.80506591583251] Fitness: 1.642647759659956
Generation 391 Best Solution: [183.86996829899306, 176.107446407694,
174.91801747974975, 75.80506591583251] Fitness: 1.9781280308359928
Generation 392 Best Solution: [183.0284474729006, 176.107446407694,
174.91801747974975, 75.80506591583251] Fitness: 1.642647759659956
Generation 393 Best Solution: [183.86996829899306, 177.38764300955037,
174.25953967264866, 75.80506591583251] Fitness: 2.640250930167616
Generation 394 Best Solution: [183.0284474729006, 177.38764300955037,
174.91801747974975, 75.17005934328199] Fitness: 1.9775973320597264
Generation 395 Best Solution: [183.86996829899306, 176.0655748947783,
175.13129639484694, 75.80506591583251] Fitness: 2.310387983426714
Generation 396 Best Solution: [183.86996829899306, 176.0655748947783,
175.13129639484694, 74.69700184492778] Fitness: 1.2856208929405823

Generation 397 Best Solution: [183.0284474729006, 177.38764300955037,
175.13129639484694, 74.69700184492778] Fitness: 2.1181773209909136
Generation 398 Best Solution: [183.0284474729006, 177.38764300955037,
174.91801747974975, 75.24840985175481] Fitness: 2.0445156644749707
Generation 399 Best Solution: [183.76504038815645, 177.38764300955037,
175.13129639484694, 74.69700184492778] Fitness: 2.3464285673797742
Generation 400 Best Solution: [183.76504038815645, 177.38764300955037,
175.13129639484694, 74.69700184492778] Fitness: 2.3464285673797742
Generation 401 Best Solution: [183.76504038815645, 176.0338041013363,
175.13129639484694, 75.80506591583251] Fitness: 2.211484937512668
Generation 402 Best Solution: [183.76504038815645, 176.0338041013363,
175.13129639484694, 75.80506591583251] Fitness: 2.211484937512668
Generation 403 Best Solution: [183.57078948263737, 177.3399719653679,
175.13129639484694, 74.69700184492778] Fitness: 2.1035701349740292
Generation 404 Best Solution: [183.86996829899306, 177.3399719653679,
175.13129639484694, 74.69700184492778] Fitness: 2.360135200245272
Generation 405 Best Solution: [183.86996829899306, 175.72575138121533,
175.13129639484694, 75.80506591583251] Fitness: 2.488021862650917
Generation 406 Best Solution: [183.86996829899306, 175.8919922404635,
175.13129639484694, 75.80506591583251] Fitness: 2.3767067254775274
Generation 407 Best Solution: [183.57078948263737, 177.38764300955037,
175.13129639484694, 74.3785559936693] Fitness: 2.2869644075202378
Generation 408 Best Solution: [183.86996829899306, 176.37993985019216,
175.13129639484694, 75.80506591583251] Fitness: 2.320076764424025
Generation 409 Best Solution: [183.86996829899306, 177.38764300955037,
173.76444554179784, 74.900548616477] Fitness: 1.74453021709211
Generation 410 Best Solution: [183.86996829899306, 176.37993985019216,
174.2839360676267, 75.80506591583251] Fitness: 1.4878674297146883
Generation 411 Best Solution: [183.7364940334175, 176.7660073964516,
174.4649865389272, 75.56571475576979] Fitness: 1.2738407239187484
Generation 412 Best Solution: [182.89857722501029, 176.7660073964516,
174.14348784942356, 75.80506591583251] Fitness: 1.4275252353956795
Generation 413 Best Solution: [182.89857722501029, 175.655333099311,
173.82928491533087, 75.80506591583251] Fitness: 1.490926277929474
Generation 414 Best Solution: [183.60680743956826, 176.18370474573828,
175.13129639484694, 75.00282232308707] Fitness: 1.111182544750728
Generation 415 Best Solution: [183.7364940334175, 176.18783143852053,
175.13129639484694, 75.80506591583251] Fitness: 2.1611053548594366
Generation 416 Best Solution: [183.86996829899306, 176.32847631646288,
175.13129639484694, 75.80506591583251] Fitness: 2.307042326702838
Generation 417 Best Solution: [183.86996829899306, 176.114990791246,
175.13129639484694, 75.86837232203207] Fitness: 2.4221267559727218
Generation 418 Best Solution: [183.86996829899306, 177.38764300955037,
174.10667583177425, 74.32450519959711] Fitness: 1.7232020814201405
Generation 419 Best Solution: [183.86996829899306, 177.38764300955037,
174.10667583177425, 75.20709247016669] Fitness: 1.814690695482812
Generation 420 Best Solution: [183.86996829899306, 177.38764300955037,
174.10667583177425, 75.20709247016669] Fitness: 1.814690695482812

Generation 421 Best Solution: [183.86996829899306, 177.38764300955037,
174.10667583177425, 75.20709247016669] Fitness: 1.814690695482812
Generation 422 Best Solution: [183.23120698942452, 176.88085627942678,
175.13129639484694, 75.80506591583251] Fitness: 2.3168362112577094
Generation 423 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.9083319046073] Fitness: 2.488749293936965
Generation 424 Best Solution: [183.86996829899306, 175.59698332346397,
175.13129639484694, 75.4840201615555] Fitness: 2.0956802186414203
Generation 425 Best Solution: [183.86996829899306, 175.59698332346397,
175.13129639484694, 75.4840201615555] Fitness: 2.0956802186414203
Generation 426 Best Solution: [183.86996829899306, 177.38764300955037,
175.13129639484694, 74.87860519822249] Fitness: 2.4792451899785943
Generation 427 Best Solution: [183.09654342400597, 177.38764300955037,
174.35004761328304, 75.86837232203207] Fitness: 2.43583788270877
Generation 428 Best Solution: [183.09654342400597, 177.38764300955037,
174.35004761328304, 75.86837232203207] Fitness: 2.43583788270877
Generation 429 Best Solution: [183.86996829899306, 177.38764300955037,
174.53146292359784, 75.65476599155235] Fitness: 2.502876967494457
Generation 430 Best Solution: [183.86996829899306, 175.87313087140524,
175.13129639484694, 75.65476599155235] Fitness: 2.126177407997194
Generation 431 Best Solution: [183.86996829899306, 176.20101823532895,
175.13129639484694, 75.86837232203207] Fitness: 2.4153510974437924
Generation 432 Best Solution: [183.86996829899306, 176.67463411382369,
175.13129639484694, 75.80506591583251] Fitness: 2.481032278972874
Generation 433 Best Solution: [183.86996829899306, 176.67463411382369,
175.13129639484694, 75.65476599155235] Fitness: 2.220225480174422
Generation 434 Best Solution: [183.86996829899306, 176.67463411382369,
175.13129639484694, 75.1392856265199] Fitness: 1.6161392824794614
Generation 435 Best Solution: [183.86996829899306, 176.67463411382369,
174.53146292359784, 75.65476599155235] Fitness: 1.5054845638142496
Generation 436 Best Solution: [183.49998130847015, 177.38764300955037,
173.98067072241483, 75.86837232203207] Fitness: 2.469807717942029
Generation 437 Best Solution: [183.86996829899306, 177.38764300955037,
174.9730895761204, 74.55455689198944] Fitness: 2.225011828749017
Generation 438 Best Solution: [183.49998130847015, 177.38764300955037,
174.9730895761204, 74.55455689198944] Fitness: 1.9298918286483082
Generation 439 Best Solution: [183.86996829899306, 176.13239988045203,
174.9730895761204, 74.55455689198944] Fitness: 1.0450377326897289
Generation 440 Best Solution: [183.49998130847015, 176.38692031072776,
175.13129639484694, 74.75039734326573] Fitness: 1.004033652434136
Generation 441 Best Solution: [183.86996829899306, 176.38692031072776,
175.13129639484694, 74.40364363518] Fitness: 1.3739870833948766
Generation 442 Best Solution: [183.86996829899306, 175.95203590308878,
175.13129639484694, 75.45576725160628] Fitness: 1.800696349379732
Generation 443 Best Solution: [183.49998130847015, 176.68415500239837,
175.13129639484694, 75.86837232203207] Fitness: 2.314857377979931
Generation 444 Best Solution: [183.86996829899306, 177.38764300955037,
174.4341010141677, 74.40364363518] Fitness: 1.7569049603094244

Generation 445 Best Solution: [183.53098705080384, 175.97229359102337,
175.13129639484694, 75.86837232203207] Fitness: 2.1817012279989862
Generation 446 Best Solution: [183.53098705080384, 175.97229359102337,
175.13129639484694, 75.86837232203207] Fitness: 2.1817012279989862
Generation 447 Best Solution: [183.86996829899306, 176.20382726778448,
175.13129639484694, 75.86837232203207] Fitness: 2.4153410066503342
Generation 448 Best Solution: [183.86996829899306, 176.20382726778448,
175.13129639484694, 75.86837232203207] Fitness: 2.4153410066503342
Generation 449 Best Solution: [183.86996829899306, 176.8277512233211,
174.46259925709555, 75.86837232203207] Fitness: 1.9861423898015063
Generation 450 Best Solution: [183.86996829899306, 176.8277512233211,
175.13129639484694, 75.5958817349978] Fitness: 2.27011756972026
Generation 451 Best Solution: [183.86996829899306, 176.8277512233211,
175.13129639484694, 75.5958817349978] Fitness: 2.27011756972026
Generation 452 Best Solution: [183.86996829899306, 176.8277512233211,
174.13174017314722, 75.86837232203207] Fitness: 1.8910144545065752
Generation 453 Best Solution: [183.86996829899306, 176.8277512233211,
174.13174017314722, 75.86837232203207] Fitness: 1.8910144545065752
Generation 454 Best Solution: [183.49998130847015, 176.8277512233211,
174.13174017314722, 75.86837232203207] Fitness: 1.5958944544058662
Generation 455 Best Solution: [183.49998130847015, 176.694676969246,
175.13129639484694, 74.30694074405636] Fitness: 1.311149780128689
Generation 456 Best Solution: [183.86996829899306, 176.694676969246,
175.13129639484694, 74.30694074405636] Fitness: 1.6062697802293981
Generation 457 Best Solution: [183.49998130847015, 177.38764300955037,
175.13129639484694, 74.98059605020124] Fitness: 2.222969690410425
Generation 458 Best Solution: [183.86996829899306, 176.26287575398223,
175.13129639484694, 75.86837232203207] Fitness: 2.418219544222777
Generation 459 Best Solution: [183.86996829899306, 176.694676969246,
174.95589836995433, 75.86837232203207] Fitness: 2.3466208899024914
Generation 460 Best Solution: [183.86996829899306, 176.694676969246,
174.95589836995433, 75.86837232203207] Fitness: 2.3466208899024914
Generation 461 Best Solution: [183.86996829899306, 176.694676969246,
174.0311679289507, 75.86837232203207] Fitness: 1.7734495392918195
Generation 462 Best Solution: [183.86996829899306, 176.694676969246,
174.0311679289507, 75.86837232203207] Fitness: 1.7734495392918195
Generation 463 Best Solution: [183.86996829899306, 176.08880996372446,
175.13129639484694, 75.86837232203207] Fitness: 2.4266745503793263
Generation 464 Best Solution: [182.9300532540492, 176.08880996372446,
175.13129639484694, 75.86837232203207] Fitness: 2.1302226640266007
Generation 465 Best Solution: [182.9300532540492, 176.08880996372446,
175.13129639484694, 75.86837232203207] Fitness: 2.1302226640266007
Generation 466 Best Solution: [183.48608902831816, 175.69737769633085,
175.13129639484694, 75.86837232203207] Fitness: 2.3312988610608647
Generation 467 Best Solution: [183.48608902831816, 175.69737769633085,
175.13129639484694, 75.86837232203207] Fitness: 2.3312988610608647
Generation 468 Best Solution: [183.329475456793, 177.43609208004503,
175.13129639484694, 74.5873704867171] Fitness: 2.2110715623392734

Generation 469 Best Solution: [183.71003414878683, 177.38764300955037,
175.13129639484694, 74.5873704867171] Fitness: 2.309229469590033
Generation 470 Best Solution: [183.86996829899306, 177.2627160124062,
175.13129639484694, 74.5873704867171] Fitness: 2.2283083530089804
Generation 471 Best Solution: [183.86996829899306, 177.2627160124062,
175.13129639484694, 74.5873704867171] Fitness: 2.2283083530089804
Generation 472 Best Solution: [182.9300532540492, 177.38764300955037,
174.15587686896782, 75.53894800729852] Fitness: 1.8939887246253588
Generation 473 Best Solution: [183.29568153930933, 176.83890430021617,
175.13129639484694, 75.86837232203207] Fitness: 2.397039063456222
Generation 474 Best Solution: [183.29568153930933, 176.83890430021617,
175.13129639484694, 75.86837232203207] Fitness: 2.397039063456222
Generation 475 Best Solution: [183.29568153930933, 176.80059642048568,
175.13129639484694, 75.86837232203207] Fitness: 2.357156117155718
Generation 476 Best Solution: [183.29568153930933, 176.80059642048568,
175.13129639484694, 75.86837232203207] Fitness: 2.357156117155718
Generation 477 Best Solution: [183.29568153930933, 176.80059642048568,
175.13129639484694, 75.13585258959525] Fitness: 1.3684531586843731
Generation 478 Best Solution: [183.71003414878683, 176.75559880224137,
175.13129639484694, 74.47046148453998] Fitness: 1.416541645900668
Generation 479 Best Solution: [182.83990338848193, 176.25106207881072,
174.49436484438496, 75.86837232203207] Fitness: 1.4319080957387316
Generation 480 Best Solution: [182.83990338848193, 176.25106207881072,
175.13129639484694, 75.86837232203207] Fitness: 2.1708600902491604
Generation 481 Best Solution: [182.8314353983359, 176.84996189508934,
175.13129639484694, 75.86837232203207] Fitness: 2.5269214819454042
Generation 482 Best Solution: [183.86996829899306, 176.8324128304914,
175.13129639484694, 74.84190281053081] Fitness: 1.61876415351113
Generation 483 Best Solution: [183.86996829899306, 175.60174785128444,
175.13129639484694, 74.59439908798255] Fitness: 1.5869595045096512
Generation 484 Best Solution: [183.13729410526315, 175.9582047337362,
175.13129639484694, 75.86837232203207] Fitness: 2.1071208086342965
Generation 485 Best Solution: [183.15761141443446, 176.91714132375364,
175.13129639484694, 75.86837232203207] Fitness: 2.483168311914012
Generation 486 Best Solution: [183.5729165560438, 176.27251696314028,
175.13129639484694, 75.86837232203207] Fitness: 2.163974301437452
Generation 487 Best Solution: [183.83560453226121, 176.34876430265865,
175.13129639484694, 75.84178826534037] Fitness: 2.344009887952411
Generation 488 Best Solution: [183.15761141443446, 177.38764300955037,
175.13129639484694, 74.53545097912112] Fitness: 2.1159603177443858
Generation 489 Best Solution: [183.86996829899306, 176.17885412522014,
174.8491599279715, 75.86837232203207] Fitness: 2.003870889345097
Generation 490 Best Solution: [183.6692296957611, 177.38764300955037,
174.8491599279715, 74.53545097912112] Fitness: 1.876908397982593
Generation 491 Best Solution: [182.99854053380585, 177.1917751074952,
175.13129639484694, 74.75179320537073] Fitness: 1.7704498036391483
Generation 492 Best Solution: [183.14790959167277, 177.38764300955037,
175.13129639484694, 75.03702918790408] Fitness: 2.1863042219317834

```
Generation 493 Best Solution: [183.83560453226121, 177.38764300955037,
174.49205568510737, 74.53545097912112] Fitness: 1.7002522854671476
Generation 494 Best Solution: [182.99854053380585, 177.38764300955037,
174.62484150460563, 75.75753546522547] Fitness: 2.422973368279332
Generation 495 Best Solution: [182.99854053380585, 175.89336257683442,
175.13129639484694, 75.86837232203207] Fitness: 2.1719269771523892
Generation 496 Best Solution: [183.86996829899306, 176.22603252037257,
175.13129639484694, 75.86837232203207] Fitness: 2.415731234362795
Generation 497 Best Solution: [182.99854053380585, 177.38764300955037,
175.13129639484694, 74.56990212131446] Fitness: 2.143598500287551
Generation 498 Best Solution: [182.99854053380585, 177.38764300955037,
175.13129639484694, 74.36579221110897] Fitness: 2.225358058772707
Generation 499 Best Solution: [183.14790959167277, 177.38764300955037,
175.13129639484694, 74.95621325988253] Fitness: 2.1463583576430127
Generation 500 Best Solution: [183.6938547679016, 176.69731400076563,
175.13129639484694, 75.75753546522547] Fitness: 2.243532358553781
Optimal Solution for Simulation 3: [183.6938547679016, 176.69731400076563,
175.13129639484694, 75.75753546522547]
--------------------------------------------------
Simulation 4/32 with conditions: {'drought': True, 'flood': True,
'policy_change': True, 'social_event': False, 'economic_crisis': False}
Generation 1 Best Solution: [183.25342273428538, 176.21278256052395,
174.1885595504075, 74.6881991985446] Fitness: 0.06798136042887012
Generation 2 Best Solution: [183.3711110256983, 176.21278256052395,
174.1885595504075, 74.6881991985446] Fitness: 0.09178918260663524
Generation 3 Best Solution: [183.00946490792643, 176.06038879172536,
174.1932980711186, 75.5129162498174] Fitness: 0.6120654768064469
Generation 4 Best Solution: [183.08941922533674, 175.74683878812237,
174.42767856465449, 74.98233833726287] Fitness: 0.3583817217777878
Generation 5 Best Solution: [183.15179368367458, 177.25608168620664,
174.3822575115534, 75.26495612616974] Fitness: 1.308327042844737
Generation 6 Best Solution: [183.3986295866654, 177.17600986957135,
174.82895475559263, 74.97135821702797] Fitness: 1.3814020937794438
Generation 7 Best Solution: [183.07008496346495, 177.17600986957135,
174.47275383334457, 74.8551664449252] Fitness: 1.0042571350890024
Generation 8 Best Solution: [183.7910561492816, 177.22798574035016,
173.8550780468571, 74.8551664449252] Fitness: 1.322234889509827
Generation 9 Best Solution: [183.8657237187036, 176.92413101508492,
174.87183836150058, 74.98233833726287] Fitness: 1.4186254753626502
Generation 10 Best Solution: [183.07008496346495, 176.2359587416272,
174.87183836150058, 75.26495612616974] Fitness: 0.7886238283710916
Generation 11 Best Solution: [183.15179368367458, 176.28047580030892,
174.47275383334457, 75.26495612616974] Fitness: 0.4148045880996668
Generation 12 Best Solution: [183.3518082765205, 176.4959185975148,
174.87183836150058, 74.98233833726287] Fitness: 0.6923809884991353
Generation 13 Best Solution: [183.7910561492816, 176.8077526587233,
174.87183836150058, 74.98233833726287] Fitness: 1.209310790325286
Generation 14 Best Solution: [183.7910561492816, 176.64887804558114,
```

174.87183836150058, 75.63177771835022] Fitness: 1.6877870617965436
Generation 15 Best Solution: [183.7910561492816, 176.64887804558114,
174.84145457770447, 74.8551664449252] Fitness: 0.9887197268824696
Generation 16 Best Solution: [183.07008496346495, 175.89056106144935,
174.02112882011627, 75.60840727797148] Fitness: 0.7864144591454616
Generation 17 Best Solution: [183.7910561492816, 176.38853590089042,
174.87183836150058, 75.63177771835022] Fitness: 1.5434782880937488
Generation 18 Best Solution: [183.8657237187036, 176.2642946814477,
174.87183836150058, 75.63177771835022] Fitness: 1.592276390026053
Generation 19 Best Solution: [183.6186534364901, 177.25608168620664,
174.30385524383578, 75.63177771835022] Fitness: 1.8647103524928683
Generation 20 Best Solution: [183.07008496346495, 177.07945723143564,
174.87183836150058, 74.55356293368716] Fitness: 1.2331976032708625
Generation 21 Best Solution: [183.60336156738688, 176.44275161747984,
174.87183836150058, 75.63177771835022] Fitness: 1.4076818718205923
Generation 22 Best Solution: [183.60336156738688, 176.43971188312815,
174.87183836150058, 75.60840727797148] Fitness: 1.3720090416156263
Generation 23 Best Solution: [183.8657237187036, 176.71607111771533,
174.08575375202855, 75.63177771835022] Fitness: 1.3446934998181779
Generation 24 Best Solution: [183.60336156738688, 176.43971188312815,
174.7367772549699, 75.63177771835022] Fitness: 1.2522301070680515
Generation 25 Best Solution: [183.52320265918783, 176.87752786925853,
174.87183836150058, 75.29696509500714] Fitness: 1.2936199478418562
Generation 26 Best Solution: [183.00193050298012, 176.73011304972368,
174.87183836150058, 75.60840727797148] Fitness: 1.452404654658302
Generation 27 Best Solution: [183.00193050298012, 176.73011304972368,
174.87183836150058, 75.63177771835022] Fitness: 1.4867794465884172
Generation 28 Best Solution: [183.00193050298012, 176.73011304972368,
174.50158354989676, 75.63177771835022] Fitness: 1.1357156828803037
Generation 29 Best Solution: [183.86563525511397, 176.73011304972368,
174.50158354989676, 75.63177771835022] Fitness: 1.482863683654896
Generation 30 Best Solution: [183.6186534364901, 177.16241190910938,
174.87183836150058, 75.63177771835022] Fitness: 2.152816116278654
Generation 31 Best Solution: [183.52320265918783, 177.3573726996208,
174.57554642432916, 75.60840727797148] Fitness: 2.105098944175611
Generation 32 Best Solution: [183.52320265918783, 177.3573726996208,
174.57554642432916, 74.32403033145603] Fitness: 1.595657020099829
Generation 33 Best Solution: [183.00193050298012, 177.3573726996208,
174.2738803812585, 75.63177771835022] Fitness: 1.920399188881458
Generation 34 Best Solution: [183.6186534364901, 177.16241190910938,
173.95577780970783, 75.63177771835022] Fitness: 1.6949437974297477
Generation 35 Best Solution: [183.6186534364901, 177.16241190910938,
174.29663300923204, 75.63177771835022] Fitness: 1.704261912272239
Generation 36 Best Solution: [183.6186534364901, 177.16241190910938,
174.29663300923204, 75.63177771835022] Fitness: 1.704261912272239
Generation 37 Best Solution: [183.52320265918783, 177.3573726996208,
174.29663300923204, 75.37860187079043] Fitness: 1.6652592203091756
Generation 38 Best Solution: [183.7780320593286, 177.3573726996208,

174.29663300923204, 75.37860187079043] Fitness: 1.858459936819853
Generation 39 Best Solution: [183.7780320593286, 175.5722171179279,
174.87183836150058, 75.21505506003604] Fitness: 1.3219060156672726
Generation 40 Best Solution: [183.6186534364901, 177.3573726996208,
174.4200764215834, 75.21505506003604] Fitness: 1.6218845521208403
Generation 41 Best Solution: [183.06737717814386, 177.3573726996208,
174.4200764215834, 75.21505506003604] Fitness: 1.4825159260367533
Generation 42 Best Solution: [183.0109791800262, 176.88230628097244,
174.87183836150058, 75.63177771835022] Fitness: 1.6403100110283835
Generation 43 Best Solution: [183.8657237187036, 176.37918167693533,
174.4200764215834, 75.63177771835022] Fitness: 1.2206825608398142
Generation 44 Best Solution: [183.06737717814386, 176.37918167693533,
174.4200764215834, 75.63177771835022] Fitness: 0.8568269831279602
Generation 45 Best Solution: [183.04129498766275, 175.71870182793683,
174.87183836150058, 74.92562339400816] Fitness: 0.7617196407131087
Generation 46 Best Solution: [183.08244164121174, 175.71870182793683,
174.87183836150058, 75.63177771835022] Fitness: 1.3957230908645173
Generation 47 Best Solution: [183.8657237187036, 176.81032445923512,
174.4200764215834, 75.63177771835022] Fitness: 1.5135307698818599
Generation 48 Best Solution: [183.8657237187036, 176.7897202715163,
174.87183836150058, 75.63177771835022] Fitness: 1.8906870484788874
Generation 49 Best Solution: [183.69899710965876, 176.64523008490883,
174.4200764215834, 75.63177771835022] Fitness: 1.2028224089989688
Generation 50 Best Solution: [183.69899710965876, 176.64523008490883,
174.4200764215834, 75.63177771835022] Fitness: 1.2028224089989688
Generation 51 Best Solution: [183.69899710965876, 176.31927496716898,
174.87183836150058, 75.63177771835022] Fitness: 1.4397650744717136
Generation 52 Best Solution: [183.08244164121174, 176.46920573406075,
174.87183836150058, 75.63177771835022] Fitness: 1.2880224010831856
Generation 53 Best Solution: [183.08244164121174, 176.46920573406075,
174.87183836150058, 75.63177771835022] Fitness: 1.2880224010831856
Generation 54 Best Solution: [183.08244164121174, 176.46920573406075,
174.87183836150058, 75.63177771835022] Fitness: 1.2880224010831856
Generation 55 Best Solution: [183.8657237187036, 176.33690400284743,
174.87183836150058, 75.29051923329739] Fitness: 1.19367132227147
Generation 56 Best Solution: [183.8657237187036, 176.33690400284743,
174.87183836150058, 75.63177771835022] Fitness: 1.6065067012713075
Generation 57 Best Solution: [183.8657237187036, 176.59155757993705,
174.87183836150058, 75.63177771835022] Fitness: 1.7248716520476892
Generation 58 Best Solution: [183.8657237187036, 176.95439064612896,
174.87183836150058, 75.63177771835022] Fitness: 2.077555878255593
Generation 59 Best Solution: [183.8657237187036, 175.5724872720747,
174.87183836150058, 75.63177771835022] Fitness: 1.8910881483757744
Generation 60 Best Solution: [183.8657237187036, 175.5724872720747,
174.87183836150058, 75.63177771835022] Fitness: 1.8910881483757744
Generation 61 Best Solution: [183.8657237187036, 176.03074576042872,
174.87183836150058, 75.63177771835022] Fitness: 1.6052389376540082
Generation 62 Best Solution: [183.4819074125179, 175.90741630237213,

174.87183836150058, 75.63177771835022] Fitness: 1.3426172655508313
Generation 63 Best Solution: [183.8657237187036, 176.1436370953886,
174.87183836150058, 75.63177771835022] Fitness: 1.58778434745198
Generation 64 Best Solution: [183.8657237187036, 175.90741630237213,
174.87183836150058, 75.63177771835022] Fitness: 1.6482380727854538
Generation 65 Best Solution: [183.60088245460446, 177.3573726996208,
173.8312886860858, 75.63177771835022] Fitness: 2.072309478940838
Generation 66 Best Solution: [183.60088245460446, 177.3573726996208,
174.6009083792066, 75.63177771835022] Fitness: 2.207007561801775
Generation 67 Best Solution: [183.60088245460446, 177.3573726996208,
174.15598903264362, 75.63177771835022] Fitness: 2.010470153340288
Generation 68 Best Solution: [183.52130163595885, 177.3573726996208,
174.15598903264362, 75.63177771835022] Fitness: 1.9618399234995851
Generation 69 Best Solution: [183.8657237187036, 176.28333042908628,
174.87183836150058, 75.63177771835022] Fitness: 1.5951694004904016
Generation 70 Best Solution: [183.20963402591042, 177.31416469897684,
174.87183836150058, 75.63177771835022] Fitness: 2.2651307978170827
Generation 71 Best Solution: [183.20963402591042, 177.31416469897684,
174.87183836150058, 75.63177771835022] Fitness: 2.2651307978170827
Generation 72 Best Solution: [183.1178981021806, 175.75225013483816,
174.87183836150058, 75.63177771835022] Fitness: 1.365969282814858
Generation 73 Best Solution: [183.20963402591042, 177.31416469897684,
174.49202903354768, 75.63177771835022] Fitness: 1.9079886250345528
Generation 74 Best Solution: [183.8657237187036, 175.60585439249027,
174.87183836150058, 75.63177771835022] Fitness: 1.8586289214293197
Generation 75 Best Solution: [183.8657237187036, 175.60585439249027,
174.87183836150058, 75.63177771835022] Fitness: 1.8586289214293197
Generation 76 Best Solution: [183.60088245460446, 176.8072406709149,
174.87183836150058, 75.63177771835022] Fitness: 1.6716840042602177
Generation 77 Best Solution: [183.60088245460446, 176.8072406709149,
174.49202903354768, 75.63177771835022] Fitness: 1.314541831477688
Generation 78 Best Solution: [183.20963402591042, 176.80412037524567,
174.87183836150058, 75.63177771835022] Fitness: 1.529563878618161
Generation 79 Best Solution: [183.20963402591042, 177.31416469897684,
174.87183836150058, 75.63177771835022] Fitness: 2.2651307978170827
Generation 80 Best Solution: [183.20963402591042, 177.31416469897684,
174.87183836150058, 75.63177771835022] Fitness: 2.2651307978170827
Generation 81 Best Solution: [183.20963402591042, 177.31416469897684,
174.87183836150058, 75.09529552989798] Fitness: 1.7021562622575552
Generation 82 Best Solution: [183.8657237187036, 176.16456968251674,
174.87183836150058, 75.63177771835022] Fitness: 1.5868489268693737
Generation 83 Best Solution: [183.8657237187036, 175.83345886025077,
174.87183836150058, 75.63177771835022] Fitness: 1.686008854965101
Generation 84 Best Solution: [183.8657237187036, 175.7292413867005,
174.87183836150058, 75.63177771835022] Fitness: 1.7544867858468158
Generation 85 Best Solution: [183.8657237187036, 175.7292413867005,
174.87183836150058, 75.63177771835022] Fitness: 1.7544867858468158
Generation 86 Best Solution: [183.8657237187036, 177.31416469897684,

174.87183836150058, 74.78949528935941] Fitness: 1.9685176036771799
Generation 87 Best Solution: [183.8657237187036, 175.8034801295762,
174.87183836150058, 75.54859704538777] Fitness: 1.5856177599007213
Generation 88 Best Solution: [183.8657237187036, 176.82457696626605,
174.87183836150058, 75.63177771835022] Fitness: 1.9265257515146075
Generation 89 Best Solution: [183.8657237187036, 175.7651637844585,
174.69575471728587, 75.63177771835022] Fitness: 1.533826290058844
Generation 90 Best Solution: [183.8657237187036, 175.7651637844585,
174.36838400385585, 75.63177771835022] Fitness: 1.3065937112062707
Generation 91 Best Solution: [183.8657237187036, 175.98944657471029,
174.36838400385585, 75.63177771835022] Fitness: 1.1945807841482905
Generation 92 Best Solution: [183.8657237187036, 175.98944657471029,
174.87183836150058, 75.63177771835022] Fitness: 1.6168551836509666
Generation 93 Best Solution: [183.8657237187036, 176.0487177118278,
174.87183836150058, 75.63177771835022] Fitness: 1.6010589569234268
Generation 94 Best Solution: [183.8657237187036, 176.0487177118278,
174.87183836150058, 75.54859704538777] Fitness: 1.4827979653089947
Generation 95 Best Solution: [183.31897838339125, 177.31416469897684,
174.87183836150058, 74.9161859525539] Fitness: 1.6329882864979977
Generation 96 Best Solution: [182.95610109210762, 177.31416469897684,
174.87183836150058, 74.9161859525539] Fitness: 1.6637443668609342
Generation 97 Best Solution: [183.31897838339125, 177.31416469897684,
174.87183836150058, 74.9161859525539] Fitness: 1.6329882864979977
Generation 98 Best Solution: [183.8657237187036, 175.79628268797128,
174.69575471728587, 75.63177771835022] Fitness: 1.513347016193901
Generation 99 Best Solution: [183.8657237187036, 176.2496268873184,
174.87183836150058, 75.63177771835022] Fitness: 1.5904532900383836
Generation 100 Best Solution: [183.31897838339125, 177.31416469897684,
174.58940772039222, 75.63177771835022] Fitness: 1.9904685550023553
Generation 101 Best Solution: [182.95610109210762, 177.31416469897684,
174.69575471728587, 75.63177771835022] Fitness: 2.114349419233523
Generation 102 Best Solution: [183.8657237187036, 177.31416469897684,
174.69575471728587, 75.11227314074006] Fitness: 1.8933724677364876
Generation 103 Best Solution: [183.8657237187036, 177.31416469897684,
173.83857920427747, 75.63177771835022] Fitness: 2.2238385262610896
Generation 104 Best Solution: [183.0489281348561, 176.8233749621106,
174.87183836150058, 75.63177771835022] Fitness: 1.565370900745577
Generation 105 Best Solution: [183.8657237187036, 177.31416469897684,
174.31419294802978, 75.5402953542207] Fitness: 2.068473524634186
Generation 106 Best Solution: [183.8657237187036, 176.62871364828518,
174.87183836150058, 75.63177771835022] Fitness: 1.7510484183561053
Generation 107 Best Solution: [183.8657237187036, 176.62871364828518,
174.87183836150058, 75.63177771835022] Fitness: 1.7510484183561053
Generation 108 Best Solution: [183.8657237187036, 176.62871364828518,
174.44068943465143, 75.52079183745535] Fitness: 1.2085551926511584
Generation 109 Best Solution: [183.8657237187036, 176.62871364828518,
174.44068943465143, 75.52079183745535] Fitness: 1.2085551926511584
Generation 110 Best Solution: [183.8657237187036, 176.62871364828518,

174.44068943465143, 75.52079183745535] Fitness: 1.2085551926511584
Generation 111 Best Solution: [183.8657237187036, 177.31416469897684,
174.31419294802978, 74.4059392591889] Fitness: 1.6056807031502593
Generation 112 Best Solution: [183.53615863885761, 176.62871364828518,
174.87183836150058, 75.54746815474212] Fitness: 1.3541517780060621
Generation 113 Best Solution: [183.53615863885761, 177.43793473892867,
174.31419294802978, 74.4059392591889] Fitness: 1.5715071940725613
Generation 114 Best Solution: [183.8657237187036, 176.62871364828518,
174.87183836150058, 75.63177771835022] Fitness: 1.7510484183561053
Generation 115 Best Solution: [183.8657237187036, 176.62871364828518,
174.87183836150058, 75.63177771835022] Fitness: 1.7510484183561053
Generation 116 Best Solution: [183.05797462337517, 176.18420471053346,
174.87183836150058, 75.63177771835022] Fitness: 1.2247235688798856
Generation 117 Best Solution: [183.8657237187036, 176.62871364828518,
174.87183836150058, 75.63177771835022] Fitness: 1.7510484183561053
Generation 118 Best Solution: [183.8657237187036, 176.62871364828518,
174.87183836150058, 75.63177771835022] Fitness: 1.7510484183561053
Generation 119 Best Solution: [183.8657237187036, 176.75935734968232,
174.87183836150058, 74.7051658034698] Fitness: 1.1856713562609673
Generation 120 Best Solution: [183.8657237187036, 176.75935734968232,
174.87183836150058, 74.7051658034698] Fitness: 1.1856713562609673
Generation 121 Best Solution: [183.8657237187036, 176.30043826646406,
174.87183836150058, 74.4059392591889] Fitness: 1.0060445434797696
Generation 122 Best Solution: [183.8657237187036, 176.52038842765674,
174.87183836150058, 75.63177771835022] Fitness: 1.6810651117246036
Generation 123 Best Solution: [183.8657237187036, 176.52038842765674,
174.87183836150058, 75.54746815474212] Fitness: 1.5612773156867037
Generation 124 Best Solution: [183.8657237187036, 176.52038842765674,
174.87183836150058, 74.4059392591889] Fitness: 1.088832322778517
Generation 125 Best Solution: [183.8657237187036, 176.4371018726073,
174.87183836150058, 74.4059392591889] Fitness: 1.0481341188949638
Generation 126 Best Solution: [183.8657237187036, 176.4371018726073,
174.87183836150058, 74.4059392591889] Fitness: 1.0481341188949638
Generation 127 Best Solution: [183.8657237187036, 176.4371018726073,
174.87183836150058, 74.4059392591889] Fitness: 1.0481341188949638
Generation 128 Best Solution: [183.8657237187036, 176.90618484309297,
174.87183836150058, 75.63177771835022] Fitness: 2.018239952515552
Generation 129 Best Solution: [183.8657237187036, 176.90618484309297,
174.87183836150058, 74.4059392591889] Fitness: 1.4260071635694656
Generation 130 Best Solution: [183.8657237187036, 176.90618484309297,
174.87183836150058, 74.37269245284357] Fitness: 1.4443302748350004
Generation 131 Best Solution: [183.8657237187036, 176.90618484309297,
174.87183836150058, 74.57350181159411] Fitness: 1.361298961965209
Generation 132 Best Solution: [183.8657237187036, 176.90618484309297,
174.87183836150058, 74.57350181159411] Fitness: 1.361298961965209
Generation 133 Best Solution: [183.8657237187036, 176.90618484309297,
174.87183836150058, 74.4059392591889] Fitness: 1.4260071635694656
Generation 134 Best Solution: [183.8657237187036, 176.12342595455416,

174.87183836150058, 75.54746815474212] Fitness: 1.469582801040354
Generation 135 Best Solution: [183.8657237187036, 176.7123998268969,
174.87183836150058, 75.63177771835022] Fitness: 1.8183130073668046
Generation 136 Best Solution: [183.38505270497137, 177.17135921138808,
174.30439369143335, 75.63177771835022] Fitness: 1.6011692493110572
Generation 137 Best Solution: [183.38505270497137, 177.17135921138808,
174.30439369143335, 75.63177771835022] Fitness: 1.6011692493110572
Generation 138 Best Solution: [183.8657237187036, 177.17135921138808,
174.30439369143335, 75.63177771835022] Fitness: 1.945670589507306
Generation 139 Best Solution: [183.8657237187036, 177.17135921138808,
174.30439369143335, 75.54746815474212] Fitness: 1.8258827934694062
Generation 140 Best Solution: [183.59566370249163, 176.26965727073375,
174.87183836150058, 75.63177771835022] Fitness: 1.3527557613344166
Generation 141 Best Solution: [183.59566370249163, 176.26965727073375,
174.87183836150058, 75.54746815474212] Fitness: 1.2329679652965169
Generation 142 Best Solution: [183.15653652055937, 176.26965727073375,
174.87183836150058, 75.54746815474212] Fitness: 1.098083697663474
Generation 143 Best Solution: [183.8657237187036, 175.68532877892278,
174.87183836150058, 75.54746815474212] Fitness: 1.6688957546277852
Generation 144 Best Solution: [182.9437455251661, 176.15775872671907,
174.87183836150058, 75.63177771835022] Fitness: 1.2605032651989563
Generation 145 Best Solution: [182.9437455251661, 176.15775872671907,
174.87183836150058, 75.54746815474212] Fitness: 1.1407154691610564
Generation 146 Best Solution: [182.9437455251661, 176.1410479736924,
174.87183836150058, 75.54746815474212] Fitness: 1.1415912593702264
Generation 147 Best Solution: [183.8657237187036, 176.55779840927258,
174.4045518427162, 75.54746815474212] Fitness: 1.1774455351154374
Generation 148 Best Solution: [183.8657237187036, 176.62431495680994,
174.87183836150058, 75.08007575787214] Fitness: 1.175782612653811
Generation 149 Best Solution: [183.8657237187036, 175.93065831149505,
174.87183836150058, 75.84158888925721] Fitness: 1.9870156802624912
Generation 150 Best Solution: [183.8657237187036, 175.93065831149505,
174.87183836150058, 75.63177771835022] Fitness: 1.6382238514988456
Generation 151 Best Solution: [183.8657237187036, 176.34130003155067,
174.87183836150058, 75.84158888925721] Fitness: 1.9564381522441647
Generation 152 Best Solution: [183.8657237187036, 177.43793473892867,
173.9325427458218, 75.54746815474212] Fitness: 2.3124068272846583
Generation 153 Best Solution: [183.82001812075814, 177.43793473892867,
174.87183836150058, 74.40621675433499] Fitness: 2.24228712744274
Generation 154 Best Solution: [183.8657237187036, 176.62160965468243,
173.9325427458218, 75.63177771835022] Fitness: 1.2943097282609157
Generation 155 Best Solution: [183.82001812075814, 176.85075143886112,
174.87183836150058, 75.54746815474212] Fitness: 1.785874118697139
Generation 156 Best Solution: [183.8657237187036, 177.43793473892867,
174.09363495612172, 75.54746815474212] Fitness: 2.286874213321136
Generation 157 Best Solution: [183.75685774203933, 176.59246046823665,
173.9325427458218, 75.63177771835022] Fitness: 1.1626482125556672
Generation 158 Best Solution: [183.8657237187036, 177.43793473892867,

174.09363495612172, 74.84436696639052] Fitness: 1.7426686758720236
Generation 159 Best Solution: [183.8657237187036, 176.49184393338953,
174.87183836150058, 75.63177771835022] Fitness: 1.665833214694898
Generation 160 Best Solution: [183.8657237187036, 176.49184393338953,
174.87183836150058, 75.54746815474212] Fitness: 1.5460454186569983
Generation 161 Best Solution: [183.833236865104, 177.43793473892867,
173.78401575880807, 75.54746815474212] Fitness: 2.338479065655563
Generation 162 Best Solution: [183.82001812075814, 177.43793473892867,
173.9325427458218, 75.54746815474212] Fitness: 2.263318913383828
Generation 163 Best Solution: [183.833236865104, 177.43793473892867,
174.87183836150058, 75.0625556037669] Fitness: 2.2664865888611505
Generation 164 Best Solution: [183.82001812075814, 177.43793473892867,
174.87183836150058, 75.0625556037669] Fitness: 2.252642378535665
Generation 165 Best Solution: [183.54852860915304, 176.01703713071774,
174.87183836150058, 75.54746815474212] Fitness: 1.2190654753388999
Generation 166 Best Solution: [183.8657237187036, 176.2440910197884,
174.98620320238027, 75.84158888925721] Fitness: 2.0926124104870274
Generation 167 Best Solution: [183.8657237187036, 176.2440910197884,
174.87183836150058, 75.88128367675932] Fitness: 2.012773366116109
Generation 168 Best Solution: [183.8657237187036, 176.10960745433007,
174.98620320238027, 75.84158888925721] Fitness: 2.093596710569174
Generation 169 Best Solution: [183.8657237187036, 176.10960745433007,
174.87183836150058, 75.63177771835022] Fitness: 1.5908413946129862
Generation 170 Best Solution: [183.63108931672483, 177.43793473892867,
173.80460244311726, 75.63177771835022] Fitness: 2.2672636931031995
Generation 171 Best Solution: [183.8657237187036, 177.43793473892867,
174.20251296172856, 75.63177771835022] Fitness: 2.4135501068206473
Generation 172 Best Solution: [183.2952220320337, 177.43793473892867,
174.87183836150058, 75.4669070495778] Fitness: 2.293626936309964
Generation 173 Best Solution: [183.8657237187036, 177.43793473892867,
173.9150567085402, 75.63177771835022] Fitness: 2.437531119937658
Generation 174 Best Solution: [183.8657237187036, 177.43793473892867,
173.9150567085402, 75.56755783541536] Fitness: 2.3452271534450992
Generation 175 Best Solution: [183.63108931672483, 177.43793473892867,
173.9150567085402, 75.56755783541536] Fitness: 2.1296425525184595
Generation 176 Best Solution: [183.5389418487574, 177.43793473892867,
174.45135658867488, 75.84158888925721] Fitness: 2.575690045282537
Generation 177 Best Solution: [183.63108931672483, 176.00481141589873,
174.87183836150058, 75.84158888925721] Fitness: 1.7454134132041617
Generation 178 Best Solution: [183.63108931672483, 176.07899874639733,
174.87183836150058, 75.8545090342928] Fitness: 1.7522659423986493
Generation 179 Best Solution: [183.5389418487574, 175.63359032896352,
174.87183836150058, 75.63177771835022] Fitness: 1.5575240508521895
Generation 180 Best Solution: [183.8657237187036, 175.83268386049627,
174.98620320238027, 75.63177771835022] Fitness: 1.8404157172315738
Generation 181 Best Solution: [183.5389418487574, 176.30108448017612,
174.98620320238027, 75.63177771835022] Fitness: 1.4768519156835074
Generation 182 Best Solution: [183.5389418487574, 176.30108448017612,

174.98620320238027, 75.63177771835022] Fitness: 1.4768519156835074
Generation 183 Best Solution: [183.5389418487574, 176.30108448017612,
174.98620320238027, 75.84158888925721] Fitness: 1.825643744447153
Generation 184 Best Solution: [183.5389418487574, 176.30108448017612,
174.98620320238027, 75.7686838698445] Fitness: 1.6962470379724117
Generation 185 Best Solution: [183.63108931672483, 177.43793473892867,
174.55844469920874, 75.44685804721051] Fitness: 2.1089278368844973
Generation 186 Best Solution: [183.5389418487574, 177.14713064692762,
174.55844469920874, 75.84158888925721] Fitness: 2.114352716776998
Generation 187 Best Solution: [183.8657237187036, 177.02400180073104,
174.45800003916614, 75.80913714334757] Fitness: 2.0824907575875753
Generation 188 Best Solution: [182.97983365193323, 177.07261718994195,
174.06341344077634, 75.84158888925721] Fitness: 1.7723858070606195
Generation 189 Best Solution: [183.8657237187036, 175.66711794341913,
174.98620320238027, 75.84158888925721] Fitness: 2.3065497791627285
Generation 190 Best Solution: [183.63108931672483, 177.14713064692762,
174.98620320238027, 75.84158888925721] Fitness: 2.6400349630882927
Generation 191 Best Solution: [183.8657237187036, 177.43793473892867,
174.98620320238027, 74.96610696646282] Fitness: 2.4098144059648847
Generation 192 Best Solution: [183.63108931672483, 177.43793473892867,
174.98620320238027, 74.6247124310416] Fitness: 2.154617722198654
Generation 193 Best Solution: [183.63108931672483, 177.14713064692762,
174.98620320238027, 75.84158888925721] Fitness: 2.6400349630882927
Generation 194 Best Solution: [183.62970770645438, 177.4214787639225,
174.98620320238027, 75.08911386674804] Fitness: 2.2206741712094784
Generation 195 Best Solution: [182.84744317800755, 177.43793473892867,
174.98620320238027, 75.08911386674804] Fitness: 2.190880305228039
Generation 196 Best Solution: [183.63108931672483, 177.35203623672757,
174.98620320238027, 75.2389216084256] Fitness: 2.1922540074851993
Generation 197 Best Solution: [183.63108931672483, 176.89804665389872,
173.79109386499115, 75.58290982430049] Fitness: 1.3281765609590144
Generation 198 Best Solution: [183.63108931672483, 176.89804665389872,
173.79109386499115, 75.2389216084256] Fitness: 0.9404266193974733
Generation 199 Best Solution: [183.63108931672483, 176.89804665389872,
173.79109386499115, 75.2389216084256] Fitness: 0.9404266193974733
Generation 200 Best Solution: [183.63108931672483, 175.97000997837486,
174.98620320238027, 75.84158888925721] Fitness: 1.9104625353057738
Generation 201 Best Solution: [183.4994445999231, 175.97000997837486,
174.98620320238027, 75.84158888925721] Fitness: 1.8291141866794198
Generation 202 Best Solution: [183.63108931672483, 175.96752099532466,
174.98620320238027, 75.72080406691096] Fitness: 1.7017050791585193
Generation 203 Best Solution: [183.63108931672483, 175.96752099532466,
174.98620320238027, 75.29531521749536] Fitness: 1.1541808072658175
Generation 204 Best Solution: [183.2609696260565, 177.43793473892867,
174.1566022661064, 75.84158888925721] Fitness: 2.3852697484984873
Generation 205 Best Solution: [183.63108931672483, 176.21059319801267,
174.98620320238027, 75.84158888925721] Fitness: 1.8744942532421875
Generation 206 Best Solution: [183.4994445999231, 176.21059319801267,

174.98620320238027, 75.84158888925721] Fitness: 1.7931459046158338
Generation 207 Best Solution: [183.4994445999231, 176.21059319801267,
174.98620320238027, 75.84158888925721] Fitness: 1.7931459046158338
Generation 208 Best Solution: [183.4994445999231, 176.21059319801267,
174.98620320238027, 75.84158888925721] Fitness: 1.7931459046158338
Generation 209 Best Solution: [183.26199558615264, 175.65065919472036,
174.98620320238027, 75.84158888925721] Fitness: 1.949004493624183
Generation 210 Best Solution: [183.63108931672483, 176.27195897922692,
174.98620320238027, 75.84158888925721] Fitness: 1.880540317004005
Generation 211 Best Solution: [183.4994445999231, 177.15477124211912,
175.05935208543445, 75.40726007713803] Fitness: 2.0386710049539607
Generation 212 Best Solution: [183.63108931672483, 177.43793473892867,
174.98620320238027, 74.55199911296994] Fitness: 2.1709149438632305
Generation 213 Best Solution: [183.26199558615264, 177.35203623672757,
174.98620320238027, 74.9759832281467] Fitness: 1.8708655751242809
Generation 214 Best Solution: [183.26199558615264, 176.8955939187036,
174.98620320238027, 75.84158888925721] Fitness: 2.136803857504371
Generation 215 Best Solution: [183.26199558615264, 176.8955939187036,
174.61559339623054, 75.84158888925721] Fitness: 1.7158795112271696
Generation 216 Best Solution: [183.26199558615264, 176.8955939187036,
174.61559339623054, 75.84158888925721] Fitness: 1.7158795112271696
Generation 217 Best Solution: [183.26199558615264, 176.8955939187036,
174.61559339623054, 75.84158888925721] Fitness: 1.7158795112271696
Generation 218 Best Solution: [183.63108931672483, 176.76040645146753,
174.61559339623054, 75.84158888925721] Fitness: 1.7283389220729835
Generation 219 Best Solution: [183.63108931672483, 176.61093112670136,
174.98620320238027, 75.84158888925721] Fitness: 2.025408204841267
Generation 220 Best Solution: [183.63108931672483, 177.2334229648919,
174.38464861622035, 75.84158888925721] Fitness: 2.213968065599053
Generation 221 Best Solution: [183.20165548695786, 176.90185138553642,
174.98620320238027, 75.84158888925721] Fitness: 2.139946634295109
Generation 222 Best Solution: [183.63108931672483, 176.22677633488212,
174.61559339623054, 75.84158888925721] Fitness: 1.4545637245298488
Generation 223 Best Solution: [183.20165548695786, 175.93791987222127,
174.61559339623054, 75.84158888925721] Fitness: 1.3412045486065842
Generation 224 Best Solution: [183.20165548695786, 176.14626632108536,
174.61559339623054, 75.27485634578862] Fitness: 0.517571680416335
Generation 225 Best Solution: [183.20165548695786, 176.14626632108536,
174.61559339623054, 75.84158888925721] Fitness: 1.2935548410389557
Generation 226 Best Solution: [183.7607703513854, 176.14626632108536,
174.61559339623054, 75.84158888925721] Fitness: 1.5618459945531655
Generation 227 Best Solution: [183.7607703513854, 176.14626632108536,
174.61559339623054, 75.27485634578862] Fitness: 0.7858628339305449
Generation 228 Best Solution: [183.26199558615264, 176.3798045245283,
174.24499943741174, 75.84158888925721] Fitness: 1.1338732506189932
Generation 229 Best Solution: [183.20165548695786, 177.43793473892867,
174.1617062548781, 75.27485634578862] Fitness: 1.6055919757848682
Generation 230 Best Solution: [183.26199558615264, 177.43793473892867,

174.21956746494521, 75.27485634578862] Fitness: 1.6175308322371755
Generation 231 Best Solution: [183.20165548695786, 177.2334229648919,
174.98620320238027, 75.12266137843517] Fitness: 1.7282604325955988
Generation 232 Best Solution: [183.20165548695786, 175.59909716805745,
174.98620320238027, 75.12266137843517] Fitness: 1.0974128312865425
Generation 233 Best Solution: [183.20165548695786, 175.59909716805745,
173.76713365285977, 75.12266137843517] Fitness: 0.5624699760089058
Generation 234 Best Solution: [183.26199558615264, 175.823227889463,
174.8998475315449, 75.84158888925721] Fitness: 1.7047826496858804
Generation 235 Best Solution: [183.26199558615264, 175.823227889463,
174.8998475315449, 75.84158888925721] Fitness: 1.7047826496858804
Generation 236 Best Solution: [183.63108931672483, 176.003230246132,
174.98620320238027, 75.84158888925721] Fitness: 1.8998374184502147
Generation 237 Best Solution: [183.26199558615264, 176.04038206079477,
174.98620320238027, 75.84158888925721] Fitness: 1.7340159467225171
Generation 238 Best Solution: [183.20165548695786, 176.62800098756807,
174.98620320238027, 75.84158888925721] Fitness: 1.8773768718409634
Generation 239 Best Solution: [183.63108931672483, 176.62800098756807,
174.98620320238027, 74.93474666357983] Fitness: 1.0493763892001613
Generation 240 Best Solution: [183.20165548695786, 176.19642043367327,
174.98620320238027, 75.84158888925721] Fitness: 1.7136584427757808
Generation 241 Best Solution: [183.20165548695786, 176.25527378867477,
174.98620320238027, 75.84158888925721] Fitness: 1.717965178095314
Generation 242 Best Solution: [183.26199558615264, 175.86642026672473,
174.98620320238027, 75.84158888925721] Fitness: 1.799149394994835
Generation 243 Best Solution: [183.63108931672483, 175.88848953651808,
174.98620320238027, 75.84158888925721] Fitness: 1.9442193093296924
Generation 244 Best Solution: [183.6382300137552, 175.6907995914385,
174.98620320238027, 75.03707531436086] Fitness: 1.132227018981963
Generation 245 Best Solution: [183.63108931672483, 175.6907995914385,
174.98620320238027, 75.03707531436086] Fitness: 1.1270004509236793
Generation 246 Best Solution: [183.77140186254232, 175.6907995914385,
174.98620320238027, 75.03707531436086] Fitness: 1.245049891349663
Generation 247 Best Solution: [183.77140186254232, 175.6907995914385,
174.98620320238027, 75.4472631892211] Fitness: 1.593706443855666
Generation 248 Best Solution: [183.77140186254232, 176.47000603389176,
173.8528068701233, 75.84158888925721] Fitness: 1.4831935128124127
Generation 249 Best Solution: [183.77140186254232, 176.17728000470325,
174.98620320238027, 75.84158888925721] Fitness: 1.991852345369004
Generation 250 Best Solution: [183.2271308219842, 176.21053424837237,
174.98620320238027, 75.03707531436086] Fitness: 0.7708091825086665
Generation 251 Best Solution: [183.77140186254232, 177.43793473892867,
173.9139669901407, 75.84158888925721] Fitness: 2.689136982815871
Generation 252 Best Solution: [183.2271308219842, 175.91009201508177,
174.98620320238027, 75.84158888925721] Fitness: 1.77494915974597
Generation 253 Best Solution: [183.1006413017216, 176.12875225148036,
174.98620320238027, 75.84158888925721] Fitness: 1.722042498486081
Generation 254 Best Solution: [183.2271308219842, 177.43793473892867,

174.98620320238027, 74.48801239582096] Fitness: 2.0331797505923643
Generation 255 Best Solution: [183.77140186254232, 177.43793473892867,
174.98620320238027, 74.48801239582096] Fitness: 2.3104907392220415
Generation 256 Best Solution: [183.2271308219842, 177.43793473892867,
174.98620320238027, 74.48801239582096] Fitness: 2.0331797505923643
Generation 257 Best Solution: [183.77140186254232, 175.6792630676629,
174.98620320238027, 75.84158888925721] Fitness: 2.1988763735103394
Generation 258 Best Solution: [183.2271308219842, 177.43793473892867,
174.78983722392866, 75.84158888925721] Fitness: 2.760531734779758
Generation 259 Best Solution: [183.77140186254232, 177.43793473892867,
174.98620320238027, 75.150227375054] Fitness: 2.4131225841109956
Generation 260 Best Solution: [183.77140186254232, 177.43793473892867,
173.78614538429912, 75.84158888925721] Fitness: 2.7436319686900608
Generation 261 Best Solution: [183.77140186254232, 176.63110880002654,
174.98620320238027, 75.84158888925721] Fitness: 2.1580337934861777
Generation 262 Best Solution: [183.24929448322564, 176.942252072406,
173.78614538429912, 75.84158888925721] Fitness: 1.646615094666131
Generation 263 Best Solution: [182.96685800446176, 176.22691678196836,
174.39747511053102, 75.84158888925721] Fitness: 1.1924604570123527
Generation 264 Best Solution: [182.96685800446176, 176.22691678196836,
174.39747511053102, 75.84158888925721] Fitness: 1.1924604570123527
Generation 265 Best Solution: [183.29177496675416, 176.0088435173622,
174.98620320238027, 75.84158888925721] Fitness: 1.7464279104581464
Generation 266 Best Solution: [183.77140186254232, 177.43793473892867,
174.78983722392866, 75.39595302228231] Fitness: 2.3833394906510055
Generation 267 Best Solution: [183.24929448322564, 177.43793473892867,
174.78983722392866, 75.39595302228231] Fitness: 2.107815670444345
Generation 268 Best Solution: [183.77140186254232, 176.33677657584775,
174.98620320238027, 75.84158888925721] Fitness: 2.0116942961552304
Generation 269 Best Solution: [183.24929448322564, 176.33677657584775,
174.98620320238027, 75.84158888925721] Fitness: 1.7361704759485699
Generation 270 Best Solution: [183.77140186254232, 175.87168819116428,
174.78983722392866, 75.62026810319722] Fitness: 1.4536193385965912
Generation 271 Best Solution: [183.77140186254232, 175.87168819116428,
174.78983722392866, 75.84158888925721] Fitness: 1.819452407921506
Generation 272 Best Solution: [183.24929448322564, 176.0692738766305,
174.98620320238027, 75.84158888925721] Fitness: 1.7266248293120212
Generation 273 Best Solution: [183.24929448322564, 176.73543442194713,
174.98620320238027, 75.84158888925721] Fitness: 1.968543193208056
Generation 274 Best Solution: [183.24929448322564, 176.18175915738638,
174.98620320238027, 75.84158888925721] Fitness: 1.7163153917632656
Generation 275 Best Solution: [183.77140186254232, 176.73543442194713,
174.63907086286824, 75.84158888925721] Fitness: 1.8431130010192884
Generation 276 Best Solution: [183.77140186254232, 176.73543442194713,
174.63907086286824, 75.84158888925721] Fitness: 1.8431130010192884
Generation 277 Best Solution: [183.77140186254232, 176.02656432170733,
174.98620320238027, 75.84158888925721] Fitness: 2.011507718510086
Generation 278 Best Solution: [183.77140186254232, 177.43793473892867,

174.98620320238027, 75.07158403756051] Fitness: 2.3632355691946723
Generation 279 Best Solution: [183.24929448322564, 176.43262662736285,
174.98620320238027, 75.84158888925721] Fitness: 1.7681990228638305
Generation 280 Best Solution: [183.24929448322564, 176.43262662736285,
174.98620320238027, 75.84158888925721] Fitness: 1.7681990228638305
Generation 281 Best Solution: [183.24929448322564, 177.43793473892867,
174.98620320238027, 74.90432187691009] Fitness: 2.015395738583373
Generation 282 Best Solution: [183.77140186254232, 175.97131901765357,
174.98620320238027, 75.84158888925721] Fitness: 2.0280589811422933
Generation 283 Best Solution: [183.77140186254232, 176.43262662736285,
174.98620320238027, 75.84158888925721] Fitness: 2.0437228430704906
Generation 284 Best Solution: [183.77140186254232, 176.43262662736285,
174.98620320238027, 75.63085289817946] Fitness: 1.6935536761855254
Generation 285 Best Solution: [183.77140186254232, 177.43793473892867,
174.71884345952287, 74.98763896353023] Fitness: 1.9948643260498036
Generation 286 Best Solution: [183.77140186254232, 176.43262662736285,
174.98620320238027, 75.84158888925721] Fitness: 2.0437228430704906
Generation 287 Best Solution: [183.77140186254232, 176.43262662736285,
174.98620320238027, 75.84158888925721] Fitness: 2.0437228430704906
Generation 288 Best Solution: [183.77140186254232, 177.013625596306,
174.98620320238027, 75.84158888925721] Fitness: 2.5608907990358984
Generation 289 Best Solution: [183.77140186254232, 177.013625596306,
174.98620320238027, 75.84158888925721] Fitness: 2.5608907990358984
Generation 290 Best Solution: [183.77140186254232, 177.013625596306,
174.98620320238027, 75.69128931337858] Fitness: 2.303684215592754
Generation 291 Best Solution: [183.77140186254232, 175.916930901648,
174.98620320238027, 75.84158888925721] Fitness: 2.0492514112707845
Generation 292 Best Solution: [183.77140186254232, 175.916930901648,
174.98620320238027, 75.84158888925721] Fitness: 2.0492514112707845
Generation 293 Best Solution: [183.08627203794592, 176.5715943148888,
174.98620320238027, 75.84158888925721] Fitness: 1.8471576181238156
Generation 294 Best Solution: [183.77140186254232, 177.43793473892867,
174.06641270777462, 75.4774949371436] Fitness: 2.1001094911945817
Generation 295 Best Solution: [183.77140186254232, 176.5715943148888,
174.98620320238027, 75.84158888925721] Fitness: 2.1169641828762114
Generation 296 Best Solution: [183.77140186254232, 176.5715943148888,
174.98620320238027, 75.4774949371436] Fitness: 1.557833838776327
Generation 297 Best Solution: [183.77140186254232, 177.43793473892867,
174.06641270777462, 75.02618328388337] Fitness: 1.7093284163709976
Generation 298 Best Solution: [183.43318869331404, 177.43793473892867,
174.06641270777462, 75.84158888925721] Fitness: 2.4296706072000847
Generation 299 Best Solution: [183.77140186254232, 176.64833716032803,
174.98620320238027, 75.84158888925721] Fitness: 2.171008752874921
Generation 300 Best Solution: [183.77140186254232, 175.8229050772228,
174.98620320238027, 75.86463368648421] Fitness: 2.1400693175298673
Generation 301 Best Solution: [183.33545290432284, 177.43793473892867,
174.93034283371645, 75.4774949371436] Fitness: 2.391046353168413
Generation 302 Best Solution: [183.33545290432284, 176.15537742854434,

174.98620320238027, 75.84158888925721] Fitness: 1.7314810583760174
Generation 303 Best Solution: [183.77140186254232, 177.43793473892867,
174.39678237527002, 74.48640395749052] Fitness: 1.7476181354399547
Generation 304 Best Solution: [183.77140186254232, 177.43793473892867,
174.98620320238027, 74.9222525284753] Fitness: 2.296472432679194
Generation 305 Best Solution: [183.77140186254232, 176.70939170033125,
174.98620320238027, 75.84158888925721] Fitness: 2.2209159632397855
Generation 306 Best Solution: [183.77140186254232, 177.43793473892867,
174.98620320238027, 74.60951143418627] Fitness: 2.2753560295529356
Generation 307 Best Solution: [183.77140186254232, 175.957755529632,
174.98620320238027, 75.84158888925721] Fitness: 2.032889174365546
Generation 308 Best Solution: [182.85826597093225, 177.43793473892867,
174.98620320238027, 75.66579978510083] Fitness: 2.803252982796522
Generation 309 Best Solution: [183.4535848078129, 176.85359563224247,
174.98620320238027, 75.84158888925721] Fitness: 2.142055057464211
Generation 310 Best Solution: [183.4535848078129, 176.9135193701913,
174.98620320238027, 74.50191056255461] Fitness: 1.2274591937521029
Generation 311 Best Solution: [183.77140186254232, 177.43793473892867,
174.5355900209753, 74.50191056255461] Fitness: 1.8230671753349539
Generation 312 Best Solution: [183.4535848078129, 177.43793473892867,
174.5355900209753, 74.50191056255461] Fitness: 1.6020175430520454
Generation 313 Best Solution: [183.4535848078129, 177.43793473892867,
174.5355900209753, 74.50191056255461] Fitness: 1.6020175430520454
Generation 314 Best Solution: [183.77140186254232, 176.86615020035163,
174.98620320238027, 74.50191056255461] Fitness: 1.3933704354319918
Generation 315 Best Solution: [183.77140186254232, 177.43793473892867,
174.92731605697082, 74.50191056255461] Fitness: 2.2232830308484317
Generation 316 Best Solution: [183.77140186254232, 177.43793473892867,
174.98620320238027, 74.89888426720294] Fitness: 2.2893399834069026
Generation 317 Best Solution: [183.77140186254232, 177.43793473892867,
174.98620320238027, 74.89888426720294] Fitness: 2.2893399834069026
Generation 318 Best Solution: [183.77140186254232, 177.43793473892867,
174.5355900209753, 74.70005362889259] Fitness: 1.7827675392739966
Generation 319 Best Solution: [183.77140186254232, 177.43793473892867,
174.5355900209753, 75.30232232175156] Fitness: 2.0562544588252796
Generation 320 Best Solution: [183.77140186254232, 177.43793473892867,
174.5355900209753, 74.74277890674921] Fitness: 1.782531250664541
Generation 321 Best Solution: [183.4535848078129, 177.43793473892867,
174.5355900209753, 74.74277890674921] Fitness: 1.5614816183816325
Generation 322 Best Solution: [183.77140186254232, 177.43793473892867,
174.5355900209753, 74.74277890674921] Fitness: 1.782531250664541
Generation 323 Best Solution: [183.77140186254232, 175.82102704457287,
174.98620320238027, 74.48557087262786] Fitness: 1.1209323133786855
Generation 324 Best Solution: [183.36943775104737, 177.43793473892867,
174.48053510994626, 74.64717642439368] Fitness: 1.499377506224366
Generation 325 Best Solution: [183.77140186254232, 177.43793473892867,
174.76106798080767, 74.74277890674921] Fitness: 1.9821044828723544
Generation 326 Best Solution: [183.77140186254232, 177.35193949675465,

174.76106798080767, 74.74277890674921] Fitness: 1.8106449963537956
Generation 327 Best Solution: [183.77140186254232, 177.43793473892867,
174.20900605020958, 74.74277890674921] Fitness: 1.6415669991339985
Generation 328 Best Solution: [182.99141440560047, 177.43793473892867,
174.98620320238027, 74.74277890674921] Fitness: 2.018321201719533
Generation 329 Best Solution: [183.77140186254232, 176.68446481241247,
174.98620320238027, 75.86463368648421] Fitness: 2.2425182306347864
Generation 330 Best Solution: [183.4535848078129, 176.14448328885592,
174.98620320238027, 75.86463368648421] Fitness: 1.8146208396733314
Generation 331 Best Solution: [183.77140186254232, 177.43793473892867,
174.03230786130817, 75.08568890676095] Fitness: 1.7450749744471925
Generation 332 Best Solution: [183.77140186254232, 176.14448328885592,
174.98620320238027, 74.92457000270805] Fitness: 1.0012081160713093
Generation 333 Best Solution: [182.85514959894718, 176.14448328885592,
174.98620320238027, 75.86463368648421] Fitness: 1.8487980866619467
Generation 334 Best Solution: [183.4535848078129, 176.48622162639344,
174.98620320238027, 75.86463368648421] Fitness: 1.8898791038469196
Generation 335 Best Solution: [182.8342779237957, 177.43793473892867,
174.83834019393404, 75.08568890676095] Fitness: 2.001380860007502
Generation 336 Best Solution: [183.52356963990513, 176.48622162639344,
174.98620320238027, 75.86463368648421] Fitness: 1.9243080227643277
Generation 337 Best Solution: [183.52356963990513, 176.48622162639344,
173.94828797914425, 75.86463368648421] Fitness: 1.314410608106293
Generation 338 Best Solution: [183.52356963990513, 176.48622162639344,
173.94828797914425, 75.86463368648421] Fitness: 1.314410608106293
Generation 339 Best Solution: [183.10870444989044, 176.88484935305064,
173.94828797914425, 75.86463368648421] Fitness: 1.5580625741348169
Generation 340 Best Solution: [183.77140186254232, 176.16820687865732,
174.98620320238027, 75.86463368648421] Fitness: 2.034697697893514
Generation 341 Best Solution: [183.8378996826098, 176.39468603389076,
174.55495620134684, 75.86463368648421] Fitness: 1.6708875532302858
Generation 342 Best Solution: [183.77140186254232, 176.16820687865732,
174.98620320238027, 75.86463368648421] Fitness: 2.034697697893514
Generation 343 Best Solution: [183.4535848078129, 177.43793473892867,
174.55495620134684, 74.83165735043103] Fitness: 1.584460539700989
Generation 344 Best Solution: [183.60653679433617, 175.94580132270065,
174.98620320238027, 75.86463368648421] Fitness: 1.9447333494922865
Generation 345 Best Solution: [183.60653679433617, 175.94580132270065,
174.26312875490535, 75.86463368648421] Fitness: 1.3328410236548356
Generation 346 Best Solution: [183.60653679433617, 175.94580132270065,
174.26312875490535, 75.86463368648421] Fitness: 1.3328410236548356
Generation 347 Best Solution: [183.60653679433617, 177.43793473892867,
173.9647522310934, 75.86463368648421] Fitness: 2.582272739253412
Generation 348 Best Solution: [183.60653679433617, 177.43793473892867,
173.9647522310934, 75.86463368648421] Fitness: 2.582272739253412
Generation 349 Best Solution: [183.77140186254232, 177.43793473892867,
174.98620320238027, 74.60706312773482] Fitness: 2.275824599126974
Generation 350 Best Solution: [183.77140186254232, 176.65646146498233,

174.98620320238027, 75.86463368648421] Fitness: 2.220014236199544
Generation 351 Best Solution: [183.60653679433617, 177.35748646385198,
174.98620320238027, 74.78612843605298] Fitness: 1.9713850317397674
Generation 352 Best Solution: [183.77140186254232, 177.35748646385198,
174.98620320238027, 74.78612843605298] Fitness: 2.106766273581209
Generation 353 Best Solution: [183.60653679433617, 177.43793473892867,
174.62574763046578, 74.4075510635494] Fitness: 1.799327156901302
Generation 354 Best Solution: [183.77140186254232, 177.43793473892867,
174.62574763046578, 74.4075510635494] Fitness: 1.9347083987427436
Generation 355 Best Solution: [183.77140186254232, 177.43793473892867,
174.98620320238027, 74.65080069955624] Fitness: 2.268937287473087
Generation 356 Best Solution: [183.77140186254232, 175.90270018670088,
174.12395354710077, 75.86463368648421] Fitness: 1.4672232997039205
Generation 357 Best Solution: [183.77140186254232, 175.90270018670088,
174.12395354710077, 75.86463368648421] Fitness: 1.4672232997039205
Generation 358 Best Solution: [183.77140186254232, 175.90270018670088,
174.98620320238027, 75.86463368648421] Fitness: 2.098316301559443
Generation 359 Best Solution: [183.6285512879032, 177.43793473892867,
174.98620320238027, 74.71668903963001] Fitness: 2.144608202890973
Generation 360 Best Solution: [183.6285512879032, 177.43793473892867,
173.8900153131499, 74.71668903963001] Fitness: 1.552939777380933
Generation 361 Best Solution: [183.77140186254232, 176.76193377487013,
174.98620320238027, 75.67246439024359] Fitness: 1.9819613818234756
Generation 362 Best Solution: [183.37670335130338, 177.43793473892867,
174.98620320238027, 74.31495474752617] Fitness: 2.1528112208932635
Generation 363 Best Solution: [183.37670335130338, 177.43793473892867,
174.98620320238027, 74.92786393805731] Fitness: 2.048722380269147
Generation 364 Best Solution: [182.87005127491219, 177.43793473892867,
174.98620320238027, 75.67246439024359] Fitness: 2.8073006949785713
Generation 365 Best Solution: [183.37670335130338, 177.43793473892867,
174.98620320238027, 75.50920338462016] Fitness: 2.52028310632038
Generation 366 Best Solution: [183.54456327592916, 177.43793473892867,
174.98620320238027, 75.67246439024359] Fitness: 2.827442940297993
Generation 367 Best Solution: [183.60653679433617, 176.4294429602267,
174.98620320238027, 75.90622347437889] Fitness: 2.0290555680960107
Generation 368 Best Solution: [182.79024374046642, 176.4294429602267,
174.98620320238027, 75.90622347437889] Fitness: 2.01663764294797
Generation 369 Best Solution: [182.79024374046642, 176.32048068611763,
174.98620320238027, 75.86463368648421] Fitness: 1.9026687350882163
Generation 370 Best Solution: [183.87593144434422, 175.82823044674507,
174.98620320238027, 75.90622347437889] Fitness: 2.3252265404956507
Generation 371 Best Solution: [183.77140186254232, 176.00024261057942,
174.98620320238027, 75.86463368648421] Fitness: 2.061485969568927
Generation 372 Best Solution: [183.77140186254232, 175.82823044674507,
174.98620320238027, 75.67246439024359] Fitness: 1.807432985599434
Generation 373 Best Solution: [183.87593144434422, 176.00024261057942,
173.90145365887778, 75.86463368648421] Fitness: 1.5747664384921938
Generation 374 Best Solution: [183.87593144434422, 175.82823044674507,

174.31245543374885, 75.67246439024359] Fitness: 1.3189506845031929
Generation 375 Best Solution: [183.14882686273143, 176.00024261057942,
174.98620320238027, 75.86463368648421] Fitness: 1.7843231243438975
Generation 376 Best Solution: [183.14882686273143, 176.00024261057942,
174.98620320238027, 75.67246439024359] Fitness: 1.454799068253506
Generation 377 Best Solution: [183.87593144434422, 176.00024261057942,
174.98620320238027, 75.86463368648421] Fitness: 2.1704530787274567
Generation 378 Best Solution: [183.77140186254232, 176.00024261057942,
174.98620320238027, 75.86463368648421] Fitness: 2.061485969568927
Generation 379 Best Solution: [183.73630473591808, 176.00024261057942,
174.98620320238027, 75.86463368648421] Fitness: 2.0289242947642316
Generation 380 Best Solution: [183.87593144434422, 177.43793473892867,
174.10270309085635, 74.69757621273807] Fitness: 1.7427920871527507
Generation 381 Best Solution: [183.87593144434422, 176.75736353118901,
174.0204143286619, 75.86463368648421] Fitness: 1.791375748312722
Generation 382 Best Solution: [183.87593144434422, 176.75736353118901,
174.0204143286619, 75.90622347437889] Fitness: 1.8706781379600173
Generation 383 Best Solution: [183.87593144434422, 176.358488317967,
174.98961036605792, 75.75758875199931] Fitness: 1.9831872131593942
Generation 384 Best Solution: [183.87593144434422, 176.7513617043316,
174.98620320238027, 75.86463368648421] Fitness: 2.4104597519867146
Generation 385 Best Solution: [183.67161194262013, 176.67154949736587,
174.98961036605792, 75.86463368648421] Fitness: 2.1496175755326727
Generation 386 Best Solution: [182.84409865857938, 177.43793473892867,
174.98620320238027, 74.82662747358839] Fitness: 2.0922529012760704
Generation 387 Best Solution: [183.87593144434422, 176.3230251805548,
174.10270309085635, 75.86463368648421] Fitness: 1.5287977429117794
Generation 388 Best Solution: [182.84409865857938, 176.3230251805548,
174.10270309085635, 75.86463368648421] Fitness: 1.2391220249356083
Generation 389 Best Solution: [183.87593144434422, 176.69105560098876,
174.98961036605792, 75.90622347437889] Fitness: 2.4411879958948277
Generation 390 Best Solution: [183.62332675448218, 177.43793473892867,
174.98961036605792, 75.11901132281686] Fitness: 2.273385069841717
Generation 391 Best Solution: [182.84409865857938, 177.43793473892867,
174.98620320238027, 75.03324968289796] Fitness: 2.161893253609071
Generation 392 Best Solution: [182.84409865857938, 177.43793473892867,
174.98620320238027, 75.03324968289796] Fitness: 2.161893253609071
Generation 393 Best Solution: [183.582412189833, 177.43793473892867,
174.2348825371808, 74.54387883398482] Fitness: 1.5215142142658786
Generation 394 Best Solution: [183.32651484248603, 177.43793473892867,
174.98620320238027, 74.7594194088397] Fitness: 2.00243369581583
Generation 395 Best Solution: [183.34779608769878, 177.43793473892867,
174.98620320238027, 74.7594194088397] Fitness: 2.0076089536912507
Generation 396 Best Solution: [183.34779608769878, 177.43793473892867,
174.98620320238027, 74.42138230761965] Fitness: 2.0822374108227537
Generation 397 Best Solution: [183.32651484248603, 177.43793473892867,
175.10684597377752, 74.67435794449035] Fitness: 2.1892410833473117
Generation 398 Best Solution: [183.32651484248603, 177.43793473892867,

174.98961036605792, 74.67435794449035] Fitness: 2.0084533262961513
Generation 399 Best Solution: [183.81956938132328, 177.43793473892867,
174.98620320238027, 74.73013354599804] Fitness: 2.3124471755352936
Generation 400 Best Solution: [183.81956938132328, 177.43793473892867,
174.98620320238027, 74.52038736264684] Fitness: 2.3467408195988204
Generation 401 Best Solution: [183.582412189833, 177.23947539989808,
174.55948207609003, 75.90622347437889] Fitness: 2.417202498065894
Generation 402 Best Solution: [182.84409865857938, 175.93523860184106,
174.98620320238027, 75.86463368648421] Fitness: 1.9035843239300307
Generation 403 Best Solution: [182.84409865857938, 175.7498305172124,
174.98620320238027, 75.74214947483763] Fitness: 1.789731182742082
Generation 404 Best Solution: [183.582412189833, 177.43793473892867,
174.98620320238027, 74.52038736264684] Fitness: 2.1473121514354236
Generation 405 Best Solution: [182.84409865857938, 177.43793473892867,
175.05493135492438, 74.52038736264684] Fitness: 2.2209116298308405
Generation 406 Best Solution: [183.32651484248603, 177.23947539989808,
173.80971234442998, 75.90622347437889] Fitness: 2.2132339821032305
Generation 407 Best Solution: [183.582412189833, 177.43793473892867,
174.98620320238027, 74.43929496275821] Fitness: 2.179944041636767
Generation 408 Best Solution: [183.582412189833, 177.43793473892867,
174.98620320238027, 75.66239671899702] Fitness: 2.8351294540877126
Generation 409 Best Solution: [183.81956938132328, 177.43793473892867,
174.98620320238027, 74.7500170107757] Fitness: 2.3129467211753445
Generation 410 Best Solution: [183.582412189833, 175.8329576062399,
174.98620320238027, 75.90622347437889] Fitness: 2.062089442353008
Generation 411 Best Solution: [183.582412189833, 175.8329576062399,
174.98620320238027, 75.90622347437889] Fitness: 2.062089442353008
Generation 412 Best Solution: [183.582412189833, 176.24996697346657,
174.98620320238027, 75.90622347437889] Fitness: 1.9662854918916073
Generation 413 Best Solution: [183.87593144434422, 177.43793473892867,
174.98620320238027, 74.6814748431331] Fitness: 2.3749490935100637
Generation 414 Best Solution: [183.61139979535542, 177.43793473892867,
174.98620320238027, 74.6814748431331] Fitness: 2.13395491784241
Generation 415 Best Solution: [183.582412189833, 177.43793473892867,
174.98620320238027, 74.97133316208111] Fitness: 2.1629273104228695
Generation 416 Best Solution: [183.582412189833, 177.43793473892867,
174.98620320238027, 74.6814748431331] Fitness: 2.1145356284663306
Generation 417 Best Solution: [183.87593144434422, 177.43793473892867,
174.98961036605792, 74.6073229918244] Fitness: 2.38965800915592
Generation 418 Best Solution: [183.582412189833, 177.43793473892867,
174.98961036605792, 74.6073229918244] Fitness: 2.129244544112187
Generation 419 Best Solution: [183.582412189833, 177.43793473892867,
174.98620320238027, 74.6073229918244] Fitness: 2.1243280419614354
Generation 420 Best Solution: [182.86452243816873, 176.55198581444515,
175.10684597377752, 75.90622347437889] Fitness: 2.2204886847127527
Generation 421 Best Solution: [182.86452243816873, 176.55198581444515,
174.98620320238027, 75.90622347437889] Fitness: 2.034784425510841
Generation 422 Best Solution: [183.8469963879906, 176.55198581444515,

174.98961036605792, 74.91380704088273] Fitness: 1.19144858798242
Generation 423 Best Solution: [183.582412189833, 175.78552380950094,
175.10684597377752, 75.90622347437889] Fitness: 2.2767874868868487
Generation 424 Best Solution: [182.86452243816873, 175.64490236153688,
175.10684597377752, 75.74203869021467] Fitness: 2.0474381283580416
Generation 425 Best Solution: [182.86452243816873, 176.9411815879118,
174.98961036605792, 75.06911542997916] Fitness: 1.3519773058501359
Generation 426 Best Solution: [183.582412189833, 177.06133450473754,
174.98620320238027, 75.1541971715317] Fitness: 1.603487491084844
Generation 427 Best Solution: [183.582412189833, 176.1761278925566,
175.10684597377752, 75.06911542997916] Fitness: 1.0989797754302408
Generation 428 Best Solution: [183.582412189833, 176.1761278925566,
174.98620320238027, 75.90622347437889] Fitness: 1.962434831504971
Generation 429 Best Solution: [182.86452243816873, 176.1761278925566,
174.98620320238027, 75.90622347437889] Fitness: 1.9219382490061603
Generation 430 Best Solution: [183.81956938132328, 176.1761278925566,
174.98620320238027, 75.90622347437889] Fitness: 2.1618634996683674
Generation 431 Best Solution: [183.42053347050913, 176.1761278925566,
174.98620320238027, 75.90622347437889] Fitness: 1.879369367220052
Generation 432 Best Solution: [183.81956938132328, 176.70228260434317,
175.10684597377752, 75.90622347437889] Fitness: 2.5704965305028376
Generation 433 Best Solution: [183.42053347050913, 177.43793473892867,
174.7360272020348, 75.90622347437889] Fitness: 2.8675919718458687
Generation 434 Best Solution: [183.01288811833976, 177.43793473892867,
174.98620320238027, 74.7654720788652] Fitness: 2.0128188559844826
Generation 435 Best Solution: [183.81956938132328, 177.43793473892867,
174.08414382086798, 74.7654720788652] Fitness: 1.6830514289621552
Generation 436 Best Solution: [183.582412189833, 177.43793473892867,
174.7360272020348, 74.44161644530539] Fitness: 1.8699700365095104
Generation 437 Best Solution: [183.4196913747686, 176.4862448876683,
174.98620320238027, 75.90622347437889] Fitness: 1.9554114080357414
Generation 438 Best Solution: [183.582412189833, 177.43793473892867,
174.17295837413636, 75.90622347437889] Fitness: 2.6315850684425337
Generation 439 Best Solution: [183.01288811833976, 177.43793473892867,
174.4105447426727, 75.90622347437889] Fitness: 2.601160009300421
Generation 440 Best Solution: [183.37382904402457, 177.43793473892867,
174.65581376859498, 75.90622347437889] Fitness: 2.774359067361081
Generation 441 Best Solution: [183.4196913747686, 177.43793473892867,
174.98620320238027, 75.48936691081092] Fitness: 2.509808207592035
Generation 442 Best Solution: [183.81956938132328, 177.43793473892867,
174.98620320238027, 75.48936691081092] Fitness: 2.7926218904551052
Generation 443 Best Solution: [183.81956938132328, 176.1201923111687,
174.98620320238027, 75.90622347437889] Fitness: 2.1649092626223294
Generation 444 Best Solution: [183.81956938132328, 175.83550132917102,
175.0748988665671, 75.90622347437889] Fitness: 2.3942685024944206
Generation 445 Best Solution: [183.81956938132328, 175.60824083698733,
174.73084122963124, 75.90622347437889] Fitness: 2.117395232660273
Generation 446 Best Solution: [183.81956938132328, 176.54695240341812,

174.98620320238027, 75.90622347437889] Fitness: 2.271665257239393
Generation 447 Best Solution: [183.81956938132328, 176.04733563206383,
174.95810109748257, 75.90622347437889] Fitness: 2.1367604816820576
Generation 448 Best Solution: [183.81956938132328, 176.04733563206383,
174.98620320238027, 75.90622347437889] Fitness: 2.176584174154729
Generation 449 Best Solution: [183.81956938132328, 176.04733563206383,
174.03917499955418, 75.70023983627875] Fitness: 1.184442135759237
Generation 450 Best Solution: [183.81956938132328, 176.04733563206383,
174.03917499955418, 75.70023983627875] Fitness: 1.184442135759237
Generation 451 Best Solution: [183.81956938132328, 176.04733563206383,
174.98620320238027, 75.90622347437889] Fitness: 2.176584174154729
Generation 452 Best Solution: [183.81956938132328, 176.04733563206383,
174.98620320238027, 75.90622347437889] Fitness: 2.176584174154729
Generation 453 Best Solution: [182.79555053384402, 176.04733563206383,
174.98620320238027, 75.90622347437889] Fitness: 1.9773482127787547
Generation 454 Best Solution: [183.56361209095706, 176.96905252076914,
175.0748988665671, 75.30313085071629] Fitness: 1.722697147106783
Generation 455 Best Solution: [183.56361209095706, 176.96905252076914,
175.0748988665671, 75.70023983627875] Fitness: 2.2295433120104406
Generation 456 Best Solution: [183.81956938132328, 177.43793473892867,
174.98620320238027, 75.02426101791839] Fitness: 2.386095381120762
Generation 457 Best Solution: [183.4196913747686, 177.43793473892867,
175.0748988665671, 74.9955152926759] Fitness: 2.224018363260306
Generation 458 Best Solution: [183.4196913747686, 177.43793473892867,
174.10715771914587, 75.90622347437889] Fitness: 2.5449037460889703
Generation 459 Best Solution: [183.81956938132328, 176.1762563909537,
175.0748988665671, 75.90622347437889] Fitness: 2.2960632291239023
Generation 460 Best Solution: [182.79555053384402, 177.43793473892867,
174.10715771914587, 75.82166996422539] Fitness: 2.4702409349738295
Generation 461 Best Solution: [183.27904672026673, 177.43793473892867,
174.10715771914587, 75.90622347437889] Fitness: 2.507879160410571
Generation 462 Best Solution: [182.79555053384402, 177.43793473892867,
174.10715771914587, 75.90622347437889] Fitness: 2.6284814675760666
Generation 463 Best Solution: [182.79555053384402, 177.43793473892867,
174.10715771914587, 75.90622347437889] Fitness: 2.6284814675760666
Generation 464 Best Solution: [182.79555053384402, 177.43793473892867,
174.27170270879188, 75.90622347437889] Fitness: 2.650070885847745
Generation 465 Best Solution: [183.81956938132328, 177.43793473892867,
175.046356353903, 75.42318894599364] Fitness: 2.8026973145315597
Generation 466 Best Solution: [183.81956938132328, 177.43793473892867,
174.10715771914587, 75.3269850296335] Fitness: 1.9790545876510435
Generation 467 Best Solution: [183.81956938132328, 177.43793473892867,
174.10715771914587, 75.3269850296335] Fitness: 1.9790545876510435
Generation 468 Best Solution: [183.81956938132328, 176.07137018457615,
175.0748988665671, 75.90622347437889] Fitness: 2.3059696892224806
Generation 469 Best Solution: [183.81956938132328, 177.43793473892867,
174.9598665802211, 74.89128296667485] Fitness: 2.2978355939061346
Generation 470 Best Solution: [182.81775894505125, 176.38336948845162,

175.0748988665671, 75.90622347437889] Fitness: 2.1163908082996232
Generation 471 Best Solution: [183.55765488641322, 176.38336948845162,
175.0748988665671, 75.90622347437889] Fitness: 2.114660459209635
Generation 472 Best Solution: [183.10220095211278, 177.43793473892867,
174.18560723594643, 75.90622347437889] Fitness: 2.512190337171381
Generation 473 Best Solution: [182.79555053384402, 177.43793473892867,
174.9598665802211, 75.90622347437889] Fitness: 3.222379755661618
Generation 474 Best Solution: [182.79555053384402, 177.43793473892867,
174.426237382681, 75.90622347437889] Fitness: 2.7108505989008367
Generation 475 Best Solution: [182.79555053384402, 176.03182917981843,
175.0748988665671, 75.90622347437889] Fitness: 2.115159371733416
Generation 476 Best Solution: [182.79555053384402, 176.03182917981843,
175.0748988665671, 75.90622347437889] Fitness: 2.115159371733416
Generation 477 Best Solution: [183.81956938132328, 177.43793473892867,
175.046356353903, 75.39409049242272] Fitness: 2.7700057973368386
Generation 478 Best Solution: [183.51994565453583, 177.43793473892867,
175.046356353903, 75.44493135749151] Fitness: 2.5914490935509926
Generation 479 Best Solution: [183.74443078711326, 176.44203899758327,
175.046356353903, 75.90622347437889] Fitness: 2.234103587375066
Generation 480 Best Solution: [183.74443078711326, 176.44203899758327,
175.046356353903, 75.90622347437889] Fitness: 2.234103587375066
Generation 481 Best Solution: [183.74443078711326, 176.44203899758327,
174.8163801838466, 75.90622347437889] Fitness: 1.9236116072765417
Generation 482 Best Solution: [183.81956938132328, 176.04607792584866,
175.0748988665671, 75.90622347437889] Fitness: 2.3110630925753775
Generation 483 Best Solution: [183.74443078711326, 176.04607792584866,
175.0748988665671, 75.90622347437889] Fitness: 2.237878091418751
Generation 484 Best Solution: [183.81956938132328, 176.04607792584866,
175.0748988665671, 75.90622347437889] Fitness: 2.3110630925753775
Generation 485 Best Solution: [183.81956938132328, 176.04607792584866,
175.046356353903, 75.90622347437889] Fitness: 2.2664665683597933
Generation 486 Best Solution: [183.81956938132328, 176.04607792584866,
175.0748988665671, 75.07711257281838] Fitness: 1.2664802119962415
Generation 487 Best Solution: [183.74443078711326, 176.04607792584866,
175.046356353903, 75.90622347437889] Fitness: 2.1932815672031665
Generation 488 Best Solution: [183.74443078711326, 176.2596875272604,
175.0748988665671, 75.90622347437889] Fitness: 2.2279040901759144
Generation 489 Best Solution: [183.74443078711326, 177.43793473892867,
175.0748988665671, 75.14399932127561] Fitness: 2.5177994991245214
Generation 490 Best Solution: [183.74443078711326, 177.43793473892867,
175.0748988665671, 74.8380182185405] Fitness: 2.3839720697782125
Generation 491 Best Solution: [183.5063628910505, 177.43793473892867,
175.0748988665671, 74.8380182185405] Fitness: 2.2133432430017015
Generation 492 Best Solution: [183.81956938132328, 176.4898881773014,
175.0748988665671, 75.90622347437889] Fitness: 2.3742619952930584
Generation 493 Best Solution: [183.23440275070624, 176.4898881773014,
175.0748988665671, 75.90622347437889] Fitness: 2.0494661130536596
Generation 494 Best Solution: [183.23440275070624, 176.4898881773014,

175.0748988665671, 75.90622347437889] Fitness: 2.0494661130536596
Generation 495 Best Solution: [183.28715630253268, 176.4898881773014,
175.0748988665671, 75.90622347437889] Fitness: 2.0556756926128927
Generation 496 Best Solution: [183.28715630253268, 176.4898881773014,
175.0748988665671, 75.90622347437889] Fitness: 2.0556756926128927
Generation 497 Best Solution: [183.81956938132328, 176.55432483269843,
175.0748988665671, 75.89723984846358] Fitness: 2.3929690143240743
Generation 498 Best Solution: [183.7477276017095, 177.4145771033229,
175.046356353903, 74.4379935104836] Fitness: 2.351647480372127
Generation 499 Best Solution: [183.81956938132328, 176.44517137403395,
175.046356353903, 75.89723984846358] Fitness: 2.2912680288179126
Generation 500 Best Solution: [183.81956938132328, 176.44517137403395,
174.05958134485934, 75.89723984846358] Fitness: 1.5724529339438873
Optimal Solution for Simulation 4: [183.81956938132328, 176.44517137403395,
174.05958134485934, 75.89723984846358]

Simulation 5/32 with conditions: {'drought': True, 'flood': True,
'policy_change': False, 'social_event': True, 'economic_crisis': True}
Generation 1 Best Solution: [183.02873558083976, 175.9932697498652,
174.34414254123575, 75.06472935626032] Fitness: 0.2610943761638593
Generation 2 Best Solution: [183.02873558083976, 175.9932697498652,
173.98978527143848, 74.64471271910519] Fitness: 0.1339869524165792
Generation 3 Best Solution: [183.07174933612617, 175.691006722134,
174.36934632230614, 74.69969638014493] Fitness: 0.32209672480407403
Generation 4 Best Solution: [182.87548486357431, 175.76906130831415,
175.09940197336377, 75.06472935626032] Fitness: 1.1217927936252061
Generation 5 Best Solution: [183.43190994755142, 177.04660881751954,
174.96556185569352, 75.12508565839747] Fitness: 1.2444799460535627
Generation 6 Best Solution: [183.764152346571, 177.04660881751954,
174.36934632230614, 74.99426574138026] Fitness: 0.8699499169110292
Generation 7 Best Solution: [182.8781873872832, 176.58172777937313,
175.09940197336377, 75.60702450848663] Fitness: 1.6013674458136566
Generation 8 Best Solution: [182.87548486357431, 176.26398394236523,
175.1267581195718, 74.9159689812525] Fitness: 0.9240327983481355
Generation 9 Best Solution: [183.764152346571, 176.26398394236523,
175.1267581195718, 74.9159689812525] Fitness: 1.0280001307950541
Generation 10 Best Solution: [183.764152346571, 177.04660881751954,
175.1267581195718, 74.9159689812525] Fitness: 1.5381319902352837
Generation 11 Best Solution: [183.6802478178355, 175.76906130831415,
175.1267581195718, 74.9159689812525] Fitness: 1.1319911227730157
Generation 12 Best Solution: [183.764152346571, 176.26398394236523,
173.92416249216194, 75.77377024701195] Fitness: 1.2153372681605732
Generation 13 Best Solution: [183.28679520118996, 176.97208182998347,
174.35498023394842, 75.77377024701195] Fitness: 1.4227777053820576
Generation 14 Best Solution: [183.764152346571, 177.04660881751954,
174.1861064444006, 75.77377024701195] Fitness: 1.687779821173849
Generation 15 Best Solution: [182.87548486357431, 176.26398394236523,
175.1267581195718, 74.92618324077338] Fitness: 0.9279219184130243

Generation 16 Best Solution: [182.87548486357431, 177.04660881751954,
175.1267581195718, 75.0555293824624] Fitness: 1.501251800554953
Generation 17 Best Solution: [183.67077500909522, 177.04660881751954,
175.1267581195718, 75.0555293824624] Fitness: 1.5369103634570285
Generation 18 Best Solution: [182.87548486357431, 177.04660881751954,
174.1861064444006, 75.77377024701195] Fitness: 1.5838124887269303
Generation 19 Best Solution: [182.87548486357431, 177.04660881751954,
174.55109737382162, 75.60702450848663] Fitness: 1.4463745778908637
Generation 20 Best Solution: [182.87548486357431, 177.04660881751954,
174.56457695808754, 75.60702450848663] Fitness: 1.4549074388088088
Generation 21 Best Solution: [182.93597183182027, 177.04660881751954,
174.94024500409955, 75.60702450848663] Fitness: 1.7740458995790018
Generation 22 Best Solution: [182.93597183182027, 177.04660881751954,
174.94024500409955, 75.60702450848663] Fitness: 1.7740458995790018
Generation 23 Best Solution: [182.93597183182027, 176.07403861738428,
175.1267581195718, 75.77377024701195] Fitness: 1.7997709097910217
Generation 24 Best Solution: [183.01701051355718, 177.13397917636487,
175.1267581195718, 74.41260582623325] Fitness: 1.4928051811082155
Generation 25 Best Solution: [182.87548486357431, 177.13397917636487,
175.1267581195718, 74.41260582623325] Fitness: 1.5579557330412812
Generation 26 Best Solution: [182.93597183182027, 177.13397917636487,
174.64615305540045, 75.34275052693961] Fitness: 1.2697795369674552
Generation 27 Best Solution: [182.93597183182027, 176.19594601637965,
175.1267581195718, 75.77377024701195] Fitness: 1.7812062901538241
Generation 28 Best Solution: [183.27263989024922, 176.4684351714604,
175.1267581195718, 75.77377024701195] Fitness: 1.7500096513825145
Generation 29 Best Solution: [183.3283001018604, 176.4684351714604,
175.1267581195718, 75.77377024701195] Fitness: 1.754857098872367
Generation 30 Best Solution: [182.93597183182027, 176.19594601637965,
175.1267581195718, 75.77377024701195] Fitness: 1.7812062901538241
Generation 31 Best Solution: [182.93597183182027, 177.13777711202346,
175.1267581195718, 75.24111476309659] Fitness: 1.7264415121411316
Generation 32 Best Solution: [182.93597183182027, 177.13777711202346,
175.1267581195718, 74.97665059751911] Fitness: 1.548985923238203
Generation 33 Best Solution: [183.53434971736266, 177.13777711202346,
175.1267581195718, 74.60619620971717] Fitness: 1.4738797496927003
Generation 34 Best Solution: [182.87548486357431, 176.92969937809127,
175.1267581195718, 74.68301336934191] Fitness: 1.252894919962596
Generation 35 Best Solution: [182.93597183182027, 177.13777711202346,
173.7588334196268, 74.78587319016896] Fitness: 0.8653652780996677
Generation 36 Best Solution: [182.87548486357431, 177.13777711202346,
175.1267581195718, 74.89369643596748] Fitness: 1.5472260655938461
Generation 37 Best Solution: [182.99056261648832, 177.13777711202346,
175.1267581195718, 74.89369643596748] Fitness: 1.4918988117737215
Generation 38 Best Solution: [183.53434971736266, 177.13777711202346,
175.1267581195718, 74.89369643596748] Fitness: 1.5073104717186858
Generation 39 Best Solution: [182.93597183182027, 177.13777711202346,
175.1267581195718, 74.89369643596748] Fitness: 1.5155934812584582

Generation 40 Best Solution: [183.53434971736266, 177.13777711202346,
175.1267581195718, 74.89369643596748] Fitness: 1.5073104717186858
Generation 41 Best Solution: [182.99056261648832, 177.13777711202346,
175.1267581195718, 74.89369643596748] Fitness: 1.4918988117737215
Generation 42 Best Solution: [182.93597183182027, 176.80324922138124,
175.1267581195718, 74.89083639949745] Fitness: 1.1336129826473247
Generation 43 Best Solution: [182.93597183182027, 176.80324922138124,
175.1267581195718, 74.61999183743883] Fitness: 1.0998302938557372
Generation 44 Best Solution: [182.93597183182027, 176.80324922138124,
174.38931665181823, 75.77377024701195] Fitness: 1.3359465022568022
Generation 45 Best Solution: [183.53904962987662, 176.80324922138124,
174.38931665181823, 75.77377024701195] Fitness: 1.3297886626057447
Generation 46 Best Solution: [183.53904962987662, 176.80324922138124,
174.38931665181823, 75.77377024701195] Fitness: 1.3297886626057447
Generation 47 Best Solution: [182.87548486357431, 177.13777711202346,
173.7889690660291, 74.56510406512304] Fitness: 0.8792968371449355
Generation 48 Best Solution: [183.21790261757226, 177.13777711202346,
173.7889690660291, 74.56510406512304] Fitness: 0.7748218679674185
Generation 49 Best Solution: [183.21790261757226, 176.80324922138124,
175.1267581195718, 75.26885774143848] Fitness: 1.2974600764702933
Generation 50 Best Solution: [182.96430481873261, 176.80324922138124,
174.5522645057079, 75.77377024701195] Fitness: 1.4043025427106337
Generation 51 Best Solution: [183.53904962987662, 177.13777711202346,
175.1267581195718, 74.53726625404389] Fitness: 1.4869746148153329
Generation 52 Best Solution: [183.79241995108234, 177.13777711202346,
174.38931665181823, 74.53726625404389] Fitness: 0.9570875845643696
Generation 53 Best Solution: [183.21790261757226, 177.13777711202346,
175.1267581195718, 75.28778651672224] Fitness: 1.6961368704311628
Generation 54 Best Solution: [183.79241995108234, 176.2125254213736,
174.38931665181823, 75.77377024701195] Fitness: 1.2444108362546467
Generation 55 Best Solution: [182.87548486357431, 175.6153341740336,
173.85469059768673, 75.77377024701195] Fitness: 1.433750008880844
Generation 56 Best Solution: [183.53904962987662, 175.6153341740336,
173.85469059768673, 75.77377024701195] Fitness: 1.3959595848943982
Generation 57 Best Solution: [182.87548486357431, 175.6153341740336,
173.85469059768673, 75.73575510138184] Fitness: 1.3702959383424278
Generation 58 Best Solution: [182.87548486357431, 175.6153341740336,
173.85469059768673, 75.73575510138184] Fitness: 1.3702959383424278
Generation 59 Best Solution: [183.79241995108234, 176.65339139255386,
173.73318325580925, 75.73575510138184] Fitness: 1.4050406343135748
Generation 60 Best Solution: [183.59805293858997, 175.6153341740336,
173.73318325580925, 75.73575510138184] Fitness: 1.4286212130224305
Generation 61 Best Solution: [183.53904962987662, 175.6153341740336,
173.73318325580925, 75.73575510138184] Fitness: 1.3990371051251165
Generation 62 Best Solution: [183.79241995108234, 176.65339139255386,
173.92787351871937, 75.73575510138184] Fitness: 1.3094474069679558
Generation 63 Best Solution: [183.53904962987662, 176.65339139255386,
174.81253445466902, 75.73575510138184] Fitness: 1.4478301273882521

Generation 64 Best Solution: [183.38684716547613, 177.13777711202346,
173.92787351871937, 75.73575510138184] Fitness: 1.5884916810043253
Generation 65 Best Solution: [183.53904962987662, 177.13777711202346,
174.5404575806296, 75.73575510138184] Fitness: 1.7213174479906974
Generation 66 Best Solution: [183.79241995108234, 176.213868213603,
174.44053272850263, 75.73575510138184] Fitness: 1.2020236017029369
Generation 67 Best Solution: [182.92632822355924, 177.13777711202346,
175.1267581195718, 74.43508661773204] Fitness: 1.5275682840744447
Generation 68 Best Solution: [182.82119258948475, 176.213868213603,
175.1267581195718, 75.73575510138184] Fitness: 1.781800347054097
Generation 69 Best Solution: [183.79241995108234, 175.95287428460557,
175.1267581195718, 75.06115131454703] Fitness: 1.1819052564172219
Generation 70 Best Solution: [182.92632822355924, 175.95287428460557,
175.1267581195718, 75.73575510138184] Fitness: 1.7821907683213407
Generation 71 Best Solution: [182.92632822355924, 175.95287428460557,
175.1267581195718, 75.77377024701195] Fitness: 1.8456448388597568
Generation 72 Best Solution: [183.79241995108234, 176.78585508745792,
175.1267581195718, 75.73575510138184] Fitness: 2.11132749828295
Generation 73 Best Solution: [183.79241995108234, 177.13777711202346,
174.00318968053995, 75.73575510138184] Fitness: 1.783792795787612
Generation 74 Best Solution: [183.79241995108234, 176.398288974107,
175.1267581195718, 75.73575510138184] Fitness: 1.896572727510473
Generation 75 Best Solution: [183.79241995108234, 176.398288974107,
175.1267581195718, 75.73575510138184] Fitness: 1.896572727510473
Generation 76 Best Solution: [183.79241995108234, 176.78585508745792,
175.1267581195718, 75.38963326381528] Fitness: 1.636328733035993
Generation 77 Best Solution: [183.55165678527493, 177.13777711202346,
175.1267581195718, 74.40041042452178] Fitness: 1.5363857282085172
Generation 78 Best Solution: [183.79241995108234, 177.05939258081037,
175.1267581195718, 74.31316938279167] Fitness: 1.6336812161728984
Generation 79 Best Solution: [183.79241995108234, 177.13777711202346,
175.1267581195718, 74.55413076659863] Fitness: 1.648705852847498
Generation 80 Best Solution: [183.55165678527493, 177.13777711202346,
174.69174382549332, 75.73575510138184] Fitness: 1.8365764037945513
Generation 81 Best Solution: [183.55165678527493, 176.398288974107,
175.1267581195718, 75.73575510138184] Fitness: 1.7373483358710429
Generation 82 Best Solution: [183.55165678527493, 177.13777711202346,
173.89241119951177, 75.73575510138184] Fitness: 1.6588834739285163
Generation 83 Best Solution: [183.55165678527493, 177.05939258081037,
175.1267581195718, 74.88725658352466] Fitness: 1.4083648118475969
Generation 84 Best Solution: [183.55165678527493, 177.13777711202346,
175.1267581195718, 74.68750223647399] Fitness: 1.4783738065886893
Generation 85 Best Solution: [183.55165678527493, 177.13777711202346,
175.1267581195718, 74.68750223647399] Fitness: 1.4783738065886893
Generation 86 Best Solution: [183.55165678527493, 177.13777711202346,
175.1267581195718, 74.68750223647399] Fitness: 1.4783738065886893
Generation 87 Best Solution: [183.55165678527493, 177.13777711202346,
175.1267581195718, 74.68750223647399] Fitness: 1.4783738065886893

Generation 88 Best Solution: [182.82054288135856, 177.22245126808144,
174.092198538022, 75.14337520205413] Fitness: 1.0999369439257738
Generation 89 Best Solution: [183.55165678527493, 176.2400670875547,
175.1267581195718, 75.35203977038317] Fitness: 1.2009618465253573
Generation 90 Best Solution: [183.41390653335455, 177.13777711202346,
174.0965321788562, 75.73575510138184] Fitness: 1.559568847340707
Generation 91 Best Solution: [182.82054288135856, 175.75961666762828,
174.18983197919772, 75.73575510138184] Fitness: 1.2169213090232227
Generation 92 Best Solution: [183.79241995108234, 176.39812228001247,
175.1267581195718, 74.59194241556514] Fitness: 1.0318672987455049
Generation 93 Best Solution: [183.41390653335455, 177.13777711202346,
174.00948208808794, 75.73575510138184] Fitness: 1.572380078292904
Generation 94 Best Solution: [183.55165678527493, 176.84899529792744,
174.00948208808794, 75.73575510138184] Fitness: 1.2840650308812762
Generation 95 Best Solution: [182.82054288135856, 176.39812228001247,
174.40657060143312, 75.73575510138184] Fitness: 1.1145383222199015
Generation 96 Best Solution: [183.55165678527493, 176.03382961478675,
175.1267581195718, 75.73575510138184] Fitness: 1.7471884733258976
Generation 97 Best Solution: [183.41390653335455, 176.03382961478675,
175.1267581195718, 75.73575510138184] Fitness: 1.6963800667064528
Generation 98 Best Solution: [183.79241995108234, 176.03382961478675,
175.1267581195718, 75.35877534489995] Fitness: 1.3980553619920455
Generation 99 Best Solution: [183.55165678527493, 176.3468576698974,
175.1267581195718, 74.41018657797976] Fitness: 0.9104085251515235
Generation 100 Best Solution: [183.41390653335455, 176.03382961478675,
175.1267581195718, 75.73575510138184] Fitness: 1.6963800667064528
Generation 101 Best Solution: [183.41390653335455, 176.03382961478675,
175.1267581195718, 75.73575510138184] Fitness: 1.6963800667064528
Generation 102 Best Solution: [183.41390653335455, 176.03382961478675,
175.1267581195718, 75.73575510138184] Fitness: 1.6963800667064528
Generation 103 Best Solution: [183.79241995108234, 177.13777711202346,
174.79908951541628, 74.51652621368996] Fitness: 1.2442501569508075
Generation 104 Best Solution: [183.41390653335455, 176.49480707744044,
175.1267581195718, 75.73575510138184] Fitness: 1.7183147105407572
Generation 105 Best Solution: [183.41390653335455, 176.49480707744044,
175.1267581195718, 75.04905641288879] Fitness: 0.9566473862989685
Generation 106 Best Solution: [183.79241995108234, 177.0016367478243,
174.0896895603983, 75.73575510138184] Fitness: 1.5942538838088396
Generation 107 Best Solution: [183.79241995108234, 177.0016367478243,
174.0896895603983, 75.73575510138184] Fitness: 1.5942538838088396
Generation 108 Best Solution: [183.41390653335455, 175.65227869136794,
175.1267581195718, 75.04601199269018] Fitness: 1.1632099734405952
Generation 109 Best Solution: [183.79241995108234, 177.0016367478243,
174.0896895603983, 75.73575510138184] Fitness: 1.5942538838088396
Generation 110 Best Solution: [183.41390653335455, 176.16232255237858,
175.1267581195718, 75.14385646386519] Fitness: 0.9696047450843917
Generation 111 Best Solution: [183.79241995108234, 177.13777711202346,
175.1267581195718, 75.14385646386519] Fitness: 1.8075506920793198

Generation 112 Best Solution: [183.79241995108234, 177.13777711202346,
175.1267581195718, 75.14385646386519] Fitness: 1.8075506920793198
Generation 113 Best Solution: [183.02659927637342, 177.13777711202346,
175.1267581195718, 75.14385646386519] Fitness: 1.6118025113883299
Generation 114 Best Solution: [183.79241995108234, 177.17643642689842,
175.1267581195718, 75.14385646386519] Fitness: 1.862730942443283
Generation 115 Best Solution: [183.79241995108234, 177.17643642689842,
175.1267581195718, 75.14385646386519] Fitness: 1.862730942443283
Generation 116 Best Solution: [183.79241995108234, 177.17643642689842,
175.1267581195718, 75.40754390011165] Fitness: 2.10764630455087
Generation 117 Best Solution: [183.79241995108234, 177.3404325215479,
175.1267581195718, 74.3895710127575] Fitness: 2.015236768659181
Generation 118 Best Solution: [183.65189111676253, 177.3404325215479,
174.27327917629495, 75.73575510138184] Fitness: 1.9907385244713163
Generation 119 Best Solution: [183.65189111676253, 176.43273137411512,
175.1267581195718, 75.73575510138184] Fitness: 1.8024387004696303
Generation 120 Best Solution: [183.65189111676253, 177.3404325215479,
174.27327917629495, 75.73575510138184] Fitness: 1.9907385244713163
Generation 121 Best Solution: [183.65189111676253, 175.58408606456942,
175.1267581195718, 75.73575510138184] Fitness: 2.0976941127425137
Generation 122 Best Solution: [183.65189111676253, 175.58408606456942,
175.1267581195718, 74.31834778642012] Fitness: 1.32586756750139
Generation 123 Best Solution: [183.0658211664124, 176.6855881965459,
174.65683533446852, 75.73575510138184] Fitness: 1.2857571105854007
Generation 124 Best Solution: [183.79241995108234, 177.13777711202346,
174.83890120689944, 75.73575510138184] Fitness: 2.136141884061688
Generation 125 Best Solution: [183.65189111676253, 175.7122351782489,
175.1267581195718, 75.73575510138184] Fitness: 1.9817626256320755
Generation 126 Best Solution: [183.79241995108234, 177.3404325215479,
174.83890120689944, 74.79787855840466] Fitness: 1.592850207567258
Generation 127 Best Solution: [183.79241995108234, 177.3404325215479,
175.1267581195718, 74.72966177658016] Fitness: 1.954746330462309
Generation 128 Best Solution: [183.65189111676253, 177.13777711202346,
173.90290771063866, 75.73575510138184] Fitness: 1.7102223433467754
Generation 129 Best Solution: [183.45818747736595, 177.13777711202346,
173.90290771063866, 75.73575510138184] Fitness: 1.6171447853517706
Generation 130 Best Solution: [183.79241995108234, 175.8912994304076,
175.1267581195718, 75.73575510138184] Fitness: 1.9660993636404671
Generation 131 Best Solution: [182.83857365246124, 175.8912994304076,
175.1267581195718, 74.43796168265581] Fitness: 1.0342311390661867
Generation 132 Best Solution: [183.79241995108234, 176.54604427890803,
175.01897650600006, 75.73575510138184] Fitness: 1.797047901962599
Generation 133 Best Solution: [183.79241995108234, 176.54604427890803,
174.65089357787414, 75.73575510138184] Fitness: 1.40641501863059
Generation 134 Best Solution: [183.79241995108234, 175.8912994304076,
174.19930324796584, 75.70344801655659] Fitness: 1.176052142768264
Generation 135 Best Solution: [183.79241995108234, 177.3404325215479,
175.1267581195718, 74.88224933923188] Fitness: 1.985697253876404

Generation 136 Best Solution: [183.57583775686095, 177.3404325215479,
175.1267581195718, 74.55981709553845] Fitness: 1.8153324968475586
Generation 137 Best Solution: [183.79241995108234, 177.3404325215479,
175.1267581195718, 74.88224933923188] Fitness: 1.985697253876404
Generation 138 Best Solution: [183.79241995108234, 176.62150927287908,
175.1267581195718, 75.73575510138184] Fitness: 1.991913873292325
Generation 139 Best Solution: [183.79241995108234, 177.3404325215479,
175.1267581195718, 74.96141557166499] Fitness: 2.015932649816703
Generation 140 Best Solution: [183.79241995108234, 176.59850573883506,
175.06450899669383, 75.73575510138184] Fitness: 1.8873869969931627
Generation 141 Best Solution: [183.79241995108234, 177.3404325215479,
174.363065077179, 74.96141557166499] Fitness: 1.3117037210822373
Generation 142 Best Solution: [183.79241995108234, 177.3404325215479,
175.1267581195718, 74.74791386673452] Fitness: 1.9565539394321867
Generation 143 Best Solution: [183.1766616228981, 175.6927086996744,
175.1267581195718, 75.73575510138184] Fitness: 1.8728161762964808
Generation 144 Best Solution: [183.1766616228981, 176.83696963855965,
175.06450899669383, 75.73575510138184] Fitness: 1.837041718449565
Generation 145 Best Solution: [183.1766616228981, 176.14932230798186,
175.06450899669383, 75.73575510138184] Fitness: 1.5611400090506464
Generation 146 Best Solution: [183.21284706160804, 176.14932230798186,
175.06450899669383, 75.73575510138184] Fitness: 1.5583794673879505
Generation 147 Best Solution: [183.21284706160804, 176.14932230798186,
175.06450899669383, 75.73575510138184] Fitness: 1.5583794673879505
Generation 148 Best Solution: [183.79241995108234, 176.14932230798186,
175.06450899669383, 75.73575510138184] Fitness: 1.7899330419521382
Generation 149 Best Solution: [183.79241995108234, 176.14932230798186,
175.06450899669383, 75.73575510138184] Fitness: 1.7899330419521382
Generation 150 Best Solution: [183.79241995108234, 176.09002458169135,
175.06450899669383, 75.73575510138184] Fitness: 1.800367023713824
Generation 151 Best Solution: [183.79241995108234, 177.3404325215479,
174.90914750366443, 74.67180708249003] Fitness: 1.659926707660467
Generation 152 Best Solution: [183.87549502917693, 177.3404325215479,
174.90914750366443, 74.67180708249003] Fitness: 1.7356555119740122
Generation 153 Best Solution: [183.79241995108234, 176.09002458169135,
175.1267581195718, 75.73575510138184] Fitness: 1.8915097588200045
Generation 154 Best Solution: [183.79241995108234, 176.09002458169135,
174.22300119777563, 75.73575510138184] Fitness: 1.1559375473225122
Generation 155 Best Solution: [183.79241995108234, 176.09002458169135,
175.1267581195718, 75.73575510138184] Fitness: 1.8915097588200045
Generation 156 Best Solution: [183.6603393236284, 177.3404325215479,
174.4992690637646, 74.88674636838455] Fitness: 1.244041140916785
Generation 157 Best Solution: [183.79241995108234, 177.3404325215479,
175.1267581195718, 74.88674636838455] Fitness: 1.9871552984580159
Generation 158 Best Solution: [183.87549502917693, 176.39737772888284,
174.90914750366443, 75.73575510138184] Fitness: 1.679577780997227
Generation 159 Best Solution: [183.6603393236284, 175.62830611059903,
175.1267581195718, 74.81061649685442] Fitness: 1.2045188375387386

Generation 160 Best Solution: [183.79241995108234, 175.9338219689898,
174.90914750366443, 75.78236164248489] Fitness: 1.7306180678530343
Generation 161 Best Solution: [183.87549502917693, 176.39737772888284,
175.1267581195718, 75.73575510138184] Fitness: 1.972070144545234
Generation 162 Best Solution: [183.6603393236284, 177.3404325215479,
174.18347238142243, 75.73575510138184] Fitness: 1.9849948127516333
Generation 163 Best Solution: [183.6603393236284, 177.30022097033378,
174.20220793709953, 75.73575510138184] Fitness: 1.9187471886261727
Generation 164 Best Solution: [183.6603393236284, 176.32102286261224,
175.1267581195718, 75.78236164248489] Fitness: 1.8611743196875048
Generation 165 Best Solution: [182.83543323479842, 177.10223896755028,
174.90914750366443, 75.73575510138184] Fitness: 2.0630016316747435
Generation 166 Best Solution: [183.6603393236284, 176.00983461381296,
175.1267581195718, 75.7867898809193] Fitness: 1.901521343562807
Generation 167 Best Solution: [183.6603393236284, 176.02959121293233,
174.90914750366443, 75.73575510138184] Fitness: 1.51680092901461
Generation 168 Best Solution: [183.6603393236284, 175.86991716916788,
174.90914750366443, 75.78236164248489] Fitness: 1.664931906295248
Generation 169 Best Solution: [183.16132118200701, 176.98509410988308,
174.90914750366443, 75.78236164248489] Fitness: 1.8706708541221782
Generation 170 Best Solution: [183.6603393236284, 176.98509410988308,
174.90914750366443, 75.27616855387774] Fitness: 1.3307209431673974
Generation 171 Best Solution: [183.63010308469723, 176.98509410988308,
174.90914750366443, 75.41544703277829] Fitness: 1.4564316009951348
Generation 172 Best Solution: [183.6603393236284, 177.30022097033378,
174.90914750366443, 74.96990891061967] Fitness: 1.561278705054788
Generation 173 Best Solution: [183.6603393236284, 176.36020076732783,
174.8182061215611, 75.78236164248489] Fitness: 1.4746132654684616
Generation 174 Best Solution: [183.87549502917693, 177.30022097033378,
173.90499597491402, 74.83190856424572] Fitness: 1.2855253721631585
Generation 175 Best Solution: [183.87549502917693, 177.30022097033378,
173.90499597491402, 74.83190856424572] Fitness: 1.2855253721631585
Generation 176 Best Solution: [183.87549502917693, 177.30022097033378,
174.90914750366443, 74.30007917640864] Fitness: 1.7765584024964667
Generation 177 Best Solution: [183.6603393236284, 175.93844848056935,
174.90914750366443, 75.78236164248489] Fitness: 1.6300501417866116
Generation 178 Best Solution: [183.6603393236284, 177.30022097033378,
173.90499597491402, 75.78236164248489] Fitness: 2.0403189470558645
Generation 179 Best Solution: [183.6603393236284, 177.30022097033378,
174.3837375255447, 75.78236164248489] Fitness: 2.0374567147016815
Generation 180 Best Solution: [183.6603393236284, 177.30022097033378,
174.3837375255447, 75.78236164248489] Fitness: 2.0374567147016815
Generation 181 Best Solution: [183.2245340018352, 177.30022097033378,
174.3837375255447, 75.78236164248489] Fitness: 1.9038854236532985
Generation 182 Best Solution: [183.6603393236284, 177.30022097033378,
174.3837375255447, 75.78236164248489] Fitness: 2.0374567147016815
Generation 183 Best Solution: [183.6603393236284, 177.30022097033378,
174.3837375255447, 75.78236164248489] Fitness: 2.0374567147016815

Generation 184 Best Solution: [183.6603393236284, 177.30022097033378,
174.3837375255447, 75.78236164248489] Fitness: 2.0374567147016815
Generation 185 Best Solution: [183.87549502917693, 177.3404325215479,
174.10722779517167, 75.78236164248489] Fitness: 2.237609183436276
Generation 186 Best Solution: [183.6603393236284, 177.3404325215479,
174.90914750366443, 75.78236164248489] Fitness: 2.5095387578945303
Generation 187 Best Solution: [183.87549502917693, 177.3404325215479,
175.1267581195718, 74.66667511980698] Fitness: 2.0281912178769836
Generation 188 Best Solution: [183.6603393236284, 177.3404325215479,
175.1267581195718, 74.56451475023181] Fitness: 1.8633535598497664
Generation 189 Best Solution: [183.62017904647678, 176.53921343868703,
175.1267581195718, 75.78236164248489] Fitness: 1.9028778793157202
Generation 190 Best Solution: [183.87549502917693, 177.3404325215479,
174.48532834382104, 75.78236164248489] Fitness: 2.3241160859910477
Generation 191 Best Solution: [183.87549502917693, 177.3404325215479,
174.69430026043543, 75.78236164248489] Fitness: 2.466726940344783
Generation 192 Best Solution: [183.87549502917693, 177.3404325215479,
175.1267581195718, 75.139543346097] Fitness: 2.1951075527588038
Generation 193 Best Solution: [183.87549502917693, 177.3404325215479,
175.1267581195718, 74.65409497822067] Fitness: 2.028469644048591
Generation 194 Best Solution: [183.87549502917693, 177.3404325215479,
173.89310217092145, 75.139543346097] Fitness: 1.505514909144714
Generation 195 Best Solution: [183.87549502917693, 175.82852897660865,
175.1267581195718, 75.4522170245042] Fitness: 1.675247990507422
Generation 196 Best Solution: [183.87549502917693, 175.82852897660865,
175.1267581195718, 75.4522170245042] Fitness: 1.675247990507422
Generation 197 Best Solution: [183.87549502917693, 175.82852897660865,
175.1267581195718, 75.4522170245042] Fitness: 1.675247990507422
Generation 198 Best Solution: [183.87549502917693, 176.09292873100887,
174.8505271135553, 74.65409497822067] Fitness: 0.7382012769496975
Generation 199 Best Solution: [183.87549502917693, 176.09292873100887,
174.8505271135553, 74.65409497822067] Fitness: 0.7382012769496975
Generation 200 Best Solution: [183.6603393236284, 176.51617363376604,
175.1267581195718, 75.4522170245042] Fitness: 1.4360622321234922
Generation 201 Best Solution: [183.87549502917693, 177.3404325215479,
174.9582770149124, 74.40737980391053] Fitness: 1.8505098517655398
Generation 202 Best Solution: [183.87549502917693, 177.3404325215479,
174.65283171060685, 75.3674171957871] Fitness: 1.8557919724950138
Generation 203 Best Solution: [182.96537744160503, 175.72135254856488,
175.1267581195718, 75.78236164248489] Fitness: 1.9841350518185357
Generation 204 Best Solution: [183.87549502917693, 177.3404325215479,
174.0863005385959, 75.36082658315847] Fitness: 1.6549841264165526
Generation 205 Best Solution: [183.6603393236284, 177.3404325215479,
174.1727997204137, 75.78236164248489] Fitness: 2.0626056827505366
Generation 206 Best Solution: [183.87549502917693, 176.32233769660834,
175.1267581195718, 75.78236164248489] Fitness: 2.035523786779919
Generation 207 Best Solution: [183.87549502917693, 177.3404325215479,
175.1267581195718, 74.87764734193235] Fitness: 2.0599663373667823

Generation 208 Best Solution: [183.87549502917693, 177.3404325215479,
174.5001030819761, 75.60118215416105] Fitness: 2.0471986159523587
Generation 209 Best Solution: [183.87549502917693, 176.37640683708128,
174.83338996535633, 75.78236164248489] Fitness: 1.6680587805749956
Generation 210 Best Solution: [183.87549502917693, 176.6148002824583,
175.1267581195718, 75.25740787744266] Fitness: 1.4560717530381413
Generation 211 Best Solution: [183.87549502917693, 176.6148002824583,
175.1267581195718, 75.25740787744266] Fitness: 1.4560717530381413
Generation 212 Best Solution: [183.87549502917693, 176.19841471419016,
175.1267581195718, 75.78236164248489] Fitness: 2.0303821603704204
Generation 213 Best Solution: [183.87549502917693, 177.3404325215479,
174.9230346094422, 74.32023638797213] Fitness: 1.8492207725436232
Generation 214 Best Solution: [183.87549502917693, 177.3404325215479,
174.9230346094422, 74.32023638797213] Fitness: 1.8492207725436232
Generation 215 Best Solution: [183.87549502917693, 177.3404325215479,
173.79719326340913, 74.32023638797213] Fitness: 1.480565362590973
Generation 216 Best Solution: [183.87549502917693, 176.34918271490181,
174.47681897418465, 75.78236164248489] Fitness: 1.3833026290072459
Generation 217 Best Solution: [183.87549502917693, 176.48233692982782,
175.1267581195718, 74.32023638797213] Fitness: 1.2262972119788402
Generation 218 Best Solution: [183.87549502917693, 176.48233692982782,
175.1267581195718, 75.78236164248489] Fitness: 2.0772651769159918
Generation 219 Best Solution: [183.87549502917693, 177.3404325215479,
175.0296302198942, 74.32023638797213] Fitness: 1.9856403913924
Generation 220 Best Solution: [183.87549502917693, 177.3404325215479,
175.0296302198942, 74.32023638797213] Fitness: 1.9856403913924
Generation 221 Best Solution: [183.87549502917693, 177.3404325215479,
175.1267581195718, 74.32023638797213] Fitness: 2.1252334996084272
Generation 222 Best Solution: [183.87549502917693, 176.21454762972192,
175.1267581195718, 75.78236164248489] Fitness: 2.0297077750560595
Generation 223 Best Solution: [183.87549502917693, 176.21454762972192,
175.1267581195718, 75.78236164248489] Fitness: 2.0297077750560595
Generation 224 Best Solution: [183.87549502917693, 176.21454762972192,
175.1267581195718, 75.78236164248489] Fitness: 2.0297077750560595
Generation 225 Best Solution: [183.87549502917693, 176.21454762972192,
175.1267581195718, 75.78236164248489] Fitness: 2.0297077750560595
Generation 226 Best Solution: [183.0695717912263, 176.21454762972192,
174.17872705052991, 75.78236164248489] Fitness: 1.0060386221019266
Generation 227 Best Solution: [183.87549502917693, 177.3404325215479,
173.85879500782852, 74.69557653956734] Fitness: 1.3533220115053144
Generation 228 Best Solution: [183.87549502917693, 177.3404325215479,
175.1267581195718, 74.71032308286448] Fitness: 2.0291216524691915
Generation 229 Best Solution: [183.87549502917693, 176.37353355972803,
175.1267581195718, 75.78236164248489] Fitness: 2.0445757072267896
Generation 230 Best Solution: [183.87549502917693, 176.37353355972803,
174.9587408659951, 75.78236164248489] Fitness: 1.81230344158423
Generation 231 Best Solution: [183.87549502917693, 176.0943534989266,
175.1267581195718, 75.78236164248489] Fitness: 2.044397032024313

Generation 232 Best Solution: [182.83238947905207, 177.11473190751013,
175.1267581195718, 75.78236164248489] Fitness: 2.452420669344757
Generation 233 Best Solution: [183.87549502917693, 177.3404325215479,
175.1267581195718, 74.82911190408049] Fitness: 2.046564142821403
Generation 234 Best Solution: [183.87549502917693, 177.3404325215479,
174.26864609012705, 75.78236164248489] Fitness: 2.247513340801677
Generation 235 Best Solution: [183.1593437999825, 177.11473190751013,
175.1267581195718, 75.78236164248489] Fitness: 2.3269811627040617
Generation 236 Best Solution: [183.87549502917693, 176.3841749138945,
175.1267581195718, 75.78236164248489] Fitness: 2.0469656799406533
Generation 237 Best Solution: [183.87549502917693, 176.40033713332772,
175.1267581195718, 75.78236164248489] Fitness: 2.050930349360601
Generation 238 Best Solution: [183.48633059867183, 176.40033713332772,
175.1267581195718, 75.78236164248489] Fitness: 1.7882260384396447
Generation 239 Best Solution: [183.48633059867183, 176.40033713332772,
175.1267581195718, 75.78236164248489] Fitness: 1.7882260384396447
Generation 240 Best Solution: [183.48633059867183, 176.40033713332772,
175.1267581195718, 75.78236164248489] Fitness: 1.7882260384396447
Generation 241 Best Solution: [183.33699756055944, 176.40033713332772,
175.1267581195718, 75.78236164248489] Fitness: 1.749558429114533
Generation 242 Best Solution: [183.33699756055944, 176.40033713332772,
175.1267581195718, 75.78236164248489] Fitness: 1.749558429114533
Generation 243 Best Solution: [183.14206539188396, 177.3404325215479,
175.1267581195718, 74.30915092472061] Fitness: 1.831410961815787
Generation 244 Best Solution: [183.87549502917693, 176.40033713332772,
174.34271593902173, 75.78236164248489] Fitness: 1.3402652867345959
Generation 245 Best Solution: [183.33699756055944, 176.10615933054476,
175.1267581195718, 75.78236164248489] Fitness: 1.7405934958185267
Generation 246 Best Solution: [183.33699756055944, 176.10615933054476,
175.1267581195718, 75.78236164248489] Fitness: 1.7405934958185267
Generation 247 Best Solution: [183.48633059867183, 176.10615933054476,
175.1267581195718, 75.78236164248489] Fitness: 1.7792611051436384
Generation 248 Best Solution: [183.33699756055944, 176.10615933054476,
175.1267581195718, 75.78236164248489] Fitness: 1.7405934958185267
Generation 249 Best Solution: [183.70778836398725, 176.47778710743424,
175.1267581195718, 74.41722854925177] Fitness: 1.0367902764947226
Generation 250 Best Solution: [183.48633059867183, 176.26202074129228,
175.1267581195718, 75.78236164248489] Fitness: 1.7673523101691204
Generation 251 Best Solution: [183.48633059867183, 177.3404325215479,
175.1267581195718, 74.41722854925177] Fitness: 1.81666469907125
Generation 252 Best Solution: [183.48633059867183, 176.23692358728357,
175.1267581195718, 75.78236164248489] Fitness: 1.766733946427142
Generation 253 Best Solution: [183.48633059867183, 177.3404325215479,
175.1267581195718, 75.07640614378495] Fitness: 1.8901267395508974
Generation 254 Best Solution: [183.70778836398725, 177.3404325215479,
175.1267581195718, 75.07640614378495] Fitness: 2.0109221094024092
Generation 255 Best Solution: [183.48633059867183, 177.3404325215479,
175.1267581195718, 74.93863202583096] Fitness: 1.819262942096476

Generation 256 Best Solution: [183.6193905694099, 176.28332899479673,
175.1267581195718, 75.78236164248489] Fitness: 1.832130728980784
Generation 257 Best Solution: [183.6193905694099, 176.3805076701041,
175.1267581195718, 75.78236164248489] Fitness: 1.8469072912182096
Generation 258 Best Solution: [183.59158828665113, 177.3404325215479,
175.1267581195718, 75.07640614378495] Fitness: 1.9380889411255922
Generation 259 Best Solution: [183.3192969829542, 176.56886431298815,
175.1267581195718, 75.78236164248489] Fitness: 1.812651560102159
Generation 260 Best Solution: [183.0884838867208, 177.3404325215479,
175.1267581195718, 74.73419791087518] Fitness: 1.7415552046630383
Generation 261 Best Solution: [183.6193905694099, 177.3404325215479,
175.1267581195718, 74.73419791087518] Fitness: 1.8316613083530553
Generation 262 Best Solution: [183.6193905694099, 177.3404325215479,
173.83713254300537, 74.79355211036275] Fitness: 1.174737360470253
Generation 263 Best Solution: [183.6193905694099, 177.3404325215479,
173.83713254300537, 75.00728729320511] Fitness: 1.249294700099727
Generation 264 Best Solution: [182.82348832330865, 176.80269526916533,
175.1267581195718, 74.83924013845326] Fitness: 1.1812928960154405
Generation 265 Best Solution: [183.6193905694099, 176.4436918006231,
175.1267581195718, 75.78236164248489] Fitness: 1.8643443621966262
Generation 266 Best Solution: [183.6193905694099, 176.4436918006231,
175.1267581195718, 75.78236164248489] Fitness: 1.8643443621966262
Generation 267 Best Solution: [183.6193905694099, 176.01720764215864,
173.83713254300537, 75.78236164248489] Fitness: 1.2012708604786528
Generation 268 Best Solution: [183.6193905694099, 176.01720764215864,
173.83713254300537, 75.78236164248489] Fitness: 1.2012708604786528
Generation 269 Best Solution: [183.6193905694099, 176.40918579068943,
174.3809827928121, 75.78236164248489] Fitness: 1.156024998726639
Generation 270 Best Solution: [183.6193905694099, 176.40918579068943,
174.3809827928121, 75.78236164248489] Fitness: 1.156024998726639
Generation 271 Best Solution: [183.6193905694099, 176.40918579068943,
174.3809827928121, 75.78236164248489] Fitness: 1.156024998726639
Generation 272 Best Solution: [183.6193905694099, 176.40918579068943,
175.1267581195718, 75.78236164248489] Fitness: 1.8540569928342074
Generation 273 Best Solution: [183.39620840108842, 177.3404325215479,
175.1267581195718, 74.30797462396465] Fitness: 1.8419033027056284
Generation 274 Best Solution: [183.5268306274832, 177.3404325215479,
174.0140705445292, 74.81826629628276] Fitness: 1.0718079920050567
Generation 275 Best Solution: [183.87549502917693, 175.77901858056512,
175.1267581195718, 75.48636387169233] Fitness: 1.7500633358589899
Generation 276 Best Solution: [183.87549502917693, 176.29675892225055,
175.1267581195718, 75.57676551290893] Fitness: 1.7132357418844966
Generation 277 Best Solution: [183.87549502917693, 176.29675892225055,
175.1267581195718, 75.15320844486043] Fitness: 1.261385884121131
Generation 278 Best Solution: [183.3436403523581, 176.29675892225055,
175.1267581195718, 75.78236164248489] Fitness: 1.7321343915938279
Generation 279 Best Solution: [183.3436403523581, 176.29675892225055,
175.1267581195718, 75.78236164248489] Fitness: 1.7321343915938279

Generation 280 Best Solution: [183.5268306274832, 177.3404325215479,
175.1267581195718, 74.40978978458467] Fitness: 1.836095295064111
Generation 281 Best Solution: [183.87549502917693, 177.3404325215479,
175.1267581195718, 74.96118962634155] Fitness: 2.0915613779308604
Generation 282 Best Solution: [183.87549502917693, 177.3404325215479,
175.1267581195718, 74.96118962634155] Fitness: 2.0915613779308604
Generation 283 Best Solution: [183.87549502917693, 175.9685177545471,
175.1267581195718, 75.78236164248489] Fitness: 2.083698902192378
Generation 284 Best Solution: [183.87549502917693, 176.57720203838971,
175.1267581195718, 75.36576694392313] Fitness: 1.5416645686840806
Generation 285 Best Solution: [183.87549502917693, 177.3404325215479,
174.5316002836128, 75.36576694392313] Fitness: 1.7708442078507938
Generation 286 Best Solution: [183.5268306274832, 175.84951985440208,
175.1267581195718, 74.48303759320738] Fitness: 0.9774291895364218
Generation 287 Best Solution: [183.5268306274832, 176.57720203838971,
175.1267581195718, 75.78236164248489] Fitness: 1.87442034538164
Generation 288 Best Solution: [183.5268306274832, 176.57720203838971,
175.1267581195718, 75.78236164248489] Fitness: 1.87442034538164
Generation 289 Best Solution: [183.87549502917693, 177.3404325215479,
175.1267581195718, 74.63747090137313] Fitness: 2.0292127252063223
Generation 290 Best Solution: [183.87549502917693, 177.3404325215479,
175.1267581195718, 74.63747090137313] Fitness: 2.0292127252063223
Generation 291 Best Solution: [183.5823633502526, 177.3404325215479,
175.1267581195718, 74.33699157200334] Fitness: 1.8966415237356895
Generation 292 Best Solution: [183.87549502917693, 177.3404325215479,
175.1267581195718, 74.63747090137313] Fitness: 2.0292127252063223
Generation 293 Best Solution: [183.87549502917693, 176.07907569211022,
175.1267581195718, 75.78236164248489] Fitness: 2.0478634995521943
Generation 294 Best Solution: [183.5029909424259, 177.3404325215479,
175.1267581195718, 74.33699157200334] Fitness: 1.860018290305131
Generation 295 Best Solution: [183.5823633502526, 177.3404325215479,
175.1267581195718, 74.33699157200334] Fitness: 1.8966415237356895
Generation 296 Best Solution: [183.5029909424259, 177.24491318301804,
175.1267581195718, 74.33699157200334] Fitness: 1.703666692633184
Generation 297 Best Solution: [183.5268306274832, 177.3404325215479,
174.96526431669668, 74.56800655143088] Fitness: 1.5665940726454664
Generation 298 Best Solution: [183.5823633502526, 177.3404325215479,
174.826072980698, 75.78236164248489] Fitness: 2.3717199204716386
Generation 299 Best Solution: [183.5866492239357, 177.3404325215479,
175.1267581195718, 74.48546543960171] Fitness: 1.8384387127434287
Generation 300 Best Solution: [183.5823633502526, 177.24491318301804,
175.1267581195718, 74.48546543960171] Fitness: 1.6798325090583885
Generation 301 Best Solution: [183.5823633502526, 176.1580021832635,
175.1267581195718, 75.78236164248489] Fitness: 1.8142071292225488
Generation 302 Best Solution: [183.5823633502526, 176.1580021832635,
175.1267581195718, 75.78236164248489] Fitness: 1.8142071292225488
Generation 303 Best Solution: [183.5823633502526, 176.1580021832635,
175.1267581195718, 75.78236164248489] Fitness: 1.8142071292225488

Generation 304 Best Solution: [183.5823633502526, 177.3404325215479,
175.1267581195718, 74.87350276592713] Fitness: 1.8390495770796182
Generation 305 Best Solution: [183.87549502917693, 176.5756746568004,
174.7318171144149, 75.78236164248489] Fitness: 1.6431616836406286
Generation 306 Best Solution: [183.60058865306053, 176.5756746568004,
174.7318171144149, 75.78236164248489] Fitness: 1.4333153677761876
Generation 307 Best Solution: [183.6845536171766, 176.48534664573305,
174.99435179430492, 75.78236164248489] Fitness: 1.7336019824186402
Generation 308 Best Solution: [183.87549502917693, 176.5756746568004,
173.9226886544238, 75.78236164248489] Fitness: 1.4193053325873322
Generation 309 Best Solution: [183.5823633502526, 177.24491318301804,
174.283020928677, 75.78236164248489] Fitness: 1.8743662793857965
Generation 310 Best Solution: [183.5823633502526, 176.4487610759994,
174.283020928677, 75.53653577632797] Fitness: 0.7456257946318217
Generation 311 Best Solution: [183.752954364249, 177.24491318301804,
174.283020928677, 75.78236164248489] Fitness: 1.9860289627285803
Generation 312 Best Solution: [183.6845536171766, 176.15642414192223,
175.1267581195718, 75.50384789265257] Fitness: 1.4590537455117298
Generation 313 Best Solution: [183.6845536171766, 175.80308559024684,
174.283020928677, 75.78236164248489] Fitness: 1.2886068911647999
Generation 314 Best Solution: [183.6845536171766, 176.15642414192223,
174.1132681299264, 75.78236164248489] Fitness: 1.136935112324979
Generation 315 Best Solution: [183.6845536171766, 175.80308559024684,
174.1132681299264, 75.78236164248489] Fitness: 1.2755134411600821
Generation 316 Best Solution: [183.87549502917693, 175.80308559024684,
174.53875269642293, 75.78236164248489] Fitness: 1.5505538863756878
Generation 317 Best Solution: [183.87549502917693, 175.80308559024684,
174.57422958428, 75.78236164248489] Fitness: 1.572937704271609
Generation 318 Best Solution: [183.87549502917693, 175.80308559024684,
174.57422958428, 75.78236164248489] Fitness: 1.572937704271609
Generation 319 Best Solution: [183.87549502917693, 175.80308559024684,
174.57422958428, 75.78236164248489] Fitness: 1.572937704271609
Generation 320 Best Solution: [183.87549502917693, 175.80308559024684,
174.57422958428, 75.78236164248489] Fitness: 1.572937704271609
Generation 321 Best Solution: [183.87549502917693, 176.05381328557735,
175.12473593701245, 75.78236164248489] Fitness: 2.051331901475354
Generation 322 Best Solution: [183.6845536171766, 176.2222405543405,
175.1267581195718, 75.78236164248489] Fitness: 1.8713864469372699
Generation 323 Best Solution: [183.6845536171766, 176.2222405543405,
175.12473593701245, 75.78236164248489] Fitness: 1.8683315358402641
Generation 324 Best Solution: [183.6845536171766, 176.2222405543405,
175.12473593701245, 75.78236164248489] Fitness: 1.8683315358402641
Generation 325 Best Solution: [183.6845536171766, 177.3640521599673,
174.76538938832, 74.46391103248158] Fitness: 1.4995962056425876
Generation 326 Best Solution: [183.6845536171766, 176.19150018741746,
175.12473593701245, 75.78236164248489] Fitness: 1.8695980492836222
Generation 327 Best Solution: [183.6845536171766, 175.937610191902,
175.1267581195718, 75.78236164248489] Fitness: 1.938954296414434

Generation 328 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.72866371271358] Fitness: 2.0712276019279416
Generation 329 Best Solution: [183.87549502917693, 176.68101606253217,
175.12473593701245, 75.35642270908586] Fitness: 1.5921641929985504
Generation 330 Best Solution: [183.87549502917693, 176.85130636061635,
175.12473593701245, 75.35642270908586] Fitness: 1.7323400543892418
Generation 331 Best Solution: [183.37471337107507, 176.68101606253217,
175.1267581195718, 75.78236164248489] Fitness: 1.8893046506829483
Generation 332 Best Solution: [183.87549502917693, 176.68101606253217,
174.76538938832, 75.78236164248489] Fitness: 1.7385876050817504
Generation 333 Best Solution: [183.37471337107507, 176.68101606253217,
174.76538938832, 75.78236164248489] Fitness: 1.4437286667261646
Generation 334 Best Solution: [183.37471337107507, 176.87483949494867,
174.76538938832, 75.78236164248489] Fitness: 1.6068006414094334
Generation 335 Best Solution: [183.37471337107507, 177.3640521599673,
175.1267581195718, 74.83662930465452] Fitness: 1.7943791955248416
Generation 336 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.68669010443239] Fitness: 2.069088827385558
Generation 337 Best Solution: [183.87549502917693, 177.3640521599673,
174.05574449143904, 75.48178467426865] Fitness: 1.8390326054879536
Generation 338 Best Solution: [183.87549502917693, 177.3640521599673,
174.05574449143904, 75.48178467426865] Fitness: 1.8390326054879536
Generation 339 Best Solution: [183.37471337107507, 177.3640521599673,
174.9827468613366, 74.59477523687033] Fitness: 1.5772987381540584
Generation 340 Best Solution: [183.87549502917693, 177.3640521599673,
174.9827468613366, 74.59477523687033] Fitness: 1.8721576765096444
Generation 341 Best Solution: [183.87549502917693, 176.78806002815546,
175.1267581195718, 75.78236164248489] Fitness: 2.2670462405064273
Generation 342 Best Solution: [183.87549502917693, 177.3640521599673,
174.9827468613366, 74.68790374503611] Fitness: 1.8673553893899328
Generation 343 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.51505947268134] Fitness: 2.0886723286226974
Generation 344 Best Solution: [183.78554154467207, 177.3640521599673,
175.1267581195718, 74.51505947268134] Fitness: 2.007151485457191
Generation 345 Best Solution: [183.78554154467207, 177.3640521599673,
174.4189649519912, 75.78236164248489] Fitness: 2.2522620712458292
Generation 346 Best Solution: [183.87549502917693, 176.27932599618148,
175.1267581195718, 75.78236164248489] Fitness: 2.0310500170837997
Generation 347 Best Solution: [183.78554154467207, 177.3640521599673,
174.9827468613366, 75.14818471752332] Fitness: 1.9589316555050313
Generation 348 Best Solution: [183.37471337107507, 176.06647859248628,
175.1267581195718, 75.78236164248489] Fitness: 1.7561341257425318
Generation 349 Best Solution: [183.78554154467207, 176.56314360375117,
174.4189649519912, 75.78236164248489] Fitness: 1.3486068500204413
Generation 350 Best Solution: [183.87549502917693, 175.87108285953425,
174.4189649519912, 75.78236164248489] Fitness: 1.4476855927451537
Generation 351 Best Solution: [183.78554154467207, 176.06647859248628,
175.01099740443712, 75.78236164248489] Fitness: 1.8047665920307074

Generation 352 Best Solution: [183.87549502917693, 176.06647859248628,
175.1267581195718, 75.78236164248489] Fitness: 2.0509930640981175
Generation 353 Best Solution: [183.87549502917693, 176.06647859248628,
175.1267581195718, 75.78236164248489] Fitness: 2.0509930640981175
Generation 354 Best Solution: [183.87549502917693, 176.3353206986477,
175.1267581195718, 75.78236164248489] Fitness: 2.037435942629159
Generation 355 Best Solution: [183.87549502917693, 177.3640521599673,
174.12308972943643, 75.78236164248489] Fitness: 2.277634813506959
Generation 356 Best Solution: [183.87549502917693, 176.26559482828085,
175.1267581195718, 75.78236164248489] Fitness: 2.0302238663777037
Generation 357 Best Solution: [183.87549502917693, 176.26559482828085,
175.1267581195718, 75.78236164248489] Fitness: 2.0302238663777037
Generation 358 Best Solution: [183.87549502917693, 176.5843750585869,
175.1267581195718, 74.44263610500606] Fitness: 1.218102629312023
Generation 359 Best Solution: [183.87549502917693, 176.5843750585869,
175.1267581195718, 74.44263610500606] Fitness: 1.218102629312023
Generation 360 Best Solution: [183.03484713255924, 176.5843750585869,
175.1267581195718, 74.44263610500606] Fitness: 0.943905572175938
Generation 361 Best Solution: [183.87549502917693, 176.15504613805442,
175.1267581195718, 74.44263610500606] Fitness: 1.1277452774070058
Generation 362 Best Solution: [183.42285709306324, 175.6003058336149,
175.1267581195718, 75.78236164248489] Fitness: 2.0559203708429052
Generation 363 Best Solution: [183.8035460142093, 177.3640521599673,
175.1267581195718, 74.76527167904163] Fitness: 2.009110695642232
Generation 364 Best Solution: [183.87549502917693, 177.3640521599673,
174.12308972943643, 75.78236164248489] Fitness: 2.277634813506959
Generation 365 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.64493648789836] Fitness: 2.0696626510012317
Generation 366 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.64493648789836] Fitness: 2.0696626510012317
Generation 367 Best Solution: [183.87549502917693, 176.34811598711465,
175.1267581195718, 75.78236164248489] Fitness: 2.0395753286203977
Generation 368 Best Solution: [183.87549502917693, 176.34811598711465,
175.1267581195718, 75.78236164248489] Fitness: 2.0395753286203977
Generation 369 Best Solution: [183.87549502917693, 177.0739367830441,
175.1267581195718, 74.80333463054917] Fitness: 1.6399196537162652
Generation 370 Best Solution: [183.87549502917693, 176.34811598711465,
175.1267581195718, 74.77567793672213] Fitness: 1.099393475791799
Generation 371 Best Solution: [183.87549502917693, 176.34811598711465,
175.1267581195718, 74.77567793672213] Fitness: 1.099393475791799
Generation 372 Best Solution: [183.87549502917693, 177.3640521599673,
174.12308972943643, 74.90780704813777] Fitness: 1.371561758921729
Generation 373 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.37386269541864] Fitness: 2.1389147831255997
Generation 374 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.37386269541864] Fitness: 2.1389147831255997
Generation 375 Best Solution: [183.87549502917693, 176.44214152323025,
175.1267581195718, 75.78236164248489] Fitness: 2.0630576834656065

Generation 376 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.67246968890414] Fitness: 2.068981711144601
Generation 377 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.67246968890414] Fitness: 2.068981711144601
Generation 378 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.67246968890414] Fitness: 2.068981711144601
Generation 379 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.67246968890414] Fitness: 2.068981711144601
Generation 380 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.89103539947351] Fitness: 2.1051402934209715
Generation 381 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.89103539947351] Fitness: 2.1051402934209715
Generation 382 Best Solution: [183.87549502917693, 177.3640521599673,
175.0069117588612, 74.89103539947351] Fitness: 1.9349999284157493
Generation 383 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.74706499999093] Fitness: 2.0730237249105796
Generation 384 Best Solution: [183.87549502917693, 177.3640521599673,
175.1267581195718, 74.89103539947351] Fitness: 2.1051402934209715
Generation 385 Best Solution: [183.87549502917693, 177.3640521599673,
174.0636610588621, 74.89103539947351] Fitness: 1.370775747179276
Generation 386 Best Solution: [183.87549502917693, 177.3640521599673,
174.63287392787063, 74.89103539947351] Fitness: 1.5467425957686982
Generation 387 Best Solution: [183.87549502917693, 177.11279625129197,
173.78841247265328, 75.78236164248489] Fitness: 1.9869878645663777
Generation 388 Best Solution: [183.45321242674697, 176.58546285263833,
175.1267581195718, 75.78236164248489] Fitness: 1.8508832813728042
Generation 389 Best Solution: [183.87549502917693, 175.873269032444,
175.1267581195718, 75.78236164248489] Fitness: 2.129719727719663
Generation 390 Best Solution: [183.45321242674697, 177.3640521599673,
175.1267581195718, 74.34131309039589] Fitness: 1.880613196779826
Generation 391 Best Solution: [183.87549502917693, 175.873269032444,
175.1267581195718, 75.78236164248489] Fitness: 2.129719727719663
Generation 392 Best Solution: [183.87549502917693, 176.98282292637842,
175.1267581195718, 75.78236164248489] Fitness: 2.463269951004921
Generation 393 Best Solution: [183.45321242674697, 177.3640521599673,
175.1267581195718, 75.36658010288615] Fitness: 2.164620747519
Generation 394 Best Solution: [183.87549502917693, 176.1897110706001,
175.1267581195718, 75.78236164248489] Fitness: 2.0309130278987526
Generation 395 Best Solution: [183.87549502917693, 176.1897110706001,
175.1267581195718, 75.78236164248489] Fitness: 2.0309130278987526
Generation 396 Best Solution: [183.87549502917693, 176.1897110706001,
175.1267581195718, 75.78236164248489] Fitness: 2.0309130278987526
Generation 397 Best Solution: [183.3752602428864, 175.81399108337877,
175.1267581195718, 75.78236164248489] Fitness: 1.8706906117075257
Generation 398 Best Solution: [183.3752602428864, 175.81399108337877,
175.1267581195718, 75.78236164248489] Fitness: 1.8706906117075257
Generation 399 Best Solution: [183.22775355051388, 175.9737390928549,
175.1267581195718, 75.78236164248489] Fitness: 1.773750433003214

Generation 400 Best Solution: [183.3752602428864, 177.3640521599673,
175.1267581195718, 74.4370948022605] Fitness: 1.8178569435263139
Generation 401 Best Solution: [183.86228725049773, 177.3640521599673,
175.1267581195718, 74.66246205593674] Fitness: 2.056340797586617
Generation 402 Best Solution: [183.86228725049773, 177.3640521599673,
175.1267581195718, 74.66246205593674] Fitness: 2.056340797586617
Generation 403 Best Solution: [183.87549502917693, 177.3640521599673,
175.0159814052512, 74.66246205593674] Fitness: 1.9110527189175872
Generation 404 Best Solution: [183.86228725049773, 177.3640521599673,
174.43504916232408, 74.66246205593674] Fitness: 1.3800153283153784
Generation 405 Best Solution: [183.86228725049773, 177.3640521599673,
175.0159814052512, 74.66246205593674] Fitness: 1.898299828890927
Generation 406 Best Solution: [183.86228725049773, 177.3640521599673,
175.0159814052512, 74.66246205593674] Fitness: 1.898299828890927
Generation 407 Best Solution: [183.86228725049773, 177.3640521599673,
175.07050047422024, 74.87615770091159] Fitness: 2.0049537125399217
Generation 408 Best Solution: [183.86228725049773, 177.3640521599673,
175.1267581195718, 75.08103799931907] Fitness: 2.183806703308376
Generation 409 Best Solution: [183.86228725049773, 177.3640521599673,
175.07050047422024, 75.05329363129239] Fitness: 2.084348231794
Generation 410 Best Solution: [183.86228725049773, 177.3640521599673,
175.0159814052512, 75.08103799931907] Fitness: 2.0257657346126865
Generation 411 Best Solution: [183.86228725049773, 176.02574724426137,
175.1267581195718, 75.78236164248489] Fitness: 2.050037751320705
Generation 412 Best Solution: [183.7133449596124, 175.60106647503184,
175.1267581195718, 75.05329363129239] Fitness: 1.3633180764229984
Generation 413 Best Solution: [183.86228725049773, 177.3640521599673,
173.945786037189, 74.88010133292985] Fitness: 1.3806032860405024
Generation 414 Best Solution: [183.86228725049773, 177.3640521599673,
173.945786037189, 74.88010133292985] Fitness: 1.3806032860405024
Generation 415 Best Solution: [183.86228725049773, 177.3640521599673,
173.945786037189, 74.88010133292985] Fitness: 1.3806032860405024
Generation 416 Best Solution: [183.86228725049773, 177.3640521599673,
173.945786037189, 74.88010133292985] Fitness: 1.3806032860405024
Generation 417 Best Solution: [182.79602814311787, 175.57565601123554,
175.07050047422024, 75.08196332485949] Fitness: 1.3084498506803603
Generation 418 Best Solution: [183.86228725049773, 175.74685333787608,
175.07050047422024, 75.78236164248489] Fitness: 2.117068994156547
Generation 419 Best Solution: [183.13512066469917, 175.74685333787608,
175.07050047422024, 75.78236164248489] Fitness: 1.8309641587161232
Generation 420 Best Solution: [183.86228725049773, 175.97862999739564,
175.07050047422024, 75.78236164248489] Fitness: 1.9842527159452128
Generation 421 Best Solution: [183.86228725049773, 175.57565601123554,
175.1267581195718, 74.6762220855017] Fitness: 1.4030525224329242
Generation 422 Best Solution: [183.83833781918483, 175.97862999739564,
174.81178844478603, 75.78236164248489] Fitness: 1.6447887380599688
Generation 423 Best Solution: [182.79602814311787, 176.02935169187296,
175.07050047422024, 75.78236164248489] Fitness: 1.826125960093956

Generation 424 Best Solution: [183.86228725049773, 175.5868945944485,
175.07050047422024, 75.39857937181542] Fitness: 1.7139947687779697
Generation 425 Best Solution: [183.0425223057563, 176.485586802152,
174.0387222764598, 75.78236164248489] Fitness: 1.0712648400313167
Generation 426 Best Solution: [183.2968830830686, 176.41496740887578,
175.07050047422024, 75.78236164248489] Fitness: 1.6663500639231628
Generation 427 Best Solution: [183.54904672980533, 175.94376065597746,
174.81178844478603, 75.78236164248489] Fitness: 1.4584659198679375
Generation 428 Best Solution: [183.86228725049773, 175.91425287589064,
174.11320784112533, 75.78236164248489] Fitness: 1.3565027220410215
Generation 429 Best Solution: [183.02473068869216, 177.3640521599673,
175.07050047422024, 75.29785012579606] Fitness: 2.0155577653108994
Generation 430 Best Solution: [183.02473068869216, 177.3640521599673,
175.07050047422024, 74.6187403565232] Fitness: 1.717921295450292
Generation 431 Best Solution: [183.02473068869216, 177.3640521599673,
175.07050047422024, 74.6187403565232] Fitness: 1.717921295450292
Generation 432 Best Solution: [183.02473068869216, 177.3640521599673,
175.07050047422024, 74.71525892355449] Fitness: 1.7167717933674753
Generation 433 Best Solution: [182.79602814311787, 177.3640521599673,
175.07050047422024, 74.89017333652426] Fitness: 1.8693353884515587
Generation 434 Best Solution: [183.02473068869216, 177.3640521599673,
174.81178844478603, 74.71525892355449] Fitness: 1.3997447598515054
Generation 435 Best Solution: [182.79602814311787, 176.14924483109715,
174.81178844478603, 75.78236164248489] Fitness: 1.4823744237165277
Generation 436 Best Solution: [183.02473068869216, 175.914137578929,
174.19255406968347, 75.78236164248489] Fitness: 1.0990360582143037
Generation 437 Best Solution: [183.86228725049773, 177.3640521599673,
175.07050047422024, 74.63254154609349] Fitness: 1.974968027670712
Generation 438 Best Solution: [183.7988230594807, 176.69404374765287,
175.07050047422024, 74.63254154609349] Fitness: 1.093716004500634
Generation 439 Best Solution: [183.86228725049773, 176.69404374765287,
175.07050047422024, 74.63254154609349] Fitness: 1.15123439489958
Generation 440 Best Solution: [183.86228725049773, 177.3640521599673,
174.81178844478603, 74.9451274899307] Fitness: 1.7130750963400418
Generation 441 Best Solution: [183.44128480698146, 175.84882476317875,
175.07050047422024, 74.63254154609349] Fitness: 0.8364758739069224
Generation 442 Best Solution: [183.02473068869216, 177.3640521599673,
175.07050047422024, 74.9451274899307] Fitness: 1.7720088862675745
Generation 443 Best Solution: [183.02473068869216, 176.69404374765287,
175.07050047422024, 75.78236164248489] Fitness: 1.8398274548985007
Generation 444 Best Solution: [183.779698355435, 176.69404374765287,
175.07050047422024, 75.78236164248489] Fitness: 2.02428986310836
Generation 445 Best Solution: [183.779698355435, 176.69404374765287,
174.15550938552218, 75.78236164248489] Fitness: 1.3670697471807978
Generation 446 Best Solution: [183.3980854945825, 177.3640521599673,
175.07050047422024, 75.78236164248489] Fitness: 2.6446875799280365
Generation 447 Best Solution: [183.47307745831793, 177.3640521599673,
174.93944028994207, 75.78236164248489] Fitness: 2.493348000905587

Generation 448 Best Solution: [183.86228725049773, 175.89401528864613,
175.07050047422024, 75.78236164248489] Fitness: 2.023117906034842
Generation 449 Best Solution: [182.96440727791446, 175.89401528864613,
174.25772030543453, 75.78236164248489] Fitness: 1.1402910121028205
Generation 450 Best Solution: [183.47307745831793, 176.06956889076892,
175.07050047422024, 75.78236164248489] Fitness: 1.7000422940562847
Generation 451 Best Solution: [183.86228725049773, 175.89401528864613,
174.93944028994207, 75.78236164248489] Fitness: 1.8495883065254597
Generation 452 Best Solution: [183.86228725049773, 175.89401528864613,
175.07050047422024, 75.02197935251692] Fitness: 1.1682478543142165
Generation 453 Best Solution: [183.47307745831793, 177.3640521599673,
175.07050047422024, 75.02197935251692] Fitness: 1.8120075486943439
Generation 454 Best Solution: [183.779698355435, 176.06956889076892,
175.07050047422024, 75.02197935251692] Fitness: 1.026318137800182
Generation 455 Best Solution: [183.55802547048248, 176.06956889076892,
175.07050047422024, 75.02197935251692] Fitness: 0.8808069673079819
Generation 456 Best Solution: [183.779698355435, 176.06956889076892,
174.12997912408903, 75.85530211471216] Fitness: 1.3531576852376168
Generation 457 Best Solution: [183.39583672795536, 177.3640521599673,
174.52312995838938, 75.02197935251692] Fitness: 1.2406290728080926
Generation 458 Best Solution: [182.8533635117031, 177.3640521599673,
174.12997912408903, 75.72741127237384] Fitness: 1.9956200665203745
Generation 459 Best Solution: [183.86228725049773, 176.2707060235651,
175.07050047422024, 75.78236164248489] Fitness: 1.9351137438237358
Generation 460 Best Solution: [183.86228725049773, 177.3640521599673,
174.58301793509077, 74.43734592687063] Fitness: 1.5059417365114545
Generation 461 Best Solution: [183.47307745831793, 177.3640521599673,
175.07050047422024, 74.46905978849381] Fitness: 1.7514963986159928
Generation 462 Best Solution: [183.86228725049773, 177.3640521599673,
173.75034855213397, 75.02197935251692] Fitness: 1.531856713595819
Generation 463 Best Solution: [183.47307745831793, 177.3640521599673,
175.07050047422024, 74.46905978849381] Fitness: 1.7514963986159928
Generation 464 Best Solution: [183.18543231822258, 177.3361706485376,
175.07050047422024, 74.46905978849381] Fitness: 1.66553619648486
Generation 465 Best Solution: [183.47307745831793, 177.3640521599673,
175.07050047422024, 74.46905978849381] Fitness: 1.7514963986159928
Generation 466 Best Solution: [183.86228725049773, 177.3640521599673,
175.07050047422024, 74.46905978849381] Fitness: 2.006273129459094
Generation 467 Best Solution: [183.86228725049773, 177.3361706485376,
175.07050047422024, 74.46905978849381] Fitness: 1.9581600484629729
Generation 468 Best Solution: [183.18543231822258, 177.3361706485376,
174.64477361711084, 75.78236164248489] Fitness: 2.114172541197968
Generation 469 Best Solution: [183.86228725049773, 177.3361706485376,
173.87990862719263, 75.78236164248489] Fitness: 2.2719159930714117
Generation 470 Best Solution: [183.86228725049773, 177.3361706485376,
173.87990862719263, 75.89434370286533] Fitness: 2.4733004119169744
Generation 471 Best Solution: [183.86228725049773, 176.2909045847467,
175.07050047422024, 75.78236164248489] Fitness: 1.9365895061040013

Generation 472 Best Solution: [183.86228725049773, 176.33174229204707,
175.07050047422024, 75.89434370286533] Fitness: 2.142883725378516
Generation 473 Best Solution: [183.86228725049773, 176.9952029769923,
175.07050047422024, 75.53372138833744] Fitness: 2.0044821480595365
Generation 474 Best Solution: [183.70672933562943, 177.3361706485376,
173.87990862719263, 75.78236164248489] Fitness: 2.142001927997215
Generation 475 Best Solution: [183.86228725049773, 177.3361706485376,
175.07050047422024, 74.96673089090768] Fitness: 1.9913782723235185
Generation 476 Best Solution: [183.86228725049773, 176.4232845551885,
175.07050047422024, 75.78236164248489] Fitness: 1.9618693591802834
Generation 477 Best Solution: [183.86228725049773, 177.29422648037644,
174.22276109135356, 75.78236164248489] Fitness: 2.1504552188077883
Generation 478 Best Solution: [183.86228725049773, 176.9952029769923,
175.07050047422024, 75.29444365002381] Fitness: 1.7310659407708848
Generation 479 Best Solution: [183.86228725049773, 177.3640521599673,
173.9478801661091, 74.71763408238286] Fitness: 1.3487762066670574
Generation 480 Best Solution: [183.86228725049773, 176.9952029769923,
175.07050047422024, 75.49437220558184] Fitness: 1.953439834992909
Generation 481 Best Solution: [183.86228725049773, 176.9952029769923,
175.07050047422024, 75.49437220558184] Fitness: 1.953439834992909
Generation 482 Best Solution: [183.86228725049773, 176.9952029769923,
175.07050047422024, 75.29444365002381] Fitness: 1.7310659407708848
Generation 483 Best Solution: [183.86228725049773, 177.3640521599673,
173.95823253858737, 75.49437220558184] Fitness: 1.8633885546101658
Generation 484 Best Solution: [183.86228725049773, 176.9952029769923,
175.07050047422024, 75.29444365002381] Fitness: 1.7310659407708848
Generation 485 Best Solution: [183.86228725049773, 176.02282127323713,
173.95823253858737, 75.78236164248489] Fitness: 1.3389885013490082
Generation 486 Best Solution: [183.86228725049773, 177.3640521599673,
175.07050047422024, 74.78068057915651] Fitness: 1.9822724679833013
Generation 487 Best Solution: [183.86228725049773, 177.3640521599673,
173.95823253858737, 75.49437220558184] Fitness: 1.8633885546101658
Generation 488 Best Solution: [183.02602697501635, 177.3640521599673,
173.95823253858737, 75.49437220558184] Fitness: 1.6048571002680712
Generation 489 Best Solution: [183.86228725049773, 176.59717782670572,
175.07050047422024, 75.49437220558184] Fitness: 1.6073303052331118
Generation 490 Best Solution: [183.86228725049773, 176.59717782670572,
175.07050047422024, 75.49437220558184] Fitness: 1.6073303052331118
Generation 491 Best Solution: [183.86228725049773, 176.4057952935009,
174.65142541482993, 75.78236164248489] Fitness: 1.4953706241814055
Generation 492 Best Solution: [182.98447230836368, 177.3640521599673,
175.07050047422024, 74.41789679390544] Fitness: 1.781365165811827
Generation 493 Best Solution: [183.41433099338232, 176.12400744356574,
175.07050047422024, 75.78236164248489] Fitness: 1.6704174736859487
Generation 494 Best Solution: [182.96833098876508, 177.3640521599673,
174.5395190032529, 74.81155453566545] Fitness: 1.2125985640450307
Generation 495 Best Solution: [183.41433099338232, 176.61635429466642,
175.07050047422024, 75.78236164248489] Fitness: 1.774395523715908

```
Generation 496 Best Solution: [183.41433099338232, 177.3640521599673,
175.07050047422024, 74.94861793757883] Fitness: 1.7586730924772505
Generation 497 Best Solution: [183.41433099338232, 176.61635429466642,
175.07050047422024, 74.94861793757883] Fitness: 0.8843114795230884
Generation 498 Best Solution: [183.41433099338232, 176.61635429466642,
175.07050047422024, 74.94861793757883] Fitness: 0.8843114795230884
Generation 499 Best Solution: [183.41433099338232, 176.61635429466642,
175.07050047422024, 74.77593188011615] Fitness: 0.834253348006108
Generation 500 Best Solution: [183.41433099338232, 176.61635429466642,
175.07050047422024, 74.94861793757883] Fitness: 0.8843114795230884
Optimal Solution for Simulation 5: [183.41433099338232, 176.61635429466642,
175.07050047422024, 74.94861793757883]
--------------------------------------------------
Simulation 6/32 with conditions: {'drought': True, 'flood': True,
'policy_change': False, 'social_event': True, 'economic_crisis': False}
Generation 1 Best Solution: [183.68435965523875, 175.9011771201634,
174.09206818693647, 74.58533660649316] Fitness: 0.29266229059053517
Generation 2 Best Solution: [183.1182302736303, 177.1348963037849,
174.07943160115764, 74.99883696348027] Fitness: 0.7720708750718809
Generation 3 Best Solution: [183.78281793695095, 176.05253080122455,
174.09206818693647, 75.63198318844063] Fitness: 0.9567551046245266
Generation 4 Best Solution: [183.04469891096153, 177.1348963037849,
174.9716668022944, 74.44210013258255] Fitness: 1.307952576137755
Generation 5 Best Solution: [183.87353699718216, 177.1348963037849,
174.87759774821356, 74.40265347798669] Fitness: 1.5199379315279224
Generation 6 Best Solution: [183.04469891096153, 177.1348963037849,
173.78912161611277, 75.8079213625765] Fitness: 1.7201246484036425
Generation 7 Best Solution: [183.04469891096153, 177.1348963037849,
173.78912161611277, 74.87680863660479] Fitness: 0.826057018733728
Generation 8 Best Solution: [183.8400547758212, 177.08839227046235,
173.78912161611277, 74.87680863660479] Fitness: 1.0385271624833181
Generation 9 Best Solution: [183.19644865682633, 177.36033962226028,
174.8996622195264, 75.47539552456517] Fitness: 1.9508430257567964
Generation 10 Best Solution: [183.04469891096153, 177.1348963037849,
174.30618628784322, 75.90356842463218] Fitness: 1.8260627540333596
Generation 11 Best Solution: [183.27542239356754, 177.2469243152209,
174.07943160115764, 75.86969937536053] Fitness: 1.8911107826809488
Generation 12 Best Solution: [183.04469891096153, 177.1348963037849,
174.4903153652255, 75.86969937536053] Fitness: 1.8415139825980722
Generation 13 Best Solution: [183.87353699718216, 177.1348963037849,
174.4903153652255, 75.86969937536053] Fitness: 2.1498502582318686
Generation 14 Best Solution: [183.87353699718216, 177.36033962226028,
174.9716668022944, 74.67270466803377] Fitness: 1.9197109175257245
Generation 15 Best Solution: [183.04469891096153, 177.36033962226028,
174.87759774821356, 74.45128615599157] Fitness: 1.5450384908925747
Generation 16 Best Solution: [183.87353699718216, 177.36033962226028,
174.8996622195264, 74.45128615599157] Fitness: 1.8786667913948731
Generation 17 Best Solution: [183.04469891096153, 177.43169872466657,
```

174.32403948951597, 75.86969937536053] Fitness: 2.2522951168390204
Generation 18 Best Solution: [183.87353699718216, 177.36033962226028,
174.9716668022944, 75.19359863851835] Fitness: 2.1001455398907978
Generation 19 Best Solution: [183.87353699718216, 177.36033962226028,
174.9716668022944, 74.90350367141889] Fitness: 1.9494431193648079
Generation 20 Best Solution: [183.87353699718216, 177.36033962226028,
174.8996622195264, 74.90350367141889] Fitness: 1.861825775736422
Generation 21 Best Solution: [183.27542239356754, 176.41003089418146,
174.87759774821356, 75.38409428371003] Fitness: 0.855317529491652
Generation 22 Best Solution: [183.04469891096153, 175.6260620788295,
174.9716668022944, 75.86969937536053] Fitness: 1.8760168036354756
Generation 23 Best Solution: [183.27542239356754, 177.43169872466657,
174.87759774821356, 75.57665831607801] Fitness: 2.1814052794327554
Generation 24 Best Solution: [183.04469891096153, 177.36033962226028,
174.87759774821356, 74.97204937471453] Fitness: 1.5524168280576713
Generation 25 Best Solution: [183.87353699718216, 177.0568607737732,
174.87759774821356, 74.97204937471453] Fitness: 1.4026314598361378
Generation 26 Best Solution: [183.04469891096153, 175.6260620788295,
174.9716668022944, 75.86969937536053] Fitness: 1.8760168036354756
Generation 27 Best Solution: [183.27542239356754, 176.5313029448797,
174.87759774821356, 75.86969937536053] Fitness: 1.5776898340508558
Generation 28 Best Solution: [183.27542239356754, 176.0573571668609,
173.86753120293892, 75.86969937536053] Fitness: 1.1362820417428128
Generation 29 Best Solution: [183.87353699718216, 176.86536315718934,
174.9716668022944, 75.48090626492281] Fitness: 1.6977090723077795
Generation 30 Best Solution: [183.87353699718216, 176.87335997544218,
174.20410960333834, 75.86969937536053] Fitness: 1.7396227194169391
Generation 31 Best Solution: [183.87353699718216, 176.67506529631515,
173.99297129196333, 75.86969937536053] Fitness: 1.577692725318626
Generation 32 Best Solution: [183.27542239356754, 176.7263460517709,
174.8964944979445, 75.86969937536053] Fitness: 1.7241146236561744
Generation 33 Best Solution: [183.87353699718216, 175.6260620788295,
174.9716668022944, 74.57702493423001] Fitness: 1.1738566726347799
Generation 34 Best Solution: [183.87353699718216, 176.0573571668609,
174.9716668022944, 74.90193472623588] Fitness: 0.9587837452117737
Generation 35 Best Solution: [183.87353699718216, 177.36033962226028,
174.8964944979445, 74.8244575292532] Fitness: 1.8389702268369348
Generation 36 Best Solution: [183.87353699718216, 175.8092198422367,
174.35266513584153, 75.86969937536053] Fitness: 1.5519450694049337
Generation 37 Best Solution: [183.27542239356754, 175.8092198422367,
174.35266513584153, 75.86969937536053] Fitness: 1.2259409842905638
Generation 38 Best Solution: [183.87353699718216, 175.8092198422367,
173.97831453347885, 75.86969937536053] Fitness: 1.5288970195777192
Generation 39 Best Solution: [183.87353699718216, 175.8092198422367,
173.97831453347885, 75.86969937536053] Fitness: 1.5288970195777192
Generation 40 Best Solution: [183.27542239356754, 175.58982673745442,
174.1425537229238, 75.86969937536053] Fitness: 1.3525564500968066
Generation 41 Best Solution: [183.27542239356754, 175.58982673745442,

174.1425537229238, 75.86969937536053] Fitness: 1.3525564500968066
Generation 42 Best Solution: [183.87353699718216, 177.36033962226028,
174.8964944979445, 74.59469770006068] Fitness: 1.8364352662727677
Generation 43 Best Solution: [183.87353699718216, 177.05873873937722,
175.08026076953033, 74.59469770006068] Fitness: 1.6188664118325022
Generation 44 Best Solution: [182.9446427904963, 177.36033962226028,
175.127559695475, 74.59469770006068] Fitness: 1.868217441093933
Generation 45 Best Solution: [183.87353699718216, 176.40429062530103,
174.9716668022944, 75.86969937536053] Fitness: 1.966006425785647
Generation 46 Best Solution: [183.27542239356754, 176.98019649275483,
174.8964944979445, 75.33897691292873] Fitness: 1.2538584811162719
Generation 47 Best Solution: [183.87353699718216, 176.98019649275483,
174.8964944979445, 75.33897691292873] Fitness: 1.5798625662306418
Generation 48 Best Solution: [183.87353699718216, 176.98019649275483,
174.8964944979445, 75.33897691292873] Fitness: 1.5798625662306418
Generation 49 Best Solution: [183.27542239356754, 176.98019649275483,
174.6991477082855, 75.33897691292873] Fitness: 1.054525398386357
Generation 50 Best Solution: [183.87353699718216, 176.98019649275483,
174.9716668022944, 74.32231642131025] Fitness: 1.4753098062015653
Generation 51 Best Solution: [183.87353699718216, 176.98019649275483,
174.8964944979445, 75.37845128674894] Fitness: 1.6187344749110988
Generation 52 Best Solution: [183.27542239356754, 176.98019649275483,
174.8964944979445, 75.37845128674894] Fitness: 1.2927303897967288
Generation 53 Best Solution: [183.87353699718216, 176.98019649275483,
174.9716668022944, 74.32231642131025] Fitness: 1.4753098062015653
Generation 54 Best Solution: [183.29137218299354, 177.36033962226028,
174.37704319412214, 74.32231642131025] Fitness: 1.2125444085621355
Generation 55 Best Solution: [183.87353699718216, 176.24300790942166,
174.8964944979445, 75.86969937536053] Fitness: 1.845336971455224
Generation 56 Best Solution: [183.29137218299354, 175.74168223412633,
174.9716668022944, 75.86969937536053] Fitness: 1.770180845462083
Generation 57 Best Solution: [183.87353699718216, 176.3642158186567,
174.9716668022944, 75.86969937536053] Fitness: 1.9550641401078654
Generation 58 Best Solution: [183.29137218299354, 177.43169872466657,
174.92733882828398, 75.37845128674894] Fitness: 2.010669771960883
Generation 59 Best Solution: [183.87353699718216, 175.9110179621438,
174.55689647093675, 75.86969937536053] Fitness: 1.600904410641737
Generation 60 Best Solution: [183.29137218299354, 177.43169872466657,
174.8964944979445, 74.59543409719637] Fitness: 1.6398246590440515
Generation 61 Best Solution: [183.35250506152113, 177.43169872466657,
174.8964944979445, 74.59543409719637] Fitness: 1.6499874662086773
Generation 62 Best Solution: [183.35250506152113, 175.92200937839354,
174.9716668022944, 75.86969937536053] Fitness: 1.680160270535103
Generation 63 Best Solution: [183.36081543699467, 177.43169872466657,
174.9716668022944, 75.16481745436916] Fitness: 1.8950381309579183
Generation 64 Best Solution: [183.87353699718216, 177.0070695469906,
174.40649851683222, 75.86969937536053] Fitness: 1.942543690365906
Generation 65 Best Solution: [183.87353699718216, 177.43169872466657,

174.8964944979445, 74.49566524071649] Fitness: 1.9875458824236931
Generation 66 Best Solution: [183.87353699718216, 177.43169872466657,
174.8964944979445, 74.49566524071649] Fitness: 1.9875458824236931
Generation 67 Best Solution: [183.87353699718216, 177.0355287857228,
174.8964944979445, 74.49566524071649] Fitness: 1.3746364014564398
Generation 68 Best Solution: [183.35250506152113, 176.66105276080816,
174.9716668022944, 75.86969937536053] Fitness: 1.7791649395044793
Generation 69 Best Solution: [183.3744849635639, 177.43169872466657,
174.9716668022944, 74.83547644719411] Fitness: 1.7510696085665145
Generation 70 Best Solution: [183.87353699718216, 175.80384722495003,
174.8964944979445, 75.69874061317469] Fitness: 1.6856508463886262
Generation 71 Best Solution: [183.87353699718216, 175.80384722495003,
174.8964944979445, 75.86969937536053] Fitness: 1.9631163460836902
Generation 72 Best Solution: [183.87353699718216, 177.43169872466657,
174.2115874055823, 75.12787476633689] Fitness: 1.6500810666254162
Generation 73 Best Solution: [183.3744849635639, 177.43169872466657,
174.2115874055823, 75.12787476633689] Fitness: 1.340993287471604
Generation 74 Best Solution: [182.935672895228, 177.43169872466657,
174.2115874055823, 75.12787476633689] Fitness: 1.3778746837993003
Generation 75 Best Solution: [183.87353699718216, 176.02305088159412,
174.8964944979445, 75.53502583870518] Fitness: 1.3660853646172126
Generation 76 Best Solution: [183.87353699718216, 176.02305088159412,
174.8964944979445, 75.53502583870518] Fitness: 1.3660853646172126
Generation 77 Best Solution: [183.87353699718216, 176.02305088159412,
174.8964944979445, 74.39901653606204] Fitness: 0.9148642329867871
Generation 78 Best Solution: [183.4748614843048, 176.24391015944775,
174.9716668022944, 75.86969937536053] Fitness: 1.6597532628432177
Generation 79 Best Solution: [183.4748614843048, 176.24391015944775,
174.8964944979445, 75.86969937536053] Fitness: 1.5684599339446608
Generation 80 Best Solution: [183.87353699718216, 176.71899454157148,
174.06085874746103, 75.86969937536053] Fitness: 1.6003659815927491
Generation 81 Best Solution: [183.87353699718216, 176.71899454157148,
174.06085874746103, 75.86969937536053] Fitness: 1.6003659815927491
Generation 82 Best Solution: [183.87353699718216, 176.24391015944775,
173.950061195035, 75.86969937536053] Fitness: 1.4211539547514311
Generation 83 Best Solution: [183.87353699718216, 176.24391015944775,
173.950061195035, 75.86969937536053] Fitness: 1.4211539547514311
Generation 84 Best Solution: [183.87353699718216, 176.24391015944775,
173.950061195035, 75.86969937536053] Fitness: 1.4211539547514311
Generation 85 Best Solution: [183.87353699718216, 176.24391015944775,
174.8964944979445, 75.86969937536053] Fitness: 1.8453927803275656
Generation 86 Best Solution: [183.2114713710877, 175.803856758423,
174.55904660450827, 75.86969937536053] Fitness: 1.3286180200606568
Generation 87 Best Solution: [183.2114713710877, 175.803856758423,
174.55904660450827, 75.86969937536053] Fitness: 1.3286180200606568
Generation 88 Best Solution: [183.47536809334235, 176.5219342698447,
173.950061195035, 75.86969937536053] Fitness: 1.219780160305773
Generation 89 Best Solution: [183.87353699718216, 176.5333449868357,

174.8964944979445, 75.86969937536053] Fitness: 1.9263210811789229
Generation 90 Best Solution: [183.30189610145655, 176.25893821859995,
175.0981658445614, 75.86969937536053] Fitness: 1.7876313503322905
Generation 91 Best Solution: [183.87353699718216, 176.5333449868357,
174.8964944979445, 75.86969937536053] Fitness: 1.9263210811789229
Generation 92 Best Solution: [183.87353699718216, 176.5333449868357,
174.80717223629605, 75.86969937536053] Fitness: 1.8288630456717727
Generation 93 Best Solution: [183.87353699718216, 176.5333449868357,
174.80717223629605, 75.86969937536053] Fitness: 1.8288630456717727
Generation 94 Best Solution: [183.87353699718216, 176.5333449868357,
174.8964944979445, 75.86969937536053] Fitness: 1.9263210811789229
Generation 95 Best Solution: [183.87353699718216, 176.25783489785636,
174.55904660450827, 75.86969937536053] Fitness: 1.541022258830391
Generation 96 Best Solution: [183.87353699718216, 176.5333449868357,
174.8964944979445, 75.86969937536053] Fitness: 1.9263210811789229
Generation 97 Best Solution: [183.87353699718216, 177.43169872466657,
174.8964944979445, 75.13113295471067] Fitness: 2.093519854799957
Generation 98 Best Solution: [183.87353699718216, 175.8555819534197,
174.8964944979445, 75.86969937536053] Fitness: 1.9342089742115978
Generation 99 Best Solution: [183.78071205488132, 176.08749763958411,
174.8964944979445, 75.86969937536053] Fitness: 1.7681885500962689
Generation 100 Best Solution: [183.87353699718216, 177.43169872466657,
174.98642694994942, 74.9911705942466] Fitness: 2.128379329754172
Generation 101 Best Solution: [183.87353699718216, 177.43169872466657,
174.61258418264006, 74.9911705942466] Fitness: 1.7498292529109631
Generation 102 Best Solution: [183.87353699718216, 177.43169872466657,
174.8964944979445, 75.06576704188251] Fitness: 2.0546701693935003
Generation 103 Best Solution: [183.87353699718216, 175.62937965217756,
175.0981658445614, 74.92086642909346] Fitness: 1.3678566743125515
Generation 104 Best Solution: [183.87353699718216, 177.43169872466657,
174.8964944979445, 75.06576704188251] Fitness: 2.0546701693935003
Generation 105 Best Solution: [183.87353699718216, 177.43169872466657,
175.0981658445614, 74.92086642909346] Fitness: 2.2554545693181147
Generation 106 Best Solution: [183.87353699718216, 175.62937965217756,
174.31709083551846, 75.86969937536053] Fitness: 1.6709893349498066
Generation 107 Best Solution: [183.87353699718216, 177.43169872466657,
174.97929414271616, 74.92086642909346] Fitness: 2.092430550510239
Generation 108 Best Solution: [183.32281868790244, 177.43169872466657,
175.0981658445614, 74.92086642909346] Fitness: 1.9357074654497957
Generation 109 Best Solution: [183.39874048512542, 177.43169872466657,
175.0981658445614, 74.92086642909346] Fitness: 1.9527521149101028
Generation 110 Best Solution: [183.87353699718216, 177.43169872466657,
175.0981658445614, 74.92086642909346] Fitness: 2.2554545693181147
Generation 111 Best Solution: [183.87353699718216, 175.87373875064137,
175.0981658445614, 75.86969937536053] Fitness: 2.1890694705235583
Generation 112 Best Solution: [183.87353699718216, 175.87373875064137,
175.0981658445614, 75.86969937536053] Fitness: 2.1890694705235583
Generation 113 Best Solution: [182.81793363081397, 177.43169872466657,

175.0981658445614, 74.50251976360441] Fitness: 2.036346763759686
Generation 114 Best Solution: [183.6079986976382, 177.43169872466657,
175.0981658445614, 74.64852442406462] Fitness: 2.0107969689162966
Generation 115 Best Solution: [183.87353699718216, 177.43169872466657,
175.0981658445614, 74.50251976360441] Fitness: 2.2495114064028447
Generation 116 Best Solution: [183.87353699718216, 176.57744829111653,
174.91405638143266, 75.86969937536053] Fitness: 1.9702546137720067
Generation 117 Best Solution: [183.87353699718216, 177.43169872466657,
175.0981658445614, 74.3651226521895] Fitness: 2.3048276925065307
Generation 118 Best Solution: [183.87353699718216, 177.43169872466657,
174.91405638143266, 74.7505899850533] Fitness: 1.9778704172717154
Generation 119 Best Solution: [183.87353699718216, 177.43169872466657,
174.91405638143266, 74.3651226521895] Fitness: 2.0613429269979364
Generation 120 Best Solution: [183.87353699718216, 176.25633991547525,
175.0981658445614, 75.86969937536053] Fitness: 2.11033997073947
Generation 121 Best Solution: [183.87353699718216, 177.43169872466657,
175.0981658445614, 74.3651226521895] Fitness: 2.3048276925065307
Generation 122 Best Solution: [183.3248752595325, 177.43169872466657,
175.0981658445614, 74.3651226521895] Fitness: 1.9854283627033
Generation 123 Best Solution: [183.87353699718216, 177.43169872466657,
174.91405638143266, 74.3651226521895] Fitness: 2.0613429269979364
Generation 124 Best Solution: [183.87353699718216, 176.6638944905555,
174.91405638143266, 74.3651226521895] Fitness: 1.0874895369618311
Generation 125 Best Solution: [183.27144445820397, 177.43169872466657,
175.0981658445614, 74.91857050679032] Fitness: 1.9283295812718246
Generation 126 Best Solution: [183.27144445820397, 177.43169872466657,
175.0981658445614, 74.76166165888873] Fitness: 1.8958744310685791
Generation 127 Best Solution: [183.27144445820397, 177.43169872466657,
174.91405638143266, 74.82003845947683] Fitness: 1.6601503799071877
Generation 128 Best Solution: [182.93930616075974, 177.43169872466657,
175.0981658445614, 74.76166165888873] Fitness: 1.9485527947666386
Generation 129 Best Solution: [183.32841387771478, 177.08101588314886,
174.91405638143266, 75.86969937536053] Fitness: 2.125081298939133
Generation 130 Best Solution: [183.20584347397582, 177.43169872466657,
175.0981658445614, 74.76166165888873] Fitness: 1.8931651574907495
Generation 131 Best Solution: [183.87353699718216, 177.08101588314886,
175.0981658445614, 74.76166165888873] Fitness: 1.6677497587093126
Generation 132 Best Solution: [183.32841387771478, 176.88447622052956,
175.0981658445614, 75.22261107546409] Fitness: 1.319311470847701
Generation 133 Best Solution: [183.27144445820397, 176.88447622052956,
175.0981658445614, 75.86969937536053] Fitness: 2.130870359527769
Generation 134 Best Solution: [183.87353699718216, 175.97467511837652,
175.0981658445614, 75.86969937536053] Fitness: 2.146976690548623
Generation 135 Best Solution: [183.87353699718216, 176.51868727227398,
174.82397302067582, 75.86969937536053] Fitness: 1.8391611710643134
Generation 136 Best Solution: [183.3466961574844, 177.43169872466657,
175.0981658445614, 75.03644142845476] Fitness: 1.988058898477653
Generation 137 Best Solution: [183.87353699718216, 176.51868727227398,

175.0981658445614, 75.86969937536053] Fitness: 2.1832560231253932
Generation 138 Best Solution: [183.27144445820397, 175.97107260359286,
175.0981658445614, 75.86969937536053] Fitness: 1.8218400805173824
Generation 139 Best Solution: [183.27144445820397, 175.97107260359286,
174.82397302067582, 75.86969937536053] Fitness: 1.4777452284563029
Generation 140 Best Solution: [183.87353699718216, 177.43169872466657,
174.82397302067582, 75.1253622505348] Fitness: 2.009791363028551
Generation 141 Best Solution: [183.87353699718216, 176.97995574426244,
175.0981658445614, 74.97637084638275] Fitness: 1.595911424447897
Generation 142 Best Solution: [183.87353699718216, 175.8930110163611,
175.0981658445614, 75.27612726657792] Fitness: 1.404625177197508
Generation 143 Best Solution: [183.3242414837363, 176.5560477722739,
174.82397302067582, 75.86969937536053] Fitness: 1.53843578504312
Generation 144 Best Solution: [183.77024366141265, 177.43169872466657,
174.82397302067582, 75.27612726657792] Fitness: 2.027896866013223
Generation 145 Best Solution: [183.87353699718216, 176.11481863432704,
174.82397302067582, 75.27612726657792] Fitness: 0.9945511881465744
Generation 146 Best Solution: [182.9068105977048, 176.11481863432704,
174.82397302067582, 75.27612726657792] Fitness: 0.7348942490883147
Generation 147 Best Solution: [183.6148641909334, 177.43169872466657,
175.0981658445614, 74.69507029746293] Fitness: 2.0127953361368904
Generation 148 Best Solution: [183.87353699718216, 177.43169872466657,
173.74128895517526, 75.27612726657792] Fitness: 1.8656319253055595
Generation 149 Best Solution: [183.87353699718216, 175.7572843183168,
175.0981658445614, 74.72669570039669] Fitness: 1.2348177669715314
Generation 150 Best Solution: [183.3099695637429, 176.5308145887397,
175.0981658445614, 74.69507029746293] Fitness: 0.8451330242330433
Generation 151 Best Solution: [183.87353699718216, 175.7572843183168,
175.0981658445614, 74.72669570039669] Fitness: 1.2348177669715314
Generation 152 Best Solution: [183.3099695637429, 176.41812493033726,
175.0981658445614, 75.19427738572075] Fitness: 0.9806641516847849
Generation 153 Best Solution: [183.87353699718216, 176.48997346382762,
175.0981658445614, 74.8418799842083] Fitness: 1.1626200505185622
Generation 154 Best Solution: [183.3948893700534, 176.48997346382762,
175.0981658445614, 75.86969937536053] Fitness: 1.8664688054234153
Generation 155 Best Solution: [183.3948893700534, 176.48997346382762,
175.0981658445614, 74.28454513739725] Fitness: 0.9750572384893863
Generation 156 Best Solution: [183.3948893700534, 175.7041512830636,
175.0981658445614, 74.69507029746293] Fitness: 0.9682104031609751
Generation 157 Best Solution: [183.87353699718216, 176.0013198876004,
175.0981658445614, 74.99280752717159] Fitness: 1.179492425504018
Generation 158 Best Solution: [183.87353699718216, 175.98000322775138,
175.0981658445614, 74.28454513739725] Fitness: 1.2537678213908936
Generation 159 Best Solution: [183.87353699718216, 175.98000322775138,
175.0981658445614, 74.55665132816304] Fitness: 1.1388203748978687
Generation 160 Best Solution: [183.87353699718216, 177.04259762207087,
174.2145295359898, 74.55665132816304] Fitness: 0.9264439922779746
Generation 161 Best Solution: [183.86590046419198, 176.28574631592548,

175.0981658445614, 75.86969937536053] Fitness: 2.1058314146472012
Generation 162 Best Solution: [183.87353699718216, 176.1678022737477,
175.0981658445614, 75.86969937536053] Fitness: 2.10903197683694
Generation 163 Best Solution: [183.61188111328187, 176.2056252197579,
175.0981658445614, 75.86969937536053] Fitness: 1.8995085175685706
Generation 164 Best Solution: [183.87353699718216, 177.399179306859,
173.8523612822388, 75.86969937536053] Fitness: 2.5270950356263824
Generation 165 Best Solution: [183.87353699718216, 175.86751196634043,
175.0981658445614, 75.86969937536053] Fitness: 2.1921666437415404
Generation 166 Best Solution: [183.61188111328187, 176.93011667955426,
174.9554435152042, 75.86969937536053] Fitness: 2.1036921559537958
Generation 167 Best Solution: [183.87353699718216, 176.38441306826093,
175.0981658445614, 75.86969937536053] Fitness: 2.133038555482422
Generation 168 Best Solution: [183.61188111328187, 177.43169872466657,
174.25982531627315, 75.86969937536053] Fitness: 2.335855146380016
Generation 169 Best Solution: [183.87353699718216, 176.57097205212048,
174.9554435152042, 75.86969937536053] Fitness: 2.016945322903066
Generation 170 Best Solution: [183.61188111328187, 176.57097205212048,
174.9554435152042, 75.86969937536053] Fitness: 1.8083017279802673
Generation 171 Best Solution: [183.61188111328187, 176.57097205212048,
174.9554435152042, 75.86969937536053] Fitness: 1.8083017279802673
Generation 172 Best Solution: [183.61188111328187, 176.57097205212048,
174.9554435152042, 75.86969937536053] Fitness: 1.8083017279802673
Generation 173 Best Solution: [183.61188111328187, 176.57097205212048,
174.9554435152042, 75.86969937536053] Fitness: 1.8083017279802673
Generation 174 Best Solution: [183.61188111328187, 176.57097205212048,
174.9554435152042, 75.86969937536053] Fitness: 1.8083017279802673
Generation 175 Best Solution: [183.61188111328187, 176.16740933714718,
174.9554435152042, 75.86969937536053] Fitness: 1.7072283529182086
Generation 176 Best Solution: [183.61188111328187, 176.57097205212048,
174.9554435152042, 75.83438162236746] Fitness: 1.7473882311582465
Generation 177 Best Solution: [183.54544251955735, 176.5817757391307,
174.9554435152042, 75.86969937536053] Fitness: 1.7777357263662446
Generation 178 Best Solution: [183.61188111328187, 176.57097205212048,
174.90843565823795, 75.86969937536053] Fitness: 1.7513638037566497
Generation 179 Best Solution: [183.61188111328187, 175.88284493274526,
175.0981658445614, 75.78046879006936] Fitness: 1.8257107905303072
Generation 180 Best Solution: [183.61188111328187, 175.88284493274526,
175.0981658445614, 75.78046879006936] Fitness: 1.8257107905303072
Generation 181 Best Solution: [183.61188111328187, 175.88284493274526,
174.9554435152042, 75.33595186020551] Fitness: 1.0617785531868955
Generation 182 Best Solution: [183.61188111328187, 175.88284493274526,
175.0981658445614, 75.86969937536053] Fitness: 1.9760012220496406
Generation 183 Best Solution: [183.54544251955735, 177.399179306859,
175.0981658445614, 74.98510429725793] Fitness: 1.9751647251037352
Generation 184 Best Solution: [183.61188111328187, 175.88284493274526,
174.83067523446695, 75.86969937536053] Fitness: 1.6389726448471982
Generation 185 Best Solution: [183.54544251955735, 175.88284493274526,

174.83067523446695, 75.86969937536053] Fitness: 1.6023433168724732
Generation 186 Best Solution: [183.61188111328187, 176.8902991999669,
174.90843565823795, 74.71289446934367] Fitness: 0.9823021667648649
Generation 187 Best Solution: [183.54544251955735, 175.88284493274526,
174.83067523446695, 75.86969937536053] Fitness: 1.6023433168724732
Generation 188 Best Solution: [183.61188111328187, 176.7591549328162,
175.0981658445614, 75.86969937536053] Fitness: 2.1321303653715535
Generation 189 Best Solution: [183.61188111328187, 176.7591549328162,
174.67028831195697, 75.86969937536053] Fitness: 1.644489487363021
Generation 190 Best Solution: [183.61188111328187, 176.7591549328162,
174.67028831195697, 75.86969937536053] Fitness: 1.644489487363021
Generation 191 Best Solution: [183.61188111328187, 176.74054186681636,
175.0981658445614, 75.86969937536053] Fitness: 2.1168408620913652
Generation 192 Best Solution: [183.61188111328187, 176.83259546623364,
174.9554435152042, 75.86969937536053] Fitness: 2.0043472202668804
Generation 193 Best Solution: [183.61188111328187, 177.399179306859,
175.0981658445614, 74.6260669286333] Fitness: 1.9561270733522858
Generation 194 Best Solution: [183.08102106063254, 176.00006393366124,
174.9554435152042, 75.86969937536053] Fitness: 1.6292005003633752
Generation 195 Best Solution: [183.54544251955735, 176.1074070311053,
175.0981658445614, 75.86969937536053] Fitness: 1.8696129446495775
Generation 196 Best Solution: [183.54544251955735, 175.7945619437866,
174.67028831195697, 75.86969937536053] Fitness: 1.499869803251806
Generation 197 Best Solution: [183.54544251955735, 175.7945619437866,
174.67028831195697, 75.86969937536053] Fitness: 1.499869803251806
Generation 198 Best Solution: [183.54544251955735, 175.7945619437866,
174.67028831195697, 75.86969937536053] Fitness: 1.499869803251806
Generation 199 Best Solution: [183.61188111328187, 175.7945619437866,
173.79814946745094, 75.86969937536053] Fitness: 1.3928912944591545
Generation 200 Best Solution: [183.61188111328187, 176.43453979110345,
175.0981658445614, 75.86969937536053] Fitness: 1.9399784218825609
Generation 201 Best Solution: [183.33560551593538, 177.399179306859,
173.79814946745094, 75.86969937536053] Fitness: 2.23394251098956
Generation 202 Best Solution: [183.33560551593538, 177.399179306859,
175.0981658445614, 75.11982271968807] Fitness: 1.9737316800533224
Generation 203 Best Solution: [183.33560551593538, 177.399179306859,
173.79814946745094, 75.86969937536053] Fitness: 2.23394251098956
Generation 204 Best Solution: [183.33560551593538, 177.399179306859,
175.0981658445614, 74.70264772789015] Fitness: 1.84293918423183
Generation 205 Best Solution: [183.61188111328187, 177.399179306859,
175.0981658445614, 74.31874308595758] Fitness: 2.0620695377587817
Generation 206 Best Solution: [183.61188111328187, 177.399179306859,
175.0981658445614, 74.93773511199888] Fitness: 1.9933770927342898
Generation 207 Best Solution: [182.86565727439978, 176.19992056670003,
175.0981658445614, 75.86969937536053] Fitness: 1.8685451244819735
Generation 208 Best Solution: [183.33560551593538, 177.399179306859,
175.0981658445614, 75.15477099331115] Fitness: 1.9965393411010992
Generation 209 Best Solution: [183.33560551593538, 177.399179306859,

175.0981658445614, 75.15477099331115] Fitness: 1.9965393411010992
Generation 210 Best Solution: [183.33560551593538, 177.399179306859,
175.0981658445614, 75.15477099331115] Fitness: 1.9965393411010992
Generation 211 Best Solution: [183.33560551593538, 177.399179306859,
174.63830005917677, 75.47241456029988] Fitness: 1.7747718974753277
Generation 212 Best Solution: [183.33560551593538, 177.399179306859,
174.63830005917677, 75.47241456029988] Fitness: 1.7747718974753277
Generation 213 Best Solution: [183.33560551593538, 176.14894785377138,
175.0981658445614, 75.86969937536053] Fitness: 1.7927901476692263
Generation 214 Best Solution: [183.33560551593538, 176.14894785377138,
175.0981658445614, 75.86969937536053] Fitness: 1.7927901476692263
Generation 215 Best Solution: [183.33560551593538, 176.14894785377138,
175.0981658445614, 75.86969937536053] Fitness: 1.7927901476692263
Generation 216 Best Solution: [183.33560551593538, 176.14894785377138,
175.0981658445614, 75.86969937536053] Fitness: 1.7927901476692263
Generation 217 Best Solution: [183.33560551593538, 176.14894785377138,
175.0981658445614, 75.86969937536053] Fitness: 1.7927901476692263
Generation 218 Best Solution: [183.61188111328187, 176.14894785377138,
175.0981658445614, 75.86969937536053] Fitness: 1.9016284510247774
Generation 219 Best Solution: [183.80751816622674, 177.399179306859,
174.13470204972776, 75.35351842342337] Fitness: 1.7052098836994847
Generation 220 Best Solution: [183.33560551593538, 176.12839916619706,
175.0981658445614, 75.86969937536053] Fitness: 1.79474891161952
Generation 221 Best Solution: [183.33560551593538, 176.12839916619706,
174.13470204972776, 75.86969937536053] Fitness: 1.0837278016848282
Generation 222 Best Solution: [183.33560551593538, 176.12839916619706,
174.13470204972776, 74.77743202188938] Fitness: 0.0657180098343646
Generation 223 Best Solution: [183.52456323809332, 176.12839916619706,
175.0981658445614, 75.86969937536053] Fitness: 1.856813945927363
Generation 224 Best Solution: [183.52456323809332, 176.12839916619706,
175.0981658445614, 75.86969937536053] Fitness: 1.856813945927363
Generation 225 Best Solution: [183.52456323809332, 176.12839916619706,
175.0981658445614, 75.86969937536053] Fitness: 1.856813945927363
Generation 226 Best Solution: [183.52456323809332, 176.12839916619706,
175.0981658445614, 75.86969937536053] Fitness: 1.856813945927363
Generation 227 Best Solution: [183.56329953440186, 177.399179306859,
175.0981658445614, 74.54097525602435] Fitness: 1.9438428580221458
Generation 228 Best Solution: [183.52456323809332, 177.399179306859,
175.13322401264276, 75.17404497396957] Fitness: 2.124093753674677
Generation 229 Best Solution: [183.79436994204238, 176.56845006447102,
175.0981658445614, 74.8046225831911] Fitness: 1.1203848566375545
Generation 230 Best Solution: [183.52456323809332, 177.399179306859,
175.0981658445614, 74.53103795290984] Fitness: 1.926982123293377
Generation 231 Best Solution: [183.52456323809332, 175.98277969524372,
175.13322401264276, 75.86969937536053] Fitness: 1.9409699693382008
Generation 232 Best Solution: [183.52456323809332, 177.399179306859,
175.0981658445614, 74.53103795290984] Fitness: 1.926982123293377
Generation 233 Best Solution: [183.52456323809332, 176.3361921348153,

175.0981658445614, 75.86969937536053] Fitness: 1.866187516336837
Generation 234 Best Solution: [183.52456323809332, 176.52538393561454,
175.0981658445614, 75.86969937536053] Fitness: 1.9310516829786852
Generation 235 Best Solution: [183.44887819354426, 175.9478081273204,
175.0981658445614, 75.86969937536053] Fitness: 1.8699821741745521
Generation 236 Best Solution: [183.79436994204238, 177.399179306859,
174.3357424193464, 75.39848710516345] Fitness: 1.7724317459874221
Generation 237 Best Solution: [183.56329953440186, 177.399179306859,
174.3357424193464, 75.39848710516345] Fitness: 1.6103161122179577
Generation 238 Best Solution: [183.79436994204238, 176.11256852234263,
175.0981658445614, 75.86969937536053] Fitness: 2.0402088736154984
Generation 239 Best Solution: [183.79436994204238, 176.11256852234263,
175.0981658445614, 75.42335042719112] Fitness: 1.407976112291811
Generation 240 Best Solution: [183.56329953440186, 176.79990784597481,
175.0981658445614, 75.50873935121551] Fitness: 1.6055862112445918
Generation 241 Best Solution: [183.52456323809332, 176.79990784597481,
175.0981658445614, 75.50873935121551] Fitness: 1.5862478740518258
Generation 242 Best Solution: [183.79436994204238, 176.79990784597481,
175.0981658445614, 75.50873935121551] Fitness: 1.7677018450140567
Generation 243 Best Solution: [183.56329953440186, 176.79990784597481,
175.0981658445614, 75.86969937536053] Fitness: 2.13998597904985
Generation 244 Best Solution: [183.79436994204238, 177.399179306859,
175.0981658445614, 74.37036636301617] Fitness: 2.169054287177635
Generation 245 Best Solution: [183.3839566096054, 177.399179306859,
174.9676032371675, 74.37036636301617] Fitness: 1.7584053525708132
Generation 246 Best Solution: [183.79436994204238, 177.399179306859,
174.9676032371675, 74.37036636301617] Fitness: 1.9911418257029934
Generation 247 Best Solution: [183.3839566096054, 177.399179306859,
174.9676032371675, 74.37036636301617] Fitness: 1.7584053525708132
Generation 248 Best Solution: [183.2597314508836, 176.14680451693806,
174.9676032371675, 74.37036636301617] Fitness: 0.6655452755858016
Generation 249 Best Solution: [183.79436994204238, 176.4365393140536,
175.0981658445614, 74.28175751591418] Fitness: 1.1856996584452382
Generation 250 Best Solution: [183.3839566096054, 176.19605755207772,
175.08184890682645, 75.86969937536053] Fitness: 1.777840018082666
Generation 251 Best Solution: [183.38136263455308, 177.10255471257238,
175.0981658445614, 74.3715723533895] Fitness: 1.4684703415996399
Generation 252 Best Solution: [183.38136263455308, 177.10255471257238,
173.9109638820946, 75.86969937536053] Fitness: 1.7318059561968662
Generation 253 Best Solution: [183.38136263455308, 177.10255471257238,
173.9109638820946, 75.86969937536053] Fitness: 1.7318059561968662
Generation 254 Best Solution: [183.38136263455308, 177.10255471257238,
174.9676032371675, 74.3715723533895] Fitness: 1.290557880124998
Generation 255 Best Solution: [183.3839566096054, 177.399179306859,
175.0981658445614, 74.71518087160301] Fitness: 1.8538475255237492
Generation 256 Best Solution: [183.64324847051736, 177.10255471257238,
174.15874591067436, 75.86969937536053] Fitness: 1.8168257733803241
Generation 257 Best Solution: [183.74920399319282, 176.69369850441095,

174.16355562542606, 75.86969937536053] Fitness: 1.467404290144256
Generation 258 Best Solution: [183.3839566096054, 176.7459582932348,
174.16355562542606, 75.86969937536053] Fitness: 1.31322798682091
Generation 259 Best Solution: [183.79436994204238, 177.399179306859,
174.36301784648484, 75.14705357559924] Fitness: 1.566452180757568
Generation 260 Best Solution: [183.79436994204238, 177.399179306859,
174.36301784648484, 75.14705357559924] Fitness: 1.566452180757568
Generation 261 Best Solution: [183.79436994204238, 175.90210512737892,
174.9676032371675, 75.86969937536053] Fitness: 1.923820775677171
Generation 262 Best Solution: [183.3839566096054, 176.6947367123513,
173.98901461531955, 75.86969937536053] Fitness: 1.2860278584753662
Generation 263 Best Solution: [183.79436994204238, 175.90210512737892,
174.9676032371675, 75.86969937536053] Fitness: 1.923820775677171
Generation 264 Best Solution: [183.79436994204238, 176.48699069191866,
175.0981658445614, 75.86969937536053] Fitness: 2.0950003451888612
Generation 265 Best Solution: [183.0949355718073, 177.399179306859,
174.23236088445535, 75.86969937536053] Fitness: 2.1613240948246206
Generation 266 Best Solution: [183.3839566096054, 176.92671623360272,
173.9070367627055, 75.86969937536053] Fitness: 1.5194622917264147
Generation 267 Best Solution: [183.79436994204238, 175.9207167647447,
174.23236088445535, 75.86969937536053] Fitness: 1.3912489708653137
Generation 268 Best Solution: [183.66198885438428, 177.399179306859,
174.23236088445535, 75.43896554593726] Fitness: 1.6885104488636287
Generation 269 Best Solution: [183.42479556651259, 177.399179306859,
174.65517226157894, 75.86969937536053] Fitness: 2.387966856127183
Generation 270 Best Solution: [183.79436994204238, 177.399179306859,
174.23236088445535, 75.86969937536053] Fitness: 2.4063446757697102
Generation 271 Best Solution: [183.42479556651259, 177.07303799388566,
174.65517226157894, 75.86969937536053] Fitness: 1.8821772404139039
Generation 272 Best Solution: [183.42479556651259, 177.399179306859,
175.0981658445614, 74.74439007193945] Fitness: 1.8668942863386127
Generation 273 Best Solution: [183.79436994204238, 177.399179306859,
174.65517226157894, 75.36807640302152] Fitness: 1.9191333844024858
Generation 274 Best Solution: [183.79436994204238, 176.6954408870518,
175.0981658445614, 75.40241116775182] Fitness: 1.562075264947575
Generation 275 Best Solution: [183.1549788222681, 176.6954408870518,
175.0981658445614, 75.86969937536053] Fitness: 1.963849000184704
Generation 276 Best Solution: [183.1549788222681, 176.6954408870518,
175.0981658445614, 75.86969937536053] Fitness: 1.963849000184704
Generation 277 Best Solution: [183.79436994204238, 177.399179306859,
174.65517226157894, 74.45685982880772] Fitness: 1.6317640699131226
Generation 278 Best Solution: [183.79436994204238, 177.399179306859,
174.65517226157894, 74.45685982880772] Fitness: 1.6317640699131226
Generation 279 Best Solution: [183.79436994204238, 176.01765566638608,
175.0981658445614, 75.86969937536053] Fitness: 2.059729144338836
Generation 280 Best Solution: [183.79436994204238, 176.73767549422993,
175.0981658445614, 75.52013018393006] Fitness: 1.7286910871106107
Generation 281 Best Solution: [183.79436994204238, 177.399179306859,

174.65517226157894, 74.45685982880772] Fitness: 1.6317640699131226
Generation 282 Best Solution: [183.79436994204238, 176.75197910077839,
175.0981658445614, 75.0155507652456] Fitness: 1.312233907908707
Generation 283 Best Solution: [183.1549788222681, 177.399179306859,
174.65517226157894, 74.45685982880772] Fitness: 1.3789840474067006
Generation 284 Best Solution: [183.79436994204238, 176.03615512591017,
174.65517226157894, 75.86969937536053] Fitness: 1.5550182326730169
Generation 285 Best Solution: [183.79436994204238, 176.00773519045163,
175.0981658445614, 75.86969937536053] Fitness: 2.062549438334126
Generation 286 Best Solution: [183.79436994204238, 176.18201157843092,
174.08473482688714, 75.14378590300085] Fitness: 0.44855118910350744
Generation 287 Best Solution: [183.1549788222681, 177.399179306859,
174.4303848275009, 74.81030444573385] Fitness: 1.2015389711878293
Generation 288 Best Solution: [183.57413336124364, 177.399179306859,
174.08473482688714, 75.14378590300085] Fitness: 1.3664702510322624
Generation 289 Best Solution: [183.65579017773624, 176.18201157843092,
173.82846606916783, 75.86969937536053] Fitness: 1.2822507182964624
Generation 290 Best Solution: [183.79436994204238, 177.399179306859,
174.08473482688714, 75.86969937536053] Fitness: 2.398792847546151
Generation 291 Best Solution: [183.2597237550973, 176.3238144850126,
175.0981658445614, 75.86969937536053] Fitness: 1.791898286938795
Generation 292 Best Solution: [183.79436994204238, 177.399179306859,
174.94064055182548, 74.362094050937] Fitness: 1.9617554738806318
Generation 293 Best Solution: [183.79436994204238, 177.399179306859,
175.0981658445614, 74.362094050937] Fitness: 2.1732234333869673
Generation 294 Best Solution: [183.4626467888715, 177.43861800361353,
173.93542042760436, 75.86969937536053] Fitness: 2.288693465788345
Generation 295 Best Solution: [183.79436994204238, 177.399179306859,
175.0981658445614, 74.67836677472972] Fitness: 2.086884865491899
Generation 296 Best Solution: [183.10143136493895, 175.69605072186764,
175.0981658445614, 75.44162975860317] Fitness: 1.3680072634809057
Generation 297 Best Solution: [183.79436994204238, 177.399179306859,
175.0981658445614, 75.09649599944521] Fitness: 2.203046912264863
Generation 298 Best Solution: [183.2597237550973, 175.57699845411048,
175.0981658445614, 75.86969937536053] Fitness: 2.0739883483198174
Generation 299 Best Solution: [183.79436994204238, 177.399179306859,
175.0981658445614, 74.93419836502409] Fitness: 2.1268177518325238
Generation 300 Best Solution: [183.79436994204238, 177.399179306859,
175.0981658445614, 74.93419836502409] Fitness: 2.1268177518325238
Generation 301 Best Solution: [183.79436994204238, 176.6026755415203,
174.94064055182548, 75.86969937536053] Fitness: 1.9429847203852293
Generation 302 Best Solution: [183.79436994204238, 177.399179306859,
174.94064055182548, 74.28250630439709] Fitness: 2.0071111829611064
Generation 303 Best Solution: [183.3794862821929, 177.43861800361353,
174.85236959946994, 74.28250630439709] Fitness: 1.7429810922904678
Generation 304 Best Solution: [183.3794862821929, 177.43861800361353,
174.85236959946994, 74.28250630439709] Fitness: 1.7429810922904678
Generation 305 Best Solution: [183.86808836144834, 177.43861800361353,

174.85236959946994, 74.28250630439709] Fitness: 2.0454327742083622
Generation 306 Best Solution: [183.79436994204238, 176.38460288354787,
175.0981658445614, 75.86969937536053] Fitness: 2.059127563408108
Generation 307 Best Solution: [183.79436994204238, 175.63557422485408,
175.0999921596799, 75.86969937536053] Fitness: 2.2776679836214724
Generation 308 Best Solution: [183.79436994204238, 176.25831472709933,
175.0981658445614, 75.86969937536053] Fitness: 2.0365403110164637
Generation 309 Best Solution: [183.3794862821929, 176.58387662268495,
175.0981658445614, 75.86969937536053] Fitness: 1.9095450960164735
Generation 310 Best Solution: [183.50179359431104, 176.34744615052006,
175.0999921596799, 75.86969937536053] Fitness: 1.8608970925781398
Generation 311 Best Solution: [183.3794862821929, 175.63557422485408,
175.0999921596799, 75.86969937536053] Fitness: 2.0437874844045987
Generation 312 Best Solution: [183.3794862821929, 177.43861800361353,
175.0999921596799, 74.70164023067085] Fitness: 1.9272239926059407
Generation 313 Best Solution: [183.79436994204238, 176.19298402707815,
175.0981658445614, 75.86969937536053] Fitness: 2.0342445423760624
Generation 314 Best Solution: [183.50179359431104, 177.43861800361353,
173.9025622904594, 75.86969937536053] Fitness: 2.3147392559140307
Generation 315 Best Solution: [183.86808836144834, 175.9958257231242,
175.0999921596799, 75.44656728091414] Fitness: 1.5306563352358025
Generation 316 Best Solution: [183.86808836144834, 175.9958257231242,
175.0981658445614, 75.65181784176411] Fitness: 1.788748056184807
Generation 317 Best Solution: [182.97748054685593, 177.399179306859,
173.9025622904594, 75.65181784176411] Fitness: 1.8743749606301014
Generation 318 Best Solution: [183.50179359431104, 175.67944887169625,
175.0981658445614, 75.86969937536053] Fitness: 2.047378597839008
Generation 319 Best Solution: [183.30469976965784, 175.9958257231242,
175.0981658445614, 75.86969937536053] Fitness: 1.8175561501808097
Generation 320 Best Solution: [183.86808836144834, 177.43861800361353,
173.9025622904594, 74.6832061676555] Fitness: 1.5531003874035985
Generation 321 Best Solution: [183.79436994204238, 175.9958257231242,
175.0981658445614, 75.86969937536053] Fitness: 2.0661301703188637
Generation 322 Best Solution: [183.79436994204238, 175.9958257231242,
175.0981658445614, 75.86969937536053] Fitness: 2.0661301703188637
Generation 323 Best Solution: [183.79436994204238, 175.9958257231242,
175.0981658445614, 75.86969937536053] Fitness: 2.0661301703188637
Generation 324 Best Solution: [183.86808836144834, 175.9958257231242,
175.0981658445614, 75.42042773776308] Fitness: 1.499313547609923
Generation 325 Best Solution: [182.97748054685593, 175.9958257231242,
175.0981658445614, 75.86969937536053] Fitness: 1.85192445284002
Generation 326 Best Solution: [182.97748054685593, 175.9958257231242,
175.0981658445614, 75.86969937536053] Fitness: 1.85192445284002
Generation 327 Best Solution: [182.97748054685593, 175.70724382090376,
175.0981658445614, 74.60883597315485] Fitness: 0.9880160963501285
Generation 328 Best Solution: [182.97748054685593, 175.70724382090376,
175.0981658445614, 74.60883597315485] Fitness: 0.9880160963501285
Generation 329 Best Solution: [182.97748054685593, 175.9958257231242,

175.0999921596799, 75.86969937536053] Fitness: 1.8545944375815748
Generation 330 Best Solution: [183.79436994204238, 175.9958257231242,
175.0999921596799, 75.86969937536053] Fitness: 2.0688001550604183
Generation 331 Best Solution: [183.79436994204238, 177.43861800361353,
175.0999921596799, 74.30077860375057] Fitness: 2.281971971895896
Generation 332 Best Solution: [183.67803099639457, 176.1807870398585,
175.0999921596799, 75.82116955745556] Fitness: 1.8623427827174617
Generation 333 Best Solution: [183.67803099639457, 176.1807870398585,
175.0999921596799, 75.86969937536053] Fitness: 1.945562605773056
Generation 334 Best Solution: [183.79436994204238, 177.17112114010928,
175.0999921596799, 75.5638646596039] Fitness: 2.2754929438403364
Generation 335 Best Solution: [182.97748054685593, 177.17112114010928,
175.0999921596799, 75.5638646596039] Fitness: 2.0612872263614928
Generation 336 Best Solution: [183.16127991986582, 177.36734492360677,
175.0999921596799, 75.29914847449294] Fitness: 2.0466914493754205
Generation 337 Best Solution: [183.79436994204238, 175.97609681966455,
175.05356549118534, 75.82116955745556] Fitness: 1.925659650233099
Generation 338 Best Solution: [182.97748054685593, 177.43861800361353,
173.86919858739464, 75.86969937536053] Fitness: 2.304243524763252
Generation 339 Best Solution: [182.97748054685593, 177.43861800361353,
173.82367424320088, 75.72910763073752] Fitness: 2.0918457789157805
Generation 340 Best Solution: [182.97748054685593, 177.2864722046916,
174.18414784798458, 75.72910763073752] Fitness: 1.7618647379263777
Generation 341 Best Solution: [182.97748054685593, 177.36734492360677,
173.77248942518517, 75.86969937536053] Fitness: 2.2199122265733298
Generation 342 Best Solution: [182.80185258105107, 177.2864722046916,
175.0999921596799, 74.40280439799285] Fitness: 1.8340113954945259
Generation 343 Best Solution: [183.09133546770897, 177.2864722046916,
175.0999921596799, 75.8040326285061] Fitness: 2.562402994185559
Generation 344 Best Solution: [183.19405974837667, 177.43861800361353,
174.44505646487812, 75.8040326285061] Fitness: 2.1799928223766942
Generation 345 Best Solution: [183.57474145434367, 176.65356336468008,
175.0999921596799, 75.86969937536053] Fitness: 2.033659261081935
Generation 346 Best Solution: [182.90993712841214, 177.43861800361353,
175.0999921596799, 74.69866262862176] Fitness: 1.9739999188137207
Generation 347 Best Solution: [182.90993712841214, 176.01727015738325,
175.0999921596799, 75.86969937536053] Fitness: 1.8753921466432075
Generation 348 Best Solution: [182.90993712841214, 176.01727015738325,
175.0999921596799, 75.86969937536053] Fitness: 1.8753921466432075
Generation 349 Best Solution: [182.90993712841214, 176.05738391417472,
175.0999921596799, 75.86969937536053] Fitness: 1.8654702985356053
Generation 350 Best Solution: [182.95075062848474, 176.44256900031755,
175.0999921596799, 74.69866262862176] Fitness: 0.8536303127316721
Generation 351 Best Solution: [183.79436994204238, 175.80699828653974,
175.0999921596799, 74.69866262862176] Fitness: 1.1317593141377489
Generation 352 Best Solution: [182.90993712841214, 177.3345583105256,
175.0999921596799, 75.13759344135609] Fitness: 1.9312843560953534
Generation 353 Best Solution: [182.90993712841214, 176.65133837795884,

175.0999921596799, 75.8040326285061] Fitness: 1.8892531391564384
Generation 354 Best Solution: [182.90993712841214, 176.69949975157996,
175.0999921596799, 74.5732625269491] Fitness: 1.0254211777589188
Generation 355 Best Solution: [182.90993712841214, 176.48845880297438,
175.0999921596799, 75.86969937536053] Fitness: 1.9111852320475708
Generation 356 Best Solution: [182.90993712841214, 175.66947053226204,
175.0999921596799, 75.86969937536053] Fitness: 2.062604969293633
Generation 357 Best Solution: [183.79436994204238, 176.35284381591475,
175.0999921596799, 75.86969937536053] Fitness: 2.053865667295335
Generation 358 Best Solution: [182.98752117729754, 176.97734086513503,
174.32963524624057, 75.86969937536053] Fitness: 1.5876158381867476
Generation 359 Best Solution: [182.90993712841214, 177.43861800361353,
175.0999921596799, 74.28575225296258] Fitness: 2.1040758422551127
Generation 360 Best Solution: [182.90993712841214, 175.7028496787829,
175.0999921596799, 75.86969937536053] Fitness: 2.0367663997828442
Generation 361 Best Solution: [182.98752117729754, 175.66883290979482,
175.0999921596799, 75.86969937536053] Fitness: 2.0325798338718943
Generation 362 Best Solution: [182.98752117729754, 175.66883290979482,
175.0999921596799, 75.86969937536053] Fitness: 2.0325798338718943
Generation 363 Best Solution: [183.79436994204238, 176.47064091043637,
175.0999921596799, 75.8040326285061] Fitness: 1.9791244142890867
Generation 364 Best Solution: [182.90993712841214, 176.79496594874152,
175.0999921596799, 75.8040326285061] Fitness: 2.0014829921447
Generation 365 Best Solution: [182.90993712841214, 176.79496594874152,
174.03980378877895, 75.86969937536053] Fitness: 1.4048485628407994
Generation 366 Best Solution: [183.79436994204238, 176.79496594874152,
174.03980378877895, 75.86969937536053] Fitness: 1.5919623848442188
Generation 367 Best Solution: [183.49266457811905, 175.70936069295485,
175.0999921596799, 75.8040326285061] Fitness: 1.9115834536196246
Generation 368 Best Solution: [182.90993712841214, 177.43861800361353,
175.0999921596799, 74.69308695466178] Fitness: 1.974053005618821
Generation 369 Best Solution: [183.5905689393788, 175.70936069295485,
175.0999921596799, 75.8040326285061] Fitness: 1.9601203701067136
Generation 370 Best Solution: [183.5905689393788, 177.43861800361353,
175.0999921596799, 74.69308695466178] Fitness: 2.0140148623029863
Generation 371 Best Solution: [183.49266457811905, 176.51655136179338,
175.0999921596799, 75.86969937536053] Fitness: 1.915263743111789
Generation 372 Best Solution: [183.49266457811905, 176.51655136179338,
175.0999921596799, 75.86969937536053] Fitness: 1.915263743111789
Generation 373 Best Solution: [183.79436994204238, 176.04827165019503,
175.0999921596799, 75.86969937536053] Fitness: 2.0546261029011017
Generation 374 Best Solution: [183.49266457811905, 177.43861800361353,
173.89286125672294, 75.87487882583906] Fitness: 2.3232110595860727
Generation 375 Best Solution: [183.79436994204238, 177.43861800361353,
175.0999921596799, 75.42136955033898] Fitness: 2.548983897440298
Generation 376 Best Solution: [183.79436994204238, 177.43861800361353,
175.0999921596799, 75.42136955033898] Fitness: 2.548983897440298
Generation 377 Best Solution: [183.79436994204238, 177.43861800361353,

175.0999921596799, 74.8053705822675] Fitness: 2.169083925922213
Generation 378 Best Solution: [183.6399695367603, 177.43861800361353,
175.0999921596799, 75.29869119880694] Fitness: 2.310732181851449
Generation 379 Best Solution: [182.90993712841214, 177.43861800361353,
175.0999921596799, 75.29869119880694] Fitness: 2.240821845804785
Generation 380 Best Solution: [183.48101003722584, 177.43861800361353,
175.0999921596799, 75.29869119880694] Fitness: 2.2274265926521335
Generation 381 Best Solution: [183.41638902788185, 177.43861800361353,
174.80907283201554, 75.29869119880694] Fitness: 1.8421638005243142
Generation 382 Best Solution: [183.3574683097881, 176.55795114802723,
175.0999921596799, 75.86969937536053] Fitness: 1.8926799947302069
Generation 383 Best Solution: [182.90993712841214, 177.43861800361353,
173.74125632801864, 74.34404596683208] Fitness: 1.4665629291293558
Generation 384 Best Solution: [183.48101003722584, 177.43861800361353,
174.58862755739665, 75.29869119880694] Fitness: 1.6752560558887741
Generation 385 Best Solution: [182.90993712841214, 176.95890718899022,
175.0999921596799, 75.44384034330903] Fitness: 1.669411701944884
Generation 386 Best Solution: [182.90993712841214, 176.95890718899022,
175.0999921596799, 75.86969937536053] Fitness: 2.2791658011265743
Generation 387 Best Solution: [183.48101003722584, 177.43861800361353,
174.22203392837696, 75.86969937536053] Fitness: 2.2762201119608667
Generation 388 Best Solution: [182.90993712841214, 176.95890718899022,
174.12400037053493, 75.44384034330903] Fitness: 0.9556418413202541
Generation 389 Best Solution: [183.35773782424968, 176.95890718899022,
174.12400037053493, 75.44384034330903] Fitness: 0.9037370601835528
Generation 390 Best Solution: [183.48101003722584, 176.0594088294809,
174.06351858360668, 75.87487882583906] Fitness: 1.1497426096770793
Generation 391 Best Solution: [183.79436994204238, 177.08880792232273,
174.22203392837696, 75.86969937536053] Fitness: 1.9197489436523072
Generation 392 Best Solution: [183.79436994204238, 177.43861800361353,
175.0999921596799, 74.40163545400897] Fitness: 2.2288685417058547
Generation 393 Best Solution: [183.0914831809205, 177.43861800361353,
175.0999921596799, 74.40163545400897] Fitness: 1.9844585250726323
Generation 394 Best Solution: [182.90993712841214, 177.43861800361353,
174.9942404957577, 74.40163545400897] Fitness: 1.895393589336686
Generation 395 Best Solution: [183.79436994204238, 177.43861800361353,
175.0999921596799, 74.75539196624193] Fitness: 2.163224504046531
Generation 396 Best Solution: [182.90993712841214, 177.43861800361353,
174.16275347747444, 74.75539196624193] Fitness: 1.2634413600260237
Generation 397 Best Solution: [182.90993712841214, 177.43861800361353,
174.16275347747444, 74.75539196624193] Fitness: 1.2634413600260237
Generation 398 Best Solution: [183.79436994204238, 177.43861800361353,
174.16275347747444, 74.75539196624193] Fitness: 1.4505551820294431
Generation 399 Best Solution: [183.84856718995715, 177.43861800361353,
175.0999921596799, 75.4373446213067] Fitness: 2.616027485462835
Generation 400 Best Solution: [183.12912954071814, 177.43861800361353,
175.0999921596799, 75.4373446213067] Fitness: 2.3163055295325754
Generation 401 Best Solution: [183.79436994204238, 177.43861800361353,

174.26303591339385, 75.86969937536053] Fitness: 2.483992103035442
Generation 402 Best Solution: [183.79436994204238, 176.60639582772694,
174.61616141606666, 75.86969937536053] Fitness: 1.6269369230806237
Generation 403 Best Solution: [183.79436994204238, 176.29898382717576,
174.55605455195447, 75.86969937536053] Fitness: 1.4698180802281222
Generation 404 Best Solution: [183.12912954071814, 177.43861800361353,
175.0999921596799, 74.49101112122698] Fitness: 1.9444594201934917
Generation 405 Best Solution: [183.79436994204238, 175.59373551150244,
175.0999921596799, 75.31164837898163] Fitness: 1.5708405798543759
Generation 406 Best Solution: [183.79436994204238, 177.43861800361353,
174.61616141606666, 74.49101112122698] Fitness: 1.6621310112850407
Generation 407 Best Solution: [183.12912954071814, 177.43861800361353,
175.0999921596799, 74.5379894126358] Fitness: 1.9312311351142624
Generation 408 Best Solution: [183.79436994204238, 177.30700374437086,
175.0999921596799, 74.5379894126358] Fitness: 1.950234312174812
Generation 409 Best Solution: [183.79436994204238, 177.43861800361353,
175.0999921596799, 74.5379894126358] Fitness: 2.181333549595678
Generation 410 Best Solution: [183.12912954071814, 177.43861800361353,
174.91815221264608, 74.5379894126358] Fitness: 1.6899401425661171
Generation 411 Best Solution: [183.79436994204238, 177.43861800361353,
174.91815221264608, 74.5379894126358] Fitness: 1.9400425570475321
Generation 412 Best Solution: [183.79436994204238, 177.43861800361353,
174.91815221264608, 74.5379894126358] Fitness: 1.9400425570475321
Generation 413 Best Solution: [183.79436994204238, 177.30700374437086,
175.0999921596799, 74.50775657255608] Fitness: 1.9583676670413805
Generation 414 Best Solution: [183.83561704674602, 177.30700374437086,
175.0999921596799, 74.70175668975556] Fitness: 1.9673677627446278
Generation 415 Best Solution: [183.79436994204238, 177.30700374437086,
175.0999921596799, 74.70175668975556] Fitness: 1.9300051629066826
Generation 416 Best Solution: [182.89013405940003, 177.30700374437086,
175.0999921596799, 74.70175668975556] Fitness: 1.7521317158060477
Generation 417 Best Solution: [183.79436994204238, 176.50826787246024,
175.0999921596799, 75.86969937536053] Fitness: 2.1070984739526835
Generation 418 Best Solution: [183.79436994204238, 177.43861800361353,
174.315785267138, 75.78719442876489] Fitness: 2.357666228810678
Generation 419 Best Solution: [183.66297724504037, 177.43861800361353,
174.315785267138, 75.86969937536053] Fitness: 2.39503899375771
Generation 420 Best Solution: [183.79436994204238, 177.43861800361353,
175.0999921596799, 74.38721553313265] Fitness: 2.2355261484155706
Generation 421 Best Solution: [183.79436994204238, 177.43861800361353,
175.0999921596799, 74.38721553313265] Fitness: 2.2355261484155706
Generation 422 Best Solution: [183.79436994204238, 177.30700374437086,
175.0999921596799, 74.38721553313265] Fitness: 2.004426910994705
Generation 423 Best Solution: [183.70200463864717, 176.82168058802225,
175.0999921596799, 74.32090407791519] Fitness: 1.3370366808466874
Generation 424 Best Solution: [183.83561704674602, 176.82168058802225,
175.0999921596799, 75.78719442876489] Fitness: 2.222630958390761
Generation 425 Best Solution: [183.83561704674602, 176.81554797118247,

175.0999921596799, 75.86969937536053] Fitness: 2.3563394745216555
Generation 426 Best Solution: [183.66297724504037, 177.43861800361353,
174.315785267138, 74.62133558106162] Fitness: 1.3776938029714483
Generation 427 Best Solution: [183.83561704674602, 176.81554797118247,
174.52457381735138, 75.78719442876489] Fitness: 1.6232684416955845
Generation 428 Best Solution: [183.66297724504037, 176.81554797118247,
174.52457381735138, 75.78719442876489] Fitness: 1.4838999462072842
Generation 429 Best Solution: [183.66297724504037, 177.43861800361353,
174.315785267138, 75.86969937536053] Fitness: 2.39503899375771
Generation 430 Best Solution: [183.10814159286642, 177.43861800361353,
175.0999921596799, 75.05476303385964] Fitness: 2.0071268990608404
Generation 431 Best Solution: [182.80312156039474, 177.43861800361353,
175.0999921596799, 75.05476303385964] Fitness: 2.124016462915488
Generation 432 Best Solution: [182.80312156039474, 177.30700374437086,
175.0999921596799, 75.05476303385964] Fitness: 1.892917225494622
Generation 433 Best Solution: [183.10814159286642, 177.43861800361353,
175.0999921596799, 75.05476303385964] Fitness: 2.0071268990608404
Generation 434 Best Solution: [183.10814159286642, 177.43861800361353,
175.0999921596799, 75.05476303385964] Fitness: 2.0071268990608404
Generation 435 Best Solution: [182.80312156039474, 177.2386004804732,
175.0999921596799, 75.86969937536053] Fitness: 2.7121088967877807
Generation 436 Best Solution: [183.3865392172211, 177.43861800361353,
175.0999921596799, 74.39121774895723] Fitness: 2.0015851919958307
Generation 437 Best Solution: [182.80312156039474, 176.7987534790593,
174.61351069837787, 75.86969937536053] Fitness: 1.6390554734704912
Generation 438 Best Solution: [182.80312156039474, 177.43861800361353,
175.0999921596799, 74.45992383400521] Fitness: 2.074834076712427
Generation 439 Best Solution: [183.10814159286642, 177.43861800361353,
175.0999921596799, 74.45992383400521] Fitness: 1.9579445128577797
Generation 440 Best Solution: [183.3865392172211, 175.8259686573447,
175.0999921596799, 75.86969937536053] Fitness: 1.910963695158277
Generation 441 Best Solution: [183.10814159286642, 175.8259686573447,
175.0999921596799, 75.86969937536053] Fitness: 1.8958344711139299
Generation 442 Best Solution: [183.10814159286642, 175.8259686573447,
175.0999921596799, 75.86969937536053] Fitness: 1.8958344711139299
Generation 443 Best Solution: [183.10814159286642, 175.8259686573447,
175.0999921596799, 75.86969937536053] Fitness: 1.8958344711139299
Generation 444 Best Solution: [183.10814159286642, 175.8259686573447,
175.0999921596799, 74.45992383400521] Fitness: 0.9176136754884053
Generation 445 Best Solution: [183.10814159286642, 175.8259686573447,
175.0999921596799, 75.8543604072745] Fitness: 1.8691490918806566
Generation 446 Best Solution: [183.83561704674602, 177.2058496973823,
175.0999921596799, 75.86969937536053] Fitness: 2.82966420044055
Generation 447 Best Solution: [183.3865392172211, 176.9815669237623,
175.0999921596799, 75.8543604072745] Fitness: 2.233636879976664
Generation 448 Best Solution: [183.83561704674602, 176.9815669237623,
175.0999921596799, 75.8543604072745] Fitness: 2.5030613623990927
Generation 449 Best Solution: [182.80312156039474, 175.87732965616198,

175.0999921596799, 75.86969937536053] Fitness: 1.985715360767204
Generation 450 Best Solution: [183.83561704674602, 175.87732965616198,
175.0999921596799, 75.86969937536053] Fitness: 2.1533795033793326
Generation 451 Best Solution: [183.06802978531448, 177.43861800361353,
175.0999921596799, 74.83503758873952] Fitness: 1.934545230423837
Generation 452 Best Solution: [183.3865392172211, 177.43861800361353,
175.0999921596799, 74.49543372362866] Fitness: 1.9611134675656388
Generation 453 Best Solution: [183.83561704674602, 177.43861800361353,
175.0999921596799, 75.04098183307214] Fitness: 2.284535393082527
Generation 454 Best Solution: [183.1655815245885, 177.43861800361353,
175.0999921596799, 75.04098183307214] Fitness: 1.9935843527867834
Generation 455 Best Solution: [182.80312156039474, 176.41121782155778,
175.0999921596799, 75.86969937536053] Fitness: 1.9393083465625165
Generation 456 Best Solution: [183.83561704674602, 176.41121782155778,
175.0999921596799, 75.86969937536053] Fitness: 2.106972489174645
Generation 457 Best Solution: [183.83561704674602, 176.41121782155778,
175.0999921596799, 75.86969937536053] Fitness: 2.106972489174645
Generation 458 Best Solution: [182.80312156039474, 176.41121782155778,
175.0999921596799, 75.86969937536053] Fitness: 1.9393083465625165
Generation 459 Best Solution: [182.80312156039474, 176.5696520419982,
174.2210745687446, 75.86969937536053] Fitness: 1.301035507667594
Generation 460 Best Solution: [182.80312156039474, 176.41121782155778,
175.0999921596799, 75.32682371219641] Fitness: 1.2096514351960614
Generation 461 Best Solution: [182.80312156039474, 176.41121782155778,
175.0040602611824, 75.32682371219641] Fitness: 1.0761744154618733
Generation 462 Best Solution: [182.80312156039474, 176.41121782155778,
175.0999921596799, 75.86969937536053] Fitness: 1.9393083465625165
Generation 463 Best Solution: [182.80312156039474, 176.41121782155778,
175.0999921596799, 75.86969937536053] Fitness: 1.9393083465625165
Generation 464 Best Solution: [182.80312156039474, 177.43861800361353,
175.0999921596799, 74.56719590588149] Fitness: 2.0444767745914163
Generation 465 Best Solution: [182.80312156039474, 177.43861800361353,
175.0999921596799, 74.56719590588149] Fitness: 2.0444767745914163
Generation 466 Best Solution: [183.86820050475364, 176.23619300803765,
175.0999921596799, 75.86969937536053] Fitness: 2.106397735180818
Generation 467 Best Solution: [183.29751014195315, 177.43861800361353,
175.0999921596799, 74.41921252741004] Fitness: 1.9716305498087983
Generation 468 Best Solution: [183.83561704674602, 176.23248673736754,
175.0999921596799, 75.86969937536053] Fitness: 2.074900074426158
Generation 469 Best Solution: [183.83561704674602, 176.65389719189832,
175.0999921596799, 75.2495705747614] Fitness: 1.4296613143879295
Generation 470 Best Solution: [183.52897778823151, 177.43861800361353,
175.0999921596799, 75.2495705747614] Fitness: 2.206433172480903
Generation 471 Best Solution: [183.19483202549634, 177.43861800361353,
173.8592094657763, 75.2495705747614] Fitness: 1.4698446975405242
Generation 472 Best Solution: [183.83561704674602, 177.43861800361353,
173.8592094657763, 75.2495705747614] Fitness: 1.7621516888016175
Generation 473 Best Solution: [183.19483202549634, 175.86971182995467,

175.0999921596799, 74.41921252741004] Fitness: 0.9026471864250903
Generation 474 Best Solution: [183.0115396809669, 177.43861800361353,
174.7747028653312, 74.98864606965694] Fitness: 1.600718623899271
Generation 475 Best Solution: [183.3396687662729, 177.43861800361353,
174.7747028653312, 74.98864606965694] Fitness: 1.5832423601274428
Generation 476 Best Solution: [183.19483202549634, 176.40491355145778,
175.0999921596799, 75.86969937536053] Fitness: 1.812718958717787
Generation 477 Best Solution: [183.19483202549634, 176.69585648437786,
175.0999921596799, 75.86969937536053] Fitness: 1.9646622160877805
Generation 478 Best Solution: [183.19483202549634, 176.69585648437786,
175.0999921596799, 75.86969937536053] Fitness: 1.9646622160877805
Generation 479 Best Solution: [183.75336692904736, 177.43861800361353,
174.25405048851397, 75.86969937536053] Fitness: 2.4475725827505896
Generation 480 Best Solution: [183.29751014195315, 177.3493317003341,
175.0999921596799, 74.77098538209371] Fitness: 1.7554951833060715
Generation 481 Best Solution: [183.83561704674602, 176.61855973093103,
175.0999921596799, 75.86969937536053] Fitness: 2.2042158096368105
Generation 482 Best Solution: [183.75336692904736, 177.14629794286134,
174.301153975317, 75.86969937536053] Fitness: 1.980347310025512
Generation 483 Best Solution: [183.8076565628434, 177.14629794286134,
175.0999921596799, 75.86969937536053] Fitness: 2.7170664527480435
Generation 484 Best Solution: [183.29751014195315, 177.43861800361353,
175.0999921596799, 74.35103802162794] Fitness: 2.0040574413508163
Generation 485 Best Solution: [183.75336692904736, 177.43861800361353,
175.0999921596799, 74.38020094060685] Fitness: 2.20426539831689
Generation 486 Best Solution: [182.8515070603696, 177.43861800361353,
175.0999921596799, 74.93865192005481] Fitness: 2.044871646387877
Generation 487 Best Solution: [183.11889501534384, 177.43861800361353,
173.78819741207414, 75.86969937536053] Fitness: 2.3056176650140854
Generation 488 Best Solution: [183.83561704674602, 177.43861800361353,
175.0999921596799, 74.9401143224943] Fitness: 2.2409113261540305
Generation 489 Best Solution: [183.11889501534384, 177.08158116738502,
175.0999921596799, 75.86969937536053] Fitness: 2.3680393431032827
Generation 490 Best Solution: [183.83561704674602, 177.3493317003341,
175.0999921596799, 74.58322986640594] Fitness: 2.049475443081066
Generation 491 Best Solution: [183.8076565628434, 177.3493317003341,
173.770505992294, 74.58322986640594] Fitness: 1.4040538647426197
Generation 492 Best Solution: [183.24044296637277, 177.43861800361353,
175.0999921596799, 74.58322986640594] Fitness: 1.9172257437495506
Generation 493 Best Solution: [183.6542729383734, 175.59808040966274,
175.0999921596799, 75.86969937536053] Fitness: 2.2027421721548213
Generation 494 Best Solution: [183.75336692904736, 176.59352028821607,
175.0999921596799, 75.86969937536053] Fitness: 2.117074032438169
Generation 495 Best Solution: [183.8076565628434, 175.91187734225787,
175.0999921596799, 75.86969937536053] Fitness: 2.111832340013985
Generation 496 Best Solution: [183.8076565628434, 176.92718748501883,
175.0999921596799, 75.57435009460164] Fitness: 1.9909912942300143
Generation 497 Best Solution: [183.75336692904736, 176.29876198485184,

175.0999921596799, 75.86969937536053] Fitness: 2.009228344426713
Generation 498 Best Solution: [183.75336692904736, 176.29876198485184,
175.0999921596799, 75.86969937536053] Fitness: 2.009228344426713
Generation 499 Best Solution: [183.75336692904736, 176.29876198485184,
175.0999921596799, 74.83553998167345] Fitness: 1.0003063858405905
Generation 500 Best Solution: [183.83561704674602, 176.0050496771763,
175.0999921596799, 75.86969937536053] Fitness: 2.1033708781583416
Optimal Solution for Simulation 6: [183.83561704674602, 176.0050496771763,
175.0999921596799, 75.86969937536053]

Simulation 7/32 with conditions: {'drought': True, 'flood': True,
'policy_change': False, 'social_event': False, 'economic_crisis': True}
Generation 1 Best Solution: [182.80156665934567, 176.26787819421045,
173.87759612863536, 74.45026174368552] Fitness: 0.27562881342668927
Generation 2 Best Solution: [182.93994285072404, 176.19254668750355,
173.99581634063424, 74.4352971895983] Fitness: 0.17278790851718406
Generation 3 Best Solution: [182.93994285072404, 176.19254668750355,
173.99581634063424, 74.44651471174905] Fitness: 0.16830492591735985
Generation 4 Best Solution: [183.41602534166546, 176.18034108374673,
174.93855447717021, 75.62248365933148] Fitness: 1.1940421389439866
Generation 5 Best Solution: [183.41602534166546, 176.5496795528641,
175.01355229195534, 74.70939935355332] Fitness: 0.7483777499955603
Generation 6 Best Solution: [183.41602534166546, 176.02270234869337,
174.17057572466015, 74.50153980408673] Fitness: 0.13438281019874138
Generation 7 Best Solution: [183.15729044669735, 175.75232827910165,
174.06120579657846, 75.46044721867554] Fitness: 0.632971020398827
Generation 8 Best Solution: [183.74355659665528, 176.02270234869337,
175.03840523310905, 74.45026174368552] Fitness: 0.9473831961842598
Generation 9 Best Solution: [183.2317297008033, 175.60859763330552,
175.01355229195534, 74.45234160425684] Fitness: 0.9484650237096515
Generation 10 Best Solution: [183.2317297008033, 175.60859763330552,
175.01355229195534, 74.45234160425684] Fitness: 0.9484650237096515
Generation 11 Best Solution: [183.2317297008033, 177.42537256464803,
175.01355229195534, 74.70939935355332] Fitness: 1.7435539865166698
Generation 12 Best Solution: [183.2317297008033, 177.42537256464803,
175.01355229195534, 74.70939935355332] Fitness: 1.7435539865166698
Generation 13 Best Solution: [183.2317297008033, 176.22031095776447,
175.01355229195534, 75.83608776773855] Fitness: 1.586527352364731
Generation 14 Best Solution: [183.2317297008033, 176.68130186447723,
175.03840523310905, 75.83608776773855] Fitness: 1.7882623729479792
Generation 15 Best Solution: [183.2317297008033, 176.68130186447723,
174.96233485999645, 75.62248365933148] Fitness: 1.3615845415289511
Generation 16 Best Solution: [183.2317297008033, 176.40219865898771,
175.01355229195534, 75.83608776773855] Fitness: 1.6148804096696145
Generation 17 Best Solution: [183.2317297008033, 176.68130186447723,
174.96233485999645, 75.62248365933148] Fitness: 1.3615845415289511
Generation 18 Best Solution: [183.2317297008033, 176.68130186447723,
175.01355229195534, 75.83608776773855] Fitness: 1.7548856646988527

Generation 19 Best Solution: [183.2317297008033, 176.68130186447723,
175.01355229195534, 75.83608776773855] Fitness: 1.7548856646988527
Generation 20 Best Solution: [183.2317297008033, 176.68130186447723,
174.42012951380485, 75.83608776773855] Fitness: 1.2331119490978388
Generation 21 Best Solution: [182.92469409850207, 176.68130186447723,
175.01355229195534, 75.83608776773855] Fitness: 1.8186514788712702
Generation 22 Best Solution: [183.2317297008033, 177.43048664396952,
175.01355229195534, 75.04145507203603] Fitness: 1.8366421266162138
Generation 23 Best Solution: [183.2317297008033, 176.45910316505302,
175.01355229195534, 75.83608776773855] Fitness: 1.6339420744359439
Generation 24 Best Solution: [183.2317297008033, 175.71562159006427,
174.6523296765042, 75.83608776773855] Fitness: 1.3872203185328893
Generation 25 Best Solution: [183.2317297008033, 176.45910316505302,
175.01355229195534, 75.07956591851837] Fitness: 0.7822949676490146
Generation 26 Best Solution: [183.2317297008033, 176.50519062565257,
175.01355229195534, 75.65460925030555] Fitness: 1.3703740805182303
Generation 27 Best Solution: [183.2317297008033, 176.50519062565257,
175.01355229195534, 75.83608776773855] Fitness: 1.6529402632536352
Generation 28 Best Solution: [183.2317297008033, 176.50519062565257,
175.01355229195534, 75.83608776773855] Fitness: 1.6529402632536352
Generation 29 Best Solution: [182.93019595761783, 177.42537256464803,
175.01355229195534, 74.80168627482837] Fitness: 1.8116065724727384
Generation 30 Best Solution: [183.2317297008033, 176.4713131294682,
174.96233485999645, 75.83608776773855] Fitness: 1.5728037552296694
Generation 31 Best Solution: [183.2317297008033, 176.4713131294682,
175.01355229195534, 75.32307728698999] Fitness: 0.9674555703448422
Generation 32 Best Solution: [182.93019595761783, 177.42537256464803,
174.92629698762346, 75.43107226767827] Fitness: 2.087933683324565
Generation 33 Best Solution: [183.2317297008033, 177.42537256464803,
175.01355229195534, 74.6202357050185] Fitness: 1.7492402828305587
Generation 34 Best Solution: [183.2317297008033, 176.86685318383536,
175.01355229195534, 75.83608776773855] Fitness: 1.9126258632174835
Generation 35 Best Solution: [182.92469409850207, 176.18423103858987,
174.96233485999645, 75.83608776773855] Fitness: 1.5847058568433443
Generation 36 Best Solution: [183.66908947356342, 177.42537256464803,
174.96233485999645, 74.72053716662845] Fitness: 1.8311648843407673
Generation 37 Best Solution: [183.2317297008033, 177.36075876292503,
174.96233485999645, 74.72053716662845] Fitness: 1.5628982478938858
Generation 38 Best Solution: [183.2317297008033, 177.42537256464803,
174.96233485999645, 74.72053716662845] Fitness: 1.6778202584385637
Generation 39 Best Solution: [183.2317297008033, 177.42537256464803,
174.96233485999645, 74.72053716662845] Fitness: 1.6778202584385637
Generation 40 Best Solution: [182.92469409850207, 177.42537256464803,
174.85597612793677, 74.93280975406553] Fitness: 1.655386402039003
Generation 41 Best Solution: [182.94367990863532, 176.87656581697752,
175.01355229195534, 74.72053716662845] Fitness: 1.022592108501722
Generation 42 Best Solution: [182.92469409850207, 177.36075876292503,
174.96233485999645, 74.72053716662845] Fitness: 1.6266640620663038

Generation 43 Best Solution: [182.92469409850207, 175.7035180869616,
174.96233485999645, 75.83608776773855] Fitness: 1.774677921683998
Generation 44 Best Solution: [183.2317297008033, 176.55114313346294,
174.96233485999645, 75.83608776773855] Fitness: 1.6091936396529865
Generation 45 Best Solution: [182.92469409850207, 175.7035180869616,
174.96233485999645, 75.83608776773855] Fitness: 1.774677921683998
Generation 46 Best Solution: [182.92469409850207, 175.7035180869616,
174.96233485999645, 75.83608776773855] Fitness: 1.774677921683998
Generation 47 Best Solution: [183.2317297008033, 177.42537256464803,
174.96233485999645, 74.3240281804201] Fitness: 1.7878816327156752
Generation 48 Best Solution: [183.28174011498308, 175.7679661355556,
174.96233485999645, 75.83608776773855] Fitness: 1.6683281505251955
Generation 49 Best Solution: [183.2317297008033, 175.7679661355556,
174.96233485999645, 75.83608776773855] Fitness: 1.6653224065164043
Generation 50 Best Solution: [183.2317297008033, 175.7679661355556,
174.96233485999645, 75.83608776773855] Fitness: 1.6653224065164043
Generation 51 Best Solution: [183.2317297008033, 175.7679661355556,
174.96233485999645, 75.83608776773855] Fitness: 1.6653224065164043
Generation 52 Best Solution: [183.49606104587656, 176.48673093298302,
174.8771325928665, 75.83608776773855] Fitness: 1.536617309159852
Generation 53 Best Solution: [183.2317297008033, 175.7679661355556,
175.01355229195534, 75.83608776773855] Fitness: 1.7311836891751127
Generation 54 Best Solution: [183.2317297008033, 175.7679661355556,
174.96233485999645, 75.34676132380989] Fitness: 1.0164085928779876
Generation 55 Best Solution: [182.92469409850207, 176.59701161136758,
175.01355229195534, 75.34676132380989] Fitness: 1.1151399091256236
Generation 56 Best Solution: [183.2317297008033, 175.68829778209616,
173.79047676787349, 75.83608776773855] Fitness: 1.2906310708235507
Generation 57 Best Solution: [183.72325218666674, 175.68829778209616,
173.79047676787349, 75.83608776773855] Fitness: 1.4829325663808133
Generation 58 Best Solution: [183.2317297008033, 175.68829778209616,
174.96233485999645, 75.83608776773855] Fitness: 1.722588183240178
Generation 59 Best Solution: [183.2317297008033, 177.42537256464803,
174.96233485999645, 74.36336295285648] Fitness: 1.7664262195123428
Generation 60 Best Solution: [183.2317297008033, 177.42537256464803,
173.79047676787349, 74.67511024721006] Fitness: 1.2465111355711531
Generation 61 Best Solution: [183.2317297008033, 177.42537256464803,
173.79047676787349, 74.67511024721006] Fitness: 1.2465111355711531
Generation 62 Best Solution: [182.92469409850207, 177.42537256464803,
174.60924918018515, 74.36336295285648] Fitness: 1.483218962521695
Generation 63 Best Solution: [182.92469409850207, 176.08038490222137,
174.60924918018515, 74.36336295285648] Fitness: 0.3825912138767713
Generation 64 Best Solution: [183.72325218666674, 176.08038490222137,
174.60924918018515, 74.36336295285648] Fitness: 0.5111268952616163
Generation 65 Best Solution: [183.72325218666674, 176.08038490222137,
174.60924918018515, 74.36336295285648] Fitness: 0.5111268952616163
Generation 66 Best Solution: [183.2317297008033, 175.75542404902376,
174.96233485999645, 75.83608776773855] Fitness: 1.6737062565880119

Generation 67 Best Solution: [182.92469409850207, 177.42537256464803,
174.96233485999645, 74.86437961323084] Fitness: 1.7599529672229521
Generation 68 Best Solution: [183.09049621243454, 177.42537256464803,
174.96233485999645, 75.13375698890718] Fitness: 1.8258364194458268
Generation 69 Best Solution: [183.72325218666674, 177.42537256464803,
174.96233485999645, 75.27123342467141] Fitness: 2.108478482843882
Generation 70 Best Solution: [183.72325218666674, 177.1087018780544,
175.01355229195534, 75.16072743926364] Fitness: 1.5866612055066946
Generation 71 Best Solution: [183.72325218666674, 177.42537256464803,
173.97749112449932, 75.27123342467141] Fitness: 1.6073215550352633
Generation 72 Best Solution: [183.72325218666674, 176.69504052520733,
174.96233485999645, 75.83608776773855] Fitness: 1.891234979965593
Generation 73 Best Solution: [183.72325218666674, 176.69504052520733,
174.96233485999645, 75.83608776773855] Fitness: 1.891234979965593
Generation 74 Best Solution: [183.72325218666674, 176.69504052520733,
174.43294832960385, 75.83608776773855] Fitness: 1.441011644929329
Generation 75 Best Solution: [183.72325218666674, 176.69504052520733,
174.96233485999645, 75.83608776773855] Fitness: 1.891234979965593
Generation 76 Best Solution: [183.72325218666674, 176.69504052520733,
174.96233485999645, 74.37444599400459] Fitness: 1.0187805355875366
Generation 77 Best Solution: [183.72325218666674, 177.42537256464803,
175.01355229195534, 74.44279410969123] Fitness: 1.988338031378142
Generation 78 Best Solution: [183.72325218666674, 177.42537256464803,
175.01355229195534, 74.44279410969123] Fitness: 1.988338031378142
Generation 79 Best Solution: [182.87968555801163, 177.43761077325823,
175.01355229195534, 74.44279410969123] Fitness: 1.9035058228904527
Generation 80 Best Solution: [183.72325218666674, 177.42537256464803,
174.07672624938803, 74.44279410969123] Fitness: 1.4059049373342651
Generation 81 Best Solution: [182.8808643214999, 177.43761077325823,
174.1783339802337, 75.83608776773855] Fitness: 2.2230790718015077
Generation 82 Best Solution: [183.64794262160908, 176.33488445929447,
174.96233485999645, 75.83608776773855] Fitness: 1.672070763690149
Generation 83 Best Solution: [182.89769625338494, 176.12319066402958,
175.01355229195534, 75.83608776773855] Fitness: 1.6678474520747801
Generation 84 Best Solution: [182.89769625338494, 176.12319066402958,
175.01355229195534, 75.83608776773855] Fitness: 1.6678474520747801
Generation 85 Best Solution: [183.72325218666674, 175.8070398805671,
175.01355229195534, 75.83608776773855] Fitness: 1.8988786701157296
Generation 86 Best Solution: [183.72325218666674, 176.46865212665728,
175.01355229195534, 75.83608776773855] Fitness: 1.8299181676782967
Generation 87 Best Solution: [183.72325218666674, 176.46865212665728,
175.01355229195534, 75.83608776773855] Fitness: 1.8299181676782967
Generation 88 Best Solution: [183.72325218666674, 176.46865212665728,
175.01355229195534, 75.83608776773855] Fitness: 1.8299181676782967
Generation 89 Best Solution: [182.89769625338494, 176.40589786227005,
175.01355229195534, 75.83608776773855] Fitness: 1.692108364590504
Generation 90 Best Solution: [183.72325218666674, 177.42537256464803,
175.01355229195534, 74.74447726825603] Fitness: 1.936887034670427

Generation 91 Best Solution: [183.72325218666674, 177.42537256464803,
175.01355229195534, 75.71471851161643] Fitness: 2.696970953812619
Generation 92 Best Solution: [183.72325218666674, 177.42537256464803,
175.01355229195534, 75.12268523455053] Fitness: 2.065240182127828
Generation 93 Best Solution: [183.5851559203701, 177.42537256464803,
175.01355229195534, 75.12268523455053] Fitness: 1.9746067628937096
Generation 94 Best Solution: [183.72325218666674, 177.42537256464803,
175.01355229195534, 75.12268523455053] Fitness: 2.065240182127828
Generation 95 Best Solution: [183.72325218666674, 177.42537256464803,
175.01355229195534, 75.12268523455053] Fitness: 2.065240182127828
Generation 96 Best Solution: [183.72325218666674, 177.42537256464803,
173.93251357122503, 75.12268523455053] Fitness: 1.5100733985466332
Generation 97 Best Solution: [182.90717563002164, 177.42537256464803,
175.01355229195534, 75.06962540925436] Fitness: 1.9136608014680394
Generation 98 Best Solution: [183.42413209304354, 177.42537256464803,
175.01355229195534, 74.62260477728034] Fitness: 1.7810462374533793
Generation 99 Best Solution: [183.42413209304354, 177.42537256464803,
173.88137828550387, 74.5384584714555] Fitness: 1.2590443327570076
Generation 100 Best Solution: [182.90717563002164, 177.0580889160963,
174.02827297155505, 75.83608776773855] Fitness: 1.6241562706821726
Generation 101 Best Solution: [182.90717563002164, 176.44218201964324,
174.91577805032279, 75.83608776773855] Fitness: 1.5771194655373928
Generation 102 Best Solution: [182.90717563002164, 176.53504364000776,
175.01355229195534, 75.83608776773855] Fitness: 1.7386149272129623
Generation 103 Best Solution: [182.90717563002164, 176.53504364000776,
175.01355229195534, 75.83608776773855] Fitness: 1.7386149272129623
Generation 104 Best Solution: [183.10437963903192, 177.42537256464803,
173.77827036810436, 75.83608776773855] Fitness: 2.2169239191866135
Generation 105 Best Solution: [183.42413209304354, 177.42537256464803,
173.77827036810436, 75.83608776773855] Fitness: 2.239749064111634
Generation 106 Best Solution: [183.42413209304354, 177.42537256464803,
173.77827036810436, 75.83608776773855] Fitness: 2.239749064111634
Generation 107 Best Solution: [183.42413209304354, 176.88239679114125,
174.72414558432666, 75.83608776773855] Fitness: 1.6398429472458964
Generation 108 Best Solution: [182.90717563002164, 176.27195077946902,
175.01355229195534, 75.83608776773855] Fitness: 1.661200954146304
Generation 109 Best Solution: [182.90717563002164, 176.27195077946902,
175.01355229195534, 75.84978591940111] Fitness: 1.6845344578139478
Generation 110 Best Solution: [182.90717563002164, 177.42537256464803,
175.01355229195534, 75.29230886937127] Fitness: 2.0718664235838196
Generation 111 Best Solution: [183.42413209304354, 177.42537256464803,
175.01355229195534, 75.29230886937127] Fitness: 2.032309324617417
Generation 112 Best Solution: [182.90717563002164, 177.42537256464803,
175.01355229195534, 75.04682799169952] Fitness: 1.901661607556044
Generation 113 Best Solution: [183.07032810809054, 177.42537256464803,
173.75285173062304, 75.83608776773855] Fitness: 2.2374495980428613
Generation 114 Best Solution: [182.90717563002164, 177.42537256464803,
173.75285173062304, 75.83608776773855] Fitness: 2.2932268940709646

Generation 115 Best Solution: [182.90717563002164, 175.99298980838577,
174.343359343979, 75.84978591940111] Fitness: 1.165173617801025
Generation 116 Best Solution: [182.90717563002164, 175.99298980838577,
174.76976574381786, 75.83608776773855] Fitness: 1.414247205708295
Generation 117 Best Solution: [182.90717563002164, 177.42537256464803,
174.2779187440644, 75.83608776773855] Fitness: 2.202431180371187
Generation 118 Best Solution: [183.72325218666674, 176.62237111631936,
175.01355229195534, 75.83608776773855] Fitness: 1.9078948048495503
Generation 119 Best Solution: [182.90717563002164, 176.62237111631936,
175.01355229195534, 74.9218444502984] Fitness: 0.8662079989510608
Generation 120 Best Solution: [182.90717563002164, 177.42537256464803,
174.343359343979, 75.52569488542296] Fitness: 1.7668192234462723
Generation 121 Best Solution: [182.90717563002164, 176.45698967875128,
173.75285173062304, 75.83608776773855] Fitness: 1.227258435410283
Generation 122 Best Solution: [182.90717563002164, 176.45698967875128,
175.01355229195534, 75.84978591940111] Fitness: 1.7281491151607087
Generation 123 Best Solution: [183.72325218666674, 177.18817597834533,
175.01355229195534, 74.83496989789369] Fitness: 1.5568951441431713
Generation 124 Best Solution: [182.90717563002164, 177.42537256464803,
175.01355229195534, 74.83496989789369] Fitness: 1.8274374597212413
Generation 125 Best Solution: [183.12298445120072, 176.04257863189062,
175.01355229195534, 75.84978591940111] Fitness: 1.636504722588981
Generation 126 Best Solution: [182.90717563002164, 177.42537256464803,
173.79265377592188, 75.84978591940111] Fitness: 2.2951918390476056
Generation 127 Best Solution: [183.0155890519885, 176.56175747585263,
175.01355229195534, 75.83608776773855] Fitness: 1.710767377008666
Generation 128 Best Solution: [182.90717563002164, 176.52409644453925,
174.32842208740436, 75.84978591940111] Fitness: 1.2013721828380988
Generation 129 Best Solution: [182.90717563002164, 176.56175747585263,
175.01355229195534, 75.84978591940111] Fitness: 1.7756152516555268
Generation 130 Best Solution: [182.90717563002164, 176.56175747585263,
175.01355229195534, 75.83608776773855] Fitness: 1.7522817479878827
Generation 131 Best Solution: [182.90717563002164, 175.6781347919289,
175.01355229195534, 75.84978591940111] Fitness: 1.8914412192235632
Generation 132 Best Solution: [183.2372518401069, 177.42537256464803,
175.01355229195534, 75.84978591940111] Fitness: 2.7225968595228567
Generation 133 Best Solution: [182.90717563002164, 177.42537256464803,
173.99841930848228, 75.84978591940111] Fitness: 2.22261932612543
Generation 134 Best Solution: [182.90717563002164, 177.42537256464803,
174.2572834031326, 75.84978591940111] Fitness: 2.221526151780329
Generation 135 Best Solution: [182.90717563002164, 176.83577934701717,
175.01355229195534, 75.84978591940111] Fitness: 1.9776111911414462
Generation 136 Best Solution: [183.71906286392957, 175.91680826538638,
175.01355229195534, 75.84978591940111] Fitness: 1.8621694363297756
Generation 137 Best Solution: [182.90717563002164, 176.83577934701717,
175.01355229195534, 75.35402288850182] Fitness: 1.3123682931547314
Generation 138 Best Solution: [182.90717563002164, 175.89157355079178,
175.01355229195534, 75.84978591940111] Fitness: 1.756181077258144

Generation 139 Best Solution: [182.90717563002164, 175.89157355079178,
175.01355229195534, 75.84978591940111] Fitness: 1.756181077258144
Generation 140 Best Solution: [182.90717563002164, 176.83577934701717,
173.84447255422194, 75.84978591940111] Fitness: 1.4544262231339444
Generation 141 Best Solution: [182.90717563002164, 175.89157355079178,
175.01355229195534, 75.84978591940111] Fitness: 1.756181077258144
Generation 142 Best Solution: [182.90717563002164, 177.42537256464803,
174.73257303151104, 75.21456457111415] Fitness: 1.6952849702212647
Generation 143 Best Solution: [182.90717563002164, 176.20240877284454,
175.01355229195534, 75.84978591940111] Fitness: 1.68142108552713
Generation 144 Best Solution: [182.90717563002164, 177.42537256464803,
175.01355229195534, 74.8875000702643] Fitness: 1.8395641800839277
Generation 145 Best Solution: [183.71906286392957, 177.1315495113139,
175.01355229195534, 75.84978591940111] Fitness: 2.4394974827313276
Generation 146 Best Solution: [183.1381902048677, 177.42537256464803,
175.01355229195534, 74.3112805445663] Fitness: 1.865642888848925
Generation 147 Best Solution: [182.90717563002164, 177.42537256464803,
175.01355229195534, 74.3112805445663] Fitness: 1.932862498879269
Generation 148 Best Solution: [182.90717563002164, 177.42537256464803,
175.01355229195534, 74.66859784672619] Fitness: 1.816344468703683
Generation 149 Best Solution: [183.1381902048677, 177.42537256464803,
175.01355229195534, 74.6834797532036] Fitness: 1.7484262751950739
Generation 150 Best Solution: [182.90717563002164, 176.70285056502675,
175.01355229195534, 75.84978591940111] Fitness: 1.8655558922339528
Generation 151 Best Solution: [182.90717563002164, 175.88481261578633,
175.01355229195534, 75.82056446131571] Fitness: 1.7099818316623696
Generation 152 Best Solution: [182.90717563002164, 175.65795096117023,
175.01355229195534, 75.84978591940111] Fitness: 1.907768640187796
Generation 153 Best Solution: [182.90717563002164, 176.592870856042,
175.01355229195534, 75.84978591940111] Fitness: 1.792882283117373
Generation 154 Best Solution: [182.90717563002164, 176.48488804940507,
175.01355229195534, 75.84978591940111] Fitness: 1.7391803623653703
Generation 155 Best Solution: [182.90717563002164, 176.01676336665793,
175.01355229195534, 75.84978591940111] Fitness: 1.7086405812041243
Generation 156 Best Solution: [182.90717563002164, 175.74769429846643,
174.80226553255486, 75.84978591940111] Fitness: 1.5935224358258653
Generation 157 Best Solution: [183.64821724173618, 175.74769429846643,
174.80226553255486, 75.84978591940111] Fitness: 1.6613608336352805
Generation 158 Best Solution: [182.90717563002164, 175.64067349617605,
173.7768165174765, 75.84978591940111] Fitness: 1.43152260089587
Generation 159 Best Solution: [182.90717563002164, 175.64067349617605,
173.7768165174765, 75.84978591940111] Fitness: 1.43152260089587
Generation 160 Best Solution: [182.90717563002164, 175.64067349617605,
173.7768165174765, 75.84978591940111] Fitness: 1.43152260089587
Generation 161 Best Solution: [182.90717563002164, 176.06228738926865,
175.01355229195534, 75.84978591940111] Fitness: 1.6971816414723366
Generation 162 Best Solution: [182.90717563002164, 176.43322229180708,
175.01355229195534, 75.84978591940111] Fitness: 1.71967226322539

Generation 163 Best Solution: [182.90717563002164, 177.42537256464803,
175.01355229195534, 74.51011099123912] Fitness: 1.844391629427509
Generation 164 Best Solution: [182.90717563002164, 175.71391675437164,
175.01355229195534, 75.84978591940111] Fitness: 1.8639978373508233
Generation 165 Best Solution: [182.90717563002164, 176.03587211312194,
175.01355229195534, 75.84978591940111] Fitness: 1.7034521095824098
Generation 166 Best Solution: [182.90717563002164, 177.42537256464803,
173.7768165174765, 75.06577233278495] Fitness: 1.4208702030629738
Generation 167 Best Solution: [182.90717563002164, 176.17935180325046,
174.5673508246449, 75.84978591940111] Fitness: 1.2403946414403084
Generation 168 Best Solution: [183.37031754770743, 176.17935180325046,
174.5673508246449, 75.84978591940111] Fitness: 1.1862625600510548
Generation 169 Best Solution: [183.37031754770743, 176.17935180325046,
174.72107912476008, 75.84978591940111] Fitness: 1.3046829226831134
Generation 170 Best Solution: [183.3362395214969, 177.42537256464803,
174.4072635906134, 75.84978591940111] Fitness: 2.2057664214544532
Generation 171 Best Solution: [183.3362395214969, 177.42537256464803,
174.30909302050807, 75.84978591940111] Fitness: 2.1722640005107063
Generation 172 Best Solution: [183.3362395214969, 177.3694049080318,
174.30909302050807, 75.84978591940111] Fitness: 2.072357082286085
Generation 173 Best Solution: [182.90717563002164, 176.4498684323297,
175.01355229195534, 75.84978591940111] Fitness: 1.725520352739009
Generation 174 Best Solution: [182.90717563002164, 177.42537256464803,
174.4072635906134, 74.47417248906368] Fitness: 1.3287562479836457
Generation 175 Best Solution: [183.52504847268602, 175.6346354474895,
175.01355229195534, 75.84978591940111] Fitness: 1.926876025782739
Generation 176 Best Solution: [182.90717563002164, 176.53942382312198,
174.5673508246449, 75.83928851598533] Fitness: 1.304613955066837
Generation 177 Best Solution: [183.3362395214969, 176.220143482188,
174.5673508246449, 75.83928851598533] Fitness: 1.1609086843325351
Generation 178 Best Solution: [183.3362395214969, 176.35042403979034,
174.5673508246449, 75.84978591940111] Fitness: 1.1940483540132316
Generation 179 Best Solution: [183.3362395214969, 176.35042403979034,
175.10370852627727, 75.83928851598533] Fitness: 1.7432298771470276
Generation 180 Best Solution: [183.3362395214969, 176.35042403979034,
174.4072635906134, 75.84978591940111] Fitness: 1.1084079573939607
Generation 181 Best Solution: [183.3362395214969, 176.35042403979034,
174.4072635906134, 75.84978591940111] Fitness: 1.1084079573939607
Generation 182 Best Solution: [182.90717563002164, 176.2242139804773,
175.01355229195534, 75.84978591940111] Fitness: 1.681616614241123
Generation 183 Best Solution: [183.48399710747816, 176.00540198143392,
175.01355229195534, 75.84978591940111] Fitness: 1.6937457473307145
Generation 184 Best Solution: [183.48399710747816, 176.00540198143392,
175.01355229195534, 75.16045135101508] Fitness: 0.8688342672318696
Generation 185 Best Solution: [183.48399710747816, 176.00540198143392,
175.01355229195534, 75.83928851598533] Fitness: 1.675839215420532
Generation 186 Best Solution: [183.52504847268602, 176.39924509019772,
174.5673508246449, 75.84978591940111] Fitness: 1.2669159861281716

Generation 187 Best Solution: [183.48399710747816, 176.00540198143392,
174.5673508246449, 75.84978591940111] Fitness: 1.2521502671965643
Generation 188 Best Solution: [183.52504847268602, 176.00540198143392,
174.5673508246449, 75.83928851598533] Fitness: 1.2519690655608517
Generation 189 Best Solution: [183.52504847268602, 176.00540198143392,
175.01355229195534, 75.84978591940111] Fitness: 1.7114710776051845
Generation 190 Best Solution: [183.52504847268602, 176.00540198143392,
174.2243403379281, 75.84978591940111] Fitness: 1.1334368746148604
Generation 191 Best Solution: [183.52504847268602, 176.00540198143392,
174.2243403379281, 75.84978591940111] Fitness: 1.1334368746148604
Generation 192 Best Solution: [182.90717563002164, 176.00540198143392,
175.01355229195534, 74.8913505899149] Fitness: 0.7584834296951957
Generation 193 Best Solution: [182.90717563002164, 176.5938347799539,
175.01355229195534, 75.83928851598533] Fitness: 1.7755338913247882
Generation 194 Best Solution: [182.90717563002164, 176.5938347799539,
175.01355229195534, 75.84978591940111] Fitness: 1.7934404232349705
Generation 195 Best Solution: [182.90717563002164, 177.15255600454827,
173.95247387549082, 75.83928851598533] Fitness: 1.7727800213932243
Generation 196 Best Solution: [182.90717563002164, 176.5938347799539,
175.01355229195534, 75.84978591940111] Fitness: 1.7934404232349705
Generation 197 Best Solution: [182.90717563002164, 177.42537256464803,
173.95247387549082, 75.14939732573225] Fitness: 1.4009838029076762
Generation 198 Best Solution: [182.90717563002164, 175.73173870604023,
174.2243403379281, 75.83928851598533] Fitness: 1.2551048492871675
Generation 199 Best Solution: [182.90717563002164, 175.73173870604023,
174.2243403379281, 75.83928851598533] Fitness: 1.2551048492871675
Generation 200 Best Solution: [183.52504847268602, 176.19919996498157,
174.5487728344732, 75.83928851598533] Fitness: 1.2095262614579918
Generation 201 Best Solution: [182.90717563002164, 176.19919996498157,
174.5487728344732, 75.83928851598533] Fitness: 1.210040284067876
Generation 202 Best Solution: [183.52504847268602, 176.31163935203242,
175.01355229195534, 75.83928851598533] Fitness: 1.671142151687697
Generation 203 Best Solution: [183.52504847268602, 177.01354913472656,
174.3478040416711, 75.84978591940111] Fitness: 1.619123022469172
Generation 204 Best Solution: [183.52504847268602, 177.42537256464803,
175.01355229195534, 74.32659912572889] Fitness: 1.9234240202819144
Generation 205 Best Solution: [182.90717563002164, 177.42537256464803,
174.2243403379281, 75.83928851598533] Fitness: 2.198176838920884
Generation 206 Best Solution: [182.90717563002164, 177.42537256464803,
175.01355229195534, 74.49383474011664] Fitness: 1.849405377240793
Generation 207 Best Solution: [182.90717563002164, 177.42537256464803,
174.2243403379281, 75.83928851598533] Fitness: 2.198176838920884
Generation 208 Best Solution: [182.90717563002164, 176.21508054265144,
175.01355229195534, 75.83928851598533] Fitness: 1.6635413795841054
Generation 209 Best Solution: [183.24876348765414, 177.42537256464803,
175.01355229195534, 74.30224438641852] Fitness: 1.8672260595614942
Generation 210 Best Solution: [183.52504847268602, 176.40079537511144,
175.01355229195534, 75.83928851598533] Fitness: 1.691053013741773

Generation 211 Best Solution: [183.52504847268602, 176.40079537511144,
175.01355229195534, 75.83928851598533] Fitness: 1.691053013741773
Generation 212 Best Solution: [183.24876348765414, 177.42537256464803,
174.20927025428193, 75.83928851598533] Fitness: 2.125163792510099
Generation 213 Best Solution: [183.52504847268602, 177.42537256464803,
175.01355229195534, 75.38050449487478] Fitness: 2.1545746772000833
Generation 214 Best Solution: [182.90717563002164, 177.42537256464803,
175.01355229195534, 75.38050449487478] Fitness: 2.155088699809968
Generation 215 Best Solution: [183.40209618434142, 177.42537256464803,
175.01355229195534, 74.44173604743848] Fitness: 1.822075132163808
Generation 216 Best Solution: [182.90717563002164, 177.42537256464803,
175.01355229195534, 74.44173604743848] Fitness: 1.8681255789676958
Generation 217 Best Solution: [182.83715567716158, 177.42537256464803,
173.84420374389936, 74.44173604743848] Fitness: 1.3812389800956062
Generation 218 Best Solution: [182.90717563002164, 177.42537256464803,
173.84420374389936, 74.44173604743848] Fitness: 1.3450560630519666
Generation 219 Best Solution: [183.52504847268602, 177.42537256464803,
175.01355229195534, 74.44173604743848] Fitness: 1.8676115563578113
Generation 220 Best Solution: [183.3585571078664, 176.63394142952166,
175.01355229195534, 75.83928851598533] Fitness: 1.7432533626001947
Generation 221 Best Solution: [183.3585571078664, 176.63394142952166,
175.01355229195534, 75.83928851598533] Fitness: 1.7432533626001947
Generation 222 Best Solution: [183.44534467703258, 177.42537256464803,
175.01355229195534, 74.44173604743848] Fitness: 1.835507276963675
Generation 223 Best Solution: [183.41891044535907, 177.42537256464803,
175.01355229195534, 74.44173604743848] Fitness: 1.8269639621598281
Generation 224 Best Solution: [183.61901489934795, 176.3052821783289,
175.01355229195534, 75.83928851598533] Fitness: 1.7202663962555225
Generation 225 Best Solution: [183.52504847268602, 176.4772563153711,
175.01355229195534, 75.83928851598533] Fitness: 1.7176261547091798
Generation 226 Best Solution: [182.90717563002164, 176.4772563153711,
175.01355229195534, 75.83928851598533] Fitness: 1.7181401773190639
Generation 227 Best Solution: [183.52504847268602, 177.42537256464803,
175.01355229195534, 74.48966244903845] Fitness: 1.850240578054066
Generation 228 Best Solution: [183.52504847268602, 175.73502396158122,
174.5473415474843, 75.83928851598533] Fitness: 1.375887646240991
Generation 229 Best Solution: [183.52504847268602, 177.42537256464803,
175.01355229195534, 74.58533596221027] Fitness: 1.8258676365724833
Generation 230 Best Solution: [183.0178210742049, 177.42537256464803,
175.01355229195534, 74.58533596221027] Fitness: 1.7841978100907614
Generation 231 Best Solution: [183.52504847268602, 176.93679224568297,
175.01355229195534, 75.21288810420138] Fitness: 1.2927623892508298
Generation 232 Best Solution: [183.52504847268602, 176.11259572130868,
175.1193234817397, 75.83928851598533] Fitness: 1.81818020687092
Generation 233 Best Solution: [183.52504847268602, 176.93679224568297,
175.01355229195534, 75.6627903045999] Fitness: 1.7857466874466508
Generation 234 Best Solution: [183.52504847268602, 176.02368159695715,
175.01355229195534, 75.83928851598533] Fitness: 1.6882783127405496

Generation 235 Best Solution: [183.52504847268602, 177.42537256464803,
175.01355229195534, 74.7762611958766] Fitness: 1.8182684053278673
Generation 236 Best Solution: [183.81945009816653, 177.42537256464803,
173.78834152943213, 75.83928851598533] Fitness: 2.4801580187375274
Generation 237 Best Solution: [183.0494741489054, 177.42537256464803,
173.78834152943213, 75.83928851598533] Fitness: 2.2286121232659735
Generation 238 Best Solution: [183.81945009816653, 177.42537256464803,
173.78834152943213, 75.83928851598533] Fitness: 2.4801580187375274
Generation 239 Best Solution: [183.144288434879, 177.42537256464803,
175.01355229195534, 74.57410006089727] Fitness: 1.7606228989295045
Generation 240 Best Solution: [182.9699215762675, 177.42537256464803,
175.01355229195534, 74.90707357032723] Fitness: 1.8189650903718102
Generation 241 Best Solution: [183.0494741489054, 177.42537256464803,
175.01355229195534, 75.10347412051046] Fitness: 1.8820412598844527
Generation 242 Best Solution: [183.52504847268602, 176.03356061028637,
175.01355229195534, 75.83928851598533] Fitness: 1.685630061991425
Generation 243 Best Solution: [183.0494741489054, 177.42537256464803,
175.01355229195534, 74.83882715803439] Fitness: 1.7773135868863532
Generation 244 Best Solution: [183.46200999976236, 177.42537256464803,
175.01355229195534, 74.90707357032723] Fitness: 1.8184477489656203
Generation 245 Best Solution: [183.0494741489054, 177.42537256464803,
175.01355229195534, 74.90707357032723] Fitness: 1.794267739141252
Generation 246 Best Solution: [182.9699215762675, 177.42537256464803,
174.25285111691713, 74.90707357032723] Fitness: 1.2455465997780437
Generation 247 Best Solution: [183.52504847268602, 177.42537256464803,
174.25285111691713, 74.90707357032723] Fitness: 1.271208748763797
Generation 248 Best Solution: [183.52504847268602, 176.0249021431258,
175.03683030048953, 75.83928851598533] Fitness: 1.7191773248467306
Generation 249 Best Solution: [183.46200999976236, 175.92372307796862,
175.01355229195534, 75.83928851598533] Fitness: 1.6971289348541454
Generation 250 Best Solution: [183.63784772277154, 177.2547727789226,
175.01355229195534, 75.17117273323508] Fitness: 1.7479183344970557
Generation 251 Best Solution: [183.47029632105233, 177.2547727789226,
175.01355229195534, 75.17117273323508] Fitness: 1.6631192241686859
Generation 252 Best Solution: [183.47029632105233, 177.42537256464803,
175.01355229195534, 74.95852070945871] Fitness: 1.8389479385703673
Generation 253 Best Solution: [183.0494741489054, 177.42537256464803,
175.01355229195534, 74.84096785364203] Fitness: 1.77773925619907
Generation 254 Best Solution: [183.0494741489054, 177.42537256464803,
175.01355229195534, 74.84096785364203] Fitness: 1.77773925619907
Generation 255 Best Solution: [183.0494741489054, 177.42537256464803,
175.01355229195534, 74.83522526030438] Fitness: 1.7766128761737452
Generation 256 Best Solution: [183.46200999976236, 175.76595373551183,
174.33959040164456, 75.86458215314698] Fitness: 1.27446311051526
Generation 257 Best Solution: [183.0494741489054, 177.42537256464803,
175.01355229195534, 74.83522526030438] Fitness: 1.7766128761737452
Generation 258 Best Solution: [183.46200999976236, 177.42537256464803,
174.24263005596643, 74.95852070945871] Fitness: 1.2606335306049974

Generation 259 Best Solution: [183.46200999976236, 177.42537256464803,
174.33959040164456, 74.46341683690882] Fitness: 1.281268644952654
Generation 260 Best Solution: [182.98105025113762, 177.42537256464803,
175.01355229195534, 74.46341683690882] Fitness: 1.829638378310029
Generation 261 Best Solution: [183.63784772277154, 176.52844035044234,
175.01355229195534, 75.83928851598533] Fitness: 1.802035237665931
Generation 262 Best Solution: [183.46200999976236, 176.32997971530548,
175.01355229195534, 75.83928851598533] Fitness: 1.648084453401258
Generation 263 Best Solution: [183.46200999976236, 176.32997971530548,
175.01355229195534, 75.83928851598533] Fitness: 1.648084453401258
Generation 264 Best Solution: [183.46200999976236, 176.32997971530548,
175.01355229195534, 75.83928851598533] Fitness: 1.648084453401258
Generation 265 Best Solution: [183.42078110933866, 176.32997971530548,
175.01355229195534, 75.83928851598533] Fitness: 1.6341864693608545
Generation 266 Best Solution: [183.185562952257, 175.81726717089686,
174.6953373004089, 75.83928851598533] Fitness: 1.3610215508703627
Generation 267 Best Solution: [183.46200999976236, 177.42537256464803,
173.85558118038537, 75.83928851598533] Fitness: 2.221656276975321
Generation 268 Best Solution: [183.42078110933866, 175.9777204575242,
175.01355229195534, 75.83928851598533] Fitness: 1.6624464226557127
Generation 269 Best Solution: [183.08388704359237, 176.11480584361914,
175.01355229195534, 75.83928851598533] Fitness: 1.611301662314656
Generation 270 Best Solution: [183.08388704359237, 176.11480584361914,
175.01355229195534, 75.83928851598533] Fitness: 1.611301662314656
Generation 271 Best Solution: [183.63784772277154, 175.68977792922203,
175.01355229195534, 75.83928851598533] Fitness: 1.9256006475820617
Generation 272 Best Solution: [183.63784772277154, 175.68977792922203,
175.01355229195534, 75.83928851598533] Fitness: 1.9256006475820617
Generation 273 Best Solution: [183.63784772277154, 177.42537256464803,
175.01355229195534, 74.83526367245496] Fitness: 1.888700354206003
Generation 274 Best Solution: [183.46200999976236, 177.42537256464803,
175.01355229195534, 74.95481663873181] Fitness: 1.8344616170980557
Generation 275 Best Solution: [183.63784772277154, 177.42537256464803,
175.01355229195534, 74.95481663873181] Fitness: 1.9223617152933914
Generation 276 Best Solution: [182.98105025113762, 176.13042309853682,
175.01355229195534, 75.83928851598533] Fitness: 1.637730548434642
Generation 277 Best Solution: [182.98105025113762, 176.13042309853682,
175.01355229195534, 75.83928851598533] Fitness: 1.637730548434642
Generation 278 Best Solution: [182.98105025113762, 177.42537256464803,
175.01355229195534, 74.47793991960425] Fitness: 1.8244829390857324
Generation 279 Best Solution: [183.05338366365572, 177.42537256464803,
175.01355229195534, 74.47793991960425] Fitness: 1.8028426219398686
Generation 280 Best Solution: [182.98105025113762, 176.5520157224173,
175.01355229195534, 75.83928851598533] Fitness: 1.7223986222807774
Generation 281 Best Solution: [183.46200999976236, 177.42537256464803,
174.70158020368336, 74.47793991960425] Fitness: 1.4878330605122099
Generation 282 Best Solution: [182.98105025113762, 177.42537256464803,
174.70158020368336, 74.47793991960425] Fitness: 1.4843243686413965

Generation 283 Best Solution: [183.18675517381033, 176.58209817729406,
175.01355229195534, 75.83928851598533] Fitness: 1.6976653780775832
Generation 284 Best Solution: [183.63614377642367, 177.42537256464803,
175.01355229195534, 74.88180466141517] Fitness: 1.8981815813885063
Generation 285 Best Solution: [183.46200999976236, 176.2477126605401,
175.01355229195534, 75.83928851598533] Fitness: 1.6380257196923669
Generation 286 Best Solution: [183.46200999976236, 176.038479309691,
175.1364160966933, 75.83928851598533] Fitness: 1.8322202796114104
Generation 287 Best Solution: [183.46200999976236, 176.80489483916597,
175.01355229195534, 75.83928851598533] Fitness: 1.9046126670828363
Generation 288 Best Solution: [183.46200999976236, 176.80489483916597,
175.01355229195534, 75.83928851598533] Fitness: 1.9046126670828363
Generation 289 Best Solution: [182.98105025113762, 177.42537256464803,
175.01355229195534, 75.46147141542313] Fitness: 2.211560120532589
Generation 290 Best Solution: [182.98105025113762, 176.1386112602753,
175.01355229195534, 75.83928851598533] Fitness: 1.6368362027663468
Generation 291 Best Solution: [182.98105025113762, 176.1386112602753,
175.01355229195534, 75.83928851598533] Fitness: 1.6368362027663468
Generation 292 Best Solution: [182.98105025113762, 176.1386112602753,
175.01355229195534, 75.83928851598533] Fitness: 1.6368362027663468
Generation 293 Best Solution: [182.98105025113762, 176.1386112602753,
175.01355229195534, 75.83928851598533] Fitness: 1.6368362027663468
Generation 294 Best Solution: [182.98105025113762, 176.1386112602753,
175.01355229195534, 75.83928851598533] Fitness: 1.6368362027663468
Generation 295 Best Solution: [183.7421187903634, 176.1386112602753,
175.01355229195534, 75.83928851598533] Fitness: 1.8022748320393454
Generation 296 Best Solution: [183.6870037218008, 177.42537256464803,
173.78087157822026, 75.21528703340782] Fitness: 1.6099582264975774
Generation 297 Best Solution: [182.98105025113762, 176.1386112602753,
175.01355229195534, 75.83928851598533] Fitness: 1.6368362027663468
Generation 298 Best Solution: [182.98105025113762, 176.09077894441944,
175.01355229195534, 75.83928851598533] Fitness: 1.6434828544536197
Generation 299 Best Solution: [182.98105025113762, 175.65409661269854,
173.96260260458692, 75.83928851598533] Fitness: 1.2994161882823974
Generation 300 Best Solution: [182.98105025113762, 175.98419558765633,
175.01355229195534, 75.83928851598533] Fitness: 1.670637000431663
Generation 301 Best Solution: [182.98105025113762, 177.42537256464803,
175.01355229195534, 74.41530800695439] Fitness: 1.8489760288240675
Generation 302 Best Solution: [182.98105025113762, 175.91934614295496,
175.01355229195534, 75.83928851598533] Fitness: 1.6954966266388225
Generation 303 Best Solution: [182.98105025113762, 177.42537256464803,
175.01355229195534, 75.09781475666445] Fitness: 1.8993737418640453
Generation 304 Best Solution: [182.98105025113762, 177.42537256464803,
175.01355229195534, 75.09781475666445] Fitness: 1.8993737418640453
Generation 305 Best Solution: [183.6870037218008, 176.83756974928409,
175.01355229195534, 75.83928851598533] Fitness: 2.0554690028892644
Generation 306 Best Solution: [182.98105025113762, 176.83756974928409,
175.01355229195534, 75.83928851598533] Fitness: 1.9311926152651355

Generation 307 Best Solution: [182.98105025113762, 176.83756974928409,
175.01355229195534, 75.83928851598533] Fitness: 1.9311926152651355
Generation 308 Best Solution: [183.7421187903634, 177.42537256464803,
175.01355229195534, 74.73997054476919] Fitness: 1.9512544553397537
Generation 309 Best Solution: [183.65735423617485, 176.61233980147003,
175.01355229195534, 75.83928851598533] Fitness: 1.8603346686859314
Generation 310 Best Solution: [183.82758535799775, 175.75542537196478,
175.01355229195534, 75.51574977444866] Fitness: 1.5539006335963608
Generation 311 Best Solution: [183.65735423617485, 177.42537256464803,
175.01355229195534, 74.54005217150232] Fitness: 1.9100220047058836
Generation 312 Best Solution: [183.65735423617485, 177.42537256464803,
175.01355229195534, 74.54005217150232] Fitness: 1.9100220047058836
Generation 313 Best Solution: [183.65735423617485, 177.42537256464803,
174.63056634708886, 74.53757712102711] Fitness: 1.5134571404492856
Generation 314 Best Solution: [183.65735423617485, 177.42537256464803,
174.63056634708886, 74.53757712102711] Fitness: 1.5134571404492856
Generation 315 Best Solution: [183.49452958291067, 177.42537256464803,
175.01355229195534, 74.54005217150232] Fitness: 1.822273704434325
Generation 316 Best Solution: [183.65735423617485, 177.42537256464803,
174.53353852365439, 74.54005217150232] Fitness: 1.4471360340644113
Generation 317 Best Solution: [183.65735423617485, 175.7103396172625,
174.20092777218014, 75.83928851598533] Fitness: 1.3416512513269034
Generation 318 Best Solution: [183.65735423617485, 176.24316925497772,
175.01355229195534, 75.83928851598533] Fitness: 1.73827514922159
Generation 319 Best Solution: [182.96650655889545, 177.42537256464803,
174.20092777218014, 75.83928851598533] Fitness: 2.1703947719335392
Generation 320 Best Solution: [183.3201821693255, 176.974846426046,
175.01355229195534, 75.83928851598533] Fitness: 2.041501693904304
Generation 321 Best Solution: [183.65735423617485, 177.42537256464803,
175.01355229195534, 74.68091736412472] Fitness: 1.8895581169501579
Generation 322 Best Solution: [183.49452958291067, 176.974846426046,
174.23534252702402, 75.83928851598533] Fitness: 1.5149734559041308
Generation 323 Best Solution: [183.67491146178523, 176.974846426046,
174.23534252702402, 75.83928851598533] Fitness: 1.6145588249263545
Generation 324 Best Solution: [183.67491146178523, 177.42537256464803,
175.01355229195534, 74.92261662251111] Fitness: 1.9356320040783042
Generation 325 Best Solution: [183.65735423617485, 176.2684285734623,
175.01355229195534, 75.43056348282569] Fitness: 1.1714159457378375
Generation 326 Best Solution: [183.65735423617485, 176.2684285734623,
175.01355229195534, 75.83928851598533] Fitness: 1.7401114605363253
Generation 327 Best Solution: [183.65735423617485, 176.2684285734623,
175.01355229195534, 75.83928851598533] Fitness: 1.7401114605363253
Generation 328 Best Solution: [183.65735423617485, 177.42537256464803,
174.23540696852365, 74.37126434381096] Fitness: 1.39735416628597
Generation 329 Best Solution: [183.60767980479588, 176.91115262844573,
174.23540696852365, 75.83928851598533] Fitness: 1.5014601222126602
Generation 330 Best Solution: [183.67491146178523, 175.88971701438376,
175.01355229195534, 75.43056348282569] Fitness: 1.2561136377609983

Generation 331 Best Solution: [183.60767980479588, 175.88971701438376,
175.01355229195534, 75.43056348282569] Fitness: 1.213290874213299
Generation 332 Best Solution: [183.67491146178523, 175.88971701438376,
174.23540696852365, 74.93868004493544] Fitness: 0.32757085425557064
Generation 333 Best Solution: [183.60767980479588, 175.99637192036116,
175.01355229195534, 75.83928851598533] Fitness: 1.739704832786893
Generation 334 Best Solution: [183.60767980479588, 177.05887577716155,
175.01355229195534, 74.80133217959381] Fitness: 1.2958046303693518
Generation 335 Best Solution: [183.34313031238955, 175.99637192036116,
175.01355229195534, 75.83928851598533] Fitness: 1.6370311674117914
Generation 336 Best Solution: [183.67491146178523, 175.6691753946336,
175.01355229195534, 75.05553809314098] Fitness: 1.0963025476663144
Generation 337 Best Solution: [183.65735423617485, 177.42537256464803,
174.35741081153986, 75.26058955773533] Fitness: 1.5725485116775886
Generation 338 Best Solution: [183.60767980479588, 175.99637192036116,
175.01355229195534, 75.83928851598533] Fitness: 1.739704832786893
Generation 339 Best Solution: [183.67491146178523, 176.9860454424479,
174.52810863882857, 75.83928851598533] Fitness: 1.7377982900943563
Generation 340 Best Solution: [183.65735423617485, 176.59135849212885,
174.52810863882857, 75.779818705405] Fitness: 1.2834500549546346
Generation 341 Best Solution: [183.65735423617485, 176.59135849212885,
174.52810863882857, 75.83928851598533] Fitness: 1.3817733016233869
Generation 342 Best Solution: [183.60767980479588, 176.92542601522464,
175.01355229195534, 75.779818705405] Fitness: 1.9947458726351155
Generation 343 Best Solution: [183.46981337831988, 176.31601695038043,
175.01355229195534, 75.779818705405] Fitness: 1.5502561318299422
Generation 344 Best Solution: [183.65735423617485, 176.92542601522464,
174.6098944167365, 75.83928851598533] Fitness: 1.7116839709226224
Generation 345 Best Solution: [183.60767980479588, 176.92542601522464,
174.6098944167365, 75.83928851598533] Fitness: 1.6806982761255878
Generation 346 Best Solution: [183.67491146178523, 176.92542601522464,
174.6098944167365, 75.83928851598533] Fitness: 1.7235210396732872
Generation 347 Best Solution: [183.65735423617485, 176.3169199232929,
175.01355229195534, 75.83928851598533] Fitness: 1.7463189047857353
Generation 348 Best Solution: [183.65735423617485, 176.3169199232929,
175.01355229195534, 75.83928851598533] Fitness: 1.7463189047857353
Generation 349 Best Solution: [183.60767980479588, 177.42537256464803,
174.11598116854987, 75.83928851598533] Fitness: 2.234587448615927
Generation 350 Best Solution: [183.60767980479588, 177.42537256464803,
174.11598116854987, 75.83928851598533] Fitness: 2.234587448615927
Generation 351 Best Solution: [183.60767980479588, 176.95596493188464,
175.09051471976716, 75.83928851598533] Fitness: 2.2330291905011643
Generation 352 Best Solution: [183.60767980479588, 177.42537256464803,
174.11598116854987, 75.83928851598533] Fitness: 2.234587448615927
Generation 353 Best Solution: [183.3243508573867, 177.42537256464803,
174.11598116854987, 75.83928851598533] Fitness: 2.1286159066719534
Generation 354 Best Solution: [183.60767980479588, 176.05894280108492,
175.01355229195534, 75.83928851598533] Fitness: 1.722841107010204

Generation 355 Best Solution: [183.60767980479588, 177.42537256464803,
175.01355229195534, 74.5929140579644] Fitness: 1.8678678975131628
Generation 356 Best Solution: [183.60767980479588, 177.42537256464803,
175.01355229195534, 74.5929140579644] Fitness: 1.8678678975131628
Generation 357 Best Solution: [183.3243508573867, 177.42537256464803,
175.01355229195534, 74.5929140579644] Fitness: 1.7618963555691893
Generation 358 Best Solution: [183.67491146178523, 177.42537256464803,
175.01355229195534, 74.5929140579644] Fitness: 1.9106906610608618
Generation 359 Best Solution: [183.67491146178523, 176.3119623374476,
175.01355229195534, 75.83928851598533] Fitness: 1.7573594793886347
Generation 360 Best Solution: [183.3243508573867, 177.17795530068787,
175.01355229195534, 75.83928851598533] Fitness: 2.3069350051538953
Generation 361 Best Solution: [183.60767980479588, 176.28830423789665,
175.0615301957506, 75.68854272265462] Fitness: 1.5375954194514088
Generation 362 Best Solution: [183.60767980479588, 176.28830423789665,
175.01355229195534, 75.83928851598533] Fitness: 1.7112435100038303
Generation 363 Best Solution: [183.3243508573867, 176.42372316013976,
175.01355229195534, 75.83928851598533] Fitness: 1.6354731703794905
Generation 364 Best Solution: [183.6110708029342, 176.52801931863095,
175.0615301957506, 75.68854272265462] Fitness: 1.6118053411732447
Generation 365 Best Solution: [183.6110708029342, 176.52801931863095,
175.0615301957506, 75.47702513401339] Fitness: 1.3340455429296376
Generation 366 Best Solution: [183.67491146178523, 176.838589984726,
174.97755160009856, 75.47702513401339] Fitness: 1.4846352746832798
Generation 367 Best Solution: [183.67491146178523, 177.01698654336383,
175.01355229195534, 75.47702513401339] Fitness: 1.7241299977727274
Generation 368 Best Solution: [183.67491146178523, 176.52801931863095,
174.97755160009856, 74.98672736770403] Fitness: 0.8771278494182558
Generation 369 Best Solution: [183.67491146178523, 176.52801931863095,
174.97755160009856, 75.83928851598533] Fitness: 1.779573981236346
Generation 370 Best Solution: [182.82120395146296, 176.5051934989612,
174.97755160009856, 75.83928851598533] Fitness: 1.7287867142097824
Generation 371 Best Solution: [182.82120395146296, 176.5051934989612,
175.07419906966015, 75.83928851598533] Fitness: 1.858566192978317
Generation 372 Best Solution: [183.67491146178523, 177.17795530068787,
173.91363648503386, 75.83928851598533] Fitness: 1.9064406149442221
Generation 373 Best Solution: [183.67491146178523, 176.53538789091974,
175.01355229195534, 75.83928851598533] Fitness: 1.8298639025770846
Generation 374 Best Solution: [183.77449741927094, 175.8221451768753,
175.01355229195534, 75.83928851598533] Fitness: 1.9363166371421878
Generation 375 Best Solution: [183.77449741927094, 175.8221451768753,
174.97755160009856, 75.83928851598533] Fitness: 1.889611934491628
Generation 376 Best Solution: [183.3243508573867, 176.4672460745904,
174.97755160009856, 75.83928851598533] Fitness: 1.6043160246707693
Generation 377 Best Solution: [183.67491146178523, 176.4672460745904,
174.97755160009856, 75.83928851598533] Fitness: 1.753110330162442
Generation 378 Best Solution: [183.60884112823183, 175.7443216582992,
175.07419906966015, 75.83928851598533] Fitness: 1.9508680795815856

Generation 379 Best Solution: [183.60884112823183, 175.7443216582992,
174.97755160009856, 74.5200190501398] Fitness: 0.8864116796797377
Generation 380 Best Solution: [183.60884112823183, 175.7443216582992,
174.17939846262036, 75.83928851598533] Fitness: 1.2849591239102427
Generation 381 Best Solution: [183.3243508573867, 176.3845269639858,
174.17939846262036, 75.83928851598533] Fitness: 1.0410687600249866
Generation 382 Best Solution: [183.77449741927094, 176.3188529724687,
174.97755160009856, 75.03758274727848] Fitness: 0.9084685749284236
Generation 383 Best Solution: [183.3243508573867, 175.7443216582992,
174.0837913941743, 75.83928851598533] Fitness: 1.178172308806336
Generation 384 Best Solution: [183.3243508573867, 177.42537256464803,
174.17939846262036, 74.5200190501398] Fitness: 1.1955583286000904
Generation 385 Best Solution: [183.3243508573867, 177.42537256464803,
174.17939846262036, 74.5200190501398] Fitness: 1.1955583286000904
Generation 386 Best Solution: [183.3243508573867, 176.7494278532973,
174.17939846262036, 75.83928851598533] Fitness: 1.2379204243983624
Generation 387 Best Solution: [183.3243508573867, 176.76952139070156,
174.78365222260953, 75.83928851598533] Fitness: 1.5725736431025827
Generation 388 Best Solution: [183.3243508573867, 177.42537256464803,
173.96228158324027, 74.6846977785022] Fitness: 1.1891116856284765
Generation 389 Best Solution: [183.3243508573867, 177.42537256464803,
174.97755160009856, 74.6846977785022] Fitness: 1.7057571012435524
Generation 390 Best Solution: [183.3243508573867, 176.74009162459126,
175.07419906966015, 75.61969281909619] Fitness: 1.5596110481130814
Generation 391 Best Solution: [183.3243508573867, 176.74009162459126,
175.07419906966015, 75.61969281909619] Fitness: 1.5596110481130814
Generation 392 Best Solution: [183.3243508573867, 176.74009162459126,
174.97755160009856, 75.83928851598533] Fitness: 1.7665235975479916
Generation 393 Best Solution: [183.3243508573867, 176.67136262954367,
175.01355229195534, 75.83928851598533] Fitness: 1.7618471433265956
Generation 394 Best Solution: [183.64989789691901, 176.4867691661279,
174.80221958349452, 75.61969281909619] Fitness: 1.2079182928244405
Generation 395 Best Solution: [183.64446864400585, 177.42537256464803,
174.97755160009856, 74.51990865910193] Fitness: 1.8602828165329497
Generation 396 Best Solution: [183.3243508573867, 175.7162212089509,
173.79126247355592, 75.83928851598533] Fitness: 1.2830292257769158
Generation 397 Best Solution: [183.3243508573867, 176.0732667929871,
175.01355229195534, 75.61969281909619] Fitness: 1.277143102247365
Generation 398 Best Solution: [183.3243508573867, 176.15723847654513,
175.01355229195534, 75.83928851598533] Fitness: 1.6022368735279784
Generation 399 Best Solution: [183.3243508573867, 176.03627302952418,
174.97755160009856, 75.61969281909619] Fitness: 1.238904242320961
Generation 400 Best Solution: [183.64446864400585, 177.42537256464803,
175.07419906966015, 74.78562998484487] Fitness: 1.9683140631468647
Generation 401 Best Solution: [183.3243508573867, 177.42537256464803,
175.01355229195534, 75.12904561611036] Fitness: 1.8854584647209884
Generation 402 Best Solution: [183.3243508573867, 176.24620464073703,
174.97755160009856, 75.83928851598533] Fitness: 1.5547832837744004

Generation 403 Best Solution: [183.3243508573867, 176.76514439031794,
175.01355229195534, 75.65886597693621] Fitness: 1.5517885112230927
Generation 404 Best Solution: [183.3243508573867, 177.42537256464803,
174.30273033133446, 74.4938979373701] Fitness: 1.223832686842876
Generation 405 Best Solution: [183.82415127504368, 177.42537256464803,
174.30273033133446, 74.4938979373701] Fitness: 1.4919139352058364
Generation 406 Best Solution: [183.3243508573867, 176.76579374594584,
175.01355229195534, 75.83928851598533] Fitness: 1.8342632768620035
Generation 407 Best Solution: [183.3243508573867, 177.36629081852846,
175.01355229195534, 74.85167555212924] Fitness: 1.6623756973335013
Generation 408 Best Solution: [183.3243508573867, 177.36629081852846,
175.01355229195534, 74.85167555212924] Fitness: 1.6623756973335013
Generation 409 Best Solution: [183.3243508573867, 175.7753330649641,
174.97755160009856, 75.83928851598533] Fitness: 1.693618223773273
Generation 410 Best Solution: [183.3243508573867, 175.7753330649641,
174.97755160009856, 75.83928851598533] Fitness: 1.693618223773273
Generation 411 Best Solution: [183.3243508573867, 177.36629081852846,
175.01355229195534, 74.68981481194412] Fitness: 1.6469798641820552
Generation 412 Best Solution: [183.3243508573867, 177.42537256464803,
174.97755160009856, 74.68981481194412] Fitness: 1.7056029989304158
Generation 413 Best Solution: [183.3243508573867, 177.42537256464803,
175.01355229195534, 74.37157111661898] Fitness: 1.8366297272669274
Generation 414 Best Solution: [182.81137305817015, 175.7372871526466,
175.01355229195534, 75.18330652867714] Fitness: 1.0894835399867793
Generation 415 Best Solution: [183.3243508573867, 175.69168322522702,
174.97755160009856, 75.83928851598533] Fitness: 1.7530713253624843
Generation 416 Best Solution: [183.3243508573867, 176.53664298497623,
175.01355229195534, 75.83928851598533] Fitness: 1.6816883875193764
Generation 417 Best Solution: [183.71809583774527, 176.91304838009665,
173.7766381799988, 75.83928851598533] Fitness: 1.6631434009730168
Generation 418 Best Solution: [183.71809583774527, 176.91304838009665,
173.7766381799988, 75.83928851598533] Fitness: 1.6631434009730168
Generation 419 Best Solution: [183.3243508573867, 176.78991668645818,
174.74614610417885, 75.83928851598533] Fitness: 1.5544032080183603
Generation 420 Best Solution: [183.3243508573867, 177.42537256464803,
174.74614610417885, 74.98007272717993] Fitness: 1.5073635433482968
Generation 421 Best Solution: [183.3243508573867, 177.42537256464803,
174.74614610417885, 75.17229697751249] Fitness: 1.6137168534377109
Generation 422 Best Solution: [183.2240138767393, 177.42537256464803,
174.74614610417885, 75.17229697751249] Fitness: 1.6050604258613288
Generation 423 Best Solution: [183.3243508573867, 176.70712388406596,
175.01355229195534, 75.52486188686431] Fitness: 1.3279704987010283
Generation 424 Best Solution: [183.2240138767393, 176.70712388406596,
175.01355229195534, 75.52486188686431] Fitness: 1.319314071124646
Generation 425 Best Solution: [183.2240138767393, 176.70712388406596,
175.01355229195534, 75.83928851598533] Fitness: 1.7790412708953411
Generation 426 Best Solution: [183.3243508573867, 176.04918082672646,
175.01355229195534, 75.83928851598533] Fitness: 1.6191139140300348

Generation 427 Best Solution: [183.3243508573867, 176.04918082672646,
175.01355229195534, 75.83928851598533] Fitness: 1.6191139140300348
Generation 428 Best Solution: [183.3243508573867, 176.05294165117792,
175.01355229195534, 75.83928851598533] Fitness: 1.6182323461595902
Generation 429 Best Solution: [183.3243508573867, 176.13793562941495,
175.01355229195534, 75.83928851598533] Fitness: 1.603966783502363
Generation 430 Best Solution: [183.3243508573867, 176.13793562941495,
175.01355229195534, 75.83928851598533] Fitness: 1.603966783502363
Generation 431 Best Solution: [183.3243508573867, 176.13793562941495,
175.01355229195534, 75.83928851598533] Fitness: 1.603966783502363
Generation 432 Best Solution: [183.46768764985768, 176.13793562941495,
175.01355229195534, 75.83928851598533] Fitness: 1.6425285203057318
Generation 433 Best Solution: [183.0754253692041, 176.40480343645177,
175.01355229195534, 75.83928851598533] Fitness: 1.635865498123117
Generation 434 Best Solution: [183.0754253692041, 176.40480343645177,
175.01355229195534, 75.83928851598533] Fitness: 1.635865498123117
Generation 435 Best Solution: [183.2240138767393, 176.40480343645177,
175.01355229195534, 75.83928851598533] Fitness: 1.6209441561633273
Generation 436 Best Solution: [183.2240138767393, 176.40480343645177,
175.01355229195534, 75.83928851598533] Fitness: 1.6209441561633273
Generation 437 Best Solution: [183.2240138767393, 176.42699416859955,
175.01355229195534, 75.08454209782731] Fitness: 0.7736094242152731
Generation 438 Best Solution: [183.0754253692041, 177.0428750386491,
175.01355229195534, 74.57462087006988] Fitness: 1.18243466967494
Generation 439 Best Solution: [183.2240138767393, 177.42537256464803,
175.01355229195534, 74.57462087006988] Fitness: 1.756630548489935
Generation 440 Best Solution: [183.81268999955023, 175.60585606270865,
175.01355229195534, 74.57462087006988] Fitness: 1.1816394259165965
Generation 441 Best Solution: [183.3243508573867, 177.42537256464803,
175.01355229195534, 74.57462087006988] Fitness: 1.765286976066317
Generation 442 Best Solution: [183.3243508573867, 176.85563449495302,
175.01355229195534, 75.83928851598533] Fitness: 1.9155755496385818
Generation 443 Best Solution: [183.18518710379485, 177.42537256464803,
175.01355229195534, 74.57462087006988] Fitness: 1.7573332845269964
Generation 444 Best Solution: [183.2732784865196, 176.0322035907862,
175.01355229195534, 75.83928851598533] Fitness: 1.6170643324760006
Generation 445 Best Solution: [183.2240138767393, 176.70251218451745,
175.01355229195534, 75.83928851598533] Fitness: 1.775599895939134
Generation 446 Best Solution: [183.0754253692041, 175.67517847499528,
175.01355229195534, 75.83928851598533] Fitness: 1.8190112618435124
Generation 447 Best Solution: [183.0754253692041, 176.4046533045359,
175.01355229195534, 75.83928851598533] Fitness: 1.6358210451804522
Generation 448 Best Solution: [182.88378550617173, 175.95412656562382,
175.01355229195534, 75.83928851598533] Fitness: 1.722850489491996
Generation 449 Best Solution: [183.0754253692041, 177.42537256464803,
175.01355229195534, 75.21870248341457] Fitness: 1.954466247761746
Generation 450 Best Solution: [183.0754253692041, 177.42537256464803,
175.01355229195534, 75.43562447973622] Fitness: 2.1561484374672615

Generation 451 Best Solution: [183.2240138767393, 177.4199674010637,
173.8347976940153, 75.83928851598533] Fitness: 2.1755979278366544
Generation 452 Best Solution: [183.2732784865196, 177.42537256464803,
175.01355229195534, 74.64973386191801] Fitness: 1.7482725618797452
Generation 453 Best Solution: [183.2240138767393, 177.4199674010637,
175.01355229195534, 74.4159103053319] Fitness: 1.7972629958969364
Generation 454 Best Solution: [183.0754253692041, 175.98027366491328,
175.01355229195534, 75.538216135921] Fitness: 1.2020695087485986
Generation 455 Best Solution: [183.85324463955928, 177.42537256464803,
175.01355229195534, 74.40748893936977] Fitness: 2.114734246061543
Generation 456 Best Solution: [183.2732784865196, 176.7931105899107,
173.8347976940153, 75.83928851598533] Fitness: 1.3324513244023168
Generation 457 Best Solution: [183.0978456840773, 177.4199674010637,
173.8347976940153, 75.83928851598533] Fitness: 2.186146261142392
Generation 458 Best Solution: [183.2732784865196, 177.42537256464803,
175.01355229195534, 74.86743624145801] Fitness: 1.7650089934417366
Generation 459 Best Solution: [183.2732784865196, 176.08644662518668,
175.01355229195534, 75.83928851598533] Fitness: 1.6050217095874306
Generation 460 Best Solution: [183.2732784865196, 177.42537256464803,
175.01355229195534, 74.86743624145801] Fitness: 1.7650089934417366
Generation 461 Best Solution: [183.2732784865196, 177.4199674010637,
175.01355229195534, 74.86743624145801] Fitness: 1.7551553533406887
Generation 462 Best Solution: [183.85324463955928, 177.42537256464803,
175.01355229195534, 74.41810558342515] Fitness: 2.110043769105726
Generation 463 Best Solution: [183.2732784865196, 177.4199674010637,
174.36076208272237, 75.83928851598533] Fitness: 2.152015203959113
Generation 464 Best Solution: [183.2732784865196, 177.42537256464803,
174.36076208272237, 75.83928851598533] Fitness: 2.1618688440601606
Generation 465 Best Solution: [183.2732784865196, 177.4199674010637,
175.01355229195534, 74.32687084455092] Fitness: 1.84449442647608
Generation 466 Best Solution: [183.2732784865196, 177.42537256464803,
174.19257233699565, 75.83928851598533] Fitness: 2.1250351527069347
Generation 467 Best Solution: [183.85324463955928, 175.60594562669317,
175.01355229195534, 75.83928851598533] Fitness: 2.1668318557747286
Generation 468 Best Solution: [183.2732784865196, 176.19980096374837,
175.01355229195534, 75.83928851598533] Fitness: 1.5941040648762033
Generation 469 Best Solution: [183.85324463955928, 175.91587148084471,
175.01355229195534, 75.83928851598533] Fitness: 1.9592939840933647
Generation 470 Best Solution: [183.85324463955928, 176.32144265523183,
175.01355229195534, 75.83928851598533] Fitness: 1.9053471785093916
Generation 471 Best Solution: [183.2732784865196, 176.32144265523183,
175.01355229195534, 75.83928851598533] Fitness: 1.603827215440631
Generation 472 Best Solution: [183.85324463955928, 177.42537256464803,
175.01355229195534, 75.00696040888187] Fitness: 2.114636538919339
Generation 473 Best Solution: [183.35315356457258, 177.4199674010637,
175.01355229195534, 75.0892993161082] Fitness: 1.8569219878107721
Generation 474 Best Solution: [183.85324463955928, 177.4199674010637,
175.01355229195534, 74.43192445315961] Fitness: 2.0943381489758712

Generation 475 Best Solution: [183.85324463955928, 177.4199674010637,
175.01355229195534, 74.43192445315961] Fitness: 2.0943381489758712
Generation 476 Best Solution: [183.62324979992198, 177.42537256464803,
175.01355229195534, 74.43192445315961] Fitness: 1.924250535583647
Generation 477 Best Solution: [183.85324463955928, 175.8093007624484,
175.01355229195534, 75.74668270673831] Fitness: 1.8635973594636965
Generation 478 Best Solution: [183.85324463955928, 176.29000315481466,
175.01355229195534, 75.83928851598533] Fitness: 1.9007072000391052
Generation 479 Best Solution: [183.85324463955928, 176.29000315481466,
175.01355229195534, 75.83928851598533] Fitness: 1.9007072000391052
Generation 480 Best Solution: [182.9425030804765, 175.8093007624484,
175.01355229195534, 75.83928851598533] Fitness: 1.7668545384306695
Generation 481 Best Solution: [183.85324463955928, 175.8093007624484,
175.01355229195534, 74.6768349978579] Fitness: 1.054109630815778
Generation 482 Best Solution: [183.6497765947496, 177.4199674010637,
175.01355229195534, 74.62095565057925] Fitness: 1.8798109592493866
Generation 483 Best Solution: [183.78651294097267, 177.4199674010637,
175.01355229195534, 74.62095565057925] Fitness: 1.9826663841517127
Generation 484 Best Solution: [183.67319830159153, 177.4199674010637,
175.01355229195534, 74.62095565057925] Fitness: 1.89543865216176
Generation 485 Best Solution: [183.85324463955928, 176.04717621639725,
175.01355229195534, 75.83928851598533] Fitness: 1.9148192020802504
Generation 486 Best Solution: [183.85324463955928, 176.04717621639725,
175.01355229195534, 75.83928851598533] Fitness: 1.9148192020802504
Generation 487 Best Solution: [183.8501741332203, 175.98720229340668,
175.01355229195534, 75.83928851598533] Fitness: 1.928999877511028
Generation 488 Best Solution: [183.85324463955928, 177.4199674010637,
173.9555320919286, 74.45545357458735] Fitness: 1.5234219891119032
Generation 489 Best Solution: [183.85324463955928, 176.79716534809867,
173.9555320919286, 75.83928851598533] Fitness: 1.5948966486783829
Generation 490 Best Solution: [183.85324463955928, 176.79716534809867,
173.9555320919286, 75.83928851598533] Fitness: 1.5948966486783829
Generation 491 Best Solution: [183.85324463955928, 176.30564739594584,
175.01355229195534, 75.83928851598533] Fitness: 1.9028307166818217
Generation 492 Best Solution: [183.7897680353415, 176.30564739594584,
175.01355229195534, 75.83928851598533] Fitness: 1.8452409640854928
Generation 493 Best Solution: [183.85324463955928, 176.99029194477245,
175.01355229195534, 74.34623609804514] Fitness: 1.4921511242713277
Generation 494 Best Solution: [183.85324463955928, 175.65477552505956,
175.01355229195534, 75.83928851598533] Fitness: 2.124571574114527
Generation 495 Best Solution: [183.85324463955928, 177.4199674010637,
175.01355229195534, 74.89659996800533] Fitness: 2.064317004767014
Generation 496 Best Solution: [183.85324463955928, 176.44553052375198,
175.01355229195534, 75.83928851598533] Fitness: 1.9381349016437934
Generation 497 Best Solution: [183.85324463955928, 176.02902774485418,
175.01355229195534, 75.83928851598533] Fitness: 1.9194261376634651
Generation 498 Best Solution: [183.85324463955928, 176.02902774485418,
175.01355229195534, 75.83928851598533] Fitness: 1.9194261376634651

```
Generation 499 Best Solution: [183.02242386916987, 176.02902774485418,
175.01355229195534, 75.83928851598533] Fitness: 1.643800199268944
Generation 500 Best Solution: [183.02242386916987, 176.02902774485418,
175.01355229195534, 75.83928851598533] Fitness: 1.643800199268944
Optimal Solution for Simulation 7: [183.02242386916987, 176.02902774485418,
175.01355229195534, 75.83928851598533]
--------------------------------------------------
Simulation 8/32 with conditions: {'drought': True, 'flood': True,
'policy_change': False, 'social_event': False, 'economic_crisis': False}
Generation 1 Best Solution: [183.1434246558506, 176.0601182512399,
174.29099774352872, 75.34264912785328] Fitness: 0.3563933137056689
Generation 2 Best Solution: [183.07474020104385, 175.94806820645772,
173.824924966504, 75.26888438036556] Fitness: 0.36203696672388264
Generation 3 Best Solution: [183.63836923582426, 175.87594799203347,
174.42296915995547, 74.62278257060832] Fitness: 0.35273170067272575
Generation 4 Best Solution: [182.94645877771873, 177.21386646391676,
174.42931241054745, 75.1342833578468] Fitness: 1.0719716964835848
Generation 5 Best Solution: [183.1434246558506, 177.21386646391676,
173.99243979536647, 74.62278257060832] Fitness: 0.8597821708149956
Generation 6 Best Solution: [183.24998339577587, 177.21692835535038,
173.99243979536647, 74.62278257060832] Fitness: 0.8668166039754872
Generation 7 Best Solution: [182.9146005960097, 177.32038641793054,
174.23291782465338, 75.1342833578468] Fitness: 1.1884024115605805
Generation 8 Best Solution: [183.5223917769105, 175.57955570962446,
173.8807671248134, 75.3657480112927] Fitness: 0.7195458237850687
Generation 9 Best Solution: [182.98292864296258, 176.06507641431986,
174.9697423345444, 75.3657480112927] Fitness: 0.9225972165501936
Generation 10 Best Solution: [183.24998339577587, 175.70555787430445,
174.9697423345444, 74.62699697255948] Fitness: 0.768746536666499
Generation 11 Best Solution: [183.24998339577587, 177.21386646391676,
174.9697423345444, 74.62699697255948] Fitness: 1.3961199126282264
Generation 12 Best Solution: [183.7141525089589, 177.21692835535038,
173.8364860454093, 75.3657480112927] Fitness: 1.391968340114632
Generation 13 Best Solution: [183.7141525089589, 177.21692835535038,
173.8364860454093, 75.1342833578468] Fitness: 1.2189958366508775
Generation 14 Best Solution: [183.24998339577587, 177.34020360406123,
173.8364860454093, 75.1342833578468] Fitness: 1.2140010184662238
Generation 15 Best Solution: [183.24998339577587, 176.89395643986077,
174.3648594660662, 75.3657480112927] Fitness: 0.7713447974818095
Generation 16 Best Solution: [183.79849748134507, 176.89395643986077,
174.3648594660662, 75.3657480112927] Fitness: 1.044995832777774
Generation 17 Best Solution: [183.24998339577587, 177.34020360406123,
173.79913990090716, 75.3657480112927] Fitness: 1.4025739993169866
Generation 18 Best Solution: [183.79849748134507, 175.64877538940237,
174.9697423345444, 75.3657480112927] Fitness: 1.362925570089992
Generation 19 Best Solution: [183.7141525089589, 175.82827134711684,
175.121376929991, 75.3657480112927] Fitness: 1.3861914226851144
Generation 20 Best Solution: [183.3456828150518, 175.82827134711684,
```

174.82773305262097, 75.3657480112927] Fitness: 0.8279180176989913
Generation 21 Best Solution: [183.7141525089589, 177.34020360406123,
174.37512252581902, 74.7694537234861] Fitness: 1.3033095077747792
Generation 22 Best Solution: [183.79849748134507, 176.73366137180582,
175.121376929991, 74.99774895900673] Fitness: 1.3594776577336876
Generation 23 Best Solution: [183.79849748134507, 175.7039495013793,
174.9697423345444, 75.71846102671972] Fitness: 1.7364418766503464
Generation 24 Best Solution: [183.3456828150518, 177.34020360406123,
174.42185459146523, 74.70455741221143] Fitness: 1.1363172551611815
Generation 25 Best Solution: [183.79849748134507, 177.34020360406123,
173.84115310897675, 74.7694537234861] Fitness: 1.371424687498356
Generation 26 Best Solution: [183.3456828150518, 177.34020360406123,
173.84115310897675, 74.7694537234861] Fitness: 1.1140019219994637
Generation 27 Best Solution: [183.3456828150518, 177.34020360406123,
175.0564120045607, 74.70455741221143] Fitness: 1.7216175971900052
Generation 28 Best Solution: [183.7141525089589, 176.43894202811148,
175.0564120045607, 74.70455741221143] Fitness: 0.9700974042702196
Generation 29 Best Solution: [183.3456828150518, 177.34020360406123,
174.9697423345444, 74.44469467663517] Fitness: 1.668231843875614
Generation 30 Best Solution: [183.79849748134507, 177.21692835535038,
175.121376929991, 75.0254255699824] Fitness: 1.9339948515570187
Generation 31 Best Solution: [183.18995561905544, 177.34020360406123,
175.121376929991, 74.58565319936068] Fitness: 1.8107447788445612
Generation 32 Best Solution: [183.79849748134507, 177.34020360406123,
174.9697423345444, 74.9500679303976] Fitness: 1.8970826826669724
Generation 33 Best Solution: [183.3167291829644, 177.34020360406123,
174.15376688338415, 75.71846102671972] Fitness: 1.7596131940865127
Generation 34 Best Solution: [183.79849748134507, 177.09823262985321,
174.15376688338415, 75.71846102671972] Fitness: 1.6553906244014334
Generation 35 Best Solution: [183.79849748134507, 177.09823262985321,
174.15376688338415, 75.24417740239271] Fitness: 1.1413918935589522
Generation 36 Best Solution: [183.7141525089589, 177.09823262985321,
173.9314021673472, 75.71846102671972] Fitness: 1.6050259912484273
Generation 37 Best Solution: [183.14460896837198, 177.34020360406123,
174.9697423345444, 74.55302088017325] Fitness: 1.612139348095993
Generation 38 Best Solution: [183.6807128934115, 177.34020360406123,
174.15376688338415, 74.55302088017325] Fitness: 1.2504808175199427
Generation 39 Best Solution: [183.6807128934115, 177.34020360406123,
174.15376688338415, 74.66190167105457] Fitness: 1.230084282003211
Generation 40 Best Solution: [183.41188394074004, 177.09823262985321,
174.9697423345444, 75.41104536495524] Fitness: 1.588633490719993
Generation 41 Best Solution: [183.41188394074004, 175.92507741817536,
174.9697423345444, 75.71846102671972] Fitness: 1.3824950445350876
Generation 42 Best Solution: [183.79849748134507, 175.92507741817536,
174.9697423345444, 75.71846102671972] Fitness: 1.6208968180663978
Generation 43 Best Solution: [183.35072026421028, 177.34020360406123,
173.9314021673472, 75.71846102671972] Fitness: 1.7873394195567338
Generation 44 Best Solution: [183.79849748134507, 176.95722026113725,

174.2595749023864, 75.71846102671972] Fitness: 1.4958798671061002
Generation 45 Best Solution: [183.79849748134507, 177.11276726084427,
173.9314021673472, 75.71846102671972] Fitness: 1.6952766475649987
Generation 46 Best Solution: [183.79849748134507, 176.29951047323917,
173.9314021673472, 75.71846102671972] Fitness: 1.0655037606310898
Generation 47 Best Solution: [183.79849748134507, 177.34020360406123,
174.79077430645805, 74.83004903099825] Fitness: 1.66841886092254
Generation 48 Best Solution: [183.79849748134507, 177.34020360406123,
174.79077430645805, 74.83004903099825] Fitness: 1.66841886092254
Generation 49 Best Solution: [182.99385594233289, 177.34020360406123,
174.9697423345444, 74.98864468712375] Fitness: 1.65861508292217
Generation 50 Best Solution: [183.79849748134507, 177.24436070659308,
174.56514611290234, 75.88432166657846] Fitness: 2.276645489176696
Generation 51 Best Solution: [183.41188394074004, 177.34020360406123,
174.79077430645805, 75.71846102671972] Fitness: 2.12524635487749
Generation 52 Best Solution: [183.41188394074004, 177.34020360406123,
174.1368080476169, 75.71846102671972] Fitness: 1.7839557021467147
Generation 53 Best Solution: [182.99385594233289, 177.34020360406123,
174.79077430645805, 75.71846102671972] Fitness: 2.112100549952152
Generation 54 Best Solution: [183.79849748134507, 177.34020360406123,
174.9697423345444, 74.58119986270951] Fitness: 1.8813171828228172
Generation 55 Best Solution: [183.41188394074004, 177.34020360406123,
174.9697423345444, 74.73076777875292] Fitness: 1.6254532345104187
Generation 56 Best Solution: [183.79849748134507, 176.13831712651378,
174.9697423345444, 75.88432166657846] Fitness: 1.8338040807300968
Generation 57 Best Solution: [183.07150782165976, 177.34020360406123,
175.121376929991, 74.97725434093792] Fitness: 1.8451965998568114
Generation 58 Best Solution: [183.79849748134507, 176.69678685328694,
175.121376929991, 75.88432166657846] Fitness: 2.2392984953027533
Generation 59 Best Solution: [183.79849748134507, 177.34020360406123,
174.79077430645805, 74.94150268819847] Fitness: 1.6926591575018035
Generation 60 Best Solution: [183.3355255242913, 177.34020360406123,
174.79077430645805, 74.58021534935287] Fitness: 1.4199497977624573
Generation 61 Best Solution: [183.79849748134507, 177.34020360406123,
174.79077430645805, 74.94150268819847] Fitness: 1.6926591575018035
Generation 62 Best Solution: [183.07150782165976, 177.34020360406123,
174.2569742543432, 74.94150268819847] Fitness: 1.0991018244582973
Generation 63 Best Solution: [183.79849748134507, 177.34020360406123,
174.2569742543432, 74.94150268819847] Fitness: 1.3674299026546426
Generation 64 Best Solution: [182.99385594233289, 176.69678685328694,
175.121376929991, 75.37510696087983] Fitness: 1.3259834003838329
Generation 65 Best Solution: [183.07150782165976, 176.69678685328694,
174.8618714826352, 75.86222729427774] Fitness: 1.5997621938607969
Generation 66 Best Solution: [183.07150782165976, 177.34020360406123,
175.121376929991, 74.53581692367376] Fitness: 1.8321713822093026
Generation 67 Best Solution: [183.07150782165976, 177.34020360406123,
174.79077430645805, 75.34049126322367] Fitness: 1.6592247797923234
Generation 68 Best Solution: [183.79849748134507, 177.34020360406123,

174.79077430645805, 74.53581692367376] Fitness: 1.691237159009366
Generation 69 Best Solution: [183.79849748134507, 176.8421755691533,
175.05457975312726, 74.9574055133969] Fitness: 1.3467501642762443
Generation 70 Best Solution: [183.16386864839404, 176.8421755691533,
174.22512207498977, 75.86222729427774] Fitness: 1.3110777058257925
Generation 71 Best Solution: [183.07150782165976, 177.34020360406123,
173.80590544269108, 75.88432166657846] Fitness: 2.0753209268700754
Generation 72 Best Solution: [183.04302455695867, 177.34020360406123,
175.121376929991, 75.12421134637772] Fitness: 1.917555757187288
Generation 73 Best Solution: [183.28856874458614, 177.34020360406123,
175.121376929991, 75.12421134637772] Fitness: 1.9120365134313437
Generation 74 Best Solution: [183.04302455695867, 176.23424874909938,
175.121376929991, 75.88432166657846] Fitness: 1.7798107633925178
Generation 75 Best Solution: [183.28856874458614, 177.34020360406123,
175.121376929991, 74.33641191407659] Fitness: 1.9188412324652895
Generation 76 Best Solution: [183.28856874458614, 177.34020360406123,
173.93740881728124, 75.88432166657846] Fitness: 2.030305085085308
Generation 77 Best Solution: [183.79849748134507, 177.34020360406123,
175.121376929991, 74.85831398030396] Fitness: 2.0821185129162294
Generation 78 Best Solution: [183.79849748134507, 176.91470959490377,
175.121376929991, 74.85831398030396] Fitness: 1.4918691389907937
Generation 79 Best Solution: [183.79849748134507, 176.15732939810692,
175.121376929991, 75.88432166657846] Fitness: 2.040536376226558
Generation 80 Best Solution: [183.79849748134507, 176.439201889904,
175.121376929991, 75.88432166657846] Fitness: 2.0915873359989794
Generation 81 Best Solution: [183.79849748134507, 177.34020360406123,
174.45262612333036, 74.42476527377944] Fitness: 1.4784421326870216
Generation 82 Best Solution: [183.79849748134507, 176.93385946635817,
173.99667561196125, 75.88432166657846] Fitness: 1.7177791936133486
Generation 83 Best Solution: [183.79849748134507, 176.93385946635817,
173.99667561196125, 75.88432166657846] Fitness: 1.7177791936133486
Generation 84 Best Solution: [183.79849748134507, 176.15225789331092,
175.121376929991, 75.88432166657846] Fitness: 2.040676212214488
Generation 85 Best Solution: [183.79849748134507, 177.34020360406123,
174.25830139049776, 74.95208125306901] Fitness: 1.3709641616612032
Generation 86 Best Solution: [183.79849748134507, 176.15225789331092,
175.121376929991, 75.86222729427774] Fitness: 2.004135350853609
Generation 87 Best Solution: [183.79849748134507, 176.15225789331092,
173.76356414850784, 74.3255536658386] Fitness: 0.5382466152396799
Generation 88 Best Solution: [183.79849748134507, 177.34020360406123,
174.71565264644667, 75.65333110312693] Fitness: 2.202855040348438
Generation 89 Best Solution: [182.93249848625788, 176.82277156082583,
175.121376929991, 75.88432166657846] Fitness: 2.11460343994468
Generation 90 Best Solution: [182.93249848625788, 176.82277156082583,
174.82807385735512, 75.88432166657846] Fitness: 1.7435588867610519
Generation 91 Best Solution: [183.43450932047497, 177.34020360406123,
174.43030009041104, 75.52108681974653] Fitness: 1.6144917009353885
Generation 92 Best Solution: [182.9855177455653, 177.34020360406123,

174.43030009041104, 75.52108681974653] Fitness: 1.5957068514944028
Generation 93 Best Solution: [183.14841783871768, 175.8622148717906,
175.121376929991, 75.88432166657846] Fitness: 1.8346221926116868
Generation 94 Best Solution: [183.10023866849886, 177.34020360406123,
175.121376929991, 74.64410980891283] Fitness: 1.8051429362840004
Generation 95 Best Solution: [183.43450932047497, 177.34020360406123,
174.22811474912754, 75.88432166657846] Fitness: 2.0591730177366276
Generation 96 Best Solution: [183.79849748134507, 175.8622148717906,
175.10740288940144, 75.88432166657846] Fitness: 2.090408368084284
Generation 97 Best Solution: [183.79849748134507, 175.8622148717906,
175.121376929991, 75.88432166657846] Fitness: 2.1109251034570438
Generation 98 Best Solution: [183.79849748134507, 177.34020360406123,
174.1188852754767, 75.88432166657846] Fitness: 2.2787291844578736
Generation 99 Best Solution: [182.93283953684139, 177.34020360406123,
174.5132457484114, 75.88432166657846] Fitness: 2.167591071876613
Generation 100 Best Solution: [182.93283953684139, 177.34020360406123,
174.1188852754767, 75.88432166657846] Fitness: 2.046520678988667
Generation 101 Best Solution: [183.79849748134507, 177.34020360406123,
174.1188852754767, 75.68094631326835] Fitness: 1.9691892373324724
Generation 102 Best Solution: [183.79849748134507, 176.02785058344313,
175.121376929991, 75.88432166657846] Fitness: 2.055821469742998
Generation 103 Best Solution: [183.4654903199101, 177.34020360406123,
175.121376929991, 74.29124285298934] Fitness: 1.9969765869807194
Generation 104 Best Solution: [183.4654903199101, 177.34020360406123,
175.121376929991, 74.29124285298934] Fitness: 1.9969765869807194
Generation 105 Best Solution: [183.4654903199101, 175.9230874036737,
175.121376929991, 75.88432166657846] Fitness: 1.8677074360565755
Generation 106 Best Solution: [183.79849748134507, 176.53834486796265,
175.121376929991, 74.29124285298934] Fitness: 1.322443715040002
Generation 107 Best Solution: [183.80866072151682, 176.27710842851553,
173.78755914763363, 75.88432166657846] Fitness: 1.3793070815060324
Generation 108 Best Solution: [182.88926473504483, 175.82242681275272,
175.121376929991, 75.22638487684537] Fitness: 1.1311421459393889
Generation 109 Best Solution: [182.98390377438955, 177.00815622572895,
175.121376929991, 74.98937011651735] Fitness: 1.385930146910246
Generation 110 Best Solution: [183.4654903199101, 176.8612913869583,
175.121376929991, 75.22638487684537] Fitness: 1.3819220305900142
Generation 111 Best Solution: [182.88926473504483, 175.82242681275272,
175.121376929991, 75.22638487684537] Fitness: 1.1311421459393889
Generation 112 Best Solution: [182.88926473504483, 175.82242681275272,
175.121376929991, 75.22638487684537] Fitness: 1.1311421459393889
Generation 113 Best Solution: [182.98390377438955, 177.00815622572895,
175.121376929991, 75.22638487684537] Fitness: 1.5140482589934996
Generation 114 Best Solution: [183.490629943276, 177.00815622572895,
174.34270704912691, 75.04061758256235] Fitness: 0.7378012929863055
Generation 115 Best Solution: [183.79849748134507, 177.34020360406123,
174.34270704912691, 75.04061758256235] Fitness: 1.4284622128536295
Generation 116 Best Solution: [183.79849748134507, 176.32671189971697,

175.121376929991, 74.38607345656298] Fitness: 1.1879420246423693
Generation 117 Best Solution: [183.79849748134507, 177.00815622572895,
174.34270704912691, 75.04061758256235] Fitness: 0.9452765685683969
Generation 118 Best Solution: [183.79849748134507, 177.00815622572895,
174.34270704912691, 75.88432166657846] Fitness: 1.8364976058668674
Generation 119 Best Solution: [183.79849748134507, 175.86146366218856,
175.121376929991, 74.34149493947965] Fitness: 1.2660026502798916
Generation 120 Best Solution: [183.79849748134507, 177.00815622572895,
174.34270704912691, 75.52544952765362] Fitness: 1.3308738552111994
Generation 121 Best Solution: [183.79849748134507, 176.22243393807997,
175.121376929991, 75.88432166657846] Fitness: 2.0420640050705168
Generation 122 Best Solution: [183.79849748134507, 175.83388339304238,
175.121376929991, 75.88432166657846] Fitness: 2.124347048487178
Generation 123 Best Solution: [183.79849748134507, 175.83388339304238,
175.121376929991, 75.33687055361212] Fitness: 1.4281117741752372
Generation 124 Best Solution: [183.79849748134507, 177.34020360406123,
174.01809693681867, 74.41660279408514] Fitness: 1.399794730759085
Generation 125 Best Solution: [183.34928961320585, 176.02289586712106,
175.121376929991, 75.88432166657846] Fitness: 1.8003401519499511
Generation 126 Best Solution: [183.79849748134507, 176.02289586712106,
175.121376929991, 75.88432166657846] Fitness: 2.0568908004011943
Generation 127 Best Solution: [183.79849748134507, 175.83388339304238,
175.121376929991, 75.88432166657846] Fitness: 2.124347048487178
Generation 128 Best Solution: [183.79849748134507, 175.83388339304238,
175.121376929991, 75.88432166657846] Fitness: 2.124347048487178
Generation 129 Best Solution: [183.79849748134507, 177.34020360406123,
175.121376929991, 74.59980489810819] Fitness: 2.084822258765407
Generation 130 Best Solution: [183.79849748134507, 177.34020360406123,
175.121376929991, 74.35288196138181] Fitness: 2.1781704676408165
Generation 131 Best Solution: [183.79849748134507, 176.02539647090225,
175.121376929991, 75.88432166657846] Fitness: 2.0563466550841767
Generation 132 Best Solution: [183.79849748134507, 176.2306588674045,
175.121376929991, 75.61079251785196] Fitness: 1.6403364103744411
Generation 133 Best Solution: [183.79849748134507, 176.2306588674045,
175.121376929991, 75.8337311500084] Fitness: 1.9600747806238383
Generation 134 Best Solution: [183.79849748134507, 175.79351272633647,
175.121376929991, 75.88432166657846] Fitness: 2.145489647584043
Generation 135 Best Solution: [182.992121038639, 177.34020360406123,
174.61431859096487, 75.88432166657846] Fitness: 2.2161749568404425
Generation 136 Best Solution: [183.79849748134507, 177.34020360406123,
173.74901317260924, 75.8337311500084] Fitness: 2.2881329251300238
Generation 137 Best Solution: [183.85898472397395, 176.04464894798303,
175.121376929991, 75.88432166657846] Fitness: 2.109428993185519
Generation 138 Best Solution: [183.79849748134507, 176.89099833405203,
175.121376929991, 75.52847377300797] Fitness: 1.912336733646649
Generation 139 Best Solution: [183.79849748134507, 176.04464894798303,
173.74901317260924, 75.88432166657846] Fitness: 1.3933378328979713
Generation 140 Best Solution: [183.79849748134507, 176.4729740705824,

175.121376929991, 75.88432166657846] Fitness: 2.105456673135194
Generation 141 Best Solution: [183.51207421939247, 177.34020360406123,
175.121376929991, 74.47723018984559] Fitness: 1.9225853184977493
Generation 142 Best Solution: [183.79849748134507, 176.20340184179864,
175.121376929991, 75.8337311500084] Fitness: 1.9583588624658674
Generation 143 Best Solution: [182.88901823215897, 177.34020360406123,
175.121376929991, 74.47723018984559] Fitness: 1.9050978581332727
Generation 144 Best Solution: [183.79849748134507, 176.1977044695528,
175.121376929991, 75.88432166657846] Fitness: 2.040757598368394
Generation 145 Best Solution: [182.88901823215897, 177.34020360406123,
174.48215864483237, 74.47723018984559] Fitness: 1.2572626432633704
Generation 146 Best Solution: [183.79849748134507, 177.34020360406123,
174.48215864483237, 74.47723018984559] Fitness: 1.472240859329804
Generation 147 Best Solution: [183.85898472397395, 176.30069797285122,
174.85367478986885, 75.88432166657846] Fitness: 1.7653857810738145
Generation 148 Best Solution: [183.85898472397395, 176.30069797285122,
174.27677255642132, 75.67658691752507] Fitness: 1.0636816625390266
Generation 149 Best Solution: [183.79849748134507, 176.63690195792827,
175.121376929991, 74.47723018984559] Fitness: 1.2865456928945151
Generation 150 Best Solution: [183.79849748134507, 176.63690195792827,
175.121376929991, 74.47723018984559] Fitness: 1.2865456928945151
Generation 151 Best Solution: [183.79849748134507, 176.63690195792827,
175.121376929991, 74.84406061792812] Fitness: 1.2462052327386002
Generation 152 Best Solution: [183.37521140140353, 177.34020360406123,
175.121376929991, 74.47723018984559] Fitness: 1.870349944750681
Generation 153 Best Solution: [183.79849748134507, 177.34020360406123,
175.121376929991, 75.15376046869658] Fitness: 2.1980569683224767
Generation 154 Best Solution: [183.23661078403086, 177.34020360406123,
175.121376929991, 75.38603121423574] Fitness: 2.1034904214745653
Generation 155 Best Solution: [183.79849748134507, 177.34020360406123,
175.121376929991, 74.72235159795804] Fitness: 2.0714229131283552
Generation 156 Best Solution: [183.79849748134507, 177.34020360406123,
174.48215864483237, 74.75380946388712] Fitness: 1.4236714576713136
Generation 157 Best Solution: [182.87675589679245, 177.34020360406123,
174.48215864483237, 74.75380946388712] Fitness: 1.2140148799704669
Generation 158 Best Solution: [183.79849748134507, 176.41672928761503,
175.121376929991, 75.38603121423574] Fitness: 1.4317482837296913
Generation 159 Best Solution: [183.79849748134507, 176.41672928761503,
175.121376929991, 75.46022197010235] Fitness: 1.5058717652716593
Generation 160 Best Solution: [183.79849748134507, 176.41672928761503,
175.121376929991, 75.46022197010235] Fitness: 1.5058717652716593
Generation 161 Best Solution: [183.85898472397395, 176.89847401116631,
175.121376929991, 75.0437613054154] Fitness: 1.589467147430713
Generation 162 Best Solution: [183.39291514391599, 176.41672928761503,
175.121376929991, 75.88432166657846] Fitness: 1.8387741783832974
Generation 163 Best Solution: [183.79849748134507, 176.4838071108059,
175.121376929991, 75.86818610575614] Fitness: 2.083501155651219
Generation 164 Best Solution: [183.79849748134507, 177.34020360406123,

175.121376929991, 75.14431991711211] Fitness: 2.192388652526588
Generation 165 Best Solution: [183.85898472397395, 177.34020360406123,
175.121376929991, 74.76911083516902] Fitness: 2.1290349653188283
Generation 166 Best Solution: [183.79849748134507, 177.34020360406123,
175.121376929991, 74.76911083516902] Fitness: 2.072067763243763
Generation 167 Best Solution: [183.79849748134507, 177.34020360406123,
175.121376929991, 74.76911083516902] Fitness: 2.072067763243763
Generation 168 Best Solution: [183.79849748134507, 177.0764581049977,
175.121376929991, 74.76911083516902] Fitness: 1.6751716995354742
Generation 169 Best Solution: [183.85898472397395, 177.34020360406123,
175.121376929991, 75.04706931393942] Fitness: 2.198510464540477
Generation 170 Best Solution: [183.79849748134507, 177.34020360406123,
175.121376929991, 75.19518099829838] Fitness: 2.22445890994278
Generation 171 Best Solution: [183.85898472397395, 177.34020360406123,
175.121376929991, 75.19518099829838] Fitness: 2.2814261120178454
Generation 172 Best Solution: [183.85898472397395, 177.34020360406123,
175.121376929991, 74.73470558827061] Fitness: 2.1282513658421625
Generation 173 Best Solution: [183.85898472397395, 177.34020360406123,
174.6951031811607, 75.19518099829838] Fitness: 1.7833895306346301
Generation 174 Best Solution: [183.85898472397395, 176.11512384243912,
175.121376929991, 75.88432166657846] Fitness: 2.0998071347989486
Generation 175 Best Solution: [183.02632611583604, 175.77605524852217,
175.121376929991, 75.88432166657846] Fitness: 1.8957352386586275
Generation 176 Best Solution: [183.0415816580041, 177.34020360406123,
173.8210837749401, 75.88432166657846] Fitness: 2.0743134859169654
Generation 177 Best Solution: [183.0415816580041, 177.34020360406123,
173.8210837749401, 75.41195248506075] Fitness: 1.4477718835293536
Generation 178 Best Solution: [183.7903198540947, 177.34020360406123,
175.121376929991, 75.19518099829838] Fitness: 2.2171655528083067
Generation 179 Best Solution: [183.85898472397395, 177.34020360406123,
175.121376929991, 74.72866374855845] Fitness: 2.1282914865718676
Generation 180 Best Solution: [183.7903198540947, 177.21155276494642,
174.6951031811607, 75.12004649958497] Fitness: 1.466839997168746
Generation 181 Best Solution: [182.8662123515099, 175.5875098464819,
175.121376929991, 75.88432166657846] Fitness: 2.085381574895453
Generation 182 Best Solution: [183.85898472397395, 176.38376253233054,
174.85688884087065, 75.88432166657846] Fitness: 1.789249318503269
Generation 183 Best Solution: [183.85898472397395, 176.74642497832954,
175.121376929991, 75.88432166657846] Fitness: 2.3358213925839575
Generation 184 Best Solution: [183.7903198540947, 177.34020360406123,
174.6951031811607, 74.75202203757765] Fitness: 1.5661334048579618
Generation 185 Best Solution: [183.85898472397395, 175.97609194648265,
173.83929183581495, 75.88432166657846] Fitness: 1.425778367138295
Generation 186 Best Solution: [183.5473036912894, 175.97609194648265,
173.83929183581495, 75.88432166657846] Fitness: 1.189175420830324
Generation 187 Best Solution: [183.85898472397395, 176.74128069813068,
174.0506661593226, 75.88432166657846] Fitness: 1.582413803450001
Generation 188 Best Solution: [183.85898472397395, 176.06802045505702,

175.121376929991, 75.88432166657846] Fitness: 2.1054374695796727
Generation 189 Best Solution: [183.85898472397395, 176.06802045505702,
175.121376929991, 75.88432166657846] Fitness: 2.1054374695796727
Generation 190 Best Solution: [183.85898472397395, 176.06802045505702,
175.121376929991, 75.88432166657846] Fitness: 2.1054374695796727
Generation 191 Best Solution: [183.33599949292847, 177.34020360406123,
174.35210056739686, 75.88432166657846] Fitness: 2.0633247058161066
Generation 192 Best Solution: [183.33599949292847, 177.34020360406123,
174.35210056739686, 75.88432166657846] Fitness: 2.0633247058161066
Generation 193 Best Solution: [183.33599949292847, 177.34020360406123,
175.121376929991, 74.45513165353701] Fitness: 1.8690864123818454
Generation 194 Best Solution: [183.85898472397395, 176.1551284297845,
175.121376929991, 74.5897495610047] Fitness: 1.1545733628371355
Generation 195 Best Solution: [183.85898472397395, 176.1551284297845,
175.121376929991, 75.46876562430073] Fitness: 1.5292037346326823
Generation 196 Best Solution: [182.9987249873881, 177.34020360406123,
175.121376929991, 74.5897495610047] Fitness: 1.8340339240728165
Generation 197 Best Solution: [183.85898472397395, 177.34020360406123,
174.62377371930094, 74.45513165353701] Fitness: 1.6301635458639252
Generation 198 Best Solution: [182.9987249873881, 177.34020360406123,
174.13687292187893, 75.88432166657846] Fitness: 2.026470282113273
Generation 199 Best Solution: [183.85898472397395, 176.09359691072612,
174.93681356336705, 75.28982673958814] Fitness: 1.1181765069924565
Generation 200 Best Solution: [183.79849748134507, 177.34020360406123,
174.43601627647996, 75.02764266285145] Fitness: 1.461434716663697
Generation 201 Best Solution: [183.27819548477981, 177.34020360406123,
174.43601627647996, 75.02764266285145] Fitness: 1.191183061239959
Generation 202 Best Solution: [183.79849748134507, 177.34020360406123,
174.43601627647996, 75.02764266285145] Fitness: 1.461434716663697
Generation 203 Best Solution: [183.85898472397395, 177.34020360406123,
174.43601627647996, 75.02764266285145] Fitness: 1.5184019187387623
Generation 204 Best Solution: [183.85898472397395, 177.34020360406123,
174.43601627647996, 75.02764266285145] Fitness: 1.5184019187387623
Generation 205 Best Solution: [183.85898472397395, 177.34020360406123,
174.35210056739686, 74.73627954375453] Fitness: 1.421367178391392
Generation 206 Best Solution: [183.85898472397395, 177.15161296900274,
175.121376929991, 74.29938875073945] Fitness: 1.972940341563847
Generation 207 Best Solution: [183.85898472397395, 176.05812330636493,
175.121376929991, 75.88432166657846] Fitness: 2.1070307754757662
Generation 208 Best Solution: [183.79849748134507, 177.34020360406123,
173.8098545979861, 75.88432166657846] Fitness: 2.341942280788702
Generation 209 Best Solution: [183.5825825487008, 175.98619158202155,
174.93681356336705, 75.88432166657846] Fitness: 1.6578990131657507
Generation 210 Best Solution: [183.5825825487008, 175.98619158202155,
174.93681356336705, 75.88432166657846] Fitness: 1.6578990131657507
Generation 211 Best Solution: [183.27819548477981, 177.34020360406123,
173.8098545979861, 75.88432166657846] Fitness: 2.0716906253649645
Generation 212 Best Solution: [183.27819548477981, 177.34020360406123,

175.121376929991, 75.21821423666748] Fitness: 1.9699686103207252
Generation 213 Best Solution: [183.27819548477981, 177.15161296900274,
174.55694588424615, 75.88432166657846] Fitness: 1.8627788069478743
Generation 214 Best Solution: [183.85898472397395, 175.77292631987257,
175.121376929991, 75.72977036964714] Fitness: 1.9734338762564756
Generation 215 Best Solution: [183.85898472397395, 177.34020360406123,
174.17475515295112, 75.09267059732544] Fitness: 1.4728891655043919
Generation 216 Best Solution: [183.27819548477981, 177.34020360406123,
174.17475515295112, 75.13508982079988] Fitness: 1.1689703519749388
Generation 217 Best Solution: [183.56217546314548, 177.34020360406123,
174.17475515295112, 75.13508982079988] Fitness: 1.267665465759235
Generation 218 Best Solution: [183.85898472397395, 175.77292631987257,
175.121376929991, 74.35706645759706] Fitness: 1.3601229191113426
Generation 219 Best Solution: [182.8727438718439, 176.65716620011943,
174.24805130400685, 75.88432166657846] Fitness: 1.2660234860553305
Generation 220 Best Solution: [183.85898472397395, 177.34020360406123,
174.17475515295112, 74.7192646630114] Fitness: 1.3807094269406044
Generation 221 Best Solution: [183.76661822639196, 175.77292631987257,
175.121376929991, 74.68039463418366] Fitness: 1.1729747499844012
Generation 222 Best Solution: [183.72259234213598, 177.34020360406123,
174.29654447237368, 75.88432166657846] Fitness: 2.241313593319176
Generation 223 Best Solution: [183.27819548477981, 177.34020360406123,
174.2487381027568, 75.53565327635306] Fitness: 1.5295283566361881
Generation 224 Best Solution: [183.79849748134507, 177.34020360406123,
175.121376929991, 74.89225623834159] Fitness: 2.0889827537262997
Generation 225 Best Solution: [183.76661822639196, 177.12992696998904,
175.121376929991, 74.89225623834159] Fitness: 1.7364895862735215
Generation 226 Best Solution: [183.85898472397395, 177.12992696998904,
175.121376929991, 74.89225623834159] Fitness: 1.8213393285359067
Generation 227 Best Solution: [183.85898472397395, 177.34020360406123,
175.121376929991, 75.05474942681562] Fitness: 2.2020255474234642
Generation 228 Best Solution: [183.85898472397395, 177.34020360406123,
175.121376929991, 75.05474942681562] Fitness: 2.2020255474234642
Generation 229 Best Solution: [183.85898472397395, 177.34020360406123,
174.29007167536005, 75.05474942681562] Fitness: 1.4756469737526816
Generation 230 Best Solution: [183.85898472397395, 177.34020360406123,
174.221688032896, 75.53356207456405] Fitness: 1.8491955056539966
Generation 231 Best Solution: [183.85898472397395, 177.34020360406123,
174.221688032896, 75.53356207456405] Fitness: 1.8491955056539966
Generation 232 Best Solution: [182.83672389775043, 177.17575722787157,
175.121376929991, 74.52686318928912] Fitness: 1.6530650419314352
Generation 233 Best Solution: [183.85898472397395, 177.34020360406123,
174.12178609238077, 75.88432166657846] Fitness: 2.3357610934582858
Generation 234 Best Solution: [183.85898472397395, 177.34020360406123,
173.87323695598263, 75.22948077987843] Fitness: 1.5929541016721185
Generation 235 Best Solution: [183.85898472397395, 177.19775943732031,
174.221688032896, 75.19221680204814] Fitness: 1.311144958467098
Generation 236 Best Solution: [183.29120533655183, 177.34020360406123,

174.12178609238077, 75.88432166657846] Fitness: 2.010499871608233
Generation 237 Best Solution: [183.29120533655183, 177.34020360406123,
175.121376929991, 74.29540857240993] Fitness: 1.9443274378627
Generation 238 Best Solution: [183.85898472397395, 176.2338689152429,
175.121376929991, 75.88432166657846] Fitness: 2.099936049431407
Generation 239 Best Solution: [183.85898472397395, 175.58535840559142,
173.87323695598263, 75.88432166657846] Fitness: 1.6368687251804839
Generation 240 Best Solution: [183.85898472397395, 175.86644166320556,
175.121376929991, 75.88432166657846] Fitness: 2.165989962081138
Generation 241 Best Solution: [183.85898472397395, 175.58535840559142,
175.121376929991, 75.45918133223603] Fitness: 1.7705917177434105
Generation 242 Best Solution: [183.4149028734079, 177.34020360406123,
174.42441208113564, 75.88432166657846] Fitness: 2.1154059035307995
Generation 243 Best Solution: [183.4149028734079, 177.34020360406123,
174.42441208113564, 75.88432166657846] Fitness: 2.1154059035307995
Generation 244 Best Solution: [183.85898472397395, 175.8697646203617,
173.89387055210753, 75.88432166657846] Fitness: 1.4456425272592843
Generation 245 Best Solution: [183.85898472397395, 175.8697646203617,
173.89387055210753, 75.88432166657846] Fitness: 1.4456425272592843
Generation 246 Best Solution: [183.85898472397395, 175.8697646203617,
175.121376929991, 75.88432166657846] Fitness: 2.164512650932392
Generation 247 Best Solution: [183.85898472397395, 177.4318629938142,
175.121376929991, 74.81778295263109] Fitness: 2.2947092072486113
Generation 248 Best Solution: [183.85898472397395, 177.4318629938142,
175.121376929991, 74.81778295263109] Fitness: 2.2947092072486113
Generation 249 Best Solution: [183.85898472397395, 175.8011647726769,
173.84702405239167, 75.88432166657846] Fitness: 1.4953796809775344
Generation 250 Best Solution: [183.85265892514795, 175.8011647726769,
173.84702405239167, 75.88432166657846] Fitness: 1.489172836387194
Generation 251 Best Solution: [183.85898472397395, 177.4318629938142,
175.121376929991, 74.81778295263109] Fitness: 2.2947092072486113
Generation 252 Best Solution: [183.85898472397395, 177.4318629938142,
175.121376929991, 74.81778295263109] Fitness: 2.2947092072486113
Generation 253 Best Solution: [183.85898472397395, 177.4318629938142,
175.121376929991, 74.9263547596793] Fitness: 2.3161580780035065
Generation 254 Best Solution: [183.85898472397395, 177.4318629938142,
174.4564325320301, 75.88432166657846] Fitness: 2.5870027999662044
Generation 255 Best Solution: [183.85898472397395, 177.4318629938142,
175.06954665109632, 74.69265509217693] Fitness: 2.216586350482483
Generation 256 Best Solution: [183.2516286740146, 177.4318629938142,
175.06954665109632, 74.69265509217693] Fitness: 1.8861345698832137
Generation 257 Best Solution: [183.85898472397395, 177.34020360406123,
175.06954665109632, 74.69265509217693] Fitness: 2.0549614088266925
Generation 258 Best Solution: [183.85898472397395, 177.4318629938142,
174.42612104599928, 74.69265509217693] Fitness: 1.6149634297861573
Generation 259 Best Solution: [183.85898472397395, 175.8817040749468,
175.121376929991, 75.07957064009514] Fitness: 1.2864648121116136
Generation 260 Best Solution: [183.85898472397395, 177.03518705986212,

174.28768989321014, 75.88432166657846] Fitness: 1.9099819093566035
Generation 261 Best Solution: [183.85898472397395, 176.2307197823265,
175.06954665109632, 74.53953677141654] Fitness: 1.0945447717230754
Generation 262 Best Solution: [183.85898472397395, 177.34020360406123,
175.121376929991, 74.97796129539329] Fitness: 2.1707400677426074
Generation 263 Best Solution: [183.19536963461192, 177.34020360406123,
173.91957401245062, 75.88432166657846] Fitness: 2.026652753067574
Generation 264 Best Solution: [183.85898472397395, 177.03518705986212,
175.121376929991, 74.79776050538842] Fitness: 1.6811538783818132
Generation 265 Best Solution: [183.85898472397395, 176.76167811291342,
175.121376929991, 75.88432166657846] Fitness: 2.348696181783366
Generation 266 Best Solution: [183.85898472397395, 177.03518705986212,
175.121376929991, 75.2942523419175] Fitness: 1.9048506575180149
Generation 267 Best Solution: [183.85898472397395, 177.03518705986212,
173.87815848380302, 75.88432166657846] Fitness: 1.9231320740034483
Generation 268 Best Solution: [183.85898472397395, 177.34020360406123,
175.121376929991, 74.71168222126971] Fitness: 2.1286885953288173
Generation 269 Best Solution: [183.85898472397395, 176.69652106141055,
175.121376929991, 75.88432166657846] Fitness: 2.2960635393792814
Generation 270 Best Solution: [183.85898472397395, 177.34020360406123,
174.0141708953057, 75.88432166657846] Fitness: 2.3415561140853045
Generation 271 Best Solution: [183.30295425211958, 177.34020360406123,
174.0141708953057, 75.88432166657846] Fitness: 2.018274346015441
Generation 272 Best Solution: [183.47266196796951, 177.34020360406123,
174.6567909192009, 75.88432166657846] Fitness: 2.2846978386298953
Generation 273 Best Solution: [183.85898472397395, 177.34020360406123,
174.0141708953057, 75.88432166657846] Fitness: 2.3415561140853045
Generation 274 Best Solution: [183.85898472397395, 177.34020360406123,
174.03567090797068, 75.70646561324915] Fitness: 2.0650516641465617
Generation 275 Best Solution: [183.6078104742063, 177.34020360406123,
174.0141708953057, 75.88432166657846] Fitness: 2.1398321010489854
Generation 276 Best Solution: [183.85898472397395, 177.34020360406123,
174.0141708953057, 75.88432166657846] Fitness: 2.3415561140853045
Generation 277 Best Solution: [183.64693821173626, 177.34020360406123,
175.121376929991, 75.88432166657846] Fitness: 2.910511235560507
Generation 278 Best Solution: [183.36457948339182, 177.4318629938142,
175.121376929991, 75.88432166657846] Fitness: 2.9388592237112783
Generation 279 Best Solution: [183.85898472397395, 177.34020360406123,
175.121376929991, 74.39878575473665] Fitness: 2.2110729888411225
Generation 280 Best Solution: [183.85898472397395, 177.34020360406123,
175.121376929991, 74.39878575473665] Fitness: 2.2110729888411225
Generation 281 Best Solution: [183.36457948339182, 177.34020360406123,
175.121376929991, 75.88432166657846] Fitness: 2.7772342820554874
Generation 282 Best Solution: [183.36457948339182, 177.34020360406123,
175.121376929991, 75.17887233438252] Fitness: 1.9611222669965684
Generation 283 Best Solution: [183.85898472397395, 176.4481276318979,
175.0572055974607, 75.88432166657846] Fitness: 2.0601849490194
Generation 284 Best Solution: [183.36457948339182, 177.34020360406123,

174.31709047099739, 75.50341177208804] Fitness: 1.5280916907994984
Generation 285 Best Solution: [183.85898472397395, 177.4318629938142,
174.31709047099739, 75.50341177208804] Fitness: 1.99932729180026
Generation 286 Best Solution: [183.36457948339182, 176.51696827465406,
175.121376929991, 75.50341177208804] Fitness: 1.3428003959333825
Generation 287 Best Solution: [183.36457948339182, 177.4318629938142,
174.71663359600402, 75.50341177208804] Fitness: 1.929071750908315
Generation 288 Best Solution: [183.85898472397395, 177.4318629938142,
174.31709047099739, 75.50341177208804] Fitness: 1.99932729180026
Generation 289 Best Solution: [183.85898472397395, 176.51432860296936,
174.31709047099739, 75.88432166657846] Fitness: 1.4630922711772714
Generation 290 Best Solution: [183.85898472397395, 176.51432860296936,
174.31709047099739, 75.50341177208804] Fitness: 0.9325240042340172
Generation 291 Best Solution: [183.33823033381233, 177.4318629938142,
174.31709047099739, 75.88432166657846] Fitness: 2.2137635362880843
Generation 292 Best Solution: [183.26319840670575, 177.34020360406123,
175.121376929991, 75.12624291035381] Fitness: 1.9096937441695887
Generation 293 Best Solution: [183.85898472397395, 177.4318629938142,
173.96976197805904, 75.50341177208804] Fitness: 1.9799141679388863
Generation 294 Best Solution: [183.85898472397395, 177.4318629938142,
174.02576104698403, 75.50341177208804] Fitness: 1.9711792509182051
Generation 295 Best Solution: [183.08993205021386, 177.4318629938142,
174.02576104698403, 75.50341177208804] Fitness: 1.6431898777350007
Generation 296 Best Solution: [183.08993205021386, 177.4318629938142,
174.77398949370237, 75.50341177208804] Fitness: 1.964112154832521
Generation 297 Best Solution: [183.6636198204305, 177.4318629938142,
174.11019746021978, 75.23186492072858] Fitness: 1.5524166724420745
Generation 298 Best Solution: [183.85898472397395, 177.4318629938142,
174.11019746021978, 75.23186492072858] Fitness: 1.7172484632914418
Generation 299 Best Solution: [183.85898472397395, 177.4318629938142,
174.11019746021978, 75.23186492072858] Fitness: 1.7172484632914418
Generation 300 Best Solution: [183.6636198204305, 176.3494070896102,
175.121376929991, 75.50341177208804] Fitness: 1.4243477219372145
Generation 301 Best Solution: [183.6636198204305, 176.3494070896102,
174.2873956287797, 75.69599101050896] Fitness: 0.9391198850496068
Generation 302 Best Solution: [183.65504675863826, 176.4128767745635,
175.121376929991, 75.88432166657846] Fitness: 1.9681006623472175
Generation 303 Best Solution: [183.85898472397395, 177.4318629938142,
173.83708657322845, 74.53066171818651] Fitness: 1.6215269961074124
Generation 304 Best Solution: [183.85898472397395, 177.4318629938142,
174.47982546911228, 75.02639669776082] Fitness: 1.701988881673415
Generation 305 Best Solution: [183.6636198204305, 177.4318629938142,
174.47982546911228, 75.02639669776082] Fitness: 1.5371570908240477
Generation 306 Best Solution: [183.6636198204305, 177.4318629938142,
173.79549758422104, 75.86462976063736] Fitness: 2.3694141025129833
Generation 307 Best Solution: [183.85898472397395, 177.02724599653448,
175.121376929991, 74.84683745002064] Fitness: 1.677392169163925
Generation 308 Best Solution: [183.85898472397395, 175.87136390639168,

175.121376929991, 75.21245826293763] Fitness: 1.3701388303241708
Generation 309 Best Solution: [183.85898472397395, 175.87136390639168,
175.121376929991, 75.21245826293763] Fitness: 1.3701388303241708
Generation 310 Best Solution: [183.85898472397395, 177.02724599653448,
175.121376929991, 74.71001382040558] Fitness: 1.668998534994937
Generation 311 Best Solution: [183.6636198204305, 176.18365898781553,
175.121376929991, 75.88432166657846] Fitness: 1.9325470979053447
Generation 312 Best Solution: [183.6636198204305, 176.0384153118415,
175.121376929991, 75.79701038707705] Fitness: 1.8055371611570152
Generation 313 Best Solution: [183.6636198204305, 177.4318629938142,
174.59160891622804, 75.88432166657846] Fitness: 2.5045589456866812
Generation 314 Best Solution: [183.85898472397395, 177.4318629938142,
174.59160891622804, 75.88432166657846] Fitness: 2.6693907365360485
Generation 315 Best Solution: [183.85898472397395, 175.80007142502322,
175.121376929991, 75.88432166657846] Fitness: 2.1988606991962762
Generation 316 Best Solution: [183.85898472397395, 177.4318629938142,
174.59160891622804, 74.61990884067741] Fitness: 1.7206392556286134
Generation 317 Best Solution: [183.85898472397395, 177.4318629938142,
175.01279709768886, 74.6198118494789] Fitness: 2.147788281638753
Generation 318 Best Solution: [183.64294831531825, 177.20370378694142,
174.4546048473366, 75.88432166657846] Fitness: 2.0273782188975873
Generation 319 Best Solution: [183.85898472397395, 176.06967824081482,
175.121376929991, 75.88432166657846] Fitness: 2.105184520042253
Generation 320 Best Solution: [182.91366093514463, 177.20370378694142,
175.121376929991, 74.36916358514873] Fitness: 1.7262298350191962
Generation 321 Best Solution: [183.85898472397395, 176.48513313348852,
175.121376929991, 75.88432166657846] Fitness: 2.167823447141329
Generation 322 Best Solution: [183.85898472397395, 176.48513313348852,
175.121376929991, 75.88432166657846] Fitness: 2.167823447141329
Generation 323 Best Solution: [183.85898472397395, 176.48513313348852,
175.121376929991, 75.41771669807542] Fitness: 1.5469713082217502
Generation 324 Best Solution: [182.91366093514463, 176.5191653756873,
175.121376929991, 75.88432166657846] Fitness: 1.9021011762250994
Generation 325 Best Solution: [183.6636198204305, 176.21114077773063,
174.63017774428155, 75.88432166657846] Fitness: 1.3827939935348217
Generation 326 Best Solution: [183.03016053590548, 176.5191653756873,
175.121376929991, 75.88432166657846] Fitness: 1.8666278078834273
Generation 327 Best Solution: [183.03016053590548, 176.5191653756873,
175.121376929991, 75.88432166657846] Fitness: 1.8666278078834273
Generation 328 Best Solution: [183.6636198204305, 176.21114077773063,
175.121376929991, 74.53685007422614] Fitness: 1.0038137604639974
Generation 329 Best Solution: [183.64294831531825, 176.21114077773063,
175.121376929991, 74.53685007422614] Fitness: 0.9896208082498491
Generation 330 Best Solution: [183.64294831531825, 176.21114077773063,
175.121376929991, 75.88432166657846] Fitness: 1.9192995145783045
Generation 331 Best Solution: [183.6636198204305, 176.98261525881315,
173.7386924465499, 75.88432166657846] Fitness: 1.7546009006486487
Generation 332 Best Solution: [183.6636198204305, 177.4318629938142,

174.21768821630613, 75.41771669807542] Fitness: 1.720732549851924
Generation 333 Best Solution: [183.64294831531825, 176.89978047209493,
175.05748241469448, 74.53685007422614] Fitness: 1.2804737992574142
Generation 334 Best Solution: [183.6636198204305, 177.4318629938142,
175.121376929991, 74.71351666315105] Fitness: 2.1254186398461843
Generation 335 Best Solution: [183.6636198204305, 177.4318629938142,
175.121376929991, 74.71351666315105] Fitness: 2.1254186398461843
Generation 336 Best Solution: [183.74178814534565, 176.77324617349623,
175.121376929991, 75.88432166657846] Fitness: 2.2531433471580535
Generation 337 Best Solution: [183.6636198204305, 177.4318629938142,
175.121376929991, 74.71351666315105] Fitness: 2.1254186398461843
Generation 338 Best Solution: [183.74178814534565, 176.2870226771078,
175.121376929991, 75.88432166657846] Fitness: 2.001096101588997
Generation 339 Best Solution: [183.74178814534565, 176.2870226771078,
175.121376929991, 75.88432166657846] Fitness: 2.001096101588997
Generation 340 Best Solution: [183.74178814534565, 176.2870226771078,
175.121376929991, 75.88432166657846] Fitness: 2.001096101588997
Generation 341 Best Solution: [183.6636198204305, 176.2870226771078,
175.121376929991, 75.88432166657846] Fitness: 1.9418071061523574
Generation 342 Best Solution: [183.07330239886454, 176.79807452941893,
175.121376929991, 74.71351666315105] Fitness: 1.0969858062161841
Generation 343 Best Solution: [183.6636198204305, 176.2870226771078,
175.121376929991, 74.71351666315105] Fitness: 0.9835876537916599
Generation 344 Best Solution: [183.85898472397395, 176.23086634862338,
175.121376929991, 75.88432166657846] Fitness: 2.0996800445660133
Generation 345 Best Solution: [183.6636198204305, 176.23086634862338,
175.121376929991, 75.88432166657846] Fitness: 1.9348482537166458
Generation 346 Best Solution: [183.6636198204305, 176.23086634862338,
175.121376929991, 75.88432166657846] Fitness: 1.9348482537166458
Generation 347 Best Solution: [183.6636198204305, 176.23086634862338,
175.121376929991, 75.88432166657846] Fitness: 1.9348482537166458
Generation 348 Best Solution: [183.6636198204305, 176.6256427427429,
175.121376929991, 75.88432166657846] Fitness: 2.080989907290204
Generation 349 Best Solution: [183.6636198204305, 176.63035854446122,
175.121376929991, 75.17289621871421] Fitness: 1.2641722204890824
Generation 350 Best Solution: [183.6636198204305, 176.63035854446122,
175.121376929991, 75.17289621871421] Fitness: 1.2641722204890824
Generation 351 Best Solution: [183.6636198204305, 176.63035854446122,
175.121376929991, 75.88432166657846] Fitness: 2.084105770512953
Generation 352 Best Solution: [183.6636198204305, 176.63035854446122,
175.121376929991, 75.17289621871421] Fitness: 1.2641722204890824
Generation 353 Best Solution: [183.6636198204305, 175.94955770412292,
175.121376929991, 75.88432166657846] Fitness: 1.9690300051716045
Generation 354 Best Solution: [183.6636198204305, 176.63035854446122,
175.121376929991, 74.84814181177812] Fitness: 1.134615753016767
Generation 355 Best Solution: [183.85898472397395, 176.64662175775408,
173.89185723947176, 75.88432166657846] Fitness: 1.5416823678744174
Generation 356 Best Solution: [183.6636198204305, 176.63035854446122,

175.121376929991, 75.88432166657846] Fitness: 2.084105770512953
Generation 357 Best Solution: [183.85898472397395, 177.4318629938142,
174.85673540066597, 74.8415544444708] Fitness: 1.9576368090355791
Generation 358 Best Solution: [183.6636198204305, 176.63035854446122,
175.121376929991, 75.88432166657846] Fitness: 2.084105770512953
Generation 359 Best Solution: [183.6636198204305, 176.63035854446122,
175.121376929991, 75.88432166657846] Fitness: 2.084105770512953
Generation 360 Best Solution: [183.0091459977109, 176.86254737378442,
175.121376929991, 75.88432166657846] Fitness: 2.129843663748648
Generation 361 Best Solution: [183.0091459977109, 176.86254737378442,
175.121376929991, 75.88432166657846] Fitness: 2.129843663748648
Generation 362 Best Solution: [183.0091459977109, 176.63035854446122,
175.121376929991, 75.88432166657846] Fitness: 1.9364251479251382
Generation 363 Best Solution: [183.85898472397395, 176.63035854446122,
174.79455463670743, 75.88432166657846] Fitness: 1.843456494589094
Generation 364 Best Solution: [183.58973231431315, 176.63035854446122,
174.79455463670743, 75.88432166657846] Fitness: 1.6307535455175193
Generation 365 Best Solution: [183.58973231431315, 176.63035854446122,
174.79455463670743, 75.88432166657846] Fitness: 1.6307535455175193
Generation 366 Best Solution: [183.85898472397395, 176.63035854446122,
174.79455463670743, 75.88432166657846] Fitness: 1.843456494589094
Generation 367 Best Solution: [183.6636198204305, 176.66296365727666,
175.121376929991, 75.88432166657846] Fitness: 2.1065338720336926
Generation 368 Best Solution: [183.85898472397395, 177.4318629938142,
174.20387563669826, 75.88432166657846] Fitness: 2.504291212303218
Generation 369 Best Solution: [183.85898472397395, 177.4318629938142,
174.85673540066597, 74.97268345345265] Fitness: 1.9902271740437547
Generation 370 Best Solution: [183.85898472397395, 177.4318629938142,
174.85673540066597, 74.97268345345265] Fitness: 1.9902271740437547
Generation 371 Best Solution: [183.6636198204305, 175.65282823270536,
174.20387563669826, 75.88432166657846] Fitness: 1.3856493723172918
Generation 372 Best Solution: [183.85898472397395, 176.72336022395842,
175.121376929991, 75.88432166657846] Fitness: 2.3169957689293286
Generation 373 Best Solution: [183.85898472397395, 176.09326091753027,
175.121376929991, 75.88432166657846] Fitness: 2.102019110105469
Generation 374 Best Solution: [183.85898472397395, 177.4318629938142,
174.85673540066597, 74.31453319656073] Fitness: 2.0789139263284717
Generation 375 Best Solution: [183.26082962850933, 176.60003392618802,
175.121376929991, 75.88432166657846] Fitness: 1.900017695213168
Generation 376 Best Solution: [183.85898472397395, 177.4318629938142,
175.121376929991, 74.91459592362385] Fitness: 2.3130071316911573
Generation 377 Best Solution: [183.85898472397395, 175.70025206313014,
175.121376929991, 75.88432166657846] Fitness: 2.260362240158705
Generation 378 Best Solution: [183.48955605430623, 175.70025206313014,
175.121376929991, 75.88432166657846] Fitness: 1.9954373434618566
Generation 379 Best Solution: [183.45142051894746, 177.4318629938142,
175.121376929991, 74.43070175398488] Fitness: 2.076803651980585
Generation 380 Best Solution: [183.07961970188774, 177.4318629938142,

175.121376929991, 74.43070175398488] Fitness: 2.0312302761158905
Generation 381 Best Solution: [182.98370195088154, 176.572800391627,
175.121376929991, 74.43070175398488] Fitness: 1.0167498572574463
Generation 382 Best Solution: [183.85898472397395, 176.55726511325713,
175.121376929991, 75.88432166657846] Fitness: 2.204277546250785
Generation 383 Best Solution: [183.85898472397395, 176.55726511325713,
174.85673540066597, 75.88432166657846] Fitness: 1.863975036284618
Generation 384 Best Solution: [183.48955605430623, 177.4318629938142,
174.85673540066597, 75.20996821674568] Fitness: 1.8478536744124792
Generation 385 Best Solution: [183.45142051894746, 176.29823301839593,
174.85673540066597, 75.88432166657846] Fitness: 1.4873058966487096
Generation 386 Best Solution: [183.85898472397395, 176.55726511325713,
173.8441711288943, 75.88432166657846] Fitness: 1.5024663725666958
Generation 387 Best Solution: [183.85898472397395, 176.55726511325713,
174.21910524553547, 75.88432166657846] Fitness: 1.462457868607736
Generation 388 Best Solution: [183.85898472397395, 176.55726511325713,
175.121376929991, 74.52484735389768] Fitness: 1.2781848387444672
Generation 389 Best Solution: [183.85898472397395, 175.69141438186307,
175.121376929991, 74.44221949783784] Fitness: 1.3707860488168881
Generation 390 Best Solution: [183.85898472397395, 177.4318629938142,
174.21910524553547, 74.44221949783784] Fitness: 1.61093048512337
Generation 391 Best Solution: [183.85898472397395, 175.9365423522254,
174.21910524553547, 75.88432166657846] Fitness: 1.3964095966558356
Generation 392 Best Solution: [183.85898472397395, 176.55726511325713,
174.73208470203917, 75.88432166657846] Fitness: 1.7389780260425347
Generation 393 Best Solution: [183.5303706222268, 177.4318629938142,
175.121376929991, 74.38235078885603] Fitness: 2.1355513032977447
Generation 394 Best Solution: [183.5303706222268, 177.4318629938142,
174.21910524553547, 75.88432166657846] Fitness: 2.2612399165288224
Generation 395 Best Solution: [183.5303706222268, 177.4318629938142,
174.21910524553547, 75.88432166657846] Fitness: 2.2612399165288224
Generation 396 Best Solution: [183.5153131283227, 176.60450434882637,
174.73208470203917, 75.85389828772823] Fitness: 1.463929610058901
Generation 397 Best Solution: [183.5153131283227, 177.3969168079661,
175.121376929991, 74.50129490466568] Fitness: 2.014068354149969
Generation 398 Best Solution: [183.01572382858205, 175.6027703594162,
174.73208470203917, 75.85389828772823] Fitness: 1.5048526505300979
Generation 399 Best Solution: [183.5303706222268, 175.6027703594162,
174.73208470203917, 75.85389828772823] Fitness: 1.57356995630446
Generation 400 Best Solution: [183.85898472397395, 175.6027703594162,
174.73208470203917, 75.85389828772823] Fitness: 1.8189802451888377
Generation 401 Best Solution: [183.85898472397395, 177.4318629938142,
175.121376929991, 74.50129490466568] Fitness: 2.330022861400474
Generation 402 Best Solution: [183.5153131283227, 177.4318629938142,
175.121376929991, 74.50129490466568] Fitness: 2.0771310924002595
Generation 403 Best Solution: [183.85898472397395, 177.4318629938142,
175.121376929991, 75.04513793035005] Fitness: 2.3592649389949356
Generation 404 Best Solution: [183.85898472397395, 176.1156660485703,

175.121376929991, 75.88432166657846] Fitness: 2.0997611122995563
Generation 405 Best Solution: [183.85898472397395, 176.61856104430856,
174.06364113180817, 75.88432166657846] Fitness: 1.4911586482162038
Generation 406 Best Solution: [183.85898472397395, 176.1156660485703,
175.121376929991, 75.74726102041066] Fitness: 1.8845427354206965
Generation 407 Best Solution: [183.5303706222268, 177.4318629938142,
174.55618229666743, 75.04513793035005] Fitness: 1.5106134872074262
Generation 408 Best Solution: [183.5303706222268, 177.4318629938142,
174.1747420842465, 75.04513793035005] Fitness: 1.3661033006539351
Generation 409 Best Solution: [183.5303706222268, 177.4318629938142,
175.0310030645839, 75.31460383773607] Fitness: 2.1600663009534498
Generation 410 Best Solution: [183.85898472397395, 177.4318629938142,
175.121376929991, 74.72633609258355] Fitness: 2.2899460532089875
Generation 411 Best Solution: [183.85898472397395, 177.4318629938142,
175.121376929991, 74.72633609258355] Fitness: 2.2899460532089875
Generation 412 Best Solution: [183.5303706222268, 176.3915330913384,
174.55618229666743, 75.88432166657846] Fitness: 1.2831501910278438
Generation 413 Best Solution: [183.85898472397395, 177.4318629938142,
175.121376929991, 74.72633609258355] Fitness: 2.2899460532089875
Generation 414 Best Solution: [183.85898472397395, 175.7371663620158,
175.121376929991, 75.78327012176771] Fitness: 2.0746077387199637
Generation 415 Best Solution: [183.85898472397395, 177.4318629938142,
173.81208944916222, 75.2983959866872] Fitness: 1.8308073999453192
Generation 416 Best Solution: [183.85898472397395, 177.4318629938142,
175.121376929991, 74.72633609258355] Fitness: 2.2899460532089875
Generation 417 Best Solution: [183.7582292392579, 177.4318629938142,
174.55618229666743, 74.72633609258355] Fitness: 1.5947635713637336
Generation 418 Best Solution: [183.85898472397395, 177.4318629938142,
174.55618229666743, 74.71412114325106] Fitness: 1.6869895453900394
Generation 419 Best Solution: [183.85898472397395, 177.4318629938142,
174.55618229666743, 74.71412114325106] Fitness: 1.6869895453900394
Generation 420 Best Solution: [183.83319029006796, 177.4318629938142,
174.55618229666743, 75.88432166657846] Fitness: 2.620284564323193
Generation 421 Best Solution: [183.85898472397395, 177.4318629938142,
174.18893668597525, 75.88432166657846] Fitness: 2.502305012826871
Generation 422 Best Solution: [183.85898472397395, 177.4318629938142,
173.90987443839506, 75.88432166657846] Fitness: 2.524871280128059
Generation 423 Best Solution: [183.4190885199687, 177.4318629938142,
173.90987443839506, 75.88432166657846] Fitness: 2.2319569568559574
Generation 424 Best Solution: [183.85898472397395, 177.4318629938142,
175.121376929991, 74.75867733359672] Fitness: 2.2902403802868037
Generation 425 Best Solution: [183.36962284196832, 176.09320196442232,
175.121376929991, 75.88432166657846] Fitness: 1.7937787211234908
Generation 426 Best Solution: [183.36962284196832, 176.09320196442232,
175.121376929991, 75.88432166657846] Fitness: 1.7937787211234908
Generation 427 Best Solution: [183.85898472397395, 177.4318629938142,
174.55618229666743, 74.40944687833115] Fitness: 1.7643063410493023
Generation 428 Best Solution: [182.8653895122221, 177.4318629938142,

174.8789180187851, 74.40944687833115] Fitness: 1.790362433146322
Generation 429 Best Solution: [183.36962284196832, 176.09320196442232,
175.121376929991, 75.88432166657846] Fitness: 1.7937787211234908
Generation 430 Best Solution: [183.36962284196832, 176.09320196442232,
175.121376929991, 75.7110732957842] Fitness: 1.5262964712350082
Generation 431 Best Solution: [183.85898472397395, 176.66036322946027,
175.121376929991, 74.64405449518885] Fitness: 1.3171066753389749
Generation 432 Best Solution: [183.85898472397395, 175.65124712797825,
174.06647217429608, 75.80333333424367] Fitness: 1.4151764766728046
Generation 433 Best Solution: [183.85898472397395, 175.65124712797825,
174.06647217429608, 75.80333333424367] Fitness: 1.4151764766728046
Generation 434 Best Solution: [183.6112627449213, 175.65124712797825,
174.06647217429608, 75.80333333424367] Fitness: 1.2156030982835033
Generation 435 Best Solution: [183.6112627449213, 175.65124712797825,
174.57342567692766, 75.80333333424367] Fitness: 1.3741036229756198
Generation 436 Best Solution: [183.85898472397395, 175.65124712797825,
174.92681153659763, 75.88432166657846] Fitness: 2.035752125917255
Generation 437 Best Solution: [183.85898472397395, 175.65124712797825,
174.06647217429608, 75.88432166657846] Fitness: 1.545650447130725
Generation 438 Best Solution: [183.85898472397395, 175.65124712797825,
174.470554664776, 75.80333333424367] Fitness: 1.5112825272808146
Generation 439 Best Solution: [183.21202151013827, 175.66480481879304,
174.470554664776, 75.80333333424367] Fitness: 1.1677468734327756
Generation 440 Best Solution: [183.85898472397395, 177.08008736942176,
174.57342567692766, 75.88432166657846] Fitness: 2.1030150997787525
Generation 441 Best Solution: [183.21202151013827, 175.92047125906828,
175.121376929991, 75.88432166657846] Fitness: 1.8105966795362187
Generation 442 Best Solution: [183.85898472397395, 177.4318629938142,
174.470554664776, 75.11971039074507] Fitness: 1.7427109512683945
Generation 443 Best Solution: [183.21202151013827, 177.4318629938142,
175.121376929991, 75.03979353453953] Fitness: 2.023519087068355
Generation 444 Best Solution: [183.87156857698352, 175.9729475865919,
175.121376929991, 75.80333333424367] Fitness: 2.0086785930891518
Generation 445 Best Solution: [183.87156857698352, 175.9729475865919,
174.30557667251352, 75.88432166657846] Fitness: 1.4171227000843043
Generation 446 Best Solution: [183.87156857698352, 175.9729475865919,
174.92681153659763, 75.80333333424367] Fitness: 1.7485710569924584
Generation 447 Best Solution: [183.85898472397395, 175.9729475865919,
174.92681153659763, 75.80333333424367] Fitness: 1.7360507794299054
Generation 448 Best Solution: [183.87156857698352, 177.4318629938142,
174.58011621603453, 74.3626628760529] Fitness: 1.8167808097096267
Generation 449 Best Solution: [183.85898472397395, 177.2897932948816,
174.17212591140083, 75.61011223523073] Fitness: 1.85192792638115
Generation 450 Best Solution: [183.21202151013827, 177.4318629938142,
174.92681153659763, 74.72222116018706] Fitness: 1.6965447371431137
Generation 451 Best Solution: [182.98867489289876, 177.34910484091063,
174.92681153659763, 74.72222116018706] Fitness: 1.5763621792425775
Generation 452 Best Solution: [182.98867489289876, 177.34910484091063,

174.92681153659763, 74.72222116018706] Fitness: 1.5763621792425775
Generation 453 Best Solution: [182.98867489289876, 177.2897932948816,
174.58011621603453, 75.53781421518804] Fitness: 1.617644531079435
Generation 454 Best Solution: [183.21202151013827, 177.2897932948816,
174.92681153659763, 74.72222116018706] Fitness: 1.451238795940236
Generation 455 Best Solution: [183.21202151013827, 177.2897932948816,
175.121376929991, 74.72222116018706] Fitness: 1.711346332036929
Generation 456 Best Solution: [183.85898472397395, 177.43415907605726,
174.44117476999423, 75.61011223523073] Fitness: 2.1803541016223136
Generation 457 Best Solution: [183.21202151013827, 177.34910484091063,
174.1286174939406, 75.88432166657846] Fitness: 2.0177563912789194
Generation 458 Best Solution: [183.21202151013827, 176.43605310472626,
175.121376929991, 75.8840759458904] Fitness: 1.8135702421544224
Generation 459 Best Solution: [183.87156857698352, 176.02827996004152,
174.1286174939406, 75.88432166657846] Fitness: 1.3743348511813571
Generation 460 Best Solution: [182.87128009155145, 176.37955543989062,
175.121376929991, 75.88432166657846] Fitness: 1.863935019018415
Generation 461 Best Solution: [183.87156857698352, 177.2897932948816,
174.1286174939406, 75.88432166657846] Fitness: 2.2648011082902015
Generation 462 Best Solution: [183.04673086782088, 175.84959828999382,
175.121376929991, 75.88432166657846] Fitness: 1.8528313861384031
Generation 463 Best Solution: [183.04673086782088, 175.84959828999382,
174.45313293181337, 75.61011223523073] Fitness: 0.7864612981073321
Generation 464 Best Solution: [183.85898472397395, 177.34910484091063,
173.8553269030175, 74.29748796728155] Fitness: 1.5774654162502875
Generation 465 Best Solution: [183.87156857698352, 177.34910484091063,
173.8553269030175, 74.29748796728155] Fitness: 1.5899856938128407
Generation 466 Best Solution: [183.21202151013827, 176.67977555672562,
175.121376929991, 75.88432166657846] Fitness: 1.9501688565918616
Generation 467 Best Solution: [183.21202151013827, 176.67977555672562,
174.6821899445732, 75.88432166657846] Fitness: 1.4411697062635018
Generation 468 Best Solution: [183.21202151013827, 177.34910484091063,
175.121376929991, 75.61011223523073] Fitness: 2.3654152511771565
Generation 469 Best Solution: [183.85898472397395, 177.12204114859787,
175.121376929991, 75.61011223523073] Fitness: 2.3480876662323475
Generation 470 Best Solution: [183.85898472397395, 177.12204114859787,
175.121376929991, 75.27956897594318] Fitness: 2.00740425056507
Generation 471 Best Solution: [182.8223396671968, 177.2897932948816,
175.121376929991, 74.93953670544668] Fitness: 1.8342542719799046
Generation 472 Best Solution: [183.85898472397395, 177.2897932948816,
174.8293618149176, 74.58315643861835] Fitness: 1.691925441510893
Generation 473 Best Solution: [183.85898472397395, 177.2897932948816,
174.8293618149176, 74.58315643861835] Fitness: 1.691925441510893
Generation 474 Best Solution: [183.21202151013827, 176.902023751245,
174.16632089325964, 75.88432166657846] Fitness: 1.4011208426465858
Generation 475 Best Solution: [183.85898472397395, 175.747845624365,
173.84560678709693, 75.88432166657846] Fitness: 1.5268764336476504
Generation 476 Best Solution: [183.01918541483334, 176.35206198893368,

175.121376929991, 75.61011223523073] Fitness: 1.402254674649515
Generation 477 Best Solution: [183.01918541483334, 176.35206198893368,
175.121376929991, 75.88432166657846] Fitness: 1.8054789258770008
Generation 478 Best Solution: [183.21202151013827, 177.2897932948816,
173.8647266990312, 75.88432166657846] Fitness: 1.960496105560051
Generation 479 Best Solution: [183.01918541483334, 177.00809721356734,
175.121376929991, 74.73149041252096] Fitness: 1.3300565327590508
Generation 480 Best Solution: [183.6147056725854, 177.2897932948816,
174.30179977681973, 75.359216122992] Fitness: 1.4062808808717244
Generation 481 Best Solution: [183.01918541483334, 175.9794891058328,
175.121376929991, 75.88432166657846] Fitness: 1.8098004746807466
Generation 482 Best Solution: [183.60880082911206, 177.2897932948816,
175.121376929991, 74.81461917890414] Fitness: 1.8479265105778984
Generation 483 Best Solution: [183.19480347048872, 175.65503027233532,
175.121376929991, 75.8840759458904] Fitness: 1.9586640951330292
Generation 484 Best Solution: [183.01918541483334, 175.65503027233532,
175.121376929991, 75.8840759458904] Fitness: 1.9776322647351745
Generation 485 Best Solution: [183.60880082911206, 177.2897932948816,
173.84860222775313, 75.8840759458904] Fitness: 2.098166821597787
Generation 486 Best Solution: [183.19480347048872, 177.2897932948816,
175.121376929991, 74.87649744179087] Fitness: 1.7249531300794652
Generation 487 Best Solution: [183.1930082527968, 177.2897932948816,
173.81585484605156, 75.86886562759581] Fitness: 1.9530885708982857
Generation 488 Best Solution: [183.21202151013827, 177.2897932948816,
174.90843167921045, 74.82994553736962] Fitness: 1.4357553827405718
Generation 489 Best Solution: [183.19480347048872, 176.27828175938757,
173.95590532522084, 75.88432166657846] Fitness: 1.03621176104051
Generation 490 Best Solution: [183.19480347048872, 176.27828175938757,
173.95590532522084, 75.6796724071653] Fitness: 0.7249225150342024
Generation 491 Best Solution: [183.19480347048872, 176.27828175938757,
173.95590532522084, 75.88432166657846] Fitness: 1.03621176104051
Generation 492 Best Solution: [183.60880082911206, 177.2897932948816,
174.01100408386347, 75.88432166657846] Fitness: 2.0571919591819565
Generation 493 Best Solution: [183.60880082911206, 175.9042681163058,
175.121376929991, 74.43293369364793] Fitness: 1.0573278946405544
Generation 494 Best Solution: [183.60880082911206, 177.2897932948816,
175.121376929991, 74.78968237263899] Fitness: 1.845536176209697
Generation 495 Best Solution: [183.60880082911206, 176.17432017109525,
175.121376929991, 75.88432166657846] Fitness: 1.8961988832637386
Generation 496 Best Solution: [183.01918541483334, 176.66180340795643,
175.121376929991, 75.88432166657846] Fitness: 1.9555887176009243
Generation 497 Best Solution: [183.08945831266814, 175.65708422624465,
175.121376929991, 75.6796724071653] Fitness: 1.6522326911221272
Generation 498 Best Solution: [183.08945831266814, 175.65708422624465,
174.2662467076265, 75.88432166657846] Fitness: 1.231142618512057
Generation 499 Best Solution: [183.19480347048872, 176.02784352456717,
175.121376929991, 75.6796724071653] Fitness: 1.467580444214544
Generation 500 Best Solution: [183.19480347048872, 177.2897932948816,

174.5744493074435, 75.88432166657846] Fitness: 2.0782329810704763
Optimal Solution for Simulation 8: [183.19480347048872, 177.2897932948816,
174.5744493074435, 75.88432166657846]

Simulation 9/32 with conditions: {'drought': True, 'flood': False,
'policy_change': True, 'social_event': True, 'economic_crisis': True}
Generation 1 Best Solution: [183.28029612887477, 176.1202555814124,
174.1722995211277, 74.44205343463376] Fitness: 0.10511626823509251
Generation 2 Best Solution: [182.96178498951, 176.52296842725235,
174.38589583303548, 74.80907125901321] Fitness: 0.21278263112963655
Generation 3 Best Solution: [182.79807286130887, 176.03724753788265,
174.33602851400403, 75.10549353854229] Fitness: 0.39095105524246365
Generation 4 Best Solution: [183.18056823559803, 175.81602302524726,
174.33602851400403, 75.10549353854229] Fitness: 0.3392112160431376
Generation 5 Best Solution: [183.18056823559803, 175.8425038200707,
174.33602851400403, 74.34527601990978] Fitness: 0.297274284349983
Generation 6 Best Solution: [182.89372184526204, 176.7542189115757,
174.54419522592138, 74.94543604195512] Fitness: 0.5029528950481754
Generation 7 Best Solution: [183.701546951796, 176.48453504131425,
174.54419522592138, 74.94543604195512] Fitness: 0.3914945480235629
Generation 8 Best Solution: [183.701546951796, 176.3100247371523,
173.82469773601684, 75.09716362738394] Fitness: 0.41044585600229233
Generation 9 Best Solution: [183.18056823559803, 176.9924888773045,
173.82469773601684, 74.63181818806586] Fitness: 0.5797611020285898
Generation 10 Best Solution: [183.23624697465414, 176.9924888773045,
173.82469773601684, 74.63181818806586] Fitness: 0.5751277666434316
Generation 11 Best Solution: [183.23624697465414, 176.9924888773045,
173.82469773601684, 75.56787242756458] Fitness: 1.1610617037685849
Generation 12 Best Solution: [182.79807286130887, 176.55183241060632,
175.09934021533476, 75.56787242756458] Fitness: 1.5540993826674203
Generation 13 Best Solution: [183.23624697465414, 177.42534077406503,
173.899555508722, 75.4199309036235] Fitness: 1.5986553812451487
Generation 14 Best Solution: [183.18056823559803, 177.42534077406503,
173.82469773601684, 74.5743840850084] Fitness: 1.2483435750894991
Generation 15 Best Solution: [183.23624697465414, 177.42534077406503,
175.09934021533476, 74.72690534948791] Fitness: 1.8200459933003121
Generation 16 Best Solution: [183.23624697465414, 177.42534077406503,
174.47509697788865, 75.56787242756458] Fitness: 1.7991533439835856
Generation 17 Best Solution: [183.18056823559803, 177.42534077406503,
175.09934021533476, 74.5743840850084] Fitness: 1.8378821110869037
Generation 18 Best Solution: [183.701546951796, 176.72255792188344,
175.09934021533476, 74.5743840850084] Fitness: 1.0668269095560685
Generation 19 Best Solution: [182.79807286130887, 175.88134800444638,
175.09934021533476, 74.61116427209866] Fitness: 1.0050534373356956
Generation 20 Best Solution: [183.701546951796, 176.72255792188344,
175.09934021533476, 74.61116427209866] Fitness: 1.0602668616080613
Generation 21 Best Solution: [183.18056823559803, 177.42534077406503,
173.8955493903248, 74.61116427209866] Fitness: 1.206909081990302

Generation 22 Best Solution: [183.23624697465414, 175.88698981836066,
175.09934021533476, 74.61116427209866] Fitness: 0.8324045708199731
Generation 23 Best Solution: [183.701546951796, 176.12041645745657,
174.9934507987574, 74.61116427209866] Fitness: 0.7481014720820105
Generation 24 Best Solution: [182.79807286130887, 176.66335709894622,
175.09934021533476, 75.42902915616277] Fitness: 1.4428919878826059
Generation 25 Best Solution: [183.51326157160585, 175.85299629019627,
175.09934021533476, 75.60155568566586] Fitness: 1.5330469694109787
Generation 26 Best Solution: [183.701546951796, 176.93865234973873,
175.09934021533476, 75.42902915616277] Fitness: 1.6682221146506508
Generation 27 Best Solution: [182.79807286130887, 176.93865234973873,
175.09934021533476, 75.42902915616277] Fitness: 1.6840716764595234
Generation 28 Best Solution: [183.701546951796, 176.9661859390564,
174.31454246351262, 75.42902915616277] Fitness: 1.0310854637570113
Generation 29 Best Solution: [183.701546951796, 176.93865234973873,
175.09934021533476, 75.83814688329323] Fitness: 2.2706839265826138
Generation 30 Best Solution: [182.79807286130887, 177.42534077406503,
174.14390383929933, 75.56787242756458] Fitness: 1.8954831771822191
Generation 31 Best Solution: [182.79807286130887, 176.93865234973873,
175.09934021533476, 75.83814688329323] Fitness: 2.2865334883914863
Generation 32 Best Solution: [183.701546951796, 177.42534077406503,
173.97276221357924, 75.60155568566586] Fitness: 1.957055723607631
Generation 33 Best Solution: [182.79807286130887, 176.75759776622712,
175.09934021533476, 75.73560005329585] Fitness: 1.938440633571452
Generation 34 Best Solution: [182.79807286130887, 176.0241617169543,
175.09934021533476, 75.56787242756458] Fitness: 1.521204301850281
Generation 35 Best Solution: [183.701546951796, 175.60671697499635,
175.09934021533476, 75.56787242756458] Fitness: 1.7922370186402188
Generation 36 Best Solution: [183.701546951796, 175.60671697499635,
175.09934021533476, 75.42902915616277] Fitness: 1.6175399312379202
Generation 37 Best Solution: [183.701546951796, 175.60671697499635,
174.08449054580183, 75.8442770391721] Fitness: 1.552564550029145
Generation 38 Best Solution: [183.701546951796, 176.57722049839984,
175.09934021533476, 75.56787242756458] Fitness: 1.5509685672834357
Generation 39 Best Solution: [182.79807286130887, 176.57722049839984,
174.85342060535717, 75.56787242756458] Fitness: 1.252435357227841
Generation 40 Best Solution: [183.70016882567242, 176.57722049839984,
174.85342060535717, 75.56787242756458] Fitness: 1.2356239839526104
Generation 41 Best Solution: [183.701546951796, 175.71312126928112,
175.09934021533476, 75.56787242756458] Fitness: 1.6928637719570387
Generation 42 Best Solution: [183.70016882567242, 176.57722049839984,
175.09934021533476, 75.83814688329323] Fitness: 1.9777714803467419
Generation 43 Best Solution: [183.70016882567242, 176.57722049839984,
175.09934021533476, 75.83814688329323] Fitness: 1.9777714803467419
Generation 44 Best Solution: [183.70016882567242, 176.57722049839984,
175.12223653276843, 75.56787242756458] Fitness: 1.584158796477761
Generation 45 Best Solution: [182.79807286130887, 177.42534077406503,
174.39086373463547, 74.56564475366534] Fitness: 1.3587042928792963

Generation 46 Best Solution: [182.79807286130887, 177.42534077406503,
173.9358788831714, 75.56787242756458] Fitness: 1.9385586708781832
Generation 47 Best Solution: [183.701546951796, 177.42534077406503,
173.9358788831714, 75.56787242756458] Fitness: 1.9227091090693107
Generation 48 Best Solution: [183.70016882567242, 177.42534077406503,
173.9358788831714, 75.56787242756458] Fitness: 1.9217472976029528
Generation 49 Best Solution: [182.79807286130887, 177.35488295248615,
174.33939388363203, 75.56787242756458] Fitness: 1.7900003292324094
Generation 50 Best Solution: [182.79807286130887, 177.35488295248615,
174.33939388363203, 74.84651985456594] Fitness: 1.2156290021423692
Generation 51 Best Solution: [183.701546951796, 175.939442121472,
174.33939388363203, 75.56787242756458] Fitness: 0.8795200869486811
Generation 52 Best Solution: [182.79807286130887, 176.11876773821396,
175.09934021533476, 75.18355142466359] Fitness: 1.0858662585690084
Generation 53 Best Solution: [182.79807286130887, 176.11876773821396,
175.09934021533476, 74.84651985456594] Fitness: 0.9202368650877533
Generation 54 Best Solution: [183.70016882567242, 176.3955581620632,
174.10587671904779, 75.83814688329323] Fitness: 1.2282430888107683
Generation 55 Best Solution: [183.70016882567242, 176.3955581620632,
175.09934021533476, 75.83814688329323] Fitness: 1.9092790841789453
Generation 56 Best Solution: [183.70016882567242, 177.42534077406503,
174.39546137535916, 75.0177422151064] Fitness: 1.4055309762926815
Generation 57 Best Solution: [183.70016882567242, 176.3955581620632,
175.12432468166386, 75.18355142466359] Fitness: 1.1100805171144266
Generation 58 Best Solution: [182.93851918741214, 176.03747290121396,
175.09934021533476, 75.83814688329323] Fitness: 1.856912350716187
Generation 59 Best Solution: [182.93851918741214, 176.03747290121396,
175.09934021533476, 75.83814688329323] Fitness: 1.856912350716187
Generation 60 Best Solution: [183.17480434558527, 176.03747290121396,
175.09934021533476, 74.46478879368647] Fitness: 0.8082504559558076
Generation 61 Best Solution: [183.70016882567242, 177.42534077406503,
173.99406313383744, 74.9499795895837] Fitness: 1.3600915094984511
Generation 62 Best Solution: [183.0761295893658, 177.42534077406503,
174.7735693910534, 74.9499795895837] Fitness: 1.4979403565556757
Generation 63 Best Solution: [183.70016882567242, 177.22933582465296,
175.09934021533476, 74.59009120097896] Fitness: 1.6474053327939753
Generation 64 Best Solution: [183.70016882567242, 177.22933582465296,
175.09934021533476, 74.59009120097896] Fitness: 1.6474053327939753
Generation 65 Best Solution: [183.0761295893658, 177.22933582465296,
175.09934021533476, 74.59009120097896] Fitness: 1.5214168991439767
Generation 66 Best Solution: [183.28227371672392, 176.54449678377136,
175.09934021533476, 74.7893030270701] Fitness: 0.7965114607289803
Generation 67 Best Solution: [183.70016882567242, 177.42534077406503,
173.99406313383744, 74.7893030270701] Fitness: 1.3181853988094339
Generation 68 Best Solution: [183.70016882567242, 177.42534077406503,
173.99406313383744, 74.7893030270701] Fitness: 1.3181853988094339
Generation 69 Best Solution: [183.28227371672392, 176.54449678377136,
175.09934021533476, 74.90353604351608] Fitness: 0.8220968936420087

Generation 70 Best Solution: [183.70016882567242, 176.41802296620557,
175.09934021533476, 75.88733500807321] Fitness: 2.005225541973278
Generation 71 Best Solution: [183.70016882567242, 176.21181060452744,
175.09934021533476, 75.6352656586426] Fitness: 1.5613068938637866
Generation 72 Best Solution: [183.70016882567242, 176.21181060452744,
175.09934021533476, 75.6352656586426] Fitness: 1.5613068938637866
Generation 73 Best Solution: [183.70016882567242, 176.21181060452744,
175.09934021533476, 75.88733500807321] Fitness: 1.9835649092712888
Generation 74 Best Solution: [183.3299312505482, 177.42534077406503,
175.09934021533476, 75.29721756114236] Fitness: 2.1009291930718095
Generation 75 Best Solution: [183.70016882567242, 177.42534077406503,
174.43817712753304, 75.88733500807321] Fitness: 2.4530455857644204
Generation 76 Best Solution: [183.70016882567242, 177.42534077406503,
175.09934021533476, 74.29451329087549] Fitness: 2.1060272468703687
Generation 77 Best Solution: [183.70016882567242, 177.42534077406503,
175.09934021533476, 74.29451329087549] Fitness: 2.1060272468703687
Generation 78 Best Solution: [183.70016882567242, 177.42534077406503,
175.09934021533476, 74.29451329087549] Fitness: 2.1060272468703687
Generation 79 Best Solution: [183.70016882567242, 176.3419242650842,
175.09934021533476, 75.83814688329323] Fitness: 1.899066273194343
Generation 80 Best Solution: [182.88279191021772, 177.42534077406503,
174.5358515344979, 75.6352656586426] Fitness: 2.0401480249153936
Generation 81 Best Solution: [183.3299312505482, 176.94546339030308,
175.09934021533476, 74.34049587779967] Fitness: 1.215468149799545
Generation 82 Best Solution: [183.70016882567242, 177.42534077406503,
174.14200845954434, 75.6352656586426] Fitness: 1.974574057573456
Generation 83 Best Solution: [183.22975694567737, 176.51273345793345,
174.8930400786337, 75.72738630189933] Fitness: 1.3356469336610246
Generation 84 Best Solution: [183.22975694567737, 176.51273345793345,
175.09934021533476, 75.6352656586426] Fitness: 1.463231412923811
Generation 85 Best Solution: [182.81928407388494, 176.63483477097552,
174.8930400786337, 75.72738630189933] Fitness: 1.552495063556282
Generation 86 Best Solution: [183.19401669440632, 175.9700295399798,
175.09934021533476, 75.88733500807321] Fitness: 1.8945626511068512
Generation 87 Best Solution: [182.81928407388494, 177.42534077406503,
174.07042506367458, 75.83814688329323] Fitness: 2.3154080820052556
Generation 88 Best Solution: [183.3299312505482, 177.42534077406503,
174.08403176565122, 75.83814688329323] Fitness: 2.1625257398231987
Generation 89 Best Solution: [183.70016882567242, 176.85204744406542,
175.09934021533476, 75.19098387088027] Fitness: 1.3500655123537448
Generation 90 Best Solution: [183.70016882567242, 177.42534077406503,
175.09934021533476, 75.19098387088027] Fitness: 2.1593961411849145
Generation 91 Best Solution: [182.94607982712716, 177.42534077406503,
175.09934021533476, 74.82173117652475] Fitness: 1.908549518911099
Generation 92 Best Solution: [182.81928407388494, 176.85204744406542,
175.09934021533476, 75.4668138655948] Fitness: 1.6245380132673246
Generation 93 Best Solution: [183.70016882567242, 176.85204744406542,
175.09934021533476, 75.07353588405624] Fitness: 1.270508860861988

Generation 94 Best Solution: [183.31582733996612, 176.85204744406542,
175.09934021533476, 75.83814688329323] Fitness: 2.0302283948321893
Generation 95 Best Solution: [183.70016882567242, 176.38381468014043,
175.09934021533476, 75.63490093898174] Fitness: 1.5741697344535583
Generation 96 Best Solution: [183.70016882567242, 176.85204744406542,
175.09934021533476, 75.6352656586426] Fitness: 1.8489440040324927
Generation 97 Best Solution: [183.3299312505482, 177.42534077406503,
175.09934021533476, 74.91912093702815] Fitness: 1.8592630583495502
Generation 98 Best Solution: [183.0143481788931, 175.62298601000947,
175.09934021533476, 75.6352656586426] Fitness: 1.7658807929824571
Generation 99 Best Solution: [183.70016882567242, 176.14411183056762,
175.09934021533476, 75.6352656586426] Fitness: 1.5689026589342545
Generation 100 Best Solution: [183.70016882567242, 177.42534077406503,
174.13754853246135, 74.91912093702815] Fitness: 1.3250527861494212
Generation 101 Best Solution: [182.94607982712716, 177.42534077406503,
175.09934021533476, 75.13592851866098] Fitness: 2.0449130396316497
Generation 102 Best Solution: [182.94607982712716, 176.1804836046942,
175.09934021533476, 75.88733500807321] Fitness: 1.9117106216849937
Generation 103 Best Solution: [183.70016882567242, 176.75000912514906,
175.09934021533476, 75.83814688329323] Fitness: 2.0914919059594537
Generation 104 Best Solution: [183.77803384606025, 176.35934626016459,
175.09934021533476, 75.83814688329323] Fitness: 1.9609497844467192
Generation 105 Best Solution: [182.94607982712716, 176.75000912514906,
175.09934021533476, 75.83814688329323] Fitness: 2.0170264230740425
Generation 106 Best Solution: [182.94607982712716, 176.75000912514906,
175.09934021533476, 75.6352656586426] Fitness: 1.685081338677339
Generation 107 Best Solution: [182.94607982712716, 176.30982081530155,
175.09934021533476, 75.83814688329323] Fitness: 1.8206707337555077
Generation 108 Best Solution: [182.94607982712716, 176.30982081530155,
175.09934021533476, 75.88733500807321] Fitness: 1.9109836647663063
Generation 109 Best Solution: [182.8090213493456, 177.42534077406503,
173.93527389947707, 75.41430505962067] Fitness: 1.7394118240342387
Generation 110 Best Solution: [183.70016882567242, 177.42534077406503,
174.0231950239422, 75.41430505962067] Fitness: 1.7038775660830456
Generation 111 Best Solution: [182.94607982712716, 175.74632000418706,
175.09934021533476, 75.83814688329323] Fitness: 2.0178719455081056
Generation 112 Best Solution: [183.51690289646953, 175.8753372996814,
175.09934021533476, 75.88733500807321] Fitness: 1.9919759133944175
Generation 113 Best Solution: [183.70016882567242, 177.42534077406503,
175.09934021533476, 75.00707259095975] Fitness: 2.0445131151237597
Generation 114 Best Solution: [183.70016882567242, 177.42534077406503,
175.09934021533476, 75.00707259095975] Fitness: 2.0445131151237597
Generation 115 Best Solution: [183.70016882567242, 175.9116588485156,
175.09934021533476, 75.69685045554594] Fitness: 1.7441804131868386
Generation 116 Best Solution: [183.70016882567242, 177.42534077406503,
173.95830557325107, 75.83814688329323] Fitness: 2.341566974703052
Generation 117 Best Solution: [182.94607982712716, 177.42534077406503,
175.09934021533476, 74.66077610240042] Fitness: 1.8993491747145566

Generation 118 Best Solution: [183.70016882567242, 177.31464466278527,
174.94612725576133, 74.66077610240042] Fitness: 1.5697974985361196
Generation 119 Best Solution: [183.70016882567242, 177.31464466278527,
174.94612725576133, 74.66077610240042] Fitness: 1.5697974985361196
Generation 120 Best Solution: [183.70016882567242, 177.31464466278527,
174.94612725576133, 74.71226278426921] Fitness: 1.5682704646140337
Generation 121 Best Solution: [183.70016882567242, 176.60645975800676,
175.09934021533476, 74.5485216916806] Fitness: 0.9951883925477941
Generation 122 Best Solution: [183.70016882567242, 177.30219466578015,
175.09934021533476, 74.7711000553456] Fitness: 1.7578779093034183
Generation 123 Best Solution: [183.70016882567242, 177.42534077406503,
174.64152830448762, 74.4255339856709] Fitness: 1.5259583265867993
Generation 124 Best Solution: [183.01716122872608, 177.42534077406503,
175.10639244872806, 74.56591109029031] Fitness: 1.8921191177596903
Generation 125 Best Solution: [183.70016882567242, 176.7438006674843,
175.09934021533476, 75.14543133590175] Fitness: 1.2223083131171917
Generation 126 Best Solution: [183.70016882567242, 176.7438006674843,
174.58599708165931, 75.83814688329323] Fitness: 1.5392067848282935
Generation 127 Best Solution: [183.70016882567242, 177.31464466278527,
174.0350388645148, 75.31358715857992] Fitness: 1.3990994195032196
Generation 128 Best Solution: [183.70016882567242, 177.31464466278527,
174.0350388645148, 75.31358715857992] Fitness: 1.3990994195032196
Generation 129 Best Solution: [182.89219374682514, 175.62571595123498,
175.09934021533476, 75.83814688329323] Fitness: 2.1547232905735068
Generation 130 Best Solution: [183.70016882567242, 176.90416018213787,
175.09934021533476, 75.83814688329323] Fitness: 2.232916964326788
Generation 131 Best Solution: [183.70016882567242, 175.62571595123498,
175.09934021533476, 75.26304887032974] Fitness: 1.4288026093969544
Generation 132 Best Solution: [183.70016882567242, 176.26414438774916,
175.09934021533476, 75.83814688329323] Fitness: 1.8923622404636897
Generation 133 Best Solution: [183.70016882567242, 175.87968810518302,
175.09934021533476, 75.88733500807321] Fitness: 2.090480062682192
Generation 134 Best Solution: [183.70016882567242, 175.95085611174616,
175.10639244872806, 75.88733500807321] Fitness: 2.0632712783473073
Generation 135 Best Solution: [182.89219374682514, 177.42534077406503,
174.82648789155806, 75.88733500807321] Fitness: 2.6922913774378663
Generation 136 Best Solution: [182.89219374682514, 177.42534077406503,
174.2057167737461, 75.88733500807321] Fitness: 2.351922417235776
Generation 137 Best Solution: [183.70016882567242, 177.42534077406503,
175.10639244872806, 74.42210014536447] Fitness: 2.04625022042759
Generation 138 Best Solution: [182.89219374682514, 177.42534077406503,
175.10639244872806, 74.42210014536447] Fitness: 2.0009941663968833
Generation 139 Best Solution: [182.89219374682514, 177.42534077406503,
175.10639244872806, 74.42210014536447] Fitness: 2.0009941663968833
Generation 140 Best Solution: [182.89219374682514, 177.42534077406503,
175.10639244872806, 74.62463127022512] Fitness: 1.9425731734012397
Generation 141 Best Solution: [182.89219374682514, 177.42534077406503,
175.10639244872806, 74.42210014536447] Fitness: 2.0009941663968833

Generation 142 Best Solution: [183.70016882567242, 177.42534077406503,
175.10639244872806, 74.63887535934333] Fitness: 1.986169374768663
Generation 143 Best Solution: [183.70016882567242, 177.42534077406503,
175.10639244872806, 74.63887535934333] Fitness: 1.986169374768663
Generation 144 Best Solution: [183.70016882567242, 177.42534077406503,
174.82355443285337, 74.81496020251913] Fitness: 1.6357689304950351
Generation 145 Best Solution: [183.57847477016642, 177.42534077406503,
174.82355443285337, 74.81496020251913] Fitness: 1.5627155579919565
Generation 146 Best Solution: [183.70016882567242, 177.42534077406503,
174.82763446696703, 74.66245503088679] Fitness: 1.6319075338096127
Generation 147 Best Solution: [183.70016882567242, 177.42534077406503,
174.82763446696703, 74.66245503088679] Fitness: 1.6319075338096127
Generation 148 Best Solution: [183.70016882567242, 177.42534077406503,
174.82763446696703, 74.66245503088679] Fitness: 1.6319075338096127
Generation 149 Best Solution: [182.89219374682514, 177.04970466167524,
174.82763446696703, 74.66245503088679] Fitness: 0.9974721308112784
Generation 150 Best Solution: [182.89219374682514, 175.63155875854693,
175.10639244872806, 74.35371195408625] Fitness: 1.2394283974715845
Generation 151 Best Solution: [182.89219374682514, 175.63155875854693,
174.90536184476824, 75.88733500807321] Fitness: 1.983317414450553
Generation 152 Best Solution: [182.89219374682514, 175.63155875854693,
173.78727603577488, 75.88733500807321] Fitness: 1.6713841484974001
Generation 153 Best Solution: [182.89219374682514, 177.42534077406503,
175.09934021533476, 74.33853260215308] Fitness: 2.033631094880147
Generation 154 Best Solution: [183.70016882567242, 177.42534077406503,
173.78727603577488, 74.33853260215308] Fitness: 1.5109817148945417
Generation 155 Best Solution: [182.89219374682514, 177.42534077406503,
173.78727603577488, 74.33853260215308] Fitness: 1.4657256608638352
Generation 156 Best Solution: [182.89219374682514, 177.42534077406503,
173.78727603577488, 74.85012011858807] Fitness: 1.37573940268884
Generation 157 Best Solution: [182.89219374682514, 176.41833557622394,
175.09934021533476, 75.88733500807321] Fitness: 1.9600535287961591
Generation 158 Best Solution: [183.70016882567242, 177.42534077406503,
173.97709175894786, 74.38816762242998] Fitness: 1.3972774245511983
Generation 159 Best Solution: [183.50126151369798, 177.42534077406503,
175.09934021533476, 74.97966975292915] Fitness: 1.9247640004930324
Generation 160 Best Solution: [183.55099194047293, 177.42534077406503,
173.78727603577488, 75.55801416539566] Fitness: 1.8948372452719282
Generation 161 Best Solution: [183.25283033757037, 177.42534077406503,
175.09934021533476, 75.55801416539566] Fitness: 2.396037632419847
Generation 162 Best Solution: [182.89219374682514, 177.42534077406503,
175.09934021533476, 74.36241402235103] Fitness: 2.0201930793259804
Generation 163 Best Solution: [183.25283033757037, 177.42534077406503,
175.09934021533476, 74.55932651912312] Fitness: 1.8361238944805556
Generation 164 Best Solution: [183.25283033757037, 177.42534077406503,
174.38047722001562, 75.55801416539566] Fitness: 1.7466861847980653
Generation 165 Best Solution: [183.25283033757037, 177.42534077406503,
175.09934021533476, 74.55932651912312] Fitness: 1.8361238944805556

Generation 166 Best Solution: [182.89219374682514, 176.03052427801737,
175.09934021533476, 75.88733500807321] Fitness: 1.9749805642238212
Generation 167 Best Solution: [182.89219374682514, 176.03052427801737,
175.09934021533476, 75.88733500807321] Fitness: 1.9749805642238212
Generation 168 Best Solution: [182.89219374682514, 176.03052427801737,
175.09934021533476, 75.88733500807321] Fitness: 1.9749805642238212
Generation 169 Best Solution: [183.25283033757037, 177.42534077406503,
173.82419764294374, 75.88733500807321] Fitness: 2.3382636753830353
Generation 170 Best Solution: [182.89219374682514, 177.0354288790641,
175.09934021533476, 75.88733500807321] Fitness: 2.4281205539206883
Generation 171 Best Solution: [182.89219374682514, 176.148919145706,
175.09934021533476, 75.88733500807321] Fitness: 1.9451254560533533
Generation 172 Best Solution: [182.89219374682514, 176.148919145706,
175.09934021533476, 75.88733500807321] Fitness: 1.9451254560533533
Generation 173 Best Solution: [182.89219374682514, 176.148919145706,
175.09934021533476, 75.88733500807321] Fitness: 1.9451254560533533
Generation 174 Best Solution: [182.89219374682514, 177.42534077406503,
174.89531262635165, 75.88733500807321] Fitness: 2.767669679214012
Generation 175 Best Solution: [182.89219374682514, 176.1216006456875,
175.09934021533476, 75.88733500807321] Fitness: 1.9500409442226423
Generation 176 Best Solution: [182.89219374682514, 177.42534077406503,
174.0485263015602, 75.88733500807321] Fitness: 2.3627219949467215
Generation 177 Best Solution: [182.89219374682514, 177.42534077406503,
175.09934021533476, 74.6179356778953] Fitness: 1.933034141318642
Generation 178 Best Solution: [182.89219374682514, 177.42534077406503,
175.09934021533476, 74.6179356778953] Fitness: 1.933034141318642
Generation 179 Best Solution: [182.89219374682514, 175.71408942010174,
173.86482104566724, 75.88733500807321] Fitness: 1.5536360520497694
Generation 180 Best Solution: [182.89219374682514, 177.42534077406503,
175.09934021533476, 75.37436135858849] Fitness: 2.2821071648881017
Generation 181 Best Solution: [183.25283033757037, 177.42534077406503,
173.99354765471273, 75.88733500807321] Fitness: 2.268001767798004
Generation 182 Best Solution: [182.89219374682514, 177.42534077406503,
175.09934021533476, 75.31457917134263] Fitness: 2.2214887852477325
Generation 183 Best Solution: [183.80064657682783, 177.42534077406503,
175.09934021533476, 74.6020789248301] Fitness: 2.0587067829131236
Generation 184 Best Solution: [182.89219374682514, 175.77005034244215,
175.09934021533476, 75.31457917134263] Fitness: 1.3051430804937818
Generation 185 Best Solution: [182.89219374682514, 177.42534077406503,
175.09934021533476, 75.153949432693] Fitness: 2.086691865176705
Generation 186 Best Solution: [182.89219374682514, 177.42534077406503,
175.09934021533476, 74.38091153137212] Fitness: 2.0104062946182113
Generation 187 Best Solution: [182.89219374682514, 177.42534077406503,
175.09934021533476, 74.86391708193825] Fitness: 1.9469669816348028
Generation 188 Best Solution: [182.89219374682514, 177.42534077406503,
175.09934021533476, 74.86391708193825] Fitness: 1.9469669816348028
Generation 189 Best Solution: [182.89219374682514, 177.42534077406503,
175.09934021533476, 74.86391708193825] Fitness: 1.9469669816348028

Generation 190 Best Solution: [183.60590793361956, 177.42534077406503,
175.09934021533476, 74.86391708193825] Fitness: 1.9335870223884677
Generation 191 Best Solution: [183.80064657682783, 177.42534077406503,
175.09934021533476, 74.76579534006596] Fitness: 2.053180552775835
Generation 192 Best Solution: [183.1237933038542, 176.83721390717406,
174.16717385657182, 75.88733500807321] Fitness: 1.4346934022739997
Generation 193 Best Solution: [183.1237933038542, 177.42534077406503,
174.16717385657182, 75.03025112276215] Fitness: 1.2335503706040765
Generation 194 Best Solution: [183.1237933038542, 177.42534077406503,
175.09934021533476, 74.60509674917313] Fitness: 1.8420446247837166
Generation 195 Best Solution: [183.80064657682783, 177.42534077406503,
174.46586103159447, 75.03025112276215] Fitness: 1.5189370320887168
Generation 196 Best Solution: [183.1237933038542, 176.9386032039601,
173.77124527878564, 75.88733500807321] Fitness: 1.6638688283754672
Generation 197 Best Solution: [183.60590793361956, 177.42534077406503,
175.09934021533476, 74.60509674917313] Fitness: 1.9215624250432872
Generation 198 Best Solution: [182.83661054772733, 177.42534077406503,
175.09934021533476, 74.40584953502447] Fitness: 2.033026142244357
Generation 199 Best Solution: [183.60590793361956, 177.42534077406503,
175.09934021533476, 74.40584953502447] Fitness: 1.9846910567341594
Generation 200 Best Solution: [183.1237933038542, 177.42534077406503,
175.09934021533476, 74.60509674917313] Fitness: 1.8420446247837166
Generation 201 Best Solution: [183.1237933038542, 177.42534077406503,
175.09934021533476, 74.60509674917313] Fitness: 1.8420446247837166
Generation 202 Best Solution: [183.80064657682783, 176.40519468492013,
175.09934021533476, 75.68811914889758] Fitness: 1.7378619588597415
Generation 203 Best Solution: [183.32187064561808, 177.42534077406503,
175.09934021533476, 74.90691850417457] Fitness: 1.8544094456685603
Generation 204 Best Solution: [183.1237933038542, 177.42534077406503,
175.09934021533476, 74.60509674917313] Fitness: 1.8420446247837166
Generation 205 Best Solution: [183.1237933038542, 177.42534077406503,
175.09934021533476, 74.60509674917313] Fitness: 1.8420446247837166
Generation 206 Best Solution: [183.60590793361956, 175.92573563023194,
175.09934021533476, 75.88733500807321] Fitness: 2.006573081778945
Generation 207 Best Solution: [183.80064657682783, 176.85850741499243,
173.7622126190345, 75.88733500807321] Fitness: 1.8032649694302423
Generation 208 Best Solution: [183.1237933038542, 175.99918830638677,
175.09934021533476, 75.88733500807321] Fitness: 1.8937058682802932
Generation 209 Best Solution: [183.1237933038542, 175.99918830638677,
175.09934021533476, 75.88733500807321] Fitness: 1.8937058682802932
Generation 210 Best Solution: [183.80064657682783, 177.07204073046847,
175.09934021533476, 74.30798781608858] Fitness: 1.6150107431388547
Generation 211 Best Solution: [183.1237933038542, 177.42534077406503,
175.09934021533476, 74.86274703517567] Fitness: 1.8537757713872198
Generation 212 Best Solution: [183.1237933038542, 177.42534077406503,
175.09934021533476, 74.36403282935208] Fitness: 1.926417160427756
Generation 213 Best Solution: [183.1237933038542, 177.42534077406503,
175.09934021533476, 74.46997132093095] Fitness: 1.877985353226585

Generation 214 Best Solution: [183.1237933038542, 177.42534077406503,
175.09934021533476, 74.67727592102841] Fitness: 1.8347136387130105
Generation 215 Best Solution: [183.60590793361956, 176.99439579348797,
175.09934021533476, 74.67727592102841] Fitness: 1.2572046814336004
Generation 216 Best Solution: [183.60590793361956, 176.85235310232025,
175.09934021533476, 74.69070881763874] Fitness: 1.1047411626105035
Generation 217 Best Solution: [183.77963825251857, 176.85235310232025,
175.09934021533476, 74.46997132093095] Fitness: 1.2674857444701564
Generation 218 Best Solution: [183.1237933038542, 176.85235310232025,
175.09934021533476, 74.46997132093095] Fitness: 1.0689473259052498
Generation 219 Best Solution: [183.1237933038542, 176.85235310232025,
175.09934021533476, 74.46997132093095] Fitness: 1.0689473259052498
Generation 220 Best Solution: [183.1237933038542, 176.54654217072897,
174.555998874017, 75.88733500807321] Fitness: 1.348251349967536
Generation 221 Best Solution: [183.77963825251857, 177.42534077406503,
174.3149932130606, 75.30813035691656] Fitness: 1.6531613183056324
Generation 222 Best Solution: [183.38947215885838, 176.85235310232025,
175.09934021533476, 75.30813035691656] Fitness: 1.3115672324983148
Generation 223 Best Solution: [183.1237933038542, 177.00069656283495,
175.09934021533476, 74.44022279136864] Fitness: 1.2402453592982323
Generation 224 Best Solution: [183.38947215885838, 175.8609615930742,
175.09934021533476, 75.88733500807321] Fitness: 1.9617799284478772
Generation 225 Best Solution: [183.38947215885838, 175.8609615930742,
175.09934021533476, 75.88733500807321] Fitness: 1.9617799284478772
Generation 226 Best Solution: [183.38947215885838, 176.85235310232025,
174.68685983272255, 75.88733500807321] Fitness: 1.6587304580089552
Generation 227 Best Solution: [183.1513325005903, 176.27152634649812,
175.09934021533476, 75.88733500807321] Fitness: 1.8393590827039432
Generation 228 Best Solution: [183.1237933038542, 177.42534077406503,
174.6539435425931, 75.88733500807321] Fitness: 2.443449683519873
Generation 229 Best Solution: [183.1237933038542, 176.89703273279758,
174.18884851046934, 75.88733500807321] Fitness: 1.4936334127464548
Generation 230 Best Solution: [183.1237933038542, 176.3013486552341,
175.09934021533476, 75.88733500807321] Fitness: 1.8465308245554213
Generation 231 Best Solution: [183.1513325005903, 177.42534077406503,
174.49074845516324, 75.61356500409838] Fitness: 1.8801256896199177
Generation 232 Best Solution: [183.1237933038542, 177.42534077406503,
175.09934021533476, 74.28195257732476] Fitness: 1.9761813859825212
Generation 233 Best Solution: [183.1513325005903, 176.87826430252773,
174.90827064725184, 75.88733500807321] Fitness: 1.9007230074468984
Generation 234 Best Solution: [183.77963825251857, 176.55989348534467,
174.49074845516324, 75.87984403699251] Fitness: 1.5027078256802393
Generation 235 Best Solution: [183.1513325005903, 176.7414133550043,
175.09934021533476, 75.88733500807321] Fitness: 2.031487991801188
Generation 236 Best Solution: [183.1513325005903, 177.42534077406503,
174.49074845516324, 75.88733500807321] Fitness: 2.3340240864045274
Generation 237 Best Solution: [182.88660688301024, 177.42534077406503,
175.09934021533476, 75.34938265638088] Fitness: 2.2593815279258025

Generation 238 Best Solution: [183.77963825251857, 176.8729997430394,
175.09934021533476, 75.73206450055105] Fitness: 2.079962769181088
Generation 239 Best Solution: [183.1237933038542, 176.66496826383053,
174.6539435425931, 75.87984403699251] Fitness: 1.4688324886337873
Generation 240 Best Solution: [183.77963825251857, 176.66496826383053,
174.6539435425931, 75.87984403699251] Fitness: 1.6673709071986942
Generation 241 Best Solution: [183.1513325005903, 177.42534077406503,
174.6539435425931, 75.88733500807321] Fitness: 2.4380632953560646
Generation 242 Best Solution: [183.77963825251857, 176.39925749939562,
175.09934021533476, 75.88733500807321] Fitness: 2.060849250550457
Generation 243 Best Solution: [183.77963825251857, 177.42534077406503,
175.09934021533476, 75.04745856484206] Fitness: 2.1255282913253555
Generation 244 Best Solution: [182.88660688301024, 177.42534077406503,
175.09934021533476, 75.04745856484206] Fitness: 2.0231795637556975
Generation 245 Best Solution: [183.77963825251857, 177.42534077406503,
175.09934021533476, 75.04745856484206] Fitness: 2.1255282913253555
Generation 246 Best Solution: [183.53446515906455, 177.42534077406503,
175.09934021533476, 75.04745856484206] Fitness: 1.971454938053392
Generation 247 Best Solution: [183.34687705031118, 176.23408699133824,
175.09934021533476, 75.88733500807321] Fitness: 1.8354658103129777
Generation 248 Best Solution: [183.34687705031118, 176.2589942813451,
175.09934021533476, 75.88733500807321] Fitness: 1.8353806696992558
Generation 249 Best Solution: [183.78074666113716, 176.25369793271227,
175.09934021533476, 75.79003994007894] Fitness: 1.8688748328047227
Generation 250 Best Solution: [183.77963825251857, 177.42534077406503,
173.95650194638358, 75.79003994007894] Fitness: 2.3179454164788442
Generation 251 Best Solution: [183.77963825251857, 177.42534077406503,
174.39018418615868, 75.79003994007894] Fitness: 2.3199460531940437
Generation 252 Best Solution: [183.262983453228, 177.42534077406503,
175.09934021533476, 75.27738847223932] Fitness: 2.0788416561398275
Generation 253 Best Solution: [183.77963825251857, 177.42534077406503,
173.79552558778616, 75.79003994007894] Fitness: 2.3931234115403868
Generation 254 Best Solution: [183.77963825251857, 177.42534077406503,
175.09934021533476, 75.07031362680681] Fitness: 2.138349792263726
Generation 255 Best Solution: [183.77963825251857, 177.42534077406503,
175.09934021533476, 74.83831429887233] Fitness: 2.046682519484419
Generation 256 Best Solution: [183.77963825251857, 176.59046368440355,
175.09934021533476, 75.88733500807321] Fitness: 2.135509247597075
Generation 257 Best Solution: [183.77963825251857, 176.72279278302256,
174.2034243970739, 75.88733500807321] Fitness: 1.5376565374869116
Generation 258 Best Solution: [183.77963825251857, 176.59046368440355,
175.09934021533476, 75.79003994007894] Fitness: 1.960580687474793
Generation 259 Best Solution: [183.77963825251857, 176.59046368440355,
174.267955356708, 75.79003994007894] Fitness: 1.2837818621909
Generation 260 Best Solution: [183.77963825251857, 176.59046368440355,
175.09934021533476, 75.30574932679771] Fitness: 1.3132485769232303
Generation 261 Best Solution: [182.91011515945715, 177.42534077406503,
174.267955356708, 75.88733500807321] Fitness: 2.3482521228924136

Generation 262 Best Solution: [183.77963825251857, 177.42534077406503,
174.45223046823878, 75.79003994007894] Fitness: 2.3446266458085474
Generation 263 Best Solution: [183.77963825251857, 176.62260736069814,
175.09934021533476, 75.88733500807321] Fitness: 2.153754257669489
Generation 264 Best Solution: [183.77963825251857, 176.62260736069814,
175.09934021533476, 75.88733500807321] Fitness: 2.153754257669489
Generation 265 Best Solution: [183.77963825251857, 177.42534077406503,
175.09934021533476, 74.99779938728338] Fitness: 2.1005258665202478
Generation 266 Best Solution: [183.42205648732738, 177.42534077406503,
175.09934021533476, 74.99779938728338] Fitness: 1.9076909105722264
Generation 267 Best Solution: [183.77963825251857, 177.42534077406503,
174.13177331547308, 75.88733500807321] Fitness: 2.4577612650284504
Generation 268 Best Solution: [183.1830423026149, 177.42534077406503,
175.09934021533476, 75.79003994007894] Fitness: 2.7573518322544235
Generation 269 Best Solution: [183.41759970112165, 177.42534077406503,
175.09934021533476, 75.79003994007894] Fitness: 2.772029998186028
Generation 270 Best Solution: [183.77963825251857, 176.30646836416312,
175.09934021533476, 75.79003994007894] Fitness: 1.8705890620651018
Generation 271 Best Solution: [183.77963825251857, 176.30646836416312,
175.09934021533476, 75.79003994007894] Fitness: 1.8705890620651018
Generation 272 Best Solution: [183.1021265112316, 177.42534077406503,
175.09934021533476, 74.79132748370185] Fitness: 1.8450586247216954
Generation 273 Best Solution: [183.77963825251857, 176.87951458504446,
175.09934021533476, 75.1337976111345] Fitness: 1.3958657963166685
Generation 274 Best Solution: [183.77963825251857, 177.42534077406503,
175.09934021533476, 74.99525304099319] Fitness: 2.0993492608493916
Generation 275 Best Solution: [183.1021265112316, 177.42534077406503,
175.09934021533476, 74.99525304099319] Fitness: 1.9058942037488564
Generation 276 Best Solution: [183.1021265112316, 176.79848243416774,
175.09934021533476, 75.79003994007894] Fitness: 1.9142146893198855
Generation 277 Best Solution: [183.77963825251857, 176.96364146240813,
174.96225288123424, 75.79003994007894] Fitness: 2.086250087745384
Generation 278 Best Solution: [183.77963825251857, 177.42534077406503,
175.09934021533476, 74.91069354621308] Fitness: 2.066118045654561
Generation 279 Best Solution: [183.77963825251857, 177.42534077406503,
175.09331944723502, 74.58443527952] Fitness: 2.035360438742327
Generation 280 Best Solution: [183.77963825251857, 176.95976694317048,
175.09331944723502, 74.58443527952] Fitness: 1.3383245309566436
Generation 281 Best Solution: [183.1021265112316, 176.95976694317048,
175.09934021533476, 75.79003994007894] Fitness: 2.0754977282909577
Generation 282 Best Solution: [183.77963825251857, 176.95976694317048,
175.09934021533476, 75.02091291733215] Fitness: 1.4146405497003118
Generation 283 Best Solution: [183.1021265112316, 175.6222264231088,
175.09331944723502, 75.79003994007894] Fitness: 1.9769087163195234
Generation 284 Best Solution: [183.22014192014242, 177.42534077406503,
175.09934021533476, 75.12208652079762] Fitness: 1.9583330750981476
Generation 285 Best Solution: [182.94910908563483, 177.42534077406503,
175.09934021533476, 74.41415133478755] Fitness: 1.9634687638851402

Generation 286 Best Solution: [183.1021265112316, 176.6804374162362,
175.09934021533476, 75.88733500807321] Fitness: 1.9972506714818685
Generation 287 Best Solution: [183.22014192014242, 176.95976694317048,
175.09934021533476, 75.79003994007894] Fitness: 2.0568275844753106
Generation 288 Best Solution: [183.22014192014242, 176.61233483439582,
175.09934021533476, 75.88733500807321] Fitness: 1.9356200998131967
Generation 289 Best Solution: [183.77963825251857, 176.4389950721139,
173.81621740607804, 75.88733500807321] Fitness: 1.4867610389785837
Generation 290 Best Solution: [183.77963825251857, 177.42534077406503,
175.09934021533476, 74.57059115855292] Fitness: 2.0470072443822
Generation 291 Best Solution: [183.1021265112316, 176.38793170921042,
175.09934021533476, 75.88733500807321] Fitness: 1.8647910770370255
Generation 292 Best Solution: [183.22014192014242, 177.42534077406503,
174.17713060122892, 75.88733500807321] Fitness: 2.2444861632073057
Generation 293 Best Solution: [183.77963825251857, 175.7428824462679,
175.09934021533476, 75.18391297368557] Fitness: 1.3192384641146035
Generation 294 Best Solution: [183.77963825251857, 175.8293629253667,
174.17713060122892, 75.88733500807321] Fitness: 1.4980238619418562
Generation 295 Best Solution: [183.77963825251857, 176.29332958905465,
175.09934021533476, 75.88733500807321] Fitness: 2.0444505031401548
Generation 296 Best Solution: [183.77963825251857, 176.29332958905465,
175.09934021533476, 75.88733500807321] Fitness: 2.0444505031401548
Generation 297 Best Solution: [183.1132638796283, 177.42534077406503,
175.09934021533476, 74.47693760991748] Fitness: 1.877801735465031
Generation 298 Best Solution: [183.0663774637446, 177.42534077406503,
175.09934021533476, 75.1498905979282] Fitness: 2.0060166957399423
Generation 299 Best Solution: [183.77963825251857, 177.42534077406503,
175.09934021533476, 74.47856184153972] Fitness: 2.073376756829375
Generation 300 Best Solution: [183.1132638796283, 177.42534077406503,
175.09934021533476, 74.47856184153972] Fitness: 1.8772156951867103
Generation 301 Best Solution: [183.1021265112316, 177.42534077406503,
175.09934021533476, 75.32933459971918] Fitness: 2.1481092751360658
Generation 302 Best Solution: [183.1021265112316, 176.7894320493182,
175.09934021533476, 75.32933459971918] Fitness: 1.2819626713949557
Generation 303 Best Solution: [182.94910908563483, 176.11375738434234,
175.09934021533476, 75.88733500807321] Fitness: 1.9209562014406092
Generation 304 Best Solution: [182.94910908563483, 176.11375738434234,
175.09934021533476, 75.88733500807321] Fitness: 1.9209562014406092
Generation 305 Best Solution: [183.1021265112316, 177.42534077406503,
175.09934021533476, 75.50255681403127] Fitness: 2.343393444083808
Generation 306 Best Solution: [183.77963825251857, 176.29658238943796,
173.8012098642078, 75.78318662866847] Fitness: 1.2817850213211592
Generation 307 Best Solution: [183.1021265112316, 176.77001456343538,
175.09934021533476, 75.78318662866847] Fitness: 1.8782756950791306
Generation 308 Best Solution: [182.94285983244907, 177.42534077406503,
173.93454901589863, 75.78318662866847] Fitness: 2.18078752587314
Generation 309 Best Solution: [183.1021265112316, 177.42534077406503,
173.93454901589863, 75.78318662866847] Fitness: 2.1205665204327016

Generation 310 Best Solution: [182.94285983244907, 177.42534077406503,
173.93199219996043, 75.2418323878036] Fitness: 1.4885277743250538
Generation 311 Best Solution: [183.76667888191358, 176.06686043733524,
174.57937216741738, 75.88733500807321] Fitness: 1.506852034601868
Generation 312 Best Solution: [182.8519521009734, 176.89003691551474,
173.93199219996043, 75.88733500807321] Fitness: 1.649015890313739
Generation 313 Best Solution: [183.1021265112316, 176.0063156825294,
175.0315315050539, 75.88733500807321] Fitness: 1.7997436132082087
Generation 314 Best Solution: [183.1021265112316, 177.42534077406503,
175.09934021533476, 74.53029551844367] Fitness: 1.863444914812204
Generation 315 Best Solution: [183.47386538642624, 177.42534077406503,
173.93199219996043, 75.88733500807321] Fitness: 2.3242077401134233
Generation 316 Best Solution: [183.1021265112316, 177.42534077406503,
175.09934021533476, 75.79378237159256] Fitness: 2.7789845444056565
Generation 317 Best Solution: [183.1021265112316, 177.42534077406503,
175.09934021533476, 74.421119998928] Fitness: 1.903190236717335
Generation 318 Best Solution: [183.1021265112316, 176.2061136111666,
175.09934021533476, 75.88733500807321] Fitness: 1.8508535137493496
Generation 319 Best Solution: [183.1021265112316, 176.2061136111666,
175.09934021533476, 75.88733500807321] Fitness: 1.8508535137493496
Generation 320 Best Solution: [183.1021265112316, 177.42534077406503,
175.09934021533476, 74.74355429640772] Fitness: 1.8403434425301288
Generation 321 Best Solution: [183.1021265112316, 176.96831179736125,
175.09934021533476, 75.79378237159256] Fitness: 2.0916443385958194
Generation 322 Best Solution: [183.1021265112316, 176.96831179736125,
175.09934021533476, 75.88733500807321] Fitness: 2.260121990389086
Generation 323 Best Solution: [183.1021265112316, 176.96831179736125,
175.0315315050539, 75.88733500807321] Fitness: 2.163856888200489
Generation 324 Best Solution: [183.1021265112316, 175.73275665465164,
175.09934021533476, 75.88733500807321] Fitness: 2.0605340598115127
Generation 325 Best Solution: [183.2978983138438, 177.1048842542982,
175.09934021533476, 75.79378237159256] Fitness: 2.2434310914122246
Generation 326 Best Solution: [183.1021265112316, 176.1829949405038,
174.9195072600435, 75.88733500807321] Fitness: 1.6135154246596373
Generation 327 Best Solution: [183.1021265112316, 177.42534077406503,
175.09934021533476, 74.88969622937961] Fitness: 1.8661690507476452
Generation 328 Best Solution: [183.1021265112316, 176.5409875401629,
175.0315315050539, 75.88733500807321] Fitness: 1.8209087668731505
Generation 329 Best Solution: [183.1021265112316, 177.42534077406503,
173.90654801647702, 75.85019583582977] Fitness: 2.2498043866027766
Generation 330 Best Solution: [183.1021265112316, 177.42534077406503,
174.9195072600435, 74.73727344641199] Fitness: 1.600669282364782
Generation 331 Best Solution: [183.1021265112316, 177.42534077406503,
174.9195072600435, 74.73727344641199] Fitness: 1.600669282364782
Generation 332 Best Solution: [183.1021265112316, 177.12093375290917,
174.96990637037214, 74.73727344641199] Fitness: 1.167913024019249
Generation 333 Best Solution: [183.1021265112316, 176.54358550733983,
174.9195072600435, 75.88733500807321] Fitness: 1.6790602244118311

Generation 334 Best Solution: [183.1021265112316, 177.42534077406503,
174.9195072600435, 74.80198141786089] Fitness: 1.607280343807076
Generation 335 Best Solution: [183.1021265112316, 177.42534077406503,
173.90654801647702, 75.1264660761326] Fitness: 1.3508016544979042
Generation 336 Best Solution: [183.1021265112316, 177.42534077406503,
174.9195072600435, 74.5304114517233] Fitness: 1.62408925113741
Generation 337 Best Solution: [183.1021265112316, 177.42534077406503,
174.09746545587285, 75.88733500807321] Fitness: 2.2673436463028707
Generation 338 Best Solution: [183.47613966919914, 176.19158591799072,
174.9195072600435, 75.88733500807321] Fitness: 1.6294463339766803
Generation 339 Best Solution: [183.1021265112316, 177.42534077406503,
175.0315315050539, 75.02971571588176] Fitness: 1.8264258306965275
Generation 340 Best Solution: [183.1021265112316, 177.42534077406503,
174.9195072600435, 74.30590193789013] Fitness: 1.7263168870311874
Generation 341 Best Solution: [183.57412123251962, 177.42534077406503,
174.9195072600435, 74.30590193789013] Fitness: 1.7841557403834138
Generation 342 Best Solution: [183.1021265112316, 176.64360858922356,
174.9195072600435, 75.88733500807321] Fitness: 1.7337813494772207
Generation 343 Best Solution: [183.1021265112316, 177.42534077406503,
173.90654801647702, 75.88733500807321] Fitness: 2.318349479474888
Generation 344 Best Solution: [183.1021265112316, 177.42534077406503,
174.9195072600435, 75.10795466296616] Fitness: 1.7284935425123849
Generation 345 Best Solution: [182.8273920467604, 177.42534077406503,
174.9195072600435, 75.10795466296616] Fitness: 1.8575341715411657
Generation 346 Best Solution: [183.1021265112316, 177.42534077406503,
173.90654801647702, 75.61503486849512] Fitness: 1.866570616864856
Generation 347 Best Solution: [183.77530529472884, 176.2357008005698,
175.0857398437494, 75.88733500807321] Fitness: 2.019561250402673
Generation 348 Best Solution: [183.1021265112316, 177.42534077406503,
175.0857398437494, 74.88145230990199] Fitness: 1.8439180234117052
Generation 349 Best Solution: [183.1021265112316, 177.42534077406503,
173.90654801647702, 74.88145230990199] Fitness: 1.234697886134538
Generation 350 Best Solution: [183.1021265112316, 177.42534077406503,
173.90654801647702, 74.88145230990199] Fitness: 1.234697886134538
Generation 351 Best Solution: [183.1021265112316, 176.06851967911086,
174.11664633949883, 75.88733500807321] Fitness: 1.1931154112835618
Generation 352 Best Solution: [183.1021265112316, 177.42534077406503,
174.11664633949883, 75.88733500807321] Fitness: 2.26541535324293
Generation 353 Best Solution: [183.1021265112316, 177.42534077406503,
175.0857398437494, 75.01350528037415] Fitness: 1.8946629043883236
Generation 354 Best Solution: [183.1021265112316, 177.42534077406503,
174.11664633949883, 75.24935654187568] Fitness: 1.3919589109019515
Generation 355 Best Solution: [183.76612089695172, 177.42534077406503,
174.11664633949883, 75.05183187287838] Fitness: 1.4348925625470286
Generation 356 Best Solution: [183.1021265112316, 175.68151902299525,
175.0857398437494, 75.65567771120489] Fitness: 1.6928410632992517
Generation 357 Best Solution: [183.1021265112316, 177.42534077406503,
175.0857398437494, 74.41298672664088] Fitness: 1.8870152765579198

Generation 358 Best Solution: [183.1021265112316, 177.42534077406503,
175.0857398437494, 74.41298672664088] Fitness: 1.8870152765579198
Generation 359 Best Solution: [183.1021265112316, 177.42534077406503,
175.0857398437494, 74.41298672664088] Fitness: 1.8870152765579198
Generation 360 Best Solution: [183.1021265112316, 177.42534077406503,
175.0857398437494, 74.71314826962922] Fitness: 1.819335109716925
Generation 361 Best Solution: [183.1021265112316, 177.42534077406503,
174.47263592947624, 75.58048154925874] Fitness: 1.8348534679410085
Generation 362 Best Solution: [183.1021265112316, 176.34042055908483,
175.0857398437494, 75.88733500807321] Fitness: 1.8361955981515754
Generation 363 Best Solution: [183.1021265112316, 177.42534077406503,
175.0857398437494, 75.16705367849224] Fitness: 1.9884484448426964
Generation 364 Best Solution: [183.1021265112316, 176.33747516194038,
175.0857398437494, 75.88733500807321] Fitness: 1.8357739389981544
Generation 365 Best Solution: [183.1021265112316, 176.33747516194038,
175.0857398437494, 75.88733500807321] Fitness: 1.8357739389981544
Generation 366 Best Solution: [183.1021265112316, 177.42534077406503,
175.0857398437494, 74.50909380044473] Fitness: 1.849791522592625
Generation 367 Best Solution: [183.1021265112316, 177.42534077406503,
175.0857398437494, 74.50909380044473] Fitness: 1.849791522592625
Generation 368 Best Solution: [182.9640427597058, 177.42534077406503,
175.0857398437494, 75.83918569404928] Fitness: 2.8890159437119673
Generation 369 Best Solution: [183.1021265112316, 177.42534077406503,
174.2547686639186, 75.88733500807321] Fitness: 2.268761462183139
Generation 370 Best Solution: [182.9640427597058, 176.0804371302053,
175.0857398437494, 75.88733500807321] Fitness: 1.901867947800571
Generation 371 Best Solution: [182.9640427597058, 176.0804371302053,
175.0857398437494, 75.88733500807321] Fitness: 1.901867947800571
Generation 372 Best Solution: [182.9640427597058, 176.2666209151586,
175.0857398437494, 75.88733500807321] Fitness: 1.8796692874996785
Generation 373 Best Solution: [183.1021265112316, 177.42534077406503,
175.0857398437494, 74.5359257686537] Fitness: 1.842015294454158
Generation 374 Best Solution: [182.9640427597058, 176.83669015352515,
175.0857398437494, 75.88733500807321] Fitness: 2.1536200815349558
Generation 375 Best Solution: [183.1021265112316, 177.42534077406503,
175.0857398437494, 74.75402196313478] Fitness: 1.8211743066208077
Generation 376 Best Solution: [183.1021265112316, 177.42534077406503,
175.0857398437494, 74.75402196313478] Fitness: 1.8211743066208077
Generation 377 Best Solution: [182.9640427597058, 176.62861567159297,
175.0857398437494, 74.28580164444612] Fitness: 1.025120478658704
Generation 378 Best Solution: [183.40823379849613, 177.42534077406503,
175.0857398437494, 74.75402196313478] Fitness: 1.8184118163244374
Generation 379 Best Solution: [182.9640427597058, 176.42074133798667,
174.3528039925957, 75.88733500807321] Fitness: 1.2648186759607158
Generation 380 Best Solution: [183.40823379849613, 177.42534077406503,
175.0857398437494, 74.75402196313478] Fitness: 1.8184118163244374
Generation 381 Best Solution: [182.9640427597058, 175.60528348478618,
175.0857398437494, 75.88733500807321] Fitness: 2.207742796053428

Generation 382 Best Solution: [182.9640427597058, 177.42534077406503,
174.74271574374185, 75.88733500807321] Fitness: 2.5727580854612135
Generation 383 Best Solution: [182.9640427597058, 177.42534077406503,
175.0857398437494, 74.4025922536486] Fitness: 1.9418108000275323
Generation 384 Best Solution: [183.5443484455594, 177.42534077406503,
175.0857398437494, 74.4025922536486] Fitness: 1.9356675183718093
Generation 385 Best Solution: [182.9640427597058, 177.32628527783086,
175.0857398437494, 74.4025922536486] Fitness: 1.7647152877663672
Generation 386 Best Solution: [182.9640427597058, 177.42534077406503,
175.05281013378146, 74.4025922536486] Fitness: 1.8948613801782976
Generation 387 Best Solution: [182.9640427597058, 177.42534077406503,
174.74271574374185, 74.29531701271547] Fitness: 1.5977296024031462
Generation 388 Best Solution: [183.5443484455594, 177.42534077406503,
174.74271574374185, 74.29531701271547] Fitness: 1.5915863207474232
Generation 389 Best Solution: [182.9640427597058, 177.42534077406503,
174.74271574374185, 74.29531701271547] Fitness: 1.5977296024031462
Generation 390 Best Solution: [183.5267457941362, 177.42534077406503,
175.0857398437494, 74.74691524174241] Fitness: 1.8567431217085313
Generation 391 Best Solution: [182.9640427597058, 176.8093336905534,
174.31623730807578, 75.88733500807321] Fitness: 1.4811099936954515
Generation 392 Best Solution: [182.9640427597058, 176.09344204460305,
174.12823869313056, 75.88733500807321] Fitness: 1.2354944261358127
Generation 393 Best Solution: [182.9640427597058, 177.42534077406503,
175.0857398437494, 74.98050230552802] Fitness: 1.9292796149011047
Generation 394 Best Solution: [182.9640427597058, 177.42534077406503,
175.0857398437494, 74.98050230552802] Fitness: 1.9292796149011047
Generation 395 Best Solution: [182.9640427597058, 177.42534077406503,
175.0857398437494, 74.8829317488245] Fitness: 1.8942249253839436
Generation 396 Best Solution: [183.4454985806254, 177.42534077406503,
175.0857398437494, 74.8829317488245] Fitness: 1.8513828419283347
Generation 397 Best Solution: [182.9640427597058, 177.42534077406503,
175.0857398437494, 74.98050230552802] Fitness: 1.9292796149011047
Generation 398 Best Solution: [182.9640427597058, 177.42534077406503,
175.0857398437494, 74.98050230552802] Fitness: 1.9292796149011047
Generation 399 Best Solution: [182.9640427597058, 177.42534077406503,
174.8801760585632, 74.85175219164671] Fitness: 1.6212699498592973
Generation 400 Best Solution: [182.9640427597058, 177.42534077406503,
174.8801760585632, 74.85175219164671] Fitness: 1.6212699498592973
Generation 401 Best Solution: [183.5433118664729, 177.42534077406503,
174.8801760585632, 74.85175219164671] Fitness: 1.6146614162238015
Generation 402 Best Solution: [182.9640427597058, 177.42534077406503,
174.8801760585632, 74.85175219164671] Fitness: 1.6212699498592973
Generation 403 Best Solution: [182.9640427597058, 175.78329175588996,
174.8801760585632, 75.88733500807321] Fitness: 1.7862645390769214
Generation 404 Best Solution: [182.9640427597058, 175.78329175588996,
174.8801760585632, 75.88733500807321] Fitness: 1.7862645390769214
Generation 405 Best Solution: [182.9640427597058, 177.42534077406503,
174.8801760585632, 74.99235452510464] Fitness: 1.6696296593627633

Generation 406 Best Solution: [183.33896908025153, 175.78329175588996,
174.8801760585632, 74.99235452510464] Fitness: 0.6783299968952797
Generation 407 Best Solution: [183.5433118664729, 175.81769084571883,
174.8801760585632, 75.88733500807321] Fitness: 1.7552001502605754
Generation 408 Best Solution: [182.9640427597058, 177.42534077406503,
174.0544704270416, 74.99509682184767] Fitness: 1.2820379449873782
Generation 409 Best Solution: [182.9640427597058, 177.42534077406503,
174.64147372930486, 75.42272852528104] Fitness: 1.7890071972436492
Generation 410 Best Solution: [182.9640427597058, 175.7363878223061,
174.64147372930486, 75.42272852528104] Fitness: 0.899117821467617
Generation 411 Best Solution: [183.33068771128464, 176.28437162621145,
173.81041295026796, 75.88733500807321] Fitness: 1.2527624480670019
Generation 412 Best Solution: [183.49818578564845, 176.5438504590368,
174.0544704270416, 75.88733500807321] Fitness: 1.2695128022930766
Generation 413 Best Solution: [182.82301200970693, 176.28437162621145,
173.81041295026796, 75.88733500807321] Fitness: 1.4009495964855692
Generation 414 Best Solution: [182.9640427597058, 176.4226560016162,
174.91869742224415, 75.88733500807321] Fitness: 1.6830228652285513
Generation 415 Best Solution: [182.9640427597058, 176.6384392857878,
175.0857398437494, 75.88733500807321] Fitness: 1.99988694448099
Generation 416 Best Solution: [182.9640427597058, 177.42534077406503,
174.8801760585632, 75.22361871316014] Fitness: 1.8173786431068992
Generation 417 Best Solution: [182.9640427597058, 176.87411815195546,
175.0857398437494, 75.88733500807321] Fitness: 2.189639485021684
Generation 418 Best Solution: [183.10198314321852, 176.59037448131448,
174.8227052954616, 75.88733500807321] Fitness: 1.5951317801680907
Generation 419 Best Solution: [182.9640427597058, 176.87411815195546,
174.30997190753135, 75.88733500807321] Fitness: 1.5406328246245724
Generation 420 Best Solution: [183.5433118664729, 176.2471505469505,
175.0857398437494, 75.88733500807321] Fitness: 1.8728078112032474
Generation 421 Best Solution: [183.5433118664729, 176.87411815195546,
174.8801760585632, 75.35845960400091] Fitness: 1.1482471929584124
Generation 422 Best Solution: [182.9640427597058, 176.7111677689697,
174.91869742224415, 75.88733500807321] Fitness: 1.8286511369394514
Generation 423 Best Solution: [182.9640427597058, 177.42534077406503,
174.8227052954616, 74.56556028186851] Fitness: 1.5576280010287586
Generation 424 Best Solution: [183.0423502264599, 177.42534077406503,
174.8227052954616, 74.56556028186851] Fitness: 1.5256219536369762
Generation 425 Best Solution: [182.9640427597058, 177.42534077406503,
174.8801760585632, 74.56556028186851] Fitness: 1.6197088530181134
Generation 426 Best Solution: [182.9640427597058, 176.00148979775747,
175.06338867676607, 75.35845960400091] Fitness: 1.125979947938998
Generation 427 Best Solution: [182.9640427597058, 176.5567522749626,
175.13142903990902, 75.35845960400091] Fitness: 1.2528221354828182
Generation 428 Best Solution: [182.9640427597058, 176.5567522749626,
175.13142903990902, 75.35845960400091] Fitness: 1.2528221354828182
Generation 429 Best Solution: [182.9640427597058, 176.5567522749626,
175.13142903990902, 75.35845960400091] Fitness: 1.2528221354828182

Generation 430 Best Solution: [182.9640427597058, 176.5567522749626,
173.76490308337728, 75.88733500807321] Fitness: 1.4206608031624677
Generation 431 Best Solution: [182.9640427597058, 175.71309062132445,
175.06338867676607, 75.35845960400091] Fitness: 1.3050325802186622
Generation 432 Best Solution: [182.9640427597058, 177.37773978143792,
175.06338867676607, 75.18158082146059] Fitness: 1.9300493524808273
Generation 433 Best Solution: [182.9640427597058, 177.42534077406503,
175.06338867676607, 74.53880057272924] Fitness: 1.859086847310443
Generation 434 Best Solution: [183.40732609297518, 177.42534077406503,
175.0857398437494, 74.53880057272924] Fitness: 1.838275860343794
Generation 435 Best Solution: [182.9640427597058, 177.37773978143792,
175.06338867676607, 75.18158082146059] Fitness: 1.9300493524808273
Generation 436 Best Solution: [182.9640427597058, 176.45465094138933,
175.06338867676607, 75.88733500807321] Fitness: 1.8810013588654277
Generation 437 Best Solution: [182.9640427597058, 176.17253640491003,
175.06338867676607, 75.88733500807321] Fitness: 1.851960012145563
Generation 438 Best Solution: [182.9640427597058, 176.54006036569953,
175.06338867676607, 75.88733500807321] Fitness: 1.9146890743149911
Generation 439 Best Solution: [183.40732609297518, 177.2826897021691,
175.12394511746723, 75.27569808328766] Fitness: 1.8809465855265262
Generation 440 Best Solution: [182.9640427597058, 177.42534077406503,
175.0857398437494, 75.18158082146059] Fitness: 2.049149534645757
Generation 441 Best Solution: [183.40732609297518, 177.42534077406503,
175.0857398437494, 74.7417965561251] Fitness: 1.8173724116979146
Generation 442 Best Solution: [183.03023680822992, 176.21983092586794,
175.12031188325912, 75.88733500807321] Fitness: 1.9035260968930947
Generation 443 Best Solution: [183.03023680822992, 176.21983092586794,
175.06338867676607, 75.88733500807321] Fitness: 1.8203296647299023
Generation 444 Best Solution: [182.9640427597058, 177.42534077406503,
175.06338867676607, 74.60372904907034] Fitness: 1.845291071462485
Generation 445 Best Solution: [183.03023680822992, 177.42534077406503,
175.0857398437494, 74.7678510497551] Fitness: 1.8445971896540658
Generation 446 Best Solution: [183.03023680822992, 177.42534077406503,
175.0857398437494, 75.18158082146059] Fitness: 2.0214585784693684
Generation 447 Best Solution: [182.9640427597058, 176.16945416021358,
175.06338867676607, 75.88733500807321] Fitness: 1.852339895297249
Generation 448 Best Solution: [182.9640427597058, 177.42534077406503,
175.12031188325912, 74.70408787936164] Fitness: 1.9203196422573985
Generation 449 Best Solution: [183.8396818708453, 176.81431303291004,
175.0069872731121, 75.88733500807321] Fitness: 2.2195584919453233
Generation 450 Best Solution: [182.82408017480387, 176.2784808659065,
175.0069872731121, 75.88733500807321] Fitness: 1.8521338282194444
Generation 451 Best Solution: [183.22625068431554, 176.2784808659065,
175.0069872731121, 75.88733500807321] Fitness: 1.7017721176247176
Generation 452 Best Solution: [183.8396818708453, 177.42534077406503,
175.0857398437494, 74.71007807140552] Fitness: 2.065026705329524
Generation 453 Best Solution: [182.9640427597058, 177.42534077406503,
175.0857398437494, 74.71007807140552] Fitness: 1.869195558724343

Generation 454 Best Solution: [183.52479175807804, 177.37773978143792,
175.0857398437494, 74.71007807140552] Fitness: 1.767516265982924
Generation 455 Best Solution: [182.9640427597058, 177.42534077406503,
175.0857398437494, 74.71007807140552] Fitness: 1.869195558724343
Generation 456 Best Solution: [183.03023680822992, 177.42534077406503,
175.0857398437494, 74.71007807140552] Fitness: 1.8415046025479545
Generation 457 Best Solution: [183.68888961546253, 176.6092541342844,
175.12031188325912, 75.34359067408697] Fitness: 1.3240032894206721
Generation 458 Best Solution: [183.48211514760155, 177.42534077406503,
174.28739243701867, 75.0063453726346] Fitness: 1.2564387810570121
Generation 459 Best Solution: [183.8396818708453, 176.6092541342844,
175.12031188325912, 75.52258632101564] Fitness: 1.651074425244368
Generation 460 Best Solution: [183.03023680822992, 177.42534077406503,
175.071271861945, 74.92286037823429] Fitness: 1.858214445252825
Generation 461 Best Solution: [182.9640427597058, 177.42534077406503,
174.08318721509414, 74.92286037823429] Fitness: 1.2483332626082042
Generation 462 Best Solution: [182.9640427597058, 177.42534077406503,
175.0857398437494, 74.92286037823429] Fitness: 1.906744917400932
Generation 463 Best Solution: [182.9640427597058, 177.2198359391731,
175.0857398437494, 75.44573426969696] Fitness: 1.954026962452875
Generation 464 Best Solution: [182.9640427597058, 177.2198359391731,
173.91909377972235, 75.88733500807321] Fitness: 2.0130542873453336
Generation 465 Best Solution: [182.9640427597058, 177.2198359391731,
173.91909377972235, 75.88733500807321] Fitness: 2.0130542873453336
Generation 466 Best Solution: [182.9640427597058, 177.42534077406503,
175.0857398437494, 75.44573426969696] Fitness: 2.304087189151437
Generation 467 Best Solution: [183.8396818708453, 177.42534077406503,
175.0857398437494, 74.65402942040983] Fitness: 2.0670860635889996
Generation 468 Best Solution: [183.03023680822992, 176.03021331401712,
175.0857398437494, 75.88733500807321] Fitness: 1.8895819172463253
Generation 469 Best Solution: [183.03023680822992, 176.03021331401712,
175.0857398437494, 75.88733500807321] Fitness: 1.8895819172463253
Generation 470 Best Solution: [183.8396818708453, 175.92121485483307,
174.39916553847007, 75.88733500807321] Fitness: 1.5373454516884657
Generation 471 Best Solution: [183.8251949444082, 175.74201071270028,
175.0857398437494, 75.88733500807321] Fitness: 2.2657245206443104
Generation 472 Best Solution: [183.8396818708453, 177.28717268686702,
175.0857398437494, 75.24036468281261] Fitness: 2.049402413625576
Generation 473 Best Solution: [183.8396818708453, 176.33305366453686,
175.0857398437494, 75.88733500807321] Fitness: 2.080889539674701
Generation 474 Best Solution: [183.8251949444082, 176.33305366453686,
175.0857398437494, 75.88733500807321] Fitness: 2.0677553091370577
Generation 475 Best Solution: [183.03023680822992, 177.42534077406503,
175.0857398437494, 75.28201315815716] Fitness: 2.1053461129034847
Generation 476 Best Solution: [183.03023680822992, 177.42534077406503,
175.0857398437494, 75.02583715787601] Fitness: 1.9230145472246707
Generation 477 Best Solution: [183.8396818708453, 177.42534077406503,
175.0857398437494, 75.02583715787601] Fitness: 2.14653665000624

Generation 478 Best Solution: [183.8251949444082, 177.42534077406503,
175.0857398437494, 74.68573515915747] Fitness: 2.0521747650209394
Generation 479 Best Solution: [183.03023680822992, 176.65272675463405,
175.0857398437494, 75.88733500807321] Fitness: 1.981190599742178
Generation 480 Best Solution: [183.03023680822992, 177.14881139520065,
175.0857398437494, 75.19426408146145] Fitness: 1.575702046654533
Generation 481 Best Solution: [183.8251949444082, 177.42534077406503,
175.0857398437494, 74.68573515915747] Fitness: 2.0521747650209394
Generation 482 Best Solution: [183.13835304838128, 177.42534077406503,
175.0857398437494, 74.68573515915747] Fitness: 1.8115052738441257
Generation 483 Best Solution: [183.8396818708453, 177.42534077406503,
175.0857398437494, 74.68573515915747] Fitness: 2.065308995558583
Generation 484 Best Solution: [183.8396818708453, 176.3673185572867,
175.0857398437494, 75.19426408146145] Fitness: 1.1678051833593759
Generation 485 Best Solution: [183.08147388316308, 177.42534077406503,
175.0857398437494, 75.19426408146145] Fitness: 2.01450773237531
Generation 486 Best Solution: [183.8251949444082, 176.47833757372547,
175.0857398437494, 75.88733500807321] Fitness: 2.1039360047002793
Generation 487 Best Solution: [183.8396818708453, 176.2397905110675,
175.0857398437494, 75.88733500807321] Fitness: 2.0753084923516276
Generation 488 Best Solution: [183.8251949444082, 175.57715859114217,
175.0857398437494, 74.42059688515114] Fitness: 1.3757348961353233
Generation 489 Best Solution: [183.8251949444082, 175.57715859114217,
175.0857398437494, 74.76597736690836] Fitness: 1.3144144709141987
Generation 490 Best Solution: [183.8396818708453, 175.57715859114217,
175.0857398437494, 74.76597736690836] Fitness: 1.3275487014518421
Generation 491 Best Solution: [183.8396818708453, 176.20463922590193,
174.45417567124983, 75.88733500807321] Fitness: 1.4761879990475164
Generation 492 Best Solution: [183.8396818708453, 176.6742192859289,
175.0857398437494, 75.88733500807321] Fitness: 2.218853127212183
Generation 493 Best Solution: [183.52472972452978, 176.6742192859289,
175.00892514684605, 75.88733500807321] Fitness: 1.9015175545704166
Generation 494 Best Solution: [183.8396818708453, 175.5893142930511,
174.72281627237948, 75.88733500807321] Fitness: 1.9976211786198712
Generation 495 Best Solution: [183.8396818708453, 175.5893142930511,
174.72281627237948, 75.88733500807321] Fitness: 1.9976211786198712
Generation 496 Best Solution: [182.87078669941286, 176.18036403900615,
175.0857398437494, 75.88733500807321] Fitness: 1.9339226265572342
Generation 497 Best Solution: [183.8396818708453, 176.51312571809993,
174.43655337193286, 75.88733500807321] Fitness: 1.5224199840004944
Generation 498 Best Solution: [183.8396818708453, 176.88252335195637,
175.0857398437494, 74.91682215883247] Fitness: 1.3210931024375074
Generation 499 Best Solution: [182.94088297006724, 177.42534077406503,
175.0857398437494, 74.30053960560738] Fitness: 2.0103883970179877
Generation 500 Best Solution: [182.87078669941286, 177.42534077406503,
175.0857398437494, 74.30053960560738] Fitness: 2.0498637103188546
Optimal Solution for Simulation 9: [182.87078669941286, 177.42534077406503,
175.0857398437494, 74.30053960560738]

```
--------------------------------------------------
Simulation 10/32 with conditions: {'drought': True, 'flood': False,
'policy_change': True, 'social_event': True, 'economic_crisis': False}
Generation 1 Best Solution: [183.5287223138628, 176.20286311903155,
174.51271425316347, 74.76638055247983] Fitness: 0.2166573205307595
Generation 2 Best Solution: [183.10304796883668, 176.55683285975755,
174.65680655165343, 75.49409352229156] Fitness: 0.8024285076218238
Generation 3 Best Solution: [183.10304796883668, 175.88121782361088,
174.99360183154485, 74.52360484278324] Fitness: 0.7323214624593324
Generation 4 Best Solution: [183.178169319364, 176.99058028308502,
174.08981681084035, 74.60519965923368] Fitness: 0.5182088930308943
Generation 5 Best Solution: [182.79673797627507, 176.63338494599637,
174.99360183154485, 74.60519965923368] Fitness: 0.8856090865613853
Generation 6 Best Solution: [183.178169319364, 177.21262361624446,
174.56337323382667, 74.4523200049362] Fitness: 0.9943565250965605
Generation 7 Best Solution: [183.43809312276687, 176.2588403760152,
174.99360183154485, 75.45116478340135] Fitness: 1.0376310330278282
Generation 8 Best Solution: [183.47761140872672, 176.39902036456598,
175.10210399119305, 75.36849961159228] Fitness: 1.1367426221383852
Generation 9 Best Solution: [183.47761140872672, 176.39902036456598,
174.98456890664696, 75.36849961159228] Fitness: 0.9744678093729551
Generation 10 Best Solution: [183.407309648623, 175.88294899657774,
174.75724300331154, 75.71730636606591] Fitness: 1.207532622895879
Generation 11 Best Solution: [183.43809312276687, 176.32124022211,
174.98456890664696, 75.76729758409265] Fitness: 1.465274439867065
Generation 12 Best Solution: [183.2631554496674, 175.88294899657774,
174.75724300331154, 75.88952688438262] Fitness: 1.4729391005088377
Generation 13 Best Solution: [183.43809312276687, 177.27743250221283,
174.2435421866138, 75.71730636606591] Fitness: 1.7069078326759817
Generation 14 Best Solution: [183.59604334770086, 176.3357623846854,
174.98456890664696, 75.72701649622202] Fitness: 1.472715463235137
Generation 15 Best Solution: [183.59604334770086, 176.3357623846854,
174.2435421866138, 75.72701649622202] Fitness: 0.9421722140491045
Generation 16 Best Solution: [183.59604334770086, 176.3357623846854,
174.98456890664696, 74.81942434622889] Fitness: 0.7017135832361925
Generation 17 Best Solution: [183.59604334770086, 175.58056731030976,
174.98456890664696, 75.72701649622202] Fitness: 1.7842477391976332
Generation 18 Best Solution: [183.25056848225057, 177.27743250221283,
174.2883926191601, 75.71730636606591] Fitness: 1.6832211538975261
Generation 19 Best Solution: [183.25056848225057, 177.27743250221283,
174.2883926191601, 75.71730636606591] Fitness: 1.6832211538975261
Generation 20 Best Solution: [183.43809312276687, 175.94694402793644,
173.90484317787792, 75.88952688438262] Fitness: 1.2367148461135589
Generation 21 Best Solution: [183.43809312276687, 175.94694402793644,
173.90484317787792, 75.88952688438262] Fitness: 1.2367148461135589
Generation 22 Best Solution: [183.59604334770086, 175.8958274906926,
174.98456890664696, 75.71730636606591] Fitness: 1.531822350669688
Generation 23 Best Solution: [183.13319642239966, 175.94694402793644,
```

174.98456890664696, 75.83566182701433] Fitness: 1.6080006380228138
Generation 24 Best Solution: [183.13319642239966, 175.94694402793644,
174.98456890664696, 75.72701649622202] Fitness: 1.4302278588784845
Generation 25 Best Solution: [183.43809312276687, 177.390196191933,
174.98456890664696, 74.86416985140738] Fitness: 1.6832425709197405
Generation 26 Best Solution: [183.59604334770086, 177.390196191933,
174.98456890664696, 74.83516473622623] Fitness: 1.746507893129384
Generation 27 Best Solution: [183.43809312276687, 176.13763033139543,
174.98456890664696, 75.83566182701433] Fitness: 1.5778891604724228
Generation 28 Best Solution: [183.65303769091787, 177.390196191933,
173.90484317787792, 75.36996068672147] Fitness: 1.6039286092664722
Generation 29 Best Solution: [183.1720747127392, 177.21262361624446,
174.51861701169736, 75.38662010758206] Fitness: 1.2498795494300394
Generation 30 Best Solution: [183.43809312276687, 177.21262361624446,
174.51861701169736, 75.38662010758206] Fitness: 1.278575453479066
Generation 31 Best Solution: [183.13319642239966, 177.21262361624446,
174.51861701169736, 75.38662010758206] Fitness: 1.2548628592705475
Generation 32 Best Solution: [182.9283475238136, 177.21262361624446,
174.51861701169736, 75.38662010758206] Fitness: 1.3197731664869352
Generation 33 Best Solution: [183.65303769091787, 177.21262361624446,
174.51861701169736, 74.54358536418201] Fitness: 1.0678377619321178
Generation 34 Best Solution: [183.13319642239966, 176.2132566618146,
174.98456890664696, 75.83566182701433] Fitness: 1.5484223486703657
Generation 35 Best Solution: [183.43809312276687, 177.390196191933,
174.10770562976475, 75.83566182701433] Fitness: 2.088314148592299
Generation 36 Best Solution: [183.13319642239966, 177.390196191933,
174.5119864149127, 75.71730636606591] Fitness: 1.9708477599124987
Generation 37 Best Solution: [182.88771614772642, 177.390196191933,
174.98456890664696, 75.00027329488276] Fitness: 1.78879493005656
Generation 38 Best Solution: [183.13319642239966, 176.1888751845682,
174.98456890664696, 75.83566182701433] Fitness: 1.549310177109494
Generation 39 Best Solution: [183.43809312276687, 176.1888751845682,
174.98456890664696, 75.83566182701433] Fitness: 1.5730227713180123
Generation 40 Best Solution: [183.43809312276687, 176.1888751845682,
174.98456890664696, 75.83566182701433] Fitness: 1.5730227713180123
Generation 41 Best Solution: [183.43809312276687, 176.1888751845682,
174.98456890664696, 75.83566182701433] Fitness: 1.5730227713180123
Generation 42 Best Solution: [183.43809312276687, 176.86211358882446,
174.98456890664696, 75.83566182701433] Fitness: 1.8867025814689833
Generation 43 Best Solution: [183.42489139654768, 175.70363794139297,
174.98456890664696, 75.83566182701433] Fitness: 1.7781313831129646
Generation 44 Best Solution: [183.43809312276687, 177.390196191933,
174.64781577549564, 75.10472393428334] Fitness: 1.4335343984992532
Generation 45 Best Solution: [183.43809312276687, 176.86211358882446,
174.64781577549564, 75.10472393428334] Fitness: 0.6963407137511448
Generation 46 Best Solution: [183.5376651869019, 176.17026450552174,
173.89076646422544, 75.83566182701433] Fitness: 1.1288637980767253
Generation 47 Best Solution: [183.13319642239966, 177.390196191933,

174.64781577549564, 74.71085123991419] Fitness: 1.2980915027932136
Generation 48 Best Solution: [183.43809312276687, 176.80448920325549,
173.89076646422544, 75.83566182701433] Fitness: 1.3480100128805583
Generation 49 Best Solution: [183.43809312276687, 175.91764170266515,
174.98456890664696, 75.83566182701433] Fitness: 1.6449748431236988
Generation 50 Best Solution: [183.43809312276687, 176.80448920325549,
174.64781577549564, 75.83566182701433] Fitness: 1.4858900303447489
Generation 51 Best Solution: [183.13319642239966, 177.390196191933,
174.64781577549564, 74.59182710423295] Fitness: 1.3115900674192378
Generation 52 Best Solution: [183.43809312276687, 176.0026926340484,
174.98456890664696, 75.83566182701433] Fitness: 1.6101534117367922
Generation 53 Best Solution: [183.43809312276687, 176.0026926340484,
174.98456890664696, 75.83566182701433] Fitness: 1.6101534117367922
Generation 54 Best Solution: [183.43809312276687, 176.50574629928965,
174.98456890664696, 75.83566182701433] Fitness: 1.6332376421854593
Generation 55 Best Solution: [183.43809312276687, 176.50574629928965,
174.5164414063124, 75.83566182701433] Fitness: 1.1991740055295312
Generation 56 Best Solution: [183.43809312276687, 177.390196191933,
174.00528047448063, 75.83566182701433] Fitness: 2.1034080927191914
Generation 57 Best Solution: [183.43809312276687, 177.390196191933,
174.00528047448063, 75.83566182701433] Fitness: 2.1034080927191914
Generation 58 Best Solution: [183.43809312276687, 177.390196191933,
174.98456890664696, 74.57167535249783] Fitness: 1.6862591809134053
Generation 59 Best Solution: [182.85938177301304, 176.09837693427363,
174.98456890664696, 75.83566182701433] Fitness: 1.6620375935088243
Generation 60 Best Solution: [183.43809312276687, 177.390196191933,
174.98456890664696, 74.41615845975723] Fitness: 1.7418037373293382
Generation 61 Best Solution: [183.43809312276687, 177.390196191933,
174.90054895035667, 74.90679166694044] Fitness: 1.5909695497081824
Generation 62 Best Solution: [183.43809312276687, 175.98639072946517,
174.98456890664696, 74.41615845975723] Fitness: 0.7338675157071076
Generation 63 Best Solution: [183.85108913038752, 175.98639072946517,
174.98456890664696, 74.41615845975723] Fitness: 0.9955254538181576
Generation 64 Best Solution: [183.41792747713785, 176.7371109341701,
174.17170299247752, 75.83566182701433] Fitness: 1.2326232096152274
Generation 65 Best Solution: [183.73319500109582, 176.11321560929596,
174.98456890664696, 75.83566182701433] Fitness: 1.7416678467544715
Generation 66 Best Solution: [183.43809312276687, 177.390196191933,
174.17170299247752, 75.83566182701433] Fitness: 2.087128766594756
Generation 67 Best Solution: [183.13082834642876, 177.390196191933,
174.98456890664696, 75.43164532029262] Fitness: 2.0318721669824162
Generation 68 Best Solution: [183.43809312276687, 175.90456266886568,
174.98456890664696, 75.83566182701433] Fitness: 1.6513232774842734
Generation 69 Best Solution: [183.43809312276687, 175.90456266886568,
174.98456890664696, 75.83566182701433] Fitness: 1.6513232774842734
Generation 70 Best Solution: [183.43809312276687, 177.390196191933,
174.98004548838227, 74.93685063792914] Fitness: 1.6972232239327107
Generation 71 Best Solution: [183.43809312276687, 177.390196191933,

174.98004548838227, 74.77166317110544] Fitness: 1.6640560774998938
Generation 72 Best Solution: [183.85108913038752, 177.390196191933,
174.53093133356842, 74.77573525831883] Fitness: 1.5061245112998987
Generation 73 Best Solution: [183.43809312276687, 176.2836056808726,
174.98004548838227, 75.83566182701433] Fitness: 1.568914820239482
Generation 74 Best Solution: [183.85108913038752, 177.390196191933,
174.53093133356842, 74.88011863254519] Fitness: 1.5228275124905055
Generation 75 Best Solution: [183.85108913038752, 177.390196191933,
174.17170299247752, 74.88011863254519] Fitness: 1.411777009361656
Generation 76 Best Solution: [183.47361719601545, 177.390196191933,
174.17170299247752, 74.88011863254519] Fitness: 1.162244294211667
Generation 77 Best Solution: [183.85108913038752, 177.390196191933,
174.17170299247752, 74.88011863254519] Fitness: 1.411777009361656
Generation 78 Best Solution: [183.47361719601545, 177.390196191933,
174.14425355151837, 74.9329424378754] Fitness: 1.1768578067811009
Generation 79 Best Solution: [183.85108913038752, 176.52002600364736,
174.98004548838227, 75.4892579469834] Fitness: 1.3924059812462763
Generation 80 Best Solution: [183.43809312276687, 177.390196191933,
174.98004548838227, 74.41255431553711] Fitness: 1.7377171847171013
Generation 81 Best Solution: [183.43809312276687, 177.390196191933,
174.14425355151837, 75.83566182701433] Fitness: 2.0868605084715584
Generation 82 Best Solution: [183.43809312276687, 177.390196191933,
174.14425355151837, 75.83566182701433] Fitness: 2.0868605084715584
Generation 83 Best Solution: [183.43809312276687, 176.64598670311975,
174.98004548838227, 75.83566182701433] Fitness: 1.703699720284614
Generation 84 Best Solution: [183.43809312276687, 176.64598670311975,
174.98004548838227, 75.83566182701433] Fitness: 1.703699720284614
Generation 85 Best Solution: [183.43809312276687, 175.61362790962707,
174.98004548838227, 75.83566182701433] Fitness: 1.855198349721077
Generation 86 Best Solution: [183.85108913038752, 176.64598670311975,
174.98004548838227, 75.8746853573903] Fitness: 2.0336719138430204
Generation 87 Best Solution: [183.43809312276687, 176.09025063766273,
174.98004548838227, 75.83566182701433] Fitness: 1.5801883445881937
Generation 88 Best Solution: [183.43809312276687, 176.64598670311975,
174.98004548838227, 75.83566182701433] Fitness: 1.703699720284614
Generation 89 Best Solution: [183.43809312276687, 176.49044344766526,
174.98004548838227, 75.83566182701433] Fitness: 1.6209390495686713
Generation 90 Best Solution: [183.43809312276687, 176.27487094203406,
174.98004548838227, 75.83566182701433] Fitness: 1.5681756249464174
Generation 91 Best Solution: [183.43809312276687, 176.27487094203406,
174.98004548838227, 75.83566182701433] Fitness: 1.5681756249464174
Generation 92 Best Solution: [183.79973125034985, 177.00483940363614,
174.98004548838227, 75.05988746017775] Fitness: 1.3835903347638423
Generation 93 Best Solution: [183.43809312276687, 176.27487094203406,
174.98004548838227, 75.83566182701433] Fitness: 1.5681756249464174
Generation 94 Best Solution: [183.79973125034985, 177.00483940363614,
174.98004548838227, 75.19209358768204] Fitness: 1.4657609002675553
Generation 95 Best Solution: [183.79973125034985, 176.43910739487444,

174.98004548838227, 75.83566182701433] Fitness: 1.8165871858946327
Generation 96 Best Solution: [183.43809312276687, 177.390196191933,
174.98004548838227, 75.02360016145904] Fitness: 1.731561537405221
Generation 97 Best Solution: [183.43809312276687, 177.390196191933,
174.98004548838227, 75.02360016145904] Fitness: 1.731561537405221
Generation 98 Best Solution: [183.79973125034985, 176.43910739487444,
174.98004548838227, 75.83566182701433] Fitness: 1.8165871858946327
Generation 99 Best Solution: [183.43809312276687, 177.390196191933,
174.16235827305564, 75.83566182701433] Fitness: 2.086906461636184
Generation 100 Best Solution: [183.85108913038752, 177.390196191933,
173.94217474283033, 75.83566182701433] Fitness: 2.382452829970825
Generation 101 Best Solution: [183.12529636796333, 176.49387161896522,
174.98004548838227, 75.83566182701433] Fitness: 1.599945486767493
Generation 102 Best Solution: [183.79973125034985, 176.69859725279895,
174.469387476767, 75.83566182701433] Fitness: 1.5018029255882577
Generation 103 Best Solution: [183.79973125034985, 176.69859725279895,
174.98004548838227, 75.83566182701433] Fitness: 1.9549112810175173
Generation 104 Best Solution: [183.79973125034985, 176.69859725279895,
174.98004548838227, 75.58408344403662] Fitness: 1.571100974833147
Generation 105 Best Solution: [183.79973125034985, 176.69859725279895,
174.80458551588185, 75.83566182701433] Fitness: 1.753691836935102
Generation 106 Best Solution: [183.85108913038752, 176.26098700234022,
174.98004548838227, 75.58408344403662] Fitness: 1.4450914247659945
Generation 107 Best Solution: [183.21984239679674, 176.26098700234022,
174.98004548838227, 75.58408344403662] Fitness: 1.1518191475772468
Generation 108 Best Solution: [183.21984239679674, 176.26098700234022,
174.98004548838227, 75.59985989792307] Fitness: 1.1730077598774082
Generation 109 Best Solution: [183.85108913038752, 176.44788222374936,
174.98004548838227, 75.83566182701433] Fitness: 1.8664788927649698
Generation 110 Best Solution: [183.85108913038752, 176.44269513014456,
173.83639985012985, 75.83566182701433] Fitness: 1.4103953222794319
Generation 111 Best Solution: [183.43809312276687, 176.82385347296562,
174.98004548838227, 75.83566182701433] Fitness: 1.8442498365588533
Generation 112 Best Solution: [183.58186232301225, 175.69299326191663,
174.98004548838227, 75.83566182701433] Fitness: 1.846115759694832
Generation 113 Best Solution: [183.21984239679674, 175.95616266569004,
174.98004548838227, 75.83566182701433] Fitness: 1.5903838494010303
Generation 114 Best Solution: [183.21984239679674, 177.390196191933,
174.98004548838227, 74.80705481539796] Fitness: 1.6359914423416448
Generation 115 Best Solution: [183.46470818465272, 177.390196191933,
174.98004548838227, 74.80705481539796] Fitness: 1.6765065718939318
Generation 116 Best Solution: [183.85108913038752, 177.390196191933,
173.83639985012985, 74.36822119467416] Fitness: 1.5680049751624858
Generation 117 Best Solution: [183.85108913038752, 176.0489883901286,
174.98004548838227, 75.83566182701433] Fitness: 1.8517469105337774
Generation 118 Best Solution: [183.2859347688357, 176.8080429342873,
174.98004548838227, 75.83566182701433] Fitness: 1.79994640828521
Generation 119 Best Solution: [183.21984239679674, 176.0489883901286,

174.98004548838227, 75.83566182701433] Fitness: 1.5584746333450297
Generation 120 Best Solution: [183.21984239679674, 175.80358255842225,
174.4147660683949, 75.83566182701433] Fitness: 1.194153000818177
Generation 121 Best Solution: [183.85108913038752, 177.390196191933,
174.4147660683949, 75.46844274498464] Fitness: 1.8747102500112567
Generation 122 Best Solution: [183.85108913038752, 177.390196191933,
174.4147660683949, 75.46844274498464] Fitness: 1.8747102500112567
Generation 123 Best Solution: [183.85108913038752, 177.390196191933,
174.4147660683949, 75.46844274498464] Fitness: 1.8747102500112567
Generation 124 Best Solution: [183.21984239679674, 176.81941910447946,
175.07720769632064, 75.46844274498464] Fitness: 1.4131137102821087
Generation 125 Best Solution: [182.8631718930186, 176.37042601553796,
175.07720769632064, 75.83566182701433] Fitness: 1.7900938664850683
Generation 126 Best Solution: [182.8631718930186, 176.37042601553796,
175.07720769632064, 75.83566182701433] Fitness: 1.7900938664850683
Generation 127 Best Solution: [183.85108913038752, 177.390196191933,
175.07720769632064, 74.29620117309534] Fitness: 2.1980052332043787
Generation 128 Best Solution: [183.85108913038752, 176.234576022122,
175.07720769632064, 74.29620117309534] Fitness: 1.1462218920660683
Generation 129 Best Solution: [183.12117298533576, 176.50914658503834,
174.98004548838227, 75.83566182701433] Fitness: 1.6072114495906353
Generation 130 Best Solution: [183.85108913038752, 177.390196191933,
173.98417709689141, 75.83566182701433] Fitness: 2.370194164033419
Generation 131 Best Solution: [183.85108913038752, 177.390196191933,
174.98004548838227, 74.7246984388408] Fitness: 1.923998017255818
Generation 132 Best Solution: [183.3455637331976, 177.390196191933,
174.98004548838227, 74.7246984388408] Fitness: 1.6399307607509837
Generation 133 Best Solution: [183.35808802642737, 177.390196191933,
174.98004548838227, 74.7246984388408] Fitness: 1.6421882210659549
Generation 134 Best Solution: [183.4662010076507, 177.390196191933,
174.98004548838227, 74.7246984388408] Fitness: 1.6717725949017492
Generation 135 Best Solution: [182.8631718930186, 177.390196191933,
174.98004548838227, 74.7246984388408] Fitness: 1.7378149893261512
Generation 136 Best Solution: [182.8631718930186, 177.390196191933,
174.98004548838227, 74.7246984388408] Fitness: 1.7378149893261512
Generation 137 Best Solution: [183.43022488975777, 175.91395626357587,
174.98004548838227, 75.83566182701433] Fitness: 1.6384978165364927
Generation 138 Best Solution: [183.85108913038752, 177.390196191933,
174.98004548838227, 74.7246984388408] Fitness: 1.923998017255818
Generation 139 Best Solution: [183.47032168007246, 177.390196191933,
174.45090223726973, 75.83566182701433] Fitness: 2.167058902203514
Generation 140 Best Solution: [183.85108913038752, 177.390196191933,
173.81958120941096, 75.5132993639066] Fitness: 1.9597210435431254
Generation 141 Best Solution: [182.96167748996027, 177.09739330776333,
173.7966840561923, 75.83566182701433] Fitness: 1.7485046419950248
Generation 142 Best Solution: [182.96167748996027, 177.09739330776333,
173.7966840561923, 75.83566182701433] Fitness: 1.7485046419950248
Generation 143 Best Solution: [182.96167748996027, 176.54414500544658,

174.98004548838227, 75.83566182701433] Fitness: 1.6714888195540527
Generation 144 Best Solution: [183.85108913038752, 176.54414500544658,
174.98004548838227, 75.83566182701433] Fitness: 1.9069361323697063
Generation 145 Best Solution: [183.85108913038752, 176.54414500544658,
174.98004548838227, 75.83566182701433] Fitness: 1.9069361323697063
Generation 146 Best Solution: [183.85108913038752, 176.54414500544658,
174.98004548838227, 75.83566182701433] Fitness: 1.9069361323697063
Generation 147 Best Solution: [182.96167748996027, 176.29808215822547,
174.98004548838227, 75.83566182701433] Fitness: 1.5966106892134433
Generation 148 Best Solution: [182.96167748996027, 175.7591189894823,
174.82108842497516, 75.83566182701433] Fitness: 1.575838299303295
Generation 149 Best Solution: [182.96167748996027, 175.7591189894823,
173.81958120941096, 75.83566182701433] Fitness: 1.3142810815669617
Generation 150 Best Solution: [182.96167748996027, 175.7591189894823,
173.81958120941096, 75.83566182701433] Fitness: 1.3142810815669617
Generation 151 Best Solution: [183.67650715819934, 176.1175772313327,
173.81958120941096, 75.83566182701433] Fitness: 1.248026321818833
Generation 152 Best Solution: [183.85108913038752, 175.7591189894823,
173.81958120941096, 75.83566182701433] Fitness: 1.549728394382615
Generation 153 Best Solution: [182.96167748996027, 177.390196191933,
173.81958120941096, 74.88063064686375] Fitness: 1.261521049832099
Generation 154 Best Solution: [183.15042812709967, 175.7591189894823,
174.98004548838227, 75.83566182701433] Fitness: 1.7077422193786025
Generation 155 Best Solution: [183.2672728979638, 175.7591189894823,
174.98004548838227, 75.83566182701433] Fitness: 1.702935845919037
Generation 156 Best Solution: [183.85108913038752, 176.53874365541725,
173.81958120941096, 75.83566182701433] Fitness: 1.4584042544408191
Generation 157 Best Solution: [183.38909915483876, 177.390196191933,
173.90606474757766, 75.83566182701433] Fitness: 2.120000567592148
Generation 158 Best Solution: [183.85108913038752, 177.390196191933,
174.98004548838227, 75.2132246896118] Fitness: 2.1088524909201207
Generation 159 Best Solution: [183.85108913038752, 177.390196191933,
175.08359826081704, 74.82592388054462] Fitness: 2.073072791837903
Generation 160 Best Solution: [182.8631718930186, 177.390196191933,
174.98004548838227, 74.82592388054462] Fitness: 1.7457658744305173
Generation 161 Best Solution: [182.8631718930186, 177.390196191933,
174.88682608117372, 74.82592388054462] Fitness: 1.6329255331180685
Generation 162 Best Solution: [183.85108913038752, 177.390196191933,
174.88682608117372, 74.82592388054462] Fitness: 1.8191085610477353
Generation 163 Best Solution: [182.8631718930186, 177.390196191933,
175.08359826081704, 74.82592388054462] Fitness: 1.8868897639082365
Generation 164 Best Solution: [182.8631718930186, 177.390196191933,
175.08359826081704, 74.82592388054462] Fitness: 1.8868897639082365
Generation 165 Best Solution: [182.8631718930186, 175.87237862329275,
174.98004548838227, 75.83566182701433] Fitness: 1.73773000417498
Generation 166 Best Solution: [182.8631718930186, 175.91795302238674,
174.73242154325064, 75.83566182701433] Fitness: 1.4443406448657334
Generation 167 Best Solution: [183.28180748014478, 177.390196191933,

174.98004548838227, 74.43773300065962] Fitness: 1.6959071463630087
Generation 168 Best Solution: [182.8631718930186, 176.59611100433543,
174.98004548838227, 75.83566182701433] Fitness: 1.7485568319521103
Generation 169 Best Solution: [182.89684341050386, 176.20406163024828,
174.98004548838227, 75.83566182701433] Fitness: 1.6234889673773958
Generation 170 Best Solution: [183.85108913038752, 176.27826595383888,
174.98004548838227, 75.83566182701433] Fitness: 1.8301068380165688
Generation 171 Best Solution: [182.8631718930186, 175.5947132135535,
174.98004548838227, 75.83566182701433] Fitness: 1.948843457091137
Generation 172 Best Solution: [182.8631718930186, 176.44344448556666,
174.98004548838227, 75.83566182701433] Fitness: 1.6787767496062989
Generation 173 Best Solution: [183.69691987460217, 175.5947132135535,
174.98004548838227, 75.43077428593698] Fitness: 1.4368145140036046
Generation 174 Best Solution: [182.8631718930186, 175.5947132135535,
174.98004548838227, 75.43077428593698] Fitness: 1.379199773802463
Generation 175 Best Solution: [183.69691987460217, 175.5947132135535,
174.98004548838227, 75.43077428593698] Fitness: 1.4368145140036046
Generation 176 Best Solution: [182.8631718930186, 175.5947132135535,
173.92213871366806, 75.83566182701433] Fitness: 1.4583698640130416
Generation 177 Best Solution: [182.8631718930186, 175.5947132135535,
174.98004548838227, 75.83566182701433] Fitness: 1.948843457091137
Generation 178 Best Solution: [183.46339638465022, 176.62438593992965,
174.98004548838227, 75.83566182701433] Fitness: 1.6984029436098202
Generation 179 Best Solution: [182.8631718930186, 176.24960018741095,
174.98004548838227, 75.83566182701433] Fitness: 1.6421772443400102
Generation 180 Best Solution: [183.69691987460217, 177.390196191933,
174.98004548838227, 74.63456699994495] Fitness: 1.8017029633549648
Generation 181 Best Solution: [183.29194566177117, 176.71125458955044,
174.98004548838227, 75.83566182701433] Fitness: 1.7202565406645665
Generation 182 Best Solution: [182.8631718930186, 176.36996432376947,
174.0069730663714, 75.83566182701433] Fitness: 1.143003996669086
Generation 183 Best Solution: [183.69691987460217, 176.01520961309092,
174.98004548838227, 75.51785897748782] Fitness: 1.2646970221373945
Generation 184 Best Solution: [182.8631718930186, 177.390196191933,
174.98004548838227, 74.86180587600852] Fitness: 1.75239306388277
Generation 185 Best Solution: [183.69691987460217, 176.01520961309092,
175.11749813521945, 74.86180587600852] Fitness: 0.9830165565808096
Generation 186 Best Solution: [183.69691987460217, 177.390196191933,
174.98004548838227, 74.86180587600852] Fitness: 1.8100078040839114
Generation 187 Best Solution: [182.8631718930186, 176.34478312936469,
174.98004548838227, 75.72235619871957] Fitness: 1.4678887951208992
Generation 188 Best Solution: [182.8631718930186, 176.3056025247561,
174.98004548838227, 75.83566182701433] Fitness: 1.6467746118684157
Generation 189 Best Solution: [183.69691987460217, 177.390196191933,
174.557354603632, 75.83566182701433] Fitness: 2.3473264798267213
Generation 190 Best Solution: [183.69691987460217, 177.390196191933,
174.557354603632, 75.83566182701433] Fitness: 2.3473264798267213
Generation 191 Best Solution: [182.8631718930186, 176.30763848577993,

174.98004548838227, 75.83566182701433] Fitness: 1.6470332304251611
Generation 192 Best Solution: [182.8631718930186, 176.30763848577993,
174.98004548838227, 75.83566182701433] Fitness: 1.6470332304251611
Generation 193 Best Solution: [183.69691987460217, 176.1108285141329,
174.98004548838227, 74.4201144138126] Fitness: 0.8253560256279058
Generation 194 Best Solution: [183.69691987460217, 177.390196191933,
174.98004548838227, 74.66096158576698] Fitness: 1.798563443215642
Generation 195 Best Solution: [183.69691987460217, 177.390196191933,
174.98004548838227, 74.66096158576698] Fitness: 1.798563443215642
Generation 196 Best Solution: [183.51135354288581, 177.0716231780958,
173.88312914056212, 75.83566182701433] Fitness: 1.6733978180968867
Generation 197 Best Solution: [183.69691987460217, 176.50966185493104,
174.98004548838227, 75.83566182701433] Fitness: 1.7622259874999058
Generation 198 Best Solution: [183.23875571161045, 177.390196191933,
174.98004548838227, 74.2863869550979] Fitness: 1.7792041741227054
Generation 199 Best Solution: [183.69691987460217, 177.390196191933,
174.98004548838227, 74.9662391999258] Fitness: 1.8406407318054427
Generation 200 Best Solution: [183.69691987460217, 177.1329073707258,
174.98004548838227, 74.9662391999258] Fitness: 1.427530451112853
Generation 201 Best Solution: [183.69691987460217, 177.390196191933,
175.0917215572543, 74.76129143579608] Fitness: 1.9493697526921232
Generation 202 Best Solution: [183.69691987460217, 177.43088232060143,
175.0917215572543, 74.76129143579608] Fitness: 2.024082457634941
Generation 203 Best Solution: [183.69691987460217, 177.390196191933,
173.83734789846184, 75.83566182701433] Fitness: 2.2963950624176412
Generation 204 Best Solution: [183.09529292494005, 177.390196191933,
173.83734789846184, 75.83566182701433] Fitness: 2.146704315625922
Generation 205 Best Solution: [183.69691987460217, 177.390196191933,
173.84394671289 97, 75.83566182701433] Fitness: 2.2932172161518594
Generation 206 Best Solution: [183.69691987460217, 176.1065223569434,
175.11749813521945, 75.83566182701433] Fitness: 1.9010292783770935
Generation 207 Best Solution: [183.69691987460217, 176.1065223569434,
175.11749813521945, 75.83566182701433] Fitness: 1.9010292783770935
Generation 208 Best Solution: [183.69691987460217, 176.1065223569434,
175.11749813521945, 75.80833873740548] Fitness: 1.8546011674623593
Generation 209 Best Solution: [183.69691987460217, 175.9097944590137,
175.11749813521945, 75.83566182701433] Fitness: 1.9669546780539497
Generation 210 Best Solution: [183.69691987460217, 175.9097944590137,
174.19670939115872, 75.83566182701433] Fitness: 1.2463341010600222
Generation 211 Best Solution: [183.69691987460217, 175.9097944590137,
174.19670939115872, 75.83566182701433] Fitness: 1.2463341010600222
Generation 212 Best Solution: [183.69691987460217, 175.9097944590137,
175.11749813521945, 75.83566182701433] Fitness: 1.9669546780539497
Generation 213 Best Solution: [183.69691987460217, 175.9097944590137,
174.98004548838227, 75.83566182701433] Fitness: 1.776023875437263
Generation 214 Best Solution: [183.69691987460217, 175.9097944590137,
174.98004548838227, 75.83566182701433] Fitness: 1.776023875437263
Generation 215 Best Solution: [183.69691987460217, 177.390196191933,

175.11749813521945, 74.9649774239572] Fitness: 2.031100639672441
Generation 216 Best Solution: [183.69691987460217, 177.390196191933,
175.11749813521945, 74.74497381592819] Fitness: 1.9866824097165006
Generation 217 Best Solution: [183.69691987460217, 176.14418331542905,
175.11749813521945, 75.73720787743237] Fitness: 1.7333690076087327
Generation 218 Best Solution: [183.53011573411973, 176.52399281566127,
175.11749813521945, 75.83566182701433] Fitness: 1.8619866771265223
Generation 219 Best Solution: [183.69691987460217, 176.2904002842065,
175.11749813521945, 75.83566182701433] Fitness: 1.8935919701521347
Generation 220 Best Solution: [183.69691987460217, 176.84904798466889,
175.11749813521945, 75.57357063071755] Fitness: 1.794424157309328
Generation 221 Best Solution: [183.69691987460217, 176.84904798466889,
175.11749813521945, 74.95025796877927] Fitness: 1.275830306958789
Generation 222 Best Solution: [183.65683769396787, 176.04578559465244,
175.11749813521945, 75.83566182701433] Fitness: 1.8875898269149745
Generation 223 Best Solution: [183.65683769396787, 176.04578559465244,
175.11749813521945, 75.72625956008622] Fitness: 1.7086426094606186
Generation 224 Best Solution: [183.65683769396787, 176.04578559465244,
175.11749813521945, 75.44508704421813] Fitness: 1.3337551262272171
Generation 225 Best Solution: [183.69691987460217, 177.43088232060143,
175.11749813521945, 74.86482373536175] Fitness: 2.0762995780765903
Generation 226 Best Solution: [183.69691987460217, 177.390196191933,
175.11749813521945, 74.86482373536175] Fitness: 2.0015868731337725
Generation 227 Best Solution: [183.69691987460217, 177.390196191933,
174.61666810005735, 74.86482373536175] Fitness: 1.4467966079016967
Generation 228 Best Solution: [183.69691987460217, 177.390196191933,
174.61666810005735, 74.86482373536175] Fitness: 1.4467966079016967
Generation 229 Best Solution: [183.53011573411973, 176.06545282553873,
175.11749813521945, 74.86482373536175] Fitness: 0.8716757854958628
Generation 230 Best Solution: [183.69691987460217, 176.23425653882424,
175.11749813521945, 75.72625956008622] Fitness: 1.7113634880904725
Generation 231 Best Solution: [183.39944330762475, 175.8575884744761,
175.11749813521945, 75.83566182701433] Fitness: 1.8504483960215232
Generation 232 Best Solution: [183.69691987460217, 176.61864152388685,
175.11749813521945, 74.64954016294747] Fitness: 1.0590779403362736
Generation 233 Best Solution: [183.69691987460217, 176.61864152388685,
175.11749813521945, 74.64954016294747] Fitness: 1.0590779403362736
Generation 234 Best Solution: [183.69691987460217, 176.14476288708428,
175.11749813521945, 75.83566182701433] Fitness: 1.8951708857267087
Generation 235 Best Solution: [183.63586453042257, 175.90428763555167,
175.11749813521945, 74.64954016294747] Fitness: 0.9775404619830117
Generation 236 Best Solution: [183.63586453042257, 175.90428763555167,
175.11749813521945, 74.64954016294747] Fitness: 0.9775404619830117
Generation 237 Best Solution: [183.00500701455147, 175.8575884744761,
175.11749813521945, 75.72625956008622] Fitness: 1.6917725608978909
Generation 238 Best Solution: [183.63586453042257, 176.59346693972148,
175.11749813521945, 74.95619909948012] Fitness: 1.0406373652738594
Generation 239 Best Solution: [183.00500701455147, 177.43088232060143,

174.2803560137905, 74.64954016294747] Fitness: 1.2322722436223035
Generation 240 Best Solution: [183.69691987460217, 177.43088232060143,
174.2803560137905, 74.64954016294747] Fitness: 1.3560631621876638
Generation 241 Best Solution: [183.67656649319966, 177.43088232060143,
174.2803560137905, 74.64954016294747] Fitness: 1.3418396461995283
Generation 242 Best Solution: [183.00500701455147, 177.43088232060143,
175.11749813521945, 74.45547941759231] Fitness: 1.9933470185610773
Generation 243 Best Solution: [183.00500701455147, 175.67884020602827,
175.11749813521945, 75.72625956008622] Fitness: 1.8180824973415612
Generation 244 Best Solution: [183.67656649319966, 176.5250373843483,
174.9869934052739, 75.72625956008622] Fitness: 1.584974346645974
Generation 245 Best Solution: [183.54036298014603, 175.67884020602827,
175.11749813521945, 74.703244561703] Fitness: 1.072536995083184
Generation 246 Best Solution: [183.24778223485285, 175.67884020602827,
175.11749813521945, 74.703244561703] Fitness: 1.000661371732405
Generation 247 Best Solution: [183.69691987460217, 175.67884020602827,
174.24228225434283, 75.83566182701433] Fitness: 1.404986382328817
Generation 248 Best Solution: [183.67656649319966, 175.67884020602827,
175.11749813521945, 75.31626765642893] Fitness: 1.4218964095615187
Generation 249 Best Solution: [183.69691987460217, 177.18134634020666,
175.11749813521945, 74.703244561703] Fitness: 1.6435456509417714
Generation 250 Best Solution: [183.00500701455147, 177.43088232060143,
174.1378874515275, 75.72625956008622] Fitness: 1.992029176159365
Generation 251 Best Solution: [183.67656649319966, 177.43088232060143,
174.7758500450293, 75.37320530123327] Fitness: 1.9519993274241947
Generation 252 Best Solution: [183.10109569627193, 176.8435359595411,
175.11749813521945, 75.83566182701433] Fitness: 2.0359099530839493
Generation 253 Best Solution: [183.10109569627193, 176.8435359595411,
175.11749813521945, 75.72625956008622] Fitness: 1.8569627356295937
Generation 254 Best Solution: [183.10109569627193, 176.8435359595411,
175.11749813521945, 75.72625956008622] Fitness: 1.8569627356295937
Generation 255 Best Solution: [183.69691987460217, 177.43088232060143,
173.8796304921664, 75.30744241809927] Fitness: 1.6591877821460197
Generation 256 Best Solution: [183.10109569627193, 176.8435359595411,
175.11749813521945, 75.83566182701433] Fitness: 2.0359099530839493
Generation 257 Best Solution: [183.69691987460217, 175.70236895130304,
175.11749813521945, 75.30744241809927] Fitness: 1.408641221778547
Generation 258 Best Solution: [183.002828779253, 175.80152665309586,
175.11749813521945, 75.83566182701433] Fitness: 1.9057909364293661
Generation 259 Best Solution: [183.002828779253, 175.80152665309586,
175.11749813521945, 75.72625956008622] Fitness: 1.7268437189750103
Generation 260 Best Solution: [183.10109569627193, 177.43088232060143,
173.87167083559103, 74.5087724817211] Fitness: 1.2847107563256972
Generation 261 Best Solution: [183.67656649319966, 175.73726851522653,
175.11749813521945, 75.83566182701433] Fitness: 2.059866927137596
Generation 262 Best Solution: [183.69691987460217, 177.43088232060143,
174.56156702795357, 75.48606721510801] Fitness: 1.9178827102400864
Generation 263 Best Solution: [183.69691987460217, 177.43088232060143,

174.56156702795357, 75.48606721510801] Fitness: 1.9178827102400864
Generation 264 Best Solution: [183.32982577152072, 177.43088232060143,
174.56156702795357, 75.48606721510801] Fitness: 1.759891382352016
Generation 265 Best Solution: [183.69691987460217, 177.43088232060143,
174.10363285746973, 75.48606721510801] Fitness: 1.789590277189376
Generation 266 Best Solution: [183.69691987460217, 176.59208938504528,
175.11749813521945, 75.83566182701433] Fitness: 1.9948013846212937
Generation 267 Best Solution: [183.32982577152072, 177.43088232060143,
175.11749813521945, 75.17703316917036] Fitness: 2.0615677213533137
Generation 268 Best Solution: [183.69691987460217, 177.43088232060143,
173.89295920317355, 75.23297729972582] Fitness: 1.590799847023748
Generation 269 Best Solution: [183.69691987460217, 176.16718045114897,
175.11749813521945, 75.83566182701433] Fitness: 1.8927893016572956
Generation 270 Best Solution: [183.69691987460217, 177.43088232060143,
174.64279908592073, 74.39172049282332] Fitness: 1.611404812523159
Generation 271 Best Solution: [183.33691834074563, 177.43088232060143,
174.24064330404565, 74.45541162019104] Fitness: 1.2441884834082315
Generation 272 Best Solution: [183.69691987460217, 177.43088232060143,
175.11749813521945, 74.99963810912608] Fitness: 2.1196449401348922
Generation 273 Best Solution: [183.67656649319966, 176.28982166580428,
175.11749813521945, 75.83566182701433] Fitness: 1.8793097462940762
Generation 274 Best Solution: [183.69691987460217, 175.66749139837214,
174.64279908592073, 75.83566182701433] Fitness: 1.5950630776918246
Generation 275 Best Solution: [183.67656649319966, 176.70854433581496,
174.64279908592073, 75.83566182701433] Fitness: 1.5218977510643632
Generation 276 Best Solution: [183.69691987460217, 176.28982166580428,
175.11749813521945, 74.81173162743931] Fitness: 0.9436747379300569
Generation 277 Best Solution: [183.69691987460217, 177.43088232060143,
175.11749813521945, 74.81173162743931] Fitness: 2.0669530914736143
Generation 278 Best Solution: [183.69691987460217, 176.71234721543462,
175.11749813521945, 75.83566182701433] Fitness: 2.074429441372991
Generation 279 Best Solution: [183.69691987460217, 176.03991394017908,
175.11749813521945, 75.83566182701433] Fitness: 1.9166404195036622
Generation 280 Best Solution: [183.69691987460217, 176.71234721543462,
174.64279908592073, 75.81529598636007] Fitness: 1.5042663187518581
Generation 281 Best Solution: [183.69691987460217, 176.4666893999079,
174.64279908592073, 75.81529598636007] Fitness: 1.3654547647825859
Generation 282 Best Solution: [183.69691987460217, 176.4666893999079,
175.11749813521945, 75.81529598636007] Fitness: 1.9009020208376648
Generation 283 Best Solution: [183.5852061924879, 176.4666893999079,
175.11749813521945, 75.81529598636007] Fitness: 1.8307349251779073
Generation 284 Best Solution: [183.67656649319966, 177.43088232060143,
174.72949874175868, 74.84639460663088] Fitness: 1.5946412987074914
Generation 285 Best Solution: [183.69691987460217, 176.89383651925294,
174.80964413154015, 75.81529598636007] Fitness: 1.8152672322125207
Generation 286 Best Solution: [183.67656649319966, 176.77923572549074,
175.11652460770645, 75.81529598636007] Fitness: 2.078016065671024
Generation 287 Best Solution: [183.67656649319966, 177.43088232060143,

175.11749813521945, 74.31787109680558] Fitness: 2.1749128153635366
Generation 288 Best Solution: [183.5852061924879, 176.70533050051537,
175.11652460770645, 75.75180515659764] Fitness: 1.8587247275571728
Generation 289 Best Solution: [183.67656649319966, 177.43088232060143,
174.42552909681172, 74.96981028361192] Fitness: 1.4292805202406909
Generation 290 Best Solution: [183.67656649319966, 177.43088232060143,
174.42552909681172, 74.96981028361192] Fitness: 1.4292805202406909
Generation 291 Best Solution: [183.67656649319966, 177.13346859600273,
174.42552909681172, 74.96981028361192] Fitness: 0.9422471065737933
Generation 292 Best Solution: [183.67656649319966, 177.13346859600273,
174.42552909681172, 74.96981028361192] Fitness: 0.9422471065737933
Generation 293 Best Solution: [183.67656649319966, 177.13346859600273,
175.11749813521945, 74.9560784684538] Fitness: 1.6013053275622142
Generation 294 Best Solution: [183.69691987460217, 177.43088232060143,
173.7657747925085, 75.81529598636007] Fitness: 2.375191578597379
Generation 295 Best Solution: [183.67656649319966, 177.43088232060143,
174.58802616522124, 75.81529598636007] Fitness: 2.393094829141493
Generation 296 Best Solution: [183.67656649319966, 176.91521957096077,
175.11652460770645, 75.83566182701433] Fitness: 2.243833199280888
Generation 297 Best Solution: [183.67656649319966, 176.2091876952023,
175.11749813521945, 75.81529598636007] Fitness: 1.841482384747278
Generation 298 Best Solution: [183.67656649319966, 177.43088232060143,
175.11749813521945, 74.90487505988386] Fitness: 2.072014954708954
Generation 299 Best Solution: [183.67656649319966, 176.37399908784639,
175.11749813521945, 75.81529598636007] Fitness: 1.858582804378175
Generation 300 Best Solution: [183.67656649319966, 177.43088232060143,
175.11749813521945, 74.54960698912917] Fitness: 2.0705529787361634
Generation 301 Best Solution: [183.85230231470103, 176.94301632422344,
174.07328827151795, 75.81529598636007] Fitness: 1.667409933551761
Generation 302 Best Solution: [182.92039697633538, 177.43088232060143,
175.11749813521945, 75.08809001005585] Fitness: 2.07531010299716
Generation 303 Best Solution: [183.67656649319966, 177.43088232060143,
175.11749813521945, 75.08809001005585] Fitness: 2.149149404277428
Generation 304 Best Solution: [183.67656649319966, 176.94301632422344,
175.11749813521945, 75.81529598636007] Fitness: 2.2408957551156625
Generation 305 Best Solution: [183.67656649319966, 176.58764749509538,
175.11749813521945, 75.81529598636007] Fitness: 1.943349656914386
Generation 306 Best Solution: [183.67656649319966, 176.94301632422344,
175.11749813521945, 75.4320563486124] Fitness: 1.7073710912772917
Generation 307 Best Solution: [183.67656649319966, 177.43088232060143,
175.11749813521945, 74.56359060577232] Fitness: 2.066915768901182
Generation 308 Best Solution: [183.67656649319966, 176.9222651132378,
175.11749813521945, 75.4320563486124] Fitness: 1.684620565040945
Generation 309 Best Solution: [183.85230231470103, 176.5190174869415,
175.11749813521945, 75.81529598636007] Fitness: 2.0523643293445826
Generation 310 Best Solution: [183.67656649319966, 176.9222651132378,
175.11749813521945, 75.50329387836675] Fitness: 1.7665860196975964
Generation 311 Best Solution: [183.67656649319966, 176.72407570141365,

175.11749813521945, 75.81529598636007] Fitness: 2.034454462241745
Generation 312 Best Solution: [183.67656649319966, 176.72407570141365,
175.11749813521945, 75.50329387836675] Fitness: 1.5828952530600258
Generation 313 Best Solution: [183.67656649319966, 176.10754124599663,
175.11749813521945, 75.16896252480123] Fitness: 1.0837641387124706
Generation 314 Best Solution: [183.67656649319966, 176.10754124599663,
175.11749813521945, 75.81529598636007] Fitness: 1.85190444245154
Generation 315 Best Solution: [183.67656649319966, 176.10754124599663,
175.11749813521945, 75.81529598636007] Fitness: 1.85190444245154
Generation 316 Best Solution: [183.67656649319966, 175.8427412035151,
175.11749813521945, 75.81529598636007] Fitness: 1.9541789419839288
Generation 317 Best Solution: [183.67656649319966, 175.8427412035151,
175.11749813521945, 75.81529598636007] Fitness: 1.9541789419839288
Generation 318 Best Solution: [183.67656649319966, 177.43088232060143,
174.01429586419766, 75.81529598636007] Fitness: 2.2602905391225376
Generation 319 Best Solution: [183.67656649319966, 176.89442202007902,
175.11749813521945, 75.81529598636007] Fitness: 2.1886670355248667
Generation 320 Best Solution: [183.67656649319966, 176.81388282577257,
175.11749813521945, 75.81529598636007] Fitness: 2.110156087546507
Generation 321 Best Solution: [183.67656649319966, 177.43088232060143,
174.46855575772275, 75.81529598636007] Fitness: 2.3234244426007633
Generation 322 Best Solution: [183.67656649319966, 175.9179108118685,
174.46855575772275, 75.81529598636007] Fitness: 1.269662472591153
Generation 323 Best Solution: [183.67656649319966, 177.43088232060143,
174.76893717347133, 75.18758774489012] Fitness: 1.7856101273922114
Generation 324 Best Solution: [183.67656649319966, 177.43088232060143,
174.61950125460726, 75.18758774489012] Fitness: 1.6600717968910432
Generation 325 Best Solution: [183.67656649319966, 177.43088232060143,
174.35442315702144, 75.81529598636007] Fitness: 2.2775081832593185
Generation 326 Best Solution: [182.83819574352594, 177.43088232060143,
174.35442315702144, 75.81529598636007] Fitness: 2.2489956353862888
Generation 327 Best Solution: [183.67656649319966, 176.18043431617374,
175.11749813521945, 75.81529598636007] Fitness: 1.8428078892964623
Generation 328 Best Solution: [183.34572156020167, 177.43088232060143,
174.20854092978945, 75.77713093070344] Fitness: 2.043609479338855
Generation 329 Best Solution: [183.67656649319966, 176.65688434579837,
175.11749813521945, 75.77713093070344] Fitness: 1.9226561857499518
Generation 330 Best Solution: [183.67656649319966, 176.65688434579837,
174.7768906938787, 75.81529598636007] Fitness: 1.5664282930785616
Generation 331 Best Solution: [183.67656649319966, 177.43088232060143,
174.35442315702144, 74.3831688065138] Fitness: 1.4467286640212105
Generation 332 Best Solution: [183.67656649319966, 175.93240622195538,
175.11749813521945, 75.77713093070344] Fitness: 1.844062610809209
Generation 333 Best Solution: [183.67656649319966, 176.65084202556642,
175.11749813521945, 75.81529598636007] Fitness: 1.9819671179324096
Generation 334 Best Solution: [183.67656649319966, 177.43088232060143,
175.11749813521945, 74.3831688065138] Fitness: 2.1370927140334715
Generation 335 Best Solution: [183.67656649319966, 176.08481894464677,

175.11749813521945, 75.81529598636007] Fitness: 1.856422043073576
Generation 336 Best Solution: [183.67656649319966, 176.42696071385254,
175.11749813521945, 75.77713093070344] Fitness: 1.8096800093119003
Generation 337 Best Solution: [183.67656649319966, 176.32226725913938,
175.11749813521945, 75.50193046677663] Fitness: 1.3954846542342207
Generation 338 Best Solution: [183.67656649319966, 175.80229072518264,
175.11749813521945, 75.50193046677663] Fitness: 1.5261599297473831
Generation 339 Best Solution: [183.67656649319966, 177.43088232060143,
174.69423715344476, 74.71433493345147] Fitness: 1.5526489396225225
Generation 340 Best Solution: [183.67656649319966, 176.104128653119,
174.69423715344476, 74.71433493345147] Fitness: 0.4373086170172471
Generation 341 Best Solution: [183.17956621953968, 177.35984567188083,
174.97637362108185, 74.71433493345147] Fitness: 1.5742708192400565
Generation 342 Best Solution: [183.67656649319966, 176.02012522137196,
175.11749813521945, 75.81529598636007] Fitness: 1.873662586455679
Generation 343 Best Solution: [183.67656649319966, 176.81325110029076,
174.28616139015926, 75.81529598636007] Fitness: 1.4013929286298834
Generation 344 Best Solution: [183.67656649319966, 177.09205740529873,
174.70840798928927, 74.55498945134474] Fitness: 1.0429115779314062
Generation 345 Best Solution: [183.67656649319966, 176.23163149171282,
175.11749813521945, 75.81529598636007] Fitness: 1.8413373402541788
Generation 346 Best Solution: [183.17956621953968, 177.43088232060143,
175.11749813521945, 74.3365393903882] Fitness: 2.0151729660545206
Generation 347 Best Solution: [183.67656649319966, 177.43088232060143,
175.11749813521945, 74.3365393903882] Fitness: 2.163426305671922
Generation 348 Best Solution: [183.17956621953968, 177.43088232060143,
175.11749813521945, 74.3365393903882] Fitness: 2.0151729660545206
Generation 349 Best Solution: [183.67656649319966, 175.59356941929042,
175.11749813521945, 75.77713093070344] Fitness: 2.0862389426563794
Generation 350 Best Solution: [183.1972116507827, 177.43088232060143,
174.46645171484482, 75.77713093070344] Fitness: 2.109530045882098
Generation 351 Best Solution: [183.67656649319966, 176.2607512342429,
175.11749813521945, 75.81529598636007] Fitness: 1.8423116167855293
Generation 352 Best Solution: [183.67656649319966, 176.9465624700098,
175.11749813521945, 74.59889150034074] Fitness: 1.3360587926422829
Generation 353 Best Solution: [183.1972116507827, 177.43088232060143,
175.11749813521945, 75.10887673508218] Fitness: 2.0116454830344215
Generation 354 Best Solution: [183.34191114976872, 177.43088232060143,
175.11749813521945, 74.6853603459956] Fitness: 1.906152690105757
Generation 355 Best Solution: [183.17956621953968, 177.43088232060143,
175.11749813521945, 74.6853603459956] Fitness: 1.899787423664293
Generation 356 Best Solution: [183.67656649319966, 176.49626296873348,
175.11749813521945, 75.81529598636007] Fitness: 1.8984419707738769
Generation 357 Best Solution: [183.67656649319966, 175.9455261680835,
175.11749813521945, 74.47437379961028] Fitness: 1.029033297080514
Generation 358 Best Solution: [183.17956621953968, 177.43088232060143,
175.11749813521945, 74.6853603459956] Fitness: 1.899787423664293
Generation 359 Best Solution: [183.17956621953968, 177.43088232060143,

175.11749813521945, 74.6853603459956] Fitness: 1.899787423664293
Generation 360 Best Solution: [183.67656649319966, 176.66206597080216,
175.11749813521945, 75.81529598636007] Fitness: 1.9894726033897676
Generation 361 Best Solution: [183.67656649319966, 177.43088232060143,
175.11749813521945, 74.6853603459956] Fitness: 2.048040763281694
Generation 362 Best Solution: [183.67656649319966, 177.43088232060143,
175.11749813521945, 74.6853603459956] Fitness: 2.048040763281694
Generation 363 Best Solution: [183.67656649319966, 177.43088232060143,
174.62971523531309, 75.81529598636007] Fitness: 2.4226077235978236
Generation 364 Best Solution: [183.77415528406954, 176.72971342866353,
175.11749813521945, 75.81529598636007] Fitness: 2.1128724616241
Generation 365 Best Solution: [183.67656649319966, 177.43088232060143,
175.11749813521945, 74.6804500992449] Fitness: 2.0483576404033017
Generation 366 Best Solution: [183.67656649319966, 176.03946848690646,
175.11749813521945, 75.81529598636007] Fitness: 1.867828563349228
Generation 367 Best Solution: [183.67656649319966, 176.03946848690646,
175.11749813521945, 74.6804500992449] Fitness: 0.9483139704809503
Generation 368 Best Solution: [183.67656649319966, 176.03946848690646,
175.11749813521945, 75.81529598636007] Fitness: 1.867828563349228
Generation 369 Best Solution: [183.67656649319966, 176.03946848690646,
175.11749813521945, 74.6804500992449] Fitness: 0.9483139704809503
Generation 370 Best Solution: [183.67656649319966, 176.03946848690646,
175.11749813521945, 74.6804500992449] Fitness: 0.9483139704809503
Generation 371 Best Solution: [183.41395005755177, 176.80557844317624,
175.11749813521945, 75.81529598636007] Fitness: 1.976640547997925
Generation 372 Best Solution: [183.1762307278853, 176.6230517208111,
175.11749813521945, 75.81529598636007] Fitness: 1.8162667739434135
Generation 373 Best Solution: [183.67656649319966, 177.43088232060143,
175.08451047486093, 74.86851909067337] Fitness: 2.014399115032597
Generation 374 Best Solution: [183.67656649319966, 175.83580535417002,
175.08451047486093, 74.86851909067337] Fitness: 1.0048438146957932
Generation 375 Best Solution: [183.67656649319966, 175.83580535417002,
175.08451047486093, 75.81529598636007] Fitness: 1.9098269805524988
Generation 376 Best Solution: [183.26551071248875, 177.43088232060143,
175.11749813521945, 74.86851909067337] Fitness: 1.9129219568165625
Generation 377 Best Solution: [183.26551071248875, 177.43088232060143,
175.08451047486093, 74.86851909067337] Fitness: 1.8644320044342857
Generation 378 Best Solution: [182.8462022140237, 177.43088232060143,
175.08451047486093, 74.34055921711148] Fitness: 2.0791462688093683
Generation 379 Best Solution: [183.77415528406954, 175.71663681082126,
175.08451047486093, 75.8603999556759] Fitness: 2.144340107673477
Generation 380 Best Solution: [183.77415528406954, 175.71663681082126,
175.08451047486093, 74.3135390049158] Fitness: 1.2763747195453725
Generation 381 Best Solution: [183.26551071248875, 175.71663681082126,
175.08451047486093, 74.3135390049158] Fitness: 1.0523744772584929
Generation 382 Best Solution: [183.77415528406954, 176.9862582097375,
174.1435646940528, 75.81529598636007] Fitness: 1.6423388516502477
Generation 383 Best Solution: [183.26551071248875, 177.43088232060143,

175.11749813521945, 74.70870669904971] Fitness: 1.897077749287626
Generation 384 Best Solution: [183.26551071248875, 177.43088232060143,
174.04994787199212, 75.24392603219451] Fitness: 1.3915146981037216
Generation 385 Best Solution: [183.25173069354426, 175.98596987234293,
175.08451047486093, 75.81529598636007] Fitness: 1.6864264990348539
Generation 386 Best Solution: [183.25173069354426, 175.98596987234293,
174.3996867809064, 75.81529598636007] Fitness: 1.0603488783234503
Generation 387 Best Solution: [183.25173069354426, 175.98596987234293,
174.92966355234893, 75.81529598636007] Fitness: 1.4813283702438182
Generation 388 Best Solution: [183.26551071248875, 175.98596987234293,
175.11749813521945, 75.54092057398098] Fitness: 1.3303166344751234
Generation 389 Best Solution: [183.26551071248875, 177.43088232060143,
175.11749813521945, 74.28261915124988] Fitness: 2.0481076363032553
Generation 390 Best Solution: [183.5752707429645, 177.43088232060143,
175.11749813521945, 74.30488237821952] Fitness: 2.121974755060212
Generation 391 Best Solution: [183.67656649319966, 177.43088232060143,
175.11749813521945, 74.28261915124988] Fitness: 2.1980747469015665
Generation 392 Best Solution: [183.26551071248875, 177.43088232060143,
175.11749813521945, 74.28261915124988] Fitness: 2.0481076363032553
Generation 393 Best Solution: [183.26551071248875, 177.43088232060143,
173.83477452889392, 75.81529598636007] Fitness: 2.173458826932902
Generation 394 Best Solution: [183.67656649319966, 177.43088232060143,
175.11749813521945, 74.99116127780725] Fitness: 2.101866846696594
Generation 395 Best Solution: [183.67656649319966, 177.43088232060143,
173.83477452889392, 75.81529598636007] Fitness: 2.323425937531213
Generation 396 Best Solution: [183.26551071248875, 175.80041038592807,
175.11749813521945, 75.81529598636007] Fitness: 1.8306267556743911
Generation 397 Best Solution: [183.5752707429645, 177.43088232060143,
175.11749813521945, 75.18279299714125] Fitness: 2.1481480745055177
Generation 398 Best Solution: [183.26551071248875, 177.43088232060143,
175.08451047486093, 75.81529598636007] Fitness: 2.7694151702909915
Generation 399 Best Solution: [183.73459757029082, 177.43088232060143,
175.08451047486093, 74.4390094946502] Fitness: 2.1037436209877747
Generation 400 Best Solution: [183.26551071248875, 177.43088232060143,
175.08451047486093, 74.4390094946502] Fitness: 1.91153000322074
Generation 401 Best Solution: [183.26365642863638, 177.43088232060143,
175.08451047486093, 74.71998449756117] Fitness: 1.8483252789722768
Generation 402 Best Solution: [183.67656649319966, 177.43088232060143,
173.83650784078174, 75.81529598636007] Fitness: 2.3225777764093953
Generation 403 Best Solution: [183.67656649319966, 177.43088232060143,
173.83650784078174, 75.81529598636007] Fitness: 2.3225777764093953
Generation 404 Best Solution: [183.26551071248875, 177.43088232060143,
175.08451047486093, 74.76621189236563] Fitness: 1.8497342764492952
Generation 405 Best Solution: [183.824268342319, 176.52361431875966,
175.11749813521945, 75.81529598636007] Fitness: 2.028307389423432
Generation 406 Best Solution: [183.26551071248875, 176.56559664510445,
175.11749813521945, 75.69405162784783] Fitness: 1.587980617938276
Generation 407 Best Solution: [183.26551071248875, 176.52361431875966,

175.11749813521945, 74.78797337364476] Fitness: 0.8426482111628723
Generation 408 Best Solution: [183.37185175527216, 177.43088232060143,
174.80657714402, 74.78797337364476] Fitness: 1.5235699448001678
Generation 409 Best Solution: [183.06124682656258, 176.56559664510445,
175.11749813521945, 75.81529598636007] Fitness: 1.8041469139473505
Generation 410 Best Solution: [183.06124682656258, 176.56559664510445,
175.11749813521945, 75.05383434236205] Fitness: 0.9670517930238419
Generation 411 Best Solution: [183.67656649319966, 175.98959785740388,
175.11749813521945, 75.73570667981471] Fitness: 1.7545384971741766
Generation 412 Best Solution: [183.06124682656258, 176.56559664510445,
175.11749813521945, 75.5739261222714] Fitness: 1.441614590062297
Generation 413 Best Solution: [183.25081391284996, 177.43088232060143,
174.1016092137578, 75.81529598636007] Fitness: 2.0971361522533876
Generation 414 Best Solution: [183.25081391284996, 176.42564366602957,
175.11749813521945, 75.81529598636007] Fitness: 1.722111136321777
Generation 415 Best Solution: [183.25081391284996, 177.43088232060143,
175.11749813521945, 75.05712660873309] Fitness: 1.9819792094890527
Generation 416 Best Solution: [183.37185175527216, 177.43088232060143,
173.84442034159994, 75.5739261222714] Fitness: 1.8199757100881109
Generation 417 Best Solution: [183.53201265973027, 175.87013474513867,
174.98600332267793, 75.81529598636007] Fitness: 1.6727391740904465
Generation 418 Best Solution: [183.67656649319966, 175.87013474513867,
174.98600332267793, 75.81529598636007] Fitness: 1.7553022602958954
Generation 419 Best Solution: [183.67656649319966, 177.43088232060143,
173.84442034159994, 75.5739261222714] Fitness: 1.956232718640862
Generation 420 Best Solution: [183.25081391284996, 177.43088232060143,
174.2642065336152, 75.5739261222714] Fitness: 1.7424692268672284
Generation 421 Best Solution: [183.37185175527216, 177.43088232060143,
173.84442034159994, 75.5739261222714] Fitness: 1.8199757100881109
Generation 422 Best Solution: [183.38042247613592, 176.43634795024212,
175.11749813521945, 75.5739261222714] Fitness: 1.3790951519312342
Generation 423 Best Solution: [183.41289831965847, 177.43088232060143,
174.19329643258288, 75.5739261222714] Fitness: 1.7582089515506745
Generation 424 Best Solution: [183.38042247613592, 175.66020218000887,
175.11749813521945, 75.5739261222714] Fitness: 1.5909778615467198
Generation 425 Best Solution: [183.73459757029082, 177.43088232060143,
173.8636113350893, 75.5739261222714] Fitness: 1.9896345056416922
Generation 426 Best Solution: [183.82961486742045, 177.43088232060143,
173.75193340934848, 74.56698848323937] Fitness: 1.5902696234588043
Generation 427 Best Solution: [183.41289831965847, 177.43088232060143,
174.5215115554894, 75.12128492630185] Fitness: 1.4261118045473695
Generation 428 Best Solution: [183.82961486742045, 176.94920237871065,
173.89373897427058, 75.81529598636007] Fitness: 1.6997974839242787
Generation 429 Best Solution: [183.82961486742045, 176.94920237871065,
174.5215115554894, 74.89482678602066] Fitness: 0.8556069979840009
Generation 430 Best Solution: [183.82961486742045, 175.66090102302888,
175.11749813521945, 75.5739261222714] Fitness: 1.8474344894601438
Generation 431 Best Solution: [183.38042247613592, 175.66090102302888,

175.11749813521945, 75.5739261222714] Fitness: 1.5903675343639883
Generation 432 Best Solution: [183.38042247613592, 176.94920237871065,
175.11749813521945, 75.81529598636007] Fitness: 2.1134172884219695
Generation 433 Best Solution: [183.82961486742045, 176.94920237871065,
175.11749813521945, 75.44958460000855] Fitness: 1.8563986269049964
Generation 434 Best Solution: [183.82961486742045, 176.94920237871065,
175.11749813521945, 75.5739261222714] Fitness: 2.0079519196330713
Generation 435 Best Solution: [183.38042247613592, 177.43088232060143,
175.11749813521945, 74.73112259182538] Fitness: 1.9124932803106014
Generation 436 Best Solution: [183.82961486742045, 177.43088232060143,
175.11749813521945, 74.73112259182538] Fitness: 2.169560235406757
Generation 437 Best Solution: [183.82961486742045, 177.43088232060143,
175.11749813521945, 74.73112259182538] Fitness: 2.169560235406757
Generation 438 Best Solution: [183.41289831965847, 175.6215766316077,
175.11749813521945, 75.81529598636007] Fitness: 1.9965275132772287
Generation 439 Best Solution: [183.38042247613592, 175.58218744446756,
175.11749813521945, 75.81529598636007] Fitness: 2.0263977145701886
Generation 440 Best Solution: [183.38042247613592, 175.58218744446756,
175.11749813521945, 75.81529598636007] Fitness: 2.0263977145701886
Generation 441 Best Solution: [183.38042247613592, 175.58218744446756,
175.11749813521945, 75.81529598636007] Fitness: 2.0263977145701886
Generation 442 Best Solution: [183.64070754712625, 176.9614434118652,
175.11749813521945, 75.81529598636007] Fitness: 2.238158470936278
Generation 443 Best Solution: [183.49637509730852, 176.9614434118652,
175.11749813521945, 75.81529598636007] Fitness: 2.1637109391215787
Generation 444 Best Solution: [183.49637509730852, 176.9614434118652,
175.11749813521945, 75.81529598636007] Fitness: 2.1637109391215787
Generation 445 Best Solution: [183.41289831965847, 175.8548702578588,
175.11749813521945, 75.81529598636007] Fitness: 1.8208402945810729
Generation 446 Best Solution: [183.41289831965847, 175.8548702578588,
175.11749813521945, 75.81529598636007] Fitness: 1.8208402945810729
Generation 447 Best Solution: [183.41289831965847, 177.43088232060143,
175.11749813521945, 74.55185668501232] Fitness: 1.9436660174776605
Generation 448 Best Solution: [183.82961486742045, 175.8548702578588,
174.92244602699054, 75.689218316799] Fitness: 1.6069373690104711
Generation 449 Best Solution: [183.82961486742045, 175.8548702578588,
174.92244602699054, 75.81529598636007] Fitness: 1.8075568065434906
Generation 450 Best Solution: [183.82961486742045, 175.8548702578588,
175.11749813521945, 75.689218316799] Fitness: 1.86917964256831
Generation 451 Best Solution: [183.41289831965847, 177.43088232060143,
173.81180324158308, 75.81529598636007] Fitness: 2.2088244576029354
Generation 452 Best Solution: [183.41289831965847, 177.43088232060143,
173.81180324158308, 75.81529598636007] Fitness: 2.2088244576029354
Generation 453 Best Solution: [183.82961486742045, 176.53164876237838,
175.11749813521945, 75.81529598636007] Fitness: 2.036974372646708
Generation 454 Best Solution: [183.82961486742045, 176.53164876237838,
175.11749813521945, 75.81529598636007] Fitness: 2.036974372646708
Generation 455 Best Solution: [183.82961486742045, 177.43088232060143,

175.11749813521945, 74.69508652029853] Fitness: 2.1702007257894023
Generation 456 Best Solution: [183.82961486742045, 176.67560783062012,
175.11749813521945, 75.87039695958111] Fitness: 2.2168727164355975
Generation 457 Best Solution: [183.82961486742045, 177.43088232060143,
175.11749813521945, 74.92070314591646] Fitness: 2.1993049210733506
Generation 458 Best Solution: [183.82961486742045, 176.02658832711043,
175.11749813521945, 75.87039695958111] Fitness: 2.089733819921851
Generation 459 Best Solution: [183.82961486742045, 176.07699065519986,
175.11749813521945, 75.87039695958111] Fitness: 2.0762485885536464
Generation 460 Best Solution: [183.82961486742045, 176.25945251197422,
175.11749813521945, 75.87039695958111] Fitness: 2.0603251776998235
Generation 461 Best Solution: [183.82961486742045, 176.07699065519986,
174.73729815039536, 75.87039695958111] Fitness: 1.6195779891374769
Generation 462 Best Solution: [183.27606889787688, 176.25945251197422,
175.11749813521945, 74.60409040518319] Fitness: 0.7830690087417057
Generation 463 Best Solution: [182.99054349264628, 176.25945251197422,
175.11749813521945, 75.87039695958111] Fitness: 1.833402367048814
Generation 464 Best Solution: [183.82961486742045, 176.6762344677761,
174.63940897332523, 75.87039695958111] Fitness: 1.6792942251956806
Generation 465 Best Solution: [183.69496320629514, 177.43088232060143,
175.11749813521945, 74.79924056634441] Fitness: 2.0639931083187983
Generation 466 Best Solution: [182.99054349264628, 177.43088232060143,
175.11749813521945, 74.79924056634441] Fitness: 1.9469194432377432
Generation 467 Best Solution: [183.82961486742045, 175.90177591700288,
175.11749813521945, 75.87039695958111] Fitness: 2.1400584177660344
Generation 468 Best Solution: [183.82961486742045, 175.90177591700288,
174.41637484789558, 75.87039695958111] Fitness: 1.4721151885242512
Generation 469 Best Solution: [183.45965305171953, 177.43088232060143,
175.11749813521945, 74.79924056634441] Fitness: 1.9394245413229787
Generation 470 Best Solution: [183.82961486742045, 176.78827731573222,
175.11749813521945, 74.79924056634441] Fitness: 1.2932696028761252
Generation 471 Best Solution: [183.4518303298083, 177.43088232060143,
175.11749813521945, 74.79924056634441] Fitness: 1.9367558320683773
Generation 472 Best Solution: [183.82961486742045, 176.78827731573222,
175.11749813521945, 74.79924056634441] Fitness: 1.2932696028761252
Generation 473 Best Solution: [183.82961486742045, 177.43088232060143,
175.11749813521945, 74.79924056634441] Fitness: 2.1738422538887527
Generation 474 Best Solution: [183.82961486742045, 176.78827731573222,
175.11749813521945, 74.72113270443283] Fitness: 1.2889636966043356
Generation 475 Best Solution: [183.82961486742045, 176.6240063370959,
175.11749813521945, 75.87039695958111] Fitness: 2.1828977271054586
Generation 476 Best Solution: [183.3190915627597, 176.6240063370959,
175.11749813521945, 75.87039695958111] Fitness: 1.9149725553303447
Generation 477 Best Solution: [182.8718825020696, 176.78827731573222,
175.11749813521945, 75.80267788004136] Fitness: 2.0177597212639644
Generation 478 Best Solution: [182.8718825020696, 176.2525723773287,
174.28288321957595, 75.87039695958111] Fitness: 1.1800984464421431
Generation 479 Best Solution: [183.82961486742045, 177.43088232060143,

175.11749813521945, 74.49210454543615] Fitness: 2.2113694934097374
Generation 480 Best Solution: [183.330206654004, 176.78827731573222,
175.11749813521945, 75.87039695958111] Fitness: 2.038995128224207
Generation 481 Best Solution: [183.6498765612611, 177.43088232060143,
175.11749813521945, 74.49210454543615] Fitness: 2.071012463047049
Generation 482 Best Solution: [183.82961486742045, 177.36090161807581,
175.11749813521945, 75.21897750790436] Fitness: 2.2318401080551755
Generation 483 Best Solution: [183.6498765612611, 177.36090161807581,
175.11749813521945, 75.21897750790436] Fitness: 2.0914830776924873
Generation 484 Best Solution: [183.82961486742045, 175.6972524649694,
175.11749813521945, 75.87039695958111] Fitness: 2.2746699403659125
Generation 485 Best Solution: [183.82961486742045, 177.43088232060143,
175.11749813521945, 75.39955904251352] Fitness: 2.522243677534501
Generation 486 Best Solution: [183.82961486742045, 177.43088232060143,
175.11749813521945, 75.32596828371805] Fitness: 2.449524955392005
Generation 487 Best Solution: [183.82961486742045, 177.43088232060143,
174.4717849728028, 75.87039695958111] Fitness: 2.5431019709965206
Generation 488 Best Solution: [183.82961486742045, 176.57109534849263,
175.11749813521945, 75.87039695958111] Fitness: 2.1523417328203394
Generation 489 Best Solution: [183.82961486742045, 175.88253083492765,
175.11749813521945, 75.87039695958111] Fitness: 2.149964443800327
Generation 490 Best Solution: [183.69496320629514, 175.88253083492765,
175.11749813521945, 75.34366511645423] Fitness: 1.3204043728861596
Generation 491 Best Solution: [183.82961486742045, 175.88253083492765,
175.11749813521945, 75.87039695958111] Fitness: 2.149964443800327
Generation 492 Best Solution: [183.70739966611836, 177.43088232060143,
175.11749813521945, 75.66207809444118] Fitness: 2.749082905235483
Generation 493 Best Solution: [183.82961486742045, 176.3568657270655,
175.11749813521945, 75.87039695958111] Fitness: 2.072931194788149
Generation 494 Best Solution: [183.82961486742045, 176.3568657270655,
175.11749813521945, 75.87039695958111] Fitness: 2.072931194788149
Generation 495 Best Solution: [183.82961486742045, 176.0092737139025,
175.11749813521945, 75.87039695958111] Fitness: 2.09527411132591
Generation 496 Best Solution: [183.82961486742045, 176.0092737139025,
175.11749813521945, 75.87039695958111] Fitness: 2.09527411132591
Generation 497 Best Solution: [183.82961486742045, 176.13491148507953,
175.11749813521945, 75.87039695958111] Fitness: 2.065609188603177
Generation 498 Best Solution: [183.82961486742045, 177.4493319195135,
174.30349750510945, 74.5703556505083] Fitness: 1.5183221296800324
Generation 499 Best Solution: [183.82961486742045, 177.4493319195135,
174.30349750510945, 74.5703556505083] Fitness: 1.5183221296800324
Generation 500 Best Solution: [182.80360094293374, 177.4493319195135,
174.30349750510945, 74.5703556505083] Fitness: 1.389336310957929
Optimal Solution for Simulation 10: [182.80360094293374, 177.4493319195135,
174.30349750510945, 74.5703556505083]

Simulation 11/32 with conditions: {'drought': True, 'flood': False,
'policy_change': True, 'social_event': False, 'economic_crisis': True}

```
Generation 1 Best Solution: [183.6300421506088, 176.57546640544854,
174.16533648857228, 75.06971587205523] Fitness: 0.34467794967742194
Generation 2 Best Solution: [183.24555690411714, 176.49653413446498,
174.16533648857228, 75.06971587205523] Fitness: 0.19068882829879402
Generation 3 Best Solution: [183.79566667874312, 176.49653413446498,
173.93061134164572, 75.44666094012233] Fitness: 0.7781450186679575
Generation 4 Best Solution: [183.0857935834714, 176.49653413446498,
174.54047690604673, 75.44666094012233] Fitness: 0.6313115394697708
Generation 5 Best Solution: [183.67607732297367, 177.07702110242533,
173.83835379055313, 75.19422542398692] Fitness: 0.9961466590893825
Generation 6 Best Solution: [183.79566667874312, 177.07702110242533,
173.83835379055313, 75.19422542398692] Fitness: 1.0878611712609283
Generation 7 Best Solution: [183.0857935834714, 175.76108849469048,
173.83835379055313, 75.19422542398692] Fitness: 0.485288445283434
Generation 8 Best Solution: [183.79566667874312, 177.07702110242533,
173.95975270348566, 74.59576127691305] Fitness: 0.8863135808621516
Generation 9 Best Solution: [183.69787999501796, 175.93185584972048,
173.93061134164572, 75.44666094012233] Fitness: 0.7166307834930574
Generation 10 Best Solution: [182.8973754495032, 175.93185584972048,
173.93061134164572, 75.44666094012233] Fitness: 0.64842885362406
Generation 11 Best Solution: [183.0857935834714, 175.66728518112984,
174.6939591386026, 75.56518500163831] Fitness: 1.0761373775044678
Generation 12 Best Solution: [183.2776733903726, 177.2349619698379,
174.6939591386026, 74.55612112428545] Fitness: 1.0768744327177835
Generation 13 Best Solution: [182.8973754495032, 175.93185584972048,
174.96205632499382, 75.44666094012233] Fitness: 1.1078428917537415
Generation 14 Best Solution: [183.67607732297367, 177.2349619698379,
174.6939591386026, 75.14948122881978] Fitness: 1.3339322257044643
Generation 15 Best Solution: [183.79566667874312, 175.91542740804744,
174.96205632499382, 74.82317849320859] Fitness: 0.8692318766168802
Generation 16 Best Solution: [183.79566667874312, 177.2349619698379,
173.95671536074346, 74.64832140861752] Fitness: 1.1054753393352996
Generation 17 Best Solution: [183.79566667874312, 177.2349619698379,
173.95671536074346, 74.64832140861752] Fitness: 1.1054753393352996
Generation 18 Best Solution: [183.2776733903726, 177.2349619698379,
174.6939591386026, 75.06155725689781] Fitness: 1.1377372922311964
Generation 19 Best Solution: [183.34431624121194, 176.336418219307,
174.02379339799128, 75.81846565573011] Fitness: 0.9970605263179373
Generation 20 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 75.02545899424885] Fitness: 1.6355439560650946
Generation 21 Best Solution: [183.2776733903726, 177.2349619698379,
174.08293918702435, 75.45562177084385] Fitness: 1.2437459520270324
Generation 22 Best Solution: [183.14876543194484, 176.69213453390108,
174.96205632499382, 75.81846565573011] Fitness: 1.6383605245467672
Generation 23 Best Solution: [183.2776733903726, 177.2349619698379,
174.08293918702435, 75.90142884275036] Fitness: 1.900536386882159
Generation 24 Best Solution: [183.2776733903726, 177.2349619698379,
174.02379339799128, 75.8016485250351] Fitness: 1.7364905964567796
```

Generation 25 Best Solution: [183.2776733903726, 177.2349619698379,
174.02379339799128, 75.8016485250351] Fitness: 1.7364905964567796
Generation 26 Best Solution: [183.79566667874312, 176.44083862527538,
174.96205632499382, 75.8016485250351] Fitness: 1.709969627107299
Generation 27 Best Solution: [183.79566667874312, 176.3848997983468,
174.02379339799128, 75.8016485250351] Fitness: 1.2082708668467195
Generation 28 Best Solution: [183.79566667874312, 176.3848997983468,
174.02379339799128, 75.8016485250351] Fitness: 1.2082708668467195
Generation 29 Best Solution: [183.79566667874312, 175.6907102223439,
174.08293918702435, 75.8016485250351] Fitness: 1.4075445692995414
Generation 30 Best Solution: [183.79566667874312, 177.2349619698379,
174.08293918702435, 75.8016485250351] Fitness: 1.9634246404998748
Generation 31 Best Solution: [183.02217489489033, 177.2349619698379,
173.8441032760533, 75.8016485250351] Fitness: 1.836611584981548
Generation 32 Best Solution: [183.02217489489033, 176.59352948520774,
173.8441032760533, 75.8016485250351] Fitness: 1.1553076438989929
Generation 33 Best Solution: [183.02217489489033, 177.2349619698379,
173.8441032760533, 75.8016485250351] Fitness: 1.836611584981548
Generation 34 Best Solution: [183.02217489489033, 177.2349619698379,
173.8441032760533, 75.90142884275036] Fitness: 2.0104681628224883
Generation 35 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 74.89401506738697] Fitness: 1.5888621064949955
Generation 36 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 75.08749294735445] Fitness: 1.6669101396003487
Generation 37 Best Solution: [183.79566667874312, 176.44485444816775,
174.96205632499382, 75.08749294735445] Fitness: 0.9190545020417101
Generation 38 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 74.94937695304323] Fitness: 1.605247650790875
Generation 39 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 74.3844078525341] Fitness: 1.6619646703525537
Generation 40 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 75.08163347030943] Fitness: 1.6636914084644223
Generation 41 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 75.08163347030943] Fitness: 1.6636914084644223
Generation 42 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 75.08163347030943] Fitness: 1.6636914084644223
Generation 43 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 75.08163347030943] Fitness: 1.6636914084644223
Generation 44 Best Solution: [183.79566667874312, 177.2349619698379,
174.7562433825266, 74.64192804101707] Fitness: 1.3506970612883884
Generation 45 Best Solution: [183.79566667874312, 177.2349619698379,
174.7562433825266, 74.64192804101707] Fitness: 1.3506970612883884
Generation 46 Best Solution: [183.79566667874312, 177.2349619698379,
174.7562433825266, 75.08163347030943] Fitness: 1.439856159487665
Generation 47 Best Solution: [183.79566667874312, 176.60254808635239,
174.96205632499382, 75.8016485250351] Fitness: 1.7829851677668707
Generation 48 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 74.68175747958288] Fitness: 1.5702209888149696

Generation 49 Best Solution: [183.656950541058, 177.2349619698379,
174.1092994641223, 75.8016485250351] Fitness: 1.8564857454290027
Generation 50 Best Solution: [183.79566667874312, 176.35691304025804,
174.96205632499382, 75.8016485250351] Fitness: 1.6881092347580817
Generation 51 Best Solution: [183.6570167330877, 177.2349619698379,
174.96205632499382, 75.20935744236886] Fitness: 1.6416821000023094
Generation 52 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 74.46148772056233] Fitness: 1.6249772020603828
Generation 53 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 74.48277081845853] Fitness: 1.6163925693959422
Generation 54 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 74.41634032945902] Fitness: 1.6455202865967735
Generation 55 Best Solution: [183.79566667874312, 176.35691304025804,
173.9276214875266, 75.8016485250351] Fitness: 1.2297690094444418
Generation 56 Best Solution: [183.79566667874312, 175.70037105289265,
173.98804018935917, 75.8016485250351] Fitness: 1.4178715623829823
Generation 57 Best Solution: [183.79566667874312, 176.35691304025804,
174.96205632499382, 75.8016485250351] Fitness: 1.6881092347580817
Generation 58 Best Solution: [183.79566667874312, 176.35691304025804,
173.9769116202608, 75.8016485250351] Fitness: 1.213841350794801
Generation 59 Best Solution: [183.79566667874312, 176.35691304025804,
174.96205632499382, 75.8016485250351] Fitness: 1.6881092347580817
Generation 60 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 74.54240201013994] Fitness: 1.5960928515429957
Generation 61 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 74.8655488981552] Fitness: 1.5822929124519483
Generation 62 Best Solution: [183.79566667874312, 177.2349619698379,
174.6832076807795, 75.32289039936302] Fitness: 1.5529635040502372
Generation 63 Best Solution: [183.79566667874312, 177.2349619698379,
174.6832076807795, 75.32289039936302] Fitness: 1.5529635040502372
Generation 64 Best Solution: [183.83895113652056, 176.73049148590218,
174.96205632499382, 75.32289039936302] Fitness: 1.2895008083854345
Generation 65 Best Solution: [183.79566667874312, 177.2349619698379,
173.7925024305423, 75.16619853280602] Fitness: 1.3200237354060929
Generation 66 Best Solution: [183.79566667874312, 177.2349619698379,
173.7925024305423, 75.32289039936302] Fitness: 1.4450920934185223
Generation 67 Best Solution: [183.0160569792269, 177.2349619698379,
173.7925024305423, 75.8016485250351] Fitness: 1.8661682162131576
Generation 68 Best Solution: [183.0160569792269, 177.2349619698379,
173.79061723249703, 75.8016485250351] Fitness: 1.8672486781682536
Generation 69 Best Solution: [183.0160569792269, 177.2349619698379,
173.7925024305423, 75.90142884275036] Fitness: 2.040024794054098
Generation 70 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 74.73648887880768] Fitness: 1.5683219583060144
Generation 71 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 75.33367803585867] Fitness: 1.8504054777359031
Generation 72 Best Solution: [183.79566667874312, 177.2349619698379,
173.7925024305423, 74.9602812610198] Fitness: 1.2137393397970846

Generation 73 Best Solution: [183.0160569792269, 177.2349619698379,
174.96205632499382, 74.47809407281788] Fitness: 1.4205384970974024
Generation 74 Best Solution: [183.0160569792269, 177.2349619698379,
174.96205632499382, 74.47809407281788] Fitness: 1.4205384970974024
Generation 75 Best Solution: [183.0160569792269, 177.2349619698379,
173.7925024305423, 75.60507472875878] Fitness: 1.568969106640031
Generation 76 Best Solution: [183.79566667874312, 175.9999615339533,
173.84750884684678, 75.90142884275036] Fitness: 1.4670001535986024
Generation 77 Best Solution: [183.79566667874312, 175.9999615339533,
173.84750884684678, 75.90142884275036] Fitness: 1.4670001535986024
Generation 78 Best Solution: [183.0160569792269, 177.2349619698379,
174.96205632499382, 74.39477166725806] Fitness: 1.458773666050227
Generation 79 Best Solution: [183.13566381663298, 177.2349619698379,
174.96205632499382, 74.43403461330597] Fitness: 1.4083405250685768
Generation 80 Best Solution: [183.0160569792269, 177.2349619698379,
174.96205632499382, 74.54997911571962] Fitness: 1.3962294701568705
Generation 81 Best Solution: [183.0160569792269, 176.1048623662078,
174.5884518422333, 75.90142884275036] Fitness: 1.3069352030493735
Generation 82 Best Solution: [183.7608579005968, 177.2349619698379,
174.96205632499382, 75.10016230963976] Fitness: 1.6450616866679755
Generation 83 Best Solution: [183.79566667874312, 177.2349619698379,
174.96205632499382, 75.12701888960055] Fitness: 1.6900178085109552
Generation 84 Best Solution: [183.0854338041907, 177.2349619698379,
174.96205632499382, 74.5170267198583] Fitness: 1.3856423919565521
Generation 85 Best Solution: [183.13566381663298, 177.2349619698379,
174.96205632499382, 74.4313169271772] Fitness: 1.4096035060476881
Generation 86 Best Solution: [182.88917370384212, 177.2349619698379,
174.96205632499382, 74.71372676240507] Fitness: 1.4281518889496425
Generation 87 Best Solution: [183.7608579005968, 177.2349619698379,
174.96205632499382, 74.71372676240507] Fitness: 1.5395551856049252
Generation 88 Best Solution: [183.13566381663298, 176.2917840335511,
174.96205632499382, 75.90142884275036] Fitness: 1.623801278463239
Generation 89 Best Solution: [183.13566381663298, 176.2917840335511,
174.96205632499382, 75.90142884275036] Fitness: 1.623801278463239
Generation 90 Best Solution: [183.7278570648207, 177.2349619698379,
174.96205632499382, 74.5085212221875] Fitness: 1.5522131079517658
Generation 91 Best Solution: [183.7278570648207, 177.2349619698379,
174.96205632499382, 74.5085212221875] Fitness: 1.5522131079517658
Generation 92 Best Solution: [182.88917370384212, 176.82536737230873,
174.96205632499382, 74.5085212221875] Fitness: 0.957641716549267
Generation 93 Best Solution: [183.7278570648207, 176.13998596757773,
174.96205632499382, 75.90142884275036] Fitness: 1.8014873208810973
Generation 94 Best Solution: [183.58115926084497, 176.37651457403288,
174.96205632499382, 75.90142884275036] Fitness: 1.7174185630568066
Generation 95 Best Solution: [182.88917370384212, 177.31273747933807,
174.96205632499382, 74.75378186863348] Fitness: 1.5544132133032635
Generation 96 Best Solution: [183.7278570648207, 176.13998596757773,
174.78663932076014, 75.90142884275036] Fitness: 1.6065625907723147

Generation 97 Best Solution: [183.277689009687, 176.37651457403288,
174.1651776616377, 75.90142884275036] Fitness: 1.129027648546213
Generation 98 Best Solution: [183.7278570648207, 176.37651457403288,
174.1651776616377, 75.90142884275036] Fitness: 1.311036721602294
Generation 99 Best Solution: [183.58115926084497, 175.92792474069122,
174.96205632499382, 75.90142884275036] Fitness: 1.7726910508172344
Generation 100 Best Solution: [183.7278570648207, 176.78017122073913,
174.1651776616377, 75.90142884275036] Fitness: 1.5288678137294849
Generation 101 Best Solution: [183.58115926084497, 175.92792474069122,
174.96205632499382, 75.67288991600914] Fitness: 1.3973734729422005
Generation 102 Best Solution: [183.7278570648207, 177.2349619698379,
174.78663932076014, 75.90142884275036] Fitness: 2.3833428933798446
Generation 103 Best Solution: [183.7278570648207, 177.2349619698379,
174.96205632499382, 74.59439735096717] Fitness: 1.5281718949793452
Generation 104 Best Solution: [183.7278570648207, 177.2349619698379,
174.3247030844391, 75.90142884275036] Fitness: 2.099026434925083
Generation 105 Best Solution: [183.7278570648207, 177.2349619698379,
174.78663932076014, 75.90142884275036] Fitness: 2.3833428933798446
Generation 106 Best Solution: [182.88267397799802, 177.39447422076893,
174.96205632499382, 74.71286557486265] Fitness: 1.7005818313027954
Generation 107 Best Solution: [183.50448338773907, 177.20545907706833,
174.96205632499382, 74.31786886075179] Fitness: 1.4714980211680972
Generation 108 Best Solution: [182.88267397799802, 177.2349619698379,
174.78663932076014, 75.90142884275036] Fitness: 2.301292772891265
Generation 109 Best Solution: [183.0756713997427, 175.6953315061992,
174.96205632499382, 75.90142884275036] Fitness: 1.8573041246157624
Generation 110 Best Solution: [183.38708842190732, 177.2349619698379,
174.96205632499382, 74.31965486423819] Fitness: 1.4786723812824163
Generation 111 Best Solution: [183.33742976430213, 176.12385314145013,
174.96205632499382, 75.90142884275036] Fitness: 1.6279882606931322
Generation 112 Best Solution: [183.73177793721442, 177.31273747933807,
174.96205632499382, 74.8293586198496] Fitness: 1.6501126964438129
Generation 113 Best Solution: [183.33742976430213, 177.31273747933807,
174.96205632499382, 74.8293586198496] Fitness: 1.4711463701109424
Generation 114 Best Solution: [182.88267397799802, 177.31273747933807,
174.78663932076014, 75.30399985273627] Fitness: 1.6176937028031393
Generation 115 Best Solution: [182.88267397799802, 176.85661857678542,
174.78663932076014, 75.47612110471682] Fitness: 1.1886424581380617
Generation 116 Best Solution: [182.88267397799802, 177.31273747933807,
174.78663932076014, 75.30399985273627] Fitness: 1.6176937028031393
Generation 117 Best Solution: [183.73177793721442, 177.2349619698379,
174.78663932076014, 75.30399985273627] Fitness: 1.5765974481519383
Generation 118 Best Solution: [183.7278570648207, 177.2349619698379,
175.09641049833547, 74.28532894731313] Fitness: 1.8500144666788538
Generation 119 Best Solution: [183.77771339554664, 176.7233593810023,
174.96205632499382, 75.90142884275036] Fitness: 2.0227498995940456
Generation 120 Best Solution: [183.77771339554664, 176.7233593810023,
174.96205632499382, 75.90142884275036] Fitness: 2.0227498995940456

Generation 121 Best Solution: [183.77771339554664, 176.3344611971221,
174.96205632499382, 75.90142884275036] Fitness: 1.8427875222436136
Generation 122 Best Solution: [183.77771339554664, 176.7233593810023,
174.78663932076014, 75.90142884275036] Fitness: 1.827825169485263
Generation 123 Best Solution: [183.7278570648207, 176.7233593810023,
174.96205632499382, 75.69489812790714] Fitness: 1.6404928609741414
Generation 124 Best Solution: [183.77771339554664, 176.59263296082273,
174.30853270101204, 75.90142884275036] Fitness: 1.4528256918406623
Generation 125 Best Solution: [183.82359373347006, 177.31273747933807,
174.78663932076014, 74.5629336225065] Fitness: 1.5461970647407979
Generation 126 Best Solution: [183.35524161017963, 176.83935356895142,
174.78663932076014, 75.90142884275036] Fitness: 1.7143753439982858
Generation 127 Best Solution: [183.82359373347006, 176.59263296082273,
174.96205632499382, 75.90142884275036] Fitness: 1.9758160399293647
Generation 128 Best Solution: [182.99122009106622, 177.3005718084477,
173.9312284045347, 75.90142884275036] Fitness: 2.0906992333213648
Generation 129 Best Solution: [182.99122009106622, 177.3005718084477,
173.9312284045347, 75.90142884275036] Fitness: 2.0906992333213648
Generation 130 Best Solution: [183.11661546181543, 177.3005718084477,
173.9312284045347, 75.90142884275036] Fitness: 2.0538448214755314
Generation 131 Best Solution: [183.11661546181543, 177.3005718084477,
173.9312284045347, 75.90142884275036] Fitness: 2.0538448214755314
Generation 132 Best Solution: [183.7278570648207, 177.3005718084477,
173.9312284045347, 75.57774259163433] Fitness: 1.716782958945914
Generation 133 Best Solution: [183.82359373347006, 177.3005718084477,
173.9312284045347, 75.57774259163433] Fitness: 1.7961396749019543
Generation 134 Best Solution: [182.99122009106622, 177.3005718084477,
173.9312284045347, 75.90142884275036] Fitness: 2.0906992333213648
Generation 135 Best Solution: [182.99122009106622, 177.31273747933807,
174.96205632499382, 74.71303700954718] Fitness: 1.5062437325445148
Generation 136 Best Solution: [182.99122009106622, 177.3005718084477,
174.96205632499382, 74.71303700954718] Fitness: 1.4858957490528681
Generation 137 Best Solution: [182.99122009106622, 177.31273747933807,
174.74232247703844, 74.71303700954718] Fitness: 1.269647713834639
Generation 138 Best Solution: [182.99122009106622, 177.31273747933807,
174.74232247703844, 74.71303700954718] Fitness: 1.269647713834639
Generation 139 Best Solution: [182.99122009106622, 177.31273747933807,
174.74232247703844, 74.65285119855912] Fitness: 1.2741869447069045
Generation 140 Best Solution: [183.82359373347006, 175.84416768499196,
174.70938230038738, 75.90142884275036] Fitness: 1.7253941096334529
Generation 141 Best Solution: [182.99122009106622, 176.53813980360846,
174.74232247703844, 75.90142884275036] Fitness: 1.4978517668078215
Generation 142 Best Solution: [182.99122009106622, 176.93906433181579,
174.96205632499382, 74.99302124117752] Fitness: 1.0392752028340901
Generation 143 Best Solution: [183.82359373347006, 176.93906433181579,
174.96205632499382, 74.99302124117752] Fitness: 1.2523348109339694
Generation 144 Best Solution: [182.99122009106622, 176.53813980360846,
174.74232247703844, 75.90142884275036] Fitness: 1.4978517668078215

Generation 145 Best Solution: [182.99122009106622, 177.31273747933807,
174.70938230038738, 74.42623860691012] Fitness: 1.3128331324194733
Generation 146 Best Solution: [182.99122009106622, 177.31273747933807,
174.70938230038738, 74.42623860691012] Fitness: 1.3128331324194733
Generation 147 Best Solution: [182.99122009106622, 176.93906433181579,
174.96205632499382, 74.42623860691012] Fitness: 1.0584942391712828
Generation 148 Best Solution: [183.46619349573731, 176.93906433181579,
174.71189018682105, 75.90142884275036] Fitness: 1.7768545166008516
Generation 149 Best Solution: [183.82359373347006, 175.91815202924096,
174.96205632499382, 75.90142884275036] Fitness: 1.9506732695706992
Generation 150 Best Solution: [183.82359373347006, 177.31273747933807,
174.96205632499382, 74.30055627409747] Fitness: 1.8634354305791234
Generation 151 Best Solution: [183.82359373347006, 177.31273747933807,
174.71189018682105, 75.8029675702101] Fitness: 2.348639123250669
Generation 152 Best Solution: [183.82359373347006, 175.6986081507522,
174.96205632499382, 75.90142884275036] Fitness: 2.095149698641042
Generation 153 Best Solution: [183.82359373347006, 177.31273747933807,
174.71189018682105, 75.90142884275036] Fitness: 2.520298419126604
Generation 154 Best Solution: [182.99122009106622, 176.15368047077334,
174.96205632499382, 75.90142884275036] Fitness: 1.665982331125627
Generation 155 Best Solution: [183.82359373347006, 176.52944101517082,
174.71189018682105, 75.90142884275036] Fitness: 1.6799736675827854
Generation 156 Best Solution: [182.99122009106622, 176.39405214080105,
174.96205632499382, 75.90142884275036] Fitness: 1.6818502874722525
Generation 157 Best Solution: [182.99122009106622, 177.3005718084477,
174.96205632499382, 74.6686641797801] Fitness: 1.4886966196277527
Generation 158 Best Solution: [182.99122009106622, 177.3005718084477,
174.96205632499382, 74.6948226528245] Fitness: 1.4866748830638543
Generation 159 Best Solution: [182.99122009106622, 177.31273747933807,
174.96205632499382, 75.13993134196137] Fitness: 1.6357719378658233
Generation 160 Best Solution: [182.99122009106622, 177.31273747933807,
174.96205632499382, 74.4882565667311] Fitness: 1.5519726903490763
Generation 161 Best Solution: [182.99122009106622, 177.31273747933807,
174.96205632499382, 74.4882565667311] Fitness: 1.5519726903490763
Generation 162 Best Solution: [182.99122009106622, 177.31273747933807,
174.96205632499382, 74.29776621759753] Fitness: 1.6522519006637253
Generation 163 Best Solution: [182.99122009106622, 177.31273747933807,
175.1240520742161, 74.82206705062364] Fitness: 1.7348941729779592
Generation 164 Best Solution: [182.99122009106622, 177.2383761572623,
173.90346493130485, 75.90142884275036] Fitness: 2.0007492161568465
Generation 165 Best Solution: [182.99122009106622, 176.2056550626178,
174.96205632499382, 75.90142884275036] Fitness: 1.6617975029263439
Generation 166 Best Solution: [182.99122009106622, 175.94828598138946,
174.96205632499382, 75.90142884275036] Fitness: 1.723635077122286
Generation 167 Best Solution: [183.82359373347006, 176.88067493862113,
173.90346493130485, 75.90142884275036] Fitness: 1.753035694576103
Generation 168 Best Solution: [182.99122009106622, 176.49023483089493,
174.96205632499382, 75.77661346241432] Fitness: 1.4983297910864084

Generation 169 Best Solution: [182.99122009106622, 176.49023483089493,
174.96205632499382, 75.90142884275036] Fitness: 1.7133769299572155
Generation 170 Best Solution: [183.82359373347006, 176.36398216378686,
174.96205632499382, 75.77661346241432] Fitness: 1.6729592241148237
Generation 171 Best Solution: [182.99122009106622, 176.88067493862113,
174.96205632499382, 75.77661346241432] Fitness: 1.7740832530437745
Generation 172 Best Solution: [183.82359373347006, 176.88067493862113,
174.96205632499382, 74.484328969303] Fitness: 1.1849792277192168
Generation 173 Best Solution: [182.99122009106622, 177.31273747933807,
173.79801353033045, 74.96746501150687] Fitness: 1.1508885826337816
Generation 174 Best Solution: [183.82359373347006, 177.1542704923598,
174.1230343635881, 74.74399858787967] Fitness: 0.972870102418911
Generation 175 Best Solution: [183.82359373347006, 176.31788534933025,
174.1230343635881, 75.90142884275036] Fitness: 1.380876919194026
Generation 176 Best Solution: [183.70193257327267, 177.31273747933807,
174.1230343635881, 75.47245543077274] Fitness: 1.5484643970530514
Generation 177 Best Solution: [183.14520329762178, 176.28951594327262,
174.96205632499382, 75.90142884275036] Fitness: 1.6220724536670006
Generation 178 Best Solution: [182.99122009106622, 177.31273747933807,
174.09533770854878, 75.90142884275036] Fitness: 2.0735931482916623
Generation 179 Best Solution: [182.99122009106622, 176.15323849134722,
175.06126116998882, 75.90142884275036] Fitness: 1.797462762985916
Generation 180 Best Solution: [182.79035892401888, 177.31273747933807,
174.96205632499382, 74.53828894359465] Fitness: 1.6449978091627542
Generation 181 Best Solution: [182.99122009106622, 177.31273747933807,
174.96205632499382, 74.53828894359465] Fitness: 1.534994028548927
Generation 182 Best Solution: [183.14520329762178, 177.31273747933807,
174.55784999108363, 74.53828894359465] Fitness: 1.1159315110139403
Generation 183 Best Solution: [183.14520329762178, 176.10397723125865,
174.1890049909814, 75.90142884275036] Fitness: 1.1323965088534629
Generation 184 Best Solution: [182.99122009106622, 176.10397723125865,
174.96205632499382, 75.90142884275036] Fitness: 1.6739149465355476
Generation 185 Best Solution: [182.99122009106622, 176.10397723125865,
173.77191213714354, 75.90142884275036] Fitness: 1.2907177116285755
Generation 186 Best Solution: [182.99122009106622, 176.5031218032903,
174.96205632499382, 75.90142884275036] Fitness: 1.718694232857604
Generation 187 Best Solution: [182.99122009106622, 176.10397723125865,
174.96205632499382, 75.90142884275036] Fitness: 1.6739149465355476
Generation 188 Best Solution: [182.99122009106622, 177.31273747933807,
175.06126116998882, 74.49546766266347] Fitness: 1.680712237309723
Generation 189 Best Solution: [183.6081737130848, 177.31273747933807,
175.06126116998882, 74.49546766266347] Fitness: 1.735261674481072
Generation 190 Best Solution: [182.99122009106622, 177.31273747933807,
175.06126116998882, 74.36047578119761] Fitness: 1.744434343399078
Generation 191 Best Solution: [183.53739120558993, 177.31273747933807,
175.06126116998882, 75.4098908945923] Fitness: 2.013649878681203
Generation 192 Best Solution: [183.53739120558993, 176.51574836809942,
175.06126116998882, 75.90142884275036] Fitness: 1.8738041146550035

Generation 193 Best Solution: [182.99122009106622, 177.03897708829376,
175.11692615914058, 75.4098908945923] Fitness: 1.6736929513547079
Generation 194 Best Solution: [183.82359373347006, 175.97752349865573,
175.06126116998882, 75.90142884275036] Fitness: 2.055908891252084
Generation 195 Best Solution: [182.99122009106622, 177.31273747933807,
175.06126116998882, 75.48241869794802] Fitness: 2.0760633859376307
Generation 196 Best Solution: [183.53739120558993, 177.31273747933807,
175.06126116998882, 75.48241869794802] Fitness: 2.0942860257077447
Generation 197 Best Solution: [183.53739120558993, 177.31273747933807,
174.36082025221302, 75.76341725118995] Fitness: 1.8835638918595754
Generation 198 Best Solution: [182.99281612737, 177.31273747933807,
174.36082025221302, 75.76341725118995] Fitness: 1.8647184882164334
Generation 199 Best Solution: [183.53739120558993, 177.31273747933807,
174.2739271765337, 75.14965387915441] Fitness: 1.1700013146424058
Generation 200 Best Solution: [182.99122009106622, 177.31273747933807,
174.56920158910728, 75.14965387915441] Fitness: 1.2719193162972773
Generation 201 Best Solution: [183.80440535573675, 176.50465444248871,
174.5004971319995, 75.90142884275036] Fitness: 1.5051898013665608
Generation 202 Best Solution: [183.53739120558993, 176.50465444248871,
174.5004971319995, 75.90142884275036] Fitness: 1.3274006270389151
Generation 203 Best Solution: [183.82359373347006, 176.08799101101206,
175.11692615914058, 75.48241869794802] Fitness: 1.4761751605399676
Generation 204 Best Solution: [183.82359373347006, 176.08799101101206,
175.11692615914058, 75.48241869794802] Fitness: 1.4761751605399676
Generation 205 Best Solution: [183.62175212118194, 176.08799101101206,
175.11692615914058, 75.76592534370215] Fitness: 1.7192336861471356
Generation 206 Best Solution: [183.75272233145895, 176.08799101101206,
175.11692615914058, 75.48241869794802] Fitness: 1.416058789439823
Generation 207 Best Solution: [183.75272233145895, 176.08799101101206,
175.11692615914058, 74.64511230645934] Fitness: 0.9831977108042395
Generation 208 Best Solution: [183.53739120558993, 177.31273747933807,
175.11692615914058, 74.6783599614285] Fitness: 1.7382707582953911
Generation 209 Best Solution: [183.53739120558993, 177.31273747933807,
175.11692615914058, 74.6783599614285] Fitness: 1.7382707582953911
Generation 210 Best Solution: [183.6081737130848, 176.08799101101206,
175.11692615914058, 75.48241869794802] Fitness: 1.3176649896114374
Generation 211 Best Solution: [183.6081737130848, 176.08799101101206,
175.11692615914058, 75.48241869794802] Fitness: 1.3176649896114374
Generation 212 Best Solution: [183.6081737130848, 177.31273747933807,
173.73007175513624, 75.48241869794802] Fitness: 1.6426090669327673
Generation 213 Best Solution: [183.6081737130848, 177.31273747933807,
175.11692615914058, 74.6783599614285] Fitness: 1.7745975556966265
Generation 214 Best Solution: [183.6081737130848, 175.99852670776409,
175.13886112351855, 75.76592534370215] Fitness: 1.770586589405176
Generation 215 Best Solution: [183.82359373347006, 175.99852670776409,
175.13886112351855, 75.76592534370215] Fitness: 1.9290967603337061
Generation 216 Best Solution: [183.6081737130848, 177.37869633393632,
174.297076989024, 75.76592534370215] Fitness: 2.0214985114079616

Generation 217 Best Solution: [183.82359373347006, 175.99852670776409,
175.13886112351855, 75.76592534370215] Fitness: 1.9290967603337061
Generation 218 Best Solution: [183.6081737130848, 177.37869633393632,
174.297076989024, 74.51490967061477] Fitness: 1.22559607118086
Generation 219 Best Solution: [183.82359373347006, 176.10509979340551,
174.6682269349309, 75.90142884275036] Fitness: 1.5873087429472648
Generation 220 Best Solution: [183.53739120558993, 176.30515140149305,
174.297076989024, 75.90142884275036] Fitness: 1.1980052249870212
Generation 221 Best Solution: [183.82359373347006, 177.08674920135408,
175.13886112351855, 74.33902890081524] Fitness: 1.7433043404131863
Generation 222 Best Solution: [183.6081737130848, 177.08674920135408,
173.77919091629138, 74.33902890081524] Fitness: 0.9523414154505413
Generation 223 Best Solution: [183.53739120558993, 177.3851660955756,
174.91391267099817, 75.53261234986616] Fitness: 2.0910996381244353
Generation 224 Best Solution: [183.39798017810944, 175.95912655379598,
174.6859999593919, 75.90142884275036] Fitness: 1.4032701383373771
Generation 225 Best Solution: [183.82359373347006, 177.37869633393632,
174.6859999593919, 74.98699681260162] Fitness: 1.5990222237682141
Generation 226 Best Solution: [183.82359373347006, 177.37869633393632,
174.6859999593919, 74.98699681260162] Fitness: 1.5990222237682141
Generation 227 Best Solution: [183.82359373347006, 177.37869633393632,
174.6859999593919, 74.98699681260162] Fitness: 1.5990222237682141
Generation 228 Best Solution: [183.82359373347006, 177.37869633393632,
174.6859999593919, 74.98699681260162] Fitness: 1.5990222237682141
Generation 229 Best Solution: [183.1503119111438, 177.3851660955756,
174.6859999593919, 74.98699681260162] Fitness: 1.354952903301653
Generation 230 Best Solution: [183.1503119111438, 177.3851660955756,
174.6859999593919, 74.98699681260162] Fitness: 1.354952903301653
Generation 231 Best Solution: [183.82359373347006, 176.46889863067284,
175.13886112351855, 75.61223333178008] Fitness: 1.7018162530745369
Generation 232 Best Solution: [183.82359373347006, 176.20095311314537,
174.80239243881664, 75.90142884275036] Fitness: 1.6956868040830455
Generation 233 Best Solution: [183.82359373347006, 176.20095311314537,
174.80239243881664, 75.90142884275036] Fitness: 1.6956868040830455
Generation 234 Best Solution: [183.82359373347006, 176.20095311314537,
174.80239243881664, 75.61850261863282] Fitness: 1.2430199276732423
Generation 235 Best Solution: [183.82359373347006, 177.3851660955756,
173.91802551683872, 75.30458352410136] Fitness: 1.6456532081941708
Generation 236 Best Solution: [182.83122301242224, 177.37869633393632,
175.11692615914058, 74.98699681260162] Fitness: 1.9655261909911033
Generation 237 Best Solution: [182.803551180447, 177.37869633393632,
174.41351740438412, 75.76592534370215] Fitness: 2.103176460155063
Generation 238 Best Solution: [183.42385545065696, 177.3851660955756,
175.13886112351855, 74.98699681260162] Fitness: 1.903822055033294
Generation 239 Best Solution: [183.1503119111438, 177.37869633393632,
174.09180399424676, 74.98699681260162] Fitness: 1.1317989561122166
Generation 240 Best Solution: [182.83122301242224, 177.37869633393632,
175.13886112351855, 74.4714984817018] Fitness: 2.0003005971203067

Generation 241 Best Solution: [183.42385545065696, 177.37869633393632,
175.13886112351855, 74.4714984817018] Fitness: 1.8939921649661453
Generation 242 Best Solution: [183.1503119111438, 177.37869633393632,
175.13886112351855, 74.4714984817018] Fitness: 1.875173234677849
Generation 243 Best Solution: [183.42385545065696, 176.35034908156,
175.13886112351855, 74.96512584138463] Fitness: 0.8711906892949974
Generation 244 Best Solution: [183.42385545065696, 177.37869633393632,
175.13886112351855, 74.4714984817018] Fitness: 1.8939921649661453
Generation 245 Best Solution: [183.0266222275512, 175.8491378827194,
175.13886112351855, 75.90142884275036] Fitness: 2.006969820979511
Generation 246 Best Solution: [183.0266222275512, 177.37869633393632,
175.13886112351855, 75.18043067222874] Fitness: 2.0091315764150313
Generation 247 Best Solution: [183.82359373347006, 176.31647091906441,
173.756704587894, 75.90142884275036] Fitness: 1.506128819547
Generation 248 Best Solution: [183.82359373347006, 175.8491378827194,
175.13886112351855, 75.90142884275036] Fitness: 2.232912281374567
Generation 249 Best Solution: [183.82359373347006, 177.37869633393632,
175.13886112351855, 74.93210177075947] Fitness: 2.1095926951264374
Generation 250 Best Solution: [183.1503119111438, 177.37869633393632,
175.13886112351855, 74.93210177075947] Fitness: 1.8539446313101102
Generation 251 Best Solution: [183.82359373347006, 177.3851660955756,
174.04366746146115, 74.43636407360273] Fitness: 1.4228494826251832
Generation 252 Best Solution: [183.1503119111438, 177.3851660955756,
174.83260560990183, 75.22409380578756] Fitness: 1.6296066583112103
Generation 253 Best Solution: [183.1503119111438, 177.3851660955756,
175.13886112351855, 74.73060799089276] Fitness: 1.8342029949886542
Generation 254 Best Solution: [183.1503119111438, 177.3851660955756,
175.13886112351855, 74.8499956939809] Fitness: 1.8451365976981509
Generation 255 Best Solution: [183.82359373347006, 177.3851660955756,
174.91825483987023, 74.73060799089276] Fitness: 1.7917920722888012
Generation 256 Best Solution: [183.82359373347006, 177.3851660955756,
174.91825483987023, 74.73060799089276] Fitness: 1.7917920722888012
Generation 257 Best Solution: [183.82359373347006, 177.3851660955756,
174.91825483987023, 74.73060799089276] Fitness: 1.7917920722888012
Generation 258 Best Solution: [183.82359373347006, 177.3851660955756,
174.91825483987023, 74.73060799089276] Fitness: 1.7917920722888012
Generation 259 Best Solution: [183.82359373347006, 176.5739759822559,
175.13886112351855, 75.32485773457842] Fitness: 1.4197067995324386
Generation 260 Best Solution: [182.84562507086872, 175.91421751283693,
173.80018546217678, 75.90142884275036] Fitness: 1.413377496036853
Generation 261 Best Solution: [182.98001480319323, 177.3851660955756,
174.91825483987023, 74.73060799089276] Fitness: 1.5832162672780137
Generation 262 Best Solution: [183.42385545065696, 177.3851660955756,
173.80018546217678, 75.3496805773081] Fitness: 1.5057692478771876
Generation 263 Best Solution: [183.42385545065696, 177.3851660955756,
173.83415709424017, 75.68502119875782] Fitness: 1.8976218229987443
Generation 264 Best Solution: [183.42385545065696, 177.3851660955756,
173.83415709424017, 75.68502119875782] Fitness: 1.8976218229987443

Generation 265 Best Solution: [183.42385545065696, 177.3851660955756,
173.83415709424017, 75.68502119875782] Fitness: 1.8976218229987443
Generation 266 Best Solution: [183.18024692899564, 177.3851660955756,
173.75066696203405, 75.90142884275036] Fitness: 2.2803672973657667
Generation 267 Best Solution: [183.42385545065696, 175.85289624445267,
175.13886112351855, 75.90142884275036] Fitness: 1.9938591326514672
Generation 268 Best Solution: [183.0262728153333, 177.3851660955756,
175.13886112351855, 74.68245152342996] Fitness: 1.865891395239792
Generation 269 Best Solution: [183.42385545065696, 175.71205969929707,
175.13886112351855, 75.84426713206472] Fitness: 1.9907218336995867
Generation 270 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.4599269570302] Fitness: 2.025925284541894
Generation 271 Best Solution: [183.42385545065696, 177.31549846777338,
175.13886112351855, 74.4599269570302] Fitness: 1.7890961510138634
Generation 272 Best Solution: [182.91548945717636, 177.3851660955756,
175.13886112351855, 74.83820471311726] Fitness: 1.9197627916082474
Generation 273 Best Solution: [183.82359373347006, 177.3851660955756,
175.13886112351855, 74.83820471311726] Fitness: 2.0987180918683768
Generation 274 Best Solution: [183.82359373347006, 176.70236713905416,
175.13886112351855, 75.63719106227101] Fitness: 1.8651338514959817
Generation 275 Best Solution: [183.42385545065696, 177.3851660955756,
175.13886112351855, 74.83820471311726] Fitness: 1.8618889583403464
Generation 276 Best Solution: [183.82359373347006, 177.3851660955756,
173.99528310387453, 75.23884726434521] Fitness: 1.5659676469287802
Generation 277 Best Solution: [183.36247055078118, 177.3851660955756,
175.13886112351855, 74.83820471311726] Fitness: 1.8475366093960839
Generation 278 Best Solution: [183.82359373347006, 177.3851660955756,
175.13886112351855, 74.81757742590831] Fitness: 2.095622910186125
Generation 279 Best Solution: [183.82359373347006, 177.3851660955756,
175.13886112351855, 74.83907459404536] Fitness: 2.0988631643416022
Generation 280 Best Solution: [183.82359373347006, 176.5534859414183,
175.13886112351855, 75.90142884275036] Fitness: 2.199914887178643
Generation 281 Best Solution: [183.82359373347006, 175.63461949877825,
174.93344021799172, 75.63719106227101] Fitness: 1.6896972120227636
Generation 282 Best Solution: [183.56790663282806, 177.31549846777338,
174.93344021799172, 75.63719106227101] Fitness: 2.146586165302592
Generation 283 Best Solution: [183.82359373347006, 177.39513245646793,
175.13886112351855, 74.88951968941899] Fitness: 2.127253292100608
Generation 284 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 75.02939136166306] Fitness: 2.0376453827867658
Generation 285 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 75.02939136166306] Fitness: 2.0376453827867658
Generation 286 Best Solution: [183.82359373347006, 176.1373939169626,
175.13886112351855, 75.63719106227101] Fitness: 1.6995133706452143
Generation 287 Best Solution: [182.91804229806723, 176.1373939169626,
175.13886112351855, 75.63719106227101] Fitness: 1.5192631339823421
Generation 288 Best Solution: [183.82359373347006, 175.67443177296656,
175.13886112351855, 75.63719106227101] Fitness: 1.9339378618446388

Generation 289 Best Solution: [182.91804229806723, 176.1373939169626,
175.13886112351855, 75.90142884275036] Fitness: 1.945870313485867
Generation 290 Best Solution: [182.91804229806723, 176.1373939169626,
175.13886112351855, 75.90142884275036] Fitness: 1.945870313485867
Generation 291 Best Solution: [183.82359373347006, 176.09144643507926,
175.13886112351855, 75.90142884275036] Fitness: 2.1344835919037024
Generation 292 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 75.06241640089509] Fitness: 2.053800697479482
Generation 293 Best Solution: [183.82359373347006, 177.3851660955756,
173.9853305414822, 75.90142884275036] Fitness: 2.4330379449166384
Generation 294 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.98784838158568] Fitness: 2.019732611979743
Generation 295 Best Solution: [183.70205526992441, 177.31549846777338,
175.13886112351855, 74.98784838158568] Fitness: 1.9214276703772541
Generation 296 Best Solution: [183.61707141120985, 177.31549846777338,
175.13886112351855, 74.42489941160281] Fitness: 1.8882806524476945
Generation 297 Best Solution: [183.70205526992441, 177.3851660955756,
174.93320495972873, 74.90042849303008] Fitness: 1.733508314461826
Generation 298 Best Solution: [183.83992115938426, 177.3851660955756,
175.13886112351855, 74.73257771393165] Fitness: 2.1048085805067567
Generation 299 Best Solution: [183.83992115938426, 176.8460496027935,
175.13886112351855, 75.90142884275036] Fitness: 2.4280126209012125
Generation 300 Best Solution: [183.70205526992441, 176.8460496027935,
175.13886112351855, 75.90142884275036] Fitness: 2.3147506609561193
Generation 301 Best Solution: [183.82359373347006, 176.8460496027935,
174.9355551808954, 75.90142884275036] Fitness: 2.135635288819027
Generation 302 Best Solution: [183.82359373347006, 176.8460496027935,
174.9355551808954, 75.90142884275036] Fitness: 2.135635288819027
Generation 303 Best Solution: [183.70205526992441, 176.1331391681502,
175.13886112351855, 75.90142884275036] Fitness: 2.0284520567566933
Generation 304 Best Solution: [182.79306006061955, 176.1847998911459,
175.13886112351855, 75.90142884275036] Fitness: 2.0159826630020294
Generation 305 Best Solution: [182.79306006061955, 176.1847998911459,
175.13886112351855, 75.90142884275036] Fitness: 2.0159826630020294
Generation 306 Best Solution: [182.79306006061955, 176.1847998911459,
175.13886112351855, 75.90142884275036] Fitness: 2.0159826630020294
Generation 307 Best Solution: [182.79306006061955, 177.3851660955756,
174.9355551808954, 75.29104417254545] Fitness: 1.951055147978293
Generation 308 Best Solution: [182.79306006061955, 177.3851660955756,
174.9355551808954, 75.01745289296467] Fitness: 1.7711087080313386
Generation 309 Best Solution: [182.79306006061955, 176.1847998911459,
175.13886112351855, 75.90142884275036] Fitness: 2.0159826630020294
Generation 310 Best Solution: [182.79306006061955, 177.31549846777338,
175.13886112351855, 74.88554812260317] Fitness: 1.8821158354999434
Generation 311 Best Solution: [182.79306006061955, 177.31549846777338,
175.13886112351855, 74.88554812260317] Fitness: 1.8821158354999434
Generation 312 Best Solution: [183.19212656443298, 177.3851660955756,
175.01615960382193, 75.21743692387656] Fitness: 1.838132783362198

Generation 313 Best Solution: [183.82359373347006, 177.3851660955756,
173.75038337818083, 74.2800067488057] Fitness: 1.633604799277868
Generation 314 Best Solution: [183.70205526992441, 177.3851660955756,
175.13886112351855, 74.88554812260317] Fitness: 2.0100200215261497
Generation 315 Best Solution: [183.70205526992441, 177.3851660955756,
175.13886112351855, 74.88554812260317] Fitness: 2.0100200215261497
Generation 316 Best Solution: [183.70205526992441, 177.01083931306837,
174.23553561919317, 75.90142884275036] Fitness: 1.7526184785481396
Generation 317 Best Solution: [183.83992115938426, 176.2601142698983,
175.04181139210573, 75.90142884275036] Fitness: 1.9943906902883661
Generation 318 Best Solution: [183.83992115938426, 176.2601142698983,
174.41275772647475, 75.90142884275036] Fitness: 1.4392869079902222
Generation 319 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.65425925091812] Fitness: 1.9732246003029936
Generation 320 Best Solution: [183.82359373347006, 176.31296394075284,
175.04181139210573, 75.90142884275036] Fitness: 1.983964341954543
Generation 321 Best Solution: [183.82359373347006, 177.31549846777338,
175.04181139210573, 75.22272767583924] Fitness: 2.015879441030422
Generation 322 Best Solution: [183.02396766856432, 177.31549846777338,
175.13886112351855, 75.21025812902113] Fitness: 1.921875910117428
Generation 323 Best Solution: [183.02396766856432, 177.31549846777338,
175.13886112351855, 75.21025812902113] Fitness: 1.921875910117428
Generation 324 Best Solution: [183.82359373347006, 176.31296394075284,
175.13886112351855, 75.90142884275036] Fitness: 2.124413716978978
Generation 325 Best Solution: [183.02396766856432, 176.31296394075284,
175.13886112351855, 75.90142884275036] Fitness: 1.8993696391963355
Generation 326 Best Solution: [183.21079832723117, 175.65972875370295,
175.13886112351855, 75.90142884275036] Fitness: 2.1119423965162003
Generation 327 Best Solution: [183.82359373347006, 176.356508734867,
173.9689830557566, 75.90142884275036] Fitness: 1.4145479206244371
Generation 328 Best Solution: [183.82359373347006, 175.65972875370295,
175.13886112351855, 74.47470767582706] Fitness: 1.3600118880205614
Generation 329 Best Solution: [183.82359373347006, 175.65972875370295,
175.13886112351855, 75.04742085757455] Fitness: 1.386413948760259
Generation 330 Best Solution: [183.82359373347006, 175.65972875370295,
175.13886112351855, 75.04742085757455] Fitness: 1.386413948760259
Generation 331 Best Solution: [183.83992115938426, 176.85113475677332,
175.13886112351855, 75.90142884275036] Fitness: 2.4328945328688967
Generation 332 Best Solution: [183.82359373347006, 176.85113475677332,
175.13886112351855, 75.90142884275036] Fitness: 2.4179375145262925
Generation 333 Best Solution: [183.83992115938426, 177.31549846777338,
175.13886112351855, 75.02211020857303] Fitness: 2.0492688376472157
Generation 334 Best Solution: [183.82359373347006, 177.31549846777338,
174.05141360392886, 75.02211020857303] Fitness: 1.2982399686301687
Generation 335 Best Solution: [183.83992115938426, 177.31549846777338,
175.13886112351855, 75.02211020857303] Fitness: 2.0492688376472157
Generation 336 Best Solution: [183.83992115938426, 177.31549846777338,
174.65085875619613, 74.6575587926459] Fitness: 1.4299509538805262

Generation 337 Best Solution: [183.82359373347006, 176.36180212531647,
174.65085875619613, 75.8647627882976] Fitness: 1.5089334893276347
Generation 338 Best Solution: [183.83992115938426, 176.36180212531647,
174.65085875619613, 75.8647627882976] Fitness: 1.5238905076702391
Generation 339 Best Solution: [183.82359373347006, 176.03015008720988,
175.13886112351855, 75.90142884275036] Fitness: 2.150753170568949
Generation 340 Best Solution: [182.98138339286805, 177.31549846777338,
175.13886112351855, 75.90142884275036] Fitness: 2.82415942873095
Generation 341 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 75.27665558579903] Fitness: 2.1998047949917727
Generation 342 Best Solution: [183.82359373347006, 175.8282439307265,
175.13886112351855, 75.90142884275036] Fitness: 2.2456769243490364
Generation 343 Best Solution: [183.82359373347006, 175.8282439307265,
175.13886112351855, 75.90142884275036] Fitness: 2.2456769243490364
Generation 344 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 75.75536543461925] Fitness: 2.784051397015116
Generation 345 Best Solution: [183.84060446816525, 177.31549846777338,
174.44988522578555, 75.90142884275036] Fitness: 2.3690020749316987
Generation 346 Best Solution: [183.82359373347006, 177.31549846777338,
174.7827813357363, 74.4520884274899] Fitness: 1.585707170853201
Generation 347 Best Solution: [183.83992115938426, 176.6788218770864,
174.7827813357363, 75.90142884275036] Fitness: 1.8463042603787527
Generation 348 Best Solution: [183.82359373347006, 176.4190901959728,
174.08161965143, 75.90142884275036] Fitness: 1.4061695771425162
Generation 349 Best Solution: [183.82359373347006, 176.4190901959728,
174.08161965143, 75.90142884275036] Fitness: 1.4061695771425162
Generation 350 Best Solution: [183.82359373347006, 176.4190901959728,
175.13886112351855, 75.38383464459002] Fitness: 1.4129818756416797
Generation 351 Best Solution: [183.82359373347006, 176.4190901959728,
175.13886112351855, 75.90142884275036] Fitness: 2.146633948326775
Generation 352 Best Solution: [183.82359373347006, 176.4190901959728,
175.13886112351855, 75.90142884275036] Fitness: 2.146633948326775
Generation 353 Best Solution: [183.82359373347006, 176.86385383888793,
174.68899857747448, 75.90142884275036] Fitness: 1.9027346823267024
Generation 354 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.87582598463706] Fitness: 1.9848099961244174
Generation 355 Best Solution: [183.82359373347006, 176.55779051324805,
175.13886112351855, 75.25316330747623] Fitness: 1.3491019618301867
Generation 356 Best Solution: [183.82359373347006, 177.31549846777338,
173.78895935810831, 74.817222942944] Fitness: 1.3361527561569417
Generation 357 Best Solution: [183.82359373347006, 176.38134617926374,
175.13886112351855, 75.90142884275036] Fitness: 2.1367238179586705
Generation 358 Best Solution: [183.82359373347006, 176.38134617926374,
175.13886112351855, 75.90142884275036] Fitness: 2.1367238179586705
Generation 359 Best Solution: [183.3304789297276, 176.38134617926374,
175.13886112351855, 75.51325823506156] Fitness: 1.2911079341039293
Generation 360 Best Solution: [182.84039280104892, 177.31549846777338,
175.13886112351855, 74.77992373434626] Fitness: 1.8340951168409216

Generation 361 Best Solution: [183.82359373347006, 176.55779051324805,
175.00930840677938, 75.51325823506156] Fitness: 1.4285956409185387
Generation 362 Best Solution: [183.82359373347006, 176.55779051324805,
175.13886112351855, 74.77992373434626] Fitness: 1.139214464296169
Generation 363 Best Solution: [183.4052357424021, 177.31549846777338,
175.13886112351855, 74.77992373434626] Fitness: 1.7286200993010847
Generation 364 Best Solution: [183.23192903533814, 177.32023074307185,
175.13886112351855, 74.87398046966857] Fitness: 1.7301791592302707
Generation 365 Best Solution: [183.74095839098908, 176.55779051324805,
174.94216579896002, 75.90142884275036] Fitness: 1.8633356536240668
Generation 366 Best Solution: [183.4052357424021, 177.31549846777338,
175.13886112351855, 75.47804775835222] Fitness: 2.160355544433622
Generation 367 Best Solution: [183.23192903533814, 177.31549846777338,
174.47024385477738, 75.90142884275036] Fitness: 2.100696246868954
Generation 368 Best Solution: [183.23192903533814, 177.31549846777338,
174.15427754257473, 75.90142884275036] Fitness: 2.0258596334959535
Generation 369 Best Solution: [183.82359373347006, 176.06970213800616,
175.13886112351855, 75.22310393463253] Fitness: 1.2629092897498195
Generation 370 Best Solution: [183.23192903533814, 176.06970213800616,
175.13886112351855, 75.22310393463253] Fitness: 1.0006927398609926
Generation 371 Best Solution: [183.23192903533814, 177.31549846777338,
175.13886112351855, 75.28877148953875] Fitness: 1.9479782915671295
Generation 372 Best Solution: [183.23192903533814, 177.31549846777338,
175.05702018594226, 75.90142884275036] Fitness: 2.6516693304162158
Generation 373 Best Solution: [183.82359373347006, 177.31549846777338,
174.3341107391363, 75.90142884275036] Fitness: 2.3116286968009154
Generation 374 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.99781519582368] Fitness: 2.0237853900141083
Generation 375 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.62320680980734] Fitness: 1.9777021469277154
Generation 376 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.79373320872105] Fitness: 1.9716120832586292
Generation 377 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.79373320872105] Fitness: 1.9716120832586292
Generation 378 Best Solution: [183.82359373347006, 176.00001750088535,
175.13886112351855, 74.79373320872105] Fitness: 1.0992126170211283
Generation 379 Best Solution: [183.82359373347006, 177.31549846777338,
174.82417640044522, 74.79597324301031] Fitness: 1.5696917910382677
Generation 380 Best Solution: [183.82359373347006, 176.00001750088535,
174.82417640044522, 75.90142884275036] Fitness: 1.7587525850056949
Generation 381 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.31037935368695] Fitness: 2.1064791978729334
Generation 382 Best Solution: [183.82359373347006, 177.31549846777338,
175.11547582806153, 74.31037935368695] Fitness: 2.071296388108167
Generation 383 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.31037935368695] Fitness: 2.1064791978729334
Generation 384 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 75.28447365987797] Fitness: 2.2064830848374126

Generation 385 Best Solution: [183.19284920677825, 176.8524983642843,
175.13886112351855, 75.90142884275036] Fitness: 2.1588890880887517
Generation 386 Best Solution: [183.82359373347006, 177.31549846777338,
174.19087287931032, 75.62763847778197] Fitness: 1.848796588559869
Generation 387 Best Solution: [183.71846272767147, 176.0219218890611,
175.11547582806153, 74.2956373372189] Fitness: 1.1148340829248555
Generation 388 Best Solution: [183.82359373347006, 177.31549846777338,
175.11547582806153, 74.59601373246025] Fitness: 1.9476722929930488
Generation 389 Best Solution: [183.82359373347006, 177.31549846777338,
174.59836924997617, 75.45054955735114] Fitness: 1.7750300136234523
Generation 390 Best Solution: [183.82359373347006, 177.31549846777338,
174.59836924997617, 75.45054955735114] Fitness: 1.7750300136234523
Generation 391 Best Solution: [183.82359373347006, 175.8401379841577,
174.59836924997617, 75.45054955735114] Fitness: 0.9800639563656994
Generation 392 Best Solution: [183.82359373347006, 177.31549846777338,
173.95160349854683, 75.09730160547056] Fitness: 1.361244136953081
Generation 393 Best Solution: [183.3873148102132, 177.31549846777338,
173.95160349854683, 75.09730160547056] Fitness: 1.115165382494041
Generation 394 Best Solution: [183.82359373347006, 175.8401379841577,
175.13886112351855, 75.09730160547056] Fitness: 1.2777425220538352
Generation 395 Best Solution: [183.82359373347006, 175.8401379841577,
175.13886112351855, 75.09730160547056] Fitness: 1.2777425220538352
Generation 396 Best Solution: [183.71846272767147, 177.31549846777338,
174.0917033260141, 75.09730160547056] Fitness: 1.2447182624448896
Generation 397 Best Solution: [183.35787087942091, 177.31549846777338,
175.13886112351855, 75.09730160547056] Fitness: 1.8206877130284171
Generation 398 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 75.09730160547056] Fitness: 2.0727085793115876
Generation 399 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.72966878561111] Fitness: 1.9685952613837212
Generation 400 Best Solution: [183.35787087942091, 177.31549846777338,
175.11547582806153, 74.68127805755942] Fitness: 1.683345345376508
Generation 401 Best Solution: [183.35787087942091, 175.86702128707742,
175.11547582806153, 75.90142884275036] Fitness: 1.9353225483823058
Generation 402 Best Solution: [183.82359373347006, 177.31549846777338,
175.11547582806153, 74.71894397816112] Fitness: 1.9335312713479007
Generation 403 Best Solution: [183.82359373347006, 176.46121602228504,
173.9347210857058, 75.90142884275036] Fitness: 1.4545415195866322
Generation 404 Best Solution: [183.82359373347006, 176.62764337374065,
174.1043259581775, 75.11935925630326] Fitness: 0.5508590103167941
Generation 405 Best Solution: [183.0109407989241, 177.31549846777338,
175.13886112351855, 74.61089854875037] Fitness: 1.7594154999382268
Generation 406 Best Solution: [183.82359373347006, 177.31549846777338,
174.4751988025481, 75.3825237278666] Fitness: 1.6303420328480422
Generation 407 Best Solution: [183.82359373347006, 177.31549846777338,
173.975761899806, 74.61089854875037] Fitness: 1.2610500142319916
Generation 408 Best Solution: [183.82359373347006, 176.7840194786552,
173.9347210857058, 75.86995496491511] Fitness: 1.5944607488817288

Generation 409 Best Solution: [183.82359373347006, 175.8078309868039,
175.13886112351855, 75.04544352138947] Fitness: 1.2707961123216671
Generation 410 Best Solution: [183.82359373347006, 176.17620205306386,
173.975761899806, 75.90142884275036] Fitness: 1.4027732405510405
Generation 411 Best Solution: [183.82359373347006, 176.5669468969613,
173.9347210857058, 75.86995496491511] Fitness: 1.4445175102296273
Generation 412 Best Solution: [183.82359373347006, 176.97271156459638,
175.11547582806153, 75.30293565000608] Fitness: 1.700787035920957
Generation 413 Best Solution: [183.82359373347006, 175.8078309868039,
173.975761899806, 75.90142884275036] Fitness: 1.5399615313937245
Generation 414 Best Solution: [183.82359373347006, 177.31549846777338,
173.9347210857058, 75.03820150567222] Fitness: 1.3360192819387606
Generation 415 Best Solution: [183.82359373347006, 175.8078309868039,
173.975761899806, 75.78162208850826] Fitness: 1.3330771446117176
Generation 416 Best Solution: [183.82359373347006, 175.8078309868039,
173.975761899806, 75.78162208850826] Fitness: 1.3330771446117176
Generation 417 Best Solution: [183.82359373347006, 177.31549846777338,
174.70943475462335, 75.30293565000608] Fitness: 1.7121826904160131
Generation 418 Best Solution: [183.82359373347006, 177.31549846777338,
174.70943475462335, 75.30293565000608] Fitness: 1.7121826904160131
Generation 419 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.98635564490449] Fitness: 2.019138929340847
Generation 420 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.74748389074276] Fitness: 1.968793343053177
Generation 421 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.74748389074276] Fitness: 1.968793343053177
Generation 422 Best Solution: [183.82359373347006, 176.050547944716,
175.11547582806153, 75.31035254057845] Fitness: 1.305481438639314
Generation 423 Best Solution: [183.82359373347006, 176.050547944716,
175.13886112351855, 75.17123382191863] Fitness: 1.2304342344170398
Generation 424 Best Solution: [183.82359373347006, 177.31549846777338,
174.06413197081872, 75.31035254057845] Fitness: 1.4911730520117898
Generation 425 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 75.17123382191863] Fitness: 2.119037364782337
Generation 426 Best Solution: [183.82359373347006, 177.31549846777338,
174.5638361539159, 75.17123382191863] Fitness: 1.500637562617348
Generation 427 Best Solution: [183.41677689803157, 177.31549846777338,
175.13886112351855, 75.31035254057845] Fitness: 1.9904842293185419
Generation 428 Best Solution: [183.82359373347006, 177.31549846777338,
174.0302106363693, 74.63961637318911] Fitness: 1.243008365686575
Generation 429 Best Solution: [183.82359373347006, 177.31549846777338,
174.0302106363693, 74.63961637318911] Fitness: 1.243008365686575
Generation 430 Best Solution: [183.82359373347006, 177.31549846777338,
174.0302106363693, 74.63961637318911] Fitness: 1.243008365686575
Generation 431 Best Solution: [183.07806025166155, 177.31549846777338,
174.2956435484641, 75.1636358045005] Fitness: 1.1417654473547865
Generation 432 Best Solution: [182.86911495534176, 177.31549846777338,
174.2956435484641, 75.1636358045005] Fitness: 1.2292850530956052

Generation 433 Best Solution: [183.82359373347006, 177.31549846777338,
174.2956435484641, 75.1636358045005] Fitness: 1.3829519813884497
Generation 434 Best Solution: [183.82359373347006, 177.31549846777338,
174.2956435484641, 75.1636358045005] Fitness: 1.3829519813884497
Generation 435 Best Solution: [183.6848465758699, 177.31549846777338,
174.2956435484641, 75.1636358045005] Fitness: 1.2725850590026189
Generation 436 Best Solution: [183.0121101591248, 176.8032197196002,
175.13886112351855, 74.3279015847872] Fitness: 1.2145845773004864
Generation 437 Best Solution: [183.56441969816674, 176.8032197196002,
175.13886112351855, 74.3279015847872] Fitness: 1.2535965934482327
Generation 438 Best Solution: [183.0121101591248, 176.8032197196002,
175.13886112351855, 75.90142884275036] Fitness: 2.1526363819245384
Generation 439 Best Solution: [183.0121101591248, 176.8032197196002,
174.0653646823404, 75.90142884275036] Fitness: 1.4143594052009525
Generation 440 Best Solution: [183.0121101591248, 176.8032197196002,
174.11610850556016, 75.90142884275036] Fitness: 1.4088921095953306
Generation 441 Best Solution: [183.0121101591248, 176.8032197196002,
174.11610850556016, 75.90142884275036] Fitness: 1.4088921095953306
Generation 442 Best Solution: [183.82359373347006, 176.8032197196002,
174.0653646823404, 75.90142884275036] Fitness: 1.6352567074482942
Generation 443 Best Solution: [183.0121101591248, 176.5333773115828,
175.13886112351855, 75.90142884275036] Fitness: 1.9692581010215409
Generation 444 Best Solution: [183.42941685770344, 176.8032197196002,
175.13886112351855, 75.90142884275036] Fitness: 2.138294437137178
Generation 445 Best Solution: [182.86911495534176, 176.8032197196002,
175.13886112351855, 75.90142884275036] Fitness: 2.219866755879036
Generation 446 Best Solution: [183.82359373347006, 175.8382166397801,
175.13886112351855, 75.4494317755099] Fitness: 1.5757658292484318
Generation 447 Best Solution: [183.31147673694105, 177.31549846777338,
175.13886112351855, 75.4494317755099] Fitness: 2.1109123899040654
Generation 448 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.62438262921636] Fitness: 1.9775052789710366
Generation 449 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.62438262921636] Fitness: 1.9775052789710366
Generation 450 Best Solution: [183.82359373347006, 175.81697751573188,
175.13886112351855, 75.65717960796357] Fitness: 1.8547083897666168
Generation 451 Best Solution: [182.93346108298596, 177.31549846777338,
175.13886112351855, 75.54833937172515] Fitness: 2.299781667036515
Generation 452 Best Solution: [182.93346108298596, 177.31549846777338,
175.13886112351855, 74.66068402410836] Fitness: 1.7846295278521134
Generation 453 Best Solution: [183.82359373347006, 177.31549846777338,
174.7179130532125, 74.99655741532587] Fitness: 1.5201193258063435
Generation 454 Best Solution: [182.86911495534176, 176.81880791084254,
175.13886112351855, 75.90142884275036] Fitness: 2.233920711290218
Generation 455 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.43586723768327] Fitness: 2.036536798729748
Generation 456 Best Solution: [182.86911495534176, 176.86639039470015,
175.13886112351855, 75.75908654632306] Fitness: 2.0358536168226715

Generation 457 Best Solution: [182.93346108298596, 176.89069703053025,
175.13886112351855, 75.75908654632306] Fitness: 2.026132237158533
Generation 458 Best Solution: [182.86911495534176, 176.86639039470015,
175.13886112351855, 75.75908654632306] Fitness: 2.0358536168226715
Generation 459 Best Solution: [182.86911495534176, 175.74151895874581,
175.13886112351855, 75.75908654632306] Fitness: 1.908947436945595
Generation 460 Best Solution: [182.93346108298596, 175.74151895874581,
175.13886112351855, 75.90142884275036] Fitness: 2.11806265301748
Generation 461 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 75.28693459563065] Fitness: 2.2086049194449213
Generation 462 Best Solution: [182.85054924049282, 177.31549846777338,
175.13886112351855, 75.28693459563065] Fitness: 2.0659997547855147
Generation 463 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.28133483424762] Fitness: 2.12615847558636
Generation 464 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.96375116948803] Fitness: 2.0105724668349256
Generation 465 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.96375116948803] Fitness: 2.0105724668349256
Generation 466 Best Solution: [183.3086607815995, 177.31549846777338,
175.13886112351855, 74.96375116948803] Fitness: 1.7516309022088248
Generation 467 Best Solution: [183.7550335014116, 176.1945678340153,
175.13886112351855, 74.96375116948803] Fitness: 1.039299762247029
Generation 468 Best Solution: [182.81793482593355, 176.1945678340153,
175.13886112351855, 75.4334384918419] Fitness: 1.3170274782059948
Generation 469 Best Solution: [183.76189940264373, 176.28177108886172,
175.13886112351855, 74.35277619732837] Fitness: 1.1152573308988365
Generation 470 Best Solution: [183.7550335014116, 177.31549846777338,
175.13886112351855, 74.96375116948803] Fitness: 1.9522932937346609
Generation 471 Best Solution: [183.7550335014116, 177.31549846777338,
175.13886112351855, 75.36436242640573] Fitness: 2.221895552229115
Generation 472 Best Solution: [183.7550335014116, 177.31549846777338,
175.13886112351855, 75.36436242640573] Fitness: 2.221895552229115
Generation 473 Best Solution: [183.7550335014116, 176.3035659436441,
175.13886112351855, 75.90142884275036] Fitness: 2.0650112647826258
Generation 474 Best Solution: [183.7550335014116, 177.31549846777338,
175.13886112351855, 74.53171782157662] Fitness: 1.941334866874804
Generation 475 Best Solution: [183.7550335014116, 176.3035659436441,
175.13886112351855, 74.53171782157662] Fitness: 1.031331958226345
Generation 476 Best Solution: [183.7550335014116, 176.3035659436441,
175.13886112351855, 75.90142884275036] Fitness: 2.0650112647826258
Generation 477 Best Solution: [183.7550335014116, 176.3035659436441,
175.13886112351855, 75.22637694355497] Fitness: 1.1908460195873474
Generation 478 Best Solution: [183.82359373347006, 176.3035659436441,
175.13886112351855, 75.22637694355497] Fitness: 1.249125192687612
Generation 479 Best Solution: [183.82359373347006, 176.3035659436441,
175.13886112351855, 75.8212989698939] Fitness: 1.982447988316823
Generation 480 Best Solution: [183.7550335014116, 177.31549846777338,
175.13886112351855, 75.11890417382483] Fitness: 2.027087168199408

```
Generation 481 Best Solution: [183.7550335014116, 177.31549846777338,
175.13886112351855, 75.16845776994589] Fitness: 2.058864973236121
Generation 482 Best Solution: [183.7550335014116, 176.3035659436441,
174.80013492903646, 75.90142884275036] Fitness: 1.6384807534110237
Generation 483 Best Solution: [183.06839781907613, 176.3035659436441,
175.13886112351855, 75.90142884275036] Fitness: 1.8846534907542043
Generation 484 Best Solution: [183.7550335014116, 175.64354055673826,
173.94582268833372, 75.90142884275036] Fitness: 1.620208972927148
Generation 485 Best Solution: [183.7550335014116, 175.64354055673826,
174.95776380798887, 75.90142884275036] Fitness: 2.0795296626527184
Generation 486 Best Solution: [183.7550335014116, 175.64354055673826,
174.95776380798887, 75.90142884275036] Fitness: 2.0795296626527184
Generation 487 Best Solution: [183.7550335014116, 175.64354055673826,
174.95776380798887, 75.90142884275036] Fitness: 2.0795296626527184
Generation 488 Best Solution: [183.7550335014116, 176.22918093507403,
174.95776380798887, 75.90142884275036] Fitness: 1.8107240113603422
Generation 489 Best Solution: [183.82359373347006, 177.31549846777338,
175.13886112351855, 74.36754272569148] Fitness: 2.071580670625023
Generation 490 Best Solution: [183.7550335014116, 177.28180648917615,
174.95776380798887, 75.90142884275036] Fitness: 2.668837458866764
Generation 491 Best Solution: [183.75753357755792, 176.12811799516834,
175.13886112351855, 75.90142884275036] Fitness: 2.0712618641269516
Generation 492 Best Solution: [183.7550335014116, 177.28180648917615,
175.13886112351855, 74.38581434442041] Fitness: 1.947285675946244
Generation 493 Best Solution: [183.7550335014116, 177.31549846777338,
175.13886112351855, 74.38581434442041] Fitness: 2.0032186082223746
Generation 494 Best Solution: [183.75753357755792, 176.12811799516834,
175.13886112351855, 74.38581434442041] Fitness: 1.0994662989182413
Generation 495 Best Solution: [183.47959552625125, 177.31549846777338,
175.13886112351855, 74.82286676632849] Fitness: 1.7563769778226488
Generation 496 Best Solution: [183.47959552625125, 177.31549846777338,
175.13886112351855, 74.33915440780434] Fitness: 1.8695577035697268
Generation 497 Best Solution: [183.79827643878315, 175.81124102709273,
175.13886112351855, 74.82286676632849] Fitness: 1.1759957501095242
Generation 498 Best Solution: [183.47959552625125, 177.3069578196883,
175.13886112351855, 74.68515900874299] Fitness: 1.737193518769278
Generation 499 Best Solution: [182.9606920196497, 177.31549846777338,
175.13886112351855, 74.82286676632849] Fitness: 1.7747108291646752
Generation 500 Best Solution: [183.66849434384804, 177.31549846777338,
175.13886112351855, 74.82286676632849] Fitness: 1.8536940614745594
Optimal Solution for Simulation 11: [183.66849434384804, 177.31549846777338,
175.13886112351855, 74.82286676632849]
--------------------------------------------------
Simulation 12/32 with conditions: {'drought': True, 'flood': False,
'policy_change': True, 'social_event': False, 'economic_crisis': False}
Generation 1 Best Solution: [182.9156822979235, 176.39336688484926,
174.35276121348912, 75.2920038379765] Fitness: 0.4061554627547509
Generation 2 Best Solution: [183.6035153555649, 176.71620865348618,
```

174.01139376109896, 74.70596479622634] Fitness: 0.38067276112703485
Generation 3 Best Solution: [183.11103444840867, 176.71620865348618,
174.01139376109896, 75.72778309196545] Fitness: 0.9967440508609691
Generation 4 Best Solution: [182.9156822979235, 176.1429731050182,
173.9661078764537, 75.10325814257948] Fitness: 0.24330203035498546
Generation 5 Best Solution: [182.9156822979235, 176.1429731050182,
173.9661078764537, 75.10325814257948] Fitness: 0.24330203035498546
Generation 6 Best Solution: [182.9156822979235, 175.66152192310298,
174.83909373589913, 75.01793038691163] Fitness: 0.7729088800759149
Generation 7 Best Solution: [183.11103444840867, 175.66152192310298,
174.83909373589913, 75.80045131946504] Fitness: 1.493894168024598
Generation 8 Best Solution: [183.11103444840867, 175.66152192310298,
174.83909373589913, 75.80045131946504] Fitness: 1.493894168024598
Generation 9 Best Solution: [183.04743965785167, 175.66152192310298,
173.82698626247347, 75.38249775136629] Fitness: 0.6779021149279424
Generation 10 Best Solution: [183.04743965785167, 176.25217841646716,
174.83909373589913, 75.38249775136629] Fitness: 0.7511377736035115
Generation 11 Best Solution: [183.6687882752458, 176.03862598349812,
173.82698626247347, 75.38249775136629] Fitness: 0.6083698672122795
Generation 12 Best Solution: [183.04743965785167, 175.66152192310298,
173.82698626247347, 75.5134115131411] Fitness: 0.8162644355616924
Generation 13 Best Solution: [183.04743965785167, 176.46153311446042,
174.83909373589913, 75.5134115131411] Fitness: 0.9374470390799295
Generation 14 Best Solution: [183.04743965785167, 175.66152192310298,
173.82698626247347, 75.5134115131411] Fitness: 0.8162644355616924
Generation 15 Best Solution: [183.04743965785167, 176.46153311446042,
174.83909373589913, 75.5134115131411] Fitness: 0.9374470390799295
Generation 16 Best Solution: [183.04743965785167, 177.42610224480856,
174.83909373589913, 75.38249775136629] Fitness: 1.8790022774442572
Generation 17 Best Solution: [182.84512593012056, 176.46312952964195,
174.83909373589913, 75.80045131946504] Fitness: 1.4152149059874737
Generation 18 Best Solution: [183.6687882752458, 176.46312952964195,
174.83909373589913, 75.80045131946504] Fitness: 1.4665123843386887
Generation 19 Best Solution: [182.9156822979235, 177.42610224480856,
175.11898055593755, 75.38249775136629] Fitness: 2.282152318927252
Generation 20 Best Solution: [182.84512593012056, 177.42610224480856,
174.35939510267212, 75.5134115131411] Fitness: 1.7658821140421679
Generation 21 Best Solution: [183.8790864663533, 177.42610224480856,
174.35939510267212, 75.5134115131411] Fitness: 1.995231415334872
Generation 22 Best Solution: [183.6687882752458, 177.42610224480856,
174.2077363758053, 75.5134115131411] Fitness: 1.7841447000373807
Generation 23 Best Solution: [182.84512593012056, 177.42610224480856,
174.2077363758053, 75.5134115131411] Fitness: 1.732847221686166
Generation 24 Best Solution: [183.6687882752458, 177.42610224480856,
174.2077363758053, 75.5134115131411] Fitness: 1.7841447000373807
Generation 25 Best Solution: [183.6687882752458, 177.42610224480856,
174.2077363758053, 75.5134115131411] Fitness: 1.7841447000373807
Generation 26 Best Solution: [183.6687882752458, 177.42610224480856,

174.35613476220294, 74.71590688642661] Fitness: 1.3765121874748851
Generation 27 Best Solution: [183.4611754200868, 177.42610224480856,
175.11898055593755, 74.36203338325389] Fitness: 2.072217106413479
Generation 28 Best Solution: [183.6687882752458, 177.42610224480856,
175.11898055593755, 75.01913303032086] Fitness: 2.1215778128300182
Generation 29 Best Solution: [183.6687882752458, 175.78644681769202,
175.11898055593755, 75.80045131946504] Fitness: 1.9061005457655908
Generation 30 Best Solution: [183.6687882752458, 176.69310078863936,
175.11898055593755, 75.80045131946504] Fitness: 1.95268528152652
Generation 31 Best Solution: [183.6687882752458, 176.71662310415536,
175.11898055593755, 75.80045131946504] Fitness: 1.9704924251732057
Generation 32 Best Solution: [183.6687882752458, 177.023935323756,
174.5073787788515, 75.80045131946504] Fitness: 1.655590761587574
Generation 33 Best Solution: [183.6687882752458, 176.601949673721,
175.11898055593755, 75.80045131946504] Fitness: 1.8916105451033323
Generation 34 Best Solution: [183.6687882752458, 176.601949673721,
175.11898055593755, 75.80045131946504] Fitness: 1.8916105451033323
Generation 35 Best Solution: [183.6687882752458, 176.601949673721,
175.11898055593755, 75.80045131946504] Fitness: 1.8916105451033323
Generation 36 Best Solution: [183.6687882752458, 176.601949673721,
175.11898055593755, 75.80045131946504] Fitness: 1.8916105451033323
Generation 37 Best Solution: [182.9156822979235, 177.023935323756,
175.11898055593755, 75.41173810968152] Fitness: 1.688956273797257
Generation 38 Best Solution: [183.6687882752458, 177.42610224480856,
175.11898055593755, 75.13349796141338] Fitness: 2.177963201412852
Generation 39 Best Solution: [183.6687882752458, 177.42610224480856,
175.11898055593755, 75.13349796141338] Fitness: 2.177963201412852
Generation 40 Best Solution: [183.6687882752458, 177.42610224480856,
175.11898055593755, 74.82552002145275] Fitness: 2.0713561529818745
Generation 41 Best Solution: [183.6687882752458, 176.2037160026911,
175.11898055593755, 75.80045131946504] Fitness: 1.772625947201349
Generation 42 Best Solution: [182.9156822979235, 177.42610224480856,
175.11898055593755, 74.50005924309464] Fitness: 2.0278805876525796
Generation 43 Best Solution: [183.87741952132217, 176.62181954575476,
175.11898055593755, 74.50005924309464] Fitness: 1.2932096259562067
Generation 44 Best Solution: [183.87741952132217, 176.62181954575476,
175.11898055593755, 75.80045131946504] Fitness: 2.0802262988076956
Generation 45 Best Solution: [183.6687882752458, 177.42610224480856,
174.12396613067446, 75.48601368339217] Fitness: 1.749750805418031
Generation 46 Best Solution: [183.6687882752458, 177.42610224480856,
174.12396613067446, 75.48601368339217] Fitness: 1.749750805418031
Generation 47 Best Solution: [183.87741952132217, 177.42610224480856,
174.5541411738403, 74.91102521255239] Fitness: 1.6659485936744076
Generation 48 Best Solution: [183.43284655118188, 175.80893260297316,
175.11898055593755, 75.80045131946504] Fitness: 1.7722829204719481
Generation 49 Best Solution: [183.70632463272088, 177.42610224480856,
174.12396613067446, 75.52827009082363] Fitness: 1.8250672709148597
Generation 50 Best Solution: [183.87741952132217, 177.09526168814068,

175.11898055593755, 75.41789692009445] Fitness: 2.0511318201610753
Generation 51 Best Solution: [183.6687882752458, 177.42610224480856,
174.09155189280773, 75.80045131946504] Fitness: 2.1768272538987956
Generation 52 Best Solution: [183.6687882752458, 176.0193845209306,
175.11898055593755, 75.80045131946504] Fitness: 1.7990157859094205
Generation 53 Best Solution: [183.6687882752458, 176.67470142008136,
175.11898055593755, 75.80045131946504] Fitness: 1.939341521049839
Generation 54 Best Solution: [183.6687882752458, 175.8223872020232,
175.11898055593755, 75.80045131946504] Fitness: 1.8842070869120615
Generation 55 Best Solution: [183.6687882752458, 175.8223872020232,
175.11898055593755, 75.78403649538531] Fitness: 1.858283703199934
Generation 56 Best Solution: [183.7491226369077, 175.8223872020232,
175.11898055593755, 75.80045131946504] Fitness: 1.9443028407286476
Generation 57 Best Solution: [183.87741952132217, 175.9123375832083,
175.11898055593755, 75.80045131946504] Fitness: 2.0143800891884953
Generation 58 Best Solution: [183.25085416740836, 176.3557566758376,
175.11898055593755, 75.80045131946504] Fitness: 1.6349902237688938
Generation 59 Best Solution: [183.25085416740836, 176.56527412015308,
175.11898055593755, 75.80045131946504] Fitness: 1.7159266500365393
Generation 60 Best Solution: [183.25085416740836, 176.56527412015308,
175.11898055593755, 75.80045131946504] Fitness: 1.7159266500365393
Generation 61 Best Solution: [183.30884912898384, 177.35097860565887,
175.11898055593755, 75.80045131946504] Fitness: 2.6182738416494358
Generation 62 Best Solution: [183.6687882752458, 176.189822436457,
175.11898055593755, 75.80045131946504] Fitness: 1.7728186147165779
Generation 63 Best Solution: [183.63503524672086, 176.189822436457,
175.11898055593755, 75.80045131946504] Fitness: 1.7504902676600866
Generation 64 Best Solution: [183.63503524672086, 176.189822436457,
175.11898055593755, 75.80045131946504] Fitness: 1.7504902676600866
Generation 65 Best Solution: [182.94760031884266, 177.42610224480856,
174.2740453707326, 75.80045131946504] Fitness: 2.087697655467974
Generation 66 Best Solution: [182.94760031884266, 177.42610224480856,
174.20938918800573, 75.80045131946504] Fitness: 2.077719288431407
Generation 67 Best Solution: [183.6687882752458, 175.5895668083546,
175.11898055593755, 75.80045131946504] Fitness: 2.0608054641609503
Generation 68 Best Solution: [183.30884912898384, 175.7379641628559,
175.11898055593755, 75.80045131946504] Fitness: 1.7896994197801854
Generation 69 Best Solution: [183.27948979259912, 177.42610224480856,
175.11898055593755, 74.46167990010622] Fitness: 1.978319251179764
Generation 70 Best Solution: [182.81261605115813, 177.42610224480856,
175.11898055593755, 74.46167990010622] Fitness: 2.098638594153961
Generation 71 Best Solution: [183.27948979259912, 177.42610224480856,
175.11898055593755, 74.46167990010622] Fitness: 1.978319251179764
Generation 72 Best Solution: [183.30884912898384, 175.83630284549898,
175.11898055593755, 75.80045131946504] Fitness: 1.7272166287808668
Generation 73 Best Solution: [182.81261605115813, 177.42610224480856,
175.11898055593755, 75.17345045170117] Fitness: 2.1701314727006524
Generation 74 Best Solution: [182.81261605115813, 176.6203951707087,

175.11898055593755, 75.80045131946504] Fitness: 1.8707456596976062
Generation 75 Best Solution: [182.81261605115813, 176.6203951707087,
175.11898055593755, 75.80045131946504] Fitness: 1.8707456596976062
Generation 76 Best Solution: [182.81261605115813, 176.43562319802962,
175.11898055593755, 75.02831278522001] Fitness: 1.00370724408913
Generation 77 Best Solution: [183.22063273840402, 177.42610224480856,
175.11898055593755, 75.1110981086506] Fitness: 2.0097807917242885
Generation 78 Best Solution: [183.27948979259912, 177.42610224480856,
174.44571452046716, 75.80045131946504] Fitness: 2.09342229448601
Generation 79 Best Solution: [183.27948979259912, 177.42610224480856,
175.11898055593755, 74.41095018675301] Fitness: 2.0025682957219173
Generation 80 Best Solution: [183.27948979259912, 177.42610224480856,
175.11898055593755, 74.41095018675301] Fitness: 2.0025682957219173
Generation 81 Best Solution: [182.81261605115813, 177.42610224480856,
173.91430867859705, 75.80045131946504] Fitness: 2.181481447794935
Generation 82 Best Solution: [183.27948979259912, 177.42610224480856,
173.91430867859705, 75.80045131946504] Fitness: 2.0611621048207383
Generation 83 Best Solution: [182.81261605115813, 176.0588218409054,
175.11898055593755, 75.80045131946504] Fitness: 1.756828103589062
Generation 84 Best Solution: [183.27948979259912, 177.42610224480856,
175.11898055593755, 74.4184566530927] Fitness: 1.9987340316655193
Generation 85 Best Solution: [182.81261605115813, 177.42610224480856,
175.11898055593755, 74.4184566530927] Fitness: 2.1190533746397158
Generation 86 Best Solution: [182.81261605115813, 177.01803397874784,
175.11898055593755, 75.80045131946504] Fitness: 2.2407614772213296
Generation 87 Best Solution: [182.81261605115813, 177.01803397874784,
175.11898055593755, 75.41911671755051] Fitness: 1.744099499055424
Generation 88 Best Solution: [182.81261605115813, 177.01803397874784,
173.80399234387838, 75.80045131946504] Fitness: 1.5993463307905706
Generation 89 Best Solution: [183.5244093159458, 177.01803397874784,
173.80399234387838, 75.80045131946504] Fitness: 1.5481600026731595
Generation 90 Best Solution: [182.81261605115813, 176.26103408108122,
175.11898055593755, 75.80045131946504] Fitness: 1.7427208732239676
Generation 91 Best Solution: [183.5244093159458, 177.42610224480856,
175.11898055593755, 74.99559144428474] Fitness: 2.0290409630164694
Generation 92 Best Solution: [183.5244093159458, 177.42610224480856,
175.11898055593755, 74.42196694030905] Fitness: 2.0661033473824695
Generation 93 Best Solution: [183.23044064429243, 177.42610224480856,
174.25750335986314, 74.42196694030905] Fitness: 1.2774738140434947
Generation 94 Best Solution: [182.81261605115813, 176.37085266353392,
173.84185905568702, 75.80045131946504] Fitness: 1.1014229301335379
Generation 95 Best Solution: [183.23044064429243, 176.04964584466836,
175.11898055593755, 75.80045131946504] Fitness: 1.6357138804570854
Generation 96 Best Solution: [182.81261605115813, 175.75356307205863,
175.06113315717607, 74.42196694030905] Fitness: 1.0596426737904392
Generation 97 Best Solution: [183.23044064429243, 176.35657200915125,
174.7510047698201, 75.51000162419191] Fitness: 0.7922427126132922
Generation 98 Best Solution: [182.81261605115813, 177.42610224480856,

174.64571976590892, 74.3347601933349] Fitness: 1.633597522333521
Generation 99 Best Solution: [182.81261605115813, 177.0006949209506,
175.11898055593755, 74.3347601933349] Fitness: 1.5163630348113832
Generation 100 Best Solution: [182.81261605115813, 176.68846385025648,
175.11898055593755, 75.80045131946504] Fitness: 1.9170673313964315
Generation 101 Best Solution: [182.81261605115813, 176.32439549236486,
175.11898055593755, 75.80045131946504] Fitness: 1.7510659001669369
Generation 102 Best Solution: [182.81261605115813, 176.68846385025648,
175.11898055593755, 74.40240339582371] Fitness: 1.1745205113321988
Generation 103 Best Solution: [182.81261605115813, 176.1258374626161,
175.11898055593755, 75.80045131946504] Fitness: 1.7452794740706008
Generation 104 Best Solution: [182.81261605115813, 176.1258374626161,
175.06113315717607, 75.80045131946504] Fitness: 1.6618944401256992
Generation 105 Best Solution: [182.81261605115813, 176.1258374626161,
175.06113315717607, 75.03355359347785] Fitness: 0.8873777681030435
Generation 106 Best Solution: [182.81261605115813, 176.65592726766042,
175.11898055593755, 75.80045131946504] Fitness: 1.8940487038112903
Generation 107 Best Solution: [182.81261605115813, 175.96742798076806,
175.11898055593755, 74.61916850016992] Fitness: 0.9618916103314848
Generation 108 Best Solution: [182.81261605115813, 175.96742798076806,
175.11898055593755, 75.80045131946504] Fitness: 1.7835608200431978
Generation 109 Best Solution: [182.81261605115813, 176.29421694464213,
175.11898055593755, 74.61916850016992] Fitness: 0.9246623281594302
Generation 110 Best Solution: [182.81261605115813, 175.96742798076806,
175.11898055593755, 75.80045131946504] Fitness: 1.7835608200431978
Generation 111 Best Solution: [183.3306196892509, 177.42610224480856,
175.11898055593755, 74.65443011559218] Fitness: 1.9287020110113509
Generation 112 Best Solution: [182.81261605115813, 176.1954648332361,
175.11898055593755, 75.80045131946504] Fitness: 1.7404983558271054
Generation 113 Best Solution: [182.81261605115813, 176.1954648332361,
175.11898055593755, 75.72080081757967] Fitness: 1.618529962473949
Generation 114 Best Solution: [182.81261605115813, 175.7217728613938,
175.11898055593755, 75.17411159448231] Fitness: 1.218878109631472
Generation 115 Best Solution: [182.81261605115813, 177.42610224480856,
175.11898055593755, 74.65443011559218] Fitness: 2.0421057558345157
Generation 116 Best Solution: [182.81261605115813, 176.94633206670625,
174.19584625247072, 75.90400119994271] Fitness: 1.6045697336833762
Generation 117 Best Solution: [183.3306196892509, 175.84834205734376,
175.11898055593755, 75.80045131946504] Fitness: 1.7240046876051907
Generation 118 Best Solution: [182.81261605115813, 176.0124543970965,
175.11898055593755, 75.80045131946504] Fitness: 1.7688067693745309
Generation 119 Best Solution: [182.81261605115813, 177.42610224480856,
174.08435725771528, 75.80045131946504] Fitness: 2.1451886867146523
Generation 120 Best Solution: [182.81261605115813, 177.42610224480856,
175.11898055593755, 75.1991913964402] Fitness: 2.187118828090135
Generation 121 Best Solution: [183.3306196892509, 177.42610224480856,
175.11898055593755, 74.95914091297956] Fitness: 1.954324644219393
Generation 122 Best Solution: [183.16904900134702, 176.64959850083807,

175.11898055593755, 75.80045131946504] Fitness: 1.7680376539317098
Generation 123 Best Solution: [183.21016596908564, 177.42610224480856,
175.11898055593755, 74.95914091297956] Fitness: 1.9443673395876506
Generation 124 Best Solution: [183.3306196892509, 177.42610224480856,
175.11898055593755, 74.95914091297956] Fitness: 1.954324644219393
Generation 125 Best Solution: [183.3306196892509, 177.42610224480856,
174.5503175058028, 74.95914091297956] Fitness: 1.3549820782406894
Generation 126 Best Solution: [182.98650891654805, 175.6284036967024,
175.11898055593755, 75.80045131946504] Fitness: 1.9099609889962577
Generation 127 Best Solution: [183.0709255160809, 175.65688432399136,
174.77676909339752, 75.80045131946504] Fitness: 1.4421829247200568
Generation 128 Best Solution: [183.21016596908564, 175.66485804029227,
174.5503175058028, 74.95914091297956] Fitness: 0.4376119440954874
Generation 129 Best Solution: [183.3306196892509, 177.42610224480856,
174.5503175058028, 75.80045131946504] Fitness: 2.1571578900840405
Generation 130 Best Solution: [182.98650891654805, 177.42610224480856,
174.5503175058028, 75.80045131946504] Fitness: 2.187097552489164
Generation 131 Best Solution: [183.3306196892509, 176.00261166353323,
175.11898055593755, 75.80045131946504] Fitness: 1.6583655334903096
Generation 132 Best Solution: [183.3306196892509, 176.18230288401017,
175.11898055593755, 75.90400119994271] Fitness: 1.8003913091326837
Generation 133 Best Solution: [183.3306196892509, 176.18230288401017,
174.5503175058028, 75.80045131946504] Fitness: 1.0280920364282686
Generation 134 Best Solution: [183.3306196892509, 177.42610224480856,
173.86400619822973, 75.56356762692819] Fitness: 1.75273908067122
Generation 135 Best Solution: [183.2788236261002, 177.42610224480856,
174.27265079895116, 75.56356762692819] Fitness: 1.701322329315679
Generation 136 Best Solution: [183.2788236261002, 177.42610224480856,
174.27265079895116, 75.56356762692819] Fitness: 1.701322329315679
Generation 137 Best Solution: [183.3306196892509, 177.42610224480856,
175.11898055593755, 74.96726919939036] Fitness: 1.9569371938847375
Generation 138 Best Solution: [182.81261605115813, 176.0649535759489,
175.11898055593755, 74.96726919939036] Fitness: 0.9559249640955069
Generation 139 Best Solution: [183.3306196892509, 177.42610224480856,
175.11898055593755, 74.96726919939036] Fitness: 1.9569371938847375
Generation 140 Best Solution: [183.3306196892509, 175.9879305190411,
175.11898055593755, 75.90400119994271] Fitness: 1.8360141679384774
Generation 141 Best Solution: [183.3306196892509, 177.42610224480856,
175.11898055593755, 74.7430257274582] Fitness: 1.921626669358483
Generation 142 Best Solution: [182.81261605115813, 176.24905116033418,
175.11898055593755, 74.96706740398739] Fitness: 0.9421982417476136
Generation 143 Best Solution: [182.79511035184692, 177.41076549856461,
175.11898055593755, 74.96706740398739] Fitness: 2.0530040559399834
Generation 144 Best Solution: [183.3306196892509, 176.5334267919808,
175.11898055593755, 75.90400119994271] Fitness: 1.8813441716583696
Generation 145 Best Solution: [183.3306196892509, 176.5334267919808,
175.11898055593755, 75.90400119994271] Fitness: 1.8813441716583696
Generation 146 Best Solution: [182.81261605115813, 176.48248631581941,

175.11898055593755, 75.90400119994271] Fitness: 1.9714029706237333
Generation 147 Best Solution: [183.3306196892509, 177.41076549856461,
175.11898055593755, 74.81188676009087] Fitness: 1.896137892688723
Generation 148 Best Solution: [183.3306196892509, 175.98728832546033,
175.11898055593755, 75.90400119994271] Fitness: 1.8362268709874252
Generation 149 Best Solution: [183.3306196892509, 175.98728832546033,
175.11898055593755, 75.90400119994271] Fitness: 1.8362268709874252
Generation 150 Best Solution: [182.81261605115813, 176.05177942385578,
175.11898055593755, 75.74380096459095] Fitness: 1.6707005150744623
Generation 151 Best Solution: [182.81261605115813, 176.05177942385578,
175.11898055593755, 75.90400119994271] Fitness: 1.9313940687061883
Generation 152 Best Solution: [183.39814545244175, 176.6739268081141,
175.11898055593755, 75.90400119994271] Fitness: 1.981348820013359
Generation 153 Best Solution: [182.8645065738076, 176.8005335670226,
175.11898055593755, 75.73691308706589] Fitness: 1.8808662387556225
Generation 154 Best Solution: [182.81261605115813, 177.42610224480856,
174.579606659431, 74.84917463345062] Fitness: 1.4617775950039655
Generation 155 Best Solution: [183.3306196892509, 177.42610224480856,
174.579606659431, 74.84917463345062] Fitness: 1.3483738501808007
Generation 156 Best Solution: [183.3306196892509, 177.42610224480856,
174.42947370841247, 75.90400119994271] Fitness: 2.2659615137223135
Generation 157 Best Solution: [182.908474533749, 176.32650503846583,
175.11898055593755, 75.90400119994271] Fitness: 1.8727209481947198
Generation 158 Best Solution: [182.908474533749, 176.32650503846583,
175.11898055593755, 75.90400119994271] Fitness: 1.8727209481947198
Generation 159 Best Solution: [182.8645065738076, 176.24598708858858,
175.00905297497584, 75.90400119994271] Fitness: 1.730768398410957
Generation 160 Best Solution: [182.81261605115813, 177.42610224480856,
175.11898055593755, 75.23615759642637] Fitness: 2.2132726412762898
Generation 161 Best Solution: [183.3306196892509, 176.57530022198355,
174.42947370841247, 75.90400119994271] Fitness: 1.2399865636414
Generation 162 Best Solution: [182.8645065738076, 177.19779796785616,
175.11898055593755, 74.3528764308184] Fitness: 1.7426073220687062
Generation 163 Best Solution: [182.95088387431883, 176.07580912943564,
175.12544315766232, 75.48471745920646] Fitness: 1.2658507513213755
Generation 164 Best Solution: [183.69175929692133, 177.41076549856461,
174.3129669515324, 75.90400119994271] Fitness: 2.358668805091492
Generation 165 Best Solution: [182.81261605115813, 177.42610224480856,
175.11898055593755, 75.51336959676411] Fitness: 2.4754731171175286
Generation 166 Best Solution: [182.81261605115813, 177.42610224480856,
175.0406956445633, 75.44422472347593] Fitness: 2.2875272063162067
Generation 167 Best Solution: [182.9345311849318, 176.18728618790783,
175.11898055593755, 75.90400119994271] Fitness: 1.8503120129419806
Generation 168 Best Solution: [182.81261605115813, 177.42610224480856,
173.9731163771151, 75.90400119994271] Fitness: 2.3369243118541894
Generation 169 Best Solution: [182.81261605115813, 176.61738005615976,
175.11898055593755, 75.90400119994271] Fitness: 2.041813131517349
Generation 170 Best Solution: [182.81261605115813, 177.42610224480856,

173.9731163771151, 75.51336959676411] Fitness: 1.7695365213600978
Generation 171 Best Solution: [182.81261605115813, 177.42610224480856,
173.9731163771151, 75.51336959676411] Fitness: 1.7695365213600978
Generation 172 Best Solution: [182.81261605115813, 177.42610224480856,
175.11898055593755, 75.23187617147344] Fitness: 2.210137347450625
Generation 173 Best Solution: [182.9345311849318, 177.41076549856461,
175.11898055593755, 74.87818600195351] Fitness: 1.9556409653502107
Generation 174 Best Solution: [182.9345311849318, 175.6099858794577,
175.11898055593755, 75.90400119994271] Fitness: 2.119453687130525
Generation 175 Best Solution: [182.81261605115813, 177.3048629458279,
175.11898055593755, 75.01093146977016] Fitness: 1.8727873797095238
Generation 176 Best Solution: [183.5715522415336, 176.56197911844777,
175.11898055593755, 75.90400119994271] Fitness: 1.982120519335143
Generation 177 Best Solution: [182.9345311849318, 177.3048629458279,
175.11898055593755, 75.1340374008759] Fitness: 1.8694433796846561
Generation 178 Best Solution: [182.88668154728575, 177.41076549856461,
174.99834366026067, 75.01093146977016] Fitness: 1.8485690192698576
Generation 179 Best Solution: [182.81261605115813, 177.3048629458279,
174.99834366026067, 75.01093146977016] Fitness: 1.7046397943484726
Generation 180 Best Solution: [182.81261605115813, 177.3048629458279,
174.99834366026067, 75.01093146977016] Fitness: 1.7046397943484726
Generation 181 Best Solution: [182.81261605115813, 176.6376527354107,
174.99834366026067, 75.01093146977016] Fitness: 0.9298627593548986
Generation 182 Best Solution: [183.53864423709595, 177.3048629458279,
175.0406956445633, 74.70471436429499] Fitness: 1.6673249101814565
Generation 183 Best Solution: [182.84694364435475, 175.57654684997763,
175.12544315766232, 74.89009872075454] Fitness: 1.2100978656675547
Generation 184 Best Solution: [183.5715522415336, 175.57654684997763,
174.3001493108958, 75.90400119994271] Fitness: 1.4788180521234362
Generation 185 Best Solution: [183.5715522415336, 176.7076717905605,
175.12544315766232, 75.90400119994271] Fitness: 2.0865637975163427
Generation 186 Best Solution: [183.5715522415336, 176.7076717905605,
175.11898055593755, 75.90400119994271] Fitness: 2.0769328893179675
Generation 187 Best Solution: [183.5715522415336, 176.18695652227507,
175.12544315766232, 75.90400119994271] Fitness: 1.8958415440032295
Generation 188 Best Solution: [182.81261605115813, 176.38609768549483,
175.11898055593755, 75.90400119994271] Fitness: 1.938003153456601
Generation 189 Best Solution: [183.5715522415336, 176.38609768549483,
175.11898055593755, 75.90400119994271] Fitness: 1.9105689825200618
Generation 190 Best Solution: [183.87798907939762, 177.42610224480856,
174.26762796270904, 75.02494211848719] Fitness: 1.586204824663914
Generation 191 Best Solution: [183.18446741627452, 177.42610224480856,
175.11898055593755, 74.29859296566674] Fitness: 2.0678597719483536
Generation 192 Best Solution: [182.81261605115813, 176.98547673608675,
175.11898055593755, 75.90400119994271] Fitness: 2.3744056101241604
Generation 193 Best Solution: [182.81261605115813, 176.98547673608675,
175.11898055593755, 75.90400119994271] Fitness: 2.3744056101241604
Generation 194 Best Solution: [182.81261605115813, 176.98547673608675,

175.11898055593755, 74.40717207475025] Fitness: 1.4563945245542147
Generation 195 Best Solution: [183.3699613758263, 176.8058933348264,
175.11898055593755, 75.90400119994271] Fitness: 2.081079010903213
Generation 196 Best Solution: [183.3699613758263, 176.8058933348264,
175.11898055593755, 75.90400119994271] Fitness: 2.081079010903213
Generation 197 Best Solution: [183.5715522415336, 176.8058933348264,
175.11898055593755, 74.34582198994562] Fitness: 1.275909824637152
Generation 198 Best Solution: [183.53864423709595, 176.8058933348264,
175.11898055593755, 75.90400119994271] Fitness: 2.142091738424347
Generation 199 Best Solution: [183.3699613758263, 177.3048629458279,
174.38535495198957, 75.90400119994271] Fitness: 2.042779746378553
Generation 200 Best Solution: [183.3699613758263, 177.3048629458279,
174.38535495198957, 75.90400119994271] Fitness: 2.042779746378553
Generation 201 Best Solution: [183.3699613758263, 177.3048629458279,
174.38535495198957, 74.64199270829225] Fitness: 1.0439711337075521
Generation 202 Best Solution: [183.5715522415336, 177.3048629458279,
174.38535495198957, 74.64199270829225] Fitness: 1.1219193671611398
Generation 203 Best Solution: [183.53864423709595, 177.41076549856461,
175.12544315766232, 74.87020016329863] Fitness: 1.9827373369730623
Generation 204 Best Solution: [183.53864423709595, 177.3048629458279,
175.11898055593755, 74.87020016329863] Fitness: 1.788018617319206
Generation 205 Best Solution: [183.53864423709595, 177.41076549856461,
175.11898055593755, 75.0209012415575] Fitness: 2.017507284310988
Generation 206 Best Solution: [183.08746886060405, 177.3048629458279,
175.11898055593755, 74.87020016329863] Fitness: 1.7215171069703847
Generation 207 Best Solution: [183.74274177520235, 175.8813365048108,
175.11898055593755, 74.87020016329863] Fitness: 1.0832922312275757
Generation 208 Best Solution: [183.53864423709595, 176.44312973613228,
175.11898055593755, 75.90400119994271] Fitness: 1.9116929768575757
Generation 209 Best Solution: [183.5021854596399, 176.75798512486568,
174.52466381627846, 75.90400119994271] Fitness: 1.4685627714227807
Generation 210 Best Solution: [183.3699613758263, 177.41076549856461,
174.52466381627846, 74.87020016329863] Fitness: 1.2972796701605311
Generation 211 Best Solution: [183.5021854596399, 176.75798512486568,
174.52466381627846, 75.90400119994271] Fitness: 1.4685627714227807
Generation 212 Best Solution: [183.0169648700174, 175.9609538620847,
175.11898055593755, 75.90400119994271] Fitness: 1.865531103654319
Generation 213 Best Solution: [183.0169648700174, 175.9609538620847,
175.11898055593755, 75.90400119994271] Fitness: 1.865531103654319
Generation 214 Best Solution: [183.0169648700174, 175.9609538620847,
175.11898055593755, 75.90400119994271] Fitness: 1.865531103654319
Generation 215 Best Solution: [183.41706797407014, 175.9609538620847,
175.11898055593755, 75.90400119994271] Fitness: 1.866203312325521
Generation 216 Best Solution: [183.41706797407014, 175.9609538620847,
175.11898055593755, 75.90400119994271] Fitness: 1.866203312325521
Generation 217 Best Solution: [183.41706797407014, 175.9609538620847,
174.85833497891363, 75.90400119994271] Fitness: 1.5305911980701763
Generation 218 Best Solution: [183.3699613758263, 176.18073507788907,

175.12544315766232, 75.90400119994271] Fitness: 1.8181015755021077
Generation 219 Best Solution: [183.47664542000945, 176.65183527738887,
175.12544315766232, 75.90400119994271] Fitness: 2.002042634570739
Generation 220 Best Solution: [183.47664542000945, 176.65183527738887,
175.12544315766232, 75.90400119994271] Fitness: 2.002042634570739
Generation 221 Best Solution: [183.47664542000945, 177.44974473622213,
175.12544315766232, 75.28567652571847] Fitness: 2.2335690107584005
Generation 222 Best Solution: [183.71429129893423, 176.1999703060735,
175.12544315766232, 75.57162927551104] Fitness: 1.490617634188578
Generation 223 Best Solution: [183.17895377865804, 176.1999703060735,
175.12544315766232, 75.57162927551104] Fitness: 1.303223811299138
Generation 224 Best Solution: [183.71429129893423, 176.1999703060735,
175.12544315766232, 74.82225463105843] Fitness: 0.984003010601819
Generation 225 Best Solution: [183.3699613758263, 175.9422013685237,
175.12544315766232, 75.90400119994271] Fitness: 1.8703766341956036
Generation 226 Best Solution: [183.68190042744496, 175.9422013685237,
175.12544315766232, 75.32769059266873] Fitness: 1.261200599375962
Generation 227 Best Solution: [183.3699613758263, 176.06513275013023,
175.12544315766232, 75.90400119994271] Fitness: 1.8326551423370798
Generation 228 Best Solution: [183.5021854596399, 176.06513275013023,
175.12544315766232, 75.90400119994271] Fitness: 1.8768235961895117
Generation 229 Best Solution: [183.68190042744496, 175.986376194924,
175.12544315766232, 75.90400119994271] Fitness: 2.000911547087611
Generation 230 Best Solution: [183.3699613758263, 176.53983756880265,
175.12544315766232, 75.90400119994271] Fitness: 1.9022132631975932
Generation 231 Best Solution: [183.3699613758263, 176.53983756880265,
175.12544315766232, 75.90400119994271] Fitness: 1.9022132631975932
Generation 232 Best Solution: [183.1673933435025, 176.10168681041978,
175.12544315766232, 75.90400119994271] Fitness: 1.8096431248019487
Generation 233 Best Solution: [183.1673933435025, 176.10168681041978,
175.12544315766232, 75.90400119994271] Fitness: 1.8096431248019487
Generation 234 Best Solution: [183.1673933435025, 176.10168681041978,
175.12544315766232, 75.90400119994271] Fitness: 1.8096431248019487
Generation 235 Best Solution: [183.71429129893423, 176.10168681041978,
175.12544315766232, 75.90400119994271] Fitness: 1.9962873677554365
Generation 236 Best Solution: [183.71429129893423, 175.81881393773466,
174.76463322054877, 75.90400119994271] Fitness: 1.661014056582636
Generation 237 Best Solution: [183.71429129893423, 176.680065389841,
175.12544315766232, 75.90400119994271] Fitness: 2.1586033010471875
Generation 238 Best Solution: [183.71429129893423, 176.00440656388594,
175.12544315766232, 75.90400119994271] Fitness: 2.0188494940388555
Generation 239 Best Solution: [183.1673933435025, 175.61227988445393,
175.12544315766232, 75.64362222933052] Fitness: 1.6648113553064465
Generation 240 Best Solution: [183.1673933435025, 175.61227988445393,
175.13331844873036, 75.90400119994271] Fitness: 2.0805592227825684
Generation 241 Best Solution: [183.71429129893423, 177.44974473622213,
175.12544315766232, 75.06605535766464] Fitness: 2.2289732924527126
Generation 242 Best Solution: [183.77038172386472, 175.6330678093376,

175.12544315766232, 75.90400119994271] Fitness: 2.2817866216154976
Generation 243 Best Solution: [183.71429129893423, 177.44974473622213,
175.12544315766232, 75.09440717315769] Fitness: 2.2431200424864133
Generation 244 Best Solution: [183.71429129893423, 177.44974473622213,
175.12544315766232, 75.13333142341625] Fitness: 2.264528714507563
Generation 245 Best Solution: [183.12408638194435, 177.44974473622213,
175.12544315766232, 75.13333142341625] Fitness: 2.0824950890359544
Generation 246 Best Solution: [183.71429129893423, 176.57856902388946,
174.5982999629828, 75.90400119994271] Fitness: 1.5160454085027424
Generation 247 Best Solution: [183.71429129893423, 176.35209841789776,
175.12544315766232, 75.90400119994271] Fitness: 2.0042597395709585
Generation 248 Best Solution: [183.12408638194435, 176.35209841789776,
175.12544315766232, 75.90400119994271] Fitness: 1.8222261140993492
Generation 249 Best Solution: [183.71429129893423, 175.60286770692082,
175.12544315766232, 74.30404731174501] Fitness: 1.4078368058132233
Generation 250 Best Solution: [183.71429129893423, 176.35209841789776,
175.12544315766232, 75.90400119994271] Fitness: 2.0042597395709585
Generation 251 Best Solution: [183.4817383612572, 177.44974473622213,
175.12544315766232, 74.55730615365853] Fitness: 2.04759300494917
Generation 252 Best Solution: [183.71429129893423, 176.03394001053354,
175.12544315766232, 75.64362222933052] Fitness: 1.606555945041646
Generation 253 Best Solution: [183.71429129893423, 176.03394001053354,
175.12544315766232, 75.60570861280837] Fitness: 1.5563200390129746
Generation 254 Best Solution: [183.71429129893423, 176.03394001053354,
175.12544315766232, 75.90400119994271] Fitness: 2.0104819808177137
Generation 255 Best Solution: [183.71429129893423, 176.03394001053354,
175.12544315766232, 75.90400119994271] Fitness: 2.0104819808177137
Generation 256 Best Solution: [183.79021805073728, 176.0333003736203,
175.12544315766232, 75.90400119994271] Fitness: 2.0724357717252815
Generation 257 Best Solution: [183.12408638194435, 176.0333003736203,
175.12544315766232, 75.90400119994271] Fitness: 1.82861555889559
Generation 258 Best Solution: [183.12408638194435, 176.0333003736203,
175.12544315766232, 75.90400119994271] Fitness: 1.82861555889559
Generation 259 Best Solution: [183.12408638194435, 176.0333003736203,
175.12544315766232, 74.64739409236209] Fitness: 0.8289327619418041
Generation 260 Best Solution: [183.12408638194435, 177.44974473622213,
175.12544315766232, 74.55440686650927] Fitness: 2.00124118333261
Generation 261 Best Solution: [183.12408638194435, 177.44974473622213,
175.12544315766232, 75.17558201898673] Fitness: 2.108335109994872
Generation 262 Best Solution: [183.45419294520337, 177.44974473622213,
174.10844408843562, 75.40584149949585] Fitness: 1.597499192194749
Generation 263 Best Solution: [183.12408638194435, 175.88684247889628,
175.12544315766232, 75.40584149949585] Fitness: 1.2003089515654068
Generation 264 Best Solution: [183.12408638194435, 175.88684247889628,
175.12367337608097, 75.90400119994271] Fitness: 1.8805999107498956
Generation 265 Best Solution: [183.12408638194435, 175.88684247889628,
175.12544315766232, 75.90400119994271] Fitness: 1.8832436326741955
Generation 266 Best Solution: [183.12408638194435, 175.88684247889628,

174.9297521476081, 75.60570861280837] Fitness: 1.1655446297901033
Generation 267 Best Solution: [183.12408638194435, 177.01149233164548,
175.12544315766232, 74.4673375108418] Fitness: 1.3516465127054942
Generation 268 Best Solution: [183.12408638194435, 177.44974473622213,
175.12544315766232, 74.4673375108418] Fitness: 2.0330113611713863
Generation 269 Best Solution: [183.12408638194435, 175.73184180251994,
175.12544315766232, 75.90400119994271] Fitness: 1.9765056465509168
Generation 270 Best Solution: [183.79021805073728, 177.44974473622213,
174.67435416951412, 75.14733963402374] Fitness: 1.8144995350089232
Generation 271 Best Solution: [183.79021805073728, 177.44974473622213,
174.67435416951412, 74.4673375108418] Fitness: 1.7567486620307382
Generation 272 Best Solution: [183.12408638194435, 175.73184180251994,
175.12544315766232, 75.90400119994271] Fitness: 1.9765056465509168
Generation 273 Best Solution: [183.12408638194435, 175.73184180251994,
175.12544315766232, 75.90400119994271] Fitness: 1.9765056465509168
Generation 274 Best Solution: [183.12408638194435, 175.73184180251994,
175.12544315766232, 75.68313218212948] Fitness: 1.6272512995823663
Generation 275 Best Solution: [183.12408638194435, 175.81054103488648,
174.67435416951412, 75.90400119994271] Fitness: 1.4045149527755398
Generation 276 Best Solution: [183.79021805073728, 175.81054103488648,
174.67435416951412, 75.90400119994271] Fitness: 1.6483351656052312
Generation 277 Best Solution: [183.79021805073728, 175.81054103488648,
174.67435416951412, 75.90400119994271] Fitness: 1.6483351656052312
Generation 278 Best Solution: [183.79021805073728, 176.1718597777771,
174.61844330944618, 74.43859057195095] Fitness: 0.5540026043635957
Generation 279 Best Solution: [183.79021805073728, 176.0875024923109,
174.7755632265804, 75.90400119994271] Fitness: 1.6302120658189765
Generation 280 Best Solution: [183.79021805073728, 176.0434271614718,
173.8830894250315, 75.90768578490494] Fitness: 1.3896915594352564
Generation 281 Best Solution: [183.3127291787859, 176.26113755490442,
175.12544315766232, 75.90400119994271] Fitness: 1.8090424877470794
Generation 282 Best Solution: [183.3127291787859, 176.26113755490442,
175.12544315766232, 74.28563591124272] Fitness: 0.9657058374505709
Generation 283 Best Solution: [183.3127291787859, 176.26113755490442,
175.12544315766232, 75.90400119994271] Fitness: 1.8090424877470794
Generation 284 Best Solution: [183.79021805073728, 175.91691986459878,
175.12544315766232, 75.90400119994271] Fitness: 2.113189636555135
Generation 285 Best Solution: [183.79021805073728, 177.44974473622213,
175.12544315766232, 75.2159832947489] Fitness: 2.3793975365369424
Generation 286 Best Solution: [183.79021805073728, 176.71469761990272,
175.12544315766232, 75.90400119994271] Fitness: 2.2462145287918744
Generation 287 Best Solution: [183.79021805073728, 176.28987832097008,
175.12544315766232, 75.90400119994271] Fitness: 2.0551825086523494
Generation 288 Best Solution: [183.79021805073728, 177.42166874785167,
175.12544315766232, 74.2928163868776] Fitness: 2.3227578416032166
Generation 289 Best Solution: [183.79021805073728, 177.44974473622213,
173.97647651811178, 75.62801742868163] Fitness: 2.082239862364627
Generation 290 Best Solution: [183.79021805073728, 177.44974473622213,

175.12544315766232, 74.71013805168806] Fitness: 2.2169223186481046
Generation 291 Best Solution: [182.95143783320398, 177.44974473622213,
175.12544315766232, 74.2928163868776] Fitness: 2.1779901294010635
Generation 292 Best Solution: [183.27439963123643, 177.44974473622213,
175.12544315766232, 74.6454709217854] Fitness: 1.9765185173732913
Generation 293 Best Solution: [183.102112268447, 177.44974473622213,
175.12544315766232, 74.2928163868776] Fitness: 2.1347522933427068
Generation 294 Best Solution: [182.95143783320398, 177.44974473622213,
175.12544315766232, 74.6454709217854] Fitness: 2.026984926427614
Generation 295 Best Solution: [182.95143783320398, 177.44974473622213,
175.12544315766232, 74.87444549613639] Fitness: 2.029970764306794
Generation 296 Best Solution: [183.79021805073728, 177.44974473622213,
174.73508021120915, 74.2928163868776] Fitness: 1.9070927318656705
Generation 297 Best Solution: [183.79021805073728, 176.8510290510545,
175.12544315766232, 74.6454709217854] Fitness: 1.3661795637896057
Generation 298 Best Solution: [183.79021805073728, 177.44974473622213,
174.73508021120915, 74.2928163868776] Fitness: 1.9070927318656705
Generation 299 Best Solution: [183.102112268447, 177.44974473622213,
175.12544315766232, 75.25233334093292] Fitness: 2.165631997738252
Generation 300 Best Solution: [183.79021805073728, 176.3134612745489,
175.12544315766232, 75.25233334093292] Fitness: 1.2411172160899486
Generation 301 Best Solution: [183.79021805073728, 176.4809753532355,
174.73508021120915, 75.90400119994271] Fitness: 1.6391798490997105
Generation 302 Best Solution: [183.79021805073728, 176.14543617175764,
174.73508021120915, 75.90400119994271] Fitness: 1.5845557648297324
Generation 303 Best Solution: [183.79021805073728, 176.14543617175764,
174.73508021120915, 75.90400119994271] Fitness: 1.5845557648297324
Generation 304 Best Solution: [183.27344344154255, 177.44974473622213,
175.12544315766232, 75.01838476348763] Fitness: 2.022017607633525
Generation 305 Best Solution: [183.79021805073728, 175.84807891524667,
175.12544315766232, 75.90400119994271] Fitness: 2.146969241976019
Generation 306 Best Solution: [183.27344344154255, 176.71599696654653,
175.12544315766232, 75.90400119994271] Fitness: 1.999513579261357
Generation 307 Best Solution: [183.79021805073728, 175.84807891524667,
173.9515087610092, 75.90400119994271] Fitness: 1.4372270372947038
Generation 308 Best Solution: [183.79021805073728, 177.44974473622213,
174.73508021120915, 74.55058093794672] Fitness: 1.7781637133619548
Generation 309 Best Solution: [183.66063135845707, 177.44974473622213,
174.73508021120915, 75.22627712622386] Fitness: 1.8185043128515357
Generation 310 Best Solution: [183.79021805073728, 176.26477733925032,
175.12544315766232, 74.887476900668] Fitness: 1.058634175888892
Generation 311 Best Solution: [183.79021805073728, 176.26477733925032,
174.5054186073577, 75.90400119994271] Fitness: 1.4170665767585908
Generation 312 Best Solution: [183.79021805073728, 176.75879738959307,
173.9515087610092, 75.90400119994271] Fitness: 1.5719907199345435
Generation 313 Best Solution: [183.79021805073728, 177.44974473622213,
173.9515087610092, 75.90400119994271] Fitness: 2.513774034350127
Generation 314 Best Solution: [183.79021805073728, 177.44974473622213,

175.05784937707745, 74.99160050907051] Fitness: 2.1618187427953623
Generation 315 Best Solution: [183.67109914674688, 176.36906044069326,
175.12544315766232, 75.90400119994271] Fitness: 1.9769950677057555
Generation 316 Best Solution: [183.67109914674688, 176.36906044069326,
173.90520011798048, 75.90400119994271] Fitness: 1.2821233741707143
Generation 317 Best Solution: [183.79021805073728, 177.44974473622213,
174.0689813170649, 75.90400119994271] Fitness: 2.4906472597049563
Generation 318 Best Solution: [183.67109914674688, 177.44974473622213,
175.12544315766232, 74.87336242762751] Fitness: 2.133892317279792
Generation 319 Best Solution: [183.66063135845707, 177.44974473622213,
175.12544315766232, 74.87336242762751] Fitness: 2.1267460408324412
Generation 320 Best Solution: [183.79021805073728, 175.86169014289314,
175.12544315766232, 74.87336242762751] Fitness: 1.1431279053004604
Generation 321 Best Solution: [183.79021805073728, 177.44974473622213,
174.07588741828027, 74.87336242762751] Fitness: 1.4933471452882598
Generation 322 Best Solution: [183.79021805073728, 176.34303843559263,
175.12544315766232, 74.87336242762751] Fitness: 1.0675069055074518
Generation 323 Best Solution: [183.79021805073728, 175.86169014289314,
175.12544315766232, 74.87336242762751] Fitness: 1.1431279053004604
Generation 324 Best Solution: [183.79021805073728, 175.86169014289314,
175.12544315766232, 75.32721257042789] Fitness: 1.3833969527297028
Generation 325 Best Solution: [183.79021805073728, 175.86169014289314,
175.12544315766232, 75.42695987249287] Fitness: 1.4780942681434688
Generation 326 Best Solution: [183.79021805073728, 176.2313741484553,
175.12544315766232, 75.20452129062508] Fitness: 1.1982264801151925
Generation 327 Best Solution: [183.79021805073728, 177.44974473622213,
175.12544315766232, 75.42695987249287] Fitness: 2.56189045518891
Generation 328 Best Solution: [183.66063135845707, 177.44974473622213,
175.12544315766232, 74.49701830116885] Fitness: 2.1645313021510955
Generation 329 Best Solution: [183.79021805073728, 176.06467175650673,
174.68223779098003, 75.90400119994271] Fitness: 1.5513223940433964
Generation 330 Best Solution: [183.66063135845707, 176.79854090761708,
174.52375766687038, 75.90400119994271] Fitness: 1.5911194263274986
Generation 331 Best Solution: [183.66063135845707, 176.79854090761708,
174.68223779098003, 75.90400119994271] Fitness: 1.702448435012746
Generation 332 Best Solution: [183.79021805073728, 175.9297993437249,
175.12544315766232, 75.90400119994271] Fitness: 2.1076682582953765
Generation 333 Best Solution: [183.79021805073728, 175.9297993437249,
175.12544315766232, 75.90400119994271] Fitness: 2.1076682582953765
Generation 334 Best Solution: [183.66063135845707, 177.44974473622213,
175.12544315766232, 74.39079443713668] Fitness: 2.214083341934133
Generation 335 Best Solution: [183.79021805073728, 176.4408907602935,
175.12544315766232, 75.90400119994271] Fitness: 2.091721893952087
Generation 336 Best Solution: [183.79021805073728, 176.73166900995625,
175.12544315766232, 75.21086678915908] Fitness: 1.4118285092512242
Generation 337 Best Solution: [183.79021805073728, 176.73166900995625,
175.12544315766232, 75.21086678915908] Fitness: 1.4118285092512242
Generation 338 Best Solution: [183.79021805073728, 177.44974473622213,

174.36627598328243, 74.87365295398075] Fitness: 1.5291247360261802
Generation 339 Best Solution: [183.79021805073728, 177.44974473622213,
174.11492122485302, 75.21086678915908] Fitness: 1.6395924974302858
Generation 340 Best Solution: [183.79021805073728, 177.44974473622213,
174.11492122485302, 75.21086678915908] Fitness: 1.6395924974302858
Generation 341 Best Solution: [183.79021805073728, 175.79741289745826,
175.12544315766232, 75.90400119994271] Fitness: 2.176423984040948
Generation 342 Best Solution: [183.67109914674688, 176.08914820333786,
175.12544315766232, 75.90400119994271] Fitness: 1.9671441822733629
Generation 343 Best Solution: [183.79021805073728, 177.44974473622213,
174.2493562132094, 74.61534773147204] Fitness: 1.502044715978605
Generation 344 Best Solution: [183.67109914674688, 177.44974473622213,
174.2493562132094, 74.61534773147204] Fitness: 1.4090129409124956
Generation 345 Best Solution: [183.79021805073728, 177.44974473622213,
174.2493562132094, 74.61534773147204] Fitness: 1.502044715978605
Generation 346 Best Solution: [183.79021805073728, 177.44974473622213,
175.12544315766232, 74.6626075770176] Fitness: 2.22160914496719
Generation 347 Best Solution: [183.79021805073728, 177.44974473622213,
175.12544315766232, 74.6626075770176] Fitness: 2.22160914496719
Generation 348 Best Solution: [183.79021805073728, 175.98099312531005,
175.12544315766232, 75.90400119994271] Fitness: 2.088210092088728
Generation 349 Best Solution: [183.10208172635197, 177.29735556569446,
175.12544315766232, 74.32725880457461] Fitness: 1.8417233504101373
Generation 350 Best Solution: [183.79021805073728, 176.23310530310644,
175.12544315766232, 75.90400119994271] Fitness: 2.050394703546787
Generation 351 Best Solution: [183.67109914674688, 177.23903118632893,
174.005205800707, 75.90643830329674] Fitness: 2.0478641179508896
Generation 352 Best Solution: [183.67109914674688, 177.44974473622213,
174.44764767349986, 75.90400119994271] Fitness: 2.465590506330186
Generation 353 Best Solution: [183.79021805073728, 175.68610415370523,
174.005205800707, 75.90400119994271] Fitness: 1.5318993088981772
Generation 354 Best Solution: [183.79021805073728, 175.68610415370523,
174.005205800707, 75.90400119994271] Fitness: 1.5318993088981772
Generation 355 Best Solution: [183.67109914674688, 176.3495275181393,
175.12544315766232, 75.90400119994271] Fitness: 1.9724493288602725
Generation 356 Best Solution: [183.79021805073728, 176.7413329217939,
175.12544315766232, 74.64479008394721] Fitness: 1.2680471002398164
Generation 357 Best Solution: [183.79021805073728, 176.7019895104523,
175.12544315766232, 74.61968881930187] Fitness: 1.241796881814457
Generation 358 Best Solution: [183.79021805073728, 175.59710047881086,
175.12544315766232, 74.74922310611024] Fitness: 1.3231350299010505
Generation 359 Best Solution: [183.5179278474272, 176.57187885719966,
175.12544315766232, 75.90400119994271] Fitness: 1.9704199878321367
Generation 360 Best Solution: [183.79021805073728, 176.57187885719966,
175.12544315766232, 74.7559975623207] Fitness: 1.1435716634053148
Generation 361 Best Solution: [183.67109914674688, 176.5623562188477,
175.12544315766232, 75.90400119994271] Fitness: 2.0531880410072803
Generation 362 Best Solution: [183.67109914674688, 176.5623562188477,

175.12544315766232, 75.90400119994271] Fitness: 2.0531880410072803
Generation 363 Best Solution: [183.5179278474272, 175.71039786401664,
175.07118270012293, 75.90400119994271] Fitness: 1.9761858920933024
Generation 364 Best Solution: [183.5179278474272, 175.71039786401664,
175.07118270012293, 75.90400119994271] Fitness: 1.9761858920933024
Generation 365 Best Solution: [183.5179278474272, 177.06877247535434,
174.6714863571698, 75.90400119994271] Fitness: 1.9112315529196184
Generation 366 Best Solution: [182.81178987849694, 177.06877247535434,
174.6714863571698, 75.90400119994271] Fitness: 1.9659258352693154
Generation 367 Best Solution: [183.5179278474272, 177.44974473622213,
175.12544315766232, 75.21165173224831] Fitness: 2.1953396588992837
Generation 368 Best Solution: [183.0336171340974, 177.41450061160666,
175.12544315766232, 75.21165173224831] Fitness: 2.085780705477399
Generation 369 Best Solution: [183.67109914674688, 177.41450061160666,
175.12544315766232, 75.21165173224831] Fitness: 2.217756148482172
Generation 370 Best Solution: [183.67109914674688, 177.41450061160666,
175.12544315766232, 75.21165173224831] Fitness: 2.217756148482172
Generation 371 Best Solution: [183.78292949681108, 176.38074809368732,
175.12544315766232, 75.90400119994271] Fitness: 2.066712939778103
Generation 372 Best Solution: [183.67109914674688, 177.41450061160666,
175.12544315766232, 74.66498591809562] Fitness: 2.0626910291028944
Generation 373 Best Solution: [182.83412720554833, 177.41450061160666,
175.12544315766232, 74.5410018091671] Fitness: 2.0439886675631675
Generation 374 Best Solution: [183.67109914674688, 177.44974473622213,
175.12544315766232, 74.5410018091671] Fitness: 2.156169116082187
Generation 375 Best Solution: [183.67109914674688, 177.12569136181588,
175.12544315766232, 75.90400119994271] Fitness: 2.598594494762208
Generation 376 Best Solution: [183.79021805073728, 175.98195547189826,
175.12544315766232, 75.90400119994271] Fitness: 2.0878823921407066
Generation 377 Best Solution: [183.79021805073728, 176.11538403691932,
175.12544315766232, 75.90400119994271] Fitness: 2.056050335675176
Generation 378 Best Solution: [183.67109914674688, 177.41450061160666,
174.91949622142724, 75.90400119994271] Fitness: 2.789167734962073
Generation 379 Best Solution: [183.79021805073728, 177.41450061160666,
174.91949622142724, 75.90400119994271] Fitness: 2.8821995100281823
Generation 380 Best Solution: [183.79021805073728, 175.7760628242524,
175.12544315766232, 75.90400119994271] Fitness: 2.190002287616376
Generation 381 Best Solution: [183.67109914674688, 177.44974473622213,
175.12544315766232, 74.55195915546503] Fitness: 2.1527656180562436
Generation 382 Best Solution: [183.53509076919107, 175.75699860034382,
175.12544315766232, 75.90400119994271] Fitness: 2.029780822645406
Generation 383 Best Solution: [183.53509076919107, 175.75699860034382,
175.12544315766232, 75.90400119994271] Fitness: 2.029780822645406
Generation 384 Best Solution: [183.67109914674688, 176.89988667446067,
175.12544315766232, 75.22789947027721] Fitness: 1.486546466824703
Generation 385 Best Solution: [183.53509076919107, 175.75699860034382,
175.03782125682068, 75.90400119994271] Fitness: 1.9045968719123993
Generation 386 Best Solution: [183.53509076919107, 175.75699860034382,

175.12544315766232, 75.90400119994271] Fitness: 2.029780822645406
Generation 387 Best Solution: [183.53509076919107, 175.75699860034382,
175.12544315766232, 75.90400119994271] Fitness: 2.029780822645406
Generation 388 Best Solution: [183.53509076919107, 176.1632473982384,
175.03782125682068, 75.90400119994271] Fitness: 1.7530998585538957
Generation 389 Best Solution: [183.67109914674688, 176.1632473982384,
175.03782125682068, 75.90400119994271] Fitness: 1.8329985809683667
Generation 390 Best Solution: [183.6006037871378, 177.44974473622213,
174.66670088827448, 75.90400119994271] Fitness: 2.559324735833183
Generation 391 Best Solution: [183.3490478880524, 177.41450061160666,
175.03782125682068, 75.73106977749798] Fitness: 2.516251948471783
Generation 392 Best Solution: [183.0694954458561, 177.41450061160666,
175.03782125682068, 75.73106977749798] Fitness: 2.519067202105042
Generation 393 Best Solution: [183.8114607810579, 175.81323018888057,
175.12544315766232, 75.90400119994271] Fitness: 2.185662932846552
Generation 394 Best Solution: [183.8114607810579, 175.81323018888057,
175.12544315766232, 75.90400119994271] Fitness: 2.185662932846552
Generation 395 Best Solution: [183.67109914674688, 177.44974473622213,
173.82934984028762, 75.90400119994271] Fitness: 2.466998833082874
Generation 396 Best Solution: [183.5098706378, 177.44974473622213,
173.82934984028762, 75.523953499972] Fitness: 1.8203028767677873
Generation 397 Best Solution: [183.79021805073728, 176.23599546458217,
175.12544315766232, 75.90400119994271] Fitness: 2.0505202976952033
Generation 398 Best Solution: [183.79021805073728, 176.23599546458217,
175.12544315766232, 75.90400119994271] Fitness: 2.0505202976952033
Generation 399 Best Solution: [183.79021805073728, 176.20856919306112,
175.12544315766232, 75.90400119994271] Fitness: 2.049838970032396
Generation 400 Best Solution: [183.85907505594002, 175.8694930790954,
175.12544315766232, 75.60185639680857] Fitness: 1.7401433895919514
Generation 401 Best Solution: [183.79021805073728, 176.23599546458217,
175.12544315766232, 75.60185639680857] Fitness: 1.5913761879402446
Generation 402 Best Solution: [183.67109914674688, 177.44974473622213,
173.80904872645127, 75.60185639680857] Fitness: 2.017735927540068
Generation 403 Best Solution: [183.67109914674688, 177.44974473622213,
174.04212177735545, 75.523953499972] Fitness: 1.8459906993085025
Generation 404 Best Solution: [183.85907505594002, 176.4248796532847,
174.09294116020823, 75.90400119994271] Fitness: 1.4147892653894523
Generation 405 Best Solution: [183.67109914674688, 177.44974473622213,
175.12544315766232, 75.01352766848454] Fitness: 2.174741215303827
Generation 406 Best Solution: [183.53890357083768, 177.44974473622213,
175.12544315766232, 75.01352766848454] Fitness: 2.0966999682995477
Generation 407 Best Solution: [183.79021805073728, 176.32452687978386,
175.12544315766232, 75.523953499972] Fitness: 1.5054412685689798
Generation 408 Best Solution: [183.67109914674688, 176.32452687978386,
175.12544315766232, 75.523953499972] Fitness: 1.4124094935028704
Generation 409 Best Solution: [183.67109914674688, 175.97140025165888,
175.12544315766232, 75.90400119994271] Fitness: 1.9985217132403057
Generation 410 Best Solution: [183.67109914674688, 175.97140025165888,

175.12544315766232, 75.2573283276186] Fitness: 1.1848455804165363
Generation 411 Best Solution: [183.53381260894193, 177.44974473622213,
173.75047111213507, 75.90400119994271] Fitness: 2.4283804067763914
Generation 412 Best Solution: [183.79021805073728, 177.44974473622213,
175.12544315766232, 74.4782300981948] Fitness: 2.2722275730790438
Generation 413 Best Solution: [183.79021805073728, 177.44974473622213,
174.76913949555865, 75.523953499972] Fitness: 2.2320293669567532
Generation 414 Best Solution: [183.79021805073728, 177.44974473622213,
174.58242239953907, 75.523953499972] Fitness: 2.0802453546921535
Generation 415 Best Solution: [183.79021805073728, 175.8266203406699,
175.12544315766232, 75.90400119994271] Fitness: 2.158968741002053
Generation 416 Best Solution: [183.79021805073728, 175.8266203406699,
175.12544315766232, 75.90400119994271] Fitness: 2.158968741002053
Generation 417 Best Solution: [183.79021805073728, 176.7016824947779,
175.12544315766232, 74.7138263819698] Fitness: 1.2294816409065616
Generation 418 Best Solution: [183.79021805073728, 176.7016824947779,
175.12544315766232, 75.23249372209503] Fitness: 1.404022899748851
Generation 419 Best Solution: [183.79021805073728, 176.7016824947779,
175.12544315766232, 75.23249372209503] Fitness: 1.404022899748851
Generation 420 Best Solution: [183.79021805073728, 176.7016824947779,
174.2475796251486, 75.90400119994271] Fitness: 1.5083805463525586
Generation 421 Best Solution: [183.79021805073728, 176.2826990680086,
175.12544315766232, 75.64185036997247] Fitness: 1.647984613907736
Generation 422 Best Solution: [183.79021805073728, 176.2826990680086,
175.12544315766232, 75.64185036997247] Fitness: 1.647984613907736
Generation 423 Best Solution: [183.3451345821694, 177.44974473622213,
175.12544315766232, 74.35420371515177] Fitness: 2.097746725876705
Generation 424 Best Solution: [183.79021805073728, 177.44974473622213,
174.002961865819, 74.60140748047408] Fitness: 1.5102432747881227
Generation 425 Best Solution: [183.79021805073728, 177.44974473622213,
174.002961865819, 74.60140748047408] Fitness: 1.5102432747881227
Generation 426 Best Solution: [183.79021805073728, 177.44974473622213,
174.002961865819, 74.60140748047408] Fitness: 1.5102432747881227
Generation 427 Best Solution: [183.79021805073728, 176.2307661925125,
174.8135720490857, 75.90400119994271] Fitness: 1.6577928158606527
Generation 428 Best Solution: [183.79021805073728, 177.30180654282918,
175.12544315766232, 74.96399142181721] Fitness: 1.9873448378050498
Generation 429 Best Solution: [183.79021805073728, 177.30180654282918,
174.62854322913563, 75.04917332127953] Fitness: 1.464697895278815
Generation 430 Best Solution: [182.9733534235358, 177.44974473622213,
175.12544315766232, 74.94164666306278] Fitness: 2.0375165303993428
Generation 431 Best Solution: [183.79021805073728, 176.40881524452857,
175.12544315766232, 75.90400119994271] Fitness: 2.081066703919644
Generation 432 Best Solution: [183.79021805073728, 176.14347739642346,
175.12544315766232, 75.90400119994271] Fitness: 2.0527904902849317
Generation 433 Best Solution: [183.79021805073728, 176.14347739642346,
173.83153647832953, 75.90400119994271] Fitness: 1.3882778557605602
Generation 434 Best Solution: [183.79021805073728, 175.95297309560254,

175.12544315766232, 75.90400119994271] Fitness: 2.098367587781418
Generation 435 Best Solution: [183.79021805073728, 176.30085730637418,
175.12544315766232, 75.90400119994271] Fitness: 2.0566726915663276
Generation 436 Best Solution: [183.79021805073728, 176.73055046827724,
175.12544315766232, 74.59476809675094] Fitness: 1.2693626525372244
Generation 437 Best Solution: [183.0736411722061, 175.71080742449934,
175.12544315766232, 75.90400119994271] Fitness: 2.0009291490810432
Generation 438 Best Solution: [183.79021805073728, 175.75724572563448,
174.08885712399615, 75.90400119994271] Fitness: 1.467832092498885
Generation 439 Best Solution: [182.91088482862386, 176.36185128645633,
175.12544315766232, 74.59476809675094] Fitness: 0.8993669656571889
Generation 440 Best Solution: [183.79021805073728, 177.44974473622213,
173.83153647832953, 74.59476809675094] Fitness: 1.5697233914438864
Generation 441 Best Solution: [183.79021805073728, 175.71080742449934,
175.12544315766232, 74.59476809675094] Fitness: 1.2465107861637543
Generation 442 Best Solution: [183.63253494455452, 177.44974473622213,
175.12544315766232, 74.59476809675094] Fitness: 2.1156987519306925
Generation 443 Best Solution: [183.79021805073728, 175.71080742449934,
175.12544315766232, 75.90400119994271] Fitness: 2.2357909992269387
Generation 444 Best Solution: [183.63253494455452, 177.44974473622213,
175.12544315766232, 74.37036772794082] Fitness: 2.2072155044335684
Generation 445 Best Solution: [182.95542625512107, 177.44974473622213,
174.08761558309686, 75.90400119994271] Fitness: 2.2901597426226363
Generation 446 Best Solution: [182.91047044159902, 177.44974473622213,
174.08761558309686, 75.90400119994271] Fitness: 2.309460161124299
Generation 447 Best Solution: [183.78972789432967, 177.44974473622213,
175.12544315766232, 74.46281691597451] Fitness: 2.278368241902671
Generation 448 Best Solution: [183.63253494455452, 177.44974473622213,
175.12544315766232, 74.9282030718861] Fitness: 2.1208194589805176
Generation 449 Best Solution: [183.79021805073728, 177.2156587283895,
174.5713710350853, 75.90400119994271] Fitness: 2.2277886306482584
Generation 450 Best Solution: [183.63253494455452, 177.44974473622213,
174.2115442615604, 75.90400119994271] Fitness: 2.372166827626027
Generation 451 Best Solution: [183.78972789432967, 176.4649705690562,
175.12544315766232, 75.90400119994271] Fitness: 2.1003198750324166
Generation 452 Best Solution: [183.63253494455452, 177.44974473622213,
174.1441918134343, 75.90400119994271] Fitness: 2.368297095854586
Generation 453 Best Solution: [183.11870062722934, 177.44974473622213,
174.5713710350853, 75.90400119994271] Fitness: 2.384938033996285
Generation 454 Best Solution: [183.79021805073728, 176.26452474222626,
174.5713710350853, 75.90400119994271] Fitness: 1.4569097414111885
Generation 455 Best Solution: [183.63253494455452, 175.76436458463834,
175.12544315766232, 75.90400119994271] Fitness: 2.0791981941754267
Generation 456 Best Solution: [183.78972789432967, 176.06228483667792,
175.12544315766232, 75.90400119994271] Fitness: 2.065055450644881
Generation 457 Best Solution: [183.79021805073728, 176.06228483667792,
174.08761558309686, 75.90400119994271] Fitness: 1.330868926206111
Generation 458 Best Solution: [183.63253494455452, 175.76436458463834,

175.12544315766232, 75.90400119994271] Fitness: 2.0791981941754267
Generation 459 Best Solution: [182.90077853045256, 177.44974473622213,
174.08761558309686, 74.57244695006749] Fitness: 1.3304239971472385
Generation 460 Best Solution: [183.79021805073728, 177.44974473622213,
174.08761558309686, 74.57244695006749] Fitness: 1.5053039021995323
Generation 461 Best Solution: [183.79021805073728, 176.50936555638648,
175.12544315766232, 74.57244695006749] Fitness: 1.1360929257248609
Generation 462 Best Solution: [182.91047044159902, 175.76436458463834,
175.12544315766232, 75.90400119994271] Fitness: 2.018292837777653
Generation 463 Best Solution: [182.91047044159902, 175.58359220151945,
173.74700305846545, 75.90400119994271] Fitness: 1.5446594948542918
Generation 464 Best Solution: [182.90077853045256, 175.86535515721508,
175.12544315766232, 75.90400119994271] Fitness: 1.9629362347539723
Generation 465 Best Solution: [182.90077853045256, 175.86535515721508,
175.12544315766232, 74.73556340481738] Fitness: 0.9552423890834478
Generation 466 Best Solution: [182.90077853045256, 177.44974473622213,
175.12544315766232, 74.3225336500005] Fitness: 2.1802597716714
Generation 467 Best Solution: [182.91047044159902, 177.44974473622213,
175.12544315766232, 74.57471443488492] Fitness: 2.0598630886998994
Generation 468 Best Solution: [183.79021805073728, 177.15676190876647,
175.12544315766232, 75.17039910729386] Fitness: 1.8610420843833326
Generation 469 Best Solution: [183.79021805073728, 177.44974473622213,
173.86414465190416, 74.85560678307552] Fitness: 1.5449095910505664
Generation 470 Best Solution: [183.79021805073728, 177.15676190876647,
175.12544315766232, 74.85560678307552] Fitness: 1.7360789799092258
Generation 471 Best Solution: [182.95616702911798, 177.44974473622213,
175.12544315766232, 75.17039910729386] Fitness: 2.1498043654270496
Generation 472 Best Solution: [182.91047044159902, 177.44974473622213,
175.12544315766232, 74.85560678307552] Fitness: 2.044434028409322
Generation 473 Best Solution: [182.91047044159902, 176.99161039272155,
175.12544315766232, 74.85560678307552] Fitness: 1.339068073533702
Generation 474 Best Solution: [182.95616702911798, 177.44974473622213,
175.12544315766232, 75.17039910729386] Fitness: 2.1498043654270496
Generation 475 Best Solution: [183.1722699271689, 175.87457500184405,
175.12544315766232, 75.90400119994271] Fitness: 1.8843449544990944
Generation 476 Best Solution: [183.79021805073728, 175.65672964977978,
174.3429819359793, 75.90400119994271] Fitness: 1.5731933665445954
Generation 477 Best Solution: [182.98908099089653, 176.7680676946312,
175.12544315766232, 75.90400119994271] Fitness: 2.0783899822619674
Generation 478 Best Solution: [182.95616702911798, 177.44974473622213,
175.12544315766232, 74.49356253929786] Fitness: 2.0670166052378502
Generation 479 Best Solution: [182.95616702911798, 176.41437321235,
175.12544315766232, 75.90400119994271] Fitness: 1.8837658032876679
Generation 480 Best Solution: [183.33771695312257, 177.44974473622213,
174.92505641648415, 74.59425319259861] Fitness: 1.7262509148000416
Generation 481 Best Solution: [183.79021805073728, 177.44974473622213,
174.92505641648415, 74.59425319259861] Fitness: 1.9652115966249502
Generation 482 Best Solution: [183.79021805073728, 175.93312349644103,

175.12544315766232, 74.59425319259861] Fitness: 1.1171263957355686
Generation 483 Best Solution: [183.79021805073728, 176.76348656832997,
175.12544315766232, 74.59425319259861] Fitness: 1.2965255192221192
Generation 484 Best Solution: [183.33771695312257, 176.76348656832997,
175.12544315766232, 74.59425319259861] Fitness: 1.0575648373972106
Generation 485 Best Solution: [183.26877513725873, 176.00910209471712,
175.12544315766232, 75.90400119994271] Fitness: 1.8311210049543152
Generation 486 Best Solution: [183.26877513725873, 176.00910209471712,
175.12544315766232, 75.90400119994271] Fitness: 1.8311210049543152
Generation 487 Best Solution: [182.91047044159902, 175.93751232595676,
174.4964568182821, 75.90400119994271] Fitness: 1.2847358519906509
Generation 488 Best Solution: [182.91047044159902, 175.86954315453826,
175.12544315766232, 75.90400119994271] Fitness: 1.956222866292149
Generation 489 Best Solution: [182.91047044159902, 175.86954315453826,
175.12544315766232, 75.90400119994271] Fitness: 1.956222866292149
Generation 490 Best Solution: [183.79021805073728, 177.44974473622213,
175.12544315766232, 74.86926954080167] Fitness: 2.2261792005904906
Generation 491 Best Solution: [183.01049415869556, 175.7055695657227,
175.12544315766232, 75.90400119994271] Fitness: 2.0215402011312897
Generation 492 Best Solution: [183.01049415869556, 175.7055695657227,
175.12544315766232, 75.90400119994271] Fitness: 2.0215402011312897
Generation 493 Best Solution: [183.79021805073728, 177.44974473622213,
175.12544315766232, 75.34230368044678] Fitness: 2.4805510397212482
Generation 494 Best Solution: [183.79021805073728, 177.44974473622213,
174.92505641648415, 75.4188507717869] Fitness: 2.2844811985953353
Generation 495 Best Solution: [183.79021805073728, 175.79904366700165,
175.12544315766232, 75.90400119994271] Fitness: 2.175415271225992
Generation 496 Best Solution: [182.91047044159902, 177.44974473622213,
175.12544315766232, 75.64064119605098] Fitness: 2.636118675981593
Generation 497 Best Solution: [183.79021805073728, 176.23885854445788,
175.12544315766232, 75.90400119994271] Fitness: 2.05065721098431
Generation 498 Best Solution: [183.79021805073728, 175.953934832178,
175.12544315766232, 75.90400119994271] Fitness: 2.09799920866108
Generation 499 Best Solution: [183.79021805073728, 175.953934832178,
173.90784504008747, 75.90400119994271] Fitness: 1.402190573675814
Generation 500 Best Solution: [183.79021805073728, 177.44974473622213,
174.92505641648415, 75.52208616633179] Fitness: 2.3971167547811714
Optimal Solution for Simulation 12: [183.79021805073728, 177.44974473622213,
174.92505641648415, 75.52208616633179]

Simulation 13/32 with conditions: {'drought': True, 'flood': False,
'policy_change': False, 'social_event': True, 'economic_crisis': True}
Generation 1 Best Solution: [183.55673940175836, 176.1026306309099,
174.56478678866898, 75.38825121761823] Fitness: 0.5139656023828366
Generation 2 Best Solution: [183.81082341795033, 176.38155735203085,
174.19700673549403, 74.33450728122735] Fitness: 0.32300270404188136
Generation 3 Best Solution: [183.60388265782314, 176.38155735203085,
174.19700673549403, 74.33450728122735] Fitness: 0.20121989127333612

Generation 4 Best Solution: [182.92174910544364, 176.12370670875603,
174.31777344676317, 75.58985272091425] Fitness: 0.6827629127808138
Generation 5 Best Solution: [182.92174910544364, 176.12370670875603,
174.84787495463496, 74.41256493097093] Fitness: 0.49057847818928046
Generation 6 Best Solution: [183.81082341795033, 177.25276562959448,
174.19700673549403, 75.40908135879775] Fitness: 1.2301636408984062
Generation 7 Best Solution: [183.75135991814122, 177.25276562959448,
174.19700673549403, 75.14174814699692] Fitness: 0.9780705582420324
Generation 8 Best Solution: [183.55673940175836, 177.43719266912447,
174.19700673549403, 75.14174814699692] Fitness: 1.153644376621225
Generation 9 Best Solution: [182.94693499948843, 177.34323838352384,
174.19700673549403, 75.14174814699692] Fitness: 1.0348984499902727
Generation 10 Best Solution: [183.4706641153969, 177.25276562959448,
174.19700673549403, 75.58985272091425] Fitness: 1.2605355745628135
Generation 11 Best Solution: [182.8139903625393, 176.8683749832804,
174.84787495463496, 75.14174814699692] Fitness: 0.8566010904835026
Generation 12 Best Solution: [183.55673940175836, 177.25276562959448,
174.6688918120096, 74.93378997757712] Fitness: 0.9404580016870266
Generation 13 Best Solution: [182.94693499948843, 177.43719266912447,
175.01379530659824, 74.93378997757712] Fitness: 1.555094606234373
Generation 14 Best Solution: [182.82229148395214, 176.7968850350658,
174.6688918120096, 75.75121322309334] Fitness: 1.2804904021853423
Generation 15 Best Solution: [182.94693499948843, 176.29381542441246,
174.8664695856456, 75.90033715107464] Fitness: 1.4088767432347842
Generation 16 Best Solution: [183.3095254202846, 177.25276562959448,
173.9401273304353, 75.53436864818302] Fitness: 1.2152036477020678
Generation 17 Best Solution: [182.81296465400155, 177.25276562959448,
175.13221850215203, 74.9367536010871] Fitness: 1.5008151542688706
Generation 18 Best Solution: [182.81296465400155, 177.25276562959448,
174.84787495463496, 74.50323081023768] Fitness: 1.1781411088005107
Generation 19 Best Solution: [183.75135991814122, 177.43719266912447,
174.84787495463496, 74.50323081023768] Fitness: 1.4509251851853826
Generation 20 Best Solution: [183.75135991814122, 177.25276562959448,
174.84787495463496, 74.50323081023768] Fitness: 1.1752522040749398
Generation 21 Best Solution: [183.55673940175836, 177.25276562959448,
175.01379530659824, 75.09373233424974] Fitness: 1.3198285451394154
Generation 22 Best Solution: [183.55673940175836, 177.25276562959448,
174.8195255239788, 75.75121322309334] Fitness: 1.7775503204503273
Generation 23 Best Solution: [183.55673940175836, 177.25276562959448,
174.3470393690662, 75.90033715107464] Fitness: 1.752547315848228
Generation 24 Best Solution: [182.94693499948843, 176.76586106820793,
175.13221850215203, 75.90033715107464] Fitness: 1.8774380151521086
Generation 25 Best Solution: [182.94693499948843, 177.43719266912447,
175.13221850215203, 74.62461864259761] Fitness: 1.6695944408557886
Generation 26 Best Solution: [182.94693499948843, 177.25276562959448,
175.13221850215203, 74.46430183851027] Fitness: 1.430426797033855
Generation 27 Best Solution: [182.94693499948843, 177.25276562959448,
175.01379530659824, 74.46430183851027] Fitness: 1.285352096062339

Generation 28 Best Solution: [182.94693499948843, 175.82548994620223,
175.13221850215203, 75.90033715107464] Fitness: 1.844663373667249
Generation 29 Best Solution: [182.81296465400155, 177.25276562959448,
175.13221850215203, 74.58648544916808] Fitness: 1.4747456885934216
Generation 30 Best Solution: [183.5701265675741, 177.25276562959448,
174.09667579941257, 75.90033715107464] Fitness: 1.7475187606153288
Generation 31 Best Solution: [183.5909344142446, 177.25276562959448,
175.13221850215203, 74.4428658441026] Fitness: 1.424183740494939
Generation 32 Best Solution: [183.5909344142446, 177.25276562959448,
175.13221850215203, 74.56908283816618] Fitness: 1.3886848208288665
Generation 33 Best Solution: [182.94693499948843, 177.25276562959448,
175.13221850215203, 75.36477747004808] Fitness: 1.6890304369235283
Generation 34 Best Solution: [183.5909344142446, 177.25276562959448,
175.13221850215203, 75.36477747004808] Fitness: 1.6751959468482276
Generation 35 Best Solution: [182.94693499948843, 177.25276562959448,
175.13221850215203, 75.2471626393571] Fitness: 1.5912002527311293
Generation 36 Best Solution: [182.94693499948843, 177.25276562959448,
175.13221850215203, 75.2471626393571] Fitness: 1.5912002527311293
Generation 37 Best Solution: [183.400828417279, 177.25276562959448,
175.13221850215203, 75.2471626393571] Fitness: 1.521423309740324
Generation 38 Best Solution: [182.93134984988535, 176.64271815914174,
175.13221850215203, 75.90033715107464] Fitness: 1.8106244906263758
Generation 39 Best Solution: [182.94693499948843, 175.9925086637316,
173.9304611977256, 75.90033715107464] Fitness: 1.194087339573585
Generation 40 Best Solution: [183.86633440371855, 175.9925086637316,
173.9304611977256, 75.79630819686945] Fitness: 1.185450018119821
Generation 41 Best Solution: [183.86633440371855, 175.9925086637316,
173.9304611977256, 75.90033715107464] Fitness: 1.3504775960126951
Generation 42 Best Solution: [182.94693499948843, 175.9925086637316,
173.9304611977256, 75.90033715107464] Fitness: 1.194087339573585
Generation 43 Best Solution: [182.94693499948843, 175.9925086637316,
173.9304611977256, 75.90033715107464] Fitness: 1.194087339573585
Generation 44 Best Solution: [182.94693499948843, 175.9925086637316,
173.9304611977256, 75.90033715107464] Fitness: 1.194087339573585
Generation 45 Best Solution: [182.94693499948843, 175.9925086637316,
173.9304611977256, 75.90033715107464] Fitness: 1.194087339573585
Generation 46 Best Solution: [182.94693499948843, 177.25276562959448,
175.10240238895622, 74.41092411164522] Fitness: 1.4121521035391846
Generation 47 Best Solution: [182.94693499948843, 176.73540403375938,
175.13221850215203, 75.90033715107464] Fitness: 1.857097693043071
Generation 48 Best Solution: [182.86094165248932, 177.25276562959448,
175.13221850215203, 74.53397090671582] Fitness: 1.455685899620041
Generation 49 Best Solution: [182.94693499948843, 176.31152248248347,
175.13221850215203, 75.90033715107464] Fitness: 1.7079867739830277
Generation 50 Best Solution: [183.75641400150084, 177.25276562959448,
175.10240238895622, 74.38871997109487] Fitness: 1.4974712475000127
Generation 51 Best Solution: [183.09107348443678, 177.25276562959448,
175.13221850215203, 74.38871997109487] Fitness: 1.4068230129537376

Generation 52 Best Solution: [183.1031954555068, 176.50772843729132,
175.02076588648652, 74.71813759766543] Fitness: 0.5637708878416825
Generation 53 Best Solution: [183.09107348443678, 177.25276562959448,
175.10240238895622, 75.04555880543535] Fitness: 1.376696748398668
Generation 54 Best Solution: [183.75641400150084, 177.0397228847706,
175.02076588648652, 74.38871997109487] Fitness: 1.1392196727851938
Generation 55 Best Solution: [183.1031954555068, 177.41480766806097,
174.47792012194165, 75.3130469707915] Fitness: 1.2693058748621513
Generation 56 Best Solution: [183.1031954555068, 175.90918926684068,
175.02076588648652, 75.04421962604103] Fitness: 0.6940506059430467
Generation 57 Best Solution: [182.94693499948843, 175.90918926684068,
175.02076588648652, 75.04421962604103] Fitness: 0.7504260679273824
Generation 58 Best Solution: [183.1031954555068, 177.25276562959448,
175.02076588648652, 75.04421962604103] Fitness: 1.2741997726183918
Generation 59 Best Solution: [183.1031954555068, 176.16084919697286,
175.13221850215203, 75.90033715107464] Fitness: 1.65883087572514
Generation 60 Best Solution: [183.1031954555068, 177.25276562959448,
174.58319293261013, 75.90033715107464] Fitness: 1.813348853933575
Generation 61 Best Solution: [183.1031954555068, 177.25276562959448,
174.61147865195545, 74.95749954519295] Fitness: 0.883762762680058
Generation 62 Best Solution: [182.94693499948843, 177.41480766806097,
175.13221850215203, 74.93595509668445] Fitness: 1.6648689399676215
Generation 63 Best Solution: [183.1031954555068, 177.25276562959448,
175.13221850215203, 74.93595509668445] Fitness: 1.368803904702668
Generation 64 Best Solution: [183.1031954555068, 177.25276562959448,
175.13221850215203, 74.93595509668445] Fitness: 1.368803904702668
Generation 65 Best Solution: [183.0147786709923, 175.9589602013145,
175.13221850215203, 75.79630819686945] Fitness: 1.580718339975622
Generation 66 Best Solution: [183.0147786709923, 175.9589602013145,
175.13221850215203, 74.93595509668445] Fitness: 0.7930758238060441
Generation 67 Best Solution: [183.1031954555068, 175.9589602013145,
175.13221850215203, 74.75953516606324] Fitness: 0.7310795100305615
Generation 68 Best Solution: [183.1031954555068, 177.41480766806097,
175.13221850215203, 74.52864437748802] Fitness: 1.5947938194803555
Generation 69 Best Solution: [183.0147786709923, 176.4673751221773,
173.79447589295887, 75.79630819686945] Fitness: 1.0359725941225044
Generation 70 Best Solution: [183.0147786709923, 176.1747634855824,
173.79447589295887, 75.60833557116611] Fitness: 0.7554266615032799
Generation 71 Best Solution: [183.1031954555068, 177.41480766806097,
175.13221850215203, 74.52864437748802] Fitness: 1.5947938194803555
Generation 72 Best Solution: [183.1031954555068, 177.41480766806097,
175.13221850215203, 74.52864437748802] Fitness: 1.5947938194803555
Generation 73 Best Solution: [183.0147786709923, 177.25276562959448,
175.13221850215203, 74.6248405562019] Fitness: 1.3652462691514804
Generation 74 Best Solution: [183.0147786709923, 177.25276562959448,
173.9020160951771, 74.6248405562019] Fitness: 0.8096358591857875
Generation 75 Best Solution: [183.19979151822497, 177.41480766806097,
174.28060227132403, 74.5192708914028] Fitness: 0.9704265606621196

Generation 76 Best Solution: [183.1031954555068, 176.3523085861069,
175.13221850215203, 75.90033715107464] Fitness: 1.6550892975053317
Generation 77 Best Solution: [183.1031954555068, 177.41480766806097,
175.13221850215203, 74.86062143505409] Fitness: 1.588562076069517
Generation 78 Best Solution: [183.05607566817793, 175.83844983596984,
175.13221850215203, 75.80707561002988] Fitness: 1.6451556292256124
Generation 79 Best Solution: [183.62614163137022, 176.84283524010988,
175.13221850215203, 75.60833557116611] Fitness: 1.511604089082308
Generation 80 Best Solution: [183.7935501083607, 176.3523085861069,
175.13221850215203, 75.60833557116611] Fitness: 1.3875710525666038
Generation 81 Best Solution: [183.1031954555068, 176.12427996748727,
173.771941341753, 75.90033715107464] Fitness: 1.1740759980559659
Generation 82 Best Solution: [183.62614163137022, 177.41480766806097,
175.13221850215203, 74.52423218492471] Fitness: 1.6543288988225588
Generation 83 Best Solution: [183.7935501083607, 175.83844983596984,
175.13221850215203, 74.99052489940034] Fitness: 1.0045963011977885
Generation 84 Best Solution: [183.1425258658695, 177.41480766806097,
175.13221850215203, 74.99052489940034] Fitness: 1.619024469104602
Generation 85 Best Solution: [183.7935501083607, 176.2159417588361,
175.13221850215203, 75.90033715107464] Fitness: 1.8100467321184057
Generation 86 Best Solution: [183.1425258658695, 177.2590739243428,
175.13221850215203, 74.38747432925275] Fitness: 1.4040293022264418
Generation 87 Best Solution: [183.1425258658695, 177.41480766806097,
175.13221850215203, 75.01759133241734] Fitness: 1.6301686136044395
Generation 88 Best Solution: [182.86260122359224, 177.41480766806097,
175.13221850215203, 75.01759133241734] Fitness: 1.7399230077690913
Generation 89 Best Solution: [183.19979151822497, 177.41480766806097,
175.13221850215203, 75.01759133241734] Fitness: 1.6211482230205172
Generation 90 Best Solution: [183.34443290748496, 177.41480766806097,
175.13221850215203, 74.91313769546296] Fitness: 1.5812966621602995
Generation 91 Best Solution: [182.86260122359224, 177.41480766806097,
175.13221850215203, 74.40250743679364] Fitness: 1.738276179275421
Generation 92 Best Solution: [182.91548651021841, 177.2590739243428,
175.13221850215203, 74.67629536062441] Fitness: 1.4138754339355781
Generation 93 Best Solution: [182.91548651021841, 177.2590739243428,
175.13221850215203, 74.67629536062441] Fitness: 1.4138754339355781
Generation 94 Best Solution: [183.19979151822497, 175.75757601015488,
175.13221850215203, 74.67629536062441] Fitness: 0.8276240512869042
Generation 95 Best Solution: [183.1425258658695, 177.41480766806097,
174.1982666917907, 75.15518880730608] Fitness: 1.0882966333265434
Generation 96 Best Solution: [182.91548651021841, 177.2590739243428,
174.15178231948343, 75.90033715107464] Fitness: 1.7887928276378688
Generation 97 Best Solution: [183.19979151822497, 176.10439034559724,
175.13221850215203, 75.90033715107464] Fitness: 1.6518128384664132
Generation 98 Best Solution: [183.19979151822497, 176.70713772593922,
175.13221850215203, 75.90033715107464] Fitness: 1.7651411164368307
Generation 99 Best Solution: [182.91548651021841, 176.16835554556172,
175.13221850215203, 75.90033715107464] Fitness: 1.7295515748078705

Generation 100 Best Solution: [182.91548651021841, 176.2807290059263,
174.12296590288744, 75.90033715107464] Fitness: 1.1114000457637405
Generation 101 Best Solution: [183.1425258658695, 177.2590739243428,
174.62673518079333, 74.62386647276834] Fitness: 0.8543054592437349
Generation 102 Best Solution: [182.91548651021841, 177.41480766806097,
174.12296590288744, 74.53031600159561] Fitness: 1.0552596031194994
Generation 103 Best Solution: [183.1425258658695, 177.41480766806097,
175.13221850215203, 74.53031600159561] Fitness: 1.5855401816139911
Generation 104 Best Solution: [183.1425258658695, 177.41480766806097,
175.13221850215203, 74.60022877515281] Fitness: 1.5716449158210881
Generation 105 Best Solution: [183.1425258658695, 177.2590739243428,
175.13221850215203, 74.60022877515281] Fitness: 1.3406030362686105
Generation 106 Best Solution: [183.1425258658695, 177.41480766806097,
175.13221850215203, 74.67138760957995] Fitness: 1.564485291834028
Generation 107 Best Solution: [182.91548651021841, 177.41480766806097,
174.0763171958051, 74.43162214939262] Fitness: 1.0924778569002038
Generation 108 Best Solution: [183.1425258658695, 175.7453703613768,
174.607809829571, 75.90033715107464] Fitness: 1.3392122664918826
Generation 109 Best Solution: [182.91548651021841, 177.41480766806097,
175.13221850215203, 74.81913819426855] Fitness: 1.6527856984333962
Generation 110 Best Solution: [182.91548651021841, 177.41480766806097,
174.0763171958051, 75.5334867075378] Fitness: 1.5131861155963588
Generation 111 Best Solution: [183.1425258658695, 175.8792418505705,
175.13221850215203, 75.90033715107464] Fitness: 1.7481701398906453
Generation 112 Best Solution: [183.11119914112672, 176.54408150284485,
175.13221850215203, 75.90033715107464] Fitness: 1.7004945854793063
Generation 113 Best Solution: [182.91548651021841, 175.8792418505705,
175.13221850215203, 75.90033715107464] Fitness: 1.8288362553965651
Generation 114 Best Solution: [182.91548651021841, 177.41480766806097,
175.13221850215203, 75.35066618811642] Fitness: 1.9314189152155592
Generation 115 Best Solution: [183.1425258658695, 177.41480766806097,
175.13221850215203, 75.35066618811642] Fitness: 1.8507527997096394
Generation 116 Best Solution: [183.59883982603196, 177.41480766806097,
175.13221850215203, 75.35066618811642] Fitness: 1.9055471996802427
Generation 117 Best Solution: [183.59883982603196, 177.41480766806097,
175.13221850215203, 75.35066618811642] Fitness: 1.9055471996802427
Generation 118 Best Solution: [183.23874897608954, 177.41480766806097,
175.13221850215203, 75.35066618811642] Fitness: 1.8382036221126328
Generation 119 Best Solution: [183.59883982603196, 175.94509653101835,
175.13221850215203, 75.90033715107464] Fitness: 1.7701214194857273
Generation 120 Best Solution: [183.23874897608954, 176.57739314522402,
175.13221850215203, 75.35066618811642] Fitness: 0.9929533383095506
Generation 121 Best Solution: [183.23874897608954, 177.41480766806097,
174.12260204603882, 75.90033715107464] Fitness: 1.9288648363213743
Generation 122 Best Solution: [183.23874897608954, 177.41480766806097,
174.12260204603882, 75.55925874298407] Fitness: 1.4440361999916749
Generation 123 Best Solution: [183.1425258658695, 177.41480766806097,
174.12260204603882, 75.55925874298407] Fitness: 1.4565853775886815

Generation 124 Best Solution: [183.1425258658695, 177.37143122140432,
175.13221850215203, 74.38521662645663] Fitness: 1.5683395812537735
Generation 125 Best Solution: [183.1425258658695, 177.41480766806097,
173.8231390476367, 74.87406238082686] Fitness: 1.063297276769463
Generation 126 Best Solution: [183.10264688005134, 176.19810821534185,
175.13221850215203, 75.90033715107464] Fitness: 1.6542444683263349
Generation 127 Best Solution: [183.1425258658695, 176.23920531167408,
174.38177931255677, 75.90033715107464] Fitness: 1.0529090769565652
Generation 128 Best Solution: [183.23874897608954, 177.37143122140432,
175.13221850215203, 74.97012951856242] Fitness: 1.5310088732516889
Generation 129 Best Solution: [183.23874897608954, 177.41480766806097,
175.13221850215203, 74.97012951856242] Fitness: 1.5987512320751087
Generation 130 Best Solution: [182.90942724323273, 176.57273993546778,
175.13221850215203, 74.97012951856242] Fitness: 0.8478806530062472
Generation 131 Best Solution: [183.23874897608954, 177.41480766806097,
174.9078543244067, 75.66081113120966] Fitness: 1.9240558880838994
Generation 132 Best Solution: [183.23874897608954, 177.41480766806097,
174.9078543244067, 74.34124090628171] Fitness: 1.3863119393007755
Generation 133 Best Solution: [183.23874897608954, 176.87535076374593,
174.9078543244067, 75.90033715107464] Fitness: 1.625132910704957
Generation 134 Best Solution: [183.175880547706, 177.41480766806097,
175.13221850215203, 74.60204011539726] Fitness: 1.5655664996331957
Generation 135 Best Solution: [183.175880547706, 176.00894938140766,
174.9078543244067, 75.90033715107464] Fitness: 1.4251121748081206
Generation 136 Best Solution: [183.175880547706, 177.41480766806097,
175.13221850215203, 74.48791663205296] Fitness: 1.5914710022924807
Generation 137 Best Solution: [183.175880547706, 177.41480766806097,
174.22806546800786, 75.90033715107464] Fitness: 1.9330863950464927
Generation 138 Best Solution: [183.23874897608954, 176.05141772047068,
175.13221850215203, 75.90033715107464] Fitness: 1.6624877878528004
Generation 139 Best Solution: [183.10264688005134, 176.18435455911387,
175.13221850215203, 75.90033715107464] Fitness: 1.6557637065270516
Generation 140 Best Solution: [183.23874897608954, 177.41480766806097,
174.8097216784435, 75.34490754260426] Fitness: 1.4837859283340293
Generation 141 Best Solution: [183.23874897608954, 177.41480766806097,
174.53325409565528, 75.64741533627279] Fitness: 1.631212072761942
Generation 142 Best Solution: [183.175880547706, 177.41480766806097,
174.9078543244067, 75.68369071539229] Fitness: 1.9614867837850092
Generation 143 Best Solution: [183.23874897608954, 177.41480766806097,
173.76890427865715, 75.45137873080026] Fitness: 1.4458066842173731
Generation 144 Best Solution: [183.23874897608954, 177.41480766806097,
173.76890427865715, 75.45137873080026] Fitness: 1.4458066842173731
Generation 145 Best Solution: [183.23874897608954, 177.41480766806097,
173.76890427865715, 75.45137873080026] Fitness: 1.4458066842173731
Generation 146 Best Solution: [183.175880547706, 177.41480766806097,
173.76890427865715, 74.53487893826629] Fitness: 1.0891300940806299
Generation 147 Best Solution: [183.23874897608954, 175.87962183544852,
175.13221850215203, 75.83088085564081] Fitness: 1.6235608601980887

Generation 148 Best Solution: [183.23874897608954, 176.9843185932799,
174.75478854191053, 75.90033715107464] Fitness: 1.5890298781255996
Generation 149 Best Solution: [183.23874897608954, 177.00915360020352,
175.13221850215203, 75.45137873080026] Fitness: 1.4040035840951628
Generation 150 Best Solution: [183.175880547706, 175.87962183544852,
174.0849475442446, 75.90033715107464] Fitness: 1.135891351583927
Generation 151 Best Solution: [183.23874897608954, 177.41480766806097,
174.77515843006253, 75.45137873080026] Fitness: 1.5571510396837809
Generation 152 Best Solution: [183.56482026205083, 177.41480766806097,
174.77515843006253, 75.45137873080026] Fitness: 1.6104156038221356
Generation 153 Best Solution: [183.23874897608954, 175.95259541304398,
175.13221850215203, 75.90033715107464] Fitness: 1.6994206482220338
Generation 154 Best Solution: [183.23874897608954, 176.6475524196326,
175.13221850215203, 75.90033715107464] Fitness: 1.7278947083900114
Generation 155 Best Solution: [183.23874897608954, 175.87962183544852,
174.98690640563015, 75.90033715107464] Fitness: 1.560117225981493
Generation 156 Best Solution: [183.23874897608954, 177.41480766806097,
174.98690640563015, 74.74848320469802] Fitness: 1.3768333823072387
Generation 157 Best Solution: [183.23874897608954, 177.12642855366911,
174.98690640563015, 74.74848320469802] Fitness: 0.9756130307118034
Generation 158 Best Solution: [183.56482026205083, 177.41480766806097,
174.98690640563015, 74.74848320469802] Fitness: 1.4300979464455934
Generation 159 Best Solution: [183.23874897608954, 177.41480766806097,
175.13221850215203, 74.6254144401375] Fitness: 1.555756219181642
Generation 160 Best Solution: [183.22864047215978, 177.41480766806097,
174.98690640563015, 74.6254144401375] Fitness: 1.381172064437335
Generation 161 Best Solution: [183.23874897608954, 177.41480766806097,
174.77515843006253, 74.79838036451441] Fitness: 1.1785214205360404
Generation 162 Best Solution: [182.95929137142568, 177.41480766806097,
174.98690640563015, 74.79838036451441] Fitness: 1.4534350363220903
Generation 163 Best Solution: [183.23874897608954, 176.4950862883903,
174.77515843006253, 75.90033715107464] Fitness: 1.2859667592776811
Generation 164 Best Solution: [183.23874897608954, 177.1698792551088,
174.70997551965877, 75.90033715107464] Fitness: 1.763577481134126
Generation 165 Best Solution: [183.8161149384969, 177.41480766806097,
174.77515843006253, 74.99172079120957] Fitness: 1.424049191042452
Generation 166 Best Solution: [183.22864047215978, 177.41480766806097,
175.13221850215203, 74.83162671025167] Fitness: 1.5623205595350438
Generation 167 Best Solution: [183.22864047215978, 176.59889114394554,
174.18927762148556, 75.90033715107464] Fitness: 1.0904327992158296
Generation 168 Best Solution: [183.47961229250603, 177.41480766806097,
174.18927762148556, 74.70676579516335] Fitness: 0.9617640087389796
Generation 169 Best Solution: [183.39469958412613, 177.33834564779795,
174.77515843006253, 75.90033715107464] Fitness: 2.050999309923219
Generation 170 Best Solution: [183.47961229250603, 177.41480766806097,
174.28742866736383, 75.90033715107464] Fitness: 1.9568033366669217
Generation 171 Best Solution: [183.23874897608954, 177.41480766806097,
173.95908520134688, 75.90033715107464] Fitness: 1.963366919286465

Generation 172 Best Solution: [183.47961229250603, 177.41480766806097,
174.16960596628007, 75.90033715107464] Fitness: 1.9508991174617023
Generation 173 Best Solution: [183.47961229250603, 177.41480766806097,
174.91418038739565, 75.51291643697863] Fitness: 1.7746333008604276
Generation 174 Best Solution: [183.47961229250603, 176.46210337868948,
174.98240401762945, 75.90033715107464] Fitness: 1.499375090958474
Generation 175 Best Solution: [183.47961229250603, 177.41480766806097,
174.91418038739565, 75.12289061156626] Fitness: 1.4437402612109027
Generation 176 Best Solution: [183.47047567743056, 177.41480766806097,
174.3577142878018, 75.90033715107464] Fitness: 1.96709910763138
Generation 177 Best Solution: [183.47961229250603, 177.33834564779795,
174.3577142878018, 75.90033715107464] Fitness: 1.8518695716925442
Generation 178 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.43563045969375] Fitness: 1.617638208537501
Generation 179 Best Solution: [183.8161149384969, 177.41480766806097,
173.83755776990063, 74.43563045969375] Fitness: 1.271228211765865
Generation 180 Best Solution: [183.8161149384969, 177.41480766806097,
173.83755776990063, 74.43563045969375] Fitness: 1.271228211765865
Generation 181 Best Solution: [183.8161149384969, 177.33834564779795,
174.3577142878018, 74.75228491138746] Fitness: 1.0346243865873794
Generation 182 Best Solution: [183.8161149384969, 175.87686196457304,
174.3577142878018, 75.90033715107464] Fitness: 1.336844673008855
Generation 183 Best Solution: [183.8161149384969, 175.87686196457304,
174.98240401762945, 74.83473505493873] Fitness: 0.7742797262365801
Generation 184 Best Solution: [183.8161149384969, 175.87686196457304,
174.98240401762945, 74.83473505493873] Fitness: 0.7742797262365801
Generation 185 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.92585738652325] Fitness: 1.5989628121162145
Generation 186 Best Solution: [183.8161149384969, 176.70667672751858,
174.98240401762945, 75.90033715107464] Fitness: 1.7763190223490737
Generation 187 Best Solution: [183.8161149384969, 177.33834564779795,
173.78950168726377, 75.47153336885046] Fitness: 1.5326794259136902
Generation 188 Best Solution: [183.8161149384969, 177.00864183277955,
174.3577142878018, 75.47153336885046] Fitness: 1.024526093431939
Generation 189 Best Solution: [183.19529615709223, 177.41480766806097,
173.78950168726377, 74.29001165422157] Fitness: 1.1753512217434312
Generation 190 Best Solution: [183.8161149384969, 176.1649075056011,
174.98240401762945, 75.5752093638096] Fitness: 1.1860585425877799
Generation 191 Best Solution: [183.8161149384969, 176.48965086219334,
174.98240401762945, 74.61786975443668] Fitness: 0.6943067895972616
Generation 192 Best Solution: [183.8161149384969, 177.41480766806097,
175.11386600210378, 74.87205255923375] Fitness: 1.741176241890668
Generation 193 Best Solution: [183.8161149384969, 177.41480766806097,
174.31676405250388, 74.87205255923375] Fitness: 1.1613684122642287
Generation 194 Best Solution: [183.8161149384969, 177.33834564779795,
174.5629044117241, 75.90033715107464] Fitness: 2.098501522397535
Generation 195 Best Solution: [183.8161149384969, 176.08527835459327,
174.98240401762945, 75.90033715107464] Fitness: 1.6679449492301828

Generation 196 Best Solution: [183.8161149384969, 177.41480766806097,
174.5629044117241, 74.76657760441856] Fitness: 1.2297526458412502
Generation 197 Best Solution: [183.8161149384969, 177.33834564779795,
174.5629044117241, 75.90033715107464] Fitness: 2.098501522397535
Generation 198 Best Solution: [183.8161149384969, 177.33834564779795,
174.5629044117241, 75.90033715107464] Fitness: 2.098501522397535
Generation 199 Best Solution: [183.8161149384969, 177.33834564779795,
174.5629044117241, 75.90033715107464] Fitness: 2.098501522397535
Generation 200 Best Solution: [183.8161149384969, 177.17532621613316,
174.98240401762945, 75.6166153371648] Fitness: 1.7988081082291898
Generation 201 Best Solution: [183.8161149384969, 177.36673722089694,
174.98240401762945, 75.26502714896402] Fitness: 1.7068171664036087
Generation 202 Best Solution: [183.8161149384969, 175.77314854671633,
174.98240401762945, 75.90033715107464] Fitness: 1.8162089093054365
Generation 203 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.50552219257982] Fitness: 1.5945388158959901
Generation 204 Best Solution: [183.8161149384969, 177.33834564779795,
174.98240401762945, 74.97119522210954] Fitness: 1.4964737362641136
Generation 205 Best Solution: [183.8161149384969, 177.33834564779795,
174.98240401762945, 74.97119522210954] Fitness: 1.4964737362641136
Generation 206 Best Solution: [183.8161149384969, 175.77314854671633,
174.98240401762945, 74.97119522210954] Fitness: 0.87557317267291
Generation 207 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.73867070223447] Fitness: 1.5666328279658748
Generation 208 Best Solution: [183.8161149384969, 177.33834564779795,
174.98240401762945, 74.97018146015942] Fitness: 1.496103403429695
Generation 209 Best Solution: [183.8161149384969, 175.77314854671633,
174.98240401762945, 75.90033715107464] Fitness: 1.8162089093054365
Generation 210 Best Solution: [183.8161149384969, 177.41480766806097,
174.94939111457597, 74.96082292434166] Fitness: 1.5748825693909194
Generation 211 Best Solution: [183.8161149384969, 175.77314854671633,
174.98240401762945, 75.6401212552948] Fitness: 1.4316852270191864
Generation 212 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 75.13677055161938] Fitness: 1.6938008063993264
Generation 213 Best Solution: [183.590971151119, 177.41480766806097,
173.75686155974344, 75.90033715107464] Fitness: 2.1214315093539415
Generation 214 Best Solution: [183.590971151119, 177.41480766806097,
174.98240401762945, 74.8461791888461] Fitness: 1.4479404848850068
Generation 215 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.8461791888461] Fitness: 1.579242163014888
Generation 216 Best Solution: [183.8161149384969, 175.71881151988453,
174.98240401762945, 75.90033715107464] Fitness: 1.8558717234031414
Generation 217 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 75.13958025578508] Fitness: 1.6954819394925338
Generation 218 Best Solution: [183.8161149384969, 176.7179336696824,
174.98240401762945, 75.90033715107464] Fitness: 1.7832365436715731
Generation 219 Best Solution: [183.8161149384969, 175.71881151988453,
173.85399518323285, 75.67895475684858] Fitness: 1.1684127721127615

Generation 220 Best Solution: [183.8161149384969, 175.71881151988453,
174.75127078879342, 75.90033715107464] Fitness: 1.6390205493799597
Generation 221 Best Solution: [183.8161149384969, 176.270661015146,
174.98240401762945, 75.90033715107464] Fitness: 1.6440469473752684
Generation 222 Best Solution: [183.590971151119, 177.41480766806097,
174.98240401762945, 74.98153168032034] Fitness: 1.4866829223012428
Generation 223 Best Solution: [183.590971151119, 177.41480766806097,
174.98240401762945, 74.98153168032034] Fitness: 1.4866829223012428
Generation 224 Best Solution: [183.8161149384969, 175.67710865257393,
174.32778022955364, 75.90033715107464] Fitness: 1.4677784512401977
Generation 225 Best Solution: [183.590971151119, 175.6289629251353,
174.98240401762945, 74.98153168032034] Fitness: 0.8623881330341477
Generation 226 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.98153168032034] Fitness: 1.617984600431124
Generation 227 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.83522001780386] Fitness: 1.577220726140183
Generation 228 Best Solution: [183.8161149384969, 176.00157257516204,
174.98240401762945, 75.90033715107464] Fitness: 1.6944046559068138
Generation 229 Best Solution: [183.8161149384969, 175.67710865257393,
174.61069517807374, 75.90033715107464] Fitness: 1.5767081670357048
Generation 230 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.74477454831784] Fitness: 1.5669181493341562
Generation 231 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.74477454831784] Fitness: 1.5669181493341562
Generation 232 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.74477454831784] Fitness: 1.5669181493341562
Generation 233 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.74477454831784] Fitness: 1.5669181493341562
Generation 234 Best Solution: [183.8161149384969, 175.80319057929552,
174.98240401762945, 74.81998639432045] Fitness: 0.8159691002319945
Generation 235 Best Solution: [183.70618359651533, 177.41480766806097,
174.60360003542183, 74.88346223911829] Fitness: 1.1979650849779562
Generation 236 Best Solution: [183.8161149384969, 177.41480766806097,
174.26192352261802, 74.88346223911829] Fitness: 1.156672045998695
Generation 237 Best Solution: [183.12677338619739, 175.95174820871682,
174.39823690215016, 75.90033715107464] Fitness: 1.1307618197520806
Generation 238 Best Solution: [183.8161149384969, 176.51105655888048,
174.98240401762945, 75.34736075092863] Fitness: 0.9797407667170953
Generation 239 Best Solution: [182.92450273004712, 175.85450391926105,
174.98240401762945, 75.90033715107464] Fitness: 1.6579025585242144
Generation 240 Best Solution: [183.8161149384969, 175.85450391926105,
174.39823690215016, 75.90033715107464] Fitness: 1.359898125376985
Generation 241 Best Solution: [183.8161149384969, 177.37042090657437,
174.39823690215016, 74.29165344975944] Fitness: 1.212748922783673
Generation 242 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.62828630214075] Fitness: 1.5704179950969097
Generation 243 Best Solution: [183.8161149384969, 177.41480766806097,
173.80940970471877, 74.62828630214075] Fitness: 1.2384596071118759

Generation 244 Best Solution: [183.8161149384969, 177.41480766806097,
173.80940970471877, 74.62828630214075] Fitness: 1.2384596071118759
Generation 245 Best Solution: [183.8161149384969, 177.41480766806097,
173.80940970471877, 74.62828630214075] Fitness: 1.2384596071118759
Generation 246 Best Solution: [183.8161149384969, 177.41480766806097,
173.80940970471877, 74.62828630214075] Fitness: 1.2384596071118759
Generation 247 Best Solution: [183.8161149384969, 176.5623416708806,
174.54864010100218, 75.90033715107464] Fitness: 1.3574285329385876
Generation 248 Best Solution: [183.8161149384969, 176.5623416708806,
174.54864010100218, 75.90033715107464] Fitness: 1.3574285329385876
Generation 249 Best Solution: [183.8161149384969, 176.0392504542058,
174.2685181699172, 75.90033715107464] Fitness: 1.2512570929299085
Generation 250 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.74646334945808] Fitness: 1.5670062464800822
Generation 251 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 75.37230467405672] Fitness: 1.8728600276215968
Generation 252 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.61898633733357] Fitness: 1.5715112010141963
Generation 253 Best Solution: [183.8161149384969, 176.73772524707232,
174.98240401762945, 75.90033715107464] Fitness: 1.795826174865999
Generation 254 Best Solution: [183.4765304726569, 177.41480766806097,
174.98240401762945, 75.41637446453694] Fitness: 1.74392717758061
Generation 255 Best Solution: [183.39969702496583, 177.30684789523468,
174.98240401762945, 75.41637446453694] Fitness: 1.5637389024415818
Generation 256 Best Solution: [183.39712794685016, 177.41480766806097,
174.98240401762945, 75.41637446453694] Fitness: 1.7270850915641938
Generation 257 Best Solution: [183.8161149384969, 176.26443908166686,
174.98240401762945, 75.41637446453694] Fitness: 1.0042457335564041
Generation 258 Best Solution: [183.8161149384969, 176.26443908166686,
174.98240401762945, 75.41637446453694] Fitness: 1.0042457335564041
Generation 259 Best Solution: [183.8161149384969, 176.26443908166686,
174.98240401762945, 75.48915478307481] Fitness: 1.0796476243619817
Generation 260 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.68287690226389] Fitness: 1.5664272116279931
Generation 261 Best Solution: [183.8161149384969, 177.41480766806097,
175.08085856103548, 74.68287690226389] Fitness: 1.681371612252613
Generation 262 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.80065864437789] Fitness: 1.5719402612993407
Generation 263 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.80065864437789] Fitness: 1.5719402612993407
Generation 264 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.49379345142782] Fitness: 1.597940635122988
Generation 265 Best Solution: [183.39969702496583, 177.41480766806097,
174.45566024274981, 75.90033715107464] Fitness: 1.9814457231252185
Generation 266 Best Solution: [183.8161149384969, 177.41480766806097,
174.2493180156556, 75.90033715107464] Fitness: 2.122957490978691
Generation 267 Best Solution: [183.8161149384969, 177.41480766806097,
174.2493180156556, 75.90033715107464] Fitness: 2.122957490978691

Generation 268 Best Solution: [183.8161149384969, 176.1524889273615,
174.9475568331385, 75.90033715107464] Fitness: 1.6163033609845505
Generation 269 Best Solution: [183.39969702496583, 175.82834920330626,
174.98240401762945, 75.90033715107464] Fitness: 1.5926797726229416
Generation 270 Best Solution: [183.8161149384969, 176.1524889273615,
174.9475568331385, 75.90033715107464] Fitness: 1.6163033609845505
Generation 271 Best Solution: [183.8161149384969, 176.1524889273615,
174.9475568331385, 75.90033715107464] Fitness: 1.6163033609845505
Generation 272 Best Solution: [183.8161149384969, 176.4904940284817,
174.98240401762945, 75.90033715107464] Fitness: 1.6776764349498001
Generation 273 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 75.23183669750023] Fitness: 1.7567830207822759
Generation 274 Best Solution: [183.8161149384969, 177.41480766806097,
174.06090704834654, 75.23183669750023] Fitness: 1.3347783932050432
Generation 275 Best Solution: [183.8161149384969, 177.41480766806097,
174.06090704834654, 74.39664519385775] Fitness: 1.2114710732299734
Generation 276 Best Solution: [183.39969702496583, 177.41480766806097,
174.98240401762945, 74.73575080543561] Fitness: 1.3790724470618765
Generation 277 Best Solution: [183.39969702496583, 177.41480766806097,
174.55105875479578, 74.34111856411245] Fitness: 1.127626577448288
Generation 278 Best Solution: [183.8161149384969, 177.41480766806097,
174.06090704834654, 74.73575080543561] Fitness: 1.144510040528447
Generation 279 Best Solution: [183.39969702496583, 177.41480766806097,
174.98240401762945, 75.18569616369201] Fitness: 1.5372018601152204
Generation 280 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.9362237230087] Fitness: 1.6021778627576673
Generation 281 Best Solution: [183.6801863160571, 177.41480766806097,
175.05119717086805, 74.62808596865187] Fitness: 1.5616275525084629
Generation 282 Best Solution: [183.8161149384969, 176.7932091123748,
174.98240401762945, 75.26220867025715] Fitness: 1.058817767330103
Generation 283 Best Solution: [183.6801863160571, 176.7932091123748,
174.98240401762945, 75.26220867025715] Fitness: 0.9711094095743966
Generation 284 Best Solution: [183.8161149384969, 176.7932091123748,
174.98240401762945, 75.26220867025715] Fitness: 1.058817767330103
Generation 285 Best Solution: [183.6801863160571, 177.41480766806097,
174.98240401762945, 75.90033715107464] Fitness: 2.4670543994240997
Generation 286 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 75.12513024082415] Fitness: 1.6869530581918308
Generation 287 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 75.35651803343895] Fitness: 1.8584453398873806
Generation 288 Best Solution: [183.8161149384969, 176.09272060379976,
174.98240401762945, 75.90033715107464] Fitness: 1.6660643198206824
Generation 289 Best Solution: [183.8161149384969, 176.14774568206013,
174.98240401762945, 75.90033715107464] Fitness: 1.6545508259702442
Generation 290 Best Solution: [183.8161149384969, 176.14774568206013,
174.98240401762945, 75.90033715107464] Fitness: 1.6545508259702442
Generation 291 Best Solution: [183.16774339159366, 176.14774568206013,
174.98240401762945, 75.90033715107464] Fitness: 1.4673199180653813

Generation 292 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.81171225941927] Fitness: 1.5734483273669169
Generation 293 Best Solution: [183.30369697751055, 177.41480766806097,
174.01557446131045, 75.90033715107464] Fitness: 1.9460533768637533
Generation 294 Best Solution: [183.73080826248884, 177.41480766806097,
174.01557446131045, 75.90033715107464] Fitness: 2.084438990110023
Generation 295 Best Solution: [183.30369697751055, 177.41480766806097,
174.98240401762945, 75.90033715107464] Fitness: 2.3583286539604877
Generation 296 Best Solution: [183.30369697751055, 175.92928907007826,
174.98240401762945, 75.90033715107464] Fitness: 1.5286634290313978
Generation 297 Best Solution: [183.03297943754126, 177.41480766806097,
174.98240401762945, 74.96441949940929] Fitness: 1.4589501824936788
Generation 298 Best Solution: [183.3523026764471, 177.41480766806097,
174.98240401762945, 74.96441949940929] Fitness: 1.4181950106005017
Generation 299 Best Solution: [183.3523026764471, 176.57504915134064,
174.3361164990538, 75.90033715107464] Fitness: 1.0953291813860568
Generation 300 Best Solution: [183.03297943754126, 176.57504915134064,
174.3361164990538, 75.90033715107464] Fitness: 1.1360843532792342
Generation 301 Best Solution: [183.3523026764471, 177.41480766806097,
174.98240401762945, 74.96441949940929] Fitness: 1.4181950106005017
Generation 302 Best Solution: [183.03297943754126, 175.80605717177087,
173.7978330320715, 75.90033715107464] Fitness: 1.3157612366736497
Generation 303 Best Solution: [183.0139228197653, 177.41480766806097,
174.98240401762945, 74.4868296983683] Fitness: 1.4542401950617214
Generation 304 Best Solution: [183.3523026764471, 177.41480766806097,
174.98240401762945, 74.4868296983683] Fitness: 1.4065670037113227
Generation 305 Best Solution: [183.8161149384969, 176.28468011092707,
174.98240401762945, 75.90033715107464] Fitness: 1.64418436987689
Generation 306 Best Solution: [183.8161149384969, 177.41480766806097,
174.30778246960764, 75.4296016341299] Fitness: 1.5032847823325564
Generation 307 Best Solution: [183.48849294829506, 177.41480766806097,
174.30778246960764, 75.4296016341299] Fitness: 1.3355747125054254
Generation 308 Best Solution: [182.96834011486584, 176.28468011092707,
174.30778246960764, 75.90033715107464] Fitness: 1.0921651539887258
Generation 309 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 75.1137935824055] Fitness: 1.6804651471491088
Generation 310 Best Solution: [183.49791102822414, 177.41480766806097,
174.30778246960764, 75.90033715107464] Fitness: 1.9649832537980654
Generation 311 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.84020755597618] Fitness: 1.5781199665176688
Generation 312 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 75.2710431068482] Fitness: 1.7864197880302675
Generation 313 Best Solution: [183.8161149384969, 177.41480766806097,
174.98240401762945, 74.76963287840788] Fitness: 1.5686155613093469
Generation 314 Best Solution: [183.03297943754126, 177.41480766806097,
174.20962340424612, 75.90033715107464] Fitness: 1.968186910596911
Generation 315 Best Solution: [183.03297943754126, 176.9905172589431,
174.98240401762945, 75.90033715107464] Fitness: 1.851836324641603

Generation 316 Best Solution: [183.60094767284642, 176.53509919434987,
174.98240401762945, 75.90033715107464] Fitness: 1.5657286817616043
Generation 317 Best Solution: [183.60094767284642, 176.35505803497026,
174.98240401762945, 75.90033715107464] Fitness: 1.5220294385736242
Generation 318 Best Solution: [183.50269112753904, 176.53509919434987,
174.98240401762945, 75.90033715107464] Fitness: 1.5291677595706588
Generation 319 Best Solution: [183.03297943754126, 176.86995842273615,
174.98240401762945, 75.90033715107464] Fitness: 1.7411968026398128
Generation 320 Best Solution: [183.03297943754126, 176.86995842273615,
174.98240401762945, 75.64378510315498] Fitness: 1.3614333351378833
Generation 321 Best Solution: [183.60094767284642, 176.14351211347272,
174.98240401762945, 75.90033715107464] Fitness: 1.528310307114153
Generation 322 Best Solution: [183.60094767284642, 177.41480766806097,
174.98240401762945, 74.4571655185321] Fitness: 1.4828196219930618
Generation 323 Best Solution: [183.06596133067953, 177.23606130109334,
174.98240401762945, 74.36086283145117] Fitness: 1.2240439000432644
Generation 324 Best Solution: [183.03297943754126, 177.41480766806097,
174.98240401762945, 74.4571655185321] Fitness: 1.4570676022579772
Generation 325 Best Solution: [183.60094767284642, 176.69776114840414,
174.72015807419402, 75.90033715107464] Fitness: 1.4036232041904775
Generation 326 Best Solution: [183.03297943754126, 176.12510288824149,
174.98240401762945, 74.99687186513962] Fitness: 0.5752705921076258
Generation 327 Best Solution: [183.03297943754126, 177.41480766806097,
174.02084761360143, 75.63020534225551] Fitness: 1.5911668585533387
Generation 328 Best Solution: [183.60094767284642, 177.41480766806097,
174.8345255904223, 74.93019315921185] Fitness: 1.3260071751666598
Generation 329 Best Solution: [183.03297943754126, 176.45152586312415,
174.98240401762945, 75.90033715107464] Fitness: 1.514083460497511
Generation 330 Best Solution: [183.03297943754126, 176.45152586312415,
174.98240401762945, 75.90033715107464] Fitness: 1.514083460497511
Generation 331 Best Solution: [183.03297943754126, 177.41480766806097,
174.3767537645896, 75.37417532537769] Fitness: 1.3114282751591215
Generation 332 Best Solution: [183.03297943754126, 177.41480766806097,
174.3767537645896, 75.37417532537769] Fitness: 1.3114282751591215
Generation 333 Best Solution: [183.03297943754126, 177.41480766806097,
174.3767537645896, 75.37417532537769] Fitness: 1.3114282751591215
Generation 334 Best Solution: [183.03297943754126, 177.41480766806097,
174.98240401762945, 74.86318478473483] Fitness: 1.4299809069559761
Generation 335 Best Solution: [183.60094767284642, 176.01285648575518,
174.98240401762945, 75.90033715107464] Fitness: 1.563292513037414
Generation 336 Best Solution: [183.18992398109128, 175.82326098884874,
174.66855525649478, 75.90033715107464] Fitness: 1.31638673181049
Generation 337 Best Solution: [183.18992398109128, 175.82326098884874,
174.98240401762945, 75.90033715107464] Fitness: 1.592783079518893
Generation 338 Best Solution: [183.60094767284642, 177.41480766806097,
174.06174304797983, 74.8682954001079] Fitness: 1.0346964353263124
Generation 339 Best Solution: [183.60094767284642, 177.41480766806097,
173.8573571216004, 75.90033715107464] Fitness: 2.071871130658258

Generation 340 Best Solution: [183.60094767284642, 177.41480766806097,
174.98240401762945, 75.09506740102215] Fitness: 1.5431631074679488
Generation 341 Best Solution: [183.03297943754126, 177.41480766806097,
174.98240401762945, 75.40803562122862] Fitness: 1.7540374919066728
Generation 342 Best Solution: [183.60094767284642, 177.41480766806097,
174.98240401762945, 75.40803562122862] Fitness: 1.7797895116417575
Generation 343 Best Solution: [183.1803310894846, 177.41480766806097,
174.98240401762945, 75.40803562122862] Fitness: 1.7176029920745954
Generation 344 Best Solution: [183.60094767284642, 177.41480766806097,
174.98240401762945, 75.40803562122862] Fitness: 1.7797895116417575
Generation 345 Best Solution: [183.1803310894846, 177.41480766806097,
174.88718984105193, 74.55042474105373] Fitness: 1.2957874179227336
Generation 346 Best Solution: [183.1803310894846, 176.24980420974057,
174.98240401762945, 75.90033715107464] Fitness: 1.4551852271196726
Generation 347 Best Solution: [183.1803310894846, 176.24980420974057,
174.88718984105193, 75.90033715107464] Fitness: 1.3568517441239287
Generation 348 Best Solution: [183.60094767284642, 176.24980420974057,
174.88718984105193, 75.90033715107464] Fitness: 1.4190382636910908
Generation 349 Best Solution: [183.1803310894846, 176.0705532357235,
174.88718984105193, 75.90033715107464] Fitness: 1.3843962349438823
Generation 350 Best Solution: [183.1803310894846, 177.41480766806097,
174.98240401762945, 75.05368407297242] Fitness: 1.4598890409128005
Generation 351 Best Solution: [183.60094767284642, 177.41480766806097,
174.07978372287482, 75.05368407297242] Fitness: 1.0968627619142697
Generation 352 Best Solution: [183.64158039617416, 176.0705532357235,
174.88718984105193, 75.05368407297242] Fitness: 0.5599175006719848
Generation 353 Best Solution: [183.60094767284642, 177.41480766806097,
174.05732576190908, 75.1733000196942] Fitness: 1.1681976082088368
Generation 354 Best Solution: [183.60094767284642, 176.20818841764145,
174.98240401762945, 75.90033715107464] Fitness: 1.519782057302201
Generation 355 Best Solution: [183.60094767284642, 177.41480766806097,
174.98240401762945, 74.97975267274978] Fitness: 1.4903333133077237
Generation 356 Best Solution: [183.60094767284642, 177.41480766806097,
174.05732576190908, 74.65211851621072] Fitness: 1.0198490358811427
Generation 357 Best Solution: [182.90217429879863, 176.20818841764145,
174.7680447722561, 74.61861785688453] Fitness: 0.3648632831168406
Generation 358 Best Solution: [183.60094767284642, 177.41480766806097,
174.25201355262107, 74.97975267274978] Fitness: 1.0587461869622883
Generation 359 Best Solution: [183.60094767284642, 177.41480766806097,
174.0240774981552, 74.97975267274978] Fitness: 1.0760152486541694
Generation 360 Best Solution: [183.60094767284642, 177.41480766806097,
174.98240401762945, 74.9336276303686] Fitness: 1.4743819002292515
Generation 361 Best Solution: [183.4658386520277, 177.41480766806097,
174.98240401762945, 74.9336276303686] Fitness: 1.4275720218390664
Generation 362 Best Solution: [183.60094767284642, 177.41480766806097,
174.98240401762945, 74.9336276303686] Fitness: 1.4743819002292515
Generation 363 Best Solution: [183.60094767284642, 177.41480766806097,
174.25201355262107, 75.84508623445943] Fitness: 1.9066761042214562

Generation 364 Best Solution: [183.60094767284642, 176.9569349537492,
174.07978372287482, 75.90033715107464] Fitness: 1.4195244071181432
Generation 365 Best Solution: [183.32153689276876, 177.41480766806097,
174.25201355262107, 75.90033715107464] Fitness: 1.9274425136736535
Generation 366 Best Solution: [183.5784818502349, 177.41480766806097,
174.07978372287482, 75.90033715107464] Fitness: 1.9930292470749231
Generation 367 Best Solution: [183.60094767284642, 177.41480766806097,
174.07978372287482, 74.75425572096715] Fitness: 1.0152745674818462
Generation 368 Best Solution: [183.60094767284642, 177.41480766806097,
174.07978372287482, 74.75425572096715] Fitness: 1.0152745674818462
Generation 369 Best Solution: [183.66541432808125, 175.6329306856501,
174.07978372287482, 75.90033715107464] Fitness: 1.4060325102767235
Generation 370 Best Solution: [183.66541432808125, 175.6329306856501,
174.25201355262107, 75.90033715107464] Fitness: 1.399658182496981
Generation 371 Best Solution: [183.60094767284642, 177.41480766806097,
174.98240401762945, 74.86474368290787] Fitness: 1.4560709266867378
Generation 372 Best Solution: [183.5784818502349, 177.25085719794032,
174.25201355262107, 75.90033715107464] Fitness: 1.744360090764944
Generation 373 Best Solution: [183.60094767284642, 177.41480766806097,
174.18922231534344, 75.90033715107464] Fitness: 1.993742431023287
Generation 374 Best Solution: [183.48003685504023, 176.988947852895,
174.6685200394642, 75.90033715107464] Fitness: 1.5565134794960016
Generation 375 Best Solution: [183.86038492555582, 176.4072160135502,
174.6685200394642, 75.90033715107464] Fitness: 1.4147209624960133
Generation 376 Best Solution: [183.60094767284642, 177.41480766806097,
174.69095071648997, 74.61395030331789] Fitness: 1.1839625589382745
Generation 377 Best Solution: [183.86038492555582, 177.41480766806097,
174.18922231534344, 74.91564204007483] Fitness: 1.1960121255603982
Generation 378 Best Solution: [183.86038492555582, 177.41480766806097,
174.18922231534344, 74.91564204007483] Fitness: 1.1960121255603982
Generation 379 Best Solution: [183.60094767284642, 175.81734081688396,
174.98240401762945, 75.90033715107464] Fitness: 1.6600034304128575
Generation 380 Best Solution: [183.60094767284642, 177.41480766806097,
174.98240401762945, 74.45689950610283] Fitness: 1.4829125523897486
Generation 381 Best Solution: [183.86038492555582, 176.41257516934814,
174.98240401762945, 75.90033715107464] Fitness: 1.692179066628844
Generation 382 Best Solution: [183.86038492555582, 176.41257516934814,
174.98240401762945, 75.90033715107464] Fitness: 1.692179066628844
Generation 383 Best Solution: [183.14440802638958, 175.75140998466796,
174.6043919947512, 75.90033715107464] Fitness: 1.3324881860751807
Generation 384 Best Solution: [183.86038492555582, 175.75140998466796,
174.6043919947512, 75.90033715107464] Fitness: 1.5496684856488652
Generation 385 Best Solution: [183.86038492555582, 176.41257516934814,
174.98240401762945, 75.90033715107464] Fitness: 1.692179066628844
Generation 386 Best Solution: [183.14440802638958, 177.41480766806097,
174.61114743436426, 75.03724066412893] Fitness: 1.1461422465339655
Generation 387 Best Solution: [183.86038492555582, 177.41480766806097,
174.61114743436426, 75.03724066412893] Fitness: 1.3633225461076501

Generation 388 Best Solution: [183.14440802638958, 175.5841291341615,
174.98240401762945, 75.90033715107464] Fitness: 1.7888260527841426
Generation 389 Best Solution: [183.14440802638958, 175.5841291341615,
174.98240401762945, 75.90033715107464] Fitness: 1.7888260527841426
Generation 390 Best Solution: [183.86038492555582, 177.41480766806097,
174.34385626844926, 75.90033715107464] Fitness: 2.1707630598426797
Generation 391 Best Solution: [183.86038492555582, 177.41480766806097,
174.61114743436426, 74.83293680122212] Fitness: 1.2988083654810632
Generation 392 Best Solution: [183.86038492555582, 176.6352866478587,
174.61114743436426, 75.90033715107464] Fitness: 1.4585412120714016
Generation 393 Best Solution: [183.86038492555582, 176.6352866478587,
174.61114743436426, 75.90033715107464] Fitness: 1.4585412120714016
Generation 394 Best Solution: [183.86038492555582, 177.41480766806097,
174.61114743436426, 75.00947471137177] Fitness: 1.351144826572916
Generation 395 Best Solution: [182.93858525949, 176.03183379889404,
174.98240401762945, 75.00947471137177] Fitness: 0.6445939068546745
Generation 396 Best Solution: [182.93858525949, 176.03183379889404,
174.71227708603737, 75.90033715107464] Fitness: 1.3240916645788725
Generation 397 Best Solution: [183.86038492555582, 176.2510624900743,
174.2621837860063, 75.90033715107464] Fitness: 1.2477549373639083
Generation 398 Best Solution: [183.86038492555582, 176.2510624900743,
174.2621837860063, 75.90033715107464] Fitness: 1.2477549373639083
Generation 399 Best Solution: [183.86038492555582, 176.2510624900743,
174.98240401762945, 74.76261466451794] Fitness: 0.6917144764277191
Generation 400 Best Solution: [183.86038492555582, 176.9296883857207,
174.98240401762945, 75.12823058661418] Fitness: 1.1143241703186493
Generation 401 Best Solution: [183.53918721424924, 176.9054769259496,
174.98240401762945, 74.5674534263293] Fitness: 0.7978817629521164
Generation 402 Best Solution: [183.26707913193968, 177.41480766806097,
174.98240401762945, 74.5674534263293] Fitness: 1.3832639490603207
Generation 403 Best Solution: [183.48325889566112, 176.08211496665407,
174.98240401762945, 75.90033715107464] Fitness: 1.4995902330348523
Generation 404 Best Solution: [182.93858525949, 176.9054769259496,
174.98240401762945, 74.5674534263293] Fitness: 0.8358930882469807
Generation 405 Best Solution: [183.05374819248922, 176.9054769259496,
174.92412177500546, 74.5674534263293] Fitness: 0.7280254321048137
Generation 406 Best Solution: [183.26707913193968, 176.9054769259496,
174.98240401762945, 75.90033715107464] Fitness: 1.7279347744668212
Generation 407 Best Solution: [183.86038492555582, 177.41480766806097,
174.98240401762945, 74.95447375012408] Fitness: 1.6423161245981126
Generation 408 Best Solution: [182.93858525949, 177.41480766806097,
174.98240401762945, 74.95447375012408] Fitness: 1.4946869839394286
Generation 409 Best Solution: [183.26707913193968, 176.70908509508754,
174.98240401762945, 75.90033715107464] Fitness: 1.581298452450993
Generation 410 Best Solution: [182.89340864288693, 177.41480766806097,
174.98240401762945, 74.62593573141031] Fitness: 1.480322821807882
Generation 411 Best Solution: [182.93858525949, 177.41480766806097,
174.98240401762945, 74.38450896874004] Fitness: 1.5253232475921237

Generation 412 Best Solution: [183.26707913193968, 177.41480766806097,
174.5184719677836, 75.90033715107464] Fitness: 1.9981444278391716
Generation 413 Best Solution: [183.26707913193968, 177.41480766806097,
174.5184719677836, 75.90033715107464] Fitness: 1.9981444278391716
Generation 414 Best Solution: [183.86038492555582, 176.1506915031686,
174.98240401762945, 75.90033715107464] Fitness: 1.6881680705135373
Generation 415 Best Solution: [183.86038492555582, 176.1506915031686,
174.98240401762945, 75.90033715107464] Fitness: 1.6881680705135373
Generation 416 Best Solution: [183.86038492555582, 177.41480766806097,
174.98240401762945, 74.96697313019904] Fitness: 1.6467089230814802
Generation 417 Best Solution: [183.86038492555582, 177.41480766806097,
174.98240401762945, 74.96697313019904] Fitness: 1.6467089230814802
Generation 418 Best Solution: [183.86038492555582, 177.41480766806097,
174.98240401762945, 74.58719103097623] Fitness: 1.6102725198510819
Generation 419 Best Solution: [183.86038492555582, 177.41480766806097,
174.3397842574594, 74.96697313019904] Fitness: 1.2277484329445918
Generation 420 Best Solution: [183.86038492555582, 176.196838164878,
174.98240401762945, 75.90033715107464] Fitness: 1.681949255470137
Generation 421 Best Solution: [183.56070391573536, 176.196838164878,
174.98240401762945, 75.90033715107464] Fitness: 1.5042589843972007
Generation 422 Best Solution: [183.20499909514493, 176.196838164878,
174.98240401762945, 75.90033715107464] Fitness: 1.4555235858581241
Generation 423 Best Solution: [183.86038492555582, 176.05297968237804,
175.13658025980604, 75.90033715107464] Fitness: 1.897089545650127
Generation 424 Best Solution: [183.86038492555582, 176.05297968237804,
174.98240401762945, 75.90033715107464] Fitness: 1.7111144639887066
Generation 425 Best Solution: [183.86038492555582, 176.05297968237804,
174.98240401762945, 75.90033715107464] Fitness: 1.7111144639887066
Generation 426 Best Solution: [183.3050068673399, 177.41480766806097,
174.98240401762945, 74.89357926427371] Fitness: 1.3935118234229538
Generation 427 Best Solution: [183.86038492555582, 175.87280380188446,
174.22185440583908, 75.90033715107464] Fitness: 1.3548052782233806
Generation 428 Best Solution: [183.03269755085347, 177.41480766806097,
174.22185440583908, 75.90033715107464] Fitness: 1.968681018251969
Generation 429 Best Solution: [182.92300564501957, 176.87816084759555,
173.95876375294853, 75.90033715107464] Fitness: 1.3989161852614522
Generation 430 Best Solution: [183.32027059776402, 177.41480766806097,
174.3129267609826, 75.90033715107464] Fitness: 1.9350030211379043
Generation 431 Best Solution: [183.32027059776402, 177.41480766806097,
174.3129267609826, 75.90033715107464] Fitness: 1.9350030211379043
Generation 432 Best Solution: [182.98883508454298, 177.41480766806097,
174.3129267609826, 74.32904435194285] Fitness: 1.10605531487766
Generation 433 Best Solution: [183.86038492555582, 177.41480766806097,
173.95876375294853, 75.323939172601] Fitness: 1.467863021716113
Generation 434 Best Solution: [183.86038492555582, 177.41480766806097,
173.95876375294853, 74.66590699774993] Fitness: 1.205292770959749
Generation 435 Best Solution: [183.86038492555582, 177.41480766806097,
174.22185440583908, 74.71670800346028] Fitness: 1.1666982361360068

Generation 436 Best Solution: [183.86038492555582, 177.41480766806097,
174.22185440583908, 74.71670800346028] Fitness: 1.1666982361360068
Generation 437 Best Solution: [183.86038492555582, 177.41480766806097,
174.22185440583908, 74.30489781623336] Fitness: 1.2797545796351164
Generation 438 Best Solution: [183.75047038569514, 177.41480766806097,
174.98240401762945, 74.71670800346028] Fitness: 1.520468079720323
Generation 439 Best Solution: [183.86038492555582, 177.41480766806097,
174.22185440583908, 74.71670800346028] Fitness: 1.1666982361360068
Generation 440 Best Solution: [183.86038492555582, 177.41480766806097,
174.98240401762945, 74.44386047884178] Fitness: 1.6486799718713512
Generation 441 Best Solution: [183.86038492555582, 176.78476196123697,
174.98240401762945, 75.90033715107464] Fitness: 1.8620482592076342
Generation 442 Best Solution: [183.86038492555582, 176.78476196123697,
174.98240401762945, 75.90033715107464] Fitness: 1.8620482592076342
Generation 443 Best Solution: [183.75047038569514, 176.49439189863875,
174.98240401762945, 75.90033715107464] Fitness: 1.6333124569425235
Generation 444 Best Solution: [183.75047038569514, 177.41480766806097,
174.98240401762945, 74.61176119396801] Fitness: 1.5268766467913293
Generation 445 Best Solution: [183.86038492555582, 176.78476196123697,
174.98240401762945, 74.61176119396801] Fitness: 0.8797290730809891
Generation 446 Best Solution: [182.9655242326178, 176.78476196123697,
174.98240401762945, 74.61176119396801] Fitness: 0.7196444661393395
Generation 447 Best Solution: [183.86038492555582, 177.41480766806097,
174.98240401762945, 74.80948946510532] Fitness: 1.6072462450143832
Generation 448 Best Solution: [183.53281473818853, 177.41480766806097,
174.42880099834773, 75.90033715107464] Fitness: 2.0057991022730164
Generation 449 Best Solution: [182.9655242326178, 177.41480766806097,
174.42880099834773, 75.90033715107464] Fitness: 2.0335858906897237
Generation 450 Best Solution: [183.86038492555582, 177.41480766806097,
174.42880099834773, 74.49108312318073] Fitness: 1.2376617067605609
Generation 451 Best Solution: [183.75047038569514, 176.80711436643776,
174.98240401762945, 75.90033715107464] Fitness: 1.79870321546383
Generation 452 Best Solution: [183.75047038569514, 176.6134335494893,
174.42880099834773, 75.90033715107464] Fitness: 1.2850243744587178
Generation 453 Best Solution: [183.86038492555582, 176.8910896149424,
174.61638336790344, 75.90033715107464] Fitness: 1.63691314686234
Generation 454 Best Solution: [183.86038492555582, 175.99532694965856,
174.42880099834773, 75.90033715107464] Fitness: 1.3356774810266339
Generation 455 Best Solution: [183.86038492555582, 175.62927172874964,
174.42880099834773, 75.90033715107464] Fitness: 1.569100091128715
Generation 456 Best Solution: [183.86038492555582, 177.41480766806097,
174.98240401762945, 74.45455282789104] Fitness: 1.6448282218366226
Generation 457 Best Solution: [183.86038492555582, 177.41480766806097,
174.98240401762945, 74.45455282789104] Fitness: 1.6448282218366226
Generation 458 Best Solution: [183.7321934278197, 177.41480766806097,
174.18689832017697, 75.45623174342585] Fitness: 1.464099028651901
Generation 459 Best Solution: [183.75047038569514, 177.41480766806097,
174.68381682262103, 74.45455282789104] Fitness: 1.299020456515407

Generation 460 Best Solution: [183.86038492555582, 177.41480766806097,
174.98240401762945, 74.84459277602217] Fitness: 1.6130540356970617
Generation 461 Best Solution: [183.86038492555582, 177.41480766806097,
174.98240401762945, 74.84459277602217] Fitness: 1.6130540356970617
Generation 462 Best Solution: [183.7494955611663, 177.41480766806097,
174.98240401762945, 74.84459277602217] Fitness: 1.5327407849706822
Generation 463 Best Solution: [183.86038492555582, 177.41480766806097,
174.98240401762945, 74.84459277602217] Fitness: 1.6130540356970617
Generation 464 Best Solution: [183.86038492555582, 175.913358381515,
174.18689832017697, 75.90033715107464] Fitness: 1.3329252286152404
Generation 465 Best Solution: [183.7494955611663, 177.41480766806097,
174.98240401762945, 75.29673708002215] Fitness: 1.7608038993049944
Generation 466 Best Solution: [183.3267178685985, 176.7119606382586,
174.98240401762945, 75.90033715107464] Fitness: 1.5840975217255255
Generation 467 Best Solution: [183.7494955611663, 177.41480766806097,
174.98240401762945, 74.56548649737955] Fitness: 1.5339388987358054
Generation 468 Best Solution: [182.99373537883693, 177.41480766806097,
174.98240401762945, 74.56548649737955] Fitness: 1.4422062051498228
Generation 469 Best Solution: [183.86038492555582, 175.65814301036968,
174.98240401762945, 74.54159410709386] Fitness: 0.9696378924832056
Generation 470 Best Solution: [183.86038492555582, 175.67884975095166,
174.98240401762945, 75.29673708002215] Fitness: 1.1740173013323292
Generation 471 Best Solution: [183.51560510464412, 175.79021483566444,
174.98240401762945, 75.90033715107464] Fitness: 1.6451001117584885
Generation 472 Best Solution: [183.7494955611663, 177.41480766806097,
173.7997331958191, 75.29673708002215] Fitness: 1.4340681813219223
Generation 473 Best Solution: [183.7494955611663, 176.19259735423762,
174.98240401762945, 75.90033715107464] Fitness: 1.6020838770678596
Generation 474 Best Solution: [183.7494955611663, 176.19259735423762,
174.98240401762945, 75.90033715107464] Fitness: 1.6020838770678596
Generation 475 Best Solution: [183.7494955611663, 176.19259735423762,
174.74950623930516, 75.90033715107464] Fitness: 1.3838630725862844
Generation 476 Best Solution: [183.4411626416996, 177.41480766806097,
174.98240401762945, 74.76759860152151] Fitness: 1.3888513126615134
Generation 477 Best Solution: [183.7494955611663, 175.67884975095166,
174.98240401762945, 75.5881927789927] Fitness: 1.3914815218220538
Generation 478 Best Solution: [183.4411626416996, 177.41480766806097,
174.74950623930516, 74.4101890458388] Fitness: 1.229920054965329
Generation 479 Best Solution: [183.4411626416996, 177.41480766806097,
174.98240401762945, 74.93375477743545] Fitness: 1.4218055298355194
Generation 480 Best Solution: [183.4411626416996, 177.41480766806097,
174.98240401762945, 74.76759860152151] Fitness: 1.3888513126615134
Generation 481 Best Solution: [183.7494955611663, 177.41480766806097,
174.74950623930516, 74.93375477743545] Fitness: 1.3369793492356368
Generation 482 Best Solution: [183.7494955611663, 177.41480766806097,
174.74950623930516, 74.93375477743545] Fitness: 1.3369793492356368
Generation 483 Best Solution: [183.7494955611663, 176.8171127356455,
173.90281569726454, 75.90033715107464] Fitness: 1.4299302561231035

```
Generation 484 Best Solution: [183.58682136293007, 176.27354321812683,
174.74950623930516, 75.81320511248957] Fitness: 1.1535250624114888
Generation 485 Best Solution: [183.7494955611663, 177.41480766806097,
174.15344121917298, 74.57163065556779] Fitness: 1.0997560778138762
Generation 486 Best Solution: [183.4411626416996, 177.41480766806097,
174.15344121917298, 74.57163065556779] Fitness: 0.9663614539321836
Generation 487 Best Solution: [183.7494955611663, 177.1034048031978,
174.74950623930516, 74.31821917353642] Fitness: 0.9790565525523122
Generation 488 Best Solution: [183.4411626416996, 177.1034048031978,
174.74950623930516, 74.31821917353642] Fitness: 0.8456619286706198
Generation 489 Best Solution: [183.20400016423696, 177.41480766806097,
174.98240401762945, 74.5426703041916] Fitness: 1.39282758864156
Generation 490 Best Solution: [183.4411626416996, 177.41480766806097,
174.74950623930516, 75.90033715107464] Fitness: 2.156948910887474
Generation 491 Best Solution: [182.97221844558035, 177.41480766806097,
174.74950623930516, 75.90033715107464] Fitness: 2.2076336760195896
Generation 492 Best Solution: [183.76664062628114, 177.41480766806097,
173.80847879003986, 75.90033715107464] Fitness: 2.1884022039497344
Generation 493 Best Solution: [183.86880077784227, 177.41480766806097,
174.74950623930516, 75.22240078195784] Fitness: 1.5726573574649725
Generation 494 Best Solution: [183.76664062628114, 177.39736663656205,
174.74950623930516, 75.22240078195784] Fitness: 1.4692969276612953
Generation 495 Best Solution: [183.86038492555582, 177.39736663656205,
174.98240401762945, 74.9939920619044] Fitness: 1.6293943910785045
Generation 496 Best Solution: [183.86038492555582, 176.6572489457587,
174.72887756168888, 75.90033715107464] Fitness: 1.5482352450443035
Generation 497 Best Solution: [183.76664062628114, 176.6572489457587,
174.72887756168888, 75.90033715107464] Fitness: 1.4792214710989278
Generation 498 Best Solution: [183.03043579411792, 176.36339831800308,
174.96561767109728, 75.90033715107464] Fitness: 1.4799476232871092
Generation 499 Best Solution: [183.76664062628114, 177.41480766806097,
174.74950623930516, 74.29094524858567] Fitness: 1.4339345815119748
Generation 500 Best Solution: [183.114663919768, 177.39736663656205,
174.02843246249228, 74.29094524858567] Fitness: 1.0675183452486992
Optimal Solution for Simulation 13: [183.114663919768, 177.39736663656205,
174.02843246249228, 74.29094524858567]
--------------------------------------------------
Simulation 14/32 with conditions: {'drought': True, 'flood': False,
'policy_change': False, 'social_event': True, 'economic_crisis': False}
Generation 1 Best Solution: [183.2472259898922, 176.63746479770268,
174.1676335345339, 74.33385446000328] Fitness: 0.26020742570432553
Generation 2 Best Solution: [183.81756765499162, 176.19998525278385,
174.1676335345339, 74.49801596422617] Fitness: 0.30109778562412365
Generation 3 Best Solution: [183.81049684082882, 176.06800455734796,
174.1676335345339, 74.49801596422617] Fitness: 0.31382817007827757
Generation 4 Best Solution: [183.81049684082882, 176.06800455734796,
174.27997730128948, 74.97607643172708] Fitness: 0.32399697817857137
Generation 5 Best Solution: [183.81049684082882, 176.06800455734796,
```

174.27997730128948, 74.78653839326813] Fitness: 0.28635887822550554
Generation 6 Best Solution: [183.53644494785686, 177.36833908575983,
174.27997730128948, 74.33018619590814] Fitness: 1.0911888479352398
Generation 7 Best Solution: [183.53644494785686, 176.06800455734796,
174.157841313704, 75.86587605648833] Fitness: 0.9858280238755884
Generation 8 Best Solution: [183.81756765499162, 175.97936563680588,
174.27997730128948, 75.86587605648833] Fitness: 1.1847212006635366
Generation 9 Best Solution: [183.80655112567158, 175.97936563680588,
174.27997730128948, 75.86587605648833] Fitness: 1.1762560910419113
Generation 10 Best Solution: [183.80655112567158, 175.9796354585573,
174.99576097011837, 75.86587605648833] Fitness: 1.6367704656884674
Generation 11 Best Solution: [183.80655112567158, 177.37831928206535,
174.27997730128948, 74.82056428413173] Fitness: 1.1558986726731875
Generation 12 Best Solution: [183.81756765499162, 177.36833908575983,
174.99576097011837, 74.82056428413173] Fitness: 1.609525396156383
Generation 13 Best Solution: [183.81756765499162, 177.37831928206535,
174.27997730128948, 74.84382869120448] Fitness: 1.1674297339029902
Generation 14 Best Solution: [183.80655112567158, 177.36833908575983,
174.99576097011837, 74.84382869120448] Fitness: 1.604126238142935
Generation 15 Best Solution: [183.80655112567158, 177.36833908575983,
174.99576097011837, 74.84382869120448] Fitness: 1.604126238142935
Generation 16 Best Solution: [183.80655112567158, 175.81490589312907,
174.99576097011837, 75.86587605648833] Fitness: 1.712498604043628
Generation 17 Best Solution: [183.80655112567158, 177.37831928206535,
174.99576097011837, 74.35559242722903] Fitness: 1.7096005259511642
Generation 18 Best Solution: [183.80655112567158, 176.5085137793312,
174.99576097011837, 74.35559242722903] Fitness: 0.8719373967365591
Generation 19 Best Solution: [183.80655112567158, 176.59874491285734,
174.99576097011837, 75.86587605648833] Fitness: 1.6821503544770282
Generation 20 Best Solution: [183.80655112567158, 176.37172416617773,
174.99576097011837, 75.86587605648833] Fitness: 1.6049816952471336
Generation 21 Best Solution: [183.80655112567158, 175.9007508628268,
174.99576097011837, 75.8107323755879] Fitness: 1.585709106287562
Generation 22 Best Solution: [183.80655112567158, 176.78103436622933,
174.99576097011837, 75.59849470232659] Fitness: 1.4323426082686674
Generation 23 Best Solution: [183.80655112567158, 176.78103436622933,
173.99762222986357, 75.86587605648833] Fitness: 1.344190576437219
Generation 24 Best Solution: [183.80655112567158, 175.75275435702466,
173.99762222986357, 75.86587605648833] Fitness: 1.299998873565265
Generation 25 Best Solution: [183.80655112567158, 176.78103436622933,
173.99762222986357, 75.86587605648833] Fitness: 1.344190576437219
Generation 26 Best Solution: [183.80655112567158, 176.78103436622933,
173.99762222986357, 75.86587605648833] Fitness: 1.344190576437219
Generation 27 Best Solution: [183.80655112567158, 176.20209506242256,
174.99576097011837, 75.86587605648833] Fitness: 1.5930742282526236
Generation 28 Best Solution: [183.80655112567158, 176.2049464301735,
174.99576097011837, 75.86587605648833] Fitness: 1.5929510155623152
Generation 29 Best Solution: [183.80655112567158, 176.2049464301735,

173.7731958168331, 75.86587605648833] Fitness: 1.227537764864052
Generation 30 Best Solution: [183.80655112567158, 176.31444652988606,
174.99576097011837, 75.86587605648833] Fitness: 1.5965850186775303
Generation 31 Best Solution: [182.89936173179274, 177.37831928206535,
174.99576097011837, 74.32146833760865] Fitness: 1.5973597490466023
Generation 32 Best Solution: [183.80655112567158, 177.1310211229624,
174.99576097011837, 74.32146833760865] Fitness: 1.385171671922733
Generation 33 Best Solution: [182.85362199803748, 177.1310211229624,
174.99576097011837, 74.32146833760865] Fitness: 1.277569083580569
Generation 34 Best Solution: [183.57989994808858, 177.36833908575983,
174.99576097011837, 75.31623668376378] Fitness: 1.6880831790100486
Generation 35 Best Solution: [183.80655112567158, 177.36833908575983,
174.99576097011837, 75.31623668376378] Fitness: 1.8256121059418475
Generation 36 Best Solution: [183.80655112567158, 176.4455623894186,
174.99576097011837, 75.86587605648833] Fitness: 1.6223894150678952
Generation 37 Best Solution: [183.34719824516765, 177.37831928206535,
174.99576097011837, 75.05754584868828] Fitness: 1.4761354949536325
Generation 38 Best Solution: [183.80655112567158, 176.35392730369125,
174.99576097011837, 75.57060390431054] Fitness: 1.2071732591335542
Generation 39 Best Solution: [183.80655112567158, 176.47551259753237,
174.99576097011837, 75.86587605648833] Fitness: 1.6315640920038372
Generation 40 Best Solution: [183.80655112567158, 177.36833908575983,
174.99576097011837, 74.4326185014482] Fitness: 1.658412525132098
Generation 41 Best Solution: [183.80655112567158, 176.42960286316062,
174.99576097011837, 75.86587605648833] Fitness: 1.6179987519469963
Generation 42 Best Solution: [183.80655112567158, 177.36833908575983,
173.765942357665, 74.4326185014482] Fitness: 1.2968971821422017
Generation 43 Best Solution: [183.54517320805877, 177.37831928206535,
174.99576097011837, 74.72341409583434] Fitness: 1.459230177648043
Generation 44 Best Solution: [183.54517320805877, 176.7836620194054,
174.99576097011837, 75.86587605648833] Fitness: 1.644377323644874
Generation 45 Best Solution: [183.54517320805877, 177.36833908575983,
174.99576097011837, 74.6359507542706] Fitness: 1.4503985703446456
Generation 46 Best Solution: [182.91053767761113, 175.97589076555022,
174.99576097011837, 74.6359507542706] Fitness: 0.6397525467408168
Generation 47 Best Solution: [182.91053767761113, 176.8424806071626,
173.74364696265997, 75.86587605648833] Fitness: 1.3580840144879485
Generation 48 Best Solution: [183.05051137346055, 177.36833908575983,
174.10727351104808, 74.6359507542706] Fitness: 0.9488337202665849
Generation 49 Best Solution: [182.91053767761113, 177.36833908575983,
174.10727351104808, 74.6359507542706] Fitness: 0.9995723157012593
Generation 50 Best Solution: [183.33034366603388, 177.13557376036442,
174.99576097011837, 74.6359507542706] Fitness: 1.0728467359008993
Generation 51 Best Solution: [183.80655112567158, 177.07780713419132,
173.74364696265997, 75.86587605648833] Fitness: 1.7259217895496444
Generation 52 Best Solution: [183.53028765745256, 177.36833908575983,
175.0361236643102, 74.67347670946508] Fitness: 1.4871288152382713
Generation 53 Best Solution: [183.54517320805877, 176.97295008482283,

174.99576097011837, 74.67347670946508] Fitness: 0.9433210160358323
Generation 54 Best Solution: [183.80655112567158, 176.97295008482283,
174.99576097011837, 74.67347670946508] Fitness: 1.0957494419630875
Generation 55 Best Solution: [183.17603746143607, 176.3507995438678,
173.74364696265997, 75.3358788729885] Fitness: 0.4189701168783514
Generation 56 Best Solution: [183.80655112567158, 176.97295008482283,
174.99576097011837, 75.00566294490453] Fitness: 1.1428584213058937
Generation 57 Best Solution: [183.80655112567158, 175.5918729933512,
174.99576097011837, 75.00566294490453] Fitness: 1.054359425056401
Generation 58 Best Solution: [183.8148196928989, 175.73995262829115,
174.5272560358934, 75.84013980997855] Fitness: 1.3461256463974243
Generation 59 Best Solution: [183.8148196928989, 175.73995262829115,
174.5272560358934, 75.19914657727756] Fitness: 0.6623687146978511
Generation 60 Best Solution: [183.80655112567158, 175.73995262829115,
174.5272560358934, 75.19914657727756] Fitness: 0.6560305920288606
Generation 61 Best Solution: [183.80655112567158, 177.07780713419132,
174.99576097011837, 74.59082160268963] Fitness: 1.2200145274400738
Generation 62 Best Solution: [183.05051137346055, 177.36833908575983,
174.5272560358934, 74.59082160268963] Fitness: 1.0434539116635722
Generation 63 Best Solution: [183.3382607206, 177.36833908575983,
174.99576097011837, 74.59082160268963] Fitness: 1.4031037567226945
Generation 64 Best Solution: [183.80655112567158, 177.07780713419132,
174.99576097011837, 74.44838513868541] Fitness: 1.2617967565889434
Generation 65 Best Solution: [183.3382607206, 177.36833908575983,
174.99576097011837, 74.79234909635714] Fitness: 1.3911241715745843
Generation 66 Best Solution: [183.80655112567158, 175.61042256756724,
175.0361236643102, 75.86587605648833] Fitness: 1.9045677229290887
Generation 67 Best Solution: [183.80655112567158, 177.37831928206535,
174.99576097011837, 74.72803493166604] Fitness: 1.6115984649620272
Generation 68 Best Solution: [183.8148196928989, 177.37831928206535,
174.99576097011837, 74.72803493166604] Fitness: 1.6179365876310177
Generation 69 Best Solution: [183.53028765745256, 175.61042256756724,
174.99576097011837, 75.84013980997855] Fitness: 1.6604062618521218
Generation 70 Best Solution: [183.80655112567158, 176.3331873799744,
174.99576097011837, 75.86587605648833] Fitness: 1.5988412494606985
Generation 71 Best Solution: [183.53028765745256, 176.71346412673307,
174.99576097011837, 75.84013980997855] Fitness: 1.5503700922254258
Generation 72 Best Solution: [183.80655112567158, 176.3331873799744,
174.24309179238602, 75.86587605648833] Fitness: 1.1333756680145197
Generation 73 Best Solution: [183.80655112567158, 176.3331873799744,
174.24309179238602, 75.86587605648833] Fitness: 1.1333756680145197
Generation 74 Best Solution: [183.76058540606726, 177.37831928206535,
174.95405433880518, 75.14463456437865] Fitness: 1.6460086730367487
Generation 75 Best Solution: [183.80655112567158, 176.5194934279865,
174.95405433880518, 75.84013980997855] Fitness: 1.5621709353065831
Generation 76 Best Solution: [183.80655112567158, 176.5194934279865,
174.95405433880518, 75.84013980997855] Fitness: 1.5621709353065831
Generation 77 Best Solution: [183.53028765745256, 176.5194934279865,

174.95405433880518, 75.84013980997855] Fitness: 1.4038580519287889
Generation 78 Best Solution: [183.52186472447286, 176.5194934279865,
174.95405433880518, 75.84013980997855] Fitness: 1.4006618552917436
Generation 79 Best Solution: [182.99291727655051, 176.5194934279865,
174.95405433880518, 75.86587605648833] Fitness: 1.432350643600862
Generation 80 Best Solution: [183.80655112567158, 176.5194934279865,
174.95405433880518, 75.86587605648833] Fitness: 1.6012923665746155
Generation 81 Best Solution: [183.75058248399316, 176.05224505273114,
174.99576097011837, 75.86587605648833] Fitness: 1.5746525001573235
Generation 82 Best Solution: [183.10321535289415, 176.5906178031099,
175.0361236643102, 75.86587605648833] Fitness: 1.5261135700921373
Generation 83 Best Solution: [183.75058248399316, 176.3906630423585,
174.95405433880518, 75.86587605648833] Fitness: 1.5223271593507683
Generation 84 Best Solution: [183.80655112567158, 177.29788199121123,
174.99576097011837, 74.77571185446364] Fitness: 1.4920322699119208
Generation 85 Best Solution: [183.80655112567158, 176.3906630423585,
174.95405433880518, 75.86587605648833] Fitness: 1.5627841398172486
Generation 86 Best Solution: [183.80655112567158, 176.3906630423585,
174.95405433880518, 75.86587605648833] Fitness: 1.5627841398172486
Generation 87 Best Solution: [183.3748402258515, 177.36833908575983,
174.25176429956988, 75.86587605648833] Fitness: 1.7990542401679561
Generation 88 Best Solution: [183.80655112567158, 177.37831928206535,
174.25176429956988, 75.49481768968768] Fitness: 1.5393484444340235
Generation 89 Best Solution: [183.80655112567158, 177.37831928206535,
174.99576097011837, 75.49481768968768] Fitness: 2.003838350240911
Generation 90 Best Solution: [183.76058540606726, 177.37831928206535,
174.99576097011837, 74.59444005820717] Fitness: 1.5915147228274806
Generation 91 Best Solution: [183.76058540606726, 177.37831928206535,
174.99576097011837, 74.59444005820717] Fitness: 1.5915147228274806
Generation 92 Best Solution: [183.80655112567158, 177.29788199121123,
174.99576097011837, 74.80050357685948] Fitness: 1.4938129723944749
Generation 93 Best Solution: [183.02051032297106, 177.37831928206535,
174.99576097011837, 74.94913736731715] Fitness: 1.4646972670481564
Generation 94 Best Solution: [183.80655112567158, 175.90477797031653,
175.0361236643102, 75.86587605648833] Fitness: 1.713335722221088
Generation 95 Best Solution: [183.02051032297106, 175.73456467538026,
174.99576097011837, 75.86587605648833] Fitness: 1.5848600893956881
Generation 96 Best Solution: [183.02051032297106, 177.37831928206535,
173.81305989583075, 75.04624152613174] Fitness: 1.1137912205663194
Generation 97 Best Solution: [183.80655112567158, 177.37831928206535,
173.81305989583075, 75.04624152613174] Fitness: 1.2917522614410748
Generation 98 Best Solution: [183.02051032297106, 175.73456467538026,
174.60227435856208, 75.86587605648833] Fitness: 1.2454133458760972
Generation 99 Best Solution: [183.02051032297106, 175.73456467538026,
174.60227435856208, 75.86587605648833] Fitness: 1.2454133458760972
Generation 100 Best Solution: [183.80655112567158, 175.6748235317883,
175.0361236643102, 75.47563379298134] Fitness: 1.3561818167375976
Generation 101 Best Solution: [183.02051032297106, 175.73456467538026,

175.0361236643102, 75.12532538080343] Fitness: 0.865845264574403
Generation 102 Best Solution: [183.02051032297106, 175.83693952497094,
174.60227435856208, 75.86587605648833] Fitness: 1.1828237027403605
Generation 103 Best Solution: [183.53601538713758, 176.28147318915447,
174.99576097011837, 74.73451218070353] Fitness: 0.5685044105605993
Generation 104 Best Solution: [183.2534269384416, 176.28147318915447,
175.0361236643102, 75.86587605648833] Fitness: 1.4276680620376636
Generation 105 Best Solution: [183.53601538713758, 176.28147318915447,
174.67320963936191, 75.47563379298134] Fitness: 0.6474019106598885
Generation 106 Best Solution: [183.02051032297106, 176.28147318915447,
175.0361236643102, 75.86587605648833] Fitness: 1.4625409577856985
Generation 107 Best Solution: [183.02051032297106, 176.28147318915447,
175.0361236643102, 75.47563379298134] Fitness: 0.9660643316048815
Generation 108 Best Solution: [183.02051032297106, 176.28147318915447,
175.0361236643102, 75.47563379298134] Fitness: 0.9660643316048815
Generation 109 Best Solution: [183.02051032297106, 176.28147318915447,
175.0361236643102, 75.86587605648833] Fitness: 1.4625409577856985
Generation 110 Best Solution: [183.73426706468354, 176.03576928575146,
175.0361236643102, 75.47563379298134] Fitness: 1.1181975349527287
Generation 111 Best Solution: [183.2534269384416, 175.95266724188343,
175.0361236643102, 75.86587605648833] Fitness: 1.4805349641335286
Generation 112 Best Solution: [183.2534269384416, 175.62907260457663,
175.0361236643102, 74.95791473550771] Fitness: 0.8406352380598026
Generation 113 Best Solution: [183.80655112567158, 175.62907260457663,
175.0361236643102, 74.95791473550771] Fitness: 1.053469174682593
Generation 114 Best Solution: [183.2534269384416, 177.37066793113604,
175.0361236643102, 74.97359994535864] Fitness: 1.4722184910605889
Generation 115 Best Solution: [183.80655112567158, 177.37831928206535,
174.02953160286262, 74.95791473550771] Fitness: 1.1880446690043547
Generation 116 Best Solution: [183.80655112567158, 177.37831928206535,
174.02953160286262, 74.95791473550771] Fitness: 1.1880446690043547
Generation 117 Best Solution: [183.80655112567158, 176.51626651627782,
175.0361236643102, 75.86587605648833] Fitness: 1.692734826520439
Generation 118 Best Solution: [183.2534269384416, 177.37831928206535,
175.0361236643102, 75.86587605648833] Fitness: 2.3146423955344124
Generation 119 Best Solution: [183.2534269384416, 177.37831928206535,
175.0361236643102, 75.86587605648833] Fitness: 2.3146423955344124
Generation 120 Best Solution: [183.80655112567158, 177.37831928206535,
174.3297916364586, 75.86587605648833] Fitness: 2.029637693895388
Generation 121 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 75.05093135600393] Fitness: 1.7260360473035505
Generation 122 Best Solution: [183.80655112567158, 177.37831928206535,
174.98490447602126, 75.05093135600393] Fitness: 1.6671186592975933
Generation 123 Best Solution: [183.009560523713, 176.15567992882745,
175.0361236643102, 75.86587605648833] Fitness: 1.4688562073597677
Generation 124 Best Solution: [182.8032731903299, 177.37831928206535,
173.79352381307578, 75.05093135600393] Fitness: 1.2244144836488289
Generation 125 Best Solution: [183.009560523713, 177.37831928206535,

174.98490447602126, 75.05093135600393] Fitness: 1.4926128475539469
Generation 126 Best Solution: [183.86577797453538, 177.37831928206535,
175.0361236643102, 75.05093135600393] Fitness: 1.7734876384358897
Generation 127 Best Solution: [183.009560523713, 175.84731335025182,
175.01906129361006, 75.86587605648833] Fitness: 1.5468827222511188
Generation 128 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 75.31053814462642] Fitness: 1.8833056478533108
Generation 129 Best Solution: [183.86577797453538, 177.37831928206535,
175.0361236643102, 74.73738609642226] Fitness: 1.7057443453294865
Generation 130 Best Solution: [183.86577797453538, 176.30534497836592,
173.73737451010655, 75.86587605648833] Fitness: 1.2976456716498523
Generation 131 Best Solution: [183.59587975243852, 177.37831928206535,
175.0361236643102, 74.73738609642226] Fitness: 1.528170949113445
Generation 132 Best Solution: [183.77155377308475, 176.4028241924438,
174.98490447602126, 75.86587605648833] Fitness: 1.573422662637301
Generation 133 Best Solution: [183.77155377308475, 176.4028241924438,
174.98490447602126, 75.86587605648833] Fitness: 1.573422662637301
Generation 134 Best Solution: [183.77155377308475, 176.4028241924438,
174.98490447602126, 75.86587605648833] Fitness: 1.573422662637301
Generation 135 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.43613918826955] Fitness: 1.7191454750661992
Generation 136 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.43613918826955] Fitness: 1.7191454750661992
Generation 137 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.75905407624519] Fitness: 1.658673662492172
Generation 138 Best Solution: [183.50983929227837, 177.37831928206535,
175.0361236643102, 74.75905407624519] Fitness: 1.4927685807398878
Generation 139 Best Solution: [182.9718303087034, 176.1197669063565,
175.0361236643102, 75.86587605648833] Fitness: 1.4867765264622403
Generation 140 Best Solution: [182.9718303087034, 176.1197669063565,
175.0361236643102, 75.86587605648833] Fitness: 1.4867765264622403
Generation 141 Best Solution: [183.80655112567158, 176.52804729857272,
175.0361236643102, 75.86587605648833] Fitness: 1.6973309031022787
Generation 142 Best Solution: [183.80655112567158, 175.66723689757694,
174.6938832889844, 75.7816870849729] Fitness: 1.4079005167845393
Generation 143 Best Solution: [182.9718303087034, 176.2582197034011,
175.0361236643102, 75.86587605648833] Fitness: 1.4780582776970803
Generation 144 Best Solution: [183.50983929227837, 177.37831928206535,
173.73737451010655, 74.37703717683847] Fitness: 1.1874667034196336
Generation 145 Best Solution: [183.38338456046074, 177.37831928206535,
174.6938832889844, 74.75905407624519] Fitness: 1.132498398704746
Generation 146 Best Solution: [183.38338456046074, 176.04567866364246,
175.09858258721545, 74.75905407624519] Fitness: 0.671148635652453
Generation 147 Best Solution: [183.80655112567158, 176.04567866364246,
175.09858258721545, 74.75905407624519] Fitness: 0.8713725477848545
Generation 148 Best Solution: [183.38338456046074, 177.37831928206535,
174.6938832889844, 75.57885312538315] Fitness: 1.6159969381568247
Generation 149 Best Solution: [183.80655112567158, 175.78193072886768,

174.6938832889844, 75.86587605648833] Fitness: 1.4528666288407939
Generation 150 Best Solution: [183.80655112567158, 175.80666738261758,
175.12402023220932, 75.08833822197994] Fitness: 1.0889730805728803
Generation 151 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.51983537597445] Fitness: 1.6898570574289504
Generation 152 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.51983537597445] Fitness: 1.6898570574289504
Generation 153 Best Solution: [183.2144661872246, 177.37831928206535,
174.6938832889844, 75.57885312538315] Fitness: 1.604081688062995
Generation 154 Best Solution: [183.76914345330206, 177.37831928206535,
175.0361236643102, 74.29912943198829] Fitness: 1.7601404213649505
Generation 155 Best Solution: [183.80655112567158, 177.37831928206535,
174.27995981282174, 74.29912943198829] Fitness: 1.280314600801265
Generation 156 Best Solution: [183.80655112567158, 177.37831928206535,
174.25559937904976, 75.15135666384445] Fitness: 1.2652502674395538
Generation 157 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 75.15135666384445] Fitness: 1.776003247058832
Generation 158 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.37701202967156] Fitness: 1.745578695365588
Generation 159 Best Solution: [183.80655112567158, 177.37831928206535,
174.11287840627986, 75.57885312538315] Fitness: 1.6276278548083183
Generation 160 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.63289895703988] Fitness: 1.6654194211247366
Generation 161 Best Solution: [183.80655112567158, 177.37831928206535,
174.6938832889844, 74.83069466203227] Fitness: 1.3385272860379067
Generation 162 Best Solution: [183.80655112567158, 176.8232162142833,
175.0361236643102, 75.144392875878] Fitness: 1.1187149865219521
Generation 163 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 75.144392875878] Fitness: 1.7720957981725614
Generation 164 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 75.144392875878] Fitness: 1.7720957981725614
Generation 165 Best Solution: [183.80655112567158, 177.37831928206535,
174.23665137509127, 75.57885312538315] Fitness: 1.6293211001298116
Generation 166 Best Solution: [183.80655112567158, 177.37831928206535,
174.23665137509127, 75.57885312538315] Fitness: 1.6293211001298116
Generation 167 Best Solution: [183.80655112567158, 176.7079119991857,
174.23665137509127, 75.57885312538315] Fitness: 0.8927866722403222
Generation 168 Best Solution: [183.80655112567158, 177.37831928206535,
174.47065454004556, 74.6502374749048] Fitness: 1.210489557038908
Generation 169 Best Solution: [183.80655112567158, 176.7079119991857,
175.0361236643102, 75.86587605648833] Fitness: 1.790941904267713
Generation 170 Best Solution: [183.80655112567158, 176.7079119991857,
175.09858258721545, 75.86587605648833] Fitness: 1.8676164937057265
Generation 171 Best Solution: [183.80655112567158, 177.05815488166346,
175.0361236643102, 75.86587605648833] Fitness: 2.0994797832728542
Generation 172 Best Solution: [183.80655112567158, 176.86102695140397,
175.0361236643102, 75.89586356290009] Fitness: 1.9520201914640127
Generation 173 Best Solution: [183.80655112567158, 176.86102695140397,

174.87356991380523, 75.89586356290009] Fitness: 1.7773412912393427
Generation 174 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.5288064981971] Fitness: 1.687283031412569
Generation 175 Best Solution: [183.80655112567158, 177.37831928206535,
174.2479531507448, 75.86587605648833] Fitness: 2.0158179189059173
Generation 176 Best Solution: [183.80655112567158, 175.92102680671093,
175.0706849067029, 75.86587605648833] Fitness: 1.7479831236518577
Generation 177 Best Solution: [183.80655112567158, 176.2359306248104,
175.0361236643102, 75.86587605648833] Fitness: 1.6390521696377902
Generation 178 Best Solution: [183.52132867118206, 177.37831928206535,
175.0361236643102, 75.34018119087887] Fitness: 1.7453846437478626
Generation 179 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.68034298845235] Fitness: 1.6603431243498596
Generation 180 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.68034298845235] Fitness: 1.6603431243498596
Generation 181 Best Solution: [183.80655112567158, 177.37831928206535,
174.58872130244146, 74.68034298845235] Fitness: 1.2662288098677061
Generation 182 Best Solution: [183.80655112567158, 177.37831928206535,
175.09858258721545, 74.68034298845235] Fitness: 1.737017713787873
Generation 183 Best Solution: [183.80655112567158, 177.37831928206535,
175.09858258721545, 74.98681044480546] Fitness: 1.7779814824225588
Generation 184 Best Solution: [183.70440656891898, 177.37831928206535,
174.60434667988156, 75.86587605648833] Fitness: 2.071910607011083
Generation 185 Best Solution: [183.32928402891253, 177.37831928206535,
175.0361236643102, 75.50166741343892] Fitness: 1.8494087944498667
Generation 186 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 75.10283488038687] Fitness: 1.7501484026092597
Generation 187 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 75.10283488038687] Fitness: 1.7501484026092597
Generation 188 Best Solution: [183.80655112567158, 177.37831928206535,
175.09858258721545, 74.77090189385369] Fitness: 1.7358265490339673
Generation 189 Best Solution: [183.80655112567158, 177.37831928206535,
175.09858258721545, 74.77090189385369] Fitness: 1.7358265490339673
Generation 190 Best Solution: [183.80655112567158, 176.64147342552877,
175.09858258721545, 75.22563697183638] Fitness: 1.1222239961524263
Generation 191 Best Solution: [183.80655112567158, 177.37831928206535,
174.84378940963967, 74.77090189385369] Fitness: 1.456366090294154
Generation 192 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.77090189385369] Fitness: 1.6591519595959539
Generation 193 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.77090189385369] Fitness: 1.6591519595959539
Generation 194 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.77090189385369] Fitness: 1.6591519595959539
Generation 195 Best Solution: [183.65215273310704, 177.37831928206535,
175.0361236643102, 74.77090189385369] Fitness: 1.557879178401929
Generation 196 Best Solution: [183.65215273310704, 176.3578783284829,
175.0361236643102, 75.4758027601186] Fitness: 1.0516909171109254
Generation 197 Best Solution: [183.65215273310704, 177.37831928206535,

175.0361236643102, 74.38453774011397] Fitness: 1.6406774218064792
Generation 198 Best Solution: [183.65215273310704, 177.37831928206535,
173.93358501422242, 74.38453774011397] Fitness: 1.1606299971888723
Generation 199 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.38453774011397] Fitness: 1.741950203000504
Generation 200 Best Solution: [183.80655112567158, 177.37831928206535,
173.99490848958544, 74.38453774011397] Fitness: 1.2451845016384555
Generation 201 Best Solution: [183.39858927384773, 176.22084385702414,
174.82242168178877, 75.89586356290009] Fitness: 1.2664565634684406
Generation 202 Best Solution: [183.65215273310704, 176.70295245960614,
175.0361236643102, 75.86587605648833] Fitness: 1.6864980538142933
Generation 203 Best Solution: [183.80655112567158, 176.70295245960614,
175.0361236643102, 75.89586356290009] Fitness: 1.8344908584464856
Generation 204 Best Solution: [183.80655112567158, 176.70295245960614,
175.0361236643102, 75.89586356290009] Fitness: 1.8344908584464856
Generation 205 Best Solution: [183.2704434074199, 177.37831928206535,
173.9975260563694, 75.86587605648833] Fitness: 1.8176211675133458
Generation 206 Best Solution: [183.80655112567158, 177.15351188375166,
173.9975260563694, 75.86587605648833] Fitness: 1.71501062059806
Generation 207 Best Solution: [183.80655112567158, 177.15351188375166,
173.9975260563694, 75.86587605648833] Fitness: 1.71501062059806
Generation 208 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 74.53362554475922] Fitness: 1.6859455232996337
Generation 209 Best Solution: [183.21295116987025, 176.71865772009167,
175.0361236643102, 75.86587605648833] Fitness: 1.5858569004156702
Generation 210 Best Solution: [183.80655112567158, 176.10052113113323,
175.0361236643102, 75.89586356290009] Fitness: 1.6981232729476494
Generation 211 Best Solution: [183.76848284081885, 176.10052113113323,
175.0361236643102, 75.10024653064133] Fitness: 0.8448050063264692
Generation 212 Best Solution: [183.80655112567158, 176.10052113113323,
175.0361236643102, 75.10024653064133] Fitness: 0.8727860749645263
Generation 213 Best Solution: [183.80655112567158, 176.10052113113323,
175.0361236643102, 75.86587605648833] Fitness: 1.651403249509482
Generation 214 Best Solution: [183.80655112567158, 175.68314052040165,
175.0361236643102, 75.89586356290009] Fitness: 1.8930859621465408
Generation 215 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 75.1588226899249] Fitness: 1.7802657653062985
Generation 216 Best Solution: [183.80655112567158, 177.37831928206535,
174.4142279990762, 75.47010010037592] Fitness: 1.5514156526902896
Generation 217 Best Solution: [183.80655112567158, 177.37831928206535,
174.1589944332676, 75.02293718201818] Fitness: 1.1982032759077272
Generation 218 Best Solution: [183.80655112567158, 177.37831928206535,
173.86782689905812, 75.47010010037592] Fitness: 1.569011226602211
Generation 219 Best Solution: [183.80655112567158, 177.37831928206535,
174.4142279990762, 75.86587605648833] Fitness: 2.053443120652672
Generation 220 Best Solution: [183.80655112567158, 177.37831928206535,
174.4142279990762, 75.86587605648833] Fitness: 2.053443120652672
Generation 221 Best Solution: [183.34128012867157, 177.10188060512183,

173.9050966596626, 75.89586356290009] Fitness: 1.5190673038330846
Generation 222 Best Solution: [183.80655112567158, 176.30797951425683,
174.34882567396627, 75.86587605648833] Fitness: 1.1493256275479329
Generation 223 Best Solution: [183.80655112567158, 177.37831928206535,
174.4142279990762, 74.69868246388083] Fitness: 1.1851680477277025
Generation 224 Best Solution: [183.80655112567158, 176.76537798434325,
175.0361236643102, 75.86587605648833] Fitness: 1.8301243500391309
Generation 225 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 75.13553102293594] Fitness: 1.767218692903564
Generation 226 Best Solution: [183.60606425083662, 175.57862122573374,
175.0361236643102, 75.32255903330127] Fitness: 1.172393127838534
Generation 227 Best Solution: [183.80655112567158, 176.8492086902998,
175.0361236643102, 75.89586356290009] Fitness: 1.9420578672902966
Generation 228 Best Solution: [183.80655112567158, 177.37831928206535,
175.0361236643102, 75.01329301621381] Fitness: 1.7108424818329568
Generation 229 Best Solution: [183.80655112567158, 176.0981881445354,
175.0361236643102, 75.89586356290009] Fitness: 1.698554590126594
Generation 230 Best Solution: [183.80655112567158, 175.76808444140974,
175.0361236643102, 75.53074428585279] Fitness: 1.3485629193809263
Generation 231 Best Solution: [183.80655112567158, 175.76808444140974,
175.0361236643102, 74.99140017390555] Fitness: 0.9629004719094728
Generation 232 Best Solution: [183.80655112567158, 177.37831928206535,
174.3420497612574, 75.05087290035347] Fitness: 1.2310266490242847
Generation 233 Best Solution: [183.80655112567158, 175.98881295233176,
175.0361236643102, 75.87007556849555] Fitness: 1.686833036357077
Generation 234 Best Solution: [183.80655112567158, 175.98881295233176,
175.0361236643102, 75.89586356290009] Fitness: 1.7270839336813575
Generation 235 Best Solution: [182.8299714152319, 177.37831928206535,
175.0361236643102, 75.09642724171353] Fitness: 1.652406012507366
Generation 236 Best Solution: [183.80655112567158, 175.98881295233176,
173.86305867177364, 75.89586356290009] Fitness: 1.2725869541598578
Generation 237 Best Solution: [183.80655112567158, 175.98881295233176,
173.86305867177364, 75.89586356290009] Fitness: 1.2725869541598578
Generation 238 Best Solution: [183.80655112567158, 175.98881295233176,
173.86305867177364, 75.89586356290009] Fitness: 1.2725869541598578
Generation 239 Best Solution: [183.80655112567158, 176.8443264371813,
175.0361236643102, 75.5183856584882] Fitness: 1.43908919499694
Generation 240 Best Solution: [183.80655112567158, 177.30848971526063,
173.86305867177364, 75.89586356290009] Fitness: 2.0144641157595884
Generation 241 Best Solution: [183.80655112567158, 176.19170089576676,
175.0361236643102, 75.89586356290009] Fitness: 1.6870641684956826
Generation 242 Best Solution: [183.80655112567158, 176.06724251219327,
175.0361236643102, 75.87007556849555] Fitness: 1.6647251328805515
Generation 243 Best Solution: [183.80655112567158, 176.05941250835912,
175.0361236643102, 75.87007556849555] Fitness: 1.6665563717858345
Generation 244 Best Solution: [183.80655112567158, 176.62045643649643,
173.86305867177364, 75.89586356290009] Fitness: 1.3321529461240909
Generation 245 Best Solution: [183.80655112567158, 176.05941250835912,

175.0361236643102, 75.89586356290009] Fitness: 1.706807269110115
Generation 246 Best Solution: [183.80655112567158, 176.05941250835912,
175.0361236643102, 75.87007556849555] Fitness: 1.6665563717858345
Generation 247 Best Solution: [183.83399177286012, 177.37831928206535,
175.0361236643102, 75.21436389154096] Fitness: 1.8357469976584833
Generation 248 Best Solution: [183.80655112567158, 175.62330878380965,
175.0361236643102, 75.21436389154096] Fitness: 1.180608390268246
Generation 249 Best Solution: [183.80046870017526, 175.62330878380965,
175.0361236643102, 75.21436389154096] Fitness: 1.1760053719489727
Generation 250 Best Solution: [183.83399177286012, 177.37831928206535,
173.86305867177364, 75.21436389154096] Fitness: 1.3812500181369833
Generation 251 Best Solution: [183.83399177286012, 177.37831928206535,
173.86305867177364, 75.15591555028583] Fitness: 1.345491906245694
Generation 252 Best Solution: [183.83399177286012, 177.37831928206535,
173.86305867177364, 75.89586356290009] Fitness: 2.1410912586543875
Generation 253 Best Solution: [183.0211207384981, 177.37831928206535,
175.0361236643102, 75.6856791372939] Fitness: 2.094338464284692
Generation 254 Best Solution: [183.0211207384981, 177.37831928206535,
174.32657714863336, 75.89586356290009] Fitness: 1.8974941776034637
Generation 255 Best Solution: [183.0211207384981, 176.93329076630496,
175.0361236643102, 75.6856791372939] Fitness: 1.5372095649147575
Generation 256 Best Solution: [183.26297023595097, 175.68702469977853,
175.0361236643102, 75.89586356290009] Fitness: 1.6774901237718125
Generation 257 Best Solution: [183.0211207384981, 177.37831928206535,
175.0361236643102, 75.6856791372939] Fitness: 2.094338464284692
Generation 258 Best Solution: [183.0211207384981, 177.37831928206535,
175.0361236643102, 75.6856791372939] Fitness: 2.094338464284692
Generation 259 Best Solution: [183.48533981627207, 177.37831928206535,
175.0361236643102, 75.6856791372939] Fitness: 2.0982347372843675
Generation 260 Best Solution: [183.80046870017526, 177.37831928206535,
174.81819207499643, 75.42939704521997] Fitness: 1.7553155665269213
Generation 261 Best Solution: [183.83399177286012, 177.37831928206535,
174.81819207499643, 75.42939704521997] Fitness: 1.781310467426712
Generation 262 Best Solution: [183.80046870017526, 177.37831928206535,
175.0361236643102, 74.9333642129301] Fitness: 1.6803643585416368
Generation 263 Best Solution: [183.10149937601122, 175.99308793680262,
175.0361236643102, 75.89586356290009] Fitness: 1.5272093480182263
Generation 264 Best Solution: [183.80046870017526, 175.64671713165308,
175.0361236643102, 75.6856791372939] Fitness: 1.6150274186546127
Generation 265 Best Solution: [183.80046870017526, 175.99308793680262,
175.0361236643102, 75.89586356290009] Fitness: 1.7210603064072432
Generation 266 Best Solution: [183.80046870017526, 176.14413848362105,
173.8721611613098, 75.89586356290009] Fitness: 1.2286444119057953
Generation 267 Best Solution: [183.83399177286012, 177.37831928206535,
175.0361236643102, 74.7882348316602] Fitness: 1.6815875098986826
Generation 268 Best Solution: [183.80046870017526, 175.87247683487558,
175.0361236643102, 75.89586356290009] Fitness: 1.7706816381213757
Generation 269 Best Solution: [183.80046870017526, 175.64671713165308,

174.46050123823832, 75.89586356290009] Fitness: 1.459862090260354
Generation 270 Best Solution: [183.80046870017526, 176.91241071784972,
175.0361236643102, 74.47920880322978] Fitness: 1.1216275953277357
Generation 271 Best Solution: [183.00169916008633, 176.91241071784972,
175.0361236643102, 75.89586356290009] Fitness: 1.82561841819965
Generation 272 Best Solution: [183.80046870017526, 176.14136570907385,
175.0361236643102, 75.89586356290009] Fitness: 1.6871682304977789
Generation 273 Best Solution: [183.80046870017526, 176.80978501882478,
174.33181015278925, 75.89586356290009] Fitness: 1.4082132830360146
Generation 274 Best Solution: [183.639067858124, 177.37831928206535,
174.33181015278925, 75.89586356290009] Fitness: 1.9684485023986236
Generation 275 Best Solution: [183.639067858124, 177.37831928206535,
174.31734209051018, 75.89586356290009] Fitness: 1.965302805291572
Generation 276 Best Solution: [183.639067858124, 177.37831928206535,
175.0361236643102, 74.74483498735559] Fitness: 1.5499865014267393
Generation 277 Best Solution: [183.00169916008633, 177.37831928206535,
174.45891552863299, 75.89586356290009] Fitness: 1.9447612192035781
Generation 278 Best Solution: [183.80046870017526, 177.37831928206535,
174.45891552863299, 75.1573173956141] Fitness: 1.3172866883740457
Generation 279 Best Solution: [183.80046870017526, 177.37831928206535,
175.0361236643102, 75.1573173956141] Fitness: 1.7747972400020549
Generation 280 Best Solution: [183.00169916008633, 177.37831928206535,
174.45891552863299, 75.87533647420821] Fitness: 1.9126483127026113
Generation 281 Best Solution: [183.25069193961363, 177.0711428157556,
175.0361236643102, 75.89586356290009] Fitness: 1.9479966520438117
Generation 282 Best Solution: [183.25069193961363, 177.0711428157556,
175.0361236643102, 75.89586356290009] Fitness: 1.9479966520438117
Generation 283 Best Solution: [183.25069193961363, 177.0711428157556,
175.0361236643102, 75.89586356290009] Fitness: 1.9479966520438117
Generation 284 Best Solution: [183.80046870017526, 176.6043479683767,
174.23313078892477, 75.89586356290009] Fitness: 1.2605982573297225
Generation 285 Best Solution: [183.6991692727377, 177.12116222582696,
174.23313078892477, 75.89586356290009] Fitness: 1.6323562477346116
Generation 286 Best Solution: [183.6991692727377, 177.0711428157556,
175.0361236643102, 74.94444274117018] Fitness: 1.2008210624317237
Generation 287 Best Solution: [183.80046870017526, 177.0711428157556,
174.02815145998522, 74.38419853737274] Fitness: 0.8204710542060357
Generation 288 Best Solution: [183.6991692727377, 177.12116222582696,
174.23313078892477, 74.84324101735815] Fitness: 0.7243733929481773
Generation 289 Best Solution: [183.6991692727377, 177.37831928206535,
174.23313078892477, 74.84324101735815] Fitness: 1.0791594635023602
Generation 290 Best Solution: [183.6991692727377, 177.37831928206535,
174.93561428225922, 74.91297963810887] Fitness: 1.4936448983796078
Generation 291 Best Solution: [183.6991692727377, 177.37831928206535,
175.0361236643102, 74.91297963810887] Fitness: 1.6058919473519695
Generation 292 Best Solution: [183.80046870017526, 177.37831928206535,
174.79888204994128, 75.06352314807415] Fitness: 1.4840571948665184
Generation 293 Best Solution: [183.80046870017526, 176.23014749363438,

175.0361236643102, 75.89586356290009] Fitness: 1.6811869109228885
Generation 294 Best Solution: [183.4833406683758, 177.37831928206535,
175.0361236643102, 74.91297963810887] Fitness: 1.5048611186580447
Generation 295 Best Solution: [183.4264151387878, 176.91080823283423,
173.83175003166468, 75.89586356290009] Fitness: 1.36409589069027
Generation 296 Best Solution: [183.4264151387878, 177.37831928206535,
175.0361236643102, 74.91297963810887] Fitness: 1.4887720314390007
Generation 297 Best Solution: [183.4264151387878, 176.91080823283423,
174.93561428225922, 75.71956671502049] Fitness: 1.435707555122478
Generation 298 Best Solution: [183.80046870017526, 177.37831928206535,
174.40184795936926, 75.59397766426783] Fitness: 1.6769460560506668
Generation 299 Best Solution: [183.80046870017526, 176.85072099664072,
175.0361236643102, 74.3563307825522] Fitness: 1.1204694697863722
Generation 300 Best Solution: [183.4264151387878, 175.9704017139747,
175.0361236643102, 75.89586356290009] Fitness: 1.542499342869061
Generation 301 Best Solution: [183.4264151387878, 176.87899380740728,
175.09436389993277, 75.89586356290009] Fitness: 1.8478712839980638
Generation 302 Best Solution: [183.18786884512386, 177.37831928206535,
174.63452792411655, 75.89586356290009] Fitness: 1.997444575565365
Generation 303 Best Solution: [182.84581148149536, 177.44590808681252,
175.0361236643102, 74.79379962638826] Fitness: 1.6654096119992374
Generation 304 Best Solution: [183.18786884512386, 177.37831928206535,
174.63452792411655, 75.89586356290009] Fitness: 1.997444575565365
Generation 305 Best Solution: [182.84581148149536, 177.35246471792013,
175.0361236643102, 75.00098451718046] Fitness: 1.5631732859056533
Generation 306 Best Solution: [182.84581148149536, 176.24702158977675,
174.93561428225922, 75.89586356290009] Fitness: 1.4702364909236698
Generation 307 Best Solution: [183.80046870017526, 176.24702158977675,
174.93561428225922, 75.89586356290009] Fitness: 1.569015373137928
Generation 308 Best Solution: [183.80046870017526, 177.37831928206535,
174.93561428225922, 74.81845623833729] Fitness: 1.5461425490264133
Generation 309 Best Solution: [183.80046870017526, 177.44590808681252,
174.93561428225922, 74.81845623833729] Fitness: 1.654316715001082
Generation 310 Best Solution: [183.18786884512386, 177.37831928206535,
174.93561428225922, 74.81845623833729] Fitness: 1.340266403169054
Generation 311 Best Solution: [183.18786884512386, 176.05438364726484,
174.93561428225922, 74.81845623833729] Fitness: 0.47409741199337424
Generation 312 Best Solution: [183.18786884512386, 176.1697960155971,
175.12795971181797, 75.89586356290009] Fitness: 1.592782467038313
Generation 313 Best Solution: [183.18786884512386, 176.1697960155971,
175.09436389993277, 74.37245129421295] Fitness: 0.7231573725535198
Generation 314 Best Solution: [183.18786884512386, 176.5033642422371,
175.09436389993277, 75.89586356290009] Fitness: 1.5954872894362708
Generation 315 Best Solution: [183.5990729982848, 177.44590808681252,
174.92617749527315, 74.47719934939443] Fitness: 1.561080361186445
Generation 316 Best Solution: [183.18786884512386, 177.37831928206535,
174.93561428225922, 74.47719934939443] Fitness: 1.3808603435045979
Generation 317 Best Solution: [183.18786884512386, 177.37831928206535,

174.6956935502338, 74.29764790016812] Fitness: 1.253407565880026
Generation 318 Best Solution: [183.18786884512386, 177.37831928206535,
175.0361236643102, 74.42458213109698] Fitness: 1.5134591323237885
Generation 319 Best Solution: [183.11628300628126, 177.37831928206535,
175.0361236643102, 74.42458213109698] Fitness: 1.5227063576904123
Generation 320 Best Solution: [183.18786884512386, 177.37831928206535,
174.93561428225922, 75.17182636644648] Fitness: 1.4651446399975012
Generation 321 Best Solution: [183.18786884512386, 176.11575784673187,
174.93561428225922, 75.89586356290009] Fitness: 1.3727621700745645
Generation 322 Best Solution: [183.18786884512386, 177.37831928206535,
175.09436389993277, 75.17182636644648] Fitness: 1.6487203426484922
Generation 323 Best Solution: [183.18786884512386, 176.48290960971661,
174.93561428225922, 75.89586356290009] Fitness: 1.4047388381315677
Generation 324 Best Solution: [183.18786884512386, 177.44590808681252,
175.12795971181797, 75.17182636644648] Fitness: 1.8001383676465883
Generation 325 Best Solution: [183.11628300628126, 176.9843921131874,
175.12795971181797, 75.51036654875845] Fitness: 1.473287768804726
Generation 326 Best Solution: [183.18786884512386, 177.37831928206535,
175.12795971181797, 75.38442022328609] Fitness: 1.8489111234824622
Generation 327 Best Solution: [183.16364696557918, 176.75476801741314,
174.93561428225922, 75.89586356290009] Fitness: 1.548894612018874
Generation 328 Best Solution: [183.18786884512386, 175.68891435132505,
175.09436389993277, 75.89586356290009] Fitness: 1.7496223844331684
Generation 329 Best Solution: [183.18786884512386, 177.37831928206535,
174.49444595249946, 75.89586356290009] Fitness: 1.9212816943600104
Generation 330 Best Solution: [183.67548519278537, 177.44590808681252,
174.93561428225922, 75.29683922403268] Fitness: 1.780595183736063
Generation 331 Best Solution: [183.16364696557918, 176.9326643899997,
174.91287775585081, 75.89586356290009] Fitness: 1.6726094297386562
Generation 332 Best Solution: [183.18786884512386, 177.37831928206535,
174.93561428225922, 75.00817674554227] Fitness: 1.3861997165657796
Generation 333 Best Solution: [183.42844181924085, 177.37831928206535,
174.93561428225922, 75.00817674554227] Fitness: 1.4061892261267122
Generation 334 Best Solution: [183.67548519278537, 175.90656989014366,
174.91287775585081, 75.89586356290009] Fitness: 1.535471066771946
Generation 335 Best Solution: [183.67548519278537, 175.90656989014366,
175.12795971181797, 75.89586356290009] Fitness: 1.785776882398232
Generation 336 Best Solution: [183.42844181924085, 175.99144478516178,
175.12795971181797, 75.89586356290009] Fitness: 1.6502892735945784
Generation 337 Best Solution: [183.45036845794047, 175.90910895418688,
175.12795971181797, 75.89586356290009] Fitness: 1.6879403841435887
Generation 338 Best Solution: [183.57594068373226, 177.37831928206535,
175.12795971181797, 74.97346974286225] Fitness: 1.67212659474141
Generation 339 Best Solution: [183.45036845794047, 176.4563029420918,
173.90476151700142, 75.89586356290009] Fitness: 1.0638100495931029
Generation 340 Best Solution: [183.16364696557918, 176.55929338273347,
175.12795971181797, 75.89586356290009] Fitness: 1.6635973997353748
Generation 341 Best Solution: [183.16364696557918, 176.55929338273347,

175.12795971181797, 75.89586356290009] Fitness: 1.6635973997353748
Generation 342 Best Solution: [183.42844181924085, 175.8797357166775,
175.12795971181797, 75.89586356290009] Fitness: 1.695821583686213
Generation 343 Best Solution: [183.16364696557918, 175.8797357166775,
175.12795971181797, 75.89586356290009] Fitness: 1.6781808515873613
Generation 344 Best Solution: [183.16364696557918, 176.54723484818473,
175.12795971181797, 75.89586356290009] Fitness: 1.6583828167378258
Generation 345 Best Solution: [183.16364696557918, 176.54723484818473,
174.38793260233516, 75.89586356290009] Fitness: 1.0613239321416157
Generation 346 Best Solution: [183.16364696557918, 176.54723484818473,
174.38793260233516, 75.89586356290009] Fitness: 1.0613239321416157
Generation 347 Best Solution: [183.67548519278537, 175.86025270385068,
175.12795971181797, 75.89586356290009] Fitness: 1.807942464532216
Generation 348 Best Solution: [183.67548519278537, 177.37831928206535,
174.14374802884598, 75.89586356290009] Fitness: 1.9700760586636514
Generation 349 Best Solution: [183.57594068373226, 176.0143639630028,
175.12795971181797, 75.89586356290009] Fitness: 1.6942248373126731
Generation 350 Best Solution: [183.57594068373226, 176.0143639630028,
175.12795971181797, 75.89586356290009] Fitness: 1.6942248373126731
Generation 351 Best Solution: [183.16364696557918, 177.4007757449964,
175.12795971181797, 74.953023006117] Fitness: 1.6322197051769518
Generation 352 Best Solution: [183.6380915090596, 176.108451804969,
175.12795971181797, 75.89586356290009] Fitness: 1.702399880433728
Generation 353 Best Solution: [183.16364696557918, 177.2821907147297,
174.6467303842104, 75.10125933942294] Fitness: 1.039708108316278
Generation 354 Best Solution: [183.16364696557918, 177.2821907147297,
175.12795971181797, 74.953023006117] Fitness: 1.4538183468571075
Generation 355 Best Solution: [183.1992034236908, 177.37831928206535,
175.12795971181797, 74.953023006117] Fitness: 1.593794063932044
Generation 356 Best Solution: [183.16364696557918, 177.37831928206535,
175.12795971181797, 74.953023006117] Fitness: 1.596967895084081
Generation 357 Best Solution: [183.16364696557918, 177.37831928206535,
175.12795971181797, 74.953023006117] Fitness: 1.596967895084081
Generation 358 Best Solution: [183.16364696557918, 176.03236944024064,
174.99221117051997, 75.89586356290009] Fitness: 1.454912174500115
Generation 359 Best Solution: [183.18261892713318, 177.37831928206535,
175.12795971181797, 74.89144045401854] Fitness: 1.5794040198324164
Generation 360 Best Solution: [183.16364696557918, 177.37831928206535,
175.12795971181797, 74.89144045401854] Fitness: 1.5813114455358401
Generation 361 Best Solution: [183.60422748102482, 177.37831928206535,
174.99221117051997, 74.89144045401854] Fitness: 1.4980243477305328
Generation 362 Best Solution: [183.16364696557918, 177.37831928206535,
174.99221117051997, 74.89144045401854] Fitness: 1.4160082476722824
Generation 363 Best Solution: [182.82706008582042, 177.37831928206535,
175.12795971181797, 74.61509383835822] Fitness: 1.6897767702141713
Generation 364 Best Solution: [183.16364696557918, 177.37831928206535,
174.2229333392295, 74.4511191522106] Fitness: 0.9910076815100641
Generation 365 Best Solution: [183.1517844770802, 177.37831928206535,

174.2229333392295, 74.4511191522106] Fitness: 0.9924490523804002
Generation 366 Best Solution: [183.16364696557918, 177.37831928206535,
174.2229333392295, 74.4511191522106] Fitness: 0.9910076815100641
Generation 367 Best Solution: [183.18261892713318, 176.35328088239277,
174.99221117051997, 75.89586356290009] Fitness: 1.4344696161258583
Generation 368 Best Solution: [183.41538430999523, 176.35328088239277,
174.99221117051997, 75.89586356290009] Fitness: 1.450912675145411
Generation 369 Best Solution: [183.41538430999523, 176.35328088239277,
174.99221117051997, 74.56789221090953] Fitness: 0.5540620087100182
Generation 370 Best Solution: [183.41538430999523, 176.35328088239277,
174.99221117051997, 74.56789221090953] Fitness: 0.5540620087100182
Generation 371 Best Solution: [183.41538430999523, 176.35328088239277,
174.99221117051997, 75.89586356290009] Fitness: 1.450912675145411
Generation 372 Best Solution: [182.9343033661149, 177.37831928206535,
175.12795971181797, 75.3056108835311] Fitness: 1.847693267299633
Generation 373 Best Solution: [183.41538430999523, 177.37831928206535,
175.12795971181797, 74.55324042146172] Fitness: 1.6018051504060726
Generation 374 Best Solution: [183.41538430999523, 177.37831928206535,
175.12795971181797, 74.55324042146172] Fitness: 1.6018051504060726
Generation 375 Best Solution: [183.16364696557918, 177.37831928206535,
175.12795971181797, 74.55324042146172] Fitness: 1.5872695170899442
Generation 376 Best Solution: [183.41538430999523, 175.71232355170974,
173.91433472563145, 75.89586356290009] Fitness: 1.2044113493870232
Generation 377 Best Solution: [183.71432936122787, 177.37831928206535,
173.91433472563145, 75.15720008725958] Fitness: 1.241199219917369
Generation 378 Best Solution: [183.41538430999523, 177.37831928206535,
173.91433472563145, 75.15720008725958] Fitness: 1.111993585432573
Generation 379 Best Solution: [182.97059175887992, 177.37831928206535,
173.91433472563145, 75.15720008725958] Fitness: 1.1447019719730964
Generation 380 Best Solution: [182.89094811841264, 177.37831928206535,
173.91433472563145, 75.15720008725958] Fitness: 1.1789609135678403
Generation 381 Best Solution: [183.71432936122787, 177.18732905987662,
175.12795971181797, 74.49604235713213] Fitness: 1.4753048506070567
Generation 382 Best Solution: [183.16364696557918, 177.37831928206535,
175.12795971181797, 74.46762154083535] Fitness: 1.6134529579458916
Generation 383 Best Solution: [183.16364696557918, 177.37831928206535,
175.12795971181797, 74.46762154083535] Fitness: 1.6134529579458916
Generation 384 Best Solution: [183.41538430999523, 177.37831928206535,
173.89736661150576, 74.46762154083535] Fitness: 1.0456449863775081
Generation 385 Best Solution: [182.9688384273822, 177.37831928206535,
173.89736661150576, 74.46762154083535] Fitness: 1.0790147075955887
Generation 386 Best Solution: [183.41538430999523, 177.37831928206535,
175.12795971181797, 74.46762154083535] Fitness: 1.62798859126202
Generation 387 Best Solution: [183.16364696557918, 175.99934746181202,
175.12795971181797, 75.89586356290009] Fitness: 1.6300702041591557
Generation 388 Best Solution: [183.16364696557918, 175.99934746181202,
175.12795971181797, 75.89586356290009] Fitness: 1.6300702041591557
Generation 389 Best Solution: [183.41538430999523, 177.37831928206535,

174.57967600320129, 75.89586356290009] Fitness: 1.9813268525271468
Generation 390 Best Solution: [183.41538430999523, 177.37831928206535,
174.3225880586968, 75.89586356290009] Fitness: 1.881180927206071
Generation 391 Best Solution: [183.41538430999523, 175.89402470364834,
174.3225880586968, 75.89586356290009] Fitness: 1.0719525619836292
Generation 392 Best Solution: [182.8279027121975, 177.37831928206535,
174.6255159164041, 75.43017329775947] Fitness: 1.5212658487019355
Generation 393 Best Solution: [183.41538430999523, 177.37831928206535,
174.6255159164041, 75.43017329775947] Fitness: 1.421062152836344
Generation 394 Best Solution: [183.41538430999523, 177.37831928206535,
174.6255159164041, 75.43017329775947] Fitness: 1.421062152836344
Generation 395 Best Solution: [182.8279027121975, 175.8521588259698,
174.17806332516858, 75.33053932485127] Fitness: 0.5009791884305651
Generation 396 Best Solution: [183.41538430999523, 177.37831928206535,
175.12795971181797, 74.91795752993707] Fitness: 1.6019563646544621
Generation 397 Best Solution: [183.16364696557918, 177.37831928206535,
175.12795971181797, 74.91795752993707] Fitness: 1.5874207313383337
Generation 398 Best Solution: [183.11800853981498, 177.37831928206535,
175.12795971181797, 74.85973451635729] Fitness: 1.5818555625971544
Generation 399 Best Solution: [182.8279027121975, 177.37831928206535,
175.12795971181797, 74.91795752993707] Fitness: 1.7021600605200538
Generation 400 Best Solution: [182.98272146997883, 177.37831928206535,
173.84121163879652, 74.91795752993707] Fitness: 1.0704216845062886
Generation 401 Best Solution: [183.41538430999523, 177.37831928206535,
175.12795971181797, 74.61004878495277] Fitness: 1.5899342850516065
Generation 402 Best Solution: [183.16364696557918, 176.2450049842702,
175.12795971181797, 75.89586356290009] Fitness: 1.5922781682001217
Generation 403 Best Solution: [183.16364696557918, 176.2450049842702,
175.12795971181797, 75.89586356290009] Fitness: 1.5922781682001217
Generation 404 Best Solution: [183.41538430999523, 177.37831928206535,
175.12795971181797, 74.53314828836977] Fitness: 1.6070543342457047
Generation 405 Best Solution: [182.8279027121975, 177.37831928206535,
175.12795971181797, 74.88770170240653] Fitness: 1.6952663223179476
Generation 406 Best Solution: [183.41538430999523, 177.37831928206535,
174.36011622827522, 75.3882191749681] Fitness: 1.2641816453442902
Generation 407 Best Solution: [183.1804418145183, 177.37831928206535,
174.36011622827522, 75.14197197130893] Fitness: 1.0705041151646306
Generation 408 Best Solution: [183.02004961844995, 177.37831928206535,
174.36011622827522, 75.3882191749681] Fitness: 1.279957448006149
Generation 409 Best Solution: [183.16364696557918, 175.7247380640838,
175.12795971181797, 74.50566980931656] Fitness: 0.8894159348623011
Generation 410 Best Solution: [183.16364696557918, 176.3439815786164,
174.98079268066158, 75.89586356290009] Fitness: 1.4221818704326006
Generation 411 Best Solution: [183.16364696557918, 176.3439815786164,
174.98079268066158, 75.89586356290009] Fitness: 1.4221818704326006
Generation 412 Best Solution: [183.16364696557918, 176.3439815786164,
174.98079268066158, 75.89586356290009] Fitness: 1.4221818704326006
Generation 413 Best Solution: [183.1804418145183, 177.37831928206535,

174.98079268066158, 74.60514776686226] Fitness: 1.3964714005943784
Generation 414 Best Solution: [183.1804418145183, 177.37831928206535,
174.98079268066158, 74.60514776686226] Fitness: 1.3964714005943784
Generation 415 Best Solution: [183.1804418145183, 177.37831928206535,
175.12795971181797, 74.83021215225989] Fitness: 1.5691449146231145
Generation 416 Best Solution: [182.8279027121975, 177.37831928206535,
174.98079268066158, 74.83021215225989] Fitness: 1.5075326593000145
Generation 417 Best Solution: [182.8279027121975, 176.1418519362109,
174.98079268066158, 75.85133575394546] Fitness: 1.4658924617957905
Generation 418 Best Solution: [183.1804418145183, 176.0926081081672,
174.98079268066158, 75.89586356290009] Fitness: 1.426279972693697
Generation 419 Best Solution: [183.1804418145183, 176.0926081081672,
174.98079268066158, 75.89586356290009] Fitness: 1.426279972693697
Generation 420 Best Solution: [183.50712779935466, 176.1418519362109,
174.98079268066158, 75.89586356290009] Fitness: 1.46134770207558
Generation 421 Best Solution: [183.50712779935466, 176.1418519362109,
174.98079268066158, 74.50693912555536] Fitness: 0.5809004063831982
Generation 422 Best Solution: [182.8279027121975, 176.47417316465055,
174.98079268066158, 75.89586356290009] Fitness: 1.5676913593635482
Generation 423 Best Solution: [183.16364696557918, 176.45690552260112,
175.12795971181797, 75.89586356290009] Fitness: 1.6256099416352154
Generation 424 Best Solution: [183.16364696557918, 177.37831928206535,
175.0453393845908, 74.93360744959315] Fitness: 1.4878821142298355
Generation 425 Best Solution: [182.8279027121975, 177.37831928206535,
173.8951426014837, 75.52423342734242] Fitness: 1.5211887600528973
Generation 426 Best Solution: [183.16364696557918, 177.37831928206535,
175.12795971181797, 74.99604811559429] Fitness: 1.6109669045124377
Generation 427 Best Solution: [182.8279027121975, 176.2625789850432,
175.12795971181797, 75.89586356290009] Fitness: 1.7074596082614537
Generation 428 Best Solution: [183.34539808058895, 176.97356318233244,
175.12795971181797, 75.89586356290009] Fitness: 1.964686956213873
Generation 429 Best Solution: [183.3957550603976, 177.37831928206535,
175.12795971181797, 74.71062945082282] Fitness: 1.5754496416986081
Generation 430 Best Solution: [182.8279027121975, 175.75782004605844,
175.12795971181797, 75.89586356290009] Fitness: 1.8619807852958563
Generation 431 Best Solution: [183.67309817211182, 176.97356318233244,
175.12795971181797, 74.7794369335233] Fitness: 1.1669558824539097
Generation 432 Best Solution: [182.8279027121975, 177.37831928206535,
175.12795971181797, 74.59414628603862] Fitness: 1.6930186764813187
Generation 433 Best Solution: [182.8279027121975, 177.37831928206535,
175.12795971181797, 74.59414628603862] Fitness: 1.6930186764813187
Generation 434 Best Solution: [182.8279027121975, 176.35332804490548,
174.98079268066158, 75.89586356290009] Fitness: 1.538362149406539
Generation 435 Best Solution: [183.60574866457014, 177.37831928206535,
174.98079268066158, 74.71049785485015] Fitness: 1.4698416041719684
Generation 436 Best Solution: [182.8279027121975, 177.37831928206535,
175.04947140666184, 74.6341374384054] Fitness: 1.5877968748428373
Generation 437 Best Solution: [182.8279027121975, 177.37831928206535,

175.12795971181797, 74.71049785485015] Fitness: 1.6798893345923516
Generation 438 Best Solution: [182.8279027121975, 177.37831928206535,
174.98079268066158, 74.70935784017473] Fitness: 1.501863674192178
Generation 439 Best Solution: [183.60574866457014, 176.20498046441313,
175.12795971181797, 75.89586356290009] Fitness: 1.6755954950495575
Generation 440 Best Solution: [183.60574866457014, 176.20498046441313,
175.12795971181797, 75.89586356290009] Fitness: 1.6755954950495575
Generation 441 Best Solution: [183.60574866457014, 176.20498046441313,
175.12795971181797, 75.89586356290009] Fitness: 1.6755954950495575
Generation 442 Best Solution: [183.61903074034257, 177.37831928206535,
174.98079268066158, 74.64896678394884] Fitness: 1.4810903401060413
Generation 443 Best Solution: [183.61903074034257, 177.37831928206535,
174.98079268066158, 74.64896678394884] Fitness: 1.4810903401060413
Generation 444 Best Solution: [182.8279027121975, 177.37831928206535,
173.95220439169114, 74.64896678394884] Fitness: 1.0843021528578116
Generation 445 Best Solution: [183.60574866457014, 177.37831928206535,
173.95220439169114, 74.64896678394884] Fitness: 1.052319411209093
Generation 446 Best Solution: [182.8279027121975, 177.37831928206535,
174.98079268066158, 74.46261166619625] Fitness: 1.5519681338034448
Generation 447 Best Solution: [183.25085989146106, 177.37831928206535,
174.98079268066158, 74.46261166619625] Fitness: 1.4325075034698085
Generation 448 Best Solution: [183.87133911102615, 176.4830839025392,
175.12795971181797, 74.46261166619625] Fitness: 1.0289036748067348
Generation 449 Best Solution: [182.8279027121975, 177.37831928206535,
175.12795971181797, 74.94951667281454] Fitness: 1.7106773195681304
Generation 450 Best Solution: [183.87133911102615, 176.4830839025392,
175.12795971181797, 74.5209147650966] Fitness: 1.0095935702868315
Generation 451 Best Solution: [182.8006970090262, 176.4830839025392,
175.12795971181797, 74.5209147650966] Fitness: 0.8800624049077427
Generation 452 Best Solution: [183.53385030905864, 175.93001407636703,
175.12795971181797, 74.48290408807193] Fitness: 0.8341956113895684
Generation 453 Best Solution: [182.8006970090262, 176.4830839025392,
175.12795971181797, 74.53310397352064] Fitness: 0.8766096012634422
Generation 454 Best Solution: [183.8123989338302, 177.37831928206535,
175.12795971181797, 74.5209147650966] Fitness: 1.8085869824158614
Generation 455 Best Solution: [183.09539449304415, 177.37831928206535,
173.75319863170304, 75.54837266923022] Fitness: 1.5093945511080316
Generation 456 Best Solution: [183.53385030905864, 177.37831928206535,
175.12795971181797, 74.53310397352064] Fitness: 1.6437293215544255
Generation 457 Best Solution: [183.8123989338302, 176.2172512307611,
175.12795971181797, 75.28665951074058] Fitness: 1.0958547126548326
Generation 458 Best Solution: [183.73019831818007, 176.2172512307611,
175.12795971181797, 75.28665951074058] Fitness: 1.0372484577187795
Generation 459 Best Solution: [182.87242828500834, 177.37831928206535,
174.48826942374654, 75.60822919720776] Fitness: 1.6138561792028159
Generation 460 Best Solution: [183.0575888907076, 177.37831928206535,
175.12795971181797, 74.4439919181125] Fitness: 1.6421147406678342
Generation 461 Best Solution: [183.73019831818007, 176.2172512307611,

175.12795971181797, 75.28665951074058] Fitness: 1.0372484577187795
Generation 462 Best Solution: [183.0575888907076, 177.37831928206535,
175.12795971181797, 74.4439919181125] Fitness: 1.6421147406678342
Generation 463 Best Solution: [183.0575888907076, 177.37831928206535,
175.12795971181797, 75.06931827217372] Fitness: 1.6602810524282703
Generation 464 Best Solution: [183.87133911102615, 175.9046786328054,
175.12795971181797, 75.89586356290009] Fitness: 1.9268250014939583
Generation 465 Best Solution: [183.87133911102615, 175.9046786328054,
175.12795971181797, 75.89586356290009] Fitness: 1.9268250014939583
Generation 466 Best Solution: [183.87133911102615, 175.9046786328054,
175.12795971181797, 75.89586356290009] Fitness: 1.9268250014939583
Generation 467 Best Solution: [183.87133911102615, 177.37831928206535,
175.12795971181797, 75.41225223234002] Fitness: 2.1366395315336124
Generation 468 Best Solution: [183.49652842066723, 177.37831928206535,
173.87849817970655, 75.89586356290009] Fitness: 1.9429859082115297
Generation 469 Best Solution: [183.87133911102615, 176.6116973875731,
175.12795971181797, 75.89586356290009] Fitness: 1.9488385209938204
Generation 470 Best Solution: [182.83713059806414, 177.37831928206535,
175.12795971181797, 74.91880232419659] Fitness: 1.697167938472568
Generation 471 Best Solution: [183.87133911102615, 177.37831928206535,
175.12795971181797, 74.91880232419659] Fitness: 1.847913553052824
Generation 472 Best Solution: [183.27988612645458, 176.7681015082345,
175.12795971181797, 75.89586356290009] Fitness: 1.7812566338472915
Generation 473 Best Solution: [182.93645345042648, 176.7681015082345,
175.12795971181797, 75.89586356290009] Fitness: 1.8461273187157752
Generation 474 Best Solution: [182.93645345042648, 177.37831928206535,
175.12795971181797, 75.3990283261528] Fitness: 1.9251735605697433
Generation 475 Best Solution: [183.87133911102615, 177.37831928206535,
175.12795971181797, 74.93623036366161] Fitness: 1.8524696299549714
Generation 476 Best Solution: [182.93645345042648, 177.37831928206535,
175.12795971181797, 74.93623036366161] Fitness: 1.6530595078564794
Generation 477 Best Solution: [182.93645345042648, 177.37831928206535,
175.12795971181797, 74.93623036366161] Fitness: 1.6530595078564794
Generation 478 Best Solution: [182.93645345042648, 177.37831928206535,
174.44050332513493, 75.89586356290009] Fitness: 1.9622911070523035
Generation 479 Best Solution: [183.27988612645458, 177.37831928206535,
175.12795971181797, 75.3990283261528] Fitness: 1.8603028757012596
Generation 480 Best Solution: [183.8535934578841, 177.37831928206535,
175.12795971181797, 74.77293689310552] Fitness: 1.8111256224046388
Generation 481 Best Solution: [182.93645345042648, 177.37831928206535,
175.12795971181797, 75.07302895092653] Fitness: 1.7031683288570463
Generation 482 Best Solution: [182.83713059806414, 177.37831928206535,
175.05224375193166, 74.65449608542592] Fitness: 1.5834165797385757
Generation 483 Best Solution: [182.83713059806414, 177.37831928206535,
175.12795971181797, 75.07302895092653] Fitness: 1.7518328363752822
Generation 484 Best Solution: [182.97728711361873, 177.37831928206535,
173.93092120845603, 75.07302895092653] Fitness: 1.0932752486347361
Generation 485 Best Solution: [182.90917549937677, 177.37831928206535,

175.11954788848328, 74.50414525459118] Fitness: 1.6629937513183761
Generation 486 Best Solution: [183.87133911102615, 177.37831928206535,
175.11954788848328, 74.8087709056627] Fitness: 1.8177140004274976
Generation 487 Best Solution: [182.93645345042648, 177.37831928206535,
175.12795971181797, 75.07302895092653] Fitness: 1.7031683288570463
Generation 488 Best Solution: [182.83713059806414, 177.37831928206535,
173.81083233149252, 74.39658362849035] Fitness: 1.2064922353295988
Generation 489 Best Solution: [182.93645345042648, 176.21167355212194,
175.12795971181797, 75.89586356290009] Fitness: 1.6534657853585402
Generation 490 Best Solution: [182.97728711361873, 177.37831928206535,
175.12795971181797, 74.77830315240362] Fitness: 1.6107468341041564
Generation 491 Best Solution: [183.63294046515256, 175.90881724256468,
175.12795971181797, 75.89586356290009] Fitness: 1.7612173357831993
Generation 492 Best Solution: [183.63294046515256, 175.90881724256468,
174.14507054520902, 75.89586356290009] Fitness: 1.130536108193491
Generation 493 Best Solution: [183.63294046515256, 177.37831928206535,
175.12795971181797, 74.57371475972982] Fitness: 1.6790092989455294
Generation 494 Best Solution: [183.63294046515256, 176.88250348436605,
175.12795971181797, 74.57371475972982] Fitness: 1.0754232367007681
Generation 495 Best Solution: [183.38409840966452, 176.88250348436605,
175.12795971181797, 75.89586356290009] Fitness: 1.8850915902206484
Generation 496 Best Solution: [182.98876440047556, 176.88250348436605,
174.78995021370417, 75.89586356290009] Fitness: 1.5525775026305566
Generation 497 Best Solution: [182.98876440047556, 176.88250348436605,
174.78995021370417, 75.89586356290009] Fitness: 1.5525775026305566
Generation 498 Best Solution: [183.5899654168486, 176.88250348436605,
175.12795971181797, 75.60349943076429] Fitness: 1.5489431864103969
Generation 499 Best Solution: [182.98876440047556, 176.88250348436605,
174.78995021370417, 75.89586356290009] Fitness: 1.5525775026305566
Generation 500 Best Solution: [183.5899654168486, 176.88250348436605,
174.78995021370417, 75.89586356290009] Fitness: 1.5871683701080745
Optimal Solution for Simulation 14: [183.5899654168486, 176.88250348436605,
174.78995021370417, 75.89586356290009]

Simulation 15/32 with conditions: {'drought': True, 'flood': False,
'policy_change': False, 'social_event': False, 'economic_crisis': True}
Generation 1 Best Solution: [183.17006854755402, 176.00874759444773,
174.53612819170388, 75.25826165087793] Fitness: 0.33958025522922963
Generation 2 Best Solution: [183.17006854755402, 176.00874759444773,
174.53612819170388, 75.0934880774985] Fitness: 0.2455455760113115
Generation 3 Best Solution: [182.89617150959137, 175.87417598867486,
174.99431568610328, 74.3488572196272] Fitness: 0.7676535480721323
Generation 4 Best Solution: [182.89617150959137, 176.7817477969183,
174.99431568610328, 74.3488572196272] Fitness: 0.8645879650740222
Generation 5 Best Solution: [182.89617150959137, 176.90736368986202,
174.89912246309777, 75.25826165087793] Fitness: 0.9347536706852567
Generation 6 Best Solution: [183.5303707486225, 177.0326364740207,
174.9612371366664, 75.12644230501753] Fitness: 0.9992611342979061

Generation 7 Best Solution: [183.5303707486225, 176.7817477969183,
174.9621616529172, 74.46382314138218] Fitness: 0.7378975399283219
Generation 8 Best Solution: [183.7665488200749, 176.7817477969183,
174.9621616529172, 75.16619076000424] Fitness: 0.9228853195317531
Generation 9 Best Solution: [183.61015711480863, 175.9482078429434,
174.9621616529172, 75.74390594568885] Fitness: 1.2482869125222358
Generation 10 Best Solution: [182.89617150959137, 176.7644035145949,
174.99431568610328, 75.60892877900044] Fitness: 1.2406041967370887
Generation 11 Best Solution: [183.61015711480863, 175.9482078429434,
174.9621616529172, 75.74390594568885] Fitness: 1.2482869125222358
Generation 12 Best Solution: [183.5303707486225, 175.9482078429434,
173.9648773387328, 75.74390594568885] Fitness: 0.8383570627083933
Generation 13 Best Solution: [183.5303707486225, 176.7817477969183,
174.9621616529172, 74.68641746926673] Fitness: 0.6866305529263338
Generation 14 Best Solution: [183.5303707486225, 176.90736368986202,
174.39704781289728, 74.68641746926673] Fitness: 0.41012559081820044
Generation 15 Best Solution: [183.5303707486225, 176.7644035145949,
174.2292511725816, 75.74390594568885] Fitness: 0.9271426425928027
Generation 16 Best Solution: [182.89617150959137, 176.7644035145949,
174.01099410598064, 75.74390594568885] Fitness: 0.9846854038247048
Generation 17 Best Solution: [183.5303707486225, 176.7644035145949,
174.9631014744338, 75.17308982301356] Fitness: 0.7934972131996497
Generation 18 Best Solution: [182.89617150959137, 176.7644035145949,
174.01099410598064, 75.74390594568885] Fitness: 0.9846854038247048
Generation 19 Best Solution: [183.61015711480863, 176.7817477969183,
173.73767002796836, 75.74390594568885] Fitness: 1.101039232801166
Generation 20 Best Solution: [183.5303707486225, 176.7644035145949,
174.9621616529172, 74.60742774669176] Fitness: 0.6851148899117185
Generation 21 Best Solution: [183.23228168029493, 176.7644035145949,
174.80509015109743, 75.17580631777832] Fitness: 0.5987647110957022
Generation 22 Best Solution: [183.5303707486225, 175.81028982108336,
175.06900789663524, 74.71179582002632] Fitness: 0.7416370229860187
Generation 23 Best Solution: [183.5303707486225, 176.7644035145949,
174.26139958436724, 75.74390594568885] Fitness: 0.9299427854824673
Generation 24 Best Solution: [183.5303707486225, 176.7817477969183,
175.06900789663524, 74.65923979172179] Fitness: 0.8083881465376787
Generation 25 Best Solution: [182.89617150959137, 175.81028982108336,
175.06900789663524, 74.65923979172179] Fitness: 0.7861233186881329
Generation 26 Best Solution: [183.5303707486225, 175.81028982108336,
175.06900789663524, 74.92650424383464] Fitness: 0.7620466014415908
Generation 27 Best Solution: [182.82835417226914, 176.7817477969183,
174.9621616529172, 74.39297770833342] Fitness: 0.844220712437737
Generation 28 Best Solution: [182.89617150959137, 176.65655191083823,
174.9621616529172, 75.11125549548795] Fitness: 0.7335989127399319
Generation 29 Best Solution: [182.82835417226914, 176.7644035145949,
175.06900789663524, 75.08714839642249] Fitness: 0.9438995486679628
Generation 30 Best Solution: [182.89617150959137, 176.7644035145949,
175.06900789663524, 75.08714839642249] Fitness: 0.9078703272847014

Generation 31 Best Solution: [182.89617150959137, 176.7817477969183,
174.80509015109743, 74.35301394148111] Fitness: 0.6808065486077142
Generation 32 Best Solution: [183.23228168029493, 175.81028982108336,
175.06900789663524, 75.08714839642249] Fitness: 0.7695783168556233
Generation 33 Best Solution: [183.8573830056034, 176.7644035145949,
174.86627426682446, 75.08714839642249] Fitness: 0.843709963748937
Generation 34 Best Solution: [182.82835417226914, 176.7644035145949,
175.06900789663524, 75.08714839642249] Fitness: 0.9438995486679628
Generation 35 Best Solution: [183.41514822604927, 176.7644035145949,
175.06900789663524, 75.08714839642249] Fitness: 0.8351759314873627
Generation 36 Best Solution: [182.82835417226914, 176.7644035145949,
175.0463348804501, 75.1565343123217] Fitness: 0.9519659212681579
Generation 37 Best Solution: [183.6954761742418, 176.7644035145949,
175.06900789663524, 75.08714839642249] Fitness: 0.9452879700131152
Generation 38 Best Solution: [182.89617150959137, 176.7644035145949,
174.43049136885585, 75.82169178111248] Fitness: 1.1148124347263597
Generation 39 Best Solution: [183.19897315083247, 176.7644035145949,
175.06900789663524, 75.74390594568885] Fitness: 1.4064804572777898
Generation 40 Best Solution: [182.89617150959137, 175.81028982108336,
174.80509015109743, 75.74390594568885] Fitness: 1.175898544967755
Generation 41 Best Solution: [182.82835417226914, 176.1568327862453,
174.60519390353096, 75.82169178111248] Fitness: 1.0571811288228061
Generation 42 Best Solution: [182.82835417226914, 175.81028982108336,
175.06900789663524, 74.96299494071326] Fitness: 0.8477733648341502
Generation 43 Best Solution: [182.89617150959137, 175.81028982108336,
175.06900789663524, 74.96299494071326] Fitness: 0.8117441434508889
Generation 44 Best Solution: [182.82835417226914, 175.81028982108336,
175.06900789663524, 74.58561928966814] Fitness: 0.8344645974614794
Generation 45 Best Solution: [182.82835417226914, 176.7644035145949,
174.80509015109743, 75.74390594568885] Fitness: 1.2621644283984503
Generation 46 Best Solution: [182.89617150959137, 176.17838972695128,
175.06900789663524, 75.74390594568885] Fitness: 1.3178704810138604
Generation 47 Best Solution: [182.82835417226914, 177.2238861989844,
174.80509015109743, 74.68561458950704] Fitness: 1.0605467497812804
Generation 48 Best Solution: [182.89617150959137, 175.75670496793498,
175.06900789663524, 75.74390594568885] Fitness: 1.4754736390526193
Generation 49 Best Solution: [182.82835417226914, 177.2238861989844,
175.06900789663524, 74.9256940204275] Fitness: 1.3454461916861085
Generation 50 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.9256940204275] Fitness: 1.309416970302847
Generation 51 Best Solution: [182.89196127339775, 177.2238861989844,
175.06900789663524, 74.9256940204275] Fitness: 1.311475202615589
Generation 52 Best Solution: [182.89196127339775, 177.2238861989844,
175.02826167044546, 74.72444797213483] Fitness: 1.243565106293613
Generation 53 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.77360736873091] Fitness: 1.2887750497085635
Generation 54 Best Solution: [182.89617150959137, 175.67177579705722,
175.06900789663524, 74.77360736873091] Fitness: 0.8749330487579134

Generation 55 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.65139955640724] Fitness: 1.2945357109426898
Generation 56 Best Solution: [183.08366496919115, 177.2238861989844,
175.06900789663524, 74.65139955640724] Fitness: 1.226839060847963
Generation 57 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.63673689471011] Fitness: 1.2965648066523683
Generation 58 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.69567534507502] Fitness: 1.2901482936333797
Generation 59 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.69567534507502] Fitness: 1.2901482936333797
Generation 60 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.69567534507502] Fitness: 1.2901482936333797
Generation 61 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 75.14967412089976] Fitness: 1.3959709228490662
Generation 62 Best Solution: [182.89617150959137, 176.0967678416716,
175.06900789663524, 75.14967412089976] Fitness: 0.7761010952011366
Generation 63 Best Solution: [183.63250349388943, 176.23512794242257,
175.06900789663524, 75.74390594568885] Fitness: 1.3183185412559746
Generation 64 Best Solution: [183.63250349388943, 176.23512794242257,
175.06900789663524, 75.74390594568885] Fitness: 1.3183185412559746
Generation 65 Best Solution: [183.63250349388943, 176.23512794242257,
175.06900789663524, 75.74390594568885] Fitness: 1.3183185412559746
Generation 66 Best Solution: [183.63250349388943, 176.23512794242257,
175.06900789663524, 75.74390594568885] Fitness: 1.3183185412559746
Generation 67 Best Solution: [183.40107092049723, 177.2238861989844,
174.55639628333208, 75.37001118602265] Fitness: 1.0384818833253804
Generation 68 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 75.0590779277473] Fitness: 1.3529052515852935
Generation 69 Best Solution: [183.60868948573994, 176.7232627820049,
175.06900789663524, 75.89066999417871] Fitness: 1.6667292140762517
Generation 70 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 75.16643571718578] Fitness: 1.4051383488781934
Generation 71 Best Solution: [183.60868948573994, 177.2238861989844,
175.06900789663524, 74.87244040252166] Fitness: 1.2908105288189569
Generation 72 Best Solution: [182.89617150959137, 176.37720284344036,
175.06900789663524, 75.74390594568885] Fitness: 1.325806873271675
Generation 73 Best Solution: [183.05433437545295, 175.81987747047577,
175.06900789663524, 75.74390594568885] Fitness: 1.37656522090393
Generation 74 Best Solution: [183.05433437545295, 175.81987747047577,
175.06900789663524, 75.74390594568885] Fitness: 1.37656522090393
Generation 75 Best Solution: [183.3537426773489, 177.2238861989844,
175.06900789663524, 74.60339612176372] Fitness: 1.2194208887149143
Generation 76 Best Solution: [182.89617150959137, 175.98218540937583,
175.06900789663524, 75.74390594568885] Fitness: 1.3617076464127469
Generation 77 Best Solution: [182.89617150959137, 175.98218540937583,
174.16233641609284, 75.74390594568885] Fitness: 0.8350256210238336
Generation 78 Best Solution: [182.89617150959137, 176.54709493613888,
175.06900789663524, 75.74390594568885] Fitness: 1.374348837717513

Generation 79 Best Solution: [183.81873435364952, 176.54709493613888,
175.06900789663524, 75.74390594568885] Fitness: 1.493345478114212
Generation 80 Best Solution: [183.60868948573994, 177.2238861989844,
175.06900789663524, 74.79099239556857] Fitness: 1.281703636444184
Generation 81 Best Solution: [183.05433437545295, 177.2238861989844,
175.06900789663524, 74.67303143289347] Fitness: 1.2318664126919585
Generation 82 Best Solution: [183.63250349388943, 177.2238861989844,
175.06900789663524, 74.67303143289347] Fitness: 1.2956227779959797
Generation 83 Best Solution: [183.60868948573994, 177.2238861989844,
175.06900789663524, 74.33859266112226] Fitness: 1.3921281470996765
Generation 84 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.65839605895535] Fitness: 1.2936685239754526
Generation 85 Best Solution: [182.89617150959137, 175.92292629930097,
175.06900789663524, 74.33859266112226] Fitness: 0.8352961674196759
Generation 86 Best Solution: [182.89617150959137, 175.89276954602184,
175.06900789663524, 75.74390594568885] Fitness: 1.3987114673829515
Generation 87 Best Solution: [182.89617150959137, 176.50615537240358,
175.06900789663524, 75.74390594568885] Fitness: 1.3591319833566495
Generation 88 Best Solution: [183.62307519849645, 175.8339468015001,
175.06900789663524, 75.74390594568885] Fitness: 1.4278105105462573
Generation 89 Best Solution: [182.89617150959137, 176.03293521274907,
175.06900789663524, 75.74390594568885] Fitness: 1.3454476237930233
Generation 90 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.6233087029978] Fitness: 1.2986745428677933
Generation 91 Best Solution: [182.89617150959137, 175.8652909259054,
173.73600871819735, 74.6233087029978] Fitness: 0.36568614952451
Generation 92 Best Solution: [182.89617150959137, 175.8652909259054,
173.73600871819735, 74.6233087029978] Fitness: 0.36568614952451
Generation 93 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.6233087029978] Fitness: 1.2986745428677933
Generation 94 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.6233087029978] Fitness: 1.2986745428677933
Generation 95 Best Solution: [182.89617150959137, 175.8652909259054,
175.06900789663524, 75.74390594568885] Fitness: 1.4122246066672761
Generation 96 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.86806123071915] Fitness: 1.2979656821126888
Generation 97 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.86806123071915] Fitness: 1.2979656821126888
Generation 98 Best Solution: [182.89617150959137, 175.8652909259054,
175.06900789663524, 75.74390594568885] Fitness: 1.4122246066672761
Generation 99 Best Solution: [182.89617150959137, 176.1562079123993,
175.06900789663524, 75.74390594568885] Fitness: 1.3202530540611104
Generation 100 Best Solution: [183.6214987866838, 177.2238861989844,
174.00482642165846, 75.41278425032208] Fitness: 1.074834880971709
Generation 101 Best Solution: [182.89617150959137, 177.2238861989844,
174.00482642165846, 75.41278425032208] Fitness: 1.0766522006274306
Generation 102 Best Solution: [182.89617150959137, 175.8652909259054,
175.06900789663524, 75.74390594568885] Fitness: 1.4122246066672761

Generation 103 Best Solution: [182.89617150959137, 175.63440773716016,
174.3575398130508, 75.74390594568885] Fitness: 1.059657011957627
Generation 104 Best Solution: [182.89617150959137, 177.2238861989844,
173.80271269276224, 75.74390594568885] Fitness: 1.517748926650457
Generation 105 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 75.09999199876135] Fitness: 1.3709989714486512
Generation 106 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.87291204468315] Fitness: 1.2987588192673385
Generation 107 Best Solution: [182.89617150959137, 175.63440773716016,
173.80271269276224, 75.74390594568885] Fitness: 1.1335405632806963
Generation 108 Best Solution: [182.89617150959137, 176.59773305634576,
174.79557959947616, 75.74390594568885] Fitness: 1.1219975716660333
Generation 109 Best Solution: [182.89617150959137, 175.63440773716016,
174.79812767269198, 75.7340777956426] Fitness: 1.2803766072160176
Generation 110 Best Solution: [182.89617150959137, 175.63440773716016,
174.79812767269198, 75.74390594568885] Fitness: 1.2933646349772818
Generation 111 Best Solution: [182.89617150959137, 176.32767665994461,
175.06900789663524, 75.74390594568885] Fitness: 1.3189007699708184
Generation 112 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.77089279143722] Fitness: 1.2886867610238668
Generation 113 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.67928898148328] Fitness: 1.2914673954563245
Generation 114 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.67928898148328] Fitness: 1.2914673954563245
Generation 115 Best Solution: [183.08874909742, 176.19642010161976,
174.82326653939796, 75.74390594568885] Fitness: 0.9965087731002092
Generation 116 Best Solution: [183.08874909742, 177.2238861989844,
175.06900789663524, 74.32932278898024] Fitness: 1.3362340641114296
Generation 117 Best Solution: [182.89617150959137, 177.2238861989844,
173.8940726533435, 75.5233189878017] Fitness: 1.216839987120887
Generation 118 Best Solution: [182.89617150959137, 177.2238861989844,
173.8940726533435, 75.5233189878017] Fitness: 1.216839987120887
Generation 119 Best Solution: [182.984635223742, 176.09236365017293,
175.06900789663524, 75.74390594568885] Fitness: 1.2929909591675885
Generation 120 Best Solution: [182.89617150959137, 175.7706717073981,
173.90772040924554, 75.7230356741978] Fitness: 0.9615646833928309
Generation 121 Best Solution: [182.89617150959137, 177.2238861989844,
174.72566349725534, 75.74390594568885] Fitness: 1.6213508866732624
Generation 122 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.81299447010227] Fitness: 1.2911615812206227
Generation 123 Best Solution: [183.3271795165172, 175.9323400769286,
174.72566349725534, 75.74390594568885] Fitness: 0.966980547853141
Generation 124 Best Solution: [182.89617150959137, 176.36345216235037,
174.72566349725534, 75.74390594568885] Fitness: 0.9951705467579348
Generation 125 Best Solution: [182.984635223742, 177.2238861989844,
175.06900789663524, 75.06947493785435] Fitness: 1.3195104795501715
Generation 126 Best Solution: [183.3271795165172, 177.2238861989844,
175.06900789663524, 75.06947493785435] Fitness: 1.2716424651809664

Generation 127 Best Solution: [182.89617150959137, 176.26822746573632,
175.06900789663524, 75.54201202925272] Fitness: 1.073975810443893
Generation 128 Best Solution: [183.3271795165172, 177.2238861989844,
174.72566349725534, 74.45882166205483] Fitness: 0.9299872085548766
Generation 129 Best Solution: [182.984635223742, 177.2238861989844,
174.72566349725534, 74.95840711030272] Fitness: 0.9517150760098894
Generation 130 Best Solution: [182.984635223742, 177.2238861989844,
174.72566349725534, 74.74012278762974] Fitness: 0.922200763649523
Generation 131 Best Solution: [182.984635223742, 177.2238861989844,
174.72566349725534, 74.74012278762974] Fitness: 0.922200763649523
Generation 132 Best Solution: [182.89617150959137, 177.2238861989844,
174.72566349725534, 74.74012278762974] Fitness: 0.9599819550921275
Generation 133 Best Solution: [182.984635223742, 177.2238861989844,
175.06900789663524, 74.95840711030272] Fitness: 1.2801060013718144
Generation 134 Best Solution: [182.89617150959137, 177.2238861989844,
174.00201608255554, 74.95840711030272] Fitness: 0.8121160158737927
Generation 135 Best Solution: [182.89617150959137, 177.2238861989844,
173.88278405433002, 74.29204468276785] Fitness: 0.9588445181890878
Generation 136 Best Solution: [182.984635223742, 175.77421432376005,
175.06900789663524, 74.95840711030272] Fitness: 0.7945758691568934
Generation 137 Best Solution: [182.84718831636076, 175.77421432376005,
175.06900789663524, 74.95840711030272] Fitness: 0.8577652550704125
Generation 138 Best Solution: [182.89617150959137, 177.2238861989844,
174.2491116381825, 74.60171453314462] Fitness: 0.7788656940375583
Generation 139 Best Solution: [182.89617150959137, 177.2238861989844,
174.2491116381825, 75.320406265414] Fitness: 0.9831689176155822
Generation 140 Best Solution: [182.89617150959137, 177.2238861989844,
174.2491116381825, 75.320406265414] Fitness: 0.9831689176155822
Generation 141 Best Solution: [182.89617150959137, 177.2238861989844,
173.966944288915, 75.11951230011725] Fitness: 0.8837898347296389
Generation 142 Best Solution: [182.97996466886156, 177.2238861989844,
174.6522173952943, 75.11951230011725] Fitness: 0.9661401308177269
Generation 143 Best Solution: [182.97996466886156, 177.2238861989844,
174.6522173952943, 75.11951230011725] Fitness: 0.9661401308177269
Generation 144 Best Solution: [182.89617150959137, 177.1268108817801,
175.06900789663524, 75.11951230011725] Fitness: 1.2603693420735613
Generation 145 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 75.11951230011725] Fitness: 1.3804180375655553
Generation 146 Best Solution: [183.0245219857805, 177.2238861989844,
175.06900789663524, 74.60300536299481] Fitness: 1.250917936989462
Generation 147 Best Solution: [183.0245219857805, 177.2238861989844,
175.06900789663524, 74.60300536299481] Fitness: 1.250917936989462
Generation 148 Best Solution: [183.0245219857805, 177.2238861989844,
175.06900789663524, 74.60300536299481] Fitness: 1.250917936989462
Generation 149 Best Solution: [183.48580344717783, 177.2238861989844,
175.06900789663524, 74.60300536299481] Fitness: 1.2475034890296974
Generation 150 Best Solution: [182.89617150959137, 177.2238861989844,
174.47131231859683, 75.12971246801061] Fitness: 0.9152272109765351

Generation 151 Best Solution: [182.96291719448035, 177.2238861989844,
175.06900789663524, 74.96393600426505] Fitness: 1.2899874978010817
Generation 152 Best Solution: [182.89966836601513, 177.2238861989844,
175.06900789663524, 74.96393600426505] Fitness: 1.3177681968570913
Generation 153 Best Solution: [182.89617150959137, 177.2238861989844,
174.47131231859683, 74.47258632258173] Fitness: 0.8685342344078905
Generation 154 Best Solution: [183.48580344717783, 177.2238861989844,
175.06900789663524, 74.54415797641991] Fitness: 1.2611776972149884
Generation 155 Best Solution: [183.1793745708078, 176.11487003972906,
175.06900789663524, 75.74390594568885] Fitness: 1.2422601885628155
Generation 156 Best Solution: [183.0245219857805, 177.2238861989844,
174.7777545116163, 74.62103511929146] Fitness: 0.9589694059293571
Generation 157 Best Solution: [183.48580344717783, 177.2238861989844,
173.86713422531082, 75.74390594568885] Fitness: 1.433290910486528
Generation 158 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.89038064362994] Fitness: 1.3018749658018862
Generation 159 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.89038064362994] Fitness: 1.3018749658018862
Generation 160 Best Solution: [182.89617150959137, 177.2238861989844,
175.06900789663524, 74.89038064362994] Fitness: 1.3018749658018862
Generation 161 Best Solution: [183.48580344717783, 177.2238861989844,
175.06900789663524, 74.89038064362994] Fitness: 1.2470574291251963
Generation 162 Best Solution: [183.55173079131333, 177.2238861989844,
174.7777545116163, 74.89038064362994] Fitness: 0.9810580236588994
Generation 163 Best Solution: [183.48580344717783, 177.2238861989844,
174.7777545116163, 74.89038064362994] Fitness: 0.9583743730437532
Generation 164 Best Solution: [182.89617150959137, 177.2238861989844,
174.7777545116163, 74.89038064362994] Fitness: 1.013191909720443
Generation 165 Best Solution: [182.89617150959137, 177.34421233409063,
174.7777545116163, 74.89038064362994] Fitness: 1.1794332343470884
Generation 166 Best Solution: [183.0245219857805, 177.34421233409063,
174.7777545116163, 74.89038064362994] Fitness: 1.1280301456301631
Generation 167 Best Solution: [183.48580344717783, 177.34421233409063,
174.7777545116163, 74.89038064362994] Fitness: 1.1246156976703985
Generation 168 Best Solution: [183.0245219857805, 177.15985939636067,
174.6965804015877, 75.52345870106093] Fitness: 1.2083922049724252
Generation 169 Best Solution: [182.89617150959137, 177.15985939636067,
174.7777545116163, 75.52063906128014] Fitness: 1.3171881344936889
Generation 170 Best Solution: [183.55173079131333, 175.6617034399805,
175.06900789663524, 74.68622930865799] Fitness: 0.8526934041849572
Generation 171 Best Solution: [183.55173079131333, 175.6617034399805,
175.06900789663524, 75.89324167858553] Fitness: 1.7247652340100579
Generation 172 Best Solution: [182.89617150959137, 175.69306205186084,
175.06900789663524, 75.76492185953914] Fitness: 1.5480566752924783
Generation 173 Best Solution: [182.89617150959137, 177.34421233409063,
173.98259798399576, 75.74390594568885] Fitness: 1.6150714804236488
Generation 174 Best Solution: [182.89617150959137, 177.15985939636067,
175.06900789663524, 74.51936417403705] Fitness: 1.2425492861475935

Generation 175 Best Solution: [183.0245219857805, 176.85691853615756,
175.06900789663524, 75.86963966603065] Fitness: 1.6880759727045662
Generation 176 Best Solution: [183.0245219857805, 176.0455264975772,
174.81883479648764, 75.89324167858553] Fitness: 1.2489202870526563
Generation 177 Best Solution: [183.0245219857805, 177.34421233409063,
175.06900789663524, 75.19323954668185] Fitness: 1.5354148932877032
Generation 178 Best Solution: [183.55173079131333, 177.2238861989844,
174.8396299957441, 75.19323954668185] Fitness: 1.1516272319868066
Generation 179 Best Solution: [183.0245219857805, 177.2238861989844,
174.8396299957441, 75.19323954668185] Fitness: 1.1323580293314248
Generation 180 Best Solution: [183.0245219857805, 177.34421233409063,
175.06900789663524, 74.75196608703008] Fitness: 1.4031824918113946
Generation 181 Best Solution: [182.88624774473723, 177.34421233409063,
175.06900789663524, 75.19323954668185] Fitness: 1.5917071517965722
Generation 182 Best Solution: [182.88624774473723, 176.85691853615756,
175.06900789663524, 75.19323954668185] Fitness: 1.037681202106203
Generation 183 Best Solution: [183.0245219857805, 177.34421233409063,
175.06900789663524, 74.74184503132814] Fitness: 1.4031953331866085
Generation 184 Best Solution: [182.90521917851535, 175.6716691734033,
174.66211678456673, 75.89324167858553] Fitness: 1.3729013214365808
Generation 185 Best Solution: [182.88624774473723, 175.6716691734033,
175.06900789663524, 74.74184503132814] Fitness: 0.879486172227977
Generation 186 Best Solution: [183.44738806675673, 175.6716691734033,
174.66211678456673, 75.89324167858553] Fitness: 1.311881540960786
Generation 187 Best Solution: [182.90521917851535, 175.6716691734033,
174.66211678456673, 75.89324167858553] Fitness: 1.3729013214365808
Generation 188 Best Solution: [182.90521917851535, 176.65138188164923,
174.66211678456673, 75.89324167858553] Fitness: 1.2601277472191605
Generation 189 Best Solution: [182.88624774473723, 177.34421233409063,
175.06900789663524, 74.90891105253172] Fitness: 1.476755735919765
Generation 190 Best Solution: [183.08792604354787, 175.6716691734033,
175.06900789663524, 75.89324167858553] Fitness: 1.6804862892415349
Generation 191 Best Solution: [183.55173079131333, 175.6716691734033,
175.06900789663524, 75.89324167858553] Fitness: 1.7170428425082842
Generation 192 Best Solution: [182.90521917851535, 175.6716691734033,
175.06900789663524, 74.64390115311322] Fitness: 0.8774343610610859
Generation 193 Best Solution: [182.88624774473723, 175.59538894437432,
174.44391214889905, 75.89324167858553] Fitness: 1.336090402324956
Generation 194 Best Solution: [182.90521917851535, 176.2189000905799,
175.06900789663524, 74.40762952871385] Fitness: 0.7266303052979087
Generation 195 Best Solution: [182.90521917851535, 176.74764127645324,
174.44391214889905, 75.89324167858553] Fitness: 1.209571076387641
Generation 196 Best Solution: [183.38558334355258, 177.0085860764924,
175.0175418933535, 75.09751172076957] Fitness: 0.9831172301166619
Generation 197 Best Solution: [183.52728569252903, 176.3525877350238,
174.96638602795025, 75.89324167858553] Fitness: 1.3793185584202177
Generation 198 Best Solution: [183.1893363623779, 177.0085860764924,
175.06900789663524, 74.33495242622678] Fitness: 1.0675126465516773

Generation 199 Best Solution: [183.52728569252903, 175.65721867355404,
173.8882636417946, 75.89324167858553] Fitness: 1.2490468399060326
Generation 200 Best Solution: [182.90521917851535, 177.34421233409063,
175.06900789663524, 74.33495242622678] Fitness: 1.563891418669894
Generation 201 Best Solution: [183.55173079131333, 177.34421233409063,
175.06900789663524, 74.33495242622678] Fitness: 1.536100642195698
Generation 202 Best Solution: [182.90521917851535, 177.34421233409063,
175.06900789663524, 74.33495242622678] Fitness: 1.563891418669894
Generation 203 Best Solution: [182.90521917851535, 177.34421233409063,
175.06900789663524, 74.33495242622678] Fitness: 1.563891418669894
Generation 204 Best Solution: [182.87146480654914, 177.34421233409063,
175.06900789663524, 74.33495242622678] Fitness: 1.58065034252853
Generation 205 Best Solution: [182.90521917851535, 177.34421233409063,
174.1133451670034, 74.61660885207694] Fitness: 0.9377871755359717
Generation 206 Best Solution: [182.90521917851535, 177.34421233409063,
174.1133451670034, 74.61660885207694] Fitness: 0.9377871755359717
Generation 207 Best Solution: [182.88624774473723, 176.66779446608942,
174.51716741431605, 75.89324167858553] Fitness: 1.199200301226366
Generation 208 Best Solution: [182.90521917851535, 175.90078883221935,
175.06900789663524, 75.89324167858553] Fitness: 1.6038095097195666
Generation 209 Best Solution: [183.40426536835912, 175.90078883221935,
175.06900789663524, 75.89324167858553] Fitness: 1.533296271110319
Generation 210 Best Solution: [182.88624774473723, 176.66779446608942,
174.12897023512443, 75.89324167858553] Fitness: 1.125148415644157
Generation 211 Best Solution: [182.88624774473723, 176.66779446608942,
174.12897023512443, 75.89324167858553] Fitness: 1.125148415644157
Generation 212 Best Solution: [182.88624774473723, 176.66779446608942,
175.06900789663524, 75.89324167858553] Fitness: 1.6503024178951962
Generation 213 Best Solution: [183.39411603855896, 176.48881800215148,
175.06900789663524, 75.89324167858553] Fitness: 1.4898261629745935
Generation 214 Best Solution: [183.39411603855896, 176.1073476616903,
175.06900789663524, 75.42842655674774] Fitness: 0.8984593436569772
Generation 215 Best Solution: [183.09464175391298, 176.48881800215148,
174.2187910967998, 75.89324167858553] Fitness: 0.9704620076865776
Generation 216 Best Solution: [182.94139471497337, 176.48881800215148,
174.2187910967998, 75.89324167858553] Fitness: 1.020051942646088
Generation 217 Best Solution: [182.88624774473723, 177.34421233409063,
173.88379282206202, 74.85681327480913] Fitness: 0.9989802218201904
Generation 218 Best Solution: [182.90521917851535, 177.0359520297141,
173.88379282206202, 74.85681327480913] Fitness: 0.6024805060045578
Generation 219 Best Solution: [182.88624774473723, 176.48881800215148,
175.06900789663524, 74.49427963296746] Fitness: 0.7397094839648607
Generation 220 Best Solution: [182.88624774473723, 177.34421233409063,
174.2187910967998, 74.38055927330541] Fitness: 1.023515117297691
Generation 221 Best Solution: [182.88624774473723, 177.34421233409063,
174.2187910967998, 74.38055927330541] Fitness: 1.023515117297691
Generation 222 Best Solution: [183.03702293587088, 177.34421233409063,
174.22883249586667, 75.89324167858553] Fitness: 1.7486421192000732

Generation 223 Best Solution: [182.88624774473723, 177.34421233409063,
175.06900789663524, 74.74415506442654] Fitness: 1.4594726316285902
Generation 224 Best Solution: [182.88624774473723, 176.61496339042247,
175.06900789663524, 75.89324167858553] Fitness: 1.6225821789338697
Generation 225 Best Solution: [182.94139471497337, 176.61496339042247,
174.54174963976504, 75.89324167858553] Fitness: 1.157427218625315
Generation 226 Best Solution: [182.88624774473723, 176.61496339042247,
174.54174963976504, 75.89324167858553] Fitness: 1.1829340199940854
Generation 227 Best Solution: [182.88624774473723, 176.1813187710059,
175.06900789663524, 75.30281427863314] Fitness: 0.8664035427939298
Generation 228 Best Solution: [182.88624774473723, 177.34421233409063,
174.22883249586667, 74.60334091562099] Fitness: 0.9480865029103352
Generation 229 Best Solution: [182.88624774473723, 176.61496339042247,
175.06900789663524, 74.34956456217644] Fitness: 0.8537363423602448
Generation 230 Best Solution: [182.88624774473723, 177.34421233409063,
174.22883249586667, 74.60334091562099] Fitness: 0.9480865029103352
Generation 231 Best Solution: [183.79524586890838, 177.34421233409063,
174.22883249586667, 74.60334091562099] Fitness: 1.0450857183241202
Generation 232 Best Solution: [182.88624774473723, 177.34421233409063,
175.06900789663524, 74.60334091562099] Fitness: 1.4733867879951663
Generation 233 Best Solution: [182.88624774473723, 177.34421233409063,
174.5870418072174, 74.60334091562099] Fitness: 1.0569521163571547
Generation 234 Best Solution: [182.88624774473723, 177.34421233409063,
174.22883249586667, 75.42365191194322] Fitness: 1.2386289229033647
Generation 235 Best Solution: [182.88624774473723, 177.34421233409063,
175.06900789663524, 75.42365191194322] Fitness: 1.7639292079881959
Generation 236 Best Solution: [182.88624774473723, 177.34421233409063,
174.22883249586667, 75.89324167858553] Fitness: 1.8087670327444412
Generation 237 Best Solution: [182.88624774473723, 176.39995942934317,
175.06900789663524, 75.89324167858553] Fitness: 1.5481609179690574
Generation 238 Best Solution: [183.86676069493916, 175.79481949495442,
174.5213196911029, 75.89324167858553] Fitness: 1.3717203552629034
Generation 239 Best Solution: [183.79524586890838, 175.8768196324577,
174.18155151276505, 75.89324167858553] Fitness: 1.1946275303702978
Generation 240 Best Solution: [183.79524586890838, 176.53266317224544,
174.5213196911029, 75.39751026633446] Fitness: 0.6413514042058532
Generation 241 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.88532966247634] Fitness: 1.5690618392195386
Generation 242 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.88532966247634] Fitness: 1.5690618392195386
Generation 243 Best Solution: [183.18804612183538, 177.34421233409063,
175.06900789663524, 74.89064750146267] Fitness: 1.383093200199393
Generation 244 Best Solution: [183.79524586890838, 177.34421233409063,
173.73028839894832, 74.9134749720458] Fitness: 1.1826858762431172
Generation 245 Best Solution: [183.79524586890838, 177.34421233409063,
173.73028839894832, 74.9134749720458] Fitness: 1.1826858762431172
Generation 246 Best Solution: [183.79524586890838, 177.34421233409063,
173.73028839894832, 74.9134749720458] Fitness: 1.1826858762431172

Generation 247 Best Solution: [183.22515371864617, 177.34421233409063,
175.06900789663524, 74.89064750146267] Fitness: 1.3804157947918434
Generation 248 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 75.47517846610795] Fitness: 1.9091269149561096
Generation 249 Best Solution: [182.97355494586964, 177.34421233409063,
175.06900789663524, 75.34806745947927] Fitness: 1.6611219749243404
Generation 250 Best Solution: [183.37759669304805, 175.90338626287104,
175.06900789663524, 75.89324167858553] Fitness: 1.527468510260359
Generation 251 Best Solution: [183.79524586890838, 176.10712383029335,
175.06900789663524, 75.89324167858553] Fitness: 1.6429406398567794
Generation 252 Best Solution: [183.79524586890838, 177.34421233409063,
173.95541578337392, 75.40107216281437] Fitness: 1.3476579717028796
Generation 253 Best Solution: [183.79524586890838, 176.5360565169253,
174.53003424368393, 75.89324167858553] Fitness: 1.2399019368032176
Generation 254 Best Solution: [183.37759669304805, 177.34421233409063,
174.35497233463434, 75.89324167858553] Fitness: 1.7432117074562046
Generation 255 Best Solution: [183.40053108677205, 177.34421233409063,
175.06900789663524, 75.40107216281437] Fitness: 1.663472434601747
Generation 256 Best Solution: [183.42323557173157, 177.34421233409063,
175.06900789663524, 75.40107216281437] Fitness: 1.6680487216355373
Generation 257 Best Solution: [183.40053108677205, 177.34421233409063,
175.06900789663524, 75.40107216281437] Fitness: 1.663472434601747
Generation 258 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 75.40107216281437] Fitness: 1.8409226183177236
Generation 259 Best Solution: [183.04087224353435, 177.34421233409063,
175.06900789663524, 75.40107216281437] Fitness: 1.6826602917029563
Generation 260 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 75.05863087773082] Fitness: 1.620849586310796
Generation 261 Best Solution: [183.40053108677205, 177.34421233409063,
175.06900789663524, 74.72369958788704] Fitness: 1.3794015847113503
Generation 262 Best Solution: [183.42323557173157, 176.59088952961454,
175.06900789663524, 75.89324167858553] Fitness: 1.5353103119074556
Generation 263 Best Solution: [183.0526672087297, 176.37939219656087,
175.06900789663524, 75.89324167858553] Fitness: 1.479644091788443
Generation 264 Best Solution: [183.42323557173157, 176.59088952961454,
174.28972612828252, 75.89324167858553] Fitness: 1.0164468338728274
Generation 265 Best Solution: [183.42323557173157, 176.37939219656087,
175.06900789663524, 75.89324167858553] Fitness: 1.4683971465663668
Generation 266 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.53424031212957] Fitness: 1.586884143699299
Generation 267 Best Solution: [183.40053108677205, 177.34421233409063,
175.06900789663524, 74.86722196494757] Fitness: 1.388511166905049
Generation 268 Best Solution: [182.97633437987386, 177.34421233409063,
175.06900789663524, 75.16141585170945] Fitness: 1.5339151921735061
Generation 269 Best Solution: [183.79524586890838, 177.28947927765174,
175.06900789663524, 74.44229414778158] Fitness: 1.5406965169918923
Generation 270 Best Solution: [183.79524586890838, 177.34421233409063,
173.9915957388878, 75.31584111900752] Fitness: 1.2683073251119088

Generation 271 Best Solution: [183.79524586890838, 176.17718691472396,
175.06900789663524, 75.89324167858553] Fitness: 1.6330662504429234
Generation 272 Best Solution: [183.79524586890838, 176.17718691472396,
174.2678826593181, 75.89324167858553] Fitness: 1.1113251314622612
Generation 273 Best Solution: [183.79524586890838, 175.69738546866435,
174.2678826593181, 75.89324167858553] Fitness: 1.310008320091745
Generation 274 Best Solution: [183.79524586890838, 176.17718691472396,
175.06900789663524, 75.89324167858553] Fitness: 1.6330662504429234
Generation 275 Best Solution: [183.17061292977846, 177.28947927765174,
175.06900789663524, 75.89324167858553] Fitness: 2.1679840047490027
Generation 276 Best Solution: [183.17061292977846, 175.78660169053083,
175.06900789663524, 75.89324167858553] Fitness: 1.5865039518201514
Generation 277 Best Solution: [183.04563948947745, 177.34421233409063,
175.06900789663524, 74.38422948805272] Fitness: 1.4849681940411577
Generation 278 Best Solution: [183.22422493305035, 177.34421233409063,
175.06900789663524, 74.38422948805272] Fitness: 1.4550177306288374
Generation 279 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.475085840783] Fitness: 1.6060654989427208
Generation 280 Best Solution: [183.04563948947745, 177.34421233409063,
174.96258463836796, 74.6833553130601] Fitness: 1.2809942544666888
Generation 281 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.6833553130601] Fitness: 1.559236367484015
Generation 282 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.38422948805272] Fitness: 1.6446127576782772
Generation 283 Best Solution: [182.82167261920836, 177.28947927765174,
175.06900789663524, 74.80030544193059] Fitness: 1.4183069514001976
Generation 284 Best Solution: [183.79524586890838, 177.28947927765174,
175.06900789663524, 74.80030544193059] Fitness: 1.4802845704026604
Generation 285 Best Solution: [183.1885463560444, 175.89273480343442,
175.06900789663524, 75.17225088900076] Fitness: 0.7722703227986609
Generation 286 Best Solution: [183.1885463560444, 177.34421233409063,
173.87580430702772, 74.91952840979185] Fitness: 0.9238995100358259
Generation 287 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.65707623090253] Fitness: 1.561956826102778
Generation 288 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 75.16268941811045] Fitness: 1.671186587847128
Generation 289 Best Solution: [183.79524586890838, 177.28947927765174,
173.75800318179836, 75.7809895550917] Fitness: 1.7818565913535835
Generation 290 Best Solution: [183.79524586890838, 177.28947927765174,
173.75800318179836, 75.7809895550917] Fitness: 1.7818565913535835
Generation 291 Best Solution: [183.1885463560444, 177.34421233409063,
174.23678897851656, 75.89324167858553] Fitness: 1.719315962871045
Generation 292 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 75.00830100089263] Fitness: 1.6016833789115115
Generation 293 Best Solution: [183.24325831190131, 177.34421233409063,
175.06900789663524, 75.89324167858553] Fitness: 2.240786998302902
Generation 294 Best Solution: [183.79524586890838, 176.01993306868667,
175.06900789663524, 75.89324167858553] Fitness: 1.6643696520709725

Generation 295 Best Solution: [183.79524586890838, 176.01993306868667,
175.06900789663524, 75.89324167858553] Fitness: 1.6643696520709725
Generation 296 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 75.03756416622478] Fitness: 1.612416149711598
Generation 297 Best Solution: [183.1885463560444, 177.34421233409063,
175.06900789663524, 75.03756416622478] Fitness: 1.4254056141388263
Generation 298 Best Solution: [183.1885463560444, 177.34421233409063,
175.06900789663524, 75.10628946986654] Fitness: 1.4551013299100481
Generation 299 Best Solution: [183.79524586890838, 176.32091023297423,
175.06900789663524, 75.10628946986654] Fitness: 0.8443398979035239
Generation 300 Best Solution: [183.79524586890838, 176.12138897555383,
175.06900789663524, 75.10628946986654] Fitness: 0.8514448601471285
Generation 301 Best Solution: [183.23366551367124, 177.34421233409063,
175.06900789663524, 74.61072493982397] Fitness: 1.379004619283088
Generation 302 Best Solution: [183.1885463560444, 177.34421233409063,
175.06900789663524, 75.10628946986654] Fitness: 1.4551013299100481
Generation 303 Best Solution: [183.1885463560444, 175.58912111339214,
174.951442837546, 75.10628946986654] Fitness: 0.8129196382380175
Generation 304 Best Solution: [183.1885463560444, 177.04131392412248,
173.84261534806473, 75.89324167858553] Fitness: 1.411437545318958
Generation 305 Best Solution: [183.1885463560444, 177.34421233409063,
175.06900789663524, 75.1528563190183] Fitness: 1.4788017296739442
Generation 306 Best Solution: [182.89015507102064, 177.34421233409063,
175.06900789663524, 74.47074633161716] Fitness: 1.5087163900315377
Generation 307 Best Solution: [183.79524586890838, 177.34421233409063,
174.55705868273043, 74.47074633161716] Fitness: 1.1755484512183496
Generation 308 Best Solution: [183.79524586890838, 175.85333340161137,
175.06900789663524, 75.89324167858553] Fitness: 1.7335027179510387
Generation 309 Best Solution: [183.57114996750124, 177.29560111752582,
174.19318137444213, 75.89324167858553] Fitness: 1.7085716922809366
Generation 310 Best Solution: [183.79524586890838, 177.34421233409063,
174.17032818109988, 75.89324167858553] Fitness: 1.9042388691104055
Generation 311 Best Solution: [183.79524586890838, 177.34421233409063,
174.17032818109988, 74.58754836690139] Fitness: 1.0467676484188737
Generation 312 Best Solution: [183.79524586890838, 177.34421233409063,
174.17032818109988, 74.58754836690139] Fitness: 1.0467676484188737
Generation 313 Best Solution: [183.79524586890838, 177.34421233409063,
174.17032818109988, 74.58754836690139] Fitness: 1.0467676484188737
Generation 314 Best Solution: [183.79524586890838, 176.24233349787434,
174.57759190603292, 75.89324167858553] Fitness: 1.2082532590913069
Generation 315 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.40424060768731] Fitness: 1.635177592280649
Generation 316 Best Solution: [183.79524586890838, 175.96828127601617,
174.67990913484823, 75.89324167858553] Fitness: 1.3215612417765088
Generation 317 Best Solution: [183.79524586890838, 176.7288050268587,
175.06900789663524, 75.58584811260086] Fitness: 1.3774927282212348
Generation 318 Best Solution: [183.79524586890838, 176.34788015080196,
174.67990913484823, 75.89324167858553] Fitness: 1.2761225329754706

Generation 319 Best Solution: [183.79524586890838, 176.34788015080196,
174.67990913484823, 75.89324167858553] Fitness: 1.2761225329754706
Generation 320 Best Solution: [183.4783214809364, 177.34421233409063,
175.06900789663524, 75.25985352298255] Fitness: 1.572308824280549
Generation 321 Best Solution: [183.79524586890838, 176.7288050268587,
174.4834923350533, 75.89324167858553] Fitness: 1.3184590125345752
Generation 322 Best Solution: [183.33243363870466, 177.34421233409063,
175.06900789663524, 74.55527460730136] Fitness: 1.394134500227545
Generation 323 Best Solution: [183.33243363870466, 176.5711303883511,
175.06900789663524, 75.89324167858553] Fitness: 1.5123535586223618
Generation 324 Best Solution: [183.79524586890838, 176.41582826293612,
175.06900789663524, 75.89324167858553] Fitness: 1.6485462611337143
Generation 325 Best Solution: [183.5729154946834, 176.07676705622183,
175.06900789663524, 75.89324167858553] Fitness: 1.5237483291431058
Generation 326 Best Solution: [183.79524586890838, 176.7288050268587,
175.06900789663524, 74.98825730454803] Fitness: 0.9478683098005614
Generation 327 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.98825730454803] Fitness: 1.5949908686132417
Generation 328 Best Solution: [183.79524586890838, 176.28994299542242,
175.06900789663524, 75.89324167858553] Fitness: 1.6309175423374227
Generation 329 Best Solution: [183.79524586890838, 176.28994299542242,
175.06900789663524, 75.89324167858553] Fitness: 1.6309175423374227
Generation 330 Best Solution: [183.79524586890838, 175.90804112571516,
174.74679187012737, 75.89324167858553] Fitness: 1.3939986691925976
Generation 331 Best Solution: [183.33243363870466, 175.90804112571516,
175.06900789663524, 75.89324167858553] Fitness: 1.5196664481250668
Generation 332 Best Solution: [183.33243363870466, 175.90804112571516,
175.06900789663524, 75.89324167858553] Fitness: 1.5196664481250668
Generation 333 Best Solution: [183.33243363870466, 175.90804112571516,
175.06900789663524, 75.75170203423131] Fitness: 1.3168656689599327
Generation 334 Best Solution: [183.79524586890838, 177.34421233409063,
174.4641910631888, 75.89324167858553] Fitness: 1.9580195466540982
Generation 335 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.46139554843022] Fitness: 1.6111695401736474
Generation 336 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.46139554843022] Fitness: 1.6111695401736474
Generation 337 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.79726407072293] Fitness: 1.5580900731333285
Generation 338 Best Solution: [183.79524586890838, 177.34421233409063,
174.4641910631888, 75.89324167858553] Fitness: 1.9580195466540982
Generation 339 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.76816984366633] Fitness: 1.5567377806620137
Generation 340 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.76816984366633] Fitness: 1.5567377806620137
Generation 341 Best Solution: [183.79524586890838, 176.1798069190278,
174.4641910631888, 75.89324167858553] Fitness: 1.1597769650665146
Generation 342 Best Solution: [183.79524586890838, 176.61906434321602,
175.10449859438384, 75.87368711282978] Fitness: 1.7348999183060148

Generation 343 Best Solution: [182.82319218213962, 176.16670353394952,
175.06900789663524, 75.89324167858553] Fitness: 1.5712617202954156
Generation 344 Best Solution: [183.79524586890838, 177.34421233409063,
174.4641910631888, 74.31174028402646] Fitness: 1.2102143332345068
Generation 345 Best Solution: [183.7208502789483, 177.34421233409063,
175.06900789663524, 74.51318755740591] Fitness: 1.543843332234776
Generation 346 Best Solution: [183.79524586890838, 177.34421233409063,
174.25765634668358, 75.802387603933] Fitness: 1.7749493509440972
Generation 347 Best Solution: [183.79524586890838, 177.34421233409063,
174.25765634668358, 75.87368711282978] Fitness: 1.8785884362452592
Generation 348 Best Solution: [183.79524586890838, 177.34421233409063,
175.10449859438384, 74.72952288925691] Fitness: 1.5995952138478775
Generation 349 Best Solution: [183.79524586890838, 176.18649235615194,
175.06900789663524, 75.89324167858553] Fitness: 1.6322471543782424
Generation 350 Best Solution: [183.79524586890838, 177.34421233409063,
174.25765634668358, 74.79975145044249] Fitness: 1.0353883681593627
Generation 351 Best Solution: [183.79524586890838, 176.0722052075743,
175.10449859438384, 75.87368711282978] Fitness: 1.663605796152902
Generation 352 Best Solution: [183.57621122546672, 177.34421233409063,
175.10449859438384, 75.2981971330695] Fitness: 1.67716339086989
Generation 353 Best Solution: [183.79524586890838, 177.34421233409063,
174.32345935868122, 75.2981971330695] Fitness: 1.245209144879468
Generation 354 Best Solution: [183.57621122546672, 177.34421233409063,
175.10449859438384, 74.43267380992413] Fitness: 1.5414743432244433
Generation 355 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.43267380992413] Fitness: 1.6226897323584135
Generation 356 Best Solution: [183.79524586890838, 176.67106984812835,
175.03257307839826, 75.87368711282978] Fitness: 1.6772315258370027
Generation 357 Best Solution: [183.57621122546672, 176.67106984812835,
175.03257307839826, 75.87368711282978] Fitness: 1.5531077295444449
Generation 358 Best Solution: [183.79524586890838, 177.34421233409063,
174.33706891845588, 75.87368711282978] Fitness: 1.8910151942217002
Generation 359 Best Solution: [183.57621122546672, 176.32609950855206,
175.10449859438384, 75.87368711282978] Fitness: 1.5229941768076318
Generation 360 Best Solution: [183.57621122546672, 176.32609950855206,
175.10449859438384, 75.87368711282978] Fitness: 1.5229941768076318
Generation 361 Best Solution: [183.57621122546672, 176.62681493836894,
175.10449859438384, 75.802387603933] Fitness: 1.5110131868328822
Generation 362 Best Solution: [183.79524586890838, 176.62681493836894,
174.33706891845588, 75.87368711282978] Fitness: 1.1854247311780712
Generation 363 Best Solution: [183.79524586890838, 177.34421233409063,
175.03257307839826, 75.14069202260046] Fitness: 1.6170394144343152
Generation 364 Best Solution: [183.79524586890838, 176.41599642050218,
175.06900789663524, 75.802387603933] Fitness: 1.515336446572078
Generation 365 Best Solution: [183.79524586890838, 176.39946897866912,
175.06900789663524, 75.89324167858553] Fitness: 1.6450608288521762
Generation 366 Best Solution: [183.79524586890838, 176.0477063559828,
175.06900789663524, 75.14069202260046] Fitness: 0.8847192933757542

Generation 367 Best Solution: [183.79524586890838, 176.0477063559828,
175.06900789663524, 75.802387603933] Fitness: 1.5231961271050305
Generation 368 Best Solution: [183.79524586890838, 177.34421233409063,
175.10449859438384, 74.51916907119332] Fitness: 1.6342366066920375
Generation 369 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.50910803269714] Fitness: 1.5944634685206984
Generation 370 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.50910803269714] Fitness: 1.5944634685206984
Generation 371 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.50910803269714] Fitness: 1.5944634685206984
Generation 372 Best Solution: [183.79524586890838, 176.7027616015639,
175.06900789663524, 74.50910803269714] Fitness: 0.9310923759918253
Generation 373 Best Solution: [183.79524586890838, 177.18433586976198,
175.1380226536278, 74.50910803269714] Fitness: 1.4627768453106347
Generation 374 Best Solution: [183.79524586890838, 177.34421233409063,
174.6034861890151, 75.89324167858553] Fitness: 2.0237368745808615
Generation 375 Best Solution: [183.11052953776675, 177.34421233409063,
174.6034861890151, 75.89324167858553] Fitness: 1.848291171303575
Generation 376 Best Solution: [183.11052953776675, 177.34421233409063,
174.6034861890151, 75.89324167858553] Fitness: 1.848291171303575
Generation 377 Best Solution: [183.79524586890838, 177.34421233409063,
174.24824373167587, 75.89324167858553] Fitness: 1.9072813109985463
Generation 378 Best Solution: [183.2984405711961, 177.34421233409063,
175.06900789663524, 74.3678865888611] Fitness: 1.4631806308447866
Generation 379 Best Solution: [183.79524586890838, 177.34421233409063,
175.0433115248061, 74.3678865888611] Fitness: 1.622695623777406
Generation 380 Best Solution: [183.79524586890838, 177.34421233409063,
174.87348455088667, 74.8404873272897] Fitness: 1.3559067988570266
Generation 381 Best Solution: [183.61992173765165, 176.24930440569344,
173.93209186036606, 75.89324167858553] Fitness: 1.0393577164013332
Generation 382 Best Solution: [182.92548014996092, 177.34421233409063,
175.06900789663524, 74.2806007443061] Fitness: 1.5864536796720379
Generation 383 Best Solution: [182.92548014996092, 176.24930440569344,
175.06900789663524, 75.89324167858553] Fitness: 1.514176734288011
Generation 384 Best Solution: [182.92548014996092, 176.24930440569344,
175.06900789663524, 75.89324167858553] Fitness: 1.514176734288011
Generation 385 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.52853926706388] Fitness: 1.5885295972142173
Generation 386 Best Solution: [183.79524586890838, 176.11835054442062,
174.67968928989302, 75.89324167858553] Fitness: 1.280487599808375
Generation 387 Best Solution: [183.79524586890838, 176.40918921165706,
175.0433115248061, 75.89324167858553] Fitness: 1.6170699196842944
Generation 388 Best Solution: [183.79524586890838, 176.40918921165706,
175.0433115248061, 75.89127194187583] Fitness: 1.6140643649704405
Generation 389 Best Solution: [183.39711841952496, 177.34421233409063,
175.0433115248061, 75.89324167858553] Fitness: 2.222969081824285
Generation 390 Best Solution: [183.39711841952496, 177.34421233409063,
175.0433115248061, 75.89324167858553] Fitness: 2.222969081824285

Generation 391 Best Solution: [183.43327044927273, 176.32306272818363,
175.06900789663524, 75.89324167858553] Fitness: 1.4628750604895358
Generation 392 Best Solution: [183.39711841952496, 177.34421233409063,
175.06900789663524, 74.40268628370828] Fitness: 1.457812707648713
Generation 393 Best Solution: [183.79524586890838, 177.34421233409063,
175.0433115248061, 74.33247065044971] Fitness: 1.6414745012243996
Generation 394 Best Solution: [183.79524586890838, 177.34421233409063,
175.0433115248061, 74.33247065044971] Fitness: 1.6414745012243996
Generation 395 Best Solution: [183.59953787408315, 176.8180642817073,
175.06900789663524, 75.1096747656293] Fitness: 0.9452166291877357
Generation 396 Best Solution: [183.49248898738827, 176.81292634782037,
175.06900789663524, 74.61726515522734] Fitness: 0.8247015549127975
Generation 397 Best Solution: [183.49248898738827, 176.81292634782037,
175.06900789663524, 75.54648717204206] Fitness: 1.238538659101338
Generation 398 Best Solution: [183.49248898738827, 176.81292634782037,
175.10901435897276, 74.71882560339992] Fitness: 0.8623823562113989
Generation 399 Best Solution: [183.81176158363118, 176.25808873372108,
175.06900789663524, 75.6438859297284] Fitness: 1.3024004955315807
Generation 400 Best Solution: [183.39711841952496, 176.81292634782037,
175.10901435897276, 75.6438859297284] Fitness: 1.3736565775469305
Generation 401 Best Solution: [183.81176158363118, 176.76138606738152,
175.06900789663524, 75.89324167858553] Fitness: 1.8174974413017826
Generation 402 Best Solution: [183.79524586890838, 177.34421233409063,
174.21082021317028, 74.54670214704979] Fitness: 1.0571814951847733
Generation 403 Best Solution: [183.81176158363118, 176.76138606738152,
174.67560501935736, 75.89324167858553] Fitness: 1.4543569712567135
Generation 404 Best Solution: [183.79524586890838, 177.34421233409063,
175.0433115248061, 75.11664417463754] Fitness: 1.6171131257142994
Generation 405 Best Solution: [183.79524586890838, 177.34421233409063,
174.4009641729392, 75.53398963209698] Fitness: 1.4741256541734482
Generation 406 Best Solution: [183.81176158363118, 176.76138606738152,
175.06900789663524, 75.13520496370921] Fitness: 1.04291917439053
Generation 407 Best Solution: [183.39711841952496, 177.34421233409063,
175.0433115248061, 75.11664417463754] Fitness: 1.4390345085733987
Generation 408 Best Solution: [183.79524586890838, 176.0902849425542,
174.01793284555023, 75.89324167858553] Fitness: 1.1369098767332857
Generation 409 Best Solution: [183.79524586890838, 176.0902849425542,
175.06900789663524, 75.89324167858553] Fitness: 1.646289395218866
Generation 410 Best Solution: [183.81176158363118, 177.34421233409063,
175.06900789663524, 74.78761905370844] Fitness: 1.5694694572286196
Generation 411 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 75.22716745786434] Fitness: 1.7096216288341057
Generation 412 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 75.22716745786434] Fitness: 1.7096216288341057
Generation 413 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.63933952571519] Fitness: 1.5643134256060163
Generation 414 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.63933952571519] Fitness: 1.5643134256060163

Generation 415 Best Solution: [183.79524586890838, 175.82051545380315,
175.06900789663524, 75.89324167858553] Fitness: 1.7514840091490065
Generation 416 Best Solution: [183.81176158363118, 177.34421233409063,
175.06900789663524, 74.39973213835962] Fitness: 1.649209463363185
Generation 417 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.28446880770933] Fitness: 1.6996152095532766
Generation 418 Best Solution: [183.79524586890838, 176.10318054510128,
175.06900789663524, 74.28446880770933] Fitness: 0.91223961794902
Generation 419 Best Solution: [183.79524586890838, 176.10318054510128,
175.06900789663524, 74.28446880770933] Fitness: 0.91223961794902
Generation 420 Best Solution: [183.79524586890838, 175.82051545380315,
175.06900789663524, 74.48692064329353] Fitness: 0.9222720905615267
Generation 421 Best Solution: [183.79524586890838, 175.82051545380315,
175.06900789663524, 74.48692064329353] Fitness: 0.9222720905615267
Generation 422 Best Solution: [183.62720941765724, 175.82051545380315,
175.06900789663524, 74.48692064329353] Fitness: 0.8213352027654575
Generation 423 Best Solution: [183.03167659300388, 175.7582254010241,
175.06900789663524, 75.89324167858553] Fitness: 1.6340523922537062
Generation 424 Best Solution: [183.79524586890838, 175.82051545380315,
175.06900789663524, 75.89324167858553] Fitness: 1.7514840091490065
Generation 425 Best Solution: [183.79524586890838, 176.4337817326244,
175.06900789663524, 75.89324167858553] Fitness: 1.652782031733405
Generation 426 Best Solution: [183.03167659300388, 176.4337817326244,
175.06900789663524, 75.89324167858553] Fitness: 1.4972715325668127
Generation 427 Best Solution: [183.03167659300388, 176.4337817326244,
175.06900789663524, 75.89324167858553] Fitness: 1.4972715325668127
Generation 428 Best Solution: [183.03167659300388, 176.4337817326244,
175.06900789663524, 75.89324167858553] Fitness: 1.4972715325668127
Generation 429 Best Solution: [183.03167659300388, 177.1295480348924,
174.77477256046345, 75.33772772705534] Fitness: 1.0587860697900195
Generation 430 Best Solution: [183.03167659300388, 176.0366621651021,
175.06900789663524, 75.8717440126338] Fitness: 1.4714390838464706
Generation 431 Best Solution: [183.76645294931546, 176.4337817326244,
175.06900789663524, 75.89324167858553] Fitness: 1.6328137533528184
Generation 432 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.63294880887929] Fitness: 1.5652653269309478
Generation 433 Best Solution: [183.03167659300388, 176.55963761449866,
174.08151639313596, 75.89324167858553] Fitness: 1.0186069258721575
Generation 434 Best Solution: [183.03167659300388, 176.55963761449866,
174.08151639313596, 75.89324167858553] Fitness: 1.0186069258721575
Generation 435 Best Solution: [183.79524586890838, 176.55963761449866,
174.08151639313596, 75.89324167858553] Fitness: 1.1741174250387498
Generation 436 Best Solution: [183.79524586890838, 176.55963761449866,
174.08151639313596, 75.89324167858553] Fitness: 1.1741174250387498
Generation 437 Best Solution: [183.79524586890838, 177.34421233409063,
175.06900789663524, 74.40914136942904] Fitness: 1.6329482935652366
Generation 438 Best Solution: [183.03167659300388, 177.34421233409063,
173.8680353816549, 75.89324167858553] Fitness: 1.813547289289395

Generation 439 Best Solution: [183.81176158363118, 177.34421233409063,
175.06900789663524, 74.94771206103364] Fitness: 1.5950434430718126
Generation 440 Best Solution: [182.99908683649576, 177.34421233409063,
173.74669224125492, 75.70096366116502] Fitness: 1.6155010551208606
Generation 441 Best Solution: [183.15405074362948, 177.34421233409063,
173.74669224125492, 75.70096366116502] Fitness: 1.5774531811378525
Generation 442 Best Solution: [183.81176158363118, 177.0488747260225,
175.06900789663524, 75.81370074611924] Fitness: 1.9521868173990415
Generation 443 Best Solution: [182.99908683649576, 177.34421233409063,
175.06900789663524, 74.96275075056334] Fitness: 1.4423988243762937
Generation 444 Best Solution: [182.99908683649576, 177.34421233409063,
174.50576809230066, 75.81370074611924] Fitness: 1.7128209134850523
Generation 445 Best Solution: [183.81176158363118, 177.0488747260225,
175.06900789663524, 75.89324167858553] Fitness: 2.069442249446372
Generation 446 Best Solution: [183.26379498776566, 177.34421233409063,
175.06900789663524, 75.81370074611924] Fitness: 2.1233347527244852
Generation 447 Best Solution: [183.81176158363118, 177.34421233409063,
175.06900789663524, 75.34369214449868] Fitness: 1.8050950169797009
Generation 448 Best Solution: [183.86224039012893, 177.37311282666496,
175.06900789663524, 75.19963205258358] Fitness: 1.7860732824631087
Generation 449 Best Solution: [183.03167659300388, 177.34421233409063,
175.06900789663524, 75.62182394473403] Fitness: 1.9101641914745946
Generation 450 Best Solution: [183.86224039012893, 177.37311282666496,
173.95082742028842, 75.62182394473403] Fitness: 1.6673415709540655
Generation 451 Best Solution: [183.79524586890838, 175.97975448143015,
175.06900789663524, 75.81370074611924] Fitness: 1.5604006556754886
Generation 452 Best Solution: [183.03167659300388, 175.85132534113112,
175.06900789663524, 75.89324167858553] Fitness: 1.579051210366796
Generation 453 Best Solution: [183.81176158363118, 176.34500328176898,
175.06900789663524, 75.89324167858553] Fitness: 1.647981051956661
Generation 454 Best Solution: [183.57420687741663, 175.622604902384,
175.06900789663524, 75.62182394473403] Fitness: 1.4000075243027488
Generation 455 Best Solution: [183.03167659300388, 177.37311282666496,
175.06900789663524, 75.11369633649414] Fitness: 1.5329815204180928
Generation 456 Best Solution: [183.03167659300388, 176.41118495344415,
175.06900789663524, 75.81370074611924] Fitness: 1.3747547809401646
Generation 457 Best Solution: [183.03167659300388, 176.41118495344415,
175.06900789663524, 75.81370074611924] Fitness: 1.3747547809401646
Generation 458 Best Solution: [183.03167659300388, 176.41118495344415,
175.06900789663524, 75.81370074611924] Fitness: 1.3747547809401646
Generation 459 Best Solution: [183.03167659300388, 176.41118495344415,
175.06900789663524, 75.81370074611924] Fitness: 1.3747547809401646
Generation 460 Best Solution: [183.79524586890838, 177.34421233409063,
174.45790777165726, 74.41336451291032] Fitness: 1.155651382623524
Generation 461 Best Solution: [183.03167659300388, 177.34421233409063,
174.45790777165726, 74.41336451291032] Fitness: 1.0001408834569314
Generation 462 Best Solution: [183.20366535732867, 177.37311282666496,
175.06900789663524, 74.52193662029262] Fitness: 1.444980242059889

Generation 463 Best Solution: [183.20366535732867, 176.79862597104807,
175.06900789663524, 74.77662429903658] Fitness: 0.7696025186757756
Generation 464 Best Solution: [183.1784472791593, 176.79004742472864,
174.45790777165726, 74.30942295894022] Fitness: 0.4178904876135071
Generation 465 Best Solution: [183.79524586890838, 177.34421233409063,
174.45790777165726, 75.24077127759651] Fitness: 1.2430373991149373
Generation 466 Best Solution: [183.79524586890838, 176.37265431366438,
175.01117545054302, 75.81370074611924] Fitness: 1.4565425797307072
Generation 467 Best Solution: [183.20366535732867, 177.37311282666496,
174.14915858664332, 75.24077127759651] Fitness: 1.0466738515319272
Generation 468 Best Solution: [183.79524586890838, 176.37265431366438,
175.06900789663524, 75.24077127759651] Fitness: 0.9274920252677394
Generation 469 Best Solution: [183.20366535732867, 176.1589832458055,
175.06900789663524, 75.81370074611924] Fitness: 1.329434055216062
Generation 470 Best Solution: [183.20366535732867, 176.1589832458055,
175.06900789663524, 75.89324167858553] Fitness: 1.4466894872633924
Generation 471 Best Solution: [183.20366535732867, 176.1589832458055,
175.06900789663524, 75.89324167858553] Fitness: 1.4466894872633924
Generation 472 Best Solution: [183.20366535732867, 176.1589832458055,
175.06900789663524, 75.89324167858553] Fitness: 1.4466894872633924
Generation 473 Best Solution: [183.20366535732867, 176.1589832458055,
175.06900789663524, 75.89324167858553] Fitness: 1.4466894872633924
Generation 474 Best Solution: [183.20366535732867, 176.1589832458055,
173.98390268045264, 75.89324167858553] Fitness: 0.9454355662326213
Generation 475 Best Solution: [183.79524586890838, 176.2105175697925,
173.81437966112213, 75.89324167858553] Fitness: 1.1928950201053272
Generation 476 Best Solution: [183.79524586890838, 175.6737040550906,
175.06900789663524, 75.16766595846644] Fitness: 1.092393713126264
Generation 477 Best Solution: [183.79524586890838, 175.77942313433732,
175.06900789663524, 75.16766595846644] Fitness: 1.0189128145569994
Generation 478 Best Solution: [183.79524586890838, 177.37311282666496,
173.98390268045264, 75.05187094895966] Fitness: 1.1596288176872256
Generation 479 Best Solution: [183.79524586890838, 175.76613949347208,
175.06900789663524, 74.73489072087972] Fitness: 0.9099462544414866
Generation 480 Best Solution: [183.79524586890838, 176.10276673723192,
174.64348639211568, 75.89324167858553] Fitness: 1.2600939438140486
Generation 481 Best Solution: [183.8399357089136, 175.8175588892857,
174.3516481698818, 75.89324167858553] Fitness: 1.2791087861010955
Generation 482 Best Solution: [183.79524586890838, 175.8175588892857,
174.74890422279188, 75.16766595846644] Fitness: 0.6849415753032044
Generation 483 Best Solution: [183.20366535732867, 176.45386094130168,
175.01117545054302, 75.89324167858553] Fitness: 1.4033938770647312
Generation 484 Best Solution: [183.79524586890838, 175.76612760233368,
175.01117545054302, 75.16766595846644] Fitness: 0.9610131606773822
Generation 485 Best Solution: [183.20366535732867, 176.45386094130168,
174.74890422279188, 75.60032320359086] Fitness: 0.7684552338261823
Generation 486 Best Solution: [183.79524586890838, 176.45386094130168,
175.01117545054302, 74.71785788351285] Fitness: 0.7177029797830798

```
Generation 487 Best Solution: [183.79524586890838, 176.45386094130168,
175.01117545054302, 74.71785788351285] Fitness: 0.7177029797830798
Generation 488 Best Solution: [183.79524586890838, 175.68205112005344,
175.06900789663524, 74.6278477543432] Fitness: 0.9781587124044645
Generation 489 Best Solution: [183.20366535732867, 177.37311282666496,
175.06900789663524, 75.16766595846644] Fitness: 1.5284464814915297
Generation 490 Best Solution: [183.79524586890838, 176.45386094130168,
174.56244795560832, 75.89324167858553] Fitness: 1.2286493719751554
Generation 491 Best Solution: [183.22049516047187, 176.45386094130168,
175.06900789663524, 75.89324167858553] Fitness: 1.4686241493978915
Generation 492 Best Solution: [183.20366535732867, 177.37311282666496,
174.3960025765916, 75.16766595846644] Fitness: 1.0326611967011257
Generation 493 Best Solution: [183.20366535732867, 177.34421233409063,
175.06900789663524, 74.71274512239907] Fitness: 1.3689716573049724
Generation 494 Best Solution: [183.20366535732867, 177.37311282666496,
175.06900789663524, 75.16766595846644] Fitness: 1.5284464814915297
Generation 495 Best Solution: [183.55931320542462, 177.37311282666496,
175.01117545054302, 74.71274512239907] Fitness: 1.402724972321645
Generation 496 Best Solution: [183.55931320542462, 176.834812683792,
175.0317390708002, 74.46392885426599] Fitness: 0.8642620971990902
Generation 497 Best Solution: [183.20366535732867, 176.63198820540478,
175.0317390708002, 74.71274512239907] Fitness: 0.6227625721830694
Generation 498 Best Solution: [183.79524586890838, 177.37311282666496,
175.06900789663524, 74.45532601468827] Fitness: 1.6563161164825384
Generation 499 Best Solution: [183.20366535732867, 175.98109489247963,
175.06900789663524, 75.8656387527033] Fitness: 1.4472185014459824
Generation 500 Best Solution: [183.20366535732867, 177.37311282666496,
175.06900789663524, 74.42684078512927] Fitness: 1.479654521668583
Optimal Solution for Simulation 15: [183.20366535732867, 177.37311282666496,
175.06900789663524, 74.42684078512927]
--------------------------------------------------
Simulation 16/32 with conditions: {'drought': True, 'flood': False,
'policy_change': False, 'social_event': False, 'economic_crisis': False}
Generation 1 Best Solution: [183.39026115069234, 176.41689285276664,
173.85733302298516, 74.74783839026252] Fitness: 0.15265446662609547
Generation 2 Best Solution: [183.27961183856516, 175.9786716152517,
173.87190509591653, 74.4629622482945] Fitness: 0.20064819880802026
Generation 3 Best Solution: [183.14903696343725, 176.74953489118334,
174.47929509944635, 74.4629622482945] Fitness: 0.3780417312144403
Generation 4 Best Solution: [183.5231504857426, 177.0810704511173,
174.3858021531582, 74.4629622482945] Fitness: 0.7054428700314818
Generation 5 Best Solution: [183.04583623966312, 177.31624020308138,
173.84504828629093, 74.6364224807169] Fitness: 0.9414718816721858
Generation 6 Best Solution: [183.14903696343725, 176.20752065036908,
174.80091034050471, 74.4629622482945] Fitness: 0.39360481288440735
Generation 7 Best Solution: [183.5231504857426, 175.81379842224882,
174.47929509944635, 74.2925309100132] Fitness: 0.43494021004748545
Generation 8 Best Solution: [183.5231504857426, 175.81379842224882,
```

174.47929509944635, 74.2925309100132] Fitness: 0.43494021004748545
Generation 9 Best Solution: [183.5231504857426, 175.76654027951636,
174.80091034050471, 74.2925309100132] Fitness: 0.6666547655691735
Generation 10 Best Solution: [183.51702791299897, 176.74953489118334,
174.88031263716297, 74.32671268128965] Fitness: 0.777506251758529
Generation 11 Best Solution: [183.51702791299897, 176.74953489118334,
174.47929509944635, 75.81252782808993] Fitness: 1.0690768179237773
Generation 12 Best Solution: [183.51702791299897, 176.65079592238217,
174.80091034050471, 75.81252782808993] Fitness: 1.2117394202742908
Generation 13 Best Solution: [183.51702791299897, 177.31624020308138,
174.47929509944635, 74.2925309100132] Fitness: 1.1295838947235604
Generation 14 Best Solution: [183.51702791299897, 175.9332002744075,
174.80091034050471, 75.81252782808993] Fitness: 1.1343656694277913
Generation 15 Best Solution: [183.51702791299897, 177.31624020308138,
174.47929509944635, 74.2925309100132] Fitness: 1.1295838947235604
Generation 16 Best Solution: [183.38220653183458, 177.31624020308138,
174.80091034050471, 75.43877641601262] Fitness: 1.429123084091261
Generation 17 Best Solution: [183.66761913764242, 177.31624020308138,
174.88031263716297, 74.83528772854432] Fitness: 1.3300399559826996
Generation 18 Best Solution: [183.8459365696003, 175.9319700487787,
174.38813730782982, 75.81252782808993] Fitness: 1.0931004577390344
Generation 19 Best Solution: [183.8459365696003, 175.9319700487787,
174.80091034050471, 75.81252782808993] Fitness: 1.332790843416704
Generation 20 Best Solution: [183.1476380488231, 177.31624020308138,
174.38813730782982, 75.81252782808993] Fitness: 1.5898952987118202
Generation 21 Best Solution: [183.30955646631588, 177.31624020308138,
174.38813730782982, 75.43877641601262] Fitness: 1.1775040473413196
Generation 22 Best Solution: [183.66761913764242, 175.99591350025085,
174.80091034050471, 75.81252782808993] Fitness: 1.1874760195611238
Generation 23 Best Solution: [183.83656165606922, 176.13567349639345,
174.80091034050471, 75.43877641601262] Fitness: 0.8643536949918884
Generation 24 Best Solution: [183.1476380488231, 177.31624020308138,
174.38813730782982, 75.65860360035235] Fitness: 1.3972425880271093
Generation 25 Best Solution: [183.51702791299897, 177.31624020308138,
174.88031263716297, 75.22932835106973] Fitness: 1.3886405524029821
Generation 26 Best Solution: [183.56028797159118, 177.31624020308138,
174.38813730782982, 75.73221032445183] Fitness: 1.5576178407650572
Generation 27 Best Solution: [182.79910967688093, 175.96089384446,
174.38813730782982, 75.81252782808993] Fitness: 0.9431942813988086
Generation 28 Best Solution: [183.83656165606922, 175.96089384446,
174.38813730782982, 75.81252782808993] Fitness: 1.0755853324680853
Generation 29 Best Solution: [183.1476380488231, 176.77242788749405,
174.88031263716297, 75.81252782808993] Fitness: 1.308897078943551
Generation 30 Best Solution: [183.83656165606922, 177.31624020308138,
174.88031263716297, 75.02407408172766] Fitness: 1.4851341507349778
Generation 31 Best Solution: [183.83656165606922, 176.065267555799,
174.88031263716297, 75.81252782808993] Fitness: 1.3601524018269138
Generation 32 Best Solution: [183.83656165606922, 176.1211994342388,

175.1243813399826, 74.72759907876669] Fitness: 0.924078414432045
Generation 33 Best Solution: [183.83656165606922, 176.065267555799,
174.88031263716297, 75.81252782808993] Fitness: 1.3601524018269138
Generation 34 Best Solution: [183.1200404158723, 177.31624020308138,
174.80091034050471, 74.98269818100239] Fitness: 1.1598616840236595
Generation 35 Best Solution: [183.20554989056578, 177.31624020308138,
173.87041793202607, 75.81252782808993] Fitness: 1.6001598747133015
Generation 36 Best Solution: [183.25908870160498, 177.31624020308138,
175.1243813399826, 74.72759907876669] Fitness: 1.4711576872477852
Generation 37 Best Solution: [183.83656165606922, 176.2644526117006,
174.88031263716297, 75.81252782808993] Fitness: 1.348597958299407
Generation 38 Best Solution: [183.798347863898, 177.31624020308138,
174.38813730782982, 74.82227158444816] Fitness: 1.1053308122899246
Generation 39 Best Solution: [183.83656165606922, 177.31624020308138,
174.80091034050471, 74.82227158444816] Fitness: 1.3748007788759355
Generation 40 Best Solution: [183.83656165606922, 177.31624020308138,
174.12085147838133, 75.81252782808993] Fitness: 1.7977404519180509
Generation 41 Best Solution: [183.83656165606922, 176.0667426642427,
173.87781940449017, 75.81252782808993] Fitness: 1.059478198511958
Generation 42 Best Solution: [183.83656165606922, 177.31624020308138,
174.67914779756796, 74.55448889742922] Fitness: 1.311590047966756
Generation 43 Best Solution: [183.83656165606922, 177.31624020308138,
174.38813730782982, 74.57119349052178] Fitness: 1.1610699454573798
Generation 44 Best Solution: [183.83656165606922, 175.8836627984569,
173.88112516927808, 75.81252782808993] Fitness: 1.1142249403065427
Generation 45 Best Solution: [183.83656165606922, 175.8836627984569,
173.88112516927808, 75.81252782808993] Fitness: 1.1142249403065427
Generation 46 Best Solution: [183.6132127903893, 175.83834239318384,
175.1243813399826, 75.81252782808993] Fitness: 1.5608492875991622
Generation 47 Best Solution: [183.83656165606922, 177.31624020308138,
175.1243813399826, 74.99458493090452] Fitness: 1.7477166472373287
Generation 48 Best Solution: [183.83656165606922, 177.31624020308138,
174.67492168902723, 75.57783512103661] Fitness: 1.6959892735501063
Generation 49 Best Solution: [183.83656165606922, 177.0481027200941,
175.1243813399826, 74.99458493090452] Fitness: 1.4074718206318717
Generation 50 Best Solution: [183.83656165606922, 177.31624020308138,
175.1243813399826, 74.99458493090452] Fitness: 1.7477166472373287
Generation 51 Best Solution: [183.83656165606922, 177.31624020308138,
175.1243813399826, 74.99458493090452] Fitness: 1.7477166472373287
Generation 52 Best Solution: [183.83656165606922, 176.97840446724197,
175.1243813399826, 74.99458493090452] Fitness: 1.3343868150956604
Generation 53 Best Solution: [183.83656165606922, 176.97840446724197,
174.38813730782982, 75.81252782808993] Fitness: 1.4210733081506082
Generation 54 Best Solution: [183.83656165606922, 176.33501549320457,
175.1243813399826, 75.81252782808993] Fitness: 1.6284756168465346
Generation 55 Best Solution: [183.83656165606922, 176.33501549320457,
175.1243813399826, 74.72692355929931] Fitness: 0.9293572855163318
Generation 56 Best Solution: [183.83656165606922, 176.33501549320457,

174.6170628900227, 75.81252782808993] Fitness: 1.151118436148806
Generation 57 Best Solution: [183.83656165606922, 176.33501549320457,
175.1243813399826, 74.92452070169955] Fitness: 0.9421724388651225
Generation 58 Best Solution: [183.83656165606922, 176.33501549320457,
175.1243813399826, 74.92452070169955] Fitness: 0.9421724388651225
Generation 59 Best Solution: [183.83656165606922, 176.22846620654107,
175.1243813399826, 74.92452070169955] Fitness: 0.9321094355133527
Generation 60 Best Solution: [183.3321547192844, 176.58009946363902,
175.1243813399826, 74.92452070169955] Fitness: 0.7818041280357011
Generation 61 Best Solution: [183.3321547192844, 176.58009946363902,
174.38813730782982, 75.81252782808993] Fitness: 0.885265168820133
Generation 62 Best Solution: [183.3321547192844, 176.58009946363902,
174.38813730782982, 74.72692355929931] Fitness: 0.18614683748992988
Generation 63 Best Solution: [183.83656165606922, 177.31624020308138,
175.1243813399826, 74.72692355929931] Fitness: 1.7181269461590538
Generation 64 Best Solution: [183.83656165606922, 177.31624020308138,
175.1243813399826, 74.72692355929931] Fitness: 1.7181269461590538
Generation 65 Best Solution: [183.83656165606922, 177.31624020308138,
174.48026704760363, 75.27718864326145] Fitness: 1.3317661203367432
Generation 66 Best Solution: [183.83656165606922, 176.30421197014542,
175.1243813399826, 74.48375903495612] Fitness: 0.9790543953193852
Generation 67 Best Solution: [183.83656165606922, 176.30421197014542,
175.1243813399826, 74.48375903495612] Fitness: 0.9790543953193852
Generation 68 Best Solution: [183.83656165606922, 177.31624020308138,
174.90432459000996, 74.48375903495612] Fitness: 1.523967902901702
Generation 69 Best Solution: [183.83656165606922, 176.22468155741996,
175.1243813399826, 74.48034324350347] Fitness: 0.97464683984432
Generation 70 Best Solution: [183.83656165606922, 177.24849189529502,
174.41565010156572, 75.81252782808993] Fitness: 1.7486455194722441
Generation 71 Best Solution: [183.83656165606922, 176.82153843525106,
174.57540048330165, 75.81252782808993] Fitness: 1.360962801095853
Generation 72 Best Solution: [183.83656165606922, 176.82153843525106,
174.57540048330165, 75.81252782808993] Fitness: 1.360962801095853
Generation 73 Best Solution: [183.83656165606922, 177.31624020308138,
174.57540048330165, 74.70774894594715] Fitness: 1.217951496393126
Generation 74 Best Solution: [183.83656165606922, 176.05582343629305,
174.57540048330165, 75.81252782808993] Fitness: 1.1318998891449188
Generation 75 Best Solution: [183.83656165606922, 177.31624020308138,
174.72864434867307, 75.81252782808993] Fitness: 2.016081780026171
Generation 76 Best Solution: [183.83656165606922, 176.79085781546274,
175.1243813399826, 75.50985454158113] Fitness: 1.4892542697598463
Generation 77 Best Solution: [183.83656165606922, 177.0908536360962,
174.89053892614518, 75.50985454158113] Fitness: 1.513759336985147
Generation 78 Best Solution: [183.83656165606922, 176.76509382819262,
174.72864434867307, 75.77877746561653] Fitness: 1.3735597907833332
Generation 79 Best Solution: [183.49745465339979, 177.31624020308138,
175.1243813399826, 74.43533143382524] Fitness: 1.5945265443586756
Generation 80 Best Solution: [183.83656165606922, 177.0908536360962,

173.74058342844756, 75.81252782808993] Fitness: 1.6138929483769684
Generation 81 Best Solution: [183.49745465339979, 177.31624020308138,
175.1243813399826, 74.28281020300824] Fitness: 1.6774707425394626
Generation 82 Best Solution: [183.49745465339979, 177.31624020308138,
174.57920949049173, 74.28281020300824] Fitness: 1.1779543698585397
Generation 83 Best Solution: [183.83656165606922, 177.31624020308138,
174.57920949049173, 74.28281020300824] Fitness: 1.3756721031465589
Generation 84 Best Solution: [183.83656165606922, 177.31624020308138,
173.74058342844756, 74.81162130229094] Fitness: 1.2063127611276347
Generation 85 Best Solution: [183.83656165606922, 177.31624020308138,
173.74058342844756, 74.81162130229094] Fitness: 1.2063127611276347
Generation 86 Best Solution: [183.83656165606922, 177.31624020308138,
173.74058342844756, 74.81162130229094] Fitness: 1.2063127611276347
Generation 87 Best Solution: [183.49745465339979, 176.52189410436918,
175.1243813399826, 75.81252782808993] Fitness: 1.484169772871966
Generation 88 Best Solution: [183.83656165606922, 177.31624020308138,
174.50814925813916, 75.18882457548747] Fitness: 1.2916351324969673
Generation 89 Best Solution: [183.83656165606922, 177.31624020308138,
174.57920949049173, 74.81162130229094] Fitness: 1.217867343137949
Generation 90 Best Solution: [183.49745465339979, 177.31624020308138,
174.57920949049173, 74.81162130229094] Fitness: 1.0201496098499299
Generation 91 Best Solution: [183.83656165606922, 176.47049687610456,
174.57920949049173, 74.81162130229094] Fitness: 0.46327837212321843
Generation 92 Best Solution: [183.83656165606922, 177.31624020308138,
174.57920949049173, 74.81162130229094] Fitness: 1.217867343137949
Generation 93 Best Solution: [183.83656165606922, 176.29000705233648,
175.1243813399826, 75.81252782808993] Fitness: 1.6224183844299482
Generation 94 Best Solution: [183.83656165606922, 176.29000705233648,
175.1243813399826, 75.81252782808993] Fitness: 1.6224183844299482
Generation 95 Best Solution: [183.83656165606922, 176.29000705233648,
175.1243813399826, 75.81252782808993] Fitness: 1.6224183844299482
Generation 96 Best Solution: [183.64494464666583, 177.31624020308138,
174.11833166535757, 75.81252782808993] Fitness: 1.6676836460599422
Generation 97 Best Solution: [183.7382682716453, 177.31624020308138,
174.11833166535757, 75.81252782808993] Fitness: 1.7250906343192374
Generation 98 Best Solution: [183.83656165606922, 175.69227705028936,
175.1243813399826, 74.38856240414414] Fitness: 1.1897381038920323
Generation 99 Best Solution: [183.83656165606922, 175.69227705028936,
173.80444277961158, 74.38856240414414] Fitness: 0.6472799813545174
Generation 100 Best Solution: [183.83656165606922, 177.31624020308138,
174.11833166535757, 74.83817590215286] Fitness: 1.099670316109008
Generation 101 Best Solution: [183.4665144012363, 177.31624020308138,
174.11833166535757, 75.81252782808993] Fitness: 1.5895476008122968
Generation 102 Best Solution: [183.83656165606922, 177.31624020308138,
173.7671227158478, 74.54308146638198] Fitness: 1.2271762203670393
Generation 103 Best Solution: [183.7382682716453, 177.31624020308138,
175.1300726049007, 74.54308146638198] Fitness: 1.6864490171073843
Generation 104 Best Solution: [183.2550412470015, 177.30891370926057,

175.1243813399826, 75.81252782808993] Fitness: 2.1595803631971595
Generation 105 Best Solution: [183.7382682716453, 176.52709369251946,
175.1243813399826, 75.81252782808993] Fitness: 1.6112773774245186
Generation 106 Best Solution: [183.7382682716453, 176.52709369251946,
175.1243813399826, 75.81252782808993] Fitness: 1.6112773774245186
Generation 107 Best Solution: [183.83656165606922, 177.31624020308138,
174.0073751105114, 75.81252782808993] Fitness: 1.8103540960258275
Generation 108 Best Solution: [183.83656165606922, 176.59325623272608,
174.0073751105114, 75.81252782808993] Fitness: 1.1074110358406468
Generation 109 Best Solution: [183.83656165606922, 176.59325623272608,
174.0073751105114, 75.81252782808993] Fitness: 1.1074110358406468
Generation 110 Best Solution: [183.83656165606922, 175.96440678426353,
175.1243813399826, 75.81252782808993] Fitness: 1.6572971445395877
Generation 111 Best Solution: [183.83656165606922, 176.11886387981005,
175.1243813399826, 75.75463568640778] Fitness: 1.547427737509739
Generation 112 Best Solution: [183.83656165606922, 176.26096385466363,
175.1243813399826, 75.81252782808993] Fitness: 1.6199123812528005
Generation 113 Best Solution: [183.84659708887745, 176.81762847684433,
174.905836798496, 75.81252782808993] Fitness: 1.620833110257284
Generation 114 Best Solution: [183.83656165606922, 176.81762847684433,
174.905836798496, 75.81252782808993] Fitness: 1.612696825873675
Generation 115 Best Solution: [183.83656165606922, 176.991480411775,
175.1243813399826, 74.88259119507866] Fitness: 1.32386461128245
Generation 116 Best Solution: [183.33387939060256, 176.56795699108116,
175.1243813399826, 75.75463568640778] Fitness: 1.3865015064737451
Generation 117 Best Solution: [183.83656165606922, 176.81762847684433,
173.86348483798994, 75.81252782808993] Fitness: 1.2927480451042892
Generation 118 Best Solution: [183.84659708887745, 176.5178029388728,
175.1243813399826, 75.81252782808993] Fitness: 1.688366787468254
Generation 119 Best Solution: [183.83656165606922, 176.3880221468381,
174.905836798496, 75.75463568640778] Fitness: 1.31611720332499
Generation 120 Best Solution: [183.84659708887745, 176.5178029388728,
175.1243813399826, 75.61618456248972] Fitness: 1.448053848756008
Generation 121 Best Solution: [183.83656165606922, 176.5178029388728,
175.1243813399826, 75.81252782808993] Fitness: 1.6802305030846447
Generation 122 Best Solution: [183.83656165606922, 177.31624020308138,
174.44707155644628, 75.75463568640778] Fitness: 1.7789413443617152
Generation 123 Best Solution: [183.83656165606922, 177.31624020308138,
174.6139274073533, 75.1224020876651] Fitness: 1.3155497381051349
Generation 124 Best Solution: [183.83656165606922, 177.31624020308138,
174.6139274073533, 74.50109132351459] Fitness: 1.2865804168452801
Generation 125 Best Solution: [183.83656165606922, 177.30891370926057,
174.6139274073533, 74.50109132351459] Fitness: 1.2760374939075791
Generation 126 Best Solution: [183.5313877857333, 177.31624020308138,
175.1243813399826, 74.50109132351459] Fitness: 1.5811576162096033
Generation 127 Best Solution: [183.83656165606922, 175.93256912198217,
174.905836798496, 74.93025871586877] Fitness: 0.736162865057868
Generation 128 Best Solution: [183.83656165606922, 176.27436683472808,

174.7452663220819, 75.75463568640778] Fitness: 1.1564247893081983
Generation 129 Best Solution: [183.10395913251344, 177.34613638940348,
174.7452663220819, 75.29372521405332] Fitness: 1.307546939135174
Generation 130 Best Solution: [183.83656165606922, 176.27436683472808,
174.6139274073533, 75.6880472519117] Fitness: 0.9834773390111375
Generation 131 Best Solution: [183.83656165606922, 177.34613638940348,
174.43522668477178, 75.29372521405332] Fitness: 1.3683828135649767
Generation 132 Best Solution: [183.83656165606922, 177.34613638940348,
174.43522668477178, 75.29372521405332] Fitness: 1.3683828135649767
Generation 133 Best Solution: [183.83656165606922, 177.30891370926057,
174.43522668477178, 75.29372521405332] Fitness: 1.3140931275870469
Generation 134 Best Solution: [182.80613490218792, 177.30891370926057,
174.905836798496, 75.27233276107363] Fitness: 1.483661827787273
Generation 135 Best Solution: [183.83656165606922, 176.32356186935178,
174.905836798496, 75.75463568640778] Fitness: 1.3038031596527144
Generation 136 Best Solution: [183.5798127358134, 177.34613638940348,
174.6139274073533, 75.75463568640778] Fitness: 1.7421554116828353
Generation 137 Best Solution: [183.83656165606922, 177.34613638940348,
175.1243813399826, 74.75498059658128] Fitness: 1.760591090691607
Generation 138 Best Solution: [183.83656165606922, 175.62455457054043,
175.1243813399826, 74.75498059658128] Fitness: 1.1407943601197732
Generation 139 Best Solution: [183.45810910205654, 175.62455457054043,
175.1243813399826, 74.75498059658128] Fitness: 0.929847187563695
Generation 140 Best Solution: [183.5798127358134, 177.34613638940348,
174.6139274073533, 74.70948061792463] Fitness: 1.1204360778660019
Generation 141 Best Solution: [183.83656165606922, 175.83166265986307,
174.7452663220819, 75.75463568640778] Fitness: 1.2466894292837365
Generation 142 Best Solution: [183.5798127358134, 177.34613638940348,
174.6139274073533, 74.85030017271119] Fitness: 1.1211506566528866
Generation 143 Best Solution: [183.09326573056632, 177.34613638940348,
175.1243813399826, 74.80705914017958] Fitness: 1.5234842347237876
Generation 144 Best Solution: [183.83656165606922, 176.3213082138717,
175.1243813399826, 75.75463568640778] Fitness: 1.5502670756732901
Generation 145 Best Solution: [183.83656165606922, 177.34613638940348,
174.17070462575052, 75.34933610030885] Fitness: 1.354945820343458
Generation 146 Best Solution: [183.83656165606922, 177.34613638940348,
174.17070462575052, 74.40403448817972] Fitness: 1.2308006408215206
Generation 147 Best Solution: [182.93448095878924, 177.34613638940348,
174.17070462575052, 74.40403448817972] Fitness: 1.0360438020122271
Generation 148 Best Solution: [182.891506307013, 176.16288481129686,
174.905836798496, 75.75463568640778] Fitness: 1.119152597058711
Generation 149 Best Solution: [182.81710688364657, 177.34613638940348,
174.58856501123003, 74.6577343980431] Fitness: 1.13315111874827
Generation 150 Best Solution: [183.83656165606922, 177.34613638940348,
174.58856501123003, 74.6577343980431] Fitness: 1.2752111703981759
Generation 151 Best Solution: [183.73684703498492, 176.22782825613365,
175.1243813399826, 75.75463568640778] Fitness: 1.468605947500913
Generation 152 Best Solution: [183.83656165606922, 177.34613638940348,

174.63442579033944, 74.6577343980431] Fitness: 1.3028569692957421
Generation 153 Best Solution: [183.83656165606922, 177.34613638940348,
175.01097381745714, 74.6577343980431] Fitness: 1.6335806423632817
Generation 154 Best Solution: [183.83656165606922, 176.09767919801297,
174.63442579033944, 75.75463568640778] Fitness: 1.0836525204359542
Generation 155 Best Solution: [183.83656165606922, 176.09767919801297,
174.63442579033944, 75.75463568640778] Fitness: 1.0836525204359542
Generation 156 Best Solution: [183.83656165606922, 177.34613638940348,
175.11398573091958, 75.08640875685423] Fitness: 1.8099915780241589
Generation 157 Best Solution: [183.3147492509953, 177.34613638940348,
175.11398573091958, 75.08640875685423] Fitness: 1.5679236289998186
Generation 158 Best Solution: [183.83656165606922, 176.48523955755945,
175.11398573091958, 75.75463568640778] Fitness: 1.5785854475819865
Generation 159 Best Solution: [183.83656165606922, 177.34613638940348,
173.88759460498048, 75.04455153105712] Fitness: 1.2319823881886471
Generation 160 Best Solution: [183.83656165606922, 176.64502094652144,
175.11398573091958, 75.08640875685423] Fitness: 1.0909714745427186
Generation 161 Best Solution: [183.83656165606922, 176.8137471396352,
175.01097381745714, 75.3198298220304] Fitness: 1.2127712838945577
Generation 162 Best Solution: [183.24512460302327, 177.34613638940348,
174.905836798496, 74.75496974242436] Fitness: 1.266283897575292
Generation 163 Best Solution: [183.51114657523, 176.1927016976737,
174.93276073631571, 75.75463568640778] Fitness: 1.129873273785999
Generation 164 Best Solution: [183.83656165606922, 176.02224194605037,
175.11398573091958, 75.75463568640778] Fitness: 1.5517673666876957
Generation 165 Best Solution: [183.83656165606922, 176.02224194605037,
173.76426522436583, 75.75463568640778] Fitness: 1.0415868451930759
Generation 166 Best Solution: [183.83656165606922, 176.02224194605037,
173.76426522436583, 75.75463568640778] Fitness: 1.0415868451930759
Generation 167 Best Solution: [183.83656165606922, 176.02224194605037,
173.76426522436583, 75.75463568640778] Fitness: 1.0415868451930759
Generation 168 Best Solution: [183.81941703182798, 177.34613638940348,
174.7706096160028, 75.75463568640778] Fitness: 2.0030068066730973
Generation 169 Best Solution: [183.81941703182798, 176.71513098724523,
174.7706096160028, 75.75463568640778] Fitness: 1.3270351186811542
Generation 170 Best Solution: [183.81941703182798, 177.34613638940348,
174.98406334167288, 75.13194060165176] Fitness: 1.6637171583741956
Generation 171 Best Solution: [183.83656165606922, 176.32580529358924,
175.11398573091958, 75.75463568640778] Fitness: 1.5377864082075137
Generation 172 Best Solution: [183.83656165606922, 176.32580529358924,
175.11398573091958, 75.75463568640778] Fitness: 1.5377864082075137
Generation 173 Best Solution: [183.83656165606922, 176.32580529358924,
175.11398573091958, 75.75463568640778] Fitness: 1.5377864082075137
Generation 174 Best Solution: [183.81941703182798, 176.32580529358924,
175.11398573091958, 75.75463568640778] Fitness: 1.5241902140622827
Generation 175 Best Solution: [183.83656165606922, 177.34613638940348,
174.9074714946343, 75.13194060165176] Fitness: 1.5977476686922207
Generation 176 Best Solution: [183.83656165606922, 176.32580529358924,

175.01097381745714, 75.75463568640778] Fitness: 1.4150929793241764
Generation 177 Best Solution: [183.81941703182798, 175.74148322144728,
175.01097381745714, 75.75463568640778] Fitness: 1.535399834961082
Generation 178 Best Solution: [183.0206575441396, 176.92873645621123,
175.11398573091958, 75.82376090028498] Fitness: 1.7358120856701647
Generation 179 Best Solution: [183.0206575441396, 177.34613638940348,
175.01097381745714, 75.75463568640778] Fitness: 2.027056855240392
Generation 180 Best Solution: [183.0206575441396, 177.34613638940348,
175.01097381745714, 75.75463568640778] Fitness: 2.027056855240392
Generation 181 Best Solution: [183.81941703182798, 176.10576960331377,
175.11398573091958, 75.82376090028498] Fitness: 1.6136911876089424
Generation 182 Best Solution: [183.0206575441396, 177.34613638940348,
175.11398573091958, 74.93633671953134] Fitness: 1.541911190810189
Generation 183 Best Solution: [183.83656165606922, 175.81806038472027,
175.11398573091958, 75.75463568640778] Fitness: 1.6287834535056742
Generation 184 Best Solution: [183.83656165606922, 175.81806038472027,
175.11398573091958, 75.75463568640778] Fitness: 1.6287834535056742
Generation 185 Best Solution: [183.0206575441396, 177.34613638940348,
175.11398573091958, 75.43561630893517] Fitness: 1.80890193736444
Generation 186 Best Solution: [183.83656165606922, 177.34613638940348,
175.11398573091958, 74.752646468488] Fitness: 1.7475079635256552
Generation 187 Best Solution: [183.83656165606922, 177.01801455842153,
175.01097381745714, 75.43031012944084] Fitness: 1.4873496391474421
Generation 188 Best Solution: [183.83656165606922, 177.04265735862842,
175.11398573091958, 74.752646468488] Fitness: 1.3575782305975508
Generation 189 Best Solution: [183.83656165606922, 177.04265735862842,
175.11398573091958, 74.752646468488] Fitness: 1.3575782305975508
Generation 190 Best Solution: [183.83656165606922, 176.22301426726906,
175.01097381745714, 75.75463568640778] Fitness: 1.4063678382986335
Generation 191 Best Solution: [183.81941703182798, 177.04265735862842,
175.13976392961263, 74.752646468488] Fitness: 1.3768506350679262
Generation 192 Best Solution: [183.81941703182798, 175.93516265287988,
174.89289264301684, 75.82376090028498] Fitness: 1.4094109580608067
Generation 193 Best Solution: [183.81941703182798, 175.93516265287988,
174.89289264301684, 75.82376090028498] Fitness: 1.4094109580608067
Generation 194 Best Solution: [183.83656165606922, 177.04265735862842,
174.1965031954079, 75.75463568640778] Fitness: 1.3763955134210235
Generation 195 Best Solution: [183.5892856083763, 177.34613638940348,
175.13976392961263, 74.93748569135354] Fitness: 1.6380434051174813
Generation 196 Best Solution: [183.83656165606922, 175.76854764064836,
175.13976392961263, 75.75463568640778] Fitness: 1.6885199911817628
Generation 197 Best Solution: [183.13814237729554, 175.62666453982283,
175.13976392961263, 75.82376090028498] Fitness: 1.6310232878146238
Generation 198 Best Solution: [183.13814237729554, 175.62666453982283,
175.01097381745714, 75.82376090028498] Fitness: 1.4754612603156807
Generation 199 Best Solution: [183.13814237729554, 175.62666453982283,
175.01097381745714, 75.75463568640778] Fitness: 1.3841079358883985
Generation 200 Best Solution: [183.83656165606922, 175.62666453982283,

175.01097381745714, 75.70534288559831] Fitness: 1.566324021306685
Generation 201 Best Solution: [183.83656165606922, 177.34613638940348,
175.01097381745714, 75.15352842175949] Fitness: 1.717411701035012
Generation 202 Best Solution: [183.83656165606922, 175.96419576763708,
175.13976392961263, 75.70534288559831] Fitness: 1.539661811752036
Generation 203 Best Solution: [183.83656165606922, 175.96419576763708,
175.13976392961263, 75.75463568640778] Fitness: 1.6009985124250554
Generation 204 Best Solution: [183.83656165606922, 175.96419576763708,
175.13976392961263, 75.75463568640778] Fitness: 1.6009985124250554
Generation 205 Best Solution: [183.83656165606922, 177.34613638940348,
174.64292239207725, 74.83377367258984] Fitness: 1.3013351291038622
Generation 206 Best Solution: [183.83656165606922, 177.34613638940348,
174.64292239207725, 74.83377367258984] Fitness: 1.3013351291038622
Generation 207 Best Solution: [183.83656165606922, 177.34613638940348,
174.64292239207725, 74.83377367258984] Fitness: 1.3013351291038622
Generation 208 Best Solution: [183.83656165606922, 175.96419576763708,
175.13976392961263, 75.75463568640778] Fitness: 1.6009985124250554
Generation 209 Best Solution: [183.83656165606922, 175.96419576763708,
174.83079821304585, 75.75463568640778] Fitness: 1.2641126099699083
Generation 210 Best Solution: [183.83656165606922, 175.96419576763708,
174.83079821304585, 75.75463568640778] Fitness: 1.2641126099699083
Generation 211 Best Solution: [183.83656165606922, 175.96419576763708,
174.83079821304585, 75.75463568640778] Fitness: 1.2641126099699083
Generation 212 Best Solution: [183.83656165606922, 175.96419576763708,
174.83079821304585, 75.75463568640778] Fitness: 1.2641126099699083
Generation 213 Best Solution: [183.83656165606922, 175.96419576763708,
174.83079821304585, 75.75463568640778] Fitness: 1.2641126099699083
Generation 214 Best Solution: [183.83656165606922, 175.96419576763708,
174.83079821304585, 75.75463568640778] Fitness: 1.2641126099699083
Generation 215 Best Solution: [183.83656165606922, 175.96419576763708,
174.83079821304585, 75.75463568640778] Fitness: 1.2641126099699083
Generation 216 Best Solution: [183.83656165606922, 177.34613638940348,
174.83079821304585, 74.8502086303738] Fitness: 1.4467264423303796
Generation 217 Best Solution: [183.83656165606922, 177.29076090491992,
174.83079821304585, 74.8502086303738] Fitness: 1.3666162712198788
Generation 218 Best Solution: [183.83656165606922, 176.33690282672828,
175.13976392961263, 75.75463568640778] Fitness: 1.5724212621182425
Generation 219 Best Solution: [182.84222672072144, 176.33690282672828,
174.61638900246405, 75.82376090028498] Fitness: 1.0114393301257592
Generation 220 Best Solution: [183.83656165606922, 176.12644091504714,
175.13976392961263, 75.75463568640778] Fitness: 1.5662889864862595
Generation 221 Best Solution: [183.83656165606922, 176.19679529180274,
175.13976392961263, 75.75463568640778] Fitness: 1.5619103231011002
Generation 222 Best Solution: [183.83656165606922, 176.19679529180274,
175.13976392961263, 75.37458621561889] Fitness: 1.1709827040694358
Generation 223 Best Solution: [183.83656165606922, 176.19679529180274,
175.13976392961263, 75.75463568640778] Fitness: 1.5619103231011002
Generation 224 Best Solution: [183.83656165606922, 177.34613638940348,

175.13976392961263, 74.9248842767038] Fitness: 1.7944773058905228
Generation 225 Best Solution: [183.768465819906, 177.34613638940348,
173.95247300140508, 75.75463568640778] Fitness: 1.7384092513072398
Generation 226 Best Solution: [183.768465819906, 177.34613638940348,
173.95247300140508, 75.75463568640778] Fitness: 1.7384092513072398
Generation 227 Best Solution: [183.83656165606922, 177.34613638940348,
175.13976392961263, 74.9248842767038] Fitness: 1.7944773058905228
Generation 228 Best Solution: [183.83656165606922, 177.34613638940348,
175.13976392961263, 74.88057657902696] Fitness: 1.7871532187370665
Generation 229 Best Solution: [183.83656165606922, 175.95501351936522,
175.13976392961263, 75.75463568640778] Fitness: 1.6039894743780696
Generation 230 Best Solution: [183.50252205505305, 177.34613638940348,
175.08991056624095, 74.3142016273374] Fitness: 1.6604519211201525
Generation 231 Best Solution: [183.83656165606922, 177.34613638940348,
175.13976392961263, 74.88057657902696] Fitness: 1.7871532187370665
Generation 232 Best Solution: [183.23389081652178, 177.34613638940348,
175.13976392961263, 74.88057657902696] Fitness: 1.539356270290641
Generation 233 Best Solution: [183.83656165606922, 177.34613638940348,
174.05121068268048, 75.75463568640778] Fitness: 1.7711533280989165
Generation 234 Best Solution: [183.83656165606922, 177.34613638940348,
174.05121068268048, 75.75463568640778] Fitness: 1.7711533280989165
Generation 235 Best Solution: [183.83656165606922, 177.34613638940348,
175.13976392961263, 75.30905842223346] Fitness: 1.9653369278455761
Generation 236 Best Solution: [183.83656165606922, 176.0444223859044,
175.13976392961263, 75.75463568640778] Fitness: 1.5795440740335924
Generation 237 Best Solution: [183.24497144752618, 176.46877353491084,
175.13976392961263, 75.75463568640778] Fitness: 1.358189005600585
Generation 238 Best Solution: [183.83656165606922, 175.80173929554277,
175.13976392961263, 75.75463568640778] Fitness: 1.6701553429860958
Generation 239 Best Solution: [183.24497144752618, 175.80173929554277,
175.13976392961263, 75.75463568640778] Fitness: 1.4226392326317865
Generation 240 Best Solution: [183.83656165606922, 177.34613638940348,
174.8677225297602, 75.07716704510295] Fitness: 1.5359981750447134
Generation 241 Best Solution: [183.83656165606922, 175.82699071533787,
175.13976392961263, 75.75463568640778] Fitness: 1.6571464155251467
Generation 242 Best Solution: [183.132054873763, 175.80173929554277,
175.13976392961263, 75.75463568640778] Fitness: 1.427276692656105
Generation 243 Best Solution: [183.83656165606922, 177.3522670902623,
174.22168061705332, 75.75463568640778] Fitness: 1.777402758095342
Generation 244 Best Solution: [182.7986524851765, 177.3522670902623,
174.80377818362743, 75.06205095349004] Fitness: 1.3499332318407642
Generation 245 Best Solution: [183.83656165606922, 176.8393314385854,
175.13976392961263, 75.09401725311446] Fitness: 1.2620065687772213
Generation 246 Best Solution: [183.132054873763, 177.34613638940348,
174.74046432646915, 75.75463568640778] Fitness: 1.7498884580784444
Generation 247 Best Solution: [182.7986524851765, 177.34613638940348,
175.13976392961263, 75.09401725311446] Fitness: 1.7138385436787622
Generation 248 Best Solution: [183.2738686519584, 177.34613638940348,

174.74046432646915, 74.33386661750917] Fitness: 1.249619941319912
Generation 249 Best Solution: [183.83656165606922, 176.03551009110896,
175.13976392961263, 75.75463568640778] Fitness: 1.581512921178473
Generation 250 Best Solution: [183.83656165606922, 177.34613638940348,
174.61704782000842, 74.28240857845822] Fitness: 1.4418273310446132
Generation 251 Best Solution: [183.2738686519584, 177.34613638940348,
174.61704782000842, 75.21337319283771] Fitness: 1.161664096907051
Generation 252 Best Solution: [183.83656165606922, 177.34613638940348,
174.4343725558988, 75.75463568640778] Fitness: 1.8178615697269014
Generation 253 Best Solution: [183.870210163933, 177.34613638940348,
174.4343725558988, 75.75463568640778] Fitness: 1.8456604707357829
Generation 254 Best Solution: [183.870210163933, 177.34613638940348,
174.74046432646915, 74.81483206077453] Fitness: 1.3969443500041625
Generation 255 Best Solution: [183.83656165606922, 177.34613638940348,
174.6182715975098, 75.71060031205737] Fitness: 1.8533435308975568
Generation 256 Best Solution: [183.83656165606922, 175.83203881440983,
175.13976392961263, 75.71060031205737] Fitness: 1.5996998153475337
Generation 257 Best Solution: [183.83656165606922, 175.92565513985244,
175.13976392961263, 75.75463568640778] Fitness: 1.6142903980822942
Generation 258 Best Solution: [183.83656165606922, 176.69561794620108,
174.74046432646915, 75.75463568640778] Fitness: 1.3041702843355072
Generation 259 Best Solution: [183.83656165606922, 175.6391134684374,
174.74046432646915, 75.75463568640778] Fitness: 1.362003742776828
Generation 260 Best Solution: [183.84453282965836, 175.92565513985244,
174.74046432646915, 75.75463568640778] Fitness: 1.2088834836817561
Generation 261 Best Solution: [183.58478207710948, 175.78421971885487,
175.13976392961263, 75.75463568640778] Fitness: 1.5185290649108873
Generation 262 Best Solution: [183.58478207710948, 175.78421971885487,
175.13976392961263, 75.75463568640778] Fitness: 1.5185290649108873
Generation 263 Best Solution: [183.58478207710948, 176.61415995371405,
175.13976392961263, 75.75463568640778] Fitness: 1.5075484175330107
Generation 264 Best Solution: [182.8532863538846, 176.90852136605034,
174.02994911909738, 75.75463568640778] Fitness: 1.0899562902864703
Generation 265 Best Solution: [182.8532863538846, 176.90852136605034,
174.02994911909738, 75.75463568640778] Fitness: 1.0899562902864703
Generation 266 Best Solution: [182.8532863538846, 176.50095736837102,
175.13976392961263, 75.75463568640778] Fitness: 1.457052495604565
Generation 267 Best Solution: [182.8532863538846, 177.34613638940348,
175.13976392961263, 74.39713377867317] Fitness: 1.7134145069722178
Generation 268 Best Solution: [183.76068631726866, 175.85662972039958,
175.13976392961263, 75.75463568640778] Fitness: 1.5856729670388159
Generation 269 Best Solution: [183.76068631726866, 175.85662972039958,
175.13976392961263, 75.75463568640778] Fitness: 1.5856729670388159
Generation 270 Best Solution: [182.8532863538846, 175.85662972039958,
175.13976392961263, 75.75463568640778] Fitness: 1.4827189700164345
Generation 271 Best Solution: [183.83656165606922, 176.22795158886987,
174.39758665474193, 75.75463568640778] Fitness: 0.9624753551650416
Generation 272 Best Solution: [183.83656165606922, 176.22795158886987,

174.39758665474193, 75.75463568640778] Fitness: 0.9624753551650416
Generation 273 Best Solution: [182.8532863538846, 177.34613638940348,
174.26075705606996, 75.75463568640778] Fitness: 1.6127519024009316
Generation 274 Best Solution: [182.8532863538846, 177.34613638940348,
174.26075705606996, 75.75463568640778] Fitness: 1.6127519024009316
Generation 275 Best Solution: [182.8532863538846, 177.34613638940348,
174.26075705606996, 75.75463568640778] Fitness: 1.6127519024009316
Generation 276 Best Solution: [183.14638001524744, 177.34613638940348,
174.26075705606996, 75.75463568640778] Fitness: 1.5285829919059717
Generation 277 Best Solution: [183.83656165606922, 176.6289398697363,
175.13976392961263, 75.67369928578005] Fitness: 1.5775942212875271
Generation 278 Best Solution: [183.83656165606922, 176.68656142666666,
174.15781061549606, 75.75463568640778] Fitness: 1.0706332897409971
Generation 279 Best Solution: [183.26061670902368, 177.34613638940348,
175.13976392961263, 74.31737243446901] Fitness: 1.6703517563246661
Generation 280 Best Solution: [183.83656165606922, 176.8994834710088,
175.13976392961263, 75.75463568640778] Fitness: 1.8724429383498258
Generation 281 Best Solution: [183.66137075550049, 176.124543585398,
175.13976392961263, 75.75463568640778] Fitness: 1.4456224754570328
Generation 282 Best Solution: [183.66137075550049, 176.124543585398,
175.13976392961263, 75.75463568640778] Fitness: 1.4456224754570328
Generation 283 Best Solution: [183.66137075550049, 176.124543585398,
175.13976392961263, 75.75463568640778] Fitness: 1.4456224754570328
Generation 284 Best Solution: [183.73993135604843, 176.0198519225845,
174.74381755086424, 75.48316912683163] Fitness: 0.805874732589119
Generation 285 Best Solution: [183.5317363847146, 177.34613638940348,
174.74381755086424, 75.46055998640564] Fitness: 1.4918333040244534
Generation 286 Best Solution: [183.83656165606922, 177.34613638940348,
174.74381755086424, 74.29448405372955] Fitness: 1.521966304266794
Generation 287 Best Solution: [183.5317363847146, 177.34613638940348,
174.74381755086424, 75.46055998640564] Fitness: 1.4918333040244534
Generation 288 Best Solution: [183.83656165606922, 176.21372770697835,
174.74381755086424, 75.75463568640778] Fitness: 1.1525543406325012
Generation 289 Best Solution: [183.83656165606922, 176.75461512012646,
173.7305839336417, 75.75463568640778] Fitness: 1.2303277045481606
Generation 290 Best Solution: [183.83656165606922, 177.34613638940348,
174.74381755086424, 74.30225680270156] Fitness: 1.5171239197931958
Generation 291 Best Solution: [183.5317363847146, 177.34613638940348,
173.9142547890892, 75.75463568640778] Fitness: 1.6163697401942907
Generation 292 Best Solution: [183.5317363847146, 177.4213055282704,
174.74381755086424, 75.21564425440609] Fitness: 1.4274116356253774
Generation 293 Best Solution: [183.83656165606922, 177.4213055282704,
174.74381755086424, 75.21564425440609] Fitness: 1.611956441394872
Generation 294 Best Solution: [183.5317363847146, 176.02842006339975,
175.13976392961263, 75.75463568640778] Fitness: 1.3986083926050665
Generation 295 Best Solution: [183.5317363847146, 175.8110900381151,
175.13976392961263, 75.75463568640778] Fitness: 1.4806962910495507
Generation 296 Best Solution: [183.83656165606922, 177.34613638940348,

174.74381755086424, 74.94529736063863] Fitness: 1.389447167184065
Generation 297 Best Solution: [183.83656165606922, 176.29508057773361,
175.13976392961263, 75.75463568640778] Fitness: 1.5666029670816335
Generation 298 Best Solution: [183.83656165606922, 176.36984694194243,
174.74381755086424, 75.75463568640778] Fitness: 1.1693447589122707
Generation 299 Best Solution: [183.51009838845368, 176.36984694194243,
174.74381755086424, 75.75463568640778] Fitness: 0.9763070192917926
Generation 300 Best Solution: [183.83656165606922, 176.27576164428027,
174.74381755086424, 75.86615346025701] Fitness: 1.3058805385727412
Generation 301 Best Solution: [183.83796016529936, 175.8032135052537,
174.74381755086424, 75.75463568640778] Fitness: 1.2612329239379603
Generation 302 Best Solution: [183.83796016529936, 175.8032135052537,
174.74381755086424, 75.75463568640778] Fitness: 1.2612329239379603
Generation 303 Best Solution: [183.83796016529936, 175.8032135052537,
175.13976392961263, 75.75463568640778] Fitness: 1.6704989768637173
Generation 304 Best Solution: [183.83796016529936, 175.8032135052537,
175.13976392961263, 74.85485810953328] Fitness: 1.0499494951648018
Generation 305 Best Solution: [183.692484907489, 176.57576714104766,
175.13976392961263, 75.75463568640778] Fitness: 1.5470474394189169
Generation 306 Best Solution: [183.83656165606922, 175.98513165132485,
175.13976392961263, 75.75463568640778] Fitness: 1.5945902307511086
Generation 307 Best Solution: [183.83656165606922, 175.94920289137966,
174.74381755086424, 75.75463568640778] Fitness: 1.1966729512702965
Generation 308 Best Solution: [183.22819693987327, 176.9042936719559,
174.74381755086424, 74.35527091575649] Fitness: 0.710481274183376
Generation 309 Best Solution: [183.36573375407983, 175.98513165132485,
175.13976392961263, 75.75463568640778] Fitness: 1.3605185251054825
Generation 310 Best Solution: [183.36573375407983, 175.98513165132485,
175.13976392961263, 75.75463568640778] Fitness: 1.3605185251054825
Generation 311 Best Solution: [183.36573375407983, 175.98513165132485,
175.13976392961263, 75.86615346025701] Fitness: 1.5109794556703768
Generation 312 Best Solution: [183.83656165606922, 177.34613638940348,
174.92725443238558, 75.75463568640778] Fitness: 2.1595445336072947
Generation 313 Best Solution: [183.0552893881656, 177.34613638940348,
175.13976392961263, 74.43453017180127] Fitness: 1.6258358443491019
Generation 314 Best Solution: [183.83656165606922, 177.34613638940348,
174.92725443238558, 74.83950210263058] Fitness: 1.5375693446905585
Generation 315 Best Solution: [183.41641594175707, 177.34613638940348,
175.13976392961263, 74.87857570632568] Fitness: 1.564120475968839
Generation 316 Best Solution: [183.0552893881656, 176.22735034595794,
175.13976392961263, 74.45950433878981] Fitness: 0.7725127689892961
Generation 317 Best Solution: [183.0552893881656, 177.09398363271873,
175.13976392961263, 74.76915057857565] Fitness: 1.2173990410334672
Generation 318 Best Solution: [183.41641594175707, 177.34613638940348,
174.92725443238558, 74.87857570632568] Fitness: 1.3190390435587305
Generation 319 Best Solution: [183.0552893881656, 177.34613638940348,
174.29628458508947, 75.86615346025701] Fitness: 1.6991889928404986
Generation 320 Best Solution: [182.95862737034258, 175.67309105887068,

174.55000880510855, 75.75463568640778] Fitness: 1.010923813016837
Generation 321 Best Solution: [183.83656165606922, 177.13541559023548,
174.26149755078757, 75.75463568640778] Fitness: 1.489583707684476
Generation 322 Best Solution: [183.83656165606922, 176.20029645210704,
174.26149755078757, 75.86615346025701] Fitness: 1.0807750634711073
Generation 323 Best Solution: [183.83656165606922, 177.13541559023548,
174.26149755078757, 75.86615346025701] Fitness: 1.6400446382493703
Generation 324 Best Solution: [183.83656165606922, 176.8794220616763,
174.1906564137878, 75.86615346025701] Fitness: 1.3664689921218416
Generation 325 Best Solution: [183.83656165606922, 176.23439479593338,
174.26149755078757, 75.86615346025701] Fitness: 1.081131418918078
Generation 326 Best Solution: [183.83656165606922, 177.34613638940348,
174.1906564137878, 74.68892823483677] Fitness: 1.1463271724369013
Generation 327 Best Solution: [183.17090298443796, 177.34613638940348,
174.92725443238558, 75.07151838929742] Fitness: 1.3455438072405883
Generation 328 Best Solution: [183.83656165606922, 177.34613638940348,
174.92725443238558, 75.07151838929742] Fitness: 1.5918945194999763
Generation 329 Best Solution: [183.83656165606922, 176.86197193896746,
175.13976392961263, 75.75463568640778] Fitness: 1.8395928972828117
Generation 330 Best Solution: [183.17090298443796, 177.34613638940348,
174.92725443238558, 75.75463568640778] Fitness: 1.9131938213479067
Generation 331 Best Solution: [183.1876841906852, 177.34613638940348,
174.1906564137878, 75.86615346025701] Fitness: 1.6692070270576889
Generation 332 Best Solution: [183.83656165606922, 177.34613638940348,
175.13976392961263, 74.73760533353112] Fitness: 1.780982277127679
Generation 333 Best Solution: [182.82372706102038, 176.22171505480986,
175.13976392961263, 75.75463568640778] Fitness: 1.4163973259102405
Generation 334 Best Solution: [183.83656165606922, 177.34613638940348,
175.13976392961263, 74.73760533353112] Fitness: 1.780982277127679
Generation 335 Best Solution: [183.83656165606922, 176.1152808944954,
175.13976392961263, 75.75463568640778] Fitness: 1.567576839687582
Generation 336 Best Solution: [183.83656165606922, 176.9024967399168,
175.13976392961263, 75.60024586008727] Fitness: 1.6936310210801173
Generation 337 Best Solution: [183.83656165606922, 176.1152808944954,
175.13976392961263, 75.75463568640778] Fitness: 1.567576839687582
Generation 338 Best Solution: [183.83656165606922, 176.81281449105526,
175.13976392961263, 75.75463568640778] Fitness: 1.7993226960836908
Generation 339 Best Solution: [183.83656165606922, 176.9024967399168,
175.13976392961263, 75.49492420039165] Fitness: 1.5876336958873591
Generation 340 Best Solution: [183.83656165606922, 176.81281449105526,
175.13976392961263, 75.49492420039165] Fitness: 1.5117949999084377
Generation 341 Best Solution: [183.83656165606922, 176.81281449105526,
175.13976392961263, 75.49492420039165] Fitness: 1.5117949999084377
Generation 342 Best Solution: [183.83656165606922, 176.9024967399168,
173.7714253005258, 75.75463568640778] Fitness: 1.3285494672023728
Generation 343 Best Solution: [182.81661187470687, 176.67533264382232,
175.13976392961263, 75.49492420039165] Fitness: 1.2741034043933195
Generation 344 Best Solution: [183.83656165606922, 176.67533264382232,

175.13976392961263, 75.75463568640778] Fitness: 1.7034308588994347
Generation 345 Best Solution: [183.83656165606922, 177.34613638940348,
174.34047604397844, 74.97050652000706] Fitness: 1.1887701504093786
Generation 346 Best Solution: [183.31573362158963, 177.34613638940348,
174.9779519899382, 74.2817711717761] Fitness: 1.5054131989730912
Generation 347 Best Solution: [183.83656165606922, 175.75855035651608,
175.13976392961263, 75.75463568640778] Fitness: 1.694332990298581
Generation 348 Best Solution: [183.61682857486113, 176.14463635482448,
175.13976392961263, 74.2817711717761] Fitness: 0.9540108447778248
Generation 349 Best Solution: [183.61682857486113, 177.34613638940348,
174.23805373648048, 74.94794748344106] Fitness: 1.0194616071027132
Generation 350 Best Solution: [183.83656165606922, 177.42491965656382,
175.13976392961263, 74.97050652000706] Fitness: 1.9255611185493569
Generation 351 Best Solution: [183.83656165606922, 177.34613638940348,
175.13976392961263, 74.63838667108003] Fitness: 1.792371371399318
Generation 352 Best Solution: [183.83656165606922, 177.34613638940348,
174.20707683235057, 74.63838667108003] Fitness: 1.1547941185510116
Generation 353 Best Solution: [183.61682857486113, 177.34613638940348,
175.13976392961263, 74.63838667108003] Fitness: 1.6471483118937698
Generation 354 Best Solution: [183.83656165606922, 176.8071872486802,
175.13976392961263, 74.35801578606535] Fitness: 1.2842506044040713
Generation 355 Best Solution: [183.83656165606922, 177.34613638940348,
175.13976392961263, 74.97050652000706] Fitness: 1.8046944309833333
Generation 356 Best Solution: [182.8864776861637, 177.01459982901895,
175.13976392961263, 74.35801578606535] Fitness: 1.2986725716237006
Generation 357 Best Solution: [183.83656165606922, 177.34613638940348,
175.13976392961263, 74.29250956167195] Fitness: 1.9324750076981523
Generation 358 Best Solution: [183.64352059802462, 177.42491965656382,
175.13976392961263, 75.30913581826124] Fitness: 1.9553148808998804
Generation 359 Best Solution: [183.83656165606922, 177.34613638940348,
175.13976392961263, 75.30913581826124] Fitness: 1.9653907447145236
Generation 360 Best Solution: [182.85936353849436, 177.42491965656382,
175.13976392961263, 74.43262801911197] Fitness: 1.8146797571794753
Generation 361 Best Solution: [183.83656165606922, 177.34613638940348,
175.13976392961263, 74.41535598801383] Fitness: 1.864845289933042
Generation 362 Best Solution: [183.83656165606922, 177.16682990444593,
175.13976392961263, 74.41535598801383] Fitness: 1.6199399608473608
Generation 363 Best Solution: [183.41291381747246, 177.34613638940348,
175.13976392961263, 74.41535598801383] Fitness: 1.6411936368267188
Generation 364 Best Solution: [183.83656165606922, 177.34613638940348,
174.24449247749865, 74.41535598801383] Fitness: 1.2309990921105116
Generation 365 Best Solution: [183.83656165606922, 176.53635498795282,
174.39750722591373, 75.86615346025701] Fitness: 1.1824300417205527
Generation 366 Best Solution: [183.83656165606922, 176.53635498795282,
174.39750722591373, 75.86615346025701] Fitness: 1.1824300417205527
Generation 367 Best Solution: [183.83656165606922, 176.96647429010366,
174.24160537611704, 75.86615346025701] Fitness: 1.451936420905329
Generation 368 Best Solution: [183.83656165606922, 176.53635498795282,

174.2368639241535, 75.86615346025701] Fitness: 1.1472591656580469
Generation 369 Best Solution: [183.83656165606922, 176.05108816105363,
175.13976392961263, 75.25239932001774] Fitness: 1.1015491301447635
Generation 370 Best Solution: [183.83656165606922, 176.53635498795282,
174.24160537611704, 75.86615346025701] Fitness: 1.1478151605820563
Generation 371 Best Solution: [183.83656165606922, 176.53635498795282,
174.24160537611704, 75.86615346025701] Fitness: 1.1478151605820563
Generation 372 Best Solution: [183.83656165606922, 176.53635498795282,
174.24160537611704, 75.86615346025701] Fitness: 1.1478151605820563
Generation 373 Best Solution: [183.83656165606922, 176.5684895471914,
175.1010850442743, 75.86615346025701] Fitness: 1.7474024732531706
Generation 374 Best Solution: [183.49555986281732, 177.34613638940348,
174.22653522120842, 75.86615346025701] Fitness: 1.7208210233473382
Generation 375 Best Solution: [183.83656165606922, 176.5684895471914,
175.1010850442743, 75.86615346025701] Fitness: 1.7474024732531706
Generation 376 Best Solution: [183.49555986281732, 176.66496770543947,
175.1010850442743, 75.86615346025701] Fitness: 1.6002684237006213
Generation 377 Best Solution: [183.31696795742258, 177.42491965656382,
174.12188042256045, 75.86615346025701] Fitness: 1.7949027615150355
Generation 378 Best Solution: [183.83656165606922, 177.42491965656382,
174.1046255009322, 75.69629121276033] Fitness: 1.8147888278040143
Generation 379 Best Solution: [182.96534754473336, 177.42491965656382,
174.1046255009322, 75.69629121276033] Fitness: 1.6091581303508602
Generation 380 Best Solution: [183.31696795742258, 175.95337017097714,
175.13976392961263, 75.86615346025701] Fitness: 1.5132067625985015
Generation 381 Best Solution: [183.83656165606922, 177.42491965656382,
174.44759017293416, 75.18203534811516] Fitness: 1.4266717450164828
Generation 382 Best Solution: [183.83656165606922, 176.91408901422656,
175.02176411548982, 75.5563232919514] Fitness: 1.514877304052125
Generation 383 Best Solution: [183.83656165606922, 176.91408901422656,
175.1010850442743, 74.50025781446199] Fitness: 1.2617198013662245
Generation 384 Best Solution: [183.83656165606922, 176.91408901422656,
175.1010850442743, 74.8343301477388] Fitness: 1.2143515317466285
Generation 385 Best Solution: [183.83656165606922, 176.43273316249065,
175.13976392961263, 74.50025781446199] Fitness: 1.0193382974640588
Generation 386 Best Solution: [183.83656165606922, 177.34613638940348,
174.22653522120842, 74.8343301477388] Fitness: 1.1463750735411327
Generation 387 Best Solution: [183.83656165606922, 176.05077682324458,
175.1010850442743, 74.50025781446199] Fitness: 0.9541934516750873
Generation 388 Best Solution: [183.21965444348172, 175.5854719330201,
175.1010850442743, 74.50025781446199] Fitness: 0.9437030540123116
Generation 389 Best Solution: [183.21965444348172, 177.42491965656382,
174.22653522120842, 74.8343301477388] Fitness: 1.019319050460078
Generation 390 Best Solution: [183.83656165606922, 175.8976098152104,
175.1010850442743, 75.86615346025701] Fitness: 1.7266492335562718
Generation 391 Best Solution: [183.83656165606922, 177.42491965656382,
174.22653522120842, 74.68521602147833] Fitness: 1.270397842381717
Generation 392 Best Solution: [183.21965444348172, 177.38328187295653,

175.1010850442743, 74.68521602147833] Fitness: 1.5444596440689942
Generation 393 Best Solution: [183.21965444348172, 177.34613638940348,
174.22653522120842, 75.86615346025701] Fitness: 1.671299424269125
Generation 394 Best Solution: [183.83656165606922, 177.42491965656382,
175.1010850442743, 74.90868562943237] Fitness: 1.8633771467007725
Generation 395 Best Solution: [183.83656165606922, 176.37799905245583,
175.1010850442743, 75.86615346025701] Fitness: 1.6818296072885495
Generation 396 Best Solution: [183.83656165606922, 176.37799905245583,
175.1010850442743, 75.86615346025701] Fitness: 1.6818296072885495
Generation 397 Best Solution: [183.83656165606922, 177.34613638940348,
175.1010850442743, 74.69366467335132] Fitness: 1.7354497828889746
Generation 398 Best Solution: [183.83656165606922, 176.4239831729357,
175.1010850442743, 75.86615346025701] Fitness: 1.693325104354337
Generation 399 Best Solution: [183.83656165606922, 177.42491965656382,
173.7775139637768, 75.23867810520099] Fitness: 1.4909101100091664
Generation 400 Best Solution: [183.83656165606922, 177.42491965656382,
173.7775139637768, 74.92028821325444] Fitness: 1.364876235738587
Generation 401 Best Solution: [183.74141021279115, 175.82470267580288,
175.1010850442743, 75.23867810520099] Fitness: 1.0536879209594208
Generation 402 Best Solution: [183.74141021279115, 175.82470267580288,
175.1010850442743, 75.86615346025701] Fitness: 1.689142402184176
Generation 403 Best Solution: [183.83656165606922, 176.57983499480312,
175.1010850442743, 75.86615346025701] Fitness: 1.7528013614264373
Generation 404 Best Solution: [183.83656165606922, 176.57983499480312,
175.1010850442743, 75.86615346025701] Fitness: 1.7528013614264373
Generation 405 Best Solution: [183.56704467760025, 177.42491965656382,
175.1010850442743, 74.98359353754833] Fitness: 1.7106258421941232
Generation 406 Best Solution: [183.83656165606922, 176.6748937377407,
175.1010850442743, 74.39576974178482] Fitness: 1.1238549427430304
Generation 407 Best Solution: [183.83656165606922, 175.98731801252524,
175.1010850442743, 75.86615346025701] Fitness: 1.6954226580709943
Generation 408 Best Solution: [183.83656165606922, 175.98731801252524,
175.1010850442743, 74.66892612009516] Fitness: 0.9278336899280832
Generation 409 Best Solution: [183.5179492063998, 177.36471469574758,
175.1010850442743, 74.68438093434638] Fitness: 1.5742488950803537
Generation 410 Best Solution: [183.83656165606922, 177.42491965656382,
174.21251537868514, 74.98359353754833] Fitness: 1.291844702450609
Generation 411 Best Solution: [183.67134602298978, 177.36471469574758,
175.1010850442743, 74.98359353754833] Fitness: 1.6718405428075331
Generation 412 Best Solution: [183.83656165606922, 175.92520308649023,
175.1010850442743, 74.57734519414056] Fitness: 0.9665976751220685
Generation 413 Best Solution: [183.83656165606922, 175.92520308649023,
175.1010850442743, 74.57734519414056] Fitness: 0.9665976751220685
Generation 414 Best Solution: [182.9924722381322, 177.36471469574758,
175.1010850442743, 74.98359353754833] Fitness: 1.5727464014717993
Generation 415 Best Solution: [182.9924722381322, 176.02243361326012,
173.81118945960523, 75.86615346025701] Fitness: 0.955708268573033
Generation 416 Best Solution: [183.83656165606922, 176.21096896061138,

175.1010850442743, 75.86615346025701] Fitness: 1.663278227159952
Generation 417 Best Solution: [183.83656165606922, 176.21096896061138,
175.1010850442743, 75.86615346025701] Fitness: 1.663278227159952
Generation 418 Best Solution: [183.65497705050754, 177.42491965656382,
175.1010850442743, 74.98359353754833] Fitness: 1.7554725984229784
Generation 419 Best Solution: [183.83656165606922, 177.36471469574758,
175.12439841551767, 74.98359353754833] Fitness: 1.8162029157307278
Generation 420 Best Solution: [183.83656165606922, 177.31826816787594,
173.93504826551433, 74.98359353754833] Fitness: 1.1575050138456533
Generation 421 Best Solution: [183.83656165606922, 177.36471469574758,
175.12439841551767, 74.98359353754833] Fitness: 1.8162029157307278
Generation 422 Best Solution: [183.83656165606922, 177.36471469574758,
173.85118165782944, 75.02730404755762] Fitness: 1.2672036687291859
Generation 423 Best Solution: [183.83656165606922, 177.42491965656382,
175.1010850442743, 74.47833463352633] Fitness: 1.9096809689660144
Generation 424 Best Solution: [183.83656165606922, 177.42491965656382,
174.88701450143432, 74.65047490498786] Fitness: 1.626307533469289
Generation 425 Best Solution: [183.83656165606922, 177.36471469574758,
174.88701450143432, 74.65047490498786] Fitness: 1.533213609592679
Generation 426 Best Solution: [183.83656165606922, 175.92334763663297,
175.1010850442743, 75.86615346025701] Fitness: 1.7166163641864323
Generation 427 Best Solution: [183.83656165606922, 177.36471469574758,
174.4974033927804, 74.42719284319577] Fitness: 1.3264021815531275
Generation 428 Best Solution: [183.83656165606922, 177.42491965656382,
174.81158691247032, 74.42719284319577] Fitness: 1.6265137381229522
Generation 429 Best Solution: [183.50828880472355, 177.42491965656382,
174.5652642420853, 75.20857193503105] Fitness: 1.3022743402805128
Generation 430 Best Solution: [183.83656165606922, 175.79544900133695,
175.12439841551767, 74.96921358576176] Fitness: 1.053570672610509
Generation 431 Best Solution: [183.16508895942053, 177.36471469574758,
175.1010850442743, 74.79497690709928] Fitness: 1.5130799141640858
Generation 432 Best Solution: [183.83656165606922, 176.47719175202707,
175.1010850442743, 75.86615346025701] Fitness: 1.7100687427176777
Generation 433 Best Solution: [183.83656165606922, 176.5980614788641,
175.1010850442743, 74.66194645339228] Fitness: 0.9952437970086446
Generation 434 Best Solution: [183.83656165606922, 176.5980614788641,
175.1010850442743, 75.86615346025701] Fitness: 1.7618262027150828
Generation 435 Best Solution: [183.83656165606922, 176.9068651392869,
174.55068140132764, 75.86615346025701] Fitness: 1.4948848994060127
Generation 436 Best Solution: [183.83656165606922, 177.42491965656382,
174.67853042372806, 74.81908081164413] Fitness: 1.4449871823163667
Generation 437 Best Solution: [183.83656165606922, 177.42491965656382,
175.1010850442743, 74.66194645339228] Fitness: 1.8603789305502954
Generation 438 Best Solution: [183.16508895942053, 177.42491965656382,
175.1010850442743, 74.66194645339228] Fitness: 1.6144225914042252
Generation 439 Best Solution: [183.00590972185572, 177.36471469574758,
174.06418452460682, 75.86615346025701] Fitness: 1.729805698342879
Generation 440 Best Solution: [183.00590972185572, 177.36471469574758,

174.06418452460682, 74.80255762692249] Fitness: 0.9551996596281503
Generation 441 Best Solution: [182.98020934905568, 177.36471469574758,
174.06418452460682, 74.80255762692249] Fitness: 0.9628072090310208
Generation 442 Best Solution: [183.83656165606922, 175.88991959207274,
175.1010850442743, 75.86615346025701] Fitness: 1.7298146159617982
Generation 443 Best Solution: [183.83656165606922, 175.68858211488373,
175.1010850442743, 75.86615346025701] Fitness: 1.8401341986209254
Generation 444 Best Solution: [183.83656165606922, 176.8032814785501,
175.1010850442743, 75.76859102014826] Fitness: 1.7608265469198323
Generation 445 Best Solution: [182.98020934905568, 176.28187572325098,
175.1010850442743, 75.86615346025701] Fitness: 1.4562854034233252
Generation 446 Best Solution: [183.83656165606922, 175.68858211488373,
174.89544597686532, 75.86615346025701] Fitness: 1.6124294752899893
Generation 447 Best Solution: [183.83656165606922, 175.68858211488373,
175.1010850442743, 75.86615346025701] Fitness: 1.8401341986209254
Generation 448 Best Solution: [183.25294578047075, 177.42491965656382,
174.56372596620952, 74.55105867334444] Fitness: 1.1589120648303575
Generation 449 Best Solution: [183.63832564751073, 177.42491965656382,
174.56265374773858, 75.86615346025701] Fitness: 2.0138203692885908
Generation 450 Best Solution: [183.25609096492926, 177.42491965656382,
174.19763620762424, 75.86615346025701] Fitness: 1.790650116952651
Generation 451 Best Solution: [182.92330085394303, 177.42491965656382,
175.1010850442743, 74.90375908927369] Fitness: 1.6720286789857393
Generation 452 Best Solution: [183.63832564751073, 177.42491965656382,
174.19763620762424, 75.86615346025701] Fitness: 1.9039286753977034
Generation 453 Best Solution: [183.0365258653793, 176.06502854604582,
175.1010850442743, 75.86615346025701] Fitness: 1.4508884523974432
Generation 454 Best Solution: [183.0365258653793, 176.06502854604582,
175.1010850442743, 75.86615346025701] Fitness: 1.4508884523974432
Generation 455 Best Solution: [183.63832564751073, 176.06502854604582,
175.1010850442743, 75.86615346025701] Fitness: 1.5430625323694827
Generation 456 Best Solution: [183.63832564751073, 176.06502854604582,
175.1010850442743, 75.86615346025701] Fitness: 1.5430625323694827
Generation 457 Best Solution: [183.63832564751073, 176.06502854604582,
175.1010850442743, 75.86615346025701] Fitness: 1.5430625323694827
Generation 458 Best Solution: [183.63832564751073, 176.06502854604582,
175.1010850442743, 75.86615346025701] Fitness: 1.5430625323694827
Generation 459 Best Solution: [183.63832564751073, 176.06502854604582,
173.8446915824751, 75.86615346025701] Fitness: 1.0128290174261696
Generation 460 Best Solution: [183.63832564751073, 176.06502854604582,
175.1010850442743, 75.86615346025701] Fitness: 1.5430625323694827
Generation 461 Best Solution: [183.63832564751073, 177.42491965656382,
175.1010850442743, 74.87723713903502] Fitness: 1.72478468171767
Generation 462 Best Solution: [183.63832564751073, 177.42491965656382,
175.1010850442743, 74.87723713903502] Fitness: 1.72478468171767
Generation 463 Best Solution: [183.63832564751073, 176.2824816344862,
174.71241063776543, 75.86615346025701] Fitness: 1.1489874457270368
Generation 464 Best Solution: [183.0365258653793, 177.42491965656382,

175.1010850442743, 75.1156835406122] Fitness: 1.7011775252953107
Generation 465 Best Solution: [183.63832564751073, 177.42491965656382,
175.1010850442743, 75.1156835406122] Fitness: 1.7933516052673504
Generation 466 Best Solution: [182.8765731868409, 177.42491965656382,
175.1010850442743, 75.1156835406122] Fitness: 1.7561422326356215
Generation 467 Best Solution: [182.88995027040826, 177.42491965656382,
175.1010850442743, 75.14571635557752] Fitness: 1.764161443826256
Generation 468 Best Solution: [183.81696908433182, 176.6581784622487,
175.1010850442743, 75.86615346025701] Fitness: 1.779158500891004
Generation 469 Best Solution: [183.47545220500197, 177.42491965656382,
174.22324622005596, 74.83521491192144] Fitness: 1.0616208065178339
Generation 470 Best Solution: [183.63832564751073, 176.4557723733217,
175.1010850442743, 74.98070662479353] Fitness: 0.8213529002740358
Generation 471 Best Solution: [183.81696908433182, 176.34182883940076,
175.1010850442743, 75.86615346025701] Fitness: 1.659219224397745
Generation 472 Best Solution: [183.47545220500197, 176.34182883940076,
175.1010850442743, 75.86615346025701] Fitness: 1.4693607948710485
Generation 473 Best Solution: [183.63832564751073, 176.34182883940076,
175.1010850442743, 74.52743739674482] Fitness: 0.8061581769264103
Generation 474 Best Solution: [183.63832564751073, 176.89712003865114,
174.22324622005596, 74.79086037465815] Fitness: 0.4758743191610558
Generation 475 Best Solution: [183.63832564751073, 176.89712003865114,
174.22324622005596, 75.86615346025701] Fitness: 1.2507963210145499
Generation 476 Best Solution: [183.47545220500197, 176.1935009715446,
175.1010850442743, 75.86615346025701] Fitness: 1.4580753195787093
Generation 477 Best Solution: [183.63832564751073, 176.1935009715446,
174.71241063776543, 75.86615346025701] Fitness: 1.1456613520650318
Generation 478 Best Solution: [183.47545220500197, 176.1935009715446,
175.1010850442743, 75.86615346025701] Fitness: 1.4580753195787093
Generation 479 Best Solution: [183.83656165606922, 176.76778836999682,
175.1010850442743, 75.11726466219332] Fitness: 1.1675584589750425
Generation 480 Best Solution: [183.51742621521785, 176.76778836999682,
175.1010850442743, 75.11726466219332] Fitness: 0.9773285096906896
Generation 481 Best Solution: [183.83656165606922, 176.1935009715446,
175.1010850442743, 75.19297709899351] Fitness: 1.0017673505803368
Generation 482 Best Solution: [183.63832564751073, 176.41704668487233,
174.86426212934782, 75.86615346025701] Fitness: 1.3002013338032676
Generation 483 Best Solution: [183.87594296021965, 177.42491965656382,
173.80075854171002, 75.19297709899351] Fitness: 1.4864534829378926
Generation 484 Best Solution: [183.63832564751073, 176.3482863589335,
175.1010850442743, 75.86615346025701] Fitness: 1.5420738834712302
Generation 485 Best Solution: [183.63832564751073, 176.3482863589335,
175.1010850442743, 75.86615346025701] Fitness: 1.5420738834712302
Generation 486 Best Solution: [183.63832564751073, 176.3482863589335,
175.1010850442743, 75.86615346025701] Fitness: 1.5420738834712302
Generation 487 Best Solution: [183.83656165606922, 176.0689345879193,
175.1010850442743, 75.86615346025701] Fitness: 1.6761320091842409
Generation 488 Best Solution: [183.63832564751073, 176.77114427943715,

175.1010850442743, 75.86615346025701] Fitness: 1.7353284024964182
Generation 489 Best Solution: [183.87594296021965, 176.57753372708888,
175.1010850442743, 75.86615346025701] Fitness: 1.7843750180373648
Generation 490 Best Solution: [183.6463649516224, 177.42491965656382,
175.1010850442743, 74.6716076394571] Fitness: 1.7296365499088115
Generation 491 Best Solution: [183.87594296021965, 176.57753372708888,
175.1010850442743, 74.6716076394571] Fitness: 1.0164162341434828
Generation 492 Best Solution: [183.87594296021965, 177.42491965656382,
174.68379270214902, 74.6716076394571] Fitness: 1.4872167190776553
Generation 493 Best Solution: [183.6463649516224, 177.42491965656382,
175.1010850442743, 75.28657313109419] Fitness: 1.892541217684206
Generation 494 Best Solution: [183.63832564751073, 177.39121721952748,
175.1010850442743, 74.7386215052234] Fitness: 1.666315728308193
Generation 495 Best Solution: [183.63832564751073, 175.7408715590088,
175.1010850442743, 75.86615346025701] Fitness: 1.6726055078276507
Generation 496 Best Solution: [183.87594296021965, 177.42491965656382,
174.32411421967356, 74.41036025028568] Fitness: 1.4009316340427278
Generation 497 Best Solution: [183.63832564751073, 177.42491965656382,
175.03265824345036, 74.41036025028568] Fitness: 1.7233988420255042
Generation 498 Best Solution: [183.6463649516224, 177.42491965656382,
175.1010850442743, 74.7309141974786] Fitness: 1.7238550051148631
Generation 499 Best Solution: [183.20336811852158, 175.79284040001576,
175.1010850442743, 75.86615346025701] Fitness: 1.528664334968802
Generation 500 Best Solution: [183.63832564751073, 177.42491965656382,
175.1010850442743, 74.89048636253958] Fitness: 1.7266487049052013
Optimal Solution for Simulation 16: [183.63832564751073, 177.42491965656382,
175.1010850442743, 74.89048636253958]

Simulation 17/32 with conditions: {'drought': False, 'flood': True,
'policy_change': True, 'social_event': True, 'economic_crisis': True}
Generation 1 Best Solution: [183.18798688543393, 176.14703457108175,
174.64770687874991, 74.9215746482145] Fitness: 0.3624499461004942
Generation 2 Best Solution: [183.08008725841728, 175.83014526158325,
174.51973668472831, 74.54826887800162] Fitness: 0.3506776104800696
Generation 3 Best Solution: [183.08008725841728, 177.211880222643,
173.75451652554815, 74.54826887800162] Fitness: 1.123044541204906
Generation 4 Best Solution: [183.08008725841728, 175.7741302140666,
174.8626396761522, 75.34746934021545] Fitness: 1.1675171403726965
Generation 5 Best Solution: [183.48054679681624, 176.38622745097234,
173.75451652554815, 75.40547113728356] Fitness: 0.7640930383808185
Generation 6 Best Solution: [183.08008725841728, 177.211880222643,
174.36783647412543, 74.8003591190063] Fitness: 1.0589185106125751
Generation 7 Best Solution: [182.90304167769133, 177.211880222643,
174.79852286966872, 74.54826887800162] Fitness: 1.493939850114672
Generation 8 Best Solution: [183.08008725841728, 177.27564980037823,
174.79852286966872, 74.8003591190063] Fitness: 1.546043973750559
Generation 9 Best Solution: [183.82845450778757, 175.74992704289204,
175.05857058228966, 75.55370519362162] Fitness: 2.114040517776748

Generation 10 Best Solution: [183.82845450778757, 175.74992704289204,
175.05857058228966, 75.28208474316158] Fitness: 1.7378660923038498
Generation 11 Best Solution: [183.82845450778757, 175.74992704289204,
175.05857058228966, 75.28208474316158] Fitness: 1.7378660923038498
Generation 12 Best Solution: [183.82845450778757, 175.74992704289204,
174.78672138801335, 75.55370519362162] Fitness: 1.7134248927897315
Generation 13 Best Solution: [183.82845450778757, 176.98228530904376,
174.42180625389685, 75.55370519362162] Fitness: 1.746371193148457
Generation 14 Best Solution: [183.82845450778757, 176.98228530904376,
174.42180625389685, 75.55370519362162] Fitness: 1.746371193148457
Generation 15 Best Solution: [183.82845450778757, 176.34702913360687,
175.05857058228966, 74.51783929134585] Fitness: 1.231739582904169
Generation 16 Best Solution: [183.0529675635468, 177.39576315057357,
175.05857058228966, 74.51783929134585] Fitness: 2.1918792027875917
Generation 17 Best Solution: [183.0529675635468, 176.34702913360687,
175.05857058228966, 75.76503979749148] Fitness: 2.0059434557494304
Generation 18 Best Solution: [183.0529675635468, 176.34702913360687,
175.05857058228966, 75.30779539339676] Fitness: 1.2729768462814983
Generation 19 Best Solution: [183.38434593439547, 177.39576315057357,
175.05857058228966, 74.51783929134585] Fitness: 2.1926507901865064
Generation 20 Best Solution: [183.82845450778757, 176.40315672361842,
175.05857058228966, 75.55370519362162] Fitness: 1.9551992715060744
Generation 21 Best Solution: [183.0529675635468, 176.40315672361842,
175.05857058228966, 75.55370519362162] Fitness: 1.6365452452117548
Generation 22 Best Solution: [183.38434593439547, 175.97274970815923,
175.05857058228966, 74.91418641975693] Fitness: 0.9845127844662052
Generation 23 Best Solution: [183.82845450778757, 176.6757202002489,
174.9289687369786, 74.51783929134585] Fitness: 1.206846448226542
Generation 24 Best Solution: [183.38434593439547, 175.7979576555752,
175.05857058228966, 75.55370519362162] Fitness: 1.7577361786698404
Generation 25 Best Solution: [183.38434593439547, 175.7979576555752,
175.05857058228966, 74.51783929134585] Fitness: 1.0514611297921614
Generation 26 Best Solution: [183.82845450778757, 175.7979576555752,
175.05857058228966, 75.55370519362162] Fitness: 2.0756186175652456
Generation 27 Best Solution: [183.82845450778757, 176.25567758939152,
175.05857058228966, 75.28624179521951] Fitness: 1.5509999666543264
Generation 28 Best Solution: [183.38434593439547, 175.7979576555752,
173.8073860978942, 74.51783929134585] Fitness: 0.3349326168844624
Generation 29 Best Solution: [183.82845450778757, 175.7979576555752,
175.05857058228966, 74.8753018720208] Fitness: 1.3899655386028515
Generation 30 Best Solution: [183.60819583924513, 176.25015085807186,
175.05857058228966, 75.75709554922167] Fitness: 2.089538323378789
Generation 31 Best Solution: [183.38434593439547, 175.7979576555752,
175.05857058228966, 75.75709554922167] Fitness: 2.1283005285905636
Generation 32 Best Solution: [183.38434593439547, 175.7979576555752,
175.05857058228966, 75.59970207025486] Fitness: 1.8348791357703729
Generation 33 Best Solution: [183.38434593439547, 177.39576315057357,
174.08608201044254, 75.75709554922167] Fitness: 2.463218989124801

Generation 34 Best Solution: [183.82845450778757, 175.7979576555752,
175.05857058228966, 75.75709554922167] Fitness: 2.4461829674859685
Generation 35 Best Solution: [183.35002353506843, 177.39576315057357,
174.4873695507801, 75.75709554922167] Fitness: 2.575580092595546
Generation 36 Best Solution: [183.11020567253115, 177.39576315057357,
175.05857058228966, 75.02174581664669] Fitness: 2.274537345097548
Generation 37 Best Solution: [183.82845450778757, 177.39576315057357,
174.3543760763678, 75.75709554922167] Fitness: 2.8297220649545536
Generation 38 Best Solution: [183.11020567253115, 177.39576315057357,
174.3543760763678, 75.75709554922167] Fitness: 2.4967653443388524
Generation 39 Best Solution: [183.11020567253115, 177.39576315057357,
174.3543760763678, 75.75709554922167] Fitness: 2.4967653443388524
Generation 40 Best Solution: [183.35002353506843, 176.80691555273913,
175.05857058228966, 75.75709554922167] Fitness: 2.2928690316873586
Generation 41 Best Solution: [183.82845450778757, 176.49455800822565,
175.05857058228966, 75.75709554922167] Fitness: 2.3661536254628937
Generation 42 Best Solution: [183.82845450778757, 177.32736043217344,
174.2434612300197, 75.75709554922167] Fitness: 2.6484617929099197
Generation 43 Best Solution: [183.82845450778757, 176.35258280751475,
174.6444897976735, 75.75709554922167] Fitness: 1.7539868378192218
Generation 44 Best Solution: [183.35002353506843, 175.6403722581157,
174.6444897976735, 75.75709554922167] Fitness: 1.7047493916657088
Generation 45 Best Solution: [183.289707198658, 177.32736043217344,
175.05857058228966, 74.60921586869378] Fitness: 2.0096197806625278
Generation 46 Best Solution: [183.82845450778757, 175.62910660402713,
175.05857058228966, 75.75709554922167] Fitness: 2.6000233093372005
Generation 47 Best Solution: [183.82845450778757, 175.62910660402713,
175.05857058228966, 75.75709554922167] Fitness: 2.6000233093372005
Generation 48 Best Solution: [182.89665461511296, 175.69961716690258,
175.05857058228966, 75.75709554922167] Fitness: 2.2805176360223243
Generation 49 Best Solution: [182.987838172642, 176.22351462547542,
175.05857058228966, 75.75709554922167] Fitness: 1.996128445139842
Generation 50 Best Solution: [183.82845450778757, 176.37105856953139,
175.05857058228966, 75.75709554922167] Fitness: 2.3152265013622415
Generation 51 Best Solution: [182.987838172642, 176.53647202264446,
175.05857058228966, 75.75709554922167] Fitness: 2.094768772443445
Generation 52 Best Solution: [183.82845450778757, 176.37105856953139,
175.05857058228966, 75.75709554922167] Fitness: 2.3152265013622415
Generation 53 Best Solution: [183.82845450778757, 176.37105856953139,
175.05857058228966, 75.75709554922167] Fitness: 2.3152265013622415
Generation 54 Best Solution: [182.987838172642, 177.39576315057357,
173.86595201995468, 75.75709554922167] Fitness: 2.5450772839909663
Generation 55 Best Solution: [183.82845450778757, 177.42260513222325,
175.05857058228966, 74.43277815594166] Fitness: 2.599547528964524
Generation 56 Best Solution: [183.3250585471276, 177.39576315057357,
175.05857058228966, 74.43277815594166] Fitness: 2.2073947842116084
Generation 57 Best Solution: [183.82845450778757, 176.0543775171913,
175.05857058228966, 75.75709554922167] Fitness: 2.312882681625787

Generation 58 Best Solution: [183.82845450778757, 175.59936719021746,
175.05857058228966, 75.75709554922167] Fitness: 2.632552402370236
Generation 59 Best Solution: [183.49134463742777, 177.39576315057357,
173.96635440865316, 75.75709554922167] Fitness: 2.5276645859516176
Generation 60 Best Solution: [183.46086008913346, 177.39576315057357,
173.94400105684625, 75.75709554922167] Fitness: 2.520003774930876
Generation 61 Best Solution: [183.82845450778757, 176.17167834129424,
175.05857058228966, 75.75709554922167] Fitness: 2.2922343805844205
Generation 62 Best Solution: [183.82845450778757, 176.66046830700907,
173.87541748383583, 75.75709554922167] Fitness: 1.7271238605551011
Generation 63 Best Solution: [183.3250585471276, 177.39576315057357,
175.05857058228966, 74.68673173711292] Fitness: 2.1581148636360252
Generation 64 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.68673173711292] Fitness: 2.4909763867918513
Generation 65 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.68673173711292] Fitness: 2.4909763867918513
Generation 66 Best Solution: [183.82845450778757, 175.61606869948116,
175.05857058228966, 74.68673173711292] Fitness: 1.5176876493488443
Generation 67 Best Solution: [183.3250585471276, 177.39576315057357,
175.00162146632258, 74.68673173711292] Fitness: 2.062931438993037
Generation 68 Best Solution: [183.82845450778757, 177.39576315057357,
173.7888749869101, 75.75709554922167] Fitness: 2.8818663672314306
Generation 69 Best Solution: [183.82845450778757, 176.19144887071872,
173.7888749869101, 75.75709554922167] Fitness: 1.5857411164880746
Generation 70 Best Solution: [183.82845450778757, 175.67977254065383,
175.05857058228966, 74.31646786643256] Fitness: 1.563426660864564
Generation 71 Best Solution: [183.3250585471276, 176.27116707487352,
175.05857058228966, 75.75709554922167] Fitness: 1.9617026629899965
Generation 72 Best Solution: [183.82845450778757, 176.10259022651636,
174.60741072098295, 75.75709554922167] Fitness: 1.7108972409362795
Generation 73 Best Solution: [183.4714593971492, 177.28372029953195,
174.1054237801017, 74.31646786643256] Fitness: 1.277935676990933
Generation 74 Best Solution: [183.3250585471276, 176.68223177912952,
175.05857058228966, 75.75709554922167] Fitness: 2.1644097450777617
Generation 75 Best Solution: [183.30346955466274, 175.75996432587925,
175.05857058228966, 75.75709554922167] Fitness: 2.139515069179244
Generation 76 Best Solution: [183.30346955466274, 177.39576315057357,
174.24211798247296, 75.75709554922167] Fitness: 2.456553823260035
Generation 77 Best Solution: [183.3250585471276, 177.39576315057357,
175.05857058228966, 74.75802647357824] Fitness: 2.1656134696188207
Generation 78 Best Solution: [183.82845450778757, 177.39576315057357,
174.24211798247296, 75.49412152092371] Fitness: 2.3285569076211963
Generation 79 Best Solution: [183.82845450778757, 177.39576315057357,
174.24211798247296, 75.33377702717499] Fitness: 2.1076556437874063
Generation 80 Best Solution: [183.82845450778757, 177.39576315057357,
174.24211798247296, 75.33377702717499] Fitness: 2.1076556437874063
Generation 81 Best Solution: [183.82845450778757, 177.39576315057357,
174.44263995988442, 75.67457841316647] Fitness: 2.71515368704491

Generation 82 Best Solution: [183.3250585471276, 177.39576315057357,
174.6484648031651, 75.73337706773674] Fitness: 2.6551807366525573
Generation 83 Best Solution: [183.21060985902346, 177.39576315057357,
174.6484648031651, 75.73337706773674] Fitness: 2.6445581892588113
Generation 84 Best Solution: [183.37494614262837, 177.39576315057357,
174.6484648031651, 75.73337706773674] Fitness: 2.6673535251985414
Generation 85 Best Solution: [183.82845450778757, 176.11899462498212,
174.6484648031651, 75.73337706773674] Fitness: 1.6990459260490305
Generation 86 Best Solution: [183.37494614262837, 176.11899462498212,
174.6484648031651, 75.73337706773674] Fitness: 1.3783571914391888
Generation 87 Best Solution: [183.37494614262837, 177.39576315057357,
175.05857058228966, 74.51877370309514] Fitness: 2.1895919079834947
Generation 88 Best Solution: [183.82845450778757, 176.19316746532942,
175.05857058228966, 75.73337706773674] Fitness: 2.2440613644134633
Generation 89 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.54936761292738] Fitness: 2.502898026540317
Generation 90 Best Solution: [183.69361646144935, 176.45101192255987,
175.05857058228966, 74.3520766287982] Fitness: 1.2035047987854688
Generation 91 Best Solution: [183.69361646144935, 176.45101192255987,
175.05857058228966, 75.75709554922167] Fitness: 2.210114311411963
Generation 92 Best Solution: [183.69361646144935, 176.45101192255987,
175.05857058228966, 75.73337706773674] Fitness: 2.1629801194216434
Generation 93 Best Solution: [183.82845450778757, 175.60779587183475,
174.6052596216409, 75.81111915313957] Fitness: 2.1420371098048343
Generation 94 Best Solution: [183.69361646144935, 175.60779587183475,
174.6052596216409, 75.81111915313957] Fitness: 2.0071578985304988
Generation 95 Best Solution: [183.82845450778757, 175.60779587183475,
175.05857058228966, 74.83172782921024] Fitness: 1.5518534131738555
Generation 96 Best Solution: [183.82845450778757, 176.2772778235824,
174.87356068735625, 75.75709554922167] Fitness: 2.0078768404062446
Generation 97 Best Solution: [183.82845450778757, 176.2772778235824,
174.80882841617571, 75.73337706773674] Fitness: 1.8750501988938428
Generation 98 Best Solution: [183.69361646144935, 176.2772778235824,
174.80882841617571, 75.73337706773674] Fitness: 1.740170987619507
Generation 99 Best Solution: [183.55883269081224, 176.2772778235824,
175.05857058228966, 75.73337706773674] Fitness: 2.011896026139296
Generation 100 Best Solution: [183.55883269081224, 176.2772778235824,
175.05857058228966, 75.73337706773674] Fitness: 2.011896026139296
Generation 101 Best Solution: [183.82845450778757, 177.39576315057357,
174.7165230372465, 74.65126212638697] Fitness: 2.008755267036005
Generation 102 Best Solution: [183.69361646144935, 175.60779587183475,
175.05857058228966, 75.75709554922167] Fitness: 2.488288612840424
Generation 103 Best Solution: [182.99934943011894, 175.60779587183475,
175.05857058228966, 75.75709554922167] Fitness: 2.3233803228084184
Generation 104 Best Solution: [183.82845450778757, 175.73982173033974,
174.7165230372465, 75.81111915313957] Fitness: 2.1224761056393096
Generation 105 Best Solution: [183.82845450778757, 175.73982173033974,
174.62836305941343, 75.73337706773674] Fitness: 1.874788329334689

Generation 106 Best Solution: [183.82845450778757, 175.73982173033974,
175.05857058228966, 75.38090001801116] Fitness: 1.8676324695924897
Generation 107 Best Solution: [183.69361646144935, 175.73982173033974,
175.05857058228966, 75.38090001801116] Fitness: 1.732753258318154
Generation 108 Best Solution: [183.69361646144935, 176.1472186559569,
174.7165230372465, 75.73337706773674] Fitness: 1.6304629343864212
Generation 109 Best Solution: [183.69361646144935, 176.1472186559569,
174.7165230372465, 75.73337706773674] Fitness: 1.6304629343864212
Generation 110 Best Solution: [183.82845450778757, 176.01191695259664,
175.05857058228966, 75.15609151839833] Fitness: 1.4514565670230486
Generation 111 Best Solution: [183.82845450778757, 176.3756055941166,
174.7165230372465, 75.81111915313957] Fitness: 1.9458509858510076
Generation 112 Best Solution: [183.55067938120206, 176.01191695259664,
175.05857058228966, 75.73337706773674] Fitness: 2.0381317251615965
Generation 113 Best Solution: [183.82845450778757, 176.01191695259664,
175.05857058228966, 74.79942388378552] Fitness: 1.2463462230789633
Generation 114 Best Solution: [183.82845450778757, 176.01191695259664,
175.05857058228966, 75.73337706773674] Fitness: 2.2794636378724595
Generation 115 Best Solution: [183.82845450778757, 177.39576315057357,
174.7165230372465, 75.73337706773674] Fitness: 3.058263984378273
Generation 116 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 75.17112842761024] Fitness: 2.726020847605945
Generation 117 Best Solution: [183.82845450778757, 177.39576315057357,
173.77886762428687, 75.17112842761024] Fitness: 2.0267359321060723
Generation 118 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.38542587367753] Fitness: 2.562571178951366
Generation 119 Best Solution: [183.82845450778757, 177.39576315057357,
174.33535556320751, 74.38542587367753] Fitness: 1.7971116852241116
Generation 120 Best Solution: [182.79806969991864, 177.39576315057357,
175.05857058228966, 74.38542587367753] Fitness: 2.3808089587469365
Generation 121 Best Solution: [183.03396761721794, 177.39576315057357,
175.05857058228966, 74.38542587367753] Fitness: 2.249997508521456
Generation 122 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.4206611819909] Fitness: 2.545573669434988
Generation 123 Best Solution: [183.82845450778757, 177.39576315057357,
175.0491402203036, 74.4206611819909] Fitness: 2.5293997127559082
Generation 124 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.4206611819909] Fitness: 2.545573669434988
Generation 125 Best Solution: [183.58667138027133, 177.39576315057357,
175.05857058228966, 74.31161942531722] Fitness: 2.38751152191448
Generation 126 Best Solution: [183.85816394844562, 176.56375717352384,
175.05857058228966, 75.73337706773674] Fitness: 2.3940387544215342
Generation 127 Best Solution: [183.85816394844562, 176.56375717352384,
175.05857058228966, 75.73337706773674] Fitness: 2.3940387544215342
Generation 128 Best Solution: [183.85816394844562, 176.56375717352384,
175.05857058228966, 75.73337706773674] Fitness: 2.3940387544215342
Generation 129 Best Solution: [183.29604799186745, 177.39576315057357,
175.05857058228966, 74.81476364917111] Fitness: 2.173291050686956

Generation 130 Best Solution: [183.82845450778757, 176.33790316642072,
175.05857058228966, 75.81111915313957] Fitness: 2.4175542381397763
Generation 131 Best Solution: [183.82845450778757, 176.33790316642072,
175.05857058228966, 75.73337706773674] Fitness: 2.2591985449925307
Generation 132 Best Solution: [183.82845450778757, 177.39576315057357,
174.13838866217588, 75.73337706773674] Fitness: 2.733052550594859
Generation 133 Best Solution: [183.29604799186745, 177.39576315057357,
174.39984972015688, 75.73337706773674] Fitness: 2.466120925368595
Generation 134 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 75.13526625640512] Fitness: 2.693820809818249
Generation 135 Best Solution: [183.82845450778757, 176.3302148722466,
175.05857058228966, 75.79042111176632] Fitness: 2.372533949703514
Generation 136 Best Solution: [183.82845450778757, 176.35744304056752,
175.05857058228966, 75.73337706773674] Fitness: 2.264195252503324
Generation 137 Best Solution: [183.06323014919548, 175.7389429710984,
175.05857058228966, 75.79042111176632] Fitness: 2.2403005866987042
Generation 138 Best Solution: [183.82845450778757, 176.41734318084596,
175.05857058228966, 75.79042111176632] Fitness: 2.3989994704474857
Generation 139 Best Solution: [183.82845450778757, 176.41734318084596,
175.05857058228966, 75.73337706773674] Fitness: 2.283890617716996
Generation 140 Best Solution: [183.82845450778757, 176.41734318084596,
175.05857058228966, 75.22241846538738] Fitness: 1.5198379249118719
Generation 141 Best Solution: [183.06323014919548, 177.39576315057357,
175.05857058228966, 74.81940407496799] Fitness: 2.1907602291350523
Generation 142 Best Solution: [183.82845450778757, 176.35382195007247,
175.05857058228966, 75.79042111176632] Fitness: 2.378325093478611
Generation 143 Best Solution: [183.82845450778757, 176.0394526003393,
175.05857058228966, 75.81111915313957] Fitness: 2.4285469766877155
Generation 144 Best Solution: [183.82845450778757, 176.41088849651254,
175.05857058228966, 75.79042111176632] Fitness: 2.396559773121814
Generation 145 Best Solution: [183.8499701773844, 176.35382195007247,
175.05857058228966, 75.81111915313957] Fitness: 2.4461891025484057
Generation 146 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.68123514878522] Fitness: 2.4907865921526646
Generation 147 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.68123514878522] Fitness: 2.4907865921526646
Generation 148 Best Solution: [183.82845450778757, 176.4299158644695,
175.05857058228966, 75.81111915313957] Fitness: 2.447218534682824
Generation 149 Best Solution: [183.60086964402433, 176.32495529266197,
175.05857058228966, 74.68123514878522] Fitness: 0.998587492273052
Generation 150 Best Solution: [183.82845450778757, 175.6493951629751,
174.72934526684008, 75.73337706773674] Fitness: 2.06384068802213
Generation 151 Best Solution: [183.82845450778757, 175.60720827109884,
174.77890246608567, 75.81111915313957] Fitness: 2.324912969062249
Generation 152 Best Solution: [183.82845450778757, 175.60720827109884,
175.05857058228966, 74.92269450841809] Fitness: 1.5879866555814275
Generation 153 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.76264538323636] Fitness: 2.4992833856052785

Generation 154 Best Solution: [183.82845450778757, 176.33686067689825,
175.05857058228966, 75.73337706773674] Fitness: 2.2589517014713034
Generation 155 Best Solution: [183.82845450778757, 176.00184314638852,
175.05857058228966, 75.81111915313957] Fitness: 2.4415601453586855
Generation 156 Best Solution: [183.82845450778757, 176.00184314638852,
175.05857058228966, 75.73337706773674] Fitness: 2.28320445221144
Generation 157 Best Solution: [183.82845450778757, 176.00184314638852,
175.05857058228966, 75.73337706773674] Fitness: 2.28320445221144
Generation 158 Best Solution: [183.82845450778757, 176.32255719099476,
175.05857058228966, 75.81111915313957] Fitness: 2.414122516961804
Generation 159 Best Solution: [183.82845450778757, 176.32255719099476,
174.82951467813635, 75.81111915313957] Fitness: 2.0675521291623573
Generation 160 Best Solution: [183.82845450778757, 176.32255719099476,
174.82951467813635, 75.81111915313957] Fitness: 2.0675521291623573
Generation 161 Best Solution: [183.82845450778757, 175.93643873789588,
175.05857058228966, 75.81111915313957] Fitness: 2.470389144123385
Generation 162 Best Solution: [183.82845450778757, 175.93643873789588,
175.05857058228966, 75.73337706773674] Fitness: 2.3120334509761395
Generation 163 Best Solution: [183.82845450778757, 176.84072757479228,
175.05857058228966, 75.81111915313957] Fitness: 2.7697024260529424
Generation 164 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 75.30843831980856] Fitness: 2.871185058918885
Generation 165 Best Solution: [182.8025095428236, 177.39576315057357,
173.82503116034894, 75.81111915313957] Fitness: 2.786976329790546
Generation 166 Best Solution: [183.82845450778757, 177.39576315057357,
173.82503116034894, 75.81111915313957] Fitness: 2.9721459833389456
Generation 167 Best Solution: [183.82845450778757, 176.2605399624225,
175.05857058228966, 75.73337706773674] Fitness: 2.2463123356984607
Generation 168 Best Solution: [183.82845450778757, 175.8680501267848,
175.05857058228966, 75.81111915313957] Fitness: 2.508951446991185
Generation 169 Best Solution: [183.82845450778757, 175.8680501267848,
175.05857058228966, 75.81111915313957] Fitness: 2.508951446991185
Generation 170 Best Solution: [183.82845450778757, 176.87073645427552,
175.05857058228966, 75.81111915313957] Fitness: 2.8054294046965107
Generation 171 Best Solution: [183.82845450778757, 177.39576315057357,
174.31880400652315, 75.73337706773674] Fitness: 2.768525320396943
Generation 172 Best Solution: [183.82845450778757, 176.10720668633672,
175.05857058228966, 75.81111915313957] Fitness: 2.411671232312732
Generation 173 Best Solution: [183.82845450778757, 176.10720668633672,
175.05857058228966, 75.81111915313957] Fitness: 2.411671232312732
Generation 174 Best Solution: [183.82845450778757, 176.27263366134306,
175.05857058228966, 75.73337706773674] Fitness: 2.247600553288913
Generation 175 Best Solution: [182.79213831607095, 177.39576315057357,
175.05857058228966, 74.41397041733677] Fitness: 2.3714721038610924
Generation 176 Best Solution: [183.82845450778757, 176.6614527218766,
174.04827058418357, 75.73337706773674] Fitness: 1.6299542782816692
Generation 177 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.63127091426439] Fitness: 2.4916107233214757

Generation 178 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.96301615364875] Fitness: 2.572139923707878
Generation 179 Best Solution: [183.82845450778757, 176.2957105789955,
175.05857058228966, 75.81111915313957] Fitness: 2.4091610926315523
Generation 180 Best Solution: [183.20800748651655, 176.2957105789955,
175.05857058228966, 75.81111915313957] Fitness: 2.0657157246754707
Generation 181 Best Solution: [183.20800748651655, 176.2957105789955,
175.05857058228966, 75.81111915313957] Fitness: 2.0657157246754707
Generation 182 Best Solution: [183.20800748651655, 175.99805108072934,
174.54719713841038, 75.81111915313957] Fitness: 1.45866373920239
Generation 183 Best Solution: [183.20800748651655, 175.99805108072934,
175.05857058228966, 75.81111915313957] Fitness: 2.0995712996256275
Generation 184 Best Solution: [183.20800748651655, 175.99805108072934,
175.05857058228966, 75.81111915313957] Fitness: 2.0995712996256275
Generation 185 Best Solution: [183.20800748651655, 175.99805108072934,
174.54719713841038, 75.81111915313957] Fitness: 1.45866373920239
Generation 186 Best Solution: [183.20800748651655, 175.99805108072934,
174.54719713841038, 75.81111915313957] Fitness: 1.45866373920239
Generation 187 Best Solution: [183.20800748651655, 175.99805108072934,
174.54719713841038, 75.81111915313957] Fitness: 1.45866373920239
Generation 188 Best Solution: [183.20800748651655, 176.74904698568494,
175.05857058228966, 74.35526160847742] Fitness: 1.2077142538901664
Generation 189 Best Solution: [183.3082466875377, 177.39576315057357,
175.05857058228966, 74.85326366459557] Fitness: 2.1871509830011555
Generation 190 Best Solution: [183.20800748651655, 177.39576315057357,
175.05857058228966, 75.0422413948444] Fitness: 2.2778805124319845
Generation 191 Best Solution: [183.20800748651655, 177.39576315057357,
175.05857058228966, 74.66397631411176] Fitness: 2.1471065981770345
Generation 192 Best Solution: [183.20800748651655, 175.9319928272201,
175.05857058228966, 75.81111915313957] Fitness: 2.1291891520553055
Generation 193 Best Solution: [183.20800748651655, 177.39576315057357,
175.05857058228966, 74.33924128047782] Fitness: 2.2448647719694623
Generation 194 Best Solution: [183.20800748651655, 176.62030154176438,
175.05857058228966, 75.81111915313957] Fitness: 2.214615710662796
Generation 195 Best Solution: [183.20800748651655, 176.3387719242644,
175.05857058228966, 75.81111915313957] Fitness: 2.074316104620964
Generation 196 Best Solution: [183.82845450778757, 176.04225807123458,
175.05857058228966, 74.29693428345253] Fitness: 1.3444188455937442
Generation 197 Best Solution: [182.86451029134673, 176.04225807123458,
175.05857058228966, 74.29693428345253] Fitness: 1.1154554746732797
Generation 198 Best Solution: [183.20800748651655, 175.95561394347527,
175.05857058228966, 75.81111915313957] Fitness: 2.117676178160484
Generation 199 Best Solution: [183.20800748651655, 176.49687531154603,
175.05857058228966, 75.81111915313957] Fitness: 2.135153569774649
Generation 200 Best Solution: [183.20800748651655, 176.49687531154603,
175.05857058228966, 75.81111915313957] Fitness: 2.135153569774649
Generation 201 Best Solution: [183.82845450778757, 177.39576315057357,
174.85743182222728, 74.31213267582639] Fitness: 2.2957491822487266

Generation 202 Best Solution: [183.4587290470107, 176.49687531154603,
175.05857058228966, 75.81111915313957] Fitness: 2.188656951771454
Generation 203 Best Solution: [183.20800748651655, 175.64002273550068,
175.05857058228966, 75.81111915313957] Fitness: 2.356267630249918
Generation 204 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.38146973261834] Fitness: 2.56462226418733
Generation 205 Best Solution: [183.82845450778757, 176.99533311466146,
175.05857058228966, 74.89291725483135] Fitness: 1.8111457102059907
Generation 206 Best Solution: [183.7963336482309, 175.64002273550068,
174.4329735554022, 75.81111915313957] Fitness: 1.946223519485626
Generation 207 Best Solution: [183.7963336482309, 175.64002273550068,
174.4329735554022, 75.81111915313957] Fitness: 1.946223519485626
Generation 208 Best Solution: [183.7963336482309, 177.39576315057357,
175.05857058228966, 75.13521650367599] Fitness: 2.658611700489414
Generation 209 Best Solution: [183.7963336482309, 177.39576315057357,
173.81598600942476, 75.81111915313957] Fitness: 2.941993305741624
Generation 210 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.86978857738708] Fitness: 2.529051969643601
Generation 211 Best Solution: [183.82845450778757, 176.21212779157887,
175.05857058228966, 75.81111915313957] Fitness: 2.402206060155389
Generation 212 Best Solution: [183.82845450778757, 176.21212779157887,
175.05857058228966, 75.81111915313957] Fitness: 2.402206060155389
Generation 213 Best Solution: [183.7963336482309, 177.16881426077987,
175.05857058228966, 74.48741830874287] Fitness: 2.036150597856455
Generation 214 Best Solution: [183.7963336482309, 175.94407056518784,
175.05857058228966, 75.81111915313957] Fitness: 2.431453463697723
Generation 215 Best Solution: [183.7963336482309, 175.94407056518784,
175.05857058228966, 75.81111915313957] Fitness: 2.431453463697723
Generation 216 Best Solution: [183.7963336482309, 175.94407056518784,
173.81598600942476, 75.81111915313957] Fitness: 1.7100187213747446
Generation 217 Best Solution: [183.82845450778757, 176.71222533479698,
173.81598600942476, 75.81111915313957] Fitness: 1.9140192505978764
Generation 218 Best Solution: [183.82845450778757, 177.39576315057357,
174.9069723670857, 74.63984141946922] Fitness: 2.2509659952493917
Generation 219 Best Solution: [183.3617982793781, 176.34530803992794,
175.05857058228966, 74.63984141946922] Fitness: 0.8875727016379286
Generation 220 Best Solution: [183.82845450778757, 175.90728857998747,
173.81598600942476, 75.22008756385932] Fitness: 0.8395458616938575
Generation 221 Best Solution: [182.9693935166303, 175.90728857998747,
173.81598600942476, 75.81111915313957] Fitness: 1.4773952969144288
Generation 222 Best Solution: [182.9693935166303, 177.39576315057357,
174.9058103680057, 74.79201879604113] Fitness: 1.9765445221009823
Generation 223 Best Solution: [182.9693935166303, 177.39576315057357,
175.05857058228966, 74.79201879604113] Fitness: 2.218399148611646
Generation 224 Best Solution: [182.9693935166303, 177.39576315057357,
174.9058103680057, 74.79201879604113] Fitness: 1.9765445221009823
Generation 225 Best Solution: [183.82845450778757, 177.39576315057357,
174.9058103680057, 74.79201879604113] Fitness: 2.2634882311154274

Generation 226 Best Solution: [183.2947950288685, 176.1733372816725,
175.05857058228966, 75.81111915313957] Fitness: 2.0653310702902137
Generation 227 Best Solution: [183.65847439680144, 176.15310870770338,
175.05857058228966, 75.81111915313957] Fitness: 2.240501510047257
Generation 228 Best Solution: [183.65847439680144, 176.15310870770338,
175.05857058228966, 75.81111915313957] Fitness: 2.240501510047257
Generation 229 Best Solution: [183.30273232935608, 177.01612085716894,
174.38168979407186, 75.81111915313957] Fitness: 1.919877395538197
Generation 230 Best Solution: [183.82845450778757, 177.01612085716894,
174.38168979407186, 75.81111915313957] Fitness: 2.256703331742899
Generation 231 Best Solution: [183.82845450778757, 176.19786166240928,
175.05857058228966, 75.81111915313957] Fitness: 2.4023032082484486
Generation 232 Best Solution: [183.38151712815792, 177.39576315057357,
175.05857058228966, 75.126587871802] Fitness: 2.3676401753828284
Generation 233 Best Solution: [183.38151712815792, 177.39576315057357,
174.31984441782302, 75.81111915313957] Fitness: 2.6085151855351976
Generation 234 Best Solution: [183.82845450778757, 177.39576315057357,
174.31984441782302, 74.28447152092487] Fitness: 1.8525857063185929
Generation 235 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.62057101114401] Fitness: 2.492384384402833
Generation 236 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.62057101114401] Fitness: 2.492384384402833
Generation 237 Best Solution: [183.38151712815792, 177.39576315057357,
174.1288704521929, 75.81111915313957] Fitness: 2.572455924325332
Generation 238 Best Solution: [183.82845450778757, 177.39576315057357,
174.1288704521929, 75.81111915313957] Fitness: 2.8912000006774985
Generation 239 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.8934629705652] Fitness: 2.538478909978278
Generation 240 Best Solution: [183.35142166654094, 177.39576315057357,
175.05857058228966, 74.69225544674902] Fitness: 2.1642237484494227
Generation 241 Best Solution: [183.82845450778757, 176.34669092893566,
175.05857058228966, 75.81111915313957] Fitness: 2.4197145036368317
Generation 242 Best Solution: [183.82845450778757, 176.34669092893566,
175.05857058228966, 75.81111915313957] Fitness: 2.4197145036368317
Generation 243 Best Solution: [183.67529447415012, 177.39576315057357,
175.05857058228966, 74.7227716077208] Fitness: 2.342972946255044
Generation 244 Best Solution: [183.82845450778757, 176.34669092893566,
175.05857058228966, 75.81111915313957] Fitness: 2.4197145036368317
Generation 245 Best Solution: [183.0198903905769, 177.39576315057357,
175.05857058228966, 74.69225544674902] Fitness: 2.1835828235640915
Generation 246 Best Solution: [182.81811113844748, 176.34669092893566,
175.05857058228966, 74.54341321611084] Fitness: 1.028464579509611
Generation 247 Best Solution: [183.82845450778757, 175.59231198313208,
175.05857058228966, 74.69225544674902] Fitness: 1.5443587361774613
Generation 248 Best Solution: [183.07619819645262, 177.39576315057357,
175.05857058228966, 74.69225544674902] Fitness: 2.1660373957187087
Generation 249 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.54341321611084] Fitness: 2.504199905609331

Generation 250 Best Solution: [183.82845450778757, 175.9808211837646,
175.05857058228966, 75.81111915313957] Fitness: 2.4499678544255716
Generation 251 Best Solution: [183.0198903905769, 177.39576315057357,
175.05857058228966, 74.54341321611084] Fitness: 2.196559608164246
Generation 252 Best Solution: [183.0198903905769, 177.39576315057357,
175.05857058228966, 74.64717810424649] Fitness: 2.1832096342445917
Generation 253 Best Solution: [183.0198903905769, 177.39576315057357,
175.05857058228966, 74.54341321611084] Fitness: 2.196559608164246
Generation 254 Best Solution: [183.82845450778757, 175.89236723107376,
175.05857058228966, 75.50804917725263] Fitness: 1.9397464052345055
Generation 255 Best Solution: [183.82845450778757, 176.23386050913018,
175.05857058228966, 75.81111915313957] Fitness: 2.4027778313645447
Generation 256 Best Solution: [182.9981146926198, 176.87454148709037,
174.40088211341222, 75.81111915313957] Fitness: 1.7750333439912955
Generation 257 Best Solution: [183.82845450778757, 176.66937356873956,
175.05857058228966, 75.81111915313957] Fitness: 2.5974414343273273
Generation 258 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.62880691564082] Fitness: 2.491770214001326
Generation 259 Best Solution: [183.82845450778757, 177.39576315057357,
173.89540487176964, 74.62880691564082] Fitness: 1.7314583192654274
Generation 260 Best Solution: [183.82845450778757, 177.39576315057357,
173.89540487176964, 74.62880691564082] Fitness: 1.7314583192654274
Generation 261 Best Solution: [183.54779837215844, 177.39576315057357,
175.05857058228966, 74.62880691564082] Fitness: 2.248679158513596
Generation 262 Best Solution: [183.82845450778757, 176.50073584915629,
175.05857058228966, 74.62880691564082] Fitness: 1.2738357864262737
Generation 263 Best Solution: [183.54779837215844, 177.39576315057357,
175.05857058228966, 74.62880691564082] Fitness: 2.248679158513596
Generation 264 Best Solution: [183.54779837215844, 176.94502507989336,
174.56104337847162, 75.81111915313957] Fitness: 2.0280189603948227
Generation 265 Best Solution: [183.54779837215844, 175.9967224908964,
175.05857058228966, 75.81111915313957] Fitness: 2.200442178867065
Generation 266 Best Solution: [183.43149510143152, 177.39576315057357,
175.05857058228966, 74.62880691564082] Fitness: 2.1904170355427284
Generation 267 Best Solution: [183.82845450778757, 175.9967224908964,
175.05857058228966, 75.81111915313957] Fitness: 2.4435332343547946
Generation 268 Best Solution: [183.82845450778757, 175.9967224908964,
175.05857058228966, 75.81111915313957] Fitness: 2.4435332343547946
Generation 269 Best Solution: [183.82845450778757, 175.9967224908964,
175.05857058228966, 75.81111915313957] Fitness: 2.4435332343547946
Generation 270 Best Solution: [183.82845450778757, 177.39576315057357,
173.84143661782804, 75.05409837962243] Fitness: 1.894523721349563
Generation 271 Best Solution: [183.82845450778757, 177.39576315057357,
173.81914932937568, 74.98719138628397] Fitness: 1.8627192472888467
Generation 272 Best Solution: [182.97070260222685, 177.39576315057357,
173.84143661782804, 75.81111915313957] Fitness: 2.6758976630588385
Generation 273 Best Solution: [183.0891143107275, 177.39576315057357,
175.12499716434294, 75.36538567846735] Fitness: 2.7317431699612906

Generation 274 Best Solution: [183.82845450778757, 176.5714279979557,
175.05857058228966, 75.81374856501269] Fitness: 2.528794365709783
Generation 275 Best Solution: [183.82845450778757, 176.5140481828792,
175.05857058228966, 75.81111915313957] Fitness: 2.4879761775084424
Generation 276 Best Solution: [183.82845450778757, 176.5140481828792,
175.05857058228966, 75.38552149726283] Fitness: 1.7572635622093289
Generation 277 Best Solution: [183.82845450778757, 176.5140481828792,
175.05857058228966, 75.8330694869967] Fitness: 2.5347008335735284
Generation 278 Best Solution: [183.82845450778757, 176.20342376435113,
175.05857058228966, 75.8330694869967] Fitness: 2.4489454483831703
Generation 279 Best Solution: [183.82845450778757, 176.20342376435113,
175.05857058228966, 75.81111915313957] Fitness: 2.4022207923180847
Generation 280 Best Solution: [183.82845450778757, 176.20342376435113,
174.1306594889941, 75.8330694869967] Fitness: 1.6415777397064932
Generation 281 Best Solution: [183.06067649079634, 176.5140481828792,
175.05857058228966, 75.81111915313957] Fitness: 2.167044568161795
Generation 282 Best Solution: [183.82845450778757, 176.5140481828792,
175.12499716434294, 74.40001374139356] Fitness: 1.4632025677488552
Generation 283 Best Solution: [183.82845450778757, 176.5140481828792,
175.12499716434294, 74.40001374139356] Fitness: 1.4632025677488552
Generation 284 Best Solution: [183.82845450778757, 176.5140481828792,
175.12499716434294, 74.40001374139356] Fitness: 1.4632025677488552
Generation 285 Best Solution: [183.82845450778757, 176.5140481828792,
175.05857058228966, 75.8330694869967] Fitness: 2.5347008335735284
Generation 286 Best Solution: [183.82845450778757, 176.5140481828792,
175.05857058228966, 75.8330694869967] Fitness: 2.5347008335735284
Generation 287 Best Solution: [183.82845450778757, 176.5140481828792,
175.05857058228966, 74.90064782708203] Fitness: 1.3309258884367599
Generation 288 Best Solution: [183.82845450778757, 176.5140481828792,
175.05857058228966, 75.81111915313957] Fitness: 2.4879761775084424
Generation 289 Best Solution: [183.45743840408613, 177.39576315057357,
175.12499716434294, 74.40001374139356] Fitness: 2.383306941052802
Generation 290 Best Solution: [183.82845450778757, 176.2925542178293,
175.05857058228966, 74.90064782708203] Fitness: 1.2516146121803582
Generation 291 Best Solution: [183.82845450778757, 177.39576315057357,
174.29012686161275, 75.20264601211888] Fitness: 1.9749188678758918
Generation 292 Best Solution: [183.82845450778757, 175.58298368096413,
174.29012686161275, 75.20264601211888] Fitness: 1.0387141950216292
Generation 293 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 75.20264601211888] Fitness: 2.7562737097552876
Generation 294 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 75.20264601211888] Fitness: 2.7562737097552876
Generation 295 Best Solution: [183.82845450778757, 175.58298368096413,
175.05857058228966, 75.81111915313957] Fitness: 2.7623894561829774
Generation 296 Best Solution: [183.82845450778757, 177.39576315057357,
174.9332506501824, 74.67407920537563] Fitness: 2.2890487535897535
Generation 297 Best Solution: [183.82845450778757, 176.36143889692866,
175.12499716434294, 75.81111915313957] Fitness: 2.5422229141971253

Generation 298 Best Solution: [183.11770052651258, 177.39576315057357,
174.0015502975909, 75.81111915313957] Fitness: 2.570059752064984
Generation 299 Best Solution: [183.82845450778757, 176.31739680019555,
174.0015502975909, 75.81111915313957] Fitness: 1.6189146988030751
Generation 300 Best Solution: [183.82845450778757, 177.39576315057357,
174.0015502975909, 74.98938064013804] Fitness: 1.7930746274725728
Generation 301 Best Solution: [183.82845450778757, 177.39576315057357,
174.0015502975909, 75.81111915313957] Fitness: 2.904442940484625
Generation 302 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.92926346091043] Fitness: 2.554693249806796
Generation 303 Best Solution: [183.82845450778757, 176.39023195389262,
174.0015502975909, 75.81111915313957] Fitness: 1.6383630289227205
Generation 304 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.92926346091043] Fitness: 2.554693249806796
Generation 305 Best Solution: [183.11770052651258, 177.39576315057357,
174.60845097106852, 75.81111915313957] Fitness: 2.7747073675158767
Generation 306 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.81964674950981] Fitness: 2.5124911016240192
Generation 307 Best Solution: [183.11770052651258, 176.04682862087984,
175.05857058228966, 75.81111915313957] Fitness: 2.091916895475687
Generation 308 Best Solution: [183.82845450778757, 176.1092846142528,
175.05857058228966, 75.81111915313957] Fitness: 2.4112871730446748
Generation 309 Best Solution: [182.93801954871535, 176.1092846142528,
175.05857058228966, 75.81111915313957] Fitness: 2.1395654851520276
Generation 310 Best Solution: [183.82845450778757, 176.52254065449782,
175.05857058228966, 75.81111915313957] Fitness: 2.492814013585961
Generation 311 Best Solution: [183.82845450778757, 176.52254065449782,
175.05857058228966, 75.81111915313957] Fitness: 2.492814013585961
Generation 312 Best Solution: [183.82845450778757, 176.52254065449782,
175.05857058228966, 75.81111915313957] Fitness: 2.492814013585961
Generation 313 Best Solution: [183.84182061732355, 176.3635532111639,
175.05857058228966, 75.81111915313957] Fitness: 2.439450247472062
Generation 314 Best Solution: [183.84182061732355, 176.3635532111639,
174.9859261069221, 75.81111915313957] Fitness: 2.3190829317261836
Generation 315 Best Solution: [183.82845450778757, 177.39576315057357,
173.89936075060763, 75.5197483709856] Fitness: 2.400410945358156
Generation 316 Best Solution: [183.82845450778757, 176.52254065449782,
175.05857058228966, 75.81111915313957] Fitness: 2.492814013585961
Generation 317 Best Solution: [183.40327443311403, 177.39576315057357,
175.05857058228966, 74.84650849073846] Fitness: 2.209049649842553
Generation 318 Best Solution: [183.14114715380393, 177.39576315057357,
175.05857058228966, 74.84650849073846] Fitness: 2.182609432140216
Generation 319 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.84650849073846] Fitness: 2.520787694797633
Generation 320 Best Solution: [183.82845450778757, 176.1534418848948,
175.05857058228966, 75.83670493817331] Fitness: 2.4595528101522532
Generation 321 Best Solution: [183.82845450778757, 176.05754944091066,
175.05857058228966, 75.81111915313957] Fitness: 2.4232127859458625

Generation 322 Best Solution: [183.82845450778757, 176.05754944091066,
175.05857058228966, 75.81111915313957] Fitness: 2.4232127859458625
Generation 323 Best Solution: [183.59927241898419, 175.7937191475182,
175.05857058228966, 75.89035574145822] Fitness: 2.5241077126538043
Generation 324 Best Solution: [183.59927241898419, 177.39576315057357,
175.05857058228966, 74.44566207017807] Fitness: 2.3255388763428315
Generation 325 Best Solution: [183.59927241898419, 175.7937191475182,
175.05857058228966, 75.09703628760657] Fitness: 1.3147710082648367
Generation 326 Best Solution: [183.82845450778757, 175.7937191475182,
174.9760262990625, 75.81111915313957] Fitness: 2.4246054079175994
Generation 327 Best Solution: [183.82845450778757, 175.7937191475182,
175.05857058228966, 75.43888963199294] Fitness: 1.9032639395801603
Generation 328 Best Solution: [183.82845450778757, 175.7937191475182,
175.05857058228966, 75.43888963199294] Fitness: 1.9032639395801603
Generation 329 Best Solution: [183.82845450778757, 175.76274603968406,
175.05857058228966, 75.41741923913204] Fitness: 1.8976560256008903
Generation 330 Best Solution: [183.82845450778757, 175.76274603968406,
175.05857058228966, 75.41741923913204] Fitness: 1.8976560256008903
Generation 331 Best Solution: [183.82845450778757, 176.3651946587676,
174.71686550192132, 75.89035574145822] Fitness: 2.1159715363278964
Generation 332 Best Solution: [182.9485425932955, 175.97103072318777,
175.05857058228966, 75.89035574145822] Fitness: 2.3499752304043136
Generation 333 Best Solution: [183.82845450778757, 176.541806808561,
175.05857058228966, 75.89035574145822] Fitness: 2.67712450095118
Generation 334 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.45096823498102] Fitness: 2.5327810339887615
Generation 335 Best Solution: [183.82845450778757, 177.39576315057357,
174.71686550192132, 75.17896857681154] Fitness: 2.251776057789448
Generation 336 Best Solution: [183.44093933690368, 177.39576315057357,
174.71686550192132, 75.89035574145822] Fitness: 3.092287304005172
Generation 337 Best Solution: [183.82845450778757, 177.39576315057357,
174.08735984987388, 75.89035574145822] Fitness: 3.06508387242625
Generation 338 Best Solution: [183.66374564856375, 176.89711494870983,
175.05857058228966, 75.89035574145822] Fitness: 2.8508131845001508
Generation 339 Best Solution: [183.82845450778757, 176.56322312168592,
175.05857058228966, 75.89035574145822] Fitness: 2.6906730503498992
Generation 340 Best Solution: [183.7176559345847, 177.39576315057357,
175.05857058228966, 74.44459785471624] Fitness: 2.4220468353472953
Generation 341 Best Solution: [183.81321700590172, 177.39576315057357,
174.05487799885492, 74.53490314970327] Fitness: 1.685926894705772
Generation 342 Best Solution: [183.81321700590172, 177.39576315057357,
174.05487799885492, 74.53490314970327] Fitness: 1.685926894705772
Generation 343 Best Solution: [183.82845450778757, 177.39576315057357,
174.08735984987388, 74.34390213856899] Fitness: 1.7791810258790544
Generation 344 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.34390213856899] Fitness: 2.5855345613914773
Generation 345 Best Solution: [183.82845450778757, 177.39576315057357,
174.71686550192132, 75.56570426656054] Fitness: 2.754957483136183

Generation 346 Best Solution: [183.02554226080417, 177.39576315057357,
174.18397388498204, 75.89035574145822] Fitness: 2.757894995346307
Generation 347 Best Solution: [183.82845450778757, 177.39576315057357,
174.71686550192132, 74.53817477209235] Fitness: 2.023799640556745
Generation 348 Best Solution: [183.82845450778757, 176.358043453309,
174.41660263414664, 75.89035574145822] Fitness: 1.8681068975957484
Generation 349 Best Solution: [183.81321700590172, 177.39576315057357,
175.05857058228966, 74.82425350744472] Fitness: 2.4969008622117976
Generation 350 Best Solution: [183.09758974774041, 177.39576315057357,
175.05857058228966, 75.25388379534121] Fitness: 2.4790342739227937
Generation 351 Best Solution: [183.44093933690368, 177.39576315057357,
174.18397388498204, 75.89035574145822] Fitness: 2.7700092493092123
Generation 352 Best Solution: [183.02554226080417, 177.39576315057357,
174.71686550192132, 75.89035574145822] Fitness: 3.080173050042266
Generation 353 Best Solution: [183.02554226080417, 177.39576315057357,
173.88822207869092, 75.89035574145822] Fitness: 2.8044994802253704
Generation 354 Best Solution: [183.81321700590172, 177.39576315057357,
174.38948651251158, 75.14685807377951] Fitness: 1.9455639542381806
Generation 355 Best Solution: [183.81321700590172, 177.39576315057357,
174.38948651251158, 75.89035574145822] Fitness: 3.1130312858405933
Generation 356 Best Solution: [183.82845450778757, 176.86657478699297,
175.05857058228966, 75.89035574145822] Fitness: 2.973219061390246
Generation 357 Best Solution: [183.82845450778757, 177.39576315057357,
173.76595594951291, 75.40538724006298] Fitness: 2.3035880641237707
Generation 358 Best Solution: [183.25677244720106, 177.39576315057357,
173.76595594951291, 75.40538724006298] Fitness: 1.9614884890729996
Generation 359 Best Solution: [183.25677244720106, 177.39576315057357,
174.09747525135478, 75.89035574145822] Fitness: 2.722438554199912
Generation 360 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.54184333705966] Fitness: 2.504554013762234
Generation 361 Best Solution: [183.5367841094414, 177.39576315057357,
175.05857058228966, 74.54184333705966] Fitness: 2.2548784570819396
Generation 362 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.54184333705966] Fitness: 2.504554013762234
Generation 363 Best Solution: [183.82845450778757, 177.39576315057357,
174.47464351377135, 74.54184333705966] Fitness: 1.8116912815883892
Generation 364 Best Solution: [183.5367841094414, 176.0754394584518,
175.05857058228966, 75.89035574145822] Fitness: 2.3416995812741863
Generation 365 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.63568338849015] Fitness: 2.491353025818776
Generation 366 Best Solution: [183.82845450778757, 176.22607459344363,
175.05857058228966, 75.89035574145822] Fitness: 2.575316367234082
Generation 367 Best Solution: [183.393363913358, 176.55469249149277,
175.05857058228966, 75.89035574145822] Fitness: 2.3701378805174453
Generation 368 Best Solution: [183.393363913358, 176.55469249149277,
174.90879464540586, 75.89035574145822] Fitness: 2.132596837822367
Generation 369 Best Solution: [183.393363913358, 176.55469249149277,
174.90879464540586, 75.89035574145822] Fitness: 2.132596837822367

Generation 370 Best Solution: [182.80093214615727, 177.39576315057357,
173.99099702850467, 75.89035574145822] Fitness: 2.8957593022239703
Generation 371 Best Solution: [182.9281265296825, 176.87892412852054,
175.05857058228966, 75.89035574145822] Fitness: 2.721762017070848
Generation 372 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.60445142297915] Fitness: 2.493947651756304
Generation 373 Best Solution: [183.3304377615681, 177.39576315057357,
175.05857058228966, 75.02841119892423] Fitness: 2.280138273484933
Generation 374 Best Solution: [183.3304377615681, 177.39576315057357,
175.05857058228966, 74.60445142297915] Fitness: 2.162178413616953
Generation 375 Best Solution: [183.3304377615681, 177.39576315057357,
175.05857058228966, 74.31079841114128] Fitness: 2.274345099362816
Generation 376 Best Solution: [183.82845450778757, 175.90210827861767,
174.72640656858204, 75.89035574145822] Fitness: 2.190446632301942
Generation 377 Best Solution: [183.82845450778757, 175.90210827861767,
174.72640656858204, 75.89035574145822] Fitness: 2.190446632301942
Generation 378 Best Solution: [183.48788864548646, 177.39576315057357,
174.72640656858204, 74.39957646147069] Fitness: 1.8081913497701687
Generation 379 Best Solution: [183.82845450778757, 176.2818871446383,
175.05857058228966, 75.89035574145822] Fitness: 2.5799669368460285
Generation 380 Best Solution: [182.81846862843244, 177.39576315057357,
175.05857058228966, 74.91201191891714] Fitness: 2.3494672625737043
Generation 381 Best Solution: [183.82845450778757, 177.08031338267796,
175.05857058228966, 74.84858194223783] Fitness: 1.924024467520663
Generation 382 Best Solution: [182.81846862843244, 177.39576315057357,
175.05857058228966, 74.84858194223783] Fitness: 2.324365080678764
Generation 383 Best Solution: [183.82845450778757, 176.15525311600499,
175.05857058228966, 74.84858194223783] Fitness: 1.2277120131403612
Generation 384 Best Solution: [183.23359178523663, 177.39576315057357,
175.05857058228966, 74.86718068605397] Fitness: 2.1847916673387022
Generation 385 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.86718068605397] Fitness: 2.5280765877143603
Generation 386 Best Solution: [183.82845450778757, 177.39576315057357,
173.8755674915524, 75.89035574145822] Fitness: 3.1197489300157493
Generation 387 Best Solution: [183.82845450778757, 176.24944699848976,
175.05857058228966, 75.7962984061731] Fitness: 2.3726761826552814
Generation 388 Best Solution: [183.13437407294674, 177.39576315057357,
175.05857058228966, 75.11110605972455] Fitness: 2.336276049680362
Generation 389 Best Solution: [183.82845450778757, 176.24944699848976,
175.05857058228966, 75.89035574145822] Fitness: 2.576566320290828
Generation 390 Best Solution: [183.23359178523663, 176.24944699848976,
175.05857058228966, 75.89035574145822] Fitness: 2.23328139991517
Generation 391 Best Solution: [183.23359178523663, 177.39576315057357,
174.2211290596705, 74.5511509534959] Fitness: 1.360883062503272
Generation 392 Best Solution: [183.13437407294674, 177.39576315057357,
175.05857058228966, 74.56719558885057] Fitness: 2.162204293280646
Generation 393 Best Solution: [183.13437407294674, 176.24944699848976,
175.05857058228966, 75.02297597276858] Fitness: 0.9762455001794138

Generation 394 Best Solution: [183.82845450778757, 176.24944699848976,
175.05857058228966, 75.02297597276858] Fitness: 1.3134313678368779
Generation 395 Best Solution: [183.82845450778757, 176.29695785033852,
175.05857058228966, 75.35935949547302] Fitness: 1.6446052412782453
Generation 396 Best Solution: [183.0977812327907, 176.40818525857443,
175.05857058228966, 75.89035574145822] Fitness: 2.2812865808295473
Generation 397 Best Solution: [183.23359178523663, 177.2891131727426,
175.05857058228966, 75.02297597276858] Fitness: 2.04253648626441
Generation 398 Best Solution: [183.0977812327907, 177.39576315057357,
174.1546005300055, 75.58522075637579] Fitness: 2.1324324758613984
Generation 399 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.87117128582975] Fitness: 2.529574195218892
Generation 400 Best Solution: [183.13437407294674, 177.39576315057357,
175.05857058228966, 74.87117128582975] Fitness: 2.192388327561428
Generation 401 Best Solution: [183.13437407294674, 177.39576315057357,
175.05857058228966, 75.13142551046215] Fitness: 2.353326690964029
Generation 402 Best Solution: [183.13437407294674, 176.43763080693623,
175.05857058228966, 75.89035574145822] Fitness: 2.2860710348640945
Generation 403 Best Solution: [183.13437407294674, 176.43763080693623,
175.05857058228966, 75.89035574145822] Fitness: 2.2860710348640945
Generation 404 Best Solution: [183.16267605450773, 176.76513531670977,
175.05857058228966, 75.89035574145822] Fitness: 2.51912147245933
Generation 405 Best Solution: [183.23359178523663, 176.76700715242478,
175.05857058228966, 75.89035574145822] Fitness: 2.5185283742741245
Generation 406 Best Solution: [183.23359178523663, 175.59790033853508,
175.05857058228966, 75.6746269253189] Fitness: 2.1314869633312323
Generation 407 Best Solution: [183.23359178523663, 175.59790033853508,
175.05857058228966, 75.89035574145822] Fitness: 2.574978823416846
Generation 408 Best Solution: [183.23359178523663, 177.39576315057357,
174.68532331613315, 75.89035574145822] Fitness: 3.0129285355330055
Generation 409 Best Solution: [183.39112198751326, 177.39576315057357,
175.05857058228966, 75.5211482393795] Fitness: 2.8488001391751245
Generation 410 Best Solution: [183.39112198751326, 175.600631291425,
175.05857058228966, 75.89035574145822] Fitness: 2.599442770051418
Generation 411 Best Solution: [183.00126015994815, 176.2539603242845,
175.05857058228966, 75.89035574145822] Fitness: 2.276372759483322
Generation 412 Best Solution: [183.00126015994815, 177.39576315057357,
175.05857058228966, 74.81153739416149] Fitness: 2.2096966072024995
Generation 413 Best Solution: [183.82845450778757, 177.39576315057357,
174.1248713137861, 75.89035574145822] Fitness: 3.064005518015658
Generation 414 Best Solution: [183.39112198751326, 177.39576315057357,
175.05857058228966, 74.73266686429685] Fitness: 2.17897740871781
Generation 415 Best Solution: [183.82845450778757, 176.17439423926817,
175.05857058228966, 75.89035574145822] Fitness: 2.5761209618155427
Generation 416 Best Solution: [183.81610862624743, 177.39576315057357,
174.01483306636146, 75.89035574145822] Fitness: 3.060770143810735
Generation 417 Best Solution: [182.9355937892086, 175.6571596246535,
175.05857058228966, 75.89035574145822] Fitness: 2.584425805188494

Generation 418 Best Solution: [182.9355937892086, 175.6571596246535,
175.05857058228966, 75.89035574145822] Fitness: 2.584425805188494
Generation 419 Best Solution: [183.7686009569562, 177.39576315057357,
175.05857058228966, 74.42702637886602] Fitness: 2.478745845944355
Generation 420 Best Solution: [183.7686009569562, 177.39576315057357,
175.05857058228966, 74.42702637886602] Fitness: 2.478745845944355
Generation 421 Best Solution: [183.45476004677553, 177.39576315057357,
175.05857058228966, 75.15934601297828] Fitness: 2.4234906230400877
Generation 422 Best Solution: [183.15579103062163, 177.09359968137846,
174.01483306636146, 75.84898764304944] Fitness: 2.066807745897746
Generation 423 Best Solution: [183.82845450778757, 177.39576315057357,
175.05857058228966, 74.46963880314503] Fitness: 2.5257414544970045
Generation 424 Best Solution: [183.23359178523663, 177.39576315057357,
175.05857058228966, 75.01430989092151] Fitness: 2.259381956132512
Generation 425 Best Solution: [183.23359178523663, 177.39576315057357,
175.1198099558817, 74.46963880314503] Fitness: 2.291469361271201
Generation 426 Best Solution: [183.23359178523663, 177.39576315057357,
175.05857058228966, 75.01430989092151] Fitness: 2.259381956132512
Generation 427 Best Solution: [183.82845450778757, 177.306680457143,
174.89827706329072, 74.77903470990394] Fitness: 2.062524008447015
Generation 428 Best Solution: [183.23359178523663, 176.3706795244832,
175.05857058228966, 75.89035574145822] Fitness: 2.2558932543639845
Generation 429 Best Solution: [183.23359178523663, 175.69148854843985,
175.05857058228966, 75.89035574145822] Fitness: 2.477856514567482
Generation 430 Best Solution: [183.23359178523663, 177.306680457143,
175.05857058228966, 75.89035574145822] Fitness: 3.3408785628542774
Generation 431 Best Solution: [183.23359178523663, 176.46810002705917,
175.05857058228966, 75.89035574145822] Fitness: 2.2936609607995795
Generation 432 Best Solution: [183.82845450778757, 176.74161783144115,
175.05857058228966, 75.89035574145822] Fitness: 2.8363240670649397
Generation 433 Best Solution: [183.23359178523663, 177.36436034265415,
175.05857058228966, 74.38756279525421] Fitness: 2.150507438018493
Generation 434 Best Solution: [183.7686009569562, 177.306680457143,
174.29704689314102, 74.84123588691389] Fitness: 1.4886129405167532
Generation 435 Best Solution: [183.7686009569562, 177.306680457143,
174.29704689314102, 75.89035574145822] Fitness: 2.840995889440183
Generation 436 Best Solution: [183.7686009569562, 176.0568363241773,
174.29704689314102, 75.89035574145822] Fitness: 1.7530872631122603
Generation 437 Best Solution: [183.50504162423917, 177.306680457143,
174.29704689314102, 75.89035574145822] Fitness: 2.6375936648127407
Generation 438 Best Solution: [183.05120350141218, 177.306680457143,
174.29704689314102, 75.89035574145822] Fitness: 2.5868792897788433
Generation 439 Best Solution: [183.00007942840233, 175.6292029608117,
175.05857058228966, 75.89035574145822] Fitness: 2.5839054546398206
Generation 440 Best Solution: [183.00007942840233, 175.6292029608117,
175.05857058228966, 75.89035574145822] Fitness: 2.5839054546398206
Generation 441 Best Solution: [183.7686009569562, 175.6292029608117,
175.05857058228966, 75.89035574145822] Fitness: 2.8199844739653788

Generation 442 Best Solution: [183.23359178523663, 176.3069592730464,
175.05857058228966, 75.89035574145822] Fitness: 2.240636861518755
Generation 443 Best Solution: [183.50504162423917, 177.36436034265415,
175.05857058228966, 74.89817428669672] Fitness: 2.2053919620656988
Generation 444 Best Solution: [183.50013347934572, 176.97300521028114,
175.05857058228966, 75.4524192215429] Fitness: 2.0317203063699836
Generation 445 Best Solution: [183.58031110948372, 177.36436034265415,
175.05857058228966, 74.78815082422946] Fitness: 2.2144188109321874
Generation 446 Best Solution: [183.66532786383567, 176.61336476062652,
175.05857058228966, 74.78815082422946] Fitness: 1.1997807908608529
Generation 447 Best Solution: [183.7686009569562, 177.36436034265415,
175.05857058228966, 74.37257957851588] Fitness: 2.4376527231068823
Generation 448 Best Solution: [183.58031110948372, 176.6274218580285,
175.05857058228966, 75.89035574145822] Fitness: 2.513991149127106
Generation 449 Best Solution: [183.50504162423917, 177.36436034265415,
175.05857058228966, 74.68226960662571] Fitness: 2.155732075967291
Generation 450 Best Solution: [183.23359178523663, 176.6274218580285,
175.05857058228966, 75.89035574145822] Fitness: 2.3930586860985654
Generation 451 Best Solution: [183.23359178523663, 176.6274218580285,
175.05857058228966, 75.89035574145822] Fitness: 2.3930586860985654
Generation 452 Best Solution: [183.50504162423917, 176.21857034858454,
175.05857058228966, 75.89035574145822] Fitness: 2.307725121201968
Generation 453 Best Solution: [183.23359178523663, 176.89285817821852,
175.05857058228966, 75.89035574145822] Fitness: 2.662385674546974
Generation 454 Best Solution: [183.23359178523663, 176.89285817821852,
175.05857058228966, 75.89035574145822] Fitness: 2.662385674546974
Generation 455 Best Solution: [183.50504162423917, 177.36436034265415,
175.05857058228966, 74.77043300386688] Fitness: 2.1656492592804724
Generation 456 Best Solution: [183.50504162423917, 177.36436034265415,
174.12249557785395, 75.89035574145822] Fitness: 2.728911029805986
Generation 457 Best Solution: [183.50504162423917, 176.78373373189456,
175.05857058228966, 75.89035574145822] Fitness: 2.6118507020216017
Generation 458 Best Solution: [183.50504162423917, 176.6893033651891,
175.05857058228966, 75.89035574145822] Fitness: 2.5201404842480053
Generation 459 Best Solution: [183.50504162423917, 177.36436034265415,
175.05857058228966, 75.05073920366759] Fitness: 2.292201462546929
Generation 460 Best Solution: [183.50504162423917, 176.28489331533348,
175.05857058228966, 75.89035574145822] Fitness: 2.3129770095044018
Generation 461 Best Solution: [183.50504162423917, 177.36436034265415,
175.05857058228966, 74.53433586966705] Fitness: 2.171224135053321
Generation 462 Best Solution: [183.50504162423917, 177.36436034265415,
175.05857058228966, 74.94058011164344] Fitness: 2.225223217696059
Generation 463 Best Solution: [183.50504162423917, 177.36436034265415,
175.05857058228966, 74.94058011164344] Fitness: 2.225223217696059
Generation 464 Best Solution: [183.50504162423917, 177.36436034265415,
175.05857058228966, 74.94058011164344] Fitness: 2.225223217696059
Generation 465 Best Solution: [183.50504162423917, 177.36436034265415,
175.05857058228966, 74.94058011164344] Fitness: 2.225223217696059

Generation 466 Best Solution: [182.93360282225865, 176.28489331533348,
175.05857058228966, 75.11659101455703] Fitness: 1.1174989869451029
Generation 467 Best Solution: [183.50504162423917, 176.28489331533348,
175.05857058228966, 75.61765221162347] Fitness: 1.766651670647867
Generation 468 Best Solution: [183.50504162423917, 176.28489331533348,
175.05857058228966, 75.89035574145822] Fitness: 2.3129770095044018
Generation 469 Best Solution: [183.50504162423917, 177.36436034265415,
175.05857058228966, 74.79373588432594] Fitness: 2.1706602029683735
Generation 470 Best Solution: [183.26081872770334, 176.28489331533348,
175.05857058228966, 75.89035574145822] Fitness: 2.2385887853048496
Generation 471 Best Solution: [183.50504162423917, 177.36436034265415,
175.05857058228966, 74.28797699967808] Fitness: 2.2863900481921045
Generation 472 Best Solution: [183.50504162423917, 176.29832448949816,
175.05857058228966, 75.40187652579137] Fitness: 1.433392561677758
Generation 473 Best Solution: [183.33769699430843, 177.36436034265415,
174.45249889949176, 75.42862977680987] Fitness: 1.9220403923732692
Generation 474 Best Solution: [183.26081872770334, 176.02575270094425,
173.82963587507578, 75.89035574145822] Fitness: 1.5350953128740619
Generation 475 Best Solution: [183.50504162423917, 176.88540161481356,
175.05857058228966, 75.42862977680987] Fitness: 1.8842196292848203
Generation 476 Best Solution: [183.50504162423917, 176.29832448949816,
175.05857058228966, 75.89035574145822] Fitness: 2.31502607454479
Generation 477 Best Solution: [183.50504162423917, 177.36436034265415,
174.74530729074067, 75.42862977680987] Fitness: 2.241928507733223
Generation 478 Best Solution: [183.18525801386792, 177.36436034265415,
175.05857058228966, 75.42862977680987] Fitness: 2.616465807357087
Generation 479 Best Solution: [183.18525801386792, 177.36436034265415,
173.74710000683748, 74.8066823504167] Fitness: 1.4203877619710465
Generation 480 Best Solution: [183.18525801386792, 177.36436034265415,
174.47916322592997, 75.89035574145822] Fitness: 2.7712689831652018
Generation 481 Best Solution: [183.50504162423917, 177.36436034265415,
173.93902300530257, 75.89035574145822] Fitness: 2.759624871162967
Generation 482 Best Solution: [183.18525801386792, 177.36436034265415,
173.7522653265891, 75.53965352996863] Fitness: 2.101907795135426
Generation 483 Best Solution: [183.18525801386792, 177.36436034265415,
175.05857058228966, 74.71808182147726] Fitness: 2.0828622907806125
Generation 484 Best Solution: [183.18525801386792, 177.36436034265415,
175.05857058228966, 74.5297350017734] Fitness: 2.097178254284479
Generation 485 Best Solution: [183.18525801386792, 177.36436034265415,
175.05857058228966, 74.5297350017734] Fitness: 2.097178254284479
Generation 486 Best Solution: [183.18525801386792, 175.94393430412907,
175.05857058228966, 75.89035574145822] Fitness: 2.2969527525257347
Generation 487 Best Solution: [183.18525801386792, 175.94393430412907,
175.05857058228966, 75.89035574145822] Fitness: 2.2969527525257347
Generation 488 Best Solution: [183.18525801386792, 175.94393430412907,
175.05857058228966, 75.89035574145822] Fitness: 2.2969527525257347
Generation 489 Best Solution: [183.18525801386792, 177.36436034265415,
175.05857058228966, 74.67040359397669] Fitness: 2.0803159596036123

```
Generation 490 Best Solution: [183.18525801386792, 177.36436034265415,
175.05857058228966, 74.50029460862397] Fitness: 2.1053147769485796
Generation 491 Best Solution: [183.46529033651493, 177.36436034265415,
175.05857058228966, 75.24909628531312] Fitness: 2.449520283729637
Generation 492 Best Solution: [183.00568276791364, 177.36436034265415,
175.05857058228966, 74.58493841263109] Fitness: 2.1265052714634556
Generation 493 Best Solution: [183.873301817507, 177.36436034265415,
175.05857058228966, 75.11695282191044] Fitness: 2.662882090647431
Generation 494 Best Solution: [183.873301817507, 177.36436034265415,
175.05857058228966, 74.88858922529714] Fitness: 2.521045761447342
Generation 495 Best Solution: [183.18525801386792, 177.36436034265415,
174.79027762841258, 75.89035574145822] Fitness: 3.0649253832640344
Generation 496 Best Solution: [183.18525801386792, 177.36436034265415,
174.79027762841258, 75.89035574145822] Fitness: 3.0649253832640344
Generation 497 Best Solution: [183.873301817507, 175.72834335886293,
174.79027762841258, 75.89035574145822] Fitness: 2.4433398049035038
Generation 498 Best Solution: [183.18525801386792, 177.36436034265415,
175.05857058228966, 74.55066890980919] Fitness: 2.092362860885124
Generation 499 Best Solution: [183.18525801386792, 177.36436034265415,
174.3067718384492, 75.69524259337621] Fitness: 2.2804241973348174
Generation 500 Best Solution: [183.8563357170477, 177.36436034265415,
175.05857058228966, 74.89570239579415] Fitness: 2.5038047059826596
Optimal Solution for Simulation 17: [183.8563357170477, 177.36436034265415,
175.05857058228966, 74.89570239579415]
--------------------------------------------------
Simulation 18/32 with conditions: {'drought': False, 'flood': True,
'policy_change': True, 'social_event': True, 'economic_crisis': False}
Generation 1 Best Solution: [183.53022611959634, 175.95956378465746,
174.15341518116117, 74.56606626782104] Fitness: 0.21263225869534982
Generation 2 Best Solution: [183.1712547306861, 175.83979906179064,
174.68209235583228, 74.38733481845037] Fitness: 0.5339915334653783
Generation 3 Best Solution: [183.75636408862613, 175.94538465208473,
174.43264752055725, 75.3738993692614] Fitness: 0.9000760833954483
Generation 4 Best Solution: [183.5719728783775, 175.94538465208473,
174.15341518116117, 74.87941133820505] Fitness: 0.26537271679870555
Generation 5 Best Solution: [183.53022611959634, 175.95956378465746,
174.15341518116117, 75.23502508751056] Fitness: 0.467485562865812
Generation 6 Best Solution: [183.12854116959082, 176.29474824058624,
174.15341518116117, 75.82977444057316] Fitness: 1.2300769317588298
Generation 7 Best Solution: [183.12854116959082, 176.29474824058624,
174.15341518116117, 75.82977444057316] Fitness: 1.2300769317588298
Generation 8 Best Solution: [183.09582927800602, 176.29474824058624,
174.15341518116117, 75.82977444057316] Fitness: 1.2348202884099302
Generation 9 Best Solution: [183.82244602101943, 175.70171103878891,
173.87548541752037, 75.15744087754858] Fitness: 0.8556042586071106
Generation 10 Best Solution: [183.82244602101943, 176.29474824058624,
173.87548541752037, 75.15744087754858] Fitness: 0.657376581521592
Generation 11 Best Solution: [183.09582927800602, 176.60769647930678,
```

173.87548541752037, 75.68578888738575] Fitness: 1.1526940854960581
Generation 12 Best Solution: [183.53022611959634, 177.16977733633337,
175.0006854627809, 74.83704333257602] Fitness: 1.7452031913422459
Generation 13 Best Solution: [183.09582927800602, 175.83979906179064,
174.0624472553843, 75.82977444057316] Fitness: 1.3305741467950802
Generation 14 Best Solution: [183.09582927800602, 175.83979906179064,
174.0624472553843, 75.84603227020634] Fitness: 1.3638601628811282
Generation 15 Best Solution: [183.09582927800602, 175.83979906179064,
174.0624472553843, 75.82977444057316] Fitness: 1.3305741467950802
Generation 16 Best Solution: [183.09582927800602, 175.83979906179064,
174.0624472553843, 75.84603227020634] Fitness: 1.3638601628811282
Generation 17 Best Solution: [183.09582927800602, 177.16469112533738,
174.9326388822895, 74.39090891889896] Fitness: 1.59730514640946
Generation 18 Best Solution: [183.1969381465504, 177.16469112533738,
175.0006854627809, 74.55451067099446] Fitness: 1.6309201880655695
Generation 19 Best Solution: [183.1969381465504, 177.16469112533738,
175.0006854627809, 74.55451067099446] Fitness: 1.6309201880655695
Generation 20 Best Solution: [183.09582927800602, 177.16469112533738,
175.0006854627809, 74.55451067099446] Fitness: 1.6394342356365863
Generation 21 Best Solution: [183.1969381465504, 177.16977733633337,
174.69281461995422, 74.43174719899885] Fitness: 1.275959289993591
Generation 22 Best Solution: [183.57794278578547, 177.16469112533738,
174.69281461995422, 74.43174719899885] Fitness: 1.3982836871559485
Generation 23 Best Solution: [183.82244602101943, 177.16977733633337,
174.16980209725827, 75.84603227020634] Fitness: 2.4619548860042006
Generation 24 Best Solution: [183.82244602101943, 177.16977733633337,
174.16980209725827, 75.31161726539739] Fitness: 1.6139453769781014
Generation 25 Best Solution: [183.05434788274798, 177.16977733633337,
175.0006854627809, 74.76334785704891] Fitness: 1.6469118328070345
Generation 26 Best Solution: [183.1969381465504, 177.16977733633337,
173.76572572077228, 75.82977444057316] Fitness: 2.1759214940685294
Generation 27 Best Solution: [183.1969381465504, 177.16977733633337,
175.0006854627809, 74.78294223099408] Fitness: 1.632674074546732
Generation 28 Best Solution: [183.1969381465504, 175.7393778242137,
175.08165226400573, 75.84603227020634] Fitness: 2.272417579860347
Generation 29 Best Solution: [183.82244602101943, 177.16469112533738,
174.13147692035173, 75.82977444057316] Fitness: 2.416689074308125
Generation 30 Best Solution: [183.82244602101943, 177.16469112533738,
175.08165226400573, 74.37120093267335] Fitness: 2.18980586196272
Generation 31 Best Solution: [183.82244602101943, 176.13762113642204,
174.13147692035173, 75.84603227020634] Fitness: 1.6011019620576386
Generation 32 Best Solution: [183.1969381465504, 177.16469112533738,
174.13147692035173, 75.82977444057316] Fitness: 2.0653570542483517
Generation 33 Best Solution: [183.1969381465504, 177.16469112533738,
174.13147692035173, 75.82977444057316] Fitness: 2.0653570542483517
Generation 34 Best Solution: [183.49209468733423, 177.16469112533738,
175.08165226400573, 75.00799342933969] Fitness: 1.9213736709536244
Generation 35 Best Solution: [183.1969381465504, 177.16469112533738,

175.08165226400573, 75.00799342933969] Fitness: 1.8422631036806145
Generation 36 Best Solution: [183.82244602101943, 176.14205616369014,
174.50877520316533, 75.84603227020634] Fitness: 1.7450323764352285
Generation 37 Best Solution: [183.1969381465504, 177.16469112533738,
175.0006854627809, 74.72405957871729] Fitness: 1.6167640222308588
Generation 38 Best Solution: [183.1969381465504, 177.16469112533738,
175.0006854627809, 74.72405957871729] Fitness: 1.6167640222308588
Generation 39 Best Solution: [183.87874659334312, 177.16469112533738,
175.08165226400573, 74.72405957871729] Fitness: 2.1685927912653047
Generation 40 Best Solution: [183.1969381465504, 177.16469112533738,
175.0006854627809, 74.72405957871729] Fitness: 1.6167640222308588
Generation 41 Best Solution: [183.1969381465504, 175.8581706346806,
175.08165226400573, 75.84603227020634] Fitness: 2.1905876495993772
Generation 42 Best Solution: [183.1969381465504, 177.28721835438873,
175.06070961513382, 75.21913787778358] Fitness: 2.193166358126018
Generation 43 Best Solution: [183.1969381465504, 175.61183674778962,
174.90997768239768, 75.84603227020634] Fitness: 2.1163244840325746
Generation 44 Best Solution: [183.1969381465504, 177.2600343264121,
174.01060283314393, 75.84603227020634] Fitness: 2.2798657740641177
Generation 45 Best Solution: [183.82244602101943, 177.28721835438873,
175.08165226400573, 74.85883765682864] Fitness: 2.3545301873672937
Generation 46 Best Solution: [183.82244602101943, 177.16469112533738,
175.08165226400573, 74.64263981839278] Fitness: 2.103268913621762
Generation 47 Best Solution: [183.44802171643252, 177.28721835438873,
175.06070961513382, 74.4787732716886] Fitness: 2.036473952793774
Generation 48 Best Solution: [182.98591111703792, 177.2600343264121,
175.06070961513382, 74.4787732716886] Fitness: 1.964851159705987
Generation 49 Best Solution: [183.82244602101943, 177.28721835438873,
175.06070961513382, 74.4787732716886] Fitness: 2.3303446915209993
Generation 50 Best Solution: [182.98591111703792, 177.2600343264121,
175.06070961513382, 74.4787732716886] Fitness: 1.964851159705987
Generation 51 Best Solution: [183.82244602101943, 177.2600343264121,
175.06070961513382, 74.4787732716886] Fitness: 2.2777948802168733
Generation 52 Best Solution: [183.49209468733423, 177.16469112533738,
175.06070961513382, 75.55536151902629] Fitness: 2.46481120148931
Generation 53 Best Solution: [183.49209468733423, 176.0846271201916,
175.03447428664393, 75.84603227020634] Fitness: 2.1032661675362903
Generation 54 Best Solution: [183.7696487406692, 176.10795389994004,
175.08165226400573, 75.62354115557388] Fitness: 1.9804216056831132
Generation 55 Best Solution: [183.0501422612471, 177.28721835438873,
174.42970830235294, 75.62354115557388] Fitness: 2.021800003936967
Generation 56 Best Solution: [183.0501422612471, 177.28721835438873,
174.42970830235294, 74.78160841785481] Fitness: 1.2489809084158612
Generation 57 Best Solution: [183.82919829581948, 177.23426168083097,
175.08165226400573, 74.78160841785481] Fitness: 2.2425655155211777
Generation 58 Best Solution: [183.0501422612471, 176.11876593643808,
175.03447428664393, 75.84603227020634] Fitness: 2.037346629139347
Generation 59 Best Solution: [183.0501422612471, 176.665394393035,

174.42970830235294, 75.82977444057316] Fitness: 1.5298724198326206
Generation 60 Best Solution: [183.0501422612471, 176.5529948155668,
175.08165226400573, 75.82977444057316] Fitness: 2.1995049296449087
Generation 61 Best Solution: [183.0501422612471, 176.5529948155668,
175.06070961513382, 75.84603227020634] Fitness: 2.1968187001058177
Generation 62 Best Solution: [183.0501422612471, 176.5529948155668,
175.06070961513382, 75.84603227020634] Fitness: 2.1968187001058177
Generation 63 Best Solution: [183.82919829581948, 176.3935945876619,
174.9791757152118, 74.91423166764409] Fitness: 1.1740137324803672
Generation 64 Best Solution: [183.0501422612471, 176.3935945876619,
174.9791757152118, 74.91423166764409] Fitness: 0.833416973342441
Generation 65 Best Solution: [183.82244602101943, 177.16469112533738,
175.08165226400573, 75.05954397219381] Fitness: 2.225443617387469
Generation 66 Best Solution: [182.8233943188883, 176.71246382141547,
175.06070961513382, 75.84603227020634] Fitness: 2.4269870325551888
Generation 67 Best Solution: [182.8233943188883, 176.71246382141547,
174.5314296979733, 75.84603227020634] Fitness: 1.7767305159812927
Generation 68 Best Solution: [183.2228601245583, 177.43429663697378,
175.08165226400573, 74.80608439963096] Fitness: 2.29733159825404
Generation 69 Best Solution: [182.8233943188883, 177.01070045188348,
175.06070961513382, 75.84603227020634] Fitness: 2.7850470874549247
Generation 70 Best Solution: [182.8233943188883, 177.43429663697378,
174.3743392236601, 75.61461940615288] Fitness: 2.3884599313562385
Generation 71 Best Solution: [182.8233943188883, 177.43429663697378,
174.3743392236601, 75.61461940615288] Fitness: 2.3884599313562385
Generation 72 Best Solution: [182.8233943188883, 177.43429663697378,
174.3743392236601, 74.45209977683415] Fitness: 1.6710879960033074
Generation 73 Best Solution: [183.82919829581948, 176.08565724966078,
175.08165226400573, 75.84603227020634] Fitness: 2.4628251641361625
Generation 74 Best Solution: [182.8233943188883, 177.43429663697378,
174.3743392236601, 75.84603227020634] Fitness: 2.817993406915624
Generation 75 Best Solution: [182.8233943188883, 177.05199596313037,
174.65341520103254, 75.84603227020634] Fitness: 2.3025367668593595
Generation 76 Best Solution: [182.8233943188883, 177.05199596313037,
174.65341520103254, 75.84603227020634] Fitness: 2.3025367668593595
Generation 77 Best Solution: [183.5819684285816, 176.8321201948083,
175.08165226400573, 75.61461940615288] Fitness: 2.1701347274669116
Generation 78 Best Solution: [182.8233943188883, 176.8321201948083,
174.87460425898652, 75.74338610080378] Fitness: 2.063963913463337
Generation 79 Best Solution: [182.8233943188883, 176.33373469455432,
175.06070961513382, 75.74338610080378] Fitness: 1.9979137619923062
Generation 80 Best Solution: [183.82244602101943, 175.7755339262418,
174.87460425898652, 75.74338610080378] Fitness: 2.0725281276994183
Generation 81 Best Solution: [182.8233943188883, 175.7755339262418,
174.87460425898652, 75.74338610080378] Fitness: 1.8431100331981227
Generation 82 Best Solution: [182.8233943188883, 175.7755339262418,
174.87460425898652, 75.84603227020634] Fitness: 2.0453839697868825
Generation 83 Best Solution: [182.85248293563106, 176.32726979228434,

173.8307387354017, 75.84603227020634] Fitness: 1.4350971191848416
Generation 84 Best Solution: [182.85248293563106, 176.32726979228434,
173.8307387354017, 75.61461940615288] Fitness: 1.005563643625456
Generation 85 Best Solution: [182.85248293563106, 176.32726979228434,
173.8307387354017, 75.30637248641652] Fitness: 0.5812811048510105
Generation 86 Best Solution: [182.85248293563106, 177.43429663697378,
173.8307387354017, 74.95464036446012] Fitness: 1.6703613855810322
Generation 87 Best Solution: [182.8233943188883, 176.2862455270194,
175.08165226400573, 75.00598382613282] Fitness: 1.1209441002814626
Generation 88 Best Solution: [182.8233943188883, 176.2862455270194,
175.08165226400573, 75.84603227020634] Fitness: 2.225713162247186
Generation 89 Best Solution: [182.86411426485927, 177.43429663697378,
175.08165226400573, 75.26794126730766] Fitness: 2.6811687232825308
Generation 90 Best Solution: [182.8233943188883, 177.43429663697378,
174.4278502176921, 75.61461940615288] Fitness: 2.4166276163891496
Generation 91 Best Solution: [182.8233943188883, 177.43429663697378,
175.08165226400573, 75.09284691130055] Fitness: 2.5526886614755107
Generation 92 Best Solution: [183.82244602101943, 177.43429663697378,
174.52603961459147, 75.09284691130055] Fitness: 2.0918176113259817
Generation 93 Best Solution: [182.8233943188883, 177.43429663697378,
174.4278502176921, 75.61461940615288] Fitness: 2.4166276163891496
Generation 94 Best Solution: [183.82244602101943, 177.43429663697378,
174.6206373898736, 74.82132278702207] Fitness: 2.0398453182549376
Generation 95 Best Solution: [182.85248293563106, 177.43429663697378,
174.6206373898736, 74.82132278702207] Fitness: 1.792027115923666
Generation 96 Best Solution: [182.85248293563106, 177.43429663697378,
175.08165226400573, 74.49931635334404] Fitness: 2.418398011399257
Generation 97 Best Solution: [182.8233943188883, 175.62145833268056,
175.08165226400573, 75.84603227020634] Fitness: 2.500371404069042
Generation 98 Best Solution: [183.80863435150667, 177.1297778737969,
175.08165226400573, 74.98953483311536] Fitness: 2.108426010350956
Generation 99 Best Solution: [182.8233943188883, 176.8186522867761,
175.08165226400573, 75.2698022992414] Fitness: 1.6793770898567273
Generation 100 Best Solution: [182.8233943188883, 176.4633973825096,
175.08165226400573, 75.84603227020634] Fitness: 2.28510122109878
Generation 101 Best Solution: [183.80863435150667, 177.43429663697378,
174.272416333504, 74.98846448508426] Fitness: 1.87812027556269
Generation 102 Best Solution: [183.59059590284807, 176.55708666708944,
175.08165226400573, 75.84603227020634] Fitness: 2.3571301118558097
Generation 103 Best Solution: [183.80863435150667, 177.01837610817503,
175.08165226400573, 75.07931703697847] Fitness: 1.9881673586496484
Generation 104 Best Solution: [183.59059590284807, 176.0596365374067,
175.08165226400573, 75.84603227020634] Fitness: 2.2491400098574665
Generation 105 Best Solution: [183.59059590284807, 177.43429663697378,
175.08165226400573, 74.75269006747158] Fitness: 2.427725616464008
Generation 106 Best Solution: [183.80863435150667, 177.43429663697378,
174.6359183320165, 74.34791347039437] Fitness: 2.124328327845435
Generation 107 Best Solution: [183.59059590284807, 176.82977972162493,

175.08165226400573, 75.69251442874437] Fitness: 2.3074268720584197
Generation 108 Best Solution: [183.80863435150667, 175.78002936378658,
175.08165226400573, 75.84603227020634] Fitness: 2.5776558600763586
Generation 109 Best Solution: [183.80863435150667, 175.78002936378658,
175.08165226400573, 75.11317637302348] Fitness: 1.5440302722523693
Generation 110 Best Solution: [183.80863435150667, 175.78002936378658,
175.08165226400573, 75.70889063540011] Fitness: 2.3116104866422034
Generation 111 Best Solution: [183.80863435150667, 175.78002936378658,
175.08165226400573, 75.84603227020634] Fitness: 2.5776558600763586
Generation 112 Best Solution: [183.59059590284807, 176.27781937427898,
175.08165226400573, 75.11317637302348] Fitness: 1.208717961052125
Generation 113 Best Solution: [183.80863435150667, 175.72030585814014,
175.08165226400573, 75.84603227020634] Fitness: 2.6239569605633726
Generation 114 Best Solution: [183.80863435150667, 175.72030585814014,
175.08165226400573, 75.70889063540011] Fitness: 2.3579115871292173
Generation 115 Best Solution: [183.59059590284807, 175.71664844458604,
175.11888110615263, 75.86727288546778] Fitness: 2.5409800509924
Generation 116 Best Solution: [183.80863435150667, 175.71664844458604,
175.11888110615263, 75.16455658020443] Fitness: 1.7005873610267872
Generation 117 Best Solution: [183.59059590284807, 177.43429663697378,
174.48855508967114, 75.74482384173754] Fitness: 2.702773756793328
Generation 118 Best Solution: [183.59059590284807, 177.43429663697378,
174.04299246723724, 75.1209095375714] Fitness: 1.747289990124435
Generation 119 Best Solution: [183.80863435150667, 177.43429663697378,
175.08165226400573, 75.12774620957492] Fitness: 2.793148474796348
Generation 120 Best Solution: [183.80863435150667, 176.24133651365955,
175.08165226400573, 75.84603227020634] Fitness: 2.433673032255495
Generation 121 Best Solution: [183.80863435150667, 177.43429663697378,
175.08165226400573, 75.04915480007786] Fitness: 2.7369537999968148
Generation 122 Best Solution: [183.80863435150667, 176.45292759675107,
175.08165226400573, 75.84603227020634] Fitness: 2.4941907359192825
Generation 123 Best Solution: [183.80863435150667, 176.45292759675107,
175.08165226400573, 75.74482384173754] Fitness: 2.294620659586851
Generation 124 Best Solution: [183.80863435150667, 176.45292759675107,
175.08165226400573, 74.89412903187193] Fitness: 1.336962982681094
Generation 125 Best Solution: [183.80863435150667, 177.43429663697378,
175.08165226400573, 74.71213933077611] Fitness: 2.6204830120221323
Generation 126 Best Solution: [183.80863435150667, 177.43429663697378,
175.08165226400573, 74.71213933077611] Fitness: 2.6204830120221323
Generation 127 Best Solution: [183.25274847562656, 177.43429663697378,
175.08165226400573, 75.21900923970657] Fitness: 2.5392031937493873
Generation 128 Best Solution: [183.80863435150667, 177.43429663697378,
175.11888110615263, 74.8167068679602] Fitness: 2.700886558311661
Generation 129 Best Solution: [183.80863435150667, 176.5603884361567,
175.08165226400573, 75.84603227020634] Fitness: 2.555401961520043
Generation 130 Best Solution: [183.80863435150667, 176.5603884361567,
175.08165226400573, 75.84603227020634] Fitness: 2.555401961520043
Generation 131 Best Solution: [183.80863435150667, 177.06633261923827,

175.11888110615263, 75.64859607699921] Fitness: 2.812900768215226
Generation 132 Best Solution: [183.80863435150667, 177.43429663697378,
175.00252258910885, 74.79308681135852] Fitness: 2.498208345343346
Generation 133 Best Solution: [183.80863435150667, 176.09247139023987,
174.82396268305754, 74.79308681135852] Fitness: 0.8649057331261242
Generation 134 Best Solution: [183.80863435150667, 176.6617360174519,
175.11888110615263, 75.6680889503284] Fitness: 2.359069024724704
Generation 135 Best Solution: [183.80863435150667, 177.43429663697378,
174.18714657699493, 75.6680889503284] Fitness: 2.6349149073629468
Generation 136 Best Solution: [183.80863435150667, 177.43429663697378,
174.18714657699493, 75.84603227020634] Fitness: 2.973659076165611
Generation 137 Best Solution: [183.19509561105366, 177.43429663697378,
174.18714657699493, 75.84603227020634] Fitness: 2.637587116113455
Generation 138 Best Solution: [183.19509561105366, 177.43429663697378,
174.18714657699493, 75.84603227020634] Fitness: 2.637587116113455
Generation 139 Best Solution: [182.96184961145866, 176.81792931267583,
175.08165226400573, 75.84603227020634] Fitness: 2.4973879251917452
Generation 140 Best Solution: [182.96184961145866, 176.81792931267583,
175.11888110615263, 75.8106334071764] Fitness: 2.4913864893467728
Generation 141 Best Solution: [183.19509561105366, 177.18917649362726,
175.11888110615263, 75.84603227020634] Fitness: 3.0548742102105098
Generation 142 Best Solution: [183.19509561105366, 177.18917649362726,
175.11888110615263, 75.8106334071764] Fitness: 2.9830014386981887
Generation 143 Best Solution: [183.80863435150667, 176.15176572773078,
175.11888110615263, 75.8106334071764] Fitness: 2.4260312773409884
Generation 144 Best Solution: [183.35743263666666, 177.43429663697378,
175.11888110615263, 75.11889694130521] Fitness: 2.5398853327482267
Generation 145 Best Solution: [183.80863435150667, 177.43429663697378,
175.11888110615263, 75.11889694130521] Fitness: 2.852143786243011
Generation 146 Best Solution: [183.35743263666666, 177.43429663697378,
175.11888110615263, 75.11889694130521] Fitness: 2.5398853327482267
Generation 147 Best Solution: [183.49923541441842, 177.43429663697378,
175.11888110615263, 74.84721166100329] Fitness: 2.4556672112979405
Generation 148 Best Solution: [182.88559935053215, 177.43429663697378,
175.11888110615263, 74.84721166100329] Fitness: 2.4571209751602145
Generation 149 Best Solution: [183.48787196166546, 177.43429663697378,
174.44021443999165, 75.05281100378163] Fitness: 1.7321636792602901
Generation 150 Best Solution: [182.82926647642418, 176.9424610672566,
175.11888110615263, 74.71127618463046] Fitness: 1.5920932773600396
Generation 151 Best Solution: [182.82926647642418, 176.45723878380906,
175.11888110615263, 74.54241117942153] Fitness: 1.1668870898908756
Generation 152 Best Solution: [182.82926647642418, 176.6939646691491,
174.0796675716884, 75.88931107146459] Fitness: 1.6857248091778834
Generation 153 Best Solution: [182.82926647642418, 176.6939646691491,
174.0796675716884, 75.88931107146459] Fitness: 1.6857248091778834
Generation 154 Best Solution: [182.82926647642418, 176.9424610672566,
175.11888110615263, 74.81553288241147] Fitness: 1.606393502931117
Generation 155 Best Solution: [183.80863435150667, 176.72137737538833,

175.11888110615263, 74.81553288241147] Fitness: 1.5706074197472732
Generation 156 Best Solution: [183.49923541441842, 177.43429663697378,
175.11888110615263, 74.3486256887545] Fitness: 2.5339618950588836
Generation 157 Best Solution: [182.82926647642418, 176.9424610672566,
175.11888110615263, 74.8692975363654] Fitness: 1.6213200246065138
Generation 158 Best Solution: [183.131523141511, 176.9424610672566,
175.11888110615263, 74.8692975363654] Fitness: 1.5066746405702196
Generation 159 Best Solution: [182.82926647642418, 175.70781823816165,
175.11888110615263, 74.3486256887545] Fitness: 1.388009400134459
Generation 160 Best Solution: [182.82926647642418, 175.70781823816165,
175.11888110615263, 75.84603227020634] Fitness: 2.48232472345977
Generation 161 Best Solution: [183.80863435150667, 176.9424610672566,
175.11888110615263, 74.3486256887545] Fitness: 1.9108474039346022
Generation 162 Best Solution: [182.82926647642418, 176.9424610672566,
175.11888110615263, 74.3486256887545] Fitness: 1.6928611579172308
Generation 163 Best Solution: [183.64320000969954, 177.11651626109946,
175.11888110615263, 75.42316153910326] Fitness: 2.39663507034286
Generation 164 Best Solution: [183.131523141511, 177.264299739038,
175.11888110615263, 75.42316153910326] Fitness: 2.4843353316747723
Generation 165 Best Solution: [183.131523141511, 177.264299739038,
175.11888110615263, 75.42316153910326] Fitness: 2.4843353316747723
Generation 166 Best Solution: [182.82926647642418, 176.40077617904228,
175.11888110615263, 75.84603227020634] Fitness: 2.319768750800865
Generation 167 Best Solution: [182.82926647642418, 176.40077617904228,
175.11888110615263, 75.88931107146459] Fitness: 2.410667441812922
Generation 168 Best Solution: [182.82926647642418, 176.00646324388424,
175.11888110615263, 75.88931107146459] Fitness: 2.398491260639 7195
Generation 169 Best Solution: [182.82926647642418, 176.00646324388424,
175.11888110615263, 75.84603227020634] Fitness: 2.307592569627663
Generation 170 Best Solution: [182.82926647642418, 176.00646324388424,
175.11888110615263, 75.84603227020634] Fitness: 2.307592569627663
Generation 171 Best Solution: [183.64320000969954, 177.43429663697378,
174.92648203551627, 74.56039095988787] Fitness: 2.2294393461639093
Generation 172 Best Solution: [183.64320000969954, 177.1169804705526,
174.4161007983615, 75.84603227020634] Fitness: 2.2826341809423147
Generation 173 Best Solution: [182.82926647642418, 177.1169804705526,
173.96911074503706, 75.84603227020634] Fitness: 2.1515453903756847
Generation 174 Best Solution: [182.82926647642418, 176.07307077871974,
174.36673715304215, 74.69419594542322] Fitness: 0.24182005592811498
Generation 175 Best Solution: [183.64320000969954, 176.6345612897871,
175.11888110615263, 75.84603227020634] Fitness: 2.5189852870824305
Generation 176 Best Solution: [183.80863435150667, 175.66096519745938,
175.11888110615263, 75.84603227020634] Fitness: 2.742112901072159
Generation 177 Best Solution: [183.80863435150667, 175.66096519745938,
175.11888110615263, 75.43624046496808] Fitness: 2.046460263968221
Generation 178 Best Solution: [183.64320000969954, 175.66096519745938,
175.01893713310486, 75.84603227020634] Fitness: 2.414333995601747
Generation 179 Best Solution: [183.80863435150667, 175.66096519745938,

175.01893713310486, 75.43624046496808] Fitness: 1.8751946513945739
Generation 180 Best Solution: [183.16987204343266, 176.6114380080456,
175.01893713310486, 75.84603227020634] Fitness: 2.1507122747097984
Generation 181 Best Solution: [183.80863435150667, 177.43429663697378,
173.9824701653918, 75.43624046496808] Fitness: 2.285353550401345
Generation 182 Best Solution: [183.80863435150667, 177.43429663697378,
174.3930432259469, 75.43624046496808] Fitness: 2.345758296752267
Generation 183 Best Solution: [183.80863435150667, 177.43429663697378,
174.3930432259469, 75.43624046496808] Fitness: 2.345758296752267
Generation 184 Best Solution: [182.96431748680303, 177.43429663697378,
173.9824701653918, 75.84603227020634] Fitness: 2.6917230874022557
Generation 185 Best Solution: [183.80863435150667, 176.32896298325937,
175.11888110615263, 75.84603227020634] Fitness: 2.514951009348289
Generation 186 Best Solution: [183.64320000969954, 177.43429663697378,
175.11888110615263, 74.74166542388495] Fitness: 2.5319749319094393
Generation 187 Best Solution: [183.80863435150667, 177.2241521700182,
175.11888110615263, 74.87118736782314] Fitness: 2.2891561957216644
Generation 188 Best Solution: [183.83572935406832, 175.7884473178442,
175.11888110615263, 75.84603227020634] Fitness: 2.6677818570476304
Generation 189 Best Solution: [183.83572935406832, 175.7884473178442,
175.11888110615263, 74.64695856363299] Fitness: 1.4735142149914604
Generation 190 Best Solution: [183.80863435150667, 177.43429663697378,
175.11888110615263, 74.42637891854112] Fitness: 2.7457878779388776
Generation 191 Best Solution: [183.80863435150667, 177.43429663697378,
175.11888110615263, 74.2954006085306] Fitness: 2.821548116013003
Generation 192 Best Solution: [183.80863435150667, 176.98912263565802,
175.11888110615263, 75.84603227020634] Fitness: 3.069828763568436
Generation 193 Best Solution: [183.80863435150667, 176.18551602498385,
175.11888110615263, 75.84603227020634] Fitness: 2.4968474921292088
Generation 194 Best Solution: [183.80863435150667, 176.01784056314156,
175.11888110615263, 75.84603227020634] Fitness: 2.522057447421344
Generation 195 Best Solution: [183.6183979752871, 177.43429663697378,
175.03843927052182, 74.65722587515266] Fitness: 2.371482573379981
Generation 196 Best Solution: [183.54839101008275, 175.75685648734424,
175.11888110615263, 74.65722587515266] Fitness: 1.2415703506487175
Generation 197 Best Solution: [183.80159241466626, 175.75685648734424,
175.11888110615263, 74.65722587515266] Fitness: 1.458194488457426
Generation 198 Best Solution: [182.86320514558332, 176.4525506226757,
175.11888110615263, 75.73308367689965] Fitness: 2.0993883372651845
Generation 199 Best Solution: [183.80863435150667, 177.43429663697378,
174.75114705708282, 74.59840890858894] Fitness: 2.1499960560057123
Generation 200 Best Solution: [183.04865883262772, 177.43429663697378,
175.11888110615263, 74.5700378364327] Fitness: 2.380402072698032
Generation 201 Best Solution: [183.6183979752871, 175.87187855489813,
175.11888110615263, 75.73308367689965] Fitness: 2.187370013702595
Generation 202 Best Solution: [183.6183979752871, 177.43429663697378,
174.46686059646464, 74.5700378364327] Fitness: 1.7246355392993375
Generation 203 Best Solution: [183.04865883262772, 176.4475997360761,

174.75114705708282, 74.5700378364327] Fitness: 0.5139437602082506
Generation 204 Best Solution: [183.6183979752871, 177.3467643385162,
175.11888110615263, 75.22084933933354] Fitness: 2.5766143078633768
Generation 205 Best Solution: [183.09155851220072, 177.43429663697378,
175.11888110615263, 74.65904571230834] Fitness: 2.359625592375366
Generation 206 Best Solution: [183.09155851220072, 177.43429663697378,
175.11888110615263, 74.65904571230834] Fitness: 2.359625592375366
Generation 207 Best Solution: [183.80863435150667, 176.91244295848753,
175.11888110615263, 75.22084933933354] Fitness: 2.0240094597455967
Generation 208 Best Solution: [183.80863435150667, 176.91244295848753,
175.11888110615263, 75.22084933933354] Fitness: 2.0240094597455967
Generation 209 Best Solution: [183.03818306726518, 176.85586475667407,
175.11888110615263, 75.30270572835074] Fitness: 1.7230025491963694
Generation 210 Best Solution: [183.03818306726518, 176.63326153308844,
175.11888110615263, 75.84603227020634] Fitness: 2.359902671531422
Generation 211 Best Solution: [183.03818306726518, 176.14599692105563,
175.11888110615263, 75.87580156718002] Fitness: 2.2458929082833325
Generation 212 Best Solution: [183.80863435150667, 176.2880872990502,
175.11888110615263, 75.77432077553615] Fitness: 2.362779398839786
Generation 213 Best Solution: [183.08126654412388, 177.43429663697378,
175.11888110615263, 74.99833486423083] Fitness: 2.4475067629356957
Generation 214 Best Solution: [183.08126654412388, 177.43429663697378,
174.17997674799742, 75.45578200830101] Fitness: 1.9785726870644194
Generation 215 Best Solution: [183.6183979752871, 177.43429663697378,
175.11888110615263, 74.4276011331372] Fitness: 2.5694406730841783
Generation 216 Best Solution: [183.2844360980485, 177.43429663697378,
175.11888110615263, 74.83464592430808] Fitness: 2.376569721515876
Generation 217 Best Solution: [183.2844360980485, 177.43429663697378,
173.79708224487612, 75.4881127650202] Fitness: 2.0990772682553045
Generation 218 Best Solution: [183.2844360980485, 177.43429663697378,
173.79708224487612, 75.4881127650202] Fitness: 2.0990772682553045
Generation 219 Best Solution: [183.80863435150667, 176.86688251185342,
175.11888110615263, 75.77432077553615] Fitness: 2.765348092807832
Generation 220 Best Solution: [183.80863435150667, 177.43429663697378,
174.49367548901975, 75.87580156718002] Fitness: 3.164111158252931
Generation 221 Best Solution: [183.6183979752871, 177.43429663697378,
174.48742536238868, 75.77432077553615] Fitness: 2.778590038978653
Generation 222 Best Solution: [183.6183979752871, 177.43429663697378,
174.48742536238868, 75.84603227020634] Fitness: 2.9218761738312016
Generation 223 Best Solution: [183.6183979752871, 177.43429663697378,
175.11888110615263, 75.17279539048837] Fitness: 2.720398167018473
Generation 224 Best Solution: [183.1055276830886, 177.43429663697378,
175.11888110615263, 75.17279539048837] Fitness: 2.5670268552724878
Generation 225 Best Solution: [183.86628433008516, 177.43429663697378,
174.14141925321772, 75.26375614414816] Fitness: 2.1356292037675537
Generation 226 Best Solution: [183.80863435150667, 177.43429663697378,
174.14141925321772, 75.26375614414816] Fitness: 2.069656173176153
Generation 227 Best Solution: [183.80863435150667, 177.43429663697378,

174.14141925321772, 75.26375614414816] Fitness: 2.069656173176153
Generation 228 Best Solution: [183.0412416808203, 177.43429663697378,
174.14141925321772, 75.26375614414816] Fitness: 1.7542570813461393
Generation 229 Best Solution: [183.80863435150667, 175.73033204059237,
174.7511954846169, 75.84603227020634] Fitness: 2.1390469178332685
Generation 230 Best Solution: [183.80863435150667, 176.4403348712408,
175.11888110615263, 75.84603227020634] Fitness: 2.554232899306209
Generation 231 Best Solution: [183.80863435150667, 177.43429663697378,
174.5890770594448, 75.84603227020634] Fitness: 3.175954666715606
Generation 232 Best Solution: [183.80863435150667, 176.3477335744679,
175.11888110615263, 75.87580156718002] Fitness: 2.5821936457283914
Generation 233 Best Solution: [183.80863435150667, 176.0206674809813,
174.5890770594448, 75.84603227020634] Fitness: 1.815774366753836
Generation 234 Best Solution: [183.86628433008516, 176.56180294244126,
175.11888110615263, 75.84603227020634] Fitness: 2.688188943926799
Generation 235 Best Solution: [183.80863435150667, 176.6357770077952,
175.11888110615263, 75.84603227020634] Fitness: 2.6764688183674235
Generation 236 Best Solution: [183.80863435150667, 176.6121212922984,
175.11888110615263, 75.84603227020634] Fitness: 2.6580615533599588
Generation 237 Best Solution: [183.86628433008516, 177.43429663697378,
174.44267800095605, 75.3345410384263] Fitness: 2.312934087589514
Generation 238 Best Solution: [183.80863435150667, 176.6121212922984,
175.11888110615263, 75.87580156718002] Fitness: 2.720228679311494
Generation 239 Best Solution: [183.80863435150667, 177.43429663697378,
174.44267800095605, 74.87495281687015] Fitness: 1.9050299304122305
Generation 240 Best Solution: [183.86628433008516, 177.43429663697378,
174.44267800095605, 74.87495281687015] Fitness: 1.9710029610036308
Generation 241 Best Solution: [183.14675303253574, 177.43429663697378,
174.44267800095605, 74.87495281687015] Fitness: 1.5709196630549438
Generation 242 Best Solution: [183.80863435150667, 176.46588944629036,
175.11888110615263, 75.87580156718002] Fitness: 2.6285236680383037
Generation 243 Best Solution: [183.86628433008516, 177.43429663697378,
174.4085274959265, 75.87580156718002] Fitness: 3.1776933488609296
Generation 244 Best Solution: [183.31868377063833, 177.43429663697378,
175.11888110615263, 75.5386905856496] Fitness: 3.009474681851917
Generation 245 Best Solution: [183.86628433008516, 176.86783962427674,
175.11888110615263, 75.59011406725422] Fitness: 2.5063220471568375
Generation 246 Best Solution: [183.86628433008516, 175.79678170304086,
175.11888110615263, 75.59011406725422] Fitness: 2.2282075035253213
Generation 247 Best Solution: [183.86628433008516, 175.79678170304086,
175.11888110615263, 75.59011406725422] Fitness: 2.2282075035253213
Generation 248 Best Solution: [183.86628433008516, 175.93375809376593,
175.13436554789237, 75.59011406725422] Fitness: 2.1781676298826107
Generation 249 Best Solution: [183.86628433008516, 175.79678170304086,
175.11888110615263, 75.87580156718002] Fitness: 2.7598187736649678
Generation 250 Best Solution: [183.86628433008516, 176.4581460969249,
175.11888110615263, 75.87580156718002] Fitness: 2.690700492053187
Generation 251 Best Solution: [182.95593187013435, 176.09595798141828,

175.11888110615263, 75.87580156718002] Fitness: 2.280460668805407
Generation 252 Best Solution: [183.86628433008516, 175.98713273534207,
175.11888110615263, 75.59011406725422] Fitness: 2.12861828641297
Generation 253 Best Solution: [183.86628433008516, 177.43429663697378,
174.2496214066301, 75.59011406725422] Fitness: 2.5827811102521747
Generation 254 Best Solution: [183.86628433008516, 176.4581460969249,
175.11888110615263, 74.3380439310994] Fitness: 1.540668967998744
Generation 255 Best Solution: [183.86628433008516, 176.5604124562652,
174.2496214066301, 75.06137036686455] Fitness: 0.7215563753011407
Generation 256 Best Solution: [183.86628433008516, 177.43429663697378,
175.11888110615263, 74.3380439310994] Fitness: 2.859507119588585
Generation 257 Best Solution: [183.4520818593915, 177.43429663697378,
175.11888110615263, 74.3380439310994] Fitness: 2.516780830196203
Generation 258 Best Solution: [183.86628433008516, 177.0820541332939,
173.8463587054252, 75.87580156718002] Fitness: 2.4840142516527286
Generation 259 Best Solution: [183.64834629561682, 175.79702583040242,
175.11888110615263, 75.84603227020634] Fitness: 2.4791318847873054
Generation 260 Best Solution: [183.64834629561682, 177.43429663697378,
174.525702420953, 75.39397829094679] Fitness: 2.222194109646482
Generation 261 Best Solution: [183.86628433008516, 176.0385744907757,
175.11888110615263, 75.87580156718002] Fitness: 2.6443721084991094
Generation 262 Best Solution: [183.86628433008516, 176.0385744907757,
175.11888110615263, 75.53655315653559] Fitness: 2.029245465862615
Generation 263 Best Solution: [183.4520818593915, 176.0385744907757,
175.11888110615263, 75.53655315653559] Fitness: 1.686519176470233
Generation 264 Best Solution: [183.86628433008516, 176.0385744907757,
175.11888110615263, 75.87580156718002] Fitness: 2.6443721084991094
Generation 265 Best Solution: [183.86628433008516, 176.0385744907757,
175.11888110615263, 75.87580156718002] Fitness: 2.6443721084991094
Generation 266 Best Solution: [183.86628433008516, 176.0385744907757,
175.11888110615263, 75.87580156718002] Fitness: 2.6443721084991094
Generation 267 Best Solution: [183.86628433008516, 176.17982046256128,
175.11888110615263, 75.87580156718002] Fitness: 2.6250239155453454
Generation 268 Best Solution: [183.86628433008516, 177.43429663697378,
175.11888110615263, 74.72699910194024] Fitness: 2.7532075843794597
Generation 269 Best Solution: [183.86628433008516, 177.43429663697378,
175.11888110615263, 74.72699910194024] Fitness: 2.7532075843794597
Generation 270 Best Solution: [183.86628433008516, 176.9878431149375,
175.11888110615263, 75.43156638669316] Fitness: 2.432111709551072
Generation 271 Best Solution: [183.86628433008516, 177.00848976483599,
175.11888110615263, 74.72699910194024] Fitness: 1.9696145379936523
Generation 272 Best Solution: [183.86628433008516, 177.00848976483599,
175.11888110615263, 75.28462166100427] Fitness: 2.2864040094461617
Generation 273 Best Solution: [183.31979422426215, 177.44154816833327,
175.11888110615263, 74.35580667456063] Fitness: 2.477003234783482
Generation 274 Best Solution: [183.86628433008516, 176.46480011949416,
175.11888110615263, 75.50399048108167] Fitness: 2.0305487098011445
Generation 275 Best Solution: [183.86628433008516, 176.46480011949416,

175.11888110615263, 74.64436339593931] Fitness: 1.4377087037567402
Generation 276 Best Solution: [183.86628433008516, 175.71470588750344,
175.11888110615263, 75.50399048108167] Fitness: 2.1592275373211085
Generation 277 Best Solution: [183.86628433008516, 175.71470588750344,
175.11888110615263, 75.50399048108167] Fitness: 2.1592275373211085
Generation 278 Best Solution: [183.86628433008516, 177.43429663697378,
174.82218055925878, 74.30462559219077] Fitness: 2.4246473679281415
Generation 279 Best Solution: [183.60145635180166, 176.4412902283029,
175.11888110615263, 75.87580156718002] Fitness: 2.428513891774176
Generation 280 Best Solution: [183.60145635180166, 177.43429663697378,
175.11888110615263, 75.50399048108167] Fitness: 3.091839628154632
Generation 281 Best Solution: [183.86628433008516, 177.43429663697378,
175.11888110615263, 74.94925278711071] Fitness: 2.8132164525741854
Generation 282 Best Solution: [183.86628433008516, 176.19670975276168,
175.11888110615263, 75.87580156718002] Fitness: 2.6250844199685983
Generation 283 Best Solution: [183.05321277586012, 175.6276771478485,
174.8302811692074, 75.87580156718002] Fitness: 2.0716803203706124
Generation 284 Best Solution: [183.05321277586012, 175.6276771478485,
174.8302811692074, 75.87580156718002] Fitness: 2.0716803203706124
Generation 285 Best Solution: [183.86628433008516, 175.6276771478485,
174.8302811692074, 75.87580156718002] Fitness: 2.456170732614316
Generation 286 Best Solution: [183.05321277586012, 177.43429663697378,
175.11888110615263, 74.74563947986408] Fitness: 2.370376390810947
Generation 287 Best Solution: [183.49281212712754, 177.43429663697378,
175.11888110615263, 74.37691131606823] Fitness: 2.514243729096474
Generation 288 Best Solution: [183.31614991147188, 176.3921543006289,
175.11888110615263, 75.87580156718002] Fitness: 2.2739435202129172
Generation 289 Best Solution: [183.31614991147188, 176.3921543006289,
175.11888110615263, 75.87580156718002] Fitness: 2.2739435202129172
Generation 290 Best Solution: [183.49281212712754, 176.19180765217834,
175.11888110615263, 75.87580156718002] Fitness: 2.3024686082431236
Generation 291 Best Solution: [182.90869947353292, 176.26474974544686,
175.11888110615263, 75.87580156718002] Fitness: 2.3006436920605395
Generation 292 Best Solution: [183.49281212712754, 177.43429663697378,
174.13119176010494, 75.87580156718002] Fitness: 2.7738647250857476
Generation 293 Best Solution: [183.50442170389468, 177.43429663697378,
175.11888110615263, 74.3176530086773] Fitness: 2.556245571189172
Generation 294 Best Solution: [183.50442170389468, 177.43429663697378,
175.11888110615263, 74.3176530086773] Fitness: 2.556245571189172
Generation 295 Best Solution: [183.49281212712754, 175.7273376699289,
174.83551248895907, 75.87580156718002] Fitness: 2.050258153403534
Generation 296 Best Solution: [183.50442170389468, 175.7273376699289,
174.83551248895907, 75.87580156718002] Fitness: 2.056550377153557
Generation 297 Best Solution: [183.49281212712754, 177.43429663697378,
174.70886612353078, 75.87580156718002] Fitness: 3.0973920911752577
Generation 298 Best Solution: [183.50442170389468, 177.43429663697378,
174.70886612353078, 74.32756384931623] Fitness: 1.9602379360025113
Generation 299 Best Solution: [183.50442170389468, 176.61693000343033,

174.9189058490226, 75.87580156718002] Fitness: 2.1487094644526326
Generation 300 Best Solution: [183.49281212712754, 176.35088741026462,
175.11888110615263, 75.87580156718002] Fitness: 2.326534909897261
Generation 301 Best Solution: [183.49281212712754, 176.35088741026462,
175.11888110615263, 75.87580156718002] Fitness: 2.326534909897261
Generation 302 Best Solution: [183.49281212712754, 176.35088741026462,
175.11888110615263, 74.88892752196324] Fitness: 1.105241796056115
Generation 303 Best Solution: [183.49281212712754, 177.23561993042404,
175.11888110615263, 74.47042711993012] Fitness: 2.064862339764558
Generation 304 Best Solution: [183.50442170389468, 176.0626164241329,
174.04906254569167, 75.87580156718002] Fitness: 1.4113639208510778
Generation 305 Best Solution: [183.49281212712754, 176.0626164241329,
174.04906254569167, 75.87580156718002] Fitness: 1.4050716971010544
Generation 306 Best Solution: [183.49281212712754, 177.43429663697378,
175.11888110615263, 74.47042711993012] Fitness: 2.4705895905942685
Generation 307 Best Solution: [183.50442170389468, 177.43429663697378,
174.81076528491315, 75.08619724531113] Fitness: 2.1066845018514595
Generation 308 Best Solution: [183.50442170389468, 177.43429663697378,
174.31684251612432, 75.08619724531113] Fitness: 1.70381571451755
Generation 309 Best Solution: [182.90574749016778, 175.84072883905029,
175.11888110615263, 75.87580156718002] Fitness: 2.402789741535485
Generation 310 Best Solution: [182.90574749016778, 175.84072883905029,
175.11888110615263, 75.87580156718002] Fitness: 2.402789741535485
Generation 311 Best Solution: [183.50442170389468, 176.9702818661826,
175.11888110615263, 75.41154927977158] Fitness: 2.0649396710587133
Generation 312 Best Solution: [183.86628433008516, 177.43429663697378,
174.07335793951256, 75.41154927977158] Fitness: 2.3064919208299965
Generation 313 Best Solution: [183.49281212712754, 177.43429663697378,
174.1525765446924, 75.41154927977158] Fitness: 1.9850679549205907
Generation 314 Best Solution: [183.50442170389468, 177.43429663697378,
175.11888110615263, 75.08619724531113] Fitness: 2.577662510762342
Generation 315 Best Solution: [183.50442170389468, 177.43429663697378,
173.79688347350907, 75.08619724531113] Fitness: 1.7483121965255073
Generation 316 Best Solution: [183.50442170389468, 177.43429663697378,
174.49759736408183, 75.50176884260144] Fitness: 2.249921156519456
Generation 317 Best Solution: [182.8295337895231, 177.43429663697378,
174.87692016381666, 75.08619724531113] Fitness: 2.225700183888372
Generation 318 Best Solution: [183.09665009824624, 177.43429663697378,
174.87692016381666, 75.08619724531113] Fitness: 2.1161667998407685
Generation 319 Best Solution: [182.8295337895231, 175.65279218195045,
174.04756731316937, 75.87580156718002] Fitness: 1.6830066209861088
Generation 320 Best Solution: [183.8249361040708, 175.65279218195045,
174.04756731316937, 75.50108677268798] Fitness: 1.2516008300012638
Generation 321 Best Solution: [182.8295337895231, 175.80471655768562,
175.11888110615263, 75.87580156718002] Fitness: 2.4702483168641143
Generation 322 Best Solution: [183.34020391972817, 177.43429663697378,
175.11888110615263, 74.78399767558346] Fitness: 2.3772435348093905
Generation 323 Best Solution: [183.34020391972817, 176.73975860304736,

174.93989302530798, 75.87580156718002] Fitness: 2.220195672390715
Generation 324 Best Solution: [183.50618213065678, 176.43841637314293,
175.11888110615263, 75.87580156718002] Fitness: 2.366230983915135
Generation 325 Best Solution: [182.8295337895231, 177.43429663697378,
174.32693371015614, 75.34664668083111] Fitness: 1.985308903370791
Generation 326 Best Solution: [182.8295337895231, 175.8300292233509,
174.93989302530798, 75.87580156718002] Fitness: 2.1595078194480015
Generation 327 Best Solution: [183.34020391972817, 177.43429663697378,
175.11888110615263, 74.96712452303939] Fitness: 2.4388754366855103
Generation 328 Best Solution: [183.86628433008516, 175.6769206050124,
174.93989302530798, 75.87580156718002] Fitness: 2.5614386354932424
Generation 329 Best Solution: [183.36653353585407, 177.43429663697378,
175.11888110615263, 74.74906175648115] Fitness: 2.379728964434963
Generation 330 Best Solution: [182.8295337895231, 177.43429663697378,
174.93989302530798, 75.34664668083111] Fitness: 2.561123829596785
Generation 331 Best Solution: [182.8295337895231, 176.50144846649673,
174.93989302530798, 75.87580156718002] Fitness: 2.135025003061683
Generation 332 Best Solution: [182.99463264841043, 176.50144846649673,
174.93989302530798, 75.87580156718002] Fitness: 2.052353244599373
Generation 333 Best Solution: [182.8295337895231, 177.43429663697378,
174.93989302530798, 75.34664668083111] Fitness: 2.561123829596785
Generation 334 Best Solution: [182.8295337895231, 177.43429663697378,
175.11888110615263, 75.34664668083111] Fitness: 2.8552647484459963
Generation 335 Best Solution: [182.8295337895231, 177.43429663697378,
173.92158242031542, 75.87580156718002] Fitness: 2.841572540492012
Generation 336 Best Solution: [182.8295337895231, 175.80025165794396,
174.93989302530798, 75.89869232238206] Fitness: 2.2280277594269196
Generation 337 Best Solution: [182.8295337895231, 175.80025165794396,
174.93989302530798, 75.89869232238206] Fitness: 2.2280277594269196
Generation 338 Best Solution: [182.8295337895231, 177.2266699769214,
175.11888110615263, 75.34664668083111] Fitness: 2.4329121442820196
Generation 339 Best Solution: [182.8295337895231, 177.2266699769214,
175.11888110615263, 75.34664668083111] Fitness: 2.4329121442820196
Generation 340 Best Solution: [183.36653353585407, 177.43429663697378,
174.71527136980984, 74.98771633198517] Fitness: 1.8743181910702071
Generation 341 Best Solution: [182.8295337895231, 177.2266699769214,
175.11888110615263, 75.42124659710568] Fitness: 2.525442119130176
Generation 342 Best Solution: [182.8295337895231, 177.2266699769214,
174.06693297409288, 75.84039215842718] Fitness: 2.316836185930318
Generation 343 Best Solution: [183.36706114575654, 177.43429663697378,
175.11888110615263, 74.8707355925332] Fitness: 2.406637604507216
Generation 344 Best Solution: [182.8295337895231, 175.57679391162364,
174.60177550128972, 75.87580156718002] Fitness: 1.9766482789291242
Generation 345 Best Solution: [182.8295337895231, 175.69560521248565,
175.11888110615263, 75.87580156718002] Fitness: 2.5548390190070016
Generation 346 Best Solution: [182.8295337895231, 177.2266699769214,
175.11888110615263, 75.17305224293834] Fitness: 2.255892275550504
Generation 347 Best Solution: [182.8295337895231, 175.69560521248565,

175.11888110615263, 75.42124659710568] Fitness: 1.77722752280832
Generation 348 Best Solution: [182.8295337895231, 175.69560521248565,
175.11888110615263, 75.87580156718002] Fitness: 2.5548390190070016
Generation 349 Best Solution: [183.86628433008516, 176.13761054560047,
175.11888110615263, 74.67344386189193] Fitness: 1.3694288255590354
Generation 350 Best Solution: [183.87258856267024, 176.6695559689133,
175.11888110615263, 75.72349042183262] Fitness: 2.538710538979767
Generation 351 Best Solution: [183.86628433008516, 177.2266699769214,
175.11888110615263, 75.42124659710568] Fitness: 2.8095745432803696
Generation 352 Best Solution: [183.86628433008516, 177.41575347937675,
175.11888110615263, 74.37681246677266] Fitness: 2.7960071358450915
Generation 353 Best Solution: [183.6876638603692, 177.41575347937675,
175.11888110615263, 74.37681246677266] Fitness: 2.6108057814643937
Generation 354 Best Solution: [183.86628433008516, 175.8821702288707,
175.11888110615263, 75.42124659710568] Fitness: 1.929566346687436
Generation 355 Best Solution: [183.6876638603692, 177.43429663697378,
175.11888110615263, 75.4339493071903] Fitness: 3.0634673541444775
Generation 356 Best Solution: [183.86628433008516, 176.46466320554762,
174.5739364104387, 75.87580156718002] Fitness: 1.9756186121617794
Generation 357 Best Solution: [183.6195478134309, 177.41575347937675,
175.11888110615263, 75.40573776168496] Fitness: 2.930151495866473
Generation 358 Best Solution: [182.8295337895231, 177.41575347937675,
174.09746153043176, 75.40573776168496] Fitness: 1.973219585788035
Generation 359 Best Solution: [182.8295337895231, 177.41575347937675,
175.11888110615263, 75.16650878837356] Fitness: 2.6317830822195463
Generation 360 Best Solution: [182.8295337895231, 176.12381964776472,
175.11888110615263, 75.84039215842718] Fitness: 2.2705514032236414
Generation 361 Best Solution: [183.87258856267024, 177.41575347937675,
174.2795477404835, 75.63424865060773] Fitness: 2.6306571268708643
Generation 362 Best Solution: [183.86628433008516, 177.41575347937675,
175.11888110615263, 75.55341659500455] Fitness: 3.3793194171979573
Generation 363 Best Solution: [183.87258856267024, 176.26510845221543,
175.11888110615263, 75.87580156718002] Fitness: 2.6380876360424477
Generation 364 Best Solution: [182.8295337895231, 176.02539617287604,
175.11888110615263, 75.87580156718002] Fitness: 2.3638538156162614
Generation 365 Best Solution: [183.87258856267024, 175.66659080503342,
175.11888110615263, 75.55341659500455] Fitness: 2.2832182990991945
Generation 366 Best Solution: [182.8295337895231, 176.62227275894753,
174.11779466287047, 75.87580156718002] Fitness: 1.5962544365980256
Generation 367 Best Solution: [182.8295337895231, 176.62227275894753,
174.11779466287047, 75.87580156718002] Fitness: 1.5962544365980256
Generation 368 Best Solution: [182.8295337895231, 176.62227275894753,
175.11888110615263, 75.37876160484352] Fitness: 1.6783260684894101
Generation 369 Best Solution: [183.87258856267024, 175.88716533657026,
175.11888110615263, 74.86746596285579] Fitness: 1.4834558801547755
Generation 370 Best Solution: [182.88063058576242, 177.41575347937675,
175.11888110615263, 74.86810224943639] Fitness: 2.4253937990314323
Generation 371 Best Solution: [182.8295337895231, 175.66659080503342,

173.74778084349552, 75.87580156718002] Fitness: 1.7805847965703818
Generation 372 Best Solution: [182.8295337895231, 175.66659080503342,
173.74778084349552, 75.87580156718002] Fitness: 1.7805847965703818
Generation 373 Best Solution: [182.8295337895231, 175.66659080503342,
173.74778084349552, 75.87580156718002] Fitness: 1.7805847965703818
Generation 374 Best Solution: [182.8295337895231, 175.66659080503342,
173.74778084349552, 75.87580156718002] Fitness: 1.7805847965703818
Generation 375 Best Solution: [183.86628433008516, 175.66659080503342,
175.11888110615263, 74.77933221099912] Fitness: 1.6149223217548307
Generation 376 Best Solution: [183.2543052011924, 175.66659080503342,
174.54353338527713, 75.87580156718002] Fitness: 1.7234465642558563
Generation 377 Best Solution: [182.89500685006143, 177.41575347937675,
174.53769798450114, 75.19492788290853] Fitness: 1.8708687309212377
Generation 378 Best Solution: [183.87258856267024, 175.66659080503342,
175.11888110615263, 74.89362488839824] Fitness: 1.6530195798515428
Generation 379 Best Solution: [183.87258856267024, 177.41575347937675,
174.0548872897064, 75.46586975299186] Fitness: 2.3472348048496103
Generation 380 Best Solution: [183.20332929058685, 177.41575347937675,
174.0548872897064, 75.46586975299186] Fitness: 1.9376969991962196
Generation 381 Best Solution: [183.20332929058685, 177.41575347937675,
174.0548872897064, 75.87580156718002] Fitness: 2.6552311962854187
Generation 382 Best Solution: [183.87258856267024, 176.87059190547225,
174.0548872897064, 75.87580156718002] Fitness: 2.1373423688595814
Generation 383 Best Solution: [183.31383272791615, 176.87059190547225,
174.0548872897064, 75.87580156718002] Fitness: 1.7405406509712023
Generation 384 Best Solution: [183.20332929058685, 177.41575347937675,
174.55723475277347, 74.9870015549019] Fitness: 1.6573854671108113
Generation 385 Best Solution: [183.87258856267024, 177.41575347937675,
174.20019636590183, 74.40462687796231] Fitness: 1.8832843052576926
Generation 386 Best Solution: [183.50281458972995, 177.41575347937675,
174.0548872897064, 75.16646577679097] Fitness: 1.6872342354990264
Generation 387 Best Solution: [183.87258856267024, 177.41575347937675,
174.17496405872888, 75.56718635972456] Fitness: 2.4989020016589514
Generation 388 Best Solution: [183.23658039546618, 176.51272576066717,
174.69202200089953, 75.87580156718002] Fitness: 1.7118835005329163
Generation 389 Best Solution: [183.23658039546618, 176.51272576066717,
174.69202200089953, 75.87580156718002] Fitness: 1.7118835005329163
Generation 390 Best Solution: [183.87258856267024, 177.41575347937675,
175.11888110615263, 75.10499756479928] Fitness: 2.874333629250082
Generation 391 Best Solution: [182.93227847903358, 177.41575347937675,
174.17496405872888, 75.87580156718002] Fitness: 2.7180394445069167
Generation 392 Best Solution: [183.87258856267024, 176.43744477782232,
175.11888110615263, 74.97494245873501] Fitness: 1.5049236944152584
Generation 393 Best Solution: [183.87258856267024, 176.3614149522444,
175.11888110615263, 75.87580156718002] Fitness: 2.6598335071615278
Generation 394 Best Solution: [183.87258856267024, 176.3614149522444,
174.91029918541562, 75.87580156718002] Fitness: 2.322546289932898
Generation 395 Best Solution: [183.87258856267024, 176.46568916740296,

175.11888110615263, 75.87580156718002] Fitness: 2.7019699407473756
Generation 396 Best Solution: [183.87258856267024, 177.33934857912828,
175.11888110615263, 74.46372912858594] Fitness: 2.600665667096913
Generation 397 Best Solution: [183.87258856267024, 177.41575347937675,
174.57966323281153, 74.97900964951613] Fitness: 2.081186052831677
Generation 398 Best Solution: [183.87258856267024, 175.971325029785,
175.11888110615263, 75.87580156718002] Fitness: 2.6736201581189896
Generation 399 Best Solution: [183.87258856267024, 177.33934857912828,
175.11888110615263, 74.30822311377823] Fitness: 2.6836779235339403
Generation 400 Best Solution: [183.87258856267024, 177.41575347937675,
175.11888110615263, 74.97900964951613] Fitness: 2.7946524642362878
Generation 401 Best Solution: [183.87258856267024, 177.33934857912828,
174.7930047341143, 74.97900964951613] Fitness: 2.1398552590261195
Generation 402 Best Solution: [183.87258856267024, 177.41575347937675,
174.98349859095333, 74.97900964951613] Fitness: 2.5669234659195874
Generation 403 Best Solution: [183.04069684605673, 177.41575347937675,
174.98349859095333, 74.97900964951613] Fitness: 2.1781265537096033
Generation 404 Best Solution: [183.13021607551482, 177.41575347937675,
175.11888110615263, 74.922429628619] Fitness: 2.362019015331467
Generation 405 Best Solution: [183.87258856267024, 177.41575347937675,
175.11888110615263, 74.6181130503267] Fitness: 2.7225941811482626
Generation 406 Best Solution: [183.87258856267024, 177.33934857912828,
175.11888110615263, 74.6181130503267] Fitness: 2.560778649937722
Generation 407 Best Solution: [183.42377280826994, 177.33934857912828,
175.11888110615263, 74.6181130503267] Fitness: 2.1981908260765506
Generation 408 Best Solution: [183.87258856267024, 177.33934857912828,
175.11888110615263, 74.6181130503267] Fitness: 2.560778649937722
Generation 409 Best Solution: [183.87258856267024, 177.33934857912828,
174.24856445878027, 74.6181130503267] Fitness: 1.665361633414695
Generation 410 Best Solution: [183.87258856267024, 176.65180781865507,
174.24856445878027, 75.87580156718002] Fitness: 1.9298043324305807
Generation 411 Best Solution: [183.87258856267024, 177.33934857912828,
175.11888110615263, 74.6181130503267] Fitness: 2.560778649937722
Generation 412 Best Solution: [183.87258856267024, 177.41575347937675,
174.8560442616172, 75.05428656259691] Fitness: 2.4265238282524604
Generation 413 Best Solution: [183.66880161491608, 177.41575347937675,
174.8560442616172, 75.05428656259691] Fitness: 2.217503599511275
Generation 414 Best Solution: [183.87258856267024, 176.37423395272387,
174.06177874207077, 75.87580156718002] Fitness: 1.7518942944598574
Generation 415 Best Solution: [183.87258856267024, 176.65180964193507,
175.11888110615263, 74.73515616255793] Fitness: 1.5695418640145302
Generation 416 Best Solution: [183.13269615374136, 177.33934857912828,
175.11888110615263, 74.92759066622796] Fitness: 2.2021194954837973
Generation 417 Best Solution: [183.87258856267024, 177.17249714401666,
175.11888110615263, 74.73515616255793] Fitness: 2.2414864407987993
Generation 418 Best Solution: [183.13269615374136, 176.43376653131915,
175.11888110615263, 75.68370100927129] Fitness: 1.9072673166478535
Generation 419 Best Solution: [183.13269615374136, 176.43376653131915,

175.11888110615263, 75.68370100927129] Fitness: 1.9072673166478535
Generation 420 Best Solution: [183.13269615374136, 176.43376653131915,
175.11888110615263, 75.76951939625124] Fitness: 2.0659909334996005
Generation 421 Best Solution: [183.13269615374136, 176.36374478882757,
175.11888110615263, 75.87580156718002] Fitness: 2.254257878309019
Generation 422 Best Solution: [183.87258856267024, 176.14691764742537,
175.11888110615263, 75.87580156718002] Fitness: 2.633935083574219
Generation 423 Best Solution: [183.87258856267024, 176.14691764742537,
175.07851440182813, 75.76951939625124] Fitness: 2.347904014265411
Generation 424 Best Solution: [183.87258856267024, 176.08704486628355,
175.11888110615263, 75.76951939625124] Fitness: 2.4265877755482004
Generation 425 Best Solution: [183.87258856267024, 177.1855064692696,
174.05846750226763, 75.68370100927129] Fitness: 2.234870566477234
Generation 426 Best Solution: [183.87258856267024, 176.08704486628355,
175.11888110615263, 75.76951939625124] Fitness: 2.4265877755482004
Generation 427 Best Solution: [183.87258856267024, 176.16228216342125,
175.11888110615263, 75.76951939625124] Fitness: 2.4183502632233704
Generation 428 Best Solution: [183.87258856267024, 176.08704486628355,
175.11888110615263, 75.76951939625124] Fitness: 2.4265877755482004
Generation 429 Best Solution: [183.87258856267024, 176.08704486628355,
175.11888110615263, 75.76951939625124] Fitness: 2.4265877755482004
Generation 430 Best Solution: [183.87258856267024, 176.08704486628355,
174.5680062837163, 75.87580156718002] Fitness: 1.9181257910801273
Generation 431 Best Solution: [183.87258856267024, 176.08704486628355,
175.11888110615263, 75.87580156718002] Fitness: 2.64130808321568
Generation 432 Best Solution: [183.87258856267024, 176.08704486628355,
175.11888110615263, 75.76951939625124] Fitness: 2.4265877755482004
Generation 433 Best Solution: [183.87258856267024, 176.08704486628355,
175.11888110615263, 75.87580156718002] Fitness: 2.64130808321568
Generation 434 Best Solution: [183.87258856267024, 176.26538675099874,
174.9740626593296, 75.76951939625124] Fitness: 2.1810197535310496
Generation 435 Best Solution: [183.87258856267024, 177.37649116849522,
174.63956148000847, 74.88976693780108] Fitness: 2.011103347315427
Generation 436 Best Solution: [183.87258856267024, 177.37649116849522,
174.47486015616488, 75.76951939625124] Fitness: 2.884964680730601
Generation 437 Best Solution: [182.8135621636003, 177.37649116849522,
174.47486015616488, 75.76951939625124] Fitness: 2.6038277889777763
Generation 438 Best Solution: [183.87258856267024, 176.87060051483203,
174.47486015616488, 75.76951939625124] Fitness: 2.0419972033706033
Generation 439 Best Solution: [183.87258856267024, 176.87060051483203,
174.47486015616488, 75.87580156718002] Fitness: 2.256717511038082
Generation 440 Best Solution: [183.87258856267024, 177.37649116849522,
174.04779745815478, 74.6378300647161] Fitness: 1.7252735704671396
Generation 441 Best Solution: [183.0275546399994, 177.33934857912828,
175.11888110615263, 74.90583691016745] Fitness: 2.2144452016920515
Generation 442 Best Solution: [183.87258856267024, 176.45613574105155,
175.11888110615263, 75.76951939625124] Fitness: 2.4825848019190864
Generation 443 Best Solution: [183.0275546399994, 177.37649116849522,

174.47486015616488, 75.76951939625124] Fitness: 2.499897217190612
Generation 444 Best Solution: [183.87258856267024, 177.37649116849522,
174.32949876618005, 74.9890564659411] Fitness: 1.8465845312721685
Generation 445 Best Solution: [183.87258856267024, 177.37649116849522,
175.11888110615263, 74.9890564659411] Fitness: 2.715522488069422
Generation 446 Best Solution: [183.87258856267024, 177.33934857912828,
175.11888110615263, 74.9890564659411] Fitness: 2.6381556382976883
Generation 447 Best Solution: [183.87258856267024, 177.33934857912828,
175.11888110615263, 74.9890564659411] Fitness: 2.6381556382976883
Generation 448 Best Solution: [183.87258856267024, 177.2491711125924,
175.11888110615263, 74.80669989267673] Fitness: 2.3919168642862005
Generation 449 Best Solution: [183.87258856267024, 176.74874357740157,
175.11888110615263, 74.9890564659411] Fitness: 1.7375002529024424
Generation 450 Best Solution: [183.87258856267024, 177.2491711125924,
174.2797425848629, 75.43046480565016] Fitness: 1.9847413743789715
Generation 451 Best Solution: [183.87258856267024, 177.2491711125924,
174.2797425848629, 75.43046480565016] Fitness: 1.9847413743789715
Generation 452 Best Solution: [182.84094642498354, 177.2491711125924,
175.11888110615263, 74.9890564659411] Fitness: 2.161545439882719
Generation 453 Best Solution: [183.87258856267024, 176.17957384941238,
175.11888110615263, 75.76951939625124] Fitness: 2.417879253264618
Generation 454 Best Solution: [183.2065283206535, 177.33934857912828,
174.09608686543376, 75.76951939625124] Fitness: 2.276597297767754
Generation 455 Best Solution: [183.87258856267024, 176.17957384941238,
175.11888110615263, 75.76951939625124] Fitness: 2.417879253264618
Generation 456 Best Solution: [183.87258856267024, 177.2491711125924,
174.09608686543376, 75.76951939625124] Fitness: 2.508440379631284
Generation 457 Best Solution: [183.81053776646272, 176.90359422314745,
175.11888110615263, 74.59043946702882] Fitness: 1.7688575568734135
Generation 458 Best Solution: [183.36503808073067, 176.50144817861414,
175.11888110615263, 75.76951939625124] Fitness: 2.122610927262119
Generation 459 Best Solution: [183.87258856267024, 176.90359422314745,
174.09608686543376, 75.76951939625124] Fitness: 1.9615794399765019
Generation 460 Best Solution: [183.38776476986507, 177.2491711125924,
174.09608686543376, 75.76951939625124] Fitness: 2.132280323415806
Generation 461 Best Solution: [183.87258856267024, 177.2491711125924,
174.09608686543376, 75.76951939625124] Fitness: 2.508440379631284
Generation 462 Best Solution: [183.87258856267024, 176.1342994146163,
174.2209720264276, 75.76951939625124] Fitness: 1.51845605845172
Generation 463 Best Solution: [183.38776476986507, 176.66040988486623,
175.11888110615263, 75.76951939625124] Fitness: 2.2415263307644135
Generation 464 Best Solution: [183.38776476986507, 177.37649116849522,
175.11888110615263, 74.33001154925596] Fitness: 2.3706638344964057
Generation 465 Best Solution: [183.87258856267024, 176.53690915009213,
173.90873459499315, 75.76951939625124] Fitness: 1.647641458016522
Generation 466 Best Solution: [183.87258856267024, 176.53690915009213,
173.90873459499315, 75.76951939625124] Fitness: 1.647641458016522
Generation 467 Best Solution: [183.87258856267024, 176.53690915009213,

173.90873459499315, 75.76951939625124] Fitness: 1.647641458016522
Generation 468 Best Solution: [182.90247156833715, 176.61148154599604,
174.18529869611237, 75.76951939625124] Fitness: 1.3363606848124765
Generation 469 Best Solution: [182.90247156833715, 176.61148154599604,
175.11888110615263, 75.76951939625124] Fitness: 2.2443669219821913
Generation 470 Best Solution: [182.90247156833715, 176.61148154599604,
175.11888110615263, 75.76951939625124] Fitness: 2.2443669219821913
Generation 471 Best Solution: [182.90247156833715, 176.61148154599604,
174.03646943729123, 75.76951939625124] Fitness: 1.3348038469705512
Generation 472 Best Solution: [183.87258856267024, 177.37649116849522,
174.03646943729123, 75.04952062363668] Fitness: 1.8417583376636866
Generation 473 Best Solution: [183.87258856267024, 176.66466895586865,
175.11888110615263, 75.76951939625124] Fitness: 2.6212926989070877
Generation 474 Best Solution: [183.87258856267024, 177.17503803233728,
174.16903976915276, 75.76951939625124] Fitness: 2.376815605732733
Generation 475 Best Solution: [183.87258856267024, 177.17503803233728,
174.16903976915276, 75.76951939625124] Fitness: 2.376815605732733
Generation 476 Best Solution: [183.87258856267024, 177.17503803233728,
175.11888110615263, 75.04952062363668] Fitness: 2.361123842642055
Generation 477 Best Solution: [183.87258856267024, 177.17503803233728,
175.11888110615263, 74.46738356371476] Fitness: 2.2864011501396653
Generation 478 Best Solution: [183.87258856267024, 176.086289162796,
174.94923458007645, 75.76951939625124] Fitness: 2.1465234548392313
Generation 479 Best Solution: [183.87258856267024, 175.80596981240387,
174.94923458007645, 75.76951939625124] Fitness: 2.266186434486894
Generation 480 Best Solution: [183.87258856267024, 177.17503803233728,
175.11888110615263, 75.04952062363668] Fitness: 2.361123842642055
Generation 481 Best Solution: [183.87258856267024, 176.42759458481908,
175.11888110615263, 75.51876141614169] Fitness: 2.0425950850748658
Generation 482 Best Solution: [183.87258856267024, 177.17503803233728,
174.2734859016069, 75.51876141614169] Fitness: 1.9713836769850517
Generation 483 Best Solution: [183.87258856267024, 177.17503803233728,
175.11888110615263, 74.86284593448487] Fitness: 2.271542134268009
Generation 484 Best Solution: [183.87258856267024, 176.61707585484953,
174.64424829343437, 75.76951939625124] Fitness: 1.9275682939568552
Generation 485 Best Solution: [183.87258856267024, 177.09886362506455,
175.11888110615263, 75.04952062363668] Fitness: 2.2323788255398567
Generation 486 Best Solution: [183.87258856267024, 177.17503803233728,
175.11888110615263, 75.04952062363668] Fitness: 2.361123842642055
Generation 487 Best Solution: [182.82962440263574, 176.6802917987993,
175.11888110615263, 75.76951939625124] Fitness: 2.343033372786242
Generation 488 Best Solution: [183.87258856267024, 177.17503803233728,
174.5018226141351, 75.51876141614169] Fitness: 2.0861220663616997
Generation 489 Best Solution: [183.15170692612054, 177.37649116849522,
174.17786198186076, 75.67299008796603] Fitness: 2.1824238589706595
Generation 490 Best Solution: [183.87258856267024, 176.176165366825,
175.11888110615263, 75.76951939625124] Fitness: 2.4179300344553787
Generation 491 Best Solution: [183.87258856267024, 176.176165366825,

```
175.11888110615263, 75.76951939625124] Fitness: 2.4179300344553787
Generation 492 Best Solution: [183.13787000846838, 176.176165366825,
175.11888110615263, 75.76951939625124] Fitness: 2.011086282709033
Generation 493 Best Solution: [183.13787000846838, 176.176165366825,
174.1484650295572, 75.76951939625124] Fitness: 1.0990278928255952
Generation 494 Best Solution: [183.13787000846838, 176.176165366825,
175.11888110615263, 75.76951939625124] Fitness: 2.011086282709033
Generation 495 Best Solution: [182.84790388106424, 177.37649116849522,
175.11888110615263, 74.59246742216237] Fitness: 2.3385368524292782
Generation 496 Best Solution: [183.13787000846838, 176.176165366825,
175.11888110615263, 75.76951939625124] Fitness: 2.011086282709033
Generation 497 Best Solution: [183.51992872920303, 177.37649116849522,
175.11888110615263, 74.65684814896507] Fitness: 2.3197389167314957
Generation 498 Best Solution: [182.84790388106424, 176.1260262172729,
175.11888110615263, 75.76951939625124] Fitness: 2.117763766005954
Generation 499 Best Solution: [182.84790388106424, 176.1260262172729,
175.11888110615263, 74.59246742216237] Fitness: 1.0824949544960283
Generation 500 Best Solution: [182.84790388106424, 176.1260262172729,
175.11888110615263, 75.76951939625124] Fitness: 2.117763766005954
Optimal Solution for Simulation 18: [182.84790388106424, 176.1260262172729,
175.11888110615263, 75.76951939625124]
--------------------------------------------------
Simulation 19/32 with conditions: {'drought': False, 'flood': True,
'policy_change': True, 'social_event': False, 'economic_crisis': True}
Generation 1 Best Solution: [183.0260828876848, 175.69124189865855,
173.8268826299391, 75.2170175593855] Fitness: 0.631574392395063
Generation 2 Best Solution: [182.98044858703807, 176.44811513800036,
174.44132667474707, 75.3440329846306] Fitness: 0.6528699207636315
Generation 3 Best Solution: [183.1261195195026, 176.44811513800036,
174.44132667474707, 75.45063032723631] Fitness: 0.7532588693192982
Generation 4 Best Solution: [183.1261195195026, 176.38236236475376,
174.67318316872246, 75.45063032723631] Fitness: 0.9167031225242779
Generation 5 Best Solution: [183.1261195195026, 175.8334372343205,
174.13156412415404, 75.45063032723631] Fitness: 0.7041454444983153
Generation 6 Best Solution: [182.98044858703807, 177.3042645630266,
174.67318316872246, 75.45063032723631] Fitness: 2.058308229458101
Generation 7 Best Solution: [182.98044858703807, 176.51280672785893,
174.13156412415404, 75.70792236785209] Fitness: 1.1428556029323196
Generation 8 Best Solution: [182.79659942831626, 177.3042645630266,
174.67318316872246, 75.08556863249083] Fitness: 1.7743114545618424
Generation 9 Best Solution: [182.79659942831626, 177.3042645630266,
174.29858814838133, 75.45063032723631] Fitness: 1.9018775512608412
Generation 10 Best Solution: [183.37606520281898, 177.4103750109877,
174.13156412415404, 74.73123825595724] Fitness: 1.4503985929899537
Generation 11 Best Solution: [182.98044858703807, 177.3042645630266,
174.29858814838133, 75.70792236785209] Fitness: 2.2178402629557468
Generation 12 Best Solution: [183.37606520281898, 177.4103750109877,
174.29858814838133, 75.5431220916206] Fitness: 2.149969579468044
```

Generation 13 Best Solution: [182.79659942831626, 177.3042645630266,
174.5196671479682, 75.26182927180186] Fitness: 1.7934688962189895
Generation 14 Best Solution: [183.39258002508603, 177.4103750109877,
174.5196671479682, 75.26182927180186] Fitness: 1.910581321309367
Generation 15 Best Solution: [182.79659942831626, 177.4103750109877,
174.67318316872246, 74.7036199281204] Fitness: 1.8566014108214455
Generation 16 Best Solution: [183.48513843271363, 177.4103750109877,
174.67318316872246, 74.7036199281204] Fitness: 1.7882413150374228
Generation 17 Best Solution: [182.79659942831626, 177.4103750109877,
174.5196671479682, 74.35409141019952] Fitness: 1.8206394652210747
Generation 18 Best Solution: [183.48513843271363, 177.10926970044864,
174.5196671479682, 75.5431220916206] Fitness: 1.7294099615630638
Generation 19 Best Solution: [182.79659942831626, 176.81258345746957,
174.5196671479682, 75.70792236785209] Fitness: 1.6610637046830172
Generation 20 Best Solution: [182.79659942831626, 175.89137059817872,
173.8244212077091, 75.8838138520851] Fitness: 1.6575383312496745
Generation 21 Best Solution: [183.48513843271363, 176.05227006836162,
173.8244212077091, 75.2966195362516] Fitness: 0.5579465630126137
Generation 22 Best Solution: [182.79659942831626, 177.4103750109877,
174.29858814838133, 75.2966195362516] Fitness: 1.9355965401874324
Generation 23 Best Solution: [183.16183352507304, 177.4103750109877,
174.29858814838133, 75.2966195362516] Fitness: 1.7915210901611978
Generation 24 Best Solution: [182.79659942831626, 176.95635563116397,
174.29858814838133, 75.8107658813908] Fitness: 1.9231800655636935
Generation 25 Best Solution: [182.79659942831626, 177.4103750109877,
174.29858814838133, 74.65964130860458] Fitness: 1.5976802068500024
Generation 26 Best Solution: [183.58392643353585, 177.4103750109877,
174.29858814838133, 75.5431220916206] Fitness: 2.2580558321352044
Generation 27 Best Solution: [182.79659942831626, 175.77693867188435,
175.06586358563305, 74.80315597706468] Fitness: 1.2044480188754092
Generation 28 Best Solution: [182.79659942831626, 177.4103750109877,
174.29858814838133, 75.49276636883596] Fitness: 2.1888571005295967
Generation 29 Best Solution: [182.79659942831626, 176.22703518790402,
175.06586358563305, 75.8107658813908] Fitness: 2.191452848188028
Generation 30 Best Solution: [182.79659942831626, 176.04125883589182,
175.06586358563305, 75.8107658813908] Fitness: 2.2095929033488693
Generation 31 Best Solution: [182.79659942831626, 176.5582484397941,
175.06586358563305, 74.80315597706468] Fitness: 1.175876857220617
Generation 32 Best Solution: [182.79659942831626, 175.72228848017267,
175.06586358563305, 75.84344376412201] Fitness: 2.4557340772752534
Generation 33 Best Solution: [182.79659942831626, 176.8138789654561,
174.4292571891109, 75.84344376412201] Fitness: 1.871089387432475
Generation 34 Best Solution: [182.79659942831626, 176.36493317321285,
175.06586358563305, 75.8107658813908] Fitness: 2.218740674597711
Generation 35 Best Solution: [182.79659942831626, 175.8866803301351,
175.06586358563305, 75.8107658813908] Fitness: 2.2727232860525595
Generation 36 Best Solution: [182.79659942831626, 175.8866803301351,
175.06586358563305, 75.8285501265564] Fitness: 2.3094869056241434

Generation 37 Best Solution: [183.54371446926774, 175.8866803301351,
175.06586358563305, 75.8285501265564] Fitness: 2.2752687719860725
Generation 38 Best Solution: [183.58392643353585, 177.2218080904185,
174.3437042520417, 75.8107658813908] Fitness: 2.370671659215688
Generation 39 Best Solution: [183.58392643353585, 176.20404885676567,
175.06586358563305, 75.84344376412201] Fitness: 2.251123991320195
Generation 40 Best Solution: [182.79659942831626, 177.4103750109877,
174.3437042520417, 75.8107658813908] Fitness: 2.766402353781675
Generation 41 Best Solution: [182.88343300337326, 177.40290816655352,
174.3437042520417, 75.8107658813908] Fitness: 2.6934548593940617
Generation 42 Best Solution: [183.60596347139804, 177.0711708286182,
174.3437042520417, 75.84344376412201] Fitness: 2.1909896777044526
Generation 43 Best Solution: [183.0769337516193, 177.40290816655352,
175.06586358563305, 74.63996849547844] Fitness: 2.2565244583311275
Generation 44 Best Solution: [182.79659942831626, 177.40290816655352,
174.49690341186133, 74.74809720865449] Fitness: 1.689095227310526
Generation 45 Best Solution: [183.31251641473125, 176.60896052089527,
175.06586358563305, 75.84344376412201] Fitness: 2.2875857237435566
Generation 46 Best Solution: [183.0769337516193, 177.4103750109877,
174.1885328404752, 74.74809720865449] Fitness: 1.4404780419122067
Generation 47 Best Solution: [182.99511892859329, 176.45534748774932,
174.49690341186133, 75.81604695128307] Fitness: 1.4567966009631108
Generation 48 Best Solution: [183.58392643353585, 176.86832421980552,
175.06586358563305, 74.74809720865449] Fitness: 1.4575215972077276
Generation 49 Best Solution: [183.31251641473125, 177.4103750109877,
175.06586358563305, 74.78212240548862] Fitness: 2.2790064186405297
Generation 50 Best Solution: [183.0769337516193, 177.4103750109877,
174.11380281271573, 74.74809720865449] Fitness: 1.4344052565858616
Generation 51 Best Solution: [183.0769337516193, 177.4103750109877,
174.11380281271573, 74.74809720865449] Fitness: 1.4344052565858616
Generation 52 Best Solution: [183.60596347139804, 176.8366459585889,
175.06586358563305, 75.03859933668743] Fitness: 1.5442814221080332
Generation 53 Best Solution: [183.31251641473125, 176.8366459585889,
175.06586358563305, 75.81604695128307] Fitness: 2.4529616847664624
Generation 54 Best Solution: [183.0769337516193, 176.8366459585889,
175.06586358563305, 75.81604695128307] Fitness: 2.453182379697533
Generation 55 Best Solution: [183.0769337516193, 177.4103750109877,
175.06586358563305, 74.7445455054] Fitness: 2.2740316560013434
Generation 56 Best Solution: [183.31251641473125, 175.91660068202603,
175.06586358563305, 74.42019644875059] Fitness: 1.0380915567149922
Generation 57 Best Solution: [183.31251641473125, 176.57223963732145,
175.06586358563305, 75.81604695128307] Fitness: 2.203424224073388
Generation 58 Best Solution: [183.31251641473125, 176.57223963732145,
175.06586358563305, 75.8107658813908] Fitness: 2.192567476478703
Generation 59 Best Solution: [183.0769337516193, 177.4103750109877,
175.1222161551159, 74.52517223419623] Fitness: 2.396772437494514
Generation 60 Best Solution: [183.68330799256427, 176.57223963732145,
175.06586358563305, 74.42019644875059] Fitness: 1.311036832529349

Generation 61 Best Solution: [183.0769337516193, 176.8366459585889,
174.7257203254379, 75.8107658813908] Fitness: 1.95217531752685
Generation 62 Best Solution: [183.09642219750353, 176.8366459585889,
174.08977566609914, 75.8107658813908] Fitness: 1.5986691125334185
Generation 63 Best Solution: [183.60596347139804, 176.8366459585889,
174.08977566609914, 75.8107658813908] Fitness: 1.7437798000338351
Generation 64 Best Solution: [183.60596347139804, 176.8366459585889,
174.08977566609914, 75.8107658813908] Fitness: 1.7437798000338351
Generation 65 Best Solution: [183.60596347139804, 175.7853477007182,
174.08977566609914, 74.52517223419623] Fitness: 0.3792513526179997
Generation 66 Best Solution: [183.31251641473125, 176.8366459585889,
174.08977566609914, 75.8107658813908] Fitness: 1.6023118114090316
Generation 67 Best Solution: [183.31251641473125, 176.8366459585889,
175.06586358563305, 74.53106804540522] Fitness: 1.3127152644646283
Generation 68 Best Solution: [183.60596347139804, 177.4103750109877,
175.06586358563305, 74.53106804540522] Fitness: 2.4346971883329886
Generation 69 Best Solution: [183.60596347139804, 176.66515404280838,
175.06586358563305, 75.8107658813908] Fitness: 2.407176341005968
Generation 70 Best Solution: [183.60596347139804, 175.63230101658803,
175.06586358563305, 75.8107658813908] Fitness: 2.4805288333151636
Generation 71 Best Solution: [183.60596347139804, 175.63230101658803,
175.06586358563305, 75.8107658813908] Fitness: 2.4805288333151636
Generation 72 Best Solution: [183.0902128473729, 177.4103750109877,
174.71859221589904, 75.02634143414225] Fitness: 1.8749593127866098
Generation 73 Best Solution: [183.0902128473729, 175.90260312893903,
174.57887946593218, 75.8107658813908] Fitness: 1.4927227298861259
Generation 74 Best Solution: [183.74507109686863, 175.63230101658803,
175.06586358563305, 75.02634143414225] Fitness: 1.5555199567758813
Generation 75 Best Solution: [183.3428695800782, 176.61136674751512,
175.06586358563305, 75.8107658813908] Fitness: 2.2287878804343606
Generation 76 Best Solution: [183.60596347139804, 176.986361667464,
174.70736581978983, 75.8107658813908] Fitness: 2.270887111920138
Generation 77 Best Solution: [183.60596347139804, 176.986361667464,
174.70736581978983, 75.8107658813908] Fitness: 2.270887111920138
Generation 78 Best Solution: [183.60596347139804, 176.986361667464,
174.70736581978983, 75.8107658813908] Fitness: 2.270887111920138
Generation 79 Best Solution: [183.60596347139804, 176.72815233108602,
174.70736581978983, 75.8107658813908] Fitness: 1.9551443125007986
Generation 80 Best Solution: [183.60596347139804, 176.1723601065358,
175.06586358563305, 75.8107658813908] Fitness: 2.199178794775647
Generation 81 Best Solution: [183.60596347139804, 177.4103750109877,
175.06586358563305, 74.57471443211105] Fitness: 2.423959099380204
Generation 82 Best Solution: [183.50939180479392, 176.72815233108602,
175.06586358563305, 75.8107658813908] Fitness: 2.4018202248204203
Generation 83 Best Solution: [183.60596347139804, 177.4103750109877,
174.07598990633312, 75.8107658813908] Fitness: 2.724892470590965
Generation 84 Best Solution: [183.60596347139804, 177.2007876130446,
175.06586358563305, 74.35917160303636] Fitness: 2.082945274202728

Generation 85 Best Solution: [182.9974216214374, 176.2414404893149,
175.06586358563305, 75.8107658813908] Fitness: 2.08331351802596
Generation 86 Best Solution: [182.9974216214374, 176.2414404893149,
175.06586358563305, 75.8107658813908] Fitness: 2.08331351802596
Generation 87 Best Solution: [183.60596347139804, 176.2414404893149,
175.06586358563305, 75.8107658813908] Fitness: 2.2016110625307768
Generation 88 Best Solution: [183.60596347139804, 175.9321650001983,
175.06586358563305, 75.8107658813908] Fitness: 2.258548474613643
Generation 89 Best Solution: [183.60596347139804, 175.9321650001983,
175.06586358563305, 75.8107658813908] Fitness: 2.258548474613643
Generation 90 Best Solution: [183.60596347139804, 177.4103750109877,
174.07598990633312, 75.8107658813908] Fitness: 2.724892470590965
Generation 91 Best Solution: [183.60596347139804, 176.22099243761252,
174.07598990633312, 75.8107658813908] Fitness: 1.3607887480399596
Generation 92 Best Solution: [183.60596347139804, 177.4103750109877,
174.07598990633312, 75.8107658813908] Fitness: 2.724892470590965
Generation 93 Best Solution: [183.60596347139804, 177.4103750109877,
174.07598990633312, 75.8107658813908] Fitness: 2.724892470590965
Generation 94 Best Solution: [183.60596347139804, 177.4103750109877,
174.07598990633312, 74.52342399456799] Fitness: 1.5977413971098153
Generation 95 Best Solution: [183.60596347139804, 176.09173410623492,
174.07598990633312, 74.52342399456799] Fitness: 0.24101519261603566
Generation 96 Best Solution: [183.60596347139804, 175.9724989178036,
175.06586358563305, 75.8107658813908] Fitness: 2.241218818008064
Generation 97 Best Solution: [183.50939180479392, 176.76380196980637,
175.06586358563305, 75.8107658813908] Fitness: 2.4381689689499546
Generation 98 Best Solution: [183.60596347139804, 176.25889870118317,
175.06586358563305, 75.8107658813908] Fitness: 2.203605187311039
Generation 99 Best Solution: [183.60596347139804, 176.25889870118317,
174.95908209647686, 75.8107658813908] Fitness: 2.0269799720111146
Generation 100 Best Solution: [183.60596347139804, 176.25889870118317,
173.94631810071186, 75.8107658813908] Fitness: 1.3870274307124406
Generation 101 Best Solution: [183.50939180479392, 177.4103750109877,
174.07598990633312, 75.8107658813908] Fitness: 2.660976887579012
Generation 102 Best Solution: [183.60596347139804, 175.92569345649102,
175.06586358563305, 75.8107658813908] Fitness: 2.261605567823495
Generation 103 Best Solution: [183.60596347139804, 177.4103750109877,
175.06586358563305, 74.74374149819343] Fitness: 2.4151959612221523
Generation 104 Best Solution: [183.60596347139804, 176.3297908288287,
175.06586358563305, 75.8107658813908] Fitness: 2.2174213852920017
Generation 105 Best Solution: [183.83464852654743, 177.16314200783557,
175.06586358563305, 74.58757719907238] Fitness: 2.144498571964582
Generation 106 Best Solution: [183.60596347139804, 177.16314200783557,
175.06586358563305, 74.58757719907238] Fitness: 1.9252308473578599
Generation 107 Best Solution: [183.60596347139804, 177.4103750109877,
175.06586358563305, 74.58757719907238] Fitness: 2.4214581995719486
Generation 108 Best Solution: [183.60596347139804, 176.32866349422315,
175.06586358563305, 75.8107658813908] Fitness: 2.217129869268359

Generation 109 Best Solution: [183.60596347139804, 176.32866349422315,
175.06586358563305, 75.8107658813908] Fitness: 2.217129869268359
Generation 110 Best Solution: [183.60596347139804, 175.95050472474153,
175.08136676901674, 75.8107658813908] Fitness: 2.2776749020340628
Generation 111 Best Solution: [183.60596347139804, 177.4103750109877,
174.80056598845093, 74.61279228854082] Fitness: 2.017005697331205
Generation 112 Best Solution: [183.60596347139804, 177.16314200783557,
175.06586358563305, 74.44547341389143] Fitness: 1.9696286241372933
Generation 113 Best Solution: [183.60596347139804, 177.16314200783557,
175.06586358563305, 74.44547341389143] Fitness: 1.9696286241372933
Generation 114 Best Solution: [183.83464852654743, 176.43551500863947,
175.08136676901674, 75.8107658813908] Fitness: 2.5017171185063543
Generation 115 Best Solution: [183.60596347139804, 176.43551500863947,
175.08136676901674, 75.8107658813908] Fitness: 2.2824493938996326
Generation 116 Best Solution: [183.60596347139804, 176.38385785420522,
175.08136676901674, 75.8107658813908] Fitness: 2.261501676339747
Generation 117 Best Solution: [183.83464852654743, 175.96547203232643,
175.06586358563305, 75.8107658813908] Fitness: 2.4632919838836367
Generation 118 Best Solution: [183.60596347139804, 177.28665380641377,
175.08136676901674, 74.63462289072308] Fitness: 2.1799828850145104
Generation 119 Best Solution: [183.5382029052793, 177.4103750109877,
175.06586358563305, 74.66561628149675] Fitness: 2.3661325825191892
Generation 120 Best Solution: [183.24057429656582, 177.4103750109877,
175.06586358563305, 74.66561628149675] Fitness: 2.260612613593487
Generation 121 Best Solution: [183.83464852654743, 177.12877125344212,
175.06586358563305, 74.34882469577298] Fitness: 2.1800120097890185
Generation 122 Best Solution: [183.83464852654743, 177.4103750109877,
173.89635804019704, 74.66278650119139] Fitness: 1.8324809124918005
Generation 123 Best Solution: [183.83464852654743, 177.4103750109877,
173.89635804019704, 74.66278650119139] Fitness: 1.8324809124918005
Generation 124 Best Solution: [183.83464852654743, 177.4103750109877,
173.89635804019704, 74.66278650119139] Fitness: 1.8324809124918005
Generation 125 Best Solution: [183.24057429656582, 176.75966635308333,
175.06586358563305, 75.8107658813908] Fitness: 2.3455993051946153
Generation 126 Best Solution: [183.5382029052793, 176.33885307758715,
175.06586358563305, 75.8107658813908] Fitness: 2.173219528675114
Generation 127 Best Solution: [183.5382029052793, 176.33885307758715,
175.06586358563305, 75.185737199525] Fitness: 1.2479986827612244
Generation 128 Best Solution: [183.83464852654743, 176.72114048186927,
175.06586358563305, 74.43395462358268] Fitness: 1.585110929032983
Generation 129 Best Solution: [183.84845482089668, 176.72114048186927,
175.06586358563305, 74.43395462358268] Fitness: 1.601405459957837
Generation 130 Best Solution: [183.83464852654743, 175.98241422372823,
175.06586358563305, 75.8107658813908] Fitness: 2.4566812965026603
Generation 131 Best Solution: [183.83464852654743, 175.98241422372823,
175.06586358563305, 74.40774616248883] Fitness: 1.376432185807513
Generation 132 Best Solution: [183.83464852654743, 177.4103750109877,
175.06586358563305, 74.61764868207287] Fitness: 2.6360579716775696

Generation 133 Best Solution: [183.4669502670223, 177.4103750109877,
175.06586358563305, 74.61764868207287] Fitness: 2.330171793294896
Generation 134 Best Solution: [183.83464852654743, 177.4103750109877,
175.06586358563305, 74.73087368387017] Fitness: 2.6332961169625086
Generation 135 Best Solution: [183.83464852654743, 177.4103750109877,
175.06586358563305, 74.52800865066311] Fitness: 2.6548480656774425
Generation 136 Best Solution: [183.83464852654743, 176.4033822893693,
175.06586358563305, 75.8107658813908] Fitness: 2.4607395526588647
Generation 137 Best Solution: [183.22331654798796, 177.4103750109877,
173.96289019022092, 75.8107658813908] Fitness: 2.5896010447333744
Generation 138 Best Solution: [183.48814509489662, 177.4103750109877,
175.06586358563305, 75.29089560605111] Fitness: 2.668923487255641
Generation 139 Best Solution: [183.83464852654743, 175.80743134407317,
175.06586358563305, 74.28803360263245] Fitness: 1.5442252592884922
Generation 140 Best Solution: [183.83464852654743, 175.80743134407317,
175.06586358563305, 74.28803360263245] Fitness: 1.5442252592884922
Generation 141 Best Solution: [183.87740551607183, 176.912124747345,
175.06586358563305, 75.8107658813908] Fitness: 2.949091286542169
Generation 142 Best Solution: [183.83464852654743, 176.23079077087297,
175.06586358563305, 75.8107658813908] Fitness: 2.4199356589702106
Generation 143 Best Solution: [183.87740551607183, 176.912124747345,
175.06586358563305, 75.38210104569082] Fitness: 2.2376902058517074
Generation 144 Best Solution: [183.39890029388417, 177.4103750109877,
173.7606661397934, 75.8107658813908] Fitness: 2.717184792663164
Generation 145 Best Solution: [183.83464852654743, 176.96292068237068,
175.06586358563305, 74.73617015295177] Fitness: 1.8174431268973659
Generation 146 Best Solution: [183.83464852654743, 176.912124747345,
173.7606661397934, 75.8107658813908] Fitness: 2.166750844894622
Generation 147 Best Solution: [183.83464852654743, 176.8008005960963,
175.06586358563305, 75.8107658813908] Fitness: 2.7615306193088798
Generation 148 Best Solution: [183.84845482089668, 176.41280675943418,
175.06586358563305, 75.8107658813908] Fitness: 2.48082844490116
Generation 149 Best Solution: [183.83464852654743, 177.4103750109877,
175.06586358563305, 75.04149815198365] Fitness: 2.7459286646112995
Generation 150 Best Solution: [182.83979461347732, 177.4103750109877,
175.06586358563305, 75.29216972333593] Fitness: 2.707348439799144
Generation 151 Best Solution: [183.87740551607183, 177.4103750109877,
174.80132308654345, 75.04149815198365] Fitness: 2.3980549931645574
Generation 152 Best Solution: [183.83464852654743, 177.4103750109877,
175.06586358563305, 75.04149815198365] Fitness: 2.7459286646112995
Generation 153 Best Solution: [183.83464852654743, 177.4103750109877,
174.53940818444346, 75.32889072265493] Fitness: 2.337985798679235
Generation 154 Best Solution: [183.87740551607183, 176.16365238259067,
175.06586358563305, 75.8107658813908] Fitness: 2.470351448891793
Generation 155 Best Solution: [183.82913994602936, 176.16365238259067,
175.06586358563305, 75.8107658813908] Fitness: 2.4123541660301555
Generation 156 Best Solution: [183.15165030600005, 176.16365238259067,
175.06586358563305, 75.8107658813908] Fitness: 2.0471991711438666

Generation 157 Best Solution: [183.82913994602936, 176.16365238259067,
175.06586358563305, 75.04149815198365] Fitness: 1.3749282449945954
Generation 158 Best Solution: [183.83464852654743, 176.16365238259067,
175.06586358563305, 74.45116632344293] Fitness: 1.31803980514221
Generation 159 Best Solution: [183.82913994602936, 176.16365238259067,
175.06586358563305, 75.30846302699263] Fitness: 1.6125175211071687
Generation 160 Best Solution: [183.87740551607183, 176.16365238259067,
175.06586358563305, 75.30846302699263] Fitness: 1.6705148039688065
Generation 161 Best Solution: [182.83979461347732, 177.4103750109877,
175.06586358563305, 74.61468223471653] Fitness: 2.378505111212155
Generation 162 Best Solution: [182.83979461347732, 177.4103750109877,
175.06586358563305, 74.61468223471653] Fitness: 2.378505111212155
Generation 163 Best Solution: [182.83979461347732, 177.4103750109877,
175.06586358563305, 74.85950588774303] Fitness: 2.4006238698973745
Generation 164 Best Solution: [183.87740551607183, 177.4103750109877,
175.06586358563305, 74.85950588774303] Fitness: 2.7101568589293583
Generation 165 Best Solution: [183.87740551607183, 177.4103750109877,
175.06586358563305, 74.85950588774303] Fitness: 2.7101568589293583
Generation 166 Best Solution: [183.15165030600005, 177.4103750109877,
175.06586358563305, 74.85950588774303] Fitness: 2.287004581181432
Generation 167 Best Solution: [183.5213401372377, 177.4103750109877,
175.06586358563305, 74.61679916594697] Fitness: 2.3599686395038577
Generation 168 Best Solution: [183.15165030600005, 177.4103750109877,
175.06586358563305, 74.85950588774303] Fitness: 2.287004581181432
Generation 169 Best Solution: [183.7565329985948, 176.31742937920424,
175.06586358563305, 75.83916872437273] Fitness: 2.406971392225686
Generation 170 Best Solution: [183.87740551607183, 177.4103750109877,
174.1877661287628, 74.32866593440528] Fitness: 1.9667757087811029
Generation 171 Best Solution: [183.7565329985948, 176.31742937920424,
175.06586358563305, 75.83571488009183] Fitness: 2.399719437125998
Generation 172 Best Solution: [183.7565329985948, 176.31742937920424,
175.06586358563305, 75.83916872437273] Fitness: 2.406971392225686
Generation 173 Best Solution: [183.5213401372377, 176.31742937920424,
175.06586358563305, 75.83916872437273] Fitness: 2.216410611291087
Generation 174 Best Solution: [183.5213401372377, 177.4103750109877,
175.06586358563305, 75.23859212959091] Fitness: 2.6325631399285627
Generation 175 Best Solution: [183.05271333275994, 175.72359545810994,
175.06586358563305, 75.83571488009183] Fitness: 2.311819379961561
Generation 176 Best Solution: [182.81331702258925, 177.4103750109877,
175.06586358563305, 75.23859212959091] Fitness: 2.668647680914453
Generation 177 Best Solution: [183.1819461238819, 177.4103750109877,
175.06586358563305, 74.85395150299622] Fitness: 2.2837159638663556
Generation 178 Best Solution: [183.38864826493338, 177.4103750109877,
175.06586358563305, 74.85395150299622] Fitness: 2.317709318736772
Generation 179 Best Solution: [183.38864826493338, 177.4103750109877,
175.06586358563305, 75.69992840337878] Fitness: 3.228108391214363
Generation 180 Best Solution: [183.87740551607183, 176.29892497384384,
175.06586358563305, 75.83916872437273] Fitness: 2.540128473935436

Generation 181 Best Solution: [183.43390682327, 176.29892497384384,
175.06586358563305, 75.83916872437273] Fitness: 2.1672509872207266
Generation 182 Best Solution: [183.43390682327, 176.29892497384384,
175.06586358563305, 74.56133691549128] Fitness: 0.9710549392262773
Generation 183 Best Solution: [183.1819461238819, 177.2130146067131,
174.17361058535857, 75.83916872437273] Fitness: 2.228136120183466
Generation 184 Best Solution: [183.43390682327, 177.2130146067131,
173.97970727742552, 74.6864663287404] Fitness: 1.0798542523099774
Generation 185 Best Solution: [182.81331702258925, 177.4103750109877,
175.06586358563305, 74.63239675933164] Fitness: 2.3942581785400376
Generation 186 Best Solution: [182.81331702258925, 176.5760699145965,
175.06586358563305, 75.83916872437273] Fitness: 2.374882562331534
Generation 187 Best Solution: [182.81331702258925, 176.5760699145965,
175.06586358563305, 75.83916872437273] Fitness: 2.374882562331534
Generation 188 Best Solution: [182.81331702258925, 176.34905160021853,
175.06586358563305, 75.83571488009183] Fitness: 2.2536520964774214
Generation 189 Best Solution: [182.8578599693065, 177.4103750109877,
175.06586358563305, 74.77252503554452] Fitness: 2.368802361822697
Generation 190 Best Solution: [182.81331702258925, 177.4103750109877,
173.90828057466334, 74.77252503554452] Fitness: 1.593937073929631
Generation 191 Best Solution: [182.8578599693065, 177.4103750109877,
173.90828057466334, 74.77252503554452] Fitness: 1.5646834294340304
Generation 192 Best Solution: [182.8578599693065, 175.96085461271122,
175.06586358563305, 74.39481127619061] Fitness: 1.122331225795894
Generation 193 Best Solution: [182.81331702258925, 176.5760699145965,
175.06586358563305, 75.60647609920287] Fitness: 1.9350070878745718
Generation 194 Best Solution: [182.81331702258925, 177.4103750109877,
173.86749773379762, 75.10963051460554] Fitness: 1.7662552111935033
Generation 195 Best Solution: [182.8578599693065, 177.4103750109877,
173.75476018756564, 74.77252503554452] Fitness: 1.6418183728771405
Generation 196 Best Solution: [182.8578599693065, 176.44425878174678,
173.86749773379762, 75.83571488009183] Fitness: 1.4729148518504762
Generation 197 Best Solution: [183.40749874793175, 177.4103750109877,
174.4395334846367, 75.03307495994662] Fitness: 1.6695050411964465
Generation 198 Best Solution: [182.8578599693065, 176.44425878174678,
174.4395334846367, 75.03307495994662] Fitness: 0.4273105050855274
Generation 199 Best Solution: [182.8578599693065, 176.44425878174678,
175.06586358563305, 75.82374844324124] Fitness: 2.2357835993795168
Generation 200 Best Solution: [182.8578599693065, 176.41206407133365,
175.06586358563305, 75.84362341280318] Fitness: 2.263236720664963
Generation 201 Best Solution: [182.8578599693065, 177.4103750109877,
175.06586358563305, 74.89056978320883] Fitness: 2.3998275557404165
Generation 202 Best Solution: [182.8578599693065, 177.4103750109877,
174.32807065002555, 75.82374844324124] Fitness: 2.7454711182305482
Generation 203 Best Solution: [182.8578599693065, 175.96623050268096,
175.06586358563305, 75.83916872437273] Fitness: 2.252606963368963
Generation 204 Best Solution: [182.8578599693065, 176.37422618695908,
175.06586358563305, 75.83571488009183] Fitness: 2.2323983156205847

Generation 205 Best Solution: [182.8578599693065, 175.6849132396336,
175.06586358563305, 75.68645538080636] Fitness: 2.1387573844271213
Generation 206 Best Solution: [182.8578599693065, 176.04385550761904,
175.06586358563305, 75.83571488009183] Fitness: 2.219474852602931
Generation 207 Best Solution: [182.8578599693065, 176.01098913169562,
175.06586358563305, 75.83571488009183] Fitness: 2.229089357258646
Generation 208 Best Solution: [182.8578599693065, 176.01098913169562,
175.06586358563305, 75.83571488009183] Fitness: 2.229089357258646
Generation 209 Best Solution: [182.8578599693065, 176.01098913169562,
175.06586358563305, 75.84362341280318] Fitness: 2.2457268828000183
Generation 210 Best Solution: [182.8578599693065, 176.5639059044618,
175.06586358563305, 75.84362341280318] Fitness: 2.3465211045436254
Generation 211 Best Solution: [182.8578599693065, 176.5639059044618,
175.06586358563305, 75.83571488009183] Fitness: 2.329883579002252
Generation 212 Best Solution: [183.5854793782747, 175.57861072324843,
175.06586358563305, 75.83571488009183] Fitness: 2.5743633873774865
Generation 213 Best Solution: [183.5854793782747, 175.57861072324843,
175.06586358563305, 75.83571488009183] Fitness: 2.5743633873774865
Generation 214 Best Solution: [182.8578599693065, 177.39407605061672,
174.77390709481392, 74.64085927673116] Fitness: 1.894508106406791
Generation 215 Best Solution: [182.8578599693065, 177.39407605061672,
175.06586358563305, 75.63640391894903] Fitness: 3.1479845899774244
Generation 216 Best Solution: [182.8578599693065, 177.4103750109877,
175.06586358563305, 74.92430410717833] Fitness: 2.413368735878898
Generation 217 Best Solution: [182.8578599693065, 177.4103750109877,
173.79599942246563, 74.92430410717833] Fitness: 1.6614367164250752
Generation 218 Best Solution: [182.8578599693065, 175.98041678301502,
175.06586358563305, 75.83571488009183] Fitness: 2.2398036040593583
Generation 219 Best Solution: [182.8578599693065, 175.98041678301502,
175.06586358563305, 75.83571488009183] Fitness: 2.2398036040593583
Generation 220 Best Solution: [182.8578599693065, 175.98041678301502,
175.06586358563305, 75.83571488009183] Fitness: 2.2398036040593583
Generation 221 Best Solution: [182.8578599693065, 175.98041678301502,
175.06586358563305, 75.83571488009183] Fitness: 2.2398036040593583
Generation 222 Best Solution: [182.8578599693065, 175.7151856819342,
174.89041760945082, 75.83571488009183] Fitness: 2.12518680700849
Generation 223 Best Solution: [182.8578599693065, 176.42856352603562,
175.06586358563305, 75.83571488009183] Fitness: 2.2536101574363254
Generation 224 Best Solution: [183.5854793782747, 176.75555663901193,
174.89041760945082, 75.86689524474129] Fitness: 2.3172842069311677
Generation 225 Best Solution: [182.8578599693065, 177.4103750109877,
174.89041760945082, 75.36626114587406] Fitness: 2.5035245581564
Generation 226 Best Solution: [182.8578599693065, 177.39407605061672,
174.54357849987494, 75.83571488009183] Fitness: 2.863813073086556
Generation 227 Best Solution: [182.8578599693065, 176.6717940799635,
174.89041760945082, 75.83571488009183] Fitness: 2.135442272904229
Generation 228 Best Solution: [182.8578599693065, 176.6717940799635,
174.54357849987494, 75.83571488009183] Fitness: 1.7488837773968207

Generation 229 Best Solution: [182.8578599693065, 176.6717940799635,
174.54357849987494, 75.83571488009183] Fitness: 1.7488837773968207
Generation 230 Best Solution: [182.8578599693065, 176.6717940799635,
174.54357849987494, 75.83571488009183] Fitness: 1.7488837773968207
Generation 231 Best Solution: [182.8578599693065, 176.7121846227596,
174.89041760945082, 75.83571488009183] Fitness: 2.1726429335221673
Generation 232 Best Solution: [182.8578599693065, 176.7121846227596,
175.11629949249232, 75.83571488009183] Fitness: 2.5425103663944437
Generation 233 Best Solution: [182.8578599693065, 176.7121846227596,
174.54357849987494, 75.83571488009183] Fitness: 1.7860844380147594
Generation 234 Best Solution: [183.5854793782747, 176.7121846227596,
174.54357849987494, 75.83571488009183] Fitness: 1.821203978958582
Generation 235 Best Solution: [183.5854793782747, 176.7121846227596,
174.54357849987494, 75.83571488009183] Fitness: 1.821203978958582
Generation 236 Best Solution: [182.8578599693065, 176.7121846227596,
174.54357849987494, 75.83571488009183] Fitness: 1.7860844380147594
Generation 237 Best Solution: [182.8578599693065, 176.7121846227596,
174.54357849987494, 75.83571488009183] Fitness: 1.7860844380147594
Generation 238 Best Solution: [182.8578599693065, 176.7121846227596,
174.89041760945082, 74.8360338998175] Fitness: 0.9892499545676432
Generation 239 Best Solution: [183.03349747510597, 177.39407605061672,
174.54357849987494, 75.83571488009183] Fitness: 2.783772055487245
Generation 240 Best Solution: [183.5854793782747, 177.39407605061672,
174.54357849987494, 74.4348538033816] Fitness: 1.7537626484309792
Generation 241 Best Solution: [183.5854793782747, 176.8733926093981,
174.54357849987494, 75.83571488009183] Fitness: 1.9993536765092248
Generation 242 Best Solution: [183.5854793782747, 177.20373580259275,
174.54357849987494, 75.83571488009183] Fitness: 2.5126739163251326
Generation 243 Best Solution: [183.5854793782747, 176.21029646276074,
173.75441357325252, 75.83571488009183] Fitness: 1.5094320572849038
Generation 244 Best Solution: [183.5854793782747, 176.21029646276074,
175.06586358563305, 75.83571488009183] Fitness: 2.2361931980775225
Generation 245 Best Solution: [182.8578599693065, 176.26451652160802,
175.06586358563305, 75.83571488009183] Fitness: 2.2060031390169987
Generation 246 Best Solution: [182.8578599693065, 176.26451652160802,
175.06586358563305, 75.83571488009183] Fitness: 2.2060031390169987
Generation 247 Best Solution: [182.8578599693065, 177.39407605061672,
174.6684770465515, 75.27990636032729] Fitness: 2.0945001442716213
Generation 248 Best Solution: [182.8578599693065, 177.39407605061672,
174.6684770465515, 75.27990636032729] Fitness: 2.0945001442716213
Generation 249 Best Solution: [183.5854793782747, 177.39407605061672,
174.3743412726924, 75.83571488009183] Fitness: 2.7900592505080257
Generation 250 Best Solution: [183.5854793782747, 177.39407605061672,
175.06586358563305, 74.53513864908581] Fitness: 2.382417536891515
Generation 251 Best Solution: [183.5854793782747, 177.39407605061672,
174.39903749304094, 75.83571488009183] Fitness: 2.802687559789707
Generation 252 Best Solution: [183.39689721291217, 177.39407605061672,
175.06586358563305, 75.04673736821853] Fitness: 2.377024953290461

Generation 253 Best Solution: [182.8578599693065, 176.15617960323252,
175.06586358563305, 75.83571488009183] Fitness: 2.2015065981607576
Generation 254 Best Solution: [182.8578599693065, 176.15617960323252,
175.06586358563305, 75.83571488009183] Fitness: 2.2015065981607576
Generation 255 Best Solution: [182.8578599693065, 177.32648192916216,
175.06586358563305, 75.12135644353826] Fitness: 2.3530940138036103
Generation 256 Best Solution: [183.5854793782747, 177.39407605061672,
173.89642601726516, 75.83571488009183] Fitness: 2.7649975181143125
Generation 257 Best Solution: [182.8578599693065, 175.8362136662142,
175.06586358563305, 75.83571488009183] Fitness: 2.3133519621501453
Generation 258 Best Solution: [182.8578599693065, 176.0710536413192,
174.88651012931456, 75.83571488009183] Fitness: 1.9282291718744
Generation 259 Best Solution: [183.5854793782747, 177.39407605061672,
173.83609084655396, 75.3700058118419] Fitness: 2.0131407260923755
Generation 260 Best Solution: [182.8578599693065, 177.39407605061672,
175.06586358563305, 74.28062393566073] Fitness: 2.4773105930626165
Generation 261 Best Solution: [183.5854793782747, 176.44080421120069,
175.06586358563305, 75.83571488009183] Fitness: 2.2942522285082783
Generation 262 Best Solution: [183.5854793782747, 177.39407605061672,
175.06586358563305, 75.45468053277962] Fitness: 2.8984085826166357
Generation 263 Best Solution: [182.8578599693065, 177.39407605061672,
175.06586358563305, 74.8593880592538] Fitness: 2.3530084633322
Generation 264 Best Solution: [182.8578599693065, 177.39407605061672,
175.06586358563305, 74.8593880592538] Fitness: 2.3530084633322
Generation 265 Best Solution: [182.8578599693065, 177.39407605061672,
173.9024005777892, 75.78774149141934] Fitness: 2.629140844976667
Generation 266 Best Solution: [183.196274585724, 177.39407605061672,
175.06586358563305, 74.83031625404399] Fitness: 2.2407360481233884
Generation 267 Best Solution: [183.196274585724, 177.39407605061672,
175.06586358563305, 74.83031625404399] Fitness: 2.2407360481233884
Generation 268 Best Solution: [183.80774835665255, 177.39407605061672,
175.06586358563305, 74.83031625404399] Fitness: 2.583280511885789
Generation 269 Best Solution: [183.80774835665255, 177.39407605061672,
175.12764043371757, 74.50634860398665] Fitness: 2.7063817674492197
Generation 270 Best Solution: [182.8578599693065, 176.70485559789108,
175.06586358563305, 75.83571488009183] Fitness: 2.444873964063087
Generation 271 Best Solution: [183.5854793782747, 176.80501220738347,
174.75342558709082, 75.59193814308338] Fitness: 1.6687982211662677
Generation 272 Best Solution: [183.5854793782747, 176.80501220738347,
174.75342558709082, 75.83571488009183] Fitness: 2.1256221905801227
Generation 273 Best Solution: [183.80774835665255, 175.89883686424463,
175.06586358563305, 74.31897918277399] Fitness: 1.4359404961580244
Generation 274 Best Solution: [183.5854793782747, 177.39407605061672,
173.98306919820897, 74.53505178166381] Fitness: 1.5563360662277606
Generation 275 Best Solution: [183.5854793782747, 177.39407605061672,
173.98306919820897, 74.53505178166381] Fitness: 1.5563360662277606
Generation 276 Best Solution: [183.5854793782747, 177.39407605061672,
173.98306919820897, 74.53505178166381] Fitness: 1.5563360662277606

Generation 277 Best Solution: [183.5854793782747, 177.39407605061672,
175.06586358563305, 74.7752707493113] Fitness: 2.3682037771665843
Generation 278 Best Solution: [183.80774835665255, 177.39407605061672,
174.66603765270463, 74.7752707493113] Fitness: 2.017337358574717
Generation 279 Best Solution: [183.5854793782747, 176.29647656784374,
175.09254569019154, 75.59193814308338] Fitness: 1.837104998172276
Generation 280 Best Solution: [183.5854793782747, 176.29647656784374,
175.06586358563305, 75.83571488009183] Fitness: 2.246543180059747
Generation 281 Best Solution: [183.5854793782747, 176.511969241098,
175.07793737038355, 75.59193814308338] Fitness: 1.8962359026343019
Generation 282 Best Solution: [183.80774835665255, 176.511969241098,
175.07793737038355, 75.59193814308338] Fitness: 2.099735545607827
Generation 283 Best Solution: [183.34011502467163, 176.65764707882087,
175.07944855535806, 75.83571488009183] Fitness: 2.3415210028415503
Generation 284 Best Solution: [183.34011502467163, 176.62687956106427,
175.07944855535806, 75.59193814308338] Fitness: 1.8591532255881589
Generation 285 Best Solution: [183.34011502467163, 176.62687956106427,
173.88898541621523, 75.83571488009183] Fitness: 1.4952246298052485
Generation 286 Best Solution: [183.34011502467163, 176.69233686161022,
173.92811006246683, 75.82118109381786] Fitness: 1.5072096760625635
Generation 287 Best Solution: [183.80774835665255, 177.39407605061672,
174.75065893452577, 74.30779943266224] Fitness: 2.2350162917438485
Generation 288 Best Solution: [183.5854793782747, 177.39407605061672,
174.75065893452577, 74.30779943266224] Fitness: 2.0315166487703236
Generation 289 Best Solution: [183.5854793782747, 177.39407605061672,
174.75065893452577, 74.30779943266224] Fitness: 2.0315166487703236
Generation 290 Best Solution: [182.8199378327902, 176.65764707882087,
175.06586358563305, 74.30779943266224] Fitness: 1.3552094267265704
Generation 291 Best Solution: [183.5854793782747, 176.2682942864531,
175.06586358563305, 75.83571488009183] Fitness: 2.2416661897763017
Generation 292 Best Solution: [183.5854793782747, 176.2682942864531,
175.06586358563305, 75.83571488009183] Fitness: 2.2416661897763017
Generation 293 Best Solution: [183.00718473038833, 177.39407605061672,
174.23160688288877, 74.30779943266224] Fitness: 1.5603495330252772
Generation 294 Best Solution: [183.5854793782747, 176.24246467391805,
175.06586358563305, 75.83571488009183] Fitness: 2.2384701254209594
Generation 295 Best Solution: [183.00718473038833, 176.2682942864531,
175.06586358563305, 75.83571488009183] Fitness: 2.1349093197741453
Generation 296 Best Solution: [183.00718473038833, 176.31818461043713,
175.06586358563305, 75.83571488009183] Fitness: 2.1445317681521647
Generation 297 Best Solution: [183.06562820513065, 176.8187787249884,
174.02768834824658, 75.83571488009183] Fitness: 1.6413796871162785
Generation 298 Best Solution: [183.00718473038833, 176.31818461043713,
174.71904045495603, 75.83571488009183] Fitness: 1.6468709625788778
Generation 299 Best Solution: [183.00718473038833, 176.31818461043713,
175.07793737038355, 75.83571488009183] Fitness: 2.1658130316924153
Generation 300 Best Solution: [183.15082472087929, 176.31818461043713,
175.06586358563305, 75.7794329294681] Fitness: 1.9989396737521015

Generation 301 Best Solution: [183.15082472087929, 176.31818461043713,
175.07793737038355, 75.83571488009183] Fitness: 2.1353251219360243
Generation 302 Best Solution: [182.96161710135766, 176.31818461043713,
174.96380726564968, 75.83571488009183] Fitness: 1.9928259218145512
Generation 303 Best Solution: [183.15082472087929, 176.31818461043713,
174.96380726564968, 75.83571488009183] Fitness: 1.9447941548513064
Generation 304 Best Solution: [183.15082472087929, 176.31818461043713,
175.06586358563305, 75.83571488009183] Fitness: 2.1140438583957737
Generation 305 Best Solution: [183.00718473038833, 177.39407605061672,
174.96380726564968, 74.53400413162737] Fitness: 2.1067280363902734
Generation 306 Best Solution: [183.15082472087929, 176.31818461043713,
175.07793737038355, 75.83571488009183] Fitness: 2.1353251219360243
Generation 307 Best Solution: [183.15082472087929, 176.20226154857104,
175.07793737038355, 75.83571488009183] Fitness: 2.119955894520488
Generation 308 Best Solution: [183.38558453875, 176.45447496279544,
175.07793737038355, 75.83571488009183] Fitness: 2.216060018202674
Generation 309 Best Solution: [183.00718473038833, 176.88887980347397,
174.96380726564968, 75.83571488009183] Fitness: 2.4087214085934567
Generation 310 Best Solution: [183.5854793782747, 177.39407605061672,
175.07793737038355, 74.31007936126088] Fitness: 2.512612052588721
Generation 311 Best Solution: [183.15082472087929, 177.39407605061672,
174.7033441234767, 75.49830307933595] Fitness: 2.3090160308089303
Generation 312 Best Solution: [183.5854793782747, 177.39407605061672,
175.06586358563305, 75.17786904037321] Fitness: 2.580634513092851
Generation 313 Best Solution: [183.00718473038833, 176.95068641140836,
174.7033441234767, 75.83571488009183] Fitness: 2.1456180101320834
Generation 314 Best Solution: [183.80774835665255, 177.19504469358847,
174.84416094899163, 75.83571488009183] Fitness: 3.0224257252355184
Generation 315 Best Solution: [183.80774835665255, 177.39407605061672,
174.7033441234767, 75.83571488009183] Fitness: 3.253204986865889
Generation 316 Best Solution: [183.80774835665255, 177.39407605061672,
175.07793737038355, 74.53637259481236] Fitness: 2.606856250733322
Generation 317 Best Solution: [183.80774835665255, 177.09352155293283,
174.2586306711188, 75.83571488009183] Fitness: 2.3695523053417684
Generation 318 Best Solution: [183.80774835665255, 177.39407605061672,
173.75120859990668, 74.53637259481236] Fitness: 1.8608848074223323
Generation 319 Best Solution: [183.80774835665255, 177.39407605061672,
175.07793737038355, 74.53637259481236] Fitness: 2.606856250733322
Generation 320 Best Solution: [183.80774835665255, 175.97116251266772,
175.07793737038355, 74.53637259481236] Fitness: 1.3206655359620791
Generation 321 Best Solution: [183.80774835665255, 175.982869245968,
175.07793737038355, 75.83571488009183] Fitness: 2.498781606264833
Generation 322 Best Solution: [183.74805276381272, 175.982869245968,
175.07793737038355, 75.83571488009183] Fitness: 2.4352659535047727
Generation 323 Best Solution: [183.55696727002476, 176.43906175882125,
175.07793737038355, 75.83571488009183] Fitness: 2.2951547696205914
Generation 324 Best Solution: [183.74805276381272, 177.39407605061672,
175.07793737038355, 74.63782112657717] Fitness: 2.5247184767786224

Generation 325 Best Solution: [183.80774835665255, 177.39407605061672,
175.07793737038355, 74.63782112657717] Fitness: 2.5882341295386824
Generation 326 Best Solution: [183.74805276381272, 175.930655996497,
175.07793737038355, 75.83571488009183] Fitness: 2.4572771750561673
Generation 327 Best Solution: [183.74805276381272, 176.5214059614082,
175.07793737038355, 75.88235495965412] Fitness: 2.5984825931403703
Generation 328 Best Solution: [183.74805276381272, 176.87276247943805,
175.07793737038355, 75.83571488009183] Fitness: 2.825591164750866
Generation 329 Best Solution: [183.80774835665255, 177.39407605061672,
174.49666661983233, 75.83571488009183] Fitness: 3.0670134866955494
Generation 330 Best Solution: [183.23467774436273, 177.39407605061672,
174.7674539865659, 75.83571488009183] Fitness: 2.985685585363571
Generation 331 Best Solution: [183.80774835665255, 176.98126284182825,
175.07793737038355, 75.83571488009183] Fitness: 3.035603684831071
Generation 332 Best Solution: [183.74805276381272, 177.39407605061672,
174.7674539865659, 75.83571488009183] Fitness: 3.2632951509823687
Generation 333 Best Solution: [183.80774835665255, 175.99784149623898,
175.07793737038355, 75.83571488009183] Fitness: 2.4933882928894047
Generation 334 Best Solution: [183.80774835665255, 175.99784149623898,
175.07793737038355, 75.83571488009183] Fitness: 2.4933882928894047
Generation 335 Best Solution: [183.21615727885143, 176.00573832641544,
175.07793737038355, 75.83571488009183] Fitness: 2.1485626956067105
Generation 336 Best Solution: [183.80774835665255, 176.3689740144424,
175.07793737038355, 75.83571488009183] Fitness: 2.4905342148310234
Generation 337 Best Solution: [183.17775447054754, 176.3689740144424,
175.07793737038355, 75.83571488009183] Fitness: 2.148267866686694
Generation 338 Best Solution: [183.74805276381272, 175.80451494980758,
175.07793737038355, 75.83571488009183] Fitness: 2.53099515516389
Generation 339 Best Solution: [183.74805276381272, 175.79212074276776,
175.07793737038355, 75.74308259339695] Fitness: 2.353435056094537
Generation 340 Best Solution: [183.138298640415, 175.79212074276776,
175.07793737038355, 75.74308259339695] Fitness: 2.0773654438427824
Generation 341 Best Solution: [183.80774835665255, 177.39407605061672,
175.07793737038355, 74.59086183079495] Fitness: 2.5945178098611383
Generation 342 Best Solution: [183.80774835665255, 177.39407605061672,
175.07793737038355, 74.53090534074755] Fitness: 2.608393538782488
Generation 343 Best Solution: [183.80774835665255, 177.39407605061672,
175.07793737038355, 74.4865949025102] Fitness: 2.622866666647794
Generation 344 Best Solution: [183.80774835665255, 177.39407605061672,
175.07793737038355, 74.4817713606716] Fitness: 2.6246585722739826
Generation 345 Best Solution: [183.80774835665255, 177.39407605061672,
175.07793737038355, 74.5313555695674] Fitness: 2.6082648805946302
Generation 346 Best Solution: [183.80774835665255, 176.14831006412533,
175.07793737038355, 75.83571488009183] Fitness: 2.4619153894409425
Generation 347 Best Solution: [183.80774835665255, 177.39407605061672,
175.07793737038355, 74.69516198050889] Fitness: 2.586021882618084
Generation 348 Best Solution: [183.80774835665255, 175.79948217808098,
175.07793737038355, 75.88235495965412] Fitness: 2.697822899780282

Generation 349 Best Solution: [183.80774835665255, 175.79948217808098,
175.07793737038355, 75.88235495965412] Fitness: 2.697822899780282
Generation 350 Best Solution: [183.24971931322244, 177.39407605061672,
173.94178323709195, 75.83571488009183] Fitness: 2.613126758771076
Generation 351 Best Solution: [183.80774835665255, 176.0873168480299,
175.07897738601753, 75.83571488009183] Fitness: 2.4715358456420358
Generation 352 Best Solution: [183.80774835665255, 177.39407605061672,
174.5123739707695, 75.64886149647668] Fitness: 2.7185609049403334
Generation 353 Best Solution: [183.86239520813717, 177.39407605061672,
174.5123739707695, 75.11402325230026] Fitness: 2.104779563983551
Generation 354 Best Solution: [183.80774835665255, 177.39407605061672,
174.5123739707695, 75.11402325230026] Fitness: 2.0409306225909276
Generation 355 Best Solution: [183.80774835665255, 175.8236031342529,
175.07793737038355, 75.88235495965412] Fitness: 2.681257876713703
Generation 356 Best Solution: [183.80774835665255, 175.89859394445918,
175.07793737038355, 75.88235495965412] Fitness: 2.636544243084498
Generation 357 Best Solution: [183.86239520813717, 176.09907594018358,
175.07793737038355, 75.88235495965412] Fitness: 2.6312797846498546
Generation 358 Best Solution: [183.86239520813717, 177.39407605061672,
174.11528985289075, 75.88235495965412] Fitness: 3.0918245738083243
Generation 359 Best Solution: [183.86239520813717, 177.39407605061672,
175.07793737038355, 75.28266990002405] Fitness: 2.9731143476341053
Generation 360 Best Solution: [183.80774835665255, 177.39407605061672,
175.07793737038355, 75.28266990002405] Fitness: 2.9092654062414818
Generation 361 Best Solution: [183.86239520813717, 177.39407605061672,
174.11528985289075, 75.28266990002405] Fitness: 2.1118473549506804
Generation 362 Best Solution: [183.15978079321565, 175.6448997815726,
175.07793737038355, 75.64886149647668] Fitness: 2.0281744598315745
Generation 363 Best Solution: [183.86239520813717, 177.39407605061672,
175.07793737038355, 74.41759403225328] Fitness: 2.7163920794376963
Generation 364 Best Solution: [183.86239520813717, 176.0416876029984,
175.07793737038355, 75.88235495965412] Fitness: 2.643565793247663
Generation 365 Best Solution: [183.86239520813717, 176.9461301259875,
175.07793737038355, 74.3571685258408] Fitness: 1.9458606221012467
Generation 366 Best Solution: [183.3331853551761, 176.19376749130484,
175.07793737038355, 75.88235495965412] Fitness: 2.2351364737402
Generation 367 Best Solution: [183.86239520813717, 176.0703068652894,
175.02636518712825, 75.88235495965412] Fitness: 2.5476457825921166
Generation 368 Best Solution: [183.86239520813717, 176.0703068652894,
175.02636518712825, 75.88235495965412] Fitness: 2.5476457825921166
Generation 369 Best Solution: [183.86239520813717, 177.27648650566803,
173.7831899821402, 75.88235495965412] Fitness: 2.940858903867369
Generation 370 Best Solution: [183.80774835665255, 175.60050515067778,
175.07793737038355, 75.88235495965412] Fitness: 2.8750012679638406
Generation 371 Best Solution: [183.09359695735574, 177.39407605061672,
174.58202379253098, 75.88235495965412] Fitness: 2.9011101935099535
Generation 372 Best Solution: [183.86239520813717, 177.42419054217459,
175.07793737038355, 74.64000662195664] Fitness: 2.719061814203209

Generation 373 Best Solution: [183.03461690116688, 177.39407605061672,
175.02636518712825, 75.14338630500266] Fitness: 2.3675998174193644
Generation 374 Best Solution: [182.80231624319964, 177.39407605061672,
175.02636518712825, 74.64000662195664] Fitness: 2.2974155932444784
Generation 375 Best Solution: [183.09359695735574, 177.1098255630776,
174.58202379253098, 75.88235495965412] Fitness: 2.3486516499670858
Generation 376 Best Solution: [183.03461690116688, 177.39407605061672,
175.02636518712825, 75.35638150091084] Fitness: 2.5862914672727313
Generation 377 Best Solution: [182.97373892579887, 177.39407605061672,
175.02636518712825, 75.35638150091084] Fitness: 2.607531336647753
Generation 378 Best Solution: [183.09359695735574, 177.39407605061672,
175.02636518712825, 75.35638150091084] Fitness: 2.5721682851609198
Generation 379 Best Solution: [182.80231624319964, 176.25401433201606,
175.07793737038355, 75.88235495965412] Fitness: 2.362714820697664
Generation 380 Best Solution: [183.19776375125244, 177.42419054217459,
175.07793737038355, 74.82865030157187] Fitness: 2.328763700306285
Generation 381 Best Solution: [183.58010773749106, 177.39407605061672,
174.4742675029207, 74.33438062256752] Fitness: 1.7546409264983398
Generation 382 Best Solution: [182.80231624319964, 176.83865260956833,
175.02636518712825, 75.88235495965412] Fitness: 2.656637169668511
Generation 383 Best Solution: [183.09359695735574, 177.39407605061672,
175.02636518712825, 74.61664020793721] Fitness: 2.168415690019354
Generation 384 Best Solution: [183.86239520813717, 177.39407605061672,
174.3552210435316, 74.61664020793721] Fitness: 1.8494910658509884
Generation 385 Best Solution: [183.09359695735574, 177.39407605061672,
174.3552210435316, 74.61664020793721] Fitness: 1.4525292597632382
Generation 386 Best Solution: [183.03888496804643, 176.23464260906715,
175.02636518712825, 75.88235495965412] Fitness: 2.1530254278217225
Generation 387 Best Solution: [183.03888496804643, 176.23464260906715,
175.02636518712825, 75.88235495965412] Fitness: 2.1530254278217225
Generation 388 Best Solution: [183.80774835665255, 177.39407605061672,
174.60951114534367, 74.9566002069458] Fitness: 2.021559922598182
Generation 389 Best Solution: [183.86239520813717, 177.39407605061672,
174.75484979732622, 75.03542709376165] Fitness: 2.28254259679573
Generation 390 Best Solution: [182.80231624319964, 176.04534918741248,
175.07793737038355, 75.88235495965412] Fitness: 2.377170471393045
Generation 391 Best Solution: [183.86239520813717, 176.3308932937977,
174.75484979732622, 75.88235495965412] Fitness: 2.165136593043317
Generation 392 Best Solution: [183.86239520813717, 176.3308932937977,
174.75484979732622, 75.88235495965412] Fitness: 2.165136593043317
Generation 393 Best Solution: [183.1258808420152, 176.84369682888888,
175.07793737038355, 75.88235495965412] Fitness: 2.6151280350051778
Generation 394 Best Solution: [183.86239520813717, 176.443270731444,
174.75484979732622, 75.88235495965412] Fitness: 2.2060685821873984
Generation 395 Best Solution: [183.86239520813717, 177.39407605061672,
174.75484979732622, 74.69929864910743] Fitness: 2.172215879348648
Generation 396 Best Solution: [183.86239520813717, 177.39407605061672,
174.75484979732622, 74.69929864910743] Fitness: 2.172215879348648

Generation 397 Best Solution: [183.86239520813717, 176.443270731444,
175.07793737038355, 74.84677168602491] Fitness: 1.4036504303821382
Generation 398 Best Solution: [183.86239520813717, 175.94047818219894,
174.75484979732622, 75.88235495965412] Fitness: 2.2021615083734636
Generation 399 Best Solution: [183.86239520813717, 175.69302246632878,
174.60702141373815, 75.88235495965412] Fitness: 2.2147256010293024
Generation 400 Best Solution: [182.9948460847523, 177.39407605061672,
175.07793737038355, 74.60365035267884] Fitness: 2.2865271718293827
Generation 401 Best Solution: [183.86239520813717, 175.7015484859861,
175.02636518712825, 74.48774607591932] Fitness: 1.4840028569196437
Generation 402 Best Solution: [183.86239520813717, 175.7015484859861,
174.60951114534367, 75.88235495965412] Fitness: 2.209371979251199
Generation 403 Best Solution: [183.86239520813717, 175.7015484859861,
175.02636518712825, 75.88235495965412] Fitness: 2.7508001868541863
Generation 404 Best Solution: [183.86239520813717, 176.15620175628382,
175.07793737038355, 75.88235495965412] Fitness: 2.6250228477251065
Generation 405 Best Solution: [183.86239520813717, 175.7015484859861,
175.07793737038355, 75.3870891166482] Fitness: 1.983282955364507
Generation 406 Best Solution: [183.00474998564195, 176.3176283229418,
175.07793737038355, 75.88235495965412] Fitness: 2.2662903222336404
Generation 407 Best Solution: [182.9597216103832, 177.39407605061672,
174.010272044935, 75.88235495965412] Fitness: 2.744490933251741
Generation 408 Best Solution: [183.86239520813717, 177.39407605061672,
174.60702141373815, 74.58520887732533] Fitness: 2.0266447302449246
Generation 409 Best Solution: [183.86239520813717, 177.39407605061672,
174.010272044935, 74.71413923838303] Fitness: 1.7976100816418048
Generation 410 Best Solution: [183.86239520813717, 175.7015484859861,
175.07793737038355, 75.88235495965412] Fitness: 2.839841475486746
Generation 411 Best Solution: [183.86239520813717, 175.7015484859861,
174.60702141373815, 75.8667001104916] Fitness: 2.1731611830115396
Generation 412 Best Solution: [183.86239520813717, 175.7015484859861,
174.32286816566827, 75.88235495965412] Fitness: 2.021185061300171
Generation 413 Best Solution: [183.02504962885948, 176.72327262279416,
175.07793737038355, 75.88235495965412] Fitness: 2.5061475151681236
Generation 414 Best Solution: [182.85787790440608, 177.39407605061672,
175.07793737038355, 74.70234434307112] Fitness: 2.3475376803009134
Generation 415 Best Solution: [182.87444787217692, 177.39407605061672,
175.07793737038355, 74.70234434307112] Fitness: 2.3375804562105653
Generation 416 Best Solution: [183.86239520813717, 176.19807517114873,
175.07793737038355, 75.88235495965412] Fitness: 2.6242214551517558
Generation 417 Best Solution: [182.85787790440608, 176.45350474388772,
174.60702141373815, 75.88235495965412] Fitness: 1.7534402495847454
Generation 418 Best Solution: [182.85787790440608, 176.45350474388772,
175.02636518712825, 75.88235495965412] Fitness: 2.297148962978976
Generation 419 Best Solution: [183.86239520813717, 177.39407605061672,
174.60702141373815, 75.8667001104916] Fitness: 3.286411274016543
Generation 420 Best Solution: [183.0420609776238, 176.76169627281465,
175.07793737038355, 75.88235495965412] Fitness: 2.540057058055824

Generation 421 Best Solution: [183.0420609776238, 176.76169627281465,
175.07793737038355, 75.88235495965412] Fitness: 2.540057058055824
Generation 422 Best Solution: [183.86239520813717, 177.39407605061672,
175.07793737038355, 75.15322519207746] Fitness: 2.8477567328949607
Generation 423 Best Solution: [183.0420609776238, 175.66432118395213,
175.02636518712825, 75.8667001104916] Fitness: 2.3662070383762344
Generation 424 Best Solution: [183.0420609776238, 177.39407605061672,
174.60702141373815, 75.25250235833104] Fitness: 1.9234433599846192
Generation 425 Best Solution: [183.10627125429625, 177.39407605061672,
175.07793737038355, 74.34101345321952] Fitness: 2.360340197153481
Generation 426 Best Solution: [182.9597216103832, 177.39407605061672,
175.07793737038355, 74.34101345321952] Fitness: 2.403758076843731
Generation 427 Best Solution: [183.0420609776238, 177.39407605061672,
175.07793737038355, 74.61739545554153] Fitness: 2.269349977725745
Generation 428 Best Solution: [182.9597216103832, 177.39407605061672,
175.07793737038355, 74.99097043610273] Fitness: 2.3780893448152334
Generation 429 Best Solution: [183.0420609776238, 176.7481504224672,
175.07793737038355, 74.99097043610273] Fitness: 1.3067717306158702
Generation 430 Best Solution: [183.0420609776238, 177.39407605061672,
175.07793737038355, 74.47183660123095] Fitness: 2.3073606271598406
Generation 431 Best Solution: [183.86239520813717, 177.39407605061672,
174.60702141373815, 74.37465006801091] Fitness: 2.106501100834697
Generation 432 Best Solution: [182.9597216103832, 176.3060894158918,
175.02636518712825, 75.8667001104916] Fitness: 2.158215149057969
Generation 433 Best Solution: [183.0420609776238, 177.39407605061672,
174.60702141373815, 74.31186382074297] Fitness: 1.7610118622402073
Generation 434 Best Solution: [183.4356277515284, 175.59108046710801,
175.02636518712825, 75.7945005404506] Fitness: 2.3217382306777665
Generation 435 Best Solution: [183.4356277515284, 175.59108046710801,
174.4445664901915, 75.8667001104916] Fitness: 1.8043862808022035
Generation 436 Best Solution: [183.2842955504532, 175.59108046710801,
175.02636518712825, 75.7945005404506] Fitness: 2.2762175747229203
Generation 437 Best Solution: [183.4356277515284, 175.82977252027882,
173.95446851204082, 75.8667001104916] Fitness: 1.5132932711987015
Generation 438 Best Solution: [183.4356277515284, 177.39407605061672,
175.02636518712825, 74.95786353749574] Fitness: 2.2738268833201016
Generation 439 Best Solution: [183.0420609776238, 175.59108046710801,
175.07793737038355, 75.54428266114202] Fitness: 1.9308491258964193
Generation 440 Best Solution: [183.0420609776238, 175.59108046710801,
173.75714677916687, 75.8667001104916] Fitness: 1.780340279676512
Generation 441 Best Solution: [183.4356277515284, 177.39407605061672,
173.75714677916687, 74.67825263676653] Fitness: 1.5464731007029682
Generation 442 Best Solution: [183.0420609776238, 177.39407605061672,
174.60702141373815, 75.39235781152712] Fitness: 2.0855666201284633
Generation 443 Best Solution: [183.18462222326534, 177.29343258474995,
174.60702141373815, 75.39235781152712] Fitness: 1.8517677275256883
Generation 444 Best Solution: [183.4356277515284, 177.39407605061672,
174.60702141373815, 75.28144178474146] Fitness: 1.9853990627568145

Generation 445 Best Solution: [183.4356277515284, 177.39407605061672,
174.60702141373815, 75.28144178474146] Fitness: 1.9853990627568145
Generation 446 Best Solution: [183.4356277515284, 176.11225682474407,
175.02636518712825, 75.8667001104916] Fitness: 2.152703478732314
Generation 447 Best Solution: [183.63753678585218, 176.55263563564822,
175.07793737038355, 75.8667001104916] Fitness: 2.4840652188915975
Generation 448 Best Solution: [183.0420609776238, 177.39407605061672,
175.07793737038355, 74.67633448613518] Fitness: 2.2649637063046963
Generation 449 Best Solution: [183.2383175562616, 177.39407605061672,
174.9889977507532, 74.67633448613518] Fitness: 2.0947121161138345
Generation 450 Best Solution: [183.0420609776238, 177.39407605061672,
175.07793737038355, 74.67633448613518] Fitness: 2.2649637063046963
Generation 451 Best Solution: [183.298045795085, 177.39407605061672,
175.07793737038355, 74.67633448613518] Fitness: 2.253191100632131
Generation 452 Best Solution: [183.0420609776238, 177.39407605061672,
175.07793737038355, 74.67633448613518] Fitness: 2.2649637063046963
Generation 453 Best Solution: [183.6427526590908, 176.1371126626099,
175.07793737038355, 75.773323046925] Fitness: 2.175891095245192
Generation 454 Best Solution: [183.12284933538552, 176.1371126626099,
175.07793737038355, 75.8667001104916] Fitness: 2.1909105019659534
Generation 455 Best Solution: [183.66700722032044, 175.71193307013385,
174.2809707652321, 75.8667001104916] Fitness: 1.760015490099095
Generation 456 Best Solution: [183.298045795085, 175.71193307013385,
174.2809707652321, 75.8667001104916] Fitness: 1.5664533001470802
Generation 457 Best Solution: [183.0420609776238, 176.39370146281493,
175.07793737038355, 75.8667001104916] Fitness: 2.243996825375169
Generation 458 Best Solution: [183.298045795085, 177.39407605061672,
173.93455124202325, 75.8667001104916] Fitness: 2.6881300906740595
Generation 459 Best Solution: [183.298045795085, 177.39407605061672,
175.07793737038355, 74.59220791343164] Fitness: 2.261386527031258
Generation 460 Best Solution: [183.298045795085, 176.557337247327,
175.07793737038355, 75.88235495965412] Fitness: 2.3521829141592336
Generation 461 Best Solution: [183.298045795085, 177.39407605061672,
174.9889977507532, 75.22917076111673] Fitness: 2.3703275293856976
Generation 462 Best Solution: [183.298045795085, 177.39407605061672,
175.07793737038355, 74.59220791343164] Fitness: 2.261386527031258
Generation 463 Best Solution: [182.92263157825093, 176.3074031764457,
175.07793737038355, 75.12563657400183] Fitness: 1.1705832497255906
Generation 464 Best Solution: [183.4302600807909, 176.77134381954127,
174.9889977507532, 75.8667001104916] Fitness: 2.3948116250064673
Generation 465 Best Solution: [182.92263157825093, 176.10102121824184,
175.07793737038355, 75.65289045706892] Fitness: 1.8398810856975825
Generation 466 Best Solution: [183.298045795085, 176.30268801119595,
175.07793737038355, 75.8667001104916] Fitness: 2.205536784704325
Generation 467 Best Solution: [183.298045795085, 177.1041754360104,
174.08885190727878, 75.773323046925] Fitness: 1.9063382381373084
Generation 468 Best Solution: [183.298045795085, 177.1041754360104,
174.08885190727878, 75.8667001104916] Fitness: 2.099427740442988

Generation 469 Best Solution: [183.298045795085, 177.1041754360104,
174.08885190727878, 75.60976297028392] Fitness: 1.606491326831887
Generation 470 Best Solution: [183.0420609776238, 177.1041754360104,
174.08885190727878, 75.60976297028392] Fitness: 1.618263932504453
Generation 471 Best Solution: [183.0420609776238, 177.1041754360104,
175.0331314185232, 75.60976297028392] Fitness: 2.40167310582581
Generation 472 Best Solution: [183.0420609776238, 177.39407605061672,
175.07793737038355, 75.02572246339795] Fitness: 2.369221017048312
Generation 473 Best Solution: [182.92263157825093, 177.39407605061672,
175.07793737038355, 74.91000807129748] Fitness: 2.356432968288538
Generation 474 Best Solution: [182.90101982901916, 176.04577518068763,
175.07793737038355, 75.773323046925] Fitness: 2.0881146478754102
Generation 475 Best Solution: [182.92263157825093, 177.39407605061672,
174.82837201473245, 74.82555231595275] Fitness: 1.9428795466346798
Generation 476 Best Solution: [182.90101982901916, 177.39407605061672,
175.07793737038355, 74.91000807129748] Fitness: 2.367617819442147
Generation 477 Best Solution: [183.0420609776238, 177.39407605061672,
175.07793737038355, 74.91000807129748] Fitness: 2.3100037179142654
Generation 478 Best Solution: [183.24767364723718, 177.1301375326829,
174.82837201473245, 75.773323046925] Fitness: 2.4185435675223133
Generation 479 Best Solution: [183.60945392862638, 175.9220305420849,
175.07793737038355, 75.773323046925] Fitness: 2.2117662453931546
Generation 480 Best Solution: [183.45480811249675, 177.39407605061672,
175.07793737038355, 74.75414227723562] Fitness: 2.3089827770979707
Generation 481 Best Solution: [183.0420609776238, 175.96197437693468,
174.82837201473245, 75.773323046925] Fitness: 1.6728501529282034
Generation 482 Best Solution: [182.92263157825093, 177.1301375326829,
175.07793737038355, 74.75414227723562] Fitness: 1.7974413245254703
Generation 483 Best Solution: [183.45480811249675, 176.6669438338619,
174.44742118542206, 75.773323046925] Fitness: 1.5066809241839803
Generation 484 Best Solution: [182.90101982901916, 176.6669438338619,
174.44742118542206, 75.773323046925] Fitness: 1.5242002146475202
Generation 485 Best Solution: [183.83817398366338, 177.39407605061672,
175.07793737038355, 74.43186451188006] Fitness: 2.68056865144571
Generation 486 Best Solution: [182.92263157825093, 177.39407605061672,
175.07793737038355, 74.66965941482326] Fitness: 2.311571189232352
Generation 487 Best Solution: [183.83817398366338, 176.72524999658773,
175.07793737038355, 74.43186451188006] Fitness: 1.6155112288954439
Generation 488 Best Solution: [183.2263551099271, 176.01996076148757,
175.07793737038355, 75.773323046925] Fitness: 2.017256538037333
Generation 489 Best Solution: [183.2263551099271, 176.01996076148757,
175.07793737038355, 75.56545779629485] Fitness: 1.644595737425286
Generation 490 Best Solution: [183.69058728991243, 177.1301375326829,
175.07793737038355, 74.70417554948177] Fitness: 1.949833110349605
Generation 491 Best Solution: [183.45480811249675, 177.1301375326829,
175.07793737038355, 74.43186451188006] Fitness: 1.846788880516464
Generation 492 Best Solution: [183.45480811249675, 176.84854963034098,
175.07793737038355, 75.773323046925] Fitness: 2.4510692967341763

```
Generation 493 Best Solution: [183.45480811249675, 177.39407605061672,
175.07793737038355, 75.27944059976555] Fitness: 2.6247382379517146
Generation 494 Best Solution: [183.45480811249675, 177.39407605061672,
173.89940014660255, 75.773323046925] Fitness: 2.5590840707455236
Generation 495 Best Solution: [183.45480811249675, 176.30506887224513,
173.89940014660255, 75.773323046925] Fitness: 1.242709078818313
Generation 496 Best Solution: [183.45480811249675, 177.39407605061672,
173.89940014660255, 75.773323046925] Fitness: 2.5590840707455236
Generation 497 Best Solution: [183.45480811249675, 176.23124068904954,
174.79607458154592, 75.773323046925] Fitness: 1.6265820018685908
Generation 498 Best Solution: [183.45480811249675, 176.23124068904954,
174.79607458154592, 75.773323046925] Fitness: 1.6265820018685908
Generation 499 Best Solution: [183.6384585431795, 177.39407605061672,
175.07793737038355, 75.00021877864465] Fitness: 2.5120080593075382
Generation 500 Best Solution: [182.92263157825093, 177.1493934884913,
174.50541425240976, 75.773323046925] Fitness: 2.1869727638997767
Optimal Solution for Simulation 19: [182.92263157825093, 177.1493934884913,
174.50541425240976, 75.773323046925]
--------------------------------------------------
Simulation 20/32 with conditions: {'drought': False, 'flood': True,
'policy_change': True, 'social_event': False, 'economic_crisis': False}
Generation 1 Best Solution: [183.07233283556204, 176.465564985026,
174.00415107123845, 75.1705013939003] Fitness: 0.33973266721294704
Generation 2 Best Solution: [183.01848481333082, 176.9679028548171,
174.00242286926158, 75.15145034950525] Fitness: 0.8253079432424333
Generation 3 Best Solution: [183.23020798666994, 176.17620211255831,
174.59621861534532, 75.15145034950525] Fitness: 0.4593827112497739
Generation 4 Best Solution: [183.54817978976266, 175.65353064916542,
174.50815290465778, 75.15145034950525] Fitness: 0.7392802895564272
Generation 5 Best Solution: [183.23020798666994, 176.17620211255831,
174.50815290465778, 75.15145034950525] Fitness: 0.38760502344608305
Generation 6 Best Solution: [183.45656258471897, 175.65353064916542,
174.65774734235407, 75.58486362534816] Fitness: 1.3181800147889815
Generation 7 Best Solution: [182.84880446788046, 175.65353064916542,
174.65774734235407, 75.58486362534816] Fitness: 1.33718886158806
Generation 8 Best Solution: [183.78118891679966, 176.6725566741304,
175.05791391295716, 74.91854840351462] Fitness: 1.4781222356289996
Generation 9 Best Solution: [182.8120775277647, 177.2288254388502,
174.88667153705399, 74.45623617774952] Fitness: 1.7848550368831106
Generation 10 Best Solution: [183.4706520929356, 177.18380571530415,
174.59011744030548, 75.65545834844467] Fitness: 2.0605779462060436
Generation 11 Best Solution: [183.48301873809555, 177.18380571530415,
174.59011744030548, 75.65545834844467] Fitness: 2.067246416372801
Generation 12 Best Solution: [183.4706520929356, 175.73098651603883,
175.05791391295716, 75.5155597679157] Fitness: 1.7000062425142577
Generation 13 Best Solution: [183.48301873809555, 177.18380571530415,
174.59011744030548, 75.68746124515977] Fitness: 2.121426336178269
Generation 14 Best Solution: [182.8120775277647, 175.9304527251491,
```

175.05791391295716, 75.87988739489586] Fitness: 2.250326640829562
Generation 15 Best Solution: [183.48301873809555, 175.9304527251491,
175.05791391295716, 75.87988739489586] Fitness: 2.223277454017058
Generation 16 Best Solution: [183.48301873809555, 175.9304527251491,
175.05791391295716, 75.87988739489586] Fitness: 2.223277454017058
Generation 17 Best Solution: [183.48301873809555, 177.2288254388502,
173.92711258605428, 75.68746124515977] Fitness: 2.006190974181889
Generation 18 Best Solution: [183.48301873809555, 177.18380571530415,
174.55189163978065, 75.65545834844467] Fitness: 2.0348245159573297
Generation 19 Best Solution: [183.48301873809555, 175.93477239310712,
175.05791391295716, 75.87988739489586] Fitness: 2.221517613101155
Generation 20 Best Solution: [183.48301873809555, 177.2288254388502,
175.05791391295716, 74.3978540634445] Fitness: 2.053018647769346
Generation 21 Best Solution: [183.48301873809555, 177.2288254388502,
174.2727141807136, 75.61791307225904] Fitness: 1.8973287826123302
Generation 22 Best Solution: [183.32658878236734, 177.2288254388502,
175.05791391295716, 74.3978540634445] Fitness: 1.988498390134558
Generation 23 Best Solution: [183.32658878236734, 177.2288254388502,
174.77561870690351, 74.44204705456968] Fitness: 1.5547519121804467
Generation 24 Best Solution: [183.4116601602215, 177.2288254388502,
174.77561870690351, 74.3978540634445] Fitness: 1.607022484781384
Generation 25 Best Solution: [183.48301873809555, 177.31283685844357,
174.77561870690351, 74.34958549422458] Fitness: 1.8339687761917995
Generation 26 Best Solution: [183.4116601602215, 177.2288254388502,
173.8282646857729, 75.62278426375032] Fitness: 1.9005583656194893
Generation 27 Best Solution: [183.48301873809555, 177.2288254388502,
175.05791391295716, 74.5205940507764] Fitness: 1.9989436243052134
Generation 28 Best Solution: [183.48301873809555, 177.2288254388502,
174.30375788374076, 75.51911711455695] Fitness: 1.758851056770496
Generation 29 Best Solution: [183.4116601602215, 177.2288254388502,
174.30375788374076, 75.51911711455695] Fitness: 1.7240767779554163
Generation 30 Best Solution: [183.33462210552779, 177.31283685844357,
174.77561870690351, 74.45335649343376] Fitness: 1.7154470533297188
Generation 31 Best Solution: [183.48301873809555, 176.38615709662483,
174.9651752124161, 75.51911711455695] Fitness: 1.4394951577646902
Generation 32 Best Solution: [183.48301873809555, 177.2288254388502,
175.05791391295716, 74.85685031780784] Fitness: 1.9866201832891337
Generation 33 Best Solution: [183.48301873809555, 175.95047083353302,
175.05791391295716, 75.87988739489586] Fitness: 2.215398581420507
Generation 34 Best Solution: [183.48301873809555, 177.2043895951457,
175.05791391295716, 75.04922647715686] Fitness: 2.023796849609026
Generation 35 Best Solution: [183.48301873809555, 177.2043895951457,
175.09682071758021, 75.04922647715686] Fitness: 2.091470041214795
Generation 36 Best Solution: [183.48301873809555, 177.2043895951457,
174.20591383434137, 75.43065883707295] Fitness: 1.5658896384248078
Generation 37 Best Solution: [183.1304182192835, 177.2043895951457,
175.05791391295716, 74.87115138058896] Fitness: 1.860679883378776
Generation 38 Best Solution: [183.48301873809555, 176.16076373119208,

175.05791391295716, 75.87988739489586] Fitness: 2.1752508108401485
Generation 39 Best Solution: [183.48301873809555, 176.04795075092247,
175.05791391295716, 75.87988739489586] Fitness: 2.187110981119574
Generation 40 Best Solution: [183.48301873809555, 176.04795075092247,
175.05791391295716, 75.87988739489586] Fitness: 2.187110981119574
Generation 41 Best Solution: [183.48301873809555, 177.31283685844357,
173.87480960608178, 74.53603348227269] Fitness: 1.37208431239293
Generation 42 Best Solution: [183.8670259027683, 177.31283685844357,
173.87480960608178, 74.53603348227269] Fitness: 1.7130977599444912
Generation 43 Best Solution: [183.8670259027683, 177.31283685844357,
175.05791391295716, 74.48460674659306] Fitness: 2.516707207301203
Generation 44 Best Solution: [183.4692471645213, 176.94862304695238,
174.87978401327013, 75.87988739489586] Fitness: 2.4336521030778533
Generation 45 Best Solution: [183.4692471645213, 176.94862304695238,
174.7651271884178, 75.87988739489586] Fitness: 2.285662033131093
Generation 46 Best Solution: [183.48301873809555, 176.0090161491865,
175.05791391295716, 75.87988739489586] Fitness: 2.1964034685611935
Generation 47 Best Solution: [183.48301873809555, 176.0090161491865,
174.5760922735804, 75.87988739489586] Fitness: 1.5791295068120874
Generation 48 Best Solution: [183.48301873809555, 176.0090161491865,
174.5760922735804, 75.87988739489586] Fitness: 1.5791295068120874
Generation 49 Best Solution: [183.48301873809555, 175.95919934664695,
174.87978401327013, 75.87988739489586] Fitness: 1.9363722187333485
Generation 50 Best Solution: [183.4692471645213, 177.2043895951457,
173.78809826176905, 74.45255414199522] Fitness: 1.2267109364362874
Generation 51 Best Solution: [183.48301873809555, 177.31283685844357,
174.33271205370994, 75.87988739489586] Fitness: 2.562749513927419
Generation 52 Best Solution: [183.48301873809555, 176.64518890387993,
174.33271205370994, 75.87988739489586] Fitness: 1.6051669754309734
Generation 53 Best Solution: [183.48301873809555, 177.31283685844357,
174.7651271884178, 75.87988739489586] Fitness: 2.912702448124808
Generation 54 Best Solution: [183.4692471645213, 177.31283685844357,
174.78501766701999, 75.00254573632964] Fitness: 1.799518784361578
Generation 55 Best Solution: [183.48301873809555, 176.54307174073594,
175.05791391295716, 75.87988739489586] Fitness: 2.301632363404746
Generation 56 Best Solution: [183.57071918557614, 177.3440426421136,
174.4000215974546, 75.87988739489586] Fitness: 2.7145590835580773
Generation 57 Best Solution: [183.8670259027683, 176.87571263034434,
174.39785308509872, 75.87988739489586] Fitness: 2.219801474244132
Generation 58 Best Solution: [183.57071918557614, 175.9824374108306,
174.33720623019644, 75.87988739489586] Fitness: 1.487482877833842
Generation 59 Best Solution: [183.8670259027683, 177.2043895951457,
175.05791391295716, 74.34520231403661] Fitness: 2.380089496844474
Generation 60 Best Solution: [183.8670259027683, 176.47241616261098,
175.05791391295716, 75.87988739489586] Fitness: 2.599911243215703
Generation 61 Best Solution: [183.46884789593418, 176.64110710231347,
175.05791391295716, 75.87988739489586] Fitness: 2.3678613349443065
Generation 62 Best Solution: [183.46884789593418, 176.64110710231347,

175.05791391295716, 75.87988739489586] Fitness: 2.3678613349443065
Generation 63 Best Solution: [183.31760286013662, 175.69118907305128,
175.05791391295716, 75.87988739489586] Fitness: 2.3051223798403617
Generation 64 Best Solution: [183.46884789593418, 177.31283685844357,
175.05791391295716, 74.53154056760759] Fitness: 2.1514326301339324
Generation 65 Best Solution: [183.57071918557614, 177.31283685844357,
175.05791391295716, 75.11436541991786] Fitness: 2.330395286632113
Generation 66 Best Solution: [183.6680485057395, 176.36862144906985,
175.05791391295716, 75.87988739489586] Fitness: 2.343968851726756
Generation 67 Best Solution: [183.8670259027683, 177.12594966749265,
175.05791391295716, 74.8885212002288] Fitness: 2.1532583282486875
Generation 68 Best Solution: [183.6680485057395, 176.36862144906985,
175.05791391295716, 75.87988739489586] Fitness: 2.343968851726756
Generation 69 Best Solution: [183.6680485057395, 176.36862144906985,
175.05791391295716, 75.1412852445692] Fitness: 1.3026639619997584
Generation 70 Best Solution: [183.6680485057395, 176.36862144906985,
175.05791391295716, 75.1412852445692] Fitness: 1.3026639619997584
Generation 71 Best Solution: [183.8670259027683, 175.93110587352083,
175.05791391295716, 74.95425139762979] Fitness: 1.4113817525652284
Generation 72 Best Solution: [183.8670259027683, 175.8561659523777,
174.78672439183464, 75.87988739489586] Fitness: 2.201999689997039
Generation 73 Best Solution: [183.55712035007986, 177.3440426421136,
175.05791391295716, 74.62849768556804] Fitness: 2.2464934773758625
Generation 74 Best Solution: [183.8670259027683, 176.32210825995793,
175.05791391295716, 75.46487105143454] Fitness: 1.834946181862476
Generation 75 Best Solution: [183.8670259027683, 177.3440426421136,
173.99715628315255, 75.46487105143454] Fitness: 2.2196386849710015
Generation 76 Best Solution: [183.8670259027683, 176.36862144906985,
175.05791391295716, 75.46487105143454] Fitness: 1.8497858494384363
Generation 77 Best Solution: [183.8670259027683, 176.36862144906985,
175.05791391295716, 75.27218413877002] Fitness: 1.6263059173081125
Generation 78 Best Solution: [183.7539159400369, 176.36862144906985,
175.05791391295716, 75.27218413877002] Fitness: 1.4988955952696206
Generation 79 Best Solution: [183.7539159400369, 177.24742190584823,
175.05791391295716, 75.1412852445692] Fitness: 2.3803126276379616
Generation 80 Best Solution: [183.7539159400369, 177.3440426421136,
175.05791391295716, 75.1412852445692] Fitness: 2.5727510661186552
Generation 81 Best Solution: [183.7539159400369, 177.24742190584823,
175.05791391295716, 75.27218413877002] Fitness: 2.494856105210613
Generation 82 Best Solution: [183.8670259027683, 177.3440426421136,
174.53337921898458, 74.96801665614034] Fitness: 1.942641728778814
Generation 83 Best Solution: [183.57071918557614, 177.3440426421136,
174.00174011120998, 74.96801665614034] Fitness: 1.4896513323567464
Generation 84 Best Solution: [183.57071918557614, 176.24487658993382,
175.05791391295716, 75.48355708330456] Fitness: 1.5578642897375854
Generation 85 Best Solution: [183.57071918557614, 176.1628808466303,
175.05791391295716, 75.04812680957522] Fitness: 1.1258005438139922
Generation 86 Best Solution: [183.57071918557614, 176.22233593486823,

175.05791391295716, 75.87988739489586] Fitness: 2.2332400129024776
Generation 87 Best Solution: [183.57071918557614, 176.22233593486823,
175.05791391295716, 75.87988739489586] Fitness: 2.2332400129024776
Generation 88 Best Solution: [183.8670259027683, 176.22233593486823,
175.05791391295716, 75.73164238784592] Fitness: 2.2332239594297167
Generation 89 Best Solution: [183.8670259027683, 176.22233593486823,
175.05791391295716, 75.73164238784592] Fitness: 2.2332239594297167
Generation 90 Best Solution: [183.57071918557614, 177.3440426421136,
174.40891475262762, 75.73164238784592] Fitness: 2.433479411407737
Generation 91 Best Solution: [183.8670259027683, 176.22233593486823,
174.7900833015674, 75.87988739489586] Fitness: 2.1256795446676744
Generation 92 Best Solution: [183.74611674031826, 176.22233593486823,
174.7900833015674, 75.87988739489586] Fitness: 1.9903138146212194
Generation 93 Best Solution: [183.55192662313803, 176.22233593486823,
174.7900833015674, 75.87988739489586] Fitness: 1.8267517200632444
Generation 94 Best Solution: [183.55192662313803, 177.3440426421136,
175.05791391295716, 74.55274981428302] Fitness: 2.2588369148864027
Generation 95 Best Solution: [183.8670259027683, 176.18369488363362,
175.05791391295716, 75.84240399114464] Fitness: 2.4406207364223858
Generation 96 Best Solution: [183.8670259027683, 176.18369488363362,
175.05791391295716, 75.87988739489586] Fitness: 2.516592714612198
Generation 97 Best Solution: [183.3028339666241, 177.3440426421136,
174.45177427198198, 74.6016432929785] Fitness: 1.42458158322728
Generation 98 Best Solution: [183.55192662313803, 176.18369488363362,
175.05791391295716, 75.87988739489586] Fitness: 2.217664890007768
Generation 99 Best Solution: [183.55192662313803, 176.18369488363362,
175.05791391295716, 75.87988739489586] Fitness: 2.217664890007768
Generation 100 Best Solution: [183.55192662313803, 176.18369488363362,
175.05791391295716, 75.87988739489586] Fitness: 2.217664890007768
Generation 101 Best Solution: [183.55192662313803, 176.18369488363362,
175.05791391295716, 75.87988739489586] Fitness: 2.217664890007768
Generation 102 Best Solution: [183.8670259027683, 175.6663405553695,
174.06546583233288, 75.84240399114464] Fitness: 1.8328146190365118
Generation 103 Best Solution: [183.8670259027683, 176.3707406095648,
175.05791391295716, 75.87988739489586] Fitness: 2.5538341235537034
Generation 104 Best Solution: [183.8670259027683, 176.3707406095648,
175.05791391295716, 75.87988739489586] Fitness: 2.5538341235537034
Generation 105 Best Solution: [183.61430070213225, 175.61099936046338,
175.05791391295716, 75.87988739489586] Fitness: 2.531828770887625
Generation 106 Best Solution: [183.42454050102106, 177.3440426421136,
174.06546583233288, 75.87988739489586] Fitness: 2.5457998484916553
Generation 107 Best Solution: [183.42454050102106, 177.3440426421136,
174.06546583233288, 75.8356805679577] Fitness: 2.4564622804504808
Generation 108 Best Solution: [183.55192662313803, 177.3440426421136,
175.05791391295716, 74.71164998667211] Fitness: 2.237206124386549
Generation 109 Best Solution: [183.8670259027683, 176.54586241296334,
175.05791391295716, 75.53256335617364] Fitness: 2.035253284091639
Generation 110 Best Solution: [183.56659805321013, 177.3440426421136,

175.05791391295716, 74.81152391593459] Fitness: 2.256393565377128
Generation 111 Best Solution: [183.55192662313803, 176.66258984268742,
174.06546583233288, 75.8356805679577] Fitness: 1.5211774840877337
Generation 112 Best Solution: [183.55192662313803, 177.3440426421136,
175.05791391295716, 74.81152391593459] Fitness: 2.2463538806145533
Generation 113 Best Solution: [183.55192662313803, 176.17556140575147,
174.36736285895606, 75.87988739489586] Fitness: 1.4596016691364089
Generation 114 Best Solution: [183.8670259027683, 175.79713116933436,
175.05791391295716, 74.81152391593459] Fitness: 1.4385702605572386
Generation 115 Best Solution: [183.7502220381888, 177.21411042873484,
174.36736285895606, 74.9306222815129] Fitness: 1.435131971134374
Generation 116 Best Solution: [183.56659805321013, 177.3440426421136,
174.36736285895606, 75.17503240488996] Fitness: 1.680077833838381
Generation 117 Best Solution: [183.55192662313803, 177.3440426421136,
174.36736285895606, 75.17503240488996] Fitness: 1.6700381490758067
Generation 118 Best Solution: [183.8670259027683, 177.3440426421136,
174.36736285895606, 75.17503240488996] Fitness: 1.968965973680237
Generation 119 Best Solution: [183.56659805321013, 177.3440426421136,
174.36736285895606, 75.17503240488996] Fitness: 1.680077833838381
Generation 120 Best Solution: [182.88098588659716, 177.3440426421136,
175.05791391295716, 74.64195531472572] Fitness: 2.1869010619779905
Generation 121 Best Solution: [183.56659805321013, 177.3440426421136,
175.05791391295716, 74.64195531472572] Fitness: 2.2512621273410565
Generation 122 Best Solution: [182.88098588659716, 177.3440426421136,
175.05791391295716, 74.64195531472572] Fitness: 2.1869010619779905
Generation 123 Best Solution: [183.8670259027683, 177.3440426421136,
174.4915318076811, 74.32341199357325] Fitness: 1.9838769994957859
Generation 124 Best Solution: [183.8670259027683, 175.76137996207655,
175.05791391295716, 74.64195531472572] Fitness: 1.4576534800150192
Generation 125 Best Solution: [183.7699038901022, 175.62069036389923,
175.05791391295716, 75.87988739489586] Fitness: 2.665337088557691
Generation 126 Best Solution: [182.88098588659716, 177.3440426421136,
175.05791391295716, 75.09590541991315] Fitness: 2.3142419863155013
Generation 127 Best Solution: [183.8670259027683, 175.93654678637364,
174.4306566903663, 75.87988739489586] Fitness: 1.8385009548453652
Generation 128 Best Solution: [183.8670259027683, 175.93654678637364,
174.4306566903663, 75.87988739489586] Fitness: 1.8385009548453652
Generation 129 Best Solution: [183.8134389714756, 175.85570691889404,
175.05791391295716, 75.87988739489586] Fitness: 2.53677435817566
Generation 130 Best Solution: [183.8134389714756, 175.85570691889404,
175.05791391295716, 75.87988739489586] Fitness: 2.53677435817566
Generation 131 Best Solution: [183.8134389714756, 175.85570691889404,
175.05791391295716, 75.87988739489586] Fitness: 2.53677435817566
Generation 132 Best Solution: [183.8134389714756, 177.3440426421136,
175.05791391295716, 74.56022114991671] Fitness: 2.492583307943293
Generation 133 Best Solution: [183.8134389714756, 177.3440426421136,
175.05791391295716, 74.56022114991671] Fitness: 2.492583307943293
Generation 134 Best Solution: [183.8670259027683, 176.0104922176943,

175.05791391295716, 75.87988739489586] Fitness: 2.5370159677313744
Generation 135 Best Solution: [183.8670259027683, 176.49912547674168,
174.90721165908846, 75.87988739489586] Fitness: 2.378052021736291
Generation 136 Best Solution: [183.8670259027683, 177.08524149164518,
175.05791391295716, 75.87988739489586] Fitness: 3.262950704943731
Generation 137 Best Solution: [183.8134389714756, 176.0677148467661,
175.05791391295716, 75.87988739489586] Fitness: 2.461259498957614
Generation 138 Best Solution: [183.8134389714756, 176.0677148467661,
175.05791391295716, 75.87988739489586] Fitness: 2.461259498957614
Generation 139 Best Solution: [183.55350842869126, 177.3440426421136,
175.05791391295716, 74.38016196441069] Fitness: 2.3337407040035902
Generation 140 Best Solution: [183.8670259027683, 176.69435065886242,
175.05791391295716, 74.43761854429616] Fitness: 1.623423579366504
Generation 141 Best Solution: [183.8134389714756, 177.3440426421136,
175.05791391295716, 74.62315382882706] Fitness: 2.479529181179735
Generation 142 Best Solution: [183.8670259027683, 176.04793938415364,
175.05791391295716, 75.87988739489586] Fitness: 2.528126752228295
Generation 143 Best Solution: [183.8670259027683, 176.04793938415364,
175.05791391295716, 75.87988739489586] Fitness: 2.528126752228295
Generation 144 Best Solution: [183.8670259027683, 177.3440426421136,
175.05791391295716, 74.42710732203493] Fitness: 2.6063287942746713
Generation 145 Best Solution: [183.8670259027683, 177.3440426421136,
173.74992340671542, 75.87988739489586] Fitness: 3.0167128699096786
Generation 146 Best Solution: [183.8670259027683, 175.7988353933975,
175.05791391295716, 75.87988739489586] Fitness: 2.6336567174275975
Generation 147 Best Solution: [183.8670259027683, 175.7988353933975,
174.82391573371805, 75.87988739489586] Fitness: 2.282844286287336
Generation 148 Best Solution: [183.0648920432156, 177.3440426421136,
174.20755315742787, 75.87988739489586] Fitness: 2.5106517060926348
Generation 149 Best Solution: [183.0648920432156, 177.3440426421136,
174.4938077587601, 75.87988739489586] Fitness: 2.642442639322738
Generation 150 Best Solution: [183.8670259027683, 177.3440426421136,
174.4938077587601, 75.31737475350262] Fitness: 2.1794081230078834
Generation 151 Best Solution: [183.8670259027683, 177.3440426421136,
174.4938077587601, 75.31737475350262] Fitness: 2.1794081230078834
Generation 152 Best Solution: [183.06476546364408, 177.3440426421136,
174.4938077587601, 75.31737475350262] Fitness: 1.76225162604982
Generation 153 Best Solution: [183.5443455658738, 177.3440426421136,
174.4938077587601, 75.87988739489586] Fitness: 2.7556555662014084
Generation 154 Best Solution: [183.8670259027683, 176.00534724474542,
173.85698445197067, 75.87988739489586] Fitness: 1.7598548688084263
Generation 155 Best Solution: [183.8670259027683, 175.71445905477265,
174.4938077587601, 75.87988739489586] Fitness: 2.0123194542954783
Generation 156 Best Solution: [183.8670259027683, 177.19805819049216,
173.85698445197067, 75.87988739489586] Fitness: 2.6784770022822744
Generation 157 Best Solution: [183.05072743492474, 177.3440426421136,
174.19437120727443, 75.87988739489586] Fitness: 2.5108644689914943
Generation 158 Best Solution: [183.8670259027683, 177.19805819049216,

173.85698445197067, 74.42360374776527] Fitness: 1.545091916335492
Generation 159 Best Solution: [183.8670259027683, 177.3440426421136,
174.19437120727443, 75.48962385817701] Fitness: 2.255399706298406
Generation 160 Best Solution: [183.55132595381875, 177.3440426421136,
174.19437120727443, 75.48962385817701] Fitness: 1.9560689153122905
Generation 161 Best Solution: [183.05072743492474, 177.20754338339742,
174.44827916512776, 75.87988739489586] Fitness: 2.347570996638317
Generation 162 Best Solution: [183.05072743492474, 177.20754338339742,
174.44827916512776, 75.87988739489586] Fitness: 2.347570996638317
Generation 163 Best Solution: [182.9891136841779, 177.19805819049216,
173.85698445197067, 75.87988739489586] Fitness: 2.2804291121642697
Generation 164 Best Solution: [182.9358724811387, 177.3440426421136,
174.44827916512776, 75.87988739489586] Fitness: 2.6504557238645017
Generation 165 Best Solution: [183.55132595381875, 177.20754338339742,
175.05791391295716, 74.84447805464269] Fitness: 1.9857473872251123
Generation 166 Best Solution: [183.80134481540213, 177.3440426421136,
174.3202716497841, 75.87988739489586] Fitness: 2.8857871889439983
Generation 167 Best Solution: [183.05072743492474, 176.03388827174564,
175.13286962599807, 75.87988739489586] Fitness: 2.2495543677997007
Generation 168 Best Solution: [183.8670259027683, 176.87813347123983,
175.05791391295716, 75.28251276072942] Fitness: 2.0484888070044933
Generation 169 Best Solution: [183.05072743492474, 176.03388827174564,
175.13286962599807, 75.87988739489586] Fitness: 2.2495543677997007
Generation 170 Best Solution: [183.8670259027683, 177.3440426421136,
175.13286962599807, 74.9587525985112] Fitness: 2.723535025355477
Generation 171 Best Solution: [183.8670259027683, 175.83097281026272,
175.13830399514217, 74.87177081389557] Fitness: 1.5744913336819701
Generation 172 Best Solution: [183.8670259027683, 175.83097281026272,
175.13830399514217, 74.87177081389557] Fitness: 1.5744913336819701
Generation 173 Best Solution: [183.58736977202565, 175.83097281026272,
175.13830399514217, 74.87177081389557] Fitness: 1.3004651859608618
Generation 174 Best Solution: [183.8670259027683, 176.57703495475474,
175.13286962599807, 75.204475522324] Fitness: 1.809292697652299
Generation 175 Best Solution: [183.4181638517725, 177.3440426421136,
175.05791391295716, 74.9587525985112] Fitness: 2.2177922032071145
Generation 176 Best Solution: [183.8670259027683, 176.57703495475474,
175.13286962599807, 74.9587525985112] Fitness: 1.6483833148311067
Generation 177 Best Solution: [183.8670259027683, 176.0627190539813,
175.13286962599807, 75.204475522324] Fitness: 1.6682757496323528
Generation 178 Best Solution: [183.8670259027683, 176.57703495475474,
174.03168187217548, 75.204475522324] Fitness: 0.8534558707853559
Generation 179 Best Solution: [183.8670259027683, 176.57703495475474,
174.03168187217548, 75.00506218618635] Fitness: 0.7147453134711804
Generation 180 Best Solution: [183.8670259027683, 176.0627190539813,
175.13286962599807, 75.204475522324] Fitness: 1.6682757496323528
Generation 181 Best Solution: [183.4181638517725, 176.0627190539813,
175.13286962599807, 75.204475522324] Fitness: 1.2952862048861673
Generation 182 Best Solution: [183.8670259027683, 176.57703495475474,

174.56785929405828, 75.87988739489586] Fitness: 2.042043582548752
Generation 183 Best Solution: [183.8670259027683, 176.57703495475474,
174.56785929405828, 75.87988739489586] Fitness: 2.042043582548752
Generation 184 Best Solution: [183.8670259027683, 176.57703495475474,
174.56785929405828, 75.87988739489586] Fitness: 2.042043582548752
Generation 185 Best Solution: [183.8670259027683, 176.56579694468263,
175.13286962599807, 75.87988739489586] Fitness: 2.7910114044417673
Generation 186 Best Solution: [183.8670259027683, 176.1441693624156,
175.13286962599807, 75.87988739489586] Fitness: 2.649357037953952
Generation 187 Best Solution: [183.8670259027683, 177.3440426421136,
174.36843523629065, 75.204475522324] Fitness: 1.9943765422545507
Generation 188 Best Solution: [183.8670259027683, 176.1441693624156,
175.13286962599807, 75.7347481613693] Fitness: 2.3689365307614376
Generation 189 Best Solution: [183.51174728295695, 177.3440426421136,
175.13286962599807, 74.70290271820852] Fitness: 2.3444389472777876
Generation 190 Best Solution: [183.8670259027683, 177.3440426421136,
174.126687587226, 74.70290271820852] Fitness: 1.7114289199546826
Generation 191 Best Solution: [183.8670259027683, 176.2571138816117,
175.13286962599807, 74.31370273037957] Fitness: 1.5891749350413322
Generation 192 Best Solution: [183.8670259027683, 176.2571138816117,
175.13286962599807, 75.03469309478099] Fitness: 1.5443209176359294
Generation 193 Best Solution: [183.73292743476006, 175.8426464946152,
175.13286962599807, 75.03469309478099] Fitness: 1.4790577711641602
Generation 194 Best Solution: [183.8670259027683, 176.31420244292033,
175.13286962599807, 75.03469309478099] Fitness: 1.5565354929073054
Generation 195 Best Solution: [183.51174728295695, 176.21590652845293,
175.13286962599807, 75.87988739489586] Fitness: 2.3268872571162644
Generation 196 Best Solution: [183.51174728295695, 176.21590652845293,
175.13286962599807, 74.31880402523356] Fitness: 1.255908431485297
Generation 197 Best Solution: [183.73292743476006, 176.21590652845293,
175.13286962599807, 74.31880402523356] Fitness: 1.4318160477932902
Generation 198 Best Solution: [183.53870466563757, 176.10148878592602,
175.13286962599807, 75.87988739489586] Fitness: 2.3448054525831363
Generation 199 Best Solution: [183.8670259027683, 177.3440426421136,
174.12391512581888, 74.59923027794461] Fitness: 1.7219225046882751
Generation 200 Best Solution: [183.73292743476006, 177.3440426421136,
175.13286962599807, 74.49967785842966] Fitness: 2.559066811704753
Generation 201 Best Solution: [183.8670259027683, 177.3440426421136,
175.13286962599807, 74.49967785842966] Fitness: 2.7076424077758294
Generation 202 Best Solution: [183.73292743476006, 177.3440426421136,
175.13286962599807, 74.49967785842966] Fitness: 2.559066811704753
Generation 203 Best Solution: [182.97306509373666, 177.3440426421136,
175.13286962599807, 74.77422463012704] Fitness: 2.279865544129
Generation 204 Best Solution: [183.8670259027683, 177.3440426421136,
175.13286962599807, 75.27408489380382] Fitness: 2.949343533885113
Generation 205 Best Solution: [183.8670259027683, 177.3440426421136,
174.22885961744274, 75.85676605498692] Fitness: 2.8855185503024163
Generation 206 Best Solution: [182.97306509373666, 176.97635167910946,

175.13286962599807, 75.87988739489586] Fitness: 2.836904072625651
Generation 207 Best Solution: [183.8670259027683, 176.02989299642417,
175.13286962599807, 75.87988739489586] Fitness: 2.6648557842870657
Generation 208 Best Solution: [182.97306509373666, 176.24502200181175,
174.63177098701703, 75.87988739489586] Fitness: 1.562496820148978
Generation 209 Best Solution: [182.97306509373666, 176.32763235250874,
175.13286962599807, 75.87988739489586] Fitness: 2.27984492840523
Generation 210 Best Solution: [183.8670259027683, 176.07283945270834,
175.13286962599807, 75.28231994906636] Fitness: 1.7397013235144179
Generation 211 Best Solution: [183.8670259027683, 176.07283945270834,
173.9766341729777, 75.87988739489586] Fitness: 1.7087272097049713
Generation 212 Best Solution: [183.8670259027683, 176.07283945270834,
173.9766341729777, 75.87988739489586] Fitness: 1.7087272097049713
Generation 213 Best Solution: [182.97306509373666, 176.07283945270834,
173.9766341729777, 75.87988739489586] Fitness: 1.3160280367461872
Generation 214 Best Solution: [183.8670259027683, 176.6849121029915,
175.026022748855, 75.87988739489586] Fitness: 2.7014751351846953
Generation 215 Best Solution: [183.8670259027683, 176.6849121029915,
175.026022748855, 75.87988739489586] Fitness: 2.7014751351846953
Generation 216 Best Solution: [183.8670259027683, 175.83859438833719,
174.5211168015812, 75.87988739489586] Fitness: 1.94775369939873
Generation 217 Best Solution: [183.8670259027683, 176.96404859854738,
175.13286962599807, 74.53364781337169] Fitness: 2.03403917851277
Generation 218 Best Solution: [183.8670259027683, 175.83859438833719,
174.8411168803273, 75.87988739489586] Fitness: 2.281182786274882
Generation 219 Best Solution: [183.8670259027683, 175.83859438833719,
175.13286962599807, 75.87988739489586] Fitness: 2.742242020914694
Generation 220 Best Solution: [183.8670259027683, 176.96404859854738,
175.13286962599807, 74.49729015460979] Fitness: 2.0464580308314204
Generation 221 Best Solution: [183.8670259027683, 175.83859438833719,
175.13286962599807, 74.49729015460979] Fitness: 1.576551829321398
Generation 222 Best Solution: [182.80837156200326, 175.83859438833719,
175.13286962599807, 74.49729015460979] Fitness: 1.2649371686674515
Generation 223 Best Solution: [183.8670259027683, 177.3531619998319,
174.6769647594587, 74.93394824032939] Fitness: 2.077595692032384
Generation 224 Best Solution: [183.8670259027683, 176.96404859854738,
175.09555178296569, 75.01893836205849] Fitness: 2.0237198337155147
Generation 225 Best Solution: [183.8670259027683, 177.3531619998319,
174.75027322711057, 75.01893836205849] Fitness: 2.1980973961396653
Generation 226 Best Solution: [183.8670259027683, 176.3061094211812,
175.13286962599807, 75.87988739489586] Fitness: 2.666756740937825
Generation 227 Best Solution: [183.8670259027683, 176.3061094211812,
175.13286962599807, 75.87988739489586] Fitness: 2.666756740937825
Generation 228 Best Solution: [183.8670259027683, 176.3844255143118,
174.55877361641342, 75.87988739489586] Fitness: 1.9271219471682648
Generation 229 Best Solution: [183.48439312378127, 177.3555628126677,
175.13286962599807, 74.98134418624373] Fitness: 2.41767811560654
Generation 230 Best Solution: [183.8670259027683, 177.3531619998319,

174.37527920141932, 74.98134418624373] Fitness: 1.8662425852365965
Generation 231 Best Solution: [183.7650879856941, 175.82687473186706,
175.13286962599807, 74.54277059473667] Fitness: 1.452388738923548
Generation 232 Best Solution: [183.7650879856941, 176.16471002159813,
175.13286962599807, 75.87988739489586] Fitness: 2.533180131129572
Generation 233 Best Solution: [183.48439312378127, 176.2716210219636,
175.13286962599807, 75.87988739489586] Fitness: 2.3189272743641953
Generation 234 Best Solution: [183.7650879856941, 176.16471002159813,
175.13286962599807, 75.87988739489586] Fitness: 2.533180131129572
Generation 235 Best Solution: [183.7650879856941, 176.16471002159813,
175.13286962599807, 75.87988739489586] Fitness: 2.533180131129572
Generation 236 Best Solution: [183.7650879856941, 177.3531619998319,
174.75152189481193, 75.17091278029551] Fitness: 2.186803356154366
Generation 237 Best Solution: [183.7650879856941, 177.3531619998319,
174.50300801977653, 75.17091278029551] Fitness: 1.951414412188374
Generation 238 Best Solution: [183.7650879856941, 177.3531619998319,
173.89848114638448, 75.17091278029551] Fitness: 1.8326185544545617
Generation 239 Best Solution: [183.06033633181642, 177.3531619998319,
173.89848114638448, 74.62456936055091] Fitness: 1.3511689880585502
Generation 240 Best Solution: [183.7650879856941, 175.94309690968723,
175.13286962599807, 75.87988739489586] Fitness: 2.5761575398650534
Generation 241 Best Solution: [183.11387155625397, 177.3531619998319,
174.75152189481193, 75.87988739489586] Fitness: 2.896482132104813
Generation 242 Best Solution: [183.8670259027683, 175.96040011164797,
175.13286962599807, 75.87988739489586] Fitness: 2.6855189421420356
Generation 243 Best Solution: [183.06033633181642, 177.3531619998319,
174.7566026466107, 75.87988739489586] Fitness: 2.9102818478732004
Generation 244 Best Solution: [183.7650879856941, 177.39029839889608,
173.88191438056904, 75.87988739489586] Fitness: 2.9353687565228626
Generation 245 Best Solution: [183.06033633181642, 176.31891398993528,
175.13286962599807, 75.87988739489586] Fitness: 2.2537861994208743
Generation 246 Best Solution: [183.7650879856941, 177.39029839889608,
175.13286962599807, 74.89594839936215] Fitness: 2.6815747293470853
Generation 247 Best Solution: [183.06595296710393, 175.9509677781025,
175.13286962599807, 75.87988739489586] Fitness: 2.2716025399435846
Generation 248 Best Solution: [183.06595296710393, 177.3531619998319,
174.43136216842183, 75.87988739489586] Fitness: 2.6202094315165754
Generation 249 Best Solution: [183.7650879856941, 177.3531619998319,
174.43136216842183, 75.87988739489586] Fitness: 2.9217575780851295
Generation 250 Best Solution: [183.7650879856941, 177.39029839889608,
174.13635356394124, 75.87988739489586] Fitness: 2.899563003747217
Generation 251 Best Solution: [183.7650879856941, 177.39029839889608,
174.13635356394124, 75.87988739489586] Fitness: 2.899563003747217
Generation 252 Best Solution: [183.06033633181642, 177.31982761037153,
174.43136216842183, 75.87988739489586] Fitness: 2.5524870118709053
Generation 253 Best Solution: [183.06595296710393, 176.2254312461566,
175.13286962599807, 75.87988739489586] Fitness: 2.234942767773929
Generation 254 Best Solution: [183.00410559804348, 175.8817426966673,

175.13286962599807, 75.87988739489586] Fitness: 2.316526924084271
Generation 255 Best Solution: [183.06033633181642, 177.3531619998319,
175.13286962599807, 74.40016771508516] Fitness: 2.355808296128614
Generation 256 Best Solution: [183.06033633181642, 177.39029839889608,
175.13286962599807, 74.40016771508516] Fitness: 2.434739675440716
Generation 257 Best Solution: [183.06033633181642, 177.3531619998319,
175.03806352039763, 75.28830065780396] Fitness: 2.400088142083728
Generation 258 Best Solution: [183.06033633181642, 175.95567462296137,
175.03806352039763, 75.87988739489586] Fitness: 2.10466330823224
Generation 259 Best Solution: [183.06033633181642, 175.95567462296137,
175.03806352039763, 75.87988739489586] Fitness: 2.10466330823224
Generation 260 Best Solution: [183.06033633181642, 175.95567462296137,
175.03806352039763, 75.87988739489586] Fitness: 2.10466330823224
Generation 261 Best Solution: [183.06033633181642, 177.3531619998319,
175.03806352039763, 74.74393883640832] Fitness: 2.1063664231708366
Generation 262 Best Solution: [183.06033633181642, 176.70955526719007,
175.13286962599807, 75.87988739489586] Fitness: 2.494520037463933
Generation 263 Best Solution: [183.06033633181642, 177.3531619998319,
175.13286962599807, 74.51050687542556] Fitness: 2.3064418539965788
Generation 264 Best Solution: [183.06033633181642, 177.3531619998319,
175.13286962599807, 74.35420056543192] Fitness: 2.3826970521286945
Generation 265 Best Solution: [183.06595296710393, 176.91189545874533,
174.3518607388844, 75.87988739489586] Fitness: 1.825762991154132
Generation 266 Best Solution: [183.2736586891627, 176.91189545874533,
174.3518607388844, 75.87988739489586] Fitness: 1.8255258447068026
Generation 267 Best Solution: [183.06033633181642, 176.17130257686134,
175.03806352039763, 75.87988739489586] Fitness: 2.06648420888088
Generation 268 Best Solution: [183.2736586891627, 176.24214205432952,
175.03806352039763, 75.87988739489586] Fitness: 2.0705014506146227
Generation 269 Best Solution: [183.06033633181642, 176.35094092280914,
175.03806352039763, 75.87988739489586] Fitness: 2.0971618180184164
Generation 270 Best Solution: [183.06033633181642, 177.39029839889608,
175.03806352039763, 74.52733438407456] Fitness: 2.2134736156307775
Generation 271 Best Solution: [183.06033633181642, 176.14975499139928,
175.03806352039763, 75.87988739489586] Fitness: 2.0666192980188267
Generation 272 Best Solution: [183.06033633181642, 177.39029839889608,
175.03806352039763, 74.8237350725893] Fitness: 2.195731402937609
Generation 273 Best Solution: [183.06033633181642, 177.39029839889608,
175.03806352039763, 74.8237350725893] Fitness: 2.195731402937609
Generation 274 Best Solution: [183.06033633181642, 177.3531619998319,
175.03806352039763, 74.86271726105694] Fitness: 2.125971677053582
Generation 275 Best Solution: [183.06033633181642, 177.3531619998319,
174.8617300443172, 74.84242912933284] Fitness: 1.8537161198429706
Generation 276 Best Solution: [183.06033633181642, 177.3531619998319,
174.8617300443172, 74.84242912933284] Fitness: 1.8537161198429706
Generation 277 Best Solution: [183.06033633181642, 176.91546708070962,
175.13286962599807, 74.65438559295086] Fitness: 1.5268519191748249
Generation 278 Best Solution: [183.06033633181642, 176.91546708070962,

175.13286962599807, 75.36135865792737] Fitness: 1.8980516199340645
Generation 279 Best Solution: [183.06033633181642, 176.91546708070962,
175.13286962599807, 74.7500374186107] Fitness: 1.5256172693388248
Generation 280 Best Solution: [183.06033633181642, 177.27956861288976,
175.13286962599807, 74.72244084030447] Fitness: 2.122478191768891
Generation 281 Best Solution: [183.06033633181642, 177.39029839889608,
174.9076865010891, 74.72244084030447] Fitness: 1.9816069414710475
Generation 282 Best Solution: [183.06033633181642, 176.34920618830034,
175.03806352039763, 75.87988739489586] Fitness: 2.096593988312265
Generation 283 Best Solution: [183.06033633181642, 176.34920618830034,
175.03806352039763, 75.87988739489586] Fitness: 2.096593988312265
Generation 284 Best Solution: [183.06033633181642, 176.34920618830034,
175.03806352039763, 75.87988739489586] Fitness: 2.096593988312265
Generation 285 Best Solution: [183.06033633181642, 176.0503177549306,
174.19884681416463, 75.87988739489586] Fitness: 1.2959471133274343
Generation 286 Best Solution: [183.06033633181642, 176.34920618830034,
175.03806352039763, 75.87988739489586] Fitness: 2.096593988312265
Generation 287 Best Solution: [183.06033633181642, 176.34920618830034,
175.03806352039763, 75.87988739489586] Fitness: 2.096593988312265
Generation 288 Best Solution: [183.06033633181642, 176.34920618830034,
175.03806352039763, 75.87988739489586] Fitness: 2.096593988312265
Generation 289 Best Solution: [183.40398703977345, 177.3531619998319,
175.03806352039763, 74.72244084030447] Fitness: 2.1427938394144026
Generation 290 Best Solution: [183.06033633181642, 176.20646274956965,
175.13286962599807, 74.72244084030447] Fitness: 1.0290824731389048
Generation 291 Best Solution: [183.06033633181642, 176.0313699867178,
175.13286962599807, 74.72244084030447] Fitness: 1.0430036070645148
Generation 292 Best Solution: [183.06033633181642, 176.0313699867178,
173.8884882343588, 75.87988739489586] Fitness: 1.3248688299218536
Generation 293 Best Solution: [183.06033633181642, 176.0313699867178,
173.8884882343588, 75.87988739489586] Fitness: 1.3248688299218536
Generation 294 Best Solution: [183.06033633181642, 177.39029839889608,
173.90348752308856, 74.72244084030447] Fitness: 1.4222348302889134
Generation 295 Best Solution: [183.40398703977345, 176.0685823270853,
175.13286962599807, 74.41684388302285] Fitness: 1.1480988126003135
Generation 296 Best Solution: [183.40398703977345, 176.0685823270853,
175.01579951924245, 75.87988739489586] Fitness: 2.075038791956477
Generation 297 Best Solution: [183.06033633181642, 177.3531619998319,
175.13286962599807, 74.30556959666448] Fitness: 2.415192375236466
Generation 298 Best Solution: [183.06033633181642, 177.3531619998319,
175.13286962599807, 74.41684388302285] Fitness: 2.3469727655110666
Generation 299 Best Solution: [183.06033633181642, 176.35201694387513,
175.13286962599807, 75.87988739489586] Fitness: 2.263770686625941
Generation 300 Best Solution: [182.86747247689735, 176.7719401067788,
175.13286962599807, 75.73126036529746] Fitness: 2.3412438335703287
Generation 301 Best Solution: [183.06033633181642, 176.7719401067788,
175.13286962599807, 75.73126036529746] Fitness: 2.2711323742250666
Generation 302 Best Solution: [183.06033633181642, 176.7719401067788,

175.13286962599807, 75.73126036529746] Fitness: 2.2711323742250666
Generation 303 Best Solution: [182.86747247689735, 176.1109040700294,
175.13286962599807, 75.87988739489586] Fitness: 2.3052932738348835
Generation 304 Best Solution: [182.86747247689735, 176.1109040700294,
175.13286962599807, 75.87988739489586] Fitness: 2.3052932738348835
Generation 305 Best Solution: [183.06033633181642, 177.3531619998319,
175.13286962599807, 75.4044944726035] Fitness: 2.696579101911751
Generation 306 Best Solution: [182.86747247689735, 175.90537374218061,
174.59972284940426, 75.4044944726035] Fitness: 0.8520881856087779
Generation 307 Best Solution: [182.86747247689735, 176.81238209226666,
175.13286962599807, 75.87988739489586] Fitness: 2.672651577968854
Generation 308 Best Solution: [182.86747247689735, 175.891571952659,
175.13286962599807, 75.87988739489586] Fitness: 2.368158486588416
Generation 309 Best Solution: [183.06033633181642, 177.39029839889608,
174.06988680032103, 75.90171057007154] Fitness: 2.6428049965783
Generation 310 Best Solution: [183.06033633181642, 175.98371938108204,
175.13286962599807, 75.87988739489586] Fitness: 2.261322256964492
Generation 311 Best Solution: [182.86747247689735, 175.6514830178784,
175.13286962599807, 75.87988739489586] Fitness: 2.5340387228847816
Generation 312 Best Solution: [182.86747247689735, 175.8888117445233,
175.13286962599807, 75.87988739489586] Fitness: 2.3694890789990826
Generation 313 Best Solution: [182.80013590813428, 175.65767323872544,
174.0117817908851, 75.87988739489586] Fitness: 1.614907284071507
Generation 314 Best Solution: [183.06033633181642, 177.3531619998319,
175.13286962599807, 74.62419242343047] Fitness: 2.277990383865935
Generation 315 Best Solution: [183.06033633181642, 177.39029839889608,
174.0117817908851, 74.62419242343047] Fitness: 1.403444130280713
Generation 316 Best Solution: [183.21412538109945, 177.3531619998319,
175.13286962599807, 74.62419242343047] Fitness: 2.268997770267679
Generation 317 Best Solution: [183.06033633181642, 177.31322431571363,
175.13286962599807, 74.62419242343047] Fitness: 2.195813807960628
Generation 318 Best Solution: [183.21412538109945, 176.11243451909002,
175.13286962599807, 75.87988739489586] Fitness: 2.226047998536506
Generation 319 Best Solution: [183.21412538109945, 176.11243451909002,
174.6220631150194, 75.87988739489586] Fitness: 1.517283117469548
Generation 320 Best Solution: [183.21412538109945, 176.11243451909002,
174.6220631150194, 75.90171057007154] Fitness: 1.562653634163627
Generation 321 Best Solution: [182.80013590813428, 177.4460488719803,
175.13286962599807, 74.40347947766762] Fitness: 2.6660063851910594
Generation 322 Best Solution: [183.21412538109945, 175.65157916943696,
175.13286962599807, 75.90171057007154] Fitness: 2.500218427807158
Generation 323 Best Solution: [183.21412538109945, 175.98956043651944,
175.13286962599807, 75.90171057007154] Fitness: 2.2958759180713093
Generation 324 Best Solution: [183.21412538109945, 177.31322431571363,
175.13286962599807, 74.39965366570442] Fitness: 2.264919241528482
Generation 325 Best Solution: [183.21412538109945, 177.4460488719803,
175.13286962599807, 75.094428821704] Fitness: 2.5949288493010916
Generation 326 Best Solution: [183.21412538109945, 176.45144157450017,

175.13286962599807, 75.87988739489586] Fitness: 2.2963617781457373
Generation 327 Best Solution: [182.80013590813428, 176.1322975795508,
174.602284708238, 75.87988739489586] Fitness: 1.6166172425811998
Generation 328 Best Solution: [182.80013590813428, 176.45144157450017,
175.13286962599807, 75.39147646588343] Fitness: 1.6192531685107734
Generation 329 Best Solution: [182.80013590813428, 177.06375178642614,
175.13286962599807, 75.55113191905376] Fitness: 2.4729034713485083
Generation 330 Best Solution: [182.80013590813428, 177.06375178642614,
175.13286962599807, 75.39147646588343] Fitness: 2.25885414309165
Generation 331 Best Solution: [183.21412538109945, 175.64616002397204,
175.13286962599807, 75.55113191905376] Fitness: 1.8777013528851239
Generation 332 Best Solution: [183.21412538109945, 175.64616002397204,
175.13286962599807, 75.39147646588343] Fitness: 1.663652024628266
Generation 333 Best Solution: [183.06033633181642, 177.06375178642614,
175.13286962599807, 75.39147646588343] Fitness: 2.148845455580393
Generation 334 Best Solution: [183.17542722978132, 177.06375178642614,
173.8719354654351, 75.87988739489586] Fitness: 2.017066807765766
Generation 335 Best Solution: [183.17542722978132, 177.06375178642614,
173.8719354654351, 75.87988739489586] Fitness: 2.017066807765766
Generation 336 Best Solution: [183.17542722978132, 177.06375178642614,
173.8719354654351, 75.87988739489586] Fitness: 2.017066807765766
Generation 337 Best Solution: [183.17542722978132, 177.06375178642614,
173.8719354654351, 75.87988739489586] Fitness: 2.017066807765766
Generation 338 Best Solution: [183.17542722978132, 175.83214648999635,
175.13286962599807, 74.59936417695202] Fitness: 1.12435484072515
Generation 339 Best Solution: [183.17542722978132, 175.83214648999635,
175.13286962599807, 74.59936417695202] Fitness: 1.12435484072515
Generation 340 Best Solution: [183.17542722978132, 175.83214648999635,
175.13286962599807, 75.90171057007154] Fitness: 2.364378459620945
Generation 341 Best Solution: [183.41215545098967, 177.4460488719803,
173.83192394730236, 74.59936417695202] Fitness: 1.6244452515555352
Generation 342 Best Solution: [183.06033633181642, 177.4460488719803,
173.83192394730236, 74.59936417695202] Fitness: 1.583707141605282
Generation 343 Best Solution: [183.06033633181642, 177.4460488719803,
173.83192394730236, 74.59936417695202] Fitness: 1.583707141605282
Generation 344 Best Solution: [183.41215545098967, 177.4460488719803,
174.3433735637327, 75.87988739489586] Fitness: 2.8178001973074434
Generation 345 Best Solution: [183.06033633181642, 176.2346845312395,
175.13286962599807, 75.87988739489586] Fitness: 2.23707787977168
Generation 346 Best Solution: [183.06033633181642, 176.2346845312395,
175.13286962599807, 75.87988739489586] Fitness: 2.23707787977168
Generation 347 Best Solution: [183.06033633181642, 176.2346845312395,
175.13286962599807, 75.63904919661064] Fitness: 1.7920415950413096
Generation 348 Best Solution: [182.80013590813428, 176.29459909456446,
174.8687628290939, 75.87988739489586] Fitness: 1.9338942008480253
Generation 349 Best Solution: [183.3055353441405, 176.2346845312395,
175.13286962599807, 75.87988739489586] Fitness: 2.242464184826613
Generation 350 Best Solution: [183.86976087277668, 176.2346845312395,

174.8687628290939, 75.87988739489586] Fitness: 2.2329573778307106
Generation 351 Best Solution: [183.3055353441405, 176.2346845312395,
175.13286962599807, 75.87988739489586] Fitness: 2.242464184826613
Generation 352 Best Solution: [183.06033633181642, 176.2346845312395,
175.13286962599807, 75.87988739489586] Fitness: 2.23707787977168
Generation 353 Best Solution: [183.3055353441405, 176.2346845312395,
174.8687628290939, 75.87988739489586] Fitness: 1.8186687626426665
Generation 354 Best Solution: [183.06033633181642, 175.7870971528227,
174.69049997041446, 75.87988739489586] Fitness: 1.7173143998884561
Generation 355 Best Solution: [183.06033633181642, 176.10882309638134,
174.8687628290939, 75.87988739489586] Fitness: 1.8115850006253371
Generation 356 Best Solution: [183.06033633181642, 176.16506344479217,
174.69049997041446, 75.87988739489586] Fitness: 1.5922458550216723
Generation 357 Best Solution: [183.3055353441405, 177.4460488719803,
175.13286962599807, 74.87677412250014] Fitness: 2.5035583624964053
Generation 358 Best Solution: [183.3055353441405, 177.4460488719803,
174.8687628290939, 75.10145680481237] Fitness: 2.1903165299047442
Generation 359 Best Solution: [183.3055353441405, 177.4460488719803,
175.13286962599807, 74.87677412250014] Fitness: 2.5035583624964053
Generation 360 Best Solution: [183.74838398607096, 177.4460488719803,
174.69049997041446, 74.95774046620105] Fitness: 2.170574988041788
Generation 361 Best Solution: [183.3055353441405, 177.4460488719803,
174.8687628290939, 74.95774046620105] Fitness: 2.109362036366302
Generation 362 Best Solution: [183.3055353441405, 176.72138942187115,
174.8687628290939, 74.87677412250014] Fitness: 0.9068685043962046
Generation 363 Best Solution: [183.3476025757799, 175.6686866230495,
175.13286962599807, 75.87988739489586] Fitness: 2.4656115484875096
Generation 364 Best Solution: [183.3055353441405, 177.4460488719803,
174.8687628290939, 74.29039162574176] Fitness: 2.2097605045495645
Generation 365 Best Solution: [183.3055353441405, 177.4460488719803,
174.8687628290939, 74.29039162574176] Fitness: 2.2097605045495645
Generation 366 Best Solution: [182.80013590813428, 176.9852555320619,
175.13286962599807, 74.83906663530111] Fitness: 1.7455215218408788
Generation 367 Best Solution: [183.3055353441405, 177.4460488719803,
174.8687628290939, 75.28053566519161] Fitness: 2.342059606388161
Generation 368 Best Solution: [183.3055353441405, 177.4460488719803,
174.8687628290939, 75.28053566519161] Fitness: 2.342059606388161
Generation 369 Best Solution: [182.80013590813428, 177.4460488719803,
174.61260256717557, 75.36650798341479] Fitness: 2.245818554199263
Generation 370 Best Solution: [183.3055353441405, 177.4460488719803,
174.2477547624313, 75.87988739489586] Fitness: 2.747701196328645
Generation 371 Best Solution: [182.80013590813428, 175.75826124372793,
174.8687628290939, 75.87988739489586] Fitness: 2.0638396919634916
Generation 372 Best Solution: [182.80013590813428, 175.75826124372793,
174.8687628290939, 75.87988739489586] Fitness: 2.0638396919634916
Generation 373 Best Solution: [183.26596525953065, 177.4460488719803,
175.13286962599807, 75.09376371702486] Fitness: 2.6008280362429104
Generation 374 Best Solution: [182.80013590813428, 177.4460488719803,

175.13286962599807, 74.51199708097128] Fitness: 2.6179134855667208
Generation 375 Best Solution: [182.80013590813428, 177.4460488719803,
175.13286962599807, 74.51199708097128] Fitness: 2.6179134855667208
Generation 376 Best Solution: [182.80013590813428, 177.4460488719803,
175.13286962599807, 74.51199708097128] Fitness: 2.6179134855667208
Generation 377 Best Solution: [182.80013590813428, 177.4460488719803,
174.99289740319696, 74.61411118262671] Fitness: 2.3516910537285285
Generation 378 Best Solution: [183.3055353441405, 176.3605555093903,
174.8687628290939, 75.87988739489586] Fitness: 1.8482498567634806
Generation 379 Best Solution: [183.26596525953065, 177.4460488719803,
174.50336847328242, 74.51199708097128] Fitness: 1.6975551840057277
Generation 380 Best Solution: [182.80013590813428, 175.62601997894464,
174.8687628290939, 74.61411118262671] Fitness: 0.9763621372776314
Generation 381 Best Solution: [183.3055353441405, 177.4460488719803,
175.13286962599807, 74.61411118262671] Fitness: 2.4869635626226745
Generation 382 Best Solution: [183.8437131443636, 176.57536885446126,
174.8687628290939, 75.87988739489586] Fitness: 2.345961601411042
Generation 383 Best Solution: [183.02013301981256, 177.4460488719803,
175.13286962599807, 74.61411118262671] Fitness: 2.4907913328526443
Generation 384 Best Solution: [183.02013301981256, 177.4460488719803,
175.13286962599807, 74.61411118262671] Fitness: 2.4907913328526443
Generation 385 Best Solution: [183.8437131443636, 176.47065579991335,
173.91913397933982, 75.87988739489586] Fitness: 1.770291595789288
Generation 386 Best Solution: [183.8437131443636, 176.47065579991335,
173.91913397933982, 75.87988739489586] Fitness: 1.770291595789288
Generation 387 Best Solution: [183.8437131443636, 176.47065579991335,
173.91913397933982, 75.74599581025609] Fitness: 1.5102771883976924
Generation 388 Best Solution: [183.8437131443636, 176.05939491160763,
175.13286962599807, 75.87988739489586] Fitness: 2.630551322711036
Generation 389 Best Solution: [183.8437131443636, 176.05939491160763,
174.37157365298805, 75.87988739489586] Fitness: 1.7420351145545805
Generation 390 Best Solution: [183.8437131443636, 176.68341218493327,
175.13286962599807, 75.87988739489586] Fitness: 2.8582366580861356
Generation 391 Best Solution: [183.39232704640295, 176.03672667126781,
175.13286962599807, 75.89234470506544] Fitness: 2.3054128585522666
Generation 392 Best Solution: [183.4823576034344, 177.4460488719803,
175.13286962599807, 75.42231462674887] Fitness: 2.995576184054258
Generation 393 Best Solution: [183.8437131443636, 176.0640009817554,
175.13286962599807, 75.89234470506544] Fitness: 2.6555174079012303
Generation 394 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 74.87538651974272] Fitness: 2.885978334805465
Generation 395 Best Solution: [183.8437131443636, 176.78230505505724,
174.9087787633439, 75.87988739489586] Fitness: 2.5897559720053467
Generation 396 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 74.84374702421432] Fitness: 2.8776243209137906
Generation 397 Best Solution: [183.8437131443636, 176.28201763924147,
175.13286962599807, 75.89234470506544] Fitness: 2.6589394816156116
Generation 398 Best Solution: [183.8437131443636, 177.19035779166728,

173.9882975313017, 75.87988739489586] Fitness: 2.597945349330227
Generation 399 Best Solution: [183.56581622336864, 177.4460488719803,
174.0049395853157, 75.89234470506544] Fitness: 2.879072058644264
Generation 400 Best Solution: [183.06429045359994, 177.4460488719803,
175.13286962599807, 74.36954015312537] Fitness: 2.5745410719499153
Generation 401 Best Solution: [183.8437131443636, 175.89768201545493,
175.13286962599807, 75.87988739489586] Fitness: 2.6833622758240425
Generation 402 Best Solution: [183.8437131443636, 175.89768201545493,
174.9087787633439, 75.89234470506544] Fitness: 2.3416828966655503
Generation 403 Best Solution: [183.8437131443636, 175.68297712556077,
174.9087787633439, 75.89234470506544] Fitness: 2.482920057868572
Generation 404 Best Solution: [183.8437131443636, 177.00236062971237,
174.42346767541713, 74.43155674821645] Fitness: 1.2415999330526721
Generation 405 Best Solution: [183.1407554446105, 177.4460488719803,
175.13286962599807, 74.70876885552511] Fitness: 2.4636683058173565
Generation 406 Best Solution: [183.06429045359994, 177.4460488719803,
175.13286962599807, 75.17290092100868] Fitness: 2.6617443932605975
Generation 407 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 74.43155674821645] Fitness: 2.9297798613132864
Generation 408 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 74.43155674821645] Fitness: 2.9297798613132864
Generation 409 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 75.63029505091394] Fitness: 3.6078217078756056
Generation 410 Best Solution: [183.8437131443636, 175.98932999206764,
175.13286962599807, 75.87988739489586] Fitness: 2.647781833793256
Generation 411 Best Solution: [183.8437131443636, 176.24547557664326,
175.13286962599807, 75.87988739489586] Fitness: 2.6267340694677923
Generation 412 Best Solution: [183.8437131443636, 177.4460488719803,
174.3201742536376, 75.63029505091394] Fitness: 2.6960761498631065
Generation 413 Best Solution: [183.8437131443636, 177.4460488719803,
174.3201742536376, 75.63029505091394] Fitness: 2.6960761498631065
Generation 414 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 74.74992481684083] Fitness: 2.863210300719848
Generation 415 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 74.72585951433621] Fitness: 2.8620096776896258
Generation 416 Best Solution: [183.36365811465572, 177.4460488719803,
175.13286962599807, 74.74992481684083] Fitness: 2.497174721489294
Generation 417 Best Solution: [183.24852451048665, 177.13033940233393,
175.13286962599807, 74.72585951433621] Fitness: 1.8438828629419506
Generation 418 Best Solution: [183.24852451048665, 177.13033940233393,
174.3201742536376, 75.87988739489586] Fitness: 2.1372393206701994
Generation 419 Best Solution: [183.24852451048665, 177.13033940233393,
174.3201742536376, 75.87988739489586] Fitness: 2.1372393206701994
Generation 420 Best Solution: [183.8437131443636, 176.81345601565047,
173.7833257223173, 75.87988739489586] Fitness: 2.1155155964971812
Generation 421 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 74.70389634574518] Fitness: 2.861803549907001
Generation 422 Best Solution: [183.8437131443636, 176.81345601565047,

175.13286962599807, 74.84974131645598] Fitness: 1.8039395253429915
Generation 423 Best Solution: [183.02268673194644, 176.81345601565047,
175.13286962599807, 75.87988739489586] Fitness: 2.6123106610912683
Generation 424 Best Solution: [182.87694004032267, 175.83812375617737,
174.95737302481646, 75.87988739489586] Fitness: 2.0960452884315526
Generation 425 Best Solution: [183.8437131443636, 176.30651798683195,
175.05423511591002, 75.6380830622158] Fitness: 2.0531264590364535
Generation 426 Best Solution: [183.58068712340256, 177.4460488719803,
174.236992051241, 75.6380830622158] Fitness: 2.4302203463511263
Generation 427 Best Solution: [183.8437131443636, 176.59589303814468,
175.05423511591002, 75.87988739489586] Fitness: 2.645969940742978
Generation 428 Best Solution: [183.8437131443636, 176.59589303814468,
175.05423511591002, 75.6380830622158] Fitness: 2.199353952476189
Generation 429 Best Solution: [183.8437131443636, 176.59589303814468,
175.05423511591002, 75.87988739489586] Fitness: 2.645969940742978
Generation 430 Best Solution: [183.02268673194644, 176.59589303814468,
175.13286962599807, 75.87988739489586] Fitness: 2.4053153336849338
Generation 431 Best Solution: [183.8437131443636, 176.59589303814468,
175.13286962599807, 75.6380830622158] Fitness: 2.338368141998634
Generation 432 Best Solution: [183.8437131443636, 176.59589303814468,
175.13286962599807, 75.87988739489586] Fitness: 2.784984130265424
Generation 433 Best Solution: [183.8437131443636, 176.59589303814468,
175.13286962599807, 75.87988739489586] Fitness: 2.784984130265424
Generation 434 Best Solution: [183.8437131443636, 176.59589303814468,
174.67256742053283, 75.87988739489586] Fitness: 2.125838460257171
Generation 435 Best Solution: [183.8437131443636, 176.59589303814468,
174.67256742053283, 75.87988739489586] Fitness: 2.125838460257171
Generation 436 Best Solution: [183.8437131443636, 176.59589303814468,
175.13286962599807, 75.87988739489586] Fitness: 2.784984130265424
Generation 437 Best Solution: [183.32621713385868, 175.81101584789957,
175.05423511591002, 75.87988739489586] Fitness: 2.2140604522897034
Generation 438 Best Solution: [183.8437131443636, 175.9094073797667,
175.05423511591002, 75.87988739489586] Fitness: 2.538971287622209
Generation 439 Best Solution: [183.8437131443636, 175.83775375300175,
175.0194631147394, 75.87988739489586] Fitness: 2.517606067193647
Generation 440 Best Solution: [183.8437131443636, 176.0860050654766,
175.13286962599807, 75.87988739489586] Fitness: 2.626271132144902
Generation 441 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 74.80597757604698] Fitness: 2.8699587225388044
Generation 442 Best Solution: [183.8437131443636, 176.76957524652158,
175.13286962599807, 75.08397169324019] Fitness: 1.8615587307520325
Generation 443 Best Solution: [183.48348597030008, 176.90745810413168,
175.05423511591002, 75.87988739489586] Fitness: 2.655520791893048
Generation 444 Best Solution: [183.86714413769195, 177.4460488719803,
174.4525061896889, 74.80597757604698] Fitness: 2.055691117381835
Generation 445 Best Solution: [183.8437131443636, 177.22968017692088,
175.13286962599807, 75.08397169324019] Fitness: 2.5381636310117166
Generation 446 Best Solution: [183.8437131443636, 177.22968017692088,

175.13286962599807, 74.82346328973118] Fitness: 2.426212024167994
Generation 447 Best Solution: [183.8437131443636, 176.09652634756492,
175.00522330345862, 75.87988739489586] Fitness: 2.404768202270567
Generation 448 Best Solution: [183.8437131443636, 176.09652634756492,
175.00522330345862, 75.87988739489586] Fitness: 2.404768202270567
Generation 449 Best Solution: [183.8437131443636, 176.07804408994463,
175.13286962599807, 74.82737487443414] Fitness: 1.434302482590036
Generation 450 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 74.67373498175857] Fitness: 2.8629039701342656
Generation 451 Best Solution: [183.8437131443636, 175.58283566397623,
175.13286962599807, 75.8105848198862] Fitness: 2.7796970973096164
Generation 452 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 74.67373498175857] Fitness: 2.8629039701342656
Generation 453 Best Solution: [183.8437131443636, 177.36567734418185,
174.18730002739875, 75.87988739489586] Fitness: 2.941208685248295
Generation 454 Best Solution: [183.8437131443636, 176.4034652996228,
174.5285927092746, 75.87988739489586] Fitness: 1.8825747918268947
Generation 455 Best Solution: [183.8437131443636, 176.4034652996228,
174.5285927092746, 75.87988739489586] Fitness: 1.8825747918268947
Generation 456 Best Solution: [183.8437131443636, 176.9219005511713,
175.04741732043834, 75.87988739489586] Fitness: 2.9757134609626634
Generation 457 Best Solution: [183.8437131443636, 176.9219005511713,
175.04741732043834, 75.8105848198862] Fitness: 2.8371903964470504
Generation 458 Best Solution: [183.8437131443636, 176.9219005511713,
175.04741732043834, 75.44030366463588] Fitness: 2.240303888721498
Generation 459 Best Solution: [183.8437131443636, 175.9495239701512,
175.13286962599807, 75.87988739489586] Fitness: 2.6614197157803394
Generation 460 Best Solution: [183.8437131443636, 175.9495239701512,
175.13286962599807, 75.87988739489586] Fitness: 2.6614197157803394
Generation 461 Best Solution: [183.8437131443636, 175.70466260749026,
175.13286962599807, 74.8937127657568] Fitness: 1.6311621218775338
Generation 462 Best Solution: [183.8437131443636, 177.36567734418185,
175.13286962599807, 74.39318674008655] Fitness: 2.774194856130318
Generation 463 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 74.8937127657568] Fitness: 2.8916229536649527
Generation 464 Best Solution: [183.8437131443636, 177.22728908491163,
175.13286962599807, 74.79832256822391] Fitness: 2.4172483231305164
Generation 465 Best Solution: [183.8437131443636, 177.4460488719803,
173.74242128022857, 74.8937127657568] Fitness: 2.038644076767286
Generation 466 Best Solution: [183.8437131443636, 176.38494941394393,
174.5285927092746, 74.79832256822391] Fitness: 0.6766751508530072
Generation 467 Best Solution: [183.8437131443636, 177.4460488719803,
173.74242128022857, 74.8937127657568] Fitness: 2.038644076767286
Generation 468 Best Solution: [183.8437131443636, 176.26682581636334,
175.13286962599807, 74.8937127657568] Fitness: 1.454704607114879
Generation 469 Best Solution: [183.54006000718803, 177.22728908491163,
175.13286962599807, 75.16288430758554] Fitness: 2.3124982236992895
Generation 470 Best Solution: [183.54006000718803, 177.4460488719803,

175.13286962599807, 74.7057036584708] Fitness: 2.583119784815482
Generation 471 Best Solution: [183.34927665288893, 177.36567734418185,
175.13286962599807, 75.45593811390182] Fitness: 2.805627902460091
Generation 472 Best Solution: [183.8437131443636, 177.22728908491163,
175.13286962599807, 75.01853539690617] Fitness: 2.4943305226694807
Generation 473 Best Solution: [183.8437131443636, 177.22728908491163,
175.13286962599807, 75.01853539690617] Fitness: 2.4943305226694807
Generation 474 Best Solution: [183.23650589642, 177.4460488719803,
174.75441565684417, 75.01853539690617] Fitness: 1.9815496893671616
Generation 475 Best Solution: [183.8437131443636, 177.4460488719803,
174.75441565684417, 74.42255313621102] Fitness: 2.3650573557494443
Generation 476 Best Solution: [183.8437131443636, 177.4460488719803,
174.8495130032469, 74.55448621828013] Fitness: 2.43304925095219
Generation 477 Best Solution: [183.8437131443636, 177.4460488719803,
174.62845228281728, 75.68143514079247] Fitness: 2.990260961541411
Generation 478 Best Solution: [183.8437131443636, 177.4460488719803,
174.62845228281728, 75.68143514079247] Fitness: 2.990260961541411
Generation 479 Best Solution: [183.8437131443636, 177.4460488719803,
174.11175070189057, 75.68143514079247] Fitness: 2.7347114033141313
Generation 480 Best Solution: [183.8437131443636, 177.4460488719803,
174.11175070189057, 75.70607793246917] Fitness: 2.7773979305587706
Generation 481 Best Solution: [183.8437131443636, 177.4460488719803,
174.11175070189057, 75.70607793246917] Fitness: 2.7773979305587706
Generation 482 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 74.95310536225013] Fitness: 2.9139783241725694
Generation 483 Best Solution: [183.8437131443636, 175.88598673110684,
175.13286962599807, 74.95310536225013] Fitness: 1.5358329535456263
Generation 484 Best Solution: [183.8437131443636, 177.4460488719803,
174.3882476222229, 75.8105848198862] Fitness: 3.0486068705611142
Generation 485 Best Solution: [183.8437131443636, 177.4460488719803,
174.75441565684417, 75.28153012651707] Fitness: 2.5804830050344525
Generation 486 Best Solution: [183.8437131443636, 176.5636495826361,
174.75441565684417, 75.8105848198862] Fitness: 2.0536728829976845
Generation 487 Best Solution: [183.8437131443636, 176.5636495826361,
174.99695791618575, 75.87988739489586] Fitness: 2.5279735721998877
Generation 488 Best Solution: [183.8437131443636, 177.4460488719803,
175.13286962599807, 74.70376614120586] Fitness: 2.8618048593755483
Generation 489 Best Solution: [183.7697022920433, 177.03631607355155,
174.21225862288046, 75.8105848198862] Fitness: 2.1236008425911495
Generation 490 Best Solution: [183.8437131443636, 177.4460488719803,
174.75441565684417, 74.60539433508309] Fitness: 2.3021222768634617
Generation 491 Best Solution: [183.8437131443636, 176.96427925722872,
175.13286962599807, 75.05490386966747] Fitness: 2.083994162690901
Generation 492 Best Solution: [183.8437131443636, 176.96427925722872,
174.21225862288046, 75.8105848198862] Fitness: 2.100460304678721
Generation 493 Best Solution: [183.8437131443636, 176.96427925722872,
174.21225862288046, 75.8105848198862] Fitness: 2.100460304678721
Generation 494 Best Solution: [183.8437131443636, 176.96427925722872,

```
174.21225862288046, 75.8105848198862] Fitness: 2.100460304678721
Generation 495 Best Solution: [183.8437131443636, 176.96427925722872,
174.21225862288046, 75.8105848198862] Fitness: 2.100460304678721
Generation 496 Best Solution: [183.8437131443636, 177.4460488719803,
174.21225862288046, 74.65279234908158] Fitness: 1.9192225949001833
Generation 497 Best Solution: [183.8437131443636, 176.96427925722872,
174.21225862288046, 75.83576144566142] Fitness: 2.149806106340236
Generation 498 Best Solution: [183.8437131443636, 176.96427925722872,
174.21225862288046, 75.29922111493472] Fitness: 1.3396384410532152
Generation 499 Best Solution: [183.8437131443636, 176.8173466807122,
175.13286962599807, 74.65279234908158] Fitness: 1.7939375305466856
Generation 500 Best Solution: [183.8437131443636, 177.4460488719803,
174.21225862288046, 75.29922111493472] Fitness: 2.2223794880388277
Optimal Solution for Simulation 20: [183.8437131443636, 177.4460488719803,
174.21225862288046, 75.29922111493472]
--------------------------------------------------
Simulation 21/32 with conditions: {'drought': False, 'flood': True,
'policy_change': False, 'social_event': True, 'economic_crisis': True}
Generation 1 Best Solution: [183.0754236901783, 176.4872873461001,
174.47286451030328, 74.448797954802] Fitness: 0.25289579161769077
Generation 2 Best Solution: [183.06468285634148, 176.50830897516798,
174.38993773340226, 75.43424701029045] Fitness: 0.705381619880355
Generation 3 Best Solution: [183.00833265683943, 176.9126434525139,
174.16455905901668, 74.85347833893637] Fitness: 0.5204683515098419
Generation 4 Best Solution: [183.69035074749792, 177.14580141458072,
174.23964989930118, 74.50761592577162] Fitness: 0.9799931306541592
Generation 5 Best Solution: [183.0942515011734, 177.38180933282678,
174.69662996728053, 74.77821719910294] Fitness: 1.4890333843991355
Generation 6 Best Solution: [183.6846304228905, 175.8591143661151,
174.80245217051407, 75.22763518253997] Fitness: 0.9894342260082337
Generation 7 Best Solution: [183.69035074749792, 175.6903292655292,
174.85698098990687, 75.22763518253997] Fitness: 1.1834941552749587
Generation 8 Best Solution: [182.79302391659888, 175.6903292655292,
174.85698098990687, 75.22763518253997] Fitness: 1.1535520295828057
Generation 9 Best Solution: [182.79302391659888, 175.6903292655292,
174.85698098990687, 75.27790590501927] Fitness: 1.2045335732013167
Generation 10 Best Solution: [182.79302391659888, 175.6903292655292,
174.85698098990687, 74.77501632228963] Fitness: 0.8862109351037732
Generation 11 Best Solution: [183.00119810369165, 177.26650049715542,
175.05769481317552, 74.46043341675495] Fitness: 1.7762288299994557
Generation 12 Best Solution: [183.7998537553656, 177.26650049715542,
174.88264958215385, 74.77501632228963] Fitness: 1.7491968809229772
Generation 13 Best Solution: [183.7136837484485, 177.3494555849748,
174.5526734370075, 75.22763518253997] Fitness: 1.7611152068937503
Generation 14 Best Solution: [183.40170599019422, 177.3494555849748,
175.05769481317552, 74.77501632228963] Fitness: 1.897987579924015
Generation 15 Best Solution: [183.46748194445098, 177.26650049715542,
175.0784529269494, 74.77501632228963] Fitness: 1.8016448310729127
```

```
Generation 16 Best Solution: [183.7136837484485, 176.095304889535,
175.05769481317552, 75.7647839804921] Fitness: 2.0310806939006767
Generation 17 Best Solution: [183.7136837484485, 176.095304889535,
175.05769481317552, 75.7647839804921] Fitness: 2.0310806939006767
Generation 18 Best Solution: [183.7136837484485, 176.095304889535,
175.05769481317552, 75.7647839804921] Fitness: 2.0310806939006767
Generation 19 Best Solution: [183.85223063248054, 175.63148163223127,
174.67098019377306, 75.78227438827031] Fitness: 1.9851195113500761
Generation 20 Best Solution: [183.7136837484485, 176.8808367098016,
174.67098019377306, 75.78227438827031] Fitness: 1.9447469149604726
Generation 21 Best Solution: [183.71355490463702, 177.26650049715542,
174.85698098990687, 74.79342078157708] Fitness: 1.6432954564438689
Generation 22 Best Solution: [183.7136837484485, 177.39481970378324,
174.85698098990687, 74.79342078157708] Fitness: 1.8849200049647388
Generation 23 Best Solution: [183.7136837484485, 175.93624886018446,
174.67098019377306, 75.7647839804921] Fitness: 1.6007817488397194
Generation 24 Best Solution: [183.7136837484485, 177.26650049715542,
174.16161650926918, 75.7647839804921] Fitness: 2.218583112666533
Generation 25 Best Solution: [183.37737208846386, 176.59100131593775,
175.0784529269494, 75.75709170813462] Fitness: 1.97498590109124
Generation 26 Best Solution: [183.7136837484485, 175.9247388076642,
174.6184583529263, 75.78227438827031] Fitness: 1.5931442483497775
Generation 27 Best Solution: [183.7136837484485, 175.9247388076642,
174.6184583529263, 75.75709170813462] Fitness: 1.5456817121288977
Generation 28 Best Solution: [183.7136837484485, 176.59100131593775,
174.92269052060882, 75.75709170813462] Fitness: 1.9279588453091887
Generation 29 Best Solution: [183.7136837484485, 177.2869377902579,
174.824462111248, 74.43492489710084] Fitness: 1.6626849867076952
Generation 30 Best Solution: [183.7136837484485, 177.2869377902579,
174.824462111248, 74.43492489710084] Fitness: 1.6626849867076952
Generation 31 Best Solution: [183.7136837484485, 176.59100131593775,
174.92269052060882, 75.78227438827031] Fitness: 1.9754213815300683
Generation 32 Best Solution: [183.7136837484485, 176.2600862345472,
175.0784529269494, 75.50052761339906] Fitness: 1.6172404055001044
Generation 33 Best Solution: [183.7136837484485, 176.59100131593775,
174.92269052060882, 75.7647839804921] Fitness: 1.9423433990783843
Generation 34 Best Solution: [183.7136837484485, 176.2600862345472,
174.92041870971977, 75.78227438827031] Fitness: 1.8539765355950406
Generation 35 Best Solution: [183.7136837484485, 177.26650049715542,
174.34658193720315, 75.50052761339906] Fitness: 1.8225130762325692
Generation 36 Best Solution: [183.7136837484485, 175.97635716491297,
174.92041870971977, 75.78227438827031] Fitness: 1.8993269778532327
Generation 37 Best Solution: [183.7136837484485, 176.05349084550338,
175.0784529269494, 75.50052761339906] Fitness: 1.636842684010972
Generation 38 Best Solution: [183.7136837484485, 175.76413827470444,
174.6170665568095, 75.75709170813462] Fitness: 1.6442449751509918
Generation 39 Best Solution: [183.31716369486537, 175.76413827470444,
174.6170665568095, 75.75709170813462] Fitness: 1.4482063143667157
```

Generation 40 Best Solution: [183.31716369486537, 176.110867047438,
175.0784529269494, 75.15749865938781] Fitness: 1.0362230378784512
Generation 41 Best Solution: [183.31716369486537, 176.17386949309838,
175.070830691362, 75.78183634403881] Fitness: 1.8777283902126212
Generation 42 Best Solution: [183.31716369486537, 176.17386949309838,
173.92338330836645, 75.78227438827031] Fitness: 1.1546189921481413
Generation 43 Best Solution: [183.7136837484485, 176.17386949309838,
174.1577388489979, 75.75709170813462] Fitness: 1.2708534911044702
Generation 44 Best Solution: [183.31716369486537, 176.2897436991911,
174.1577388489979, 75.75709170813462] Fitness: 1.078460666314831
Generation 45 Best Solution: [183.5255162744671, 176.17386949309838,
174.1577388489979, 75.7647839804921] Fitness: 1.159193563618533
Generation 46 Best Solution: [183.5255162744671, 176.17386949309838,
175.0784529269494, 75.78227438827031] Fitness: 1.9607803646508042
Generation 47 Best Solution: [183.5255162744671, 176.17386949309838,
175.0784529269494, 75.75709170813462] Fitness: 1.9133178284299244
Generation 48 Best Solution: [182.91317706259088, 177.26650049715542,
174.06123746967992, 75.75709170813462] Fitness: 2.0826335633891118
Generation 49 Best Solution: [182.83757421819936, 175.6364605124623,
175.0784529269494, 75.75709170813462] Fitness: 2.238783094701583
Generation 50 Best Solution: [183.30085610647416, 177.26650049715542,
175.0784529269494, 74.61734062965584] Fitness: 1.7440026350860292
Generation 51 Best Solution: [183.31716369486537, 177.26650049715542,
175.0784529269494, 74.6543156488736] Fitness: 1.7454743755232214
Generation 52 Best Solution: [183.31716369486537, 177.26650049715542,
175.0784529269494, 74.6543156488736] Fitness: 1.7454743755232214
Generation 53 Best Solution: [183.7136837484485, 177.17214091572458,
174.45469422890596, 74.6543156488736] Fitness: 1.1049813274914284
Generation 54 Best Solution: [183.31716369486537, 177.26650049715542,
175.0784529269494, 75.75709170813462] Fitness: 2.776425916846314
Generation 55 Best Solution: [183.31716369486537, 176.39679928084757,
175.0784529269494, 75.75709170813462] Fitness: 1.8704354355304815
Generation 56 Best Solution: [183.5255162744671, 176.16036630066952,
175.0784529269494, 75.78227438827031] Fitness: 1.961826668257491
Generation 57 Best Solution: [183.30085610647416, 177.26650049715542,
175.1278769729932, 74.5167010371758] Fitness: 1.839804435331198
Generation 58 Best Solution: [183.7136837484485, 176.39679928084757,
175.1278769729932, 74.5167010371758] Fitness: 1.1322458107484954
Generation 59 Best Solution: [183.30085610647416, 176.39679928084757,
175.0784529269494, 75.75709170813462] Fitness: 1.8680422395816279
Generation 60 Best Solution: [183.5255162744671, 177.26650049715542,
174.40305613043705, 75.78227438827031] Fitness: 2.1906116799922684
Generation 61 Best Solution: [183.5255162744671, 177.26650049715542,
174.40305613043705, 75.66026276607916] Fitness: 1.970601642704474
Generation 62 Best Solution: [183.5255162744671, 177.26650049715542,
174.40305613043705, 75.66026276607916] Fitness: 1.970601642704474
Generation 63 Best Solution: [183.5255162744671, 177.26650049715542,
175.1278769729932, 75.11741150167447] Fitness: 2.0805741352950546

Generation 64 Best Solution: [183.5255162744671, 176.63441693229376,
175.0784529269494, 75.75709170813462] Fitness: 2.061482794656203
Generation 65 Best Solution: [183.5255162744671, 176.63441693229376,
175.0784529269494, 75.75709170813462] Fitness: 2.061482794656203
Generation 66 Best Solution: [183.30085610647416, 176.63441693229376,
175.0784529269494, 75.75709170813462] Fitness: 1.9890954191782053
Generation 67 Best Solution: [183.5255162744671, 176.1750317630536,
175.0784529269494, 75.75709170813462] Fitness: 1.913242123064831
Generation 68 Best Solution: [183.5255162744671, 176.1750317630536,
175.0784529269494, 75.78227438827031] Fitness: 1.9607046592857107
Generation 69 Best Solution: [183.5255162744671, 177.26650049715542,
175.0784529269494, 74.31564094319974] Fitness: 1.909953352880785
Generation 70 Best Solution: [183.4017814879283, 177.26650049715542,
175.0784529269494, 74.92053443338784] Fitness: 1.8264208095208574
Generation 71 Best Solution: [183.30085610647416, 176.21113848578025,
174.94502540345465, 74.31564094319974] Fitness: 0.7033394923639772
Generation 72 Best Solution: [183.30085610647416, 177.26650049715542,
174.94502540345465, 74.92053443338784] Fitness: 1.604588145576668
Generation 73 Best Solution: [183.5255162744671, 177.26650049715542,
174.94502540345465, 74.92053443338784] Fitness: 1.6769755210546655
Generation 74 Best Solution: [183.30085610647416, 177.26650049715542,
174.40305613043705, 74.81891558202112] Fitness: 1.0636486262190568
Generation 75 Best Solution: [183.5255162744671, 175.80985046885763,
175.0784529269494, 74.62151550486396] Fitness: 1.0187225771817456
Generation 76 Best Solution: [183.30085610647416, 177.26650049715542,
175.0784529269494, 74.60594992747929] Fitness: 1.7447504353348915
Generation 77 Best Solution: [183.5255162744671, 177.26650049715542,
174.40305613043705, 74.68079196141959] Fitness: 1.1129524863827658
Generation 78 Best Solution: [183.30085610647416, 177.26650049715542,
174.04292432534803, 74.64446741231592] Fitness: 0.9788795664797592
Generation 79 Best Solution: [183.30085610647416, 177.26650049715542,
174.04108049926884, 74.60594992747929] Fitness: 0.9807783398032432
Generation 80 Best Solution: [183.4424152042983, 177.26650049715542,
174.04108049926884, 74.60594992747929] Fitness: 1.0164835747776058
Generation 81 Best Solution: [183.30085610647416, 177.26650049715542,
174.04108049926884, 74.60594992747929] Fitness: 0.9807783398032432
Generation 82 Best Solution: [183.4424152042983, 177.26650049715542,
174.04108049926884, 74.60594992747929] Fitness: 1.0164835747776058
Generation 83 Best Solution: [183.30085610647416, 177.26650049715542,
174.1478288726951, 75.78227438827031] Fitness: 2.052480986170013
Generation 84 Best Solution: [183.5255162744671, 177.14013015898422,
174.40305613043705, 75.78227438827031] Fitness: 1.9798652604351592
Generation 85 Best Solution: [183.5255162744671, 177.14013015898422,
174.40305613043705, 75.78227438827031] Fitness: 1.9798652604351592
Generation 86 Best Solution: [182.93311287837676, 177.14013015898422,
175.0784529269494, 75.39819743999291] Fitness: 2.0679107430507937
Generation 87 Best Solution: [183.30085610647416, 176.77342407042678,
175.0784529269494, 75.39819743999291] Fitness: 1.5436588995419878

Generation 88 Best Solution: [183.4424152042983, 177.26650049715542,
173.94388263493684, 75.78227438827031] Fitness: 2.1145126306686715
Generation 89 Best Solution: [183.4424152042983, 176.2142073795289,
175.0784529269494, 75.78227438827031] Fitness: 1.922801530861562
Generation 90 Best Solution: [183.4424152042983, 176.2142073795289,
175.0784529269494, 75.78227438827031] Fitness: 1.922801530861562
Generation 91 Best Solution: [183.4424152042983, 176.2142073795289,
175.0784529269494, 75.78227438827031] Fitness: 1.922801530861562
Generation 92 Best Solution: [183.4050665579205, 176.77636937045688,
173.74584088566593, 75.78227438827031] Fitness: 1.527172313476385
Generation 93 Best Solution: [183.4424152042983, 177.14013015898422,
175.0784529269494, 75.78227438827031] Fitness: 2.6464540725355934
Generation 94 Best Solution: [183.30085610647416, 175.9040480700514,
175.0784529269494, 75.78227438827031] Fitness: 1.967540402902784
Generation 95 Best Solution: [183.30085610647416, 175.9040480700514,
175.0784529269494, 75.75709170813462] Fitness: 1.9200778666819043
Generation 96 Best Solution: [183.4424152042983, 177.26650049715542,
174.0263364228554, 74.69114005287035] Fitness: 1.0183900218296422
Generation 97 Best Solution: [183.12415930710844, 177.26650049715542,
174.2718007164095, 75.78227438827031] Fitness: 2.0734995070843527
Generation 98 Best Solution: [183.4424152042983, 175.9040480700514,
173.81957161749799, 75.78227438827031] Fitness: 1.3109674116726364
Generation 99 Best Solution: [183.4424152042983, 175.9040480700514,
174.58990858541407, 75.78227438827031] Fitness: 1.4176665112882065
Generation 100 Best Solution: [182.88984670187355, 176.23468124086406,
175.0784529269494, 74.76317867653069] Fitness: 0.9073121002489701
Generation 101 Best Solution: [183.4424152042983, 177.26650049715542,
175.0784529269494, 74.48023043290539] Fitness: 1.8032248883224147
Generation 102 Best Solution: [183.4424152042983, 177.26650049715542,
175.0784529269494, 74.48023043290539] Fitness: 1.8032248883224147
Generation 103 Best Solution: [182.88984670187355, 176.5171491316094,
175.0784529269494, 75.78227438827031] Fitness: 2.0525146341038347
Generation 104 Best Solution: [182.88984670187355, 176.5171491316094,
175.0784529269494, 75.78227438827031] Fitness: 2.0525146341038347
Generation 105 Best Solution: [183.4424152042983, 175.79892038048678,
175.0784529269494, 75.78227438827031] Fitness: 2.067276726887373
Generation 106 Best Solution: [183.3506674284776, 177.26650049715542,
175.0784529269494, 74.69589405145729] Fitness: 1.7535107716451148
Generation 107 Best Solution: [183.4424152042983, 177.26650049715542,
175.0784529269494, 74.34192435586405] Fitness: 1.8590242813283833
Generation 108 Best Solution: [183.505740668282, 177.26650049715542,
175.0784529269494, 74.97349838462803] Fitness: 1.8934603829944554
Generation 109 Best Solution: [183.62843886284708, 176.01684456682514,
175.0784529269494, 75.78227438827031] Fitness: 2.053480292767304
Generation 110 Best Solution: [183.3506674284776, 176.18758363005145,
175.0784529269494, 75.78227438827031] Fitness: 1.8963600133496685
Generation 111 Best Solution: [183.3506674284776, 176.18758363005145,
175.0784529269494, 75.78227438827031] Fitness: 1.8963600133496685

Generation 112 Best Solution: [183.75164417233466, 177.11442079504576,
175.0784529269494, 74.70470912411717] Fitness: 1.726292130756207
Generation 113 Best Solution: [183.76613196601105, 177.26650049715542,
175.0784529269494, 74.70470912411717] Fitness: 1.9897246334388605
Generation 114 Best Solution: [182.85910073238688, 177.26650049715542,
175.0784529269494, 74.73478943671242] Fitness: 1.8546072305894077
Generation 115 Best Solution: [183.75164417233466, 176.68839129822507,
175.0784529269494, 75.31311394294757] Fitness: 1.5995640895837124
Generation 116 Best Solution: [183.76613196601105, 176.18758363005145,
175.0784529269494, 74.87959829457208] Fitness: 1.0975613498322043
Generation 117 Best Solution: [183.6062594506395, 177.26650049715542,
175.0784529269494, 74.87959829457208] Fitness: 1.9063894233231133
Generation 118 Best Solution: [183.76613196601105, 176.68839129822507,
175.0784529269494, 75.66937157970995] Fitness: 2.1170489428370383
Generation 119 Best Solution: [183.6062594506395, 177.26650049715542,
175.0784529269494, 74.37154045149417] Fitness: 1.9278289311862777
Generation 120 Best Solution: [183.76613196601105, 176.68839129822507,
175.0784529269494, 75.01727772131447] Fitness: 1.3562823107331128
Generation 121 Best Solution: [183.3506674284776, 177.26650049715542,
175.0784529269494, 74.28605967500529] Fitness: 1.863707478313453
Generation 122 Best Solution: [183.32165943259278, 177.26650049715542,
175.0784529269494, 75.14980668412659] Fitness: 1.956039201204965
Generation 123 Best Solution: [183.6062594506395, 177.26650049715542,
175.0784529269494, 74.28605967500529] Fitness: 1.9741613025326306
Generation 124 Best Solution: [183.37095646889267, 177.26650049715542,
175.0784529269494, 74.28605967500529] Fitness: 1.8684550949315901
Generation 125 Best Solution: [183.4024848626501, 177.26650049715542,
175.0784529269494, 74.67232612700153] Fitness: 1.76568263248438
Generation 126 Best Solution: [183.76613196601105, 176.6449502867905,
175.0784529269494, 75.78227438827031] Fitness: 2.288316958319138
Generation 127 Best Solution: [183.76613196601105, 177.26650049715542,
175.0784529269494, 74.77219412503297] Fitness: 1.9997069152744482
Generation 128 Best Solution: [183.4024848626501, 177.26650049715542,
175.0784529269494, 74.77219412503297] Fitness: 1.7777315699565028
Generation 129 Best Solution: [183.76613196601105, 177.26650049715542,
175.0784529269494, 74.77219412503297] Fitness: 1.9997069152744482
Generation 130 Best Solution: [183.76613196601105, 177.26650049715542,
175.0784529269494, 74.77219412503297] Fitness: 1.9997069152744482
Generation 131 Best Solution: [183.76613196601105, 177.26650049715542,
175.0784529269494, 74.77219412503297] Fitness: 1.9997069152744482
Generation 132 Best Solution: [183.76613196601105, 177.26650049715542,
175.0784529269494, 74.60198395808062] Fitness: 1.9892537774198962
Generation 133 Best Solution: [183.76613196601105, 175.80195154724913,
174.93005743721704, 75.78227438827031] Fitness: 2.0532799442532754
Generation 134 Best Solution: [182.92690875441497, 177.26650049715542,
174.5232470559724, 75.78227438827031] Fitness: 2.255195699376908
Generation 135 Best Solution: [183.31540855056775, 177.26650049715542,
175.0784529269494, 75.29257397012537] Fitness: 2.0922141221072983

Generation 136 Best Solution: [182.99155890117547, 177.26650049715542,
174.5232470559724, 75.78227438827031] Fitness: 2.2265981577712015
Generation 137 Best Solution: [183.76613196601105, 175.81452090450813,
175.0784529269494, 75.78227438827031] Fitness: 2.2650850817980386
Generation 138 Best Solution: [183.76613196601105, 176.75806419361447,
175.0784529269494, 75.66937157970995] Fitness: 2.176906349581148
Generation 139 Best Solution: [183.76613196601105, 177.26650049715542,
175.0784529269494, 74.64457526213391] Fitness: 1.987292218151893
Generation 140 Best Solution: [182.92690875441497, 177.26650049715542,
175.1169638717581, 74.80741933765802] Fitness: 1.8950393336595006
Generation 141 Best Solution: [183.76613196601105, 176.47513926199701,
173.9257780877133, 75.78227438827031] Fitness: 1.452133301400177
Generation 142 Best Solution: [183.76613196601105, 176.47513926199701,
173.9257780877133, 75.78227438827031] Fitness: 1.452133301400177
Generation 143 Best Solution: [183.76613196601105, 176.47513926199701,
175.0784529269494, 74.9758044122829] Fitness: 1.1997188171639526
Generation 144 Best Solution: [183.7835501765319, 176.58655205025926,
173.9257780877133, 75.78227438827031] Fitness: 1.5279730043526576
Generation 145 Best Solution: [183.17515361280562, 176.47513926199701,
175.0784529269494, 75.78227438827031] Fitness: 1.941492232548992
Generation 146 Best Solution: [183.76613196601105, 177.26650049715542,
174.55037986737744, 74.66942742917809] Fitness: 1.3721782092075092
Generation 147 Best Solution: [183.17515361280562, 176.47513926199701,
175.0784529269494, 75.66937157970995] Fitness: 1.7370410794526945
Generation 148 Best Solution: [183.4531339995848, 177.26650049715542,
174.10671404805433, 75.78227438827031] Fitness: 2.0919339160191965
Generation 149 Best Solution: [183.76613196601105, 177.26650049715542,
174.55037986737744, 75.78227438827031] Fitness: 2.45030604636926
Generation 150 Best Solution: [183.76613196601105, 176.90515557077538,
174.55037986737744, 75.78227438827031] Fitness: 1.9191956235315915
Generation 151 Best Solution: [183.68195673878373, 177.26650049715542,
175.1169638717581, 75.02465751998703] Fitness: 2.09713279219758
Generation 152 Best Solution: [182.92690875441497, 177.26650049715542,
174.55037986737744, 75.78227438827031] Fitness: 2.2741287335128146
Generation 153 Best Solution: [183.76613196601105, 177.26650049715542,
173.95172188901873, 75.78227438827031] Fitness: 2.3206925577658013
Generation 154 Best Solution: [183.76613196601105, 177.26650049715542,
175.0784529269494, 75.27452472530872] Fitness: 2.3150516016567835
Generation 155 Best Solution: [183.68195673878373, 177.26650049715542,
175.0784529269494, 75.24416752581959] Fitness: 2.212737011363484
Generation 156 Best Solution: [183.46476510141952, 177.26650049715542,
175.0784529269494, 74.59378556043151] Fitness: 1.7902177582598862
Generation 157 Best Solution: [182.92690875441497, 176.37105596334973,
173.73126829428506, 75.78227438827031] Fitness: 1.3354536327354118
Generation 158 Best Solution: [183.76613196601105, 177.26650049715542,
175.0784529269494, 74.44293506516759] Fitness: 2.02358570000658
Generation 159 Best Solution: [183.76613196601105, 177.26650049715542,
175.0784529269494, 74.44293506516759] Fitness: 2.02358570000658

Generation 160 Best Solution: [183.76613196601105, 177.26650049715542,
175.0784529269494, 74.44293506516759] Fitness: 2.02358570000658
Generation 161 Best Solution: [182.92690875441497, 177.26650049715542,
175.0784529269494, 74.44293506516759] Fitness: 1.8474083871501348
Generation 162 Best Solution: [183.76613196601105, 177.26650049715542,
174.16887037468965, 74.7930726275037] Fitness: 1.2366050100330737
Generation 163 Best Solution: [183.5702686329515, 177.26650049715542,
175.0784529269494, 74.87942646751705] Fitness: 1.8841141050060408
Generation 164 Best Solution: [183.68195673878373, 176.18806141527912,
175.0784529269494, 75.78227438827031] Fitness: 2.060624045711195
Generation 165 Best Solution: [183.24695625208676, 177.26650049715542,
174.16887037468965, 75.78227438827031] Fitness: 2.049028113650611
Generation 166 Best Solution: [183.24695625208676, 177.26650049715542,
174.15828969704006, 75.78227438827031] Fitness: 2.04829606376881
Generation 167 Best Solution: [183.76613196601105, 177.42144600044364,
174.9200474805769, 74.34124485823922] Fitness: 2.129064984572749
Generation 168 Best Solution: [183.68195673878373, 177.26650049715542,
174.16887037468965, 75.4551657266435] Fitness: 1.693408391555036
Generation 169 Best Solution: [182.80323412280774, 177.26650049715542,
174.16887037468965, 75.78227438827031] Fitness: 2.196085894576979
Generation 170 Best Solution: [183.76613196601105, 175.991254419507,
174.16887037468965, 75.78227438827031] Fitness: 1.4049971177147778
Generation 171 Best Solution: [183.76613196601105, 175.991254419507,
174.16887037468965, 75.78227438827031] Fitness: 1.4049971177147778
Generation 172 Best Solution: [183.76613196601105, 175.991254419507,
174.16887037468965, 75.78227438827031] Fitness: 1.4049971177147778
Generation 173 Best Solution: [182.94714183527134, 175.991254419507,
174.16887037468965, 75.78227438827031] Fitness: 1.21911304760974
Generation 174 Best Solution: [183.76613196601105, 177.26650049715542,
173.9226198491511, 74.60297582049014] Fitness: 1.2532626780592226
Generation 175 Best Solution: [182.94714183527134, 177.26650049715542,
173.9226198491511, 74.60297582049014] Fitness: 1.067378607954185
Generation 176 Best Solution: [183.76613196601105, 177.26650049715542,
173.9226198491511, 74.57306909726864] Fitness: 1.256415919103865
Generation 177 Best Solution: [183.19457168470134, 177.26650049715542,
173.9226198491511, 74.60297582049014] Fitness: 1.0044456894063745
Generation 178 Best Solution: [183.76613196601105, 175.94227281747686,
174.14876257575014, 75.78227438827031] Fitness: 1.4240908958414427
Generation 179 Best Solution: [183.76613196601105, 175.94227281747686,
174.9200474805769, 75.78227438827031] Fitness: 1.959158062906305
Generation 180 Best Solution: [183.0750726378891, 177.26650049715542,
173.9226198491511, 75.78227438827031] Fitness: 2.0984796735408215
Generation 181 Best Solution: [183.76613196601105, 176.08424445549008,
174.77992567130607, 75.78227438827031] Fitness: 1.7396850687954095
Generation 182 Best Solution: [183.6423555520241, 175.80701899536328,
173.9226198491511, 75.78227438827031] Fitness: 1.4337008188913347
Generation 183 Best Solution: [183.6423555520241, 177.26650049715542,
174.77992567130607, 74.47600935275491] Fitness: 1.5067909880572492

Generation 184 Best Solution: [183.76613196601105, 177.26650049715542,
174.77992567130607, 74.55026723062615] Fitness: 1.59015335271572
Generation 185 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 74.47600935275491] Fitness: 1.7790311740811418
Generation 186 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 74.81867601308565] Fitness: 1.7771368740696019
Generation 187 Best Solution: [183.76613196601105, 176.48180216650522,
174.9200474805769, 75.69652650281778] Fitness: 1.80093164660958
Generation 188 Best Solution: [183.76613196601105, 176.48180216650522,
174.9200474805769, 75.07323408041906] Fitness: 1.0301429992463405
Generation 189 Best Solution: [183.76613196601105, 176.81168654663173,
175.03078160071664, 75.07323408041906] Fitness: 1.4301278310195764
Generation 190 Best Solution: [183.76613196601105, 176.81168654663173,
175.03078160071664, 75.07323408041906] Fitness: 1.4301278310195764
Generation 191 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 74.5610197419564] Fitness: 1.7601162081916604
Generation 192 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 74.5610197419564] Fitness: 1.7601162081916604
Generation 193 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 74.5610197419564] Fitness: 1.7601162081916604
Generation 194 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 74.68844196510449] Fitness: 1.7545591728911594
Generation 195 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 74.68844196510449] Fitness: 1.7545591728911594
Generation 196 Best Solution: [183.76613196601105, 176.5144101068749,
174.9200474805769, 75.26233129318285] Fitness: 1.2104979045854654
Generation 197 Best Solution: [182.9416287253978, 177.26650049715542,
174.9200474805769, 74.91602381626588] Fitness: 1.6293623208511319
Generation 198 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 74.91602381626588] Fitness: 1.8126698329548367
Generation 199 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 74.60389052493143] Fitness: 1.755194110415555
Generation 200 Best Solution: [183.76613196601105, 176.35785210296967,
174.9200474805769, 75.78227438827031] Fitness: 1.9150191404275225
Generation 201 Best Solution: [183.56148757851787, 176.35785210296967,
174.9200474805769, 74.31772180074944] Fitness: 0.7776004124567316
Generation 202 Best Solution: [182.9416287253978, 176.80860879456785,
174.9200474805769, 75.78227438827031] Fitness: 2.0126842727523857
Generation 203 Best Solution: [182.9416287253978, 177.1514318684404,
174.9200474805769, 74.32203843397895] Fitness: 1.4679749027872768
Generation 204 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 74.32203843397895] Fitness: 1.844276232697299
Generation 205 Best Solution: [182.9416287253978, 177.1514318684404,
174.9200474805769, 74.88443557005753] Fitness: 1.4230892196724354
Generation 206 Best Solution: [183.76613196601105, 177.26650049715542,
174.3618597154593, 74.32203843397895] Fitness: 1.3568750450006273
Generation 207 Best Solution: [182.9416287253978, 177.26650049715542,
174.9200474805769, 74.56207455244453] Fitness: 1.576650447240136

Generation 208 Best Solution: [183.76613196601105, 177.26650049715542,
174.16299237648752, 75.13606147361465] Fitness: 1.4175285167604346
Generation 209 Best Solution: [182.9416287253978, 177.26650049715542,
174.16299237648752, 75.13606147361465] Fitness: 1.2342210046567297
Generation 210 Best Solution: [183.0445620833669, 177.26650049715542,
174.9200474805769, 74.56207455244453] Fitness: 1.5369888859572096
Generation 211 Best Solution: [183.28861938532492, 177.26650049715542,
174.5776786473645, 74.56207455244453] Fitness: 1.1530904100073136
Generation 212 Best Solution: [183.76613196601105, 177.26650049715542,
174.5776786473645, 74.56207455244453] Fitness: 1.3987837422844944
Generation 213 Best Solution: [183.76613196601105, 176.97030955670468,
174.9200474805769, 74.56207455244453] Fitness: 1.3083609192635313
Generation 214 Best Solution: [183.0445620833669, 177.26650049715542,
174.5776786473645, 74.56207455244453] Fitness: 1.1758146688978632
Generation 215 Best Solution: [183.76613196601105, 177.26650049715542,
174.77584232397663, 74.77058445372921] Fitness: 1.5892828412818365
Generation 216 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 75.1034442063513] Fitness: 1.9260245293466622
Generation 217 Best Solution: [182.9416287253978, 177.26650049715542,
174.9200474805769, 75.2958192829545] Fitness: 1.9205952733411042
Generation 218 Best Solution: [182.9416287253978, 176.30149021079137,
174.9200474805769, 75.78227438827031] Fitness: 1.720648450339994
Generation 219 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 75.2958192829545] Fitness: 2.1039027854448094
Generation 220 Best Solution: [183.76613196601105, 176.97030955670468,
174.9200474805769, 75.2958192829545] Fitness: 1.6523057453644996
Generation 221 Best Solution: [182.97517929139266, 177.26650049715542,
174.36484421469672, 75.78227438827031] Fitness: 2.147394606439784
Generation 222 Best Solution: [183.76613196601105, 176.39776551340145,
174.9200474805769, 75.77363452832421] Fitness: 1.909685454187147
Generation 223 Best Solution: [183.76613196601105, 176.0237202183697,
174.9200474805769, 75.78227438827031] Fitness: 1.9275902873499875
Generation 224 Best Solution: [183.76613196601105, 175.7438709834759,
174.36484421469672, 75.77363452832421] Fitness: 1.5802167783616943
Generation 225 Best Solution: [183.76613196601105, 176.72299203737325,
174.36484421469672, 75.6924535156634] Fitness: 1.4657084155138627
Generation 226 Best Solution: [183.76613196601105, 175.7438709834759,
174.36484421469672, 75.77363452832421] Fitness: 1.5802167783616943
Generation 227 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 75.29259115652539] Fitness: 2.100403739858968
Generation 228 Best Solution: [183.76613196601105, 176.97030955670468,
174.84037207415813, 75.78227438827031] Fitness: 2.2785061717198127
Generation 229 Best Solution: [183.76613196601105, 176.97030955670468,
174.84037207415813, 74.66390160675277] Fitness: 1.2002290604055323
Generation 230 Best Solution: [183.76613196601105, 176.0692769551617,
174.9200474805769, 75.77363452832421] Fitness: 1.8984013622415867
Generation 231 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 74.73196336445808] Fitness: 1.758926167992635

Generation 232 Best Solution: [183.76613196601105, 176.0692769551617,
174.9200474805769, 75.77363452832421] Fitness: 1.8984013622415867
Generation 233 Best Solution: [183.76613196601105, 176.39829919540404,
174.9200474805769, 75.86786526430134] Fitness: 2.095555621584051
Generation 234 Best Solution: [183.76613196601105, 175.73551069256942,
174.9200474805769, 75.78227438827031] Fitness: 2.089477660283017
Generation 235 Best Solution: [183.50872233063905, 177.26650049715542,
174.9200474805769, 74.38887209597605] Fitness: 1.6308961884351147
Generation 236 Best Solution: [182.796816160816, 177.26650049715542,
174.9200474805769, 74.74512883603708] Fitness: 1.6635783088009644
Generation 237 Best Solution: [183.76613196601105, 177.26650049715542,
174.9200474805769, 74.74512883603708] Fitness: 1.7608756799867469
Generation 238 Best Solution: [183.1301356459632, 176.99409446557127,
174.9200474805769, 75.15284535634449] Fitness: 1.3026088534400824
Generation 239 Best Solution: [183.76613196601105, 176.99409446557127,
174.9200474805769, 75.15284535634449] Fitness: 1.5449663574168975
Generation 240 Best Solution: [183.1301356459632, 176.99409446557127,
174.83853702795918, 75.33313318567949] Fitness: 1.3785828987678954
Generation 241 Best Solution: [183.1301356459632, 176.27909826668986,
174.83853702795918, 75.78227438827031] Fitness: 1.5547951768446369
Generation 242 Best Solution: [183.76613196601105, 177.35116534312564,
174.9200474805769, 74.92321863010805] Fitness: 1.9721701614384257
Generation 243 Best Solution: [183.76613196601105, 176.99409446557127,
174.9200474805769, 75.02643194983183] Fitness: 1.4514973698085787
Generation 244 Best Solution: [183.62804302697063, 175.9889684088353,
174.9200474805769, 75.77363452832421] Fitness: 1.8127682998756496
Generation 245 Best Solution: [183.76613196601105, 177.35116534312564,
174.34390050929284, 74.92321863010805] Fitness: 1.4778006471146963
Generation 246 Best Solution: [183.31281816127134, 177.4189166973826,
174.9200474805769, 75.06309780999564] Fitness: 1.9441106584792358
Generation 247 Best Solution: [183.76613196601105, 177.4189166973826,
174.9200474805769, 75.44358599350781] Fitness: 2.5728250203392165
Generation 248 Best Solution: [183.31281816127134, 177.35116534312564,
174.9200474805769, 75.86786526430134] Fitness: 2.9148409718732875
Generation 249 Best Solution: [183.76613196601105, 177.4189166973826,
174.9200474805769, 74.61063778686699] Fitness: 2.044666569803843
Generation 250 Best Solution: [183.1301356459632, 177.27284285408922,
174.9200474805769, 75.86786526430134] Fitness: 2.7700085998235946
Generation 251 Best Solution: [183.76613196601105, 177.4189166973826,
174.9200474805769, 75.12678128593575] Fitness: 2.2342462173170947
Generation 252 Best Solution: [183.76613196601105, 177.4189166973826,
174.27451487818118, 74.39604959771493] Fitness: 1.5817075783498118
Generation 253 Best Solution: [182.97350282479746, 177.35116534312564,
174.9200474805769, 74.93630584166759] Fitness: 1.7808264548983819
Generation 254 Best Solution: [183.76613196601105, 177.4189166973826,
174.9200474805769, 75.25782548022437] Fitness: 2.3537978927264245
Generation 255 Best Solution: [183.65055183872227, 177.35116534312564,
174.9200474805769, 75.25782548022437] Fitness: 2.125377170292815

Generation 256 Best Solution: [183.76613196601105, 176.54759419312168,
174.27451487818118, 75.86786526430134] Fitness: 1.6446981040411621
Generation 257 Best Solution: [183.76613196601105, 176.54759419312168,
174.27451487818118, 75.86786526430134] Fitness: 1.6446981040411621
Generation 258 Best Solution: [183.76613196601105, 176.54759419312168,
174.27451487818118, 75.25782548022437] Fitness: 0.7074505774291995
Generation 259 Best Solution: [183.76613196601105, 177.4189166973826,
174.2694425838023, 75.25782548022437] Fitness: 1.8363327467558443
Generation 260 Best Solution: [183.76613196601105, 177.4189166973826,
174.2694425838023, 75.25782548022437] Fitness: 1.8363327467558443
Generation 261 Best Solution: [183.76613196601105, 177.4189166973826,
174.1433000810294, 75.25782548022437] Fitness: 1.818514237649602
Generation 262 Best Solution: [183.76613196601105, 175.751841221034,
174.1433000810294, 75.25782548022437] Fitness: 0.7733345385004236
Generation 263 Best Solution: [183.76613196601105, 175.751841221034,
174.2694425838023, 75.86786526430134] Fitness: 1.7284005742186288
Generation 264 Best Solution: [183.34075264164747, 177.4189166973826,
175.1179704268427, 74.44129092643426] Fitness: 2.1415233384079913
Generation 265 Best Solution: [183.76613196601105, 177.36838320289124,
174.2694425838023, 75.86786526430134] Fitness: 2.673106924206277
Generation 266 Best Solution: [183.5511556883051, 175.7281737221494,
175.04080192785054, 75.86786526430134] Fitness: 2.281067059734784
Generation 267 Best Solution: [183.5511556883051, 177.4189166973826,
174.9200474805769, 74.50034881507459] Fitness: 1.9041984276402397
Generation 268 Best Solution: [183.5511556883051, 175.7281737221494,
174.9200474805769, 75.86786526430134] Fitness: 2.1065858412762934
Generation 269 Best Solution: [183.76613196601105, 175.7281737221494,
173.9102840459316, 75.86786526430134] Fitness: 1.7670014208531128
Generation 270 Best Solution: [183.76613196601105, 175.7281737221494,
173.9102840459316, 75.86786526430134] Fitness: 1.7670014208531128
Generation 271 Best Solution: [182.7978464520026, 177.4189166973826,
174.9200474805769, 74.78111931843775] Fitness: 1.9596260261369878
Generation 272 Best Solution: [183.25043562026232, 175.70294233625506,
174.9200474805769, 75.86786526430134] Fitness: 2.0371109325066312
Generation 273 Best Solution: [183.76613196601105, 176.00080995226213,
174.74543828158443, 75.86786526430134] Fitness: 1.8957731506292765
Generation 274 Best Solution: [183.76613196601105, 176.1018819957304,
174.9200474805769, 75.86786526430134] Fitness: 2.0771013254622797
Generation 275 Best Solution: [183.25043562026232, 176.22207555614804,
174.9200474805769, 75.86786526430134] Fitness: 1.8179898900388234
Generation 276 Best Solution: [183.76613196601105, 176.1018819957304,
174.9200474805769, 74.7371430397147] Fitness: 0.8356794392809396
Generation 277 Best Solution: [183.76613196601105, 175.933974484713,
174.9200474805769, 74.75063677308428] Fitness: 0.8929990079086989
Generation 278 Best Solution: [183.76613196601105, 177.4189166973826,
174.9200474805769, 74.58311079872792] Fitness: 2.0471585194116426
Generation 279 Best Solution: [183.65328560136524, 177.24113577201274,
174.9200474805769, 74.58311079872792] Fitness: 1.6200191440989606

Generation 280 Best Solution: [183.5730732611121, 177.4189166973826,
174.9200474805769, 75.4446777507013] Fitness: 2.4287047737944025
Generation 281 Best Solution: [183.76613196601105, 177.36838320289124,
174.9200474805769, 75.21749197339548] Fitness: 2.213447273958031
Generation 282 Best Solution: [183.41859775547675, 177.4189166973826,
174.9200474805769, 75.3606074607196] Fitness: 2.250948940150069
Generation 283 Best Solution: [183.76613196601105, 177.4189166973826,
174.9200474805769, 75.3606074607196] Fitness: 2.4678043528062257
Generation 284 Best Solution: [183.7818116569675, 176.12542363033344,
174.9200474805769, 75.86786526430134] Fitness: 2.08773706357332
Generation 285 Best Solution: [183.76613196601105, 176.59644711617682,
174.72777000840134, 75.86786526430134] Fitness: 1.9632744943439153
Generation 286 Best Solution: [183.76613196601105, 177.36838320289124,
174.72777000840134, 75.21535083901846] Fitness: 1.9842657219783602
Generation 287 Best Solution: [182.7978464520026, 177.4189166973826,
174.9200474805769, 75.0133533930422] Fitness: 2.0560833903259925
Generation 288 Best Solution: [183.41859775547675, 176.44859561522333,
174.9200474805769, 75.86786526430134] Fitness: 1.8965165693406267
Generation 289 Best Solution: [183.41859775547675, 177.36838320289124,
174.9200474805769, 74.3377064020403] Fitness: 1.8084486052366142
Generation 290 Best Solution: [183.41859775547675, 176.44859561522333,
174.9200474805769, 75.86786526430134] Fitness: 1.8965165693406267
Generation 291 Best Solution: [183.41859775547675, 177.4189166973826,
174.9200474805769, 74.7622440865352] Fitness: 1.8369562883180526
Generation 292 Best Solution: [183.41859775547675, 176.44859561522333,
174.9200474805769, 75.44422117389051] Fitness: 1.179144801910943
Generation 293 Best Solution: [183.41859775547675, 176.44859561522333,
174.9200474805769, 75.44422117389051] Fitness: 1.179144801910943
Generation 294 Best Solution: [183.76613196601105, 176.24244693878705,
174.9200474805769, 75.44422117389051] Fitness: 1.350032522323378
Generation 295 Best Solution: [183.41859775547675, 176.63030947541645,
174.9200474805769, 75.86786526430134] Fitness: 1.9963872858836575
Generation 296 Best Solution: [183.41859775547675, 176.0615485601934,
174.7894727374437, 75.78311544679973] Fitness: 1.5404411219685223
Generation 297 Best Solution: [183.41859775547675, 176.0615485601934,
174.9200474805769, 75.86786526430134] Fitness: 1.8691725747897192
Generation 298 Best Solution: [183.41859775547675, 176.0615485601934,
174.9200474805769, 75.86786526430134] Fitness: 1.8691725747897192
Generation 299 Best Solution: [183.46791264088787, 176.1651564584237,
174.59227791981928, 75.86786526430134] Fitness: 1.520346934590131
Generation 300 Best Solution: [183.41859775547675, 176.0615485601934,
174.9200474805769, 75.86786526430134] Fitness: 1.8691725747897192
Generation 301 Best Solution: [183.41859775547675, 177.4189166973826,
174.9200474805769, 74.35165493388588] Fitness: 1.9017346293313768
Generation 302 Best Solution: [183.76613196601105, 177.4189166973826,
174.7894727374437, 74.58952593950417] Fitness: 1.8854284786020887
Generation 303 Best Solution: [183.41859775547675, 177.4189166973826,
174.7894727374437, 74.58952593950417] Fitness: 1.6685730659459324

Generation 304 Best Solution: [183.5222682360801, 176.5121825189864,
174.96937064388064, 75.86786526430134] Fitness: 2.0368392190090123
Generation 305 Best Solution: [183.41859775547675, 177.4189166973826,
174.09492946509678, 75.86786526430134] Fitness: 2.539182133520373
Generation 306 Best Solution: [183.5222682360801, 176.45937524328855,
174.9200474805769, 75.86786526430134] Fitness: 1.9442883802427942
Generation 307 Best Solution: [183.41859775547675, 177.4189166973826,
174.09492946509678, 75.86786526430134] Fitness: 2.539182133520373
Generation 308 Best Solution: [183.41859775547675, 177.4189166973826,
174.7894727374437, 75.86786526430134] Fitness: 2.913154763497885
Generation 309 Best Solution: [183.41859775547675, 177.4189166973826,
173.73560207025434, 75.86786526430134] Fitness: 2.6645967639596355
Generation 310 Best Solution: [183.41859775547675, 176.3529508588,
174.9200474805769, 75.86786526430134] Fitness: 1.86628908315579
Generation 311 Best Solution: [183.50664239001816, 176.11687633841632,
174.7894727374437, 75.86786526430134] Fitness: 1.7322897624665021
Generation 312 Best Solution: [183.41859775547675, 177.4189166973826,
174.9200474805769, 75.1753866454386] Fitness: 2.058359205408145
Generation 313 Best Solution: [183.4979722351741, 177.4189166973826,
174.69506878801033, 74.36423822451073] Fitness: 1.6675596050958779
Generation 314 Best Solution: [183.31770696495317, 177.4189166973826,
174.0930890103509, 74.81650038721655] Fitness: 1.2898596728462968
Generation 315 Best Solution: [183.5222682360801, 177.4189166973826,
174.7894727374437, 75.682845941177] Fitness: 2.6060747332900918
Generation 316 Best Solution: [182.92115434603406, 175.97680959626138,
174.77661453578818, 75.86786526430134] Fitness: 1.7650769657740286
Generation 317 Best Solution: [183.4979722351741, 175.65049836076318,
174.4826975361407, 75.86786526430134] Fitness: 1.721604945391967
Generation 318 Best Solution: [182.92115434603406, 175.97680959626138,
174.77661453578818, 75.86786526430134] Fitness: 1.7650769657740286
Generation 319 Best Solution: [183.76613196601105, 176.96149225316256,
174.9200474805769, 74.86075257799922] Fitness: 1.3277609276572704
Generation 320 Best Solution: [183.5222682360801, 177.4189166973826,
174.77661453578818, 74.86075257799922] Fitness: 1.73170565629448
Generation 321 Best Solution: [183.76613196601105, 176.96149225316256,
174.77661453578818, 74.86075257799922] Fitness: 1.1524210052613417
Generation 322 Best Solution: [183.5222682360801, 177.4189166973826,
174.77661453578818, 74.86075257799922] Fitness: 1.73170565629448
Generation 323 Best Solution: [183.0016065528096, 176.45616555809002,
174.9200474805769, 75.10967855781625] Fitness: 0.8375666379349067
Generation 324 Best Solution: [183.76613196601105, 176.45616555809002,
174.77661453578818, 74.62176359127614] Fitness: 0.6940585203336268
Generation 325 Best Solution: [183.0016065528096, 176.45616555809002,
174.77661453578818, 75.86786526430134] Fitness: 1.7324950878275271
Generation 326 Best Solution: [183.0016065528096, 176.45616555809002,
174.77661453578818, 75.86786526430134] Fitness: 1.7324950878275271
Generation 327 Best Solution: [183.0016065528096, 175.95740312469593,
174.9200474805769, 74.35480134895397] Fitness: 0.7391299458984943

Generation 328 Best Solution: [183.58385405365482, 176.00388913762532,
174.69506878801033, 75.5874879474686] Fitness: 1.1962669548267004
Generation 329 Best Solution: [183.76613196601105, 176.45616555809002,
174.77661453578818, 74.35480134895397] Fitness: 0.767051069053353
Generation 330 Best Solution: [183.76613196601105, 175.74184555070605,
174.9200474805769, 74.35480134895397] Fitness: 1.0796841337230554
Generation 331 Best Solution: [183.76613196601105, 177.2914862518066,
173.9470646072479, 74.50509291981923] Fitness: 1.3063133692347344
Generation 332 Best Solution: [183.76613196601105, 177.4189166973826,
174.77661453578818, 75.12092927996389] Fitness: 2.054242132492724
Generation 333 Best Solution: [183.0016065528096, 176.45616555809002,
174.9200474805769, 75.86786526430134] Fitness: 1.9078350102234558
Generation 334 Best Solution: [183.8217227034495, 177.2914862518066,
173.9470646072479, 75.86786526430134] Fitness: 2.5897625456193767
Generation 335 Best Solution: [183.76613196601105, 177.4189166973826,
174.33900914715724, 74.94856879902338] Fitness: 1.6219008916485085
Generation 336 Best Solution: [183.76613196601105, 177.4189166973826,
174.33900914715724, 74.94856879902338] Fitness: 1.6219008916485085
Generation 337 Best Solution: [183.76613196601105, 177.4189166973826,
174.33900914715724, 74.94856879902338] Fitness: 1.6219008916485085
Generation 338 Best Solution: [183.76613196601105, 177.4189166973826,
174.33900914715724, 74.94856879902338] Fitness: 1.6219008916485085
Generation 339 Best Solution: [183.0016065528096, 177.4189166973826,
174.33900914715724, 74.94856879902338] Fitness: 1.413313560091554
Generation 340 Best Solution: [183.76613196601105, 177.4189166973826,
174.33900914715724, 74.94856879902338] Fitness: 1.6219008916485085
Generation 341 Best Solution: [183.76613196601105, 177.4189166973826,
174.33900914715724, 74.94856879902338] Fitness: 1.6219008916485085
Generation 342 Best Solution: [183.76613196601105, 177.4189166973826,
174.33900914715724, 74.47817865600544] Fitness: 1.572189088853227
Generation 343 Best Solution: [183.15312011237063, 177.4189166973826,
174.33900914715724, 74.47817865600544] Fitness: 1.3267251642396454
Generation 344 Best Solution: [183.76613196601105, 176.57462971764164,
174.77661453578818, 75.65016551137242] Fitness: 1.5949976369328511
Generation 345 Best Solution: [183.15312011237063, 177.4189166973826,
174.77661453578818, 74.47817865600544] Fitness: 1.6475585243462851
Generation 346 Best Solution: [183.38958628277226, 177.4189166973826,
174.77661453578818, 74.4984775671052] Fitness: 1.6617151050415104
Generation 347 Best Solution: [183.38958628277226, 177.4189166973826,
174.77661453578818, 74.41479202451006] Fitness: 1.689056874297896
Generation 348 Best Solution: [183.76613196601105, 177.4189166973826,
174.1311756481054, 74.4984775671052] Fitness: 1.5272292200925375
Generation 349 Best Solution: [182.98030584452005, 177.4189166973826,
174.1311756481054, 74.4984775671052] Fitness: 1.326926072158003
Generation 350 Best Solution: [183.0546816177829, 177.4189166973826,
174.1311756481054, 74.4984775671052] Fitness: 1.301324362834125
Generation 351 Best Solution: [182.98030584452005, 177.4189166973826,
174.77661453578818, 74.58585945027752] Fitness: 1.6712092672348189

Generation 352 Best Solution: [182.98030584452005, 177.4189166973826,
174.77661453578818, 75.22643861076236] Fitness: 1.9468865202102594
Generation 353 Best Solution: [183.3213056021995, 176.01094743491166,
174.77661453578818, 75.86786526430134] Fitness: 1.68464271795336
Generation 354 Best Solution: [183.76613196601105, 177.4189166973826,
174.77661453578818, 75.22643861076236] Fitness: 2.147189668144794
Generation 355 Best Solution: [183.76613196601105, 176.93146777804563,
174.75398283451332, 75.86786526430134] Fitness: 2.301379158962748
Generation 356 Best Solution: [183.12451062730008, 176.93420643323446,
174.77661453578818, 75.86786526430134] Fitness: 2.0877383322252836
Generation 357 Best Solution: [183.76613196601105, 175.96220052175568,
174.77661453578818, 75.86786526430134] Fitness: 1.944361898041727
Generation 358 Best Solution: [183.76613196601105, 175.96220052175568,
174.77661453578818, 75.86786526430134] Fitness: 1.944361898041727
Generation 359 Best Solution: [182.98030584452005, 176.08783614280136,
174.08995285351412, 75.86786526430134] Fitness: 1.344839966756402
Generation 360 Best Solution: [182.98030584452005, 176.39727794564254,
174.08995285351412, 75.86786526430134] Fitness: 1.3601789222517149
Generation 361 Best Solution: [182.98030584452005, 175.96220052175568,
174.77661453578818, 75.86786526430134] Fitness: 1.7440587501071925
Generation 362 Best Solution: [183.76613196601105, 177.4189166973826,
174.46697695256773, 74.67375891834934] Fitness: 1.6080506439426603
Generation 363 Best Solution: [183.76613196601105, 177.4189166973826,
174.77661453578818, 74.67375891834934] Fitness: 1.8684308166407977
Generation 364 Best Solution: [183.76613196601105, 176.34967884972573,
174.50045915698536, 75.86786526430134] Fitness: 1.6670323473153457
Generation 365 Best Solution: [183.76613196601105, 176.34967884972573,
174.77661453578818, 75.86786526430134] Fitness: 1.9070430433829741
Generation 366 Best Solution: [183.76613196601105, 176.34967884972573,
174.77661453578818, 75.86786526430134] Fitness: 1.9070430433829741
Generation 367 Best Solution: [183.76613196601105, 176.74032065491207,
174.77661453578818, 75.86786526430134] Fitness: 2.1253910894320223
Generation 368 Best Solution: [183.76613196601105, 177.4189166973826,
175.06904170411656, 75.06125841984127] Fitness: 2.4041445000217054
Generation 369 Best Solution: [183.76613196601105, 177.4189166973826,
174.11040707647783, 74.2969273229174] Fitness: 1.6131443168221484
Generation 370 Best Solution: [183.7250748999603, 177.4189166973826,
175.06904170411656, 74.90744750512948] Fitness: 2.281477121427146
Generation 371 Best Solution: [183.76613196601105, 177.4189166973826,
175.06904170411656, 74.90744750512948] Fitness: 2.3176922841483987
Generation 372 Best Solution: [182.98030584452005, 177.4189166973826,
175.06904170411656, 74.90744750512948] Fitness: 2.117389136213864
Generation 373 Best Solution: [182.98030584452005, 177.4189166973826,
175.06904170411656, 74.5687432417825] Fitness: 2.0674917241614876
Generation 374 Best Solution: [183.63019191653044, 177.4189166973826,
175.06904170411656, 74.5687432417825] Fitness: 2.1587481851683368
Generation 375 Best Solution: [182.98030584452005, 177.4189166973826,
175.06904170411656, 74.5687432417825] Fitness: 2.0674917241614876

Generation 376 Best Solution: [183.63019191653044, 177.4189166973826,
175.06904170411656, 74.567932827515] Fitness: 2.15886049932539
Generation 377 Best Solution: [183.7250748999603, 177.4189166973826,
174.77661453578818, 74.49392748911787] Fitness: 1.8524421227519334
Generation 378 Best Solution: [183.76613196601105, 176.42360790054553,
174.77661453578818, 75.86786526430134] Fitness: 1.9286481781595153
Generation 379 Best Solution: [183.7250748999603, 177.4189166973826,
174.07506927245848, 74.9355315168751] Fitness: 1.5417065113220196
Generation 380 Best Solution: [183.7250748999603, 176.42360790054553,
174.07506927245848, 75.86786526430134] Fitness: 1.5340194137420329
Generation 381 Best Solution: [183.7250748999603, 176.42360790054553,
174.08663935494098, 75.86786526430134] Fitness: 1.533207886962185
Generation 382 Best Solution: [183.3469529514435, 176.42360790054553,
174.08663935494098, 75.86786526430134] Fitness: 1.333152606639861
Generation 383 Best Solution: [183.76613196601105, 176.3435055619689,
174.07506927245848, 75.86786526430134] Fitness: 1.547241767635719
Generation 384 Best Solution: [183.7250748999603, 176.3435055619689,
174.07506927245848, 75.86786526430134] Fitness: 1.5110266049144663
Generation 385 Best Solution: [183.7250748999603, 176.3435055619689,
175.06904170411656, 75.86786526430134] Fitness: 2.2636089406759456
Generation 386 Best Solution: [183.13748614747723, 176.30679551482348,
175.06904170411656, 75.86786526430134] Fitness: 2.049450131006903
Generation 387 Best Solution: [183.13748614747723, 177.4189166973826,
174.7139237306834, 74.8623902485076] Fitness: 1.5965414515561074
Generation 388 Best Solution: [183.76613196601105, 176.71394057052154,
173.93284931606365, 75.86786526430134] Fitness: 1.7725342752061604
Generation 389 Best Solution: [183.13748614747723, 176.71394057052154,
175.06904170411656, 75.47509545999151] Fitness: 1.5779720246420377
Generation 390 Best Solution: [183.7250748999603, 176.8728490488968,
175.06904170411656, 75.86786526430134] Fitness: 2.615808292047277
Generation 391 Best Solution: [183.7250748999603, 177.4189166973826,
175.06904170411656, 75.64915237923468] Fitness: 3.0655605975873974
Generation 392 Best Solution: [183.13748614747723, 177.4189166973826,
174.97021446925527, 74.79012506141126] Fitness: 1.8857654842529703
Generation 393 Best Solution: [183.76613196601105, 177.4189166973826,
174.97021446925527, 74.79012506141126] Fitness: 2.1292132035131357
Generation 394 Best Solution: [183.13748614747723, 177.4189166973826,
175.06904170411656, 75.63001293852149] Fitness: 2.8264491417138293
Generation 395 Best Solution: [183.13748614747723, 177.4189166973826,
174.23669064176573, 75.86786526430134] Fitness: 2.522930347566486
Generation 396 Best Solution: [182.91959418003077, 175.91369659179443,
175.06904170411656, 75.86786526430134] Fitness: 2.188511538993751
Generation 397 Best Solution: [183.76613196601105, 176.03025983228906,
175.06904170411656, 75.63001293852149] Fitness: 1.8736914311509587
Generation 398 Best Solution: [183.7250748999603, 176.89434925926173,
175.06904170411656, 75.63001293852149] Fitness: 2.200109381562119
Generation 399 Best Solution: [183.7250748999603, 176.89434925926173,
175.06904170411656, 75.63001293852149] Fitness: 2.200109381562119

Generation 400 Best Solution: [183.76613196601105, 176.06942837932127,
175.06904170411656, 75.58355967974227] Fitness: 1.788112847384196
Generation 401 Best Solution: [183.13748614747723, 177.4189166973826,
175.06904170411656, 74.52024507341122] Fitness: 2.033016039512494
Generation 402 Best Solution: [183.7250748999603, 175.81418735762628,
175.06904170411656, 74.59123903759935] Fitness: 1.138560505355704
Generation 403 Best Solution: [183.7250748999603, 177.00953811725174,
175.06904170411656, 75.26192060610069] Fitness: 1.850373757006807
Generation 404 Best Solution: [183.7250748999603, 177.4189166973826,
174.34609763970744, 75.26192060610069] Fitness: 1.8282385746885261
Generation 405 Best Solution: [183.13748614747723, 177.4189166973826,
174.3050359447283, 75.58496342549081] Fitness: 2.0274308113670987
Generation 406 Best Solution: [183.7250748999603, 177.00953811725174,
175.06904170411656, 75.40504881148046] Fitness: 2.0149825732516
Generation 407 Best Solution: [183.13748614747723, 177.4189166973826,
175.06904170411656, 74.55811814157312] Fitness: 2.0259074867535176
Generation 408 Best Solution: [183.7250748999603, 176.2697251802208,
175.06904170411656, 74.63256682015853] Fitness: 1.004540920704976
Generation 409 Best Solution: [183.7250748999603, 176.7095150501803,
174.81954735164766, 75.63001293852149] Fitness: 1.671494984356599
Generation 410 Best Solution: [183.8789635482229, 176.7095150501803,
174.81954735164766, 75.63001293852149] Fitness: 1.8218572756287141
Generation 411 Best Solution: [183.7250748999603, 177.4189166973826,
173.90200799577485, 74.63256682015853] Fitness: 1.5126719214385556
Generation 412 Best Solution: [183.13748614747723, 175.65413272893997,
173.90200799577485, 75.86786526430134] Fitness: 1.5916595794725061
Generation 413 Best Solution: [183.13748614747723, 175.65413272893997,
173.90200799577485, 74.63256682015853] Fitness: 0.34420883153378684
Generation 414 Best Solution: [183.77978150108078, 177.4189166973826,
173.8424503909358, 74.63256682015853] Fitness: 1.5866597818494408
Generation 415 Best Solution: [183.7250748999603, 176.61848031188822,
175.06904170411656, 74.63256682015853] Fitness: 1.1402119509996858
Generation 416 Best Solution: [183.77978150108078, 176.61848031188822,
175.06904170411656, 75.63001293852149] Fitness: 1.996569139117857
Generation 417 Best Solution: [183.7250748999603, 176.3931644882384,
175.06904170411656, 74.63256682015853] Fitness: 1.0291393517989689
Generation 418 Best Solution: [183.7250748999603, 176.61848031188822,
175.06904170411656, 75.63001293852149] Fitness: 1.947685328904691
Generation 419 Best Solution: [183.77978150108078, 176.61848031188822,
175.06904170411656, 75.63001293852149] Fitness: 1.996569139117857
Generation 420 Best Solution: [183.18191492314705, 176.60182648821933,
175.06904170411656, 74.67962805851744] Fitness: 0.9176247223968494
Generation 421 Best Solution: [183.77978150108078, 176.61848031188822,
175.06904170411656, 74.63256682015853] Fitness: 1.1890957612128519
Generation 422 Best Solution: [183.7250748999603, 177.4189166973826,
175.06904170411656, 74.60996097710888] Fitness: 2.2273261855254223
Generation 423 Best Solution: [183.26082304233748, 177.4189166973826,
175.06904170411656, 74.29013279465623] Fitness: 2.123417030188814

Generation 424 Best Solution: [183.24606408991903, 177.4189166973826,
175.06904170411656, 74.29013279465623] Fitness: 2.122632324562868
Generation 425 Best Solution: [183.7250748999603, 176.39191317816056,
175.06904170411656, 75.63001293852149] Fitness: 1.8362346205038305
Generation 426 Best Solution: [183.69115612873765, 177.4189166973826,
175.06904170411656, 74.64133696655074] Fitness: 2.1982292114427984
Generation 427 Best Solution: [183.77978150108078, 176.08308886035056,
175.06904170411656, 74.64133696655074] Fitness: 1.0647645093771487
Generation 428 Best Solution: [183.7250748999603, 176.21171190305088,
175.06904170411656, 75.81062891179016] Fitness: 2.134714726137616
Generation 429 Best Solution: [183.69115612873765, 176.21171190305088,
175.06904170411656, 75.81062891179016] Fitness: 2.106937579585378
Generation 430 Best Solution: [183.47624963082825, 176.21171190305088,
175.06904170411656, 75.86786526430134] Fitness: 2.0905556517434043
Generation 431 Best Solution: [183.47624963082825, 176.21171190305088,
175.06904170411656, 75.81062891179016] Fitness: 1.9759747796019902
Generation 432 Best Solution: [183.77978150108078, 176.68423157864783,
175.06904170411656, 75.42817686754594] Fitness: 1.746476214631182
Generation 433 Best Solution: [183.77978150108078, 176.68423157864783,
175.06904170411656, 75.42817686754594] Fitness: 1.746476214631182
Generation 434 Best Solution: [183.7250748999603, 177.4189166973826,
174.5834708988175, 74.70198510031774] Fitness: 1.6526207279015857
Generation 435 Best Solution: [183.56848992353423, 177.4189166973826,
173.85776480055623, 75.81062891179016] Fitness: 2.551562654325452
Generation 436 Best Solution: [183.77978150108078, 177.4189166973826,
174.49669534144837, 75.6692577606125] Fitness: 2.5120332329302877
Generation 437 Best Solution: [183.56848992353423, 176.80815361279952,
175.06904170411656, 75.81062891179016] Fitness: 2.320810573803892
Generation 438 Best Solution: [183.56848992353423, 176.80815361279952,
174.49669534144837, 75.86786526430134] Fitness: 1.7988280404124968
Generation 439 Best Solution: [183.77978150108078, 176.51730183394062,
173.85776480055623, 75.81062891179016] Fitness: 1.5660533385872553
Generation 440 Best Solution: [183.77978150108078, 177.4189166973826,
173.85776480055623, 74.59276146451282] Fitness: 1.5821789676746283
Generation 441 Best Solution: [183.77978150108078, 177.4189166973826,
173.92481651717114, 74.98422872005406] Fitness: 1.646968261867176
Generation 442 Best Solution: [183.77978150108078, 177.35488137052982,
174.9700286050481, 74.83166848530492] Fitness: 2.0262400687703312
Generation 443 Best Solution: [183.77978150108078, 177.4189166973826,
175.06904170411656, 74.65192852481218] Fitness: 2.2748189527232174
Generation 444 Best Solution: [183.77978150108078, 177.4189166973826,
175.06904170411656, 74.65192852481218] Fitness: 2.2748189527232174
Generation 445 Best Solution: [183.77978150108078, 176.4145721967551,
175.06904170411656, 75.81062891179016] Fitness: 2.2177703058139406
Generation 446 Best Solution: [183.56848992353423, 176.4145721967551,
175.06904170411656, 75.81062891179016] Fitness: 2.0568290353031555
Generation 447 Best Solution: [183.56848992353423, 177.4189166973826,
173.92481651717114, 75.81062891179016] Fitness: 2.5254523667960322

Generation 448 Best Solution: [183.77978150108078, 177.4189166973826,
173.92481651717114, 75.81062891179016] Fitness: 2.686393637306817
Generation 449 Best Solution: [183.77978150108078, 177.4189166973826,
173.92481651717114, 75.81062891179016] Fitness: 2.686393637306817
Generation 450 Best Solution: [183.77978150108078, 175.61414069621057,
175.06904170411656, 75.81062891179016] Fitness: 2.485729230534465
Generation 451 Best Solution: [183.77978150108078, 175.62812677659437,
174.06440124772405, 75.81062891179016] Fitness: 1.7201502769224901
Generation 452 Best Solution: [183.77978150108078, 175.61414069621057,
175.06904170411656, 75.81062891179016] Fitness: 2.485729230534465
Generation 453 Best Solution: [183.77978150108078, 176.94176235615524,
175.06904170411656, 75.81062891179016] Fitness: 2.6306809784071223
Generation 454 Best Solution: [183.77978150108078, 176.39412917769585,
175.06904170411656, 75.86786526430134] Fitness: 2.325767211018816
Generation 455 Best Solution: [183.77978150108078, 177.4189166973826,
175.06904170411656, 74.70385423316617] Fitness: 2.2772035018487107
Generation 456 Best Solution: [182.8809513356125, 177.4189166973826,
175.06904170411656, 74.4612384022075] Fitness: 2.140785155028624
Generation 457 Best Solution: [183.68403597671355, 177.4189166973826,
175.06904170411656, 74.4612384022075] Fitness: 2.2227757432033046
Generation 458 Best Solution: [183.77978150108078, 177.4189166973826,
175.06904170411656, 74.4612384022075] Fitness: 2.3050215499834117
Generation 459 Best Solution: [183.68403597671355, 177.4189166973826,
175.06904170411656, 74.4612384022075] Fitness: 2.2227757432033046
Generation 460 Best Solution: [182.91693163133706, 176.4696311579326,
175.06904170411656, 75.86786526430134] Fitness: 2.1697787401618003
Generation 461 Best Solution: [182.91693163133706, 176.4696311579326,
175.06904170411656, 74.4612384022075] Fitness: 0.9522574116455903
Generation 462 Best Solution: [182.91693163133706, 177.4189166973826,
175.06904170411656, 74.4612384022075] Fitness: 2.1212159029461857
Generation 463 Best Solution: [182.91693163133706, 177.4189166973826,
175.06904170411656, 74.98005653085531] Fitness: 2.182401460923666
Generation 464 Best Solution: [183.68403597671355, 177.39569344627478,
175.06904170411656, 74.8382343917969] Fitness: 2.1755037542474467
Generation 465 Best Solution: [183.77978150108078, 177.39569344627478,
175.06904170411656, 74.54862322434631] Fitness: 2.2368713112137426
Generation 466 Best Solution: [183.68403597671355, 176.48892431061142,
175.06904170411656, 75.39427140842544] Fitness: 1.497954134443368
Generation 467 Best Solution: [183.77978150108078, 176.48892431061142,
175.06904170411656, 75.39427140842544] Fitness: 1.5801999412234757
Generation 468 Best Solution: [183.14663850800733, 176.48892431061142,
175.06904170411656, 75.86786526430134] Fitness: 2.104886043161593
Generation 469 Best Solution: [183.77978150108078, 176.3919832659854,
175.06904170411656, 75.86786526430134] Fitness: 2.3251169095181474
Generation 470 Best Solution: [183.77978150108078, 176.48892431061142,
175.06904170411656, 74.75197746209001] Fitness: 1.123157701903599
Generation 471 Best Solution: [183.77978150108078, 176.48892431061142,
175.06904170411656, 74.75197746209001] Fitness: 1.123157701903599

Generation 472 Best Solution: [183.46100636591692, 176.2136122410447,
175.06904170411656, 75.86786526430134] Fitness: 2.084219313147303
Generation 473 Best Solution: [183.77978150108078, 176.19166741636167,
175.06904170411656, 75.86786526430134] Fitness: 2.2985654556943933
Generation 474 Best Solution: [183.77978150108078, 176.663167775184,
175.06904170411656, 75.86786526430134] Fitness: 2.4687364633573194
Generation 475 Best Solution: [183.77978150108078, 177.4189166973826,
175.06904170411656, 75.44382871718577] Fitness: 2.80464668980144
Generation 476 Best Solution: [182.81403583369755, 177.4189166973826,
175.06904170411656, 74.36278648129324] Fitness: 2.222543108762322
Generation 477 Best Solution: [183.77978150108078, 177.4189166973826,
175.06904170411656, 74.36278648129324] Fitness: 2.3445867958174174
Generation 478 Best Solution: [183.77978150108078, 177.4189166973826,
175.06904170411656, 74.36278648129324] Fitness: 2.3445867958174174
Generation 479 Best Solution: [182.81403583369755, 177.4189166973826,
175.06904170411656, 74.43936892764239] Fitness: 2.1903562584521437
Generation 480 Best Solution: [182.81403583369755, 177.4189166973826,
175.06904170411656, 74.43936892764239] Fitness: 2.1903562584521437
Generation 481 Best Solution: [182.81403583369755, 176.93643469635285,
175.06904170411656, 74.43936892764239] Fitness: 1.4065611421247677
Generation 482 Best Solution: [182.81403583369755, 177.4189166973826,
175.06904170411656, 74.7454552779144] Fitness: 2.1603467035059225
Generation 483 Best Solution: [183.46100636591692, 177.4189166973826,
175.06904170411656, 74.43936892764239] Fitness: 2.098441331795316
Generation 484 Best Solution: [182.81403583369755, 177.4189166973826,
175.06904170411656, 74.43936892764239] Fitness: 2.1903562584521437
Generation 485 Best Solution: [182.92116280012263, 177.4189166973826,
175.06904170411656, 74.43936892764239] Fitness: 2.126436293470929
Generation 486 Best Solution: [183.77978150108078, 177.4189166973826,
173.77264839595594, 75.86786526430134] Fitness: 2.871136518264107
Generation 487 Best Solution: [182.81403583369755, 177.4189166973826,
173.77264839595594, 75.86786526430134] Fitness: 2.749092831209012
Generation 488 Best Solution: [182.81403583369755, 177.4189166973826,
173.77264839595594, 75.86786526430134] Fitness: 2.749092831209012
Generation 489 Best Solution: [183.77978150108078, 176.45580144907964,
174.69322814798477, 75.86786526430134] Fitness: 1.8675935418523353
Generation 490 Best Solution: [183.77978150108078, 176.27057130549375,
174.69322814798477, 75.86786526430134] Fitness: 1.8207794105487365
Generation 491 Best Solution: [183.77978150108078, 176.27690959307915,
175.06904170411656, 75.86786526430134] Fitness: 2.301603810573279
Generation 492 Best Solution: [183.77978150108078, 175.80111731812647,
174.69322814798477, 75.86786526430134] Fitness: 1.960948043734352
Generation 493 Best Solution: [183.46100636591692, 177.4189166973826,
174.68908528348004, 75.86786526430134] Fitness: 2.824439158113409
Generation 494 Best Solution: [182.81403583369755, 177.4189166973826,
175.06904170411656, 74.44194303507622] Fitness: 2.1894459716112897
Generation 495 Best Solution: [183.77978150108078, 176.26043562813982,
175.06904170411656, 75.86786526430134] Fitness: 2.3000626073907195

```
Generation 496 Best Solution: [183.77978150108078, 176.46413800743056,
175.06904170411656, 75.86786526430134] Fitness: 2.351236699487015
Generation 497 Best Solution: [183.77978150108078, 177.4189166973826,
175.06904170411656, 75.22136291137834] Fitness: 2.549126375298017
Generation 498 Best Solution: [182.81403583369755, 177.4189166973826,
175.06904170411656, 74.71419477527944] Fitness: 2.156176880112032
Generation 499 Best Solution: [182.81403583369755, 177.4189166973826,
175.06904170411656, 74.71419477527944] Fitness: 2.156176880112032
Generation 500 Best Solution: [182.81403583369755, 177.4189166973826,
175.06904170411656, 74.73329302908954] Fitness: 2.1585287847052705
Optimal Solution for Simulation 21: [182.81403583369755, 177.4189166973826,
175.06904170411656, 74.73329302908954]
--------------------------------------------------
Simulation 22/32 with conditions: {'drought': False, 'flood': True,
'policy_change': False, 'social_event': True, 'economic_crisis': False}
Generation 1 Best Solution: [183.79038729958873, 175.99725117445226,
174.14124981651793, 74.48927991699071] Fitness: 0.3977835033289787
Generation 2 Best Solution: [183.156948343988, 175.755130833384,
173.99535136949348, 74.78928967689245] Fitness: 0.21147069361088472
Generation 3 Best Solution: [182.9871310077194, 176.5828727654687,
174.46118231761736, 74.37112474406514] Fitness: 0.39308744151608577
Generation 4 Best Solution: [183.68379211646072, 176.5828727654687,
174.46118231761736, 74.37112474406514] Fitness: 0.5591479992671062
Generation 5 Best Solution: [183.5185951942329, 177.04875905309467,
174.5565511714847, 74.37112474406514] Fitness: 0.9907143921553874
Generation 6 Best Solution: [182.9871310077194, 177.04875905309467,
174.63795432650144, 75.33038358657055] Fitness: 1.2641638435476072
Generation 7 Best Solution: [183.5185951942329, 176.39686081363087,
174.36947038977468, 75.62612296296562] Fitness: 0.953757304819201
Generation 8 Best Solution: [183.52125307672006, 177.04875905309467,
174.63795432650144, 75.62396122243524] Fitness: 1.6995509393910448
Generation 9 Best Solution: [183.52125307672006, 177.04875905309467,
174.63795432650144, 75.8349238635394] Fitness: 2.057595191450507
Generation 10 Best Solution: [183.68379211646072, 177.04875905309467,
174.63795432650144, 75.8349238635394] Fitness: 2.16661839387445
Generation 11 Best Solution: [183.68379211646072, 177.22430760437555,
174.63795432650144, 75.12614373942087] Fitness: 1.5208497949590587
Generation 12 Best Solution: [183.68379211646072, 177.13266431531065,
174.63795432650144, 75.12614373942087] Fitness: 1.3728741501222725
Generation 13 Best Solution: [183.49951629887386, 175.96739841718653,
174.63795432650144, 75.8349238635394] Fitness: 1.4739257230545573
Generation 14 Best Solution: [183.52125307672006, 177.04875905309467,
174.36947038977468, 74.99649055193147] Fitness: 0.8844961298644496
Generation 15 Best Solution: [183.52125307672006, 177.22430760437555,
175.08389278808448, 74.34415184228301] Fitness: 1.8937273392959324
Generation 16 Best Solution: [182.85820515890987, 177.22430760437555,
174.08493627306066, 74.34415184228301] Fitness: 1.1067491387286534
Generation 17 Best Solution: [182.94052069962143, 176.90975759418095,
```

174.36947038977468, 75.8349238635394] Fitness: 1.6618840054430295
Generation 18 Best Solution: [183.52125307672006, 176.90975759418095,
174.36947038977468, 75.8349238635394] Fitness: 1.70198645755975
Generation 19 Best Solution: [183.68379211646072, 177.04875905309467,
173.84092394954627, 75.8349238635394] Fitness: 1.9851637751940807
Generation 20 Best Solution: [183.31776589550284, 177.42031474410652,
175.08389278808448, 74.51910166167241] Fitness: 2.1089805908005603
Generation 21 Best Solution: [183.52125307672006, 177.30013409471152,
175.08389278808448, 74.86803658125845] Fitness: 1.96275924187235
Generation 22 Best Solution: [182.94052069962143, 177.22430760437555,
175.08389278808448, 74.86803658125845] Fitness: 1.7899405402688335
Generation 23 Best Solution: [182.94052069962143, 177.22430760437555,
175.08389278808448, 75.29038756369779] Fitness: 2.0621198133705954
Generation 24 Best Solution: [182.94052069962143, 177.22430760437555,
175.08389278808448, 74.46990995210973] Fitness: 1.7978065543067274
Generation 25 Best Solution: [182.94052069962143, 177.30013409471152,
175.08389278808448, 74.91379943018417] Fitness: 1.9381970081515951
Generation 26 Best Solution: [182.94052069962143, 177.22430760437555,
174.2779926868619, 75.8349238635394] Fitness: 2.079714840387034
Generation 27 Best Solution: [182.94052069962143, 177.22430760437555,
174.2779926868619, 75.8349238635394] Fitness: 2.079714840387034
Generation 28 Best Solution: [182.94052069962143, 177.22430760437555,
174.2779926868619, 75.8349238635394] Fitness: 2.079714840387034
Generation 29 Best Solution: [182.94052069962143, 176.36691293156392,
175.08389278808448, 75.8349238635394] Fitness: 1.9863243916736892
Generation 30 Best Solution: [183.36417191203296, 177.22430760437555,
175.08389278808448, 74.94907569995021] Fitness: 1.7951574916313078
Generation 31 Best Solution: [183.08670537447082, 177.22430760437555,
175.08389278808448, 74.62014555236988] Fitness: 1.7233603448731787
Generation 32 Best Solution: [182.85820515890987, 177.22430760437555,
175.08389278808448, 74.62014555236988] Fitness: 1.8031887550898509
Generation 33 Best Solution: [182.85820515890987, 177.22430760437555,
175.08389278808448, 74.62014555236988] Fitness: 1.8031887550898509
Generation 34 Best Solution: [182.85820515890987, 176.8968971085532,
175.08389278808448, 75.63312771676264] Fitness: 2.0675953176897943
Generation 35 Best Solution: [183.36417191203296, 177.22430760437555,
175.08389278808448, 75.27615570186906] Fitness: 2.023629629991653
Generation 36 Best Solution: [182.96992404541183, 177.22430760437555,
175.08389278808448, 74.80321121459386] Fitness: 1.762637778320658
Generation 37 Best Solution: [183.25802987491923, 177.22430760437555,
175.08389278808448, 74.80321121459386] Fitness: 1.7279994543855266
Generation 38 Best Solution: [183.36417191203296, 177.30013409471152,
175.08389278808448, 74.80321121459386] Fitness: 1.881830035788588
Generation 39 Best Solution: [183.36417191203296, 177.30013409471152,
175.08389278808448, 74.80321121459386] Fitness: 1.881830035788588
Generation 40 Best Solution: [182.85820515890987, 177.22430760437555,
175.08389278808448, 74.80321121459386] Fitness: 1.8122292788051384
Generation 41 Best Solution: [182.85820515890987, 177.22430760437555,

174.69328007158853, 74.41911328505385] Fitness: 1.3563069026425185
Generation 42 Best Solution: [183.42079111179612, 176.897485566793,
174.12950305777028, 75.79112142160855] Fitness: 1.5028321783974874
Generation 43 Best Solution: [183.81715754472327, 176.897485566793,
174.12950305777028, 75.79112142160855] Fitness: 1.7792529789928535
Generation 44 Best Solution: [182.96992404541183, 175.96308690248523,
175.08389278808448, 75.79112142160855] Fitness: 1.9055396269315479
Generation 45 Best Solution: [183.42079111179612, 176.09347540658243,
175.08389278808448, 75.8349238635394] Fitness: 1.9587302601940837
Generation 46 Best Solution: [183.42079111179612, 176.09347540658243,
175.08389278808448, 75.79112142160855] Fitness: 1.8784617224996039
Generation 47 Best Solution: [183.0792430217852, 177.22430760437555,
173.99537966691713, 75.79112142160855] Fitness: 1.9420638881087047
Generation 48 Best Solution: [183.0792430217852, 177.22430760437555,
173.99537966691713, 75.79112142160855] Fitness: 1.9420638881087047
Generation 49 Best Solution: [182.96992404541183, 177.22430760437555,
173.99537966691713, 75.79112142160855] Fitness: 1.9710290433475686
Generation 50 Best Solution: [183.81715754472327, 177.22430760437555,
174.2436271999259, 74.81095917560043] Fitness: 1.2772805949168022
Generation 51 Best Solution: [183.81715754472327, 177.22430760437555,
173.955764633474, 75.73237273406951] Fitness: 2.157773558504329
Generation 52 Best Solution: [183.81715754472327, 177.22430760437555,
175.08389278808448, 74.45513538343756] Fitness: 2.0735641276363084
Generation 53 Best Solution: [183.81715754472327, 177.22430760437555,
173.955764633474, 75.73237273406951] Fitness: 2.157773558504329
Generation 54 Best Solution: [183.0792430217852, 177.22430760437555,
173.955764633474, 75.73237273406951] Fitness: 1.8471885112692203
Generation 55 Best Solution: [183.81715754472327, 177.22430760437555,
173.955764633474, 74.97473276744036] Fitness: 1.332999708794657
Generation 56 Best Solution: [183.81715754472327, 177.22430760437555,
173.955764633474, 74.97473276744036] Fitness: 1.332999708794657
Generation 57 Best Solution: [183.81715754472327, 177.22430760437555,
173.955764633474, 74.97473276744036] Fitness: 1.332999708794657
Generation 58 Best Solution: [183.05566636045745, 175.74901633016992,
174.60358566150924, 75.79112142160855] Fitness: 1.414593723759707
Generation 59 Best Solution: [183.81715754472327, 176.3343072032184,
175.08389278808448, 75.42616821659563] Fitness: 1.6196192699292427
Generation 60 Best Solution: [183.0294482316664, 177.22430760437555,
173.955764633474, 75.79112142160855] Fitness: 1.960763265417186
Generation 61 Best Solution: [183.0294482316664, 177.22430760437555,
173.955764633474, 75.73237273406951] Fitness: 1.8579827145145995
Generation 62 Best Solution: [183.32353592567182, 177.22430760437555,
173.955764633474, 75.73237273406951] Fitness: 1.8523912766503208
Generation 63 Best Solution: [183.0294482316664, 176.5610429219751,
175.08389278808448, 75.73237273406951] Fitness: 1.8628977830810012
Generation 64 Best Solution: [183.0294482316664, 177.22430760437555,
174.56237228369778, 75.01783106061892] Fitness: 1.21257717093952
Generation 65 Best Solution: [183.0294482316664, 177.22430760437555,

174.56237228369778, 75.73237273406951] Fitness: 2.0153635334757487
Generation 66 Best Solution: [183.52511465190588, 177.22430760437555,
175.08389278808448, 75.18007969385884] Fitness: 2.0053652741892023
Generation 67 Best Solution: [183.81715754472327, 177.22430760437555,
174.86092781280365, 75.18007969385884] Fitness: 1.9183390279868084
Generation 68 Best Solution: [183.81715754472327, 177.22430760437555,
174.86092781280365, 75.18007969385884] Fitness: 1.9183390279868084
Generation 69 Best Solution: [183.81715754472327, 177.22430760437555,
174.86092781280365, 74.40285070006416] Fitness: 1.7795549204590062
Generation 70 Best Solution: [183.81715754472327, 177.22430760437555,
174.86092781280365, 75.18007969385884] Fitness: 1.9183390279868084
Generation 71 Best Solution: [183.81715754472327, 175.66072441268756,
174.82338260874852, 75.73237273406951] Fitness: 1.9046093917740148
Generation 72 Best Solution: [183.81715754472327, 176.81487667441206,
174.48154233936611, 75.73237273406951] Fitness: 1.7040040746571865
Generation 73 Best Solution: [183.81715754472327, 176.81487667441206,
174.48154233936611, 75.73237273406951] Fitness: 1.7040040746571865
Generation 74 Best Solution: [183.81715754472327, 176.13330931178186,
174.1888440574458, 75.79112142160855] Fitness: 1.3715403644735034
Generation 75 Best Solution: [183.81715754472327, 176.13330931178186,
175.08389278808448, 75.01414190871147] Fitness: 1.242944218724027
Generation 76 Best Solution: [183.81715754472327, 177.22430760437555,
175.08389278808448, 74.76135116717788] Fitness: 2.037405458644917
Generation 77 Best Solution: [182.9592101906671, 177.22430760437555,
175.08389278808448, 75.01414190871147] Fitness: 1.8442104171761322
Generation 78 Best Solution: [183.11351445483504, 177.22430760437555,
175.08389278808448, 75.73237273406951] Fitness: 2.611054761087785
Generation 79 Best Solution: [183.81715754472327, 177.22430760437555,
175.08389278808448, 74.73910205024416] Fitness: 2.034918108344391
Generation 80 Best Solution: [183.68537958557673, 177.22430760437555,
173.75006911103296, 75.79112142160855] Fitness: 2.222322765575499
Generation 81 Best Solution: [183.81715754472327, 177.22430760437555,
174.86092781280365, 74.56150362329193] Fitness: 1.7284386370931593
Generation 82 Best Solution: [182.98111750241154, 177.22430760437555,
174.4199877713515, 75.73237273406951] Fitness: 1.9392306291604444
Generation 83 Best Solution: [183.81715754472327, 177.22430760437555,
174.86092781280365, 74.78517740241523] Fitness: 1.7256009351167283
Generation 84 Best Solution: [183.81715754472327, 177.22430760437555,
174.4199877713515, 75.79112142160855] Fitness: 2.3274860419828336
Generation 85 Best Solution: [182.98111750241154, 177.22430760437555,
175.08389278808448, 74.5564680391009] Fitness: 1.7593168627005131
Generation 86 Best Solution: [183.81715754472327, 177.22430760437555,
174.86092781280365, 75.33221128160547] Fitness: 2.059958660483452
Generation 87 Best Solution: [183.81715754472327, 177.22430760437555,
174.86092781280365, 75.33221128160547] Fitness: 2.059958660483452
Generation 88 Best Solution: [183.81715754472327, 177.22430760437555,
174.86092781280365, 75.33221128160547] Fitness: 2.059958660483452
Generation 89 Best Solution: [183.81715754472327, 177.22430760437555,

174.86092781280365, 75.33221128160547] Fitness: 2.059958660483452
Generation 90 Best Solution: [183.81715754472327, 176.07812760766083,
175.08389278808448, 74.88868667874074] Fitness: 1.1947956099117751
Generation 91 Best Solution: [182.98111750241154, 175.58799854376588,
175.08389278808448, 75.73237273406951] Fitness: 2.0452397672300022
Generation 92 Best Solution: [183.81715754472327, 176.07812760766083,
174.95979822240037, 75.79112142160855] Fitness: 1.971803507610708
Generation 93 Best Solution: [183.26293861025397, 176.07812760766083,
175.08389278808448, 75.79112142160855] Fitness: 1.8415684590084205
Generation 94 Best Solution: [183.6790544156476, 176.07812760766083,
175.08389278808448, 75.79112142160855] Fitness: 2.032067888460835
Generation 95 Best Solution: [182.8705545287005, 176.07812760766083,
175.08389278808448, 75.79112142160855] Fitness: 1.9187488502304404
Generation 96 Best Solution: [183.81715754472327, 176.07812760766083,
175.08389278808448, 75.79112142160855] Fitness: 2.157252478137302
Generation 97 Best Solution: [182.93835918260615, 176.07812760766083,
174.05254918735645, 75.79112142160855] Fitness: 1.1037986995903513
Generation 98 Best Solution: [183.81715754472327, 176.07812760766083,
175.08389278808448, 74.85142903427982] Fitness: 1.1834019257333477
Generation 99 Best Solution: [183.75443368385066, 176.32311940020787,
175.08389278808448, 74.85142903427982] Fitness: 1.1304316632529865
Generation 100 Best Solution: [183.81715754472327, 176.07812760766083,
175.08389278808448, 74.85142903427982] Fitness: 1.1834019257333477
Generation 101 Best Solution: [183.81715754472327, 176.07812760766083,
173.78549808121983, 74.85142903427982] Fitness: 0.4773444812450243
Generation 102 Best Solution: [183.75443368385066, 176.07812760766083,
175.08389278808448, 74.65090169908949] Fitness: 1.0999951054846664
Generation 103 Best Solution: [183.75443368385066, 176.07812760766083,
175.08389278808448, 75.79112142160855] Fitness: 2.0965684841801613
Generation 104 Best Solution: [183.81715754472327, 177.4307033824268,
174.43237383160053, 75.03476698469136] Fitness: 1.8210096824081643
Generation 105 Best Solution: [183.81715754472327, 177.4307033824268,
174.43237383160053, 74.29702347154654] Fitness: 1.8392622383678412
Generation 106 Best Solution: [183.75443368385066, 176.07812760766083,
175.08389278808448, 75.79112142160855] Fitness: 2.0965684841801613
Generation 107 Best Solution: [183.6492285150228, 176.07812760766083,
174.88587854867563, 75.79112142160855] Fitness: 1.7250199587783266
Generation 108 Best Solution: [183.81087115343183, 176.07812760766083,
174.88587854867563, 75.79112142160855] Fitness: 1.866816686441088
Generation 109 Best Solution: [182.97022332063588, 175.59572377540695,
175.08389278808448, 75.79112142160855] Fitness: 2.144398708047886
Generation 110 Best Solution: [183.4983319921489, 175.59572377540695,
173.84144570958054, 75.79112142160855] Fitness: 1.4518613208206539
Generation 111 Best Solution: [183.4983319921489, 177.4307033824268,
174.32196502556388, 75.79112142160855] Fitness: 2.424999145331188
Generation 112 Best Solution: [183.81715754472327, 177.4307033824268,
175.08389278808448, 74.68451679502178] Fitness: 2.4152734580615918
Generation 113 Best Solution: [183.4983319921489, 177.4307033824268,

175.08389278808448, 74.68451679502178] Fitness: 2.172915601013577
Generation 114 Best Solution: [182.97022332063588, 176.10243229706285,
174.88587854867563, 75.79112142160855] Fitness: 1.5878834258624184
Generation 115 Best Solution: [182.97022332063588, 176.10243229706285,
174.88587854867563, 75.65488216980484] Fitness: 1.3580798498935485
Generation 116 Best Solution: [183.81715754472327, 177.4307033824268,
174.52343765943053, 74.68451679502178] Fitness: 1.7756976083236258
Generation 117 Best Solution: [183.81715754472327, 176.08054798762586,
174.88587854867563, 75.79112142160855] Fitness: 1.872786768650866
Generation 118 Best Solution: [182.97022332063588, 176.58697590608938,
174.59794066311252, 75.79112142160855] Fitness: 1.4163735399338009
Generation 119 Best Solution: [182.97022332063588, 177.4307033824268,
175.04963341854534, 74.80833918644407] Fitness: 2.092939876440417
Generation 120 Best Solution: [182.97022332063588, 177.4307033824268,
174.88587854867563, 75.23830231265548] Fitness: 2.100739642934817
Generation 121 Best Solution: [182.97022332063588, 175.9951769182506,
175.08389278808448, 75.19413237853264] Fitness: 1.1106262344461695
Generation 122 Best Solution: [182.97022332063588, 177.4307033824268,
173.77718646628995, 75.28749285575103] Fitness: 1.7293504803028934
Generation 123 Best Solution: [182.97022332063588, 176.9611434273021,
175.08389278808448, 75.79112142160855] Fitness: 2.3594030470763396
Generation 124 Best Solution: [182.97022332063588, 177.4307033824268,
174.6462691106054, 75.28749285575103] Fitness: 1.8882165035367882
Generation 125 Best Solution: [182.97022332063588, 176.9611434273021,
175.08389278808448, 75.79112142160855] Fitness: 2.3594030470763396
Generation 126 Best Solution: [182.97022332063588, 176.9611434273021,
175.08389278808448, 75.28749285575103] Fitness: 1.6596844755793596
Generation 127 Best Solution: [182.8713087122073, 177.4307033824268,
174.26117508254723, 75.28749285575103] Fitness: 1.7097768941562739
Generation 128 Best Solution: [182.8713087122073, 176.70300003256074,
174.26117508254723, 75.79112142160855] Fitness: 1.3660049679939448
Generation 129 Best Solution: [183.4983319921489, 176.05365945774878,
175.08389278808448, 75.28749285575103] Fitness: 1.2197429946884375
Generation 130 Best Solution: [183.81715754472327, 175.80795687352492,
175.08389278808448, 75.28749285575103] Fitness: 1.5616984109824694
Generation 131 Best Solution: [182.97022332063588, 177.4307033824268,
174.26117508254723, 75.79112142160855] Fitness: 2.366660939863767
Generation 132 Best Solution: [183.4983319921489, 177.4307033824268,
175.08389278808448, 75.28749285575103] Fitness: 2.470503354325123
Generation 133 Best Solution: [182.97022332063588, 175.70772034214548,
175.08389278808448, 75.79112142160855] Fitness: 2.048393917738581
Generation 134 Best Solution: [182.97022332063588, 177.4307033824268,
174.12554788737165, 75.79112142160855] Fitness: 2.3459016444613296
Generation 135 Best Solution: [183.4983319921489, 177.4307033824268,
175.08389278808448, 74.62966538027202] Fitness: 2.1750545528797742
Generation 136 Best Solution: [182.97022332063588, 177.4307033824268,
175.08389278808448, 74.62966538027202] Fitness: 2.13568680972148
Generation 137 Best Solution: [183.4983319921489, 177.4307033824268,

175.08389278808448, 74.62966538027202] Fitness: 2.1750545528797742
Generation 138 Best Solution: [183.31016945445143, 177.4307033824268,
175.08389278808448, 74.74871934526604] Fitness: 2.11079182059168
Generation 139 Best Solution: [183.4983319921489, 177.4307033824268,
175.08389278808448, 74.91908044420938] Fitness: 2.218724240501959
Generation 140 Best Solution: [183.7119903002766, 176.81479373299345,
175.08389278808448, 74.43428031511617] Fitness: 1.4275540252792964
Generation 141 Best Solution: [183.81715754472327, 177.4307033824268,
175.08389278808448, 75.07168270827376] Fitness: 2.5387128574124342
Generation 142 Best Solution: [183.81715754472327, 177.4307033824268,
175.08389278808448, 74.6374767627895] Fitness: 2.4168103444332267
Generation 143 Best Solution: [183.81715754472327, 175.9453377234852,
175.08389278808448, 75.56040311112632] Fitness: 1.8221602466875433
Generation 144 Best Solution: [183.81715754472327, 175.9453377234852,
175.08389278808448, 75.56040311112632] Fitness: 1.8221602466875433
Generation 145 Best Solution: [183.81715754472327, 176.64960583570846,
175.08389278808448, 75.79112142160855] Fitness: 2.3262867178162128
Generation 146 Best Solution: [183.81715754472327, 176.64960583570846,
175.08389278808448, 75.79112142160855] Fitness: 2.3262867178162128
Generation 147 Best Solution: [182.9721386342175, 176.47000166261273,
175.08389278808448, 75.79112142160855] Fitness: 1.9337878768362828
Generation 148 Best Solution: [183.14555054084397, 177.4307033824268,
175.08389278808448, 74.92077217091551] Fitness: 2.1430093369165473
Generation 149 Best Solution: [183.81715754472327, 176.79528355205926,
175.08389278808448, 75.79112142160855] Fitness: 2.453949786192439
Generation 150 Best Solution: [183.81715754472327, 176.62041421225524,
175.08389278808448, 75.79112142160855] Fitness: 2.304837351204906
Generation 151 Best Solution: [182.86014562630342, 177.4307033824268,
174.24821487879817, 75.79112142160855] Fitness: 2.4120536582536456
Generation 152 Best Solution: [183.81715754472327, 177.4307033824268,
174.24821487879817, 74.60883922173639] Fitness: 1.6520364800516512
Generation 153 Best Solution: [182.86014562630342, 177.4307033824268,
173.9186560597211, 74.60883922173639] Fitness: 1.4271849071758587
Generation 154 Best Solution: [183.81715754472327, 177.4307033824268,
173.9186560597211, 74.60883922173639] Fitness: 1.6602471091790454
Generation 155 Best Solution: [183.81715754472327, 177.4307033824268,
173.9186560597211, 74.60883922173639] Fitness: 1.6602471091790454
Generation 156 Best Solution: [183.29476714152136, 177.4307033824268,
173.9186560597211, 74.60883922173639] Fitness: 1.3492327869159382
Generation 157 Best Solution: [183.0812024198891, 177.4307033824268,
173.9186560597211, 74.60883922173639] Fitness: 1.34931940730281
Generation 158 Best Solution: [183.3340509021327, 177.2359775447793,
175.08389278808448, 74.55688310681337] Fitness: 1.7615380366647089
Generation 159 Best Solution: [183.81715754472327, 176.2035737030801,
175.08389278808448, 74.55744992118197] Fitness: 1.1641392188176558
Generation 160 Best Solution: [183.81715754472327, 177.2359775447793,
174.95203789753452, 74.55744992118197] Fitness: 1.8681958246541963
Generation 161 Best Solution: [183.81715754472327, 177.2359775447793,

174.95203789753452, 74.55744992118197] Fitness: 1.8681958246541963
Generation 162 Best Solution: [183.81715754472327, 177.2359775447793,
174.95203789753452, 74.55744992118197] Fitness: 1.8681958246541963
Generation 163 Best Solution: [182.86014562630342, 177.2359775447793,
175.08389278808448, 74.70134404425158] Fitness: 1.8192882746975136
Generation 164 Best Solution: [183.22782533100963, 177.2359775447793,
175.08389278808448, 74.55744992118197] Fitness: 1.7454808066602756
Generation 165 Best Solution: [182.86014562630342, 176.18338242198112,
174.95203789753452, 74.55744992118197] Fitness: 0.734457076070897
Generation 166 Best Solution: [183.81715754472327, 176.81592264565433,
174.95203789753452, 74.55744992118197] Fitness: 1.2936746727965218
Generation 167 Best Solution: [182.86014562630342, 176.34549384591625,
175.08389278808448, 74.55744992118197] Fitness: 0.9525282351379777
Generation 168 Best Solution: [183.81715754472327, 177.2359775447793,
174.4204771251226, 74.74270401137005] Fitness: 1.3533092611997963
Generation 169 Best Solution: [183.81715754472327, 176.34627868002934,
175.08389278808448, 74.74270401137005] Fitness: 1.1764718953584754
Generation 170 Best Solution: [182.86014562630342, 177.2359775447793,
174.20928120481574, 75.79112142160855] Fitness: 2.0406205979360044
Generation 171 Best Solution: [182.86014562630342, 176.12611415213271,
175.08389278808448, 75.79112142160855] Fitness: 1.9180482496089784
Generation 172 Best Solution: [182.86014562630342, 177.22452462517913,
175.08389278808448, 74.68748239023569] Fitness: 1.7995384025960066
Generation 173 Best Solution: [182.86014562630342, 177.2359775447793,
174.18407672926907, 75.79112142160855] Fitness: 2.036627387008119
Generation 174 Best Solution: [183.31105989547453, 177.2359775447793,
174.18407672926907, 75.79112142160855] Fitness: 1.9617003343283357
Generation 175 Best Solution: [183.31105989547453, 177.2359775447793,
174.18407672926907, 75.79112142160855] Fitness: 1.9617003343283357
Generation 176 Best Solution: [182.86014562630342, 176.58824255536996,
174.18407672926907, 75.79112142160855] Fitness: 1.2700872608892226
Generation 177 Best Solution: [182.86014562630342, 177.2359775447793,
174.18407672926907, 75.47244452448858] Fitness: 1.5461590422496267
Generation 178 Best Solution: [182.86014562630342, 177.2359775447793,
174.18407672926907, 75.47244452448858] Fitness: 1.5461590422496267
Generation 179 Best Solution: [183.31105989547453, 177.4307033824268,
174.15403954296573, 75.79112142160855] Fitness: 2.3215279179063457
Generation 180 Best Solution: [182.86014562630342, 177.2359775447793,
175.08389278808448, 74.81340996890933] Fitness: 1.8331141051671833
Generation 181 Best Solution: [182.86014562630342, 177.2359775447793,
174.73455996490694, 75.79112142160855] Fitness: 2.357921483418278
Generation 182 Best Solution: [182.86014562630342, 176.3503632974713,
174.46414408568484, 75.75824215611678] Fitness: 1.203062803791401
Generation 183 Best Solution: [182.86014562630342, 176.2279094784905,
175.08389278808448, 75.79112142160855] Fitness: 1.9173618666059409
Generation 184 Best Solution: [182.86014562630342, 176.2279094784905,
175.08389278808448, 75.79112142160855] Fitness: 1.9173618666059409
Generation 185 Best Solution: [183.14101266267215, 177.2359775447793,

174.73455996490694, 75.79112142160855] Fitness: 2.272587268918337
Generation 186 Best Solution: [183.14101266267215, 177.4307033824268,
174.73455996490694, 74.59820456471572] Fitness: 1.644088153812036
Generation 187 Best Solution: [183.31105989547453, 176.20893877967166,
175.08389278808448, 75.79112142160855] Fitness: 1.8412907720321994
Generation 188 Best Solution: [183.67911123253745, 176.20893877967166,
175.08389278808448, 75.79112142160855] Fitness: 2.0241405897349547
Generation 189 Best Solution: [183.14101266267215, 176.64389655425725,
175.08389278808448, 75.05256434026236] Fitness: 1.1179281270215822
Generation 190 Best Solution: [183.67911123253745, 177.20356699847326,
174.01449322543112, 75.79112142160855] Fitness: 2.089923952846579
Generation 191 Best Solution: [183.31105989547453, 177.20356699847326,
174.01449322543112, 75.79112142160855] Fitness: 1.9070741351438238
Generation 192 Best Solution: [183.31105989547453, 177.20356699847326,
174.01449322543112, 75.79112142160855] Fitness: 1.9070741351438238
Generation 193 Best Solution: [183.31105989547453, 177.20356699847326,
174.01449322543112, 75.79112142160855] Fitness: 1.9070741351438238
Generation 194 Best Solution: [183.67911123253745, 176.84429152343262,
174.01449322543112, 75.79112142160855] Fitness: 1.5997080523987823
Generation 195 Best Solution: [183.31105989547453, 176.84429152343262,
174.01449322543112, 75.79112142160855] Fitness: 1.4168582346960268
Generation 196 Best Solution: [183.67911123253745, 177.20356699847326,
174.01449322543112, 75.79112142160855] Fitness: 2.089923952846579
Generation 197 Best Solution: [183.31105989547453, 177.4307033824268,
173.886514886293, 75.19441458371166] Fitness: 1.5714731184002146
Generation 198 Best Solution: [183.67911123253745, 177.20356699847326,
174.01449322543112, 75.33028720960398] Fitness: 1.4336972039213993
Generation 199 Best Solution: [183.32566054730938, 177.4307033824268,
173.886514886293, 75.19441458371166] Fitness: 1.574549161151534
Generation 200 Best Solution: [183.68909695647528, 177.20356699847326,
173.886514886293, 75.19441458371166] Fitness: 1.3445971171508642
Generation 201 Best Solution: [183.31105989547453, 177.20356699847326,
173.886514886293, 75.19441458371166] Fitness: 1.1537304087670508
Generation 202 Best Solution: [183.67911123253745, 176.05976802365936,
174.4808888466671, 75.79112142160855] Fitness: 1.3680877024305091
Generation 203 Best Solution: [183.44826361229266, 177.20356699847326,
175.08389278808448, 74.40062956620709] Fitness: 1.7957706495961514
Generation 204 Best Solution: [183.45954877441577, 177.20356699847326,
175.08389278808448, 75.19441458371166] Fitness: 1.9501493281787474
Generation 205 Best Solution: [183.67911123253745, 175.6848351621117,
175.08389278808448, 75.79112142160855] Fitness: 2.2229466434400638
Generation 206 Best Solution: [182.8000579478997, 175.6848351621117,
175.08389278808448, 75.41492322880404] Fitness: 1.5883843859923452
Generation 207 Best Solution: [183.68909695647528, 175.71712334353222,
175.08389278808448, 75.79112142160855] Fitness: 2.205860694783725
Generation 208 Best Solution: [183.82820112511675, 177.40123495141432,
175.08389278808448, 74.68646972064178] Fitness: 2.3677180802243556
Generation 209 Best Solution: [183.67911123253745, 176.59777189688276,

175.08389278808448, 75.79112142160855] Fitness: 2.1640109258238054
Generation 210 Best Solution: [183.82820112511675, 176.26481911784404,
175.08389278808448, 75.79112142160855] Fitness: 2.165663504855898
Generation 211 Best Solution: [183.82820112511675, 176.26481911784404,
175.07686132966822, 75.79112142160855] Fitness: 2.154489238376083
Generation 212 Best Solution: [183.68909695647528, 177.4307033824268,
175.08389278808448, 74.89603249583693] Fitness: 2.335512017234433
Generation 213 Best Solution: [183.68909695647528, 176.2650033007221,
175.08389278808448, 75.79112142160855] Fitness: 2.0372220165092996
Generation 214 Best Solution: [183.44698360217575, 177.4307033824268,
174.8706816652849, 74.89603249583693] Fitness: 1.883386347414997
Generation 215 Best Solution: [183.82820112511675, 177.4307033824268,
174.8706816652849, 74.89603249583693] Fitness: 2.1607337200196906
Generation 216 Best Solution: [183.82820112511675, 177.4307033824268,
174.53267794753543, 74.89603249583693] Fitness: 1.8308241705721542
Generation 217 Best Solution: [183.82820112511675, 175.73710741932175,
173.79676933132907, 75.79112142160855] Fitness: 1.6079635481345669
Generation 218 Best Solution: [183.36381800210688, 175.73710741932175,
173.79676933132907, 75.79112142160855] Fitness: 1.3013750425380795
Generation 219 Best Solution: [183.73839915664632, 175.79626947458323,
175.08389278808448, 75.79112142160855] Fitness: 2.193415044695689
Generation 220 Best Solution: [183.82820112511675, 177.4307033824268,
175.08389278808448, 74.92948857058565] Fitness: 2.4765226842146695
Generation 221 Best Solution: [183.68909695647528, 175.5963855012376,
175.08389278808448, 75.79112142160855] Fitness: 2.3083748825342596
Generation 222 Best Solution: [183.82820112511675, 176.32859843699669,
175.08389278808448, 75.79112142160855] Fitness: 2.1775935735976204
Generation 223 Best Solution: [183.82820112511675, 176.32859843699669,
175.08389278808448, 75.79112142160855] Fitness: 2.1775935735976204
Generation 224 Best Solution: [183.83424229758612, 177.4307033824268,
175.08389278808448, 74.36104001801507] Fitness: 2.5158633013605707
Generation 225 Best Solution: [182.98482909498918, 177.4307033824268,
175.08389278808448, 74.36104001801507] Fitness: 2.211522042614413
Generation 226 Best Solution: [182.98482909498918, 177.4307033824268,
175.08389278808448, 74.36104001801507] Fitness: 2.211522042614413
Generation 227 Best Solution: [183.82820112511675, 177.4307033824268,
175.08389278808448, 74.57090738050601] Fitness: 2.4364515577756864
Generation 228 Best Solution: [183.36381800210688, 176.29620407334514,
175.08389278808448, 75.79112142160855] Fitness: 1.864122588223824
Generation 229 Best Solution: [182.98482909498918, 177.4307033824268,
175.08389278808448, 74.30390956628294] Fitness: 2.243775874514677
Generation 230 Best Solution: [183.40013597635377, 177.4307033824268,
174.17860252095122, 75.79112142160855] Fitness: 2.3483489763883734
Generation 231 Best Solution: [182.88966624893123, 177.4307033824268,
175.08389278808448, 74.30390956628294] Fitness: 2.2824463354334297
Generation 232 Best Solution: [183.40013597635377, 177.4307033824268,
175.08389278808448, 74.66454423747949] Fitness: 2.131635885633709
Generation 233 Best Solution: [183.40013597635377, 177.4307033824268,

175.08389278808448, 74.66454423747949] Fitness: 2.131635885633709
Generation 234 Best Solution: [183.82820112511675, 176.24144206188032,
174.2291053300682, 75.79112142160855] Fitness: 1.3911497804901594
Generation 235 Best Solution: [183.82820112511675, 176.24144206188032,
175.08389278808448, 75.79112142160855] Fitness: 2.1629401445661376
Generation 236 Best Solution: [183.82820112511675, 176.24144206188032,
175.08389278808448, 75.76405785344035] Fitness: 2.1148983834017394
Generation 237 Best Solution: [182.92496700648024, 176.47784236817233,
175.08389278808448, 75.79112142160855] Fitness: 1.9558094243433095
Generation 238 Best Solution: [183.40013597635377, 175.79658201754523,
175.08389278808448, 75.79112142160855] Fitness: 1.9845426190308095
Generation 239 Best Solution: [183.40013597635377, 177.38961244813697,
175.08389278808448, 74.62069656709664] Fitness: 2.052611139517037
Generation 240 Best Solution: [182.98482909498918, 176.1652614887937,
175.08389278808448, 75.54509930835826] Fitness: 1.4690357939501464
Generation 241 Best Solution: [183.69910237002082, 175.6556296170789,
175.08389278808448, 74.30226458526083] Fitness: 1.382229619404067
Generation 242 Best Solution: [183.82820112511675, 175.6556296170789,
175.08389278808448, 74.30226458526083] Fitness: 1.502501458702956
Generation 243 Best Solution: [183.82820112511675, 177.03686915769592,
175.08389278808448, 74.29834563210738] Fitness: 1.8740694173198458
Generation 244 Best Solution: [183.69910237002082, 177.03686915769592,
175.08389278808448, 74.29834563210738] Fitness: 1.753797578020957
Generation 245 Best Solution: [182.98482909498918, 177.03686915769592,
175.08389278808448, 74.29834563210738] Fitness: 1.5760171739368767
Generation 246 Best Solution: [183.40013597635377, 177.4307033824268,
173.89037236967124, 75.79112142160855] Fitness: 2.378371946607612
Generation 247 Best Solution: [182.98482909498918, 177.38961244813697,
173.89037236967124, 75.79112142160855] Fitness: 2.2937541965757458
Generation 248 Best Solution: [183.4176837482122, 177.38961244813697,
175.08389278808448, 74.29214011385659] Fitness: 2.1784404135874036
Generation 249 Best Solution: [183.4176837482122, 177.38961244813697,
173.89037236967124, 75.79112142160855] Fitness: 2.30287956281967
Generation 250 Best Solution: [183.4176837482122, 175.93332782181847,
175.08389278808448, 75.79112142160855] Fitness: 1.9208013241154107
Generation 251 Best Solution: [183.69910237002082, 177.38961244813697,
174.89699559747365, 75.0425457824036] Fitness: 2.0604881597492617
Generation 252 Best Solution: [183.69910237002082, 177.4307033824268,
174.13399842636622, 75.0425457824036] Fitness: 1.6275213240433437
Generation 253 Best Solution: [182.98482909498918, 177.38961244813697,
174.13399842636622, 75.0425457824036] Fitness: 1.3679790220178836
Generation 254 Best Solution: [183.4176837482122, 175.69777103842813,
175.08389278808448, 75.79112142160855] Fitness: 2.0602433053279476
Generation 255 Best Solution: [183.4176837482122, 177.38961244813697,
175.08389278808448, 74.72514438862541] Fitness: 2.0574845157910904
Generation 256 Best Solution: [183.4176837482122, 177.38961244813697,
175.08389278808448, 74.72514438862541] Fitness: 2.0574845157910904
Generation 257 Best Solution: [183.4176837482122, 177.38961244813697,

175.08389278808448, 74.72514438862541] Fitness: 2.0574845157910904
Generation 258 Best Solution: [183.08630520992605, 176.4519438095266,
175.06972355551906, 75.79112142160855] Fitness: 1.8737797810339734
Generation 259 Best Solution: [183.4176837482122, 177.02618575984897,
175.06972355551906, 74.72514438862541] Fitness: 1.43092005188967
Generation 260 Best Solution: [182.98482909498918, 177.38961244813697,
173.8364683639713, 74.72514438862541] Fitness: 1.3185481687696237
Generation 261 Best Solution: [183.4176837482122, 177.38961244813697,
173.91144252544214, 75.79112142160855] Fitness: 2.296127935632785
Generation 262 Best Solution: [183.69910237002082, 176.1893338174298,
174.89699559747365, 75.79112142160855] Fitness: 1.7699816725681934
Generation 263 Best Solution: [183.69910237002082, 176.1893338174298,
174.89699559747365, 75.79112142160855] Fitness: 1.7699816725681934
Generation 264 Best Solution: [183.69910237002082, 177.4307033824268,
174.89699559747365, 74.57778332641736] Fitness: 2.0451900656227227
Generation 265 Best Solution: [183.69910237002082, 177.38961244813697,
175.08389278808448, 74.41319126933931] Fitness: 2.282711417184691
Generation 266 Best Solution: [183.69910237002082, 177.38961244813697,
173.95778929460755, 74.47773312633765] Fitness: 1.488651404471158
Generation 267 Best Solution: [183.69910237002082, 176.23550488881813,
175.08389278808448, 74.63127596326929] Fitness: 1.0468179401133912
Generation 268 Best Solution: [183.69910237002082, 176.23550488881813,
174.89699559747365, 75.79112142160855] Fitness: 1.7723171768652297
Generation 269 Best Solution: [183.69910237002082, 177.38961244813697,
174.56509805550786, 75.09413302736769] Fitness: 1.7530946630441324
Generation 270 Best Solution: [183.69910237002082, 177.38961244813697,
174.89699559747365, 74.67132411301306] Fitness: 1.9548528226194508
Generation 271 Best Solution: [183.69910237002082, 176.13389159811084,
173.77183002751138, 75.79112142160855] Fitness: 1.3425655814942044
Generation 272 Best Solution: [183.44409039941908, 176.13389159811084,
174.2784940376211, 75.79112142160855] Fitness: 1.1238592918434986
Generation 273 Best Solution: [183.69910237002082, 177.38961244813697,
174.89699559747365, 74.34629893517146] Fitness: 2.0455566422008484
Generation 274 Best Solution: [183.69910237002082, 175.64833511375332,
174.89699559747365, 75.79112142160855] Fitness: 1.9997677333865311
Generation 275 Best Solution: [183.69910237002082, 175.64833511375332,
174.89699559747365, 75.79112142160855] Fitness: 1.9997677333865311
Generation 276 Best Solution: [182.96186483416702, 175.96578737921857,
173.78269449135072, 75.79112142160855] Fitness: 1.2028643057396105
Generation 277 Best Solution: [183.69910237002082, 177.38961244813697,
173.78269449135072, 75.79112142160855] Fitness: 2.517261105709345
Generation 278 Best Solution: [183.69910237002082, 175.61009859133492,
174.89699559747365, 75.79112142160855] Fitness: 2.03393775979222
Generation 279 Best Solution: [183.69910237002082, 177.38961244813697,
173.78269449135072, 75.79112142160855] Fitness: 2.517261105709345
Generation 280 Best Solution: [183.69910237002082, 176.0689292246018,
175.10345763491028, 75.79112142160855] Fitness: 2.0814410506420744
Generation 281 Best Solution: [183.69910237002082, 177.38961244813697,

175.10345763491028, 74.82005001075892] Fitness: 2.271700301543662
Generation 282 Best Solution: [183.69910237002082, 177.38961244813697,
175.10345763491028, 74.82005001075892] Fitness: 2.271700301543662
Generation 283 Best Solution: [183.69910237002082, 177.38961244813697,
174.35207660235432, 74.71929809353283] Fitness: 1.492140879020196
Generation 284 Best Solution: [183.69910237002082, 177.38961244813697,
174.35207660235432, 74.71929809353283] Fitness: 1.492140879020196
Generation 285 Best Solution: [183.69910237002082, 177.38961244813697,
174.35207660235432, 74.71929809353283] Fitness: 1.492140879020196
Generation 286 Best Solution: [183.69910237002082, 175.81082158541273,
175.10345763491028, 75.79112142160855] Fitness: 2.1822780051473605
Generation 287 Best Solution: [183.2139579637901, 175.81082158541273,
175.10345763491028, 75.79112142160855] Fitness: 1.971539922784721
Generation 288 Best Solution: [183.69910237002082, 175.81082158541273,
175.10345763491028, 75.79112142160855] Fitness: 2.1822780051473605
Generation 289 Best Solution: [183.43867452266716, 177.38961244813697,
174.51145371116735, 75.79112142160855] Fitness: 2.4136912254456555
Generation 290 Best Solution: [183.69910237002082, 176.1461155270009,
175.10345763491028, 75.79112142160855] Fitness: 2.0722366236931595
Generation 291 Best Solution: [183.43867452266716, 177.38961244813697,
173.86474633591274, 75.79112142160855] Fitness: 2.320214269043392
Generation 292 Best Solution: [183.69910237002082, 176.6206268900349,
175.10345763491028, 75.79112142160855] Fitness: 2.2275740438632847
Generation 293 Best Solution: [183.69910237002082, 176.6206268900349,
174.93670286210707, 75.79112142160855] Fitness: 1.978849545475712
Generation 294 Best Solution: [183.69910237002082, 177.38961244813697,
174.62978903852772, 75.79112142160855] Fitness: 2.6645837558908303
Generation 295 Best Solution: [183.69910237002082, 175.84314832286663,
174.8540155383811, 75.79112142160855] Fitness: 1.8083796587999446
Generation 296 Best Solution: [183.69910237002082, 176.09696012554647,
175.10345763491028, 75.83536101244965] Fitness: 2.15806823567013
Generation 297 Best Solution: [183.69910237002082, 176.83751276475044,
174.8540155383811, 75.79112142160855] Fitness: 2.0646055226333773
Generation 298 Best Solution: [183.69910237002082, 176.26474778483745,
174.19024014143105, 75.83536101244965] Fitness: 1.3477017789016617
Generation 299 Best Solution: [183.69910237002082, 176.26474778483745,
174.19024014143105, 75.83536101244965] Fitness: 1.3477017789016617
Generation 300 Best Solution: [183.69910237002082, 177.38961244813697,
174.8540155383811, 75.79112142160855] Fitness: 2.8980428118520942
Generation 301 Best Solution: [183.87029599965422, 177.38961244813697,
175.10345763491028, 75.01660781847791] Fitness: 2.512493130986824
Generation 302 Best Solution: [183.87029599965422, 176.7110902377516,
175.10345763491028, 75.79112142160855] Fitness: 2.463885026083946
Generation 303 Best Solution: [183.87029599965422, 176.18314858102542,
175.10345763491028, 75.79112142160855] Fitness: 2.236566926371346
Generation 304 Best Solution: [183.87029599965422, 176.18314858102542,
175.10345763491028, 75.79112142160855] Fitness: 2.236566926371346
Generation 305 Best Solution: [183.69910237002082, 176.7110902377516,

175.10345763491028, 75.79112142160855] Fitness: 2.2985627486783216
Generation 306 Best Solution: [182.80471704508147, 175.8888028946107,
175.10345763491028, 75.79112142160855] Fitness: 2.0482375749565587
Generation 307 Best Solution: [183.805993354288, 175.8888028946107,
174.6984629143579, 75.79112142160855] Fitness: 1.7121366298784868
Generation 308 Best Solution: [183.805993354288, 175.8888028946107,
174.8540155383811, 75.83536101244965] Fitness: 1.9638270867238194
Generation 309 Best Solution: [183.48910888656727, 177.38961244813697,
174.6984629143579, 74.62191995684384] Fitness: 1.5949731493968764
Generation 310 Best Solution: [183.21002328863145, 177.38961244813697,
174.6984629143579, 74.62191995684384] Fitness: 1.5220753668832405
Generation 311 Best Solution: [183.48910888656727, 177.38961244813697,
174.30750251004207, 75.83536101244965] Fitness: 2.414705173344208
Generation 312 Best Solution: [183.48910888656727, 177.38961244813697,
174.30750251004207, 75.83536101244965] Fitness: 2.414705173344208
Generation 313 Best Solution: [183.71484762452397, 177.38961244813697,
173.79722608594042, 75.83536101244965] Fitness: 2.604303525273626
Generation 314 Best Solution: [183.87029599965422, 176.40842626892632,
174.7324258473732, 75.79112142160855] Fitness: 1.7863104254831186
Generation 315 Best Solution: [183.71484762452397, 176.12849582808605,
174.7324258473732, 75.83536101244965] Fitness: 1.6757381205367174
Generation 316 Best Solution: [183.66897414489378, 177.38961244813697,
173.97738720832578, 74.46598688042543] Fitness: 1.464283251106209
Generation 317 Best Solution: [183.87029599965422, 177.38961244813697,
174.7324258473732, 74.3283544702791] Fitness: 2.0301233007275883
Generation 318 Best Solution: [182.87218880374456, 177.38961244813697,
174.7324258473732, 75.83536101244965] Fitness: 2.7120152961450517
Generation 319 Best Solution: [183.87029599965422, 176.5982839537435,
175.10345763491028, 75.83536101244965] Fitness: 2.4584889138111263
Generation 320 Best Solution: [183.71484762452397, 177.38961244813697,
174.50203308494477, 75.83536101244965] Fitness: 2.662279133464077
Generation 321 Best Solution: [183.71484762452397, 177.38961244813697,
174.50203308494477, 75.11617357068062] Fitness: 1.7370586023730152
Generation 322 Best Solution: [183.71484762452397, 176.18088581963886,
173.74725239490843, 75.83536101244965] Fitness: 1.4485656631294546
Generation 323 Best Solution: [183.87029599965422, 176.18088581963886,
173.74725239490843, 75.83536101244965] Fitness: 1.6006640636779388
Generation 324 Best Solution: [183.87029599965422, 176.18088581963886,
175.10345763491028, 75.83536101244965] Fitness: 2.3176491206471015
Generation 325 Best Solution: [183.87029599965422, 176.33338180608249,
175.10345763491028, 75.83536101244965] Fitness: 2.336402856574436
Generation 326 Best Solution: [183.50991570591418, 177.38961244813697,
175.10345763491028, 75.05784962057034] Fitness: 2.2434397604695495
Generation 327 Best Solution: [183.71484762452397, 177.38961244813697,
175.10345763491028, 74.98072200863899] Fitness: 2.3419483868061293
Generation 328 Best Solution: [183.71484762452397, 177.38961244813697,
174.5011566807756, 75.41095451596749] Fitness: 2.0144399422070354
Generation 329 Best Solution: [183.13572691280467, 177.38961244813697,

174.12432478236914, 75.83536101244965] Fitness: 2.3089292373033032
Generation 330 Best Solution: [183.4174243338073, 176.58791670677073,
175.10345763491028, 75.83536101244965] Fitness: 2.1175011143226605
Generation 331 Best Solution: [183.71484762452397, 177.38961244813697,
175.10345763491028, 74.67001307580358] Fitness: 2.269412699391318
Generation 332 Best Solution: [182.85715054071238, 176.58791670677073,
175.10345763491028, 75.83536101244965] Fitness: 2.1637164222464884
Generation 333 Best Solution: [182.85715054071238, 176.58791670677073,
175.10345763491028, 75.83536101244965] Fitness: 2.1637164222464884
Generation 334 Best Solution: [182.85715054071238, 176.58791670677073,
175.10345763491028, 75.36847724976994] Fitness: 1.4677205591351872
Generation 335 Best Solution: [182.85715054071238, 176.4050205300611,
175.10345763491028, 75.74157620957827] Fitness: 1.9022193403972691
Generation 336 Best Solution: [183.71484762452397, 176.58791670677073,
174.3598283068348, 75.83536101244965] Fitness: 1.5375799026556207
Generation 337 Best Solution: [183.4174243338073, 177.38961244813697,
174.44573508325064, 75.36847724976994] Fitness: 1.7501215916936472
Generation 338 Best Solution: [183.71484762452397, 177.38961244813697,
174.44573508325064, 75.64633534947232] Fitness: 2.3037891120651333
Generation 339 Best Solution: [183.71484762452397, 177.10852886130303,
175.10345763491028, 74.30004466174202] Fitness: 1.900918901488753
Generation 340 Best Solution: [183.55324660638823, 175.6686727529857,
175.10345763491028, 75.83536101244965] Fitness: 2.2615381338926674
Generation 341 Best Solution: [183.7442284206414, 177.38961244813697,
173.81240153689174, 75.36847724976994] Fitness: 1.9268330884308222
Generation 342 Best Solution: [183.71484762452397, 176.64291767070387,
174.44573508325064, 75.64633534947232] Fitness: 1.2942217886770806
Generation 343 Best Solution: [183.71484762452397, 177.38961244813697,
175.10345763491028, 74.30004466174202] Fitness: 2.3869044224213907
Generation 344 Best Solution: [183.71484762452397, 176.17939852199316,
175.10345763491028, 75.83536101244965] Fitness: 2.165553210698377
Generation 345 Best Solution: [182.85715054071238, 177.38961244813697,
174.44573508325064, 75.83536101244965] Fitness: 2.4923327627287755
Generation 346 Best Solution: [183.71484762452397, 177.38961244813697,
174.44573508325064, 75.83536101244965] Fitness: 2.628092684356183
Generation 347 Best Solution: [183.71484762452397, 177.38961244813697,
175.10345763491028, 75.0147098977362] Fitness: 2.3593669249109914
Generation 348 Best Solution: [183.0348218406203, 177.42693033073613,
174.44573508325064, 75.83536101244965] Fitness: 2.496788172932008
Generation 349 Best Solution: [182.96337952038942, 176.47797545756072,
175.10345763491028, 75.83536101244965] Fitness: 2.053293346420447
Generation 350 Best Solution: [182.96337952038942, 176.47797545756072,
175.10345763491028, 75.83536101244965] Fitness: 2.053293346420447
Generation 351 Best Solution: [183.71484762452397, 176.0545413601545,
175.10345763491028, 75.83536101244965] Fitness: 2.1785325407739515
Generation 352 Best Solution: [183.71484762452397, 177.38961244813697,
174.20173186894536, 75.06780430583244] Fitness: 1.5818492391195724
Generation 353 Best Solution: [183.71484762452397, 177.38961244813697,

174.20173186894536, 74.70448428189029] Fitness: 1.461280520471724
Generation 354 Best Solution: [183.4000832483027, 177.38961244813697,
174.20173186894536, 74.70448428189029] Fitness: 1.2731140040745355
Generation 355 Best Solution: [183.71484762452397, 176.55854334652034,
174.1138158531548, 75.83536101244965] Fitness: 1.4639694292902432
Generation 356 Best Solution: [183.4000832483027, 177.42693033073613,
175.10345763491028, 74.66332200065975] Fitness: 2.1555437350033047
Generation 357 Best Solution: [183.71484762452397, 177.42693033073613,
175.10345763491028, 74.66332200065975] Fitness: 2.3437102514004935
Generation 358 Best Solution: [183.71484762452397, 177.38961244813697,
174.74706077943407, 74.98365690366005] Fitness: 1.8665059296625373
Generation 359 Best Solution: [183.71484762452397, 177.38961244813697,
175.10345763491028, 74.70703454424837] Fitness: 2.269853095392986
Generation 360 Best Solution: [183.71484762452397, 177.38961244813697,
175.10345763491028, 74.70703454424837] Fitness: 2.269853095392986
Generation 361 Best Solution: [183.71484762452397, 177.38961244813697,
175.10345763491028, 74.35603142706849] Fitness: 2.35489410534464
Generation 362 Best Solution: [183.71484762452397, 176.387514119179,
174.08904718376056, 75.83536101244965] Fitness: 1.3830496505640957
Generation 363 Best Solution: [183.71484762452397, 177.1771967619229,
174.08904718376056, 75.83536101244965] Fitness: 2.1516772850362247
Generation 364 Best Solution: [182.92354674128507, 177.1771967619229,
174.08904718376056, 75.83536101244965] Fitness: 1.983894962126631
Generation 365 Best Solution: [183.71484762452397, 177.42693033073613,
174.08904718376056, 75.83536101244965] Fitness: 2.6048861616310424
Generation 366 Best Solution: [183.71484762452397, 177.38961244813697,
174.48373230209324, 75.83536101244965] Fitness: 2.650603223913537
Generation 367 Best Solution: [183.71484762452397, 176.1073419421008,
175.10345763491028, 75.83536101244965] Fitness: 2.1699638011949425
Generation 368 Best Solution: [183.71484762452397, 176.79289307952743,
175.10345763491028, 75.83536101244965] Fitness: 2.468472202382408
Generation 369 Best Solution: [183.71484762452397, 175.86221608465053,
174.9679353403119, 75.83536101244965] Fitness: 2.042342246431594
Generation 370 Best Solution: [183.71484762452397, 176.79289307952743,
173.7438335194041, 75.83979768011423] Fitness: 1.7617493128597004
Generation 371 Best Solution: [183.71484762452397, 176.22378041117605,
175.10345763491028, 75.83536101244965] Fitness: 2.1670200149407988
Generation 372 Best Solution: [183.71484762452397, 176.22378041117605,
175.10345763491028, 75.83536101244965] Fitness: 2.1670200149407988
Generation 373 Best Solution: [183.132192333245, 176.22378041117605,
175.10345763491028, 75.83536101244965] Fitness: 1.9450649929296373
Generation 374 Best Solution: [183.62340546092068, 175.93630782574172,
175.10345763491028, 75.83536101244965] Fitness: 2.1428946551908417
Generation 375 Best Solution: [183.51584720114937, 177.38961244813697,
175.10345763491028, 75.83536101244965] Fitness: 3.2101006973193806
Generation 376 Best Solution: [183.71484762452397, 177.38961244813697,
174.68488469642594, 75.83536101244965] Fitness: 2.808713689595085
Generation 377 Best Solution: [183.71484762452397, 176.71418345675113,

175.10345763491028, 75.83536101244965] Fitness: 2.3955334900009078
Generation 378 Best Solution: [183.71484762452397, 176.0114153550312,
175.10345763491028, 75.83536101244965] Fitness: 2.1888801859039644
Generation 379 Best Solution: [183.71484762452397, 176.0114153550312,
175.10345763491028, 75.83536101244965] Fitness: 2.1888801859039644
Generation 380 Best Solution: [183.62340546092068, 177.38961244813697,
175.10345763491028, 74.97959005258018] Fitness: 2.27020487678419
Generation 381 Best Solution: [183.71484762452397, 177.38961244813697,
175.10345763491028, 74.35832780288315] Fitness: 2.3536895038936536
Generation 382 Best Solution: [183.62340546092068, 175.77906099064313,
175.10345763491028, 75.83536101244965] Fitness: 2.225252536224305
Generation 383 Best Solution: [183.71484762452397, 176.90382370035968,
175.10345763491028, 75.83536101244965] Fitness: 2.5882993811325345
Generation 384 Best Solution: [183.71484762452397, 176.90382370035968,
175.10345763491028, 75.83536101244965] Fitness: 2.5882993811325345
Generation 385 Best Solution: [183.71484762452397, 176.90382370035968,
175.10345763491028, 75.83536101244965] Fitness: 2.5882993811325345
Generation 386 Best Solution: [183.71484762452397, 177.38961244813697,
175.10345763491028, 75.40562329983021] Fitness: 2.6941686984039794
Generation 387 Best Solution: [183.62340546092068, 177.38961244813697,
175.10345763491028, 75.40562329983021] Fitness: 2.6229731254596844
Generation 388 Best Solution: [183.62340546092068, 176.38114533084052,
175.10345763491028, 75.83536101244965] Fitness: 2.1267259485864476
Generation 389 Best Solution: [183.71484762452397, 176.38114533084052,
175.10345763491028, 75.83536101244965] Fitness: 2.197921521530742
Generation 390 Best Solution: [183.71484762452397, 176.38114533084052,
175.10345763491028, 75.83536101244965] Fitness: 2.197921521530742
Generation 391 Best Solution: [183.71484762452397, 176.38114533084052,
175.10345763491028, 75.83536101244965] Fitness: 2.197921521530742
Generation 392 Best Solution: [183.62340546092068, 176.38699690464924,
174.28980703444157, 75.83536101244965] Fitness: 1.3411312737090657
Generation 393 Best Solution: [183.71484762452397, 177.38961244813697,
175.10345763491028, 74.95808114210853] Fitness: 2.331383022598602
Generation 394 Best Solution: [183.71484762452397, 176.37108595033885,
175.10345763491028, 75.83536101244965] Fitness: 2.194746622115759
Generation 395 Best Solution: [183.71484762452397, 177.38961244813697,
175.10345763491028, 74.62609168860209] Fitness: 2.2717681803911676
Generation 396 Best Solution: [183.62340546092068, 177.38961244813697,
175.10345763491028, 74.68644192155824] Fitness: 2.1981386864967183
Generation 397 Best Solution: [183.62340546092068, 177.38961244813697,
175.09123070763272, 74.68644192155824] Fitness: 2.1783719943401603
Generation 398 Best Solution: [183.51868014828085, 177.38961244813697,
175.10345763491028, 74.62609168860209] Fitness: 2.1356655773825173
Generation 399 Best Solution: [183.71484762452397, 175.90880125054923,
175.10345763491028, 75.83536101244965] Fitness: 2.2256078149665446
Generation 400 Best Solution: [183.71484762452397, 175.90880125054923,
173.99969344715268, 75.83536101244965] Fitness: 1.4165187903819727
Generation 401 Best Solution: [182.83610535552006, 175.79295559133573,

175.10345763491028, 75.83536101244965] Fitness: 2.1634381169279644
Generation 402 Best Solution: [183.71484762452397, 175.90880125054923,
175.09123070763272, 75.83536101244965] Fitness: 2.2058411228099866
Generation 403 Best Solution: [183.71484762452397, 176.2145005223727,
175.10345763491028, 75.65183722826742] Fitness: 1.8507680002028741
Generation 404 Best Solution: [183.71484762452397, 177.38961244813697,
175.10345763491028, 75.65183722826742] Fitness: 3.03203082543698
Generation 405 Best Solution: [183.0479214821802, 176.20723193054818,
175.09123070763272, 75.83536101244965] Fitness: 1.9377738265422098
Generation 406 Best Solution: [183.5929831901635, 176.0521701063584,
175.10345763491028, 75.83536101244965] Fitness: 2.0871425675259307
Generation 407 Best Solution: [183.71484762452397, 176.23483065331263,
175.10345763491028, 75.65183722826742] Fitness: 1.85219946407146
Generation 408 Best Solution: [183.0479214821802, 176.51720244521314,
174.5607412832708, 75.83536101244965] Fitness: 1.4040795281502172
Generation 409 Best Solution: [183.71484762452397, 176.51720244521314,
174.5607412832708, 75.83536101244965] Fitness: 1.6126392363485307
Generation 410 Best Solution: [183.71484762452397, 176.86570360033414,
174.5607412832708, 75.83536101244965] Fitness: 1.900557660515405
Generation 411 Best Solution: [183.71484762452397, 175.78061719265213,
175.10345763491028, 75.83536101244965] Fitness: 2.295436913492248
Generation 412 Best Solution: [182.91191714578784, 177.38961244813697,
174.5607412832708, 75.83536101244965] Fitness: 2.540705850116322
Generation 413 Best Solution: [183.71484762452397, 176.7603984229307,
174.5607412832708, 75.83536101244965] Fitness: 1.7928267813817524
Generation 414 Best Solution: [183.71484762452397, 176.7603984229307,
174.5607412832708, 75.83536101244965] Fitness: 1.7928267813817524
Generation 415 Best Solution: [183.71484762452397, 176.7603984229307,
174.5607412832708, 75.83536101244965] Fitness: 1.7928267813817524
Generation 416 Best Solution: [183.71484762452397, 176.43733579744676,
175.09123070763272, 75.83536101244965] Fitness: 2.1989029696752436
Generation 417 Best Solution: [183.36325872783186, 176.43733579744676,
175.09123070763272, 75.83536101244965] Fitness: 1.99920369104778
Generation 418 Best Solution: [183.36325872783186, 175.64132265910843,
173.80891738801216, 75.83536101244965] Fitness: 1.4527735819039584
Generation 419 Best Solution: [183.36325872783186, 175.74704806816703,
175.09123070763272, 75.83536101244965] Fitness: 2.0986535850423955
Generation 420 Best Solution: [183.36325872783186, 177.38961244813697,
175.10345763491028, 74.67341231540506] Fitness: 2.0696613365869037
Generation 421 Best Solution: [183.5929831901635, 176.75790832761706,
175.10345763491028, 74.67341231540506] Fitness: 1.2645816347486627
Generation 422 Best Solution: [183.36325872783186, 176.19806131116385,
175.10345763491028, 75.83536101244965] Fitness: 1.9660821665747692
Generation 423 Best Solution: [183.5929831901635, 176.75790832761706,
175.10345763491028, 75.83536101244965] Fitness: 2.342933466540309
Generation 424 Best Solution: [183.8557283458629, 176.7603984229307,
173.79906496722663, 75.83536101244965] Fitness: 1.8290235147965512
Generation 425 Best Solution: [183.36325872783186, 175.86478005614666,

175.10345763491028, 75.83536101244965] Fitness: 2.046890122999236
Generation 426 Best Solution: [183.36325872783186, 175.86478005614666,
175.10345763491028, 75.83536101244965] Fitness: 2.046890122999236
Generation 427 Best Solution: [183.36325872783186, 175.86478005614666,
173.933414403442, 75.83536101244965] Fitness: 1.2519948078276013
Generation 428 Best Solution: [183.36325872783186, 176.47040886340153,
175.10345763491028, 75.67190454512176] Fitness: 1.7497534753148205
Generation 429 Best Solution: [183.8557283458629, 175.86478005614666,
174.48720380376696, 75.83536101244965] Fitness: 1.6878366228592494
Generation 430 Best Solution: [183.36325872783186, 175.86478005614666,
175.10345763491028, 75.83536101244965] Fitness: 2.046890122999236
Generation 431 Best Solution: [183.36325872783186, 175.86478005614666,
175.10345763491028, 74.42550193737091] Fitness: 1.0214076313082163
Generation 432 Best Solution: [183.8557283458629, 177.38961244813697,
174.69988209179039, 74.42550193737091] Fitness: 1.9338270186574231
Generation 433 Best Solution: [183.36325872783186, 176.29779675189226,
175.10345763491028, 75.83536101244965] Fitness: 1.9768611316649038
Generation 434 Best Solution: [183.36325872783186, 176.29779675189226,
175.10345763491028, 75.33214780900191] Fitness: 1.2415072914264287
Generation 435 Best Solution: [183.5929831901635, 176.01577897535904,
175.10345763491028, 75.83536101244965] Fitness: 2.0958155541962618
Generation 436 Best Solution: [183.5929831901635, 176.17845749508305,
175.10345763491028, 75.83536101244965] Fitness: 2.0736759372341194
Generation 437 Best Solution: [183.36325872783186, 177.38961244813697,
174.6360465782425, 75.83536101244965] Fitness: 2.5646046736400407
Generation 438 Best Solution: [183.36325872783186, 177.38961244813697,
175.10345763491028, 75.13006863804274] Fitness: 2.2327349665981955
Generation 439 Best Solution: [183.36325872783186, 177.38961244813697,
175.10345763491028, 74.58063924698327] Fitness: 2.0777949875781543
Generation 440 Best Solution: [183.36325872783186, 177.38961244813697,
175.10345763491028, 75.08716320893795] Fitness: 2.2030424862712996
Generation 441 Best Solution: [183.36325872783186, 177.38961244813697,
175.10345763491028, 74.74800588199365] Fitness: 2.07322800792464
Generation 442 Best Solution: [183.36325872783186, 175.98766463585022,
175.10345763491028, 75.83536101244965] Fitness: 1.9961654737090995
Generation 443 Best Solution: [183.36325872783186, 177.38961244813697,
175.10345763491028, 74.74800588199365] Fitness: 2.07322800792464
Generation 444 Best Solution: [183.35491855121245, 177.38961244813697,
175.10345763491028, 74.35826147649205] Fitness: 2.151717840948895
Generation 445 Best Solution: [183.36325872783186, 176.18970912465818,
173.77552135750972, 75.83536101244965] Fitness: 1.2334811849123748
Generation 446 Best Solution: [183.36325872783186, 176.36691727695714,
175.10345763491028, 75.83536101244965] Fitness: 1.993779658872779
Generation 447 Best Solution: [183.5929831901635, 176.96908681872523,
175.10345763491028, 75.34593415054466] Fitness: 1.8555547174468316
Generation 448 Best Solution: [183.36325872783186, 176.96908681872523,
175.10345763491028, 75.40938022527905] Fitness: 1.8192801808166141
Generation 449 Best Solution: [183.36325872783186, 176.36691727695714,

175.10345763491028, 75.34593415054466] Fitness: 1.273109843568798
Generation 450 Best Solution: [183.5929831901635, 176.96908681872523,
174.3604584211806, 75.83536101244965] Fitness: 1.814594686142303
Generation 451 Best Solution: [183.36325872783186, 177.38961244813697,
175.10345763491028, 75.00795860937451] Fitness: 2.1560588046873073
Generation 452 Best Solution: [183.1848452772338, 177.38961244813697,
175.10345763491028, 75.00795860937451] Fitness: 2.131269801244977
Generation 453 Best Solution: [182.98897161749304, 177.38961244813697,
173.89853042207903, 75.83536101244965] Fitness: 2.370789725882336
Generation 454 Best Solution: [183.5929831901635, 176.1670508695622,
175.10345763491028, 75.83536101244965] Fitness: 2.0738314789579926
Generation 455 Best Solution: [183.5929831901635, 176.03612506570056,
175.10345763491028, 75.83536101244965] Fitness: 2.090702260707716
Generation 456 Best Solution: [183.79364225846817, 176.9264716178215,
173.89853042207903, 75.83536101244965] Fitness: 1.902851712234561
Generation 457 Best Solution: [183.79364225846817, 176.2836276480728,
173.89853042207903, 75.83536101244965] Fitness: 1.4616866447933585
Generation 458 Best Solution: [182.98897161749304, 176.2836276480728,
173.89853042207903, 75.83536101244965] Fitness: 1.1971248898287494
Generation 459 Best Solution: [182.98897161749304, 176.9264716178215,
173.89853042207903, 75.83536101244965] Fitness: 1.6382899572699516
Generation 460 Best Solution: [182.98897161749304, 176.9264716178215,
174.48396635919548, 75.83536101244965] Fitness: 1.7258947381418412
Generation 461 Best Solution: [182.98897161749304, 176.7545659212609,
175.10345763491028, 75.83536101244965] Fitness: 2.239347593772996
Generation 462 Best Solution: [182.98897161749304, 176.7545659212609,
174.19470234017922, 75.83536101244965] Fitness: 1.4297453093538197
Generation 463 Best Solution: [182.98897161749304, 177.38961244813697,
174.18237414788683, 75.83536101244965] Fitness: 2.3439648405986984
Generation 464 Best Solution: [182.98897161749304, 176.76135767233126,
175.1273691337886, 75.83536101244965] Fitness: 2.285046134921137
Generation 465 Best Solution: [182.98897161749304, 176.76135767233126,
175.10345763491028, 75.83536101244965] Fitness: 2.2456900223504066
Generation 466 Best Solution: [182.98897161749304, 175.7978245419364,
175.10345763491028, 75.83536101244965] Fitness: 2.092162596820258
Generation 467 Best Solution: [182.98897161749304, 175.7978245419364,
175.10345763491028, 75.83536101244965] Fitness: 2.092162596820258
Generation 468 Best Solution: [183.2719668506857, 177.09240915945912,
173.94101771726469, 75.83536101244965] Fitness: 1.822027951145841
Generation 469 Best Solution: [183.5929831901635, 177.38961244813697,
174.09207013631558, 75.83536101244965] Fitness: 2.4388246374101117
Generation 470 Best Solution: [182.98897161749304, 177.38961244813697,
175.10345763491028, 74.7636391874517] Fitness: 2.0824620354761327
Generation 471 Best Solution: [183.83994896189708, 176.13847608390518,
175.10345763491028, 75.83536101244965] Fitness: 2.2863587079676977
Generation 472 Best Solution: [183.83994896189708, 176.13847608390518,
175.10345763491028, 75.83536101244965] Fitness: 2.2863587079676977
Generation 473 Best Solution: [183.5929831901635, 176.13847608390518,

175.10345763491028, 75.83536101244965] Fitness: 2.0751459759634736
Generation 474 Best Solution: [183.5929831901635, 176.13847608390518,
175.10345763491028, 75.83536101244965] Fitness: 2.0751459759634736
Generation 475 Best Solution: [183.5929831901635, 177.38961244813697,
175.10345763491028, 74.53288397426819] Fitness: 2.1952330827383486
Generation 476 Best Solution: [183.83994896189708, 176.46812502339134,
175.10345763491028, 75.83536101244965] Fitness: 2.3515384475800545
Generation 477 Best Solution: [183.5929831901635, 177.38961244813697,
175.10345763491028, 74.5230343353159] Fitness: 2.1976764445584567
Generation 478 Best Solution: [183.83994896189708, 176.362697512646,
175.10345763491028, 75.83536101244965] Fitness: 2.311556403999915
Generation 479 Best Solution: [183.83994896189708, 175.70484606085253,
175.10345763491028, 75.83536101244965] Fitness: 2.4685560753807323
Generation 480 Best Solution: [183.83994896189708, 177.38961244813697,
173.761069708581, 74.3179109299179] Fitness: 1.7707746473229042
Generation 481 Best Solution: [183.5929831901635, 176.2914256772294,
174.91551633612795, 75.83536101244965] Fitness: 1.8064075628970753
Generation 482 Best Solution: [183.5929831901635, 176.2914256772294,
174.91551633612795, 75.83536101244965] Fitness: 1.8064075628970753
Generation 483 Best Solution: [182.9960517156051, 176.2914256772294,
175.1273691337886, 74.3179109299179] Fitness: 1.0485738599573928
Generation 484 Best Solution: [183.83994896189708, 177.38961244813697,
174.1422216933909, 75.83536101244965] Fitness: 2.6515251060530796
Generation 485 Best Solution: [183.5929831901635, 177.06613708029056,
174.1422216933909, 75.83536101244965] Fitness: 1.8921334250554809
Generation 486 Best Solution: [183.2135884722625, 177.38961244813697,
175.1273691337886, 74.74123923418846] Fitness: 2.087611077295481
Generation 487 Best Solution: [183.81037465982004, 177.38961244813697,
175.1273691337886, 74.35349100122086] Fitness: 2.484427374706047
Generation 488 Best Solution: [183.5929831901635, 177.06613708029056,
175.1273691337886, 75.13635876088226] Fitness: 1.8363342605943853
Generation 489 Best Solution: [183.5929831901635, 177.06613708029056,
174.35097551515705, 75.83536101244965] Fitness: 1.9420787962809567
Generation 490 Best Solution: [183.5929831901635, 176.14514855449727,
175.1273691337886, 75.83536101244965] Fitness: 2.1140768357320407
Generation 491 Best Solution: [183.5929831901635, 176.14514855449727,
173.9649582769503, 75.83536101244965] Fitness: 1.2721832821164474
Generation 492 Best Solution: [183.5929831901635, 176.68504772387783,
175.1273691337886, 75.83536101244965] Fitness: 2.318553021237472
Generation 493 Best Solution: [183.83994896189708, 176.68504772387783,
174.35097551515705, 75.83536101244965] Fitness: 1.7248356380869714
Generation 494 Best Solution: [183.83994896189708, 175.90284052946538,
175.1273691337886, 75.83536101244965] Fitness: 2.3869533320938414
Generation 495 Best Solution: [183.83994896189708, 176.68504772387783,
175.1273691337886, 75.83536101244965] Fitness: 2.529765753241696
Generation 496 Best Solution: [183.83994896189708, 177.38961244813697,
175.1273691337886, 75.25041295342369] Fitness: 2.690310101205824
Generation 497 Best Solution: [183.2135884722625, 176.29553221990506,

173.7480683858605, 75.83536101244965] Fitness: 1.2347107616250355
Generation 498 Best Solution: [183.83994896189708, 177.38961244813697,
174.88770294109892, 74.86079577765621] Fitness: 2.1015091010005182
Generation 499 Best Solution: [183.5929831901635, 176.27321531510702,
175.1273691337886, 75.6006710958944] Fitness: 1.7259114217691174
Generation 500 Best Solution: [183.26216858930994, 176.27321531510702,
175.1273691337886, 75.6006710958944] Fitness: 1.5977194997374788
Optimal Solution for Simulation 22: [183.26216858930994, 176.27321531510702,
175.1273691337886, 75.6006710958944]

Simulation 23/32 with conditions: {'drought': False, 'flood': True,
'policy_change': False, 'social_event': False, 'economic_crisis': True}
Generation 1 Best Solution: [182.85856419676415, 175.79894107700977,
173.83988330079708, 75.1231133903243] Fitness: 0.47350992780818296
Generation 2 Best Solution: [183.43153539887143, 176.30540638650288,
174.41890641041968, 75.29450978272592] Fitness: 0.5000583112242545
Generation 3 Best Solution: [183.43153539887143, 176.30540638650288,
173.9102540324673, 75.29450978272592] Fitness: 0.44595426829685525
Generation 4 Best Solution: [183.38962479628825, 175.77559273566504,
174.29553674223337, 74.8244839827691] Fitness: 0.26398619782004257
Generation 5 Best Solution: [182.98597808338795, 175.58229033290377,
173.97002070768153, 74.84177709085492] Fitness: 0.4119088686128457
Generation 6 Best Solution: [183.38962479628825, 176.7014783890757,
173.97002070768153, 74.7212486101275] Fitness: 0.31698414998035834
Generation 7 Best Solution: [183.34116788087476, 176.96356913747908,
173.97002070768153, 74.4186553177014] Fitness: 0.6395989189247185
Generation 8 Best Solution: [182.98597808338795, 176.30540638650288,
173.97002070768153, 75.77782800329612] Fitness: 1.0957789726524303
Generation 9 Best Solution: [183.76513786710322, 176.30540638650288,
173.97002070768153, 75.8030420981809] Fitness: 1.3808526370376504
Generation 10 Best Solution: [183.34116788087476, 176.30540638650288,
174.29553674223337, 75.8030420981809] Fitness: 1.144337042012232
Generation 11 Best Solution: [183.34116788087476, 176.30540638650288,
174.29553674223337, 75.8030420981809] Fitness: 1.144337042012232
Generation 12 Best Solution: [183.67148592582657, 176.9561098593953,
173.97002070768153, 75.82400802084122] Fitness: 1.8191212440533528
Generation 13 Best Solution: [183.68848624655612, 176.24390738994114,
174.9200816740307, 75.82400802084122] Fitness: 1.8826263969900632
Generation 14 Best Solution: [183.68848624655612, 176.82335045255132,
174.86921709611974, 75.89330680142784] Fitness: 2.285789634346959
Generation 15 Best Solution: [183.68848624655612, 176.82335045255132,
174.86921709611974, 74.65108383231208] Fitness: 1.0785338627719339
Generation 16 Best Solution: [182.98597808338795, 175.8660379822799,
174.9200816740307, 74.65108383231208] Fitness: 0.7176195813461257
Generation 17 Best Solution: [182.98597808338795, 176.82335045255132,
174.86921709611974, 75.89330680142784] Fitness: 2.114604898998177
Generation 18 Best Solution: [182.98597808338795, 175.8660379822799,
174.9200816740307, 75.89330680142784] Fitness: 1.9248753529211506

Generation 19 Best Solution: [183.4583640783307, 176.98896555047753,
174.9200816740307, 75.89330680142784] Fitness: 2.404259844507607
Generation 20 Best Solution: [183.68848624655612, 176.03810456747507,
174.9200816740307, 75.88869811614926] Fitness: 2.0216807483344406
Generation 21 Best Solution: [183.68848624655612, 177.44250723912145,
174.9200816740307, 75.45321624372363] Fitness: 2.6035453890978584
Generation 22 Best Solution: [183.68848624655612, 176.20213523309516,
174.9200816740307, 75.89330680142784] Fitness: 2.014100302656827
Generation 23 Best Solution: [183.68848624655612, 177.44250723912145,
174.9200816740307, 75.45321624372363] Fitness: 2.6035453890978584
Generation 24 Best Solution: [183.68848624655612, 177.44250723912145,
174.9200816740307, 75.45321624372363] Fitness: 2.6035453890978584
Generation 25 Best Solution: [183.68848624655612, 177.44250723912145,
174.9200816740307, 74.43134375791281] Fitness: 2.165712700247287
Generation 26 Best Solution: [183.68848624655612, 177.44250723912145,
174.9200816740307, 74.43134375791281] Fitness: 2.165712700247287
Generation 27 Best Solution: [183.68848624655612, 177.44250723912145,
174.9200816740307, 74.43134375791281] Fitness: 2.165712700247287
Generation 28 Best Solution: [183.8629907433212, 177.44250723912145,
175.09464035764924, 74.39368169807653] Fitness: 2.611870000340548
Generation 29 Best Solution: [183.8629907433212, 177.44250723912145,
175.09464035764924, 74.80193420862615] Fitness: 2.5551618701290395
Generation 30 Best Solution: [183.68848624655612, 175.99943637955712,
174.9200816740307, 75.89330680142784] Fitness: 2.0412619183025886
Generation 31 Best Solution: [183.68848624655612, 177.44250723912145,
175.09464035764924, 74.80193420862615] Fitness: 2.38660409799956
Generation 32 Best Solution: [183.81040090894194, 177.44250723912145,
173.95445033485663, 75.88869811614926] Fitness: 2.8888613472970186
Generation 33 Best Solution: [183.74229826185694, 177.44250723912145,
175.09464035764924, 74.9329162016785] Fitness: 2.473205634651433
Generation 34 Best Solution: [182.98597808338795, 177.44250723912145,
175.09464035764924, 74.9329162016785] Fitness: 2.255391258777341
Generation 35 Best Solution: [182.79848859135, 177.44250723912145,
175.09464035764924, 74.9329162016785] Fitness: 2.3471228810826266
Generation 36 Best Solution: [183.68848624655612, 177.44250723912145,
175.09464035764924, 74.74681965856607] Fitness: 2.37823137391309
Generation 37 Best Solution: [183.68848624655612, 177.0675166209532,
175.09464035764924, 74.74681965856607] Fitness: 1.7155935249700087
Generation 38 Best Solution: [183.68848624655612, 176.3842412975562,
175.09464035764924, 75.88869811614926] Fitness: 2.2991049457432347
Generation 39 Best Solution: [183.68848624655612, 177.44250723912145,
175.09464035764924, 75.88869811614926] Fitness: 3.573660495178468
Generation 40 Best Solution: [183.74229826185694, 177.44250723912145,
175.09464035764924, 74.49441835283896] Fitness: 2.450437697510944
Generation 41 Best Solution: [183.68848624655612, 177.44250723912145,
175.09464035764924, 75.12221263141295] Fitness: 2.5342723413350394
Generation 42 Best Solution: [183.74229826185694, 176.53584516486208,
175.09464035764924, 75.89330680142784] Fitness: 2.42430262725425

Generation 43 Best Solution: [183.68848624655612, 177.44250723912145,
175.09464035764924, 74.75487057250534] Fitness: 2.379142388451509
Generation 44 Best Solution: [183.68848624655612, 176.59935233690737,
175.09464035764924, 75.7994129959958] Fitness: 2.23794713205739
Generation 45 Best Solution: [183.68848624655612, 176.59935233690737,
175.09464035764924, 75.7994129959958] Fitness: 2.23794713205739
Generation 46 Best Solution: [183.74229826185694, 176.43063651700652,
175.09464035764924, 75.88869811614926] Fitness: 2.3629507316361353
Generation 47 Best Solution: [183.74229826185694, 177.44250723912145,
174.40848516893465, 75.31255140771206] Fitness: 2.012343199454313
Generation 48 Best Solution: [183.74229826185694, 176.43063651700652,
175.09464035764924, 75.7994129959958] Fitness: 2.192082354101456
Generation 49 Best Solution: [183.06972528962848, 177.44250723912145,
175.09464035764924, 75.04792858355925] Fitness: 2.2915216596930326
Generation 50 Best Solution: [183.74229826185694, 176.43063651700652,
175.09464035764924, 75.88869811614926] Fitness: 2.3629507316361353
Generation 51 Best Solution: [183.68848624655612, 177.44250723912145,
175.09464035764924, 75.01784567982651] Fitness: 2.4675952651610955
Generation 52 Best Solution: [183.68848624655612, 177.44250723912145,
174.07774462485247, 75.88869811614926] Fitness: 2.759342968661815
Generation 53 Best Solution: [183.68848624655612, 175.68300686178836,
175.09464035764924, 75.76096385053962] Fitness: 2.230234914565619
Generation 54 Best Solution: [183.48224446622302, 177.44250723912145,
175.09464035764924, 74.49030827117645] Fitness: 2.2705475563292277
Generation 55 Best Solution: [183.60518311940388, 177.44250723912145,
175.09464035764924, 74.5312350159095] Fitness: 2.3307627829105386
Generation 56 Best Solution: [182.8192042969524, 177.44250723912145,
175.09464035764924, 74.5312350159095] Fitness: 2.301107162569348
Generation 57 Best Solution: [182.8192042969524, 177.44250723912145,
175.09464035764924, 74.5312350159095] Fitness: 2.301107162569348
Generation 58 Best Solution: [182.8192042969524, 177.44250723912145,
175.09464035764924, 74.96822677298893] Fitness: 2.3497537287482655
Generation 59 Best Solution: [183.68848624655612, 176.81789271001594,
175.09464035764924, 74.96822677298893] Fitness: 1.4668481890094895
Generation 60 Best Solution: [183.60518311940388, 176.33127197338757,
175.09464035764924, 75.89330680142784] Fitness: 2.230181875780338
Generation 61 Best Solution: [183.60518311940388, 176.33127197338757,
175.09464035764924, 75.89330680142784] Fitness: 2.230181875780338
Generation 62 Best Solution: [183.17143954341938, 177.44250723912145,
175.09464035764924, 74.74361701561368] Fitness: 2.172936442782522
Generation 63 Best Solution: [183.60518311940388, 176.33127197338757,
175.09464035764924, 75.89330680142784] Fitness: 2.230181875780338
Generation 64 Best Solution: [183.68848624655612, 177.44250723912145,
173.93772153917422, 74.74361701561368] Fitness: 1.5848603020431073
Generation 65 Best Solution: [183.68848624655612, 176.24963085748655,
175.09464035764924, 75.89330680142784] Fitness: 2.278395946014987
Generation 66 Best Solution: [183.68848624655612, 176.42658672077246,
175.09464035764924, 75.88869811614926] Fitness: 2.314677082206839

Generation 67 Best Solution: [183.11426275664516, 176.42658672077246,
175.09464035764924, 75.88869811614926] Fitness: 2.114087777416073
Generation 68 Best Solution: [183.11426275664516, 175.6314208729263,
173.79015036132688, 75.88869811614926] Fitness: 1.5789906965550822
Generation 69 Best Solution: [183.09304135404815, 176.11137999018368,
174.14407227650784, 75.88869811614926] Fitness: 1.2588706322976535
Generation 70 Best Solution: [183.11426275664516, 175.7899302324896,
175.09464035764924, 75.89330680142784] Fitness: 2.2007399105452787
Generation 71 Best Solution: [183.11426275664516, 175.7899302324896,
175.09464035764924, 75.88869811614926] Fitness: 2.1915637283923726
Generation 72 Best Solution: [183.11426275664516, 175.7899302324896,
175.09464035764924, 75.88869811614926] Fitness: 2.1915637283923726
Generation 73 Best Solution: [183.11426275664516, 175.7899302324896,
175.09464035764924, 75.89330680142784] Fitness: 2.2007399105452787
Generation 74 Best Solution: [183.11426275664516, 175.7899302324896,
175.09464035764924, 75.89330680142784] Fitness: 2.2007399105452787
Generation 75 Best Solution: [183.11426275664516, 175.7899302324896,
175.09464035764924, 75.89330680142784] Fitness: 2.2007399105452787
Generation 76 Best Solution: [183.11426275664516, 175.7899302324896,
175.09464035764924, 75.88869811614926] Fitness: 2.1915637283923726
Generation 77 Best Solution: [183.11426275664516, 175.7899302324896,
175.09464035764924, 75.88869811614926] Fitness: 2.1915637283923726
Generation 78 Best Solution: [183.11426275664516, 175.7899302324896,
175.09464035764924, 75.88869811614926] Fitness: 2.1915637283923726
Generation 79 Best Solution: [183.11426275664516, 177.44250723912145,
175.09464035764924, 74.7750438051222] Fitness: 2.1813047141166595
Generation 80 Best Solution: [183.11426275664516, 176.21083948062133,
175.09464035764924, 75.88869811614926] Fitness: 2.065557760490223
Generation 81 Best Solution: [183.74026162282777, 177.44250723912145,
175.09464035764924, 74.2846374447226] Fitness: 2.5496908106719354
Generation 82 Best Solution: [183.73356328891225, 177.44250723912145,
175.09464035764924, 74.2846374447226] Fitness: 2.5436490912758387
Generation 83 Best Solution: [183.74026162282777, 177.44250723912145,
175.09464035764924, 74.2846374447226] Fitness: 2.5496908106719354
Generation 84 Best Solution: [183.68848624655612, 176.91970357427832,
175.09464035764924, 75.41845885336683] Fitness: 1.9610771570046606
Generation 85 Best Solution: [183.68848624655612, 177.44250723912145,
174.8561579476286, 75.27976858483052] Fitness: 2.3250633453417997
Generation 86 Best Solution: [183.68848624655612, 176.7180013636056,
174.8561579476286, 75.89330680142784] Fitness: 2.1672648596841033
Generation 87 Best Solution: [183.74026162282777, 176.7180013636056,
174.8561579476286, 75.89330680142784] Fitness: 2.2120428552867875
Generation 88 Best Solution: [183.68848624655612, 176.43472244324016,
175.09464035764924, 75.88869811614926] Fitness: 2.318007142669006
Generation 89 Best Solution: [183.68848624655612, 177.27737921324342,
174.8561579476286, 75.52994385056296] Fitness: 2.3025268983886673
Generation 90 Best Solution: [183.8004954096513, 176.39164044577726,
175.09464035764924, 75.89330680142784] Fitness: 2.413216546979858

Generation 91 Best Solution: [183.4636828612953, 176.39164044577726,
175.09464035764924, 75.89330680142784] Fitness: 2.1675713138088932
Generation 92 Best Solution: [183.4636828612953, 176.9055164762406,
174.8561579476286, 75.89330680142784] Fitness: 2.218502946399645
Generation 93 Best Solution: [183.68848624655612, 176.723311716427,
175.09464035764924, 75.8262383894457] Fitness: 2.3857393740355937
Generation 94 Best Solution: [183.68848624655612, 177.44250723912145,
175.09464035764924, 74.8298944887029] Fitness: 2.392764572960647
Generation 95 Best Solution: [182.9360292785274, 177.44250723912145,
175.09464035764924, 74.8298944887029] Fitness: 2.2403602854121236
Generation 96 Best Solution: [183.68848624655612, 176.0769667124787,
175.09464035764924, 75.89330680142784] Fitness: 2.283754974075242
Generation 97 Best Solution: [183.68848624655612, 176.601296338239,
175.09464035764924, 75.89330680142784] Fitness: 2.4193306605646736
Generation 98 Best Solution: [183.68848624655612, 176.601296338239,
174.8561579476286, 75.89330680142784] Fitness: 2.0754786376094856
Generation 99 Best Solution: [183.68848624655612, 176.601296338239,
175.09464035764924, 75.88869811614926] Fitness: 2.4101544784117674
Generation 100 Best Solution: [183.55007507488818, 175.72451848722812,
174.8561579476286, 75.88869811614926] Fitness: 1.9960539262457635
Generation 101 Best Solution: [183.68848624655612, 175.93251422755696,
175.09464035764924, 75.89330680142784] Fitness: 2.3259628586773147
Generation 102 Best Solution: [183.68848624655612, 177.263271624712,
175.09464035764924, 74.42154264356597] Fitness: 2.085096425166582
Generation 103 Best Solution: [183.55007507488818, 177.44250723912145,
175.09464035764924, 74.42154264356597] Fitness: 2.3326878701309712
Generation 104 Best Solution: [183.68848624655612, 177.263271624712,
175.09464035764924, 74.42154264356597] Fitness: 2.085096425166582
Generation 105 Best Solution: [183.68848624655612, 177.263271624712,
175.09464035764924, 74.42154264356597] Fitness: 2.085096425166582
Generation 106 Best Solution: [183.68848624655612, 177.263271624712,
175.09464035764924, 74.42154264356597] Fitness: 2.085096425166582
Generation 107 Best Solution: [183.55007507488818, 176.0385807543307,
174.50817362748475, 75.88869811614926] Fitness: 1.5068946524637896
Generation 108 Best Solution: [183.68848624655612, 177.263271624712,
175.09464035764924, 74.88852068198572] Fitness: 2.0642441130364535
Generation 109 Best Solution: [183.68848624655612, 176.0519346242781,
175.09464035764924, 74.42154264356597] Fitness: 1.1364850607997001
Generation 110 Best Solution: [183.68848624655612, 177.263271624712,
175.09464035764924, 74.42154264356597] Fitness: 2.085096425166582
Generation 111 Best Solution: [183.86750291211385, 177.263271624712,
175.09464035764924, 74.42154264356597] Fitness: 2.2586778101166436
Generation 112 Best Solution: [183.68848624655612, 177.263271624712,
175.09464035764924, 74.42154264356597] Fitness: 2.085096425166582
Generation 113 Best Solution: [183.6439055717389, 176.15145181822533,
175.09464035764924, 75.89330680142784] Fitness: 2.2404027073512296
Generation 114 Best Solution: [183.6439055717389, 176.32319067592312,
175.09464035764924, 75.89330680142784] Fitness: 2.256010687491208

Generation 115 Best Solution: [183.6439055717389, 176.3777382641471,
175.09464035764924, 75.89330680142784] Fitness: 2.2711332207158437
Generation 116 Best Solution: [183.68848624655612, 176.42628558934692,
175.09464035764924, 75.89330680142784] Fitness: 2.3237320994836588
Generation 117 Best Solution: [183.68848624655612, 176.47301462149167,
175.09464035764924, 75.89330680142784] Fitness: 2.344320919160282
Generation 118 Best Solution: [183.6439055717389, 175.8661118704498,
175.09464035764924, 75.89330680142784] Fitness: 2.3218774735585774
Generation 119 Best Solution: [183.6439055717389, 175.8661118704498,
173.80394372433295, 75.89330680142784] Fitness: 1.5793338430974773
Generation 120 Best Solution: [183.6439055717389, 175.8661118704498,
174.9452346007853, 75.09834759508695] Fitness: 1.030106016236072
Generation 121 Best Solution: [183.6439055717389, 175.8661118704498,
175.09464035764924, 74.69243621257323] Fitness: 1.113914529159793
Generation 122 Best Solution: [183.6439055717389, 175.8661118704498,
175.09464035764924, 75.89330680142784] Fitness: 2.3218774735585774
Generation 123 Best Solution: [183.86750291211385, 176.0545442160811,
175.09464035764924, 75.32376626741812] Fitness: 1.592615564590226
Generation 124 Best Solution: [183.6439055717389, 176.55872587152254,
175.09464035764924, 75.89330680142784] Fitness: 2.356415623580189
Generation 125 Best Solution: [182.89592964982452, 176.55872587152254,
175.09464035764924, 75.89330680142784] Fitness: 2.25708032479874
Generation 126 Best Solution: [183.68848624655612, 176.55872587152254,
175.09464035764924, 75.15840312716735] Fitness: 1.3701796130475903
Generation 127 Best Solution: [183.6439055717389, 176.0816480260387,
175.09464035764924, 75.89330680142784] Fitness: 2.247943989523197
Generation 128 Best Solution: [183.6439055717389, 176.55872587152254,
175.09464035764924, 75.89330680142784] Fitness: 2.356415623580189
Generation 129 Best Solution: [183.6439055717389, 176.55872587152254,
175.09464035764924, 75.89330680142784] Fitness: 2.356415623580189
Generation 130 Best Solution: [183.68848624655612, 176.55872587152254,
175.09464035764924, 75.89330680142784] Fitness: 2.391433705509373
Generation 131 Best Solution: [183.68848624655612, 176.55872587152254,
175.09464035764924, 75.89330680142784] Fitness: 2.391433705509373
Generation 132 Best Solution: [183.6439055717389, 176.39879158724673,
175.09464035764924, 75.89330680142784] Fitness: 2.27828072225717
Generation 133 Best Solution: [183.6439055717389, 176.39879158724673,
175.09464035764924, 75.89330680142784] Fitness: 2.27828072225717
Generation 134 Best Solution: [183.68848624655612, 175.75180821495704,
175.09464035764924, 74.81689735748792] Fitness: 1.2340568456049408
Generation 135 Best Solution: [183.6439055717389, 176.57120151024557,
175.09464035764924, 75.89330680142784] Fitness: 2.364281868502974
Generation 136 Best Solution: [183.6439055717389, 176.57120151024557,
175.09464035764924, 75.89330680142784] Fitness: 2.364281868502974
Generation 137 Best Solution: [183.86750291211385, 176.57120151024557,
175.09464035764924, 75.37865413757123] Fitness: 1.7643482820641112
Generation 138 Best Solution: [183.86750291211385, 176.56199728622175,
174.992067696503, 75.31664084324123] Fitness: 1.5311723661707741

Generation 139 Best Solution: [183.86750291211385, 177.1385714099675,
175.09464035764924, 74.28704288936324] Fitness: 2.122060603612048
Generation 140 Best Solution: [183.86750291211385, 176.3147466989821,
174.992067696503, 75.89330680142784] Fitness: 2.3033339638259758
Generation 141 Best Solution: [183.6439055717389, 176.3147466989821,
174.992067696503, 75.89330680142784] Fitness: 2.0947344969467303
Generation 142 Best Solution: [183.86750291211385, 175.77577717007483,
175.09464035764924, 75.52988476803901] Fitness: 1.9680110748389847
Generation 143 Best Solution: [183.86750291211385, 176.3147466989821,
174.992067696503, 75.89330680142784] Fitness: 2.3033339638259758
Generation 144 Best Solution: [183.86750291211385, 176.97743994486711,
174.09566621084676, 75.89330680142784] Fitness: 2.1543967636136108
Generation 145 Best Solution: [183.6439055717389, 176.40717051598045,
174.09566621084676, 75.89330680142784] Fitness: 1.4666184645572728
Generation 146 Best Solution: [183.86750291211385, 175.6855489582051,
174.992067696503, 75.74685313009518] Fitness: 2.217449388402622
Generation 147 Best Solution: [183.86750291211385, 176.6404679509819,
174.09566621084676, 75.74685313009518] Fitness: 1.5320179512675947
Generation 148 Best Solution: [183.86750291211385, 175.6855489582051,
175.09464035764924, 75.44321360376696] Fitness: 1.920274997073267
Generation 149 Best Solution: [183.86750291211385, 175.6855489582051,
175.09464035764924, 75.44321360376696] Fitness: 1.920274997073267
Generation 150 Best Solution: [183.86750291211385, 176.60915718262726,
174.992067696503, 75.74685313009518] Fitness: 2.164525874984595
Generation 151 Best Solution: [183.86750291211385, 176.58728684624282,
174.992067696503, 75.89330680142784] Fitness: 2.4240286827945767
Generation 152 Best Solution: [183.86750291211385, 176.58728684624282,
175.09464035764924, 75.74685313009518] Fitness: 2.3089112379949017
Generation 153 Best Solution: [183.86750291211385, 176.58728684624282,
175.09464035764924, 74.76205501562465] Fitness: 1.380610757312676
Generation 154 Best Solution: [182.85896612598555, 176.58728684624282,
175.09464035764924, 75.74685313009518] Fitness: 2.019961501629553
Generation 155 Best Solution: [183.86750291211385, 176.9160666511791,
173.90595842774596, 75.89330680142784] Fitness: 2.108097962932426
Generation 156 Best Solution: [183.86750291211385, 176.9160666511791,
173.90595842774596, 75.89330680142784] Fitness: 2.108097962932426
Generation 157 Best Solution: [183.86750291211385, 176.9160666511791,
173.90595842774596, 75.89330680142784] Fitness: 2.108097962932426
Generation 158 Best Solution: [183.86750291211385, 176.9160666511791,
173.90595842774596, 75.89330680142784] Fitness: 2.108097962932426
Generation 159 Best Solution: [183.86750291211385, 176.9160666511791,
173.90595842774596, 75.89330680142784] Fitness: 2.108097962932426
Generation 160 Best Solution: [183.86750291211385, 176.9160666511791,
173.90595842774596, 75.89330680142784] Fitness: 2.108097962932426
Generation 161 Best Solution: [183.86750291211385, 176.9160666511791,
174.04833699168518, 75.89330680142784] Fitness: 2.079288379715628
Generation 162 Best Solution: [183.86750291211385, 176.9160666511791,
174.04833699168518, 75.89330680142784] Fitness: 2.079288379715628

Generation 163 Best Solution: [183.86750291211385, 176.9160666511791,
174.04833699168518, 75.89330680142784] Fitness: 2.079288379715628
Generation 164 Best Solution: [183.86750291211385, 177.11606791626767,
174.992067696503, 75.02829761481311] Fitness: 1.8977764078951251
Generation 165 Best Solution: [183.86750291211385, 176.42440024688818,
175.09464035764924, 75.89330680142784] Fitness: 2.496558282596907
Generation 166 Best Solution: [183.86750291211385, 177.44250723912145,
174.4707114650667, 74.58054299548623] Fitness: 1.8553185174037625
Generation 167 Best Solution: [183.86750291211385, 177.43391693661036,
174.4707114650667, 74.58054299548623] Fitness: 1.837546728540815
Generation 168 Best Solution: [183.86750291211385, 177.43391693661036,
174.4707114650667, 74.58054299548623] Fitness: 1.837546728540815
Generation 169 Best Solution: [183.86750291211385, 177.44250723912145,
174.4707114650667, 74.58054299548623] Fitness: 1.8553185174037625
Generation 170 Best Solution: [183.86750291211385, 175.6114529181065,
175.09464035764924, 75.49527772606616] Fitness: 2.0528971121590835
Generation 171 Best Solution: [183.69328595282516, 176.4333543953099,
175.09464035764924, 75.89330680142784] Fitness: 2.3305810907474958
Generation 172 Best Solution: [183.0483402628095, 177.44250723912145,
175.09464035764924, 75.89330680142784] Fitness: 3.393972105814831
Generation 173 Best Solution: [183.2172209488193, 177.44250723912145,
175.09464035764924, 75.89330680142784] Fitness: 3.3782548469038045
Generation 174 Best Solution: [183.86750291211385, 175.86751747078804,
175.09464035764924, 75.89330680142784] Fitness: 2.5297469221226576
Generation 175 Best Solution: [183.86750291211385, 176.46980616511,
175.09464035764924, 74.32670700938057] Fitness: 1.4123637584345865
Generation 176 Best Solution: [183.86750291211385, 177.44250723912145,
173.76197094600622, 75.25807077042221] Fitness: 2.1009008189242224
Generation 177 Best Solution: [183.86750291211385, 177.44250723912145,
173.76197094600622, 75.25807077042221] Fitness: 2.1009008189242224
Generation 178 Best Solution: [183.86750291211385, 177.44250723912145,
173.76197094600622, 75.25807077042221] Fitness: 2.1009008189242224
Generation 179 Best Solution: [183.86750291211385, 177.44250723912145,
173.76197094600622, 75.10950283098856] Fitness: 1.9781484779427685
Generation 180 Best Solution: [183.68899621075857, 177.44250723912145,
175.09464035764924, 74.61315211329426] Fitness: 2.379125762233531
Generation 181 Best Solution: [183.68899621075857, 176.30076898335443,
175.09464035764924, 75.89330680142784] Fitness: 2.2866534231290645
Generation 182 Best Solution: [183.68899621075857, 175.7421749344852,
173.81300480124025, 75.89330680142784] Fitness: 1.6875822381125762
Generation 183 Best Solution: [183.68899621075857, 175.7421749344852,
173.81300480124025, 75.89330680142784] Fitness: 1.6875822381125762
Generation 184 Best Solution: [183.68899621075857, 176.1693802992022,
174.13629943817244, 75.89330680142784] Fitness: 1.4615584269509192
Generation 185 Best Solution: [183.86750291211385, 177.44250723912145,
174.22293684148312, 74.90709664002075] Fitness: 1.7877628755720745
Generation 186 Best Solution: [183.68899621075857, 175.7655481266509,
173.81300480124025, 75.89330680142784] Fitness: 1.671094682293961

Generation 187 Best Solution: [183.50145011036062, 177.44250723912145,
173.95030779374542, 74.38004014755965] Fitness: 1.528700431514471
Generation 188 Best Solution: [183.67272907138496, 175.7655481266509,
173.81300480124025, 75.89330680142784] Fitness: 1.657923893431335
Generation 189 Best Solution: [183.86750291211385, 177.44250723912145,
173.81300480124025, 74.66418557689165] Fitness: 1.8017390233605104
Generation 190 Best Solution: [183.86750291211385, 177.44250723912145,
173.81300480124025, 74.66418557689165] Fitness: 1.8017390233605104
Generation 191 Best Solution: [183.86750291211385, 177.17941005801103,
173.81300480124025, 75.89330680142784] Fitness: 2.520366542669179
Generation 192 Best Solution: [183.18113119212327, 177.44250723912145,
173.9268967600846, 75.89330680142784] Fitness: 2.5876183606161574
Generation 193 Best Solution: [183.86750291211385, 177.44250723912145,
173.81300480124025, 75.28021292845054] Fitness: 2.096039896767536
Generation 194 Best Solution: [183.86750291211385, 176.82464195445812,
173.81300480124025, 75.89330680142784] Fitness: 2.0412891945355764
Generation 195 Best Solution: [183.86750291211385, 176.82464195445812,
173.81300480124025, 75.28021292845054] Fitness: 1.1278054265127624
Generation 196 Best Solution: [183.86750291211385, 177.44250723912145,
173.81300480124025, 74.48554172035567] Fitness: 1.8333045294197983
Generation 197 Best Solution: [183.86750291211385, 177.44250723912145,
173.81300480124025, 74.91169250422354] Fitness: 1.8448677334694432
Generation 198 Best Solution: [183.86750291211385, 175.8360764278884,
173.81300480124025, 74.91169250422354] Fitness: 0.6353036377843241
Generation 199 Best Solution: [183.68198170920027, 177.44250723912145,
173.81300480124025, 74.91169250422354] Fitness: 1.6659730783900284
Generation 200 Best Solution: [183.86750291211385, 176.4727650601835,
173.81300480124025, 75.89330680142784] Fitness: 1.7708883973526222
Generation 201 Best Solution: [183.7052715971439, 176.4727650601835,
173.81300480124025, 75.89330680142784] Fitness: 1.6113401676059969
Generation 202 Best Solution: [183.86750291211385, 177.44250723912145,
173.81300480124025, 74.91169250422354] Fitness: 1.8448677334694432
Generation 203 Best Solution: [183.23954630947895, 177.44250723912145,
173.81300480124025, 74.91169250422354] Fitness: 1.4681421946453568
Generation 204 Best Solution: [183.29904879526, 177.44250723912145,
173.81300480124025, 74.91169250422354] Fitness: 1.4759836823642436
Generation 205 Best Solution: [183.86750291211385, 177.44250723912145,
173.87679413936695, 74.91169250422354] Fitness: 1.8180656493245677
Generation 206 Best Solution: [183.86750291211385, 175.653382968534,
173.81300480124025, 74.91169250422354] Fitness: 0.766924264185856
Generation 207 Best Solution: [183.86750291211385, 177.44250723912145,
174.23189808317102, 75.73847736783786] Fitness: 2.6668986661405065
Generation 208 Best Solution: [183.86750291211385, 177.17094493336953,
174.50096135437218, 75.89330680142784] Fitness: 2.571088904702226
Generation 209 Best Solution: [183.86750291211385, 177.17094493336953,
174.50096135437218, 75.73847736783786] Fitness: 2.2819679035075238
Generation 210 Best Solution: [183.86750291211385, 176.73178280435963,
174.50096135437218, 75.89330680142784] Fitness: 2.014691455217987

Generation 211 Best Solution: [183.86750291211385, 177.44250723912145,
175.1331483707064, 74.988997527742] Fitness: 2.6902174300848096
Generation 212 Best Solution: [183.79434656144946, 176.12683033457733,
175.1331483707064, 75.89330680142784] Fitness: 2.437739348367483
Generation 213 Best Solution: [183.86750291211385, 176.12683033457733,
175.1331483707064, 75.89330680142784] Fitness: 2.5150522390281167
Generation 214 Best Solution: [183.41737659266798, 176.6792897536226,
175.1331483707064, 75.89330680142784] Fitness: 2.380110721414894
Generation 215 Best Solution: [182.79543747957652, 177.44250723912145,
175.07105441388524, 74.988997527742] Fitness: 2.3366671811088886
Generation 216 Best Solution: [183.4186632920022, 177.44250723912145,
175.07105441388524, 74.988997527742] Fitness: 2.2522534738416904
Generation 217 Best Solution: [183.4186632920022, 177.33791322873546,
175.1331483707064, 74.988997527742] Fitness: 2.146624015442658
Generation 218 Best Solution: [183.4186632920022, 177.44250723912145,
175.1331483707064, 74.45817572114383] Fitness: 2.3185067199763725
Generation 219 Best Solution: [183.86750291211385, 177.44250723912145,
174.31790896082705, 74.988997527742] Fitness: 1.850290525991628
Generation 220 Best Solution: [183.79434656144946, 176.08897213174276,
175.1331483707064, 75.89330680142784] Fitness: 2.4423686615949096
Generation 221 Best Solution: [183.79434656144946, 176.08897213174276,
175.1331483707064, 75.89330680142784] Fitness: 2.4423686615949096
Generation 222 Best Solution: [183.79434656144946, 176.08897213174276,
175.1331483707064, 74.51349380105938] Fitness: 1.257741700332926
Generation 223 Best Solution: [183.86750291211385, 175.71786217827812,
175.07105441388524, 75.89330680142784] Fitness: 2.5875625151611286
Generation 224 Best Solution: [183.7983913457997, 175.71786217827812,
175.07105441388524, 75.89330680142784] Fitness: 2.514294011677061
Generation 225 Best Solution: [183.4186632920022, 176.82396311922508,
175.1331483707064, 75.0103516734174] Fitness: 1.3969564091201632
Generation 226 Best Solution: [183.86750291211385, 176.82396311922508,
174.57860440631748, 75.0103516734174] Fitness: 1.0420408576928448
Generation 227 Best Solution: [183.79434656144946, 176.82396311922508,
174.2761480166267, 75.89330680142784] Fitness: 1.92097986492568
Generation 228 Best Solution: [183.51114873438954, 176.82396311922508,
174.2761480166267, 75.89330680142784] Fitness: 1.7048010678173149
Generation 229 Best Solution: [182.79543747957652, 176.82396311922508,
174.64741430615402, 75.89330680142784] Fitness: 1.9684580666004032
Generation 230 Best Solution: [182.79543747957652, 176.00733650817523,
175.1331483707064, 75.89330680142784] Fitness: 2.2866345382379563
Generation 231 Best Solution: [183.79434656144946, 177.44250723912145,
175.1331483707064, 74.84705174269106] Fitness: 2.557757017265317
Generation 232 Best Solution: [183.4012772837464, 177.44250723912145,
175.1331483707064, 74.84705174269106] Fitness: 2.293273791060991
Generation 233 Best Solution: [183.4012772837464, 176.49286876615878,
175.1331483707064, 75.89330680142784] Fitness: 2.2502488054182708
Generation 234 Best Solution: [183.79434656144946, 176.13072615007482,
175.1331483707064, 74.57056683494409] Fitness: 1.2395990165849213

Generation 235 Best Solution: [183.4012772837464, 175.61935397796734,
175.1331483707064, 75.89330680142784] Fitness: 2.431574114115811
Generation 236 Best Solution: [183.4012772837464, 176.49286876615878,
175.1331483707064, 75.89330680142784] Fitness: 2.2502488054182708
Generation 237 Best Solution: [183.4012772837464, 176.49286876615878,
175.1331483707064, 75.89330680142784] Fitness: 2.2502488054182708
Generation 238 Best Solution: [183.14799829766477, 176.49286876615878,
175.1331483707064, 75.89330680142784] Fitness: 2.214642873270245
Generation 239 Best Solution: [183.4012772837464, 176.57923425976597,
175.1331483707064, 75.89330680142784] Fitness: 2.3005919748916748
Generation 240 Best Solution: [183.4012772837464, 177.44250723912145,
174.24688883295985, 75.89330680142784] Fitness: 2.6176652650971715
Generation 241 Best Solution: [183.79434656144946, 177.44250723912145,
175.1331483707064, 74.79843989619] Fitness: 2.5464994202428075
Generation 242 Best Solution: [183.14799829766477, 177.44250723912145,
175.1331483707064, 74.30088846696273] Fitness: 2.354976069088499
Generation 243 Best Solution: [183.4012772837464, 176.35797902163318,
175.106996707554, 75.89330680142784] Fitness: 2.1522608119337265
Generation 244 Best Solution: [183.4012772837464, 176.5202116337282,
175.1331483707064, 75.89330680142784] Fitness: 2.2648581407332067
Generation 245 Best Solution: [183.4012772837464, 176.74801334973657,
175.1331483707064, 75.89330680142784] Fitness: 2.434438425063134
Generation 246 Best Solution: [183.4012772837464, 177.44250723912145,
175.1331483707064, 75.08666132799732] Fitness: 2.405636352132494
Generation 247 Best Solution: [183.86750291211385, 177.44250723912145,
175.1331483707064, 75.08666132799732] Fitness: 2.747432468997453
Generation 248 Best Solution: [183.15204444905237, 177.44250723912145,
174.9793023768509, 75.89330680142784] Fitness: 3.200759357921628
Generation 249 Best Solution: [182.9359284552328, 177.44250723912145,
175.1331483707064, 74.53829272122536] Fitness: 2.303771646807089
Generation 250 Best Solution: [183.86750291211385, 177.44250723912145,
175.1331483707064, 74.53829272122536] Fitness: 2.629715254690079
Generation 251 Best Solution: [182.9359284552328, 176.79095117111487,
175.1331483707064, 75.89330680142784] Fitness: 2.4918282330347945
Generation 252 Best Solution: [183.39748613034484, 176.26874410022447,
175.1331483707064, 75.89330680142784] Fitness: 2.1756009474724016
Generation 253 Best Solution: [183.39748613034484, 176.26874410022447,
175.1331483707064, 75.89330680142784] Fitness: 2.1756009474724016
Generation 254 Best Solution: [183.14799829766477, 177.2190926260278,
174.09526646263117, 75.89330680142784] Fitness: 2.1416296476203853
Generation 255 Best Solution: [183.44220579464553, 177.22541033889465,
174.09526646263117, 75.89330680142784] Fitness: 2.203728814794197
Generation 256 Best Solution: [183.14799829766477, 177.44250723912145,
174.1812119225495, 75.89330680142784] Fitness: 2.5697220776328087
Generation 257 Best Solution: [183.86750291211385, 177.44250723912145,
174.95184847473035, 75.07621749145152] Fitness: 2.459122968415263
Generation 258 Best Solution: [183.86750291211385, 177.44250723912145,
174.95184847473035, 75.07621749145152] Fitness: 2.459122968415263

Generation 259 Best Solution: [182.9286710428987, 177.44250723912145,
175.1331483707064, 74.47699843994253] Fitness: 2.324459921598422
Generation 260 Best Solution: [183.14799829766477, 176.18347771249682,
175.1331483707064, 75.89330680142784] Fitness: 2.135132089639106
Generation 261 Best Solution: [183.15204444905237, 176.88819975988037,
175.1331483707064, 74.80865180820024] Fitness: 1.3505418199023533
Generation 262 Best Solution: [183.14799829766477, 176.18347771249682,
175.1331483707064, 75.89330680142784] Fitness: 2.135132089639106
Generation 263 Best Solution: [183.14799829766477, 176.18347771249682,
175.1331483707064, 74.5274177188077] Fitness: 0.9467970754600532
Generation 264 Best Solution: [183.14799829766477, 177.44250723912145,
174.98084943699857, 74.5274177188077] Fitness: 2.0149285316979393
Generation 265 Best Solution: [183.15204444905237, 177.22541033889465,
175.1331483707064, 74.73771777953944] Fitness: 1.8256974929793297
Generation 266 Best Solution: [183.74360276582928, 177.44250723912145,
175.1331483707064, 74.97588745049427] Fitness: 2.5579708964609824
Generation 267 Best Solution: [183.15204444905237, 177.22541033889465,
173.75101733788927, 75.89330680142784] Fitness: 2.2524692686828836
Generation 268 Best Solution: [182.93383108298923, 176.4088879284435,
175.1331483707064, 75.89330680142784] Fitness: 2.2298082947919724
Generation 269 Best Solution: [183.74360276582928, 175.595997169328,
175.1331483707064, 75.89330680142784] Fitness: 2.6697078659404054
Generation 270 Best Solution: [182.94572932499602, 177.22541033889465,
175.1331483707064, 75.44247075140818] Fitness: 2.3469444534597375
Generation 271 Best Solution: [183.34571582977946, 177.22541033889465,
174.283359267588, 75.89330680142784] Fitness: 2.198870014833007
Generation 272 Best Solution: [183.34571582977946, 177.22541033889465,
174.283359267588, 75.89330680142784] Fitness: 2.198870014833007
Generation 273 Best Solution: [183.34571582977946, 177.22541033889465,
174.283359267588, 75.89330680142784] Fitness: 2.198870014833007
Generation 274 Best Solution: [183.74360276582928, 177.44250723912145,
175.1331483707064, 75.44247075140818] Fitness: 2.962993054663724
Generation 275 Best Solution: [183.86750291211385, 177.22541033889465,
175.1331483707064, 75.16170140236987] Fitness: 2.390210007480719
Generation 276 Best Solution: [183.30290030701826, 177.22541033889465,
175.1331483707064, 75.16170140236987] Fitness: 2.022034473099954
Generation 277 Best Solution: [182.94572932499602, 177.22541033889465,
175.1331483707064, 75.16170140236987] Fitness: 2.060255658226344
Generation 278 Best Solution: [182.94572932499602, 177.08412072798558,
175.1331483707064, 75.89330680142784] Fitness: 2.852572269816311
Generation 279 Best Solution: [182.94572932499602, 177.44250723912145,
173.76433342084954, 75.89330680142784] Fitness: 2.7045270399750327
Generation 280 Best Solution: [183.30290030701826, 177.22541033889465,
175.1331483707064, 75.89330680142784] Fitness: 3.040692345035248
Generation 281 Best Solution: [183.86750291211385, 176.88374246339555,
175.1331483707064, 75.89330680142784] Fitness: 2.9179084421327746
Generation 282 Best Solution: [183.86750291211385, 177.44250723912145,
174.05755915144277, 75.89330680142784] Fitness: 2.943176002508677

Generation 283 Best Solution: [183.86750291211385, 176.22039811005232,
174.7831973508242, 75.89330680142784] Fitness: 2.0190956948537044
Generation 284 Best Solution: [183.86750291211385, 176.22039811005232,
175.1331483707064, 75.89330680142784] Fitness: 2.5137378772114385
Generation 285 Best Solution: [183.69235774821945, 177.44250723912145,
175.1331483707064, 74.64391793042249] Fitness: 2.4434915265971036
Generation 286 Best Solution: [183.30290030701826, 176.22039811005232,
175.1331483707064, 75.89330680142784] Fitness: 2.145562342830674
Generation 287 Best Solution: [183.86750291211385, 176.0766710954602,
175.1331483707064, 75.89330680142784] Fitness: 2.52169385332449
Generation 288 Best Solution: [183.86750291211385, 176.0766710954602,
175.1331483707064, 75.21817832427442] Fitness: 1.550271671513774
Generation 289 Best Solution: [183.30290030701826, 176.94610574648178,
174.7831973508242, 75.44582954080005] Fitness: 1.4026055227551533
Generation 290 Best Solution: [183.56247905620378, 176.94610574648178,
174.7831973508242, 75.44582954080005] Fitness: 1.506670812565904
Generation 291 Best Solution: [183.2603660431929, 176.0766710954602,
175.1331483707064, 75.89330680142784] Fitness: 2.1470487851002424
Generation 292 Best Solution: [183.2603660431929, 176.0766710954602,
175.1331483707064, 75.89330680142784] Fitness: 2.1470487851002424
Generation 293 Best Solution: [183.05969506972087, 176.94610574648178,
174.7831973508242, 75.89330680142784] Fitness: 2.1335588710284616
Generation 294 Best Solution: [183.86750291211385, 176.94610574648178,
174.7831973508242, 75.89330680142784] Fitness: 2.498534615251742
Generation 295 Best Solution: [183.05969506972087, 176.94610574648178,
174.7831973508242, 75.89330680142784] Fitness: 2.1335588710284616
Generation 296 Best Solution: [183.03730642927025, 177.01452390548965,
175.1331483707064, 75.89330680142784] Fitness: 2.7233375438478973
Generation 297 Best Solution: [183.56247905620378, 177.44250723912145,
174.20353287433525, 75.89330680142784] Fitness: 2.686409853596667
Generation 298 Best Solution: [183.05969506972087, 177.047137446453,
175.1331483707064, 75.7217180678687] Fitness: 2.4456862339131717
Generation 299 Best Solution: [183.56247905620378, 177.44250723912145,
175.1331483707064, 74.8197593493022] Fitness: 2.364160076140694
Generation 300 Best Solution: [183.65580154847936, 177.047137446453,
175.1331483707064, 75.48166167819261] Fitness: 2.2475096579087595
Generation 301 Best Solution: [183.56247905620378, 176.36907856295767,
175.1331483707064, 75.89330680142784] Fitness: 2.2772006274806254
Generation 302 Best Solution: [183.56247905620378, 177.047137446453,
175.1331483707064, 75.89330680142784] Fitness: 2.8646000121855035
Generation 303 Best Solution: [183.43876781835544, 176.8804363776086,
175.1331483707064, 75.89330680142784] Fitness: 2.5865486894694687
Generation 304 Best Solution: [183.56247905620378, 177.44250723912145,
175.1331483707064, 74.9196879898367] Fitness: 2.39503369238376
Generation 305 Best Solution: [183.43876781835544, 176.8804363776086,
175.1331483707064, 74.9196879898367] Fitness: 1.4249683519299383
Generation 306 Best Solution: [183.56247905620378, 176.8804363776086,
175.12851584178196, 74.82445680882924] Fitness: 1.4507504240681608

Generation 307 Best Solution: [183.56247905620378, 176.8804363776086,
175.1331483707064, 75.185281679687] Fitness: 1.650411945365044
Generation 308 Best Solution: [183.56247905620378, 177.44250723912145,
175.1331483707064, 74.82445680882924] Fitness: 2.3652429859656885
Generation 309 Best Solution: [183.56247905620378, 177.44250723912145,
175.1331483707064, 74.82445680882924] Fitness: 2.3652429859656885
Generation 310 Best Solution: [183.56247905620378, 177.44250723912145,
173.94449748979517, 74.82445680882924] Fitness: 1.5061251416723824
Generation 311 Best Solution: [182.98786027320423, 177.44250723912145,
175.1331483707064, 75.03194348356068] Fitness: 2.368048055003826
Generation 312 Best Solution: [183.05969506972087, 175.67764656744234,
175.1331483707064, 75.14218229627993] Fitness: 1.3233972615768719
Generation 313 Best Solution: [183.56247905620378, 177.44250723912145,
174.46286543832983, 75.14218229627993] Fitness: 1.7522561113898845
Generation 314 Best Solution: [183.56247905620378, 177.44250723912145,
174.83376273748604, 75.14218229627993] Fitness: 2.087215523168821
Generation 315 Best Solution: [182.98786027320423, 175.7403070400936,
175.1331483707064, 75.89330680142784] Fitness: 2.3279076960540297
Generation 316 Best Solution: [182.98786027320423, 177.44250723912145,
175.1331483707064, 75.03194348356068] Fitness: 2.368048055003826
Generation 317 Best Solution: [183.56247905620378, 177.44250723912145,
175.1331483707064, 75.03194348356068] Fitness: 2.4493312826461557
Generation 318 Best Solution: [183.56247905620378, 177.44250723912145,
175.1331483707064, 75.03194348356068] Fitness: 2.4493312826461557
Generation 319 Best Solution: [183.56247905620378, 176.02202763995876,
174.60825693363674, 75.89330680142784] Fitness: 1.603241845778696
Generation 320 Best Solution: [183.56247905620378, 176.01256509517935,
175.1331483707064, 74.90449299752737] Fitness: 1.1047690992053085
Generation 321 Best Solution: [183.56247905620378, 176.35797328155482,
175.1331483707064, 75.8984758668974] Fitness: 2.2842164974877974
Generation 322 Best Solution: [183.7360336940649, 177.44250723912145,
173.92562551639995, 75.90577190868336] Fitness: 2.8591018656777143
Generation 323 Best Solution: [183.7360336940649, 175.87094884797096,
175.1331483707064, 75.90577190868336] Fitness: 2.4846532797279433
Generation 324 Best Solution: [183.7360336940649, 175.87094884797096,
175.1331483707064, 75.72628435371134] Fitness: 2.1494466754428134
Generation 325 Best Solution: [183.59236232581392, 177.44250723912145,
175.1331483707064, 74.65473627732344] Fitness: 2.368286210503714
Generation 326 Best Solution: [183.59236232581392, 177.44250723912145,
175.1331483707064, 74.65473627732344] Fitness: 2.368286210503714
Generation 327 Best Solution: [183.56247905620378, 177.44250723912145,
174.69280095596795, 74.68876539359717] Fitness: 1.758967233322625
Generation 328 Best Solution: [183.56247905620378, 177.44250723912145,
174.69280095596795, 74.68876539359717] Fitness: 1.758967233322625
Generation 329 Best Solution: [183.56247905620378, 177.44250723912145,
175.1331483707064, 74.44887581070155] Fitness: 2.393374959939332
Generation 330 Best Solution: [183.7360336940649, 177.44250723912145,
175.1331483707064, 74.68876539359717] Fitness: 2.480084722506392

Generation 331 Best Solution: [183.56247905620378, 177.44250723912145,
175.1331483707064, 74.68876539359717] Fitness: 2.3485999483199773
Generation 332 Best Solution: [183.56247905620378, 175.83861012387138,
175.1331483707064, 75.89330680142784] Fitness: 2.3456110351327966
Generation 333 Best Solution: [182.88133586075847, 176.57625333049847,
175.1331483707064, 75.89330680142784] Fitness: 2.339737761898156
Generation 334 Best Solution: [182.88133586075847, 176.57625333049847,
174.53889410979664, 75.89330680142784] Fitness: 1.6193409837859425
Generation 335 Best Solution: [182.88133586075847, 176.57625333049847,
174.34467583941793, 75.89330680142784] Fitness: 1.5100077587817975
Generation 336 Best Solution: [183.67841628853762, 176.57625333049847,
175.1331483707064, 75.89330680142784] Fitness: 2.458668261270378
Generation 337 Best Solution: [183.7360336940649, 177.44250723912145,
175.1331483707064, 74.62151577445421] Fitness: 2.483075604408007
Generation 338 Best Solution: [183.7360336940649, 177.44250723912145,
175.1331483707064, 74.55864213131011] Fitness: 2.4926094282763778
Generation 339 Best Solution: [183.7360336940649, 177.44250723912145,
175.1331483707064, 74.55864213131011] Fitness: 2.4926094282763778
Generation 340 Best Solution: [183.7360336940649, 177.44250723912145,
175.1331483707064, 75.17161689180546] Fitness: 2.6773537587815204
Generation 341 Best Solution: [183.7360336940649, 176.5296716089336,
175.1331483707064, 75.89330680142784] Fitness: 2.4793699812540084
Generation 342 Best Solution: [183.7360336940649, 176.5296716089336,
175.1331483707064, 75.90577190868336] Fitness: 2.504364062532328
Generation 343 Best Solution: [183.8121312607021, 177.44250723912145,
175.1331483707064, 74.29603661247226] Fitness: 2.6761155187862165
Generation 344 Best Solution: [183.27914316120382, 177.44250723912145,
175.1331483707064, 74.98197473410818] Fitness: 2.3145522668365475
Generation 345 Best Solution: [183.27914316120382, 177.44250723912145,
175.1331483707064, 74.98197473410818] Fitness: 2.3145522668365475
Generation 346 Best Solution: [183.01851974774118, 177.07210994223223,
175.1331483707064, 75.64799467286322] Fitness: 2.3702066045563694
Generation 347 Best Solution: [183.27914316120382, 177.44250723912145,
175.1331483707064, 74.7832775762971] Fitness: 2.2489409184620777
Generation 348 Best Solution: [183.27914316120382, 176.8925285468883,
174.08369387302994, 75.90577190868336] Fitness: 1.7021500099346007
Generation 349 Best Solution: [182.98781583555422, 176.8925285468883,
174.08369387302994, 75.64799467286322] Fitness: 1.264129066167648
Generation 350 Best Solution: [183.27914316120382, 177.44250723912145,
174.08369387302994, 75.64799467286322] Fitness: 2.1299506627876856
Generation 351 Best Solution: [182.98781583555422, 176.8925285468883,
175.1331483707064, 75.20596800973792] Fitness: 1.600666605784373
Generation 352 Best Solution: [183.27914316120382, 177.44250723912145,
175.1331483707064, 75.25122483782667] Fitness: 2.5072212108437544
Generation 353 Best Solution: [182.98781583555422, 177.44250723912145,
175.1331483707064, 74.77002393449898] Fitness: 2.2736555331983306
Generation 354 Best Solution: [183.27914316120382, 175.85551015012499,
175.1331483707064, 75.90577190868336] Fitness: 2.2532318149354165

Generation 355 Best Solution: [183.27914316120382, 177.44250723912145,
175.1331483707064, 74.77002393449898] Fitness: 2.24687785568232
Generation 356 Best Solution: [183.27914316120382, 177.44250723912145,
175.1331483707064, 75.57294131254528] Fitness: 2.8940062828270925
Generation 357 Best Solution: [183.31896700708842, 176.57510752110042,
175.1331483707064, 75.57294131254528] Fitness: 1.7201851182232477
Generation 358 Best Solution: [183.27914316120382, 177.44250723912145,
174.04378011460344, 75.57294131254528] Fitness: 2.0175775578088437
Generation 359 Best Solution: [183.27914316120382, 177.44250723912145,
175.1331483707064, 75.2010158029806] Fitness: 2.462235779347676
Generation 360 Best Solution: [183.7360336940649, 177.44250723912145,
175.1331483707064, 74.3372264112263] Fitness: 2.5780217197678437
Generation 361 Best Solution: [183.27914316120382, 177.44250723912145,
175.1331483707064, 74.48435392474413] Fitness: 2.272734519935859
Generation 362 Best Solution: [183.27914316120382, 177.44250723912145,
175.1331483707064, 74.48435392474413] Fitness: 2.272734519935859
Generation 363 Best Solution: [183.15831847788795, 177.44250723912145,
175.1331483707064, 74.73660733699096] Fitness: 2.2371008360822344
Generation 364 Best Solution: [183.27914316120382, 177.44250723912145,
174.5005450293333, 75.43946357208821] Fitness: 1.9659327289892263
Generation 365 Best Solution: [183.7360336940649, 177.44250723912145,
174.04378011460344, 74.93061092874952] Fitness: 1.6544616059977806
Generation 366 Best Solution: [183.7360336940649, 177.44250723912145,
174.04378011460344, 74.93061092874952] Fitness: 1.6544616059977806
Generation 367 Best Solution: [183.15831847788795, 177.44250723912145,
175.1331483707064, 74.50595503317815] Fitness: 2.260222058227886
Generation 368 Best Solution: [183.09170831521055, 176.5120171003734,
175.1331483707064, 75.90577190868336] Fitness: 2.256175997720966
Generation 369 Best Solution: [183.27914316120382, 176.5120171003734,
174.18099898294687, 75.90577190868336] Fitness: 1.3813557670280776
Generation 370 Best Solution: [183.38718941987403, 177.44250723912145,
174.64354130485896, 74.3126239667826] Fitness: 1.742840168249916
Generation 371 Best Solution: [183.27914316120382, 176.5120171003734,
175.1331483707064, 74.3126239667826] Fitness: 1.1343882524984505
Generation 372 Best Solution: [183.15831847788795, 176.36423962954987,
174.64354130485896, 75.89330680142784] Fitness: 1.5260936850331228
Generation 373 Best Solution: [183.15831847788795, 177.44250723912145,
175.1331483707064, 74.68404115866085] Fitness: 2.234660902892199
Generation 374 Best Solution: [183.7360336940649, 177.44250723912145,
175.1331483707064, 74.68404115866085] Fitness: 2.480051571790739
Generation 375 Best Solution: [183.15831847788795, 176.51464114484247,
175.1331483707064, 75.89330680142784] Fitness: 2.2255619177688355
Generation 376 Best Solution: [183.15831847788795, 177.44250723912145,
174.14817513877955, 74.45568300530564] Fitness: 1.3997715197557532
Generation 377 Best Solution: [182.98090576354053, 176.90274041860584,
175.1331483707064, 74.60583010280986] Fitness: 1.393858568898854
Generation 378 Best Solution: [182.98090576354053, 176.87827785520537,
175.1331483707064, 75.90577190868336] Fitness: 2.5935659294503717

Generation 379 Best Solution: [183.15831847788795, 176.6291408004218,
175.1331483707064, 75.89330680142784] Fitness: 2.29906360598142
Generation 380 Best Solution: [183.28221815784264, 177.44250723912145,
175.1331483707064, 74.66616146812416] Fitness: 2.241190890430037
Generation 381 Best Solution: [182.88080125791555, 175.9938498330215,
174.28035949463754, 75.89330680142784] Fitness: 1.389051074451012
Generation 382 Best Solution: [182.98090576354053, 175.97744559719033,
175.1331483707064, 75.89330680142784] Fitness: 2.203992933928768
Generation 383 Best Solution: [183.1652317604233, 176.49215425498858,
175.1331483707064, 75.90577190868336] Fitness: 2.238344357760293
Generation 384 Best Solution: [183.15831847788795, 177.44250723912145,
174.28035949463754, 75.90577190868336] Fitness: 2.6154596211822163
Generation 385 Best Solution: [182.98090576354053, 175.9822257685002,
175.1331483707064, 75.90577190868336] Fitness: 2.22739420564179
Generation 386 Best Solution: [182.98090576354053, 175.9822257685002,
175.1331483707064, 75.90577190868336] Fitness: 2.22739420564179
Generation 387 Best Solution: [182.98090576354053, 177.141516661943,
175.1331483707064, 74.3016799208431] Fitness: 1.8386185251599947
Generation 388 Best Solution: [183.1652317604233, 175.59812844360548,
175.1331483707064, 75.10419812344186] Fitness: 1.3537324193171978
Generation 389 Best Solution: [182.98090576354053, 177.44250723912145,
174.84463699646273, 75.04049883227668] Fitness: 1.9529869443991923
Generation 390 Best Solution: [183.1652317604233, 177.44250723912145,
174.84463699646273, 75.04049883227668] Fitness: 1.9177039704894248
Generation 391 Best Solution: [183.37422550618632, 177.44250723912145,
174.84463699646273, 75.04049883227668] Fitness: 1.9453944337273996
Generation 392 Best Solution: [182.98090576354053, 176.84803382710052,
175.1331483707064, 75.10419812344186] Fitness: 1.4732937038813443
Generation 393 Best Solution: [183.1652317604233, 176.8281463667215,
175.1331483707064, 75.10419812344186] Fitness: 1.4165246421618654
Generation 394 Best Solution: [183.1652317604233, 176.8281463667215,
175.1331483707064, 75.10419812344186] Fitness: 1.4165246421618654
Generation 395 Best Solution: [183.1652317604233, 176.8281463667215,
174.90923329724873, 75.90577190868336] Fitness: 2.1631367209997077
Generation 396 Best Solution: [182.98090576354053, 176.36458448143063,
175.1331483707064, 75.90577190868336] Fitness: 2.2218914677111803
Generation 397 Best Solution: [182.98090576354053, 176.84803382710052,
175.1331483707064, 75.01404939834927] Fitness: 1.4173194838806582
Generation 398 Best Solution: [183.1652317604233, 176.63066766504178,
174.0666098628433, 75.90577190868336] Fitness: 1.4467570016553344
Generation 399 Best Solution: [183.1652317604233, 176.3923946309916,
174.17063601741197, 75.90577190868336] Fitness: 1.320679467021893
Generation 400 Best Solution: [182.98090576354053, 176.74920860661004,
175.1331483707064, 74.5009708299241] Fitness: 1.253284062926533
Generation 401 Best Solution: [183.1652317604233, 176.3923946309916,
175.1331483707064, 74.81683861702048] Fitness: 0.9774991148157698
Generation 402 Best Solution: [183.1652317604233, 177.44250723912145,
173.96902684810112, 74.31309912803398] Fitness: 1.481599012099346

Generation 403 Best Solution: [183.1652317604233, 177.44250723912145,
175.1331483707064, 74.36076541471692] Fitness: 2.3194016510598643
Generation 404 Best Solution: [183.1652317604233, 176.74920860661004,
175.1331483707064, 75.90577190868336] Fitness: 2.424014414462839
Generation 405 Best Solution: [183.1652317604233, 177.44250723912145,
175.1331483707064, 74.36076541471692] Fitness: 2.3194016510598643
Generation 406 Best Solution: [183.1652317604233, 177.44250723912145,
175.1331483707064, 74.71647346389898] Fitness: 2.2353152875798736
Generation 407 Best Solution: [183.1652317604233, 177.44250723912145,
175.1331483707064, 74.71647346389898] Fitness: 2.2353152875798736
Generation 408 Best Solution: [183.1652317604233, 177.44250723912145,
174.64261090790765, 74.36549830441965] Fitness: 1.6803521776825427
Generation 409 Best Solution: [183.1652317604233, 177.44250723912145,
175.1331483707064, 75.59658094233906] Fitness: 2.92297751671863
Generation 410 Best Solution: [183.52491970551128, 177.44250723912145,
174.29322820823884, 75.90577190868336] Fitness: 2.711399808996834
Generation 411 Best Solution: [183.1652317604233, 176.21620001109918,
175.1331483707064, 75.90577190868336] Fitness: 2.160152564888462
Generation 412 Best Solution: [183.1652317604233, 176.21620001109918,
175.1331483707064, 75.59658094233906] Fitness: 1.6157411167924192
Generation 413 Best Solution: [183.1652317604233, 177.44250723912145,
174.75804468545385, 75.59658094233906] Fitness: 2.4005529008510176
Generation 414 Best Solution: [183.1652317604233, 176.02708313246646,
175.1331483707064, 75.90577190868336] Fitness: 2.178997905990558
Generation 415 Best Solution: [183.1652317604233, 176.43112890101878,
175.1331483707064, 75.90577190868336] Fitness: 2.210251285143244
Generation 416 Best Solution: [183.1652317604233, 176.43112890101878,
175.1331483707064, 75.90577190868336] Fitness: 2.210251285143244
Generation 417 Best Solution: [183.1652317604233, 176.43112890101878,
175.1331483707064, 75.90577190868336] Fitness: 2.210251285143244
Generation 418 Best Solution: [183.1652317604233, 176.43112890101878,
175.1331483707064, 75.90577190868336] Fitness: 2.210251285143244
Generation 419 Best Solution: [183.4305112284699, 177.44250723912145,
174.93330202105517, 74.70407937982552] Fitness: 1.9750062701749902
Generation 420 Best Solution: [183.1652317604233, 177.44250723912145,
174.93330202105517, 74.70407937982552] Fitness: 1.9275617887656875
Generation 421 Best Solution: [183.1652317604233, 177.44250723912145,
174.93330202105517, 74.70407937982552] Fitness: 1.9275617887656875
Generation 422 Best Solution: [183.1652317604233, 176.1572670734546,
175.1331483707064, 75.90577190868336] Fitness: 2.159706965376639
Generation 423 Best Solution: [183.4305112284699, 176.1572670734546,
173.80416087267452, 75.90577190868336] Fitness: 1.4001957566146244
Generation 424 Best Solution: [183.1652317604233, 176.1572670734546,
174.09553828379785, 75.90577190868336] Fitness: 1.2806917250386565
Generation 425 Best Solution: [183.1652317604233, 176.1572670734546,
175.1331483707064, 75.90577190868336] Fitness: 2.159706965376639
Generation 426 Best Solution: [183.75872681441209, 176.27161579414067,
175.1331483707064, 75.72364547031565] Fitness: 2.092614693836195

Generation 427 Best Solution: [183.4305112284699, 176.44757079379121,
174.34346521468086, 75.90577190868336] Fitness: 1.434444456037606
Generation 428 Best Solution: [183.1652317604233, 176.31911638061783,
175.1331483707064, 75.90577190868336] Fitness: 2.1746482113334977
Generation 429 Best Solution: [183.1652317604233, 176.3305454447207,
174.8703536491289, 75.90577190868336] Fitness: 1.787021787203727
Generation 430 Best Solution: [183.1652317604233, 176.31911638061783,
175.1331483707064, 75.72364547031565] Fitness: 1.8349090892514734
Generation 431 Best Solution: [183.1652317604233, 176.39720783947305,
175.1331483707064, 75.72364547031565] Fitness: 1.8575489014209765
Generation 432 Best Solution: [183.1652317604233, 177.44250723912145,
174.2379728345653, 75.90577190868336] Fitness: 2.60403445073147
Generation 433 Best Solution: [183.1652317604233, 176.58505949382183,
175.1331483707064, 75.90577190868336] Fitness: 2.2928906219048897
Generation 434 Best Solution: [183.1652317604233, 177.44250723912145,
174.646888307988, 75.72173789943741] Fitness: 2.4916557215171764
Generation 435 Best Solution: [183.1652317604233, 176.31877893101966,
175.1331483707064, 75.90577190868336] Fitness: 2.1745721753169382
Generation 436 Best Solution: [183.1652317604233, 176.02962013253799,
174.55764820751904, 75.90577190868336] Fitness: 1.4718051912604153
Generation 437 Best Solution: [183.1652317604233, 176.31877893101966,
175.1331483707064, 75.90577190868336] Fitness: 2.1745721753169382
Generation 438 Best Solution: [183.1652317604233, 176.31877893101966,
175.1331483707064, 75.90577190868336] Fitness: 2.1745721753169382
Generation 439 Best Solution: [183.1652317604233, 176.21638632696917,
175.1331483707064, 75.90577190868336] Fitness: 2.1601630447038724
Generation 440 Best Solution: [183.10724000214853, 176.21638632696917,
175.1331483707064, 75.90577190868336] Fitness: 2.165230143913963
Generation 441 Best Solution: [182.94690880302804, 175.6969743881061,
174.2379728345653, 75.90577190868336] Fitness: 1.5375951915327688
Generation 442 Best Solution: [183.1652317604233, 176.56546247227016,
175.1331483707064, 75.90577190868336] Fitness: 2.2802017723284793
Generation 443 Best Solution: [183.1652317604233, 176.56546247227016,
175.1331483707064, 74.55353407717551] Fitness: 1.0607788473373398
Generation 444 Best Solution: [183.1652317604233, 175.6969743881061,
174.2379728345653, 75.90577190868336] Fitness: 1.4896921537797385
Generation 445 Best Solution: [183.0002195264557, 175.6969743881061,
174.2379728345653, 75.90577190868336] Fitness: 1.5186535672843355
Generation 446 Best Solution: [183.1652317604233, 175.6969743881061,
175.1331483707064, 74.87937599107516] Fitness: 1.1520384294426862
Generation 447 Best Solution: [183.1652317604233, 177.44250723912145,
174.2379728345653, 74.94374802900732] Fitness: 1.4273502708733734
Generation 448 Best Solution: [183.0002195264557, 177.44250723912145,
174.8654040650088, 74.77536194538804] Fitness: 1.8738900237717488
Generation 449 Best Solution: [183.0002195264557, 177.44250723912145,
174.66420922300614, 74.45685489718113] Fitness: 1.6882282258406476
Generation 450 Best Solution: [182.94690880302804, 175.8443791559309,
175.1331483707064, 75.51317616030676] Fitness: 1.6367507865474438

Generation 451 Best Solution: [183.1652317604233, 175.8443791559309,
175.1331483707064, 75.45818172988012] Fitness: 1.5160642037618004
Generation 452 Best Solution: [183.1652317604233, 175.8443791559309,
175.1331483707064, 75.90577190868336] Fitness: 2.253149152752642
Generation 453 Best Solution: [182.94690880302804, 177.44250723912145,
175.1331483707064, 74.9464878737135] Fitness: 2.339794546577424
Generation 454 Best Solution: [183.06282487119807, 176.94235560366423,
175.1331483707064, 75.51317616030676] Fitness: 1.9835265699457434
Generation 455 Best Solution: [182.94690880302804, 175.61302983179544,
175.1331483707064, 75.90577190868336] Fitness: 2.4738330404287123
Generation 456 Best Solution: [183.0002195264557, 177.44250723912145,
174.21155474869337, 75.90577190868336] Fitness: 2.627567124488583
Generation 457 Best Solution: [183.0002195264557, 177.44250723912145,
175.1331483707064, 75.51317616030676] Fitness: 2.832048974361041
Generation 458 Best Solution: [182.94690880302804, 177.44250723912145,
175.1331483707064, 75.51317616030676] Fitness: 2.8509905986094743
Generation 459 Best Solution: [182.94690880302804, 175.71730233740217,
175.1331483707064, 75.90577190868336] Fitness: 2.385045924217997
Generation 460 Best Solution: [183.1652317604233, 177.44250723912145,
175.1331483707064, 74.80116395064775] Fitness: 2.2460108182824623
Generation 461 Best Solution: [182.94690880302804, 175.7843331749265,
175.1331483707064, 74.80116395064775] Fitness: 1.1160478766629658
Generation 462 Best Solution: [182.94690880302804, 177.44250723912145,
175.1331483707064, 74.72146411692987] Fitness: 2.283521030789017
Generation 463 Best Solution: [182.94690880302804, 176.77929116496713,
175.1331483707064, 75.90577190868336] Fitness: 2.5007595417723794
Generation 464 Best Solution: [183.06832340896293, 177.44250723912145,
175.1331483707064, 74.50700811742772] Fitness: 2.2711774183742293
Generation 465 Best Solution: [182.94690880302804, 177.44250723912145,
175.1331483707064, 74.72600495017032] Fitness: 2.283832095746027
Generation 466 Best Solution: [182.94690880302804, 176.10131378081323,
175.1331483707064, 75.90577190868336] Fitness: 2.212480796631242
Generation 467 Best Solution: [182.94690880302804, 176.10131378081323,
175.1331483707064, 75.90577190868336] Fitness: 2.212480796631242
Generation 468 Best Solution: [182.94690880302804, 176.10131378081323,
175.1331483707064, 75.90577190868336] Fitness: 2.212480796631242
Generation 469 Best Solution: [183.71012837041997, 176.10131378081323,
173.95592732243682, 75.90577190868336] Fitness: 1.5255682320495585
Generation 470 Best Solution: [183.22400034495953, 176.10131378081323,
175.1331483707064, 75.90577190868336] Fitness: 2.1650937106670165
Generation 471 Best Solution: [182.94690880302804, 176.10131378081323,
174.7209198624938, 75.90577190868336] Fitness: 1.6509539070102996
Generation 472 Best Solution: [183.79414567614464, 176.10131378081323,
174.7209198624938, 75.90577190868336] Fitness: 1.9038672063566797
Generation 473 Best Solution: [183.79414567614464, 176.15823617799336,
175.1331483707064, 75.90577190868336] Fitness: 2.4604843698709864
Generation 474 Best Solution: [183.06832340896293, 177.44250723912145,
175.1331483707064, 75.08549641945174] Fitness: 2.3799013518101453

Generation 475 Best Solution: [183.79414567614464, 177.44250723912145,
174.8728394991366, 74.67962710324045] Fitness: 2.148012979729761
Generation 476 Best Solution: [182.81019072888284, 177.44250723912145,
175.1331483707064, 74.67962710324045] Fitness: 2.352225296524633
Generation 477 Best Solution: [183.79414567614464, 177.44250723912145,
174.24397825344016, 74.68842245009084] Fitness: 1.6732334093950565
Generation 478 Best Solution: [183.79414567614464, 175.57235247310285,
174.41594252957677, 75.90577190868336] Fitness: 1.9694112363218483
Generation 479 Best Solution: [183.06832340896293, 176.89537826099323,
174.8728394991366, 75.90577190868336] Fitness: 2.2025507197125718
Generation 480 Best Solution: [183.6205322562303, 177.44250723912145,
174.8728394991366, 74.2944349721442] Fitness: 2.1236604586546117
Generation 481 Best Solution: [183.06832340896293, 177.44250723912145,
174.61495709283113, 74.2944349721442] Fitness: 1.7090613419262037
Generation 482 Best Solution: [183.6205322562303, 177.44250723912145,
174.8728394991366, 74.41033874941787] Fitness: 2.0607099271490013
Generation 483 Best Solution: [183.57848103526536, 177.44250723912145,
174.8728394991366, 74.2944349721442] Fitness: 2.095248214599674
Generation 484 Best Solution: [183.06832340896293, 176.89537826099323,
174.8728394991366, 74.41033874941787] Fitness: 1.0303467103497137
Generation 485 Best Solution: [183.57848103526536, 175.6812018571651,
175.1331483707064, 75.2059506271199] Fitness: 1.48303262078198
Generation 486 Best Solution: [183.14801775089506, 176.41950931158016,
175.1331483707064, 75.90577190868336] Fitness: 2.2065233616458366
Generation 487 Best Solution: [183.6205322562303, 175.69644048386039,
175.1331483707064, 74.76810268499473] Fitness: 1.2791926241882674
Generation 488 Best Solution: [183.62134375797515, 177.44250723912145,
174.8728394991366, 74.76810268499473] Fitness: 2.0065317324949077
Generation 489 Best Solution: [183.6205322562303, 176.65889632194947,
174.8728394991366, 75.90577190868336] Fitness: 2.1119055172075942
Generation 490 Best Solution: [183.6205322562303, 176.65889632194947,
174.8728394991366, 75.90577190868336] Fitness: 2.1119055172075942
Generation 491 Best Solution: [183.71012837041997, 177.44250723912145,
174.8728394991366, 74.82223297545384] Fitness: 2.08628155367539
Generation 492 Best Solution: [183.0696915716474, 175.728924981489,
174.8728394991366, 75.90577190868336] Fitness: 1.9525046100722059
Generation 493 Best Solution: [183.0696915716474, 175.728924981489,
174.24778021562136, 75.90577190868336] Fitness: 1.478611281239696
Generation 494 Best Solution: [183.0696915716474, 177.44250723912145,
175.1331483707064, 74.45782750121914] Fitness: 2.28708477960333
Generation 495 Best Solution: [183.71012837041997, 176.724373763781,
174.8728394991366, 75.90577190868336] Fitness: 2.2371017380226874
Generation 496 Best Solution: [182.87268600291506, 177.44250723912145,
175.1331483707064, 74.45782750121914] Fitness: 2.3578528181922853
Generation 497 Best Solution: [183.37010503801847, 177.44250723912145,
174.24778021562136, 74.84666226146199] Fitness: 1.4220375942249075
Generation 498 Best Solution: [183.71012837041997, 177.44250723912145,
174.1074539128224, 75.71274646328817] Fitness: 2.4529936181452667

```
Generation 499 Best Solution: [183.71012837041997, 177.44250723912145,
174.1074539128224, 75.40600678821203] Fitness: 2.0097990750773005
Generation 500 Best Solution: [183.6205322562303, 177.44250723912145,
174.55234910961946, 75.17307368178628] Fitness: 1.8749829848074613
Optimal Solution for Simulation 23: [183.6205322562303, 177.44250723912145,
174.55234910961946, 75.17307368178628]
--------------------------------------------------
Simulation 24/32 with conditions: {'drought': False, 'flood': True,
'policy_change': False, 'social_event': False, 'economic_crisis': False}
Generation 1 Best Solution: [182.94690713376212, 176.38638881847038,
173.98211902091677, 74.53488543189941] Fitness: 0.16282265559163345
Generation 2 Best Solution: [183.0539433762363, 176.20506745209835,
174.63857936743756, 74.8902428335024] Fitness: 0.33779545956226137
Generation 3 Best Solution: [183.08635397726428, 175.9335572139024,
174.78973149672782, 74.62390691002527] Fitness: 0.5020592004721548
Generation 4 Best Solution: [183.2509537383267, 176.11681729604513,
174.22347916301987, 75.71651645964964] Fitness: 0.8702091615273021
Generation 5 Best Solution: [183.2509537383267, 176.2460825580347,
173.81483788702104, 75.73274111046024] Fitness: 0.9387246048679742
Generation 6 Best Solution: [183.2509537383267, 176.16202501690344,
173.81483788702104, 75.73274111046024] Fitness: 0.9312850097957485
Generation 7 Best Solution: [183.0539433762363, 176.30014304729107,
174.55065806500815, 75.71651645964964] Fitness: 1.0493773863730615
Generation 8 Best Solution: [183.0539433762363, 176.30014304729107,
174.55065806500815, 75.71651645964964] Fitness: 1.0493773863730615
Generation 9 Best Solution: [183.2509537383267, 176.12080287340268,
174.55065806500815, 75.73274111046024] Fitness: 1.0572848512370285
Generation 10 Best Solution: [183.0539433762363, 175.68945445522874,
174.55065806500815, 75.71651645964964] Fitness: 1.1971009467340274
Generation 11 Best Solution: [183.66758089059542, 176.2460825580347,
174.89071162983447, 75.73274111046024] Fitness: 1.612413442124576
Generation 12 Best Solution: [183.0539433762363, 177.01498284435266,
174.55065806500815, 75.71651645964964] Fitness: 1.6245182381331773
Generation 13 Best Solution: [183.0539433762363, 177.01498284435266,
174.89071162983447, 75.64492762983205] Fitness: 1.8621845716904106
Generation 14 Best Solution: [183.0539433762363, 177.01498284435266,
174.43163296608523, 75.71651645964964] Fitness: 1.5462475494164696
Generation 15 Best Solution: [183.0539433762363, 175.86376827059215,
174.89071162983447, 75.64492762983205] Fitness: 1.3325491166380574
Generation 16 Best Solution: [183.0539433762363, 176.12020267299204,
174.89071162983447, 75.64492762983205] Fitness: 1.2693793919432745
Generation 17 Best Solution: [183.0539433762363, 176.12020267299204,
174.89071162983447, 75.64492762983205] Fitness: 1.2693793919432745
Generation 18 Best Solution: [183.0539433762363, 177.132599205016,
174.28742398459033, 75.64492762983205] Fitness: 1.544115600533696
Generation 19 Best Solution: [183.0539433762363, 177.132599205016,
174.5525274576704, 74.48441810988632] Fitness: 1.0473110196589681
Generation 20 Best Solution: [183.41691525021827, 177.132599205016,
```

174.5525274576704, 74.48441810988632] Fitness: 1.0943613328197501
Generation 21 Best Solution: [183.0539433762363, 177.132599205016,
174.28742398459033, 74.92199469591749] Fitness: 0.8948520101049834
Generation 22 Best Solution: [183.0539433762363, 177.132599205016,
174.28742398459033, 74.92199469591749] Fitness: 0.8948520101049834
Generation 23 Best Solution: [183.0539433762363, 177.01498284435266,
175.1064718754669, 74.92199469591749] Fitness: 1.5276783025741238
Generation 24 Best Solution: [183.0539433762363, 177.132599205016,
174.89071162983447, 74.92199469591749] Fitness: 1.3848496724061437
Generation 25 Best Solution: [183.0539433762363, 177.01498284435266,
175.06753820276606, 74.92199469591749] Fitness: 1.4654455906970858
Generation 26 Best Solution: [183.20201240493088, 177.01498284435266,
174.89071162983447, 74.92199469591749] Fitness: 1.2069930510826334
Generation 27 Best Solution: [183.0539433762363, 176.86927099716416,
174.89071162983447, 75.0156253634513] Fitness: 1.0677871584240461
Generation 28 Best Solution: [183.0539433762363, 177.01498284435266,
174.0808560295541, 74.92199469591749] Fitness: 0.6872175020259694
Generation 29 Best Solution: [183.78853166947727, 175.84868072441074,
174.89071162983447, 74.92199469591749] Fitness: 1.0009720383714846
Generation 30 Best Solution: [183.0539433762363, 176.86927099716416,
174.0808560295541, 75.80979371727453] Fitness: 1.4173945632431408
Generation 31 Best Solution: [183.0539433762363, 176.86927099716416,
174.0808560295541, 75.80979371727453] Fitness: 1.4173945632431408
Generation 32 Best Solution: [183.0539433762363, 176.86927099716416,
174.0808560295541, 75.73274111046024] Fitness: 1.2888463917688087
Generation 33 Best Solution: [183.2317456211364, 176.86927099716416,
174.0808560295541, 74.89988509126464] Fitness: 0.4947151404622725
Generation 34 Best Solution: [183.78853166947727, 175.98865877449506,
174.92505985275912, 74.88207710887731] Fitness: 0.9836652996045369
Generation 35 Best Solution: [183.6630740569263, 177.01498284435266,
174.92505985275912, 74.88207710887731] Fitness: 1.4436350143193764
Generation 36 Best Solution: [183.21491353119052, 177.01498284435266,
174.92505985275912, 74.88207710887731] Fitness: 1.2414954212765235
Generation 37 Best Solution: [183.21491353119052, 177.01498284435266,
174.92505985275912, 74.63419876020409] Fitness: 1.2244813379680803
Generation 38 Best Solution: [183.21491353119052, 177.01498284435266,
174.92505985275912, 74.88207710887731] Fitness: 1.2414954212765235
Generation 39 Best Solution: [183.21491353119052, 177.01498284435266,
175.06753820276606, 74.88207710887731] Fitness: 1.4488311972585453
Generation 40 Best Solution: [183.21491353119052, 177.01498284435266,
175.06753820276606, 74.88207710887731] Fitness: 1.4488311972585453
Generation 41 Best Solution: [183.10381052555886, 177.132599205016,
174.55068178839997, 75.2345257384578] Fitness: 1.2115631213028788
Generation 42 Best Solution: [183.21491353119052, 177.132599205016,
175.06753820276606, 74.87000428926942] Fitness: 1.6176837206409205
Generation 43 Best Solution: [183.0539433762363, 177.132599205016,
175.06753820276606, 74.87000428926942] Fitness: 1.6224886491700923
Generation 44 Best Solution: [183.02516806755642, 177.01498284435266,

175.06753820276606, 74.32403430648687] Fitness: 1.55714824929171
Generation 45 Best Solution: [183.21491353119052, 177.132599205016,
174.92505985275912, 74.40539435804946] Fitness: 1.4668107740390597
Generation 46 Best Solution: [183.78853166947727, 176.50004139090066,
174.55068178839997, 75.80979371727453] Fitness: 1.5941197791361787
Generation 47 Best Solution: [183.78853166947727, 176.58466432436657,
174.57605465569833, 74.32403430648687] Fitness: 0.8400401695626801
Generation 48 Best Solution: [183.6630740569263, 176.50004139090066,
174.55068178839997, 75.80979371727453] Fitness: 1.4807187362801666
Generation 49 Best Solution: [183.78853166947727, 176.58466432436657,
174.92505985275912, 74.67316715877489] Fitness: 1.0933879777807427
Generation 50 Best Solution: [182.8457067466042, 177.01498284435266,
174.55068178839997, 74.67316715877489] Fitness: 0.9004110325303702
Generation 51 Best Solution: [183.6630740569263, 176.50004139090066,
174.55068178839997, 74.67316715877489] Fitness: 0.5352600143187238
Generation 52 Best Solution: [183.45803631273398, 177.132599205016,
175.06753820276606, 74.67316715877489] Fitness: 1.670297576204253
Generation 53 Best Solution: [182.8314548285018, 177.132599205016,
174.56598418584144, 74.67316715877489] Fitness: 1.0911088001531815
Generation 54 Best Solution: [182.8314548285018, 177.132599205016,
174.43219997706265, 75.73274111046024] Fitness: 1.8183644796360718
Generation 55 Best Solution: [182.9709367777926, 177.132599205016,
174.56598418584144, 75.73274111046024] Fitness: 1.8524899282196658
Generation 56 Best Solution: [183.09303156579463, 175.9655362306323,
175.06753820276606, 75.80979371727453] Fitness: 1.8062693219496977
Generation 57 Best Solution: [182.8314548285018, 177.374432967493,
174.56598418584144, 74.69331327602825] Fitness: 1.5122424743942668
Generation 58 Best Solution: [183.78853166947727, 177.132599205016,
175.06753820276606, 75.23953863832065] Fitness: 2.1316671009861574
Generation 59 Best Solution: [182.8314548285018, 175.81058576471062,
175.06753820276606, 75.80979371727453] Fitness: 1.9487889070141762
Generation 60 Best Solution: [183.36251319731443, 177.132599205016,
174.04885936964715, 74.49063079697869] Fitness: 0.8920772874996661
Generation 61 Best Solution: [182.8314548285018, 177.132599205016,
174.69496305633854, 75.73274111046024] Fitness: 2.0212099083181068
Generation 62 Best Solution: [183.78853166947727, 177.132599205016,
174.46665548454635, 75.23953863832065] Fitness: 1.4747155321543248
Generation 63 Best Solution: [182.8314548285018, 177.132599205016,
173.9299031927467, 75.73274111046024] Fitness: 1.733990581219227
Generation 64 Best Solution: [183.78853166947727, 177.132599205016,
174.69496305633854, 74.45427274167183] Fitness: 1.4932103641886767
Generation 65 Best Solution: [182.8314548285018, 175.6640139310572,
174.69496305633854, 75.80979371727453] Fitness: 1.5693424906444666
Generation 66 Best Solution: [182.8314548285018, 177.132599205016,
174.05535710283442, 75.80979371727453] Fitness: 1.846288538909888
Generation 67 Best Solution: [182.8314548285018, 177.132599205016,
174.05535710283442, 75.80979371727453] Fitness: 1.846288538909888
Generation 68 Best Solution: [182.8314548285018, 177.132599205016,

173.8240749815799, 75.73274111046024] Fitness: 1.7670219464191568
Generation 69 Best Solution: [182.8314548285018, 177.132599205016,
173.79647436448798, 75.73274111046024] Fitness: 1.7785441288061516
Generation 70 Best Solution: [183.6630740569263, 176.30525105127413,
174.43219997706265, 75.80979371727453] Fitness: 1.3244301237618652
Generation 71 Best Solution: [182.8314548285018, 177.132599205016,
173.8240749815799, 75.73274111046024] Fitness: 1.7670219464191568
Generation 72 Best Solution: [182.8314548285018, 177.132599205016,
175.06753820276606, 74.78317395202477] Fitness: 1.681139343474381
Generation 73 Best Solution: [183.507949345283, 177.132599205016,
173.8240749815799, 74.368701449727] Fitness: 1.0607296587494353
Generation 74 Best Solution: [183.1439424385361, 177.132599205016,
174.0548399839469, 75.73274111046024] Fitness: 1.636031937270913
Generation 75 Best Solution: [183.6630740569263, 176.01323679759074,
174.87283815534923, 75.73274111046024] Fitness: 1.5927352824715912
Generation 76 Best Solution: [183.6630740569263, 177.132599205016,
175.06753820276606, 74.61431755122214] Fitness: 1.8087234462953168
Generation 77 Best Solution: [182.8314548285018, 175.94384575592642,
173.79647436448798, 75.73274111046024] Fitness: 1.046104676849676
Generation 78 Best Solution: [183.6630740569263, 177.1962337665953,
175.06753820276606, 74.78514810601791] Fitness: 1.906820385691345
Generation 79 Best Solution: [183.1439424385361, 175.95220560429368,
175.06753820276606, 75.80979371727453] Fitness: 1.8074490356299084
Generation 80 Best Solution: [183.6630740569263, 176.496916572986,
175.06753820276606, 75.80979371727453] Fitness: 2.078360677128553
Generation 81 Best Solution: [183.1439424385361, 177.1962337665953,
175.06753820276606, 74.89530900246513] Fitness: 1.7235966518537522
Generation 82 Best Solution: [183.70031270686366, 176.496916572986,
174.87283815534923, 74.6324638732881] Fitness: 0.8926412378080848
Generation 83 Best Solution: [182.84419536449704, 176.496916572986,
174.87283815534923, 74.6324638732881] Fitness: 0.7318884867885559
Generation 84 Best Solution: [183.48962866631578, 176.496916572986,
174.87283815534923, 74.6324638732881] Fitness: 0.7457265924803078
Generation 85 Best Solution: [183.6630740569263, 177.1962337665953,
174.09122832743938, 75.7409874334156] Fitness: 1.9564365049373562
Generation 86 Best Solution: [183.6630740569263, 177.1962337665953,
173.95844644354042, 75.80979371727453] Fitness: 2.082122130491774
Generation 87 Best Solution: [183.15300355843857, 177.30868138038468,
173.95844644354042, 75.80979371727453] Fitness: 2.0730840311261414
Generation 88 Best Solution: [183.70031270686366, 177.1962337665953,
173.95844644354042, 75.80979371727453] Fitness: 2.1131885750537887
Generation 89 Best Solution: [183.6630740569263, 176.94032735555402,
174.3773713018067, 75.80979371727453] Fitness: 1.7719470500710417
Generation 90 Best Solution: [183.6630740569263, 176.58088251082492,
174.98058143220393, 75.7409874334156] Fitness: 1.8845948043255758
Generation 91 Best Solution: [183.70031270686366, 176.58088251082492,
174.98058143220393, 75.7409874334156] Fitness: 1.9156612488875906
Generation 92 Best Solution: [183.70031270686366, 176.58088251082492,

174.98058143220393, 75.80979371727453] Fitness: 2.030899886292564
Generation 93 Best Solution: [183.69539408418325, 177.30868138038468,
174.74156913311617, 75.50258181679605] Fitness: 2.1888118220606807
Generation 94 Best Solution: [183.70031270686366, 177.30868138038468,
173.82585937500264, 75.50258181679605] Fitness: 1.8907876742325804
Generation 95 Best Solution: [183.02720342834982, 177.30868138038468,
173.82585937500264, 75.80979371727453] Fitness: 2.1238523776798335
Generation 96 Best Solution: [183.69539408418325, 176.0876373141777,
175.06753820276606, 75.7409874334156] Fitness: 1.8967234591380138
Generation 97 Best Solution: [183.69539408418325, 176.0876373141777,
175.06753820276606, 75.80979371727453] Fitness: 2.011962096542987
Generation 98 Best Solution: [183.69539408418325, 176.25561741090132,
175.06753820276606, 75.50258181679605] Fitness: 1.5615238440059507
Generation 99 Best Solution: [183.70031270686366, 175.92767800781326,
174.98058143220393, 75.80979371727453] Fitness: 1.92128502739497
Generation 100 Best Solution: [183.69539408418325, 175.7597114343764,
173.82585937500264, 75.7409874334156] Fitness: 1.2834883571984066
Generation 101 Best Solution: [183.70031270686366, 177.30868138038468,
173.82585937500264, 74.34884846171681] Fitness: 1.5063239968136164
Generation 102 Best Solution: [183.70031270686366, 177.30868138038468,
174.98058143220393, 74.77173429343195] Fitness: 2.002279456538729
Generation 103 Best Solution: [183.70031270686366, 177.30868138038468,
174.0189500857533, 74.87538787190266] Fitness: 1.3746457987161964
Generation 104 Best Solution: [183.06564713080178, 175.76201441060687,
174.74156913311617, 75.80979371727453] Fitness: 1.4736715016287563
Generation 105 Best Solution: [183.70031270686366, 177.2103452389449,
174.74156913311617, 75.23351768867578] Fitness: 1.7428766772943824
Generation 106 Best Solution: [183.06564713080178, 176.47617363280864,
174.74156913311617, 75.80979371727453] Fitness: 1.44107654745901
Generation 107 Best Solution: [183.06564713080178, 177.30868138038468,
174.22430413686615, 75.53102133592228] Fitness: 1.6654824647809823
Generation 108 Best Solution: [183.70031270686366, 177.1962337665953,
174.74156913311617, 74.94990576713425] Fitness: 1.5505195778763952
Generation 109 Best Solution: [183.70031270686366, 177.1962337665953,
174.74156913311617, 74.94990576713425] Fitness: 1.5505195778763952
Generation 110 Best Solution: [183.70031270686366, 175.9637299403935,
174.27681908115724, 75.80979371727453] Fitness: 1.2941253254677385
Generation 111 Best Solution: [183.70031270686366, 175.9637299403935,
174.27681908115724, 75.80979371727453] Fitness: 1.2941253254677385
Generation 112 Best Solution: [183.70031270686366, 177.30868138038468,
174.27681908115724, 74.94711495251796] Fitness: 1.4267211890412403
Generation 113 Best Solution: [183.5104750657877, 177.1962337665953,
173.7474545955851, 75.80979371727453] Fitness: 2.051549928625504
Generation 114 Best Solution: [183.25013507041564, 177.1962337665953,
173.7474545955851, 75.80979371727453] Fitness: 1.958250212237334
Generation 115 Best Solution: [183.25013507041564, 177.1962337665953,
173.7474545955851, 75.80979371727453] Fitness: 1.958250212237334
Generation 116 Best Solution: [183.70031270686366, 177.30868138038468,

174.74156913311617, 75.2495354929084] Fitness: 1.9287836285902369
Generation 117 Best Solution: [183.70031270686366, 177.30868138038468,
174.74156913311617, 74.67798565475807] Fitness: 1.7039895512345251
Generation 118 Best Solution: [183.70031270686366, 177.1962337665953,
175.06753820276606, 74.32514722730842] Fitness: 2.0537482199993615
Generation 119 Best Solution: [183.25013507041564, 177.1962337665953,
175.06753820276606, 75.2495354929084] Fitness: 1.931019332274443
Generation 120 Best Solution: [183.70031270686366, 177.30868138038468,
174.74156913311617, 75.2495354929084] Fitness: 1.9287836285902369
Generation 121 Best Solution: [183.70031270686366, 177.1962337665953,
175.06753820276606, 75.2495354929084] Fitness: 2.1598213260366617
Generation 122 Best Solution: [183.70031270686366, 177.1962337665953,
175.06753820276606, 75.2495354929084] Fitness: 2.1598213260366617
Generation 123 Best Solution: [183.70031270686366, 177.1962337665953,
175.06753820276606, 74.77799101097482] Fitness: 1.9371307823236494
Generation 124 Best Solution: [183.29793393078356, 177.1962337665953,
175.06753820276606, 74.77799101097482] Fitness: 1.7174383331515446
Generation 125 Best Solution: [183.70031270686366, 177.1962337665953,
175.05556119670425, 74.77799101097482] Fitness: 1.9184677758983426
Generation 126 Best Solution: [182.9984528261059, 177.1962337665953,
175.06753820276606, 74.73153329022725] Fitness: 1.7169276614233902
Generation 127 Best Solution: [183.57573201295364, 177.1962337665953,
175.05556119670425, 74.73153329022725] Fitness: 1.820185186334364
Generation 128 Best Solution: [183.70031270686366, 177.1962337665953,
174.72105696820915, 74.41857795792306] Fitness: 1.5547769025303992
Generation 129 Best Solution: [183.29793393078356, 177.1962337665953,
175.06753820276606, 74.41857795792306] Fitness: 1.7834845575173826
Generation 130 Best Solution: [183.29793393078356, 177.1962337665953,
175.06753820276606, 74.41857795792306] Fitness: 1.7834845575173826
Generation 131 Best Solution: [183.29793393078356, 177.1962337665953,
174.70631592085766, 75.41247249011943] Fitness: 1.6354198235090762
Generation 132 Best Solution: [182.87658558434788, 177.1962337665953,
175.06753820276606, 74.41857795792306] Fitness: 1.827442439453343
Generation 133 Best Solution: [183.70031270686366, 177.1962337665953,
175.05556119670425, 74.8394095598794] Fitness: 1.9275868418128372
Generation 134 Best Solution: [183.70051704664982, 176.08488936261926,
175.06753820276606, 75.80979371727453] Fitness: 2.016635352241224
Generation 135 Best Solution: [183.70031270686366, 177.1962337665953,
174.01300602830497, 75.66010822346242] Fitness: 1.8643870285298005
Generation 136 Best Solution: [183.26246075673353, 177.1962337665953,
175.06753820276606, 74.85923073434752] Fitness: 1.7236659059178985
Generation 137 Best Solution: [183.70031270686366, 175.64371688744214,
174.88073295316812, 74.85923073434752] Fitness: 1.0184559372979116
Generation 138 Best Solution: [183.70031270686366, 177.20911614105606,
174.11830136841766, 75.66010822346242] Fitness: 1.884341282413636
Generation 139 Best Solution: [183.70031270686366, 177.20911614105606,
174.11830136841766, 75.66010822346242] Fitness: 1.884341282413636
Generation 140 Best Solution: [182.8089947947865, 176.73149607381052,

175.06753820276606, 75.78568777618302] Fitness: 2.098288001189637
Generation 141 Best Solution: [183.70031270686366, 177.20911614105606,
174.11830136841766, 75.80979371727453] Fitness: 2.1254809031952493
Generation 142 Best Solution: [183.70031270686366, 177.1962337665953,
175.06753820276606, 74.85647141897444] Fitness: 1.9498402153723426
Generation 143 Best Solution: [183.70031270686366, 177.20911614105606,
174.11830136841766, 75.80979371727453] Fitness: 2.1254809031952493
Generation 144 Best Solution: [183.70031270686366, 177.1962337665953,
175.06753820276606, 74.41419119719619] Fitness: 2.005243046932684
Generation 145 Best Solution: [183.70031270686366, 177.1962337665953,
174.37343979054984, 75.80979371727453] Fitness: 2.1729977113364694
Generation 146 Best Solution: [183.70031270686366, 177.10674881154455,
174.42387122012096, 75.80979371727453] Fitness: 2.057011154604221
Generation 147 Best Solution: [183.34890239538302, 176.19363504969365,
174.6348635342996, 75.80979371727453] Fitness: 1.2784486341608226
Generation 148 Best Solution: [183.70031270686366, 177.20911614105606,
174.37343979054984, 74.88124329100359] Fitness: 1.269538198321887
Generation 149 Best Solution: [183.70031270686366, 177.20911614105606,
174.37343979054984, 75.80979371727453] Fitness: 2.194450423477889
Generation 150 Best Solution: [183.34890239538302, 177.1962337665953,
174.6348635342996, 75.81823674090481] Fitness: 2.1589289947799726
Generation 151 Best Solution: [183.70031270686366, 176.75122650026623,
173.79407526057824, 75.81823674090481] Fitness: 1.599143493380847
Generation 152 Best Solution: [183.70031270686366, 176.75122650026623,
173.79407526057824, 75.80979371727453] Fitness: 1.5844879788169792
Generation 153 Best Solution: [183.86096830193293, 176.75122650026623,
173.79407526057824, 75.80979371727453] Fitness: 1.743614937014869
Generation 154 Best Solution: [183.70031270686366, 176.75122650026623,
173.79407526057824, 75.80979371727453] Fitness: 1.5844879788169792
Generation 155 Best Solution: [183.70031270686366, 175.75777218977436,
174.11532373284183, 75.81823674090481] Fitness: 1.3712358563619127
Generation 156 Best Solution: [183.70031270686366, 177.20911614105606,
174.30900358594153, 75.01437775697799] Fitness: 1.2914282336327643
Generation 157 Best Solution: [183.70031270686366, 175.8011703902854,
174.96336643624164, 75.81823674090481] Fitness: 1.968194265083436
Generation 158 Best Solution: [183.70031270686366, 176.1725588836687,
173.83429052497402, 75.80979371727453] Fitness: 1.2811485686601158
Generation 159 Best Solution: [183.70031270686366, 176.1725588836687,
173.83429052497402, 75.80979371727453] Fitness: 1.2811485686601158
Generation 160 Best Solution: [183.70031270686366, 177.20911614105606,
174.11532373284183, 75.51398033085559] Fitness: 1.6828609715787592
Generation 161 Best Solution: [183.70031270686366, 177.20911614105606,
174.80271810613544, 75.44768718073513] Fitness: 2.019826215080333
Generation 162 Best Solution: [183.70031270686366, 177.20911614105606,
175.06753820276606, 74.57686048807716] Fitness: 1.9704102604719518
Generation 163 Best Solution: [183.70031270686366, 175.8664570632212,
175.06753820276606, 74.57686048807716] Fitness: 1.1440673345037973
Generation 164 Best Solution: [183.70031270686366, 175.8664570632212,

175.06753820276606, 74.57686048807716] Fitness: 1.1440673345037973
Generation 165 Best Solution: [183.70031270686366, 176.01393810328872,
174.60511479900003, 75.81823674090481] Fitness: 1.4860388082111056
Generation 166 Best Solution: [183.70031270686366, 177.20911614105606,
175.06753820276606, 74.57686048807716] Fitness: 1.9704102604719518
Generation 167 Best Solution: [183.70031270686366, 177.20911614105606,
174.4529228483517, 74.57686048807716] Fitness: 1.3051080381981974
Generation 168 Best Solution: [183.70031270686366, 177.20911614105606,
174.4529228483517, 74.57686048807716] Fitness: 1.3051080381981974
Generation 169 Best Solution: [183.70031270686366, 177.20911614105606,
174.89559573622535, 75.05253517574509] Fitness: 1.7993275525400176
Generation 170 Best Solution: [183.70031270686366, 176.7424376299141,
174.6296667752055, 75.81823674090481] Fitness: 1.7718546527560084
Generation 171 Best Solution: [183.70031270686366, 177.20911614105606,
174.18751563041775, 75.80979371727453] Fitness: 2.134031614844824
Generation 172 Best Solution: [183.70031270686366, 177.20911614105606,
175.06753820276606, 74.86781443160405] Fitness: 1.973934230246589
Generation 173 Best Solution: [183.70031270686366, 175.58081193337034,
175.06753820276606, 75.81823674090481] Fitness: 2.2819454215355846
Generation 174 Best Solution: [183.70031270686366, 175.58081193337034,
175.06753820276606, 74.46199789591441] Fitness: 1.3708727365035867
Generation 175 Best Solution: [183.70031270686366, 177.20911614105606,
175.06753820276606, 74.46199789591441] Fitness: 2.005818916937226
Generation 176 Best Solution: [183.70031270686366, 177.18646245018886,
174.86317988695495, 75.81823674090481] Fitness: 2.591941081192049
Generation 177 Best Solution: [183.70031270686366, 177.18646245018886,
174.86317988695495, 74.31551163809954] Fitness: 1.7562498928313417
Generation 178 Best Solution: [183.56668574104938, 177.20911614105606,
174.20222155407117, 75.81823674090481] Fitness: 2.0501680282861496
Generation 179 Best Solution: [183.56668574104938, 177.20911614105606,
174.86317988695495, 74.33234822919363] Fitness: 1.682102234717568
Generation 180 Best Solution: [182.97960288455553, 177.18646245018886,
174.86317988695495, 74.31551163809954] Fitness: 1.5438781826457402
Generation 181 Best Solution: [183.70031270686366, 177.20911614105606,
174.86317988695495, 74.31551163809954] Fitness: 1.7937997910082815
Generation 182 Best Solution: [183.70031270686366, 177.20911614105606,
174.8898967248768, 74.31551163809954] Fitness: 1.8276263298818713
Generation 183 Best Solution: [182.99146096357208, 175.99386444151688,
175.06753820276606, 75.81823674090481] Fitness: 1.8312125503405636
Generation 184 Best Solution: [182.99146096357208, 175.99386444151688,
174.36417752543548, 75.81823674090481] Fitness: 1.1191251688790025
Generation 185 Best Solution: [182.99146096357208, 175.99386444151688,
174.36417752543548, 75.81823674090481] Fitness: 1.1191251688790025
Generation 186 Best Solution: [183.01494900802876, 175.99386444151688,
175.06753820276606, 75.81823674090481] Fitness: 1.8256448324967827
Generation 187 Best Solution: [183.70031270686366, 176.67131539744142,
175.06753820276606, 74.88178724097556] Fitness: 1.3047499582794224
Generation 188 Best Solution: [183.70031270686366, 177.20911614105606,

174.86317988695495, 74.71631800005777] Fitness: 1.6680192568542382
Generation 189 Best Solution: [183.8541956381273, 177.20911614105606,
174.32877445122872, 75.32002743203806] Fitness: 1.669044098039651
Generation 190 Best Solution: [183.53993475046283, 177.20911614105606,
174.32877445122872, 75.32002743203806] Fitness: 1.399243586549203
Generation 191 Best Solution: [183.01494900802876, 176.38533707404673,
175.06753820276606, 75.81823674090481] Fitness: 1.851358871703224
Generation 192 Best Solution: [183.70031270686366, 177.25389527171157,
175.1329186462169, 75.15304695316091] Fitness: 2.2896157748976775
Generation 193 Best Solution: [183.01494900802876, 176.67131539744142,
175.06753820276606, 75.81823674090481] Fitness: 2.023093175768124
Generation 194 Best Solution: [183.31447264860847, 176.82477120120444,
174.60968606070443, 75.81823674090481] Fitness: 1.6215887882559779
Generation 195 Best Solution: [183.01494900802876, 177.20911614105606,
175.1329186462169, 75.57290349315174] Fitness: 2.4217105701501467
Generation 196 Best Solution: [183.31447264860847, 175.76688074964233,
175.1329186462169, 75.81823674090481] Fitness: 2.032813097023552
Generation 197 Best Solution: [183.31447264860847, 175.76688074964233,
175.1329186462169, 75.81823674090481] Fitness: 2.032813097023552
Generation 198 Best Solution: [183.31447264860847, 175.76688074964233,
175.06753820276606, 75.81823674090481] Fitness: 1.926942019371757
Generation 199 Best Solution: [183.31447264860847, 175.64443047499705,
175.06753820276606, 75.81823674090481] Fitness: 2.012466006360046
Generation 200 Best Solution: [183.70031270686366, 177.20911614105606,
174.3378265073991, 75.55051075395363] Fitness: 1.7833871104507062
Generation 201 Best Solution: [183.31447264860847, 177.20911614105606,
175.1329186462169, 74.38424030134718] Fitness: 1.9317843517294178
Generation 202 Best Solution: [183.70031270686366, 176.0668107904577,
175.1329186462169, 75.81823674090481] Fitness: 2.1390449608865993
Generation 203 Best Solution: [183.70031270686366, 176.0668107904577,
175.07406962227202, 75.78005800298568] Fitness: 1.9780719131500613
Generation 204 Best Solution: [183.31447264860847, 177.20911614105606,
175.06753820276606, 75.81823674090481] Fitness: 2.7011911629111016
Generation 205 Best Solution: [183.57413106851277, 177.20911614105606,
174.9326730108407, 75.81823674090481] Fitness: 2.623416837477923
Generation 206 Best Solution: [183.31447264860847, 177.20911614105606,
175.06753820276606, 74.42450204525866] Fitness: 1.8061874250923085
Generation 207 Best Solution: [183.70031270686366, 177.20911614105606,
174.9326730108407, 74.76676702071063] Fitness: 1.760493079009241
Generation 208 Best Solution: [183.31447264860847, 177.20911614105606,
174.1425963002511, 75.81823674090481] Fitness: 1.9265758060078524
Generation 209 Best Solution: [183.31447264860847, 176.08895112923003,
175.1329186462169, 75.81823674090481] Fitness: 1.92089327029672
Generation 210 Best Solution: [183.31447264860847, 177.20911614105606,
174.9326730108407, 75.81823674090481] Fitness: 2.504123530895182
Generation 211 Best Solution: [183.31447264860847, 177.20911614105606,
174.8720806948279, 75.81823674090481] Fitness: 2.424934777473174
Generation 212 Best Solution: [183.70031270686366, 177.20911614105606,

175.06753820276606, 75.07167094622116] Fitness: 2.056037770547834
Generation 213 Best Solution: [183.31447264860847, 177.20911614105606,
175.06753820276606, 75.07167094622116] Fitness: 1.8403373314897125
Generation 214 Best Solution: [183.70031270686366, 177.20911614105606,
174.8720806948279, 74.90424463061764] Fitness: 1.7075348710011473
Generation 215 Best Solution: [183.15530876255525, 177.20911614105606,
175.06753820276606, 75.07167094622116] Fitness: 1.8198410341686089
Generation 216 Best Solution: [183.70031270686366, 177.20911614105606,
174.02239444367703, 75.07167094622116] Fitness: 1.2799365848372384
Generation 217 Best Solution: [183.70031270686366, 177.20911614105606,
174.02239444367703, 75.07167094622116] Fitness: 1.2799365848372384
Generation 218 Best Solution: [183.58544729851727, 177.20911614105606,
174.9326730108407, 75.02840177341415] Fitness: 1.747270432288481
Generation 219 Best Solution: [183.70031270686366, 177.20911614105606,
174.7394203249021, 75.02840177341415] Fitness: 1.6037334551741527
Generation 220 Best Solution: [183.58544729851727, 177.20911614105606,
174.9326730108407, 75.02840177341415] Fitness: 1.747270432288481
Generation 221 Best Solution: [183.70031270686366, 177.20911614105606,
175.06753820276606, 75.02840177341415] Fitness: 2.0331253981154354
Generation 222 Best Solution: [183.70031270686366, 177.20911614105606,
174.9326730108407, 75.04255764326936] Fitness: 1.8432283782034238
Generation 223 Best Solution: [183.15530876255525, 176.07022345862362,
174.9326730108407, 75.81823674090481] Fitness: 1.5994812274520576
Generation 224 Best Solution: [182.96240611773752, 177.20911614105606,
174.9326730108407, 74.64897215413774] Fitness: 1.5543363069085414
Generation 225 Best Solution: [183.15530876255525, 175.82927517345144,
175.06753820276606, 75.72622459724948] Fitness: 1.7183269090792561
Generation 226 Best Solution: [183.58544729851727, 176.17185904105818,
174.9326730108407, 75.72622459724948] Fitness: 1.5889189997224247
Generation 227 Best Solution: [183.15530876255525, 176.17185904105818,
174.9326730108407, 75.72622459724948] Fitness: 1.441509597154234
Generation 228 Best Solution: [183.58544729851727, 177.20911614105606,
174.05266055993843, 75.33851833295846] Fitness: 1.3960103531461079
Generation 229 Best Solution: [183.7414856176197, 177.20911614105606,
174.30359501828505, 75.04414721662386] Fitness: 1.3411348926019355
Generation 230 Best Solution: [183.58544729851727, 176.17185904105818,
174.43085049558061, 75.81823674090481] Fitness: 1.2615317590639392
Generation 231 Best Solution: [183.353961244083, 177.20911614105606,
174.9326730108407, 75.33851833295846] Fitness: 1.861183798599522
Generation 232 Best Solution: [183.7414856176197, 177.20911614105606,
174.30359501828505, 75.45348993484241] Fitness: 1.6863736206001958
Generation 233 Best Solution: [183.58544729851727, 177.20911614105606,
174.30359501828505, 75.45348993484241] Fitness: 1.5606889122283851
Generation 234 Best Solution: [183.353961244083, 177.20911614105606,
174.30359501828505, 75.81823674090481] Fitness: 1.9755858860343478
Generation 235 Best Solution: [183.353961244083, 177.20911614105606,
173.85022467989114, 75.81823674090481] Fitness: 1.9739801086025328
Generation 236 Best Solution: [183.58544729851727, 177.20911614105606,

174.30359501828505, 75.81823674090481] Fitness: 2.091220823141092
Generation 237 Best Solution: [183.58544729851727, 176.55643147986854,
175.06753820276606, 75.88735604396393] Fitness: 2.195090275376434
Generation 238 Best Solution: [183.58544729851727, 176.55643147986854,
174.9326730108407, 75.45348993484241] Fitness: 1.3432801007068544
Generation 239 Best Solution: [183.58544729851727, 176.55643147986854,
175.06753820276606, 75.81823674090481] Fitness: 2.0708796436354806
Generation 240 Best Solution: [183.78292639802774, 176.55643147986854,
175.06753820276606, 74.89718670483809] Fitness: 1.3012309097810835
Generation 241 Best Solution: [183.6639353977647, 177.20911614105606,
175.06753820276606, 75.0253638341308] Fitness: 2.0012551711141215
Generation 242 Best Solution: [183.7414856176197, 177.20911614105606,
175.06753820276606, 75.0253638341308] Fitness: 2.0685251510954616
Generation 243 Best Solution: [183.6639353977647, 177.41973803029617,
175.06753820276606, 74.77392362391589] Fitness: 2.3157252842752456
Generation 244 Best Solution: [183.78292639802774, 177.41973803029617,
175.06753820276606, 74.77392362391589] Fitness: 2.4228355149934457
Generation 245 Best Solution: [183.78292639802774, 177.20911614105606,
174.40995931523153, 75.81823674090481] Fitness: 2.304124442781551
Generation 246 Best Solution: [183.78292639802774, 177.41973803029617,
174.9326730108407, 74.81556280895646] Fitness: 2.231032602356362
Generation 247 Best Solution: [183.4147816821585, 177.41973803029617,
175.06753820276606, 74.81556280895646] Fitness: 2.169127454366046
Generation 248 Best Solution: [183.70777586371796, 175.64002440480982,
174.9326730108407, 74.81556280895646] Fitness: 1.0876699812999662
Generation 249 Best Solution: [183.6907382285409, 177.11741739863146,
175.06753820276606, 74.53879343480814] Fitness: 1.8246426802972529
Generation 250 Best Solution: [183.82911019209166, 175.89763009053397,
175.06753820276606, 74.72822700599863] Fitness: 1.2404620192619558
Generation 251 Best Solution: [183.7414856176197, 177.20911614105606,
174.9326730108407, 74.37074506782538] Fitness: 1.8886280477210984
Generation 252 Best Solution: [183.7414856176197, 177.41973803029617,
175.06753820276606, 74.72822700599863] Fitness: 2.38036828727266
Generation 253 Best Solution: [183.82911019209166, 177.41973803029617,
175.06753820276606, 74.72822700599863] Fitness: 2.4678034872637142
Generation 254 Best Solution: [183.4186678850166, 177.41973803029617,
175.06753820276606, 74.72822700599863] Fitness: 2.16285195685609
Generation 255 Best Solution: [183.82911019209166, 177.41973803029617,
175.06753820276606, 74.72822700599863] Fitness: 2.4678034872637142
Generation 256 Best Solution: [183.82911019209166, 176.5776542212016,
175.06753820276606, 75.81823674090481] Fitness: 2.298041506773683
Generation 257 Best Solution: [183.82911019209166, 176.5776542212016,
174.60781289459288, 75.81823674090481] Fitness: 1.7441870434754616
Generation 258 Best Solution: [182.88854663434583, 177.41973803029617,
175.06753820276606, 74.68882500854575] Fitness: 2.16300470433139
Generation 259 Best Solution: [183.7414856176197, 176.5776542212016,
175.06753820276606, 75.81823674090481] Fitness: 2.21060630678263
Generation 260 Best Solution: [183.7414856176197, 176.5776542212016,

175.06753820276606, 75.81823674090481] Fitness: 2.21060630678263
Generation 261 Best Solution: [183.6907382285409, 176.70617551237058,
175.06753820276606, 75.81823674090481] Fitness: 2.2657416210159855
Generation 262 Best Solution: [182.94236651048425, 177.41973803029617,
174.22819612944855, 75.81823674090481] Fitness: 2.3440874916374
Generation 263 Best Solution: [183.6907382285409, 177.20911614105606,
175.06753820276606, 74.3661162894576] Fitness: 2.043132162251359
Generation 264 Best Solution: [183.6907382285409, 177.20911614105606,
175.06753820276606, 75.30429683614477] Fitness: 2.2216924553878323
Generation 265 Best Solution: [183.82911019209166, 176.21652779582405,
175.06753820276606, 75.81823674090481] Fitness: 2.1560099086190556
Generation 266 Best Solution: [183.78292639802774, 175.63688663464347,
174.30532657477306, 75.81823674090481] Fitness: 1.5746920010413268
Generation 267 Best Solution: [183.82911019209166, 175.63688663464347,
174.30532657477306, 75.81823674090481] Fitness: 1.6222869502955202
Generation 268 Best Solution: [183.82911019209166, 175.63688663464347,
174.30532657477306, 75.81823674090481] Fitness: 1.6222869502955202
Generation 269 Best Solution: [182.8165645488788, 175.63688663464347,
174.30532657477306, 75.81823674090481] Fitness: 1.3512913066242302
Generation 270 Best Solution: [183.82911019209166, 176.2521826115362,
175.06753820276606, 75.81823674090481] Fitness: 2.160871454208003
Generation 271 Best Solution: [183.82911019209166, 176.2521826115362,
175.06753820276606, 75.81823674090481] Fitness: 2.160871454208003
Generation 272 Best Solution: [183.82911019209166, 176.2521826115362,
175.06753820276606, 75.81823674090481] Fitness: 2.160871454208003
Generation 273 Best Solution: [183.82911019209166, 176.8680225015513,
175.06753820276606, 75.81823674090481] Fitness: 2.5615912327819785
Generation 274 Best Solution: [183.82911019209166, 176.35000799033475,
175.06753820276606, 75.81823674090481] Fitness: 2.1845187265337245
Generation 275 Best Solution: [183.82911019209166, 175.90968115912824,
173.811609593456, 75.81823674090481] Fitness: 1.473513814685171
Generation 276 Best Solution: [183.82911019209166, 175.90968115912824,
175.08201402431425, 75.81823674090481] Fitness: 2.2200004003641762
Generation 277 Best Solution: [183.82911019209166, 176.99870471770896,
175.06753820276606, 74.74671881012371] Fitness: 1.7629806105227284
Generation 278 Best Solution: [183.82911019209166, 177.41973803029617,
175.06753820276606, 74.95790314818187] Fitness: 2.514366726582477
Generation 279 Best Solution: [183.82911019209166, 176.86746755367713,
175.08201402431425, 75.81823674090481] Fitness: 2.583819630006041
Generation 280 Best Solution: [183.82911019209166, 176.86746755367713,
174.41728140357102, 75.81823674090481] Fitness: 1.8753744923622093
Generation 281 Best Solution: [183.82911019209166, 176.86746755367713,
174.41728140357102, 75.81823674090481] Fitness: 1.8753744923622093
Generation 282 Best Solution: [183.82911019209166, 176.6261667141121,
175.06753820276606, 75.81823674090481] Fitness: 2.3328104462645154
Generation 283 Best Solution: [183.82911019209166, 176.6261667141121,
175.06753820276606, 75.81823674090481] Fitness: 2.3328104462645154
Generation 284 Best Solution: [183.0443692771912, 176.6261667141121,

175.06753820276606, 75.81823674090481] Fitness: 1.9816516062889862
Generation 285 Best Solution: [183.0443692771912, 176.6261667141121,
175.06753820276606, 74.81373379091488] Fitness: 1.0279258586727962
Generation 286 Best Solution: [183.0443692771912, 176.6261667141121,
175.06753820276606, 75.81823674090481] Fitness: 1.9816516062889862
Generation 287 Best Solution: [183.00463996308986, 176.6261667141121,
175.06753820276606, 75.81823674090481] Fitness: 1.9897330953601062
Generation 288 Best Solution: [183.0443692771912, 176.58946973050996,
175.06753820276606, 75.81823674090481] Fitness: 1.9550085403456594
Generation 289 Best Solution: [183.0443692771912, 176.58946973050996,
175.06753820276606, 75.81823674090481] Fitness: 1.9550085403456594
Generation 290 Best Solution: [183.29001510457002, 176.6261667141121,
173.86324979097915, 75.81823674090481] Fitness: 1.244055776699427
Generation 291 Best Solution: [183.0443692771912, 176.58946973050996,
175.06753820276606, 75.81823674090481] Fitness: 1.9550085403456594
Generation 292 Best Solution: [183.0443692771912, 176.58946973050996,
175.06753820276606, 75.81823674090481] Fitness: 1.9550085403456594
Generation 293 Best Solution: [183.0443692771912, 176.58946973050996,
175.06753820276606, 75.81823674090481] Fitness: 1.9550085403456594
Generation 294 Best Solution: [183.0443692771912, 176.58946973050996,
175.06753820276606, 75.81823674090481] Fitness: 1.9550085403456594
Generation 295 Best Solution: [183.57993891133626, 176.58946973050996,
174.5195811499448, 75.81823674090481] Fitness: 1.467330248677146
Generation 296 Best Solution: [183.0443692771912, 176.58946973050996,
175.06753820276606, 75.81823674090481] Fitness: 1.9550085403456594
Generation 297 Best Solution: [183.1740526641003, 177.41973803029617,
174.08087491420935, 75.42508846015203] Fitness: 1.7276836121306847
Generation 298 Best Solution: [183.82911019209166, 177.41973803029617,
174.13259757268978, 74.64088009123031] Fitness: 1.6963477072017015
Generation 299 Best Solution: [182.9236882117486, 177.41973803029617,
174.13259757268978, 74.64088009123031] Fitness: 1.3774496959130533
Generation 300 Best Solution: [182.94652810754928, 177.41973803029617,
175.06753820276606, 74.38059134859816] Fitness: 2.2290007035148323
Generation 301 Best Solution: [183.63951880447246, 176.83250739464683,
174.08087491420935, 75.81823674090481] Fitness: 1.5700632658667124
Generation 302 Best Solution: [183.1740526641003, 177.41973803029617,
173.7742149824393, 74.28439715754888] Fitness: 1.5466299790004259
Generation 303 Best Solution: [183.1740526641003, 177.41973803029617,
173.7742149824393, 74.38059134859816] Fitness: 1.4885835061052535
Generation 304 Best Solution: [182.9236882117486, 177.03718676325045,
175.06753820276606, 75.21570809098712] Fitness: 1.6943218772807156
Generation 305 Best Solution: [182.9236882117486, 175.8988309397241,
175.06753820276606, 74.86592724135919] Fitness: 0.9390082245677258
Generation 306 Best Solution: [183.0443692771912, 177.41973803029617,
173.75517608589368, 75.29259505190925] Fitness: 1.682511236503391
Generation 307 Best Solution: [183.0443692771912, 177.18637834670994,
174.83469394093152, 74.86592724135919] Fitness: 1.386741033312992
Generation 308 Best Solution: [183.0443692771912, 175.71872740096933,

175.06753820276606, 75.81823674090481] Fitness: 1.9466238887577725
Generation 309 Best Solution: [183.0443692771912, 175.71872740096933,
173.95246195546932, 75.81823674090481] Fitness: 1.180155550001821
Generation 310 Best Solution: [182.9236882117486, 176.4475349588605,
175.06753820276606, 75.81823674090481] Fitness: 1.904237002155981
Generation 311 Best Solution: [183.40138548919896, 176.4134048739715,
175.06753820276606, 75.81823674090481] Fitness: 1.895956643506134
Generation 312 Best Solution: [182.9236882117486, 176.4475349588605,
175.06753820276606, 75.40335434320396] Fitness: 1.3172031656246799
Generation 313 Best Solution: [183.0443692771912, 176.2075267561058,
175.06753820276606, 75.81823674090481] Fitness: 1.803941102242456
Generation 314 Best Solution: [182.9236882117486, 177.41973803029617,
175.06753820276606, 74.79733194776385] Fitness: 2.1541552122088126
Generation 315 Best Solution: [182.9236882117486, 175.66804241265413,
175.06753820276606, 75.77796831493093] Fitness: 1.9463803551971608
Generation 316 Best Solution: [182.9236882117486, 175.66804241265413,
175.06753820276606, 75.81823674090481] Fitness: 2.0152670764228007
Generation 317 Best Solution: [182.883661232588, 175.66804241265413,
175.06753820276606, 75.81823674090481] Fitness: 2.0310456238328594
Generation 318 Best Solution: [182.9236882117486, 175.609498221179,
175.06753820276606, 75.81823674090481] Fitness: 2.062339350219526
Generation 319 Best Solution: [182.9236882117486, 176.45515491298053,
175.06753820276606, 74.75764193266953] Fitness: 0.9478708814106237
Generation 320 Best Solution: [182.9236882117486, 177.29607268390615,
173.81661490890684, 75.81823674090481] Fitness: 2.1483000430284567
Generation 321 Best Solution: [182.9236882117486, 177.41973803029617,
174.1729296337999, 75.81823674090481] Fitness: 2.33959869923663
Generation 322 Best Solution: [182.9236882117486, 176.17004374033166,
174.3352520071507, 75.81823674090481] Fitness: 1.109138765882608
Generation 323 Best Solution: [183.82911019209166, 176.65440824130252,
175.06753820276606, 75.81823674090481] Fitness: 2.3547624728758514
Generation 324 Best Solution: [182.9236882117486, 177.41973803029617,
175.06753820276606, 74.97939859734814] Fitness: 2.204088985701021
Generation 325 Best Solution: [183.81735359370663, 176.65440824130252,
175.06753820276606, 75.81823674090481] Fitness: 2.342327112365018
Generation 326 Best Solution: [182.9236882117486, 177.00798100112135,
175.06753820276606, 74.53565672814044] Fitness: 1.481097881140688
Generation 327 Best Solution: [182.9236882117486, 176.4116593821693,
175.06753820276606, 75.81823674090481] Fitness: 1.888285748124136
Generation 328 Best Solution: [182.9236882117486, 176.4116593821693,
175.06753820276606, 75.81823674090481] Fitness: 1.888285748124136
Generation 329 Best Solution: [183.40138548919896, 177.41973803029617,
174.58827032942546, 75.81823674090481] Fitness: 2.5471719375403223
Generation 330 Best Solution: [183.40138548919896, 177.41973803029617,
175.06753820276606, 74.57635802802028] Fitness: 2.1708042024619165
Generation 331 Best Solution: [183.72703329490645, 177.41973803029617,
175.06753820276606, 74.42607461686721] Fitness: 2.4327183068534795
Generation 332 Best Solution: [183.66806878251361, 176.41837680116743,

175.06753820276606, 75.36125528298443] Fitness: 1.4272433604214116
Generation 333 Best Solution: [183.74300887129402, 176.41837680116743,
175.06753820276606, 75.36125528298443] Fitness: 1.4925838794422568
Generation 334 Best Solution: [183.74300887129402, 176.41837680116743,
175.06753820276606, 75.81823674090481] Fitness: 2.123997092601936
Generation 335 Best Solution: [183.40138548919896, 176.7259521607941,
175.06753820276606, 75.54930611610752] Fitness: 1.6925448447016695
Generation 336 Best Solution: [183.66806878251361, 176.41837680116743,
175.06753820276606, 75.81823674090481] Fitness: 2.058656573581091
Generation 337 Best Solution: [183.66806878251361, 176.41837680116743,
175.06753820276606, 75.81823674090481] Fitness: 2.058656573581091
Generation 338 Best Solution: [183.66806878251361, 176.14803088456048,
174.82007972538366, 75.81823674090481] Fitness: 1.661348516149872
Generation 339 Best Solution: [183.66806878251361, 176.81402468326252,
174.77047547456684, 75.22689585547839] Fitness: 1.2007774476683764
Generation 340 Best Solution: [183.66806878251361, 177.41973803029617,
174.02234299298104, 74.56688502756211] Fitness: 1.5574437210753784
Generation 341 Best Solution: [183.66806878251361, 176.81402468326252,
174.77047547456684, 75.49079612353586] Fitness: 1.46919707289155
Generation 342 Best Solution: [183.81735359370663, 176.53660842196405,
174.58033912405392, 75.7719595055785] Fitness: 1.603758047248881
Generation 343 Best Solution: [183.81735359370663, 176.7023359347474,
174.48109453781154, 75.49079612353586] Fitness: 1.2486998286857296
Generation 344 Best Solution: [183.81735359370663, 176.53660842196405,
174.58033912405392, 75.81823674090481] Fitness: 1.6827044379381555
Generation 345 Best Solution: [183.81735359370663, 176.53660842196405,
174.82007972538366, 75.81823674090481] Fitness: 1.9194960121058644
Generation 346 Best Solution: [183.81735359370663, 176.53660842196405,
174.82007972538366, 75.81823674090481] Fitness: 1.9194960121058644
Generation 347 Best Solution: [183.66806878251361, 176.75408281490735,
174.65418543891158, 75.81823674090481] Fitness: 1.7778256926108602
Generation 348 Best Solution: [183.4534755534837, 176.75408281490735,
174.65418543891158, 75.81823674090481] Fitness: 1.6397726160376793
Generation 349 Best Solution: [183.4534755534837, 176.75408281490735,
174.82007972538366, 75.81823674090481] Fitness: 1.8132978429231332
Generation 350 Best Solution: [183.05583116648216, 176.52891740979817,
174.82007972538366, 75.81823674090481] Fitness: 1.5742318019640806
Generation 351 Best Solution: [183.4534755534837, 176.81939878917942,
174.82007972538366, 75.81823674090481] Fitness: 1.8791714920696365
Generation 352 Best Solution: [183.81735359370663, 176.81939878917942,
174.82007972538366, 75.81823674090481] Fitness: 2.156148508282338
Generation 353 Best Solution: [183.81735359370663, 176.81939878917942,
174.2906517037641, 75.81823674090481] Fitness: 1.7543134357645611
Generation 354 Best Solution: [183.81735359370663, 176.81939878917942,
174.2906517037641, 75.81823674090481] Fitness: 1.7543134357645611
Generation 355 Best Solution: [183.81735359370663, 176.54420149983687,
173.83453501524676, 75.81823674090481] Fitness: 1.531060515863305
Generation 356 Best Solution: [183.4534755534837, 176.47639856386363,

174.65418543891158, 75.81823674090481] Fitness: 1.4349114883179497
Generation 357 Best Solution: [183.4534755534837, 176.47639856386363,
174.65418543891158, 75.81823674090481] Fitness: 1.4349114883179497
Generation 358 Best Solution: [183.4534755534837, 176.47639856386363,
174.65418543891158, 75.81823674090481] Fitness: 1.4349114883179497
Generation 359 Best Solution: [183.4534755534837, 177.41973803029617,
174.82007972538366, 74.45267632276068] Fitness: 1.8928398987559467
Generation 360 Best Solution: [183.81735359370663, 175.77177901577193,
174.82007972538366, 75.81823674090481] Fitness: 1.9120163200154594
Generation 361 Best Solution: [183.4534755534837, 177.41973803029617,
174.82007972538366, 74.59967698847089] Fitness: 1.8490901770124673
Generation 362 Best Solution: [183.81735359370663, 177.41973803029617,
174.82007972538366, 74.59967698847089] Fitness: 2.1260671932251682
Generation 363 Best Solution: [183.81735359370663, 177.41973803029617,
174.72488529981896, 74.96307493057088] Fitness: 2.0594574687069747
Generation 364 Best Solution: [183.81735359370663, 176.33616950271042,
174.72488529981896, 75.7719595055785] Fitness: 1.6443929626704157
Generation 365 Best Solution: [183.81735359370663, 175.91855032114157,
174.82007972538366, 75.81823674090481] Fitness: 1.8418359985586419
Generation 366 Best Solution: [183.59641929718023, 175.91855032114157,
174.82007972538366, 75.7719595055785] Fitness: 1.569786133005091
Generation 367 Best Solution: [183.4534755534837, 177.41973803029617,
174.44505323539917, 74.96307493057088] Fitness: 1.5570082993882175
Generation 368 Best Solution: [183.4534755534837, 177.41973803029617,
173.81121340967704, 74.96307493057088] Fitness: 1.5034988849048005
Generation 369 Best Solution: [183.4534755534837, 177.11873146650913,
174.03183459519678, 74.96307493057088] Fitness: 0.9172585665605929
Generation 370 Best Solution: [183.81735359370663, 177.11873146650913,
174.03183459519678, 74.96307493057088] Fitness: 1.1942355827732947
Generation 371 Best Solution: [183.4534755534837, 177.09198554122474,
174.72488529981896, 74.82862004889876] Fitness: 1.1706934462463845
Generation 372 Best Solution: [183.4534755534837, 177.09198554122474,
174.82007972538366, 75.19400039294386] Fitness: 1.446718555482205
Generation 373 Best Solution: [183.4534755534837, 176.77167144213556,
174.03183459519678, 74.96307493057088] Fitness: 0.4804005305062304
Generation 374 Best Solution: [183.4534755534837, 176.3359555729825,
174.82007972538366, 75.7719595055785] Fitness: 1.4722390197779354
Generation 375 Best Solution: [183.4534755534837, 176.3359555729825,
174.82007972538366, 75.81823674090481] Fitness: 1.5511854104672098
Generation 376 Best Solution: [183.4534755534837, 176.40992884500383,
174.82007972538366, 75.7719595055785] Fitness: 1.4985122480120372
Generation 377 Best Solution: [183.81735359370663, 177.11873146650913,
174.72488529981896, 75.19400039294386] Fitness: 1.659238614449283
Generation 378 Best Solution: [182.85850786894454, 175.96725686087845,
174.85045325317844, 75.81823674090481] Fitness: 1.5830656894924926
Generation 379 Best Solution: [182.85850786894454, 175.96725686087845,
174.85045325317844, 75.7719595055785] Fitness: 1.5041192988032182
Generation 380 Best Solution: [182.85850786894454, 177.11873146650913,

174.69567218808427, 75.7719595055785] Fitness: 2.05158339950139
Generation 381 Best Solution: [182.85850786894454, 176.0254107982626,
174.72488529981896, 75.7719595055785] Fitness: 1.3488294091689652
Generation 382 Best Solution: [182.85850786894454, 176.0254107982626,
174.72488529981896, 75.7719595055785] Fitness: 1.3488294091689652
Generation 383 Best Solution: [182.85850786894454, 176.7125168876031,
174.6958300199888, 75.7719595055785] Fitness: 1.5593906022887745
Generation 384 Best Solution: [182.85850786894454, 176.1880700476275,
174.6958300199888, 75.81823674090481] Fitness: 1.3880156448157677
Generation 385 Best Solution: [182.85850786894454, 176.98295981160274,
174.69567218808427, 75.7719595055785] Fitness: 1.8580346944387949
Generation 386 Best Solution: [182.85850786894454, 177.41973803029617,
174.6958300199888, 74.77255753845441] Fitness: 1.7047496868687257
Generation 387 Best Solution: [182.85850786894454, 176.64318289103545,
174.72488529981896, 75.7719595055785] Fitness: 1.5305488695834002
Generation 388 Best Solution: [183.81735359370663, 176.98295981160274,
174.72488529981896, 75.7719595055785] Fitness: 2.1668272720715183
Generation 389 Best Solution: [183.81735359370663, 177.41973803029617,
174.6958300199888, 75.0868999946617] Fitness: 2.0909658572645795
Generation 390 Best Solution: [183.4534755534837, 176.89663570748883,
174.51865859753158, 75.0868999946617] Fitness: 0.8306383631480443
Generation 391 Best Solution: [183.4534755534837, 175.65307976801626,
174.6958300199888, 75.7719595055785] Fitness: 1.503677458211793
Generation 392 Best Solution: [183.34799462888375, 177.41973803029617,
173.75092351726585, 74.33224801856474] Fitness: 1.556983384384495
Generation 393 Best Solution: [182.85850786894454, 176.41780105241466,
174.6958300199888, 75.7719595055785] Fitness: 1.3652695166269053
Generation 394 Best Solution: [182.81677920671572, 177.41973803029617,
174.6958300199888, 74.42791999924793] Fitness: 1.787817258392946
Generation 395 Best Solution: [183.81735359370663, 177.41973803029617,
173.82992621700643, 74.42791999924793] Fitness: 1.7891658391027305
Generation 396 Best Solution: [183.81735359370663, 176.55946027581487,
174.6958300199888, 75.7719595055785] Fitness: 1.7209335711719729
Generation 397 Best Solution: [183.81735359370663, 176.55946027581487,
174.6958300199888, 75.7719595055785] Fitness: 1.7209335711719729
Generation 398 Best Solution: [183.01912822665537, 177.41973803029617,
173.82992621700643, 74.80243366332374] Fitness: 1.3964559698122105
Generation 399 Best Solution: [182.85850786894454, 176.46391682556916,
174.6958300199888, 75.7719595055785] Fitness: 1.386593858306091
Generation 400 Best Solution: [182.81677920671572, 176.71460065618126,
174.72488529981896, 75.7719595055785] Fitness: 1.6112147909271513
Generation 401 Best Solution: [182.81677920671572, 176.71460065618126,
174.6958300199888, 75.7719595055785] Fitness: 1.5820513493050006
Generation 402 Best Solution: [183.72356643925272, 177.41973803029617,
174.6958300199888, 74.4464311631004] Fitness: 1.9469939023725542
Generation 403 Best Solution: [183.72356643925272, 177.14543014744467,
174.50922097213555, 74.4464311631004] Fitness: 1.3000540188628702
Generation 404 Best Solution: [183.64507018860076, 175.98036688116153,

174.72488529981896, 75.7719595055785] Fitness: 1.4814261174782843
Generation 405 Best Solution: [183.81735359370663, 177.41973803029617,
174.6958300199888, 75.23283115299293] Fitness: 2.1935475291747095
Generation 406 Best Solution: [183.64507018860076, 176.32143704402037,
174.61576148524279, 75.7719595055785] Fitness: 1.3803568102402515
Generation 407 Best Solution: [183.64507018860076, 176.32143704402037,
174.61576148524279, 75.7719595055785] Fitness: 1.3803568102402515
Generation 408 Best Solution: [183.72356643925272, 176.70982268710242,
174.72488529981896, 75.7719595055785] Fitness: 1.7742454828490581
Generation 409 Best Solution: [183.64507018860076, 175.65612660233555,
174.6958300199888, 75.7719595055785] Fitness: 1.6210826049075495
Generation 410 Best Solution: [183.64507018860076, 175.65612660233555,
174.6958300199888, 75.7719595055785] Fitness: 1.6210826049075495
Generation 411 Best Solution: [183.64507018860076, 175.65612660233555,
174.6958300199888, 74.98881087309098] Fitness: 0.7978880512510618
Generation 412 Best Solution: [183.64507018860076, 175.74286270385423,
174.10844971815868, 75.7719595055785] Fitness: 1.2559464252675974
Generation 413 Best Solution: [183.81735359370663, 175.74286270385423,
174.10844971815868, 75.7719595055785] Fitness: 1.4131446658133284
Generation 414 Best Solution: [183.64507018860076, 175.74286270385423,
175.01062403263285, 75.7719595055785] Fitness: 1.946637132903049
Generation 415 Best Solution: [183.64507018860076, 177.41973803029617,
174.10844971815868, 75.67344139758143] Fitness: 2.2463902882147493
Generation 416 Best Solution: [183.64507018860076, 176.14737002668252,
174.6958300199888, 75.7719595055785] Fitness: 1.4300780976994505
Generation 417 Best Solution: [183.64507018860076, 175.96558046202153,
174.72488529981896, 75.7719595055785] Fitness: 1.4855124532452033
Generation 418 Best Solution: [183.64507018860076, 177.41973803029617,
174.10844971815868, 75.7719595055785] Fitness: 2.403194890246506
Generation 419 Best Solution: [183.81735359370663, 176.51775126076882,
174.72488529981896, 75.7719595055785] Fitness: 1.724373146688714
Generation 420 Best Solution: [183.81735359370663, 176.51775126076882,
174.72488529981896, 75.7719595055785] Fitness: 1.724373146688714
Generation 421 Best Solution: [183.7431928530168, 175.97552059311832,
175.01062403263285, 75.7719595055785] Fitness: 1.9243282133263733
Generation 422 Best Solution: [183.64507018860076, 177.41973803029617,
174.72488529981896, 75.67344139758143] Fitness: 2.579266322159747
Generation 423 Best Solution: [183.7431928530168, 175.64067859048163,
175.01062403263285, 75.67344139758143] Fitness: 1.9472282429452363
Generation 424 Best Solution: [183.7431928530168, 175.64067859048163,
175.01062403263285, 75.7719595055785] Fitness: 2.104032844976993
Generation 425 Best Solution: [183.81735359370663, 176.19811756549097,
174.72488529981896, 75.67344139758143] Fitness: 1.4616178411284926
Generation 426 Best Solution: [183.6573316347965, 177.41973803029617,
174.38898360220554, 75.67344139758143] Fitness: 2.3331242311999283
Generation 427 Best Solution: [183.64507018860076, 175.58654799023216,
175.01062403263285, 75.67344139758143] Fitness: 1.9091157918271113
Generation 428 Best Solution: [183.64507018860076, 175.58654799023216,

175.01062403263285, 75.67344139758143] Fitness: 1.9091157918271113
Generation 429 Best Solution: [183.64507018860076, 175.58654799023216,
175.01062403263285, 75.7719595055785] Fitness: 2.0659203938588684
Generation 430 Best Solution: [183.64507018860076, 175.58654799023216,
175.01062403263285, 75.67344139758143] Fitness: 1.9091157918271113
Generation 431 Best Solution: [183.64507018860076, 176.27078897942766,
175.01062403263285, 75.00345112434418] Fitness: 1.0122707280523446
Generation 432 Best Solution: [183.64507018860076, 175.96855032077693,
175.01062403263285, 75.00345112434418] Fitness: 1.0257905074505365
Generation 433 Best Solution: [183.64507018860076, 175.96855032077693,
175.01062403263285, 75.00345112434418] Fitness: 1.0257905074505365
Generation 434 Best Solution: [183.64507018860076, 177.41973803029617,
173.99387956169173, 75.00345112434418] Fitness: 1.5907582409391219
Generation 435 Best Solution: [183.64507018860076, 175.58654799023216,
175.01062403263285, 75.00345112434418] Fitness: 1.249232222107705
Generation 436 Best Solution: [183.77318625892033, 175.58654799023216,
174.54676156711193, 75.7719595055785] Fitness: 1.6627124348987699
Generation 437 Best Solution: [183.64507018860076, 175.58654799023216,
174.54676156711193, 75.7719595055785] Fitness: 1.5502815023747138
Generation 438 Best Solution: [183.5279111387659, 175.58654799023216,
175.01062403263285, 75.67344139758143] Fitness: 1.8289868951121435
Generation 439 Best Solution: [183.01576737305095, 175.58654799023216,
175.01062403263285, 75.67344139758143] Fitness: 1.7331573117745709
Generation 440 Best Solution: [183.64507018860076, 176.5559200773765,
175.01062403263285, 75.7719595055785] Fitness: 1.9484233651124245
Generation 441 Best Solution: [183.64507018860076, 177.41973803029617,
175.01062403263285, 74.68955178484173] Fitness: 2.211856079519991
Generation 442 Best Solution: [183.64507018860076, 176.617536814945,
175.01062403263285, 74.68955178484173] Fitness: 1.1090775554834933
Generation 443 Best Solution: [183.77318625892033, 177.41973803029617,
175.01062403263285, 74.68955178484173] Fitness: 2.324287012044047
Generation 444 Best Solution: [183.64507018860076, 176.1522133359783,
175.01062403263285, 75.7719595055785] Fitness: 1.8170699145912805
Generation 445 Best Solution: [183.64507018860076, 177.41973803029617,
174.77656019135415, 75.7719595055785] Fitness: 2.791231647165011
Generation 446 Best Solution: [183.77318625892033, 176.96917117865678,
174.82456192286733, 75.7719595055785] Fitness: 2.2142095761856586
Generation 447 Best Solution: [183.64507018860076, 175.97736768373645,
175.01062403263285, 75.7719595055785] Fitness: 1.8400417318176485
Generation 448 Best Solution: [183.64507018860076, 175.97736768373645,
175.01062403263285, 75.7719595055785] Fitness: 1.8400417318176485
Generation 449 Best Solution: [183.77318625892033, 175.65346171680534,
175.01062403263285, 75.67344139758143] Fitness: 1.9657623198296772
Generation 450 Best Solution: [183.77318625892033, 177.41973803029617,
175.01062403263285, 75.16506175211406] Fitness: 2.4839383086541136
Generation 451 Best Solution: [183.64507018860076, 177.41973803029617,
175.01062403263285, 74.4635806678203] Fitness: 2.261129833357568
Generation 452 Best Solution: [183.09487865559322, 177.41973803029617,

174.45785642612842, 75.7719595055785] Fitness: 2.330014557385698
Generation 453 Best Solution: [183.77318625892033, 176.18538911321156,
175.01062403263285, 75.7719595055785] Fitness: 1.9305904736311368
Generation 454 Best Solution: [183.77318625892033, 176.18538911321156,
175.01062403263285, 74.3645645823787] Fitness: 1.144829938532611
Generation 455 Best Solution: [183.77318625892033, 176.5211196933295,
175.01062403263285, 75.67344139758143] Fitness: 1.8825914187811303
Generation 456 Best Solution: [183.00502935929907, 176.64367654875988,
175.01062403263285, 75.05280905699597] Fitness: 1.0452319290122367
Generation 457 Best Solution: [183.64507018860076, 176.64367654875988,
175.01062403263285, 75.05280905699597] Fitness: 1.2187669812074564
Generation 458 Best Solution: [183.67627272890226, 176.18538911321156,
175.01062403263285, 75.7719595055785] Fitness: 1.8431546542200048
Generation 459 Best Solution: [183.64507018860076, 176.64367654875988,
175.01062403263285, 75.05280905699597] Fitness: 1.2187669812074564
Generation 460 Best Solution: [183.64507018860076, 176.64367654875988,
175.01062403263285, 75.05280905699597] Fitness: 1.2187669812074564
Generation 461 Best Solution: [183.64507018860076, 176.64367654875988,
175.01062403263285, 75.67344139758143] Fitness: 1.8542212247625063
Generation 462 Best Solution: [183.77318625892033, 175.68639020619852,
175.01062403263285, 75.7719595055785] Fitness: 2.0977107283258953
Generation 463 Best Solution: [183.35757731776508, 177.41973803029617,
173.9002214924094, 75.67344139758143] Fitness: 2.1115484155319555
Generation 464 Best Solution: [183.64507018860076, 177.41973803029617,
175.01062403263285, 74.6139944520173] Fitness: 2.2193593459778813
Generation 465 Best Solution: [183.77318625892033, 177.41973803029617,
175.01062403263285, 74.55238998042317] Fitness: 2.34457882699859
Generation 466 Best Solution: [183.77318625892033, 176.2611333307516,
175.01062403263285, 75.7719595055785] Fitness: 1.9395914247418227
Generation 467 Best Solution: [183.77318625892033, 176.28787782301174,
175.01062403263285, 75.7719595055785] Fitness: 1.944933520670945
Generation 468 Best Solution: [183.77318625892033, 176.28787782301174,
175.01062403263285, 75.7719595055785] Fitness: 1.944933520670945
Generation 469 Best Solution: [183.77318625892033, 176.88515485187935,
175.01062403263285, 74.99603553151496] Fitness: 1.5384585952792695
Generation 470 Best Solution: [183.77318625892033, 176.88515485187935,
175.01062403263285, 74.97986392837005] Fitness: 1.5314805554179178
Generation 471 Best Solution: [183.77318625892033, 176.88515485187935,
175.01062403263285, 74.97986392837005] Fitness: 1.5314805554179178
Generation 472 Best Solution: [183.77318625892033, 175.76206339957645,
175.01062403263285, 75.7719595055785] Fitness: 2.0470766160730385
Generation 473 Best Solution: [183.77318625892033, 176.88515485187935,
175.01062403263285, 75.53374810265585] Fitness: 2.0056106694972327
Generation 474 Best Solution: [183.23467603600383, 175.86239843604542,
174.56919011755969, 75.7719595055785] Fitness: 1.1972535362961223
Generation 475 Best Solution: [183.77318625892033, 175.5736910361877,
175.01062403263285, 75.7719595055785] Fitness: 2.1898795579430304
Generation 476 Best Solution: [183.64507018860076, 175.8157657131525,

175.01062403263285, 75.7719595055785] Fitness: 1.9041977060310173
Generation 477 Best Solution: [183.6724975649235, 176.33178400930424,
175.01062403263285, 75.7719595055785] Fitness: 1.865610841757224
Generation 478 Best Solution: [183.6724975649235, 176.33178400930424,
175.01062403263285, 75.7719595055785] Fitness: 1.865610841757224
Generation 479 Best Solution: [183.64507018860076, 177.41973803029617,
174.66825758507565, 75.7719595055785] Fitness: 2.6804650617066446
Generation 480 Best Solution: [183.77318625892033, 175.61792766085688,
175.01062403263285, 75.7719595055785] Fitness: 2.1513105306873843
Generation 481 Best Solution: [183.77318625892033, 176.762646597399,
175.01062403263285, 75.7719595055785] Fitness: 2.2277424425054155
Generation 482 Best Solution: [183.77318625892033, 177.41973803029617,
174.0213285029161, 75.7719595055785] Fitness: 2.5169708215685906
Generation 483 Best Solution: [183.77318625892033, 177.41973803029617,
175.01062403263285, 74.61728177163899] Fitness: 2.331276270325998
Generation 484 Best Solution: [183.77318625892033, 175.8845739686748,
175.01062403263285, 75.7719595055785] Fitness: 1.984270995888767
Generation 485 Best Solution: [183.64507018860076, 176.52553110172602,
175.01062403263285, 75.7719595055785] Fitness: 1.9295793729363284
Generation 486 Best Solution: [183.64507018860076, 176.52553110172602,
175.01062403263285, 75.7719595055785] Fitness: 1.9295793729363284
Generation 487 Best Solution: [183.64507018860076, 176.52553110172602,
175.01062403263285, 75.7719595055785] Fitness: 1.9295793729363284
Generation 488 Best Solution: [183.77318625892033, 176.52553110172602,
175.01062403263285, 75.7719595055785] Fitness: 2.0420103054603844
Generation 489 Best Solution: [183.77318625892033, 176.09245158046824,
175.01062403263285, 75.7719595055785] Fitness: 1.9319228534672395
Generation 490 Best Solution: [183.77318625892033, 176.063029155749,
175.01062403263285, 75.7719595055785] Fitness: 1.9351868672516472
Generation 491 Best Solution: [183.77318625892033, 177.41973803029617,
173.7321871202838, 75.59324182922572] Fitness: 2.3341867023815572
Generation 492 Best Solution: [183.77318625892033, 176.4974506333935,
173.7321871202838, 75.59324182922572] Fitness: 1.153764171442762
Generation 493 Best Solution: [183.77318625892033, 176.4974506333935,
173.7321871202838, 75.7719595055785] Fitness: 1.4269014064021577
Generation 494 Best Solution: [183.77318625892033, 176.52679556401955,
175.01062403263285, 75.59324182922572] Fitness: 1.7696280809355383
Generation 495 Best Solution: [183.77318625892033, 176.64117255242994,
175.01062403263285, 75.61981471423832] Fitness: 1.885784732703807
Generation 496 Best Solution: [183.77318625892033, 176.12135480612093,
174.18784015871663, 75.7719595055785] Fitness: 1.2485673701836664
Generation 497 Best Solution: [183.77318625892033, 176.12135480612093,
175.01062403263285, 75.7719595055785] Fitness: 1.9300473319759577
Generation 498 Best Solution: [183.37715260654483, 177.41973803029617,
175.01062403263285, 74.81779161424423] Fitness: 2.068403918801375
Generation 499 Best Solution: [183.37715260654483, 177.41973803029617,
174.05861246160896, 74.65593878000934] Fitness: 1.3713426738950947
Generation 500 Best Solution: [183.77318625892033, 177.41973803029617,

```
174.21076856110292, 74.65593878000934] Fitness: 1.6495441768953174
Optimal Solution for Simulation 24: [183.77318625892033, 177.41973803029617,
174.21076856110292, 74.65593878000934]
------------------------------------------------
Simulation 25/32 with conditions: {'drought': False, 'flood': False,
'policy_change': True, 'social_event': True, 'economic_crisis': True}
Generation 1 Best Solution: [183.2488392693337, 175.93190276010225,
174.54105327545648, 75.13632000813533] Fitness: 0.41469381819777096
Generation 2 Best Solution: [183.67414322991033, 175.94013398465916,
174.4735729432641, 75.25450751216852] Fitness: 0.6268034105972751
Generation 3 Best Solution: [183.2488392693337, 175.93190276010225,
174.54105327545648, 74.80994433596531] Fitness: 0.25780985829832564
Generation 4 Best Solution: [183.30678041996723, 175.94013398465916,
174.4735729432641, 75.25450751216852] Fitness: 0.4744846122701536
Generation 5 Best Solution: [183.69506033831638, 176.59255088752397,
174.4735729432641, 75.25450751216852] Fitness: 0.677455159027744
Generation 6 Best Solution: [183.31570396159603, 175.98581819939307,
174.27397052547374, 75.72880610569864] Fitness: 1.024003140995899
Generation 7 Best Solution: [183.31570396159603, 175.98581819939307,
173.75609305545032, 75.67641337071258] Fitness: 1.0548172964254243
Generation 8 Best Solution: [183.30678041996723, 177.19802232326373,
174.77107068149004, 75.65077483492306] Fitness: 1.9350783436385293
Generation 9 Best Solution: [183.33683849020207, 177.19802232326373,
175.0246373157612, 74.61167525391268] Fitness: 1.4812233570312465
Generation 10 Best Solution: [183.69506033831638, 176.27789560638763,
174.77107068149004, 75.67641337071258] Fitness: 1.3613401195926484
Generation 11 Best Solution: [183.68263610920354, 176.62493481780436,
175.0246373157612, 75.69261300971975] Fitness: 1.82641714553022
Generation 12 Best Solution: [183.59673975391928, 176.62493481780436,
175.0246373157612, 75.67641337071258] Fitness: 1.7418208964947377
Generation 13 Best Solution: [183.68263610920354, 177.37897541576012,
174.68147121470923, 75.24899180308506] Fitness: 1.812109969184578
Generation 14 Best Solution: [182.91135542368204, 177.37897541576012,
174.68147121470923, 75.24899180308506] Fitness: 1.7448360875198432
Generation 15 Best Solution: [183.69506033831638, 176.57099797431408,
174.68147121470923, 75.69261300971975] Fitness: 1.3958577901375882
Generation 16 Best Solution: [182.91135542368204, 177.06043339321732,
174.68147121470923, 75.50789514122948] Fitness: 1.5158011195579175
Generation 17 Best Solution: [183.61947398098644, 175.93523646033424,
174.68147121470923, 75.69261300971975] Fitness: 1.3231991898695337
Generation 18 Best Solution: [182.91135542368204, 176.22311485078117,
174.68147121470923, 75.69261300971975] Fitness: 1.2235869719235801
Generation 19 Best Solution: [183.69506033831638, 177.06043339321732,
174.68147121470923, 75.03863877237448] Fitness: 1.1284881444503962
Generation 20 Best Solution: [183.69506033831638, 176.22311485078117,
174.68147121470923, 75.69261300971975] Fitness: 1.3001834799130774
Generation 21 Best Solution: [183.53214318039696, 176.57099797431408,
174.68147121470923, 75.52654882443862] Fitness: 1.0356705391791703
```

Generation 22 Best Solution: [183.69506033831638, 177.06043339321732,
174.68147121470923, 75.52654882443862] Fitness: 1.6185398801255262
Generation 23 Best Solution: [183.69506033831638, 176.29559968303596,
174.40388967088143, 75.69261300971975] Fitness: 1.120045241140681
Generation 24 Best Solution: [183.69506033831638, 176.57099797431408,
174.40388967088143, 75.01867318223755] Fitness: 0.4524641660025824
Generation 25 Best Solution: [183.69506033831638, 177.19802232326373,
174.40388967088143, 75.01867318223755] Fitness: 1.1421534862688574
Generation 26 Best Solution: [182.8180872416071, 177.151511724956,
175.00710058047235, 75.33132788066568] Fitness: 1.8730205289769795
Generation 27 Best Solution: [183.69506033831638, 177.19802232326373,
175.00710058047235, 75.33132788066568] Fitness: 1.9635277607214214
Generation 28 Best Solution: [183.69506033831638, 176.28529715672147,
175.0246373157612, 75.72880610569864] Fitness: 1.77153612595349
Generation 29 Best Solution: [183.69506033831638, 176.58856171298666,
175.0246373157612, 75.72880610569864] Fitness: 1.875403013310136
Generation 30 Best Solution: [183.69506033831638, 177.21487993863371,
175.0246373157612, 75.14864606009168] Fitness: 1.8465588439461234
Generation 31 Best Solution: [183.69506033831638, 177.19802232326373,
174.0426306482542, 75.72880610569864] Fitness: 1.9218181977567057
Generation 32 Best Solution: [183.69506033831638, 177.19802232326373,
174.0426306482542, 75.72880610569864] Fitness: 1.9218181977567057
Generation 33 Best Solution: [183.69506033831638, 177.19802232326373,
174.0426306482542, 75.72880610569864] Fitness: 1.9218181977567057
Generation 34 Best Solution: [183.69506033831638, 177.19802232326373,
174.0426306482542, 75.72880610569864] Fitness: 1.9218181977567057
Generation 35 Best Solution: [183.69506033831638, 177.06043339321732,
175.0246373157612, 74.35123134345542] Fitness: 1.5283993588801534
Generation 36 Best Solution: [183.69506033831638, 177.06043339321732,
175.0246373157612, 74.68438686450688] Fitness: 1.4318303943712751
Generation 37 Best Solution: [183.69506033831638, 177.37897541576012,
175.0246373157612, 74.91001257753568] Fitness: 2.004180113508679
Generation 38 Best Solution: [183.69506033831638, 177.37897541576012,
175.0246373157612, 74.91001257753568] Fitness: 2.004180113508679
Generation 39 Best Solution: [183.69506033831638, 177.37897541576012,
175.0246373157612, 74.28444403595296] Fitness: 2.1014308667295123
Generation 40 Best Solution: [183.67051056288767, 175.64867694019122,
175.0246373157612, 75.72880610569864] Fitness: 2.0416228742619302
Generation 41 Best Solution: [183.19908326814922, 175.76975775201143,
175.0246373157612, 75.72880610569864] Fitness: 1.7829226486387282
Generation 42 Best Solution: [183.19908326814922, 175.76975775201143,
175.0246373157612, 75.72880610569864] Fitness: 1.7829226486387282
Generation 43 Best Solution: [183.69506033831638, 175.76975775201143,
175.0246373157612, 75.72880610569864] Fitness: 1.952149249390727
Generation 44 Best Solution: [183.19908326814922, 177.37897541576012,
175.0246373157612, 74.30753873182087] Fitness: 1.9168406269337483
Generation 45 Best Solution: [183.69506033831638, 176.45716140120246,
174.77220294770987, 75.72880610569864] Fitness: 1.4927327601860676

Generation 46 Best Solution: [183.69506033831638, 175.76975775201143,
173.83554278122216, 75.72880610569864] Fitness: 1.3940083004359063
Generation 47 Best Solution: [183.19908326814922, 176.7886850678748,
175.0246373157612, 75.14912610612383] Fitness: 1.1243338598653496
Generation 48 Best Solution: [183.19908326814922, 176.7886850678748,
175.0246373157612, 74.67693595778708] Fitness: 0.9453853186078753
Generation 49 Best Solution: [182.83282219863034, 176.7886850678748,
175.0246373157612, 75.72880610569864] Fitness: 2.002210407879288
Generation 50 Best Solution: [182.83282219863034, 177.37897541576012,
174.77220294770987, 75.72880610569864] Fitness: 2.5320484349549788
Generation 51 Best Solution: [182.83282219863034, 175.60595022856506,
175.0246373157612, 75.72880610569864] Fitness: 2.0767237707800534
Generation 52 Best Solution: [182.83282219863034, 175.77272720570545,
175.0246373157612, 75.72880610569864] Fitness: 1.9228457327184683
Generation 53 Best Solution: [182.83282219863034, 176.0679226825517,
175.0246373157612, 75.72880610569864] Fitness: 1.765873239259816
Generation 54 Best Solution: [182.83282219863034, 176.33022198418794,
175.0246373157612, 75.72880610569864] Fitness: 1.7501270355638958
Generation 55 Best Solution: [182.83282219863034, 176.30698863624022,
175.0246373157612, 75.72880610569864] Fitness: 1.7468219723905065
Generation 56 Best Solution: [182.83282219863034, 176.30698863624022,
175.0246373157612, 75.72880610569864] Fitness: 1.7468219723905065
Generation 57 Best Solution: [182.83282219863034, 176.30698863624022,
175.0246373157612, 75.72880610569864] Fitness: 1.7468219723905065
Generation 58 Best Solution: [182.83282219863034, 175.8544741692311,
175.0246373157612, 75.72880610569864] Fitness: 1.8646118062990327
Generation 59 Best Solution: [182.83282219863034, 176.65904361965113,
174.10577930975347, 75.72880610569864] Fitness: 1.2534530741264995
Generation 60 Best Solution: [183.46304864666766, 176.81303070845482,
174.10577930975347, 74.88619245429327] Fitness: 0.39830424460731817
Generation 61 Best Solution: [182.83282219863034, 176.65904361965113,
174.10577930975347, 75.72880610569864] Fitness: 1.2534530741264995
Generation 62 Best Solution: [182.83282219863034, 176.65904361965113,
174.10577930975347, 75.72880610569864] Fitness: 1.2534530741264995
Generation 63 Best Solution: [182.83282219863034, 176.65904361965113,
174.10577930975347, 75.72880610569864] Fitness: 1.2534530741264995
Generation 64 Best Solution: [182.83282219863034, 176.65904361965113,
174.10577930975347, 75.72880610569864] Fitness: 1.2534530741264995
Generation 65 Best Solution: [182.83282219863034, 176.65904361965113,
174.10577930975347, 75.72880610569864] Fitness: 1.2534530741264995
Generation 66 Best Solution: [182.83282219863034, 176.65904361965113,
175.0246373157612, 75.06014126477076] Fitness: 1.0966554257726797
Generation 67 Best Solution: [182.83282219863034, 175.64124360676408,
174.10577930975347, 75.72880610569864] Fitness: 1.3988294044672613
Generation 68 Best Solution: [182.83282219863034, 177.37897541576012,
173.86375486242713, 74.49042043827131] Fitness: 1.3966937647487987
Generation 69 Best Solution: [183.64044715969962, 177.37897541576012,
174.90165922394135, 74.67341702663721] Fitness: 1.7553982139056652

Generation 70 Best Solution: [183.6383215928679, 175.64124360676408,
174.13500295707343, 75.79109487597874] Fitness: 1.496639844510828
Generation 71 Best Solution: [183.6383215928679, 175.64124360676408,
174.13500295707343, 75.79109487597874] Fitness: 1.496639844510828
Generation 72 Best Solution: [183.6383215928679, 177.37897541576012,
174.90165922394135, 74.67341702663721] Fitness: 1.7539812037507725
Generation 73 Best Solution: [182.83282219863034, 177.37897541576012,
175.0246373157612, 74.67341702663721] Fitness: 1.936114966463154
Generation 74 Best Solution: [183.6383215928679, 177.34213142640593,
174.13500295707343, 75.72880610569864] Fitness: 2.123922924858527
Generation 75 Best Solution: [183.6383215928679, 176.47664093097262,
175.0246373157612, 75.72880610569864] Fitness: 1.7785030456774946
Generation 76 Best Solution: [183.6383215928679, 177.34213142640593,
175.0246373157612, 75.24033201822667] Fitness: 2.1093454220388415
Generation 77 Best Solution: [183.64889115386939, 177.34213142640593,
175.0246373157612, 75.24033201822667] Fitness: 2.1164671436331366
Generation 78 Best Solution: [183.6383215928679, 175.8037307293363,
175.0246373157612, 75.72880610569864] Fitness: 1.8859570599279516
Generation 79 Best Solution: [183.6383215928679, 175.8037307293363,
175.0246373157612, 75.72880610569864] Fitness: 1.8859570599279516
Generation 80 Best Solution: [183.6383215928679, 177.30495651119443,
174.42269366990556, 74.5851539305825] Fitness: 1.2112580481666562
Generation 81 Best Solution: [183.64889115386939, 177.34213142640593,
174.42269366990556, 74.5851539305825] Fitness: 1.286890185416565
Generation 82 Best Solution: [183.46158693534824, 175.8037307293363,
174.43020671284788, 75.72880610569864] Fitness: 1.2167647910975723
Generation 83 Best Solution: [183.64044715969962, 177.0748835567914,
175.0246373157612, 74.86352802569193] Fitness: 1.4389018838077219
Generation 84 Best Solution: [183.6383215928679, 176.56996216493943,
175.0246373157612, 75.72880610569864] Fitness: 1.8241057879048155
Generation 85 Best Solution: [183.64044715969962, 175.88884862954376,
175.0246373157612, 75.72880610569864] Fitness: 1.8314475040021847
Generation 86 Best Solution: [183.64044715969962, 175.8969976908646,
174.73694233268048, 75.72880610569864] Fitness: 1.4722656133796104
Generation 87 Best Solution: [183.6383215928679, 175.8969976908646,
174.73694233268048, 75.72880610569864] Fitness: 1.4708486032247179
Generation 88 Best Solution: [183.64044715969962, 177.1765507394991,
175.0246373157612, 74.44620163538204] Fitness: 1.6164018797092023
Generation 89 Best Solution: [183.6383215928679, 177.37897541576012,
175.0246373157612, 74.56606815589356] Fitness: 1.9352417276645362
Generation 90 Best Solution: [183.6383215928679, 176.19534456520773,
175.0246373157612, 74.56606815589356] Fitness: 0.8283174591714398
Generation 91 Best Solution: [183.6383215928679, 177.37897541576012,
174.0066322797786, 74.92691135427252] Fitness: 1.3450837654950925
Generation 92 Best Solution: [183.6383215928679, 176.140457421348,
175.0246373157612, 75.72880610569864] Fitness: 1.736400408841277
Generation 93 Best Solution: [183.06787776719221, 176.7136761954277,
175.0246373157612, 75.46042262747197] Fitness: 1.4079637913092315

Generation 94 Best Solution: [183.06787776719221, 176.38011009780632,
175.0246373157612, 75.73997596983334] Fitness: 1.6626811363725134
Generation 95 Best Solution: [183.06787776719221, 177.37897541576012,
174.43587615761916, 75.46042262747197] Fitness: 1.7465098073503822
Generation 96 Best Solution: [183.64044715969962, 176.7136761954277,
175.0246373157612, 74.32796324392876] Fitness: 1.1202259807587687
Generation 97 Best Solution: [183.06787776719221, 176.02596187344346,
175.0246373157612, 75.72880610569864] Fitness: 1.6618064875366731
Generation 98 Best Solution: [183.41782858895147, 177.37897541576012,
175.0246373157612, 74.56051651457553] Fitness: 1.830966018552095
Generation 99 Best Solution: [183.6383215928679, 176.7940678999005,
175.0246373157612, 74.73363954148958] Fitness: 1.0807916929969201
Generation 100 Best Solution: [183.6383215928679, 176.7940678999005,
175.0246373157612, 74.38603953052348] Fitness: 1.156823318491789
Generation 101 Best Solution: [183.6383215928679, 176.49859119575143,
175.0246373157612, 75.73997596983334] Fitness: 1.8076629072427224
Generation 102 Best Solution: [183.62239792702044, 176.7940678999005,
173.98999578375674, 75.73997596983334] Fitness: 1.3823364196290409
Generation 103 Best Solution: [183.6383215928679, 176.5787669203149,
175.0246373157612, 75.73997596983334] Fitness: 1.848928361436723
Generation 104 Best Solution: [183.6383215928679, 176.5787669203149,
175.0246373157612, 75.72880610569864] Fitness: 1.8291692130382398
Generation 105 Best Solution: [183.06787776719221, 177.37897541576012,
175.0246373157612, 74.56770604188466] Fitness: 1.8309859982482872
Generation 106 Best Solution: [183.06787776719221, 177.37897541576012,
175.01376602697783, 74.56770604188466] Fitness: 1.8150449786296194
Generation 107 Best Solution: [183.6383215928679, 175.99065835584966,
175.0246373157612, 75.73997596983334] Fitness: 1.7989989378637736
Generation 108 Best Solution: [183.06787776719221, 175.99065835584966,
175.0246373157612, 75.73997596983334] Fitness: 1.6950819338316285
Generation 109 Best Solution: [183.06787776719221, 175.99065835584966,
175.0246373157612, 75.73997596983334] Fitness: 1.6950819338316285
Generation 110 Best Solution: [183.06787776719221, 176.84726671339703,
175.0246373157612, 75.24601259652854] Fitness: 1.2904078239113834
Generation 111 Best Solution: [183.06787776719221, 176.84726671339703,
174.48527717639456, 75.72880610569864] Fitness: 1.3929686418689298
Generation 112 Best Solution: [183.06787776719221, 176.84726671339703,
174.48527717639456, 75.73997596983334] Fitness: 1.412727790267413
Generation 113 Best Solution: [183.06787776719221, 176.84726671339703,
175.0246373157612, 75.05224457029213] Fitness: 1.1395545141237022
Generation 114 Best Solution: [183.06787776719221, 176.27998026194552,
175.0246373157612, 75.73997596983334] Fitness: 1.6464986609812824
Generation 115 Best Solution: [183.6383215928679, 176.39435501295236,
175.0246373157612, 75.73997596983334] Fitness: 1.7702789346735819
Generation 116 Best Solution: [183.06787776719221, 177.37897541576012,
175.0246373157612, 74.85920136985717] Fitness: 1.8430041606813479
Generation 117 Best Solution: [183.6383215928679, 177.37897541576012,
175.0246373157612, 74.85920136985717] Fitness: 1.9469211647134927

Generation 118 Best Solution: [183.06787776719221, 177.37897541576012,
175.0246373157612, 74.5897247919663] Fitness: 1.8268731314562234
Generation 119 Best Solution: [183.6383215928679, 175.71363685650584,
175.0246373157612, 75.72880610569864] Fitness: 1.9585100476664272
Generation 120 Best Solution: [183.06787776719221, 175.71363685650584,
175.0246373157612, 75.72880610569864] Fitness: 1.854593043634282
Generation 121 Best Solution: [183.6383215928679, 176.18324185337303,
175.0246373157612, 75.73997596983334] Fitness: 1.750896058510932
Generation 122 Best Solution: [183.06787776719221, 176.18324185337303,
175.0246373157612, 75.73997596983334] Fitness: 1.6469790544787872
Generation 123 Best Solution: [183.8611900417896, 176.18324185337303,
175.0246373157612, 75.73997596983334] Fitness: 1.9410993300140564
Generation 124 Best Solution: [183.6383215928679, 176.18324185337303,
175.0246373157612, 75.73997596983334] Fitness: 1.750896058510932
Generation 125 Best Solution: [183.6383215928679, 176.18324185337303,
175.0246373157612, 75.72880610569864] Fitness: 1.7311369101124487
Generation 126 Best Solution: [183.06787776719221, 177.37897541576012,
175.0246373157612, 74.76658291006079] Fitness: 1.823599687979367
Generation 127 Best Solution: [183.6383215928679, 176.0044858326753,
175.0246373157612, 75.73997596983334] Fitness: 1.7934536793832083
Generation 128 Best Solution: [183.6383215928679, 176.6361587645379,
174.47452646176473, 75.73997596983334] Fitness: 1.32950195451918
Generation 129 Best Solution: [183.6383215928679, 177.37897541576012,
173.84058937751917, 75.72880610569864] Fitness: 2.2763060647474314
Generation 130 Best Solution: [183.2618652364703, 176.80935312913286,
175.0246373157612, 75.72348177242446] Fitness: 1.8687666806650505
Generation 131 Best Solution: [183.6383215928679, 177.37897541576012,
173.76676463777932, 74.4026180156049] Fitness: 1.4746566266365309
Generation 132 Best Solution: [183.6383215928679, 175.64123601589796,
173.76676463777932, 74.4026180156049] Fitness: 0.6642918337508772
Generation 133 Best Solution: [183.6383215928679, 177.37897541576012,
174.13273781260042, 75.72880610569864] Fitness: 2.194203344886152
Generation 134 Best Solution: [183.6383215928679, 176.14234691776701,
175.13414612400686, 75.73997596983334] Fitness: 1.9275935847523082
Generation 135 Best Solution: [183.6383215928679, 176.45270199670972,
175.0246373157612, 75.72880610569864] Fitness: 1.769180170985825
Generation 136 Best Solution: [183.06787776719221, 176.0238798146696,
175.13414612400686, 75.1898612871052] Fitness: 1.1317689672353053
Generation 137 Best Solution: [183.6383215928679, 176.53480999200835,
175.0246373157612, 75.72880610569864] Fitness: 1.8051979918516785
Generation 138 Best Solution: [183.6383215928679, 176.14234691776701,
175.0246373157612, 75.72880610569864] Fitness: 1.736102573004687
Generation 139 Best Solution: [183.6383215928679, 175.67742139748452,
175.13414612400686, 75.73997596983334] Fitness: 2.1830361303285746
Generation 140 Best Solution: [183.6383215928679, 175.67742139748452,
175.0246373157612, 75.72880610569864] Fitness: 1.991545118580953
Generation 141 Best Solution: [183.6383215928679, 176.79210388151228,
175.13414612400686, 75.73997596983334] Fitness: 2.1834458249272224

Generation 142 Best Solution: [183.6383215928679, 176.79210388151228,
175.13414612400686, 75.55387791582167] Fitness: 1.8817895724123042
Generation 143 Best Solution: [183.6383215928679, 177.37897541576012,
174.8899279892107, 74.74020116060906] Fitness: 1.741231709642123
Generation 144 Best Solution: [182.90151255786571, 176.52729981911492,
175.0246373157612, 75.73997596983334] Fitness: 1.7906861320500986
Generation 145 Best Solution: [182.90151255786571, 176.52729981911492,
175.0246373157612, 75.73997596983334] Fitness: 1.7906861320500986
Generation 146 Best Solution: [183.6383215928679, 176.52729981911492,
175.0246373157612, 75.73997596983334] Fitness: 1.8211886464565692
Generation 147 Best Solution: [183.6383215928679, 177.37897541576012,
173.7635363921863, 74.55205921515149] Fitness: 1.4232217709428003
Generation 148 Best Solution: [183.6383215928679, 176.72164970610697,
175.0246373157612, 75.87820594627784] Fitness: 2.2114353034996745
Generation 149 Best Solution: [182.90151255786571, 177.37897541576012,
174.3939144581967, 75.73997596983334] Fitness: 2.232343708039969
Generation 150 Best Solution: [182.90151255786571, 175.95636294559904,
175.13414612400686, 75.87820594627784] Fitness: 2.217377307853006
Generation 151 Best Solution: [182.90151255786571, 177.37897541576012,
175.0246373157612, 74.64393897777344] Fitness: 1.8936580663405165
Generation 152 Best Solution: [183.6383215928679, 177.37897541576012,
175.0246373157612, 74.64393897777344] Fitness: 1.924160580746987
Generation 153 Best Solution: [182.90151255786571, 177.37897541576012,
175.0246373157612, 74.64393897777344] Fitness: 1.8936580663405165
Generation 154 Best Solution: [182.90151255786571, 177.37897541576012,
175.0246373157612, 74.64393897777344] Fitness: 1.8936580663405165
Generation 155 Best Solution: [183.6383215928679, 177.37897541576012,
173.98039481287924, 74.32297011093019] Fitness: 1.4177181231083724
Generation 156 Best Solution: [183.6383215928679, 176.05687711227228,
174.03247876356554, 74.64393897777344] Fitness: 0.211611775351412
Generation 157 Best Solution: [183.06138958537312, 177.37897541576012,
175.13414612400686, 74.98764379896397] Fitness: 2.0676567334114972
Generation 158 Best Solution: [182.90151255786571, 176.05687711227228,
175.13414612400686, 75.87820594627784] Fitness: 2.178606988405786
Generation 159 Best Solution: [182.90151255786571, 176.04415027735905,
175.13414612400686, 75.73997596983334] Fitness: 1.9205716308404015
Generation 160 Best Solution: [182.90151255786571, 176.33877524069686,
173.98039481287924, 75.87820594627784] Fitness: 1.3712182621749665
Generation 161 Best Solution: [182.90151255786571, 176.42581234077016,
175.13414612400686, 75.87820594627784] Fitness: 2.182852126953473
Generation 162 Best Solution: [183.6383215928679, 176.35008886900783,
174.0619924590651, 75.87820594627784] Fitness: 1.3856614213485465
Generation 163 Best Solution: [183.6383215928679, 175.83289806952624,
175.13414612400686, 75.60553987610925] Fitness: 1.8331114962620445
Generation 164 Best Solution: [183.6383215928679, 175.83289806952624,
175.13414612400686, 75.60553987610925] Fitness: 1.8331114962620445
Generation 165 Best Solution: [183.6383215928679, 176.35157552225724,
175.13414612400686, 75.00127561283578] Fitness: 1.0799973381505434

Generation 166 Best Solution: [183.31118418665406, 177.37897541576012,
175.13414612400686, 75.00127561283578] Fitness: 2.0496894496757103
Generation 167 Best Solution: [183.6383215928679, 176.15431143982232,
174.9677469809484, 75.87820594627784] Fitness: 1.9349096431628872
Generation 168 Best Solution: [183.59285684141238, 175.83958517771362,
174.00137728524362, 75.87820594627784] Fitness: 1.4902978130934241
Generation 169 Best Solution: [183.6383215928679, 175.83958517771362,
175.13414612400686, 75.87820594627784] Fitness: 2.314394378559859
Generation 170 Best Solution: [183.67508954462068, 177.37897541576012,
174.0619924590651, 75.41439616634214] Fitness: 1.7568560114695597
Generation 171 Best Solution: [183.67508954462068, 177.37897541576012,
175.13414612400686, 74.48195091199614] Fitness: 2.156062441608366
Generation 172 Best Solution: [183.6383215928679, 176.46391694002392,
175.13414612400686, 75.41439616634214] Fitness: 1.4755942884586815
Generation 173 Best Solution: [183.67508954462068, 177.37897541576012,
175.13414612400686, 74.67662892937899] Fitness: 2.1198879452976827
Generation 174 Best Solution: [183.14297460823283, 177.37897541576012,
175.13414612400686, 74.67662892937899] Fitness: 1.9725860187799336
Generation 175 Best Solution: [183.6383215928679, 176.46391694002392,
175.13414612400686, 74.67662892937899] Fitness: 1.0305914755481125
Generation 176 Best Solution: [183.6383215928679, 176.46391694002392,
175.13414612400686, 74.67662892937899] Fitness: 1.0305914755481125
Generation 177 Best Solution: [182.87122667284166, 177.37897541576012,
175.13414612400686, 74.67662892937899] Fitness: 2.082200479870809
Generation 178 Best Solution: [183.14297460823283, 177.37897541576012,
175.13414612400686, 74.58888630616055] Fitness: 1.980950809288472
Generation 179 Best Solution: [182.95121451968464, 175.82670512259617,
175.13414612400686, 75.5183233248918] Fitness: 1.6511829028893188
Generation 180 Best Solution: [183.14297460823283, 176.62409382839064,
175.13414612400686, 75.87820594627784] Fitness: 2.189089093535258
Generation 181 Best Solution: [183.67508954462068, 176.49643982734932,
174.95155531941953, 75.87820594627784] Fitness: 1.9909774929943356
Generation 182 Best Solution: [183.67508954462068, 176.65506758141856,
175.13414612400686, 75.05085915070818] Fitness: 1.2719063076403376
Generation 183 Best Solution: [183.67508954462068, 176.65506758141856,
175.13414612400686, 75.05085915070818] Fitness: 1.2719063076403376
Generation 184 Best Solution: [183.6383215928679, 177.37897541576012,
175.13414612400686, 74.776129642877] Fitness: 2.1005776354077623
Generation 185 Best Solution: [183.6383215928679, 177.37897541576012,
175.13414612400686, 74.776129642877] Fitness: 2.1005776354077623
Generation 186 Best Solution: [183.6383215928679, 177.37727985426721,
175.13414612400686, 74.776129642877] Fitness: 2.0972962272537674
Generation 187 Best Solution: [183.67508954462068, 177.37897541576012,
175.13414612400686, 74.776129642877] Fitness: 2.1261667793475443
Generation 188 Best Solution: [183.67508954462068, 176.13275480286012,
175.13414612400686, 74.776129642877] Fitness: 1.026709502970004
Generation 189 Best Solution: [183.3370846799852, 177.37897541576012,
174.3627994692917, 75.87820594627784] Fitness: 2.3894695872078513

Generation 190 Best Solution: [183.67508954462068, 176.29920887012247,
175.13414612400686, 75.87820594627784] Fitness: 2.211526992301872
Generation 191 Best Solution: [183.3370846799852, 177.37897541576012,
175.13414612400686, 74.31537274305315] Fitness: 2.0888508663032956
Generation 192 Best Solution: [183.3370846799852, 177.0203940591657,
175.13414612400686, 75.87820594627784] Fitness: 2.58145404863006
Generation 193 Best Solution: [183.67508954462068, 177.37727985426721,
175.13414612400686, 75.06703054063972] Fitness: 2.2373383606610764
Generation 194 Best Solution: [183.67508954462068, 177.37727985426721,
175.13414612400686, 75.06703054063972] Fitness: 2.2373383606610764
Generation 195 Best Solution: [183.67508954462068, 176.73000675573803,
175.13414612400686, 75.87820594627784] Fitness: 2.415704375403324
Generation 196 Best Solution: [183.3370846799852, 176.3704164266631,
175.13414612400686, 75.87820594627784] Fitness: 2.0745257262059487
Generation 197 Best Solution: [183.3370846799852, 176.3704164266631,
173.76222048802757, 75.87820594627784] Fitness: 1.3885594538786517
Generation 198 Best Solution: [183.67508954462068, 176.11339410209138,
175.13414612400686, 75.87820594627784] Fitness: 2.220408470898875
Generation 199 Best Solution: [183.6383215928679, 176.11339410209138,
175.13414612400686, 75.87820594627784] Fitness: 2.1948193269590934
Generation 200 Best Solution: [183.6383215928679, 176.11339410209138,
175.13414612400686, 75.87820594627784] Fitness: 2.1948193269590934
Generation 201 Best Solution: [183.67508954462068, 176.05901280731442,
175.13414612400686, 75.87820594627784] Fitness: 2.2340603678579
Generation 202 Best Solution: [183.67508954462068, 176.05901280731442,
175.13414612400686, 75.87820594627784] Fitness: 2.2340603678579
Generation 203 Best Solution: [183.67508954462068, 176.05901280731442,
175.13414612400686, 75.87820594627784] Fitness: 2.2340603678579
Generation 204 Best Solution: [183.67508954462068, 177.37727985426721,
175.13414612400686, 74.79013847309278] Fitness: 2.1251148757418497
Generation 205 Best Solution: [183.47902010989395, 177.37727985426721,
175.13414612400686, 74.33658978749277] Fitness: 2.1115874742131653
Generation 206 Best Solution: [183.47902010989395, 176.46013641864437,
175.13414612400686, 75.87820594627784] Fitness: 2.1410225482769842
Generation 207 Best Solution: [183.67508954462068, 176.3446626515843,
175.13414612400686, 75.87820594627784] Fitness: 2.218249210568459
Generation 208 Best Solution: [183.67508954462068, 177.37727985426721,
175.13414612400686, 74.66636582753368] Fitness: 2.1169121073227544
Generation 209 Best Solution: [183.67508954462068, 177.07522389597736,
175.13414612400686, 74.66636582753368] Fitness: 1.609979928045077
Generation 210 Best Solution: [183.67508954462068, 176.09707312325565,
175.13414612400686, 74.66636582753368] Fitness: 1.027960112645132
Generation 211 Best Solution: [183.67508954462068, 176.09707312325565,
175.13414612400686, 74.66636582753368] Fitness: 1.027960112645132
Generation 212 Best Solution: [183.67508954462068, 176.10976882902165,
173.75911441535627, 75.87820594627784] Fitness: 1.5372574274435893
Generation 213 Best Solution: [183.02307999757667, 176.11428183309556,
175.13414612400686, 75.87820594627784] Fitness: 2.105881459380421

Generation 214 Best Solution: [183.67508954462068, 176.32534957251957,
173.75911441535627, 75.87820594627784] Fitness: 1.5310603503053906
Generation 215 Best Solution: [183.02307999757667, 176.11428183309556,
175.13414612400686, 75.87820594627784] Fitness: 2.105881459380421
Generation 216 Best Solution: [183.02307999757667, 176.11428183309556,
175.13414612400686, 75.87820594627784] Fitness: 2.105881459380421
Generation 217 Best Solution: [183.02307999757667, 176.11428183309556,
175.13414612400686, 75.87820594627784] Fitness: 2.105881459380421
Generation 218 Best Solution: [183.60782103957519, 175.9423018058471,
175.13414612400686, 75.87820594627784] Fitness: 2.2351755000615374
Generation 219 Best Solution: [183.60782103957519, 175.9423018058471,
175.13414612400686, 75.31913578278149] Fitness: 1.374604860307424
Generation 220 Best Solution: [183.3331858611432, 175.9423018058471,
175.13414612400686, 75.31913578278149] Fitness: 1.2700095052768787
Generation 221 Best Solution: [183.60782103957519, 175.9423018058471,
174.49724988728576, 75.87820594627784] Fitness: 1.520609467234459
Generation 222 Best Solution: [183.3331858611432, 175.9423018058471,
175.13414612400686, 75.87820594627784] Fitness: 2.1305801450309922
Generation 223 Best Solution: [183.3331858611432, 175.9423018058471,
175.13414612400686, 75.87820594627784] Fitness: 2.1305801450309922
Generation 224 Best Solution: [183.60782103957519, 177.37727985426721,
173.75911441535627, 74.6231432387088] Fitness: 1.3911693628763981
Generation 225 Best Solution: [183.60782103957519, 177.37727985426721,
174.27570805784327, 75.87820594627784] Fitness: 2.4656482361846868
Generation 226 Best Solution: [183.60782103957519, 177.37727985426721,
174.27570805784327, 74.58644228824046] Fitness: 1.2781069934679485
Generation 227 Best Solution: [183.60782103957519, 177.37727985426721,
174.27570805784327, 74.58644228824046] Fitness: 1.2781069934679485
Generation 228 Best Solution: [183.60782103957519, 176.17510732938095,
175.13414612400686, 75.87820594627784] Fitness: 2.165897957423385
Generation 229 Best Solution: [183.60782103957519, 176.17510732938095,
175.13414612400686, 75.87820594627784] Fitness: 2.165897957423385
Generation 230 Best Solution: [183.60782103957519, 177.37727985426721,
175.13414612400686, 74.79224274329897] Fitness: 2.0803981278101844
Generation 231 Best Solution: [183.60782103957519, 176.17510732938095,
175.13414612400686, 74.51828272505696] Fitness: 0.9941279193278185
Generation 232 Best Solution: [183.03093278528848, 176.17510732938095,
175.13414612400686, 74.51828272505696] Fitness: 0.9219595504630488
Generation 233 Best Solution: [183.6677761426819, 176.17510732938095,
175.13414612400686, 74.51828272505696] Fitness: 1.0339361200706805
Generation 234 Best Solution: [183.67508954462068, 176.17510732938095,
175.13414612400686, 74.48464378251222] Fitness: 1.049889414026707
Generation 235 Best Solution: [183.60782103957519, 177.37727985426721,
175.13414612400686, 74.48464378251222] Fitness: 2.106762865758162
Generation 236 Best Solution: [183.60782103957519, 176.33166943026774,
175.13414612400686, 75.87820594627784] Fitness: 2.1708904139854877
Generation 237 Best Solution: [183.67508954462068, 177.37727985426721,
175.13414612400686, 74.33009757185027] Fitness: 2.2255242346585398

Generation 238 Best Solution: [183.60782103957519, 177.37727985426721,
175.13414612400686, 74.76553403077071] Fitness: 2.076339339307747
Generation 239 Best Solution: [183.60782103957519, 177.37727985426721,
175.13414612400686, 74.90011521946349] Fitness: 2.10907509283673
Generation 240 Best Solution: [183.6677761426819, 177.37727985426721,
174.44909209504223, 75.87820594627784] Fitness: 2.5669784506489575
Generation 241 Best Solution: [183.60782103957519, 176.6858633066336,
175.11018920471494, 75.87820594627784] Fitness: 2.2959562902648374
Generation 242 Best Solution: [183.60782103957519, 177.37727985426721,
175.13414612400686, 74.53102148595488] Fitness: 2.0925367476510104
Generation 243 Best Solution: [183.33100803282557, 177.37727985426721,
175.11018920471494, 74.53102148595488] Fitness: 1.9483185507037308
Generation 244 Best Solution: [183.19349405122932, 177.37727985426721,
175.13414612400686, 74.53102148595488] Fitness: 1.9837132542577216
Generation 245 Best Solution: [183.60782103957519, 177.37727985426721,
175.13414612400686, 74.53102148595488] Fitness: 2.0925367476510104
Generation 246 Best Solution: [183.33100803282557, 177.37727985426721,
175.11018920471494, 74.53102148595488] Fitness: 1.9483185507037308
Generation 247 Best Solution: [183.60782103957519, 177.37727985426721,
175.11018920471494, 74.53102148595488] Fitness: 2.053233231123683
Generation 248 Best Solution: [183.19349405122932, 177.37727985426721,
175.13414612400686, 74.34670308104913] Fitness: 2.061765431555161
Generation 249 Best Solution: [183.33100803282557, 177.37727985426721,
175.11018920471494, 74.30015243139513] Fitness: 2.0551769480821336
Generation 250 Best Solution: [183.67508954462068, 177.06181399461036,
175.13414612400686, 74.53102148595488] Fitness: 1.6117589738609774
Generation 251 Best Solution: [183.55426253472865, 177.37727985426721,
174.05048352809328, 74.53102148595488] Fitness: 1.2559645105993746
Generation 252 Best Solution: [183.60782103957519, 176.3985136153179,
174.05048352809328, 75.87820594627784] Fitness: 1.3795024478566549
Generation 253 Best Solution: [183.67508954462068, 177.37727985426721,
175.13414612400686, 74.54262015276406] Fitness: 2.134628244584211
Generation 254 Best Solution: [183.55426253472865, 177.37727985426721,
174.77071237610932, 75.45806697895866] Fitness: 2.049479175216629
Generation 255 Best Solution: [183.60782103957519, 176.3985136153179,
174.77071237610932, 75.45806697895866] Fitness: 0.9977021185916816
Generation 256 Best Solution: [183.60782103957519, 176.3985136153179,
175.13414612400686, 74.29401158396274] Fitness: 1.1212751461572479
Generation 257 Best Solution: [183.60782103957519, 177.37727985426721,
175.11018920471494, 74.73775884463475] Fitness: 2.0340954526325286
Generation 258 Best Solution: [183.60782103957519, 177.37727985426721,
175.11018920471494, 74.73775884463475] Fitness: 2.0340954526325286
Generation 259 Best Solution: [183.67508954462068, 177.37727985426721,
175.13414612400686, 74.73775884463475] Fitness: 2.1184793015382537
Generation 260 Best Solution: [183.55426253472865, 176.9682284172138,
175.13414612400686, 74.65320380768371] Fitness: 1.392601350118066
Generation 261 Best Solution: [183.60782103957519, 176.9682284172138,
175.13414612400686, 75.87820594627784] Fitness: 2.6183853257967593

Generation 262 Best Solution: [183.60782103957519, 176.30175023858553,
175.13414612400686, 75.87820594627784] Fitness: 2.166730227809755
Generation 263 Best Solution: [183.60782103957519, 177.37727985426721,
175.13414612400686, 75.42286373875847] Fitness: 2.526983120982212
Generation 264 Best Solution: [183.67508954462068, 177.37727985426721,
175.11018920471494, 75.27904994919193] Fitness: 2.3716768705120836
Generation 265 Best Solution: [183.67508954462068, 176.49350688262243,
175.13414612400686, 75.87820594627784] Fitness: 2.2647328617705913
Generation 266 Best Solution: [183.67508954462068, 176.49350688262243,
175.13414612400686, 75.87820594627784] Fitness: 2.2647328617705913
Generation 267 Best Solution: [183.67508954462068, 176.49350688262243,
175.13414612400686, 75.65250319917018] Fitness: 1.853644812472801
Generation 268 Best Solution: [183.60782103957519, 177.37727985426721,
175.13414612400686, 75.08289173250323] Fitness: 2.202615563488422
Generation 269 Best Solution: [183.60782103957519, 177.37727985426721,
174.7754810151227, 75.65250319917018] Fitness: 2.369919990868226
Generation 270 Best Solution: [183.55426253472865, 177.37727985426721,
175.13414612400686, 74.37442554942957] Fitness: 2.124759338912793
Generation 271 Best Solution: [183.41510639378313, 177.37727985426721,
175.11018920471494, 75.0439656437713] Fitness: 2.0518983458931763
Generation 272 Best Solution: [183.60782103957519, 177.37727985426721,
173.93742920207404, 75.0439656437713] Fitness: 1.4021771134974585
Generation 273 Best Solution: [183.30345659019943, 177.37727985426721,
175.13414612400686, 74.98639151832074] Fitness: 2.037922992217392
Generation 274 Best Solution: [183.77159769683945, 175.6292764713397,
174.7061953547245, 75.87820594627784] Fitness: 2.04022612807911
Generation 275 Best Solution: [183.77159769683945, 177.37727985426721,
174.7061953547245, 74.55919043658221] Fitness: 1.6530015281617454
Generation 276 Best Solution: [183.55426253472865, 175.6292764713397,
174.7061953547245, 75.87820594627784] Fitness: 1.8866794621862655
Generation 277 Best Solution: [183.55426253472865, 177.37727985426721,
174.7061953547245, 74.54863433886217] Fitness: 1.5018697137294248
Generation 278 Best Solution: [183.30345659019943, 177.37727985426721,
174.7061953547245, 74.97206122269334] Fitness: 1.4750855710651187
Generation 279 Best Solution: [183.60782103957519, 176.56446187609228,
175.13414612400686, 75.87820594627784] Fitness: 2.2550080727553143
Generation 280 Best Solution: [183.77159769683945, 176.89641227947828,
174.7061953547245, 75.87820594627784] Fitness: 2.1009098720077954
Generation 281 Best Solution: [183.30345659019943, 176.39848783186392,
174.25623175922948, 75.87820594627784] Fitness: 1.2714534263673973
Generation 282 Best Solution: [183.55426253472865, 176.111560120183,
175.13414612400686, 75.87820594627784] Fitness: 2.1452902188970384
Generation 283 Best Solution: [182.83075262510897, 176.10522200462813,
175.13414612400686, 75.87820594627784] Fitness: 2.2114716941293135
Generation 284 Best Solution: [182.83075262510897, 176.15774787817665,
174.8965535159461, 75.87820594627784] Fitness: 1.8554675515453547
Generation 285 Best Solution: [182.83075262510897, 176.15774787817665,
174.8965535159461, 75.87820594627784] Fitness: 1.8554675515453547

Generation 286 Best Solution: [182.83075262510897, 177.37727985426721,
174.87791044847071, 74.34530511858328] Fitness: 1.835795572035154
Generation 287 Best Solution: [183.30345659019943, 177.37727985426721,
174.87791044847071, 74.34530511858328] Fitness: 1.6931226565750888
Generation 288 Best Solution: [183.77159769683945, 177.37727985426721,
174.87791044847071, 74.34530511858328] Fitness: 1.9245138999674904
Generation 289 Best Solution: [182.83075262510897, 176.66398033152956,
175.13414612400686, 74.34530511858328] Fitness: 1.2569118807900455
Generation 290 Best Solution: [183.60782103957519, 176.66398033152956,
175.13414612400686, 74.34530511858328] Fitness: 1.2225003212941106
Generation 291 Best Solution: [183.30242153307944, 177.37727985426721,
174.87791044847071, 74.34530511858328] Fitness: 1.69302196467733
Generation 292 Best Solution: [183.60782103957519, 177.37727985426721,
175.13414612400686, 74.43806088062408] Fitness: 2.1247160906722797
Generation 293 Best Solution: [183.55413896238665, 177.37727985426721,
175.13414612400686, 74.43806088062408] Fitness: 2.094234746453025
Generation 294 Best Solution: [182.93569311385895, 176.4379563497198,
175.13414612400686, 75.87820594627784] Fitness: 2.1680013012656616
Generation 295 Best Solution: [182.93569311385895, 177.37727985426721,
175.13414612400686, 74.43806088062408] Fitness: 2.0947983485854715
Generation 296 Best Solution: [183.60782103957519, 177.37727985426721,
175.13414612400686, 74.43806088062408] Fitness: 2.1247160906722797
Generation 297 Best Solution: [183.77159769683945, 176.65082517145265,
175.13414612400686, 74.43806088062408] Fitness: 1.289531811824381
Generation 298 Best Solution: [183.78267262252993, 177.37727985426721,
175.13414612400686, 74.43806088062408] Fitness: 2.2578108329052187
Generation 299 Best Solution: [183.77159769683945, 175.66950625054383,
175.13414612400686, 75.66082139374615] Fitness: 2.1587218232577534
Generation 300 Best Solution: [183.77159769683945, 177.37727985426721,
174.04230811077792, 74.51694394164548] Fitness: 1.4149157674666872
Generation 301 Best Solution: [183.77159769683945, 177.31600179649908,
174.04230811077792, 75.87820594627784] Fitness: 2.4709715328193615
Generation 302 Best Solution: [183.60782103957519, 177.31600179649908,
174.04230811077792, 75.87820594627784] Fitness: 2.3478416453910906
Generation 303 Best Solution: [183.77159769683945, 176.12246769340797,
175.13414612400686, 75.87820594627784] Fitness: 2.2966673734673275
Generation 304 Best Solution: [183.77159769683945, 176.12246769340797,
175.13414612400686, 75.87820594627784] Fitness: 2.2966673734673275
Generation 305 Best Solution: [183.77159769683945, 175.8114221224819,
175.13414612400686, 75.87820594627784] Fitness: 2.4375282369707683
Generation 306 Best Solution: [182.9980090394177, 175.7640553180525,
175.13414612400686, 75.87820594627784] Fitness: 2.2909153945246903
Generation 307 Best Solution: [183.77159769683945, 176.39213299013653,
175.13414612400686, 75.87820594627784] Fitness: 2.307051717709853
Generation 308 Best Solution: [182.9980090394177, 176.39213299013653,
175.13414612400686, 75.87820594627784] Fitness: 2.1246232015401936
Generation 309 Best Solution: [182.9980090394177, 176.39213299013653,
175.13414612400686, 75.87820594627784] Fitness: 2.1246232015401936

Generation 310 Best Solution: [183.60782103957519, 177.37727985426721,
175.13414612400686, 74.57792703348761] Fitness: 2.0818510639428935
Generation 311 Best Solution: [182.9980090394177, 177.37727985426721,
175.13414612400686, 74.57792703348761] Fitness: 2.022552435201505
Generation 312 Best Solution: [182.9980090394177, 177.37727985426721,
175.13414612400686, 74.57792703348761] Fitness: 2.022552435201505
Generation 313 Best Solution: [183.77159769683945, 177.37727985426721,
174.56193854313085, 74.57792703348761] Fitness: 1.531671714752857
Generation 314 Best Solution: [183.77159769683945, 177.37727985426721,
175.13414612400686, 74.57792703348761] Fitness: 2.2049809513711645
Generation 315 Best Solution: [183.82647206107055, 177.37727985426721,
175.13414612400686, 75.08521655635778] Fitness: 2.3787072165830785
Generation 316 Best Solution: [183.60782103957519, 177.37727985426721,
174.84421981780022, 75.08521655635778] Fitness: 1.7937670711511797
Generation 317 Best Solution: [182.81960289932715, 177.37727985426721,
175.13414612400686, 75.08521655635778] Fitness: 2.2465111382465324
Generation 318 Best Solution: [183.60782103957519, 176.86346256996995,
174.95167416820482, 75.87820594627784] Fitness: 2.2227062114593568
Generation 319 Best Solution: [183.77159769683945, 176.86346256996995,
174.18147256570913, 75.87820594627784] Fitness: 1.8064216104898951
Generation 320 Best Solution: [183.77159769683945, 176.03038624060312,
175.13414612400686, 75.78066916501686] Fitness: 2.133078823441508
Generation 321 Best Solution: [183.77159769683945, 176.86346256996995,
175.13414612400686, 75.71387282882114] Fitness: 2.3128786022070025
Generation 322 Best Solution: [183.77159769683945, 176.71326055855792,
175.13414612400686, 75.30388138680539] Fitness: 1.6033115540070275
Generation 323 Best Solution: [183.67573180677806, 177.37727985426721,
175.13414612400686, 75.11942788723663] Fitness: 2.2736416088493163
Generation 324 Best Solution: [183.22261048721907, 177.37727985426721,
173.86424882414522, 75.71387282882114] Fitness: 2.1047562830096123
Generation 325 Best Solution: [183.22261048721907, 177.37727985426721,
173.86424882414522, 75.71387282882114] Fitness: 2.1047562830096123
Generation 326 Best Solution: [183.22261048721907, 177.37727985426721,
173.86424882414522, 75.71387282882114] Fitness: 2.1047562830096123
Generation 327 Best Solution: [183.76658044209032, 177.37727985426721,
175.04547745183032, 75.30388138680539] Fitness: 2.369283237690116
Generation 328 Best Solution: [183.86541883351055, 176.11242169800667,
175.13414612400686, 75.87820594627784] Fitness: 2.389644357385488
Generation 329 Best Solution: [183.77159769683945, 177.37727985426721,
174.52409533539455, 75.30388138680539] Fitness: 1.816038890316327
Generation 330 Best Solution: [183.22261048721907, 175.7647495164109,
173.75335522891177, 75.87820594627784] Fitness: 1.5589541520217844
Generation 331 Best Solution: [183.22261048721907, 176.5133361054951,
175.13414612400686, 75.87820594627784] Fitness: 2.1180087900358173
Generation 332 Best Solution: [183.22261048721907, 175.92507961795397,
175.13414612400686, 75.87820594627784] Fitness: 2.13327753207918
Generation 333 Best Solution: [183.22261048721907, 175.92507961795397,
175.0307160404382, 75.87820594627784] Fitness: 1.970546336723452

Generation 334 Best Solution: [183.22261048721907, 175.92507961795397,
175.0307160404382, 75.87820594627784] Fitness: 1.970546336723452
Generation 335 Best Solution: [183.22261048721907, 175.92507961795397,
175.13414612400686, 75.87820594627784] Fitness: 2.13327753207918
Generation 336 Best Solution: [183.22261048721907, 175.92507961795397,
175.13414612400686, 75.87820594627784] Fitness: 2.13327753207918
Generation 337 Best Solution: [183.77159769683945, 177.37727985426721,
174.24510639902365, 74.99635150790145] Fitness: 1.4666293134001407
Generation 338 Best Solution: [183.22261048721907, 175.92507961795397,
173.9957013431362, 75.87820594627784] Fitness: 1.3391403005474056
Generation 339 Best Solution: [183.0483011608314, 177.37727985426721,
174.17756980200207, 75.46989454821407] Fitness: 1.6945589061542083
Generation 340 Best Solution: [183.77159769683945, 175.92507961795397,
175.0307160404382, 75.3643448447371] Fitness: 1.3937047042338981
Generation 341 Best Solution: [182.86877512475627, 177.37727985426721,
174.17756980200207, 74.59928790096227] Fitness: 1.2726904484166783
Generation 342 Best Solution: [183.0483011608314, 175.92507961795397,
175.0307160404382, 75.3643448447371] Fitness: 1.1923563391962304
Generation 343 Best Solution: [183.0483011608314, 175.92507961795397,
174.01040446306914, 75.87820594627784] Fitness: 1.3678640468723449
Generation 344 Best Solution: [183.77159769683945, 176.14296522367337,
175.00924228766468, 75.87820594627784] Fitness: 2.0988878162691686
Generation 345 Best Solution: [183.77159769683945, 176.14296522367337,
174.5109191942299, 75.46989454821407] Fitness: 0.9061013211325891
Generation 346 Best Solution: [183.77159769683945, 177.37727985426721,
174.77057808096373, 74.95947302230829] Fitness: 1.764393500179069
Generation 347 Best Solution: [183.77159769683945, 177.37727985426721,
175.0307160404382, 74.6118257944371] Fitness: 2.036844940456709
Generation 348 Best Solution: [183.22261048721907, 177.37727985426721,
174.24580167252205, 75.46989454821407] Fitness: 1.668918825555099
Generation 349 Best Solution: [183.6744381288192, 177.37727985426721,
174.24580167252205, 75.46989454821407] Fitness: 1.8241923410779313
Generation 350 Best Solution: [183.22261048721907, 176.6818963402404,
174.24580167252205, 74.97699672825432] Fitness: 0.2875865016406841
Generation 351 Best Solution: [183.22261048721907, 176.46427107813648,
174.35097474371918, 75.87820594627784] Fitness: 1.3151515642693552
Generation 352 Best Solution: [183.22261048721907, 176.46427107813648,
174.35097474371918, 75.87820594627784] Fitness: 1.3151515642693552
Generation 353 Best Solution: [183.22261048721907, 176.46427107813648,
175.0307160404382, 75.87820594627784] Fitness: 1.9341657474378469
Generation 354 Best Solution: [183.77159769683945, 177.37727985426721,
175.0307160404382, 75.29728542654765] Fitness: 2.344786279252223
Generation 355 Best Solution: [183.0483011608314, 177.37727985426721,
174.24580167252205, 75.87820594627784] Fitness: 2.381962395058636
Generation 356 Best Solution: [182.89821217298172, 176.46246501123605,
175.13414612400686, 75.87820594627784] Fitness: 2.197782915158811
Generation 357 Best Solution: [183.49300039517775, 177.37727985426721,
175.13414612400686, 75.32791971965409] Fitness: 2.3574554831600922

Generation 358 Best Solution: [183.77159769683945, 175.99092457793324,
175.0307160404382, 75.87820594627784] Fitness: 2.1735274686328943
Generation 359 Best Solution: [183.77159769683945, 177.37727985426721,
174.8875645785927, 74.96632244059272] Fitness: 1.9014679522871216
Generation 360 Best Solution: [183.77159769683945, 175.99092457793324,
175.0307160404382, 75.14678639124554] Fitness: 1.1543271924734595
Generation 361 Best Solution: [183.77159769683945, 176.54795588930574,
174.35097474371918, 75.87820594627784] Fitness: 1.5874074900835236
Generation 362 Best Solution: [183.49300039517775, 176.54795588930574,
174.35097474371918, 75.87820594627784] Fitness: 1.4050210017470102
Generation 363 Best Solution: [183.77159769683945, 176.54795588930574,
174.35097474371918, 75.87820594627784] Fitness: 1.5874074900835236
Generation 364 Best Solution: [183.77159769683945, 177.37727985426721,
175.13414612400686, 74.90460467783622] Fitness: 2.233825301377034
Generation 365 Best Solution: [183.2353303469225, 177.37727985426721,
175.0307160404382, 75.10144746572826] Fitness: 1.9415205612694932
Generation 366 Best Solution: [183.77159769683945, 176.54943745435622,
174.24580167252205, 75.87820594627784] Fitness: 1.562269659758702
Generation 367 Best Solution: [183.77159769683945, 176.54943745435622,
174.24580167252205, 75.87820594627784] Fitness: 1.562269659758702
Generation 368 Best Solution: [183.71661906687726, 176.36424241152946,
175.0307160404382, 75.87820594627784] Fitness: 2.091145557548775
Generation 369 Best Solution: [183.71661906687726, 176.36424241152946,
174.13941973740833, 75.87820594627784] Fitness: 1.4390256311616199
Generation 370 Best Solution: [183.77159769683945, 176.36424241152946,
175.0307160404382, 75.87820594627784] Fitness: 2.137540688665638
Generation 371 Best Solution: [183.77159769683945, 176.36424241152946,
175.0307160404382, 75.60755904847272] Fitness: 1.654885355230977
Generation 372 Best Solution: [183.32379986095754, 176.50866703656544,
175.0307160404382, 75.87820594627784] Fitness: 1.9578452926420007
Generation 373 Best Solution: [182.86082249692654, 177.37727985426721,
175.13414612400686, 74.37602440619851] Fitness: 2.1684003879093936
Generation 374 Best Solution: [183.77159769683945, 176.6596843655929,
174.64010925973443, 75.87820594627784] Fitness: 1.8249703731848896
Generation 375 Best Solution: [183.14759227570295, 177.37727985426721,
174.67699992776252, 74.43760515357341] Fitness: 1.4394799016609046
Generation 376 Best Solution: [182.86082249692654, 177.37727985426721,
174.67699992776252, 75.41389653091585] Fitness: 1.9475638881426158
Generation 377 Best Solution: [183.77159769683945, 177.37727985426721,
174.67699992776252, 75.41389653091585] Fitness: 2.0566202599071115
Generation 378 Best Solution: [183.77159769683945, 177.37727985426721,
175.0307160404382, 74.45962930619487] Fitness: 2.076345711369991
Generation 379 Best Solution: [182.86082249692654, 177.37727985426721,
173.9721770104444, 75.87820594627784] Fitness: 2.494507548827642
Generation 380 Best Solution: [183.77159769683945, 175.7327510336387,
175.0307160404382, 75.87820594627784] Fitness: 2.336366822682923
Generation 381 Best Solution: [183.2353303469225, 177.37727985426721,
175.11566800905462, 75.60755904847272] Fitness: 2.6437744201777775

Generation 382 Best Solution: [183.77159769683945, 177.37727985426721,
175.0307160404382, 75.04743017423765] Fitness: 2.1404455897673125
Generation 383 Best Solution: [183.77159769683945, 176.83071127601417,
175.0307160404382, 75.60755904847272] Fitness: 1.9413866049542154
Generation 384 Best Solution: [182.99853710525537, 177.37727985426721,
175.0307160404382, 74.93106665051394] Fitness: 1.8986882980121178
Generation 385 Best Solution: [182.99853710525537, 177.37727985426721,
175.11566800905462, 75.04743017423765] Fitness: 2.0901266752866277
Generation 386 Best Solution: [183.77159769683945, 177.37727985426721,
174.08779726811483, 74.33698711579193] Fitness: 1.487994704952128
Generation 387 Best Solution: [183.77159769683945, 177.37727985426721,
175.0307160404382, 74.49108017077278] Fitness: 2.0649697019963296
Generation 388 Best Solution: [183.77159769683945, 177.37727985426721,
175.0307160404382, 74.49108017077278] Fitness: 2.0649697019963296
Generation 389 Best Solution: [183.77159769683945, 175.89583285263032,
175.0307160404382, 74.49108017077278] Fitness: 1.0571025867867503
Generation 390 Best Solution: [182.86082249692654, 175.89583285263032,
175.0307160404382, 74.49108017077278] Fitness: 0.9480462150222548
Generation 391 Best Solution: [183.77159769683945, 177.37727985426721,
174.47483992432558, 74.3323767191972] Fitness: 1.574986586121654
Generation 392 Best Solution: [182.86082249692654, 175.89583285263032,
175.0307160404382, 75.59452728567084] Fitness: 1.6085596438679421
Generation 393 Best Solution: [183.77159769683945, 175.89583285263032,
175.0307160404382, 75.59452728567084] Fitness: 1.7176160156324376
Generation 394 Best Solution: [183.77159769683945, 175.89583285263032,
175.0307160404382, 74.47808419341641] Fitness: 1.0616003865771289
Generation 395 Best Solution: [183.77159769683945, 177.37727985426721,
174.47483992432558, 74.59040137829885] Fitness: 1.4755589486124825
Generation 396 Best Solution: [183.72734321496583, 176.72350491234684,
175.0406855631274, 75.60755904847272] Fitness: 1.8201017794190046
Generation 397 Best Solution: [183.19111431029083, 177.37727985426721,
175.0307160404382, 74.47808419341641] Fitness: 1.8377281709784332
Generation 398 Best Solution: [183.77159769683945, 176.4767925854783,
175.0307160404382, 75.60755904847272] Fitness: 1.6903073621338351
Generation 399 Best Solution: [183.19111431029083, 176.28013956261205,
175.0406855631274, 75.87820594627784] Fitness: 1.9082240412678038
Generation 400 Best Solution: [183.19111431029083, 176.28013956261205,
175.0406855631274, 75.87820594627784] Fitness: 1.9082240412678038
Generation 401 Best Solution: [183.34153674173086, 177.37727985426721,
175.0307160404382, 74.70134941878338] Fitness: 1.8054942484493213
Generation 402 Best Solution: [183.77159769683945, 176.14720599849792,
175.0307160404382, 75.87820594627784] Fitness: 2.12976189476175
Generation 403 Best Solution: [182.86082249692654, 177.37727985426721,
175.0307160404382, 74.5029769380988] Fitness: 1.9520465406011236
Generation 404 Best Solution: [182.86082249692654, 177.37727985426721,
175.0307160404382, 74.73622736701459] Fitness: 1.92461713984255
Generation 405 Best Solution: [182.86082249692654, 177.37727985426721,
175.0307160404382, 74.73622736701459] Fitness: 1.92461713984255

Generation 406 Best Solution: [183.77159769683945, 176.1270843202638,
175.0307160404382, 75.87820594627784] Fitness: 2.1330785765142597
Generation 407 Best Solution: [182.86082249692654, 176.1270843202638,
175.0307160404382, 75.87820594627784] Fitness: 2.024022204749764
Generation 408 Best Solution: [183.77159769683945, 176.1270843202638,
173.80737477298055, 75.87820594627784] Fitness: 1.5817281766221414
Generation 409 Best Solution: [183.77159769683945, 176.1270843202638,
173.80737477298055, 75.87820594627784] Fitness: 1.5817281766221414
Generation 410 Best Solution: [182.97248862169982, 177.37727985426721,
175.0406855631274, 74.40478281608651] Fitness: 1.9438954674845799
Generation 411 Best Solution: [182.97248862169982, 176.28115210773365,
175.0307160404382, 75.87820594627784] Fitness: 1.9539546311100722
Generation 412 Best Solution: [182.97248862169982, 176.28115210773365,
175.0307160404382, 74.40478281608651] Fitness: 0.8258932250270482
Generation 413 Best Solution: [182.86082249692654, 177.37727985426721,
175.0307160404382, 74.91300764741577] Fitness: 1.9651621752580106
Generation 414 Best Solution: [182.86082249692654, 177.37727985426721,
175.023371572026, 74.91300764741577] Fitness: 1.9542952072905717
Generation 415 Best Solution: [182.86082249692654, 176.68068049059445,
175.023371572026, 75.87820594627784] Fitness: 2.1717994156363813
Generation 416 Best Solution: [182.86082249692654, 177.02982936632972,
175.0307160404382, 75.87820594627784] Fitness: 2.5494233320251167
Generation 417 Best Solution: [183.77159769683945, 175.68312684528743,
175.0307160404382, 75.87820594627784] Fitness: 2.3805909195761448
Generation 418 Best Solution: [183.77159769683945, 177.37727985426721,
175.0406855631274, 74.38032720004692] Fitness: 2.1273586768246266
Generation 419 Best Solution: [182.97248862169982, 177.37727985426721,
175.0307160404382, 74.65562685007782] Fitness: 1.8615504001441727
Generation 420 Best Solution: [183.77159769683945, 177.37727985426721,
174.05371800143658, 74.65562685007782] Fitness: 1.3887659830459038
Generation 421 Best Solution: [183.50618801480792, 177.37727985426721,
175.0307160404382, 74.65562685007782] Fitness: 1.8560263241139903
Generation 422 Best Solution: [183.50618801480792, 175.62582584123894,
175.0307160404382, 74.65562685007782] Fitness: 1.0646166512280362
Generation 423 Best Solution: [183.77159769683945, 177.37727985426721,
175.0307160404382, 74.92545407788126] Fitness: 2.079066028595143
Generation 424 Best Solution: [183.58312058405957, 177.42712077217863,
175.0406855631274, 75.35730278756726] Fitness: 2.385230665089203
Generation 425 Best Solution: [183.58312058405957, 176.54814761770848,
175.0406855631274, 75.87820594627784] Fitness: 2.0836596452460165
Generation 426 Best Solution: [183.77159769683945, 176.23385577228947,
175.0307160404382, 75.87820594627784] Fitness: 2.1233076829054642
Generation 427 Best Solution: [183.77159769683945, 176.23385577228947,
174.4078900776072, 75.87820594627784] Fitness: 1.5261293435550123
Generation 428 Best Solution: [183.77159769683945, 177.42712077217863,
174.3393843919705, 75.02847415451423] Fitness: 1.6049525100787179
Generation 429 Best Solution: [183.77159769683945, 177.42712077217863,
174.3393843919705, 75.02847415451423] Fitness: 1.6049525100787179

Generation 430 Best Solution: [183.77159769683945, 177.42712077217863,
174.3393843919705, 75.48731311229544] Fitness: 2.046625947198792
Generation 431 Best Solution: [182.86082249692654, 176.61760988053857,
175.0307160404382, 75.48731311229544] Fitness: 1.4810914589154764
Generation 432 Best Solution: [182.86082249692654, 176.61760988053857,
175.0406855631274, 75.64696698953347] Fitness: 1.733222923503926
Generation 433 Best Solution: [183.77159769683945, 177.37727985426721,
174.3393843919705, 75.87820594627784] Fitness: 2.6054614061683994
Generation 434 Best Solution: [183.77159769683945, 177.42712077217863,
175.0406855631274, 75.48731311229544] Fitness: 2.6843120624747487
Generation 435 Best Solution: [183.77159769683945, 177.42712077217863,
175.0307160404382, 74.8634690580935] Fitness: 2.156010327310757
Generation 436 Best Solution: [183.77159769683945, 177.42712077217863,
175.0307160404382, 74.8634690580935] Fitness: 2.156010327310757
Generation 437 Best Solution: [183.77159769683945, 177.42712077217863,
174.07014121205123, 74.8634690580935] Fitness: 1.509511711395828
Generation 438 Best Solution: [183.77159769683945, 177.42712077217863,
175.0307160404382, 74.8634690580935] Fitness: 2.156010327310757
Generation 439 Best Solution: [183.77159769683945, 177.42712077217863,
175.0307160404382, 74.8634690580935] Fitness: 2.156010327310757
Generation 440 Best Solution: [183.77159769683945, 177.42712077217863,
173.88519479251252, 74.8648759584715] Fitness: 1.5647197089472433
Generation 441 Best Solution: [183.77159769683945, 177.42712077217863,
175.0307160404382, 74.8634690580935] Fitness: 2.156010327310757
Generation 442 Best Solution: [183.77159769683945, 177.42712077217863,
174.38020822609718, 74.92739779396985] Fitness: 1.5698505692305593
Generation 443 Best Solution: [183.77159769683945, 177.42712077217863,
175.0307160404382, 74.7081081321575] Fitness: 2.1305867177076436
Generation 444 Best Solution: [182.84716794621863, 176.42549692922017,
175.0307160404382, 75.87820594627784] Fitness: 2.0541494893212655
Generation 445 Best Solution: [182.84716794621863, 176.42549692922017,
175.0307160404382, 75.87820594627784] Fitness: 2.0541494893212655
Generation 446 Best Solution: [183.69312580743116, 177.42712077217863,
175.0307160404382, 74.45715256529942] Fitness: 2.1111392961367748
Generation 447 Best Solution: [183.822739803386, 177.16459250932104,
175.0307160404382, 74.45715256529942] Fitness: 1.752031896677982
Generation 448 Best Solution: [183.39565848176198, 177.16459250932104,
173.89524665570121, 75.87820594627784] Fitness: 2.0438992304395223
Generation 449 Best Solution: [183.39565848176198, 177.42712077217863,
175.0307160404382, 75.19267193646127] Fitness: 2.1297124741660687
Generation 450 Best Solution: [182.79821835446745, 176.67520639569022,
175.0406855631274, 75.87820594627784] Fitness: 2.2375222342711227
Generation 451 Best Solution: [182.79821835446745, 176.67520639569022,
175.0307160404382, 75.87820594627784] Fitness: 2.2226251464325473
Generation 452 Best Solution: [183.77159769683945, 176.67520639569022,
174.4603235802974, 75.87820594627784] Fitness: 1.7154046688689517
Generation 453 Best Solution: [183.822739803386, 176.67520639569022,
174.4603235802974, 74.67918312005563] Fitness: 0.5667806461649195

Generation 454 Best Solution: [183.69312580743116, 177.06156981992294,
175.0307160404382, 74.71796149382426] Fitness: 1.4416382814033217
Generation 455 Best Solution: [182.79821835446745, 176.90206856963965,
175.0307160404382, 75.53688534843678] Fitness: 1.8470732020031446
Generation 456 Best Solution: [183.63418553556224, 175.9777980554755,
175.0406855631274, 75.87820594627784] Fitness: 2.0876080861342636
Generation 457 Best Solution: [183.21495808400994, 177.42712077217863,
175.0307160404382, 75.54961703489987] Fitness: 2.523411082724425
Generation 458 Best Solution: [183.21495808400994, 177.42712077217863,
175.0307160404382, 75.54961703489987] Fitness: 2.523411082724425
Generation 459 Best Solution: [183.21495808400994, 177.42712077217863,
175.0307160404382, 75.54961703489987] Fitness: 2.523411082724425
Generation 460 Best Solution: [183.69312580743116, 177.42712077217863,
175.0307160404382, 75.23352160058629] Fitness: 2.3164241582702703
Generation 461 Best Solution: [183.39565848176198, 176.63237759173836,
175.0307160404382, 75.87820594627784] Fitness: 2.042080755598042
Generation 462 Best Solution: [183.39565848176198, 176.63237759173836,
175.0307160404382, 75.87820594627784] Fitness: 2.042080755598042
Generation 463 Best Solution: [183.39565848176198, 177.42712077217863,
175.0307160404382, 75.36897276569069] Fitness: 2.3062586561559093
Generation 464 Best Solution: [183.77159769683945, 176.86160019517672,
175.0307160404382, 75.62227365748566] Fitness: 1.9964122264104072
Generation 465 Best Solution: [183.77159769683945, 176.86160019517672,
173.88183971267787, 75.87820594627784] Fitness: 1.8658340387935048
Generation 466 Best Solution: [183.77159769683945, 176.86160019517672,
175.0406855631274, 74.50043341685328] Fitness: 1.3045691359592684
Generation 467 Best Solution: [182.79821835446745, 176.92976056462444,
175.0307160404382, 75.87820594627784] Fitness: 2.4672837484746246
Generation 468 Best Solution: [182.79821835446745, 176.27911327277383,
175.0307160404382, 75.87820594627784] Fitness: 2.0599979525157397
Generation 469 Best Solution: [182.79821835446745, 177.42712077217863,
175.0307160404382, 74.62331292847853] Fitness: 2.068951756948904
Generation 470 Best Solution: [182.79821835446745, 177.42712077217863,
175.0307160404382, 75.40358816779482] Fitness: 2.497257563839802
Generation 471 Best Solution: [183.77159769683945, 176.92976056462444,
175.0307160404382, 75.87820594627784] Fitness: 2.5322737836612914
Generation 472 Best Solution: [183.39565848176198, 177.42712077217863,
175.0307160404382, 74.30767040150882] Fitness: 2.0382328801079677
Generation 473 Best Solution: [183.77159769683945, 176.45944493474042,
175.0307160404382, 75.87820594627784] Fitness: 2.16610555258499
Generation 474 Best Solution: [183.1740383250884, 176.09091137669387,
175.0307160404382, 75.87820594627784] Fitness: 1.9108417804322984
Generation 475 Best Solution: [183.39565848176198, 177.42712077217863,
175.0307160404382, 74.43389771529122] Fitness: 1.970237617656624
Generation 476 Best Solution: [182.79821835446745, 177.42712077217863,
175.0307160404382, 74.43389771529122] Fitness: 2.120395222926645
Generation 477 Best Solution: [183.5928623326448, 176.62123796672526,
175.0307160404382, 75.87820594627784] Fitness: 2.1177198336608956

Generation 478 Best Solution: [183.1740383250884, 175.63797046644873,
175.0307160404382, 75.87820594627784] Fitness: 2.1945325995958074
Generation 479 Best Solution: [183.1740383250884, 175.63797046644873,
175.0307160404382, 75.30223889154843] Fitness: 1.3161875722114964
Generation 480 Best Solution: [183.1740383250884, 175.63797046644873,
175.0307160404382, 75.87820594627784] Fitness: 2.1945325995958074
Generation 481 Best Solution: [183.1740383250884, 175.63797046644873,
175.0307160404382, 75.87820594627784] Fitness: 2.1945325995958074
Generation 482 Best Solution: [182.79821835446745, 177.31028807839138,
173.9137883570217, 75.07717615357063] Fitness: 1.3645263514041153
Generation 483 Best Solution: [182.79821835446745, 175.6651988654342,
173.9137883570217, 75.87820594627784] Fitness: 1.7286413794813507
Generation 484 Best Solution: [183.63121489115022, 177.42712077217863,
173.9137883570217, 75.17461014767025] Fitness: 1.617461454195857
Generation 485 Best Solution: [182.79821835446745, 177.3092815073849,
175.0307160404382, 74.75652234752383] Fitness: 1.8430641771956793
Generation 486 Best Solution: [183.1740383250884, 176.68297275714738,
175.0307160404382, 75.87820594627784] Fitness: 2.063535570538773
Generation 487 Best Solution: [183.1740383250884, 177.31028807839138,
174.24240496899387, 75.87820594627784] Fitness: 2.227108075172109
Generation 488 Best Solution: [183.1740383250884, 177.42712077217863,
175.0307160404382, 74.63935310669504] Fitness: 1.902451671935152
Generation 489 Best Solution: [183.1740383250884, 176.0513630468816,
175.0307160404382, 75.87820594627784] Fitness: 1.9217783579499752
Generation 490 Best Solution: [183.1740383250884, 177.3092815073849,
174.5024336820368, 75.36391955245516] Fitness: 1.5108278446222099
Generation 491 Best Solution: [183.1740383250884, 176.0513630468816,
174.5024336820368, 75.36391955245516] Fitness: 0.5618647925645962
Generation 492 Best Solution: [183.1740383250884, 176.05210351002876,
174.5024336820368, 75.87820594627784] Fitness: 1.3727595284421643
Generation 493 Best Solution: [183.54699661354346, 177.31028807839138,
174.5024336820368, 74.86061190471472] Fitness: 1.2248428264672921
Generation 494 Best Solution: [183.58161164881577, 177.42712077217863,
174.5024336820368, 74.86061190471472] Fitness: 1.4677631666672852
Generation 495 Best Solution: [183.1182825556105, 177.42712077217863,
175.0307160404382, 74.76826874270436] Fitness: 1.9150320493965833
Generation 496 Best Solution: [183.1182825556105, 177.3092815073849,
175.0307160404382, 74.76826874270436] Fitness: 1.6889580990500819
Generation 497 Best Solution: [182.79821835446745, 176.28210636070864,
175.0307160404382, 75.80377329365115] Fitness: 1.915134718495786
Generation 498 Best Solution: [182.79821835446745, 177.42712077217863,
175.0307160404382, 74.76260575899016] Fitness: 2.0698636488446693
Generation 499 Best Solution: [183.4278284865359, 176.64249595538274,
175.0307160404382, 75.06940443937101] Fitness: 0.9839620230084725
Generation 500 Best Solution: [183.8552763649043, 177.0555762759213,
175.0307160404382, 75.08473800766178] Fitness: 1.7101216422320884
Optimal Solution for Simulation 25: [183.8552763649043, 177.0555762759213,
175.0307160404382, 75.08473800766178]

```
--------------------------------------------------
Simulation 26/32 with conditions: {'drought': False, 'flood': False,
'policy_change': True, 'social_event': True, 'economic_crisis': False}
Generation 1 Best Solution: [183.3088227451412, 175.99709828992349,
174.51007946015994, 75.08620414300549] Fitness: 0.32683261770020544
Generation 2 Best Solution: [183.41820063635447, 176.60747475523834,
174.65765346323403, 74.66157819737904] Fitness: 0.4399976565946618
Generation 3 Best Solution: [182.81013750749906, 176.63633912936115,
173.80534359360118, 75.4100306762747] Fitness: 0.8287847485562418
Generation 4 Best Solution: [183.26285640020515, 177.33661111253681,
173.80534359360118, 74.80323062083924] Fitness: 1.1905251535620145
Generation 5 Best Solution: [182.98708218734598, 177.1440895412551,
174.86849914966191, 74.42929112439236] Fitness: 1.3196613586320536
Generation 6 Best Solution: [182.98708218734598, 177.33661111253681,
174.5257794153617, 74.89318388965768] Fitness: 1.2912651466112897
Generation 7 Best Solution: [182.92734913084453, 177.08154744493694,
173.74536174495577, 75.2014612305018] Fitness: 1.0670766664272378
Generation 8 Best Solution: [183.26285640020515, 177.33661111253681,
174.33540245410322, 75.37261161649953] Fitness: 1.4887056182680425
Generation 9 Best Solution: [182.98708218734598, 177.40197843761018,
173.79696155621411, 74.89318388965768] Fitness: 1.3828848647694916
Generation 10 Best Solution: [183.6680535261486, 177.40197843761018,
174.33540245410322, 74.89318388965768] Fitness: 1.4437979675595083
Generation 11 Best Solution: [182.92734913084453, 177.40197843761018,
174.86849914966191, 74.44750709175509] Fitness: 1.7876590013146267
Generation 12 Best Solution: [182.92734913084453, 177.33758007067263,
174.2352608091985, 74.31487555344968] Fitness: 1.3017837085999737
Generation 13 Best Solution: [183.03258629665842, 175.9042122596934,
174.9344212904535, 75.77219137243556] Fitness: 1.609252201140043
Generation 14 Best Solution: [183.26285640020515, 177.41165717953118,
174.58928007217105, 74.44750709175509] Fitness: 1.4651609281712807
Generation 15 Best Solution: [183.4031203621943, 177.41165717953118,
174.86849914966191, 74.44750709175509] Fitness: 1.7642233413423951
Generation 16 Best Solution: [182.92734913084453, 176.71738597022213,
174.86849914966191, 75.77219137243556] Fitness: 1.7022106431900237
Generation 17 Best Solution: [182.92734913084453, 177.41165717953118,
174.18117100434623, 75.77219137243556] Fitness: 2.2354459468179457
Generation 18 Best Solution: [182.92734913084453, 175.57229489035828,
174.86849914966191, 75.77219137243556] Fitness: 1.8247901979740644
Generation 19 Best Solution: [182.92734913084453, 176.6501949977424,
174.86849914966191, 75.77219137243556] Fitness: 1.6499112005100707
Generation 20 Best Solution: [183.7004994205213, 175.67724270092842,
174.67893614233154, 75.77219137243556] Fitness: 1.6444894442426743
Generation 21 Best Solution: [183.7004994205213, 175.67724270092842,
174.67893614233154, 75.77219137243556] Fitness: 1.6444894442426743
Generation 22 Best Solution: [182.92734913084453, 177.41165717953118,
175.1051810425243, 74.64135154997979] Fitness: 2.085609415022938
Generation 23 Best Solution: [182.92734913084453, 177.41165717953118,
```

174.9344212904535, 75.0970491831031] Fitness: 1.9562725194146344
Generation 24 Best Solution: [182.92734913084453, 176.39762761210145,
174.18117100434623, 75.77219137243556] Fitness: 1.0782372077177227
Generation 25 Best Solution: [182.92734913084453, 176.56401196238738,
174.86849914966191, 75.77219137243556] Fitness: 1.593686742293872
Generation 26 Best Solution: [183.4031203621943, 177.40197843761018,
174.0541276332099, 75.77219137243556] Fitness: 2.1773440576674687
Generation 27 Best Solution: [182.92734913084453, 175.66397179488132,
175.1051810425243, 75.77219137243556] Fitness: 2.0672954396662124
Generation 28 Best Solution: [182.92734913084453, 175.85119273987613,
174.0541276332099, 75.77219137243556] Fitness: 1.1584668968671212
Generation 29 Best Solution: [182.92734913084453, 176.1233857261998,
174.86849914966191, 75.77219137243556] Fitness: 1.4969040261149118
Generation 30 Best Solution: [182.92734913084453, 176.1233857261998,
174.86849914966191, 75.77219137243556] Fitness: 1.4969040261149118
Generation 31 Best Solution: [182.92734913084453, 177.41165717953118,
175.0635956423225, 74.37798645087899] Fitness: 2.107351475803318
Generation 32 Best Solution: [182.90962490766898, 175.58028424813966,
175.1051810425243, 75.77219137243556] Fitness: 2.1568869139279654
Generation 33 Best Solution: [182.92734913084453, 176.5189655241648,
174.86849914966191, 75.77219137243556] Fitness: 1.569154963329888
Generation 34 Best Solution: [182.92734913084453, 176.2233062858295,
174.86849914966191, 75.77219137243556] Fitness: 1.4908870822142966
Generation 35 Best Solution: [183.26289461358417, 176.62881438415843,
174.86849914966191, 75.77219137243556] Fitness: 1.56622491824582
Generation 36 Best Solution: [183.26289461358417, 177.3588808271084,
174.46161740910605, 75.77219137243556] Fitness: 2.1513244403377145
Generation 37 Best Solution: [183.26289461358417, 176.4533335294598,
174.86849914966191, 75.77219137243556] Fitness: 1.4707796153233632
Generation 38 Best Solution: [183.26289461358417, 176.4533335294598,
174.86849914966191, 75.77219137243556] Fitness: 1.4707796153233632
Generation 39 Best Solution: [183.4031203621943, 177.41165717953118,
174.62798891046052, 75.77219137243556] Fitness: 2.3916651482666493
Generation 40 Best Solution: [182.92734913084453, 177.40197843761018,
174.86849914966191, 74.63272813223611] Fitness: 1.7359651990783234
Generation 41 Best Solution: [183.4031203621943, 177.3588808271084,
174.86849914966191, 74.9919161419321] Fitness: 1.6710087458271878
Generation 42 Best Solution: [182.92734913084453, 176.75572534894528,
174.86849914966191, 75.77219137243556] Fitness: 1.7353763423033635
Generation 43 Best Solution: [183.4031203621943, 176.75572534894528,
174.86849914966191, 75.77219137243556] Fitness: 1.6929103333312103
Generation 44 Best Solution: [183.26289461358417, 177.3588808271084,
174.86849914966191, 75.02416576007073] Fitness: 1.660665612238752
Generation 45 Best Solution: [183.4031203621943, 176.22368805454548,
174.86849914966191, 75.77219137243556] Fitness: 1.4484295385520458
Generation 46 Best Solution: [183.4031203621943, 176.22368805454548,
174.86849914966191, 75.71502319853967] Fitness: 1.3513540282772807
Generation 47 Best Solution: [182.92734913084453, 176.17231434882285,

174.86849914966191, 75.77219137243556] Fitness: 1.4919082420152825
Generation 48 Best Solution: [183.4031203621943, 177.41165717953118,
174.86849914966191, 74.55627966058414] Fitness: 1.7270351472366405
Generation 49 Best Solution: [183.4031203621943, 176.2811059373938,
174.86849914966191, 75.77219137243556] Fitness: 1.4524288208518463
Generation 50 Best Solution: [182.92734913084453, 176.2811059373938,
174.86849914966191, 75.55049986802611] Fitness: 1.148407646928768
Generation 51 Best Solution: [183.26289461358417, 176.2811059373938,
175.1051810425243, 75.77219137243556] Fitness: 1.757942522679096
Generation 52 Best Solution: [183.26289461358417, 176.2811059373938,
175.1051810425243, 75.77219137243556] Fitness: 1.757942522679096
Generation 53 Best Solution: [183.26289461358417, 176.82194862174228,
174.86849914966191, 75.77219137243556] Fitness: 1.7297513263639692
Generation 54 Best Solution: [183.26289461358417, 176.82194862174228,
174.86849914966191, 75.77219137243556] Fitness: 1.7297513263639692
Generation 55 Best Solution: [183.26289461358417, 177.41165717953118,
174.86849914966191, 74.92810172081703] Fitness: 1.7205650527749345
Generation 56 Best Solution: [183.85331534386788, 176.82194862174228,
174.04554259612758, 75.77219137243556] Fitness: 1.6212852770938047
Generation 57 Best Solution: [182.92734913084453, 176.94991625597964,
174.86849914966191, 75.77219137243556] Fitness: 1.9404541250095877
Generation 58 Best Solution: [183.26289461358417, 177.41165717953118,
174.72450937318013, 75.65075216123026] Fitness: 2.2517283222889404
Generation 59 Best Solution: [182.92734913084453, 177.41165717953118,
174.86849914966191, 74.29659544061876] Fitness: 1.8904757240745993
Generation 60 Best Solution: [182.92734913084453, 176.49515162635416,
174.86849914966191, 75.77219137243556] Fitness: 1.5575331905609535
Generation 61 Best Solution: [183.26289461358417, 177.41165717953118,
174.72450937318013, 74.64879150885531] Fitness: 1.527829485223312
Generation 62 Best Solution: [183.26289461358417, 177.41165717953118,
174.55510660587254, 74.64879150885531] Fitness: 1.387036621792475
Generation 63 Best Solution: [183.26289461358417, 177.41165717953118,
174.86849914966191, 74.74341460383945] Fitness: 1.682408196628762
Generation 64 Best Solution: [182.92734913084453, 177.41165717953118,
174.72450937318013, 75.77219137243556] Fitness: 2.5201290298613053
Generation 65 Best Solution: [183.26289461358417, 177.41165717953118,
174.72450937318013, 75.77219137243556] Fitness: 2.4515292260315844
Generation 66 Best Solution: [183.41343548305355, 177.41165717953118,
174.33213776665613, 75.77219137243556] Fitness: 2.226619832094737
Generation 67 Best Solution: [183.0592494694584, 177.41165717953118,
174.33213776665613, 75.77219137243556] Fitness: 2.2168575306636202
Generation 68 Best Solution: [182.92734913084453, 177.41165717953118,
174.86849914966191, 75.32747133248861] Fitness: 2.0632776128155172
Generation 69 Best Solution: [183.6955397040247, 177.41165717953118,
174.72450937318013, 74.62828841225732] Fitness: 1.714790389841108
Generation 70 Best Solution: [183.57856838618562, 176.33921523042528,
174.86849914966191, 75.77219137243556] Fitness: 1.5401831074812837
Generation 71 Best Solution: [183.261974008619, 176.33921523042528,

174.86849914966191, 75.77219137243556] Fitness: 1.435792046712986
Generation 72 Best Solution: [183.0592494694584, 177.41165717953118,
174.72450937318013, 74.96507514817516] Fitness: 1.5977663691037436
Generation 73 Best Solution: [183.26289461358417, 176.70334278910175,
174.86849914966191, 75.77219137243556] Fitness: 1.6220669625728161
Generation 74 Best Solution: [183.0592494694584, 177.41165717953118,
174.86849914966191, 74.96507514817516] Fitness: 1.7545053913550723
Generation 75 Best Solution: [183.57856838618562, 177.41165717953118,
175.1051810425243, 74.65703635919452] Fitness: 2.1197689916326388
Generation 76 Best Solution: [183.6073786198221, 175.7853826502544,
175.1051810425243, 75.77219137243556] Fitness: 2.023892868099288
Generation 77 Best Solution: [183.26289461358417, 177.41165717953118,
175.1051810425243, 74.90585815154867] Fitness: 2.0446488802743383
Generation 78 Best Solution: [183.26289461358417, 176.66855115248183,
175.1051810425243, 74.65703635919452] Fitness: 1.0020374525238653
Generation 79 Best Solution: [183.22332482307635, 177.41165717953118,
174.86849914966191, 74.65703635919452] Fitness: 1.6822647009086233
Generation 80 Best Solution: [183.6073786198221, 176.22284186300757,
175.1051810425243, 74.90585815154867] Fitness: 0.9806579839070455
Generation 81 Best Solution: [183.26289461358417, 177.41165717953118,
175.1051810425243, 74.65703635919452] Fitness: 2.0154427670437696
Generation 82 Best Solution: [183.26289461358417, 177.41165717953118,
175.1051810425243, 74.65703635919452] Fitness: 2.0154427670437696
Generation 83 Best Solution: [183.26289461358417, 177.41165717953118,
175.1051810425243, 74.65703635919452] Fitness: 2.0154427670437696
Generation 84 Best Solution: [183.26289461358417, 176.84020712548707,
175.1051810425243, 75.77219137243556] Fitness: 2.0800287453107567
Generation 85 Best Solution: [183.26289461358417, 176.84020712548707,
175.1051810425243, 74.58252700767973] Fitness: 1.1665992309598445
Generation 86 Best Solution: [183.26289461358417, 176.84020712548707,
175.1051810425243, 75.77219137243556] Fitness: 2.0800287453107567
Generation 87 Best Solution: [183.70698619331174, 177.41165717953118,
174.00475239390582, 75.77219137243556] Fitness: 2.3723272638340647
Generation 88 Best Solution: [183.70698619331174, 175.92414714263117,
174.00475239390582, 75.77219137243556] Fitness: 1.2533213072549079
Generation 89 Best Solution: [183.6073786198221, 176.72144382150915,
175.1051810425243, 75.43546366542512] Fitness: 1.5961930060171348
Generation 90 Best Solution: [183.22332482307635, 177.41165717953118,
175.1051810425243, 74.57133062897114] Fitness: 2.0274033792336494
Generation 91 Best Solution: [183.06057809578317, 176.01903347055998,
175.1051810425243, 75.77219137243556] Fitness: 1.802966361029808
Generation 92 Best Solution: [183.06057809578317, 176.85560171042954,
175.1051810425243, 75.77219137243556] Fitness: 2.1153830439688437
Generation 93 Best Solution: [182.9828271682245, 177.41165717953118,
175.1051810425243, 74.34126347928587] Fitness: 2.1693201121018597
Generation 94 Best Solution: [183.70698619331174, 177.41165717953118,
175.1051810425243, 74.85412855727529] Fitness: 2.223814277552934
Generation 95 Best Solution: [183.06057809578317, 176.88544220239814,

175.1051810425243, 75.77219137243556] Fitness: 2.1477642513342268
Generation 96 Best Solution: [183.70698619331174, 177.28459751381638,
175.1051810425243, 75.16919893235784] Fitness: 2.1423864395126
Generation 97 Best Solution: [182.9828271682245, 176.43493357642015,
175.1051810425243, 75.77219137243556] Fitness: 1.8398443125647934
Generation 98 Best Solution: [182.9828271682245, 177.41165717953118,
175.1051810425243, 75.16919893235784] Fitness: 2.2308426788416904
Generation 99 Best Solution: [183.06057809578317, 177.41165717953118,
175.1051810425243, 75.16919893235784] Fitness: 2.2055689653067105
Generation 100 Best Solution: [183.06057809578317, 177.41165717953118,
175.1051810425243, 75.16919893235784] Fitness: 2.2055689653067105
Generation 101 Best Solution: [183.06057809578317, 175.91870220344333,
175.1051810425243, 75.16919893235784] Fitness: 1.0891443892595345
Generation 102 Best Solution: [183.06057809578317, 177.41165717953118,
175.1051810425243, 74.55052674981437] Fitness: 2.0532498701303594
Generation 103 Best Solution: [183.84344433424212, 176.4185155119413,
175.1051810425243, 75.77219137243556] Fitness: 2.1076824778890697
Generation 104 Best Solution: [182.95601610290856, 177.41165717953118,
175.1051810425243, 75.16919893235784] Fitness: 2.241860690694929
Generation 105 Best Solution: [183.6073786198221, 177.41165717953118,
174.58876998445322, 75.16919893235784] Fitness: 1.7033932418834419
Generation 106 Best Solution: [183.06057809578317, 177.41165717953118,
175.1051810425243, 74.46534661473007] Fitness: 2.0815263669940443
Generation 107 Best Solution: [183.79223837866436, 177.41165717953118,
173.9940260540967, 75.16919893235784] Fitness: 1.6953669381120624
Generation 108 Best Solution: [183.2851078503801, 175.99740010840287,
175.1051810425243, 75.16919893235784] Fitness: 1.0393356217307745
Generation 109 Best Solution: [182.9828271682245, 175.59408762636724,
175.1312057974694, 75.77219137243556] Fitness: 2.1519742317025665
Generation 110 Best Solution: [183.2851078503801, 176.32876471566095,
175.1051810425243, 75.77219137243556] Fitness: 1.7673621755689157
Generation 111 Best Solution: [183.06057809578317, 175.8876288591876,
175.1051810425243, 75.77219137243556] Fitness: 1.8583757736167144
Generation 112 Best Solution: [183.06057809578317, 175.8876288591876,
175.1051810425243, 75.77219137243556] Fitness: 1.8583757736167144
Generation 113 Best Solution: [183.06057809578317, 175.8876288591876,
175.1051810425243, 74.73826980066065] Fitness: 0.9322550382208429
Generation 114 Best Solution: [183.6073786198221, 176.2325706677964,
175.1051810425243, 75.77219137243556] Fitness: 1.8762078484855527
Generation 115 Best Solution: [182.95601610290856, 175.99467041355487,
175.1051810425243, 75.77219137243556] Fitness: 1.8473890919516707
Generation 116 Best Solution: [183.2851078503801, 177.41165717953118,
174.50768424105502, 75.77219137243556] Fitness: 2.28175578045452
Generation 117 Best Solution: [183.2851078503801, 177.41165717953118,
175.1051810425243, 75.1537649903853] Fitness: 2.1768868474368444
Generation 118 Best Solution: [183.2851078503801, 177.41165717953118,
173.79700520325545, 74.73883665771191] Fitness: 1.3341418373757008
Generation 119 Best Solution: [183.2851078503801, 177.41165717953118,

174.1393546975737, 75.77219137243556] Fitness: 2.1670225011549
Generation 120 Best Solution: [183.06057809578317, 177.41165717953118,
175.1051810425243, 74.35644613615685] Fitness: 2.135038377045144
Generation 121 Best Solution: [183.2851078503801, 177.41165717953118,
174.86444332174625, 74.57476950786463] Fitness: 1.6936181423853933
Generation 122 Best Solution: [183.06057809578317, 177.41165717953118,
174.86444332174625, 74.57476950786463] Fitness: 1.7108525911802899
Generation 123 Best Solution: [183.2851078503801, 177.41165717953118,
174.86444332174625, 74.57476950786463] Fitness: 1.6936181423853933
Generation 124 Best Solution: [183.2851078503801, 176.349666479389,
173.83127421271064, 75.77219137243556] Fitness: 1.0721300071766715
Generation 125 Best Solution: [183.6073786198221, 177.41165717953118,
173.83127421271064, 74.57476950786463] Fitness: 1.450491034318344
Generation 126 Best Solution: [183.2851078503801, 177.41165717953118,
173.83127421271064, 74.57476950786463] Fitness: 1.3304775285178227
Generation 127 Best Solution: [183.2851078503801, 176.0716232684281,
174.86444332174625, 75.77219137243556] Fitness: 1.4349760521384898
Generation 128 Best Solution: [183.2851078503801, 175.9718494844401,
174.86444332174625, 75.77219137243556] Fitness: 1.4658351096056812
Generation 129 Best Solution: [183.86033325379, 176.0716232684281,
174.86444332174625, 75.77219137243556] Fitness: 1.7687124919176473
Generation 130 Best Solution: [183.34390078385306, 177.41165717953118,
173.7997304241632, 74.40179428961144] Fitness: 1.4191546293312918
Generation 131 Best Solution: [183.86033325379, 176.0716232684281,
174.86444332174625, 75.77219137243556] Fitness: 1.7687124919176473
Generation 132 Best Solution: [183.2851078503801, 177.41165717953118,
174.86444332174625, 74.73475501585746] Fitness: 1.6790998234768
Generation 133 Best Solution: [183.2851078503801, 177.41165717953118,
174.86444332174625, 74.73475501585746] Fitness: 1.6790998234768
Generation 134 Best Solution: [183.60847238940826, 175.90406474527714,
175.1051810425243, 75.77219137243556] Fitness: 1.9533699721538902
Generation 135 Best Solution: [183.2851078503801, 175.84584988900136,
174.86444332174625, 75.77219137243556] Fitness: 1.5281730242990454
Generation 136 Best Solution: [183.2851078503801, 175.84584988900136,
174.86444332174625, 75.77219137243556] Fitness: 1.5281730242990454
Generation 137 Best Solution: [183.31944615115447, 176.40800170746925,
174.86444332174625, 75.77219137243556] Fitness: 1.4561170939864159
Generation 138 Best Solution: [183.34390078385306, 176.15861797455153,
173.7997304241632, 75.87028223015488] Fitness: 1.2630209603920663
Generation 139 Best Solution: [183.80105574086056, 175.90406474527714,
175.00951044926614, 74.44252512893016] Fitness: 1.093731851037662
Generation 140 Best Solution: [183.2851078503801, 176.98060758387956,
174.86444332174625, 74.49467702651238] Fitness: 1.0183486304858156
Generation 141 Best Solution: [183.34390078385306, 176.98060758387956,
174.86444332174625, 75.27622406779399] Fitness: 1.2527919088027977
Generation 142 Best Solution: [183.60847238940826, 175.90406474527714,
175.00951044926614, 75.77219137243556] Fitness: 1.8082311755176952
Generation 143 Best Solution: [183.80105574086056, 175.90406474527714,

174.86444332174625, 75.87028223015488] Fitness: 1.9494269046188695
Generation 144 Best Solution: [183.80105574086056, 176.98060758387956,
175.00951044926614, 74.46933803447533] Fitness: 1.4934513508474638
Generation 145 Best Solution: [183.80105574086056, 175.90406474527714,
174.86444332174625, 75.77219137243556] Fitness: 1.770351909456612
Generation 146 Best Solution: [183.5941498439832, 175.90406474527714,
174.86444332174625, 75.77219137243556] Fitness: 1.607858892052035
Generation 147 Best Solution: [183.80105574086056, 175.90406474527714,
174.86444332174625, 75.87028223015488] Fitness: 1.9494269046188695
Generation 148 Best Solution: [182.9766480261019, 175.90406474527714,
174.86444332174625, 75.77219137243556] Fitness: 1.5410727400853772
Generation 149 Best Solution: [183.66755373142945, 175.75948535816931,
175.00951044926614, 75.77219137243556] Fitness: 1.9386838755806766
Generation 150 Best Solution: [183.66755373142945, 177.41165717953118,
175.00951044926614, 75.01057602921034] Fitness: 2.1055258528059917
Generation 151 Best Solution: [183.80105574086056, 177.41940265041845,
175.00951044926614, 75.01057602921034] Fitness: 2.2337609677910004
Generation 152 Best Solution: [182.801531988342, 177.41940265041845,
175.00951044926614, 75.01057602921034] Fitness: 2.099556420977112
Generation 153 Best Solution: [182.801531988342, 177.3705132699534,
175.00951044926614, 75.01057602921034] Fitness: 2.004382650072513
Generation 154 Best Solution: [183.66755373142945, 177.41165717953118,
175.00951044926614, 75.01057602921034] Fitness: 2.1055258528059917
Generation 155 Best Solution: [183.80105574086056, 177.41165717953118,
174.2262271844432, 75.77219137243556] Fitness: 2.4482202392573944
Generation 156 Best Solution: [183.80105574086056, 177.41165717953118,
174.2262271844432, 75.77219137243556] Fitness: 2.4482202392573944
Generation 157 Best Solution: [183.80105574086056, 176.5770834090403,
175.00951044926614, 75.77219137243556] Fitness: 1.9955466990755044
Generation 158 Best Solution: [182.801531988342, 176.31948369778118,
175.1051810425243, 75.77219137243556] Fitness: 1.9056394621265704
Generation 159 Best Solution: [182.801531988342, 176.31948369778118,
174.86444332174625, 75.77219137243556] Fitness: 1.5691108233549538
Generation 160 Best Solution: [183.80105574086056, 177.41165717953118,
175.1051810425243, 74.59355155982497] Fitness: 2.3004857791325097
Generation 161 Best Solution: [183.80105574086056, 176.53255141020807,
174.68323935775462, 75.77219137243556] Fitness: 1.5884284350782263
Generation 162 Best Solution: [183.80105574086056, 176.53255141020807,
174.68323935775462, 75.77219137243556] Fitness: 1.5884284350782263
Generation 163 Best Solution: [183.80105574086056, 176.02281606804007,
174.9423096867507, 75.77219137243556] Fitness: 1.8206825545586693
Generation 164 Best Solution: [183.80105574086056, 177.41165717953118,
173.98172670669015, 75.77219137243556] Fitness: 2.460244951796912
Generation 165 Best Solution: [183.80105574086056, 177.41165717953118,
173.74499983507664, 75.77219137243556] Fitness: 2.565464039617553
Generation 166 Best Solution: [183.80105574086056, 176.62314085842254,
175.1051810425243, 75.77219137243556] Fitness: 2.1702036461840413
Generation 167 Best Solution: [183.80105574086056, 175.65682466975255,

173.74499983507664, 75.77219137243556] Fitness: 1.6306971926224798
Generation 168 Best Solution: [183.80105574086056, 175.65682466975255,
173.74499983507664, 75.77219137243556] Fitness: 1.6306971926224798
Generation 169 Best Solution: [183.80105574086056, 177.41165717953118,
175.1051810425243, 74.99388397144565] Fitness: 2.35554602950983
Generation 170 Best Solution: [183.80105574086056, 177.41165717953118,
175.1051810425243, 74.63615646086666] Fitness: 2.2938261529970974
Generation 171 Best Solution: [183.80105574086056, 176.45655224598426,
175.1051810425243, 75.77219137243556] Fitness: 2.0799308156604903
Generation 172 Best Solution: [183.80105574086056, 176.45655224598426,
173.86063643347674, 75.77219137243556] Fitness: 1.3663710774772733
Generation 173 Best Solution: [183.80105574086056, 176.3569633689406,
175.1051810425243, 75.77219137243556] Fitness: 2.047739212147679
Generation 174 Best Solution: [183.80105574086056, 176.53098139138876,
174.86444332174625, 75.77219137243556] Fitness: 1.7781001770036475
Generation 175 Best Solution: [183.80105574086056, 176.53098139138876,
174.86444332174625, 75.77219137243556] Fitness: 1.7781001770036475
Generation 176 Best Solution: [183.80105574086056, 176.14465159739095,
175.1051810425243, 75.54980489879347] Fitness: 1.6860592328728805
Generation 177 Best Solution: [183.80105574086056, 176.53098139138876,
174.86444332174625, 75.77219137243556] Fitness: 1.7781001770036475
Generation 178 Best Solution: [183.80105574086056, 176.69399890830658,
174.08240673554496, 75.77219137243556] Fitness: 1.4496965678621592
Generation 179 Best Solution: [182.9609552607657, 176.3437496584089,
174.55736510306218, 75.77219137243556] Fitness: 1.1942656778791174
Generation 180 Best Solution: [183.80105574086056, 176.3437496584089,
174.55736510306218, 75.77219137243556] Fitness: 1.4170799186658773
Generation 181 Best Solution: [183.80105574086056, 177.41165717953118,
175.1051810425243, 74.67360116279245] Fitness: 2.290435301668578
Generation 182 Best Solution: [183.80105574086056, 177.41165717953118,
174.51266068401566, 75.54980489879347] Fitness: 2.211607436110432
Generation 183 Best Solution: [183.80105574086056, 176.95737314398608,
174.01368617348584, 75.54980489879347] Fitness: 1.3783431996330004
Generation 184 Best Solution: [183.80105574086056, 177.41165717953118,
174.51266068401566, 75.29441425369838] Fitness: 1.9128273751615539
Generation 185 Best Solution: [183.80105574086056, 175.88142446476257,
175.1051810425243, 75.46170442722861] Fitness: 1.6560596317922966
Generation 186 Best Solution: [183.80105574086056, 176.23374898925292,
175.1051810425243, 75.77219137243556] Fitness: 2.0304611050635293
Generation 187 Best Solution: [183.80105574086056, 177.41165717953118,
174.035638753043, 75.77219137243556] Fitness: 2.449153329163987
Generation 188 Best Solution: [183.80105574086056, 176.34505616961212,
175.1051810425243, 75.77219137243556] Fitness: 2.0449808033697625
Generation 189 Best Solution: [183.80105574086056, 176.34505616961212,
175.1051810425243, 75.77219137243556] Fitness: 2.0449808033697625
Generation 190 Best Solution: [183.80105574086056, 176.3479533978275,
175.1051810425243, 75.77219137243556] Fitness: 2.0456305295662847
Generation 191 Best Solution: [183.80105574086056, 176.02594670099631,

174.1546714248701, 75.77219137243556] Fitness: 1.2832740585152018
Generation 192 Best Solution: [183.80105574086056, 176.02594670099631,
174.1546714248701, 75.77219137243556] Fitness: 1.2832740585152018
Generation 193 Best Solution: [183.80105574086056, 176.02594670099631,
174.1546714248701, 74.45179156332853] Fitness: 0.4112628069834523
Generation 194 Best Solution: [183.11291710542218, 176.05665260724098,
175.1051810425243, 75.86193921498817] Fitness: 1.944135075794084
Generation 195 Best Solution: [183.80105574086056, 176.47692990878377,
175.1051810425243, 75.77219137243556] Fitness: 2.0885258878990247
Generation 196 Best Solution: [183.80105574086056, 176.47692990878377,
174.11650240198423, 75.77219137243556] Fitness: 1.3138707772820986
Generation 197 Best Solution: [183.80105574086056, 176.45040089340705,
174.11650240198423, 75.77219137243556] Fitness: 1.3028151811033366
Generation 198 Best Solution: [183.80105574086056, 176.45040089340705,
174.11650240198423, 75.02130027305311] Fitness: 0.4556408600857407
Generation 199 Best Solution: [183.80105574086056, 176.45040089340705,
175.1301940903716, 74.3670017336355] Fitness: 1.2877584717935204
Generation 200 Best Solution: [183.43677700587338, 177.41165717953118,
174.11650240198423, 75.02130027305311] Fitness: 1.355299922728436
Generation 201 Best Solution: [183.43677700587338, 177.41165717953118,
174.11650240198423, 75.77219137243556] Fitness: 2.202474243746032
Generation 202 Best Solution: [183.43677700587338, 177.41165717953118,
174.11650240198423, 75.77219137243556] Fitness: 2.202474243746032
Generation 203 Best Solution: [183.43677700587338, 176.16108826642136,
175.1051810425243, 75.77219137243556] Fitness: 1.7929680955395892
Generation 204 Best Solution: [183.11291710542218, 176.14009631117256,
175.1051810425243, 75.77219137243556] Fitness: 1.765603867521846
Generation 205 Best Solution: [183.11291710542218, 176.14009631117256,
175.1051810425243, 75.77219137243556] Fitness: 1.765603867521846
Generation 206 Best Solution: [183.80105574086056, 177.41165717953118,
174.11650240198423, 75.77219137243556] Fitness: 2.4414691066055454
Generation 207 Best Solution: [183.80105574086056, 176.40267892841112,
175.1051810425243, 75.77219137243556] Fitness: 2.060493482354665
Generation 208 Best Solution: [183.54680742438777, 177.41165717953118,
175.1051810425243, 74.61554089601114] Fitness: 2.1068902746338836
Generation 209 Best Solution: [183.21167091006953, 177.41165717953118,
175.1051810425243, 74.77821091929249] Fitness: 2.015469435360598
Generation 210 Best Solution: [183.80105574086056, 175.5750862864421,
175.1051810425243, 75.77219137243556] Fitness: 2.361128334027368
Generation 211 Best Solution: [183.43677700587338, 175.8997505639207,
175.1051810425243, 75.77219137243556] Fitness: 1.870069315640834
Generation 212 Best Solution: [183.37581825932938, 177.41165717953118,
175.1051810425243, 74.91857630009434] Fitness: 2.0673870367525287
Generation 213 Best Solution: [183.21167091006953, 177.41165717953118,
175.1051810425243, 74.91857630009434] Fitness: 2.0473830389090755
Generation 214 Best Solution: [183.70718395384105, 176.18839511488704,
175.1051810425243, 75.77219137243556] Fitness: 1.9479433149177305
Generation 215 Best Solution: [183.43677700587338, 177.41165717953118,

175.1051810425243, 74.91857630009434] Fitness: 2.08608763684666
Generation 216 Best Solution: [182.91611057119422, 177.41165717953118,
175.1051810425243, 74.91857630009434] Fitness: 2.1229728692962726
Generation 217 Best Solution: [183.43677700587338, 175.8800738698778,
175.1051810425243, 75.77219137243556] Fitness: 1.880416472471091
Generation 218 Best Solution: [183.43677700587338, 177.41165717953118,
175.1051810425243, 74.91857630009434] Fitness: 2.08608763684666
Generation 219 Best Solution: [183.43677700587338, 176.2257261162508,
175.1051810425243, 75.77219137243556] Fitness: 1.79120609268835
Generation 220 Best Solution: [183.70718395384105, 176.2257261162508,
175.1051810425243, 75.77219137243556] Fitness: 1.9477630439336788
Generation 221 Best Solution: [183.43677700587338, 176.84855180509854,
175.1051810425243, 75.1903095892742] Fitness: 1.3886632855273793
Generation 222 Best Solution: [183.43677700587338, 176.3914754308618,
175.1051810425243, 75.1903095892742] Fitness: 1.0807794732728468
Generation 223 Best Solution: [183.70718395384105, 175.88140911742997,
175.1051810425243, 75.77219137243556] Fitness: 2.0362511553495324
Generation 224 Best Solution: [183.70718395384105, 175.88140911742997,
175.1051810425243, 75.77219137243556] Fitness: 2.0362511553495324
Generation 225 Best Solution: [183.57111723247237, 176.93391863938817,
175.1051810425243, 75.1903095892742] Fitness: 1.5469670669185533
Generation 226 Best Solution: [183.57111723247237, 175.88140911742997,
175.1051810425243, 75.1903095892742] Fitness: 1.2051819507776425
Generation 227 Best Solution: [183.43677700587338, 177.41165717953118,
175.1051810425243, 75.1903095892742] Fitness: 2.2398535031865565
Generation 228 Best Solution: [183.57111723247237, 177.41165717953118,
175.1051810425243, 74.71553007328] Fitness: 2.1131408668835365
Generation 229 Best Solution: [183.57111723247237, 177.41165717953118,
175.00590956902917, 75.1903095892742] Fitness: 2.1523091558880205
Generation 230 Best Solution: [183.72784572386922, 176.5421715484712,
175.1051810425243, 75.1903095892742] Fitness: 1.3178213486226515
Generation 231 Best Solution: [183.72784572386922, 176.5421715484712,
175.1051810425243, 75.1903095892742] Fitness: 1.3178213486226515
Generation 232 Best Solution: [183.43677700587338, 177.41165717953118,
175.1051810425243, 75.1903095892742] Fitness: 2.2398535031865565
Generation 233 Best Solution: [183.57111723247237, 175.615906488344,
175.1051810425243, 75.77219137243556] Fitness: 2.143687375065583
Generation 234 Best Solution: [183.72784572386922, 175.615906488344,
175.1051810425243, 75.1903095892742] Fitness: 1.5171074467831622
Generation 235 Best Solution: [183.72784572386922, 175.615906488344,
175.1051810425243, 75.77219137243556] Fitness: 2.2543832979595635
Generation 236 Best Solution: [183.81929649417802, 176.6748115892719,
175.1051810425243, 75.77219137243556] Fitness: 2.225166105547819
Generation 237 Best Solution: [183.72784572386922, 177.41165717953118,
174.8355740946472, 75.1903095892742] Fitness: 2.042821388551885
Generation 238 Best Solution: [182.95071957875555, 177.41165717953118,
175.1051810425243, 74.57656840766421] Fitness: 2.085592260571379
Generation 239 Best Solution: [183.72784572386922, 177.41165717953118,

174.55642852186023, 75.1903095892742] Fitness: 1.7850497910922902
Generation 240 Best Solution: [183.72784572386922, 175.9922375222285,
175.00590956902917, 75.77219137243556] Fitness: 1.853107071063639
Generation 241 Best Solution: [182.83992993568512, 177.41165717953118,
175.00590956902917, 74.78565904348204] Fitness: 1.9843772163798064
Generation 242 Best Solution: [183.45357956653248, 177.41165717953118,
175.00590956902917, 74.78565904348204] Fitness: 1.910974169066248
Generation 243 Best Solution: [183.45357956653248, 177.41165717953118,
175.00590956902917, 74.78565904348204] Fitness: 1.910974169066248
Generation 244 Best Solution: [183.72784572386922, 175.9600257298474,
175.00590956902917, 74.78565904348204] Fitness: 0.9434081288685002
Generation 245 Best Solution: [183.72784572386922, 177.41165717953118,
174.90402291877996, 74.78565904348204] Fitness: 1.9407736567053142
Generation 246 Best Solution: [183.64006579934795, 177.41165717953118,
174.90402291877996, 74.78565904348204] Fitness: 1.8738039539085234
Generation 247 Best Solution: [183.72784572386922, 177.41165717953118,
174.90402291877996, 74.78565904348204] Fitness: 1.9407736567053142
Generation 248 Best Solution: [183.64006579934795, 176.65646307007256,
174.90402291877996, 74.78565904348204] Fitness: 0.8514123040462171
Generation 249 Best Solution: [183.64006579934795, 176.65646307007256,
174.90402291877996, 75.77219137243556] Fitness: 1.7734874140021537
Generation 250 Best Solution: [182.83992993568512, 175.78067588531565,
175.1051810425243, 74.56085852940932] Fitness: 1.113532891226777
Generation 251 Best Solution: [183.72784572386922, 176.65646307007256,
174.90402291877996, 75.77219137243556] Fitness: 1.8404571167989445
Generation 252 Best Solution: [183.72784572386922, 176.65646307007256,
175.1051810425243, 74.85095355337762] Fitness: 1.2177400833967766
Generation 253 Best Solution: [183.72784572386922, 176.44508590076032,
175.00590956902917, 75.77219137243556] Fitness: 1.859551082929901
Generation 254 Best Solution: [183.72784572386922, 176.04977666648162,
175.1051810425243, 75.77219137243556] Fitness: 1.9855497825408297
Generation 255 Best Solution: [183.72784572386922, 176.04977666648162,
174.25799855803265, 75.77470579579808] Fitness: 1.2276863184072377
Generation 256 Best Solution: [183.05025946137485, 177.41165717953118,
175.00590956902917, 75.3478145237913] Fitness: 2.2189973970748653
Generation 257 Best Solution: [183.05025946137485, 177.40308051214143,
175.00590956902917, 75.3478145237913] Fitness: 2.2021261835268366
Generation 258 Best Solution: [182.83992993568512, 177.41165717953118,
174.31049450901145, 75.77219137243556] Fitness: 2.307469136626127
Generation 259 Best Solution: [183.05025946137485, 177.41165717953118,
174.71617088687748, 74.82779598967738] Fitness: 1.5501202554032907
Generation 260 Best Solution: [183.05025946137485, 177.40308051214143,
174.71617088687748, 74.82779598967738] Fitness: 1.533249041855262
Generation 261 Best Solution: [183.06663779664996, 177.41165717953118,
174.31049450901145, 75.77219137243556] Fitness: 2.208293012507013
Generation 262 Best Solution: [183.05025946137485, 176.2894346704408,
175.1051810425243, 75.7999673909102] Fitness: 1.8300797362756311
Generation 263 Best Solution: [183.05025946137485, 177.40308051214143,

174.187637882578, 75.7999673909102] Fitness: 2.221618654162324
Generation 264 Best Solution: [183.81119162917315, 177.40308051214143,
174.187637882578, 75.23794588299114] Fitness: 1.7386640908408033
Generation 265 Best Solution: [183.72784572386922, 176.02749667781956,
175.08029363929194, 75.7999673909102] Fitness: 2.0017229338239506
Generation 266 Best Solution: [183.72784572386922, 176.29427959673197,
175.1051810425243, 75.7999673909102] Fitness: 2.0194008008990445
Generation 267 Best Solution: [183.05025946137485, 177.41165717953118,
175.1051810425243, 75.0080640410749] Fitness: 2.1081196470062373
Generation 268 Best Solution: [183.72784572386922, 176.11103450435326,
175.1051810425243, 75.7999673909102] Fitness: 2.02161136285533
Generation 269 Best Solution: [183.72784572386922, 176.25564354661427,
175.1051810425243, 75.0080640410749] Fitness: 1.1123740677526297
Generation 270 Best Solution: [183.72784572386922, 176.25564354661427,
175.1051810425243, 75.0080640410749] Fitness: 1.1123740677526297
Generation 271 Best Solution: [183.05025946137485, 176.25564354661427,
175.1051810425243, 75.0080640410749] Fitness: 0.9237045467655391
Generation 272 Best Solution: [183.72784572386922, 175.93992989251635,
175.1051810425243, 75.7999673909102] Fitness: 2.0734793969065954
Generation 273 Best Solution: [183.72784572386922, 175.63000697838177,
175.1051810425243, 75.7999673909102] Fitness: 2.2898880744030623
Generation 274 Best Solution: [183.72784572386922, 175.63000697838177,
175.1051810425243, 75.7999673909102] Fitness: 2.2898880744030623
Generation 275 Best Solution: [183.75493466401448, 175.8979700058672,
175.1051810425243, 75.7999673909102] Fitness: 2.1167655820520355
Generation 276 Best Solution: [183.75493466401448, 175.8979700058672,
175.1051810425243, 75.7999673909102] Fitness: 2.1167655820520355
Generation 277 Best Solution: [183.75493466401448, 176.15307970678793,
175.1051810425243, 75.7999673909102] Fitness: 2.039450426996286
Generation 278 Best Solution: [183.72784572386922, 177.41165717953118,
173.8851244598057, 75.10530934997817] Fitness: 1.628114721166299
Generation 279 Best Solution: [183.75493466401448, 176.43631022302472,
175.1051810425243, 75.7999673909102] Fitness: 2.0788596104649084
Generation 280 Best Solution: [183.75493466401448, 177.41165717953118,
175.1051810425243, 75.10530934997817] Fitness: 2.375398798576891
Generation 281 Best Solution: [183.72784572386922, 177.41165717953118,
175.1051810425243, 74.47182756811819] Fitness: 2.2704081143758854
Generation 282 Best Solution: [183.75493466401448, 176.5897994032888,
175.1051810425243, 75.7999673909102] Fitness: 2.1552781534270355
Generation 283 Best Solution: [183.75493466401448, 176.05938457537368,
175.1051810425243, 75.7999673909102] Fitness: 2.0554231765030155
Generation 284 Best Solution: [183.75493466401448, 177.41165717953118,
173.80060020475537, 75.42434770430256] Fitness: 1.9825580452848413
Generation 285 Best Solution: [183.68121169634009, 176.05938457537368,
175.1051810425243, 75.7999673909102] Fitness: 1.9950457345176051
Generation 286 Best Solution: [183.75493466401448, 176.05938457537368,
175.1051810425243, 74.32664204187626] Fitness: 1.2002621568105833
Generation 287 Best Solution: [183.68121169634009, 176.05938457537368,

175.1051810425243, 75.3808049233191] Fitness: 1.388790650008175
Generation 288 Best Solution: [183.68121169634009, 176.05938457537368,
175.1051810425243, 75.7999673909102] Fitness: 1.9950457345176051
Generation 289 Best Solution: [183.7215469293046, 176.312873426918,
175.1051810425243, 74.73679218079617] Fitness: 1.0417409222104237
Generation 290 Best Solution: [183.7215469293046, 176.54961043530307,
175.1051810425243, 75.7999673909102] Fitness: 2.103078713504108
Generation 291 Best Solution: [183.49218802911426, 177.41165717953118,
175.1051810425243, 74.76061064500433] Fitness: 2.074751594681863
Generation 292 Best Solution: [183.40881362601388, 176.59354083177377,
175.1051810425243, 75.7999673909102] Fitness: 1.9516053487776888
Generation 293 Best Solution: [183.7215469293046, 176.59354083177377,
175.1051810425243, 75.7999673909102] Fitness: 2.129174120755265
Generation 294 Best Solution: [183.7215469293046, 175.89531861519097,
175.1051810425243, 75.7999673909102] Fitness: 2.0896804957222375
Generation 295 Best Solution: [183.40881362601388, 176.2542253139403,
173.99309280536752, 75.7999673909102] Fitness: 1.0737619849084064
Generation 296 Best Solution: [183.66980495671476, 176.1543017510206,
173.99309280536752, 75.7999673909102] Fitness: 1.2118008214320117
Generation 297 Best Solution: [183.35747888318568, 175.95139456727233,
175.1051810425243, 75.7999673909102] Fitness: 1.8719090503005487
Generation 298 Best Solution: [183.75493466401448, 175.97082984570028,
175.1051810425243, 75.63135093703573] Fitness: 1.8051963132006168
Generation 299 Best Solution: [183.75493466401448, 176.31753042761719,
174.9039633962038, 75.68445348697686] Fitness: 1.5625861516832855
Generation 300 Best Solution: [183.66980495671476, 175.6917496864789,
175.1051810425243, 75.7999673909102] Fitness: 2.1884894792105007
Generation 301 Best Solution: [183.66980495671476, 177.41165717953118,
175.1051810425243, 74.61749111424072] Fitness: 2.1851426262844615
Generation 302 Best Solution: [183.75493466401448, 177.3257486963826,
174.43605238075932, 75.7999673909102] Fitness: 2.359464119775662
Generation 303 Best Solution: [183.75493466401448, 176.28462310525566,
175.1051810425243, 75.7999673909102] Fitness: 2.041363277901605
Generation 304 Best Solution: [183.66980495671476, 177.41165717953118,
174.06878826344072, 75.7999673909102] Fitness: 2.3825735833225377
Generation 305 Best Solution: [182.801542276427, 177.41165717953118,
174.06878826344072, 75.7999673909102] Fitness: 2.359596054574212
Generation 306 Best Solution: [183.24126315656014, 177.41165717953118,
174.1112547536299, 75.7999673909102] Fitness: 2.2133815902120655
Generation 307 Best Solution: [183.75493466401448, 175.91008711036835,
175.1051810425243, 75.7999673909102] Fitness: 2.1106747090148845
Generation 308 Best Solution: [183.15395796352757, 177.41165717953118,
174.06878826344072, 75.7999673909102] Fitness: 2.2195926991993282
Generation 309 Best Solution: [183.66980495671476, 177.41165717953118,
175.1051810425243, 74.44749577085767] Fitness: 2.234716314019577
Generation 310 Best Solution: [183.75493466401448, 175.95964796330244,
175.1051810425243, 75.7999673909102] Fitness: 2.0882730383349704
Generation 311 Best Solution: [183.75493466401448, 177.41165717953118,

174.1756003584442, 75.7999673909102] Fitness: 2.4494388685229285
Generation 312 Best Solution: [183.66980495671476, 176.77740851680596,
175.1051810425243, 75.7999673909102] Fitness: 2.2323283314255606
Generation 313 Best Solution: [183.66980495671476, 176.77740851680596,
175.1051810425243, 75.7999673909102] Fitness: 2.2323283314255606
Generation 314 Best Solution: [183.66980495671476, 177.41165717953118,
175.1051810425243, 74.60405321880283] Fitness: 2.1873332337679328
Generation 315 Best Solution: [183.66980495671476, 177.2658659988608,
174.69587552729993, 74.83054077469596] Fitness: 1.404198590869474
Generation 316 Best Solution: [183.66980495671476, 177.2658659988608,
174.69587552729993, 75.7999673909102] Fitness: 2.368144062380746
Generation 317 Best Solution: [183.70025719104996, 176.01391250775742,
175.1051810425243, 75.7999673909102] Fitness: 2.022738227779433
Generation 318 Best Solution: [183.70025719104996, 176.01391250775742,
173.947844639098, 75.7999673909102] Fitness: 1.2762731662854263
Generation 319 Best Solution: [183.75493466401448, 176.01391250775742,
173.947844639098, 75.7999673909102] Fitness: 1.3219081816164437
Generation 320 Best Solution: [183.75493466401448, 177.41165717953118,
174.13188953057823, 74.49714250246997] Fitness: 1.509351377730016
Generation 321 Best Solution: [183.70025719104996, 176.1992722170266,
175.1051810425243, 75.7999673909102] Fitness: 1.9912485833655564
Generation 322 Best Solution: [183.70025719104996, 176.1992722170266,
175.1051810425243, 75.7999673909102] Fitness: 1.9912485833655564
Generation 323 Best Solution: [183.75493466401448, 176.1992722170266,
175.1051810425243, 75.7999673909102] Fitness: 2.036883598696574
Generation 324 Best Solution: [183.75493466401448, 175.88842111003268,
175.1051810425243, 75.7999673909102] Fitness: 2.1217354427800412
Generation 325 Best Solution: [183.74017447085828, 175.65668998049742,
175.1051810425243, 75.7999673909102] Fitness: 2.2754678909737835
Generation 326 Best Solution: [183.75493466401448, 177.41165717953118,
173.86696134600282, 75.18686609532757] Fitness: 1.71746200895759
Generation 327 Best Solution: [183.75493466401448, 177.25053165621168,
174.90821436475528, 75.18686609532757] Fitness: 1.8546450800126157
Generation 328 Best Solution: [183.75493466401448, 175.65668998049742,
174.13188953057823, 75.7999673909102] Fitness: 1.5133702987938595
Generation 329 Best Solution: [183.75493466401448, 175.65668998049742,
174.13188953057823, 75.18686609532757] Fitness: 0.7242833084861352
Generation 330 Best Solution: [183.62232574309704, 177.41165717953118,
174.13188953057823, 75.18686609532757] Fitness: 1.5567384053039275
Generation 331 Best Solution: [183.71568943281056, 177.41165717953118,
175.1051810425243, 74.64165134032056] Fitness: 2.217619496665969
Generation 332 Best Solution: [183.62232574309704, 177.41165717953118,
175.1051810425243, 74.4832673547075] Fitness: 2.187072997143221
Generation 333 Best Solution: [183.43913423472068, 177.41165717953118,
174.12041237662504, 75.7999673909102] Fitness: 2.252324387440968
Generation 334 Best Solution: [183.75493466401448, 176.29513961651097,
175.1051810425243, 75.7999673909102] Fitness: 2.042743398209378
Generation 335 Best Solution: [183.43913423472068, 177.41165717953118,

174.04320264998785, 75.7999673909102] Fitness: 2.258933690015225
Generation 336 Best Solution: [183.75493466401448, 177.41165717953118,
174.04320264998785, 75.24107482333724] Fitness: 1.7105655918744964
Generation 337 Best Solution: [183.71568943281056, 177.41165717953118,
174.04320264998785, 75.24107482333724] Fitness: 1.677313171631589
Generation 338 Best Solution: [182.94706821166025, 177.41165717953118,
175.1051810425243, 75.47344991864608] Fitness: 2.5513656131009026
Generation 339 Best Solution: [182.94706821166025, 175.6921196045029,
175.1051810425243, 75.7999673909102] Fitness: 2.0826529353662973
Generation 340 Best Solution: [183.75493466401448, 176.11912312326217,
175.1051810425243, 75.7999673909102] Fitness: 2.0435729198450123
Generation 341 Best Solution: [182.94706821166025, 176.46221017520017,
175.1051810425243, 75.7999673909102] Fitness: 1.9145967558658636
Generation 342 Best Solution: [183.4439798787036, 176.58358021131903,
175.1051810425243, 75.7999673909102] Fitness: 1.957359936207528
Generation 343 Best Solution: [183.75493466401448, 175.7035344955785,
175.1051810425243, 75.47344991864608] Fitness: 1.7503849914863567
Generation 344 Best Solution: [183.75493466401448, 175.72163556143343,
174.51311454143024, 75.7999673909102] Fitness: 1.5759118401595171
Generation 345 Best Solution: [183.75493466401448, 177.41165717953118,
174.07925705863062, 75.47344991864608] Fitness: 1.9533586941702707
Generation 346 Best Solution: [183.75493466401448, 176.4192450788233,
175.1051810425243, 75.47344991864608] Fitness: 1.5756475809287005
Generation 347 Best Solution: [183.32544612104164, 177.41165717953118,
173.92909944127928, 75.00115392051816] Fitness: 1.3499918770441512
Generation 348 Best Solution: [183.75493466401448, 176.4192450788233,
175.1051810425243, 75.47344991864608] Fitness: 1.5756475809287005
Generation 349 Best Solution: [183.75493466401448, 176.2945136587433,
175.1051810425243, 75.47344991864608] Fitness: 1.5455496622342508
Generation 350 Best Solution: [183.43312157299633, 177.41165717953118,
174.07925705863062, 75.49935343819328] Fitness: 1.788423184499421
Generation 351 Best Solution: [183.43312157299633, 177.41165717953118,
174.07925705863062, 75.10921086668026] Fitness: 1.407518924171743
Generation 352 Best Solution: [183.4439798787036, 177.41165717953118,
175.04751905800524, 74.83866350681022] Fitness: 1.9772858312661683
Generation 353 Best Solution: [183.75493466401448, 177.41165717953118,
174.16922496944673, 75.08631063048172] Fitness: 1.5894769819803345
Generation 354 Best Solution: [183.75493466401448, 177.23962374329264,
174.16922496944673, 75.08631063048172] Fitness: 1.2741675798218333
Generation 355 Best Solution: [183.75493466401448, 175.70591328545615,
175.04751905800524, 75.7999673909102] Fitness: 2.156239432553883
Generation 356 Best Solution: [183.75493466401448, 177.41165717953118,
174.13634964902712, 75.49935343819328] Fitness: 1.9839501172737335
Generation 357 Best Solution: [183.75493466401448, 176.2699122508533,
175.04751905800524, 75.7999673909102] Fitness: 1.9504601263582373
Generation 358 Best Solution: [183.75493466401448, 176.11129255159094,
174.13634964902712, 75.7999673909102] Fitness: 1.2698931408662637
Generation 359 Best Solution: [182.8594429559054, 177.41165717953118,

174.13634964902712, 75.7999673909102] Fitness: 2.319107482241702
Generation 360 Best Solution: [183.75493466401448, 175.82570930109577,
174.13634964902712, 75.7999673909102] Fitness: 1.3831978608854107
Generation 361 Best Solution: [182.8594429559054, 177.41165717953118,
174.24860413312152, 75.23143443338556] Fitness: 1.5773682134416387
Generation 362 Best Solution: [183.75493466401448, 176.00436472301257,
175.1051810425243, 75.38099564970067] Fitness: 1.4654790165939855
Generation 363 Best Solution: [182.8594429559054, 176.61845234519564,
175.1051810425243, 75.7999673909102] Fitness: 2.044921960539183
Generation 364 Best Solution: [183.75493466401448, 175.99911218041802,
175.1051810425243, 75.7999673909102] Fitness: 2.07332100954557
Generation 365 Best Solution: [183.75493466401448, 177.41165717953118,
175.1051810425243, 74.74319659576739] Fitness: 2.2479398068332714
Generation 366 Best Solution: [183.75493466401448, 177.41165717953118,
175.1051810425243, 74.74319659576739] Fitness: 2.2479398068332714
Generation 367 Best Solution: [183.75493466401448, 177.41165717953118,
174.1883037680136, 74.74466682935105] Fitness: 1.4755477282875613
Generation 368 Best Solution: [183.75493466401448, 177.41165717953118,
174.1883037680136, 74.74466682935105] Fitness: 1.4755477282875613
Generation 369 Best Solution: [183.71936041418707, 176.4078167277214,
175.1051810425243, 75.7999673909102] Fitness: 2.038683485711996
Generation 370 Best Solution: [183.75493466401448, 177.41165717953118,
174.81159414509497, 75.06804799691145] Fitness: 1.9546787205369938
Generation 371 Best Solution: [183.75493466401448, 177.22426923312676,
174.81159414509497, 74.70223120025592] Fitness: 1.5083342200025487
Generation 372 Best Solution: [183.75493466401448, 177.41165717953118,
174.1125148572511, 75.76739636655077] Fitness: 2.3909352576841565
Generation 373 Best Solution: [183.75493466401448, 177.41165717953118,
174.81159414509497, 75.01884412236747] Fitness: 1.927724629396228
Generation 374 Best Solution: [183.75493466401448, 177.41165717953118,
174.1125148572511, 75.7999673909102] Fitness: 2.448387481141824
Generation 375 Best Solution: [183.75493466401448, 176.38249446675968,
175.1051810425243, 75.7999673909102] Fitness: 2.06123013019333
Generation 376 Best Solution: [183.75493466401448, 175.72883516734797,
175.1051810425243, 75.7999673909102] Fitness: 2.226965875469123
Generation 377 Best Solution: [183.75493466401448, 177.2291348642534,
175.1051810425243, 75.02429302844241] Fitness: 1.9952141542627047
Generation 378 Best Solution: [183.75493466401448, 177.41165717953118,
174.71367618374873, 74.82232015824638] Fitness: 1.7585963877953044
Generation 379 Best Solution: [182.80683202480714, 177.41165717953118,
174.71367618374873, 74.82232015824638] Fitness: 1.6630874452707072
Generation 380 Best Solution: [182.80683202480714, 177.41165717953118,
174.1125148572511, 75.7999673909102] Fitness: 2.352878538617227
Generation 381 Best Solution: [183.75493466401448, 177.27852248816927,
175.1051810425243, 75.29943661540914] Fitness: 2.2841597470132236
Generation 382 Best Solution: [183.75493466401448, 177.41165717953118,
174.91376738656976, 75.29943661540914] Fitness: 2.257094447615101
Generation 383 Best Solution: [183.75493466401448, 177.41165717953118,

174.07365577204507, 75.7999673909102] Fitness: 2.4509939203166637
Generation 384 Best Solution: [183.75493466401448, 177.3755965970954,
174.24297455913646, 75.7999673909102] Fitness: 2.3876608948169977
Generation 385 Best Solution: [183.18308529801126, 177.41165717953118,
173.9641386805906, 75.7999673909102] Fitness: 2.2373425960486295
Generation 386 Best Solution: [183.75493466401448, 177.41165717953118,
173.96462745027742, 75.62792275411557] Fitness: 2.187791730293358
Generation 387 Best Solution: [183.75493466401448, 177.41165717953118,
173.96462745027742, 75.62792275411557] Fitness: 2.187791730293358
Generation 388 Best Solution: [183.75493466401448, 176.32928976413834,
175.1051810425243, 75.7999673909102] Fitness: 2.0484780321792546
Generation 389 Best Solution: [183.75493466401448, 176.23206192299344,
175.1051810425243, 75.7999673909102] Fitness: 2.0371888769726025
Generation 390 Best Solution: [183.75493466401448, 176.23206192299344,
175.1051810425243, 75.57747283596028] Fitness: 1.67944047630374
Generation 391 Best Solution: [183.75493466401448, 176.70135921665744,
175.1051810425243, 75.57747283596028] Fitness: 1.8773613506253435
Generation 392 Best Solution: [183.75493466401448, 176.6630518779179,
175.1051810425243, 75.7999673909102] Fitness: 2.205392210533012
Generation 393 Best Solution: [183.75493466401448, 176.6630518779179,
175.1051810425243, 74.94809637353868] Fitness: 1.2760780071911417
Generation 394 Best Solution: [183.75493466401448, 177.12484639604745,
175.1051810425243, 74.94809637353868] Fitness: 1.79496451641791
Generation 395 Best Solution: [183.75493466401448, 175.68783627571682,
175.1051810425243, 75.7999673909102] Fitness: 2.260755634463877
Generation 396 Best Solution: [183.75493466401448, 176.21729205086646,
175.1051810425243, 75.7999673909102] Fitness: 2.0368327429680466
Generation 397 Best Solution: [183.75493466401448, 175.70512523307036,
175.1051810425243, 75.02772514332544] Fitness: 1.3532117188176922
Generation 398 Best Solution: [183.75493466401448, 175.70512523307036,
175.1051810425243, 75.02772514332544] Fitness: 1.3532117188176922
Generation 399 Best Solution: [183.75493466401448, 175.70512523307036,
174.88034400232874, 75.7999673909102] Fitness: 1.9289323555618902
Generation 400 Best Solution: [183.75493466401448, 177.41165717953118,
175.1051810425243, 74.30630738103251] Fitness: 2.380900931214719
Generation 401 Best Solution: [183.22621617887265, 175.80035550045108,
175.1051810425243, 75.7999673909102] Fitness: 1.939229970006076
Generation 402 Best Solution: [183.22621617887265, 175.80035550045108,
175.1051810425243, 75.7999673909102] Fitness: 1.939229970006076
Generation 403 Best Solution: [183.22621617887265, 175.80035550045108,
175.1051810425243, 74.92015121950972] Fitness: 0.9996262664960102
Generation 404 Best Solution: [183.22621617887265, 175.80035550045108,
175.1051810425243, 75.52202047625151] Fitness: 1.5049803926451977
Generation 405 Best Solution: [183.75493466401448, 177.41165717953118,
173.73350271490045, 75.52202047625151] Fitness: 2.145460140342855
Generation 406 Best Solution: [183.75493466401448, 177.29089181815422,
175.1051810425243, 74.43760909374566] Fitness: 2.0815528267364494
Generation 407 Best Solution: [183.75493466401448, 177.41165717953118,

175.1051810425243, 74.78621205254719] Fitness: 2.251805665712092
Generation 408 Best Solution: [183.75493466401448, 175.7412390657625,
175.1051810425243, 75.7999673909102] Fitness: 2.2172871610970355
Generation 409 Best Solution: [183.75493466401448, 175.7412390657625,
175.1051810425243, 75.7999673909102] Fitness: 2.2172871610970355
Generation 410 Best Solution: [183.75493466401448, 175.7412390657625,
175.1051810425243, 75.7999673909102] Fitness: 2.2172871610970355
Generation 411 Best Solution: [183.75493466401448, 175.7412390657625,
175.1051810425243, 75.52202047625151] Fitness: 1.7830375837361572
Generation 412 Best Solution: [183.75493466401448, 175.7412390657625,
175.1051810425243, 75.7999673909102] Fitness: 2.2172871610970355
Generation 413 Best Solution: [183.75493466401448, 175.7412390657625,
175.1051810425243, 75.7999673909102] Fitness: 2.2172871610970355
Generation 414 Best Solution: [183.75493466401448, 175.73956391181537,
175.1051810425243, 75.7999673909102] Fitness: 2.2185795143392735
Generation 415 Best Solution: [183.75493466401448, 175.73956391181537,
175.1051810425243, 75.7999673909102] Fitness: 2.2185795143392735
Generation 416 Best Solution: [183.75493466401448, 175.83826897586783,
175.1051810425243, 75.7999673909102] Fitness: 2.1502973124973246
Generation 417 Best Solution: [183.75493466401448, 177.41165717953118,
174.47615785851164, 75.82721987554653] Fitness: 2.593636562091006
Generation 418 Best Solution: [183.2379873505875, 177.41165717953118,
175.1051810425243, 75.7999673909102] Fitness: 2.9877560242436783
Generation 419 Best Solution: [183.5976128961998, 177.41165717953118,
175.1051810425243, 74.35190316260284] Fitness: 2.2343295026667733
Generation 420 Best Solution: [183.5976128961998, 177.41165717953118,
175.1051810425243, 74.35190316260284] Fitness: 2.2343295026667733
Generation 421 Best Solution: [183.67322351768894, 177.41165717953118,
175.1051810425243, 74.35190316260284] Fitness: 2.2859857004331468
Generation 422 Best Solution: [183.5976128961998, 176.2169982301701,
175.1051810425243, 75.7999673909102] Fitness: 1.9187896822099964
Generation 423 Best Solution: [183.5976128961998, 177.41165717953118,
175.1051810425243, 74.35190316260284] Fitness: 2.2343295026667733
Generation 424 Best Solution: [183.5976128961998, 177.41165717953118,
175.1051810425243, 74.7498426503292] Fitness: 2.1302989352425734
Generation 425 Best Solution: [183.6845572546929, 177.41165717953118,
175.1051810425243, 74.7498426503292] Fitness: 2.1905076348115773
Generation 426 Best Solution: [183.5976128961998, 175.66019931608102,
174.79324034488977, 75.82721987554653] Fitness: 1.7986427079536251
Generation 427 Best Solution: [183.5976128961998, 175.66019931608102,
174.79324034488977, 75.7999673909102] Fitness: 1.7492326810758243
Generation 428 Best Solution: [183.5976128961998, 176.03985440949526,
174.94206727926968, 75.82721987554653] Fitness: 1.7535212326873215
Generation 429 Best Solution: [183.67322351768894, 176.03985440949526,
174.79324034488977, 75.82721987554653] Fitness: 1.6257767910199905
Generation 430 Best Solution: [183.14120543938134, 176.44512717159702,
175.03633651149119, 75.36110544443794] Fitness: 1.118202392498403
Generation 431 Best Solution: [182.84566814080793, 176.03985440949526,

174.94206727926968, 75.7999673909102] Fitness: 1.7015475078729954
Generation 432 Best Solution: [183.5976128961998, 176.31043780589056,
175.1051810425243, 75.36110544443794] Fitness: 1.2993899599093504
Generation 433 Best Solution: [183.2982689300814, 176.03985440949526,
174.94206727926968, 74.7498426503292] Fitness: 0.6172238222722628
Generation 434 Best Solution: [183.67322351768894, 177.41165717953118,
175.07005813489027, 74.7498426503292] Fitness: 2.1269244345335987
Generation 435 Best Solution: [183.67322351768894, 177.41165717953118,
175.1051810425243, 74.65042688526691] Fitness: 2.183566380934524
Generation 436 Best Solution: [183.6121944336198, 177.41165717953118,
175.1051810425243, 74.65042688526691] Fitness: 2.141141113226208
Generation 437 Best Solution: [183.17166383751243, 177.41165717953118,
175.1051810425243, 74.65042688526691] Fitness: 2.016396375277495
Generation 438 Best Solution: [183.67322351768894, 176.38896902057837,
175.1051810425243, 75.82721987554653] Fitness: 2.0461259606873545
Generation 439 Best Solution: [183.57813123983095, 177.41165717953118,
175.1051810425243, 74.65042688526691] Fitness: 2.120122309414622
Generation 440 Best Solution: [183.57813123983095, 177.41165717953118,
175.1051810425243, 74.39779929130165] Fitness: 2.197271149050093
Generation 441 Best Solution: [183.5976128961998, 177.41165717953118,
175.1051810425243, 74.39779929130165] Fitness: 2.209059022803621
Generation 442 Best Solution: [182.84489406138965, 176.32292727241483,
175.1051810425243, 75.82721987554653] Fitness: 1.9765470837407675
Generation 443 Best Solution: [183.6121944336198, 175.89236480255863,
175.1051810425243, 75.82721987554653] Fitness: 2.0602660783903697
Generation 444 Best Solution: [183.5976128961998, 176.19883894693794,
175.1051810425243, 75.82721987554653] Fitness: 1.9682618040606055
Generation 445 Best Solution: [183.6121944336198, 176.19883894693794,
175.1051810425243, 75.82721987554653] Fitness: 1.977492734118663
Generation 446 Best Solution: [183.6121944336198, 177.41165717953118,
175.1051810425243, 74.63504116714968] Fitness: 2.1428413640953314
Generation 447 Best Solution: [183.6121944336198, 177.41165717953118,
175.1051810425243, 74.63504116714968] Fitness: 2.1428413640953314
Generation 448 Best Solution: [183.5976128961998, 177.41165717953118,
173.8913103424278, 75.7999673909102] Fitness: 2.3783178096752144
Generation 449 Best Solution: [183.57813123983095, 176.7292727758611,
175.1051810425243, 75.82721987554653] Fitness: 2.177865068622634
Generation 450 Best Solution: [183.5976128961998, 177.41165717953118,
175.1051810425243, 75.21472430222728] Fitness: 2.338251122488716
Generation 451 Best Solution: [183.57813123983095, 177.41165717953118,
173.8913103424278, 75.7999673909102] Fitness: 2.3665299359216863
Generation 452 Best Solution: [183.5976128961998, 177.41165717953118,
173.8913103424278, 75.55675417273086] Fitness: 1.9913951476451957
Generation 453 Best Solution: [183.6121944336198, 177.41165717953118,
174.33794002569132, 75.55675417273086] Fitness: 1.9864736346683165
Generation 454 Best Solution: [183.22737251554267, 176.56721366216757,
175.1051810425243, 75.82721987554653] Fitness: 1.9556265386702456
Generation 455 Best Solution: [183.22737251554267, 176.56721366216757,

175.07005813489027, 75.7999673909102] Fitness: 1.851185813317096
Generation 456 Best Solution: [183.57813123983095, 176.11611667241218,
175.1051810425243, 75.82721987554653] Fitness: 1.9636117421696027
Generation 457 Best Solution: [183.57813123983095, 176.7314636818161,
175.07005813489027, 75.7999673909102] Fitness: 2.075297328311012
Generation 458 Best Solution: [183.57813123983095, 176.7314636818161,
175.07005813489027, 75.7999673909102] Fitness: 2.075297328311012
Generation 459 Best Solution: [183.57813123983095, 176.14803837654648,
175.1051810425243, 75.82721987554653] Fitness: 1.959525269397701
Generation 460 Best Solution: [183.57813123983095, 176.7314636818161,
175.07005813489027, 74.97183586844568] Fitness: 1.1557319533614994
Generation 461 Best Solution: [183.57813123983095, 177.41165717953118,
175.1051810425243, 74.85711778193647] Fitness: 2.1349858012912946
Generation 462 Best Solution: [183.57813123983095, 176.7314636818161,
175.1051810425243, 75.7999673909102] Fitness: 2.1303280267863607
Generation 463 Best Solution: [183.22737251554267, 177.41165717953118,
174.39519542650032, 74.80385314067276] Fitness: 1.3000554818824084
Generation 464 Best Solution: [183.6121944336198, 177.41165717953118,
174.39519542650032, 74.80385314067276] Fitness: 1.4266349530692335
Generation 465 Best Solution: [183.22737251554267, 177.41165717953118,
174.39519542650032, 74.80385314067276] Fitness: 1.3000554818824084
Generation 466 Best Solution: [183.6121944336198, 177.41165717953118,
175.07005813489027, 74.80385314067276] Fitness: 2.090430682191867
Generation 467 Best Solution: [183.6121944336198, 176.7314636818161,
174.37517389114566, 75.84133947975359] Fitness: 1.4997448179015318
Generation 468 Best Solution: [183.22737251554267, 176.7314636818161,
174.37517389114566, 75.84133947975359] Fitness: 1.3731653467147067
Generation 469 Best Solution: [183.39708259074757, 176.5894752548448,
175.02898651139935, 75.71181386060655] Fitness: 1.6772419976785566
Generation 470 Best Solution: [183.22737251554267, 176.37886571953098,
175.1051810425243, 75.7999673909102] Fitness: 1.8248244656585717
Generation 471 Best Solution: [183.22737251554267, 177.41165717953118,
175.07005813489027, 74.65851406209363] Fitness: 1.9587931740451499
Generation 472 Best Solution: [182.94587588574336, 176.5507191950401,
175.1051810425243, 75.84133947975359] Fitness: 2.0337293093869433
Generation 473 Best Solution: [183.6121944336198, 175.87578033395693,
175.07005813489027, 75.7999673909102] Fitness: 1.9647055433899179
Generation 474 Best Solution: [183.22737251554267, 177.41165717953118,
175.07005813489027, 74.91998947599185] Fitness: 1.9928370990158804
Generation 475 Best Solution: [183.24067357669097, 177.41165717953118,
175.07005813489027, 74.91998947599185] Fitness: 1.9931530277780833
Generation 476 Best Solution: [183.22737251554267, 176.5507191950401,
175.1051810425243, 75.82721987554653] Fitness: 1.946170227844564
Generation 477 Best Solution: [183.22737251554267, 177.41165717953118,
175.07005813489027, 74.91998947599185] Fitness: 1.9928370990158804
Generation 478 Best Solution: [183.22737251554267, 177.41165717953118,
175.07005813489027, 74.91998947599185] Fitness: 1.9928370990158804
Generation 479 Best Solution: [183.6121944336198, 176.7554020238285,

175.1051810425243, 75.14954780123423] Fitness: 1.3551462629801438
Generation 480 Best Solution: [183.6121944336198, 176.7554020238285,
175.07005813489027, 75.84133947975359] Fitness: 2.1927839653699097
Generation 481 Best Solution: [183.3043051002313, 177.41165717953118,
175.07005813489027, 74.96239049292923] Fitness: 2.0147900032608757
Generation 482 Best Solution: [183.6121944336198, 177.41165717953118,
175.1051810425243, 74.71937581895085] Fitness: 2.1382981078542778
Generation 483 Best Solution: [183.3043051002313, 177.41165717953118,
175.07005813489027, 74.71937581895085] Fitness: 1.9625364224172523
Generation 484 Best Solution: [183.65206584579147, 176.9327544424062,
175.07005813489027, 74.51809163624253] Fitness: 1.383438205877331
Generation 485 Best Solution: [183.5089064199001, 175.62277665615434,
175.1051810425243, 75.61184207045706] Fitness: 1.8455763870809065
Generation 486 Best Solution: [183.5089064199001, 175.62277665615434,
175.1051810425243, 75.06726654151552] Fitness: 1.282339459142487
Generation 487 Best Solution: [183.6121944336198, 176.53335947253612,
175.1051810425243, 75.82721987554653] Fitness: 2.0632800886039404
Generation 488 Best Solution: [183.6121944336198, 176.53335947253612,
175.1051810425243, 75.82721987554653] Fitness: 2.0632800886039404
Generation 489 Best Solution: [183.5089064199001, 176.53335947253612,
175.1051810425243, 75.82721987554653] Fitness: 2.0054191775596353
Generation 490 Best Solution: [183.5089064199001, 176.53335947253612,
175.1051810425243, 75.82721987554653] Fitness: 2.0054191775596353
Generation 491 Best Solution: [183.3043051002313, 177.41165717953118,
174.05958191180613, 75.82721987554653] Fitness: 2.2724202107675975
Generation 492 Best Solution: [183.3043051002313, 177.41165717953118,
174.05958191180613, 75.82721987554653] Fitness: 2.2724202107675975
Generation 493 Best Solution: [183.65206584579147, 177.41165717953118,
175.1051810425243, 74.4375806950929] Fitness: 2.226210238861923
Generation 494 Best Solution: [183.5089064199001, 177.41165717953118,
175.1051810425243, 74.95733056885608] Fitness: 2.1306137539214904
Generation 495 Best Solution: [183.14509771659692, 176.20339369560674,
175.07005813489027, 74.36203911021616] Fitness: 0.8744861846987492
Generation 496 Best Solution: [183.6121944336198, 177.002641869002,
175.1051810425243, 74.95733056885608] Fitness: 1.5184363283104922
Generation 497 Best Solution: [183.49271755204975, 177.002641869002,
175.1051810425243, 74.95733056885608] Fitness: 1.4530953790012968
Generation 498 Best Solution: [183.6121944336198, 175.61566872989542,
175.1051810425243, 74.95733056885608] Fitness: 1.2925011890594487
Generation 499 Best Solution: [183.5089064199001, 175.61566872989542,
175.1051810425243, 74.95733056885608] Fitness: 1.2346402780151433
Generation 500 Best Solution: [183.6121944336198, 176.16043166000563,
175.1051810425243, 74.54143456388218] Fitness: 0.9774580792872156
Optimal Solution for Simulation 26: [183.6121944336198, 176.16043166000563,
175.1051810425243, 74.54143456388218]

Simulation 27/32 with conditions: {'drought': False, 'flood': False,
'policy_change': True, 'social_event': False, 'economic_crisis': True}

Generation 1 Best Solution: [183.07510574692617, 176.01899927172326,
174.25584233017432, 74.49271873880932] Fitness: 0.15448998093859276
Generation 2 Best Solution: [183.29871092827747, 175.9134319242196,
174.1844990163992, 75.1810951217879] Fitness: 0.3143970252436755
Generation 3 Best Solution: [183.13295729336568, 176.45508720829173,
174.31132878462492, 75.25733751105597] Fitness: 0.3760602131411027
Generation 4 Best Solution: [183.23224816031754, 176.77177533322242,
173.88616964681296, 74.97971120553908] Fitness: 0.4218568150416145
Generation 5 Best Solution: [183.51729501145925, 176.10786698691192,
174.66353277410545, 75.25733751105597] Fitness: 0.6056711647318607
Generation 6 Best Solution: [183.13295729336568, 177.12652558171297,
174.4562118118779, 75.13652226985906] Fitness: 0.9822472332666394
Generation 7 Best Solution: [183.18511157055298, 177.12652558171297,
174.66353277410545, 75.65995188044214] Fitness: 1.7176379956954952
Generation 8 Best Solution: [183.18511157055298, 176.9787369647066,
174.66353277410545, 75.65995188044214] Fitness: 1.5113173833138984
Generation 9 Best Solution: [183.51729501145925, 177.12652558171297,
174.66353277410545, 75.1827497756487] Fitness: 1.2257222316859089
Generation 10 Best Solution: [183.13295729336568, 177.26983492606476,
175.00148334171797, 74.76997440733041] Fitness: 1.6066660955079228
Generation 11 Best Solution: [183.18511157055298, 176.6373279937084,
175.00148334171797, 75.47810300241136] Fitness: 1.3050889678502564
Generation 12 Best Solution: [183.15127028694627, 177.26983492606476,
175.00148334171797, 74.50905874587252] Fitness: 1.6369609500351563
Generation 13 Best Solution: [182.88768686879843, 177.12652558171297,
174.14863104043505, 75.47810300241136] Fitness: 1.3239456014553304
Generation 14 Best Solution: [183.6519133692411, 177.12652558171297,
175.02004021748863, 75.47810300241136] Fitness: 2.026338239700518
Generation 15 Best Solution: [183.18511157055298, 175.81696325955608,
174.3866327011416, 75.90494777674915] Fitness: 1.4154189451747001
Generation 16 Best Solution: [183.51729501145925, 176.67897783003443,
175.00148334171797, 75.90494777674915] Fitness: 2.100430049946041
Generation 17 Best Solution: [183.6519133692411, 177.12652558171297,
175.00148334171797, 75.65995188044214] Fitness: 2.258247096135545
Generation 18 Best Solution: [183.6519133692411, 177.12652558171297,
175.00148334171797, 75.47810300241136] Fitness: 1.9994072094172948
Generation 19 Best Solution: [183.51729501145925, 177.40105976142596,
175.02004021748863, 74.34905412761864] Fitness: 2.0518635810901142
Generation 20 Best Solution: [183.18511157055298, 177.40105976142596,
175.02004021748863, 75.21847212554528] Fitness: 2.0816323417300415
Generation 21 Best Solution: [183.51729501145925, 177.40105976142596,
174.96976104073445, 74.88287295247179] Fitness: 1.892111536908397
Generation 22 Best Solution: [183.18511157055298, 177.40105976142596,
174.96976104073445, 74.88287295247179] Fitness: 1.822240464474881
Generation 23 Best Solution: [183.18511157055298, 176.12634498677033,
174.9839886035163, 75.90494777674915] Fitness: 1.8328485603584803
Generation 24 Best Solution: [183.18511157055298, 176.12634498677033,
175.00148334171797, 75.90494777674915] Fitness: 1.857712548018609

Generation 25 Best Solution: [183.18511157055298, 176.12634498677033,
175.00148334171797, 74.97413708229932] Fitness: 0.733692811378896
Generation 26 Best Solution: [183.51729501145925, 176.48602817672565,
175.02004021748863, 75.65995188044214] Fitness: 1.573310405876474
Generation 27 Best Solution: [183.82247727210301, 177.40105976142596,
174.9839886035163, 75.65995188044214] Fitness: 2.858868889641667
Generation 28 Best Solution: [183.51729501145925, 177.40105976142596,
175.00148334171797, 75.42190720806782] Fitness: 2.3297670668946395
Generation 29 Best Solution: [183.24806993907018, 177.40105976142596,
175.00148334171797, 75.42190720806782] Fitness: 2.2590136071985842
Generation 30 Best Solution: [183.82247727210301, 177.40105976142596,
175.00148334171797, 74.28293961379138] Fitness: 2.295184644784794
Generation 31 Best Solution: [183.49512423117508, 177.40105976142596,
175.02004021748863, 74.93667125936433] Fitness: 1.9708239913061019
Generation 32 Best Solution: [183.09877800264647, 177.40105976142596,
175.02004021748863, 75.11968197495804] Fitness: 2.018819196285897
Generation 33 Best Solution: [183.18511157055298, 176.57031655693183,
175.02004021748863, 75.8558847442255] Fitness: 1.8880039386979799
Generation 34 Best Solution: [183.18511157055298, 176.57031655693183,
174.41234325930265, 75.90494777674915] Fitness: 1.3997187539262415
Generation 35 Best Solution: [183.49512423117508, 177.40105976142596,
174.96976104073445, 74.34145267757725] Fitness: 1.9745173706263726
Generation 36 Best Solution: [183.09877800264647, 177.40105976142596,
174.96976104073445, 75.37263672874572] Fitness: 2.171111835004833
Generation 37 Best Solution: [183.51729501145925, 177.40105976142596,
175.02004021748863, 75.19937407169307] Fitness: 2.135781797603097
Generation 38 Best Solution: [183.18511157055298, 177.40105976142596,
174.07174647508444, 74.30848241705979] Fitness: 1.3665991111504023
Generation 39 Best Solution: [183.09877800264647, 177.40105976142596,
175.02004021748863, 75.33942073849695] Fitness: 2.207263994460183
Generation 40 Best Solution: [183.49512423117508, 177.40105976142596,
175.02004021748863, 75.33942073849695] Fitness: 2.2547932508368658
Generation 41 Best Solution: [183.1355385901177, 176.82359902813965,
175.02004021748863, 75.90494777674915] Fitness: 2.1920980139152917
Generation 42 Best Solution: [183.18511157055298, 176.50162597430463,
175.02004021748863, 75.90494777674915] Fitness: 1.9465558597812687
Generation 43 Best Solution: [183.3682580467886, 177.40105976142596,
175.02004021748863, 75.03765280516923] Fitness: 1.972926035603737
Generation 44 Best Solution: [183.09877800264647, 177.40105976142596,
175.02004021748863, 75.03765280516923] Fitness: 1.969100306142257
Generation 45 Best Solution: [183.09877800264647, 177.40105976142596,
175.02004021748863, 75.03765280516923] Fitness: 1.969100306142257
Generation 46 Best Solution: [183.49512423117508, 176.5424558705527,
174.45117352156356, 75.90494777674915] Fitness: 1.4622284327832347
Generation 47 Best Solution: [183.38385926891365, 177.40105976142596,
174.3663959521279, 74.91233556465994] Fitness: 1.3198712310109413
Generation 48 Best Solution: [183.09877800264647, 176.20561637829925,
175.02004021748863, 75.90494777674915] Fitness: 1.8901179506331933

Generation 49 Best Solution: [183.09877800264647, 177.40105976142596,
174.3663959521279, 75.05391143793379] Fitness: 1.3753840334932952
Generation 50 Best Solution: [183.09877800264647, 176.22531212191132,
174.71685472930852, 75.90494777674915] Fitness: 1.522042018825084
Generation 51 Best Solution: [183.37613691682458, 177.40105976142596,
174.13483876299844, 75.90494777674915] Fitness: 2.422714562548899
Generation 52 Best Solution: [183.09877800264647, 176.46564819005334,
175.02004021748863, 75.90494777674915] Fitness: 1.942398336317187
Generation 53 Best Solution: [183.09877800264647, 176.46564819005334,
175.02004021748863, 75.90494777674915] Fitness: 1.942398336317187
Generation 54 Best Solution: [183.09877800264647, 176.5825053613214,
174.17742927699751, 75.90494777674915] Fitness: 1.3583698711522607
Generation 55 Best Solution: [183.49512423117508, 176.5825053613214,
174.17742927699751, 75.90494777674915] Fitness: 1.4058991275289436
Generation 56 Best Solution: [183.09877800264647, 176.5825053613214,
174.17742927699751, 75.90494777674915] Fitness: 1.3583698711522607
Generation 57 Best Solution: [183.09877800264647, 176.05898607555255,
175.02004021748863, 75.90494777674915] Fitness: 1.9103281941875865
Generation 58 Best Solution: [183.49512423117508, 177.40105976142596,
174.55084842834566, 74.47331192607342] Fitness: 1.473824511745131
Generation 59 Best Solution: [183.49512423117508, 176.46338086243819,
175.02004021748863, 74.85181564383554] Fitness: 0.8246037149075287
Generation 60 Best Solution: [183.49512423117508, 176.46338086243819,
175.02004021748863, 75.90494777674915] Fitness: 1.98898470740097
Generation 61 Best Solution: [183.49512423117508, 176.46338086243819,
175.02004021748863, 75.90494777674915] Fitness: 1.98898470740097
Generation 62 Best Solution: [183.0667327324486, 177.40105976142596,
174.55084842834566, 75.90494777674915] Fitness: 2.5651945234302787
Generation 63 Best Solution: [183.49512423117508, 177.40105976142596,
174.18044849474853, 75.90494777674915] Fitness: 2.465826375049505
Generation 64 Best Solution: [183.49512423117508, 177.40105976142596,
175.02004021748863, 75.15824834585915] Fitness: 2.0935997820383565
Generation 65 Best Solution: [183.49512423117508, 176.38619254853887,
175.02004021748863, 75.90494777674915] Fitness: 1.9619962372757183
Generation 66 Best Solution: [183.49512423117508, 176.38619254853887,
175.02004021748863, 75.90494777674915] Fitness: 1.9619962372757183
Generation 67 Best Solution: [183.05296941061007, 177.40105976142596,
175.02004021748863, 74.70808126761729] Fitness: 1.893771612179038
Generation 68 Best Solution: [183.49512423117508, 177.40105976142596,
175.02004021748863, 74.70808126761729] Fitness: 1.9299154259974463
Generation 69 Best Solution: [183.49512423117508, 177.40105976142596,
175.02004021748863, 74.61115750255733] Fitness: 1.9388615888452345
Generation 70 Best Solution: [183.49512423117508, 175.70054951000066,
175.02004021748863, 74.45959930952026] Fitness: 1.032501420019552
Generation 71 Best Solution: [183.49512423117508, 177.40105976142596,
175.02004021748863, 74.82101535582527] Fitness: 1.9392415816749258
Generation 72 Best Solution: [183.05296941061007, 177.40105976142596,
175.02004021748863, 74.82101535582527] Fitness: 1.9030977678565175

Generation 73 Best Solution: [183.15950369583993, 176.41207785616407,
175.02004021748863, 75.90494777674915] Fitness: 1.9127091191971248
Generation 74 Best Solution: [183.05296941061007, 176.41207785616407,
175.02004021748863, 75.90494777674915] Fitness: 1.9337964173864712
Generation 75 Best Solution: [183.49512423117508, 176.41207785616407,
175.02004021748863, 75.90494777674915] Fitness: 1.9699402312048793
Generation 76 Best Solution: [183.49512423117508, 177.40105976142596,
175.02004021748863, 75.14154999865322] Fitness: 2.081496286408513
Generation 77 Best Solution: [183.05296941061007, 176.46434075361208,
175.02004021748863, 75.90494777674915] Fitness: 1.9532390257378134
Generation 78 Best Solution: [183.05296941061007, 177.40105976142596,
175.02004021748863, 75.14154999865322] Fitness: 2.0453524725901047
Generation 79 Best Solution: [183.05296941061007, 177.40105976142596,
175.02004021748863, 75.14154999865322] Fitness: 2.0453524725901047
Generation 80 Best Solution: [182.92742612779372, 177.40105976142596,
174.35535851470186, 75.90494777674915] Fitness: 2.5160983069487317
Generation 81 Best Solution: [183.49512423117508, 177.40105976142596,
175.02004021748863, 74.28275077746225] Fitness: 2.0855750272696953
Generation 82 Best Solution: [183.05296941061007, 177.40105976142596,
175.02004021748863, 74.94022296592186] Fitness: 1.9360028751130867
Generation 83 Best Solution: [183.49512423117508, 177.40105976142596,
175.02004021748863, 74.94022296592186] Fitness: 1.972146688931495
Generation 84 Best Solution: [183.05296941061007, 176.25592946819998,
174.35535851470186, 75.90494777674915] Fitness: 1.2960831219148612
Generation 85 Best Solution: [183.05296941061007, 176.25592946819998,
174.35535851470186, 75.90494777674915] Fitness: 1.2960831219148612
Generation 86 Best Solution: [183.05296941061007, 176.25592946819998,
174.35535851470186, 75.90494777674915] Fitness: 1.2960831219148612
Generation 87 Best Solution: [183.05296941061007, 177.40105976142596,
174.6513966581861, 74.90214555421217] Fitness: 1.495464401520758
Generation 88 Best Solution: [183.05296941061007, 177.40105976142596,
175.02004021748863, 75.25600191449064] Fitness: 2.1376342810330025
Generation 89 Best Solution: [183.63674662855811, 177.40105976142596,
174.73313487558542, 74.6631995938771] Fitness: 1.6603669220941637
Generation 90 Best Solution: [183.05296941061007, 177.40105976142596,
174.73313487558542, 74.6631995938771] Fitness: 1.5437496200582175
Generation 91 Best Solution: [183.05296941061007, 175.8241074537543,
175.02004021748863, 75.90494777674915] Fitness: 2.028760835153327
Generation 92 Best Solution: [183.05296941061007, 175.8534652700361,
174.73313487558542, 75.90494777674915] Fitness: 1.6581346472349212
Generation 93 Best Solution: [183.05296941061007, 175.8534652700361,
175.02004021748863, 75.24633216997324] Fitness: 1.0656346199181836
Generation 94 Best Solution: [183.05296941061007, 175.8241074537543,
174.63890049879694, 75.90494777674915] Fitness: 1.5907866045720003
Generation 95 Best Solution: [183.05296941061007, 177.23048189677183,
175.02004021748863, 75.17239336937901] Fitness: 1.755127070922065
Generation 96 Best Solution: [183.05296941061007, 177.23048189677183,
175.02004021748863, 75.17239336937901] Fitness: 1.755127070922065

Generation 97 Best Solution: [182.8621455625521, 176.43684987909458,
174.58135471978883, 75.90494777674915] Fitness: 1.5444378430833845
Generation 98 Best Solution: [183.45214721307346, 177.40105976142596,
175.02004021748863, 74.69853592521852] Fitness: 1.9122920535305594
Generation 99 Best Solution: [183.05296941061007, 177.40105976142596,
175.02004021748863, 75.23678763258175] Fitness: 2.120616999393613
Generation 100 Best Solution: [182.8621455625521, 177.40105976142596,
175.02004021748863, 75.23678763258175] Fitness: 2.205674348635108
Generation 101 Best Solution: [182.84860313721416, 177.40105976142596,
175.02004021748863, 75.23678763258175] Fitness: 2.214017061584595
Generation 102 Best Solution: [183.05296941061007, 177.40105976142596,
175.02004021748863, 75.23678763258175] Fitness: 2.120616999393613
Generation 103 Best Solution: [183.45657591707865, 177.40105976142596,
174.72381722409241, 75.90494777674915] Fitness: 2.7323820213182035
Generation 104 Best Solution: [182.84860313721416, 177.40105976142596,
175.02004021748863, 74.96955143660988] Fitness: 2.041128811888662
Generation 105 Best Solution: [183.76025654813924, 177.40105976142596,
175.02004021748863, 74.82702625025996] Fitness: 2.1182774959537225
Generation 106 Best Solution: [182.84860313721416, 177.40105976142596,
175.02004021748863, 74.70852181512768] Fitness: 1.9871667560545672
Generation 107 Best Solution: [183.76025654813924, 177.40105976142596,
175.02004021748863, 74.70852181512768] Fitness: 2.107854234396999
Generation 108 Best Solution: [183.76025654813924, 177.40105976142596,
175.02004021748863, 74.70852181512768] Fitness: 2.107854234396999
Generation 109 Best Solution: [183.76025654813924, 175.81077062948418,
175.02004021748863, 74.70852181512768] Fitness: 1.0717405896585248
Generation 110 Best Solution: [182.85632143451662, 177.40105976142596,
174.03378071729, 75.90494777674915] Fitness: 2.5263290747753
Generation 111 Best Solution: [183.76025654813924, 177.40105976142596,
174.5017072333194, 75.56852076594394] Fitness: 2.178535125444035
Generation 112 Best Solution: [183.76025654813924, 176.69426141587974,
174.6403871830045, 75.90494777674915] Fitness: 1.870190934485352
Generation 113 Best Solution: [183.76025654813924, 177.40105976142596,
175.02004021748863, 74.6397217605022] Fitness: 2.1125416064492053
Generation 114 Best Solution: [182.9021298207963, 177.40105976142596,
175.02004021748863, 74.76961898524627] Fitness: 1.9584266068663194
Generation 115 Best Solution: [182.9021298207963, 177.40105976142596,
175.02004021748863, 74.76961898524627] Fitness: 1.9584266068663194
Generation 116 Best Solution: [182.9021298207963, 175.82622205049253,
175.02004021748863, 75.90494777674915] Fitness: 2.089595785358301
Generation 117 Best Solution: [183.76025654813924, 175.93227415082768,
175.02004021748863, 75.90494777674915] Fitness: 2.182129144919872
Generation 118 Best Solution: [183.76025654813924, 175.93227415082768,
175.02004021748863, 74.46343756004934] Fitness: 1.0548857697562581
Generation 119 Best Solution: [183.76025654813924, 176.5281206834719,
175.00343538704192, 75.90494777674915] Fitness: 2.173096261441955
Generation 120 Best Solution: [183.76025654813924, 176.5281206834719,
175.00343538704192, 75.38736153934708] Fitness: 1.3698415349443889

Generation 121 Best Solution: [182.9021298207963, 176.95784564047494,
175.02004021748863, 75.38736153934708] Fitness: 1.6202338518185604
Generation 122 Best Solution: [182.9021298207963, 177.40105976142596,
175.02004021748863, 74.61821584716371] Fitness: 1.963746555775407
Generation 123 Best Solution: [182.9021298207963, 177.40105976142596,
175.02004021748863, 74.61821584716371] Fitness: 1.963746555775407
Generation 124 Best Solution: [182.9021298207963, 177.40105976142596,
175.02004021748863, 74.37109545002052] Fitness: 2.054499195155698
Generation 125 Best Solution: [182.9021298207963, 177.40105976142596,
175.02004021748863, 74.61821584716371] Fitness: 1.963746555775407
Generation 126 Best Solution: [182.9021298207963, 177.40105976142596,
175.02004021748863, 75.1423393389258] Fitness: 2.1081230044031574
Generation 127 Best Solution: [182.9021298207963, 177.40105976142596,
175.02004021748863, 74.61821584716371] Fitness: 1.963746555775407
Generation 128 Best Solution: [183.76025654813924, 176.71566018899966,
175.02004021748863, 75.46500988294675] Fitness: 1.6131840050752868
Generation 129 Best Solution: [182.9021298207963, 176.334696491031,
175.02004021748863, 74.61821584716371] Fitness: 0.803404466018196
Generation 130 Best Solution: [183.76025654813924, 177.40105976142596,
175.02004021748863, 74.61821584716371] Fitness: 2.1156252388644967
Generation 131 Best Solution: [183.76025654813924, 177.23503965995602,
175.02004021748863, 74.61821584716371] Fitness: 1.8104111973723223
Generation 132 Best Solution: [183.76025654813924, 177.40105976142596,
174.50367466296368, 74.45381887049312] Fitness: 1.6294893107432973
Generation 133 Best Solution: [183.76025654813924, 177.40105976142596,
174.50367466296368, 74.45381887049312] Fitness: 1.6294893107432973
Generation 134 Best Solution: [183.76025654813924, 177.40105976142596,
175.02004021748863, 74.5363689746122] Fitness: 2.1344101313568777
Generation 135 Best Solution: [183.76025654813924, 176.33142986117866,
175.02004021748863, 75.8926715523066] Fitness: 2.1025929620080737
Generation 136 Best Solution: [183.76025654813924, 177.40105976142596,
175.02004021748863, 74.5363689746122] Fitness: 2.1344101313568777
Generation 137 Best Solution: [183.76025654813924, 177.40105976142596,
175.02004021748863, 74.5363689746122] Fitness: 2.1344101313568777
Generation 138 Best Solution: [183.76025654813924, 176.95752090558634,
174.07522102234273, 75.90494777674915] Fitness: 1.933069912412727
Generation 139 Best Solution: [183.76025654813924, 176.95752090558634,
174.07522102234273, 75.8926715523066] Fitness: 1.9088487238477032
Generation 140 Best Solution: [183.76025654813924, 175.8467635823551,
173.90845178405857, 75.90494777674915] Fitness: 1.6278024327515164
Generation 141 Best Solution: [183.76025654813924, 175.8467635823551,
173.90845178405857, 75.8926715523066] Fitness: 1.6035812441864925
Generation 142 Best Solution: [183.16698497429755, 176.29271514005552,
174.50367466296368, 75.90494777674915] Fitness: 1.3494294650609266
Generation 143 Best Solution: [183.76025654813924, 175.85093920818025,
175.02004021748863, 75.41760748124543] Fitness: 1.4573690746382044
Generation 144 Best Solution: [183.76025654813924, 176.79074636353533,
174.53466869155977, 74.55552282998443] Fitness: 0.7173789927531473

Generation 145 Best Solution: [183.50685622378285, 177.0120297009452,
174.50367466296368, 75.90494777674915] Fitness: 1.9371743724549284
Generation 146 Best Solution: [183.76025654813924, 176.79074636353533,
174.50367466296368, 75.09085512118756] Fitness: 0.7943206500456322
Generation 147 Best Solution: [183.76025654813924, 176.79074636353533,
174.50367466296368, 75.09085512118756] Fitness: 0.7943206500456322
Generation 148 Best Solution: [183.76025654813924, 177.40105976142596,
174.50367466296368, 74.92786711527604] Fitness: 1.6104033327061806
Generation 149 Best Solution: [182.85963465737197, 175.85093920818025,
175.02004021748863, 74.72826439166512] Fitness: 0.9186281851814985
Generation 150 Best Solution: [183.50685622378285, 176.79074636353533,
175.02004021748863, 75.57309447841882] Fitness: 1.6528322702496663
Generation 151 Best Solution: [183.76025654813924, 177.40105976142596,
174.50367466296368, 74.82551702819624] Fitness: 1.582821336669001
Generation 152 Best Solution: [183.50685622378285, 177.40105976142596,
174.37361761392546, 75.90494777674915] Fitness: 2.515340491237839
Generation 153 Best Solution: [183.76025654813924, 177.40105976142596,
173.75079782003823, 75.75867321005869] Fitness: 2.4963703635674483
Generation 154 Best Solution: [183.76025654813924, 177.40105976142596,
173.75079782003823, 75.75867321005869] Fitness: 2.4963703635674483
Generation 155 Best Solution: [183.50685622378285, 177.40105976142596,
173.75079782003823, 75.75867321005869] Fitness: 2.3238231623667738
Generation 156 Best Solution: [183.50685622378285, 177.40105976142596,
175.02004021748863, 74.87705932512382] Fitness: 1.9571580815132825
Generation 157 Best Solution: [183.50685622378285, 177.40105976142596,
175.02004021748863, 74.91050266043003] Fitness: 1.9671231359929735
Generation 158 Best Solution: [183.50685622378285, 177.40105976142596,
173.90774013565348, 74.52241430499878] Fitness: 1.3657413192202366
Generation 159 Best Solution: [183.50685622378285, 177.40105976142596,
174.9862635453215, 74.41078752616122] Fitness: 1.963802421615108
Generation 160 Best Solution: [183.76025654813924, 176.6424726304203,
173.85074978341277, 75.90494777674915] Fitness: 1.6921240177253356
Generation 161 Best Solution: [183.50685622378285, 176.35250610361112,
175.02004021748863, 75.90494777674915] Fitness: 1.9587269626692805
Generation 162 Best Solution: [183.50685622378285, 176.6424726304203,
174.39468138082464, 75.90494777674915] Fitness: 1.503907581477145
Generation 163 Best Solution: [183.50685622378285, 175.87517117876664,
174.9862635453215, 75.90494777674915] Fitness: 1.99061604846363
Generation 164 Best Solution: [183.50685622378285, 177.40105976142596,
175.02004021748863, 74.44149569502757] Fitness: 1.9976092314956657
Generation 165 Best Solution: [183.50685622378285, 176.35250610361112,
175.02004021748863, 75.90494777674915] Fitness: 1.9587269626692805
Generation 166 Best Solution: [183.50685622378285, 176.35250610361112,
174.9862635453215, 75.90494777674915] Fitness: 1.9101362991215054
Generation 167 Best Solution: [182.86169263047293, 176.24449518560633,
174.1039115529664, 75.90494777674915] Fitness: 1.3431794151051575
Generation 168 Best Solution: [182.86169263047293, 176.24449518560633,
175.02004021748863, 75.90494777674915] Fitness: 1.9874864220952682

Generation 169 Best Solution: [183.50685622378285, 177.40105976142596,
174.1039115529664, 75.90494777674915] Fitness: 2.470944813298732
Generation 170 Best Solution: [183.50685622378285, 177.40105976142596,
175.02004021748863, 74.53874958252808] Fitness: 1.9611589021794766
Generation 171 Best Solution: [183.28267486700517, 177.40105976142596,
174.9862635453215, 75.90494777674915] Fitness: 3.0032307062401804
Generation 172 Best Solution: [183.50685622378285, 177.40105976142596,
175.02004021748863, 75.22361269671649] Fitness: 2.1508444568479743
Generation 173 Best Solution: [183.466153625516, 177.40105976142596,
174.9862635453215, 75.90494777674915] Fitness: 3.048921273586388
Generation 174 Best Solution: [183.76567842551077, 177.40105976142596,
174.4856021481732, 75.90494777674915] Fitness: 2.7468020614156616
Generation 175 Best Solution: [183.76567842551077, 176.73211730865162,
175.02004021748863, 75.90494777674915] Fitness: 2.343220805423003
Generation 176 Best Solution: [183.50685622378285, 177.40105976142596,
175.02004021748863, 75.00849865967669] Fitness: 2.0070566134903034
Generation 177 Best Solution: [183.466153625516, 177.40105976142596,
175.02004021748863, 75.00849865967669] Fitness: 1.9893167303356236
Generation 178 Best Solution: [183.466153625516, 177.40105976142596,
174.9862635453215, 74.98927753071514] Fitness: 1.9316316506405005
Generation 179 Best Solution: [183.466153625516, 176.22751501734288,
175.02004021748863, 75.90494777674915] Fitness: 1.9253609937885854
Generation 180 Best Solution: [183.50685622378285, 176.47100659531233,
174.9862635453215, 74.94377577295133] Fitness: 0.8126313751317679
Generation 181 Best Solution: [183.466153625516, 177.40105976142596,
175.02004021748863, 74.63377874603768] Fitness: 1.9230295877216204
Generation 182 Best Solution: [183.31909417297288, 177.40105976142596,
175.02004021748863, 74.98927753071514] Fitness: 1.9391378494282239
Generation 183 Best Solution: [183.466153625516, 177.40105976142596,
175.02004021748863, 74.98927753071514] Fitness: 1.9802223141882758
Generation 184 Best Solution: [183.466153625516, 177.40105976142596,
174.9862635453215, 74.98927753071514] Fitness: 1.9316316506405005
Generation 185 Best Solution: [183.466153625516, 176.38894434942074,
175.02004021748863, 75.73172084154928] Fitness: 1.6318989681941876
Generation 186 Best Solution: [182.8086650695212, 176.04079769146585,
175.02004021748863, 75.90494777674915] Fitness: 2.0465022620192563
Generation 187 Best Solution: [183.466153625516, 177.40105976142596,
175.02004021748863, 75.17619621549271] Fitness: 2.0947838019629446
Generation 188 Best Solution: [183.50685622378285, 176.43865469059438,
174.9862635453215, 75.90787203587423] Fitness: 1.9418708613507056
Generation 189 Best Solution: [183.50685622378285, 177.11024906614043,
175.02004021748863, 75.73172084154928] Fitness: 2.2923173961237806
Generation 190 Best Solution: [183.466153625516, 177.06753734140372,
175.02004021748863, 75.90494777674915] Fitness: 2.5309137378124404
Generation 191 Best Solution: [183.466153625516, 177.06753734140372,
175.02004021748863, 75.90494777674915] Fitness: 2.5309137378124404
Generation 192 Best Solution: [183.31909417297288, 177.40105976142596,
174.9862635453215, 75.21648866403204] Fitness: 2.0374327827949887

Generation 193 Best Solution: [183.50685622378285, 175.95558661356,
174.9862635453215, 75.90494777674915] Fitness: 1.9504591128998043
Generation 194 Best Solution: [182.82816183558728, 175.95558661356,
174.9862635453215, 75.90494777674915] Fitness: 2.015490444440906
Generation 195 Best Solution: [183.31909417297288, 176.7147866087721,
175.02004021748863, 74.30319854985429] Fitness: 1.0536388220290902
Generation 196 Best Solution: [183.31909417297288, 176.7147866087721,
174.9862635453215, 75.49996420075776] Fitness: 1.377205126922285
Generation 197 Best Solution: [183.50685622378285, 175.79700341234258,
175.02004021748863, 75.28641552406022] Fitness: 1.1806879697606942
Generation 198 Best Solution: [183.50685622378285, 175.79700341234258,
174.45655740740995, 75.90494777674915] Fitness: 1.5266861514622954
Generation 199 Best Solution: [183.50685622378285, 176.2137570840838,
174.9862635453215, 75.90494777674915] Fitness: 1.8943809793795583
Generation 200 Best Solution: [182.88370773492557, 176.32775588904374,
174.8544894166159, 75.90494777674915] Fitness: 1.7644501154667336
Generation 201 Best Solution: [183.50685622378285, 176.2137570840838,
174.9862635453215, 75.90494777674915] Fitness: 1.8943809793795583
Generation 202 Best Solution: [183.50685622378285, 176.2137570840838,
174.9862635453215, 75.90494777674915] Fitness: 1.8943809793795583
Generation 203 Best Solution: [183.50685622378285, 176.2137570840838,
174.9862635453215, 75.90494777674915] Fitness: 1.8943809793795583
Generation 204 Best Solution: [183.50685622378285, 176.08897151549812,
175.02004021748863, 75.39244005736049] Fitness: 1.1586641045874932
Generation 205 Best Solution: [183.50685622378285, 177.40105976142596,
175.02004021748863, 74.51918343714969] Fitness: 1.9672254993690463
Generation 206 Best Solution: [183.39543992096768, 177.40105976142596,
174.28126648868223, 75.90494777674915] Fitness: 2.444495380456774
Generation 207 Best Solution: [183.50685622378285, 177.40105976142596,
173.85392404874258, 75.82447006277891] Fitness: 2.3838491322078417
Generation 208 Best Solution: [183.50685622378285, 176.44988071102736,
174.9862635453215, 75.90494777674915] Fitness: 1.9403538156901772
Generation 209 Best Solution: [183.50685622378285, 176.55349313337433,
174.9862635453215, 75.90494777674915] Fitness: 1.9898610264642218
Generation 210 Best Solution: [183.50685622378285, 177.40105976142596,
173.77719731868694, 75.82447006277891] Fitness: 2.425337745274598
Generation 211 Best Solution: [183.50685622378285, 177.40105976142596,
173.80734964967172, 74.42848109364577] Fitness: 1.4505041898590554
Generation 212 Best Solution: [183.39543992096768, 177.40105976142596,
173.80734964967172, 74.42848109364577] Fitness: 1.4085098863376715
Generation 213 Best Solution: [183.305636090004, 176.36607623735597,
174.9862635453215, 75.82447006277891] Fitness: 1.698405847372939
Generation 214 Best Solution: [183.50685622378285, 176.36607623735597,
174.9862635453215, 75.82447006277891] Fitness: 1.7591898072679373
Generation 215 Best Solution: [183.50685622378285, 175.69177457754412,
174.9862635453215, 75.90494777674915] Fitness: 2.1225170609330246
Generation 216 Best Solution: [183.50685622378285, 177.40105976142596,
174.9862635453215, 74.90932585179569] Fitness: 1.9181501778126155

Generation 217 Best Solution: [183.32991455306322, 177.40105976142596,
174.36096735260313, 75.90494777674915] Fitness: 2.453518419520784
Generation 218 Best Solution: [183.32991455306322, 177.40105976142596,
173.9813106008129, 75.90494777674915] Fitness: 2.4338148816023946
Generation 219 Best Solution: [183.50685622378285, 176.18042866241802,
174.9862635453215, 75.90494777674915] Fitness: 1.8953756731647522
Generation 220 Best Solution: [183.50685622378285, 176.18042866241802,
174.9862635453215, 75.90494777674915] Fitness: 1.8953756731647522
Generation 221 Best Solution: [183.50685622378285, 176.76886172610574,
174.9862635453215, 75.90494777674915] Fitness: 2.150015368383601
Generation 222 Best Solution: [183.39543992096768, 176.48705705818858,
174.36096735260313, 75.90494777674915] Fitness: 1.3579517432303274
Generation 223 Best Solution: [183.39543992096768, 176.48705705818858,
174.36096735260313, 75.90494777674915] Fitness: 1.3579517432303274
Generation 224 Best Solution: [182.98099369961662, 177.40105976142596,
174.36096735260313, 75.90494777674915] Fitness: 2.493961762172723
Generation 225 Best Solution: [182.9223676882511, 177.40105976142596,
174.38377332645993, 75.29602915019382] Fitness: 1.6308494848804755
Generation 226 Best Solution: [183.50685622378285, 177.31987602593645,
174.38377332645993, 75.29602915019382] Fitness: 1.4657410597021212
Generation 227 Best Solution: [182.9223676882511, 177.31987602593645,
174.38377332645993, 75.29602915019382] Fitness: 1.4758605473932536
Generation 228 Best Solution: [182.9223676882511, 175.67064457388105,
174.9862635453215, 75.90494777674915] Fitness: 2.1514348713250024
Generation 229 Best Solution: [182.9223676882511, 176.85563375306586,
174.9778293140483, 75.53940501279187] Fitness: 1.6210620885639737
Generation 230 Best Solution: [183.53479348483603, 176.85563375306586,
174.9778293140483, 75.90494777674915] Fitness: 2.2383268997358505
Generation 231 Best Solution: [182.9223676882511, 176.34424543722048,
174.38377332645993, 75.90494777674915] Fitness: 1.371139037700338
Generation 232 Best Solution: [183.50685622378285, 177.0304745574188,
174.9862635453215, 75.90494777674915] Fitness: 2.448545320177416
Generation 233 Best Solution: [183.50685622378285, 176.45832524856186,
174.9862635453215, 75.90494777674915] Fitness: 1.9437190058359883
Generation 234 Best Solution: [183.53479348483603, 177.40105976142596,
174.9862635453215, 74.74813430662205] Fitness: 1.9013705261133897
Generation 235 Best Solution: [183.53479348483603, 177.40105976142596,
174.9862635453215, 74.81340127113525] Fitness: 1.9085246326848142
Generation 236 Best Solution: [183.50685622378285, 177.40105976142596,
174.9862635453215, 74.63996718367636] Fitness: 1.8913730001870623
Generation 237 Best Solution: [183.53479348483603, 176.36351211077587,
174.9862635453215, 74.63996718367636] Fitness: 0.7512456349662648
Generation 238 Best Solution: [182.9223676882511, 177.40105976142596,
174.9862635453215, 74.63996718367636] Fitness: 1.9014924878781947
Generation 239 Best Solution: [183.2667206905743, 176.2838018611954,
174.9862635453215, 75.90494777674915] Fitness: 1.8335725097135935
Generation 240 Best Solution: [183.53479348483603, 176.73665792840845,
174.2027866799635, 75.90494777674915] Fitness: 1.5414098967411254

Generation 241 Best Solution: [183.53479348483603, 176.73665792840845,
174.9862635453215, 75.38544428146491] Fitness: 1.3295262668394636
Generation 242 Best Solution: [183.53479348483603, 177.40105976142596,
174.9862635453215, 75.38544428146491] Fitness: 2.2750352363414605
Generation 243 Best Solution: [183.6791814391591, 176.1709347934249,
174.9862635453215, 75.90494777674915] Fitness: 2.0016960180312537
Generation 244 Best Solution: [183.6791814391591, 176.1709347934249,
174.9862635453215, 75.90494777674915] Fitness: 2.0016960180312537
Generation 245 Best Solution: [183.6791814391591, 177.40105976142596,
174.2027866799635, 75.07584786228257] Fitness: 1.5073724035376377
Generation 246 Best Solution: [183.6791814391591, 176.1709347934249,
174.2027866799635, 75.90494777674915] Fitness: 1.4081795005652038
Generation 247 Best Solution: [182.88761366555477, 176.26880037295265,
174.2027866799635, 74.98741689309259] Fitness: 0.21379345801941124
Generation 248 Best Solution: [183.53479348483603, 176.29709584122074,
174.9862635453215, 75.90494777674915] Fitness: 1.9137703582691339
Generation 249 Best Solution: [183.53479348483603, 176.1709347934249,
174.9862635453215, 75.90494777674915] Fitness: 1.9097720376731067
Generation 250 Best Solution: [183.53479348483603, 177.40105976142596,
174.11930207339898, 75.90494777674915] Fitness: 2.483990730397073
Generation 251 Best Solution: [183.53479348483603, 177.40105976142596,
174.11930207339898, 75.90494777674915] Fitness: 2.483990730397073
Generation 252 Best Solution: [183.2667206905743, 177.40105976142596,
174.40845141860598, 75.90494777674915] Fitness: 2.4651627076980076
Generation 253 Best Solution: [183.53479348483603, 177.40105976142596,
173.956180104004, 75.90494777674915] Fitness: 2.5117918553957375
Generation 254 Best Solution: [183.53479348483603, 176.28349477009215,
173.956180104004, 75.90494777674915] Fitness: 1.3434200019425622
Generation 255 Best Solution: [183.53479348483603, 177.40105976142596,
173.956180104004, 75.90494777674915] Fitness: 2.5117918553957375
Generation 256 Best Solution: [183.66346456659628, 176.0195695122672,
174.9862635453215, 75.90494777674915] Fitness: 2.0202141260219273
Generation 257 Best Solution: [183.66346456659628, 176.0195695122672,
174.9862635453215, 75.90494777674915] Fitness: 2.0202141260219273
Generation 258 Best Solution: [183.53479348483603, 176.02595643578195,
175.10781824872916, 75.90494777674915] Fitness: 2.121693621985469
Generation 259 Best Solution: [183.53479348483603, 176.02595643578195,
175.10781824872916, 75.90494777674915] Fitness: 2.121693621985469
Generation 260 Best Solution: [183.03090480607702, 176.39716888905002,
174.9862635453215, 75.90494777674915] Fitness: 1.887225956643401
Generation 261 Best Solution: [183.66346456659628, 176.90383005810438,
173.956180104004, 75.90494777674915] Fitness: 1.8151487422189843
Generation 262 Best Solution: [183.28580144611186, 176.02595643578195,
174.9862635453215, 75.90494777674915] Fitness: 1.861039153793271
Generation 263 Best Solution: [183.6791814391591, 176.32067866835592,
173.956180104004, 75.90494777674915] Fitness: 1.4407410218136303
Generation 264 Best Solution: [183.66346456659628, 177.12693582465877,
173.956180104004, 75.02878967120921] Fitness: 1.0148390303150852

Generation 265 Best Solution: [183.03090480607702, 177.12693582465877,
174.70188013839456, 75.02878967120921] Fitness: 1.1209567835666194
Generation 266 Best Solution: [183.03090480607702, 177.40105976142596,
173.956180104004, 74.47169724812528] Fitness: 1.3325597376763119
Generation 267 Best Solution: [183.03090480607702, 177.12693582465877,
173.956180104004, 74.5541618168755] Fitness: 0.8255279732623632
Generation 268 Best Solution: [183.03090480607702, 177.40105976142596,
173.956180104004, 74.62586153437188] Fitness: 1.2898509351635834
Generation 269 Best Solution: [183.03090480607702, 176.02595643578195,
174.9862635453215, 74.47169724812528] Fitness: 0.7587030255718126
Generation 270 Best Solution: [183.42405204443767, 176.32983348120408,
174.8080543345926, 75.90494777674915] Fitness: 1.647296458465423
Generation 271 Best Solution: [183.03090480607702, 177.40105976142596,
174.8080543345926, 74.63587143612727] Fitness: 1.6322030213052685
Generation 272 Best Solution: [183.53479348483603, 177.12693582465877,
174.8080543345926, 74.63587143612727] Fitness: 1.2015265626095528
Generation 273 Best Solution: [183.8524765661301, 176.32983348120408,
174.9862635453215, 74.77051398590154] Fitness: 0.9898603001822056
Generation 274 Best Solution: [183.6791814391591, 177.40105976142596,
174.8080543345926, 75.11941052889776] Fitness: 1.9037816726681376
Generation 275 Best Solution: [183.07645315282343, 177.40105976142596,
174.03144798422392, 74.84172718836736] Fitness: 1.2652685365670278
Generation 276 Best Solution: [183.53479348483603, 177.12693582465877,
174.22791313434416, 74.8915513699572] Fitness: 0.8568108808057561
Generation 277 Best Solution: [183.53479348483603, 176.32080431029848,
174.9862635453215, 74.84172718836736] Fitness: 0.750886103113493
Generation 278 Best Solution: [183.6791814391591, 176.32983348120408,
174.50819440359027, 75.90494777674915] Fitness: 1.527237781922946
Generation 279 Best Solution: [183.53479348483603, 175.77667907399504,
174.26466784915567, 75.90494777674915] Fitness: 1.4844083511867243
Generation 280 Best Solution: [183.66346456659628, 176.32983348120408,
174.8080543345926, 75.54146765406153] Fitness: 1.1637152347865602
Generation 281 Best Solution: [183.8524765661301, 176.32983348120408,
174.8080543345926, 74.57585003905639] Fitness: 0.7785398958720436
Generation 282 Best Solution: [183.8524765661301, 177.40105976142596,
174.8080543345926, 74.57585003905639] Fitness: 1.9398324202623267
Generation 283 Best Solution: [183.53479348483603, 177.38539540898654,
174.26466784915567, 75.90494777674915] Fitness: 2.4658251671891525
Generation 284 Best Solution: [183.53479348483603, 177.38539540898654,
174.26466784915567, 75.90494777674915] Fitness: 2.4658251671891525
Generation 285 Best Solution: [183.33719647135752, 177.40105976142596,
174.26466784915567, 75.90494777674915] Fitness: 2.427098824885193
Generation 286 Best Solution: [183.8524765661301, 175.88176067405135,
175.10781824872916, 75.90494777674915] Fitness: 2.4325818028038135
Generation 287 Best Solution: [183.8524765661301, 175.88176067405135,
175.10781824872916, 74.69663300353834] Fitness: 1.2528831654444463
Generation 288 Best Solution: [183.8524765661301, 177.38539540898654,
174.9862635453215, 74.85603071804472] Fitness: 2.134399088326246

Generation 289 Best Solution: [183.53479348483603, 177.38539540898654,
175.10781824872916, 74.85603071804472] Fitness: 2.0700285355241723
Generation 290 Best Solution: [183.8524765661301, 176.21485525278848,
174.5814871795392, 75.90494777674915] Fitness: 1.7219070859016057
Generation 291 Best Solution: [183.8524765661301, 176.21485525278848,
174.5814871795392, 75.90494777674915] Fitness: 1.7219070859016057
Generation 292 Best Solution: [183.78632542623515, 177.38539540898654,
173.96095498450526, 74.83746619366585] Fitness: 1.494701444732798
Generation 293 Best Solution: [183.0926212968959, 175.7878045287723,
173.96095498450526, 75.90494777674915] Fitness: 1.424773267197286
Generation 294 Best Solution: [183.6010783075034, 177.40105976142596,
173.77679790374717, 75.28386405867151] Fitness: 1.7211414425159142
Generation 295 Best Solution: [182.88685171790726, 177.40105976142596,
174.02024389203424, 75.11637490309084] Fitness: 1.4651669638186227
Generation 296 Best Solution: [183.8524765661301, 175.74417981736886,
175.10781824872916, 75.28386405867151] Fitness: 1.614459971030544
Generation 297 Best Solution: [183.8524765661301, 176.41285276006678,
175.10781824872916, 75.90494777674915] Fitness: 2.372662858059922
Generation 298 Best Solution: [183.53479348483603, 177.40105976142596,
174.0492079869909, 75.17528731136713] Fitness: 1.487076966293727
Generation 299 Best Solution: [183.8524765661301, 177.40105976142596,
174.02024389203424, 74.87560701258145] Fitness: 1.5852297805013795
Generation 300 Best Solution: [183.3458780509345, 177.40105976142596,
174.02024389203424, 74.87560701258145] Fitness: 1.2693003322974643
Generation 301 Best Solution: [183.53479348483603, 177.38539540898654,
174.02024389203424, 74.87560701258145] Fitness: 1.306340462090888
Generation 302 Best Solution: [183.53479348483603, 177.33818127070708,
173.77679790374717, 75.36190798358611] Fitness: 1.6413240600532726
Generation 303 Best Solution: [183.53479348483603, 176.30350773566863,
175.10781824872916, 75.36190798358611] Fitness: 1.2672022702672252
Generation 304 Best Solution: [183.8524765661301, 176.2773806428475,
175.10781824872916, 75.90494777674915] Fitness: 2.343284216339942
Generation 305 Best Solution: [183.8524765661301, 176.2773806428475,
175.10781824872916, 75.90494777674915] Fitness: 2.343284216339942
Generation 306 Best Solution: [183.53479348483603, 177.40105976142596,
175.10781824872916, 74.35609925010641] Fitness: 2.1915555598066656
Generation 307 Best Solution: [183.8524765661301, 176.68207714460783,
175.10781824872916, 74.35609925010641] Fitness: 1.4492040090873166
Generation 308 Best Solution: [183.8524765661301, 176.68207714460783,
175.10781824872916, 75.01525640339129] Fitness: 1.416990808673398
Generation 309 Best Solution: [183.8524765661301, 176.49828730376663,
175.10781824872916, 75.90494777674915] Fitness: 2.4069178429010236
Generation 310 Best Solution: [183.8524765661301, 176.49828730376663,
174.522641207659, 75.90494777674915] Fitness: 1.7483696189223283
Generation 311 Best Solution: [183.8524765661301, 176.49828730376663,
174.47209714863774, 75.90494777674915] Fitness: 1.7182648042011421
Generation 312 Best Solution: [183.20959273110495, 177.40105976142596,
175.10781824872916, 75.14966823151833] Fitness: 2.161882313060874

Generation 313 Best Solution: [183.613572049145, 176.49828730376663,
174.76069624724946, 75.90494777674915] Fitness: 1.7451287304932843
Generation 314 Best Solution: [183.20959273110495, 177.18619811484982,
174.42854364661298, 75.90494777674915] Fitness: 2.0869760913446793
Generation 315 Best Solution: [183.20959273110495, 176.90985238915493,
175.10781824872916, 75.90494777674915] Fitness: 2.414483417177136
Generation 316 Best Solution: [183.20959273110495, 176.51842488159426,
175.10781824872916, 75.90494777674915] Fitness: 2.0888557638906113
Generation 317 Best Solution: [183.8524765661301, 177.40105976142596,
174.402240795447, 75.17721073939352] Fitness: 1.7871238342696565
Generation 318 Best Solution: [183.20959273110495, 176.90985238915493,
175.10781824872916, 75.17721073939352] Fitness: 1.4125359640275035
Generation 319 Best Solution: [183.8524765661301, 177.40105976142596,
175.10781824872916, 74.79281054516153] Fitness: 2.3373885351897674
Generation 320 Best Solution: [183.8524765661301, 177.38539540898654,
174.61468596757115, 74.90276654632352] Fitness: 1.7381752484553377
Generation 321 Best Solution: [183.81816434850626, 177.40105976142596,
175.10781824872916, 74.90276654632352] Fitness: 2.3268209504200748
Generation 322 Best Solution: [182.9491391781154, 176.04803730774566,
175.10781824872916, 75.90494777674915] Fitness: 2.0985821730159264
Generation 323 Best Solution: [182.9491391781154, 176.04803730774566,
175.10781824872916, 75.90494777674915] Fitness: 2.0985821730159264
Generation 324 Best Solution: [182.9491391781154, 176.04803730774566,
175.10781824872916, 75.90494777674915] Fitness: 2.0985821730159264
Generation 325 Best Solution: [182.9491391781154, 176.04803730774566,
175.10781824872916, 75.90494777674915] Fitness: 2.0985821730159264
Generation 326 Best Solution: [183.81816434850626, 177.40105976142596,
174.7425418291192, 74.96360282637121] Fitness: 1.8709306026089358
Generation 327 Best Solution: [183.81816434850626, 177.40105976142596,
174.7425418291192, 74.96360282637121] Fitness: 1.8709306026089358
Generation 328 Best Solution: [183.8524765661301, 177.40105976142596,
174.7425418291192, 74.96360282637121] Fitness: 1.9058330165285693
Generation 329 Best Solution: [182.9491391781154, 177.38539540898654,
173.97788070099048, 75.90494777674915] Fitness: 2.4582149874410124
Generation 330 Best Solution: [182.9491391781154, 177.38539540898654,
173.97788070099048, 75.90494777674915] Fitness: 2.4582149874410124
Generation 331 Best Solution: [183.8524765661301, 177.38539540898654,
174.20556327382818, 75.90494777674915] Fitness: 2.7045845905704295
Generation 332 Best Solution: [183.8524765661301, 177.38539540898654,
174.20556327382818, 75.64363329102684] Fitness: 2.2432379887480067
Generation 333 Best Solution: [183.8524765661301, 176.95260821906504,
175.10781824872916, 74.36641841398918] Fitness: 1.714707508794761
Generation 334 Best Solution: [183.13734812003136, 177.40105976142596,
175.10781824872916, 74.34866148604027] Fitness: 2.1224763992660174
Generation 335 Best Solution: [182.9491391781154, 177.40105976142596,
175.10781824872916, 74.34866148604027] Fitness: 2.179490016756697
Generation 336 Best Solution: [182.9491391781154, 177.40105976142596,
175.10781824872916, 74.34866148604027] Fitness: 2.179490016756697

Generation 337 Best Solution: [183.83302075921418, 177.40105976142596,
175.10781824872916, 74.87336537842214] Fitness: 2.3332114641376225
Generation 338 Best Solution: [183.13734812003136, 177.40105976142596,
175.10781824872916, 74.34866148604027] Fitness: 2.1224763992660174
Generation 339 Best Solution: [183.13734812003136, 177.40105976142596,
175.10781824872916, 75.27604949591186] Fitness: 2.272953590976634
Generation 340 Best Solution: [183.83302075921418, 177.38539540898654,
175.10781824872916, 75.67205554481497] Fitness: 3.0448476128720126
Generation 341 Best Solution: [183.13734812003136, 176.75102277229385,
175.10781824872916, 75.90494777674915] Fitness: 2.257771782035883
Generation 342 Best Solution: [183.83302075921418, 175.89391166127342,
175.10781824872916, 75.90494777674915] Fitness: 2.4059248657208605
Generation 343 Best Solution: [183.48664139553046, 176.26238013865571,
175.10781824872916, 75.90494777674915] Fitness: 2.07085330946467
Generation 344 Best Solution: [183.83302075921418, 175.89391166127342,
175.10781824872916, 75.50625381849459] Fitness: 1.7476806106308644
Generation 345 Best Solution: [183.83302075921418, 176.26645894669363,
175.10781824872916, 75.90494777674915] Fitness: 2.3222167839947114
Generation 346 Best Solution: [183.83302075921418, 176.10484200482753,
175.10781824872916, 75.2895565586025] Fitness: 1.4252386909517853
Generation 347 Best Solution: [183.78947196317745, 176.10484200482753,
175.10781824872916, 75.2895565586025] Fitness: 1.3826881522667978
Generation 348 Best Solution: [183.21144763134225, 177.38539540898654,
175.10781824872916, 74.52578137579414] Fitness: 2.003466608660445
Generation 349 Best Solution: [183.78947196317745, 177.38539540898654,
175.10781824872916, 74.52578137579414] Fitness: 2.2688373840914227
Generation 350 Best Solution: [183.21144763134225, 177.38539540898654,
175.10781824872916, 75.12381881453379] Fitness: 2.1129075548024954
Generation 351 Best Solution: [183.78947196317745, 177.38539540898654,
175.10781824872916, 74.583828875177] Fitness: 2.2533391976255652
Generation 352 Best Solution: [183.21144763134225, 177.38539540898654,
175.10781824872916, 74.52578137579414] Fitness: 2.003466608660445
Generation 353 Best Solution: [183.78947196317745, 176.54719971047302,
175.10781824872916, 75.90494777674915] Fitness: 2.3694234323460077
Generation 354 Best Solution: [183.78947196317745, 177.02520328984176,
174.9148113239755, 75.90494777674915] Fitness: 2.5442010560289474
Generation 355 Best Solution: [183.78947196317745, 177.02520328984176,
174.9148113239755, 74.51849507119731] Fitness: 1.3963997857401866
Generation 356 Best Solution: [183.78947196317745, 177.02520328984176,
174.9148113239755, 74.51849507119731] Fitness: 1.3963997857401866
Generation 357 Best Solution: [183.78947196317745, 177.02520328984176,
175.10781824872916, 74.49395416072899] Fitness: 1.6852219947633413
Generation 358 Best Solution: [183.78947196317745, 176.9962321017699,
174.03426139222907, 74.49395416072899] Fitness: 0.875557591901305
Generation 359 Best Solution: [182.88349397194224, 177.38539540898654,
175.10781824872916, 74.49395416072899] Fitness: 2.111383505173955
Generation 360 Best Solution: [183.78947196317745, 177.02520328984176,
175.10781824872916, 74.49395416072899] Fitness: 1.6852219947633413

Generation 361 Best Solution: [183.6985690748774, 175.59484073712258,
174.68633139491234, 75.90494777674915] Fitness: 1.987494903725044
Generation 362 Best Solution: [183.78947196317745, 175.59484073712258,
174.68633139491234, 75.90494777674915] Fitness: 2.066129010836007
Generation 363 Best Solution: [183.78947196317745, 175.819239104325,
174.9148113239755, 75.90494777674915] Fitness: 2.1276988190915214
Generation 364 Best Solution: [183.78947196317745, 177.40105976142596,
174.9148113239755, 74.47048692532641] Fitness: 2.0393004886118264
Generation 365 Best Solution: [183.62823362453477, 176.48286317303038,
174.9148113239755, 75.90494777674915] Fitness: 1.9269427788102391
Generation 366 Best Solution: [183.78947196317745, 177.02520328984176,
175.10781824872916, 74.9181191407316] Fitness: 1.6788854299904443
Generation 367 Best Solution: [182.88349397194224, 177.40105976142596,
175.10781824872916, 74.9181191407316] Fitness: 2.135807227315245
Generation 368 Best Solution: [183.78947196317745, 177.02520328984176,
174.29041465846814, 75.90494777674915] Fitness: 2.0627777074634652
Generation 369 Best Solution: [183.62823362453477, 177.40105976142596,
174.9148113239755, 75.02019057097138] Fitness: 1.9370699883331224
Generation 370 Best Solution: [183.62823362453477, 177.02520328984176,
174.9148113239755, 75.02019057097138] Fitness: 1.3118129993302992
Generation 371 Best Solution: [183.78947196317745, 177.40105976142596,
174.9148113239755, 75.02019057097138] Fitness: 2.0670960227147512
Generation 372 Best Solution: [183.62823362453477, 177.38539540898654,
174.9148113239755, 75.13122879715519] Fitness: 1.9730639738951177
Generation 373 Best Solution: [183.78947196317745, 177.40105976142596,
174.9148113239755, 75.13122879715519] Fitness: 2.1338502951909333
Generation 374 Best Solution: [183.78947196317745, 177.40105976142596,
174.9148113239755, 75.13122879715519] Fitness: 2.1338502951909333
Generation 375 Best Solution: [183.62823362453477, 177.38539540898654,
174.9148113239755, 75.13122879715519] Fitness: 1.9730639738951177
Generation 376 Best Solution: [183.36114792784434, 176.82243492382472,
175.10781824872916, 75.90494777674915] Fitness: 2.3348635990847386
Generation 377 Best Solution: [183.36114792784434, 176.62688320350622,
175.10781824872916, 75.90494777674915] Fitness: 2.16875575462525
Generation 378 Best Solution: [183.36114792784434, 176.62688320350622,
175.10781824872916, 75.82249435744582] Fitness: 2.0108957882786447
Generation 379 Best Solution: [183.78947196317745, 176.62688320350622,
175.10781824872916, 75.20779249873112] Fitness: 1.4412233166023363
Generation 380 Best Solution: [183.78947196317745, 176.43965842347328,
175.10781824872916, 75.90494777674915] Fitness: 2.3195190653452893
Generation 381 Best Solution: [183.35552040119993, 177.252099823964,
175.10781824872916, 74.86075449412981] Fitness: 1.7601692959397797
Generation 382 Best Solution: [183.78947196317745, 175.9239211647375,
174.73986388708673, 74.49099310505169] Fitness: 0.729205214638643
Generation 383 Best Solution: [183.78947196317745, 175.9239211647375,
174.73986388708673, 75.90494777674915] Fitness: 1.8673690666435425
Generation 384 Best Solution: [183.78947196317745, 175.9239211647375,
175.10781824872916, 75.05487479182844] Fitness: 1.264347935203996

Generation 385 Best Solution: [183.78947196317745, 175.9239211647375,
174.73986388708673, 75.90494777674915] Fitness: 1.8673690666435425
Generation 386 Best Solution: [183.35552040119993, 177.40105976142596,
175.10781824872916, 74.44227266038457] Fitness: 2.0804172476250518
Generation 387 Best Solution: [183.35552040119993, 177.40105976142596,
175.10781824872916, 75.21132575495488] Fitness: 2.223716458673247
Generation 388 Best Solution: [183.35552040119993, 177.40105976142596,
175.10781824872916, 75.21132575495488] Fitness: 2.223716458673247
Generation 389 Best Solution: [183.78947196317745, 177.07201161179395,
175.10781824872916, 75.21132575495488] Fitness: 1.9148199313064942
Generation 390 Best Solution: [183.78947196317745, 177.40105976142596,
174.403929729636, 75.54232963419983] Fitness: 2.1176454839151297
Generation 391 Best Solution: [183.250219543767, 177.21544030093025,
175.10781824872916, 74.49287951193182] Fitness: 1.70766791242171
Generation 392 Best Solution: [183.78947196317745, 177.38539540898654,
174.403929729636, 74.36814664999461] Fitness: 1.616745965988213
Generation 393 Best Solution: [183.78947196317745, 177.38539540898654,
175.10781824872916, 74.49287951193182] Fitness: 2.280115570319163
Generation 394 Best Solution: [183.35552040119993, 177.38539540898654,
175.10781824872916, 74.57706953670508] Fitness: 2.0035274698200682
Generation 395 Best Solution: [183.35552040119993, 176.53810700979005,
175.10781824872916, 75.90494777674915] Fitness: 2.113130490723584
Generation 396 Best Solution: [183.35552040119993, 176.1664237932861,
175.10781824872916, 75.90494777674915] Fitness: 2.028098414126288
Generation 397 Best Solution: [183.35552040119993, 176.1664237932861,
175.10781824872916, 75.90494777674915] Fitness: 2.028098414126288
Generation 398 Best Solution: [183.35552040119993, 176.64004638358526,
175.10781824872916, 75.90494777674915] Fitness: 2.1766857310764474
Generation 399 Best Solution: [183.35552040119993, 176.64004638358526,
175.10781824872916, 75.90494777674915] Fitness: 2.1766857310764474
Generation 400 Best Solution: [183.78947196317745, 176.64004638358526,
175.10781824872916, 75.74148339355851] Fitness: 2.126090287697413
Generation 401 Best Solution: [182.91446493449394, 177.40105976142596,
175.10781824872916, 74.71416251468025] Fitness: 2.0845980404478994
Generation 402 Best Solution: [183.58788348479052, 176.57496816747405,
175.10781824872916, 75.90494777674915] Fitness: 2.229653649243156
Generation 403 Best Solution: [183.78947196317745, 176.9825708378067,
175.10781824872916, 75.07423295070363] Fitness: 1.6959908864242403
Generation 404 Best Solution: [182.91446493449394, 177.40105976142596,
174.15046618002572, 74.44069274702197] Fitness: 1.3668323287861024
Generation 405 Best Solution: [183.78947196317745, 177.40105976142596,
174.15046618002572, 74.44069274702197] Fitness: 1.5519961373732323
Generation 406 Best Solution: [183.58788348479052, 177.40105976142596,
174.15046618002572, 74.44069274702197] Fitness: 1.3962093259734907
Generation 407 Best Solution: [183.78947196317745, 177.40105976142596,
174.15046618002572, 74.44069274702197] Fitness: 1.5519961373732323
Generation 408 Best Solution: [183.78947196317745, 176.9825708378067,
175.10781824872916, 74.80390375926225] Fitness: 1.5950371910406744

Generation 409 Best Solution: [183.78947196317745, 176.19563250921576,
175.10781824872916, 75.90494777674915] Fitness: 2.277772142396774
Generation 410 Best Solution: [183.78947196317745, 177.40105976142596,
174.53887136113113, 75.12276035752679] Fitness: 1.760340188990143
Generation 411 Best Solution: [183.78947196317745, 177.40105976142596,
175.10781824872916, 74.72653486350546] Fitness: 2.26987214211372
Generation 412 Best Solution: [182.91446493449394, 175.69560234620005,
175.10781824872916, 75.90494777674915] Fitness: 2.3171073530376614
Generation 413 Best Solution: [183.78947196317745, 176.49971013578585,
175.10781824872916, 75.90494777674915] Fitness: 2.345009443830037
Generation 414 Best Solution: [183.78947196317745, 176.49971013578585,
175.10781824872916, 75.33320961181903] Fitness: 1.4835156211260176
Generation 415 Best Solution: [183.03559053834124, 176.42854459988598,
175.10781824872916, 75.90494777674915] Fitness: 2.0798334285581372
Generation 416 Best Solution: [182.91446493449394, 176.60250505991834,
175.10781824872916, 75.58111293395179] Fitness: 1.6625758127524382
Generation 417 Best Solution: [183.3241636504898, 176.48549778970784,
175.10781824872916, 75.90494777674915] Fitness: 2.081104464815607
Generation 418 Best Solution: [183.3241636504898, 176.2036521576577,
175.10781824872916, 75.90494777674915] Fitness: 2.02023756422311
Generation 419 Best Solution: [182.91446493449394, 177.40105976142596,
174.43099239739706, 75.90494777674915] Fitness: 2.5545808927631612
Generation 420 Best Solution: [182.91446493449394, 177.40105976142596,
174.43099239739706, 75.90494777674915] Fitness: 2.5545808927631612
Generation 421 Best Solution: [183.10819873671025, 177.40105976142596,
174.43099239739706, 75.90494777674915] Fitness: 2.4855896495941847
Generation 422 Best Solution: [183.28684102549616, 176.26175752655305,
175.10781824872916, 75.90494777674915] Fitness: 2.016944543906016
Generation 423 Best Solution: [183.78947196317745, 176.16470135026722,
175.10781824872916, 75.90494777674915] Fitness: 2.2795678544514932
Generation 424 Best Solution: [183.78947196317745, 177.40105976142596,
175.10781824872916, 75.05111261343276] Fitness: 2.363904338640816
Generation 425 Best Solution: [183.78947196317745, 177.40105976142596,
175.10781824872916, 75.05111261343276] Fitness: 2.363904338640816
Generation 426 Best Solution: [182.91446493449394, 177.40105976142596,
174.43099239739706, 75.90494777674915] Fitness: 2.5545808927631612
Generation 427 Best Solution: [183.10819873671025, 177.40105976142596,
174.43099239739706, 75.90494777674915] Fitness: 2.4855896495941847
Generation 428 Best Solution: [183.10819873671025, 176.77720029391858,
175.0991156978956, 75.90494777674915] Fitness: 2.2727211157735354
Generation 429 Best Solution: [183.10819873671025, 176.77720029391858,
175.0991156978956, 75.90494777674915] Fitness: 2.2727211157735354
Generation 430 Best Solution: [183.78947196317745, 177.40105976142596,
174.38410431023445, 75.90494777674915] Fitness: 2.718836650460413
Generation 431 Best Solution: [183.10819873671025, 176.48662121812018,
175.10781824872916, 75.90494777674915] Fitness: 2.084786246002175
Generation 432 Best Solution: [183.10819873671025, 176.48662121812018,
173.88194403255102, 75.90494777674915] Fitness: 1.3597201791378857

Generation 433 Best Solution: [183.78947196317745, 176.48662121812018,
175.10781824872916, 75.48421956098221] Fitness: 1.6520438157644557
Generation 434 Best Solution: [183.83858255858863, 176.17226953114374,
175.01950542657906, 75.90494777674915] Fitness: 2.191241629954271
Generation 435 Best Solution: [183.83858255858863, 176.17226953114374,
175.01950542657906, 75.90494777674915] Fitness: 2.191241629954271
Generation 436 Best Solution: [183.83858255858863, 176.17226953114374,
175.10781824872916, 75.90494777674915] Fitness: 2.3271936026135758
Generation 437 Best Solution: [183.4723927453889, 176.17226953114374,
175.10781824872916, 75.90494777674915] Fitness: 2.064459947835983
Generation 438 Best Solution: [183.10819873671025, 176.17226953114374,
175.10781824872916, 75.90494777674915] Fitness: 2.0248260850318323
Generation 439 Best Solution: [183.16289441263464, 176.17226953114374,
175.10781824872916, 75.90494777674915] Fitness: 2.016671559095095
Generation 440 Best Solution: [183.10819873671025, 176.17226953114374,
175.10781824872916, 74.55449159182348] Fitness: 0.8663154497860799
Generation 441 Best Solution: [182.88759400405817, 177.40105976142596,
175.01950542657906, 74.68769039626578] Fitness: 1.9637759780078439
Generation 442 Best Solution: [183.83858255858863, 177.40105976142596,
175.0991156978956, 74.38867963848823] Fitness: 2.392736845137411
Generation 443 Best Solution: [183.83858255858863, 177.40105976142596,
175.0991156978956, 75.05929920116202] Fitness: 2.402784313123774
Generation 444 Best Solution: [183.83858255858863, 176.79549663302402,
175.10781824872916, 75.16430368577238] Fitness: 1.5947340514818473
Generation 445 Best Solution: [182.9143615268956, 177.40105976142596,
175.10781824872916, 74.82743982137336] Fitness: 2.095186578875063
Generation 446 Best Solution: [183.83858255858863, 177.40105976142596,
175.10781824872916, 74.82743982137336] Fitness: 2.3285093249013435
Generation 447 Best Solution: [182.88759400405817, 177.18526774962527,
175.10781824872916, 75.16430368577238] Fitness: 1.879569397235165
Generation 448 Best Solution: [183.83858255858863, 177.40105976142596,
175.01950542657906, 74.47127475131147] Fitness: 2.2315234217568802
Generation 449 Best Solution: [183.66815868278945, 177.40105976142596,
175.10781824872916, 74.82743982137336] Fitness: 2.1784310431742733
Generation 450 Best Solution: [183.78947196317745, 177.40105976142596,
175.0991156978956, 74.82743982137336] Fitness: 2.2663224918895257
Generation 451 Best Solution: [183.83858255858863, 177.40105976142596,
175.10781824872916, 74.66886666181236] Fitness: 2.319747301119561
Generation 452 Best Solution: [183.83858255858863, 175.6814143035834,
175.10781824872916, 75.52555082015242] Fitness: 1.930447538539581
Generation 453 Best Solution: [183.83858255858863, 177.40105976142596,
174.18961544792003, 75.90494777674915] Fitness: 2.719444764950441
Generation 454 Best Solution: [182.88759400405817, 177.40105976142596,
174.18961544792003, 75.90494777674915] Fitness: 2.5005774731948747
Generation 455 Best Solution: [183.83858255858863, 177.40105976142596,
174.18961544792003, 75.90494777674915] Fitness: 2.719444764950441
Generation 456 Best Solution: [183.83858255858863, 177.40105976142596,
175.10781824872916, 74.404436904056] Fitness: 2.398348260844549

Generation 457 Best Solution: [182.88759400405817, 177.40105976142596,
174.18961544792003, 74.404436904056] Fitness: 1.4009722441715895
Generation 458 Best Solution: [183.83858255858863, 177.40105976142596,
175.0991156978956, 74.7281162944112] Fitness: 2.3041427275056687
Generation 459 Best Solution: [182.88759400405817, 177.40105976142596,
175.0991156978956, 74.7281162944112] Fitness: 2.085275435750102
Generation 460 Best Solution: [182.88759400405817, 177.40105976142596,
175.0991156978956, 74.7281162944112] Fitness: 2.085275435750102
Generation 461 Best Solution: [183.74514674461415, 177.40105976142596,
175.0991156978956, 74.7281162944112] Fitness: 2.2158669606127748
Generation 462 Best Solution: [182.88759400405817, 177.40105976142596,
174.20109292416254, 75.90494777674915] Fitness: 2.5016363504535617
Generation 463 Best Solution: [183.42340855180083, 177.40105976142596,
174.20109292416254, 75.90494777674915] Fitness: 2.4395723065620727
Generation 464 Best Solution: [183.6367545129881, 177.40105976142596,
174.20109292416254, 75.26561012259829] Fitness: 1.620715324275242
Generation 465 Best Solution: [183.83858255858863, 177.40105976142596,
175.0991156978956, 74.53517873744832] Fitness: 2.3309457685778336
Generation 466 Best Solution: [183.83858255858863, 177.40105976142596,
175.10781824872916, 75.26561012259829] Fitness: 2.570616652965705
Generation 467 Best Solution: [183.83858255858863, 176.93546107681897,
174.9481457687116, 75.90494777674915] Fitness: 2.52191501191005
Generation 468 Best Solution: [183.83858255858863, 176.63438132341648,
175.10781824872916, 75.90494777674915] Fitness: 2.4722392704984735
Generation 469 Best Solution: [183.83858255858863, 175.74263159589643,
174.9481457687116, 75.90494777674915] Fitness: 2.2753048002302325
Generation 470 Best Solution: [183.83858255858863, 175.74263159589643,
174.9481457687116, 75.90494777674915] Fitness: 2.2753048002302325
Generation 471 Best Solution: [183.83858255858863, 176.2051192842445,
174.9481457687116, 75.90494777674915] Fitness: 2.0894428996873007
Generation 472 Best Solution: [182.98484131157267, 175.74263159589643,
175.10781824872916, 75.90494777674915] Fitness: 2.245942482321614
Generation 473 Best Solution: [183.83858255858863, 177.40105976142596,
174.23759192769364, 74.38892683104989] Fitness: 1.6339535081150742
Generation 474 Best Solution: [182.98484131157267, 175.74263159589643,
175.10781824872916, 75.90494777674915] Fitness: 2.245942482321614
Generation 475 Best Solution: [183.83858255858863, 177.40105976142596,
174.9481457687116, 75.04812228838006] Fitness: 2.174138630316676
Generation 476 Best Solution: [183.83858255858863, 177.40105976142596,
174.9481457687116, 75.04812228838006] Fitness: 2.174138630316676
Generation 477 Best Solution: [183.83858255858863, 177.40105976142596,
174.9481457687116, 74.92477711496187] Fitness: 2.118335279438475
Generation 478 Best Solution: [183.34487765470746, 177.40105976142596,
174.9481457687116, 74.92477711496187] Fitness: 1.8165745260167996
Generation 479 Best Solution: [182.98484131157267, 177.40105976142596,
174.9481457687116, 75.12900406125827] Fitness: 1.958823249604792
Generation 480 Best Solution: [182.9515056096969, 177.17136496422484,
175.10781824872916, 75.12900406125827] Fitness: 1.7993163763629554

```
Generation 481 Best Solution: [183.83858255858863, 177.17136496422484,
174.9481457687116, 74.92477711496187] Fitness: 1.7082487687076235
Generation 482 Best Solution: [183.83858255858863, 177.30084599403622,
174.9481457687116, 75.03168443619525] Fitness: 1.9755072454862228
Generation 483 Best Solution: [183.83858255858863, 177.30084599403622,
174.89571747808128, 75.03168443619525] Fitness: 1.9071816814787634
Generation 484 Best Solution: [183.34487765470746, 177.30084599403622,
174.89571747808128, 75.03168443619525] Fitness: 1.6054209280570884
Generation 485 Best Solution: [183.83858255858863, 177.30084599403622,
174.89571747808128, 75.03168443619525] Fitness: 1.9071816814787634
Generation 486 Best Solution: [183.83858255858863, 176.97491645463785,
175.0991156978956, 75.03168443619525] Fitness: 1.696519606926328
Generation 487 Best Solution: [183.34487765470746, 177.40105976142596,
174.89571747808128, 75.03168443619525] Fitness: 1.795151123814822
Generation 488 Best Solution: [182.91730194702245, 176.73492332060283,
175.0991156978956, 75.90494777674915] Fitness: 2.302126653710723
Generation 489 Best Solution: [182.91730194702245, 176.62389707626676,
175.0991156978956, 75.90494777674915] Fitness: 2.21619045343808
Generation 490 Best Solution: [183.83858255858863, 177.30084599403622,
175.0991156978956, 74.8113214596147] Fitness: 2.122000687083957
Generation 491 Best Solution: [183.83858255858863, 175.6756440274539,
175.0991156978956, 74.8113214596147] Fitness: 1.3818701390478307
Generation 492 Best Solution: [183.07101077239875, 175.62832696634757,
174.89571747808128, 75.90494777674915] Fitness: 2.013881290583308
Generation 493 Best Solution: [182.85049070016584, 175.62832696634757,
174.6102551535153, 75.90494777674915] Fitness: 1.8193609040626606
Generation 494 Best Solution: [183.83858255858863, 175.62832696634757,
175.0991156978956, 74.97247203959682] Fitness: 1.4737821481191933
Generation 495 Best Solution: [183.83858255858863, 176.81830931526966,
174.01429692774767, 75.90494777674915] Fitness: 1.8615980498827116
Generation 496 Best Solution: [182.85049070016584, 176.81830931526966,
174.01429692774767, 75.90494777674915] Fitness: 1.6647427660977947
Generation 497 Best Solution: [182.85049070016584, 176.81830931526966,
174.01429692774767, 75.90494777674915] Fitness: 1.6647427660977947
Generation 498 Best Solution: [182.85049070016584, 176.81830931526966,
174.01429692774767, 75.90494777674915] Fitness: 1.6647427660977947
Generation 499 Best Solution: [183.83858255858863, 177.40105976142596,
175.0991156978956, 74.56618520429214] Fitness: 2.3224542327077438
Generation 500 Best Solution: [183.83858255858863, 177.40105976142596,
174.01429692774767, 75.00738830369649] Fitness: 1.6218237469917904
Optimal Solution for Simulation 27: [183.83858255858863, 177.40105976142596,
174.01429692774767, 75.00738830369649]
--------------------------------------------------
Simulation 28/32 with conditions: {'drought': False, 'flood': False,
'policy_change': True, 'social_event': False, 'economic_crisis': False}
Generation 1 Best Solution: [183.28822286673307, 176.13776173134832,
174.55663297084712, 74.61203054338179] Fitness: 0.2337110213719353
Generation 2 Best Solution: [183.15928500184515, 176.33042702957502,
```

173.9181109964828, 74.58060872527703] Fitness: 0.12834681580750862
Generation 3 Best Solution: [183.67486069898015, 176.86827036979204,
174.4200553058884, 74.58023752921865] Fitness: 0.7074908839665451
Generation 4 Best Solution: [183.83517610143255, 176.86827036979204,
174.4200553058884, 74.58023752921865] Fitness: 0.8518411710643565
Generation 5 Best Solution: [183.18199599322563, 176.86827036979204,
174.4200553058884, 74.58023752921865] Fitness: 0.5237283534561094
Generation 6 Best Solution: [183.61368588820508, 176.86827036979204,
174.4200553058884, 74.58023752921865] Fitness: 0.6633521325384197
Generation 7 Best Solution: [183.61368588820508, 177.17367358241182,
174.1247464344868, 74.44587730122933] Fitness: 1.0496087386094164
Generation 8 Best Solution: [183.6510649491312, 176.56074321680015,
175.08693076600966, 74.44587730122933] Fitness: 1.1595384700618498
Generation 9 Best Solution: [183.61368588820508, 177.17367358241182,
174.5460814902415, 74.95758961762745] Fitness: 1.167347179409918
Generation 10 Best Solution: [183.14007924495493, 177.17367358241182,
173.73728796458497, 74.87251370359414] Fitness: 0.9773535126862927
Generation 11 Best Solution: [182.8526248696446, 177.29967958538384,
174.16906457431912, 75.1111114312962] Fitness: 1.2660377064245607
Generation 12 Best Solution: [183.61368588820508, 177.29967958538384,
174.16906457431912, 75.1111114312962] Fitness: 1.3097262946160437
Generation 13 Best Solution: [183.50381432882776, 177.17367358241182,
175.08693076600966, 74.52108402376665] Fitness: 1.7098922997843327
Generation 14 Best Solution: [183.6510649491312, 176.2796103277167,
175.1390612937386, 75.1111114312962] Fitness: 1.1831434616334229
Generation 15 Best Solution: [183.61368588820508, 176.2796103277167,
175.1390612937386, 75.1111114312962] Fitness: 1.1568921937479766
Generation 16 Best Solution: [183.43024897294012, 177.29967958538384,
175.08693076600966, 74.58023752921865] Fitness: 1.8733453215984683
Generation 17 Best Solution: [183.43024897294012, 176.65600548690017,
175.08693076600966, 75.31684225551082] Fitness: 1.305279106811505
Generation 18 Best Solution: [183.43024897294012, 176.65600548690017,
174.09239549914295, 75.31684225551082] Fitness: 0.5507385278285211
Generation 19 Best Solution: [182.87318961722949, 176.65600548690017,
174.09239549914295, 75.31684225551082] Fitness: 0.5920421138326768
Generation 20 Best Solution: [183.516503338081, 175.88352902239555,
175.08693076600966, 75.31684225551082] Fitness: 1.242975718084093
Generation 21 Best Solution: [183.516503338081, 175.88352902239555,
175.08693076600966, 75.31684225551082] Fitness: 1.242975718084093
Generation 22 Best Solution: [183.516503338081, 175.88352902239555,
175.08693076600966, 75.31684225551082] Fitness: 1.242975718084093
Generation 23 Best Solution: [183.6510649491312, 175.88352902239555,
175.08693076600966, 75.31684225551082] Fitness: 1.326915944729854
Generation 24 Best Solution: [183.52008854459217, 176.50379687583546,
175.08693076600966, 75.31684225551082] Fitness: 1.2494573418526027
Generation 25 Best Solution: [183.516503338081, 176.6132590796547,
175.08693076600966, 75.31684225551082] Fitness: 1.3129329780307513
Generation 26 Best Solution: [183.516503338081, 176.6132590796547,

175.08693076600966, 75.24773207905926] Fitness: 1.2520201695276891
Generation 27 Best Solution: [183.61368588820508, 176.67586512382633,
175.08693076600966, 74.85695925290985] Fitness: 1.156504734995775
Generation 28 Best Solution: [183.52008854459217, 175.77758454662515,
175.08693076600966, 74.85695925290985] Fitness: 1.0460007461756131
Generation 29 Best Solution: [183.55106359148328, 177.29967958538384,
174.02402561381308, 75.31684225551082] Fitness: 1.4349266891136985
Generation 30 Best Solution: [183.55106359148328, 177.29967958538384,
174.02402561381308, 75.31684225551082] Fitness: 1.4349266891136985
Generation 31 Best Solution: [183.09278763557523, 177.26199242510714,
174.3182019221468, 75.31684225551082] Fitness: 1.3007259392982355
Generation 32 Best Solution: [183.516503338081, 177.29967958538384,
174.2105228484677, 75.31684225551082] Fitness: 1.415338089749942
Generation 33 Best Solution: [183.55106359148328, 175.6040049678816,
174.2105228484677, 75.31684225551082] Fitness: 0.7142671386849115
Generation 34 Best Solution: [183.516503338081, 175.6040049678816,
174.2105228484677, 75.31684225551082] Fitness: 0.6954997174324737
Generation 35 Best Solution: [183.516503338081, 176.18087414105187,
174.23512697211265, 75.31684225551082] Fitness: 0.42318765778317363
Generation 36 Best Solution: [183.55106359148328, 176.18087414105187,
175.08693076600966, 75.31684225551082] Fitness: 1.1864057950937696
Generation 37 Best Solution: [183.516503338081, 176.18087414105187,
175.08693076600966, 75.30403292283766] Fitness: 1.1557659065246961
Generation 38 Best Solution: [183.61368588820508, 176.20945441398896,
175.08693076600966, 75.04561904284513] Fitness: 1.0308329963035947
Generation 39 Best Solution: [183.55106359148328, 177.29967958538384,
174.03266835971002, 75.31684225551082] Fitness: 1.433647088514754
Generation 40 Best Solution: [183.516503338081, 176.510874275872,
175.08693076600966, 75.31684225551082] Fitness: 1.2512390437376537
Generation 41 Best Solution: [183.516503338081, 176.510874275872,
175.08693076600966, 74.4578726922725] Fitness: 1.042145570338591
Generation 42 Best Solution: [183.55106359148328, 177.29967958538384,
173.76066678079758, 75.31684225551082] Fitness: 1.5317764918824224
Generation 43 Best Solution: [183.55106359148328, 176.85615306434283,
174.27368474131774, 75.31684225551082] Fitness: 0.809719718940497
Generation 44 Best Solution: [183.55106359148328, 176.68556218536378,
175.08693076600966, 75.31684225551082] Fitness: 1.3854695676035027
Generation 45 Best Solution: [183.55106359148328, 177.2408041342181,
174.3751699545631, 75.31684225551082] Fitness: 1.3772485942957418
Generation 46 Best Solution: [183.2481610213578, 177.29967958538384,
174.19263323472686, 75.31684225551082] Fitness: 1.3329118790984547
Generation 47 Best Solution: [183.55106359148328, 177.29967958538384,
173.79326640174313, 75.31684225551082] Fitness: 1.513712034880795
Generation 48 Best Solution: [183.55106359148328, 177.29967958538384,
173.79326640174313, 75.31684225551082] Fitness: 1.513712034880795
Generation 49 Best Solution: [183.55106359148328, 177.29967958538384,
173.79326640174313, 75.31935522256568] Fitness: 1.5160723050725355
Generation 50 Best Solution: [183.46965807466844, 177.26199242510714,

173.79326640174313, 75.31935522256568] Fitness: 1.4084791110690413
Generation 51 Best Solution: [183.55106359148328, 177.29967958538384,
174.3110520414778, 75.31684225551082] Fitness: 1.4569684504493339
Generation 52 Best Solution: [183.55106359148328, 177.29967958538384,
174.3110520414778, 75.31684225551082] Fitness: 1.4569684504493339
Generation 53 Best Solution: [183.46965807466844, 177.26199242510714,
175.08693076600966, 75.31684225551082] Fitness: 2.0750498217445665
Generation 54 Best Solution: [183.55106359148328, 176.97050248653056,
175.08693076600966, 75.31684225551082] Fitness: 1.6795866792312668
Generation 55 Best Solution: [183.55106359148328, 175.97811635653574,
175.08693076600966, 75.31684225551082] Fitness: 1.222287680747944
Generation 56 Best Solution: [183.55106359148328, 177.29967958538384,
174.89304522409432, 75.31684225551082] Fitness: 1.9101630179411129
Generation 57 Best Solution: [183.55106359148328, 177.2408041342181,
175.08693076600966, 75.31684225551082] Fitness: 2.0798144178122864
Generation 58 Best Solution: [183.55106359148328, 176.3307310561251,
175.08693076600966, 75.43403707146614] Fitness: 1.3234947305622389
Generation 59 Best Solution: [183.55106359148328, 176.3307310561251,
175.08693076600966, 75.31684225551082] Fitness: 1.2025656105780147
Generation 60 Best Solution: [183.55106359148328, 175.7100951109894,
175.08693076600966, 75.31684225551082] Fitness: 1.3716328213971045
Generation 61 Best Solution: [183.55106359148328, 175.7100951109894,
175.08693076600966, 75.31684225551082] Fitness: 1.3716328213971045
Generation 62 Best Solution: [183.55106359148328, 175.94672375050217,
175.08693076600966, 75.31684225551082] Fitness: 1.2337802213249227
Generation 63 Best Solution: [183.55106359148328, 176.5775728439221,
175.08693076600966, 75.31684225551082] Fitness: 1.308274484980136
Generation 64 Best Solution: [183.55106359148328, 176.5775728439221,
175.08693076600966, 75.31684225551082] Fitness: 1.308274484980136
Generation 65 Best Solution: [183.55106359148328, 176.5775728439221,
175.08693076600966, 75.31684225551082] Fitness: 1.308274484980136
Generation 66 Best Solution: [183.55106359148328, 176.5775728439221,
175.08693076600966, 74.92392449074256] Fitness: 1.0647243832311553
Generation 67 Best Solution: [182.88863238708814, 176.5775728439221,
175.08693076600966, 74.92392449074256] Fitness: 1.0409376111839022
Generation 68 Best Solution: [182.88863238708814, 176.55255336194546,
175.08693076600966, 74.92392449074256] Fitness: 1.0257405369483183
Generation 69 Best Solution: [183.55106359148328, 175.99863186762886,
175.08693076600966, 74.95291988691359] Fitness: 0.9815371402656485
Generation 70 Best Solution: [182.9543068818245, 176.4483995768154,
175.08693076600966, 75.31684225551082] Fitness: 1.187596653960116
Generation 71 Best Solution: [183.55106359148328, 176.5775728439221,
174.6227039392402, 75.31684225551082] Fitness: 0.7572313707379681
Generation 72 Best Solution: [182.7938946304764, 176.5775728439221,
175.08693076600966, 74.42810091506612] Fitness: 1.1440540557562209
Generation 73 Best Solution: [183.55106359148328, 177.29967958538384,
173.74921054704384, 74.87573978480434] Fitness: 1.2822820134316308
Generation 74 Best Solution: [183.65291968349956, 176.5035279549988,

175.08693076600966, 75.31684225551082] Fitness: 1.3327650523040577
Generation 75 Best Solution: [183.55106359148328, 177.29967958538384,
174.70736113912685, 75.31684225551082] Fitness: 1.706135247729122
Generation 76 Best Solution: [182.9543068818245, 177.29967958538384,
174.1997113295301, 75.31684225551082] Fitness: 1.379535617246096
Generation 77 Best Solution: [183.55106359148328, 175.678247395169,
175.08693076600966, 75.31684225551082] Fitness: 1.3970923844172338
Generation 78 Best Solution: [182.9543068818245, 176.6741546322011,
175.08693076600966, 75.31684225551082] Fitness: 1.323341765516529
Generation 79 Best Solution: [183.55106359148328, 175.8857186061197,
175.08693076600966, 75.31684225551082] Fitness: 1.260666386292498
Generation 80 Best Solution: [183.29428893886563, 176.6741546322011,
175.08693076600966, 74.94142994434152] Fitness: 1.0450663586208782
Generation 81 Best Solution: [183.45719240186574, 175.8857186061197,
174.3840715740686, 75.67955588193716] Fitness: 0.9615539597198361
Generation 82 Best Solution: [183.29428893886563, 176.6741546322011,
175.08693076600966, 74.76452611017326] Fitness: 1.0118283248853117
Generation 83 Best Solution: [182.9543068818245, 176.6741546322011,
175.08693076600966, 74.76452611017326] Fitness: 1.0520964243141628
Generation 84 Best Solution: [182.9543068818245, 176.14891856351704,
175.08693076600966, 75.31684225551082] Fitness: 1.134569152617642
Generation 85 Best Solution: [183.29428893886563, 176.6741546322011,
175.08693076600966, 74.5161361385995] Fitness: 1.0504825121760315
Generation 86 Best Solution: [183.29428893886563, 176.6741546322011,
175.08693076600966, 74.43652237041128] Fitness: 1.083963688332425
Generation 87 Best Solution: [183.13778128405235, 176.40950296468768,
175.08693076600966, 74.5161361385995] Fitness: 0.8950075868298071
Generation 88 Best Solution: [183.13778128405235, 176.40950296468768,
175.08693076600966, 75.67955588193716] Fitness: 1.5737973378967531
Generation 89 Best Solution: [183.55106359148328, 176.77162472145037,
174.06570905464633, 75.70786219291669] Fitness: 1.1976639849981685
Generation 90 Best Solution: [183.55106359148328, 176.77162472145037,
174.2174864919062, 75.70786219291669] Fitness: 1.2029593040940292
Generation 91 Best Solution: [183.38924390279396, 176.77162472145037,
174.2174864919062, 75.70786219291669] Fitness: 1.1317184514094738
Generation 92 Best Solution: [183.38924390279396, 176.77162472145037,
174.2174864919062, 75.70786219291669] Fitness: 1.1317184514094738
Generation 93 Best Solution: [183.2535379695825, 176.77162472145037,
175.06679140162947, 75.70786219291669] Fitness: 1.8209361759852252
Generation 94 Best Solution: [183.13778128405235, 177.0508661730768,
174.2174864919062, 75.81711923589184] Fitness: 1.6117724880475714
Generation 95 Best Solution: [183.2535379695825, 176.77162472145037,
174.95141811818849, 75.67955588193716] Fitness: 1.6114704249650877
Generation 96 Best Solution: [183.2535379695825, 176.77162472145037,
175.08693076600966, 75.70786219291669] Fitness: 1.8520655648001083
Generation 97 Best Solution: [183.13778128405235, 176.77162472145037,
175.08693076600966, 75.70786219291669] Fitness: 1.8524319599700023
Generation 98 Best Solution: [183.13778128405235, 176.1320386138852,

175.08693076600966, 75.70786219291669] Fitness: 1.5809293782368603
Generation 99 Best Solution: [183.13778128405235, 176.1320386138852,
174.287619470386, 75.86561500554033] Fitness: 1.1160578561951002
Generation 100 Best Solution: [182.9543068818245, 176.1320386138852,
174.287619470386, 74.72773170859284] Fitness: 0.13162287188001323
Generation 101 Best Solution: [183.17958675967978, 175.87657557709323,
175.02742647166656, 75.70786219291669] Fitness: 1.5644533040594064
Generation 102 Best Solution: [182.9543068818245, 177.29967958538384,
174.287619470386, 74.77493579297682] Fitness: 1.1264074350012139
Generation 103 Best Solution: [182.92535222276666, 175.72233105205797,
174.95141811818849, 74.37433717731741] Fitness: 0.9585910528380794
Generation 104 Best Solution: [183.5368105525998, 175.64746814005144,
173.77141944875618, 75.67955588193716] Fitness: 1.204464995981668
Generation 105 Best Solution: [183.42579758504002, 177.29967958538384,
174.87268586498504, 74.32910160182746] Fitness: 1.6885201821039901
Generation 106 Best Solution: [182.9784525472525, 177.29967958538384,
174.95141811818849, 74.60529552141755] Fitness: 1.6657736941443892
Generation 107 Best Solution: [183.56920332935022, 177.2393387546933,
174.07442659101605, 74.60529552141755] Fitness: 1.076537825461764
Generation 108 Best Solution: [183.70950016862417, 177.29967958538384,
174.07442659101605, 74.60529552141755] Fitness: 1.2819657205201809
Generation 109 Best Solution: [183.38474066000725, 176.11651580529275,
174.95141811818849, 75.67955588193716] Fitness: 1.367347327687697
Generation 110 Best Solution: [182.9784525472525, 176.11012978908818,
175.08693076600966, 75.67955588193716] Fitness: 1.5754780336416894
Generation 111 Best Solution: [182.9784525472525, 176.11012978908818,
174.95141811818849, 75.67955588193716] Fitness: 1.3786442106000436
Generation 112 Best Solution: [182.9784525472525, 176.11012978908818,
174.95141811818849, 75.67955588193716] Fitness: 1.3786442106000436
Generation 113 Best Solution: [183.49798664227458, 176.11012978908818,
174.95141811818849, 75.67955588193716] Fitness: 1.4127181135908464
Generation 114 Best Solution: [183.21165220254747, 176.40644782429365,
175.08693076600966, 75.67955588193716] Fitness: 1.5699839489534746
Generation 115 Best Solution: [183.21165220254747, 176.40644782429365,
174.95141811818849, 75.67955588193716] Fitness: 1.373150125911829
Generation 116 Best Solution: [183.2362262132931, 177.2393387546933,
174.03810523579614, 75.67955588193716] Fitness: 1.6740905203714438
Generation 117 Best Solution: [182.9784525472525, 176.15466416023406,
175.08693076600966, 75.67955588193716] Fitness: 1.5714025784664583
Generation 118 Best Solution: [182.9784525472525, 176.15466416023406,
174.95141811818849, 75.67955588193716] Fitness: 1.3745687554248125
Generation 119 Best Solution: [182.9784525472525, 176.15466416023406,
174.95141811818849, 75.67955588193716] Fitness: 1.3745687554248125
Generation 120 Best Solution: [182.9784525472525, 176.15466416023406,
174.1062190699209, 75.67955588193716] Fitness: 0.8163999232100834
Generation 121 Best Solution: [182.83347142412458, 175.7694344332041,
174.95141811818849, 75.67955588193716] Fitness: 1.5841346533654832
Generation 122 Best Solution: [183.56920332935022, 177.2393387546933,

174.95141811818849, 75.27301351985244] Fitness: 1.8515902537710547
Generation 123 Best Solution: [183.72118179452193, 177.29967958538384,
174.95141811818849, 74.67672764938555] Fitness: 1.8373423235310322
Generation 124 Best Solution: [183.72118179452193, 177.2393387546933,
175.02742647166656, 74.93981958973006] Fitness: 1.8692084048629152
Generation 125 Best Solution: [183.56920332935022, 177.2393387546933,
174.95141811818849, 74.97538852775573] Fitness: 1.6652719376491445
Generation 126 Best Solution: [182.83347142412458, 176.2435398714762,
175.02742647166656, 75.7031051885689] Fitness: 1.5873849997762228
Generation 127 Best Solution: [183.07790924045656, 176.2435398714762,
175.02742647166656, 75.7031051885689] Fitness: 1.491856764042319
Generation 128 Best Solution: [183.07790924045656, 176.2435398714762,
175.02742647166656, 75.67955588193716] Fitness: 1.4555402166882287
Generation 129 Best Solution: [182.83347142412458, 176.85309470151077,
174.95141811818849, 75.67955588193716] Fitness: 1.798078858822239
Generation 130 Best Solution: [183.56920332935022, 176.00261184028548,
174.95141811818849, 74.50609531201994] Fitness: 0.7992818955252752
Generation 131 Best Solution: [182.9784525472525, 177.2393387546933,
175.02742647166656, 74.50609531201994] Fitness: 1.6962072099332968
Generation 132 Best Solution: [183.72118179452193, 177.29967958538384,
173.96795780144694, 74.50609531201994] Fitness: 1.3377290673877675
Generation 133 Best Solution: [182.88497011260387, 175.59816154029122,
174.95141811818849, 74.50609531201994] Fitness: 1.0206400981652937
Generation 134 Best Solution: [182.83347142412458, 176.31444336320183,
175.02742647166656, 75.67955588193716] Fitness: 1.5613703150614482
Generation 135 Best Solution: [183.72118179452193, 176.31444336320183,
175.02742647166656, 75.67955588193716] Fitness: 1.6756704544154637
Generation 136 Best Solution: [183.72118179452193, 176.31444336320183,
175.02742647166656, 75.67955588193716] Fitness: 1.6756704544154637
Generation 137 Best Solution: [183.32820174529417, 175.59816154029122,
175.02742647166656, 75.67955588193716] Fitness: 1.7373427345877266
Generation 138 Best Solution: [183.09138782651982, 177.29967958538384,
174.2995947848334, 75.67955588193716] Fitness: 1.8079525879754912
Generation 139 Best Solution: [182.9784525472525, 177.29967958538384,
174.2995947848334, 75.67955588193716] Fitness: 1.8376343777247421
Generation 140 Best Solution: [183.09138782651982, 177.29967958538384,
175.02742647166656, 74.46599178217524] Fitness: 1.7880779743916733
Generation 141 Best Solution: [183.22030545725374, 177.2393387546933,
175.02742647166656, 75.08931205230388] Fitness: 1.7153304049818552
Generation 142 Best Solution: [183.22030545725374, 177.29967958538384,
175.02742647166656, 74.46599178217524] Fitness: 1.7793788058441162
Generation 143 Best Solution: [183.09138782651982, 177.06919032238304,
174.61137701791097, 75.67955588193716] Fitness: 1.6063080845965323
Generation 144 Best Solution: [183.72118179452193, 177.29967958538384,
175.02742647166656, 75.08288972711786] Fitness: 2.0380089833971398
Generation 145 Best Solution: [183.72118179452193, 177.12035161662018,
175.02742647166656, 74.58634557317423] Fitness: 1.66368381766461
Generation 146 Best Solution: [183.5090018241872, 177.29967958538384,

175.02742647166656, 75.08288972711786] Fitness: 1.8949192509192543
Generation 147 Best Solution: [183.5090018241872, 176.17813685268817,
174.45421707937555, 75.67955588193716] Fitness: 0.9451244659202335
Generation 148 Best Solution: [183.5090018241872, 177.29967958538384,
175.02742647166656, 75.08288972711786] Fitness: 1.8949192509192543
Generation 149 Best Solution: [183.09138782651982, 177.0880451719051,
175.02742647166656, 75.08288972711786] Fitness: 1.4822038877640504
Generation 150 Best Solution: [183.5090018241872, 177.2393387546933,
175.02742647166656, 74.62109584151015] Fitness: 1.7035723498866142
Generation 151 Best Solution: [183.09138782651982, 176.7830054116158,
175.02742647166656, 75.67955588193716] Fitness: 1.735624557143781
Generation 152 Best Solution: [183.09138782651982, 176.28349819537198,
175.02742647166656, 75.67955588193716] Fitness: 1.4578874837717266
Generation 153 Best Solution: [183.72118179452193, 176.03001961424394,
175.02742647166656, 75.67955588193716] Fitness: 1.6833966885253435
Generation 154 Best Solution: [183.72118179452193, 176.03001961424394,
175.02742647166656, 75.67955588193716] Fitness: 1.6833966885253435
Generation 155 Best Solution: [182.956762557274, 176.47490690026166,
174.78889291556933, 75.67955588193716] Fitness: 1.250727592353629
Generation 156 Best Solution: [183.09138782651982, 177.29967958538384,
175.02742647166656, 74.43300255163696] Fitness: 1.8033813218881674
Generation 157 Best Solution: [183.09138782651982, 177.29967958538384,
175.02742647166656, 74.43300255163696] Fitness: 1.8033813218881674
Generation 158 Best Solution: [183.09138782651982, 177.29967958538384,
175.02742647166656, 74.43300255163696] Fitness: 1.8033813218881674
Generation 159 Best Solution: [183.12872362880975, 177.29967958538384,
175.02742647166656, 74.43300255163696] Fitness: 1.7981002309008671
Generation 160 Best Solution: [183.09138782651982, 177.29967958538384,
175.02742647166656, 74.43300255163696] Fitness: 1.8033813218881674
Generation 161 Best Solution: [182.82293676561883, 177.2393387546933,
174.10123659060403, 75.67955588193716] Fitness: 1.781143361100857
Generation 162 Best Solution: [183.09138782651982, 176.74056131235722,
175.02742647166656, 75.67955588193716] Fitness: 1.696355764284792
Generation 163 Best Solution: [183.09138782651982, 177.29967958538384,
175.02742647166656, 74.57093697578864] Fitness: 1.7510868276311211
Generation 164 Best Solution: [183.09138782651982, 177.29967958538384,
174.73326879888793, 75.67955588193716] Fitness: 2.085673010742874
Generation 165 Best Solution: [183.09138782651982, 177.29967958538384,
174.00301713263028, 75.67955588193716] Fitness: 1.792954310855969
Generation 166 Best Solution: [183.09138782651982, 175.70110580848998,
174.00301713263028, 75.67955588193716] Fitness: 0.9889643401902197
Generation 167 Best Solution: [183.71883934880105, 177.2393387546933,
174.00301713263028, 75.67955588193716] Fitness: 1.8979491698912638
Generation 168 Best Solution: [183.09138782651982, 177.2393387546933,
174.00301713263028, 75.67955588193716] Fitness: 1.687637780253235
Generation 169 Best Solution: [183.09138782651982, 177.2393387546933,
174.00301713263028, 75.67955588193716] Fitness: 1.687637780253235
Generation 170 Best Solution: [183.72118179452193, 176.4659260995058,

175.02742647166656, 75.67955588193716] Fitness: 1.7248892854179867
Generation 171 Best Solution: [183.6261184368604, 177.2393387546933,
174.00301713263028, 75.32667671951315] Fitness: 1.389921139216308
Generation 172 Best Solution: [183.09138782651982, 177.29967958538384,
175.02742647166656, 74.77947997902336] Fitness: 1.7303829674313125
Generation 173 Best Solution: [183.23338618546555, 177.29967958538384,
175.02742647166656, 74.95240869169814] Fitness: 1.758414621536228
Generation 174 Best Solution: [183.72118179452193, 176.337275986918,
175.02742647166656, 75.67955588193716] Fitness: 1.6807165576531597
Generation 175 Best Solution: [183.09138782651982, 177.29967958538384,
174.39976371610433, 75.67955588193716] Fitness: 1.8451172976790344
Generation 176 Best Solution: [183.09138782651982, 176.63938708720192,
175.02742647166656, 75.67955588193716] Fitness: 1.614486684902475
Generation 177 Best Solution: [183.09138782651982, 177.29967958538384,
174.10317056597393, 75.67955588193716] Fitness: 1.782129811256873
Generation 178 Best Solution: [183.72118179452193, 176.09249257087362,
175.02742647166656, 75.67955588193716] Fitness: 1.6705002420383028
Generation 179 Best Solution: [183.09138782651982, 176.09249257087362,
175.02742647166656, 75.67955588193716] Fitness: 1.4582121485854773
Generation 180 Best Solution: [183.72118179452193, 177.29967958538384,
174.6348798267694, 75.39529292662418] Fitness: 1.8406062188685428
Generation 181 Best Solution: [183.3570284141467, 176.10198726418267,
175.02742647166656, 75.67955588193716] Fitness: 1.4682137231661971
Generation 182 Best Solution: [183.40646584238195, 177.29967958538384,
175.02742647166656, 74.99849986051339] Fitness: 1.8103893756490894
Generation 183 Best Solution: [183.72118179452193, 175.85720654176959,
175.02742647166656, 75.67955588193716] Fitness: 1.7519120915792423
Generation 184 Best Solution: [183.72118179452193, 175.88911662082182,
175.02742647166656, 75.67955588193716] Fitness: 1.735629097951596
Generation 185 Best Solution: [183.72118179452193, 175.85720654176959,
175.02742647166656, 75.67955588193716] Fitness: 1.7519120915792423
Generation 186 Best Solution: [183.72118179452193, 175.85720654176959,
175.02742647166656, 75.67955588193716] Fitness: 1.7519120915792423
Generation 187 Best Solution: [183.72118179452193, 177.29574120163207,
175.02742647166656, 74.99199366994563] Fitness: 1.9867922385556103
Generation 188 Best Solution: [183.72118179452193, 175.9365927503538,
175.02742647166656, 75.67955588193716] Fitness: 1.7144472517069884
Generation 189 Best Solution: [183.09138782651982, 175.9365927503538,
175.02742647166656, 75.67955588193716] Fitness: 1.502159158254163
Generation 190 Best Solution: [183.09138782651982, 175.9365927503538,
175.02742647166656, 74.99199366994563] Fitness: 0.8367433095207012
Generation 191 Best Solution: [183.09138782651982, 177.29574120163207,
175.02742647166656, 74.99199366994563] Fitness: 1.7745041451027845
Generation 192 Best Solution: [183.09138782651982, 177.29574120163207,
175.02742647166656, 74.99199366994563] Fitness: 1.7745041451027845
Generation 193 Best Solution: [183.09138782651982, 176.7316904696951,
175.02742647166656, 75.67955588193716] Fitness: 1.6885162296104737
Generation 194 Best Solution: [183.09138782651982, 176.04395591010652,

175.02742647166656, 75.67955588193716] Fitness: 1.4676853509003571
Generation 195 Best Solution: [183.58270177731893, 176.6385025608685,
175.02742647166656, 75.67955588193716] Fitness: 1.7245006324047156
Generation 196 Best Solution: [183.58270177731893, 175.82989206784805,
175.02742647166656, 75.67955588193716] Fitness: 1.6655253350907029
Generation 197 Best Solution: [182.8050711071226, 177.1987022781246,
175.02742647166656, 74.93469653273185] Fitness: 1.7029947895125206
Generation 198 Best Solution: [183.58270177731893, 177.29574120163207,
174.75671299042162, 74.2806026702471] Fitness: 1.6628703721479936
Generation 199 Best Solution: [183.58270177731893, 177.29574120163207,
174.22566666303436, 75.43352904037687] Fitness: 1.5686811806837024
Generation 200 Best Solution: [183.58270177731893, 177.29574120163207,
174.22566666303436, 74.2806026702471] Fitness: 1.3444473814406366
Generation 201 Best Solution: [183.58270177731893, 177.29574120163207,
174.22566666303436, 74.2806026702471] Fitness: 1.3444473814406366
Generation 202 Best Solution: [183.44400310590413, 177.29574120163207,
175.02742647166656, 74.2806026702471] Fitness: 1.9297607975419666
Generation 203 Best Solution: [183.44400310590413, 175.68144886693153,
175.02742647166656, 75.67955588193716] Fitness: 1.6987054793899854
Generation 204 Best Solution: [183.58270177731893, 177.01184958154374,
174.81301506941008, 75.67955588193716] Fitness: 1.8309871171692853
Generation 205 Best Solution: [183.58270177731893, 177.29574120163207,
173.95783623254542, 74.64207362151026] Fitness: 1.1959007469160452
Generation 206 Best Solution: [183.58270177731893, 177.29574120163207,
173.95783623254542, 74.64207362151026] Fitness: 1.1959007469160452
Generation 207 Best Solution: [183.58270177731893, 177.29574120163207,
173.95783623254542, 74.64207362151026] Fitness: 1.1959007469160452
Generation 208 Best Solution: [183.58270177731893, 177.29574120163207,
173.95783623254542, 74.64207362151026] Fitness: 1.1959007469160452
Generation 209 Best Solution: [183.58270177731893, 175.8881649562532,
175.02742647166656, 74.42361775181553] Fitness: 0.9955423185601555
Generation 210 Best Solution: [183.44400310590413, 175.8881649562532,
175.02742647166656, 74.42361775181553] Fitness: 0.9248017148963922
Generation 211 Best Solution: [183.44400310590413, 175.8881649562532,
175.02742647166656, 75.67955588193716] Fitness: 1.5637190007757311
Generation 212 Best Solution: [183.58270177731893, 177.29574120163207,
174.23466327915577, 75.14523385676779] Fitness: 1.3128787307500458
Generation 213 Best Solution: [183.58270177731893, 177.29574120163207,
174.23466327915577, 75.14523385676779] Fitness: 1.3128787307500458
Generation 214 Best Solution: [183.58270177731893, 176.45822014128788,
174.23466327915577, 75.67955588193716] Fitness: 0.9654063374146356
Generation 215 Best Solution: [183.58270177731893, 175.58087959047796,
175.02742647166656, 75.67955588193716] Fitness: 1.8600787374504555
Generation 216 Best Solution: [183.58270177731893, 175.89480871405365,
175.02742647166656, 75.67955588193716] Fitness: 1.631266115411711
Generation 217 Best Solution: [182.82010696335084, 177.1566255986863,
175.02742647166656, 75.67955588193716] Fitness: 2.3128431480665435
Generation 218 Best Solution: [183.58270177731893, 177.1566255986863,

175.02742647166656, 74.64072890334104] Fitness: 1.6068694426498162
Generation 219 Best Solution: [182.94190646996046, 177.29574120163207,
174.34435901730825, 75.67955588193716] Fitness: 1.859203776961642
Generation 220 Best Solution: [183.50548723833984, 177.29574120163207,
174.27838966319052, 75.67955588193716] Fitness: 1.8625510293502052
Generation 221 Best Solution: [183.58270177731893, 176.70813396376144,
175.02742647166656, 75.67955588193716] Fitness: 1.778972119303341
Generation 222 Best Solution: [183.58270177731893, 177.29574120163207,
174.07466160994, 75.22363950956988] Fitness: 1.3606321677275994
Generation 223 Best Solution: [182.94190646996046, 177.29574120163207,
174.07466160994, 75.22363950956988] Fitness: 1.2936746479911874
Generation 224 Best Solution: [183.58270177731893, 176.32858921210388,
175.02742647166656, 75.67955588193716] Fitness: 1.5770661839729874
Generation 225 Best Solution: [183.22717781218225, 177.29574120163207,
174.81860450503854, 74.3377921967017] Fitness: 1.571065082534465
Generation 226 Best Solution: [183.22717781218225, 177.29574120163207,
174.81860450503854, 74.3377921967017] Fitness: 1.571065082534465
Generation 227 Best Solution: [183.37740678640827, 177.29574120163207,
174.04003177404442, 75.67955588193716] Fitness: 1.797005538514843
Generation 228 Best Solution: [183.37740678640827, 177.29574120163207,
174.04003177404442, 74.3377921967017] Fitness: 1.2074392377781005
Generation 229 Best Solution: [182.82010696335084, 176.77432920434094,
174.04003177404442, 75.67955588193716] Fitness: 1.173447362095164
Generation 230 Best Solution: [182.82010696335084, 176.77432920434094,
174.04003177404442, 75.67955588193716] Fitness: 1.173447362095164
Generation 231 Best Solution: [182.82010696335084, 176.77432920434094,
174.04003177404442, 75.67955588193716] Fitness: 1.173447362095164
Generation 232 Best Solution: [182.82010696335084, 177.29574120163207,
175.02742647166656, 74.3377921967017] Fitness: 1.956347398102509
Generation 233 Best Solution: [183.7586559751969, 177.29574120163207,
173.983504952809, 74.3377921967017] Fitness: 1.4454457939502903
Generation 234 Best Solution: [183.04787382888964, 176.5944563619182,
173.983504952809, 74.3377921967017] Fitness: 0.3528374073834115
Generation 235 Best Solution: [183.04787382888964, 176.5944563619182,
173.983504952809, 75.66455609448766] Fitness: 0.9197388264955364
Generation 236 Best Solution: [182.82010696335084, 176.208766031574,
175.02742647166656, 75.66455609448766] Fitness: 1.5343249991853334
Generation 237 Best Solution: [183.58270177731893, 176.3165626667695,
175.02742647166656, 75.66455609448766] Fitness: 1.5518071813968133
Generation 238 Best Solution: [182.82010696335084, 176.208766031574,
173.75985451779422, 75.66455609448766] Fitness: 0.9740142801204207
Generation 239 Best Solution: [182.94190646996046, 177.29574120163207,
175.02742647166656, 74.81649832013505] Fitness: 1.7706959278297683
Generation 240 Best Solution: [183.58270177731893, 176.73567577824338,
175.02742647166656, 75.67955588193716] Fitness: 1.8026792706227677
Generation 241 Best Solution: [183.7586559751969, 176.73567577824338,
175.02742647166656, 75.67955588193716] Fitness: 1.937138818055471
Generation 242 Best Solution: [183.58270177731893, 177.29574120163207,

174.49696751034048, 75.52806177316761] Fitness: 1.7880731131605554
Generation 243 Best Solution: [182.94190646996046, 177.29574120163207,
174.03935821342378, 75.66455609448766] Fitness: 1.801134614990421
Generation 244 Best Solution: [183.58270177731893, 177.29574120163207,
175.02742647166656, 74.80583213671848] Fitness: 1.836369838196635
Generation 245 Best Solution: [183.58270177731893, 176.02537824375003,
175.02742647166656, 74.80583213671848] Fitness: 0.8687681671278211
Generation 246 Best Solution: [183.7586559751969, 176.80508702032773,
174.68341212600032, 75.67955588193716] Fitness: 1.5936374740817327
Generation 247 Best Solution: [183.58270177731893, 176.80508702032773,
174.68341212600032, 75.67955588193716] Fitness: 1.4591779266490292
Generation 248 Best Solution: [182.94190646996046, 176.80508702032773,
174.68341212600032, 75.67955588193716] Fitness: 1.3922204069126172
Generation 249 Best Solution: [182.94190646996046, 176.80508702032773,
174.68341212600032, 75.67955588193716] Fitness: 1.3922204069126172
Generation 250 Best Solution: [183.76371311810948, 175.66365347004165,
175.02742647166656, 74.8801412168586] Fitness: 1.221912421449211
Generation 251 Best Solution: [183.58270177731893, 177.29574120163207,
175.02742647166656, 74.8801412168586] Fitness: 1.849132252443216
Generation 252 Best Solution: [183.50540944088803, 175.9364837485928,
175.02742647166656, 75.67955588193716] Fitness: 1.5696054250799927
Generation 253 Best Solution: [183.50540944088803, 175.9364837485928,
175.02742647166656, 75.67955588193716] Fitness: 1.5696054250799927
Generation 254 Best Solution: [183.50540944088803, 177.29574120163207,
175.02742647166656, 74.8801412168586] Fitness: 1.8058772742503997
Generation 255 Best Solution: [183.7586559751969, 177.3894833703733,
174.28950417532403, 74.8801412168586] Fitness: 1.516409784386115
Generation 256 Best Solution: [183.7586559751969, 177.29574120163207,
175.02742647166656, 74.8801412168586] Fitness: 1.9835917998759194
Generation 257 Best Solution: [183.58270177731893, 177.29574120163207,
174.6407609108829, 75.67955588193716] Fitness: 2.1045441471125703
Generation 258 Best Solution: [183.7586559751969, 175.73471245850703,
175.02742647166656, 75.67955588193716] Fitness: 1.8625226629490004
Generation 259 Best Solution: [183.7586559751969, 175.58149380738794,
175.02742647166656, 75.67955588193716] Fitness: 1.9939351673772312
Generation 260 Best Solution: [183.58270177731893, 176.02334978500542,
175.02742647166656, 75.71078203961433] Fitness: 1.6318637483813028
Generation 261 Best Solution: [183.58270177731893, 176.3249688385942,
174.99377451147268, 75.71078203961433] Fitness: 1.5761958992979155
Generation 262 Best Solution: [183.7586559751969, 176.3249688385942,
174.3645398995612, 75.71078203961433] Fitness: 1.142296799697535
Generation 263 Best Solution: [183.7586559751969, 176.3249688385942,
174.3645398995612, 75.71078203961433] Fitness: 1.142296799697535
Generation 264 Best Solution: [183.58270177731893, 177.29574120163207,
175.02742647166656, 75.04292457157086] Fitness: 1.9082625644829914
Generation 265 Best Solution: [183.58270177731893, 176.78885436798612,
175.02742647166656, 75.5888874368417] Fitness: 1.720461007942219
Generation 266 Best Solution: [183.46599891834398, 176.81245159300184,

175.02742647166656, 75.71078203961433] Fitness: 1.861988039361684
Generation 267 Best Solution: [183.46599891834398, 176.81245159300184,
175.02742647166656, 75.5888874368417] Fitness: 1.6821791588770716
Generation 268 Best Solution: [183.46599891834398, 176.81245159300184,
175.02742647166656, 75.67955588193716] Fitness: 1.8136390225028136
Generation 269 Best Solution: [183.46599891834398, 176.81245159300184,
175.02742647166656, 75.67955588193716] Fitness: 1.8136390225028136
Generation 270 Best Solution: [183.46599891834398, 176.81245159300184,
175.02742647166656, 75.71078203961433] Fitness: 1.861988039361684
Generation 271 Best Solution: [183.58270177731893, 176.81245159300184,
174.57085450423418, 75.71078203961433] Fitness: 1.4226906032234514
Generation 272 Best Solution: [183.7586559751969, 176.81245159300184,
174.57085450423418, 75.71078203961433] Fitness: 1.557150150656155
Generation 273 Best Solution: [183.7586559751969, 176.81245159300184,
174.57085450423418, 75.67955588193716] Fitness: 1.5088011337972844
Generation 274 Best Solution: [183.58270177731893, 176.093433798495,
174.57085450423418, 75.67955588193716] Fitness: 1.067830018930502
Generation 275 Best Solution: [183.58270177731893, 176.36137415497055,
175.02742647166656, 75.67955588193716] Fitness: 1.5853244976114826
Generation 276 Best Solution: [183.58270177731893, 176.6148661943925,
175.02742647166656, 75.07796760228517] Fitness: 1.0838051472511068
Generation 277 Best Solution: [183.58270177731893, 176.1690126479925,
175.02742647166656, 75.54876498027002] Fitness: 1.3762683375763172
Generation 278 Best Solution: [183.2805382489999, 176.55393723479096,
175.02742647166656, 75.71078203961433] Fitness: 1.6030093806515713
Generation 279 Best Solution: [183.2805382489999, 176.55393723479096,
175.02742647166656, 75.67955588193716] Fitness: 1.5546603637927008
Generation 280 Best Solution: [183.58270177731893, 176.03193961804539,
175.02742647166656, 75.71078203961433] Fitness: 1.6296241401976679
Generation 281 Best Solution: [183.4524381596957, 176.55393723479096,
175.02742647166656, 75.67955588193716] Fitness: 1.6015527062680799
Generation 282 Best Solution: [183.4524381596957, 176.55393723479096,
175.02742647166656, 74.74386705395301] Fitness: 0.8835237539364176
Generation 283 Best Solution: [183.10322340408118, 177.29574120163207,
174.20794326458505, 75.71078203961433] Fitness: 1.8275255611272105
Generation 284 Best Solution: [183.4524381596957, 176.24111101584148,
175.02742647166656, 75.71078203961433] Fitness: 1.5445513184260347
Generation 285 Best Solution: [183.4524381596957, 176.24111101584148,
175.02742647166656, 75.67955588193716] Fitness: 1.4962023015671644
Generation 286 Best Solution: [183.4524381596957, 176.24111101584148,
175.02742647166656, 75.71078203961433] Fitness: 1.5445513184260347
Generation 287 Best Solution: [183.4524381596957, 175.97867429040488,
175.02742647166656, 75.71078203961433] Fitness: 1.5781079777525244
Generation 288 Best Solution: [183.4524381596957, 175.97867429040488,
175.02742647166656, 74.64578530693348] Fitness: 0.8183592045376638
Generation 289 Best Solution: [183.51819289714408, 175.7602123867028,
174.31576681344092, 75.71078203961433] Fitness: 1.0873374737358639
Generation 290 Best Solution: [183.58270177731893, 175.7602123867028,

174.31576681344092, 75.71078203961433] Fitness: 1.124104528735487
Generation 291 Best Solution: [183.55926432079775, 177.29574120163207,
175.02742647166656, 74.66388950482965] Fitness: 1.822648153838015
Generation 292 Best Solution: [183.58270177731893, 177.29574120163207,
175.02742647166656, 74.66388950482965] Fitness: 1.8367839041115883
Generation 293 Best Solution: [183.56657959715207, 177.29574120163207,
174.66461104656395, 74.43020116442881] Fitness: 1.4730365699734285
Generation 294 Best Solution: [183.4524381596957, 177.29574120163207,
174.66461104656395, 74.43020116442881] Fitness: 1.4155295987474394
Generation 295 Best Solution: [183.56657959715207, 176.50815144339,
175.02742647166656, 75.61640109632079] Fitness: 1.540793665303269
Generation 296 Best Solution: [183.10322340408118, 176.66318608595228,
175.02742647166656, 75.71078203961433] Fitness: 1.6786883710875653
Generation 297 Best Solution: [183.10322340408118, 176.96084072973687,
175.02742647166656, 74.43020116442881] Fitness: 1.2876664844456462
Generation 298 Best Solution: [183.4524381596957, 175.96647035403046,
174.66461104656395, 75.71078203961433] Fitness: 1.156865821336663
Generation 299 Best Solution: [183.4524381596957, 175.96647035403046,
174.66461104656395, 75.71078203961433] Fitness: 1.156865821336663
Generation 300 Best Solution: [183.56657959715207, 176.11358934658944,
175.02742647166656, 75.71078203961433] Fitness: 1.6044679214031712
Generation 301 Best Solution: [183.56657959715207, 176.11358934658944,
175.02742647166656, 75.61640109632079] Fitness: 1.4631475385194697
Generation 302 Best Solution: [183.10322340408118, 176.80167097777908,
174.62408872848172, 75.71078203961433] Fitness: 1.3404128906984212
Generation 303 Best Solution: [183.2038903966882, 177.29574120163207,
175.02742647166656, 74.81942616644982] Fitness: 1.7182980600012838
Generation 304 Best Solution: [183.56657959715207, 176.49042200879074,
175.02742647166656, 75.53949527510872] Fitness: 1.427393972143356
Generation 305 Best Solution: [183.2038903966882, 177.29574120163207,
175.02742647166656, 74.43883200792202] Fitness: 1.7844128035416236
Generation 306 Best Solution: [183.2038903966882, 177.29574120163207,
173.83731483989214, 75.61640109632079] Fitness: 1.73733668437622
Generation 307 Best Solution: [183.2038903966882, 176.04144618125372,
175.02742647166656, 75.71078203961433] Fitness: 1.5075445830009258
Generation 308 Best Solution: [183.56657959715207, 177.29574120163207,
174.0505458070851, 75.71078203961433] Fitness: 1.9279307746321264
Generation 309 Best Solution: [183.10322340408118, 177.29574120163207,
175.02742647166656, 74.43496569763994] Fitness: 1.7934502330685467
Generation 310 Best Solution: [183.56657959715207, 175.90161233723435,
175.02742647166656, 75.71078203961433] Fitness: 1.6665997251463507
Generation 311 Best Solution: [183.2038903966882, 175.863700097516,
175.02742647166656, 75.61640109632079] Fitness: 1.4341227128821064
Generation 312 Best Solution: [183.56657959715207, 175.99203997459185,
174.53798991534916, 75.71078203961433] Fitness: 1.107261283242569
Generation 313 Best Solution: [183.56657959715207, 177.29574120163207,
175.02742647166656, 74.69566834970703] Fitness: 1.8240445974672979
Generation 314 Best Solution: [183.10322340408118, 177.29574120163207,

175.0555780075834, 75.05928699080974] Fitness: 1.845904845308977
Generation 315 Best Solution: [183.10322340408118, 176.529621597283,
175.0555780075834, 75.71078203961433] Fitness: 1.6327038275699637
Generation 316 Best Solution: [183.10322340408118, 177.29574120163207,
175.02742647166656, 75.05928699080974] Fitness: 1.8039990167458115
Generation 317 Best Solution: [183.10322340408118, 177.29574120163207,
175.02742647166656, 75.05928699080974] Fitness: 1.8039990167458115
Generation 318 Best Solution: [183.10322340408118, 177.29574120163207,
173.89303978905266, 75.61640109632079] Fitness: 1.7215634594325486
Generation 319 Best Solution: [183.10322340408118, 177.29574120163207,
175.0555780075834, 74.93879771342287] Fitness: 1.794836953646059
Generation 320 Best Solution: [183.00423144155465, 177.29574120163207,
175.02742647166656, 75.05928699080974] Fitness: 1.8270087398945867
Generation 321 Best Solution: [183.10322340408118, 176.02512914006456,
175.02742647166656, 75.05928699080974] Fitness: 0.8364632560684628
Generation 322 Best Solution: [183.10322340408118, 176.02512914006456,
175.02742647166656, 75.47134361456281] Fitness: 1.1883476110152387
Generation 323 Best Solution: [183.10322340408118, 176.02512914006456,
175.02742647166656, 75.47134361456281] Fitness: 1.1883476110152387
Generation 324 Best Solution: [183.00423144155465, 176.02512914006456,
175.02742647166656, 75.47134361456281] Fitness: 1.211357334164014
Generation 325 Best Solution: [183.7994872141529, 176.02512914006456,
175.02742647166656, 75.47134361456281] Fitness: 1.473733372543148
Generation 326 Best Solution: [183.10322340408118, 177.29574120163207,
174.33571111336403, 75.47134361456281] Fitness: 1.5283940694579
Generation 327 Best Solution: [183.7994872141529, 177.29574120163207,
174.65461278362991, 75.05928699080974] Fitness: 1.6551597231093154
Generation 328 Best Solution: [183.7994872141529, 176.02512914006456,
175.02742647166656, 75.71078203961433] Fitness: 1.8042011180393933
Generation 329 Best Solution: [183.7994872141529, 176.02512914006456,
175.02742647166656, 75.79949068754543] Fitness: 1.950146384594111
Generation 330 Best Solution: [183.7994872141529, 176.02512914006456,
175.02742647166656, 75.47134361456281] Fitness: 1.473733372543148
Generation 331 Best Solution: [183.7994872141529, 176.59352038994084,
175.02742647166656, 75.47134361456281] Fitness: 1.584168297446233
Generation 332 Best Solution: [183.7994872141529, 176.59352038994084,
175.02742647166656, 74.597392749998] Fitness: 1.163883186983881
Generation 333 Best Solution: [183.7994872141529, 176.59352038994084,
175.02742647166656, 75.47134361456281] Fitness: 1.584168297446233
Generation 334 Best Solution: [183.07131350399263, 177.29574120163207,
175.02742647166656, 74.597392749998] Fitness: 1.7412864769790264
Generation 335 Best Solution: [183.07131350399263, 176.14629684844255,
175.02742647166656, 75.79949068754543] Fitness: 1.6502228993280461
Generation 336 Best Solution: [183.10049632183225, 176.59352038994084,
175.02742647166656, 75.52158321112597] Fitness: 1.360866653501656
Generation 337 Best Solution: [183.11829960161114, 176.59352038994084,
175.02742647166656, 74.96870987731464] Fitness: 0.9042045469764673
Generation 338 Best Solution: [183.10322340408118, 176.59352038994084,

175.02742647166656, 74.95679926775203] Fitness: 0.90196988322997
Generation 339 Best Solution: [183.10049632183225, 176.59352038994084,
175.02742647166656, 75.28547482336664] Fitness: 1.1065214470779723
Generation 340 Best Solution: [183.10322340408118, 176.59352038994084,
175.02742647166656, 75.28547482336664] Fitness: 1.106099600414438
Generation 341 Best Solution: [183.10322340408118, 176.59352038994084,
175.02742647166656, 75.28547482336664] Fitness: 1.106099600414438
Generation 342 Best Solution: [183.10322340408118, 176.59352038994084,
173.8694179315249, 75.28547482336664] Fitness: 0.49174344667446845
Generation 343 Best Solution: [183.7994872141529, 176.88379100381746,
175.02742647166656, 75.18251604777234] Fitness: 1.566481129144097
Generation 344 Best Solution: [183.00671939980623, 176.88379100381746,
175.02742647166656, 75.18251604777234] Fitness: 1.3033328637061377
Generation 345 Best Solution: [183.35870994746094, 176.88379100381746,
174.53733935477848, 75.79949068754543] Fitness: 1.5222741522488028
Generation 346 Best Solution: [183.35870994746094, 176.88379100381746,
174.53733935477848, 75.79949068754543] Fitness: 1.5222741522488028
Generation 347 Best Solution: [182.7954248389961, 175.75376168226904,
174.53733935477848, 75.79949068754543] Fitness: 1.3951558049329547
Generation 348 Best Solution: [182.7954248389961, 175.75376168226904,
174.53733935477848, 75.506474597036] Fitness: 0.9614325878696861
Generation 349 Best Solution: [182.7954248389961, 177.29574120163207,
173.7494051495384, 75.10069071561824] Fitness: 1.3962735331446
Generation 350 Best Solution: [182.7954248389961, 177.29574120163207,
173.7494051495384, 75.10069071561824] Fitness: 1.3962735331446
Generation 351 Best Solution: [183.79824193157293, 177.29574120163207,
173.7494051495384, 75.10069071561824] Fitness: 1.556994341331687
Generation 352 Best Solution: [183.7994872141529, 177.29574120163207,
173.7494051495384, 75.10069071561824] Fitness: 1.558203813036006
Generation 353 Best Solution: [182.7954248389961, 175.90185363134043,
175.02742647166656, 75.79949068754543] Fitness: 1.833132777598237
Generation 354 Best Solution: [182.7954248389961, 177.29574120163207,
175.02742647166656, 74.3276169701387] Fitness: 1.9785311673926242
Generation 355 Best Solution: [182.7954248389961, 177.29574120163207,
174.22535578710873, 75.25713634981162] Fitness: 1.4060744605475852
Generation 356 Best Solution: [183.3229424151652, 177.29574120163207,
175.02742647166656, 75.25713634981162] Fitness: 1.9442145842687764
Generation 357 Best Solution: [183.3229424151652, 175.9141067690133,
175.02742647166656, 75.79949068754543] Fitness: 1.7096025790736145
Generation 358 Best Solution: [183.7994872141529, 176.24875045867392,
175.02742647166656, 75.79949068754543] Fitness: 1.9313232456500273
Generation 359 Best Solution: [183.7994872141529, 177.29574120163207,
174.45045904624135, 75.08010008747273] Fitness: 1.523711833485099
Generation 360 Best Solution: [182.7954248389961, 177.29574120163207,
174.76526480924974, 75.87497951611329] Fitness: 2.5611829209607704
Generation 361 Best Solution: [183.7994872141529, 177.29574120163207,
175.02742647166656, 74.85347417154073] Fitness: 2.0163371826021494
Generation 362 Best Solution: [183.7994872141529, 177.29574120163207,

175.02742647166656, 74.85347417154073] Fitness: 2.0163371826021494
Generation 363 Best Solution: [183.2333303704041, 177.29574120163207,
175.02742647166656, 75.79949068754543] Fitness: 2.6261294599811946
Generation 364 Best Solution: [183.7994872141529, 176.0798516882586,
175.02742647166656, 75.79949068754543] Fitness: 1.9380750059219571
Generation 365 Best Solution: [183.7994872141529, 176.47947229113328,
175.02742647166656, 75.79949068754543] Fitness: 1.9965702949972943
Generation 366 Best Solution: [183.7994872141529, 176.80536092571364,
175.02742647166656, 74.69604757655732] Fitness: 1.3242073285273297
Generation 367 Best Solution: [183.7994872141529, 177.29574120163207,
174.56399050551806, 75.34858364613638] Fitness: 1.8018974305108184
Generation 368 Best Solution: [183.7994872141529, 177.29574120163207,
174.56399050551806, 75.34858364613638] Fitness: 1.8018974305108184
Generation 369 Best Solution: [183.7994872141529, 177.29574120163207,
174.56399050551806, 75.34858364613638] Fitness: 1.8018974305108184
Generation 370 Best Solution: [183.7994872141529, 176.92031438946307,
174.83682005816817, 75.79949068754543] Fitness: 2.1103037052682367
Generation 371 Best Solution: [183.7994872141529, 176.92031438946307,
174.83682005816817, 75.79949068754543] Fitness: 2.1103037052682367
Generation 372 Best Solution: [183.7994872141529, 177.29574120163207,
175.02742647166656, 74.59156563725088] Fitness: 2.022322437463493
Generation 373 Best Solution: [183.7994872141529, 177.29574120163207,
175.02742647166656, 74.59156563725088] Fitness: 2.022322437463493
Generation 374 Best Solution: [183.7994872141529, 176.27615503266244,
175.04185221614824, 75.79949068754543] Fitness: 1.9558867674597185
Generation 375 Best Solution: [183.7994872141529, 176.1053001149776,
175.04185221614824, 75.79949068754543] Fitness: 1.955423105981147
Generation 376 Best Solution: [183.7994872141529, 177.29574120163207,
174.07074883695182, 75.79949068754543] Fitness: 2.254569083240228
Generation 377 Best Solution: [183.7994872141529, 177.29574120163207,
174.4488911308972, 74.73949353694935] Fitness: 1.4276187988540214
Generation 378 Best Solution: [183.79182810855085, 177.29574120163207,
174.4488911308972, 74.73949353694935] Fitness: 1.4202196250978456
Generation 379 Best Solution: [183.79182810855085, 177.29574120163207,
174.46278196750166, 74.575459994297] Fitness: 1.448738999210524
Generation 380 Best Solution: [183.7994872141529, 177.29574120163207,
174.46278196750166, 74.575459994297] Fitness: 1.4561381729666998
Generation 381 Best Solution: [183.7994872141529, 177.29574120163207,
174.0574267380782, 75.79949068754543] Fitness: 2.2557723315850233
Generation 382 Best Solution: [183.7994872141529, 175.75396426066013,
174.7544390247594, 75.79949068754543] Fitness: 1.7413727326191535
Generation 383 Best Solution: [182.88994807974092, 177.29574120163207,
174.0574267380782, 75.79949068754543] Fitness: 2.039646901829376
Generation 384 Best Solution: [183.7994872141529, 177.29574120163207,
175.02742647166656, 74.86850469498768] Fitness: 2.0193557488656126
Generation 385 Best Solution: [182.88994807974092, 177.29574120163207,
175.02742647166656, 74.86850469498768] Fitness: 1.8032303191099652
Generation 386 Best Solution: [183.7994872141529, 176.41643234509624,

174.7544390247594, 75.79949068754543] Fitness: 1.6302393473416048
Generation 387 Best Solution: [183.7994872141529, 176.41643234509624,
174.7544390247594, 75.79949068754543] Fitness: 1.6302393473416048
Generation 388 Best Solution: [183.7994872141529, 176.41643234509624,
174.7544390247594, 75.79949068754543] Fitness: 1.6302393473416048
Generation 389 Best Solution: [183.7994872141529, 176.08463383715343,
174.56996585981244, 75.79949068754543] Fitness: 1.4357105505891126
Generation 390 Best Solution: [183.7994872141529, 175.57025812156007,
174.7544390247594, 74.79153745128838] Fitness: 0.9878110079717756
Generation 391 Best Solution: [183.7994872141529, 177.0894736994665,
175.02742647166656, 74.79153745128838] Fitness: 1.6733613463665897
Generation 392 Best Solution: [183.7994872141529, 177.29574120163207,
174.56996585981244, 74.62549988015125] Fitness: 1.5137591529729413
Generation 393 Best Solution: [183.7994872141529, 177.29574120163207,
175.02742647166656, 74.62549988015125] Fitness: 2.0152985397488585
Generation 394 Best Solution: [183.7994872141529, 175.69913430663806,
174.56996585981244, 75.79949068754543] Fitness: 1.6207632095260331
Generation 395 Best Solution: [183.7994872141529, 176.6275170325976,
174.56996585981244, 75.89733758318921] Fitness: 1.757911342256421
Generation 396 Best Solution: [183.7994872141529, 177.29574120163207,
173.99160846699732, 74.79153745128838] Fitness: 1.3559908855094123
Generation 397 Best Solution: [183.7994872141529, 176.61363654283724,
174.56996585981244, 75.64792262376058] Fitness: 1.3308446568618062
Generation 398 Best Solution: [183.7994872141529, 177.2550618052614,
175.04185221614824, 74.62549988015125] Fitness: 1.9652250626197776
Generation 399 Best Solution: [183.7994872141529, 177.2550618052614,
175.04185221614824, 74.62549988015125] Fitness: 1.9652250626197776
Generation 400 Best Solution: [183.7994872141529, 177.29574120163207,
174.64901888505494, 74.67729933598605] Fitness: 1.5691330282343288
Generation 401 Best Solution: [183.7994872141529, 176.42412052120108,
175.04185221614824, 75.79949068754543] Fitness: 1.9943907260837128
Generation 402 Best Solution: [183.57651730080963, 176.42412052120108,
175.04185221614824, 75.79949068754543] Fitness: 1.817763438547762
Generation 403 Best Solution: [183.7994872141529, 175.82069451692024,
174.814944365161, 75.89733758318921] Fitness: 1.9387592851374207
Generation 404 Best Solution: [183.7994872141529, 177.29574120163207,
175.04185221614824, 75.22484124986423] Fitness: 2.2191157166732944
Generation 405 Best Solution: [183.10804297293376, 177.29574120163207,
175.04185221614824, 75.22484124986423] Fitness: 1.9330138031599629
Generation 406 Best Solution: [183.7994872141529, 175.65816807690166,
174.20022592341803, 75.79949068754543] Fitness: 1.496204906682762
Generation 407 Best Solution: [183.7994872141529, 177.29574120163207,
175.04185221614824, 75.17476071585136] Fitness: 2.1816484634913547
Generation 408 Best Solution: [183.7994872141529, 176.0479607519588,
175.04185221614824, 75.89733758318921] Fitness: 2.141558977956482
Generation 409 Best Solution: [183.7994872141529, 176.59738280413862,
174.6058542022094, 75.79949068754543] Fitness: 1.5887039637881248
Generation 410 Best Solution: [182.89105338249396, 177.29574120163207,

174.6058542022094, 75.79949068754543] Fitness: 2.226595343381909
Generation 411 Best Solution: [182.86786069169293, 177.29574120163207,
174.62209139310826, 75.79949068754543] Fitness: 2.2514718848229065
Generation 412 Best Solution: [182.86786069169293, 176.28718243173668,
174.62209139310826, 75.79949068754543] Fitness: 1.2700124567569946
Generation 413 Best Solution: [183.7994872141529, 177.29574120163207,
174.74081229400377, 74.64065706506702] Fitness: 1.6589809525828416
Generation 414 Best Solution: [183.10804297293376, 177.29574120163207,
175.04185221614824, 74.5275743622413] Fitness: 1.775841054309413
Generation 415 Best Solution: [183.7994872141529, 176.46039712711564,
175.02742647166656, 75.79949068754543] Fitness: 1.987915117029087
Generation 416 Best Solution: [183.10804297293376, 177.29574120163207,
175.02742647166656, 75.79949068754543] Fitness: 2.631580231758128
Generation 417 Best Solution: [183.7994872141529, 176.28330023381614,
175.04185221614824, 75.89733758318921] Fitness: 2.1326566347965237
Generation 418 Best Solution: [183.10804297293376, 177.29574120163207,
175.04185221614824, 75.27788970835864] Fitness: 1.9771202301115887
Generation 419 Best Solution: [183.7994872141529, 176.28330023381614,
175.04185221614824, 75.89733758318921] Fitness: 2.1326566347965237
Generation 420 Best Solution: [183.81079786134399, 176.39906619815403,
175.04185221614824, 75.89733758318921] Fitness: 2.172207078402801
Generation 421 Best Solution: [182.83803147802655, 176.75643882849658,
175.04185221614824, 75.89733758318921] Fitness: 2.197358571305961
Generation 422 Best Solution: [183.7994872141529, 177.29574120163207,
174.76232276147135, 75.37333782679205] Fitness: 2.000879487549568
Generation 423 Best Solution: [183.7994872141529, 176.28514740223613,
175.02742647166656, 75.89733758318921] Fitness: 2.1116266742596537
Generation 424 Best Solution: [183.7994872141529, 176.78727137494243,
174.23324438905433, 75.89733758318921] Fitness: 1.7385035074573791
Generation 425 Best Solution: [183.7994872141529, 176.28514740223613,
175.02742647166656, 75.2021648230758] Fitness: 1.1985560088880882
Generation 426 Best Solution: [183.7994872141529, 176.28514740223613,
175.02742647166656, 75.79949068754543] Fitness: 1.9359034581368788
Generation 427 Best Solution: [183.4383275363651, 177.29574120163207,
175.02742647166656, 74.54412816010822] Fitness: 1.789474481432515
Generation 428 Best Solution: [183.81079786134399, 177.29574120163207,
175.02742647166656, 74.72977354118872] Fitness: 2.016455501204593
Generation 429 Best Solution: [183.4383275363651, 177.29574120163207,
174.4843667538968, 75.79949068754543] Fitness: 2.114057066316276
Generation 430 Best Solution: [183.4383275363651, 176.86557553294134,
175.02742647166656, 75.79949068754543] Fitness: 2.052332414634498
Generation 431 Best Solution: [183.85888706257936, 176.86557553294134,
173.8518331517738, 75.89733758318921] Fitness: 1.9274531241915551
Generation 432 Best Solution: [182.83803147802655, 177.29574120163207,
174.02897957135846, 75.79949068754543] Fitness: 2.071155868064433
Generation 433 Best Solution: [183.1414972007886, 176.19985600847477,
175.02742647166656, 75.89733758318921] Fitness: 1.8142230788718638
Generation 434 Best Solution: [183.7994872141529, 177.29574120163207,

175.02742647166656, 75.07678939041092] Fitness: 2.0987536808428615
Generation 435 Best Solution: [183.81079786134399, 177.29574120163207,
175.04185221614824, 74.62238542157583] Fitness: 2.0482797046459065
Generation 436 Best Solution: [183.1414972007886, 177.29574120163207,
175.02742647166656, 75.07678939041092] Fitness: 1.8087149042236979
Generation 437 Best Solution: [183.7994872141529, 176.14667042318737,
175.02742647166656, 75.79949068754543] Fitness: 1.9298947591523432
Generation 438 Best Solution: [183.7994872141529, 176.14667042318737,
175.02742647166656, 75.79949068754543] Fitness: 1.9298947591523432
Generation 439 Best Solution: [183.7994872141529, 176.45218742947202,
175.02742647166656, 75.79949068754543] Fitness: 1.9843709666571665
Generation 440 Best Solution: [183.7994872141529, 176.14667042318737,
175.02742647166656, 75.79949068754543] Fitness: 1.9298947591523432
Generation 441 Best Solution: [183.7994872141529, 175.58602370003496,
175.04185221614824, 75.79949068754543] Fitness: 2.2434657310477686
Generation 442 Best Solution: [183.7994872141529, 175.59167427840464,
175.02742647166656, 75.79949068754543] Fitness: 2.2166732479608644
Generation 443 Best Solution: [183.7994872141529, 176.59445137427997,
175.02742647166656, 75.89733758318921] Fitness: 2.236913511374262
Generation 444 Best Solution: [183.7994872141529, 177.29574120163207,
175.02742647166656, 74.73892453440217] Fitness: 2.0053211518409166
Generation 445 Best Solution: [183.7994872141529, 177.29574120163207,
175.02742647166656, 74.28024621130143] Fitness: 2.173575198621339
Generation 446 Best Solution: [183.7994872141529, 175.59167427840464,
175.02742647166656, 75.89733758318921] Fitness: 2.3923964640836393
Generation 447 Best Solution: [183.81079786134399, 176.78484153474133,
175.02742647166656, 75.89733758318921] Fitness: 2.4019744425913703
Generation 448 Best Solution: [183.7994872141529, 176.9360285941667,
175.02742647166656, 74.74939910879147] Fitness: 1.466886731841362
Generation 449 Best Solution: [183.7994872141529, 176.4269795704215,
175.02742647166656, 75.89733758318921] Fitness: 2.149892264577724
Generation 450 Best Solution: [183.7994872141529, 176.4269795704215,
175.02742647166656, 75.89733758318921] Fitness: 2.149892264577724
Generation 451 Best Solution: [183.7994872141529, 176.28012628441567,
175.02742647166656, 75.89561312796178] Fitness: 2.1076367377794245
Generation 452 Best Solution: [183.7994872141529, 176.4269795704215,
175.02742647166656, 75.46943556142914] Fitness: 1.495494526503592
Generation 453 Best Solution: [183.7994872141529, 176.4269795704215,
175.02742647166656, 75.89733758318921] Fitness: 2.149892264577724
Generation 454 Best Solution: [183.7994872141529, 176.81735059473496,
174.62736243330227, 75.89733758318921] Fitness: 1.9658570304770053
Generation 455 Best Solution: [182.83788913337972, 177.29574120163207,
173.82617462618322, 75.80203188074933] Fitness: 2.138645961161569
Generation 456 Best Solution: [183.7994872141529, 176.4269795704215,
175.02742647166656, 75.46943556142914] Fitness: 1.495494526503592
Generation 457 Best Solution: [183.7994872141529, 176.4269795704215,
175.02742647166656, 75.46943556142914] Fitness: 1.495494526503592
Generation 458 Best Solution: [183.7994872141529, 176.09776002454404,

173.8532601636819, 75.89733758318921] Fitness: 1.5032935012966626
Generation 459 Best Solution: [183.81079786134399, 177.29574120163207,
173.97195658829986, 75.2258917814934] Fitness: 1.5623609900263078
Generation 460 Best Solution: [183.7994872141529, 176.90498457915578,
173.97195658829986, 75.89733758318921] Fitness: 1.8707927076642394
Generation 461 Best Solution: [183.81079786134399, 177.29574120163207,
173.97195658829986, 75.89733758318921] Fitness: 2.457135260532663
Generation 462 Best Solution: [183.81079786134399, 175.61007845320495,
175.02742647166656, 75.89733758318921] Fitness: 2.386010327414341
Generation 463 Best Solution: [183.81079786134399, 175.7895441866695,
173.97195658829986, 75.89733758318921] Fitness: 1.5968075082909055
Generation 464 Best Solution: [183.81079786134399, 177.29574120163207,
173.97195658829986, 74.81693991065079] Fitness: 1.3742515359944798
Generation 465 Best Solution: [183.81079786134399, 176.95402881581305,
175.02742647166656, 75.89733758318921] Fitness: 2.5879260837439615
Generation 466 Best Solution: [183.354135013063, 176.95402881581305,
175.02742647166656, 75.89733758318921] Fitness: 2.304029498612894
Generation 467 Best Solution: [183.7994872141529, 176.95402881581305,
175.02742647166656, 75.33407324738282] Fitness: 1.7769987086922137
Generation 468 Best Solution: [183.81079786134399, 177.29574120163207,
175.02742647166656, 74.90321734089804] Fitness: 2.0388218031303427
Generation 469 Best Solution: [183.354135013063, 177.10836921638796,
175.02742647166656, 75.89733758318921] Fitness: 2.5139939342500544
Generation 470 Best Solution: [183.81079786134399, 177.29574120163207,
175.02742647166656, 74.41949811154915] Fitness: 2.09766765200098
Generation 471 Best Solution: [183.81079786134399, 177.29574120163207,
175.02742647166656, 74.41949811154915] Fitness: 2.09766765200098
Generation 472 Best Solution: [183.354135013063, 177.10836921638796,
173.84793495093524, 75.89733758318921] Fitness: 1.9087065641834862
Generation 473 Best Solution: [183.81079786134399, 177.10836921638796,
173.84793495093524, 75.89733758318921] Fitness: 2.1926031493145537
Generation 474 Best Solution: [183.354135013063, 177.29574120163207,
174.62736243330227, 75.89733758318921] Fitness: 2.363450041807371
Generation 475 Best Solution: [183.81079786134399, 177.29574120163207,
174.65481835577256, 75.89733758318921] Fitness: 2.670457962855449
Generation 476 Best Solution: [183.354135013063, 177.29574120163207,
175.02742647166656, 75.89733758318921] Fitness: 2.820608865570881
Generation 477 Best Solution: [183.81079786134399, 175.89062212181094,
175.02742647166656, 75.89733758318921] Fitness: 2.1871988708948207
Generation 478 Best Solution: [183.81079786134399, 177.29574120163207,
175.02742647166656, 75.24004340247649] Fitness: 2.221076773608042
Generation 479 Best Solution: [183.81079786134399, 177.29574120163207,
175.02742647166656, 75.27252815624956] Fitness: 2.248343979574676
Generation 480 Best Solution: [183.81079786134399, 177.29574120163207,
174.0531823149822, 75.69390683096093] Fitness: 2.0950464007696956
Generation 481 Best Solution: [183.81079786134399, 177.29574120163207,
175.02742647166656, 75.27252815624956] Fitness: 2.248343979574676
Generation 482 Best Solution: [183.0144159482946, 177.29574120163207,

175.02742647166656, 75.27252815624956] Fitness: 1.971770036421596
Generation 483 Best Solution: [183.81079786134399, 177.29574120163207,
175.02742647166656, 74.2838514102782] Fitness: 2.1820276922055597
Generation 484 Best Solution: [183.81079786134399, 177.29574120163207,
175.02742647166656, 74.2838514102782] Fitness: 2.1820276922055597
Generation 485 Best Solution: [183.81079786134399, 177.29574120163207,
175.02742647166656, 74.2838514102782] Fitness: 2.1820276922055597
Generation 486 Best Solution: [183.32036607514922, 177.29574120163207,
175.02742647166656, 74.2838514102782] Fitness: 1.8905142336860872
Generation 487 Best Solution: [183.81079786134399, 177.18443991758352,
175.02742647166656, 75.03930336774937] Fitness: 1.901421216520025
Generation 488 Best Solution: [183.32036607514922, 177.29574120163207,
174.10619768863774, 75.89733758318921] Fitness: 2.1480735979006362
Generation 489 Best Solution: [183.81079786134399, 177.29574120163207,
175.02742647166656, 74.2838514102782] Fitness: 2.1820276922055597
Generation 490 Best Solution: [183.81079786134399, 175.82177369537536,
175.02742647166656, 75.89733758318921] Fitness: 2.2242173135787366
Generation 491 Best Solution: [183.81079786134399, 177.29574120163207,
174.8157512315812, 75.37747183567478] Fitness: 2.0739880092237635
Generation 492 Best Solution: [183.81079786134399, 177.29574120163207,
174.2107915289821, 75.37747183567478] Fitness: 1.6896664644257289
Generation 493 Best Solution: [183.81079786134399, 177.29574120163207,
174.8157512315812, 75.37747183567478] Fitness: 2.0739880092237635
Generation 494 Best Solution: [183.81079786134399, 177.29574120163207,
174.8157512315812, 75.37747183567478] Fitness: 2.0739880092237635
Generation 495 Best Solution: [183.81079786134399, 176.41890664139078,
174.10619768863774, 75.89733758318921] Fitness: 1.4930237553738785
Generation 496 Best Solution: [183.4682940383323, 177.25933281190734,
174.10619768863774, 75.0286366051521] Fitness: 1.1118264128257482
Generation 497 Best Solution: [183.81079786134399, 177.25933281190734,
174.10619768863774, 75.0286366051521] Fitness: 1.3563336992516621
Generation 498 Best Solution: [183.81079786134399, 177.25933281190734,
174.10619768863774, 75.0286366051521] Fitness: 1.3563336992516621
Generation 499 Best Solution: [183.81079786134399, 177.29574120163207,
174.10619768863774, 75.0286366051521] Fitness: 1.4203516271461019
Generation 500 Best Solution: [183.81079786134399, 177.29574120163207,
174.10619768863774, 75.0286366051521] Fitness: 1.4203516271461019
Optimal Solution for Simulation 28: [183.81079786134399, 177.29574120163207,
174.10619768863774, 75.0286366051521]

Simulation 29/32 with conditions: {'drought': False, 'flood': False,
'policy_change': False, 'social_event': True, 'economic_crisis': True}
Generation 1 Best Solution: [183.0963841761125, 176.2479323710547,
174.1543579921261, 74.9118942519101] Fitness: 0.08906255063666957
Generation 2 Best Solution: [183.0963841761125, 175.91481916401617,
173.76238710754214, 74.9118942519101] Fitness: 0.29723700095367495
Generation 3 Best Solution: [183.0963841761125, 176.42813955835013,
174.4626740501534, 75.04134343964225] Fitness: 0.2341736838866354

Generation 4 Best Solution: [183.67167586581527, 177.0554337075307,
173.78936657818628, 75.31668310826825] Fitness: 1.0463828146404237
Generation 5 Best Solution: [183.86433944345876, 177.0554337075307,
173.78936657818628, 75.31668310826825] Fitness: 1.1922313550711747
Generation 6 Best Solution: [183.86433944345876, 176.73791946391904,
174.05565157840488, 75.85007508327558] Fitness: 1.5058660141682285
Generation 7 Best Solution: [183.60868357595066, 177.0554337075307,
173.78936657818628, 75.65262939471883] Fitness: 1.4139054846545809
Generation 8 Best Solution: [183.60868357595066, 177.0554337075307,
174.96585334484092, 75.3611554943012] Fitness: 1.4243244717363082
Generation 9 Best Solution: [183.87744006655052, 175.69701412155942,
174.96585334484092, 75.3611554943012] Fitness: 1.363238971584415
Generation 10 Best Solution: [183.06362455832735, 175.69701412155942,
174.96585334484092, 75.3611554943012] Fitness: 1.110306357858187
Generation 11 Best Solution: [183.06362455832735, 175.69701412155942,
174.96585334484092, 75.3611554943012] Fitness: 1.110306357858187
Generation 12 Best Solution: [183.86433944345876, 177.0554337075307,
173.78936657818628, 75.3611554943012] Fitness: 1.2358756186658888
Generation 13 Best Solution: [183.87744006655052, 177.20525355149104,
174.921735839756, 74.60856285216634] Fitness: 1.4250362768301816
Generation 14 Best Solution: [183.87744006655052, 177.20525355149104,
174.0717730025565, 74.60856285216634] Fitness: 1.0058927160084168
Generation 15 Best Solution: [183.86433944345876, 176.44483691302412,
174.0717730025565, 75.85007508327558] Fitness: 1.353852294225868
Generation 16 Best Solution: [183.50772298630366, 177.0554337075307,
174.05565157840488, 75.85007508327558] Fitness: 1.5871129947008606
Generation 17 Best Solution: [183.50772298630366, 177.0554337075307,
174.05565157840488, 75.85007508327558] Fitness: 1.5871129947008606
Generation 18 Best Solution: [183.87744006655052, 177.20525355149104,
174.62050699729247, 75.22319276211938] Fitness: 1.3656838503561968
Generation 19 Best Solution: [183.87744006655052, 177.415100631462,
174.05565157840488, 75.1760953784218] Fitness: 1.5184903129114566
Generation 20 Best Solution: [183.87744006655052, 177.415100631462,
174.05565157840488, 75.1760953784218] Fitness: 1.5184903129114566
Generation 21 Best Solution: [183.87744006655052, 177.415100631462,
174.05565157840488, 75.1760953784218] Fitness: 1.5184903129114566
Generation 22 Best Solution: [183.50772298630366, 176.73791946391904,
174.96585334484092, 74.96390299667335] Fitness: 0.7875624396200889
Generation 23 Best Solution: [183.87744006655052, 176.73791946391904,
174.96585334484092, 74.96390299667335] Fitness: 1.0256130225978877
Generation 24 Best Solution: [183.66466597922144, 177.20525355149104,
174.96585334484092, 74.96390299667335] Fitness: 1.3684669223512864
Generation 25 Best Solution: [183.66466597922144, 176.87019887202877,
174.61508948156364, 75.83413729190654] Fitness: 1.5782116278409108
Generation 26 Best Solution: [183.66466597922144, 176.87019887202877,
174.61508948156364, 75.85007508327558] Fitness: 1.605882172251074
Generation 27 Best Solution: [183.2931155209386, 176.87019887202877,
174.61508948156364, 75.85007508327558] Fitness: 1.4857464867833268

Generation 28 Best Solution: [183.24903779129292, 176.87019887202877,
174.61508948156364, 75.85007508327558] Fitness: 1.485234524651081
Generation 29 Best Solution: [183.87744006655052, 177.20525355149104,
173.82717153532658, 75.24967666701824] Fitness: 1.3211544039162424
Generation 30 Best Solution: [183.24903779129292, 176.87019887202877,
174.61508948156364, 75.85007508327558] Fitness: 1.485234524651081
Generation 31 Best Solution: [183.87744006655052, 177.20525355149104,
173.82717153532658, 75.85007508327558] Fitness: 2.1003597616704646
Generation 32 Best Solution: [183.87744006655052, 176.4726714140701,
173.82717153532658, 75.85007508327558] Fitness: 1.4546298342011124
Generation 33 Best Solution: [183.87744006655052, 177.20525355149104,
174.96585334484092, 74.71569810835534] Fitness: 1.4727538294798947
Generation 34 Best Solution: [183.87744006655052, 177.20525355149104,
174.96585334484092, 74.71569810835534] Fitness: 1.4727538294798947
Generation 35 Best Solution: [183.87744006655052, 177.20525355149104,
174.96585334484092, 74.61251808999867] Fitness: 1.475935679536349
Generation 36 Best Solution: [183.87744006655052, 177.20525355149104,
174.96585334484092, 74.88585030056967] Fitness: 1.5023877576127513
Generation 37 Best Solution: [183.87744006655052, 177.20525355149104,
174.96585334484092, 74.65468777094632] Fitness: 1.4727056673413945
Generation 38 Best Solution: [183.1720432042933, 177.20525355149104,
174.96585334484092, 75.83413729190654] Fitness: 2.1863987613250493
Generation 39 Best Solution: [183.87744006655052, 177.20525355149104,
174.04180868089037, 75.85007508327558] Fitness: 2.0251799166857074
Generation 40 Best Solution: [183.87744006655052, 175.9631454619421,
174.96585334484092, 75.85007508327558] Fitness: 1.87048054978235
Generation 41 Best Solution: [183.87744006655052, 176.00631538753512,
174.96585334484092, 75.83413729190654] Fitness: 1.8255461637818113
Generation 42 Best Solution: [183.87744006655052, 177.20525355149104,
174.96585334484092, 74.62951311949467] Fitness: 1.4743130520241758
Generation 43 Best Solution: [183.87744006655052, 177.415100631462,
174.96585334484092, 74.81608928877763] Fitness: 1.818278452089558
Generation 44 Best Solution: [183.87744006655052, 177.415100631462,
174.96585334484092, 74.81608928877763] Fitness: 1.818278452089558
Generation 45 Best Solution: [183.87744006655052, 177.415100631462,
174.96585334484092, 74.81608928877763] Fitness: 1.818278452089558
Generation 46 Best Solution: [183.87744006655052, 176.72425440415242,
174.96585334484092, 74.81608928877763] Fitness: 0.9702534633627988
Generation 47 Best Solution: [183.87744006655052, 176.7118519896105,
174.96585334484092, 75.83413729190654] Fitness: 1.9395788930439177
Generation 48 Best Solution: [183.87744006655052, 176.7118519896105,
174.96585334484092, 75.83413729190654] Fitness: 1.9395788930439177
Generation 49 Best Solution: [183.87744006655052, 176.7118519896105,
174.96585334484092, 75.83413729190654] Fitness: 1.9395788930439177
Generation 50 Best Solution: [183.87744006655052, 175.6244403463099,
174.2393453208944, 75.85007508327558] Fitness: 1.6293276219587154
Generation 51 Best Solution: [183.87744006655052, 176.38237722445825,
174.96585334484092, 75.83413729190654] Fitness: 1.7934001945102982

Generation 52 Best Solution: [183.87744006655052, 177.415100631462,
174.1814011386815, 75.04408750118772] Fitness: 1.42491924808661
Generation 53 Best Solution: [183.87744006655052, 177.415100631462,
174.1814011386815, 74.97388309696025] Fitness: 1.3907663535345751
Generation 54 Best Solution: [183.87744006655052, 177.415100631462,
174.1814011386815, 74.97388309696025] Fitness: 1.3907663535345751
Generation 55 Best Solution: [183.4721920482438, 177.415100631462,
174.1814011386815, 75.04408750118772] Fitness: 1.174790410345135
Generation 56 Best Solution: [183.87744006655052, 177.415100631462,
174.0870936014446, 75.04408750118772] Fitness: 1.4296939391935441
Generation 57 Best Solution: [183.87744006655052, 177.415100631462,
174.1814011386815, 75.41329280184932] Fitness: 1.7262037795241665
Generation 58 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.56925079103438] Fitness: 1.589488223834877
Generation 59 Best Solution: [183.87744006655052, 177.415100631462,
174.1814011386815, 75.41329280184932] Fitness: 1.7262037795241665
Generation 60 Best Solution: [183.53905878258374, 176.2776468213923,
174.96585334484092, 75.85007508327558] Fitness: 1.5828837452962001
Generation 61 Best Solution: [183.87744006655052, 177.415100631462,
174.96585334484092, 74.49383951110991] Fitness: 1.8326357753610765
Generation 62 Best Solution: [183.87744006655052, 176.69389766250904,
174.96585334484092, 74.48149589911061] Fitness: 0.9674149394284509
Generation 63 Best Solution: [183.87744006655052, 177.415100631462,
174.96585334484092, 74.48149589911061] Fitness: 1.8362832934796518
Generation 64 Best Solution: [183.87744006655052, 177.415100631462,
174.96585334484092, 74.55289978182095] Fitness: 1.8183464068441224
Generation 65 Best Solution: [183.87744006655052, 176.69389766250904,
174.96585334484092, 75.85007508327558] Fitness: 1.9550887519636502
Generation 66 Best Solution: [183.87744006655052, 176.41569962883716,
174.96585334484092, 75.85007508327558] Fitness: 1.82845357219283
Generation 67 Best Solution: [183.87744006655052, 177.415100631462,
174.96585334484092, 75.0171713045825] Fitness: 1.8882440235643247
Generation 68 Best Solution: [183.53905878258374, 176.134124487865,
173.90540601455044, 75.85007508327558] Fitness: 1.1669571116381738
Generation 69 Best Solution: [183.53905878258374, 176.134124487865,
173.90540601455044, 75.83413729190654] Fitness: 1.1392865672280106
Generation 70 Best Solution: [183.53905878258374, 176.134124487865,
173.90540601455044, 75.83413729190654] Fitness: 1.1392865672280106
Generation 71 Best Solution: [183.53905878258374, 177.415100631462,
174.30602846197223, 74.857763031579] Fitness: 1.138830791563729
Generation 72 Best Solution: [183.87744006655052, 175.84489006315366,
174.78599468937935, 75.17828433806623] Fitness: 0.9052102899570479
Generation 73 Best Solution: [183.87744006655052, 177.18851922874273,
174.96585334484092, 74.54353621526859] Fitness: 1.4632351848920098
Generation 74 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.58869847956645] Fitness: 1.5864050252753785
Generation 75 Best Solution: [183.53905878258374, 176.14624861578037,
174.96585334484092, 75.85007508327558] Fitness: 1.5906465439532196

Generation 76 Best Solution: [183.87744006655052, 176.68214791902815,
174.96585334484092, 74.58869847956645] Fitness: 0.9356671048794308
Generation 77 Best Solution: [183.87744006655052, 177.415100631462,
174.96585334484092, 74.40530290005867] Fitness: 1.863857608457719
Generation 78 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.40530290005867] Fitness: 1.6380306911439548
Generation 79 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.87288910705072] Fitness: 1.6060683708720007
Generation 80 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.87288910705072] Fitness: 1.6060683708720007
Generation 81 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.87288910705072] Fitness: 1.6060683708720007
Generation 82 Best Solution: [183.87744006655052, 177.415100631462,
174.96585334484092, 74.87288910705072] Fitness: 1.8318952881857649
Generation 83 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.87288910705072] Fitness: 1.6060683708720007
Generation 84 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.87288910705072] Fitness: 1.6060683708720007
Generation 85 Best Solution: [183.53905878258374, 176.38305463286736,
174.96585334484092, 75.85007508327558] Fitness: 1.5953773211190265
Generation 86 Best Solution: [183.87744006655052, 176.23336205442175,
174.96585334484092, 75.85007508327558] Fitness: 1.808433588018881
Generation 87 Best Solution: [183.4207859849447, 176.10828625192403,
174.96585334484092, 75.85007508327558] Fitness: 1.5592864067819538
Generation 88 Best Solution: [183.87744006655052, 176.23336205442175,
174.96585334484092, 75.85007508327558] Fitness: 1.808433588018881
Generation 89 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.4116649192501] Fitness: 1.6353950711432743
Generation 90 Best Solution: [183.87744006655052, 176.23336205442175,
174.96585334484092, 75.24956088316812] Fitness: 1.0291301117692886
Generation 91 Best Solution: [183.87744006655052, 177.415100631462,
174.96585334484092, 74.60190876551145] Fitness: 1.8104611755403774
Generation 92 Best Solution: [182.89741479859012, 176.03927309643984,
174.96585334484092, 75.85007508327558] Fitness: 1.659495491637529
Generation 93 Best Solution: [183.87744006655052, 177.415100631462,
173.84675003999175, 75.85007508327558] Fitness: 2.423930788713444
Generation 94 Best Solution: [182.9183757296914, 177.415100631462,
174.04143157697484, 75.83413729190654] Fitness: 2.137275267375372
Generation 95 Best Solution: [183.41820738573952, 177.415100631462,
174.96585334484092, 75.83413729190654] Fitness: 2.531430730849206
Generation 96 Best Solution: [183.54563370488248, 177.415100631462,
174.96585334484092, 74.9351611858112] Fitness: 1.6293099659743258
Generation 97 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.92070273009229] Fitness: 1.621282373002001
Generation 98 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.61556883161141] Fitness: 1.5830784886808937
Generation 99 Best Solution: [183.53905878258374, 177.3644185901486,
174.96585334484092, 74.92070273009229] Fitness: 1.5347357215613706

Generation 100 Best Solution: [183.53905878258374, 177.415100631462,
174.4964406522015, 75.83413729190654] Fitness: 2.173937120899448
Generation 101 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.58649816894282] Fitness: 1.5867253960329182
Generation 102 Best Solution: [183.53905878258374, 177.415100631462,
174.42029412306456, 75.85007508327558] Fitness: 2.1684419621953896
Generation 103 Best Solution: [183.14814139578397, 177.415100631462,
174.38878117466075, 75.85007508327558] Fitness: 2.110194458443163
Generation 104 Best Solution: [183.53905878258374, 176.47829916628044,
174.96585334484092, 75.83413729190654] Fitness: 1.593329018139693
Generation 105 Best Solution: [183.14814139578397, 176.47829916628044,
174.96585334484092, 75.83413729190654] Fitness: 1.5462625124737195
Generation 106 Best Solution: [183.53905878258374, 177.415100631462,
174.4964406522015, 74.58943478790718] Fitness: 1.1897769152314712
Generation 107 Best Solution: [183.53905878258374, 177.415100631462,
174.4964406522015, 74.58943478790718] Fitness: 1.1897769152314712
Generation 108 Best Solution: [183.53905878258374, 177.415100631462,
174.4964406522015, 74.58943478790718] Fitness: 1.1897769152314712
Generation 109 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.67031262559533] Fitness: 1.57965213444521
Generation 110 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.67031262559533] Fitness: 1.57965213444521
Generation 111 Best Solution: [183.3875600062093, 177.415100631462,
174.96585334484092, 74.73963896170703] Fitness: 1.5363198776575255
Generation 112 Best Solution: [183.53905878258374, 176.5864811860192,
174.96585334484092, 74.76007755431439] Fitness: 0.6522400262804652
Generation 113 Best Solution: [183.30926401793678, 177.415100631462,
174.96585334484092, 74.76007755431439] Fitness: 1.528326535373617
Generation 114 Best Solution: [183.61068790712116, 177.2769476481717,
174.96585334484092, 74.57223658167281] Fitness: 1.3955988049636985
Generation 115 Best Solution: [183.10236352721915, 177.415100631462,
174.96585334484092, 74.76007755431439] Fitness: 1.5461777733928361
Generation 116 Best Solution: [183.53905878258374, 177.2769476481717,
174.96585334484092, 74.9091802551324] Fitness: 1.3904501680839858
Generation 117 Best Solution: [183.3875600062093, 175.96432488291072,
174.96585334484092, 75.83413729190654] Fitness: 1.5710300962732808
Generation 118 Best Solution: [183.53905878258374, 176.3022900316447,
174.96585334484092, 75.83413729190654] Fitness: 1.5566413399727619
Generation 119 Best Solution: [183.3875600062093, 175.96432488291072,
174.96585334484092, 74.43507018405354] Fitness: 0.6267918606558897
Generation 120 Best Solution: [183.30926401793678, 175.96432488291072,
174.96585334484092, 74.43507018405354] Fitness: 0.616799881806621
Generation 121 Best Solution: [183.30926401793678, 175.66282483035192,
174.96585334484092, 75.83413729190654] Fitness: 1.759014423024586
Generation 122 Best Solution: [183.30926401793678, 175.97958112891297,
174.96585334484092, 75.83413729190654] Fitness: 1.5546446749190141
Generation 123 Best Solution: [183.30926401793678, 175.97958112891297,
174.96585334484092, 75.83413729190654] Fitness: 1.5546446749190141

Generation 124 Best Solution: [183.30926401793678, 175.66282483035192,
174.0642642712784, 75.83413729190654] Fitness: 1.289657779179251
Generation 125 Best Solution: [183.53905878258374, 176.2652782649264,
174.96585334484092, 75.83413729190654] Fitness: 1.5548397469835167
Generation 126 Best Solution: [183.53905878258374, 176.2652782649264,
174.96585334484092, 75.83413729190654] Fitness: 1.5548397469835167
Generation 127 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.41856433917634] Fitness: 1.6326054420957261
Generation 128 Best Solution: [183.53905878258374, 175.66282483035192,
174.0642642712784, 75.83413729190654] Fitness: 1.3450939508613997
Generation 129 Best Solution: [183.53905878258374, 175.66282483035192,
174.0642642712784, 75.83413729190654] Fitness: 1.3450939508613997
Generation 130 Best Solution: [183.53905878258374, 177.415100631462,
174.6645320600069, 75.18017639489435] Fitness: 1.471125618705746
Generation 131 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.47458121845408] Fitness: 1.6125995272369142
Generation 132 Best Solution: [183.87744006655052, 177.415100631462,
173.97431777252837, 74.54370723527623] Fitness: 1.3709338278844965
Generation 133 Best Solution: [183.53905878258374, 177.415100631462,
173.97431777252837, 74.54370723527623] Fitness: 1.1451069105707323
Generation 134 Best Solution: [183.87744006655052, 176.4087692800457,
174.96585334484092, 75.83413729190654] Fitness: 1.7991103764473435
Generation 135 Best Solution: [183.53905878258374, 176.4087692800457,
174.96585334484092, 75.83413729190654] Fitness: 1.5732834591335796
Generation 136 Best Solution: [183.53905878258374, 177.415100631462,
173.83032057999745, 75.83413729190654] Fitness: 2.1785527174054833
Generation 137 Best Solution: [183.53905878258374, 177.415100631462,
174.85190996714402, 74.98410154033955] Fitness: 1.520115700716345
Generation 138 Best Solution: [183.53905878258374, 176.8744202126465,
174.96585334484092, 75.83413729190654] Fitness: 1.8458718455248253
Generation 139 Best Solution: [183.4447627153796, 177.415100631462,
174.85190996714402, 75.83413729190654] Fitness: 2.411500879733756
Generation 140 Best Solution: [183.87744006655052, 177.415100631462,
174.15761658599982, 75.83413729190654] Fitness: 2.3189399531640267
Generation 141 Best Solution: [183.02615008993163, 177.415100631462,
174.15761658599982, 75.83413729190654] Fitness: 2.0782865615399175
Generation 142 Best Solution: [183.4752175830343, 176.94877162116057,
175.0469891896705, 75.83413729190654] Fitness: 1.9982008608342876
Generation 143 Best Solution: [183.4752175830343, 177.415100631462,
174.96585334484092, 74.9882517025911] Fitness: 1.6252733200320841
Generation 144 Best Solution: [183.87744006655052, 177.415100631462,
174.85190996714402, 74.75353513588497] Fitness: 1.68225383489084
Generation 145 Best Solution: [183.87744006655052, 177.415100631462,
174.85190996714402, 75.07849301372177] Fitness: 1.7950210790775614
Generation 146 Best Solution: [183.35658755609256, 177.415100631462,
173.7610983242956, 75.83413729190654] Fitness: 2.16671277467739
Generation 147 Best Solution: [183.53905878258374, 176.01920351692874,
175.0469891896705, 75.83413729190654] Fitness: 1.6971461166565205

Generation 148 Best Solution: [183.87744006655052, 176.47840898895748,
173.96005625843068, 75.83413729190654] Fitness: 1.3741963133302806
Generation 149 Best Solution: [183.87744006655052, 176.7579748803448,
175.0469891896705, 75.83413729190654] Fitness: 2.0750740880908785
Generation 150 Best Solution: [183.87744006655052, 176.7579748803448,
174.85190996714402, 75.83413729190654] Fitness: 1.8464072237918665
Generation 151 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 74.98231717494387] Fitness: 1.645944474382163
Generation 152 Best Solution: [183.87744006655052, 177.415100631462,
174.32336885381235, 75.17397613004033] Fitness: 1.5255694767437735
Generation 153 Best Solution: [183.87744006655052, 176.31561444732205,
174.32336885381235, 75.17397613004033] Fitness: 0.512902779810237
Generation 154 Best Solution: [183.87744006655052, 177.415100631462,
174.32336885381235, 75.17397613004033] Fitness: 1.5255694767437735
Generation 155 Best Solution: [183.87744006655052, 177.415100631462,
174.85190996714402, 74.59889337957861] Fitness: 1.6842144779349018
Generation 156 Best Solution: [183.87744006655052, 175.8171565476898,
174.42959217296726, 75.83413729190654] Fitness: 1.4957983081661652
Generation 157 Best Solution: [183.87744006655052, 175.8171565476898,
174.42959217296726, 75.83413729190654] Fitness: 1.4957983081661652
Generation 158 Best Solution: [183.87744006655052, 177.415100631462,
174.50881685006746, 74.96833574545367] Fitness: 1.4753652566446125
Generation 159 Best Solution: [183.53905878258374, 177.415100631462,
174.50881685006746, 74.96833574545367] Fitness: 1.2495383393308483
Generation 160 Best Solution: [183.87744006655052, 176.39428763222583,
174.42959217296726, 75.83413729190654] Fitness: 1.3697431459512257
Generation 161 Best Solution: [183.1311863569122, 177.415100631462,
174.96585334484092, 74.5141382554901] Fitness: 1.557386284847385
Generation 162 Best Solution: [183.05833495824663, 177.415100631462,
173.88169900632053, 74.5141382554901] Fitness: 1.1598561419865914
Generation 163 Best Solution: [182.96183690246406, 176.04887994644486,
174.96585334484092, 75.83413729190654] Fitness: 1.5965911123895542
Generation 164 Best Solution: [182.96183690246406, 176.04887994644486,
174.96585334484092, 75.83413729190654] Fitness: 1.5965911123895542
Generation 165 Best Solution: [183.87744006655052, 176.04887994644486,
174.96585334484092, 75.83413729190654] Fitness: 1.8112613488563922
Generation 166 Best Solution: [183.87744006655052, 176.04887994644486,
174.96585334484092, 75.83413729190654] Fitness: 1.8112613488563922
Generation 167 Best Solution: [182.96183690246406, 177.415100631462,
174.85190996714402, 74.43014918490037] Fitness: 1.5126108102781373
Generation 168 Best Solution: [183.53905878258374, 177.28758417656746,
174.96585334484092, 74.74855751680049] Fitness: 1.3721552505543382
Generation 169 Best Solution: [182.96183690246406, 177.415100631462,
174.7755478572644, 74.74855751680049] Fitness: 1.393124146213481
Generation 170 Best Solution: [183.53905878258374, 177.415100631462,
174.96585334484092, 75.13551652801212] Fitness: 1.7319468036803143
Generation 171 Best Solution: [183.19970755965284, 177.28758417656746,
174.37937065017283, 75.83413729190654] Fitness: 1.8621542394177963

Generation 172 Best Solution: [183.41619105341564, 177.26249660074086,
174.85190996714402, 75.30056417463865] Fitness: 1.448945414571052
Generation 173 Best Solution: [183.41619105341564, 177.26249660074086,
174.85190996714402, 75.30056417463865] Fitness: 1.448945414571052
Generation 174 Best Solution: [183.19970755965284, 176.7032006980925,
174.96585334484092, 75.83413729190654] Fitness: 1.6538497744305343
Generation 175 Best Solution: [183.19970755965284, 176.7032006980925,
174.96585334484092, 75.79871150841711] Fitness: 1.593709667786603
Generation 176 Best Solution: [183.19970755965284, 176.7032006980925,
174.0365682380043, 75.83413729190654] Fitness: 1.1893780355711208
Generation 177 Best Solution: [183.87744006655052, 176.28837661987785,
174.37937065017283, 75.83413729190654] Fitness: 1.3376306661784216
Generation 178 Best Solution: [183.87744006655052, 176.28837661987785,
174.37937065017283, 75.55253254943055] Fitness: 0.9115620203105583
Generation 179 Best Solution: [183.87744006655052, 176.28837661987785,
174.37937065017283, 75.83413729190654] Fitness: 1.3376306661784216
Generation 180 Best Solution: [183.87744006655052, 176.28837661987785,
174.37937065017283, 75.55253254943055] Fitness: 0.9115620203105583
Generation 181 Best Solution: [183.41619105341564, 176.28837661987785,
174.96585334484092, 75.83413729190654] Fitness: 1.5162287743724974
Generation 182 Best Solution: [183.87744006655052, 176.09758720333713,
174.96585334484092, 75.55253254943055] Fitness: 1.3721805769581747
Generation 183 Best Solution: [183.87744006655052, 176.24613240354876,
174.96585334484092, 75.83413729190654] Fitness: 1.7805411057123974
Generation 184 Best Solution: [183.87744006655052, 176.28837661987785,
174.96585334484092, 75.83413729190654] Fitness: 1.7815499722679435
Generation 185 Best Solution: [183.87744006655052, 176.28837661987785,
174.96585334484092, 75.83413729190654] Fitness: 1.7815499722679435
Generation 186 Best Solution: [183.87744006655052, 176.28837661987785,
174.96585334484092, 75.83413729190654] Fitness: 1.7815499722679435
Generation 187 Best Solution: [183.41619105341564, 176.95985750691813,
174.37937065017283, 75.83413729190654] Fitness: 1.447785192889967
Generation 188 Best Solution: [183.41619105341564, 176.28837661987785,
174.96585334484092, 75.83413729190654] Fitness: 1.5162287743724974
Generation 189 Best Solution: [183.41619105341564, 177.26249660074086,
174.37937065017283, 75.83413729190654] Fitness: 1.8381186794901536
Generation 190 Best Solution: [183.41619105341564, 177.26249660074086,
174.37937065017283, 75.83413729190654] Fitness: 1.8381186794901536
Generation 191 Best Solution: [183.41619105341564, 176.96922729456523,
174.72156813179348, 75.83413729190654] Fitness: 1.654129212741498
Generation 192 Best Solution: [183.41619105341564, 176.96922729456523,
174.72156813179348, 75.83413729190654] Fitness: 1.654129212741498
Generation 193 Best Solution: [183.87744006655052, 177.26249660074086,
174.72156813179348, 75.30611230044187] Fitness: 1.5984439390594019
Generation 194 Best Solution: [183.41619105341564, 175.61096327447433,
174.37937065017283, 75.83413729190654] Fitness: 1.3788356736867786
Generation 195 Best Solution: [183.41619105341564, 176.31262000355395,
174.37937065017283, 75.83413729190654] Fitness: 1.0740973564345082

Generation 196 Best Solution: [183.41619105341564, 176.31262000355395,
174.37937065017283, 75.83413729190654] Fitness: 1.0740973564345082
Generation 197 Best Solution: [183.87744006655052, 177.26249660074086,
174.96585334484092, 74.77876862383309] Fitness: 1.5630385285150137
Generation 198 Best Solution: [183.41619105341564, 176.31262000355395,
174.96585334484092, 75.83413729190654] Fitness: 1.5180166625240303
Generation 199 Best Solution: [183.1641548653307, 176.0319773266562,
174.96585334484092, 75.83413729190654] Fitness: 1.5411405782579273
Generation 200 Best Solution: [183.87744006655052, 175.61096327447433,
174.96585334484092, 74.90077513652567] Fitness: 1.1321412920133247
Generation 201 Best Solution: [183.1641548653307, 177.26249660074086,
174.96585334484092, 75.12758167451328] Fitness: 1.4280593894394202
Generation 202 Best Solution: [183.87744006655052, 176.66722433791216,
174.66783063959215, 75.83413729190654] Fitness: 1.6201903455275555
Generation 203 Best Solution: [183.87744006655052, 177.26249660074086,
174.96585334484092, 75.12758167451328] Fitness: 1.7035274228899269
Generation 204 Best Solution: [183.3547999046008, 175.61096327447433,
174.96585334484092, 75.83413729190654] Fitness: 1.8115054680792997
Generation 205 Best Solution: [183.3547999046008, 177.33127020330554,
174.96585334484092, 75.12758167451328] Fitness: 1.5348160806257984
Generation 206 Best Solution: [183.87744006655052, 176.66722433791216,
174.96585334484092, 75.83413729190654] Fitness: 1.9102448142266155
Generation 207 Best Solution: [183.87744006655052, 176.01567083622513,
174.96585334484092, 75.83413729190654] Fitness: 1.8221734278668875
Generation 208 Best Solution: [183.87744006655052, 177.26249660074086,
174.96585334484092, 74.4996218843443] Fitness: 1.5820783233566498
Generation 209 Best Solution: [183.87744006655052, 177.17206609686372,
175.0698544335836, 75.04017997287745] Fitness: 1.6527412739162297
Generation 210 Best Solution: [183.87744006655052, 177.26249660074086,
175.0698544335836, 75.04017997287745] Fitness: 1.7837680225532067
Generation 211 Best Solution: [183.87744006655052, 176.01034077448415,
175.0698544335836, 75.83413729190654] Fitness: 1.9566574731503499
Generation 212 Best Solution: [183.87744006655052, 176.28733067767226,
175.0698544335836, 75.83413729190654] Fitness: 1.9140712696844315
Generation 213 Best Solution: [183.87744006655052, 177.26249660074086,
175.0698544335836, 74.35453380685681] Fitness: 1.7707165182281863
Generation 214 Best Solution: [183.87744006655052, 177.26249660074086,
175.0698544335836, 74.35453380685681] Fitness: 1.7707165182281863
Generation 215 Best Solution: [183.87744006655052, 177.26249660074086,
174.4885345302734, 74.35453380685681] Fitness: 1.2377672425205801
Generation 216 Best Solution: [183.87744006655052, 177.26249660074086,
174.4885345302734, 74.35453380685681] Fitness: 1.2377672425205801
Generation 217 Best Solution: [183.87744006655052, 177.34068996244835,
174.4885345302734, 74.35453380685681] Fitness: 1.3609522765525066
Generation 218 Best Solution: [183.87744006655052, 177.26249660074086,
174.25202710528137, 75.13519607101831] Fitness: 1.236497648808279
Generation 219 Best Solution: [183.87744006655052, 177.26249660074086,
174.25202710528137, 75.13519607101831] Fitness: 1.236497648808279

Generation 220 Best Solution: [183.87744006655052, 176.27611434106564,
175.0698544335836, 75.35683283530989] Fitness: 1.2614534668273738
Generation 221 Best Solution: [183.87744006655052, 176.27611434106564,
175.0698544335836, 74.36072532396246] Fitness: 1.0013014432607428
Generation 222 Best Solution: [183.87744006655052, 176.27611434106564,
175.0698544335836, 75.35683283530989] Fitness: 1.2614534668273738
Generation 223 Best Solution: [183.11510622098643, 177.26249660074086,
174.05360155442062, 75.83413729190654] Fitness: 1.813380114621974
Generation 224 Best Solution: [183.87744006655052, 176.58469313380692,
174.18038675349032, 75.83413729190654] Fitness: 1.3865549897294074
Generation 225 Best Solution: [183.48257695972765, 177.26249660074086,
173.79804029599367, 75.32507593361674] Fitness: 1.2424208496395892
Generation 226 Best Solution: [183.87744006655052, 177.26249660074086,
174.88154180151702, 74.4171679887577] Fitness: 1.5144927587491088
Generation 227 Best Solution: [183.87744006655052, 177.26249660074086,
174.88154180151702, 74.41710152565572] Fitness: 1.5145194302499783
Generation 228 Best Solution: [183.87744006655052, 177.26249660074086,
173.8391466748301, 74.4171679887577] Fitness: 1.2137447139714777
Generation 229 Best Solution: [183.87744006655052, 177.26249660074086,
173.8391466748301, 74.4171679887577] Fitness: 1.2137447139714777
Generation 230 Best Solution: [183.87744006655052, 177.26249660074086,
175.01100410752773, 74.61039468847983] Fitness: 1.6160988638920148
Generation 231 Best Solution: [183.23143239113864, 177.26249660074086,
175.0698544335836, 74.60391026184917] Fitness: 1.4117848798204586
Generation 232 Best Solution: [183.87744006655052, 177.26249660074086,
175.01100410752773, 74.4171679887577] Fitness: 1.665628275846459
Generation 233 Best Solution: [183.87744006655052, 177.26249660074086,
175.01100410752773, 74.4171679887577] Fitness: 1.665628275846459
Generation 234 Best Solution: [183.87744006655052, 176.68314265839896,
175.0698544335836, 75.83413729190654] Fitness: 2.052943882995064
Generation 235 Best Solution: [183.87744006655052, 176.09759543876552,
175.01100410752773, 75.83413729190654] Fitness: 1.8538117977051134
Generation 236 Best Solution: [183.87744006655052, 176.09759543876552,
175.01100410752773, 75.83413729190654] Fitness: 1.8538117977051134
Generation 237 Best Solution: [183.30900608541032, 176.49103495820935,
175.0698544335836, 75.83413729190654] Fitness: 1.6749112551856875
Generation 238 Best Solution: [183.87744006655052, 176.09759543876552,
175.0698544335836, 74.99874349724867] Fitness: 1.0138470711687548
Generation 239 Best Solution: [183.30900608541032, 177.26249660074086,
175.01100410752773, 74.53817774543407] Fitness: 1.3467750622417873
Generation 240 Best Solution: [183.87744006655052, 177.26249660074086,
174.47177724765973, 75.83413729190654] Fitness: 2.1391421757165268
Generation 241 Best Solution: [183.87744006655052, 177.26249660074086,
174.32913183867583, 75.83413729190654] Fitness: 2.0894043156460276
Generation 242 Best Solution: [183.04909646389657, 177.26249660074086,
174.0000987000867, 75.63433626514438] Fitness: 1.5280948367742966
Generation 243 Best Solution: [183.87744006655052, 177.26249660074086,
174.47177724765973, 75.63433626514438] Fitness: 1.8245843967041322

Generation 244 Best Solution: [183.87744006655052, 177.26249660074086,
175.10020050051838, 75.19417379911786] Fitness: 1.92541614324174
Generation 245 Best Solution: [183.87744006655052, 177.26249660074086,
174.0768976881302, 75.53822880339001] Fitness: 1.6316206392095285
Generation 246 Best Solution: [183.87744006655052, 176.43379043104466,
175.10020050051838, 75.19417379911786] Fitness: 1.1835457226049755
Generation 247 Best Solution: [183.59575363040094, 177.26249660074086,
174.68482345732744, 75.19417379911786] Fitness: 1.2740298084894812
Generation 248 Best Solution: [183.87744006655052, 175.58719032615673,
175.0698544335836, 75.37315034688797] Fitness: 1.608459158302314
Generation 249 Best Solution: [183.87744006655052, 175.94358658387935,
175.0698544335836, 75.83413729190654] Fitness: 1.984130459434461
Generation 250 Best Solution: [183.87744006655052, 177.26249660074086,
175.0698544335836, 74.4375111587983] Fitness: 1.7347901231857616
Generation 251 Best Solution: [183.79846121799528, 175.94358658387935,
175.0698544335836, 74.4375111587983] Fitness: 0.9709088843897543
Generation 252 Best Solution: [183.59575363040094, 177.26249660074086,
175.0698544335836, 74.4375111587983] Fitness: 1.5348222537756298
Generation 253 Best Solution: [183.87744006655052, 176.4230258033319,
174.3025488849034, 75.83413729190654] Fitness: 1.3387796285662603
Generation 254 Best Solution: [183.59575363040094, 175.94358658387935,
175.01100410752773, 75.83413729190654] Fitness: 1.707148469474994
Generation 255 Best Solution: [183.25477941940736, 175.97965564926494,
175.0698544335836, 75.83413729190654] Fitness: 1.6856655533459861
Generation 256 Best Solution: [183.00919489157107, 175.97965564926494,
175.0698544335836, 75.83413729190654] Fitness: 1.7340504369494072
Generation 257 Best Solution: [183.00919489157107, 175.94358658387935,
175.0698544335836, 75.78586332232132] Fitness: 1.6682384273851802
Generation 258 Best Solution: [183.59575363040094, 177.26249660074086,
175.10017089997945, 74.60843255349926] Fitness: 1.5350674755267648
Generation 259 Best Solution: [182.93970574601755, 176.3625686327819,
175.0698544335836, 75.83413729190654] Fitness: 1.7180851835842494
Generation 260 Best Solution: [183.59575363040094, 176.3625686327819,
175.0698544335836, 75.83413729190654] Fitness: 1.7224115125572648
Generation 261 Best Solution: [183.1328726176758, 177.26249660074086,
175.01100410752773, 75.34761377279152] Fitness: 1.6715054568840553
Generation 262 Best Solution: [183.1856599121597, 175.94358658387935,
175.01100410752773, 75.83413729190654] Fitness: 1.6287958993157172
Generation 263 Best Solution: [183.00919489157107, 175.94358658387935,
175.01100410752773, 75.83413729190654] Fitness: 1.6727107897341176
Generation 264 Best Solution: [183.59575363040094, 177.26249660074086,
175.01100410752773, 74.61153395499764] Fitness: 1.4160050445610923
Generation 265 Best Solution: [183.79846121799528, 177.4328741195285,
175.109091482039, 74.79657970107604] Fitness: 1.9647196404168772
Generation 266 Best Solution: [183.87744006655052, 177.15599769653514,
175.109091482039, 74.72863675132658] Fitness: 1.5916344268060185
Generation 267 Best Solution: [183.59575363040094, 177.26249660074086,
175.0698544335836, 74.72863675132658] Fitness: 1.490457896232129

Generation 268 Best Solution: [183.87744006655052, 177.26249660074086,
174.260810381142, 75.83413729190654] Fitness: 2.076392038705528
Generation 269 Best Solution: [182.9622198467914, 176.82988473055602,
175.0698544335836, 75.83413729190654] Fitness: 1.9492973915021272
Generation 270 Best Solution: [183.87744006655052, 175.71019271359773,
175.10017089997945, 74.38745220827002] Fitness: 1.2497154533692703
Generation 271 Best Solution: [183.79846121799528, 177.26249660074086,
175.109091482039, 74.33119263246448] Fitness: 1.7688435575389116
Generation 272 Best Solution: [182.9622198467914, 176.82988473055602,
175.0698544335836, 75.83413729190654] Fitness: 1.9492973915021272
Generation 273 Best Solution: [182.9622198467914, 176.82988473055602,
175.0698544335836, 75.83413729190654] Fitness: 1.9492973915021272
Generation 274 Best Solution: [182.9622198467914, 177.4328741195285,
175.109091482039, 74.65899216353318] Fitness: 1.8090512391731908
Generation 275 Best Solution: [182.9622198467914, 176.82988473055602,
175.0698544335836, 75.83413729190654] Fitness: 1.9492973915021272
Generation 276 Best Solution: [182.9622198467914, 177.26249660074086,
174.6646226129829, 75.83413729190654] Fitness: 2.040063917360256
Generation 277 Best Solution: [183.87744006655052, 177.26249660074086,
174.53476583387237, 75.3095313473095] Fitness: 1.4726994593609433
Generation 278 Best Solution: [182.9622198467914, 177.26249660074086,
174.53476583387237, 75.3095313473095] Fitness: 1.2578561483569943
Generation 279 Best Solution: [182.9622198467914, 176.82988473055602,
175.0698544335836, 75.83413729190654] Fitness: 1.9492973915021272
Generation 280 Best Solution: [183.87744006655052, 176.20460043084736,
175.0698544335836, 75.83413729190654] Fitness: 1.914737386215913
Generation 281 Best Solution: [183.87744006655052, 176.20460043084736,
175.0698544335836, 75.83413729190654] Fitness: 1.914737386215913
Generation 282 Best Solution: [183.87744006655052, 176.82988473055602,
174.51694045673128, 75.83413729190654] Fitness: 1.6454542197331534
Generation 283 Best Solution: [183.87744006655052, 177.26249660074086,
174.43546606727529, 75.83413729190654] Fitness: 2.1235852864275166
Generation 284 Best Solution: [182.87199414210488, 177.26249660074086,
174.62329653083947, 75.83413729190654] Fitness: 2.057417373922058
Generation 285 Best Solution: [183.87744006655052, 176.30083076965624,
175.10017089997945, 75.83413729190654] Fitness: 1.9566377857823154
Generation 286 Best Solution: [183.87744006655052, 176.5515694714671,
175.0698544335836, 75.83413729190654] Fitness: 1.9807057170659488
Generation 287 Best Solution: [183.87744006655052, 177.26249660074086,
174.72312403860582, 75.83413729190654] Fitness: 2.3010537372482607
Generation 288 Best Solution: [183.65227026611174, 177.4328741195285,
175.10017089997945, 74.65611480602712] Fitness: 1.842087271977162
Generation 289 Best Solution: [183.87744006655052, 176.32676186869352,
175.10017089997945, 75.83413729190654] Fitness: 1.9590673865903474
Generation 290 Best Solution: [183.87744006655052, 176.32676186869352,
175.10017089997945, 75.83413729190654] Fitness: 1.9590673865903474
Generation 291 Best Solution: [183.87744006655052, 176.32676186869352,
175.10017089997945, 75.83413729190654] Fitness: 1.9590673865903474

Generation 292 Best Solution: [183.87744006655052, 177.4328741195285,
174.43546606727529, 75.45588848353881] Fitness: 1.858901172891486
Generation 293 Best Solution: [183.87744006655052, 177.26249660074086,
174.10569725314167, 75.83413729190654] Fitness: 2.0728429377693063
Generation 294 Best Solution: [183.87744006655052, 177.26249660074086,
173.92647135117858, 75.83413729190654] Fitness: 2.113683779720863
Generation 295 Best Solution: [183.4500348353602, 177.26249660074086,
173.92647135117858, 75.83413729190654] Fitness: 1.8569815580281652
Generation 296 Best Solution: [183.4500348353602, 177.22643600571405,
173.92647135117858, 75.83413729190654] Fitness: 1.8032621092754768
Generation 297 Best Solution: [183.87744006655052, 177.25016952427086,
173.92647135117858, 75.5412118413452] Fitness: 1.6543909054318984
Generation 298 Best Solution: [183.87744006655052, 177.4328741195285,
173.92647135117858, 74.41138234360794] Fitness: 1.4589255889923938
Generation 299 Best Solution: [182.9547671645653, 177.22643600571405,
173.92647135117858, 75.83413729190654] Fitness: 1.8485288315122839
Generation 300 Best Solution: [183.87744006655052, 177.22643600571405,
173.92647135117858, 75.83413729190654] Fitness: 2.059964330968174
Generation 301 Best Solution: [183.87744006655052, 176.58282618884144,
175.10017089997945, 75.83413729190654] Fitness: 2.0372159235700957
Generation 302 Best Solution: [183.87744006655052, 177.4328741195285,
175.10017089997945, 74.71566634946387] Fitness: 2.0115879572455313
Generation 303 Best Solution: [183.17423936914625, 177.22643600571405,
175.0698544335836, 75.57884520673493] Fitness: 1.9580307277736861
Generation 304 Best Solution: [183.87744006655052, 177.2560906039716,
175.10017089997945, 74.90638454411993] Fitness: 1.7578602424754914
Generation 305 Best Solution: [183.87744006655052, 175.91499545861996,
175.10017089997945, 75.57884520673493] Fitness: 1.648347070527382
Generation 306 Best Solution: [183.87744006655052, 177.2560906039716,
174.01754232148144, 75.03275341958692] Fitness: 1.1771356149501622
Generation 307 Best Solution: [183.87744006655052, 177.22643600571405,
175.10017089997945, 74.49987033835603] Fitness: 1.7025694283942097
Generation 308 Best Solution: [183.51198058953972, 177.2560906039716,
175.10017089997945, 74.49987033835603] Fitness: 1.510127175483728
Generation 309 Best Solution: [183.87744006655052, 176.40226442265956,
174.83766317789485, 75.83413729190654] Fitness: 1.6565150727191178
Generation 310 Best Solution: [183.87744006655052, 177.4328741195285,
175.0698544335836, 74.28056666081025] Fitness: 2.091638957499679
Generation 311 Best Solution: [183.87744006655052, 177.4328741195285,
175.0698544335836, 74.62180284720802] Fitness: 1.9721302412975124
Generation 312 Best Solution: [183.87744006655052, 177.4328741195285,
175.0698544335836, 74.28056666081025] Fitness: 2.091638957499679
Generation 313 Best Solution: [183.87744006655052, 176.4887455512773,
175.0698544335836, 75.83413729190654] Fitness: 1.9553728340066454
Generation 314 Best Solution: [183.87744006655052, 176.4887455512773,
175.0698544335836, 75.83413729190654] Fitness: 1.9553728340066454
Generation 315 Best Solution: [183.87744006655052, 176.4887455512773,
175.10017089997945, 75.83413729190654] Fitness: 1.9970737095522155

Generation 316 Best Solution: [182.88787080703796, 177.4328741195285,
175.10017089997945, 74.62676860458342] Fitness: 1.836265132842189
Generation 317 Best Solution: [183.87744006655052, 177.4328741195285,
175.0698544335836, 74.62676860458342] Fitness: 1.9716804856445618
Generation 318 Best Solution: [183.39917508446783, 177.4328741195285,
175.0698544335836, 74.62676860458342] Fitness: 1.7026749118210083
Generation 319 Best Solution: [183.87744006655052, 176.69692150032895,
175.10017089997945, 74.80422488854774] Fitness: 1.1234704871527967
Generation 320 Best Solution: [183.61663625909983, 177.2560906039716,
175.0698544335836, 74.62676860458342] Fitness: 1.4925763014172426
Generation 321 Best Solution: [183.87744006655052, 177.4328741195285,
175.10017089997945, 74.88532870465302] Fitness: 2.041066162112014
Generation 322 Best Solution: [183.87744006655052, 177.4328741195285,
175.10017089997945, 74.75940199357683] Fitness: 2.0150562488116623
Generation 323 Best Solution: [183.87744006655052, 177.12127018332626,
175.10017089997945, 74.75940199357683] Fitness: 1.5356198689613714
Generation 324 Best Solution: [183.87744006655052, 177.12127018332626,
175.10017089997945, 75.34530129044997] Fitness: 1.8587594217510754
Generation 325 Best Solution: [182.88234581368374, 176.05417512260112,
175.10017089997945, 75.83413729190654] Fitness: 1.8099721164342948
Generation 326 Best Solution: [182.88234581368374, 176.05417512260112,
175.10017089997945, 75.83413729190654] Fitness: 1.8099721164342948
Generation 327 Best Solution: [182.88234581368374, 176.05417512260112,
175.10017089997945, 75.83413729190654] Fitness: 1.8099721164342948
Generation 328 Best Solution: [182.88234581368374, 176.05417512260112,
175.10017089997945, 75.83413729190654] Fitness: 1.8099721164342948
Generation 329 Best Solution: [183.87744006655052, 176.05417512260112,
175.10017089997945, 75.83413729190654] Fitness: 1.9839538086747566
Generation 330 Best Solution: [183.29577264739214, 177.4328741195285,
174.62778983259741, 75.83413729190654] Fitness: 2.226621260241445
Generation 331 Best Solution: [183.65147984477778, 177.4328741195285,
174.04159739622898, 75.83413729190654] Fitness: 2.192253510238967
Generation 332 Best Solution: [183.29577264739214, 177.4328741195285,
174.04159739622898, 75.83413729190654] Fitness: 2.0800494563015466
Generation 333 Best Solution: [183.11411248513778, 177.4328741195285,
175.10017089997945, 75.83413729190654] Fitness: 2.735684097027537
Generation 334 Best Solution: [183.11411248513778, 176.48828030083035,
175.10017089997945, 75.83413729190654] Fitness: 1.7307630240000638
Generation 335 Best Solution: [183.87744006655052, 177.4328741195285,
174.29483562261836, 75.83413729190654] Fitness: 2.362187777009782
Generation 336 Best Solution: [183.7569032412279, 176.0112183267032,
175.10017089997945, 75.83413729190654] Fitness: 1.8979048315844238
Generation 337 Best Solution: [183.87744006655052, 176.77345256648124,
175.10017089997945, 75.83413729190654] Fitness: 2.159256658847625
Generation 338 Best Solution: [183.8781324748228, 177.4328741195285,
175.10017089997945, 74.73204682874265] Fitness: 2.013189060259264
Generation 339 Best Solution: [183.87744006655052, 177.4328741195285,
175.10017089997945, 74.73204682874265] Fitness: 2.0125508818141458

Generation 340 Best Solution: [183.87744006655052, 177.4328741195285,
175.10017089997945, 74.68822249631638] Fitness: 2.0108767049387586
Generation 341 Best Solution: [183.27682020215212, 177.40484577157392,
175.10017089997945, 74.68822249631638] Fitness: 1.6790820536244395
Generation 342 Best Solution: [183.8781324748228, 177.4328741195285,
175.10017089997945, 74.68822249631638] Fitness: 2.0115148833838767
Generation 343 Best Solution: [183.27682020215212, 177.40484577157392,
174.10841033625226, 75.83413729190654] Fitness: 2.0209908653408974
Generation 344 Best Solution: [183.27682020215212, 177.4328741195285,
173.75044935477288, 75.83413729190654] Fitness: 2.19817483851911
Generation 345 Best Solution: [183.27682020215212, 176.76144344711304,
173.75044935477288, 75.83413729190654] Fitness: 1.3463051125723666
Generation 346 Best Solution: [183.87744006655052, 175.879580404508,
175.10017089997945, 75.83413729190654] Fitness: 2.0584504686980702
Generation 347 Best Solution: [183.8781324748228, 177.4328741195285,
173.75044935477288, 75.83413729190654] Fitness: 2.4815219050783455
Generation 348 Best Solution: [183.7569032412279, 177.13970675651598,
175.10017089997945, 75.83413729190654] Fitness: 2.4465689235406067
Generation 349 Best Solution: [183.87744006655052, 177.4328741195285,
175.10017089997945, 74.65415977393172] Fitness: 2.011565224873641
Generation 350 Best Solution: [183.8781324748228, 177.4328741195285,
175.10017089997945, 74.65415977393172] Fitness: 2.0122034033187592
Generation 351 Best Solution: [183.87744006655052, 177.0718685527477,
173.89863534208484, 75.83413729190654] Fitness: 1.8624698724788897
Generation 352 Best Solution: [183.87744006655052, 177.0718685527477,
173.89863534208484, 75.83413729190654] Fitness: 1.8624698724788897
Generation 353 Best Solution: [183.8781324748228, 177.0718685527477,
173.89863534208484, 75.83413729190654] Fitness: 1.8631080509240077
Generation 354 Best Solution: [183.8781324748228, 177.4328741195285,
175.10017089997945, 74.65415977393172] Fitness: 2.0122034033187592
Generation 355 Best Solution: [183.8781324748228, 177.4328741195285,
175.10017089997945, 74.65415977393172] Fitness: 2.0122034033187592
Generation 356 Best Solution: [183.87744006655052, 176.75142773531246,
175.10017089997945, 74.78565385599674] Fitness: 1.159057867388123
Generation 357 Best Solution: [183.87744006655052, 176.75142773531246,
175.10017089997945, 74.46261496803939] Fitness: 1.1883892757289742
Generation 358 Best Solution: [183.8781324748228, 176.1981218161356,
175.10017089997945, 75.83413729190654] Fitness: 1.9575620662473923
Generation 359 Best Solution: [183.8781324748228, 176.1981218161356,
175.10017089997945, 75.83413729190654] Fitness: 1.9575620662473923
Generation 360 Best Solution: [183.8781324748228, 176.1981218161356,
175.10017089997945, 74.78565385599674] Fitness: 0.9742488368369623
Generation 361 Best Solution: [182.84514868526432, 177.4328741195285,
175.10017089997945, 75.1034171895086] Fitness: 1.9906947123541157
Generation 362 Best Solution: [183.8781324748228, 176.10878405913138,
175.10017089997945, 75.83413729190654] Fitness: 1.9706786887571568
Generation 363 Best Solution: [182.84514868526432, 177.4328741195285,
174.11432405091983, 75.83413729190654] Fitness: 2.2005966413417135

Generation 364 Best Solution: [183.87744006655052, 177.4328741195285,
175.10017089997945, 75.11081276955383] Fitness: 2.1470675977498264
Generation 365 Best Solution: [183.85675775503597, 177.4328741195285,
175.10017089997945, 74.73304408347279] Fitness: 1.99389160120583
Generation 366 Best Solution: [183.85675775503597, 177.4328741195285,
175.10017089997945, 75.11081276955383] Fitness: 2.1283366961176458
Generation 367 Best Solution: [182.84514868526432, 177.4200012771352,
174.43321789532666, 75.11081276955383] Fitness: 1.3737391468185995
Generation 368 Best Solution: [183.85675775503597, 177.4328741195285,
174.7408228046411, 74.48335950269067] Fitness: 1.6169172851255615
Generation 369 Best Solution: [183.72127997960823, 177.4328741195285,
175.10017089997945, 75.11081276955383] Fitness: 2.0215086360386283
Generation 370 Best Solution: [182.84514868526432, 175.610355130782,
175.10017089997945, 75.2227182338626] Fitness: 1.3374161488177636
Generation 371 Best Solution: [183.1163465913444, 176.14450944185216,
175.10017089997945, 75.11081276955383] Fitness: 0.8419556504145834
Generation 372 Best Solution: [183.85675775503597, 177.2783180660718,
175.10017089997945, 74.4025521999776] Fitness: 1.7958220084760488
Generation 373 Best Solution: [182.84514868526432, 177.2783180660718,
175.10017089997945, 74.4025521999776] Fitness: 1.6628664001816598
Generation 374 Best Solution: [183.1163465913444, 177.4200012771352,
173.89184632674122, 74.4025521999776] Fitness: 1.1867154665701454
Generation 375 Best Solution: [182.84514868526432, 177.4200012771352,
175.10017089997945, 75.0028137208146] Fitness: 1.9124034731866548
Generation 376 Best Solution: [183.1163465913444, 177.2783180660718,
173.89184632674122, 75.83413729190654] Fitness: 1.8846707348565541
Generation 377 Best Solution: [183.47390817656276, 176.85672778047885,
175.10017089997945, 75.39479513631817] Fitness: 1.3673379235603476
Generation 378 Best Solution: [183.87999408834338, 177.4328741195285,
174.13139043452748, 75.39479513631817] Fitness: 1.7407348649294314
Generation 379 Best Solution: [183.1163465913444, 176.666810512022,
175.10017089997945, 75.83413729190654] Fitness: 1.8176247700421029
Generation 380 Best Solution: [183.87999408834338, 176.1351761389102,
173.89184632674122, 75.83413729190654] Fitness: 1.3727659125971323
Generation 381 Best Solution: [182.84514868526432, 176.1351761389102,
173.89184632674122, 75.83413729190654] Fitness: 1.2187218474304047
Generation 382 Best Solution: [182.84514868526432, 176.1351761389102,
173.89184632674122, 75.83413729190654] Fitness: 1.2187218474304047
Generation 383 Best Solution: [182.84514868526432, 176.1351761389102,
175.10017089997945, 75.83413729190654] Fitness: 1.8132331490660318
Generation 384 Best Solution: [183.40400671465017, 177.4200012771352,
175.00400017596561, 74.30989631794682] Fitness: 1.6979781536815113
Generation 385 Best Solution: [183.87999408834338, 177.4200012771352,
173.89184632674122, 75.55222481190141] Fitness: 1.9605159271928643
Generation 386 Best Solution: [183.87999408834338, 176.50246592141247,
175.10017089997945, 75.83413729190654] Fitness: 2.004458526467059
Generation 387 Best Solution: [183.87999408834338, 176.50246592141247,
175.10017089997945, 75.83413729190654] Fitness: 2.004458526467059

Generation 388 Best Solution: [183.87999408834338, 176.11194487133065,
175.10017089997945, 75.83413729190654] Fitness: 1.9717297013751742
Generation 389 Best Solution: [183.40400671465017, 177.4328741195285,
175.10017089997945, 75.19316098272012] Fitness: 1.9367885144107027
Generation 390 Best Solution: [183.87999408834338, 176.2696024835339,
173.89184632674122, 75.83413729190654] Fitness: 1.362896863621729
Generation 391 Best Solution: [183.87999408834338, 177.4200012771352,
175.10017089997945, 75.19316098272012] Fitness: 2.184459229400126
Generation 392 Best Solution: [183.87999408834338, 176.4646652678511,
175.10017089997945, 75.83413729190654] Fitness: 1.9912907295843874
Generation 393 Best Solution: [183.87999408834338, 176.50887865121754,
175.10017089997945, 75.83413729190654] Fitness: 2.006905036943625
Generation 394 Best Solution: [183.87999408834338, 176.53737378672125,
175.07252277286372, 75.19316098272012] Fitness: 1.1833991578737562
Generation 395 Best Solution: [183.87999408834338, 176.2099538010418,
173.89184632674122, 75.83413729190654] Fitness: 1.3639307415044413
Generation 396 Best Solution: [183.87999408834338, 176.2099538010418,
173.89184632674122, 75.83413729190654] Fitness: 1.3639307415044413
Generation 397 Best Solution: [183.6685496082396, 176.2099538010418,
173.89184632674122, 75.89266944009307] Fitness: 1.305369035251354
Generation 398 Best Solution: [183.87999408834338, 176.92151997976742,
175.10017089997945, 75.83413729190654] Fitness: 2.294020449664494
Generation 399 Best Solution: [183.87999408834338, 177.4200012771352,
175.10017089997945, 74.84631968666015] Fitness: 2.010133151460886
Generation 400 Best Solution: [183.87999408834338, 177.4200012771352,
175.10017089997945, 74.51628154177116] Fitness: 2.0117995400206787
Generation 401 Best Solution: [182.947782455432, 177.4200012771352,
175.10017089997945, 74.51628154177116] Fitness: 1.8012759423517295
Generation 402 Best Solution: [183.87999408834338, 175.83474643154563,
175.0046851082748, 75.83413729190654] Fitness: 1.960640866656624
Generation 403 Best Solution: [182.95160414900948, 175.80781863631123,
175.10017089997945, 75.83413729190654] Fitness: 1.89236525691876
Generation 404 Best Solution: [183.2195149316201, 175.83474643154563,
175.10017089997945, 75.83413729190654] Fitness: 1.8035535030957317
Generation 405 Best Solution: [182.82990171004013, 177.4200012771352,
175.0046851082748, 74.80168174096286] Fitness: 1.7298226747215277
Generation 406 Best Solution: [183.1445980633532, 175.80543466025804,
175.10017089997945, 75.83413729190654] Fitness: 1.8316477470340646
Generation 407 Best Solution: [183.87999408834338, 177.4200012771352,
175.0046851082748, 74.80168174096286] Fitness: 1.874128310745775
Generation 408 Best Solution: [183.87999408834338, 177.4200012771352,
175.0046851082748, 74.80168174096286] Fitness: 1.874128310745775
Generation 409 Best Solution: [183.87999408834338, 176.69513955375226,
175.10017089997945, 75.83413729190654] Fitness: 2.104880814832462
Generation 410 Best Solution: [183.02535916493858, 176.8512619219456,
175.10017089997945, 75.83413729190654] Fitness: 1.9843644909864895
Generation 411 Best Solution: [183.77612162821652, 176.69513955375226,
174.24829335219448, 75.89266944009307] Fitness: 1.473511139748196

Generation 412 Best Solution: [183.87999408834338, 177.4200012771352,
175.10017089997945, 74.63632934632791] Fitness: 1.9922867497092727
Generation 413 Best Solution: [183.87999408834338, 177.4200012771352,
175.10017089997945, 74.69067436138855] Fitness: 1.990561446603028
Generation 414 Best Solution: [183.87999408834338, 177.4200012771352,
174.24829335219448, 74.63632934632791] Fitness: 1.3454150764378758
Generation 415 Best Solution: [183.87999408834338, 176.25550844924464,
174.55997266782157, 74.63632934632791] Fitness: 0.4314862404143502
Generation 416 Best Solution: [183.3270036916898, 177.4200012771352,
174.24829335219448, 75.83413729190654] Fitness: 2.052462313574505
Generation 417 Best Solution: [183.87999408834338, 177.4200012771352,
174.24829335219448, 75.22687282650739] Fitness: 1.564153343861648
Generation 418 Best Solution: [183.87999408834338, 176.27730348216298,
175.10017089997945, 75.83413729190654] Fitness: 1.9576636803432432
Generation 419 Best Solution: [183.87999408834338, 176.04631165832592,
175.10017089997945, 75.83413729190654] Fitness: 1.9886832592877037
Generation 420 Best Solution: [183.87999408834338, 176.04631165832592,
175.10017089997945, 75.77497728611078] Fitness: 1.8893041557251475
Generation 421 Best Solution: [183.87999408834338, 176.04631165832592,
175.10017089997945, 74.43546280611947] Fitness: 1.0442981436543983
Generation 422 Best Solution: [183.87999408834338, 177.4200012771352,
175.10017089997945, 74.78570259168283] Fitness: 1.9981905709140295
Generation 423 Best Solution: [183.87999408834338, 176.70182748146027,
175.10017089997945, 75.89266944009307] Fitness: 2.212857563730428
Generation 424 Best Solution: [183.87999408834338, 177.4200012771352,
175.10017089997945, 74.85608535175574] Fitness: 2.012572651504375
Generation 425 Best Solution: [183.87999408834338, 176.70182748146027,
175.10017089997945, 75.58200849745194] Fitness: 1.7223213092338059
Generation 426 Best Solution: [183.87999408834338, 177.4200012771352,
174.0825088764685, 75.89266944009307] Fitness: 2.438762011836191
Generation 427 Best Solution: [183.87999408834338, 177.4200012771352,
175.10017089997945, 74.43366844286162] Fitness: 2.0377844585710903
Generation 428 Best Solution: [183.87999408834338, 175.90849186778684,
175.10017089997945, 74.43366844286162] Fitness: 1.101601089292067
Generation 429 Best Solution: [183.19705614059296, 177.4200012771352,
175.10017089997945, 74.43366844286162] Fitness: 1.7558760932020707
Generation 430 Best Solution: [183.87999408834338, 177.35264713157127,
175.10017089997945, 74.9795736008294] Fitness: 1.941092441658446
Generation 431 Best Solution: [183.87999408834338, 175.69548586060682,
175.10017089997945, 75.58200849745194] Fitness: 1.8018373858136882
Generation 432 Best Solution: [183.87999408834338, 176.32130206571,
175.10017089997945, 74.39614015041784] Fitness: 1.0323031857203573
Generation 433 Best Solution: [183.48689121288035, 176.32130206571,
175.10017089997945, 74.39614015041784] Fitness: 0.7845839147738517
Generation 434 Best Solution: [183.87999408834338, 175.69548586060682,
175.10017089997945, 74.47582051543561] Fitness: 1.2306331510189028
Generation 435 Best Solution: [183.87999408834338, 176.07389266417033,
174.61258049945644, 75.89266944009307] Fitness: 1.5807946492090692

Generation 436 Best Solution: [182.7978291088807, 175.62801357438386,
175.10017089997945, 74.47582051543561] Fitness: 1.1676217548994416
Generation 437 Best Solution: [183.87999408834338, 177.4200012771352,
174.66538390054853, 75.00925533033829] Fitness: 1.6033888460615127
Generation 438 Best Solution: [183.87999408834338, 175.69548586060682,
175.10017089997945, 75.24086767006382] Fitness: 1.4299402819908908
Generation 439 Best Solution: [183.87999408834338, 177.4200012771352,
174.43493697284862, 74.82003817833039] Fitness: 1.4060130324625375
Generation 440 Best Solution: [183.87999408834338, 177.4200012771352,
174.43493697284862, 74.82003817833039] Fitness: 1.4060130324625375
Generation 441 Best Solution: [183.87999408834338, 175.9566157166696,
175.10017089997945, 75.58200849745194] Fitness: 1.6352564312903117
Generation 442 Best Solution: [183.87999408834338, 175.69548586060682,
175.10017089997945, 75.58200849745194] Fitness: 1.8018373858136882
Generation 443 Best Solution: [183.87999408834338, 175.69548586060682,
175.10017089997945, 75.58200849745194] Fitness: 1.8018373858136882
Generation 444 Best Solution: [183.87999408834338, 177.4200012771352,
174.06163728184936, 75.58200849745194] Fitness: 1.951228979046854
Generation 445 Best Solution: [183.87999408834338, 177.4200012771352,
174.06163728184936, 75.58200849745194] Fitness: 1.951228979046854
Generation 446 Best Solution: [183.87999408834338, 177.4200012771352,
175.10017089997945, 75.0100026478341] Fitness: 2.0699170027877583
Generation 447 Best Solution: [183.87999408834338, 177.4200012771352,
175.10017089997945, 75.0100026478341] Fitness: 2.0699170027877583
Generation 448 Best Solution: [183.87999408834338, 175.8230705216341,
175.10017089997945, 75.73136718418598] Fitness: 1.9254390620983648
Generation 449 Best Solution: [183.87999408834338, 175.8230705216341,
175.10017089997945, 75.59481300097761] Fitness: 1.725027344461693
Generation 450 Best Solution: [183.87999408834338, 177.4200012771352,
175.0229051820313, 75.27363881464107] Fitness: 2.14714073324494
Generation 451 Best Solution: [183.87999408834338, 177.4200012771352,
175.0229051820313, 75.27363881464107] Fitness: 2.14714073324494
Generation 452 Best Solution: [183.87999408834338, 177.37546397503007,
175.0229051820313, 75.28344910481195] Fitness: 2.0792935986497136
Generation 453 Best Solution: [183.09947336278964, 177.37546397503007,
175.0229051820313, 75.28344910481195] Fitness: 1.8142283010324702
Generation 454 Best Solution: [183.6542041836883, 177.4200012771352,
175.0229051820313, 75.36733255220898] Fitness: 2.0658812745559434
Generation 455 Best Solution: [183.54078474948275, 175.9688397031727,
175.10017089997945, 75.90073402870145] Fitness: 1.907694266384545
Generation 456 Best Solution: [183.54078474948275, 177.4200012771352,
175.0229051820313, 75.36733255220898] Fitness: 2.009030724555516
Generation 457 Best Solution: [183.87999408834338, 175.7558122566796,
174.43247724861268, 75.83413729190654] Fitness: 1.5420679181255592
Generation 458 Best Solution: [183.87999408834338, 176.51866789426336,
175.0229051820313, 75.83413729190654] Fitness: 1.907198882092925
Generation 459 Best Solution: [183.87999408834338, 176.51866789426336,
174.01858139012123, 75.83413729190654] Fitness: 1.3757961701285155

Generation 460 Best Solution: [183.87999408834338, 176.51866789426336,
175.10017089997945, 75.83413729190654] Fitness: 2.0107586694713038
Generation 461 Best Solution: [182.8210820181619, 175.9115475493623,
175.0229051820313, 75.83413729190654] Fitness: 1.801675644984774
Generation 462 Best Solution: [182.8210820181619, 175.9115475493623,
175.0229051820313, 75.83413729190654] Fitness: 1.801675644984774
Generation 463 Best Solution: [183.6542041836883, 175.89845974695433,
175.10017089997945, 75.59534360868795] Fitness: 1.5109671518392145
Generation 464 Best Solution: [183.87999408834338, 175.71091683336465,
175.10017089997945, 75.36733255220898] Fitness: 1.5347578336948071
Generation 465 Best Solution: [183.87999408834338, 177.4200012771352,
175.0229051820313, 74.41482088454681] Fitness: 1.9415871495236505
Generation 466 Best Solution: [183.87999408834338, 177.1924666533366,
174.3805292760516, 75.83413729190654] Fitness: 2.003626368193726
Generation 467 Best Solution: [183.07373795642786, 176.4171409210657,
175.10017089997945, 75.90073402870145] Fitness: 1.8376867341512186
Generation 468 Best Solution: [183.87999408834338, 175.89041222014845,
174.82479466113543, 75.90073402870145] Fitness: 1.8448361944608298
Generation 469 Best Solution: [183.43987533165028, 177.4200012771352,
174.4203249705369, 74.8015773590898] Fitness: 1.1350018805993722
Generation 470 Best Solution: [183.6542041836883, 175.89041222014845,
174.82479466113543, 75.90073402870145] Fitness: 1.6742183456265716
Generation 471 Best Solution: [183.87999408834338, 177.4200012771352,
174.08575639364147, 75.36733255220898] Fitness: 1.693424978782355
Generation 472 Best Solution: [183.87999408834338, 176.4833445680779,
174.7055204735126, 75.83413729190654] Fitness: 1.562519008163878
Generation 473 Best Solution: [183.87999408834338, 176.6752375340874,
174.70638303067278, 75.83413729190654] Fitness: 1.6576141981442052
Generation 474 Best Solution: [183.87999408834338, 177.4200012771352,
174.4203249705369, 75.24847761790362] Fitness: 1.6249904468481366
Generation 475 Best Solution: [183.87999408834338, 177.4200012771352,
174.4203249705369, 75.24847761790362] Fitness: 1.6249904468481366
Generation 476 Best Solution: [183.56755628997126, 177.4200012771352,
174.4203249705369, 75.24847761790362] Fitness: 1.4092012872801059
Generation 477 Best Solution: [183.48648749148623, 177.4200012771352,
174.56661520948361, 75.73278070072124] Fitness: 2.034050741088641
Generation 478 Best Solution: [183.87999408834338, 176.70886954655398,
174.16754367944088, 75.83413729190654] Fitness: 1.4624555997717574
Generation 479 Best Solution: [183.87999408834338, 176.24680875240338,
175.10017089997945, 75.83413729190654] Fitness: 1.957173197745012
Generation 480 Best Solution: [183.48648749148623, 176.24680875240338,
175.10017089997945, 75.83413729190654] Fitness: 1.709318666393533
Generation 481 Best Solution: [183.48648749148623, 176.24680875240338,
175.10017089997945, 75.83413729190654] Fitness: 1.709318666393533
Generation 482 Best Solution: [183.87999408834338, 177.4200012771352,
173.74219153096624, 75.28557930797326] Fitness: 1.745634078316973
Generation 483 Best Solution: [183.87999408834338, 177.4200012771352,
175.10017089997945, 74.32794412303394] Fitness: 2.085972607157972

```
Generation 484 Best Solution: [183.56755628997126, 176.02409953836724,
175.10017089997945, 75.83413729190654] Fitness: 1.7800950824835589
Generation 485 Best Solution: [183.06169222360222, 176.68264653519051,
175.10017089997945, 75.54430142276371] Fitness: 1.4052395369163013
Generation 486 Best Solution: [183.87296744579157, 176.52305183003355,
173.74219153096624, 75.83413729190654] Fitness: 1.4903461973340155
Generation 487 Best Solution: [183.87296744579157, 176.52305183003355,
173.74219153096624, 75.83413729190654] Fitness: 1.4903461973340155
Generation 488 Best Solution: [183.04467746981902, 176.60249619846837,
173.90644183937462, 75.83413729190654] Fitness: 1.1999018506335979
Generation 489 Best Solution: [183.41430558798962, 176.60249619846837,
173.90644183937462, 75.83413729190654] Fitness: 1.1811385269037105
Generation 490 Best Solution: [183.87296744579157, 176.60249619846837,
173.90644183937462, 75.83413729190654] Fitness: 1.4427843739512733
Generation 491 Best Solution: [183.87296744579157, 176.60249619846837,
174.82898881113064, 75.83413729190654] Fitness: 1.7191545267388992
Generation 492 Best Solution: [183.87296744579157, 176.60249619846837,
174.82898881113064, 75.83413729190654] Fitness: 1.7191545267388992
Generation 493 Best Solution: [183.87296744579157, 176.60249619846837,
174.82898881113064, 75.83413729190654] Fitness: 1.7191545267388992
Generation 494 Best Solution: [183.87999408834338, 175.62186937711667,
175.10017089997945, 75.4198151054744] Fitness: 1.668696879596592
Generation 495 Best Solution: [183.87999408834338, 175.62186937711667,
173.77853203986638, 75.83413729190654] Fitness: 1.716383526904143
Generation 496 Best Solution: [183.87296744579157, 177.12361179490244,
175.10017089997945, 74.76539395449133] Fitness: 1.5352732105186055
Generation 497 Best Solution: [183.41430558798962, 175.95698781864388,
175.10017089997945, 75.53160885598888] Fitness: 1.3010501718871534
Generation 498 Best Solution: [183.87999408834338, 175.73337691808445,
175.10017089997945, 74.76539395449133] Fitness: 1.1722926124763033
Generation 499 Best Solution: [183.6526222044894, 175.73337691808445,
175.10017089997945, 74.76539395449133] Fitness: 1.0007491163036057
Generation 500 Best Solution: [183.87296744579157, 177.12361179490244,
174.72178509086723, 74.9626640834558] Fitness: 1.1666489671487894
Optimal Solution for Simulation 29: [183.87296744579157, 177.12361179490244,
174.72178509086723, 74.9626640834558]
--------------------------------------------------
Simulation 30/32 with conditions: {'drought': False, 'flood': False,
'policy_change': False, 'social_event': True, 'economic_crisis': False}
Generation 1 Best Solution: [183.04194436756086, 176.51312565558698,
174.03259413639537, 75.15029788907924] Fitness: 0.2800607584438967
Generation 2 Best Solution: [183.5880696710334, 176.3278037694118,
174.59543947954202, 74.8353931839911] Fitness: 0.31137375614169854
Generation 3 Best Solution: [182.8621075761065, 176.22231685135642,
174.04853099740106, 75.28161065457107] Fitness: 0.3802426617767105
Generation 4 Best Solution: [183.08583610315017, 176.22231685135642,
174.04853099740106, 74.59052266676191] Fitness: 0.07633186285579108
Generation 5 Best Solution: [183.50808400952116, 176.4569197876022,
```

174.82329914607362, 75.30757696857654] Fitness: 0.7406414013035503
Generation 6 Best Solution: [182.8621075761065, 176.4569197876022,
174.1099865636507, 74.94677672805693] Fitness: 0.22291807489677626
Generation 7 Best Solution: [183.32794429240863, 176.1159083911712,
174.1099865636507, 74.59052266676191] Fitness: 0.07162697245917025
Generation 8 Best Solution: [182.9698253905478, 176.1159083911712,
174.1099865636507, 75.71321826849166] Fitness: 0.8246899972082516
Generation 9 Best Solution: [183.00344071150045, 176.1159083911712,
174.1099865636507, 75.71321826849166] Fitness: 0.8130350423074167
Generation 10 Best Solution: [182.8621075761065, 176.80867868293822,
174.17986785100516, 75.83792356101529] Fitness: 1.3151760431429753
Generation 11 Best Solution: [183.00344071150045, 176.87033232839698,
174.09576550860686, 75.84134079846648] Fitness: 1.3169631007138567
Generation 12 Best Solution: [183.00344071150045, 176.1159083911712,
175.0281089484976, 75.16002508525983] Fitness: 0.8053581790347119
Generation 13 Best Solution: [182.85947084324226, 176.74220406543967,
174.17986785100516, 75.22951454563719] Fitness: 0.5378511667433759
Generation 14 Best Solution: [182.8621075761065, 176.1159083911712,
173.98019303290033, 75.64453146147713] Fitness: 0.7947287729956316
Generation 15 Best Solution: [182.9698253905478, 176.1159083911712,
173.98019303290033, 75.64453146147713] Fitness: 0.7463092602609229
Generation 16 Best Solution: [183.00344071150045, 176.87033232839698,
175.0281089484976, 74.85463411678323] Fitness: 0.9806975781337915
Generation 17 Best Solution: [183.00344071150045, 177.24638688109025,
175.0281089484976, 74.57957110449041] Fitness: 1.4411642959606725
Generation 18 Best Solution: [183.00344071150045, 176.87033232839698,
174.3899130410565, 75.83792356101529] Fitness: 1.3553729679534603
Generation 19 Best Solution: [183.00344071150045, 176.87033232839698,
174.3899130410565, 75.83792356101529] Fitness: 1.3553729679534603
Generation 20 Best Solution: [183.00344071150045, 177.24638688109025,
175.0281089484976, 74.34546976545128] Fitness: 1.5271363748746218
Generation 21 Best Solution: [182.9698253905478, 177.24638688109025,
175.0281089484976, 74.34546976545128] Fitness: 1.5387913297754565
Generation 22 Best Solution: [183.00344071150045, 176.87033232839698,
175.0281089484976, 75.33192512276112] Fitness: 1.2433149748785146
Generation 23 Best Solution: [183.00344071150045, 176.0498007685546,
174.1239143275721, 75.83792356101529] Fitness: 1.0176511833682182
Generation 24 Best Solution: [183.00344071150045, 176.8822036249425,
175.0281089484976, 74.57685290398278] Fitness: 0.9918155054518596
Generation 25 Best Solution: [182.9698253905478, 176.8822036249425,
174.3899130410565, 75.70274951238311] Fitness: 1.1709493714535673
Generation 26 Best Solution: [183.00344071150045, 176.36981256324492,
175.0281089484976, 75.83792356101529] Fitness: 1.5886729282499243
Generation 27 Best Solution: [183.00344071150045, 176.0121550339327,
174.9193135391291, 75.83792356101529] Fitness: 1.4692338288963296
Generation 28 Best Solution: [183.00344071150045, 176.0121550339327,
175.0281089484976, 75.75102310742508] Fitness: 1.4647207142541876
Generation 29 Best Solution: [183.00344071150045, 176.0121550339327,

175.0281089484976, 74.96017638725702] Fitness: 0.7278244100721308
Generation 30 Best Solution: [182.9698253905478, 176.8822036249425,
174.3899130410565, 75.75102310742508] Fitness: 1.2420001204758664
Generation 31 Best Solution: [183.1536930619517, 177.24638688109025,
175.0281089484976, 74.31759866441456] Fitness: 1.5106792027821538
Generation 32 Best Solution: [183.00344071150045, 175.94624314493575,
175.06223497489017, 75.83792356101529] Fitness: 1.668704318733381
Generation 33 Best Solution: [183.1536930619517, 175.94624314493575,
175.06223497489017, 75.41035445822767] Fitness: 1.0712909803968267
Generation 34 Best Solution: [183.3342989612627, 176.20038363104314,
175.06223497489017, 75.41035445822767] Fitness: 1.0239055073908199
Generation 35 Best Solution: [183.3342989612627, 177.24638688109025,
174.35010258147096, 75.83792356101529] Fitness: 1.7763495127579578
Generation 36 Best Solution: [183.3342989612627, 177.24638688109025,
175.0281089484976, 75.01891361207188] Fitness: 1.4679862904889032
Generation 37 Best Solution: [183.3342989612627, 177.02737216747056,
174.35010258147096, 75.83792356101529] Fitness: 1.482669250936687
Generation 38 Best Solution: [183.3342989612627, 176.8822036249425,
175.06223497489017, 75.01891361207188] Fitness: 1.063027470969285
Generation 39 Best Solution: [183.3342989612627, 177.24638688109025,
174.35010258147096, 75.83792356101529] Fitness: 1.7763495127579578
Generation 40 Best Solution: [183.3342989612627, 177.24638688109025,
175.06223497489017, 74.63984189361226] Fitness: 1.4498518191160996
Generation 41 Best Solution: [183.3342989612627, 176.66759070248588,
174.7204229355548, 75.83792356101529] Fitness: 1.3645874290262496
Generation 42 Best Solution: [183.3342989612627, 176.66759070248588,
174.7204229355548, 75.83792356101529] Fitness: 1.3645874290262496
Generation 43 Best Solution: [183.3342989612627, 176.66759070248588,
174.7204229355548, 75.83792356101529] Fitness: 1.3645874290262496
Generation 44 Best Solution: [183.3342989612627, 177.24638688109025,
174.32357216624712, 75.83792356101529] Fitness: 1.7687536185615191
Generation 45 Best Solution: [183.3342989612627, 177.24638688109025,
174.32357216624712, 75.83792356101529] Fitness: 1.7687536185615191
Generation 46 Best Solution: [183.3342989612627, 177.24638688109025,
175.06223497489017, 74.31170522374033] Fitness: 1.5640321359179477
Generation 47 Best Solution: [183.3342989612627, 176.5351680516665,
174.32357216624712, 75.83792356101529] Fitness: 1.0696617515235962
Generation 48 Best Solution: [183.6879847093985, 177.24638688109025,
173.8205497281289, 74.31170522374033] Fitness: 1.1668973762754467
Generation 49 Best Solution: [183.82622062175113, 176.24921809154574,
173.8205497281289, 74.31170522374033] Fitness: 0.5010961938652483
Generation 50 Best Solution: [183.3342989612627, 177.24638688109025,
175.06223497489017, 74.529086372916] Fitness: 1.4708810087740902
Generation 51 Best Solution: [183.3342989612627, 175.79481576979654,
174.5504192445059, 75.83792356101529] Fitness: 1.221510785737332
Generation 52 Best Solution: [183.3342989612627, 175.79481576979654,
174.5504192445059, 75.83792356101529] Fitness: 1.221510785737332
Generation 53 Best Solution: [183.3342989612627, 175.79481576979654,

175.06223497489017, 75.83792356101529] Fitness: 1.7175641622099826
Generation 54 Best Solution: [183.3342989612627, 175.79481576979654,
174.5066957242199, 75.83792356101529] Fitness: 1.1967993660285092
Generation 55 Best Solution: [183.3342989612627, 175.79481576979654,
174.5066957242199, 75.83792356101529] Fitness: 1.1967993660285092
Generation 56 Best Solution: [183.6879847093985, 176.54550363091508,
174.46826489205836, 75.83792356101529] Fitness: 1.2756201123099802
Generation 57 Best Solution: [183.11048726220423, 177.07495548735892,
174.97424032635675, 74.71877727754806] Fitness: 1.0983874889466814
Generation 58 Best Solution: [183.6879847093985, 177.24638688109025,
174.5504192445059, 75.83792356101529] Fitness: 2.0139563780115863
Generation 59 Best Solution: [183.3342989612627, 175.8412520724071,
174.97424032635675, 75.83792356101529] Fitness: 1.5783440162299542
Generation 60 Best Solution: [183.6879847093985, 177.24638688109025,
173.88582972362263, 74.37453424160718] Fitness: 1.1056786921117285
Generation 61 Best Solution: [183.6879847093985, 175.8412520724071,
174.22968799107267, 75.83792356101529] Fitness: 1.22524990119472
Generation 62 Best Solution: [183.51208100703732, 175.8412520724071,
175.06223497489017, 75.83792356101529] Fitness: 1.7420019569276444
Generation 63 Best Solution: [183.706510757484, 177.24638688109025,
175.06223497489017, 74.87918836731224] Fitness: 1.6252812195848947
Generation 64 Best Solution: [183.706510757484, 176.21455904486885,
175.10908220002113, 75.83792356101529] Fitness: 1.8135312278511462
Generation 65 Best Solution: [183.3342989612627, 175.9228105005168,
175.10908220002113, 75.83792356101529] Fitness: 1.7156898915072183
Generation 66 Best Solution: [183.33722270968852, 175.9761971927227,
175.06223497489017, 75.83792356101529] Fitness: 1.631105738274397
Generation 67 Best Solution: [183.706510757484, 176.48442184381125,
175.10908220002113, 75.83792356101529] Fitness: 1.866322453655672
Generation 68 Best Solution: [183.6879847093985, 177.01690183370866,
173.94626455671334, 75.83792356101529] Fitness: 1.6127761002988148
Generation 69 Best Solution: [183.6879847093985, 177.01690183370866,
173.94626455671334, 75.83792356101529] Fitness: 1.6127761002988148
Generation 70 Best Solution: [183.06237869669442, 176.12795698922343,
175.06223497489017, 75.83792356101529] Fitness: 1.6052158885874883
Generation 71 Best Solution: [183.6879847093985, 176.12795698922343,
175.06223497489017, 75.83792356101529] Fitness: 1.741877090658777
Generation 72 Best Solution: [183.6879847093985, 177.24638688109025,
174.43525353725542, 75.83792356101529] Fitness: 1.954851428921443
Generation 73 Best Solution: [183.6879847093985, 175.8871542393599,
174.94446475339385, 75.83792356101529] Fitness: 1.6666425798129494
Generation 74 Best Solution: [183.706510757484, 177.24638688109025,
174.20336112221747, 75.83792356101529] Fitness: 1.9070970645453174
Generation 75 Best Solution: [183.36374607149705, 177.24638688109025,
175.06223497489017, 74.31181612232335] Fitness: 1.569279242887259
Generation 76 Best Solution: [183.6879847093985, 176.1850094982577,
175.06223497489017, 75.83792356101529] Fitness: 1.7370210502731083
Generation 77 Best Solution: [183.36374607149705, 175.62948185231096,

173.90112896590225, 75.83792356101529] Fitness: 1.2667444861405603
Generation 78 Best Solution: [183.36374607149705, 175.62948185231096,
173.90112896590225, 75.83792356101529] Fitness: 1.2667444861405603
Generation 79 Best Solution: [183.6879847093985, 175.92796355039394,
175.06223497489017, 75.83792356101529] Fitness: 1.7962868635150713
Generation 80 Best Solution: [183.3277943106468, 177.24638688109025,
175.06223497489017, 75.40710731766411] Fitness: 1.7931130595354974
Generation 81 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 75.40710731766411] Fitness: 2.1351384329053
Generation 82 Best Solution: [183.6879847093985, 177.36836707774805,
174.2270846420715, 75.83792356101529] Fitness: 2.0907829206820896
Generation 83 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 74.6493784891575] Fitness: 1.789897446375142
Generation 84 Best Solution: [183.6879847093985, 176.2338881910062,
175.06223497489017, 75.83792356101529] Fitness: 1.7366263837747313
Generation 85 Best Solution: [183.36374607149705, 177.24638688109025,
175.06223497489017, 75.26150382676273] Fitness: 1.6682677472130332
Generation 86 Best Solution: [183.8400205651405, 176.70117083485115,
175.06223497489017, 75.26150382676273] Fitness: 1.3276153874782548
Generation 87 Best Solution: [183.6879847093985, 176.70117083485115,
175.06223497489017, 75.26150382676273] Fitness: 1.2084186933626824
Generation 88 Best Solution: [183.6879847093985, 176.14243929627418,
174.88457635997335, 75.83792356101529] Fitness: 1.5248335829646853
Generation 89 Best Solution: [183.6879847093985, 176.14243929627418,
174.88457635997335, 75.83792356101529] Fitness: 1.5248335829646853
Generation 90 Best Solution: [183.6879847093985, 176.70117083485115,
175.06223497489017, 75.84159376760562] Fitness: 1.9142711184837795
Generation 91 Best Solution: [183.706510757484, 177.39039443057231,
174.1010921940586, 75.13295707590848] Fitness: 1.3463866266606432
Generation 92 Best Solution: [183.6879847093985, 177.39039443057231,
174.1010921940586, 75.83792356101529] Fitness: 2.1236749453117607
Generation 93 Best Solution: [183.82927598012216, 175.77270548961334,
175.06223497489017, 75.84159376760562] Fitness: 1.9943125528649368
Generation 94 Best Solution: [183.6879847093985, 175.77270548961334,
175.06223497489017, 75.84159376760562] Fitness: 1.8846437390352773
Generation 95 Best Solution: [183.82927598012216, 176.514024501769,
175.06223497489017, 75.84159376760562] Fitness: 1.9170698028576802
Generation 96 Best Solution: [183.82927598012216, 177.36836707774805,
175.06223497489017, 74.70947016458416] Fitness: 1.8964567303098738
Generation 97 Best Solution: [183.58906088798858, 176.514024501769,
175.06223497489017, 75.84159376760562] Fitness: 1.7478995265016302
Generation 98 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 74.70947016458416] Fitness: 1.786787916480214
Generation 99 Best Solution: [183.6879847093985, 176.514024501769,
175.06223497489017, 74.92833942949883] Fitness: 0.9174374944615222
Generation 100 Best Solution: [183.82927598012216, 177.36836707774805,
174.90194981455886, 74.64114384496511] Fitness: 1.7040744201661748
Generation 101 Best Solution: [183.82927598012216, 176.514024501769,

174.04169016380942, 75.84159376760562] Fitness: 1.3055406937530545
Generation 102 Best Solution: [183.6879847093985, 177.36836707774805,
174.41837248079733, 74.64114384496511] Fitness: 1.2285319468157714
Generation 103 Best Solution: [183.6879847093985, 177.39039443057231,
175.06223497489017, 74.64114384496511] Fitness: 1.8280392648032362
Generation 104 Best Solution: [183.6879847093985, 177.39039443057231,
175.06223497489017, 75.66520465588354] Fitness: 2.480718651295041
Generation 105 Best Solution: [183.82927598012216, 177.36836707774805,
174.07139079324617, 75.83792356101529] Fitness: 2.198359726051145
Generation 106 Best Solution: [183.6879847093985, 177.39039443057231,
175.06223497489017, 75.39116595659036] Fitness: 2.1565815696870327
Generation 107 Best Solution: [183.6879847093985, 177.36836707774805,
174.07139079324617, 75.83792356101529] Fitness: 2.088690912221485
Generation 108 Best Solution: [183.35944492264704, 177.39039443057231,
174.95282563517358, 74.77971006731127] Fitness: 1.5463093895656337
Generation 109 Best Solution: [182.93452148348527, 177.39039443057231,
174.95282563517358, 74.77971006731127] Fitness: 1.5945292668972877
Generation 110 Best Solution: [183.6879847093985, 177.39039443057231,
174.95282563517358, 75.130818581774] Fitness: 1.811761828873534
Generation 111 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 74.67352206690205] Fitness: 1.7880168909197423
Generation 112 Best Solution: [183.82927598012216, 175.9385798573945,
174.2501329180669, 75.66520465588354] Fitness: 1.0314550567966205
Generation 113 Best Solution: [183.6879847093985, 176.10883961183416,
174.6595541774202, 75.83792356101529] Fitness: 1.3223227173166532
Generation 114 Best Solution: [183.6879847093985, 176.10883961183416,
174.6595541774202, 74.67352206690205] Fitness: 0.4065140591717914
Generation 115 Best Solution: [183.6879847093985, 176.10883961183416,
174.6595541774202, 75.83792356101529] Fitness: 1.3223227173166532
Generation 116 Best Solution: [183.37852872631106, 176.10883961183416,
174.6595541774202, 75.83792356101529] Fitness: 1.1835708836191068
Generation 117 Best Solution: [183.82927598012216, 175.96669545231867,
174.6595541774202, 74.67352206690205] Fitness: 0.5528265969168734
Generation 118 Best Solution: [183.6879847093985, 177.39039443057231,
174.23271632374738, 74.67352206690205] Fitness: 1.2130171957676663
Generation 119 Best Solution: [183.6879847093985, 175.7951212413735,
174.17331189863285, 75.83792356101529] Fitness: 1.24683274149553
Generation 120 Best Solution: [183.82927598012216, 177.07919638767493,
175.06223497489017, 74.67352206690205] Fitness: 1.4733800905220011
Generation 121 Best Solution: [183.82927598012216, 177.07919638767493,
175.06223497489017, 74.67352206690205] Fitness: 1.4733800905220011
Generation 122 Best Solution: [183.43098348326058, 177.05638211723277,
175.06223497489017, 74.67352206690205] Fitness: 1.2103751493963766
Generation 123 Best Solution: [183.1478896084982, 176.44991448299882,
175.06223497489017, 75.83792356101529] Fitness: 1.6249312724320784
Generation 124 Best Solution: [183.39855866618026, 177.05638211723277,
175.06223497489017, 74.52025367445225] Fitness: 1.2277563331130195
Generation 125 Best Solution: [183.6879847093985, 177.05638211723277,

175.06223497489017, 74.52025367445225] Fitness: 1.3617434098625105
Generation 126 Best Solution: [183.6879847093985, 175.83969579626205,
175.06223497489017, 74.52025367445225] Fitness: 0.9493284121492621
Generation 127 Best Solution: [183.6879847093985, 175.83969579626205,
174.07924167241777, 75.83792356101529] Fitness: 1.2229323525338618
Generation 128 Best Solution: [183.6879847093985, 175.9213978193788,
174.50831633678766, 75.83792356101529] Fitness: 1.2791609668583976
Generation 129 Best Solution: [183.6879847093985, 176.39442095426418,
175.06223497489017, 75.83792356101529] Fitness: 1.759779194734426
Generation 130 Best Solution: [183.5793497297398, 176.39442095426418,
175.06223497489017, 75.83792356101529] Fitness: 1.6952038418834012
Generation 131 Best Solution: [183.6879847093985, 175.97171579180662,
175.06223497489017, 75.83792356101529] Fitness: 1.779412159917856
Generation 132 Best Solution: [183.6879847093985, 175.97171579180662,
173.91348626968474, 75.65787297108798] Fitness: 0.9276617867074417
Generation 133 Best Solution: [183.6879847093985, 175.97171579180662,
173.91348626968474, 75.65787297108798] Fitness: 0.9276617867074417
Generation 134 Best Solution: [183.6879847093985, 175.97171579180662,
173.91348626968474, 75.83792356101529] Fitness: 1.198169338907424
Generation 135 Best Solution: [183.6879847093985, 175.64026802547855,
175.06223497489017, 75.65787297108798] Fitness: 1.7060897979586356
Generation 136 Best Solution: [183.6879847093985, 175.83969579626205,
175.06223497489017, 75.83792356101529] Fitness: 1.8388056431084718
Generation 137 Best Solution: [183.6879847093985, 175.83969579626205,
175.06223497489017, 75.83792356101529] Fitness: 1.8388056431084718
Generation 138 Best Solution: [183.5145074462844, 177.36836707774805,
174.884907449516, 75.83792356101529] Fitness: 2.3938833525199015
Generation 139 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 75.12238607412945] Fitness: 1.9074663414364932
Generation 140 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 75.12238607412945] Fitness: 1.9074663414364932
Generation 141 Best Solution: [183.6879847093985, 176.55949012321528,
174.884907449516, 75.83792356101529] Fitness: 1.6076710678962827
Generation 142 Best Solution: [183.6879847093985, 176.55949012321528,
174.884907449516, 75.83792356101529] Fitness: 1.6076710678962827
Generation 143 Best Solution: [183.6879847093985, 177.4049804833171,
174.31406333691834, 75.83792356101529] Fitness: 2.169700917058063
Generation 144 Best Solution: [183.6879847093985, 177.36836707774805,
173.90035477686322, 75.83792356101529] Fitness: 2.1270346283821286
Generation 145 Best Solution: [183.6879847093985, 177.36836707774805,
173.92341745068663, 75.83792356101529] Fitness: 2.119382386705623
Generation 146 Best Solution: [183.6879847093985, 177.4049804833171,
174.884907449516, 74.98730149786383] Fitness: 1.68808625883326
Generation 147 Best Solution: [183.6879847093985, 176.77459546247317,
175.06223497489017, 75.83792356101529] Fitness: 1.9641102513100572
Generation 148 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 75.20753139791341] Fitness: 1.963192715644446
Generation 149 Best Solution: [183.6782198464847, 177.36836707774805,

175.06223497489017, 75.20753139791341] Fitness: 1.9566860888159299
Generation 150 Best Solution: [183.6879847093985, 176.25177698770895,
175.06223497489017, 75.83792356101529] Fitness: 1.7373505896405184
Generation 151 Best Solution: [183.6879847093985, 176.25177698770895,
175.06223497489017, 75.83792356101529] Fitness: 1.7373505896405184
Generation 152 Best Solution: [183.6879847093985, 176.25177698770895,
175.06223497489017, 75.83792356101529] Fitness: 1.7373505896405184
Generation 153 Best Solution: [183.6879847093985, 176.25177698770895,
175.06223497489017, 75.83792356101529] Fitness: 1.7373505896405184
Generation 154 Best Solution: [182.91618301930254, 177.36836707774805,
175.06223497489017, 74.55242849132993] Fitness: 1.7194511912306167
Generation 155 Best Solution: [183.69820939672047, 177.36836707774805,
175.06223497489017, 74.74564428898633] Fitness: 1.7944103058442034
Generation 156 Best Solution: [183.46790347468357, 177.36836707774805,
175.06223497489017, 74.55242849132993] Fitness: 1.6930033151915322
Generation 157 Best Solution: [182.91618301930254, 177.36836707774805,
173.92341745068663, 75.83792356101529] Fitness: 2.0328464495640004
Generation 158 Best Solution: [183.46790347468357, 175.99502744635285,
175.06223497489017, 75.83792356101529] Fitness: 1.658574323756616
Generation 159 Best Solution: [183.69820939672047, 177.36836707774805,
173.92341745068663, 74.48772629598163] Fitness: 1.246850244642131
Generation 160 Best Solution: [183.6879847093985, 176.18503012042524,
175.06223497489017, 75.83792356101529] Fitness: 1.7370201509909053
Generation 161 Best Solution: [183.6879847093985, 176.18503012042524,
175.06223497489017, 75.83792356101529] Fitness: 1.7370201509909053
Generation 162 Best Solution: [183.6879847093985, 176.18503012042524,
175.06223497489017, 75.83792356101529] Fitness: 1.7370201509909053
Generation 163 Best Solution: [183.6879847093985, 176.18503012042524,
175.06223497489017, 75.83792356101529] Fitness: 1.7370201509909053
Generation 164 Best Solution: [183.6879847093985, 176.9827928269339,
175.06223497489017, 75.27357791234526] Fitness: 1.47495439948991
Generation 165 Best Solution: [183.6879847093985, 177.36836707774805,
173.89450076322365, 75.27357791234526] Fitness: 1.4389560847910217
Generation 166 Best Solution: [183.6879847093985, 176.18503012042524,
175.06223497489017, 74.9846722923549] Fitness: 0.8728498392677683
Generation 167 Best Solution: [183.6879847093985, 176.18503012042524,
175.06223497489017, 75.83792356101529] Fitness: 1.7370201509909053
Generation 168 Best Solution: [183.6879847093985, 176.18503012042524,
175.06223497489017, 75.83792356101529] Fitness: 1.7370201509909053
Generation 169 Best Solution: [183.6879847093985, 176.18503012042524,
175.06223497489017, 75.83792356101529] Fitness: 1.7370201509909053
Generation 170 Best Solution: [183.6879847093985, 176.18503012042524,
175.06223497489017, 75.83792356101529] Fitness: 1.7370201509909053
Generation 171 Best Solution: [183.6879847093985, 176.0918084328013,
175.06223497489017, 75.83792356101529] Fitness: 1.7474041281161052
Generation 172 Best Solution: [183.6879847093985, 177.36836707774805,
174.57363770704058, 75.83792356101529] Fitness: 2.222025048742225
Generation 173 Best Solution: [183.86156460656017, 176.4435475875837,

175.06223497489017, 75.83792356101529] Fitness: 1.9131623061159133
Generation 174 Best Solution: [183.69820939672047, 176.08618457551134,
175.06223497489017, 75.83792356101529] Fitness: 1.755396523290689
Generation 175 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 75.83792356101529] Fitness: 2.703825549064604
Generation 176 Best Solution: [183.69820939672047, 176.56855019270012,
175.06223497489017, 75.83792356101529] Fitness: 1.8342360325067557
Generation 177 Best Solution: [183.6879847093985, 176.16619560710964,
175.06223497489017, 75.83792356101529] Fitness: 1.738099187415393
Generation 178 Best Solution: [183.6879847093985, 176.16619560710964,
175.06223497489017, 75.83792356101529] Fitness: 1.738099187415393
Generation 179 Best Solution: [183.6879847093985, 176.16619560710964,
175.06223497489017, 75.83792356101529] Fitness: 1.738099187415393
Generation 180 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 74.74428019524673] Fitness: 1.7873891897558218
Generation 181 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 74.74428019524673] Fitness: 1.7873891897558218
Generation 182 Best Solution: [183.20487684104634, 176.7659000649123,
174.827126020097, 75.83792356101529] Fitness: 1.5263056832176523
Generation 183 Best Solution: [183.20487684104634, 176.485460123474,
175.06223497489017, 75.90122454198696] Fitness: 1.7402724602463424
Generation 184 Best Solution: [183.6879847093985, 176.485460123474,
175.06223497489017, 75.90122454198696] Fitness: 1.8958723872358338
Generation 185 Best Solution: [183.4592306516078, 177.36836707774805,
174.49406932727578, 75.83792356101529] Fitness: 2.0604487466083445
Generation 186 Best Solution: [183.6879847093985, 177.36836707774805,
174.13450991797328, 75.83792356101529] Fitness: 2.085289397857318
Generation 187 Best Solution: [183.12502327918682, 176.22968611012368,
175.06223497489017, 75.83792356101529] Fitness: 1.5878986904211376
Generation 188 Best Solution: [183.12502327918682, 176.22968611012368,
175.06223497489017, 75.83792356101529] Fitness: 1.5878986904211376
Generation 189 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 74.95932352939577] Fitness: 1.8301794858887113
Generation 190 Best Solution: [183.6879847093985, 177.14042844625155,
175.06223497489017, 74.96798378874189] Fitness: 1.4887023974874596
Generation 191 Best Solution: [183.6879847093985, 177.14042844625155,
175.06223497489017, 74.96798378874189] Fitness: 1.4887023974874596
Generation 192 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 74.73660695778017] Fitness: 1.7871052121491509
Generation 193 Best Solution: [183.6879847093985, 177.36836707774805,
173.87079856161776, 74.73660695778017] Fitness: 1.221252168888342
Generation 194 Best Solution: [183.6879847093985, 177.36836707774805,
174.07582523588124, 75.83792356101529] Fitness: 2.0882626773838915
Generation 195 Best Solution: [183.6879847093985, 176.68273211061853,
175.06223497489017, 75.83792356101529] Fitness: 1.895474793625041
Generation 196 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 75.04197574934777] Fitness: 1.8645206885397139
Generation 197 Best Solution: [183.6879847093985, 177.36836707774805,

174.1808158515664, 75.83792356101529] Fitness: 2.0864790653034486
Generation 198 Best Solution: [183.6879847093985, 175.60420488954384,
175.06223497489017, 75.83792356101529] Fitness: 2.007691040645767
Generation 199 Best Solution: [183.6879847093985, 176.90863240296395,
174.44735841184414, 75.83792356101529] Fitness: 1.53642346821307
Generation 200 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 74.86910292139791] Fitness: 1.8040367392465018
Generation 201 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 74.86910292139791] Fitness: 1.8040367392465018
Generation 202 Best Solution: [183.6879847093985, 176.84203312301292,
175.06223497489017, 74.90121902487034] Fitness: 1.1304692746374596
Generation 203 Best Solution: [183.6879847093985, 177.36836707774805,
174.44735841184414, 74.90121902487034] Fitness: 1.2621324914362917
Generation 204 Best Solution: [183.6879847093985, 177.36836707774805,
174.65014574581315, 75.23259567875049] Fitness: 1.5523195824016882
Generation 205 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 74.71081021051327] Fitness: 1.7867784439392809
Generation 206 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 74.71081021051327] Fitness: 1.7867784439392809
Generation 207 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 74.71081021051327] Fitness: 1.7867784439392809
Generation 208 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 74.71081021051327] Fitness: 1.7867784439392809
Generation 209 Best Solution: [183.6879847093985, 176.2788294893245,
175.06223497489017, 75.83792356101529] Fitness: 1.7393299764641412
Generation 210 Best Solution: [183.6879847093985, 176.2788294893245,
175.06223497489017, 75.83792356101529] Fitness: 1.7393299764641412
Generation 211 Best Solution: [183.6879847093985, 176.2788294893245,
175.06223497489017, 75.83792356101529] Fitness: 1.7393299764641412
Generation 212 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 75.20743080791497] Fitness: 1.9631206593922663
Generation 213 Best Solution: [183.6879847093985, 177.36836707774805,
175.06223497489017, 75.20743080791497] Fitness: 1.9631206593922663
Generation 214 Best Solution: [183.6879847093985, 176.10480601960063,
175.06223497489017, 75.83792356101529] Fitness: 1.7451979834523828
Generation 215 Best Solution: [183.6879847093985, 176.10480601960063,
175.06223497489017, 75.83792356101529] Fitness: 1.7451979834523828
Generation 216 Best Solution: [183.6879847093985, 176.10480601960063,
173.8836243500113, 75.83792356101529] Fitness: 1.1744424379952543
Generation 217 Best Solution: [183.6879847093985, 176.10480601960063,
175.06223497489017, 75.83792356101529] Fitness: 1.7451979834523828
Generation 218 Best Solution: [183.6879847093985, 177.36836707774805,
174.07256665132923, 75.5523643387656] Fitness: 1.6814630905594177
Generation 219 Best Solution: [183.6879847093985, 176.2561469244371,
173.95745200633905, 75.83792356101529] Fitness: 1.1432756395638977
Generation 220 Best Solution: [183.6879847093985, 177.4298835856369,
174.07256665132923, 74.34004501386453] Fitness: 1.3797167024927985
Generation 221 Best Solution: [183.6879847093985, 175.92354540923645,

175.06223497489017, 75.5523643387656] Fitness: 1.3910341849898258
Generation 222 Best Solution: [183.6879847093985, 175.92354540923645,
175.06223497489017, 75.83792356101529] Fitness: 1.798145667205758
Generation 223 Best Solution: [183.16677612518296, 175.92354540923645,
175.06223497489017, 75.83792356101529] Fitness: 1.6447167060871286
Generation 224 Best Solution: [183.16677612518296, 176.10437778404543,
175.06223497489017, 75.83792356101529] Fitness: 1.5918377942220763
Generation 225 Best Solution: [183.16677612518296, 176.10437778404543,
175.06223497489017, 75.83792356101529] Fitness: 1.5918377942220763
Generation 226 Best Solution: [183.51799914820586, 175.9494235454036,
174.5472851743939, 75.83792356101529] Fitness: 1.1962857390906843
Generation 227 Best Solution: [183.51799914820586, 175.9494235454036,
174.5472851743939, 74.94937743022946] Fitness: 0.31917697975810627
Generation 228 Best Solution: [183.6879847093985, 177.4298835856369,
173.94574187071683, 75.37174292963842] Fitness: 1.615286883061413
Generation 229 Best Solution: [183.6879847093985, 175.99816197334275,
175.06223497489017, 75.37174292963842] Fitness: 1.167183414896126
Generation 230 Best Solution: [183.6879847093985, 175.99816197334275,
173.89184072219473, 75.83792356101529] Fitness: 1.1967919432622087
Generation 231 Best Solution: [183.6879847093985, 175.99816197334275,
173.89184072219473, 75.83792356101529] Fitness: 1.1967919432622087
Generation 232 Best Solution: [183.6879847093985, 176.16979001915834,
175.01258282450158, 75.83792356101529] Fitness: 1.673041376729414
Generation 233 Best Solution: [183.6879847093985, 176.16979001915834,
175.01258282450158, 75.71035191366693] Fitness: 1.476509091752801
Generation 234 Best Solution: [183.6879847093985, 176.16979001915834,
175.06223497489017, 75.83792356101529] Fitness: 1.7378534227625253
Generation 235 Best Solution: [182.9280388329236, 175.7466152007492,
175.01258282450158, 75.83792356101529] Fitness: 1.7393479013956419
Generation 236 Best Solution: [182.9280388329236, 175.7466152007492,
175.06223497489017, 75.83792356101529] Fitness: 1.804159947428753
Generation 237 Best Solution: [183.6879847093985, 175.7466152007492,
175.01258282450158, 75.71035191366693] Fitness: 1.634574821062989
Generation 238 Best Solution: [183.6879847093985, 177.4298835856369,
175.01258282450158, 74.32643503904886] Fitness: 1.9377130780224738
Generation 239 Best Solution: [183.58513388015353, 176.07143090071895,
175.06223497489017, 75.83792356101529] Fitness: 1.6897877605067357
Generation 240 Best Solution: [183.4076197109119, 177.4298835856369,
175.06223497489017, 74.32643503904886] Fitness: 1.8708852096386535
Generation 241 Best Solution: [183.6879847093985, 175.7182484404722,
175.01258282450158, 75.71035191366693] Fitness: 1.6544859423168785
Generation 242 Best Solution: [183.58513388015353, 175.7182484404722,
175.01258282450158, 75.71035191366693] Fitness: 1.592916175406057
Generation 243 Best Solution: [183.6879847093985, 175.7182484404722,
174.94660906828497, 75.83792356101529] Fitness: 1.7704490661475987
Generation 244 Best Solution: [183.6879847093985, 177.4298835856369,
174.0596109023148, 75.83792356101529] Fitness: 2.1959212354157787
Generation 245 Best Solution: [182.8719553626973, 177.4298835856369,

175.06223497489017, 75.47252781324947] Fitness: 2.2448170031529995
Generation 246 Best Solution: [183.6879847093985, 177.06609556019066,
173.97170906055604, 75.83792356101529] Fitness: 1.6652164784692876
Generation 247 Best Solution: [183.6879847093985, 177.13569226236473,
175.01258282450158, 75.47252781324947] Fitness: 1.7883296411467517
Generation 248 Best Solution: [183.58513388015353, 177.4298835856369,
175.01258282450158, 75.47252781324947] Fitness: 2.1836819721789777
Generation 249 Best Solution: [183.34609290785562, 177.4298835856369,
175.01258282450158, 75.83792356101529] Fitness: 2.5997372570773183
Generation 250 Best Solution: [183.6879847093985, 176.59645758631584,
175.06223497489017, 75.83792356101529] Fitness: 1.8421922938826687
Generation 251 Best Solution: [183.6879847093985, 176.59645758631584,
175.06223497489017, 74.96587629683694] Fitness: 0.9709061792872907
Generation 252 Best Solution: [183.6879847093985, 176.33891567602592,
175.01258282450158, 75.83792356101529] Fitness: 1.6827222148365095
Generation 253 Best Solution: [183.6879847093985, 176.75500986797962,
174.59914651158758, 75.83792356101529] Fitness: 1.4832097030098161
Generation 254 Best Solution: [183.6879847093985, 176.75500986797962,
174.59914651158758, 75.83792356101529] Fitness: 1.4832097030098161
Generation 255 Best Solution: [183.34609290785562, 175.64491495187448,
175.01258282450158, 75.83792356101529] Fitness: 1.7626862283109994
Generation 256 Best Solution: [183.6879847093985, 176.3817911393749,
175.01258282450158, 75.83792356101529] Fitness: 1.6917870798612515
Generation 257 Best Solution: [183.6879847093985, 176.3817911393749,
175.01258282450158, 75.83792356101529] Fitness: 1.6917870798612515
Generation 258 Best Solution: [183.6879847093985, 176.74142714392624,
175.01258282450158, 75.83792356101529] Fitness: 1.8731007189073314
Generation 259 Best Solution: [183.6879847093985, 177.4298835856369,
174.13184122826368, 74.71541541801639] Fitness: 1.2742099679383025
Generation 260 Best Solution: [183.6879847093985, 176.74142714392624,
175.01258282450158, 75.83792356101529] Fitness: 1.8731007189073314
Generation 261 Best Solution: [183.6879847093985, 177.30210783037282,
175.06223497489017, 74.9806116285769] Fitness: 1.7301092088295773
Generation 262 Best Solution: [183.6879847093985, 177.4298835856369,
175.01258282450158, 74.98832637610212] Fitness: 1.8822400445636833
Generation 263 Best Solution: [183.6879847093985, 177.4298835856369,
174.90465090534175, 74.89475378521614] Fitness: 1.722890471364572
Generation 264 Best Solution: [183.6879847093985, 177.4298835856369,
174.90465090534175, 74.89475378521614] Fitness: 1.722890471364572
Generation 265 Best Solution: [183.6879847093985, 177.4298835856369,
174.90465090534175, 74.89475378521614] Fitness: 1.722890471364572
Generation 266 Best Solution: [183.6879847093985, 175.9666519490232,
175.06223497489017, 75.53391798149873] Fitness: 1.3518913490134148
Generation 267 Best Solution: [183.6879847093985, 177.4298835856369,
175.01258282450158, 74.59667038335179] Fitness: 1.8380906556931307
Generation 268 Best Solution: [183.6879847093985, 177.4298835856369,
175.01258282450158, 74.86839144727078] Fitness: 1.8450194105438324
Generation 269 Best Solution: [183.68851394866007, 177.3882745996018,

175.06223497489017, 74.86839144727078] Fitness: 1.8379198194034347
Generation 270 Best Solution: [183.43626529887428, 177.4298835856369,
175.01258282450158, 74.65943886399147] Fitness: 1.7067185298968117
Generation 271 Best Solution: [183.6254054381116, 177.4298835856369,
175.01258282450158, 74.86839144727078] Fitness: 1.8057246278038566
Generation 272 Best Solution: [183.6254054381116, 175.7945139247431,
175.01258282450158, 75.77061582547452] Fitness: 1.6542062625176301
Generation 273 Best Solution: [183.6879847093985, 176.07996677215192,
175.01258282450158, 75.77061582547452] Fitness: 1.5781740875398653
Generation 274 Best Solution: [183.6879847093985, 177.00873767254856,
175.01258282450158, 74.81264690353419] Fitness: 1.219625481705162
Generation 275 Best Solution: [183.68851394866007, 177.4298835856369,
174.11390972488388, 74.82309530543554] Fitness: 1.2835109852838296
Generation 276 Best Solution: [183.77701948603925, 176.38343452852416,
175.01258282450158, 75.83792356101529] Fitness: 1.7579118639015396
Generation 277 Best Solution: [183.68851394866007, 176.38343452852416,
175.01258282450158, 75.83792356101529] Fitness: 1.6925443494545678
Generation 278 Best Solution: [183.77701948603925, 175.68026008803298,
175.01258282450158, 75.77061582547452] Fitness: 1.8385984312908814
Generation 279 Best Solution: [182.81448177723365, 177.4298835856369,
175.01258282450158, 74.47881926341076] Fitness: 1.8351450645994116
Generation 280 Best Solution: [183.68851394866007, 177.4298835856369,
175.01258282450158, 74.93272812232904] Fitness: 1.8627408204512783
Generation 281 Best Solution: [183.6254054381116, 177.4298835856369,
175.01258282450158, 75.42412453934585] Fitness: 2.154327301408692
Generation 282 Best Solution: [183.77701948603925, 176.35771416459713,
175.01258282450158, 75.83792356101529] Fitness: 1.7520916038124479
Generation 283 Best Solution: [183.77701948603925, 176.35771416459713,
175.01258282450158, 75.83792356101529] Fitness: 1.7520916038124479
Generation 284 Best Solution: [182.81448177723365, 176.71919151217102,
175.01258282450158, 75.77061582547452] Fitness: 1.716461708200693
Generation 285 Best Solution: [182.9493175854067, 176.35771416459713,
175.01258282450158, 75.77061582547452] Fitness: 1.4791044968000435
Generation 286 Best Solution: [182.81448177723365, 176.96223787367427,
174.16031346235144, 75.83792356101529] Fitness: 1.490861535790867
Generation 287 Best Solution: [183.68851394866007, 176.07531521704638,
175.01258282450158, 75.83792356101529] Fitness: 1.6861019117743095
Generation 288 Best Solution: [183.77701948603925, 177.4298835856369,
175.01258282450158, 75.3259216008809] Fitness: 2.1650680928400394
Generation 289 Best Solution: [183.68851394866007, 176.94181803655354,
175.01258282450158, 75.83792356101529] Fitness: 2.0561038169893933
Generation 290 Best Solution: [182.81448177723365, 177.4298835856369,
173.94066526315964, 75.83792356101529] Fitness: 2.186788362958313
Generation 291 Best Solution: [183.6254054381116, 177.4298835856369,
173.94066526315964, 75.83792356101529] Fitness: 2.1808242201554924
Generation 292 Best Solution: [182.81448177723365, 175.89423945509617,
175.01258282450158, 75.83792356101529] Fitness: 1.7130513802826595
Generation 293 Best Solution: [183.6254054381116, 177.4298835856369,

175.01258282450158, 75.09183684942396] Fitness: 1.8918897732496385
Generation 294 Best Solution: [183.77701948603925, 176.48066823348026,
175.01258282450158, 75.83792356101529] Fitness: 1.7886096238152587
Generation 295 Best Solution: [183.6254054381116, 177.4298835856369,
173.94066526315964, 74.56569990174208] Fitness: 1.2799756350329916
Generation 296 Best Solution: [183.6254054381116, 177.4298835856369,
173.94066526315964, 74.56569990174208] Fitness: 1.2799756350329916
Generation 297 Best Solution: [182.81448177723365, 177.4298835856369,
173.94066526315964, 75.07120628681245] Fitness: 1.3620069774175543
Generation 298 Best Solution: [182.81448177723365, 177.4298835856369,
173.94066526315964, 75.07120628681245] Fitness: 1.3620069774175543
Generation 299 Best Solution: [183.6254054381116, 177.4298835856369,
175.01258282450158, 75.03454835671499] Fitness: 1.8628753304827137
Generation 300 Best Solution: [183.6254054381116, 177.4298835856369,
175.01258282450158, 75.03454835671499] Fitness: 1.8628753304827137
Generation 301 Best Solution: [182.95112048635013, 177.4298835856369,
175.01258282450158, 74.9831175690047] Fitness: 1.778847222597234
Generation 302 Best Solution: [183.4646527378722, 177.4298835856369,
175.01258282450158, 74.42224749901104] Fitness: 1.7761129991782214
Generation 303 Best Solution: [183.10122179639896, 177.4298835856369,
175.01258282450158, 74.9831175690047] Fitness: 1.735437081280591
Generation 304 Best Solution: [182.95112048635013, 177.4298835856369,
175.01258282450158, 74.9831175690047] Fitness: 1.778847222597234
Generation 305 Best Solution: [183.4480692498864, 177.4298835856369,
175.01258282450158, 74.9831175690047] Fitness: 1.7604835151410894
Generation 306 Best Solution: [182.88538988777057, 176.45853112732235,
175.01258282450158, 75.87632551262486] Fitness: 1.7064692146850375
Generation 307 Best Solution: [183.15882348349746, 177.4298835856369,
175.01258282450158, 74.82244704432098] Fitness: 1.6835942624980693
Generation 308 Best Solution: [182.95112048635013, 177.4298835856369,
175.01258282450158, 74.9831175690047] Fitness: 1.778847222597234
Generation 309 Best Solution: [182.95112048635013, 176.0713639588417,
175.01258282450158, 75.83792356101529] Fitness: 1.5852177525928244
Generation 310 Best Solution: [183.4480692498864, 177.4298835856369,
174.9506422898982, 74.9831175690047] Fitness: 1.6846581599092822
Generation 311 Best Solution: [182.95112048635013, 177.4298835856369,
175.01258282450158, 74.83486164369023] Fitness: 1.7370143373747835
Generation 312 Best Solution: [182.95112048635013, 177.4298835856369,
175.01258282450158, 74.83486164369023] Fitness: 1.7370143373747835
Generation 313 Best Solution: [182.81448177723365, 176.388791610778,
175.01258282450158, 75.83792356101529] Fitness: 1.6601904325079695
Generation 314 Best Solution: [182.95112048635013, 177.4298835856369,
175.01258282450158, 74.83486164369023] Fitness: 1.7370143373747835
Generation 315 Best Solution: [183.2516840564227, 177.4298835856369,
175.01258282450158, 74.85958908414743] Fitness: 1.6877352395540968
Generation 316 Best Solution: [182.95112048635013, 177.4298835856369,
175.01258282450158, 74.98030162685242] Fitness: 1.777754802525634
Generation 317 Best Solution: [183.6499293341389, 176.41761462219728,

175.01258282450158, 75.87632551262486] Fitness: 1.7406329862024985
Generation 318 Best Solution: [182.9617884626809, 177.4298835856369,
175.01258282450158, 74.98030162685242] Fitness: 1.7735877664389468
Generation 319 Best Solution: [182.95112048635013, 177.4298835856369,
175.01258282450158, 74.98030162685242] Fitness: 1.777754802525634
Generation 320 Best Solution: [182.95112048635013, 177.4298835856369,
175.01258282450158, 74.98079619621878] Fitness: 1.7779458314084178
Generation 321 Best Solution: [182.95112048635013, 176.30886696461323,
175.01258282450158, 75.86249215018866] Fitness: 1.6171891021109348
Generation 322 Best Solution: [182.95112048635013, 176.30886696461323,
175.01258282450158, 74.59298909739846] Fitness: 0.6703879570993311
Generation 323 Best Solution: [182.95112048635013, 176.30886696461323,
175.01258282450158, 75.81325664117172] Fitness: 1.5367741312987149
Generation 324 Best Solution: [182.95112048635013, 176.30886696461323,
175.01258282450158, 75.81325664117172] Fitness: 1.5367741312987149
Generation 325 Best Solution: [183.0081994981436, 177.4298835856369,
175.01258282450158, 74.71320125330764] Fitness: 1.7061989857163402
Generation 326 Best Solution: [183.0081994981436, 177.4298835856369,
175.01258282450158, 74.40973666063175] Fitness: 1.7739680554418598
Generation 327 Best Solution: [182.95112048635013, 177.06169108388585,
175.01258282450158, 74.70212635631125] Fitness: 1.174645306069707
Generation 328 Best Solution: [183.6499293341389, 176.49782050778848,
175.01258282450158, 75.87632551262486] Fitness: 1.7689491712916514
Generation 329 Best Solution: [183.80555215537174, 176.49782050778848,
175.01258282450158, 75.87632551262486] Fitness: 1.8827498011789574
Generation 330 Best Solution: [183.7973164457019, 177.4298835856369,
175.01258282450158, 74.74373588595583] Fitness: 1.9108289290382972
Generation 331 Best Solution: [183.17468251449606, 177.4298835856369,
173.87794726963824, 74.74373588595583] Fitness: 1.1706518986393648
Generation 332 Best Solution: [182.96525846353805, 177.4298835856369,
173.87794726963824, 74.74373588595583] Fitness: 1.2178764923965812
Generation 333 Best Solution: [183.6499293341389, 176.50538313656736,
173.87794726963824, 75.81325664117172] Fitness: 1.1646557006416605
Generation 334 Best Solution: [183.6499293341389, 176.50538313656736,
173.87794726963824, 75.87632551262486] Fitness: 1.2682988252930107
Generation 335 Best Solution: [183.7973164457019, 176.43089102167752,
175.01258282450158, 75.87632551262486] Fitness: 1.8515692939275579
Generation 336 Best Solution: [183.6499293341389, 176.43089102167752,
175.01258282450158, 75.87632551262486] Fitness: 1.744673893920913
Generation 337 Best Solution: [183.7973164457019, 175.78147041947892,
173.87794726963824, 75.87632551262486] Fitness: 1.450558247978993
Generation 338 Best Solution: [183.6499293341389, 177.4298835856369,
175.01258282450158, 74.82472043296713] Fitness: 1.8120883088635276
Generation 339 Best Solution: [183.7973164457019, 177.4298835856369,
175.01258282450158, 74.77499804430371] Fitness: 1.9128463938954638
Generation 340 Best Solution: [183.6499293341389, 175.96617326476473,
175.01258282450158, 75.87632551262486] Fitness: 1.7558053927768251
Generation 341 Best Solution: [183.6646876157276, 177.4298835856369,

173.87794726963824, 75.87632551262486] Fitness: 2.2896661133364633
Generation 342 Best Solution: [183.6646876157276, 177.4298835856369,
174.2189902557097, 75.87632551262486] Fitness: 2.2442608966143056
Generation 343 Best Solution: [183.6499293341389, 176.445777690301,
175.01258282450158, 75.87632551262486] Fitness: 1.7495098278115986
Generation 344 Best Solution: [183.27645778985766, 177.4298835856369,
175.01258282450158, 74.76409131492859] Fitness: 1.6756914717116893
Generation 345 Best Solution: [183.6499293341389, 177.4298835856369,
175.01258282450158, 74.76409131492859] Fitness: 1.8050856788932905
Generation 346 Best Solution: [183.7973164457019, 177.4298835856369,
175.01258282450158, 74.76409131492859] Fitness: 1.911981078899935
Generation 347 Best Solution: [183.6499293341389, 177.4298835856369,
175.01258282450158, 74.76409131492859] Fitness: 1.8050856788932905
Generation 348 Best Solution: [183.6499293341389, 177.4298835856369,
175.01258282450158, 74.76409131492859] Fitness: 1.8050856788932905
Generation 349 Best Solution: [183.6408157018628, 177.4298835856369,
173.75793375457948, 75.87632551262486] Fitness: 2.3310337358027367
Generation 350 Best Solution: [183.1509160329144, 177.4298835856369,
174.55613507269277, 75.87632551262486] Fitness: 2.229055204238198
Generation 351 Best Solution: [183.1509160329144, 176.41945623297065,
175.01258282450158, 75.87632551262486] Fitness: 1.6138498307405371
Generation 352 Best Solution: [183.1509160329144, 177.4298835856369,
175.01258282450158, 75.04379474348495] Fitness: 1.7546271489364325
Generation 353 Best Solution: [183.1509160329144, 177.4298835856369,
175.01258282450158, 74.87467434184707] Fitness: 1.6945470850098565
Generation 354 Best Solution: [182.98938255646243, 177.4298835856369,
175.01258282450158, 74.87467434184707] Fitness: 1.730930463569885
Generation 355 Best Solution: [183.09602818860193, 175.87432577313857,
175.01258282450158, 75.87632551262486] Fitness: 1.6759602828156293
Generation 356 Best Solution: [182.98938255646243, 177.4298835856369,
173.75793375457948, 74.57277246406866] Fitness: 1.2795170867025272
Generation 357 Best Solution: [183.75697112143956, 175.87432577313857,
174.50004258276263, 75.87632551262486] Fitness: 1.4102071349593457
Generation 358 Best Solution: [183.75697112143956, 177.4298835856369,
174.88993723438583, 75.04379474348495] Fitness: 1.8117252651325366
Generation 359 Best Solution: [183.6646876157276, 177.13335123668585,
174.88993723438583, 75.87632551262486] Fitness: 2.1886923601005903
Generation 360 Best Solution: [183.7973164457019, 177.13335123668585,
174.88993723438583, 75.87632551262486] Fitness: 2.2863075707163416
Generation 361 Best Solution: [183.75697112143956, 177.4298835856369,
174.68439090631227, 74.85546986909591] Fitness: 1.5539662863236088
Generation 362 Best Solution: [183.75697112143956, 177.4298835856369,
174.50004258276263, 75.87632551262486] Fitness: 2.3992130969974013
Generation 363 Best Solution: [183.6408157018628, 177.4298835856369,
174.50004258276263, 75.87632551262486] Fitness: 2.319147223448304
Generation 364 Best Solution: [183.6408157018628, 176.22836580823736,
175.01258282450158, 75.87632551262486] Fitness: 1.7053338724611313
Generation 365 Best Solution: [183.75697112143956, 176.19529654129053,

175.01258282450158, 75.87632551262486] Fitness: 1.7855521688867948
Generation 366 Best Solution: [183.6408157018628, 176.84258271200778,
174.68439090631227, 75.87632551262486] Fitness: 1.6534372098004393
Generation 367 Best Solution: [183.6408157018628, 175.80663892895004,
174.68439090631227, 75.87632551262486] Fitness: 1.4882726308462844
Generation 368 Best Solution: [183.75697112143956, 177.4298835856369,
174.68439090631227, 74.38432354955224] Fitness: 1.6191409268740864
Generation 369 Best Solution: [183.1264687477107, 177.4298835856369,
174.2427177394019, 75.87632551262486] Fitness: 2.11385702879193
Generation 370 Best Solution: [183.75697112143956, 177.4298835856369,
175.01258282450158, 74.48758889167053] Fitness: 1.9154378170926756
Generation 371 Best Solution: [183.75697112143956, 176.61115658864014,
174.2427177394019, 75.87632551262486] Fitness: 1.3533354160217503
Generation 372 Best Solution: [183.75697112143956, 177.4298835856369,
174.3879458245643, 75.87632551262486] Fitness: 2.349646893736662
Generation 373 Best Solution: [183.75697112143956, 176.56613281476052,
175.01258282450158, 75.87632551262486] Fitness: 1.8749383705715428
Generation 374 Best Solution: [183.75697112143956, 176.1893030907578,
175.01258282450158, 75.87632551262486] Fitness: 1.7857500633300578
Generation 375 Best Solution: [182.87036036093482, 177.4298835856369,
174.68439090631227, 75.06570987918695] Fitness: 1.514723838895324
Generation 376 Best Solution: [183.1264687477107, 175.62847147099941,
174.3879458245643, 75.87632551262486] Fitness: 1.327675055419319
Generation 377 Best Solution: [183.1264687477107, 176.1893030907578,
175.01258282450158, 75.87632551262486] Fitness: 1.5869974332442571
Generation 378 Best Solution: [183.1264687477107, 176.2108989415039,
174.68439090631227, 75.87632551262486] Fitness: 1.2483190047326447
Generation 379 Best Solution: [183.1264687477107, 176.2108989415039,
174.68439090631227, 74.51859426783305] Fitness: 0.2955177431723773
Generation 380 Best Solution: [183.75697112143956, 177.4003187407746,
174.68439090631227, 75.20706436201033] Fitness: 1.6641000629187412
Generation 381 Best Solution: [183.1264687477107, 176.25758444753757,
174.68439090631227, 75.85575372837202] Fitness: 1.2151835421963304
Generation 382 Best Solution: [183.75697112143956, 176.61825754564236,
174.68439090631227, 75.87632551262486] Fitness: 1.5653260562269637
Generation 383 Best Solution: [183.1264687477107, 175.91576692024356,
175.01258282450158, 75.49503749653745] Fitness: 1.1132968184129421
Generation 384 Best Solution: [183.1264687477107, 176.58379055254187,
174.68439090631227, 75.87632551262486] Fitness: 1.34722057052417
Generation 385 Best Solution: [183.75697112143956, 176.58379055254187,
174.68439090631227, 75.87632551262486] Fitness: 1.5459732006099705
Generation 386 Best Solution: [183.75697112143956, 177.4003187407746,
174.68439090631227, 74.55335491099572] Fitness: 1.507010461783152
Generation 387 Best Solution: [183.1264687477107, 176.01165224279185,
175.01258282450158, 75.87632551262486] Fitness: 1.6165901454020621
Generation 388 Best Solution: [183.1264687477107, 177.4003187407746,
175.01258282450158, 74.54132758945852] Fitness: 1.64940132124189
Generation 389 Best Solution: [183.75697112143956, 177.02766799728514,

174.28464329269892, 75.87632551262486] Fitness: 1.7270517645478454
Generation 390 Best Solution: [183.1264687477107, 176.80739696583368,
175.01258282450158, 75.87632551262486] Fitness: 1.841742039778321
Generation 391 Best Solution: [183.75697112143956, 177.4298835856369,
174.79466564427378, 74.54132758945852] Fitness: 1.6577160819427834
Generation 392 Best Solution: [183.75697112143956, 177.4003187407746,
174.28464329269892, 75.31059590183014] Fitness: 1.5456763284922057
Generation 393 Best Solution: [183.12621268205493, 177.4298835856369,
175.01258282450158, 74.7101992588554] Fitness: 1.6791271953217717
Generation 394 Best Solution: [183.1264687477107, 177.4298835856369,
174.3360783489307, 74.54132758945852] Fitness: 1.1752249886766108
Generation 395 Best Solution: [183.20596582472788, 176.42561904961016,
175.01258282450158, 75.87632551262486] Fitness: 1.6119821205640226
Generation 396 Best Solution: [183.1264687477107, 175.7157521261274,
175.01258282450158, 74.43272518537948] Fitness: 0.8448812391477905
Generation 397 Best Solution: [183.20596582472788, 177.4003187407746,
175.01258282450158, 74.43272518537948] Fitness: 1.6786164899228728
Generation 398 Best Solution: [183.75697112143956, 177.4003187407746,
175.01258282450158, 74.43272518537948] Fitness: 1.8841662893536966
Generation 399 Best Solution: [183.1264687477107, 176.38753122353896,
175.01258282450158, 75.87632551262486] Fitness: 1.6081659364815268
Generation 400 Best Solution: [183.75697112143956, 176.38753122353896,
175.01258282450158, 75.87632551262486] Fitness: 1.8069185665673273
Generation 401 Best Solution: [183.75697112143956, 177.1526120218257,
175.01258282450158, 75.15514899076818] Fitness: 1.5846665434233667
Generation 402 Best Solution: [183.75697112143956, 177.1526120218257,
175.01258282450158, 75.15514899076818] Fitness: 1.5846665434233667
Generation 403 Best Solution: [183.75697112143956, 177.4003187407746,
174.136624918031, 75.15514899076818] Fitness: 1.413375502930681
Generation 404 Best Solution: [183.20596582472788, 177.4298835856369,
175.01258282450158, 74.544473041504] Fitness: 1.693425204121674
Generation 405 Best Solution: [183.75697112143956, 177.4298835856369,
175.01258282450158, 74.544473041504] Fitness: 1.8989750035524977
Generation 406 Best Solution: [183.1264687477107, 176.31833800297778,
175.01258282450158, 75.87632551262486] Fitness: 1.5942835508550446
Generation 407 Best Solution: [183.1264687477107, 177.4003187407746,
175.01258282450158, 75.36366621194732] Fitness: 1.9334782291246544
Generation 408 Best Solution: [183.75697112143956, 177.4003187407746,
175.01258282450158, 74.55725506240859] Fitness: 1.8443149398021854
Generation 409 Best Solution: [183.75697112143956, 175.73232467046316,
175.01258282450158, 75.87632551262486] Fitness: 1.9547065306747127
Generation 410 Best Solution: [183.75697112143956, 175.73232467046316,
175.01258282450158, 75.87632551262486] Fitness: 1.9547065306747127
Generation 411 Best Solution: [183.75697112143956, 175.73232467046316,
175.01258282450158, 75.87632551262486] Fitness: 1.9547065306747127
Generation 412 Best Solution: [183.75697112143956, 177.1526120218257,
174.62679536222342, 75.87632551262486] Fitness: 2.04322272616244
Generation 413 Best Solution: [183.1264687477107, 177.4298835856369,

174.29878550329272, 75.87632551262486] Fitness: 2.1245202302959907
Generation 414 Best Solution: [183.28292232873812, 177.1526120218257,
174.29878550329272, 75.87632551262486] Fitness: 1.685844423249327
Generation 415 Best Solution: [183.75697112143956, 177.1526120218257,
174.29878550329272, 75.87632551262486] Fitness: 1.8892177162523986
Generation 416 Best Solution: [183.20596582472788, 177.4298835856369,
175.01258282450158, 74.85746790028087] Fitness: 1.6870380470573443
Generation 417 Best Solution: [183.75697112143956, 176.2588988864514,
175.01258282450158, 75.87632551262486] Fitness: 1.786671358666108
Generation 418 Best Solution: [183.75697112143956, 176.2588988864514,
175.01258282450158, 75.87632551262486] Fitness: 1.786671358666108
Generation 419 Best Solution: [183.5794499170461, 175.95808360314263,
175.01258282450158, 75.87632551262486] Fitness: 1.7188316440412563
Generation 420 Best Solution: [183.1264687477107, 177.1526120218257,
175.01258282450158, 74.69360835023512] Fitness: 1.2453513450505913
Generation 421 Best Solution: [183.75697112143956, 177.1526120218257,
174.11993957366482, 75.65702144833512] Fitness: 1.5357028417060865
Generation 422 Best Solution: [183.20596582472788, 176.60933872852962,
174.11993957366482, 75.65702144833512] Fitness: 0.8038888454362124
Generation 423 Best Solution: [183.75697112143956, 177.1526120218257,
175.01258282450158, 74.38721530623992] Fitness: 1.521911395342735
Generation 424 Best Solution: [183.20596582472788, 177.4298835856369,
175.01258282450158, 74.38721530623992] Fitness: 1.750416740041304
Generation 425 Best Solution: [183.75697112143956, 177.4298835856369,
173.7352529524347, 74.37795385267297] Fitness: 1.5258404935537828
Generation 426 Best Solution: [183.75697112143956, 177.4298835856369,
173.7352529524347, 74.37795385267297] Fitness: 1.5258404935537828
Generation 427 Best Solution: [183.75697112143956, 177.4298835856369,
173.7352529524347, 74.37795385267297] Fitness: 1.5258404935537828
Generation 428 Best Solution: [183.75697112143956, 177.1526120218257,
175.01258282450158, 74.87493891944699] Fitness: 1.4623751426085438
Generation 429 Best Solution: [183.20596582472788, 175.8966975763763,
175.01258282450158, 74.87493891944699] Fitness: 0.6911527100232955
Generation 430 Best Solution: [182.90797942397964, 175.98340256768034,
175.01258282450158, 75.87632551262486] Fitness: 1.6915579259388305
Generation 431 Best Solution: [183.75697112143956, 176.17544581497563,
175.01258282450158, 75.87632551262486] Fitness: 1.7864076650815774
Generation 432 Best Solution: [183.75697112143956, 177.4298835856369,
175.01258282450158, 74.80384816938347] Fitness: 1.8835674061916987
Generation 433 Best Solution: [182.90797942397964, 175.752166900399,
175.01258282450158, 75.87632551262486] Fitness: 1.8083412767964369
Generation 434 Best Solution: [182.90797942397964, 175.752166900399,
175.01258282450158, 75.87632551262486] Fitness: 1.8083412767964369
Generation 435 Best Solution: [183.75697112143956, 177.4298835856369,
174.1637647868696, 75.03584023170542] Fitness: 1.3993553600519184
Generation 436 Best Solution: [183.39246605993515, 176.4845222665227,
175.01258282450158, 75.87632551262486] Fitness: 1.652683538378394
Generation 437 Best Solution: [183.75697112143956, 177.21064006549403,

175.01258282450158, 74.85515067943706] Fitness: 1.5396439364541468
Generation 438 Best Solution: [183.75697112143956, 177.4298835856369,
173.85023604317217, 75.22371936475821] Fitness: 1.57335923160157
Generation 439 Best Solution: [182.90797942397964, 177.4298835856369,
173.85023604317217, 75.22371936475821] Fitness: 1.440638338201283
Generation 440 Best Solution: [182.90797942397964, 177.4298835856369,
173.85023604317217, 75.22371936475821] Fitness: 1.440638338201283
Generation 441 Best Solution: [183.75697112143956, 177.4298835856369,
173.85023604317217, 75.22371936475821] Fitness: 1.57335923160157
Generation 442 Best Solution: [183.75697112143956, 177.4298835856369,
173.85023604317217, 75.22371936475821] Fitness: 1.57335923160157
Generation 443 Best Solution: [183.75697112143956, 177.21064006549403,
173.85023604317217, 75.22371936475821] Fitness: 1.2208916049593814
Generation 444 Best Solution: [182.99086534243372, 176.59031525679893,
175.01258282450158, 75.87632551262486] Fitness: 1.7217696812714731
Generation 445 Best Solution: [183.75697112143956, 177.21064006549403,
175.01258282450158, 74.98998817176629] Fitness: 1.580353959483016
Generation 446 Best Solution: [183.75697112143956, 175.752166900399,
174.0506791300368, 75.87632551262486] Fitness: 1.3931184845209654
Generation 447 Best Solution: [182.99086534243372, 177.4298835856369,
174.38519746846532, 74.98998817176629] Fitness: 1.2568556055831357
Generation 448 Best Solution: [182.85080048547852, 175.85612553959987,
174.98221606254134, 75.87632551262486] Fitness: 1.737087998015736
Generation 449 Best Solution: [183.75697112143956, 176.53335128961783,
175.01258282450158, 75.87632551262486] Fitness: 1.858977126934951
Generation 450 Best Solution: [183.75697112143956, 176.62730279158006,
174.98221606254134, 75.87632551262486] Fitness: 1.8710303332846725
Generation 451 Best Solution: [183.75697112143956, 176.62730279158006,
174.66125690263945, 75.87632551262486] Fitness: 1.5527618696822256
Generation 452 Best Solution: [183.75697112143956, 176.62730279158006,
174.66125690263945, 75.87632551262486] Fitness: 1.5527618696822256
Generation 453 Best Solution: [183.75697112143956, 177.4298835856369,
174.0855641375839, 75.70550371348182] Fitness: 2.0397609238064747
Generation 454 Best Solution: [183.75697112143956, 177.4298835856369,
175.01258282450158, 74.7616838456451] Fitness: 1.879411322003641
Generation 455 Best Solution: [183.75697112143956, 177.4298835856369,
174.0855641375839, 74.29529019958022] Fitness: 1.4543490766407499
Generation 456 Best Solution: [183.75697112143956, 177.4298835856369,
175.01258282450158, 74.42147182454538] Fitness: 1.9404872553429149
Generation 457 Best Solution: [183.75697112143956, 177.4298835856369,
175.01258282450158, 74.54029629879508] Fitness: 1.9000236869276264
Generation 458 Best Solution: [183.75697112143956, 175.90735473330764,
174.0855641375839, 75.87632551262486] Fitness: 1.3025120680394724
Generation 459 Best Solution: [183.75697112143956, 175.80987691667204,
173.87891267230117, 75.87632551262486] Fitness: 1.4004629277448362
Generation 460 Best Solution: [182.88812003334414, 177.4298835856369,
175.01258282450158, 74.31106409225623] Fitness: 1.873214721282873
Generation 461 Best Solution: [182.88812003334414, 177.4090659188166,

175.01258282450158, 74.31106409225623] Fitness: 1.836742933522396
Generation 462 Best Solution: [183.75697112143956, 176.1419508776944,
175.01258282450158, 75.87632551262486] Fitness: 1.789150676511888
Generation 463 Best Solution: [182.88812003334414, 176.1419508776944,
175.01258282450158, 75.87632551262486] Fitness: 1.6658741949761642
Generation 464 Best Solution: [182.88812003334414, 176.1419508776944,
175.01258282450158, 75.87632551262486] Fitness: 1.6658741949761642
Generation 465 Best Solution: [183.75697112143956, 176.66492271963602,
175.01258282450158, 75.87632551262486] Fitness: 1.9324919706028365
Generation 466 Best Solution: [182.88812003334414, 176.91578876288793,
175.01258282450158, 74.75667519849263] Fitness: 1.0395677162971588
Generation 467 Best Solution: [183.4033955827868, 176.0263321089553,
175.01258282450158, 75.87632551262486] Fitness: 1.628489664696476
Generation 468 Best Solution: [183.75697112143956, 177.4298835856369,
175.01258282450158, 74.69998451958858] Fitness: 1.877990291609292
Generation 469 Best Solution: [183.75697112143956, 177.4298835856369,
175.01258282450158, 75.18763192729034] Fitness: 2.0402851818218006
Generation 470 Best Solution: [183.75697112143956, 177.4298835856369,
175.01258282450158, 75.03708192994442] Fitness: 1.9532712406600117
Generation 471 Best Solution: [182.83527444708963, 177.4298835856369,
175.01258282450158, 75.03708192994442] Fitness: 1.8579205006931871
Generation 472 Best Solution: [182.90439475285928, 177.4298835856369,
175.01258282450158, 74.66650822259137] Fitness: 1.7484778830803838
Generation 473 Best Solution: [182.83527444708963, 177.4090659188166,
175.01258282450158, 74.66650822259137] Fitness: 1.74771393684509
Generation 474 Best Solution: [183.75697112143956, 176.9011084706066,
175.01258282450158, 74.75738627398218] Fitness: 1.1480803558698498
Generation 475 Best Solution: [182.85447839534424, 177.4090659188166,
175.01258282450158, 74.55664616490363] Fitness: 1.753622836873355
Generation 476 Best Solution: [183.75697112143956, 176.29343971853723,
175.01258282450158, 75.87632551262486] Fitness: 1.7897445784570372
Generation 477 Best Solution: [182.85234794467976, 176.29343971853723,
175.01258282450158, 75.87632551262486] Fitness: 1.6849273277579246
Generation 478 Best Solution: [183.75697112143956, 177.4298835856369,
175.01258282450158, 74.69243924543748] Fitness: 1.8781964901507062
Generation 479 Best Solution: [183.75697112143956, 176.87861049851094,
175.01258282450158, 75.00008368406762] Fitness: 1.1837968438664175
Generation 480 Best Solution: [183.75697112143956, 176.87861049851094,
175.01258282450158, 75.00008368406762] Fitness: 1.1837968438664175
Generation 481 Best Solution: [183.75697112143956, 177.4090659188166,
174.7753082740223, 74.87584687424237] Fitness: 1.5999636454688937
Generation 482 Best Solution: [183.75697112143956, 177.4090659188166,
175.01258282450158, 74.87584687424237] Fitness: 1.8601703237044924
Generation 483 Best Solution: [183.75697112143956, 177.4090659188166,
175.01258282450158, 75.00008368406762] Fitness: 1.9004619527721476
Generation 484 Best Solution: [183.75697112143956, 176.11039527494975,
175.01258282450158, 75.87632551262486] Fitness: 1.793227748681502
Generation 485 Best Solution: [183.75697112143956, 177.4298835856369,

175.01258282450158, 74.88941628307536] Fitness: 1.8999507099000854
Generation 486 Best Solution: [183.06228234250278, 177.4298835856369,
174.8220868122133, 74.52506603799442] Fitness: 1.5021174487489475
Generation 487 Best Solution: [183.75697112143956, 177.4298835856369,
174.8220868122133, 74.52506603799442] Fitness: 1.6886747127375237
Generation 488 Best Solution: [183.75697112143956, 177.4090659188166,
175.01258282450158, 75.13196464751087] Fitness: 1.967797814846547
Generation 489 Best Solution: [183.75697112143956, 177.4298835856369,
174.84019840337643, 74.80475806014593] Fitness: 1.6865052466528774
Generation 490 Best Solution: [182.83106866744612, 176.95714941901568,
175.01258282450158, 75.87632551262486] Fitness: 2.0928875540374774
Generation 491 Best Solution: [182.83106866744612, 177.4090659188166,
174.8220868122133, 75.87632551262486] Fitness: 2.513868857393241
Generation 492 Best Solution: [182.83106866744612, 177.4298835856369,
174.84019840337643, 75.87632551262486] Fitness: 2.5685481976595983
Generation 493 Best Solution: [182.83106866744612, 177.4298835856369,
174.84019840337643, 75.87632551262486] Fitness: 2.5685481976595983
Generation 494 Best Solution: [182.83106866744612, 177.4298835856369,
174.84019840337643, 75.87632551262486] Fitness: 2.5685481976595983
Generation 495 Best Solution: [182.83106866744612, 177.4298835856369,
175.01258282450158, 74.98669131427414] Fitness: 1.8385570743889477
Generation 496 Best Solution: [183.75697112143956, 177.4298835856369,
175.01258282450158, 74.46586180490365] Fitness: 1.9229680193212033
Generation 497 Best Solution: [182.83106866744612, 176.27731867065506,
174.84019840337643, 75.87632551262486] Fitness: 1.497960190432143
Generation 498 Best Solution: [183.75697112143956, 176.76089982630015,
175.01258282450158, 74.69214912000045] Fitness: 1.021525703805168
Generation 499 Best Solution: [183.75697112143956, 177.4090659188166,
175.01258282450158, 74.65411642223094] Fitness: 1.8440503914051605
Generation 500 Best Solution: [182.81951931764857, 175.80232074453352,
175.01258282450158, 75.87632551262486] Fitness: 1.822888351666258
Optimal Solution for Simulation 30: [182.81951931764857, 175.80232074453352,
175.01258282450158, 75.87632551262486]

Simulation 31/32 with conditions: {'drought': False, 'flood': False,
'policy_change': False, 'social_event': False, 'economic_crisis': True}
Generation 1 Best Solution: [183.09658923461095, 176.22045748156128,
174.20102680218986, 75.21523799297995] Fitness: 0.23426326690642718
Generation 2 Best Solution: [183.23929719578067, 176.39500780795026,
173.87794550551143, 74.38077954147772] Fitness: 0.20421250485171846
Generation 3 Best Solution: [183.23929719578067, 176.39891302691248,
174.20284778608033, 75.21523799297995] Fitness: 0.24278114182342048
Generation 4 Best Solution: [183.09658923461095, 176.22045748156128,
174.44605237082732, 75.21523799297995] Fitness: 0.29541115696903986
Generation 5 Best Solution: [183.23929719578067, 177.03631579336906,
173.83055917418133, 74.8081594062418] Fitness: 0.6002259600591321
Generation 6 Best Solution: [183.3714199029566, 175.60298615701015,
174.43073005466294, 74.3825625080653] Fitness: 0.47691602174240133

Generation 7 Best Solution: [183.43554533183834, 176.38724714471414,
174.48203044212977, 75.54296740874767] Fitness: 0.6565551828786553
Generation 8 Best Solution: [183.43554533183834, 176.38724714471414,
174.48203044212977, 75.52627168597051] Fitness: 0.6370025360373848
Generation 9 Best Solution: [183.23929719578067, 176.45210142873952,
175.04520360256166, 74.98170162618378] Fitness: 0.7077306449071482
Generation 10 Best Solution: [183.71986193548724, 177.14800532682983,
174.96507504569544, 74.98170162618378] Fitness: 1.358086784998387
Generation 11 Best Solution: [183.23929719578067, 176.45210142873952,
175.04520360256166, 75.80627404800434] Fitness: 1.505902262611845
Generation 12 Best Solution: [183.23929719578067, 176.45210142873952,
175.04520360256166, 75.52627168597051] Fitness: 1.124739798789253
Generation 13 Best Solution: [183.2935477735748, 177.14800532682983,
175.04520360256166, 74.84076751725996] Fitness: 1.2512765151728027
Generation 14 Best Solution: [183.23929719578067, 176.45210142873952,
175.04520360256166, 75.80627404800434] Fitness: 1.505902262611845
Generation 15 Best Solution: [183.1830842205612, 175.64879749454215,
174.15407045053337, 75.8899860635627] Fitness: 1.2671675377785474
Generation 16 Best Solution: [183.71986193548724, 176.67648106936315,
175.04520360256166, 75.80627404800434] Fitness: 1.7862835123937264
Generation 17 Best Solution: [183.23929719578067, 176.99315004979414,
174.6482512226327, 75.80627404800434] Fitness: 1.4966603463704398
Generation 18 Best Solution: [183.23929719578067, 176.67648106936315,
175.04520360256166, 75.8899860635627] Fitness: 1.7521872909304825
Generation 19 Best Solution: [183.23929719578067, 176.67648106936315,
175.04520360256166, 75.8899860635627] Fitness: 1.7521872909304825
Generation 20 Best Solution: [183.23929719578067, 176.67648106936315,
175.04520360256166, 75.80627404800434] Fitness: 1.6162418543323755
Generation 21 Best Solution: [183.27438710094202, 176.67648106936315,
175.04520360256166, 75.80627404800434] Fitness: 1.6173684271562165
Generation 22 Best Solution: [183.23929719578067, 176.67648106936315,
175.04520360256166, 75.68708739088335] Fitness: 1.4401521849325893
Generation 23 Best Solution: [183.23929719578067, 175.77139770529394,
175.04520360256166, 75.80627404800434] Fitness: 1.6161199662256598
Generation 24 Best Solution: [183.23929719578067, 175.77139770529394,
175.04520360256166, 75.80627404800434] Fitness: 1.6161199662256598
Generation 25 Best Solution: [183.23929719578067, 175.77139770529394,
175.04520360256166, 75.80627404800434] Fitness: 1.6161199662256598
Generation 26 Best Solution: [183.10630212965626, 175.77139770529394,
175.04520360256166, 75.8899860635627] Fitness: 1.7639404492988457
Generation 27 Best Solution: [183.23929719578067, 176.90695592033236,
175.04520360256166, 75.80627404800434] Fitness: 1.8052911666773181
Generation 28 Best Solution: [183.23929719578067, 177.3937402550454,
175.04520360256166, 75.68708739088335] Fitness: 2.2806555147428247
Generation 29 Best Solution: [183.23929719578067, 177.3937402550454,
175.01268405633272, 75.62694811815095] Fitness: 2.1583427023928543
Generation 30 Best Solution: [183.23929719578067, 177.3937402550454,
175.04520360256166, 74.77394962230689] Fitness: 1.6123664480037676

Generation 31 Best Solution: [183.23929719578067, 177.3937402550454,
175.04520360256166, 74.36657544588712] Fitness: 1.7027395599914754
Generation 32 Best Solution: [183.23929719578067, 177.44611955956603,
175.04520360256166, 74.73057937770805] Fitness: 1.7010806501034113
Generation 33 Best Solution: [183.60890828040255, 176.14475542696547,
175.04520360256166, 75.8899860635627] Fitness: 1.7099013450742133
Generation 34 Best Solution: [183.60890828040255, 176.14475542696547,
175.04520360256166, 75.8899860635627] Fitness: 1.7099013450742133
Generation 35 Best Solution: [183.23929719578067, 177.44611955956603,
175.04520360256166, 74.56734559728955] Fitness: 1.7187368802526548
Generation 36 Best Solution: [183.71986193548724, 176.65238485461146,
175.04520360256166, 75.68708739088335] Fitness: 1.59485893560566
Generation 37 Best Solution: [182.9910838034768, 177.3937402550454,
174.6866085487105, 75.58367555341172] Fitness: 1.8168946388762635
Generation 38 Best Solution: [182.9910838034768, 177.3937402550454,
174.6866085487105, 75.58367555341172] Fitness: 1.8168946388762635
Generation 39 Best Solution: [183.29694531736223, 177.3937402550454,
174.6866085487105, 75.8899860635627] Fitness: 2.2250607537846268
Generation 40 Best Solution: [182.9910838034768, 175.61781869801914,
174.72564193368189, 75.8899860635627] Fitness: 1.5731790251576043
Generation 41 Best Solution: [182.9910838034768, 176.3185895621329,
175.04520360256166, 75.68708739088335] Fitness: 1.34148482705604
Generation 42 Best Solution: [183.23929719578067, 176.46342122083638,
175.04520360256166, 75.8899860635627] Fitness: 1.6456724019148075
Generation 43 Best Solution: [183.21286955493062, 176.46342122083638,
175.0721530948323, 75.8899860635627] Fitness: 1.6813403150506048
Generation 44 Best Solution: [183.71986193548724, 176.46342122083638,
175.0721530948323, 75.8899860635627] Fitness: 1.8510562785126452
Generation 45 Best Solution: [183.32180061419683, 177.44611955956603,
175.0721530948323, 74.92272410222867] Fitness: 1.770429746993145
Generation 46 Best Solution: [183.71986193548724, 176.46342122083638,
173.736792156203, 75.8899860635627] Fitness: 1.3616992959036707
Generation 47 Best Solution: [182.9910838034768, 176.30863568213076,
175.0721530948323, 75.8899860635627] Fitness: 1.687571504868652
Generation 48 Best Solution: [182.9910838034768, 176.46342122083638,
174.96873492339535, 75.8899860635627] Fitness: 1.5939184922944591
Generation 49 Best Solution: [182.9910838034768, 176.46342122083638,
175.04520360256166, 75.20476192832564] Fitness: 0.873388194245476
Generation 50 Best Solution: [182.9910838034768, 176.46342122083638,
174.96873492339535, 75.8899860635627] Fitness: 1.5939184922944591
Generation 51 Best Solution: [183.05242598588612, 177.3937402550454,
175.04520360256166, 74.3831210369861] Fitness: 1.718355668542299
Generation 52 Best Solution: [182.9910838034768, 177.3937402550454,
174.4047941314998, 75.00831508201628] Fitness: 1.1806319196314359
Generation 53 Best Solution: [183.7523119313714, 176.46342122083638,
174.60937229343725, 75.8899860635627] Fitness: 1.4133275321679835
Generation 54 Best Solution: [183.78799506523842, 176.25379583017215,
174.1496671658506, 75.8899860635627] Fitness: 1.2482308343489854

Generation 55 Best Solution: [183.05242598588612, 176.25379583017215,
174.1496671658506, 75.8899860635627] Fitness: 1.0510380998532816
Generation 56 Best Solution: [183.7523119313714, 176.48579056486426,
174.2781107335466, 75.8899860635627] Fitness: 1.2812459078767497
Generation 57 Best Solution: [183.78799506523842, 177.44611955956603,
174.96873492339535, 74.61275390080516] Fitness: 1.8365394460307207
Generation 58 Best Solution: [182.9910838034768, 177.44611955956603,
173.82759606148784, 74.61275390080516] Fitness: 1.2504211449474372
Generation 59 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.34090520870112] Fitness: 1.8497747532519555
Generation 60 Best Solution: [183.78799506523842, 177.44611955956603,
174.48900692731237, 74.32220078327097] Fitness: 1.5435442760800062
Generation 61 Best Solution: [183.78799506523842, 176.68518316478747,
175.04520360256166, 75.64662699729085] Fitness: 1.6119337709471766
Generation 62 Best Solution: [183.78799506523842, 176.5591666227547,
175.04520360256166, 75.8899860635627] Fitness: 1.9065757254877251
Generation 63 Best Solution: [182.9910838034768, 177.3339901432296,
175.04520360256166, 75.05757108948325] Fitness: 1.6354162828303138
Generation 64 Best Solution: [182.9910838034768, 177.44611955956603,
174.81860792307384, 75.8899860635627] Fitness: 2.470332783017928
Generation 65 Best Solution: [182.9910838034768, 177.44611955956603,
174.16902476157682, 75.8899860635627] Fitness: 2.148456210298881
Generation 66 Best Solution: [182.9910838034768, 175.84732138314612,
175.09345416298552, 75.64662699729085] Fitness: 1.4462990819263377
Generation 67 Best Solution: [182.9910838034768, 176.12958965778236,
174.16902476157682, 75.8899860635627] Fitness: 1.0759071054856726
Generation 68 Best Solution: [183.3594274911834, 175.84732138314612,
175.04520360256166, 74.36913542583981] Fitness: 0.8265772450539892
Generation 69 Best Solution: [182.9910838034768, 175.84732138314612,
174.96873492339535, 75.8899860635627] Fitness: 1.654855768569763
Generation 70 Best Solution: [182.9910838034768, 175.84732138314612,
174.96873492339535, 75.8899860635627] Fitness: 1.654855768569763
Generation 71 Best Solution: [182.9910838034768, 177.44611955956603,
174.49107517656978, 75.8899860635627] Fitness: 2.231854729212648
Generation 72 Best Solution: [182.9910838034768, 177.44611955956603,
174.96873492339535, 75.35083884463081] Fitness: 1.9332138779230437
Generation 73 Best Solution: [182.9910838034768, 177.44611955956603,
173.82279219869946, 75.8899860635627] Fitness: 2.2259041935812682
Generation 74 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.49494002092997] Fitness: 1.7817083784094476
Generation 75 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.49494002092997] Fitness: 1.7817083784094476
Generation 76 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.49494002092997] Fitness: 1.7817083784094476
Generation 77 Best Solution: [182.9910838034768, 177.44611955956603,
173.82279219869946, 74.38558523801503] Fitness: 1.3264039744639011
Generation 78 Best Solution: [182.9910838034768, 176.8163858121,
175.04520360256166, 75.8899860635627] Fitness: 1.900611607024346

Generation 79 Best Solution: [182.9910838034768, 176.8163858121,
175.04520360256166, 75.8899860635627] Fitness: 1.900611607024346
Generation 80 Best Solution: [183.78799506523842, 177.44611955956603,
175.04520360256166, 74.65709921138507] Fitness: 1.925415128144244
Generation 81 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.65709921138507] Fitness: 1.747082977094805
Generation 82 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.65709921138507] Fitness: 1.747082977094805
Generation 83 Best Solution: [182.9910838034768, 176.90168517143383,
175.04520360256166, 75.8899860635627] Fitness: 1.978873403307019
Generation 84 Best Solution: [182.9910838034768, 176.71382469347736,
175.04520360256166, 75.8899860635627] Fitness: 1.8204274067489012
Generation 85 Best Solution: [182.9910838034768, 176.36437973984295,
175.04520360256166, 75.8899860635627] Fitness: 1.6613007041612258
Generation 86 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 75.02240209917878] Fitness: 1.8082411146588178
Generation 87 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.69415298353294] Fitness: 1.7445021376758887
Generation 88 Best Solution: [183.41763590455398, 177.44611955956603,
175.04520360256166, 74.69415298353294] Fitness: 1.7258607730196478
Generation 89 Best Solution: [183.41763590455398, 177.44611955956603,
174.0045066647516, 75.8899860635627] Fitness: 2.1450244656186914
Generation 90 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.55647048485883] Fitness: 1.7640981428244515
Generation 91 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 75.01800935958663] Fitness: 1.8063606929107474
Generation 92 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 75.01800935958663] Fitness: 1.8063606929107474
Generation 93 Best Solution: [183.1703454427052, 177.44611955956603,
175.04520360256166, 75.01800935958663] Fitness: 1.7665106333261205
Generation 94 Best Solution: [183.36175506384427, 176.19653064001278,
175.04520360256166, 75.8899860635627] Fitness: 1.6164169875962657
Generation 95 Best Solution: [183.36175506384427, 176.19653064001278,
175.04520360256166, 75.8899860635627] Fitness: 1.6164169875962657
Generation 96 Best Solution: [183.36175506384427, 176.19653064001278,
175.04520360256166, 75.8899860635627] Fitness: 1.6164169875962657
Generation 97 Best Solution: [182.9910838034768, 177.44611955956603,
174.7453399847617, 74.4451774450696] Fitness: 1.4774793216934556
Generation 98 Best Solution: [182.9910838034768, 177.44611955956603,
174.103268142901, 74.95184293129213] Fitness: 1.2052508378357216
Generation 99 Best Solution: [183.1703454427052, 175.91442188741757,
174.5818958576333, 75.77518199044795] Fitness: 1.0489275845923283
Generation 100 Best Solution: [182.9910838034768, 177.44611955956603,
173.77997070121552, 74.73455539110905] Fitness: 1.2654605930744927
Generation 101 Best Solution: [182.96307627038877, 175.91442188741757,
174.5818958576333, 75.77518199044795] Fitness: 1.0991963205136543
Generation 102 Best Solution: [182.9910838034768, 175.91442188741757,
174.5818958576333, 75.77518199044795] Fitness: 1.0887776441769552

Generation 103 Best Solution: [182.9910838034768, 175.91442188741757,
175.04520360256166, 75.77518199044795] Fitness: 1.5323251331593317
Generation 104 Best Solution: [182.9910838034768, 177.0945627414063,
174.2419109559504, 75.8899860635627] Fitness: 1.6228002332667262
Generation 105 Best Solution: [182.9910838034768, 177.0945627414063,
174.2419109559504, 74.75120724403735] Fitness: 0.6412037012053596
Generation 106 Best Solution: [183.41385944367426, 176.2254130589346,
174.5818958576333, 75.8899860635627] Fitness: 1.183861903247093
Generation 107 Best Solution: [183.812433579318, 176.40608332521592,
175.04520360256166, 75.8899860635627] Fitness: 1.869320155738592
Generation 108 Best Solution: [183.1703454427052, 177.34263522772025,
173.93398835492786, 75.8899860635627] Fitness: 1.9673376042587698
Generation 109 Best Solution: [182.9910838034768, 177.44611955956603,
174.89601925401413, 74.56417773904951] Fitness: 1.5856111326926743
Generation 110 Best Solution: [182.9910838034768, 177.44611955956603,
174.89601925401413, 74.56417773904951] Fitness: 1.5856111326926743
Generation 111 Best Solution: [182.9910838034768, 175.96928743286367,
174.74840010033037, 75.8899860635627] Fitness: 1.3740008951484293
Generation 112 Best Solution: [182.9910838034768, 175.8871755706537,
175.04520360256166, 75.69701749502677] Fitness: 1.430751166368038
Generation 113 Best Solution: [182.9910838034768, 177.44611955956603,
174.89601925401413, 74.56417773904951] Fitness: 1.5856111326926743
Generation 114 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.56417773904951] Fitness: 1.762277700907305
Generation 115 Best Solution: [182.9910838034768, 176.60143807298397,
174.5818958576333, 75.8899860635627] Fitness: 1.3064608591713909
Generation 116 Best Solution: [182.9910838034768, 176.26180280162214,
175.04520360256166, 74.75672687406005] Fitness: 0.6667211665828534
Generation 117 Best Solution: [183.6741896467961, 176.6929139100775,
174.5818958576333, 75.8899860635627] Fitness: 1.4591146616974318
Generation 118 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.60284331747673] Fitness: 1.7544399169314753
Generation 119 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.9864828900583] Fitness: 1.7936828499527755
Generation 120 Best Solution: [182.9910838034768, 176.0388220933915,
175.04520360256166, 75.8899860635627] Fitness: 1.6717615254946785
Generation 121 Best Solution: [182.9910838034768, 175.84822978536627,
175.04520360256166, 75.8899860635627] Fitness: 1.7489336806723212
Generation 122 Best Solution: [182.9910838034768, 176.42658577761728,
175.04520360256166, 75.8899860635627] Fitness: 1.6767227964555922
Generation 123 Best Solution: [182.9910838034768, 176.56336503799153,
175.04520360256166, 75.8899860635627] Fitness: 1.7302898147543722
Generation 124 Best Solution: [182.9910838034768, 175.84822978536627,
175.04520360256166, 75.5207297748066] Fitness: 1.2254245591265969
Generation 125 Best Solution: [182.9910838034768, 176.3990070148188,
175.04520360256166, 75.8899860635627] Fitness: 1.6691957763560215
Generation 126 Best Solution: [182.9910838034768, 176.6705569042599,
175.04520360256166, 75.5207297748066] Fitness: 1.2676477200959084

Generation 127 Best Solution: [182.9910838034768, 176.6705569042599,
175.04520360256166, 75.8899860635627] Fitness: 1.7911568416416326
Generation 128 Best Solution: [183.50951264054743, 175.87469833478707,
175.04520360256166, 75.8899860635627] Fitness: 1.7468229238212465
Generation 129 Best Solution: [182.9910838034768, 176.1057670958829,
174.29205834140845, 75.8899860635627] Fitness: 1.0937371292841434
Generation 130 Best Solution: [182.9910838034768, 176.1057670958829,
174.29205834140845, 75.5207297748066] Fitness: 0.5702280077384191
Generation 131 Best Solution: [182.9910838034768, 177.44611955956603,
173.82714980062497, 74.6188565016369] Fitness: 1.2496772391649154
Generation 132 Best Solution: [182.9910838034768, 177.44611955956603,
173.82714980062497, 74.6188565016369] Fitness: 1.2496772391649154
Generation 133 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.6188565016369] Fitness: 1.7518262962981677
Generation 134 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.60280866160511] Fitness: 1.7544459750170978
Generation 135 Best Solution: [182.9910838034768, 176.3951993877581,
175.04520360256166, 75.8899860635627] Fitness: 1.668242878923814
Generation 136 Best Solution: [183.50951264054743, 177.44611955956603,
175.04520360256166, 74.60280866160511] Fitness: 1.766189936905573
Generation 137 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.60280866160511] Fitness: 1.7544459750170978
Generation 138 Best Solution: [183.51822641348275, 177.44611955956603,
175.04520360256166, 74.49556402468095] Fitness: 1.7967611488210007
Generation 139 Best Solution: [182.9910838034768, 176.00516711621168,
175.04520360256166, 75.8899860635627] Fitness: 1.6815740833389548
Generation 140 Best Solution: [183.50951264054743, 176.95619778655228,
174.02167020670183, 75.8899860635627] Fitness: 1.4803867877260062
Generation 141 Best Solution: [182.9910838034768, 176.95619778655228,
174.02167020670183, 75.8899860635627] Fitness: 1.4686428258375308
Generation 142 Best Solution: [182.9910838034768, 176.95619778655228,
175.04520360256166, 75.8899860635627] Fitness: 2.0343929223137085
Generation 143 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.37069799359423] Fitness: 1.8339363156447128
Generation 144 Best Solution: [183.33160848374942, 176.26913176751262,
175.04520360256166, 75.8899860635627] Fitness: 1.612479296807753
Generation 145 Best Solution: [183.51822641348275, 176.59639460275744,
175.04520360256166, 75.8899860635627] Fitness: 1.7625347683164367
Generation 146 Best Solution: [182.9910838034768, 177.44611955956603,
175.04520360256166, 74.79299422826068] Fitness: 1.7473186395545472
Generation 147 Best Solution: [182.86145466164936, 176.92365016288088,
175.04520360256166, 74.79299422826068] Fitness: 1.079779232303873
Generation 148 Best Solution: [183.51822641348275, 176.59639460275744,
175.04520360256166, 75.8899860635627] Fitness: 1.7625347683164367
Generation 149 Best Solution: [183.51822641348275, 176.59639460275744,
175.04520360256166, 75.8899860635627] Fitness: 1.7625347683164367
Generation 150 Best Solution: [183.50951264054743, 177.44611955956603,
174.88366870667795, 75.8899860635627] Fitness: 2.5478951114078416

Generation 151 Best Solution: [183.50951264054743, 177.44611955956603,
174.88366870667795, 74.89866861853638] Fitness: 1.587831103999302
Generation 152 Best Solution: [183.0007391137006, 177.44611955956603,
174.34245236252215, 75.8899860635627] Fitness: 2.171422615289047
Generation 153 Best Solution: [183.0007391137006, 176.92861769184393,
175.04520360256166, 75.8899860635627] Fitness: 2.002437828818395
Generation 154 Best Solution: [183.38917948923677, 177.44611955956603,
174.34245236252215, 75.8899860635627] Fitness: 2.14917235864601
Generation 155 Best Solution: [183.51822641348275, 176.23348216529666,
175.04520360256166, 75.8899860635627] Fitness: 1.6623629114884433
Generation 156 Best Solution: [183.38917948923677, 177.44611955956603,
174.34245236252215, 74.84827615865235] Fitness: 1.178217190968814
Generation 157 Best Solution: [183.51822641348275, 176.23348216529666,
175.04520360256166, 74.84827615865235] Fitness: 0.6914077438112474
Generation 158 Best Solution: [183.51822641348275, 176.23348216529666,
174.34245236252215, 75.74116107702599] Fitness: 0.8764240358658055
Generation 159 Best Solution: [182.86734820857353, 177.44611955956603,
174.34245236252215, 75.21654024450544] Fitness: 1.4225159043828743
Generation 160 Best Solution: [182.86145466164936, 177.36298082302926,
175.04520360256166, 74.7919717086692] Fitness: 1.6631627389972663
Generation 161 Best Solution: [182.86145466164936, 176.96097920691568,
173.9726885814944, 75.8899860635627] Fitness: 1.5423193709037055
Generation 162 Best Solution: [183.38917948923677, 175.72073009661227,
173.9726885814944, 75.8899860635627] Fitness: 1.2493873132978304
Generation 163 Best Solution: [183.38917948923677, 176.26045660666693,
175.04520360256166, 75.8834854748091] Fitness: 1.6115064673008088
Generation 164 Best Solution: [183.38917948923677, 176.62993738549454,
175.04520360256166, 75.8899860635627] Fitness: 1.7405594154781094
Generation 165 Best Solution: [182.86145466164936, 176.08376201294007,
174.5322272596919, 75.81139476282321] Fitness: 1.118328434590628
Generation 166 Best Solution: [182.86145466164936, 176.08376201294007,
174.22979572278243, 75.73901174736687] Fitness: 0.9078253308061255
Generation 167 Best Solution: [182.86145466164936, 176.08163320421804,
175.04520360256166, 75.81139476282321] Fitness: 1.5914587936792313
Generation 168 Best Solution: [182.86145466164936, 177.36298082302926,
175.04520360256166, 74.33726236405263] Fitness: 1.7677448033886667
Generation 169 Best Solution: [183.46684034977542, 177.36298082302926,
175.04520360256166, 74.33726236405263] Fitness: 1.7061244489791014
Generation 170 Best Solution: [183.51822641348275, 177.36298082302926,
175.04520360256166, 74.33726236405263] Fitness: 1.7252682354386084
Generation 171 Best Solution: [183.38917948923677, 175.90809744036207,
175.04520360256166, 75.81139476282321] Fitness: 1.565541029005432
Generation 172 Best Solution: [183.38917948923677, 175.90809744036207,
174.34245236252215, 75.8899860635627] Fitness: 1.142210393789258
Generation 173 Best Solution: [183.46684034977542, 177.36298082302926,
175.04520360256166, 74.48505270163794] Fitness: 1.6393732668526106
Generation 174 Best Solution: [183.46684034977542, 177.44611955956603,
175.04520360256166, 74.80975430749461] Fitness: 1.7453105426266133

Generation 175 Best Solution: [183.51822641348275, 177.44611955956603,
175.04520360256166, 74.88945614117803] Fitness: 1.7789323258377927
Generation 176 Best Solution: [182.86145466164936, 175.77679341339055,
175.04520360256166, 75.81139476282321] Fitness: 1.7211927484675758
Generation 177 Best Solution: [182.86145466164936, 175.77679341339055,
175.04520360256166, 75.8899860635627] Fitness: 1.8491129739061982
Generation 178 Best Solution: [182.86145466164936, 175.77679341339055,
175.04520360256166, 75.8899860635627] Fitness: 1.8491129739061982
Generation 179 Best Solution: [182.86145466164936, 175.77679341339055,
175.04520360256166, 75.81139476282321] Fitness: 1.7211927484675758
Generation 180 Best Solution: [182.86145466164936, 175.77679341339055,
175.04520360256166, 75.8899860635627] Fitness: 1.8491129739061982
Generation 181 Best Solution: [182.89027096355096, 177.44611955956603,
174.06172134175367, 74.55474426406373] Fitness: 1.235260284184301
Generation 182 Best Solution: [183.13204638409374, 176.1317984377985,
175.04520360256166, 75.8899860635627] Fitness: 1.6179210490059046
Generation 183 Best Solution: [182.86145466164936, 176.45939842790085,
175.04520360256166, 75.70154787547311] Fitness: 1.4530803156021321
Generation 184 Best Solution: [182.86145466164936, 177.12231938187864,
174.06172134175367, 75.81139476282321] Fitness: 1.5878088989051613
Generation 185 Best Solution: [183.46684034977542, 176.88004498026237,
175.04520360256166, 75.81139476282321] Fitness: 1.8262184891432804
Generation 186 Best Solution: [183.46684034977542, 177.44611955956603,
175.04520360256166, 74.69904910007291] Fitness: 1.7404244803578957
Generation 187 Best Solution: [183.13204638409374, 176.68731054865432,
174.64732791903162, 75.81139476282321] Fitness: 1.2393060420515083
Generation 188 Best Solution: [182.86145466164936, 176.98725128283783,
174.06172134175367, 75.8899860635627] Fitness: 1.5536142489680014
Generation 189 Best Solution: [183.01740646216493, 176.66122541208324,
175.04520360256166, 75.81139476282321] Fitness: 1.6485193205226487
Generation 190 Best Solution: [183.46684034977542, 176.30590715481102,
175.04520360256166, 75.81139476282321] Fitness: 1.5200952894300284
Generation 191 Best Solution: [183.46684034977542, 176.1637795456394,
174.94720367805522, 74.62789562709078] Fitness: 0.5505939280436236
Generation 192 Best Solution: [183.46684034977542, 176.1637795456394,
174.94720367805522, 75.8899860635627] Fitness: 1.526081995312414
Generation 193 Best Solution: [183.46684034977542, 176.1637795456394,
174.94720367805522, 74.62789562709078] Fitness: 0.5505939280436236
Generation 194 Best Solution: [183.63113948086843, 175.85406252648502,
175.04520360256166, 74.62789562709078] Fitness: 0.8410282820142654
Generation 195 Best Solution: [183.13204638409374, 175.66938596311473,
174.06890483364356, 75.8899860635627] Fitness: 1.2608848999471198
Generation 196 Best Solution: [183.46684034977542, 176.12320070762513,
175.04520360256166, 75.81139476282321] Fitness: 1.5225475935163484
Generation 197 Best Solution: [183.13204638409374, 177.44611955956603,
174.12441190932597, 74.79950952857757] Fitness: 1.1354795053615512
Generation 198 Best Solution: [183.13204638409374, 177.44611955956603,
174.94720367805522, 74.94196246925218] Fitness: 1.6233138228953345

Generation 199 Best Solution: [183.13204638409374, 176.49110518692524,
175.04520360256166, 75.58810237116094] Fitness: 1.220706864298247
Generation 200 Best Solution: [183.46684034977542, 176.89960171293316,
174.94720367805522, 75.58810237116094] Fitness: 1.4105943996756314
Generation 201 Best Solution: [183.46684034977542, 177.02284933659914,
173.82029108705018, 75.8899860635627] Fitness: 1.605314152125878
Generation 202 Best Solution: [183.12815654177348, 177.02284933659914,
173.82029108705018, 75.8899860635627] Fitness: 1.5745511332416269
Generation 203 Best Solution: [183.12815654177348, 176.12074392413868,
175.04520360256166, 75.81139476282321] Fitness: 1.4921460930228259
Generation 204 Best Solution: [183.12815654177348, 176.89960171293316,
175.04520360256166, 74.94196246925218] Fitness: 0.9944109445962154
Generation 205 Best Solution: [183.12815654177348, 176.12074392413868,
175.04520360256166, 75.81139476282321] Fitness: 1.4921460930228259
Generation 206 Best Solution: [183.13204638409374, 177.01257560489057,
175.04520360256166, 75.46231766621923] Fitness: 1.4728116187884168
Generation 207 Best Solution: [183.12815654177348, 175.708107934202,
175.04520360256166, 75.8899860635627] Fitness: 1.8044899817172295
Generation 208 Best Solution: [182.918712176605, 177.44611955956603,
173.7916312856946, 75.46231766621923] Fitness: 1.6823281312586138
Generation 209 Best Solution: [183.46684034977542, 175.93821023678998,
175.04520360256166, 75.8899860635627] Fitness: 1.702076642648159
Generation 210 Best Solution: [183.46684034977542, 175.93821023678998,
174.94720367805522, 75.8899860635627] Fitness: 1.5824007973115095
Generation 211 Best Solution: [183.46684034977542, 175.93821023678998,
174.94720367805522, 75.8899860635627] Fitness: 1.5824007973115095
Generation 212 Best Solution: [182.79422351494276, 175.93821023678998,
174.94720367805522, 75.8899860635627] Fitness: 1.6835238716221828
Generation 213 Best Solution: [182.918712176605, 175.93821023678998,
174.94720367805522, 75.8899860635627] Fitness: 1.615526532369268
Generation 214 Best Solution: [182.918712176605, 175.708107934202,
175.04520360256166, 75.8899860635627] Fitness: 1.868378735659239
Generation 215 Best Solution: [182.918712176605, 177.44611955956603,
175.04520360256166, 75.72554926425308] Fitness: 2.4977904908489865
Generation 216 Best Solution: [183.63113948086843, 177.44611955956603,
174.94720367805522, 75.20645630576307] Fitness: 1.8631268384212998
Generation 217 Best Solution: [182.918712176605, 177.44611955956603,
175.04520360256166, 74.75905684481701] Fitness: 1.7740015312754633
Generation 218 Best Solution: [183.63113948086843, 177.44611955956603,
174.94720367805522, 74.31524099605403] Fitness: 1.8154931168176154
Generation 219 Best Solution: [182.918712176605, 176.24039329114203,
174.94720367805522, 75.20645630576307] Fitness: 0.7428771215035039
Generation 220 Best Solution: [183.49687448416307, 177.44611955956603,
174.94720367805522, 74.96788113685828] Fitness: 1.674041109400605
Generation 221 Best Solution: [183.57965817448857, 176.84263524845449,
174.94720367805522, 75.72554926425308] Fitness: 1.5895977354428252
Generation 222 Best Solution: [183.63113948086843, 175.90062739217734,
174.94720367805522, 75.8899860635627] Fitness: 1.6735347083238923

Generation 223 Best Solution: [183.63113948086843, 175.90062739217734,
174.94720367805522, 75.8899860635627] Fitness: 1.6735347083238923
Generation 224 Best Solution: [182.79422351494276, 176.71501051555816,
175.0149015458712, 75.8899860635627] Fitness: 1.8800198699378194
Generation 225 Best Solution: [182.79422351494276, 176.71501051555816,
175.0149015458712, 75.8899860635627] Fitness: 1.8800198699378194
Generation 226 Best Solution: [183.29637260620913, 177.23588625896897,
174.08009946975926, 75.8899860635627] Fitness: 1.7724963331995953
Generation 227 Best Solution: [183.45340980440812, 176.67682374680143,
175.04520360256166, 75.8899860635627] Fitness: 1.7869690883063039
Generation 228 Best Solution: [183.29637260620913, 175.93772100263345,
175.11059358373709, 75.8899860635627] Fitness: 1.7536542785168048
Generation 229 Best Solution: [183.45492353770163, 176.37334980217707,
174.08009946975926, 75.8899860635627] Fitness: 1.081219812477324
Generation 230 Best Solution: [183.29637260620913, 175.93772100263345,
174.41272506793996, 75.8899860635627] Fitness: 1.1378573305728912
Generation 231 Best Solution: [182.83145178091823, 175.87155644373985,
174.08009946975926, 75.8899860635627] Fitness: 1.2370478571528958
Generation 232 Best Solution: [183.45340980440812, 175.78704527710292,
175.11059358373709, 74.35245321720669] Fitness: 0.981623908173675
Generation 233 Best Solution: [182.83145178091823, 175.93772100263345,
174.08009946975926, 75.8899860635627] Fitness: 1.2065408743765724
Generation 234 Best Solution: [182.83145178091823, 175.93772100263345,
175.04520360256166, 74.35245321720669] Fitness: 0.8982018583081273
Generation 235 Best Solution: [183.45340980440812, 175.78704527710292,
175.04520360256166, 75.8899860635627] Fitness: 1.7765736427870484
Generation 236 Best Solution: [183.45340980440812, 175.78704527710292,
174.8573994755191, 75.8899860635627] Fitness: 1.5594111397631552
Generation 237 Best Solution: [183.29637260620913, 175.78704527710292,
174.8573994755191, 75.8899860635627] Fitness: 1.5275921637478755
Generation 238 Best Solution: [183.45340980440812, 175.78704527710292,
174.8573994755191, 75.8899860635627] Fitness: 1.5594111397631552
Generation 239 Best Solution: [183.79669650922207, 175.78704527710292,
174.8573994755191, 75.8899860635627] Fitness: 1.7530130565735367
Generation 240 Best Solution: [183.44310793876093, 176.3985885051598,
175.04520360256166, 75.8899860635627] Fitness: 1.6576511092019646
Generation 241 Best Solution: [183.44310793876093, 176.3985885051598,
175.04520360256166, 75.45498546678886] Fitness: 1.0615884023495112
Generation 242 Best Solution: [183.79669650922207, 176.3985885051598,
175.04520360256166, 75.45498546678886] Fitness: 1.2583694352142263
Generation 243 Best Solution: [183.79669650922207, 176.3985885051598,
175.04520360256166, 75.8899860635627] Fitness: 1.8544321420666794
Generation 244 Best Solution: [183.45340980440812, 177.44611955956603,
175.04520360256166, 74.30301371273852] Fitness: 1.8635110899646323
Generation 245 Best Solution: [183.79669650922207, 176.3985885051598,
174.33341290004404, 75.8899860635627] Fitness: 1.3007374886373415
Generation 246 Best Solution: [183.79669650922207, 176.3985885051598,
174.33341290004404, 75.8899860635627] Fitness: 1.3007374886373415

Generation 247 Best Solution: [183.4530004778033, 177.44611955956603,
175.04520360256166, 74.61158794865806] Fitness: 1.7445777096974135
Generation 248 Best Solution: [183.45340980440812, 177.44611955956603,
175.04520360256166, 74.61158794865806] Fitness: 1.7447069507904531
Generation 249 Best Solution: [183.4530004778033, 175.7708831880689,
175.04520360256166, 75.8899860635627] Fitness: 1.786830276791062
Generation 250 Best Solution: [183.79669650922207, 176.68723446482844,
174.020688482432, 75.8899860635627] Fitness: 1.4218946926075913
Generation 251 Best Solution: [183.45340980440812, 176.8485707973719,
173.95849419204342, 75.8899860635627] Fitness: 1.3695600512267254
Generation 252 Best Solution: [183.45340980440812, 176.8485707973719,
173.95849419204342, 75.8899860635627] Fitness: 1.3695600512267254
Generation 253 Best Solution: [183.45340980440812, 176.8485707973719,
174.71016783122002, 75.8899860635627] Fitness: 1.568862403876766
Generation 254 Best Solution: [183.4530004778033, 177.44611955956603,
175.04520360256166, 74.4438573458329] Fitness: 1.7920940299056467
Generation 255 Best Solution: [183.4530004778033, 176.68723446482844,
175.04520360256166, 75.59940099845349] Fitness: 1.3652455311533984
Generation 256 Best Solution: [183.79669650922207, 177.44611955956603,
174.8623653283324, 75.141109958792] Fitness: 1.8428768345908553
Generation 257 Best Solution: [183.79669650922207, 177.44611955956603,
175.04520360256166, 74.68466033228879] Fitness: 1.9303164849190542
Generation 258 Best Solution: [183.79669650922207, 177.44611955956603,
174.1483102410051, 75.04824973190478] Fitness: 1.4274297961651552
Generation 259 Best Solution: [183.76847606229381, 177.44611955956603,
174.1483102410051, 75.04824973190478] Fitness: 1.4050929307717095
Generation 260 Best Solution: [183.70103870458306, 176.11044600372793,
174.1483102410051, 75.04824973190478] Fitness: 0.2866962906282259
Generation 261 Best Solution: [183.79669650922207, 177.05921504795054,
175.11059358373709, 75.13913883209644] Fitness: 1.5663680039268204
Generation 262 Best Solution: [183.79669650922207, 176.74581860227192,
174.47426023407812, 75.8899860635627] Fitness: 1.5269433147091012
Generation 263 Best Solution: [183.79669650922207, 177.44611955956603,
174.47426023407812, 74.44255209792635] Fitness: 1.484145219019939
Generation 264 Best Solution: [183.79669650922207, 177.44611955956603,
174.60776686701385, 74.67020310407263] Fitness: 1.5043325154454652
Generation 265 Best Solution: [183.79669650922207, 177.44611955956603,
175.11059358373709, 74.67020310407263] Fitness: 2.018855145291069
Generation 266 Best Solution: [183.79669650922207, 176.38349747446034,
174.47426023407812, 75.8899860635627] Fitness: 1.3485784810622914
Generation 267 Best Solution: [183.79669650922207, 177.44611955956603,
173.78578550205873, 74.28491681829956] Fitness: 1.5867724975321316
Generation 268 Best Solution: [183.71784379444924, 175.71501477829347,
175.04520360256166, 75.8899860635627] Fitness: 1.959839742225764
Generation 269 Best Solution: [183.5339361547535, 177.44611955956603,
175.04520360256166, 75.60859492475494] Fitness: 2.331079467893256
Generation 270 Best Solution: [183.45340980440812, 175.7294900046119,
175.04520360256166, 75.8899860635627] Fitness: 1.8152796957152015

Generation 271 Best Solution: [183.45340980440812, 175.7294900046119,
175.04520360256166, 75.8899860635627] Fitness: 1.8152796957152015
Generation 272 Best Solution: [183.79669650922207, 177.01623064673882,
173.78578550205873, 75.8899860635627] Fitness: 1.804265555187792
Generation 273 Best Solution: [183.79669650922207, 176.63454793489112,
175.04520360256166, 75.8899860635627] Fitness: 1.9542006880078695
Generation 274 Best Solution: [183.5339361547535, 177.44611955956603,
175.04520360256166, 74.55031327905776] Fitness: 1.7874868302877134
Generation 275 Best Solution: [183.59410682499944, 177.44611955956603,
175.04520360256166, 74.3927900524402] Fitness: 1.873522026229298
Generation 276 Best Solution: [183.79669650922207, 177.44611955956603,
175.04520360256166, 74.36398488661648] Fitness: 2.022735364893124
Generation 277 Best Solution: [183.79669650922207, 177.44611955956603,
174.012670004786, 75.8899860635627] Fitness: 2.3473314318939855
Generation 278 Best Solution: [183.79669650922207, 176.39316551482128,
175.04520360256166, 75.8899860635627] Fitness: 1.8530845885421905
Generation 279 Best Solution: [183.79669650922207, 176.915966318963,
175.04520360256166, 75.8899860635627] Fitness: 2.178345510791327
Generation 280 Best Solution: [183.79669650922207, 176.915966318963,
175.04520360256166, 75.8899860635627] Fitness: 2.178345510791327
Generation 281 Best Solution: [183.79669650922207, 176.915966318963,
173.8147204475136, 75.8899860635627] Fitness: 1.6821181336074662
Generation 282 Best Solution: [183.38452197680195, 176.915966318963,
175.04520360256166, 74.86010208232406] Fitness: 0.9976711997799766
Generation 283 Best Solution: [183.38452197680195, 176.915966318963,
175.04520360256166, 74.86010208232406] Fitness: 0.9976711997799766
Generation 284 Best Solution: [183.79669650922207, 176.915966318963,
175.04520360256166, 74.93966289631702] Fitness: 1.229847149950278
Generation 285 Best Solution: [183.38452197680195, 176.915966318963,
175.04520360256166, 75.2662140164018] Fitness: 1.1967136366545068
Generation 286 Best Solution: [183.79669650922207, 176.915966318963,
175.04520360256166, 75.2662140164018] Fitness: 1.4086592841742713
Generation 287 Best Solution: [183.38452197680195, 176.915966318963,
175.04520360256166, 75.00568822930815] Fitness: 1.0416323446088398
Generation 288 Best Solution: [183.79669650922207, 176.84245848937317,
175.04520360256166, 74.38333174009605] Fitness: 1.21036455431103
Generation 289 Best Solution: [183.79669650922207, 176.20539818039884,
175.04520360256166, 75.8899860635627] Fitness: 1.8326249586756778
Generation 290 Best Solution: [183.38452197680195, 176.6422763040393,
175.04520360256166, 75.8899860635627] Fitness: 1.7468829232260576
Generation 291 Best Solution: [183.79669650922207, 177.44611955956603,
174.05370156317116, 75.2662140164018] Fitness: 1.570676292298314
Generation 292 Best Solution: [183.79669650922207, 175.90121505931023,
175.04520360256166, 75.8899860635627] Fitness: 1.9075558720469432
Generation 293 Best Solution: [183.79669650922207, 175.90121505931023,
175.04520360256166, 75.8899860635627] Fitness: 1.9075558720469432
Generation 294 Best Solution: [183.79669650922207, 176.6422763040393,
175.04520360256166, 75.2662140164018] Fitness: 1.1891423441287663

Generation 295 Best Solution: [183.09515479479043, 177.44611955956603,
175.04520360256166, 74.74130047179042] Fitness: 1.7152987888649769
Generation 296 Best Solution: [183.79669650922207, 176.04066375669612,
175.04520360256166, 75.8899860635627] Fitness: 1.85661390410441
Generation 297 Best Solution: [183.09515479479043, 177.44611955956603,
175.04520360256166, 75.58473191377212] Fitness: 2.250335194396606
Generation 298 Best Solution: [183.62863599520566, 176.3817807107347,
175.04520360256166, 75.8899860635627] Fitness: 1.7343449929567296
Generation 299 Best Solution: [183.62863599520566, 175.65051044868824,
175.04520360256166, 75.8899860635627] Fitness: 1.953734803242374
Generation 300 Best Solution: [183.0415854981605, 176.21788129291554,
175.04520360256166, 75.7604462508993] Fitness: 1.4250591206631242
Generation 301 Best Solution: [183.0255406847133, 176.47740029940485,
175.04520360256166, 75.7604462508993] Fitness: 1.4761244338353277
Generation 302 Best Solution: [183.79669650922207, 177.23556210781547,
175.04520360256166, 75.7604462508993] Fitness: 2.365542853541892
Generation 303 Best Solution: [183.79669650922207, 177.23556210781547,
174.0487242170102, 75.8899860635627] Fitness: 2.0013571465286866
Generation 304 Best Solution: [183.79669650922207, 177.44611955956603,
175.04520360256166, 74.98408288705087] Fitness: 1.9781186612009007
Generation 305 Best Solution: [183.2467543400796, 176.15317139313206,
175.04520360256166, 75.7604462508993] Fitness: 1.4018369024599922
Generation 306 Best Solution: [183.79669650922207, 176.15317139313206,
175.04520360256166, 75.7604462508993] Fitness: 1.6299058954150882
Generation 307 Best Solution: [183.2467543400796, 177.23556210781547,
174.0487242170102, 75.8899860635627] Fitness: 1.7732881535735905
Generation 308 Best Solution: [183.2467543400796, 177.23556210781547,
174.86181714723682, 75.7604462508993] Fitness: 1.9248345174145438
Generation 309 Best Solution: [183.7787476010307, 176.5807767052523,
175.04520360256166, 75.20368278229112] Fitness: 1.0941994426139712
Generation 310 Best Solution: [183.0415854981605, 177.44611955956603,
175.04520360256166, 75.02642649033723] Fitness: 1.7940654189186478
Generation 311 Best Solution: [183.0415854981605, 177.23556210781547,
175.04520360256166, 74.81057919116773] Fitness: 1.3936543349149473
Generation 312 Best Solution: [183.7787476010307, 176.5807767052523,
174.86181714723682, 75.8899860635627] Fitness: 1.6974105598568152
Generation 313 Best Solution: [183.2467543400796, 177.23556210781547,
174.99518015009696, 75.7604462508993] Fitness: 2.0746527727018176
Generation 314 Best Solution: [183.0415854981605, 177.23556210781547,
175.04520360256166, 75.7604462508993] Fitness: 2.164277825960426
Generation 315 Best Solution: [183.0415854981605, 177.2485085852807,
175.04520360256166, 75.7604462508993] Fitness: 2.1833184398162486
Generation 316 Best Solution: [183.0415854981605, 177.2485085852807,
175.04520360256166, 75.7604462508993] Fitness: 2.1833184398162486
Generation 317 Best Solution: [183.2467543400796, 177.44611955956603,
175.04520360256166, 74.40171856446096] Fitness: 1.7760807429648364
Generation 318 Best Solution: [183.59289340192527, 177.44611955956603,
175.04520360256166, 74.39306840849004] Fitness: 1.8727599492735894

Generation 319 Best Solution: [183.2467543400796, 176.38695423668963,
173.7437873787266, 75.8899860635627] Fitness: 1.1653620167410055
Generation 320 Best Solution: [183.2467543400796, 176.38695423668963,
173.7437873787266, 75.7604462508993] Fitness: 0.9592813022020893
Generation 321 Best Solution: [183.59289340192527, 177.44611955956603,
175.04520360256166, 74.94952992892557] Fitness: 1.8305238516983633
Generation 322 Best Solution: [183.59289340192527, 177.44611955956603,
175.04520360256166, 74.94952992892557] Fitness: 1.8305238516983633
Generation 323 Best Solution: [183.59289340192527, 176.36686335370624,
175.04520360256166, 75.8899860635627] Fitness: 1.711682736683371
Generation 324 Best Solution: [183.59289340192527, 176.83320268192486,
175.04520360256166, 75.86374174163889] Fitness: 1.9213735462164383
Generation 325 Best Solution: [183.59289340192527, 177.44611955956603,
174.56863503292783, 75.8899860635627] Fitness: 2.324197483715273
Generation 326 Best Solution: [183.59289340192527, 175.6876388855424,
175.04520360256166, 75.8899860635627] Fitness: 1.9045625198298883
Generation 327 Best Solution: [183.59289340192527, 176.02217105152974,
175.04520360256166, 75.8899860635627] Fitness: 1.726285295388969
Generation 328 Best Solution: [183.59289340192527, 177.44611955956603,
175.04520360256166, 75.35987970456148] Fitness: 2.085905609860521
Generation 329 Best Solution: [183.59289340192527, 176.31799515599965,
175.04520360256166, 75.8899860635627] Fitness: 1.7033122642921963
Generation 330 Best Solution: [183.59289340192527, 177.44611955956603,
175.04520360256166, 75.11908360419172] Fitness: 1.906559028680604
Generation 331 Best Solution: [183.59289340192527, 177.44611955956603,
175.04520360256166, 75.35987970456148] Fitness: 2.085905609860521
Generation 332 Best Solution: [183.48761436553013, 177.44611955956603,
175.04520360256166, 74.84078197287404] Fitness: 1.7571360827662708
Generation 333 Best Solution: [183.48761436553013, 177.44611955956603,
175.04520360256166, 74.7076202296187] Fitness: 1.7474505734878845
Generation 334 Best Solution: [182.84047704141096, 175.9506863303132,
175.04520360256166, 75.8899860635627] Fitness: 1.7702869285295382
Generation 335 Best Solution: [183.48761436553013, 177.44611955956603,
175.04520360256166, 75.35987970456148] Fitness: 2.039427249112366
Generation 336 Best Solution: [183.3305001062595, 177.44611955956603,
175.04520360256166, 75.35987970456148] Fitness: 1.9998389236407703
Generation 337 Best Solution: [183.6554575698037, 176.19260045903957,
175.04520360256166, 75.8899860635627] Fitness: 1.7328202897251037
Generation 338 Best Solution: [183.6554575698037, 176.19260045903957,
175.04520360256166, 75.8899860635627] Fitness: 1.7328202897251037
Generation 339 Best Solution: [183.6554575698037, 176.19260045903957,
175.04520360256166, 75.62071589231296] Fitness: 1.3316208177067321
Generation 340 Best Solution: [183.59289340192527, 176.19260045903957,
175.04520360256166, 75.62071589231296] Fitness: 1.2964160903230924
Generation 341 Best Solution: [182.84047704141096, 177.44611955956603,
173.86405161269278, 74.41799981985733] Fitness: 1.3623684703889078
Generation 342 Best Solution: [182.84047704141096, 177.44611955956603,
174.51018396259937, 75.69906744423521] Fitness: 2.0156087508340406

Generation 343 Best Solution: [183.59289340192527, 177.44611955956603,
174.7017382814947, 75.71254986171914] Fitness: 2.1411920053393096
Generation 344 Best Solution: [183.6554575698037, 177.44611955956603,
175.04520360256166, 74.7080770569882] Fitness: 1.8291231054742818
Generation 345 Best Solution: [183.6554575698037, 177.44611955956603,
175.06351617329233, 75.09906622275464] Fitness: 1.9545265736624977
Generation 346 Best Solution: [183.29437326535609, 177.44611955956603,
175.03238560371528, 74.3709760814014] Fitness: 1.7770985980843275
Generation 347 Best Solution: [183.59289340192527, 177.44611955956603,
174.51018396259937, 75.38624576168114] Fitness: 1.6261394247985121
Generation 348 Best Solution: [183.59289340192527, 177.44611955956603,
174.51018396259937, 75.38624576168114] Fitness: 1.6261394247985121
Generation 349 Best Solution: [183.48761436553013, 177.44611955956603,
174.51018396259937, 75.38624576168114] Fitness: 1.5796610640503574
Generation 350 Best Solution: [183.48761436553013, 177.44611955956603,
174.51018396259937, 75.38624576168114] Fitness: 1.5796610640503574
Generation 351 Best Solution: [183.48761436553013, 177.44611955956603,
174.51018396259937, 75.38624576168114] Fitness: 1.5796610640503574
Generation 352 Best Solution: [182.82141297913233, 177.44611955956603,
174.13255570666448, 75.79861948148026] Fitness: 2.0810602248602375
Generation 353 Best Solution: [182.82141297913233, 177.44611955956603,
174.13255570666448, 75.79861948148026] Fitness: 2.0810602248602375
Generation 354 Best Solution: [183.48761436553013, 177.44611955956603,
174.13255570666448, 75.79861948148026] Fitness: 2.0039790767385752
Generation 355 Best Solution: [183.6554575698037, 176.9846368326431,
174.13255570666448, 75.79861948148026] Fitness: 1.4247206272028965
Generation 356 Best Solution: [182.82141297913233, 176.97262885129237,
175.04520360256166, 75.8899860635627] Fitness: 2.1324455746683837
Generation 357 Best Solution: [183.6554575698037, 177.44611955956603,
175.04520360256166, 74.37982852633606] Fitness: 1.9144171625345854
Generation 358 Best Solution: [182.82141297913233, 177.44611955956603,
174.24291963969765, 75.8899860635627] Fitness: 2.2348246211041323
Generation 359 Best Solution: [182.82141297913233, 177.44611955956603,
175.04520360256166, 75.39644346989988] Fitness: 2.1510656269591784
Generation 360 Best Solution: [183.48761436553013, 177.44611955956603,
175.04520360256166, 75.8899860635627] Fitness: 2.7293976351221625
Generation 361 Best Solution: [182.84047704141096, 177.44611955956603,
175.04520360256166, 74.74729788217348] Fitness: 1.813622922287816
Generation 362 Best Solution: [183.6554575698037, 177.44611955956603,
175.04520360256166, 74.73199743290657] Fitness: 1.8289915013714968
Generation 363 Best Solution: [183.48761436553013, 177.44611955956603,
175.04520360256166, 74.73199743290657] Fitness: 1.747308413239702
Generation 364 Best Solution: [183.48761436553013, 176.46534827413524,
175.04520360256166, 74.73199743290657] Fitness: 0.7104652435209747
Generation 365 Best Solution: [182.82141297913233, 177.18615234380596,
175.04520360256166, 74.73199743290657] Fitness: 1.4142257849037883
Generation 366 Best Solution: [183.81538277743303, 177.44611955956603,
173.78038435422806, 74.73199743290657] Fitness: 1.46596991824299

Generation 367 Best Solution: [182.82141297913233, 177.44611955956603,
175.04520360256166, 74.55624617220728] Fitness: 1.844628618941107
Generation 368 Best Solution: [182.84047704141096, 177.44611955956603,
175.04520360256166, 74.55624617220728] Fitness: 1.8335127563816789
Generation 369 Best Solution: [182.82141297913233, 177.44611955956603,
175.04520360256166, 74.55624617220728] Fitness: 1.844628618941107
Generation 370 Best Solution: [182.82141297913233, 177.44611955956603,
175.04520360256166, 74.55624617220728] Fitness: 1.844628618941107
Generation 371 Best Solution: [183.20463444970318, 177.44611955956603,
175.04520360256166, 74.55624617220728] Fitness: 1.7219682951242201
Generation 372 Best Solution: [183.20463444970318, 177.44611955956603,
175.04520360256166, 74.55624617220728] Fitness: 1.7219682951242201
Generation 373 Best Solution: [183.20463444970318, 177.44611955956603,
175.04520360256166, 74.55624617220728] Fitness: 1.7219682951242201
Generation 374 Best Solution: [182.82141297913233, 175.99635845189732,
175.04520360256166, 75.8899860635627] Fitness: 1.7648887216162046
Generation 375 Best Solution: [182.82141297913233, 175.99635845189732,
175.04520360256166, 75.8899860635627] Fitness: 1.7648887216162046
Generation 376 Best Solution: [183.06885471661465, 176.32303384606288,
175.04520360256166, 75.8899860635627] Fitness: 1.6311533268278193
Generation 377 Best Solution: [183.5006989920331, 175.99635845189732,
175.04520360256166, 75.8899860635627] Fitness: 1.692712908122479
Generation 378 Best Solution: [183.5006989920331, 177.44611955956603,
173.76568459963462, 75.8899860635627] Fitness: 2.263614373083474
Generation 379 Best Solution: [182.82141297913233, 176.71179373192666,
175.04520360256166, 75.8899860635627] Fitness: 1.8994691881393528
Generation 380 Best Solution: [182.82141297913233, 177.44611955956603,
175.04520360256166, 74.6964647695449] Fitness: 1.8248830535350928
Generation 381 Best Solution: [183.06885471661465, 177.44611955956603,
175.04520360256166, 74.6964647695449] Fitness: 1.7214177660388665
Generation 382 Best Solution: [183.81538277743303, 177.44611955956603,
175.04520360256166, 74.30502605276658] Fitness: 2.071316162192244
Generation 383 Best Solution: [183.81538277743303, 177.44611955956603,
175.04520360256166, 74.63761511835162] Fitness: 1.9500012559358013
Generation 384 Best Solution: [183.50286299426315, 177.44611955956603,
175.04520360256166, 74.63761511835162] Fitness: 1.7583711560346864
Generation 385 Best Solution: [183.81538277743303, 177.44611955956603,
175.04520360256166, 74.68101024752355] Fitness: 1.9459561783827044
Generation 386 Best Solution: [182.82141297913233, 177.44611955956603,
174.7423257525876, 75.22952375356061] Fitness: 1.6839676524359302
Generation 387 Best Solution: [183.81538277743303, 177.44611955956603,
173.8051576472421, 75.33304106086044] Fitness: 1.721141430278207
Generation 388 Best Solution: [183.11383622102198, 176.17821496002287,
175.04520360256166, 75.8899860635627] Fitness: 1.6162430597856807
Generation 389 Best Solution: [183.81538277743303, 176.78925614260703,
173.8051576472421, 75.33304106086044] Fitness: 0.8730620025996925
Generation 390 Best Solution: [183.11383622102198, 176.17821496002287,
175.04520360256166, 75.8899860635627] Fitness: 1.6162430597856807

Generation 391 Best Solution: [183.11383622102198, 176.17821496002287,
174.11565959939028, 75.8899860635627] Fitness: 1.0393462781209097
Generation 392 Best Solution: [183.11383622102198, 176.17821496002287,
175.04520360256166, 75.79536963701636] Fitness: 1.4633343774173195
Generation 393 Best Solution: [183.06885471661465, 177.44611955956603,
174.2881038255332, 75.8899860635627] Fitness: 2.138848123035296
Generation 394 Best Solution: [183.06885471661465, 177.44611955956603,
175.04520360256166, 75.43700798878884] Fitness: 2.088198366814608
Generation 395 Best Solution: [183.5080047012496, 175.6179735517338,
175.04520360256166, 75.8899860635627] Fitness: 1.9232983770643999
Generation 396 Best Solution: [183.81538277743303, 176.2419892019412,
175.04520360256166, 75.8899860635627] Fitness: 1.8480403705135873
Generation 397 Best Solution: [183.81538277743303, 176.49339518112524,
175.04520360256166, 75.79536963701636] Fitness: 1.7473682216378894
Generation 398 Best Solution: [183.81538277743303, 176.3013355313822,
175.04520360256166, 75.8899860635627] Fitness: 1.852139294662346
Generation 399 Best Solution: [183.1671809720024, 177.44611955956603,
175.04520360256166, 75.29196232623167] Fitness: 1.9374178548658496
Generation 400 Best Solution: [183.81538277743303, 177.44611955956603,
173.84035495563603, 75.29196232623167] Fitness: 1.6695366347069989
Generation 401 Best Solution: [183.54103557716627, 177.44611955956603,
175.04520360256166, 75.3735111581366] Fitness: 2.0736683753846585
Generation 402 Best Solution: [183.81538277743303, 177.44611955956603,
174.37312224394762, 75.40046446505556] Fitness: 1.7331934414854582
Generation 403 Best Solution: [183.81538277743303, 176.59653565494398,
175.04520360256166, 75.8899860635627] Fitness: 1.9481174717606267
Generation 404 Best Solution: [183.54103557716627, 177.44611955956603,
175.04520360256166, 74.94105623532] Fitness: 1.802918532163786
Generation 405 Best Solution: [183.81538277743303, 175.98197761205685,
175.04520360256166, 75.8899860635627] Fitness: 1.8900529192397193
Generation 406 Best Solution: [183.54103557716627, 177.44611955956603,
175.04520360256166, 74.89146041688039] Fitness: 1.78913443115391
Generation 407 Best Solution: [183.54103557716627, 177.44611955956603,
175.04520360256166, 74.89146041688039] Fitness: 1.78913443115391
Generation 408 Best Solution: [183.81538277743303, 175.98197761205685,
175.04520360256166, 75.58449113920426] Fitness: 1.442872781642405
Generation 409 Best Solution: [183.81538277743303, 175.98197761205685,
175.04520360256166, 75.58449113920426] Fitness: 1.442872781642405
Generation 410 Best Solution: [183.81538277743303, 175.98197761205685,
175.04520360256166, 75.58449113920426] Fitness: 1.442872781642405
Generation 411 Best Solution: [183.35262109629474, 176.31114631890148,
175.04520360256166, 75.58449113920426] Fitness: 1.17258509495473
Generation 412 Best Solution: [183.81538277743303, 177.44611955956603,
175.04520360256166, 74.55195401139488] Fitness: 1.9659703961025987
Generation 413 Best Solution: [183.54103557716627, 177.44611955956603,
175.0251490021786, 75.3822721387086] Fitness: 2.056325869069751
Generation 414 Best Solution: [183.22155711513918, 177.44611955956603,
174.62354170233905, 75.3822721387086] Fitness: 1.5978960423975546

Generation 415 Best Solution: [183.03632357713437, 177.44611955956603,
175.04520360256166, 75.09895656988158] Fitness: 1.8310519125005762
Generation 416 Best Solution: [183.03632357713437, 177.44611955956603,
174.65801995257476, 75.09895656988158] Fitness: 1.4390950147933972
Generation 417 Best Solution: [183.22155711513918, 177.44611955956603,
174.15667115050334, 75.58449113920426] Fitness: 1.6583504017758701
Generation 418 Best Solution: [183.81538277743303, 177.44611955956603,
174.15667115050334, 75.58449113920426] Fitness: 1.9018261351125634
Generation 419 Best Solution: [183.54103557716627, 176.27121493888353,
175.0251490021786, 75.8899860635627] Fitness: 1.6480168387612828
Generation 420 Best Solution: [183.54103557716627, 176.27121493888353,
175.0251490021786, 75.7770314759294] Fitness: 1.4669680297075307
Generation 421 Best Solution: [183.81538277743303, 177.44611955956603,
175.0251490021786, 74.61447035561109] Fitness: 1.9276516652981857
Generation 422 Best Solution: [183.81538277743303, 176.54627525864925,
175.0251490021786, 75.8899860635627] Fitness: 1.8972659514174701
Generation 423 Best Solution: [183.81538277743303, 176.54627525864925,
175.0251490021786, 75.8899860635627] Fitness: 1.8972659514174701
Generation 424 Best Solution: [183.33083751008354, 177.44611955956603,
175.0251490021786, 74.45834830486977] Fitness: 1.7330084253388542
Generation 425 Best Solution: [183.21653191036464, 177.44611955956603,
175.04520360256166, 75.61107659780905] Fitness: 2.269785595769564
Generation 426 Best Solution: [183.81538277743303, 177.44611955956603,
174.26585395943346, 75.73017695770885] Fitness: 2.1078050381819367
Generation 427 Best Solution: [183.21653191036464, 177.44611955956603,
175.04520360256166, 75.73017695770885] Fitness: 2.4326639146255946
Generation 428 Best Solution: [183.81538277743303, 176.95944553638608,
175.0251490021786, 75.8899860635627] Fitness: 2.2129824520617483
Generation 429 Best Solution: [183.81538277743303, 176.95944553638608,
175.0251490021786, 75.8899860635627] Fitness: 2.2129824520617483
Generation 430 Best Solution: [183.81538277743303, 176.95944553638608,
175.0251490021786, 75.73017695770885] Fitness: 1.9622409060061836
Generation 431 Best Solution: [183.81538277743303, 176.82011213519093,
175.04520360256166, 75.8899860635627] Fitness: 2.1045765568520967
Generation 432 Best Solution: [183.81538277743303, 177.44611955956603,
175.04520360256166, 74.36440522760311] Fitness: 2.0379404694740177
Generation 433 Best Solution: [183.13067950207108, 177.24712239504873,
175.04520360256166, 74.62088316292692] Fitness: 1.3937597668253139
Generation 434 Best Solution: [183.13067950207108, 177.24712239504873,
175.04520360256166, 74.66029305011594] Fitness: 1.3890202697937146
Generation 435 Best Solution: [183.81538277743303, 176.10978827285774,
175.04520360256166, 75.8899860635627] Fitness: 1.8571981699734788
Generation 436 Best Solution: [183.07244873849422, 175.8210128238536,
175.0251490021786, 75.8899860635627] Fitness: 1.714775866355006
Generation 437 Best Solution: [183.81538277743303, 177.44611955956603,
175.04520360256166, 74.39518453917795] Fitness: 2.0226442921113605
Generation 438 Best Solution: [183.81538277743303, 177.44611955956603,
175.04520360256166, 74.39518453917795] Fitness: 2.0226442921113605

Generation 439 Best Solution: [183.81538277743303, 177.24712239504873,
175.0251490021786, 74.39518453917795] Fitness: 1.6742942169400123
Generation 440 Best Solution: [183.81538277743303, 177.24712239504873,
175.0251490021786, 74.59864048168907] Fitness: 1.607602774939564
Generation 441 Best Solution: [183.81538277743303, 177.24712239504873,
175.0251490021786, 74.59864048168907] Fitness: 1.607602774939564
Generation 442 Best Solution: [183.0737886319639, 177.24712239504873,
175.0251490021786, 74.39518453917795] Fitness: 1.4493757856903835
Generation 443 Best Solution: [183.81538277743303, 177.01293206953568,
175.04520360256166, 74.59864048168907] Fitness: 1.3266837363400095
Generation 444 Best Solution: [183.38197821195283, 176.3107253943221,
175.04520360256166, 75.8899860635627] Fitness: 1.6253387214679347
Generation 445 Best Solution: [183.81538277743303, 177.44611955956603,
174.26583914336615, 74.59864048168907] Fitness: 1.3877288007463335
Generation 446 Best Solution: [183.81538277743303, 176.826630058775,
174.9832122258478, 74.59864048168907] Fitness: 1.0620906545663429
Generation 447 Best Solution: [183.81538277743303, 175.92183461112666,
174.97787170650724, 75.8899860635627] Fitness: 1.829961315052941
Generation 448 Best Solution: [183.81538277743303, 175.92183461112666,
175.0251490021786, 75.8899860635627] Fitness: 1.8880579389759151
Generation 449 Best Solution: [183.81538277743303, 175.92183461112666,
175.0251490021786, 75.8899860635627] Fitness: 1.8880579389759151
Generation 450 Best Solution: [183.81538277743303, 176.31722977949167,
174.21708849553994, 75.8899860635627] Fitness: 1.2794902316328653
Generation 451 Best Solution: [183.81538277743303, 177.44611955956603,
174.9832122258478, 74.39028094410575] Fitness: 1.9476745825812949
Generation 452 Best Solution: [183.81538277743303, 177.2876890317349,
174.9832122258478, 74.39028094410575] Fitness: 1.6860924757234903
Generation 453 Best Solution: [183.81538277743303, 176.89412057512905,
174.9832122258478, 74.7879520311563] Fitness: 1.1157896584748501
Generation 454 Best Solution: [183.81538277743303, 176.04257854448235,
174.9832122258478, 75.8899860635627] Fitness: 1.7942183992961216
Generation 455 Best Solution: [183.81538277743303, 176.826630058775,
173.945709465792, 75.8899860635627] Fitness: 1.5628029849028526
Generation 456 Best Solution: [183.81538277743303, 176.826630058775,
173.945709465792, 75.8899860635627] Fitness: 1.5628029849028526
Generation 457 Best Solution: [183.81538277743303, 176.04257854448235,
174.9832122258478, 75.8899860635627] Fitness: 1.7942183992961216
Generation 458 Best Solution: [183.81538277743303, 176.63805463856872,
174.9832122258478, 75.30759410310688] Fitness: 1.1583668061835053
Generation 459 Best Solution: [183.81538277743303, 176.04257854448235,
174.2133038339259, 75.8899860635627] Fitness: 1.2965706256836442
Generation 460 Best Solution: [183.81538277743303, 176.5734271441193,
174.68464397202865, 75.8899860635627] Fitness: 1.5641254928582895
Generation 461 Best Solution: [183.81538277743303, 176.5734271441193,
174.68464397202865, 75.8899860635627] Fitness: 1.5641254928582895
Generation 462 Best Solution: [183.81538277743303, 177.44611955956603,
174.9832122258478, 74.70750403378923] Fitness: 1.8675088661704284

Generation 463 Best Solution: [183.39305235786034, 176.89412057512905,
174.642783776089, 75.8899860635627] Fitness: 1.5438516421818338
Generation 464 Best Solution: [183.40644003081962, 177.44611955956603,
174.9832122258478, 74.70750403378923] Fitness: 1.6452326594081528
Generation 465 Best Solution: [183.81538277743303, 177.44611955956603,
173.95285404068366, 74.70750403378923] Fitness: 1.3953265368007515
Generation 466 Best Solution: [183.81538277743303, 177.0609054889587,
174.9832122258478, 74.70750403378923] Fitness: 1.2945815705147008
Generation 467 Best Solution: [183.81538277743303, 177.44611955956603,
173.95285404068366, 74.70750403378923] Fitness: 1.3953265368007515
Generation 468 Best Solution: [183.81538277743303, 176.40950509008522,
174.9832122258478, 75.8899860635627] Fitness: 1.7953821119190305
Generation 469 Best Solution: [183.81538277743303, 176.40950509008522,
174.9832122258478, 75.8899860635627] Fitness: 1.7953821119190305
Generation 470 Best Solution: [183.81538277743303, 176.40950509008522,
174.9832122258478, 75.8899860635627] Fitness: 1.7953821119190305
Generation 471 Best Solution: [183.81538277743303, 176.20643584704996,
174.9832122258478, 75.8899860635627] Fitness: 1.770706579491538
Generation 472 Best Solution: [183.81538277743303, 177.44611955956603,
174.9685006865834, 75.52069901523919] Fitness: 2.308375563433415
Generation 473 Best Solution: [183.21272032705164, 177.44611955956603,
174.9832122258478, 75.8899860635627] Fitness: 2.6062005378763535
Generation 474 Best Solution: [183.81538277743303, 177.44611955956603,
174.2154040007697, 75.8899860635627] Fitness: 2.351998309980658
Generation 475 Best Solution: [183.81538277743303, 177.44611955956603,
174.2154040007697, 75.8899860635627] Fitness: 2.351998309980658
Generation 476 Best Solution: [183.81538277743303, 176.26141430214906,
174.9832122258478, 75.8899860635627] Fitness: 1.7715069800912173
Generation 477 Best Solution: [183.81538277743303, 176.30086705197118,
174.9832122258478, 75.8899860635627] Fitness: 1.77477204057755
Generation 478 Best Solution: [183.81538277743303, 176.53985427682107,
174.9832122258478, 75.8899860635627] Fitness: 1.8426095903273212
Generation 479 Best Solution: [183.40644003081962, 177.44611955956603,
174.2154040007697, 74.89924179740127] Fitness: 1.1698030729782183
Generation 480 Best Solution: [183.81538277743303, 177.44611955956603,
174.9832122258478, 74.63650698258274] Fitness: 1.8728251915780567
Generation 481 Best Solution: [183.81538277743303, 176.70286284981978,
174.9832122258478, 75.8899860635627] Fitness: 1.9362066012195227
Generation 482 Best Solution: [183.81538277743303, 176.70286284981978,
174.9832122258478, 75.8899860635627] Fitness: 1.9362066012195227
Generation 483 Best Solution: [183.81538277743303, 177.44611955956603,
174.45723885985203, 75.8899860635627] Fitness: 2.416814851133032
Generation 484 Best Solution: [183.81538277743303, 177.44611955956603,
174.45723885985203, 75.8899860635627] Fitness: 2.416814851133032
Generation 485 Best Solution: [183.81538277743303, 175.757079234839,
174.9832122258478, 75.8899860635627] Fitness: 1.9278391957502559
Generation 486 Best Solution: [183.81538277743303, 175.757079234839,
174.9832122258478, 75.8899860635627] Fitness: 1.9278391957502559

```
Generation 487 Best Solution: [183.81538277743303, 175.757079234839,
174.9832122258478, 75.8899860635627] Fitness: 1.9278391957502559
Generation 488 Best Solution: [183.00608934354648, 175.757079234839,
174.9832122258478, 75.8899860635627] Fitness: 1.7219577529459993
Generation 489 Best Solution: [183.09318219565282, 177.44611955956603,
174.9832122258478, 74.50296003142203] Fitness: 1.673401674805412
Generation 490 Best Solution: [183.38404031982844, 176.69027038549785,
174.9832122258478, 74.50296003142203] Fitness: 0.7532354941524518
Generation 491 Best Solution: [183.81538277743303, 176.15962392049622,
174.9832122258478, 74.50296003142203] Fitness: 0.8265669664223245
Generation 492 Best Solution: [183.38404031982844, 176.27363588811858,
174.9832122258478, 75.8899860635627] Fitness: 1.544804817951103
Generation 493 Best Solution: [183.38404031982844, 176.27363588811858,
174.9832122258478, 75.8899860635627] Fitness: 1.544804817951103
Generation 494 Best Solution: [183.58972115290518, 176.12845233838092,
174.9832122258478, 75.8899860635627] Fitness: 1.6245332819173084
Generation 495 Best Solution: [183.81538277743303, 175.6755229052902,
174.4157388565306, 75.8899860635627] Fitness: 1.5378641779573379
Generation 496 Best Solution: [183.81538277743303, 177.44611955956603,
174.9832122258478, 74.89208400266368] Fitness: 1.8877577448108918
Generation 497 Best Solution: [183.81538277743303, 176.4702669609443,
174.9832122258478, 75.8899860635627] Fitness: 1.8143433150496502
Generation 498 Best Solution: [183.58972115290518, 176.4702669609443,
174.9832122258478, 75.8899860635627] Fitness: 1.6618167287975536
Generation 499 Best Solution: [183.81538277743303, 176.53589849058255,
174.9832122258478, 75.0466028865318] Fitness: 0.9339133072704964
Generation 500 Best Solution: [183.81538277743303, 175.77844259435486,
174.9832122258478, 75.8899860635627] Fitness: 1.913765232620548
Optimal Solution for Simulation 31: [183.81538277743303, 175.77844259435486,
174.9832122258478, 75.8899860635627]
--------------------------------------------------
Simulation 32/32 with conditions: {'drought': False, 'flood': False,
'policy_change': False, 'social_event': False, 'economic_crisis': False}
Generation 1 Best Solution: [183.10357611802087, 176.60573309808777,
174.13161916321988, 74.62367860203702] Fitness: 0.19615876922779568
Generation 2 Best Solution: [183.14069507901092, 176.60573309808777,
174.13161916321988, 74.9775435128789] Fitness: 0.2147259638662416
Generation 3 Best Solution: [183.15615332755038, 177.20230079225482,
174.25614780659498, 74.79865639072088] Fitness: 0.7946060544860423
Generation 4 Best Solution: [183.16196140179179, 176.9454292712335,
174.75064623325892, 74.9775435128789] Fitness: 0.7730004284816298
Generation 5 Best Solution: [183.14069507901092, 177.13574616599584,
174.3727902698971, 75.29700647137727] Fitness: 0.939377187449892
Generation 6 Best Solution: [183.14069507901092, 176.58258536622975,
174.5324877608208, 75.6803082559278] Fitness: 0.8844171307864811
Generation 7 Best Solution: [183.14069507901092, 177.24812194188308,
174.24459599323387, 74.83900802098276] Fitness: 0.8637998569713164
Generation 8 Best Solution: [183.14069507901092, 177.20230079225482,
```

174.25614780659498, 74.35509047689402] Fitness: 0.9066264403465589
Generation 9 Best Solution: [183.14069507901092, 176.4370963788618,
174.18474681165438, 75.82931592620035] Fitness: 0.9103612929998945
Generation 10 Best Solution: [183.14069507901092, 176.4370963788618,
174.18474681165438, 75.2364316735563] Fitness: 0.2652894001633843
Generation 11 Best Solution: [183.14069507901092, 177.3914216357619,
174.18474681165438, 75.29803006697294] Fitness: 1.2846861704596877
Generation 12 Best Solution: [183.14069507901092, 177.3914216357619,
174.18474681165438, 75.29803006697294] Fitness: 1.2846861704596877
Generation 13 Best Solution: [183.14069507901092, 177.13574616599584,
174.18474681165438, 75.6803082559278] Fitness: 1.290893393441982
Generation 14 Best Solution: [183.27530546035538, 177.13574616599584,
174.8015774306124, 74.94287219088893] Fitness: 1.0421039855599852
Generation 15 Best Solution: [183.14069507901092, 177.20230079225482,
174.18474681165438, 74.44747483996603] Fitness: 0.8507440712167802
Generation 16 Best Solution: [183.14069507901092, 177.24812194188308,
174.596495054753, 75.04152387626979] Fitness: 1.064190712946184
Generation 17 Best Solution: [183.27530546035538, 177.24812194188308,
174.596495054753, 75.04152387626979] Fitness: 1.0670445038537006
Generation 18 Best Solution: [183.14069507901092, 177.24812194188308,
174.8015774306124, 75.09277766250057] Fitness: 1.252788111189431
Generation 19 Best Solution: [183.34350207815677, 177.13574616599584,
174.8015774306124, 74.36670570092923] Fitness: 1.1346410238881566
Generation 20 Best Solution: [183.14069507901092, 177.24812194188308,
174.51326647556465, 74.36670570092923] Fitness: 1.0622667913535355
Generation 21 Best Solution: [183.14069507901092, 177.13574616599584,
174.51326647556465, 74.36670570092923] Fitness: 0.9037374217674191
Generation 22 Best Solution: [183.14069507901092, 177.13574616599584,
174.51326647556465, 74.36670570092923] Fitness: 0.9037374217674191
Generation 23 Best Solution: [183.27530546035538, 177.13574616599584,
174.51326647556465, 74.36670570092923] Fitness: 0.9065912126749357
Generation 24 Best Solution: [183.27530546035538, 175.96195394376295,
174.05877964650844, 75.3975134786841] Fitness: 0.384266865342871
Generation 25 Best Solution: [183.27530546035538, 175.96195394376295,
174.18474681165438, 75.6803082559278] Fitness: 0.6946552818602484
Generation 26 Best Solution: [183.27530546035538, 177.13574616599584,
174.18474681165438, 74.59462961050795] Fitness: 0.713499168331295
Generation 27 Best Solution: [183.27530546035538, 175.96195394376295,
174.18474681165438, 75.80662703510662] Fitness: 0.8693357998163606
Generation 28 Best Solution: [183.27530546035538, 175.96195394376295,
174.43612065419, 75.80662703510662] Fitness: 0.9363538826866373
Generation 29 Best Solution: [183.27530546035538, 175.96195394376295,
174.67603927033923, 75.82931592620035] Fitness: 1.1165690771276628
Generation 30 Best Solution: [183.27530546035538, 177.13574616599584,
174.51326647556465, 74.84085683077859] Fitness: 0.8057363144354269
Generation 31 Best Solution: [183.26290779213733, 177.24812194188308,
174.51326647556465, 74.5552435261652] Fitness: 0.9859997326238534
Generation 32 Best Solution: [183.27530546035538, 177.24812194188308,

174.51326647556465, 74.5552435261652] Fitness: 0.9873231747394542
Generation 33 Best Solution: [183.26290779213733, 177.13574616599584,
174.51326647556465, 74.5552435261652] Fitness: 0.827470363037737
Generation 34 Best Solution: [183.34350207815677, 177.13574616599584,
174.5295148622374, 75.82931592620035] Fitness: 1.6277542262118587
Generation 35 Best Solution: [183.27530546035538, 177.13574616599584,
174.92783971168996, 74.97266805639278] Fitness: 1.1821726296637278
Generation 36 Best Solution: [183.34350207815677, 177.24812194188308,
174.8015774306124, 74.33543079904635] Fitness: 1.310887764564773
Generation 37 Best Solution: [183.34350207815677, 177.13574616599584,
174.92783971168996, 75.24158029526252] Fitness: 1.3255977381308255
Generation 38 Best Solution: [183.34350207815677, 177.13574616599584,
174.92783971168996, 74.30880787647938] Fitness: 1.3001594296974592
Generation 39 Best Solution: [183.34350207815677, 177.13574616599584,
174.92783971168996, 74.30880787647938] Fitness: 1.3001594296974592
Generation 40 Best Solution: [183.34350207815677, 177.24812194188308,
173.80711114632285, 75.59760931950522] Fitness: 1.4267382402786688
Generation 41 Best Solution: [183.34350207815677, 177.13574616599584,
174.30590505766307, 74.29641039928144] Fitness: 0.8753927182083918
Generation 42 Best Solution: [183.5273588142779, 176.4747966562965,
173.80711114632285, 75.59760931950522] Fitness: 0.7560308322305391
Generation 43 Best Solution: [183.27530546035538, 176.18178040553252,
174.30590505766307, 75.82931592620035] Fitness: 0.8897381936131021
Generation 44 Best Solution: [183.27530546035538, 177.31766600142592,
174.30590505766307, 74.30880787647938] Fitness: 1.121892812019944
Generation 45 Best Solution: [183.16332999410113, 176.8575797564706,
173.83497136735548, 75.80662703510662] Fitness: 1.2018475250410987
Generation 46 Best Solution: [183.34350207815677, 176.6885175665443,
173.80711114632285, 75.82931592620035] Fitness: 1.1231730736299084
Generation 47 Best Solution: [183.03285230685506, 176.44183649459816,
173.83497136735548, 75.80662703510662] Fitness: 0.9482635876008841
Generation 48 Best Solution: [183.27530546035538, 176.37128157001354,
174.92783971168996, 75.82931592620035] Fitness: 1.3468420315773244
Generation 49 Best Solution: [183.60603734253985, 176.37128157001354,
174.20424222078583, 75.82931592620035] Fitness: 1.0097303646091451
Generation 50 Best Solution: [183.60603734253985, 176.37128157001354,
174.20424222078583, 75.82931592620035] Fitness: 1.0097303646091451
Generation 51 Best Solution: [183.27530546035538, 177.24812194188308,
173.80711114632285, 75.82931592620035] Fitness: 1.7262941844202397
Generation 52 Best Solution: [183.27530546035538, 176.92368338198477,
174.92783971168996, 75.82931592620035] Fitness: 1.7050610542349212
Generation 53 Best Solution: [183.03285230685506, 176.6458800098844,
174.92783971168996, 75.82931592620035] Fitness: 1.4846759978862574
Generation 54 Best Solution: [183.13213435066476, 176.23528112329782,
174.92783971168996, 75.82931592620035] Fitness: 1.3220078614573205
Generation 55 Best Solution: [183.13213435066476, 176.23528112329782,
174.92783971168996, 75.82931592620035] Fitness: 1.3220078614573205
Generation 56 Best Solution: [183.13213435066476, 177.4276126736358,

174.92783971168996, 74.63297040589762] Fitness: 1.6060027430364423
Generation 57 Best Solution: [183.34350207815677, 176.24673241486448,
174.85211200672663, 75.82931592620035] Fitness: 1.2545899432831646
Generation 58 Best Solution: [183.13213435066476, 177.24812194188308,
174.503174420127, 75.30519117344517] Fitness: 1.1628761555498008
Generation 59 Best Solution: [183.6931698322475, 176.49520657315878,
174.92783971168996, 74.7481372174049] Fitness: 0.7537014570049664
Generation 60 Best Solution: [183.13213435066476, 176.6276135068153,
174.92783971168996, 75.82931592620035] Fitness: 1.4577651613795053
Generation 61 Best Solution: [183.34350207815677, 177.31766600142592,
174.19867777465706, 75.50635439025612] Fitness: 1.3677754428352786
Generation 62 Best Solution: [183.27530546035538, 177.31766600142592,
174.19867777465706, 75.50635439025612] Fitness: 1.3566481228040805
Generation 63 Best Solution: [183.04058767016954, 176.5408283665442,
174.92783971168996, 75.82931592620035] Fitness: 1.4228224676293997
Generation 64 Best Solution: [183.27530546035538, 176.7116896371121,
174.92783971168996, 75.82931592620035] Fitness: 1.5170732450046198
Generation 65 Best Solution: [183.27530546035538, 176.7116896371121,
174.92783971168996, 75.82931592620035] Fitness: 1.5170732450046198
Generation 66 Best Solution: [183.27530546035538, 176.68582466109027,
174.92783971168996, 75.82931592620035] Fitness: 1.4984437371490293
Generation 67 Best Solution: [183.27530546035538, 176.68582466109027,
174.2749544270279, 75.82931592620035] Fitness: 1.0584278494195924
Generation 68 Best Solution: [183.34350207815677, 176.53940318933752,
174.92783971168996, 75.82931592620035] Fitness: 1.4217680313279741
Generation 69 Best Solution: [183.13213435066476, 176.30859086252207,
174.92783971168996, 75.82931592620035] Fitness: 1.3310037281537446
Generation 70 Best Solution: [183.13213435066476, 176.30859086252207,
174.92783971168996, 75.82931592620035] Fitness: 1.3310037281537446
Generation 71 Best Solution: [183.34350207815677, 177.4276126736358,
174.92783971168996, 74.35685852807858] Fitness: 1.7201212891209923
Generation 72 Best Solution: [183.27530546035538, 177.4276126736358,
174.92783971168996, 74.28705380266483] Fitness: 1.7513836821708895
Generation 73 Best Solution: [183.34350207815677, 176.80642613356176,
174.92783971168996, 75.82931592620035] Fitness: 1.6044332274748336
Generation 74 Best Solution: [183.803059586425, 176.20966451389737,
174.92783971168996, 75.82931592620035] Fitness: 1.5787001829191036
Generation 75 Best Solution: [183.34350207815677, 176.80642613356176,
174.82031217590074, 75.82931592620035] Fitness: 1.4909157282268954
Generation 76 Best Solution: [183.803059586425, 176.20966451389737,
174.92783971168996, 75.82931592620035] Fitness: 1.5787001829191036
Generation 77 Best Solution: [183.34350207815677, 175.84438178317984,
174.92783971168996, 75.82931592620035] Fitness: 1.4143664142338164
Generation 78 Best Solution: [183.34350207815677, 176.78793536840638,
174.86861204286808, 75.82931592620035] Fitness: 1.5240376636716202
Generation 79 Best Solution: [183.34350207815677, 176.20837941858858,
174.92783971168996, 75.82931592620035] Fitness: 1.3338984911778804
Generation 80 Best Solution: [183.34350207815677, 176.20837941858858,

174.92783971168996, 75.82931592620035] Fitness: 1.3338984911778804
Generation 81 Best Solution: [183.59683198633746, 176.20837941858858,
174.92783971168996, 75.82931592620035] Fitness: 1.4322497355213455
Generation 82 Best Solution: [183.34350207815677, 177.4276126736358,
174.86861204286808, 74.88311389730427] Fitness: 1.5555931047255465
Generation 83 Best Solution: [183.34350207815677, 177.4276126736358,
174.86861204286808, 74.88311389730427] Fitness: 1.5555931047255465
Generation 84 Best Solution: [183.803059586425, 177.4276126736358,
174.09306990832934, 74.50393888807676] Fitness: 1.4415760929016508
Generation 85 Best Solution: [183.34350207815677, 177.4276126736358,
174.92783971168996, 74.88311389730427] Fitness: 1.6201226110648803
Generation 86 Best Solution: [183.20136417205035, 177.4276126736358,
174.92783971168996, 74.88311389730427] Fitness: 1.604287542174864
Generation 87 Best Solution: [183.01969920510655, 177.4276126736358,
174.04148721681568, 75.38736373181696] Fitness: 1.44064218682511
Generation 88 Best Solution: [183.803059586425, 176.38451773752786,
174.86861204286808, 75.82931592620035] Fitness: 1.541784822343109
Generation 89 Best Solution: [183.34350207815677, 177.4276126736358,
174.04148721681568, 75.38736373181696] Fitness: 1.435539344412886
Generation 90 Best Solution: [183.34350207815677, 177.4276126736358,
174.04148721681568, 74.88311389730427] Fitness: 1.1676140912029072
Generation 91 Best Solution: [183.803059586425, 176.81466428807775,
174.92756136756495, 75.17750210625941] Fitness: 1.1737977188916713
Generation 92 Best Solution: [183.34350207815677, 177.4276126736358,
174.04148721681568, 74.88311389730427] Fitness: 1.1676140912029072
Generation 93 Best Solution: [183.71133637689243, 175.848370358332,
174.92756136756495, 75.77654300620297] Fitness: 1.5070831084430667
Generation 94 Best Solution: [183.71133637689243, 175.76759596624396,
174.8535921111154, 75.82931592620035] Fitness: 1.5472088225644167
Generation 95 Best Solution: [183.71133637689243, 175.76759596624396,
174.8535921111154, 75.82931592620035] Fitness: 1.5472088225644167
Generation 96 Best Solution: [183.71133637689243, 175.76759596624396,
174.8535921111154, 75.82931592620035] Fitness: 1.5472088225644167
Generation 97 Best Solution: [183.803059586425, 176.61168237773802,
174.92783971168996, 75.19566254328335] Fitness: 1.0347706508599814
Generation 98 Best Solution: [183.803059586425, 176.5674538559921,
174.92665674775185, 75.19566254328335] Fitness: 1.0083386689191776
Generation 99 Best Solution: [183.71133637689243, 175.76759596624396,
174.04784156388934, 75.82931592620035] Fitness: 1.1741434406522446
Generation 100 Best Solution: [183.71133637689243, 175.76759596624396,
174.04784156388934, 75.82931592620035] Fitness: 1.1741434406522446
Generation 101 Best Solution: [183.4745744856742, 176.15628013925237,
174.8535921111154, 75.82931592620035] Fitness: 1.2935973604239626
Generation 102 Best Solution: [183.4745744856742, 175.69949329574717,
174.92783971168996, 75.82931592620035] Fitness: 1.537678254216402
Generation 103 Best Solution: [183.4745744856742, 175.70937530543884,
175.11981218011883, 75.08091151797635] Fitness: 1.0411461065704901
Generation 104 Best Solution: [183.803059586425, 175.70937530543884,

174.8535921111154, 75.82931592620035] Fitness: 1.6560139219872807
Generation 105 Best Solution: [183.803059586425, 175.70937530543884,
175.11981218011883, 75.82931592620035] Fitness: 1.9790408290035264
Generation 106 Best Solution: [183.803059586425, 175.70937530543884,
174.8535921111154, 75.74258497802131] Fitness: 1.530919218603406
Generation 107 Best Solution: [183.803059586425, 175.70937530543884,
174.8269309866506, 75.82931592620035] Fitness: 1.6291297460645149
Generation 108 Best Solution: [183.803059586425, 175.70937530543884,
174.8535921111154, 75.82931592620035] Fitness: 1.6560139219872807
Generation 109 Best Solution: [183.803059586425, 175.70937530543884,
174.8269309866506, 75.79804723020804] Fitness: 1.5828159853703707
Generation 110 Best Solution: [183.4745744856742, 177.4276126736358,
174.8535921111154, 74.8673800716108] Fitness: 1.5770458270534466
Generation 111 Best Solution: [183.803059586425, 177.4276126736358,
174.8535921111154, 74.8673800716108] Fitness: 1.782133479678053
Generation 112 Best Solution: [183.803059586425, 177.4276126736358,
174.66866691867213, 74.8673800716108] Fitness: 1.6161480811861437
Generation 113 Best Solution: [183.4745744856742, 177.4276126736358,
173.77449275949346, 75.82931592620035] Fitness: 2.0820041987797766
Generation 114 Best Solution: [183.4745744856742, 177.4276126736358,
174.66866691867213, 74.38850130502283] Fitness: 1.496723716375429
Generation 115 Best Solution: [183.803059586425, 177.4276126736358,
174.66866691867213, 75.35259372796594] Fitness: 1.8568051557760774
Generation 116 Best Solution: [183.803059586425, 177.4276126736358,
174.66866691867213, 74.90892887659696] Fitness: 1.623851905369766
Generation 117 Best Solution: [183.803059586425, 177.4276126736358,
174.63836235563568, 74.90892887659696] Fitness: 1.6012168062362653
Generation 118 Best Solution: [183.803059586425, 176.58391085843746,
173.77449275949346, 75.47698885695442] Fitness: 0.8741102482627898
Generation 119 Best Solution: [183.803059586425, 177.4276126736358,
174.63836235563568, 74.90892887659696] Fitness: 1.6012168062362653
Generation 120 Best Solution: [183.8193404492062, 176.7149995906421,
175.11981218011883, 75.47698885695442] Fitness: 1.5897485394545912
Generation 121 Best Solution: [183.803059586425, 175.94243943676892,
175.11981218011883, 75.82931592620035] Fitness: 1.8621056368658624
Generation 122 Best Solution: [183.803059586425, 175.94243943676892,
173.77449275949346, 75.82931592620035] Fitness: 1.2459312372758022
Generation 123 Best Solution: [183.803059586425, 175.94243943676892,
173.77449275949346, 75.82931592620035] Fitness: 1.2459312372758022
Generation 124 Best Solution: [183.803059586425, 176.4019876327907,
175.11981218011883, 75.82931592620035] Fitness: 1.8543387892262782
Generation 125 Best Solution: [183.803059586425, 176.4019876327907,
175.11981218011883, 75.82931592620035] Fitness: 1.8543387892262782
Generation 126 Best Solution: [183.803059586425, 176.4019876327907,
175.11981218011883, 75.82931592620035] Fitness: 1.8543387892262782
Generation 127 Best Solution: [183.803059586425, 176.4019876327907,
175.11981218011883, 75.82931592620035] Fitness: 1.8543387892262782
Generation 128 Best Solution: [182.903288732005, 176.0547241352769,

175.11981218011883, 75.82931592620035] Fitness: 1.6309652331294973
Generation 129 Best Solution: [182.903288732005, 177.4276126736358,
175.11981218011883, 74.36398517441333] Fitness: 2.0019000906238884
Generation 130 Best Solution: [183.803059586425, 177.4276126736358,
175.11981218011883, 74.76946247019323] Fitness: 2.096564222079906
Generation 131 Best Solution: [183.803059586425, 175.87286228723102,
175.11981218011883, 75.82931592620035] Fitness: 1.8890520871446974
Generation 132 Best Solution: [182.903288732005, 177.4276126736358,
175.11981218011883, 75.51248258611935] Fitness: 2.2953726853240233
Generation 133 Best Solution: [183.803059586425, 176.8828242168819,
174.59960715470572, 75.82931592620035] Fitness: 1.6240111683599285
Generation 134 Best Solution: [182.903288732005, 175.91076720936212,
175.11981218011883, 74.69363696022515] Fitness: 0.867446730295618
Generation 135 Best Solution: [182.903288732005, 177.4276126736358,
173.96470452185417, 75.82931592620035] Fitness: 2.018462843706514
Generation 136 Best Solution: [182.903288732005, 177.4276126736358,
175.11981218011883, 74.51920275257214] Fitness: 1.9336402966919166
Generation 137 Best Solution: [183.803059586425, 176.97962665361,
174.69249256518265, 75.82931592620035] Fitness: 1.793646287510238
Generation 138 Best Solution: [183.803059586425, 177.4276126736358,
173.96470452185417, 74.29803508823201] Fitness: 1.5602056897549939
Generation 139 Best Solution: [183.803059586425, 175.68468600461773,
174.66485745572336, 75.82931592620035] Fitness: 1.5039544865123484
Generation 140 Best Solution: [183.803059586425, 175.68468600461773,
174.66485745572336, 75.82931592620035] Fitness: 1.5039544865123484
Generation 141 Best Solution: [183.803059586425, 176.97962665361,
175.11981218011883, 75.06880118442396] Fitness: 1.5257419949199134
Generation 142 Best Solution: [183.53873965676817, 175.608645056985,
175.11981218011883, 75.82931592620035] Fitness: 1.876217801843315
Generation 143 Best Solution: [182.903288732005, 177.30396411716652,
173.7571811107475, 75.82931592620035] Fitness: 1.889199904934324
Generation 144 Best Solution: [183.53873965676817, 177.30396411716652,
173.7571811107475, 75.82931592620035] Fitness: 1.914249299140511
Generation 145 Best Solution: [182.96498298590757, 176.86919531499066,
175.11981218011883, 75.82931592620035] Fitness: 1.9252394407867108
Generation 146 Best Solution: [183.01818617037213, 177.4276126736358,
174.54283507816842, 75.82931592620035] Fitness: 2.089395875010318
Generation 147 Best Solution: [183.01818617037213, 177.4276126736358,
175.11981218011883, 75.32483249698247] Fitness: 2.083872838699208
Generation 148 Best Solution: [183.53873965676817, 176.34114573243914,
174.54283507816842, 75.82931592620035] Fitness: 1.0869034065845824
Generation 149 Best Solution: [183.53873965676817, 176.3901099188467,
175.11981218011883, 75.82931592620035] Fitness: 1.6739188076352134
Generation 150 Best Solution: [182.96498298590757, 177.30396411716652,
175.11981218011883, 74.49491614773515] Fitness: 1.7152210665279337
Generation 151 Best Solution: [182.96498298590757, 176.54081205515647,
175.11981218011883, 75.82931592620035] Fitness: 1.6858463502045251
Generation 152 Best Solution: [183.01818617037213, 177.4276126736358,

175.11981218011883, 74.40426741202968] Fitness: 1.9436063953371152
Generation 153 Best Solution: [182.96498298590757, 177.4276126736358,
175.11981218011883, 74.73991346626889] Fitness: 1.8723124058914755
Generation 154 Best Solution: [182.96498298590757, 177.4276126736358,
175.11981218011883, 74.73991346626889] Fitness: 1.8723124058914755
Generation 155 Best Solution: [183.803059586425, 175.88962167943242,
175.11981218011883, 74.40426741202968] Fitness: 1.1615678227047779
Generation 156 Best Solution: [182.903288732005, 175.88962167943242,
175.11981218011883, 74.40426741202968] Fitness: 0.9596198124038435
Generation 157 Best Solution: [183.803059586425, 177.40056878728606,
175.11981218011883, 75.01425899507066] Fitness: 2.0966614800660217
Generation 158 Best Solution: [183.803059586425, 177.40056878728606,
175.11981218011883, 75.01425899507066] Fitness: 2.0966614800660217
Generation 159 Best Solution: [183.803059586425, 175.85768289442524,
175.11981218011883, 75.82931592620035] Fitness: 1.8958314828603224
Generation 160 Best Solution: [183.54738770683213, 176.54645527869653,
175.11981218011883, 75.01425899507066] Fitness: 0.9802785049282263
Generation 161 Best Solution: [182.96498298590757, 175.85768289442524,
175.11981218011883, 75.82931592620035] Fitness: 1.6715372093488172
Generation 162 Best Solution: [182.96498298590757, 175.85768289442524,
175.11981218011883, 75.82931592620035] Fitness: 1.6715372093488172
Generation 163 Best Solution: [182.96498298590757, 175.85768289442524,
175.11981218011883, 75.82931592620035] Fitness: 1.6715372093488172
Generation 164 Best Solution: [182.96498298590757, 175.85768289442524,
175.11981218011883, 75.82931592620035] Fitness: 1.6715372093488172
Generation 165 Best Solution: [182.95055041679444, 175.85768289442524,
175.1256391863583, 75.82931592620035] Fitness: 1.6844673963494516
Generation 166 Best Solution: [182.95055041679444, 175.85768289442524,
175.11981218011883, 75.82931592620035] Fitness: 1.6762873520116388
Generation 167 Best Solution: [182.95055041679444, 175.85768289442524,
175.11981218011883, 75.82931592620035] Fitness: 1.6762873520116388
Generation 168 Best Solution: [182.95055041679444, 175.85768289442524,
175.11981218011883, 75.82931592620035] Fitness: 1.6762873520116388
Generation 169 Best Solution: [182.95055041679444, 175.85768289442524,
175.11981218011883, 75.74198951836047] Fitness: 1.550370198465477
Generation 170 Best Solution: [182.95055041679444, 175.85768289442524,
175.11981218011883, 75.82931592620035] Fitness: 1.6762873520116388
Generation 171 Best Solution: [183.803059586425, 176.4676409889784,
175.11981218011883, 75.82931592620035] Fitness: 1.8773660496890623
Generation 172 Best Solution: [183.803059586425, 176.4676409889784,
175.11981218011883, 75.82931592620035] Fitness: 1.8773660496890623
Generation 173 Best Solution: [183.06766975522495, 177.4276126736358,
173.93075699628622, 75.82931592620035] Fitness: 1.9789256161730795
Generation 174 Best Solution: [183.803059586425, 177.4276126736358,
173.93075699628622, 74.6655682925768] Fitness: 1.4274522865899555
Generation 175 Best Solution: [183.803059586425, 176.18084214462732,
173.93075699628622, 74.6655682925768] Fitness: 0.3453572378176576
Generation 176 Best Solution: [183.803059586425, 176.04421838216825,

175.11981218011883, 75.82931592620035] Fitness: 1.8348961017703036
Generation 177 Best Solution: [182.95055041679444, 176.20925106020994,
175.11981218011883, 75.82931592620035] Fitness: 1.6020550586010485
Generation 178 Best Solution: [183.803059586425, 177.4276126736358,
175.1256391863583, 74.77722280637616] Fitness: 2.104935789136935
Generation 179 Best Solution: [183.803059586425, 175.59867499505424,
175.1256391863583, 75.82931592620035] Fitness: 2.06940088597473
Generation 180 Best Solution: [183.803059586425, 177.4276126736358,
173.7762806774593, 75.82931592620035] Fitness: 2.286229255016151
Generation 181 Best Solution: [183.803059586425, 176.34561308566634,
175.11981218011883, 75.82931592620035] Fitness: 1.8393814660313585
Generation 182 Best Solution: [183.66352293390142, 176.34561308566634,
175.11981218011883, 75.82931592620035] Fitness: 1.7338069513375565
Generation 183 Best Solution: [183.803059586425, 175.90072012893594,
175.11981218011883, 75.74198951836047] Fitness: 1.7515323567919208
Generation 184 Best Solution: [183.803059586425, 177.4276126736358,
175.11981218011883, 74.40164516035911] Fitness: 2.184187375147223
Generation 185 Best Solution: [183.803059586425, 176.03810670475934,
175.11981218011883, 75.74198951836047] Fitness: 1.7102035599338674
Generation 186 Best Solution: [183.803059586425, 177.4276126736358,
175.11981218011883, 74.40164516035911] Fitness: 2.184187375147223
Generation 187 Best Solution: [183.70787471740485, 177.4276126736358,
173.74480629727168, 75.74198951836047] Fitness: 2.101178496755165
Generation 188 Best Solution: [183.66352293390142, 175.74789435386,
175.11981218011883, 74.40164516035911] Fitness: 1.1298157045786708
Generation 189 Best Solution: [183.70787471740485, 177.4276126736358,
174.91348457761526, 75.0133333190395] Fitness: 1.8089118312576573
Generation 190 Best Solution: [183.66352293390142, 175.74789435386,
175.11981218011883, 75.5540236535206] Fitness: 1.4881687354169544
Generation 191 Best Solution: [183.803059586425, 177.4276126736358,
173.8538259370168, 75.31014510778674] Fitness: 1.6614643542048486
Generation 192 Best Solution: [183.07132572472793, 177.4276126736358,
175.00645962165785, 75.31014510778674] Fitness: 1.9116337789416198
Generation 193 Best Solution: [183.803059586425, 177.4276126736358,
175.11981218011883, 74.68216724892987] Fitness: 2.100218329910216
Generation 194 Best Solution: [183.7259599558394, 177.4276126736358,
174.91348457761526, 74.6805081235905] Fitness: 1.7796825665822396
Generation 195 Best Solution: [182.90644952373498, 177.4276126736358,
175.11981218011883, 74.68216724892987] Fitness: 1.8969959414187338
Generation 196 Best Solution: [183.7259599558394, 177.4276126736358,
175.11981218011883, 74.68216724892987] Fitness: 2.0385144408615865
Generation 197 Best Solution: [183.7259599558394, 177.4276126736358,
174.79486086332247, 74.3456743690898] Fitness: 1.7714898900168043
Generation 198 Best Solution: [183.7259599558394, 177.4276126736358,
173.90967829983583, 74.84195560233394] Fitness: 1.3711618949445425
Generation 199 Best Solution: [183.7259599558394, 177.4276126736358,
173.90967829983583, 74.93575510752957] Fitness: 1.3886458879238857
Generation 200 Best Solution: [183.7259599558394, 177.4276126736358,

174.4094077212741, 75.12512153350873] Fitness: 1.4891439280672745
Generation 201 Best Solution: [182.97417564469728, 177.4276126736358,
174.4094077212741, 75.12512153350873] Fitness: 1.3236800199667715
Generation 202 Best Solution: [182.97417564469728, 177.4276126736358,
174.4094077212741, 75.12512153350873] Fitness: 1.3236800199667715
Generation 203 Best Solution: [183.30402695709247, 177.4276126736358,
174.4094077212741, 75.67286651908505] Fitness: 1.792150564056287
Generation 204 Best Solution: [182.97417564469728, 177.4276126736358,
174.4094077212741, 75.67286651908505] Fitness: 1.8169745730845102
Generation 205 Best Solution: [183.7259599558394, 177.4276126736358,
174.41888905209672, 74.80854684453416] Fitness: 1.3994791086822864
Generation 206 Best Solution: [182.97417564469728, 177.4276126736358,
174.41888905209672, 74.80854684453416] Fitness: 1.2340152005817835
Generation 207 Best Solution: [183.7259599558394, 177.4276126736358,
174.4094077212741, 74.69848433175483] Fitness: 1.3959134151022663
Generation 208 Best Solution: [183.7259599558394, 176.1256286996059,
175.11981218011883, 75.74198951836047] Fitness: 1.6359503106667639
Generation 209 Best Solution: [183.7259599558394, 176.1256286996059,
175.11981218011883, 75.74198951836047] Fitness: 1.6359503106667639
Generation 210 Best Solution: [183.7259599558394, 176.1256286996059,
175.11981218011883, 75.74198951836047] Fitness: 1.6359503106667639
Generation 211 Best Solution: [183.7259599558394, 176.45640701887314,
174.9689809597954, 75.74198951836047] Fitness: 1.4901778542010229
Generation 212 Best Solution: [182.85623615534988, 176.1153782628433,
175.11981218011883, 75.74198951836047] Fitness: 1.5172461807265065
Generation 213 Best Solution: [183.7259599558394, 176.1153782628433,
175.11981218011883, 75.74198951836047] Fitness: 1.6368656247119615
Generation 214 Best Solution: [183.7259599558394, 175.916153228087,
175.11981218011883, 75.74198951836047] Fitness: 1.6838683800155292
Generation 215 Best Solution: [183.7259599558394, 176.46406701029912,
175.11981218011883, 75.74198951836047] Fitness: 1.6883361604262166
Generation 216 Best Solution: [183.7259599558394, 175.82499664267033,
175.11981218011883, 75.74198951836047] Fitness: 1.7239039047945335
Generation 217 Best Solution: [183.7259599558394, 175.82499664267033,
174.10922276313633, 75.74198951836047] Fitness: 1.0242481321188508
Generation 218 Best Solution: [183.7259599558394, 175.58187919588443,
175.11981218011883, 75.68932777461085] Fitness: 1.816794106778441
Generation 219 Best Solution: [183.7259599558394, 177.4276126736358,
173.95893759363602, 74.36719062328943] Fitness: 1.458815161873418
Generation 220 Best Solution: [183.7259599558394, 177.4276126736358,
173.95893759363602, 74.36719062328943] Fitness: 1.458815161873418
Generation 221 Best Solution: [183.18223565493574, 177.4276126736358,
173.95893759363602, 75.74198951836047] Fitness: 1.8352068454177626
Generation 222 Best Solution: [183.7259599558394, 176.60029727288224,
175.11981218011883, 75.50316939127391] Fitness: 1.4648530009342071
Generation 223 Best Solution: [183.7259599558394, 177.4276126736358,
174.67858563810003, 74.75781914557477] Fitness: 1.5534065892366313
Generation 224 Best Solution: [183.7259599558394, 177.4276126736358,

174.67858563810003, 74.75781914557477] Fitness: 1.5534065892366313
Generation 225 Best Solution: [183.7259599558394, 177.4276126736358,
174.41439540285597, 74.75781914557477] Fitness: 1.3956621059284096
Generation 226 Best Solution: [183.7259599558394, 177.4276126736358,
174.67858563810003, 75.74198951836047] Fitness: 2.234320672104012
Generation 227 Best Solution: [183.7259599558394, 176.1433295374877,
175.11981218011883, 75.74198951836047] Fitness: 1.634716044799498
Generation 228 Best Solution: [183.7259599558394, 177.4276126736358,
173.95152614549357, 75.74198951836047] Fitness: 2.0357765392964557
Generation 229 Best Solution: [183.7259599558394, 177.4276126736358,
173.95152614549357, 75.74198951836047] Fitness: 2.0357765392964557
Generation 230 Best Solution: [183.7259599558394, 175.72721924114032,
175.11981218011883, 75.74198951836047] Fitness: 1.7797786915184615
Generation 231 Best Solution: [183.7259599558394, 176.72908151108902,
174.67858563810003, 75.74198951836047] Fitness: 1.3599790447615017
Generation 232 Best Solution: [183.7259599558394, 175.72721924114032,
175.11981218011883, 75.21150853378404] Fitness: 1.244285384947992
Generation 233 Best Solution: [183.7259599558394, 175.73371334196133,
175.11981218011883, 75.74198951836047] Fitness: 1.7756526827983758
Generation 234 Best Solution: [183.7259599558394, 176.72908151108902,
175.11981218011883, 75.74198951836047] Fitness: 1.8413035810571423
Generation 235 Best Solution: [183.37119833298962, 175.59716185993892,
173.95152614549357, 75.74198951836047] Fitness: 1.0182981636656885
Generation 236 Best Solution: [183.7259599558394, 177.4276126736358,
174.84066794622728, 75.09970396787415] Fitness: 1.781374780474304
Generation 237 Best Solution: [183.7259599558394, 177.4276126736358,
174.84066794622728, 75.09970396787415] Fitness: 1.781374780474304
Generation 238 Best Solution: [183.37119833298962, 177.4276126736358,
174.84066794622728, 75.09970396787415] Fitness: 1.6046996600961034
Generation 239 Best Solution: [183.7259599558394, 176.2533465292138,
175.11981218011883, 75.74198951836047] Fitness: 1.6368804362179303
Generation 240 Best Solution: [183.13350637541168, 177.4276126736358,
174.84066794622728, 75.09970396787415] Fitness: 1.584901513545049
Generation 241 Best Solution: [183.7259599558394, 176.2533465292138,
175.11981218011883, 75.74198951836047] Fitness: 1.6368804362179303
Generation 242 Best Solution: [183.7259599558394, 176.2533465292138,
175.11981218011883, 75.74198951836047] Fitness: 1.6368804362179303
Generation 243 Best Solution: [183.7259599558394, 177.4276126736358,
174.49713307646368, 75.74198951836047] Fitness: 2.115468681693952
Generation 244 Best Solution: [183.7259599558394, 177.4276126736358,
174.49713307646368, 75.74198951836047] Fitness: 2.115468681693952
Generation 245 Best Solution: [183.13350637541168, 177.4276126736358,
174.64359599606368, 75.74198951836047] Fitness: 2.011341801576366
Generation 246 Best Solution: [183.13350637541168, 177.4276126736358,
173.98495466798244, 75.7962415334243] Fitness: 1.9091716168874138
Generation 247 Best Solution: [183.13350637541168, 177.4276126736358,
174.64359599606368, 75.74198951836047] Fitness: 2.011341801576366
Generation 248 Best Solution: [183.13350637541168, 177.4276126736358,

173.9824900972257, 75.74198951836047] Fitness: 1.8326711459633207
Generation 249 Best Solution: [183.13350637541168, 177.4276126736358,
173.9824900972257, 75.74198951836047] Fitness: 1.8326711459633207
Generation 250 Best Solution: [183.7259599558394, 177.4276126736358,
175.11981218011883, 74.92555871427108] Fitness: 2.054921015165998
Generation 251 Best Solution: [183.13350637541168, 177.4276126736358,
175.00370211710504, 75.74198951836047] Fitness: 2.3660854668537685
Generation 252 Best Solution: [183.7259599558394, 176.1115121427293,
175.11981218011883, 75.74198951836047] Fitness: 1.6372490538089126
Generation 253 Best Solution: [183.13350637541168, 176.1115121427293,
174.99452643201383, 75.74198951836047] Fitness: 1.2763962462234688
Generation 254 Best Solution: [183.13350637541168, 176.1115121427293,
175.11981218011883, 75.74198951836047] Fitness: 1.4407757868796578
Generation 255 Best Solution: [183.13350637541168, 176.1115121427293,
175.11981218011883, 74.49238089120762] Fitness: 0.8083997363714369
Generation 256 Best Solution: [183.7259599558394, 177.4276126736358,
175.00370211710504, 74.2942244784382] Fitness: 2.030722539164924
Generation 257 Best Solution: [183.7259599558394, 176.67572343991387,
175.00370211710504, 75.74198951836047] Fitness: 1.6495131110097287
Generation 258 Best Solution: [183.7259599558394, 176.67572343991387,
175.00370211710504, 75.7962415334243] Fitness: 1.7264837940967048
Generation 259 Best Solution: [183.7259599558394, 176.67572343991387,
175.00370211710504, 75.7962415334243] Fitness: 1.7264837940967048
Generation 260 Best Solution: [183.7259599558394, 176.67572343991387,
175.11981218011883, 75.74198951836047] Fitness: 1.802599585626358
Generation 261 Best Solution: [183.13350637541168, 177.4276126736358,
174.48705315270448, 75.74198951836047] Fitness: 1.9137444764478304
Generation 262 Best Solution: [183.13350637541168, 177.4276126736358,
175.11981218011883, 74.65393540261942] Fitness: 1.84550601602932
Generation 263 Best Solution: [183.21091370427973, 177.4276126736358,
173.96622739149012, 75.74198951836047] Fitness: 1.833749543270007
Generation 264 Best Solution: [182.81361032011873, 177.4276126736358,
173.96622739149012, 75.74198951836047] Fitness: 1.9342010224727473
Generation 265 Best Solution: [183.13350637541168, 176.96989047172437,
175.11981218011883, 75.74198951836047] Fitness: 1.8690909764148753
Generation 266 Best Solution: [182.97593463788422, 176.35480316555913,
175.11981218011883, 75.74198951836047] Fitness: 1.4878949345070045
Generation 267 Best Solution: [182.97593463788422, 176.35480316555913,
175.01086774412687, 75.74198951836047] Fitness: 1.3437095939152184
Generation 268 Best Solution: [183.7259599558394, 176.96989047172437,
174.71255046821346, 75.74198951836047] Fitness: 1.6116084066503533
Generation 269 Best Solution: [182.97593463788422, 177.4276126736358,
174.66353218718257, 75.74198951836047] Fitness: 2.0567069332602017
Generation 270 Best Solution: [183.7259599558394, 176.91065451659105,
175.11981218011883, 75.7962415334243] Fitness: 2.079840687867676
Generation 271 Best Solution: [183.7259599558394, 175.87426327429415,
175.11981218011883, 75.55896736612841] Fitness: 1.4715552718802374
Generation 272 Best Solution: [182.8502548451968, 177.26572420503723,

175.11981218011883, 75.74198951836047] Fitness: 2.3354227130089935
Generation 273 Best Solution: [183.7259599558394, 176.75768201167628,
175.11981218011883, 75.74198951836047] Fitness: 1.8636901786098048
Generation 274 Best Solution: [183.7259599558394, 177.4276126736358,
173.7840085415098, 75.74198951836047] Fitness: 2.094931313664743
Generation 275 Best Solution: [183.7259599558394, 176.16195215048668,
175.11981218011883, 75.74198951836047] Fitness: 1.6338910105573357
Generation 276 Best Solution: [183.7259599558394, 176.16195215048668,
175.11981218011883, 75.74198951836047] Fitness: 1.6338910105573357
Generation 277 Best Solution: [183.7259599558394, 176.34892932709081,
175.11981218011883, 75.74198951836047] Fitness: 1.6525171223732205
Generation 278 Best Solution: [183.52045497651469, 177.10898580256543,
174.08369005216673, 75.74198951836047] Fitness: 1.4094620599357706
Generation 279 Best Solution: [183.7259599558394, 177.06999644388148,
175.11981218011883, 74.3910427334342] Fitness: 1.5948528610294492
Generation 280 Best Solution: [183.7259599558394, 177.4276126736358,
174.08369005216673, 75.74198951836047] Fitness: 2.016831020158037
Generation 281 Best Solution: [183.11666182516473, 176.63151942806587,
175.11981218011883, 75.74198951836047] Fitness: 1.5786794372861355
Generation 282 Best Solution: [183.11666182516473, 176.63151942806587,
175.11981218011883, 75.74198951836047] Fitness: 1.5786794372861355
Generation 283 Best Solution: [183.7259599558394, 177.4276126736358,
174.08369005216673, 75.74198951836047] Fitness: 2.016831020158037
Generation 284 Best Solution: [183.7259599558394, 177.4276126736358,
174.08369005216673, 75.74198951836047] Fitness: 2.016831020158037
Generation 285 Best Solution: [183.7259599558394, 177.4276126736358,
174.08369005216673, 75.7962415334243] Fitness: 2.093801703245013
Generation 286 Best Solution: [183.67340216377832, 177.4276126736358,
174.08369005216673, 75.7962415334243] Fitness: 2.0565091387908008
Generation 287 Best Solution: [183.67340216377832, 177.4276126736358,
175.11981218011883, 74.53473759699311] Fitness: 2.0316169476442587
Generation 288 Best Solution: [183.67340216377832, 177.4276126736358,
175.02964432906703, 74.53473759699311] Fitness: 1.9110968345671433
Generation 289 Best Solution: [183.7259599558394, 176.16600558956452,
175.11981218011883, 74.53473759699311] Fitness: 0.9870400766776541
Generation 290 Best Solution: [183.7259599558394, 176.42397691746962,
175.02964432906703, 75.7962415334243] Fitness: 1.6302087399354424
Generation 291 Best Solution: [183.67340216377832, 177.4276126736358,
175.11981218011883, 74.65585379286986] Fitness: 2.0044159423953283
Generation 292 Best Solution: [183.7259599558394, 177.4276126736358,
175.11981218011883, 74.65585379286986] Fitness: 2.041708506849541
Generation 293 Best Solution: [183.70559973551468, 177.4276126736358,
175.11981218011883, 74.65585379286986] Fitness: 2.0268029568304566
Generation 294 Best Solution: [183.7259599558394, 177.4276126736358,
174.98784629848163, 74.65585379286986] Fitness: 1.8691814684415504
Generation 295 Best Solution: [183.67340216377832, 177.4276126736358,
175.11981218011883, 75.13660556343105] Fitness: 2.0989901755820335
Generation 296 Best Solution: [183.67340216377832, 176.42331997502015,

175.11981218011883, 75.7962415334243] Fitness: 1.713216142003123
Generation 297 Best Solution: [183.34761478765043, 176.42331997502015,
175.11981218011883, 75.7962415334243] Fitness: 1.568334630046002
Generation 298 Best Solution: [183.34761478765043, 176.42331997502015,
175.11981218011883, 75.7962415334243] Fitness: 1.568334630046002
Generation 299 Best Solution: [183.67340216377832, 176.42331997502015,
175.11981218011883, 75.7962415334243] Fitness: 1.713216142003123
Generation 300 Best Solution: [183.67340216377832, 177.4276126736358,
175.11981218011883, 74.68998029672623] Fitness: 2.000460129889334
Generation 301 Best Solution: [183.67340216377832, 175.67134628278927,
175.11981218011883, 75.74198951836047] Fitness: 1.7804240967214744
Generation 302 Best Solution: [183.67340216377832, 175.67134628278927,
175.11981218011883, 75.74198951836047] Fitness: 1.7804240967214744
Generation 303 Best Solution: [183.67340216377832, 177.4276126736358,
174.4297724226175, 74.39635206648653] Fitness: 1.4552684938095926
Generation 304 Best Solution: [183.7259599558394, 175.9238781360086,
175.11981218011883, 74.68998029672623] Fitness: 1.0031178196277617
Generation 305 Best Solution: [183.7259599558394, 175.63661958797465,
174.4297724226175, 75.7962415334243] Fitness: 1.2879034629401829
Generation 306 Best Solution: [183.7259599558394, 175.9238781360086,
175.11981218011883, 75.7962415334243] Fitness: 1.7579810167708443
Generation 307 Best Solution: [183.8745450839778, 177.4276126736358,
174.58703683926967, 74.63935514025047] Fitness: 1.6235048372474525
Generation 308 Best Solution: [183.67340216377832, 177.4276126736358,
174.58703683926967, 74.63935514025047] Fitness: 1.4598623848914143
Generation 309 Best Solution: [183.67340216377832, 176.51318863321407,
174.58703683926967, 74.63935514025047] Fitness: 0.45348302645443556
Generation 310 Best Solution: [183.67340216377832, 176.51318863321407,
174.58703683926967, 75.7962415334243] Fitness: 1.201893278200167
Generation 311 Best Solution: [183.0260738989163, 177.23315242292827,
175.11981218011883, 74.63935514025047] Fitness: 1.552719879872157
Generation 312 Best Solution: [183.67340216377832, 177.23315242292827,
175.11981218011883, 74.63935514025047] Fitness: 1.6948946865430563
Generation 313 Best Solution: [183.0260738989163, 177.23315242292827,
175.11981218011883, 75.00709393557759] Fitness: 1.5874764742147875
Generation 314 Best Solution: [183.0260738989163, 175.9400268349982,
175.11981218011883, 75.7962415334243] Fitness: 1.5728088509463776
Generation 315 Best Solution: [183.67340216377832, 175.69664926229643,
175.11981218011883, 74.63935514025047] Fitness: 1.0912622423100806
Generation 316 Best Solution: [183.0260738989163, 177.4276126736358,
174.35598609017399, 75.7962415334243] Fitness: 1.952394047830666
Generation 317 Best Solution: [183.0260738989163, 177.4276126736358,
175.11981218011883, 74.63935514025047] Fitness: 1.8647382686157854
Generation 318 Best Solution: [183.67340216377832, 177.26255713826504,
175.11981218011883, 75.00709393557759] Fitness: 1.7734348047758968
Generation 319 Best Solution: [183.0260738989163, 177.4276126736358,
175.11981218011883, 75.00709393557759] Fitness: 1.8994948629584159
Generation 320 Best Solution: [183.67340216377832, 176.3150019802136,

175.11981218011883, 74.63935514025047] Fitness: 0.9367704453186813
Generation 321 Best Solution: [183.62831175641477, 176.7454200193185,
175.11981218011883, 75.7962415334243] Fitness: 1.8647182156744946
Generation 322 Best Solution: [183.0260738989163, 177.4276126736358,
175.11981218011883, 75.00709393557759] Fitness: 1.8994948629584159
Generation 323 Best Solution: [183.67340216377832, 177.4276126736358,
174.35598609017399, 74.39625200948227] Fitness: 1.4271301768664169
Generation 324 Best Solution: [183.62831175641477, 177.4276126736358,
174.35598609017399, 74.39625200948227] Fitness: 1.3982182192019605
Generation 325 Best Solution: [183.0260738989163, 177.4276126736358,
175.11981218011883, 75.00709393557759] Fitness: 1.8994948629584159
Generation 326 Best Solution: [183.0260738989163, 177.4276126736358,
175.11981218011883, 74.52609211740601] Fitness: 1.892169091227653
Generation 327 Best Solution: [183.0260738989163, 175.72238039022588,
175.11981218011883, 75.7962415334243] Fitness: 1.680394740745201
Generation 328 Best Solution: [183.0260738989163, 175.72238039022588,
175.11981218011883, 75.7962415334243] Fitness: 1.680394740745201
Generation 329 Best Solution: [183.62831175641477, 176.21001952774918,
175.11981218011883, 75.7962415334243] Fitness: 1.644771580284028
Generation 330 Best Solution: [183.0260738989163, 177.4276126736358,
174.35598609017399, 75.7962415334243] Fitness: 1.952394047830666
Generation 331 Best Solution: [183.0260738989163, 175.61871989714894,
175.11981218011883, 75.7962415334243] Fitness: 1.7549504563283413
Generation 332 Best Solution: [183.0260738989163, 176.00829988075296,
175.11981218011883, 75.7962415334243] Fitness: 1.5527248803917784
Generation 333 Best Solution: [183.0260738989163, 176.00829988075296,
175.11981218011883, 75.7962415334243] Fitness: 1.5527248803917784
Generation 334 Best Solution: [183.0260738989163, 176.00829988075296,
175.11981218011883, 75.7962415334243] Fitness: 1.5527248803917784
Generation 335 Best Solution: [183.62831175641477, 177.4276126736358,
175.11981218011883, 74.37093468231885] Fitness: 2.0721622423414847
Generation 336 Best Solution: [183.29163659657428, 177.4276126736358,
175.11981218011883, 74.43375065423412] Fitness: 1.915181570111826
Generation 337 Best Solution: [183.62831175641477, 177.4276126736358,
174.38415555990937, 75.7962415334243] Fitness: 2.07551900438991
Generation 338 Best Solution: [183.0260738989163, 177.32538656459232,
175.11981218011883, 75.7962415334243] Fitness: 2.4425229762681426
Generation 339 Best Solution: [183.62831175641477, 176.26689875404185,
175.11981218011883, 75.7962415334243] Fitness: 1.6490854599116909
Generation 340 Best Solution: [183.0260738989163, 176.13219278639932,
175.11981218011883, 75.7962415334243] Fitness: 1.5329447420638687
Generation 341 Best Solution: [183.0260738989163, 176.13219278639932,
175.11981218011883, 75.7962415334243] Fitness: 1.5329447420638687
Generation 342 Best Solution: [183.76805666870482, 176.69266467740516,
175.11981218011883, 75.7962415334243] Fitness: 1.9240860808216822
Generation 343 Best Solution: [183.76805666870482, 175.62277779911128,
175.11981218011883, 74.37623219835392] Fitness: 1.3067922498647708
Generation 344 Best Solution: [183.72994025940324, 175.62277779911128,

175.11981218011883, 74.37623219835392] Fitness: 1.2771148955452043
Generation 345 Best Solution: [183.76805666870482, 176.42291210589613,
174.81750810171806, 75.7962415334243] Fitness: 1.4238566595769653
Generation 346 Best Solution: [183.76805666870482, 176.42291210589613,
174.81750810171806, 74.84036267889817] Fitness: 0.6709845376461164
Generation 347 Best Solution: [183.76805666870482, 176.69266467740516,
175.11981218011883, 74.7339758867048] Fitness: 1.1665290041568501
Generation 348 Best Solution: [183.62831175641477, 177.32538656459232,
175.11981218011883, 74.61382743076851] Fitness: 1.8119902737401885
Generation 349 Best Solution: [183.62831175641477, 176.22406050911854,
175.11981218011883, 75.7962415334243] Fitness: 1.6454154438880626
Generation 350 Best Solution: [183.62831175641477, 176.44746702271672,
175.11981218011883, 74.29372391088691] Fitness: 1.0843098078948024
Generation 351 Best Solution: [183.27384907106656, 175.8381848902637,
175.11981218011883, 75.7962415334243] Fitness: 1.6000153465238838
Generation 352 Best Solution: [183.62831175641477, 176.69266467740516,
175.11981218011883, 75.7962415334243] Fitness: 1.8252224389749485
Generation 353 Best Solution: [183.62831175641477, 175.88522274617048,
175.11981218011883, 75.7962415334243] Fitness: 1.7069150811037923
Generation 354 Best Solution: [183.62831175641477, 176.69266467740516,
175.11981218011883, 75.7962415334243] Fitness: 1.8252224389749485
Generation 355 Best Solution: [183.62831175641477, 176.69266467740516,
175.11981218011883, 75.7962415334243] Fitness: 1.8252224389749485
Generation 356 Best Solution: [183.11902569031727, 175.7094805321898,
174.60887989407226, 75.7962415334243] Fitness: 1.140762655297632
Generation 357 Best Solution: [183.11902569031727, 175.7094805321898,
175.09915039159418, 75.7962415334243] Fitness: 1.644575012269283
Generation 358 Best Solution: [183.62831175641477, 177.36369403933963,
174.53492079380928, 75.7962415334243] Fitness: 2.0387852266547584
Generation 359 Best Solution: [183.11902569031727, 176.0615734288786,
175.11981218011883, 75.7962415334243] Fitness: 1.5259300874090145
Generation 360 Best Solution: [183.62831175641477, 175.7250997501651,
174.53492079380928, 74.54373178045984] Fitness: 0.4862463938768713
Generation 361 Best Solution: [183.33412830957406, 176.8195264224943,
175.11981218011883, 75.7962415334243] Fitness: 1.8080153254848619
Generation 362 Best Solution: [183.62831175641477, 177.4276126736358,
175.09915039159418, 74.5020460348267] Fitness: 1.9849445748564845
Generation 363 Best Solution: [183.62831175641477, 177.4276126736358,
175.09915039159418, 74.5020460348267] Fitness: 1.9849445748564845
Generation 364 Best Solution: [183.62831175641477, 177.36369403933963,
175.09915039159418, 74.5020460348267] Fitness: 1.8765440234083324
Generation 365 Best Solution: [183.7882219189468, 177.36369403933963,
173.7781418558993, 75.54986118398939] Fitness: 1.7990482218084807
Generation 366 Best Solution: [183.62831175641477, 175.6356091980287,
175.11981218011883, 75.54986118398939] Fitness: 1.538620803488726
Generation 367 Best Solution: [183.62831175641477, 175.80481370166575,
173.87429995704113, 75.7962415334243] Fitness: 1.0876282066457803
Generation 368 Best Solution: [183.62831175641477, 175.80481370166575,

173.87429995704113, 74.69630971169465] Fitness: 0.33221057177701885
Generation 369 Best Solution: [183.62831175641477, 175.6356091980287,
174.24458374669237, 75.7962415334243] Fitness: 1.1661678636177828
Generation 370 Best Solution: [183.62831175641477, 177.36369403933963,
174.6258957557488, 75.54986118398939] Fitness: 1.781005742008384
Generation 371 Best Solution: [183.68433353269782, 177.36369403933963,
174.67461621976415, 75.2968324585189] Fitness: 1.6170213184281157
Generation 372 Best Solution: [183.33412830957406, 177.36369403933963,
174.6258957557488, 75.54986118398939] Fitness: 1.6622415174622338
Generation 373 Best Solution: [183.62831175641477, 175.72488012404176,
175.11981218011883, 75.54986118398939] Fitness: 1.4756261072875514
Generation 374 Best Solution: [183.62831175641477, 177.36369403933963,
174.58204871836298, 75.54986118398939] Fitness: 1.751296693727752
Generation 375 Best Solution: [183.68433353269782, 175.95398630382564,
175.11981218011883, 75.7962415334243] Fitness: 1.7177843177075873
Generation 376 Best Solution: [183.62831175641477, 176.60668625198377,
175.11981218011883, 75.05271428745694] Fitness: 1.073062003314309
Generation 377 Best Solution: [183.62831175641477, 177.36369403933963,
174.3175640045514, 75.54986118398939] Fitness: 1.6291764561552093
Generation 378 Best Solution: [183.1503410279452, 175.6886662549702,
175.11981218011883, 74.32696633898594] Fitness: 1.055575397500573
Generation 379 Best Solution: [183.62831175641477, 176.25638364481398,
175.11981218011883, 75.7962415334243] Fitness: 1.647946697985788
Generation 380 Best Solution: [183.62831175641477, 177.36369403933963,
174.30023573216428, 75.54986118398939] Fitness: 1.6245938244332956
Generation 381 Best Solution: [183.62831175641477, 176.25638364481398,
175.11981218011883, 75.57049666598664] Fitness: 1.3547680597685092
Generation 382 Best Solution: [183.33412830957406, 176.7175069058534,
175.11981218011883, 75.57049666598664] Fitness: 1.431392544507649
Generation 383 Best Solution: [183.62831175641477, 177.2896611734778,
174.4959119098011, 75.7962415334243] Fitness: 1.8987862118155996
Generation 384 Best Solution: [183.62831175641477, 176.59675151962372,
175.11981218011883, 75.54986118398939] Fitness: 1.4469782882455635
Generation 385 Best Solution: [183.33412830957406, 177.36369403933963,
174.01293485254322, 75.7962415334243] Fitness: 1.807533578573843
Generation 386 Best Solution: [183.01213336758968, 177.36369403933963,
174.6711645901387, 75.54986118398939] Fitness: 1.7046303185598657
Generation 387 Best Solution: [183.04560685490415, 177.36369403933963,
174.01293485254322, 75.38719362618575] Fitness: 1.329989594854233
Generation 388 Best Solution: [183.62831175641477, 176.43493211058123,
174.01293485254322, 75.7962415334243] Fitness: 0.9965715362586012
Generation 389 Best Solution: [182.98392803464742, 176.79140835427012,
175.11981218011883, 75.041342186206] Fitness: 1.0993684471331147
Generation 390 Best Solution: [183.62831175641477, 177.36369403933963,
174.59217726067175, 74.89039203082234] Fitness: 1.3291484961671607
Generation 391 Best Solution: [183.12839959900424, 176.47993617014748,
175.11981218011883, 75.7962415334243] Fitness: 1.5756682924547984
Generation 392 Best Solution: [183.33412830957406, 177.26365676458707,

174.47613884445315, 75.7962415334243] Fitness: 1.7301204369992884
Generation 393 Best Solution: [183.33412830957406, 177.26365676458707,
174.47613884445315, 75.7962415334243] Fitness: 1.7301204369992884
Generation 394 Best Solution: [183.2646679940486, 177.36369403933963,
173.93178563824722, 75.041342186206] Fitness: 1.113166688773821
Generation 395 Best Solution: [183.2646679940486, 176.12700411752593,
174.47613884445315, 75.7962415334243] Fitness: 0.9065274380306838
Generation 396 Best Solution: [183.62831175641477, 175.9381153515721,
175.11981218011883, 75.7962415334243] Fitness: 1.686727913110257
Generation 397 Best Solution: [183.33412830957406, 175.9381153515721,
175.11981218011883, 75.7962415334243] Fitness: 1.567963688564107
Generation 398 Best Solution: [183.67576165051517, 177.36369403933963,
174.55295247807905, 75.041342186206] Fitness: 1.3792109060044913
Generation 399 Best Solution: [183.62831175641477, 177.36369403933963,
174.55295247807905, 75.041342186206] Fitness: 1.34870767587102
Generation 400 Best Solution: [183.67576165051517, 176.0516229062008,
175.11981218011883, 75.7962415334243] Fitness: 1.6871308444630206
Generation 401 Best Solution: [183.67576165051517, 176.0516229062008,
175.11981218011883, 75.18337421883781] Fitness: 1.057268090083597
Generation 402 Best Solution: [183.62831175641477, 177.36369403933963,
174.25857514233073, 75.7962415334243] Fitness: 1.931715890110421
Generation 403 Best Solution: [183.62831175641477, 176.52886623828732,
175.11981218011883, 75.7962415334243] Fitness: 1.7274230361401215
Generation 404 Best Solution: [183.67576165051517, 175.83396616046406,
175.11981218011883, 75.78504912694068] Fitness: 1.744501186356521
Generation 405 Best Solution: [183.62831175641477, 176.70511850380575,
175.11981218011883, 75.7962415334243] Fitness: 1.8341947761227204
Generation 406 Best Solution: [183.62831175641477, 177.36369403933963,
175.11981218011883, 74.85226250128488] Fitness: 1.8666480412118356
Generation 407 Best Solution: [183.62831175641477, 177.36369403933963,
175.11981218011883, 74.85226250128488] Fitness: 1.8666480412118356
Generation 408 Best Solution: [183.62831175641477, 177.36369403933963,
175.11981218011883, 74.85226250128488] Fitness: 1.8666480412118356
Generation 409 Best Solution: [183.00068044057835, 177.36369403933963,
174.10931041321234, 75.7962415334243] Fitness: 1.8114715432471011
Generation 410 Best Solution: [183.62831175641477, 176.27894227304145,
174.82995066620427, 75.7962415334243] Fitness: 1.3036638118070472
Generation 411 Best Solution: [183.62831175641477, 175.78691445138728,
175.11981218011883, 75.7962415334243] Fitness: 1.754840764279519
Generation 412 Best Solution: [183.00068044057835, 177.36369403933963,
175.11981218011883, 74.85226250128488] Fitness: 1.7597658563129193
Generation 413 Best Solution: [183.00068044057835, 175.76053773403544,
175.11981218011883, 75.7962415334243] Fitness: 1.6631194805126848
Generation 414 Best Solution: [183.62831175641477, 175.76053773403544,
175.11981218011883, 75.7962415334243] Fitness: 1.770001665411601
Generation 415 Best Solution: [183.62831175641477, 176.4255262728412,
175.11981218011883, 75.7962415334243] Fitness: 1.685045924839784
Generation 416 Best Solution: [183.62831175641477, 175.76053773403544,

175.11981218011883, 75.7962415334243] Fitness: 1.770001665411601
Generation 417 Best Solution: [183.62831175641477, 177.36369403933963,
174.6171017317782, 75.7962415334243] Fitness: 2.0912508332758577
Generation 418 Best Solution: [183.62831175641477, 175.99533787147695,
175.11981218011883, 75.7962415334243] Fitness: 1.6692989209867282
Generation 419 Best Solution: [183.62831175641477, 177.36369403933963,
174.6171017317782, 74.73858095073687] Fitness: 1.3335684518341921
Generation 420 Best Solution: [183.62831175641477, 177.36369403933963,
174.6171017317782, 74.73858095073687] Fitness: 1.3335684518341921
Generation 421 Best Solution: [183.62831175641477, 177.36369403933963,
174.6171017317782, 74.73858095073687] Fitness: 1.3335684518341921
Generation 422 Best Solution: [183.2859016931159, 177.36369403933963,
174.6171017317782, 74.73858095073687] Fitness: 1.20689404933915
Generation 423 Best Solution: [183.62831175641477, 176.91688739498508,
174.6171017317782, 74.73858095073687] Fitness: 0.7355593095841255
Generation 424 Best Solution: [183.85651718025363, 176.14452671392405,
175.11981218011883, 75.7962415334243] Fitness: 1.820991311890039
Generation 425 Best Solution: [183.85651718025363, 176.14452671392405,
174.6171017317782, 75.7962415334243] Fitness: 1.2942313272460888
Generation 426 Best Solution: [183.62831175641477, 176.14452671392405,
174.6171017317782, 75.7962415334243] Fitness: 1.1186545799081484
Generation 427 Best Solution: [183.62831175641477, 176.14452671392405,
174.6171017317782, 75.7962415334243] Fitness: 1.1186545799081484
Generation 428 Best Solution: [183.62831175641477, 177.36369403933963,
175.11981218011883, 74.42921599298974] Fitness: 1.9347437452673577
Generation 429 Best Solution: [183.62831175641477, 176.06626819466751,
175.11981218011883, 75.7962415334243] Fitness: 1.6540577143310502
Generation 430 Best Solution: [183.62831175641477, 176.63360688085092,
175.11981218011883, 74.7483060769485] Fitness: 1.0277812272128304
Generation 431 Best Solution: [183.62831175641477, 176.06626819466751,
175.11981218011883, 75.7962415334243] Fitness: 1.6540577143310502
Generation 432 Best Solution: [183.62831175641477, 176.94382204033477,
175.11981218011883, 75.7962415334243] Fitness: 2.048134821286928
Generation 433 Best Solution: [183.85651718025363, 176.23243756106456,
175.11981218011883, 75.7962415334243] Fitness: 1.8215077874569565
Generation 434 Best Solution: [183.62831175641477, 177.36369403933963,
175.11981218011883, 74.39070223084903] Fitness: 1.9533866107987967
Generation 435 Best Solution: [183.85651718025363, 176.46886505791915,
175.11981218011883, 75.7962415334243] Fitness: 1.8765745491891908
Generation 436 Best Solution: [183.62831175641477, 177.36369403933963,
175.11981218011883, 74.39070223084903] Fitness: 1.9533866107987967
Generation 437 Best Solution: [183.62831175641477, 177.36369403933963,
174.19302860596198, 74.39070223084903] Fitness: 1.2573827580411485
Generation 438 Best Solution: [183.62831175641477, 177.36369403933963,
175.11981218011883, 74.61794577079412] Fitness: 1.8734090652532431
Generation 439 Best Solution: [183.62831175641477, 176.62107197224276,
175.11981218011883, 75.7962415334243] Fitness: 1.7778555844733486
Generation 440 Best Solution: [183.62831175641477, 175.70705152335344,

173.78955322816796, 75.7962415334243] Fitness: 1.180434222421793
Generation 441 Best Solution: [183.62831175641477, 176.62107197224276,
175.11981218011883, 75.7962415334243] Fitness: 1.7778555844733486
Generation 442 Best Solution: [183.62831175641477, 176.62107197224276,
173.78955322816796, 75.7962415334243] Fitness: 1.1545550559235263
Generation 443 Best Solution: [183.62831175641477, 176.62107197224276,
173.78955322816796, 75.7962415334243] Fitness: 1.1545550559235263
Generation 444 Best Solution: [182.81276185955548, 177.36369403933963,
174.3478698974096, 75.7962415334243] Fitness: 1.9230172169055315
Generation 445 Best Solution: [182.81276185955548, 177.36369403933963,
174.3478698974096, 75.7962415334243] Fitness: 1.9230172169055315
Generation 446 Best Solution: [183.62831175641477, 177.36369403933963,
175.11981218011883, 75.12195785338773] Fitness: 1.9540169962644496
Generation 447 Best Solution: [182.81276185955548, 177.36369403933963,
175.11981218011883, 75.12195785338773] Fitness: 1.922413189008126
Generation 448 Best Solution: [183.65671427911042, 177.36369403933963,
174.23174697756536, 75.7962415334243] Fitness: 1.944894817625654
Generation 449 Best Solution: [183.3122555105238, 177.36369403933963,
174.3578156690403, 75.7962415334243] Fitness: 1.8351078205817761
Generation 450 Best Solution: [183.65671427911042, 177.36369403933963,
174.25504314218605, 75.50278029413748] Fitness: 1.581706769305275
Generation 451 Best Solution: [183.62831175641477, 177.36369403933963,
174.25504314218605, 75.50278029413748] Fitness: 1.5638268636361512
Generation 452 Best Solution: [182.81276185955548, 176.73561280081282,
175.11981218011883, 75.7962415334243] Fitness: 1.8254773020558654
Generation 453 Best Solution: [183.3122555105238, 177.36369403933963,
174.11403261037705, 75.7962415334243] Fitness: 1.7955433426902578
Generation 454 Best Solution: [182.81276185955548, 176.73561280081282,
175.11981218011883, 75.7962415334243] Fitness: 1.8254773020558654
Generation 455 Best Solution: [183.62831175641477, 177.36369403933963,
175.11981218011883, 75.10316002987474] Fitness: 1.9446259322555801
Generation 456 Best Solution: [182.81276185955548, 177.36369403933963,
174.25504314218605, 75.7962415334243] Fitness: 1.899435588979748
Generation 457 Best Solution: [183.65671427911042, 177.36369403933963,
175.11981218011883, 74.89145784716385] Fitness: 1.8909012764625612
Generation 458 Best Solution: [182.81276185955548, 177.36369403933963,
174.15698177803367, 75.7962415334243] Fitness: 1.8876272901382825
Generation 459 Best Solution: [182.81276185955548, 175.7044354497293,
175.11981218011883, 75.7962415334243] Fitness: 1.7738836060953558
Generation 460 Best Solution: [182.81276185955548, 175.7044354497293,
175.11981218011883, 75.7962415334243] Fitness: 1.7738836060953558
Generation 461 Best Solution: [183.5383120556373, 176.27327559439064,
175.11981218011883, 75.7962415334243] Fitness: 1.6006543224147554
Generation 462 Best Solution: [183.64591886834515, 177.36369403933963,
174.00688891504925, 75.7962415334243] Fitness: 1.93818056798063
Generation 463 Best Solution: [183.62831175641477, 177.39381764158404,
175.11981218011883, 74.89641613700473] Fitness: 1.9243552984937704
Generation 464 Best Solution: [182.81276185955548, 175.9448462269328,

175.11981218011883, 75.7962415334243] Fitness: 1.6528361002916196
Generation 465 Best Solution: [182.81276185955548, 175.9448462269328,
175.11981218011883, 75.50054459226952] Fitness: 1.2832887629269591
Generation 466 Best Solution: [183.32326057763342, 177.0947617054179,
175.11981218011883, 74.43640238346258] Fitness: 1.4172378222927875
Generation 467 Best Solution: [182.81276185955548, 176.2626038964067,
175.11981218011883, 75.50054459226952] Fitness: 1.2474504916097966
Generation 468 Best Solution: [183.62831175641477, 177.36369403933963,
174.0623568803976, 75.7962415334243] Fitness: 1.9205996515262815
Generation 469 Best Solution: [183.0993886760773, 176.1170557590921,
175.11981218011883, 75.7962415334243] Fitness: 1.5208528703574835
Generation 470 Best Solution: [183.0993886760773, 176.1170557590921,
175.11981218011883, 74.86977648369064] Fitness: 0.7720721326036432
Generation 471 Best Solution: [183.43602003503398, 177.39381764158404,
174.09790087502753, 75.7962415334243] Fitness: 1.8776452713146896
Generation 472 Best Solution: [183.0993886760773, 177.36369403933963,
175.11981218011883, 74.85870185302079] Fitness: 1.740928401644208
Generation 473 Best Solution: [183.0993886760773, 177.36369403933963,
173.990199125216, 74.29360872775042] Fitness: 1.1950396871276805
Generation 474 Best Solution: [183.34059168791418, 176.6472555023216,
175.11981218011883, 75.7962415334243] Fitness: 1.676890079853267
Generation 475 Best Solution: [183.62831175641477, 176.06075977047766,
175.11981218011883, 75.7962415334243] Fitness: 1.6549890810308372
Generation 476 Best Solution: [183.62831175641477, 176.06075977047766,
175.11981218011883, 75.7962415334243] Fitness: 1.6549890810308372
Generation 477 Best Solution: [183.62831175641477, 176.06075977047766,
175.11981218011883, 75.7962415334243] Fitness: 1.6549890810308372
Generation 478 Best Solution: [183.62831175641477, 176.06075977047766,
175.11981218011883, 75.00152602879216] Fitness: 0.939397602896446
Generation 479 Best Solution: [183.34317396539174, 176.90280946979263,
175.11981218011883, 75.14766797653365] Fitness: 1.2384351093664556
Generation 480 Best Solution: [183.34317396539174, 176.53494142960787,
175.11981218011883, 75.7962415334243] Fitness: 1.6134617200247185
Generation 481 Best Solution: [183.34317396539174, 176.53494142960787,
175.11981218011883, 74.78021814960618] Fitness: 0.8559941603077825
Generation 482 Best Solution: [183.0993886760773, 176.813375130058,
175.01031595728082, 75.7962415334243] Fitness: 1.649843084462886
Generation 483 Best Solution: [183.34317396539174, 177.36369403933963,
173.99538448809975, 75.14766797653365] Fitness: 1.1618776775561064
Generation 484 Best Solution: [183.62831175641477, 177.39381764158404,
174.69395844077354, 75.14766797653365] Fitness: 1.5488995052018903
Generation 485 Best Solution: [183.62831175641477, 177.39381764158404,
174.69395844077354, 75.14766797653365] Fitness: 1.5488995052018903
Generation 486 Best Solution: [183.62831175641477, 177.39381764158404,
174.69395844077354, 75.05150202633757] Fitness: 1.5026033894311628
Generation 487 Best Solution: [183.34317396539174, 177.39381764158404,
174.4425094666607, 75.7962415334243] Fitness: 1.924538929151114
Generation 488 Best Solution: [183.62831175641477, 177.39381764158404,

```
175.11981218011883, 74.88770652074736] Fitness: 1.9226927559004965
Generation 489 Best Solution: [183.62831175641477, 176.1870541471008,
175.11981218011883, 75.7962415334243] Fitness: 1.6443133848294433
Generation 490 Best Solution: [183.69218849314043, 176.1870541471008,
175.11981218011883, 75.7962415334243] Fitness: 1.6861111386644207
Generation 491 Best Solution: [183.62831175641477, 175.89165736267205,
175.11981218011883, 74.37795174457722] Fitness: 1.0462552753745142
Generation 492 Best Solution: [183.62831175641477, 175.89165736267205,
175.11981218011883, 74.94370746889733] Fitness: 0.9711011049803874
Generation 493 Best Solution: [183.76861450622184, 176.04309378482648,
175.11981218011883, 75.7962415334243] Fitness: 1.7575757072363904
Generation 494 Best Solution: [183.34317396539174, 177.36369403933963,
174.65633826898966, 75.7962415334243] Fitness: 2.002716956875797
Generation 495 Best Solution: [183.8738185512436, 177.39381764158404,
174.65633826898966, 74.37795174457722] Fitness: 1.7038760108229445
Generation 496 Best Solution: [183.34317396539174, 177.39381764158404,
175.11981218011883, 75.2148642390454] Fitness: 1.9411525652427415
Generation 497 Best Solution: [183.34317396539174, 177.36369403933963,
174.65633826898966, 75.7962415334243] Fitness: 2.002716956875797
Generation 498 Best Solution: [183.34317396539174, 177.39381764158404,
174.65633826898966, 75.2148642390454] Fitness: 1.4427766055309426
Generation 499 Best Solution: [183.41398966573848, 177.39381764158404,
174.25805967467406, 75.7962415334243] Fitness: 1.8834857234081248
Generation 500 Best Solution: [183.8738185512436, 176.1068929850371,
175.11981218011883, 75.32713244880328] Fitness: 1.3110374022405116
Optimal Solution for Simulation 32: [183.8738185512436, 176.1068929850371,
175.11981218011883, 75.32713244880328]
------------------------------------------------

Final DataFrame of All 32 Simulations:
    drought  flood  policy_change  social_event  economic_crisis  \
0      True   True           True          True             True
1      True   True           True          True            False
2      True   True           True         False             True
3      True   True           True         False            False
4      True   True          False          True             True
5      True   True          False          True            False
6      True   True          False         False             True
7      True   True          False         False            False
8      True  False           True          True             True
9      True  False           True          True            False
10     True  False           True         False             True
11     True  False           True         False            False
12     True  False          False          True             True
13     True  False          False          True            False
14     True  False          False         False             True
15     True  False          False         False            False
16    False   True           True          True             True
```

17	False	True	True	True	False
18	False	True	True	False	True
19	False	True	True	False	False
20	False	True	False	True	True
21	False	True	False	True	False
22	False	True	False	False	True
23	False	True	False	False	False
24	False	False	True	True	True
25	False	False	True	True	False
26	False	False	True	False	True
27	False	False	True	False	False
28	False	False	False	True	True
29	False	False	False	True	False
30	False	False	False	False	True
31	False	False	False	False	False

	Lake Superior	Lake Michigan-Huron	Lake Erie	Lake Ontario
0	183.589041	175.979960	175.072284	75.335307
1	183.767943	177.388178	174.319516	75.111143
2	183.693855	176.697314	175.131296	75.757535
3	183.819569	176.445171	174.059581	75.897240
4	183.414331	176.616354	175.070500	74.948618
5	183.835617	176.005050	175.099992	75.869699
6	183.022424	176.029028	175.013552	75.839289
7	183.194803	177.289793	174.574449	75.884322
8	182.870787	177.425341	175.085740	74.300540
9	182.803601	177.449332	174.303498	74.570356
10	183.668494	177.315498	175.138861	74.822867
11	183.790218	177.449745	174.925056	75.522086
12	183.114664	177.397367	174.028432	74.290945
13	183.589965	176.882503	174.789950	75.895864
14	183.203665	177.373113	175.069008	74.426841
15	183.638326	177.424920	175.101085	74.890486
16	183.856336	177.364360	175.058571	74.895702
17	182.847904	176.126026	175.118881	75.769519
18	182.922632	177.149393	174.505414	75.773323
19	183.843713	177.446049	174.212259	75.299221
20	182.814036	177.418917	175.069042	74.733293
21	183.262169	176.273215	175.127369	75.600671
22	183.620532	177.442507	174.552349	75.173074
23	183.773186	177.419738	174.210769	74.655939
24	183.855276	177.055576	175.030716	75.084738
25	183.612194	176.160432	175.105181	74.541435
26	183.838583	177.401060	174.014297	75.007388
27	183.810798	177.295741	174.106198	75.028637
28	183.872967	177.123612	174.721785	74.962664
29	182.819519	175.802321	175.012583	75.876326
30	183.815383	175.778443	174.983212	75.889986

31 183.873819 176.106893 175.119812 75.327132

